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MCAT[®] General Chemistry Review

2019-2020

Edited by Alexander Stone Macnow, MD



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Published by Kaplan Publishing, a division of Kaplan, Inc.
750 Third Avenue
New York, NY 10017

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ISBN: 978-1-5062-3543-1
10 9 8 7 6 5 4 3 2 1

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About Scientific American

As the world's premier science and technology magazine, and the oldest continuously published magazine in the United States, Scientific American is committed to bringing the most important developments in modern science, medicine, and technology to our worldwide audience in an understandable, credible, and provocative format.

Founded in 1845 and on the “cutting edge” ever since, Scientific American boasts over 150 Nobel laureate authors including Albert Einstein, Francis Crick, Stanley Prusiner, and Richard Axel. Scientific American is a forum where scientific theories and discoveries are explained to a broader audience.

Scientific American published its first foreign edition in 1890, and in 1979 was the first Western magazine published in the People's Republic of China. Today, Scientific American is published in 14 foreign language editions. Scientific American is also a leading online destination (www.ScientificAmerican.com), providing the latest science news and exclusive features to millions of visitors each month.

The knowledge that fills our pages has the power to spark new ideas, paradigms and visions for the future. As science races forward, Scientific American continues to cover the promising strides, inevitable setbacks and challenges, and new medical discoveries as they unfold.

Preface

And now it starts: your long, yet fruitful journey toward wearing a white coat. Proudly wearing that white coat, though, is hopefully only part of your motivation. You are reading this book because you want to be a healer.

If you're serious about going to medical school, then you are likely already familiar with the importance of the MCAT in medical school admissions. While the holistic review process puts additional weight on your experiences, extracurricular activities, and personal attributes, the fact remains: along with your GPA, your MCAT score remains one of the two most important components of your application portfolio—at least early in the admissions process. Each additional point you score on the MCAT pushes you in front of thousands of other students and makes you an even more attractive applicant. But the MCAT is not simply an obstacle to overcome; it is an opportunity to show schools that you will be a strong student and a future leader in medicine.

We at Kaplan take our jobs very seriously and aim to help students see success not only on the MCAT, but as future physicians. We work with our Learning Science experts to ensure that we're using the most up-to-date teaching techniques in our resources. Multiple members of our team hold advanced degrees in medicine or associated biomedical sciences, and are committed to the highest level of medical education. Kaplan has been working with the MCAT for over 50 years and our commitment to premed

students is unflagging; in fact, Stanley Kaplan created this company when he had difficulty being accepted to medical school due to unfair quota systems that existed at the time.

We stand now at the beginning of a new era in medical education. As citizens of this 21st-century world of healthcare, we are charged with creating a patient-oriented, culturally competent, cost-conscious, universally available, technically advanced, and research-focused healthcare system, run by compassionate providers. Suffice it to say, this is no easy task. Problem-based learning, integrated curricula, and classes in interpersonal skills are some of the responses to this demand for an excellent workforce—a workforce of which you'll soon be a part.

We're thrilled that you've chosen us to help you on this journey. Please reach out to us to share your challenges, concerns, and successes. Together, we will shape the future of medicine in the United States and abroad; we look forward to helping you become the doctor you deserve to be.

Good luck!

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About the MCAT

ANATOMY OF THE MCAT

Here is a general overview of the structure of Test Day:

Section	Number of Questions	Time Allotted
Test-Day Certification		4 minutes
Tutorial (optional)		10 minutes
Chemical and Physical Foundations of Biological Systems	59	95 minutes
Break (optional)		10 minutes
Critical Analysis and Reasoning Skills (CARS)	53	90 minutes
Lunch Break (optional)		30 minutes
Biological and Biochemical Foundations of Living Systems	59	95 minutes
Break (optional)		10 minutes
Psychological, Social, and Biological Foundations of Behavior	59	95 minutes
Void Question		3 minutes
Satisfaction Survey (optional)		5 minutes

The structure of the four sections of the MCAT is shown below.

Chemical and Physical Foundations of Biological Systems	
Time	95 minutes
Format	• 59 questions • 10 passages • 44 questions are passage-based, and 15 are discrete (stand-alone) questions. • Score between 118 and 132
What It Tests	• Biochemistry: 25% • Biology: 5% • General Chemistry: 30% • Organic Chemistry: 15% • Physics: 25%
Critical Analysis and Reasoning Skills (CARS)	
Time	90 minutes
Format	• 53 questions • 9 passages • All questions are passage-based. There are no discrete (stand-alone) questions. • Score between 118 and 132
What It Tests	Disciplines: • Humanities: 50% • Social Sciences: 50% Skills: • Foundations of Comprehension: 30% • Reasoning Within the Text: 30% • Reasoning Beyond the Text: 40%

Chemical and Physical Foundations of Biological Systems

Biological and Biochemical Foundations of Living Systems

Time 95 minutes

Format

- 59 questions
- 10 passages
- 44 questions are passage-based, and 15 are discrete (stand-alone) questions.
- Score between 118 and 132

What It Tests

- Biochemistry: 25%
- Biology: 65%
- General Chemistry: 5%
- Organic Chemistry: 5%

Psychological, Social, and Biological Foundations of Behavior

Time 95 minutes

Format

- 59 questions
- 10 passages
- 44 questions are passage-based, and 15 are discrete (stand-alone) questions.
- Score between 118 and 132

What It Tests

- Biology: 5%
- Psychology: 65%
- Sociology: 30%

Total

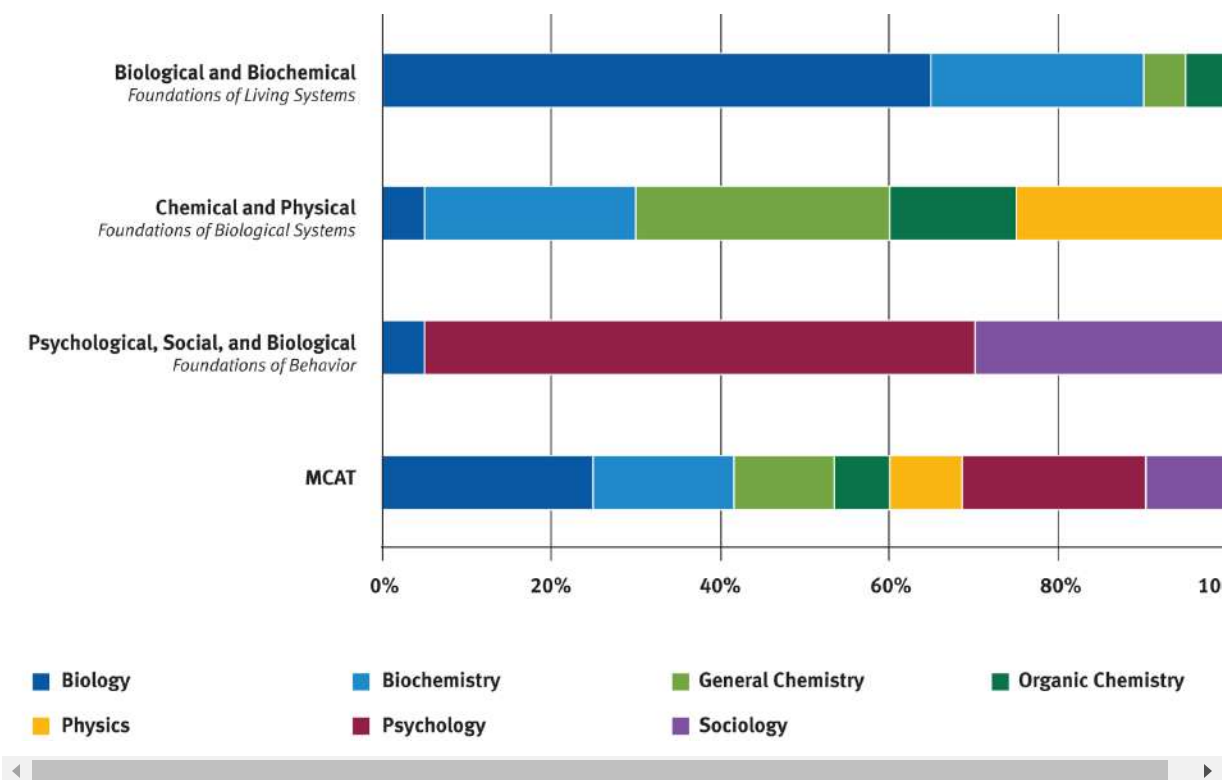
Testing Time 375 minutes (6 hours, 15 minutes)

Total Seat Time 447 minutes (7 hours, 27 minutes)

Questions 230

Chemical and Physical Foundations of Biological Systems

Score 472 to 528



SCIENTIFIC INQUIRY AND REASONING SKILLS (SIRS)

The AAMC has defined four Scientific Inquiry and Reasoning Skills (SIRS) that will be tested in the three science sections of the MCAT:

Knowledge of Scientific Concepts and Principles (35% of questions)

Scientific Reasoning and Problem-Solving (45% of questions)

Reasoning About the Design and Execution of Research (10% of questions)

Data-Based and Statistical Reasoning (10% of questions)

Let's see how each one breaks down into more specific Test Day behaviors. Note that the bullet points of specific objectives for each of the SIRS are taken directly from the Official Guide to the MCAT Exam; the descriptions of what these behaviors mean and sample question stems, however, are written by Kaplan.

Skill 1: Knowledge of Scientific Concepts and Principles

This is probably the least surprising of the four SIRS; the testing of science knowledge is, after all, one of the signature qualities of the MCAT. Skill 1 questions will require you to do the following:

- Recognize correct scientific principles
- Identify the relationships among closely related concepts
- Identify the relationships between different representations of concepts (verbal, symbolic, graphic)
- Identify examples of observations that illustrate scientific principles
- Use mathematical equations to solve problems

At Kaplan, we simply call these Science Knowledge or Skill 1 questions. Another way to think of Skill 1 questions is as “one-step” problems. The single step is either to realize which scientific concept the question stem is suggesting or to take the concept stated in the question stem and identify which answer choice is an accurate application of it. Skill 1 questions are particularly prominent among discrete questions (those not associated with a passage). These questions are an opportunity to gain quick points on Test Day—if you know the science concept attached to the question, then that's it! On Test Day, 35% of the questions in each science section will be Skill 1 questions.

Here are some sample Skill 1 question stems:

- How would a proponent of the James–Lange theory of emotion interpret the findings of the study cited the passage?
- Which of the following most accurately describes the function of FSH in the human female menstrual cycle?
- If the products of Reaction 1 and Reaction 2 were combined in solution, the resulting reaction would form:
- Ionic bonds are maintained by which of the following forces?

Skill 2: Scientific Reasoning and Problem-Solving

The MCAT science sections do, of course, move beyond testing straightforward science knowledge; Skill 2 questions are the most common way in which it does so. At Kaplan, we also call these Critical Thinking questions. Skill 2 questions will require you to do the following:

- Reason about scientific principles, theories, and models
- Analyze and evaluate scientific explanations and predictions
- Evaluate arguments about causes and consequences
- Bring together theory, observations, and evidence to draw conclusions
- Recognize scientific findings that challenge or invalidate a scientific theory or model
- Determine and use scientific formulas to solve problems

Just as Skill 1 questions can be thought of as “one-step” problems, many Skill 2 questions are “two-step” problems, and more difficult Skill 2 questions may require three or more steps. These questions can require a wide spectrum of reasoning skills, including integration of multiple facts

from a passage, combination of multiple science content areas, and prediction of an experiment's results. Skill 2 questions also tend to ask about science content without actually mentioning it by name. For example, a question might describe the results of one experiment and ask you to predict the results of a second experiment without actually telling you what underlying scientific principles are at work—part of the question's difficulty will be figuring out which principles to apply in order to get the correct answer. On Test Day, 45% of the questions in each science section will be Skill 2 questions.

Here are some sample Skill 2 question stems:

- Which of the following experimental conditions would most likely yield results similar to those in Figure 2?
- All of the following conclusions are supported by the information in the passage EXCEPT:
- The most likely cause of the anomalous results found by the experimenter is:
- An impact to a man's chest quickly reduces the volume of one of his lungs to 70% of its initial value while not allowing any air to escape from the man's mouth. By what percentage is the force of outward air pressure increased on a 2 cm^2 portion of the inner surface of the compressed lung?

Skill 3: Reasoning About the Design and Execution of Research

The MCAT is interested in your ability to critically appraise and analyze research, as this is an important day-to-day task of a physician. We call

these questions Skill 3 or Experimental and Research Design questions for short. Skill 3 questions will require you to do the following:

- Identify the role of theory, past findings, and observations in scientific questioning
- Identify testable research questions and hypotheses
- Distinguish between samples and populations and distinguish results that support generalizations about populations
- Identify independent and dependent variables
- Reason about the features of research studies that suggest associations between variables or causal relationships between them (such as temporality and random assignment)
- Identify conclusions that are supported by research results
- Determine the implications of results for real-world situations
- Reason about ethical issues in scientific research

Over the years, the AAMC has received input from medical schools to require more practical research skills of MCAT test-takers, and Skill 3 questions are the response to these demands. This skill is unique in that the outside knowledge you need to answer Skill 3 questions is not taught in any one undergraduate course; instead, the research design principles needed to answer these questions are learned gradually throughout your science classes and especially through any laboratory work you have completed. It should be noted that Skill 3 comprises 10% of the questions in each science section on Test Day.

Here are some sample Skill 3 question stems:

- What is the dependent variable in the study described in the passage?
- The major flaw in the method used to measure disease susceptibility in Experiment 1 is:
- Which of the following procedures is most important for the experimenters to follow in order for their study to maintain a proper, randomized sample of research subjects?
- A researcher would like to test the hypothesis that individuals who move to an urban area during adulthood are more likely to own a car than are those who have lived in an urban area since birth. Which of the following studies would best test this hypothesis?

Skill 4: Data-Based and Statistical Reasoning

Lastly, the science sections of the MCAT test your ability to analyze the visual and numerical results of experiments and studies. We call these Data and Statistical Analysis questions. Skill 4 questions will require you to do the following:

- Use, analyze, and interpret data in figures, graphs, and tables
- Evaluate whether representations make sense for particular scientific observations and data
- Use measures of central tendency (mean, median, and mode) and measures of dispersion (range, interquartile range, and standard deviation) to describe data
- Reason about random and systematic error
- Reason about statistical significance and uncertainty (interpreting statistical significance levels and interpreting a confidence interval)
- Use data to explain relationships between variables or make predictions

- Use data to answer research questions and draw conclusions

Skill 4 is included in the MCAT because physicians and researchers spend much of their time examining the results of their own studies and the studies of others, and it's very important for them to make legitimate conclusions and sound judgments based on that data. The MCAT tests Skill 4 on all three science sections with graphical representations of data (charts and bar graphs) as well as numerical ones (tables, lists, and results summarized in sentence or paragraph form). On Test Day, 10% of the questions in each science section will be Skill 4 questions.

Here are some sample Skill 4 question stems:

- According to the information in the passage, there is an inverse correlation between:
- What conclusion is best supported by the findings displayed in Figure 2?
- A medical test for a rare type of heavy metal poisoning returns a positive result for 98% of affected individuals and 13% of unaffected individuals. Which of the following types of error is most prevalent in this test?
- If a fourth trial of Experiment 1 was run and yielded a result of 54% compliance, which of the following would be true?

SIRS Summary

Discussing the SIRS tested on the MCAT is a daunting prospect given that the very nature of the skills tends to make the conversation rather abstract. Nevertheless, with enough practice, you'll be able to identify

each of the four skills quickly, and you'll also be able to apply the proper strategies to solve those problems on Test Day. If you need a quick reference to remind you of the four SIRS, these guidelines may help:

Skill 1 (Science Knowledge) questions ask:

- Do you remember this science content?

Skill 2 (Critical Thinking) questions ask:

- Do you remember this science content? And if you do, could you please apply it to this novel situation?
- Could you answer this question that cleverly combines multiple content areas at the same time?

Skill 3 (Experimental and Research Design) questions ask:

- Let's forget about the science content for a while. Could you give some insight into the experimental or research methods involved in this situation?

Skill 4 (Data and Statistical Analysis) questions ask:

- Let's forget about the science content for a while. Could you accurately read some graphs and tables for a moment? Could you make some conclusions or extrapolations based on the information presented?

CRITICAL ANALYSIS AND REASONING SKILLS (CARS)

The Critical Analysis and Reasoning Skills (CARS) section of the MCAT tests three discrete families of textual reasoning skills; each of these families requires a higher level of reasoning than the last. Those three skills are as follows:

Foundations of Comprehension (30% of questions)

Reasoning Within the Text (30% of questions)

Reasoning Beyond the Text (40% of questions)

These three skills are tested through nine humanities- and social sciences-themed passages, with approximately 5 to 7 questions per passage. Let's take a more in-depth look into these three skills. Again, the bullet points of specific objectives for each of the CARS are taken directly from the Official Guide to the MCAT Exam; the descriptions of what these behaviors mean and sample question stems, however, are written by Kaplan.

Foundations of Comprehension

Questions in this skill will ask for basic facts and simple inferences about the passage; the questions themselves will be similar to those seen on reading comprehension sections of other standardized exams like the SAT® and ACT®. Foundations of Comprehension questions will require you to do the following:

- Understand the basic components of the text
- Infer meaning from rhetorical devices, word choice, and text structure

This admittedly covers a wide range of potential question types including Main Idea, Detail, Function, and Definition-in-Context questions, but finding the correct answer to all Foundations of Comprehension questions will follow from a basic understanding of the passage and the point of view of its author (and occasionally that of other voices in the passage).

Here are some sample Foundations of Comprehension question stems:

- Main Idea—The author's primary purpose in this passage is:
- Detail—Based on the information in the second paragraph, which of the following is the most accurate summary of the opinion held by Schubert's critics?
- (Scattered) Detail—According to the passage, which of the following is FALSE about literary reviews in the 1920s?
- Function—The author's discussion of the effect of socioeconomic status on social mobility primarily serves which of the following functions?
- Definition-in-Context—The word “obscure” (paragraph 3), when used in reference to the historian's actions, most nearly means:

Reasoning Within the Text

While Foundations of Comprehension questions will usually depend on interpreting a single piece of information in the passage or understanding the passage as a whole, Reasoning Within the Text questions will typically require you to infer unstated parts of arguments or bring together two disparate pieces of the passage. Reasoning Within the Text questions will require you to:

- Integrate different components of the text to increase comprehension

In other words, questions in this skill often ask either How do these two details relate to one another? or What else must be true that the author didn't say? The CARS section will also ask you to judge certain parts of the passage or even judge the author. These questions, which fall under the Reasoning Within the Text skill, can ask you to identify authorial bias, evaluate the credibility of cited sources, determine the logical soundness of an argument, or search for relevant evidence in the passage to support a given conclusion. In all, this category includes Inference and Strengthen–Weaken (Within the Passage) questions, as well as a smattering of related—but rare—question types.

Here are some sample Reasoning Within the Text question stems:

- Inference (Implication)—Which of the following phrases, as used in the passage, is most suggestive that the author has a personal bias toward narrative records of history?
- Inference (Assumption)—In putting together her argument in the passage, the author most likely assumes:
- Strengthen–Weaken (Within the Passage)—Which of the following facts is used in the passage as the most prominent piece of evidence in favor of the author's conclusions?
- Strengthen–Weaken (Within the Passage)—Based on the role it plays in the author's argument, The Possessed can be considered:

Reasoning Beyond the Text

The distinguishing factor of Reasoning Beyond the Text questions is in the title of the skill: the word *Beyond*. Questions that test this skill, which make up a larger share of the CARS section than questions from either of the other two skills, will always introduce a completely new situation that was not present in the passage itself; these questions will ask you to determine how one influences the other. Reasoning Beyond the Text questions will require you to:

- Apply or extrapolate ideas from the passage to new contexts
- Assess the impact of introducing new factors, information, or conditions to ideas from the passage

The Reasoning Beyond the Text skill is further divided into Apply and Strengthen–Weaken (Beyond the Passage) questions, and a few other rarely appearing question types.

Here are some sample Reasoning Beyond the Text question stems:

- Apply—If a document were located that demonstrated Berlioz intended to include a chorus of at least 700 in his *Grande Messe des Morts*, how would the author likely respond?
- Apply—Which of the following is the best example of a “virtuous rebellion,” as it is defined in the passage?
- Strengthen–Weaken (Beyond the Text)—Suppose Jane Austen had written in a letter to her sister, “My strongest characters were those forced by circumstance to confront basic questions about the society in which they lived.” What relevance would this have to the passage?

- Strengthen–Weaken (Beyond the Text)—Which of the following sentences, if added to the end of the passage, would most WEAKEN the author's conclusions in the last paragraph?

CARS Summary

Through the Foundations of Comprehension skill, the CARS section tests many of the reading skills you have been building on since grade school, albeit in the context of very challenging doctorate-level passages. But through the two other skills (Reasoning Within the Text and Reasoning Beyond the Text), the MCAT demands that you understand the deep structure of passages and the arguments within them at a very advanced level. And, of course, all of this is tested under very tight timing restrictions: only 102 seconds per question—and that doesn't even include the time spent reading the passages.

Here's a quick reference guide to the three CARS skills:

Foundations of Comprehension questions ask:

- Did you understand the passage and its main ideas?
- What does the passage have to say about this particular detail?

Reasoning Within the Text questions ask:

- What must be true that the author did not say?
- What's the logical relationship between these two ideas from the passage?
- How well argued is the author's thesis?

Reasoning Beyond the Text questions ask:

- How does this principle from the passage apply to this new situation?
- How does this new piece of information influence the arguments in the passage?

Scoring

Each of the four sections of the MCAT is scored between 118 and 132, with the median at 125. This means the total score ranges from 472 to 528, with the median at 500. Why such peculiar numbers? The AAMC stresses that this scale emphasizes the importance of the central portion of the score distribution, where most students score (around 125 per section, or 500 total), rather than putting undue focus on the high end of the scale.

Note that there is no wrong answer penalty on the MCAT, so you should select an answer for every question—even if it is only a guess.

The AAMC has released the 2017–2018 correlation between scaled score and percentile, as shown on the following page. It should be noted that the percentile scale is adjusted and renormalized over time and thus can shift slightly from year to year.

Total Score	Percentile	Total Score	Percentile
528	>99	499	47
527	>99	498	43
526	>99	497	40

Total Score	Percentile	Total Score	Percentile
525	>99	496	37
524	>99	495	33
523	>99	494	30
522	99	493	27
521	99	492	24
520	98	491	22
519	97	490	19
518	97	489	17
517	95	488	15
516	94	487	12
515	93	486	11
514	91	485	9
513	89	484	7
512	87	483	6
511	85	482	5
510	82	481	4
509	80	480	3
508	77	479	2
507	74	478	2
506	71	477	1
505	67	476	1

Total Score	Percentile	Total Score	Percentile
504	64	475	<1
503	61	474	<1
502	57	473	<1
501	54	472	<1
500	50		

Source: AAMC. 2018. *Summary of MCAT Total and Section Scores*. Accessed January 2018. <https://students-residents.aamc.org/advisors/article/percentile-ranks-for-the-mcat-exam/>.

Further information on score reporting is included at the end of the next section (see After Your Test).

MCAT POLICIES AND PROCEDURES

We strongly encourage you to download the latest copy of MCAT[®] Essentials, available on the AAMC's website, to ensure that you have the latest information about registration and Test Day policies and procedures; this document is updated annually. A brief summary of some of the most important rules is provided here.

MCAT Registration

The only way to register for the MCAT is online. You can access AAMC's registration system at: www.aamc.org/mcat .

You will be able to access the site approximately six months before Test Day. The AAMC designates three registration “Zones”—Gold, Silver, and Bronze. Registering during the Gold Zone (from the opening of registration until approximately one month before Test Day) provides the most flexibility and lowest test fees. The Silver Zone runs until approximately two to three weeks before Test Day and has less flexibility and higher fees; the Bronze Zone runs until approximately one to two weeks before Test Day and has the least flexibility and highest fees.

Fees and the Fee Assistance Program (FAP)

Payment for test registration must be made by MasterCard or VISA. As described earlier, the fees for registering for the MCAT—as well as rescheduling the exam or changing your testing center—increase as one approaches Test Day. In addition, it is not uncommon for test centers to fill up well in advance of the registration deadline. For these reasons, we recommend identifying your preferred Test Day as soon as possible and registering. There are ancillary benefits to having a set Test Day, as well: when you know the date you're working toward, you'll study harder and are less likely to keep pushing back the exam. The AAMC offers a Fee Assistance Program (FAP) for students with financial hardship to help reduce the cost of taking the MCAT, as well as for the American Medical College Application Service (AMCAS®) application. Further information on the FAP can be found at: www.aamc.org/students/applying/fap .

Testing Security

On Test Day, you will be required to present a qualifying form of ID. Generally, a current driver's license or United States passport will be

sufficient (consult the AAMC website for the full list of qualifying criteria). When registering, take care to spell your first and last names (middle names, suffixes, and prefixes are not required and will not be verified on Test Day) precisely the same as they appear on this ID; failure to provide this ID at the test center or differences in spelling between your registration and ID will be considered a “no-show,” and you will not receive a refund for the exam.

During Test Day registration other identity data collected may include: a digital palm vein scan, a Test Day photo, a digitization of your valid ID, and signatures. Some testing centers may use a metal detection wand to ensure that no prohibited items are brought into the testing room. Prohibited items include all electronic devices, including watches and timers, calculators, cell phones, and any and all forms of recording equipment; food, drinks (including water), and cigarettes or other smoking paraphernalia; hats and scarves (except for religious purposes); and books, notes, or other study materials. If you require a medical device, such as an insulin pump or pacemaker, you must apply for accommodated testing. During breaks, you are allowed to access food and drink, but not electronic devices, including cell phones.

Testing centers are under video surveillance and the AAMC does not take potential violations of testing security lightly. The bottom line: know the rules and don't break them.

Accommodations

Students with disabilities or medical conditions can apply for accommodated testing. Documentation of the disability or condition is

required, and requests may take two months—or more—to be approved. For this reason, it is recommended that you begin the process of applying for accommodated testing as early as possible. More information on applying for accommodated testing can be found at: www.aamc.org/students/applying/mcat/accommodations .

After Your Test

When your MCAT is all over, no matter how you feel you did, be good to yourself when you leave the test center. Celebrate! Take a nap. Watch a movie. Ride your bike. Plan a trip. Call up all of your neglected friends or stalk them on Facebook. Totally consume a cheesesteak and drink dirty martinis at night (assuming you're over 21). Whatever you do, make sure that it has absolutely nothing to do with thinking too hard—you deserve some rest and relaxation.

Perhaps most importantly, do not discuss specific details about the test with anyone. For one, it is important to let go of the stress of Test Day, and reliving your exam only inhibits you from being able to do so. But more significantly, the Examinee Agreement you sign at the beginning of your exam specifically prohibits you from discussing or disclosing exam content. The AAMC is known to seek out individuals who violate this agreement and retains the right to prosecute these individuals at their discretion. This means that you should not, under any circumstances, discuss the exam in person or over the phone with other individuals—including us at Kaplan—or post information or questions about exam content to Facebook, Student Doctor Network, or other online social media. You are permitted to comment on your “general exam experience,” including how you felt about the exam overall or an

individual section, but this is a fine line. In summary: if you're not certain whether you can discuss an aspect of the test or not, just don't do it! Do not let a silly Facebook post stop you from becoming the doctor you deserve to be.

Scores are released approximately one month after Test Day. The release is staggered during the afternoon and evening, ending at 5 p.m. Eastern. This means that not all examinees receive their scores at exactly the same time. Your score report will include a scaled score for each section between 118 and 132, as well as your total combined score between 472 and 528. These scores are given as confidence intervals. For each section, the confidence interval is approximately the given score ± 1 ; for the total score, it is approximately the given score ± 2 . You will also be given the corresponding percentile rank for each of these section scores and the total score.

AAMC CONTACT INFORMATION

For further questions, contact the MCAT team at the Association of American Medical Colleges:

MCAT Resource Center
Association of American Medical Colleges

www.aamc.org/mcat 

(202) 828-0690

mcataamc.org 

How This Book Was Created

The Kaplan MCAT Review project began shortly after the release of the Preview Guide for the MCAT 2015 Exam, 2nd edition. Through thorough analysis by our staff psychometricians, we were able to analyze the relative yield of the different topics on the MCAT, and we began constructing tables of contents for the books of the Kaplan MCAT Review series. A dedicated staff of 30 writers, 7 editors, and 32 proofreaders worked over 5,000 combined hours to produce these books. The format of the books was heavily influenced by weekly meetings with Kaplan's learning-science team.

In the years since this book was created, a number of opportunities for expansion and improvement have occurred. The current edition represents the culmination of the wisdom accumulated during that time frame, and it also includes several new features designed to improve the reading and learning experience in these texts.

These books were submitted for publication in April 2018. For any updates after this date, please visit www.kaptest.com/pages/retail-book-corrections-and-updates .

If you have any questions about the content presented here, email KaplanMCATfeedback@kaplan.com . For other questions not related to content, email booksupport@kaplan.com .

Each book has been vetted through at least ten rounds of review. To that end, the information presented in these books is true and accurate to the best of our knowledge. Still, your feedback helps us improve our prep materials. Please notify us of any inaccuracies or errors in the books by sending an email to KaplanMCATfeedback@kaplan.com .

Using This Book

Kaplan MCAT General Chemistry Review, and the other six books in the Kaplan MCAT Review series, bring the Kaplan classroom experience to you—right in your home, at your convenience. This book offers the same Kaplan content review, strategies, and practice that make Kaplan the #1 choice for MCAT prep.

This book is designed to help you review the general chemistry topics covered on the MCAT. Please understand that content review—no matter how thorough—is not sufficient preparation for the MCAT! The MCAT tests not only your science knowledge but also your critical reading, reasoning, and problem-solving skills. Do not assume that simply memorizing the contents of this book will earn you high scores on Test Day; to maximize your scores, you must also improve your reading and test-taking skills through MCAT-style questions and practice tests.

LEARNING GOALS

At the beginning of each section, you'll find a short list of objectives describing the skills covered within that section. Learning goals for these texts were developed in conjunction with Kaplan's learning science team, and have been designed specifically to focus your attention on tasks and concepts that are likely to show up on your MCAT. These learning goals will function as a means to guide your study and indicate what

information and relationships you should be focused on within each section. Before starting each section, read these learning goals carefully. They will not only allow you to assess your existing familiarity with the content, but also provide a goal-oriented focus for your studying experience of the section.

MCAT CONCEPT CHECKS

At the end of each section, you'll find a few open-ended questions that you can use to assess your mastery of the material. These MCAT Concept Checks were introduced after numerous conversations with Kaplan's learning science team. Research has demonstrated repeatedly that introspection and self-analysis improve mastery, retention, and recall of material. Complete these MCAT Concept Checks to ensure that you've got the key points from each section before moving on!

PRACTICE QUESTIONS

At the end of each chapter, you'll find 15 MCAT-style practice questions. These are designed to help you assess your understanding of the chapter you just read. Most of these questions focus on the first of the Scientific Inquiry and Reasoning Skills (Knowledge of Scientific Concepts and Principles), although there are occasional questions that fall into the second or fourth SIRS (Scientific Reasoning and Problem-Solving, and Data-Based and Statistical Reasoning, respectively).

SIDEBARS

The following is a guide to the five types of sidebars you'll find in Kaplan MCAT General Chemistry Review:

- **Bridge:** These sidebars create connections between science topics that appear in multiple chapters throughout the Kaplan MCAT Review series.
- **Key Concept:** These sidebars draw attention to the most important takeaways in a given topic, and they sometimes offer synopses or overviews of complex information. If you understand nothing else, make sure you grasp the Key Concepts for any given subject.
- **MCAT Expertise:** These sidebars point out how information may be tested on the MCAT or offer key strategy points and test-taking tips that you should apply on Test Day.
- **Mnemonic:** These sidebars present memory devices to help recall certain facts.
- **Real World:** These sidebars illustrate how a concept in the text relates to the practice of medicine or the world at large. While this is not information you need to know for Test Day, many of the topics in Real World sidebars are excellent examples of how a concept may appear in a passage or discrete (stand-alone) question on the MCAT.

WHAT THIS BOOK COVERS

The information presented in the Kaplan MCAT Review series covers everything listed on the official MCAT content lists. Every topic in these lists is covered in the same level of detail as is common to the undergraduate and postbaccalaureate classes that are considered prerequisites for the MCAT. Note that your premedical classes may include topics not discussed in these books, or they may go into more depth than these books do. Additional exposure to science content is never a bad thing, but all of the content knowledge you are expected to have walking in on Test Day is covered in these books.

Chapter Profiles, on the first page of each chapter, represent a holistic look at the content within the chapter and will include a pie chart as well as text information. The pie chart analysis is based directly on data released by the AAMC and will give a rough estimate of the importance of the chapter in relation to the book as a whole. Further, the text portion of the Chapter Profiles includes which AAMC content categories are covered within the chapter. These are referenced directly from the AAMC MCAT exam content listing, available on the testmaker's website.

You'll also see new High-Yield badges scattered throughout the sections of this book:

In This Chapter

1.1 Amino Acids Found in Proteins	HY	1.3 Peptide Bond Formation and Hydrolysis	HY
A Note on Terminology	4	Peptide Bond Formation	1
Stereochemistry of Amino Acids	5	Peptide Bond Hydrolysis	1
Structures of the Amino Acids	6	1.4 Primary and Secondary Protein Structure	HY
Hydrophobic and Hydrophilic Amino Acids	9	Primary Structure	2
Amino Acid Abbreviations	10	Secondary Structure	2
1.2 Acid–Base Chemistry of Amino Acids		1.5 Tertiary and Quaternary Protein Structure	HY
Protonation and Deprotonation	12	Tertiary Structure	2
Titration of Amino Acids	14	Folding and the Solvation Layer	2
		Quaternary Structure	2
		Conjugated Proteins	2
		1.6 Denaturation	
		Concept Summary	2

1.1 Amino Acids Found in Proteins

High-Yield

LEARNING GOALS

After Chapter 1.1, you will be able to:

These badges represent the top 100 topics most tested by the AAMC. In other words, according to the testmaker and all our experience with their resources, a High-Yield badge means more questions on Test Day.

This book also contains a thorough glossary and index for easy navigation of the text.

In the end, this is your book, so write in the margins, draw diagrams, highlight the key points—do whatever is necessary to help you get that higher score. We look forward to working with you as you achieve your dreams and become the doctor you deserve to be!

STUDYING WITH THIS BOOK

In addition to providing you with the best practice questions and test strategies, Kaplan's team of learning scientists are dedicated to researching and testing the best methods for getting the most out of your study time. Here are their top four tips for improving retention:

Review multiple topics in one study session. This may seem counterintuitive—we're used to practicing one skill at a time in order to improve each skill. But research shows that weaving topics together leads to increased learning. Beyond that consideration, the MCAT often

includes more than one topic in a single question. Studying in an integrated manner is the most effective way to prepare for this test.

Customize the content. Drawing attention to difficult or critical content can ensure you don't overlook it as you read and re-read sections. The best way to do this is to make it more visual—highlight, make tabs, use stickies, whatever works. We recommend highlighting only the most important or difficult sections of text. Selective highlighting of up to about 10 percent of text in a given chapter is great for emphasizing parts of the text, but over-highlighting can have the opposite effect.

Repeat topics over time. Many people try to memorize concepts by repeating them over and over again in succession. Our research shows that retention is improved by spacing out the repeats over time and mixing up the order in which you study content. For example, try reading chapters in a different order the second (or third!) time around. Revisit practice questions that you answered incorrectly in a new sequence. Perhaps information you reviewed more recently will help you better understand those questions and solutions you struggled with in the past.

Take a moment to reflect. When you finish reading a section for the first time, stop and think about what you just read. Jot down a few thoughts in the margins or in your notes about why the content is important or what topics came to mind when you read it. Associating learning with a memory is a fantastic way to retain information! This also works when answering questions. After answering a question, take a moment to think through each step you took to arrive at a solution. What led you to the

answer you chose? Understanding the steps you took will help you make good decisions when answering future questions.

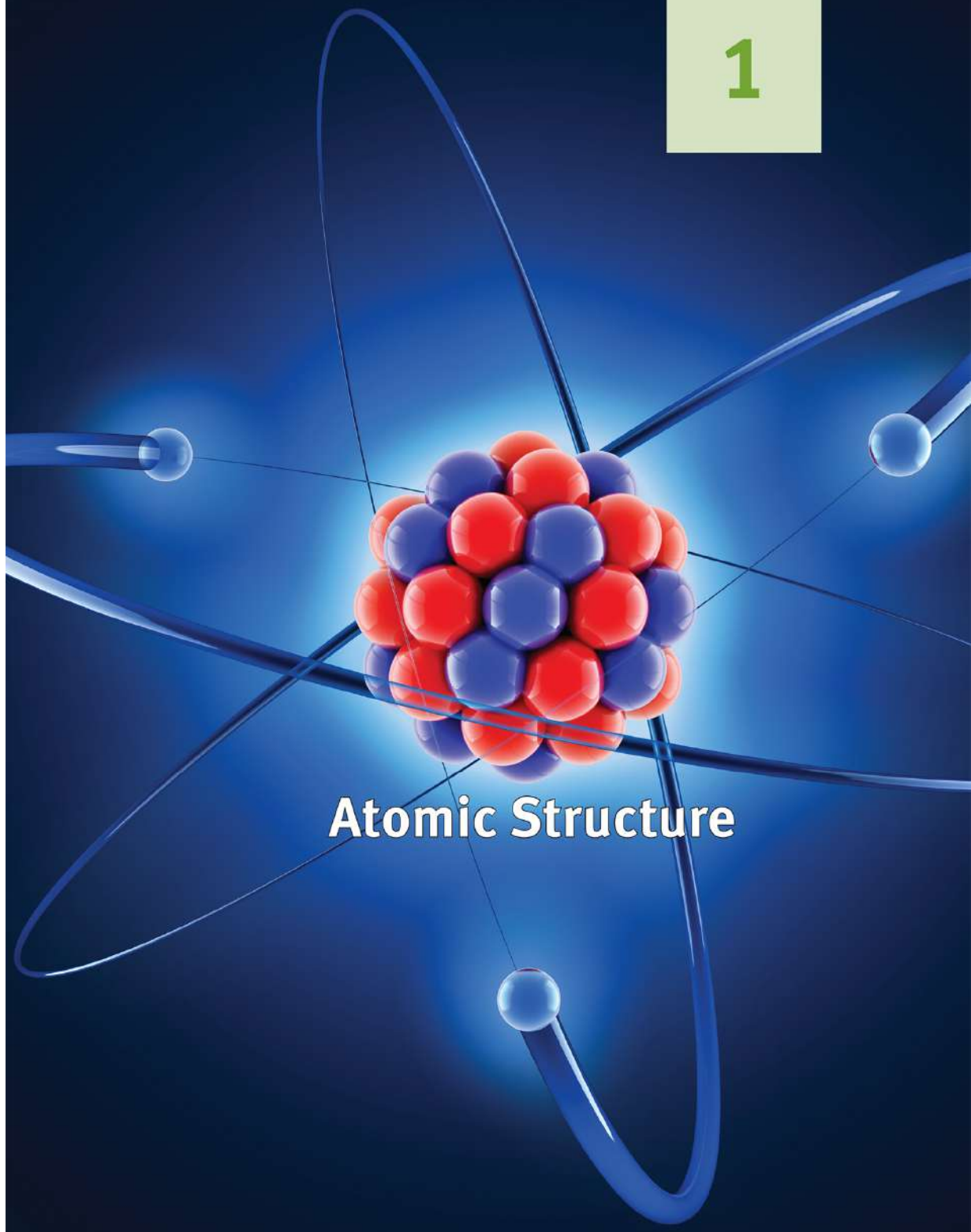
ONLINE RESOURCES

In addition to the resources located within this text, you also have additional online resources awaiting you at www.kaptest.com/booksonline. Make sure to log on and take advantage of free practice and access to online versions of the book!

Please note that access to the online resources is limited to the original owner of this book.

1

Atomic Structure



In This Chapter

1.1 Subatomic Particles

Protons 

Neutrons 

Electrons 

1.2 Atomic Mass vs. Atomic Weight

Atomic Mass 

Atomic Weight 

1.3 Rutherford, Planck, and Bohr

Bohr Model 

Applications of the Bohr Model 



1.4 Quantum Mechanical Model of Atoms

Quantum Numbers 

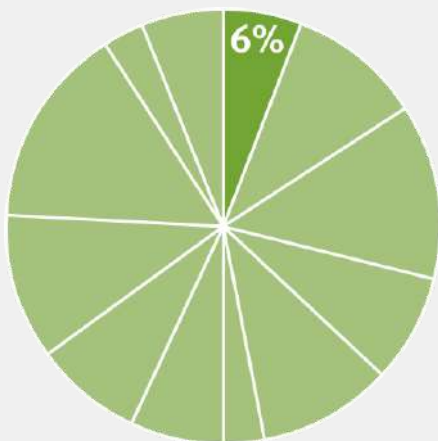
Electron Configurations 

Hund's Rule 

Valence Electrons 

Concept Summary 

Chapter Profile



The content in this chapter should be relevant to about 6% of all questions about general chemistry on the MCAT.

This chapter covers material from the following AAMC content category:

4E: Atoms, nuclear decay, electronic structure, and atomic chemical behavior

Introduction

Chemistry is the investigation of the atoms and molecules that make up our bodies, our possessions, the food we eat, and the world around us. There are different branches of chemistry, three of which are tested

directly on the MCAT: general (inorganic) chemistry, organic chemistry, and biochemistry. Ultimately, all investigations in chemistry are seeking to answer the questions that confront us in the form—the shape, structure, mode, and essence—of the physical world that surrounds us.

Many students feel similarly about general chemistry and physics: But I'm premed, they say. Why do I need to know any of this? What good will this be when I'm a doctor? Do I only need to know this for the MCAT? Recognize that to be an effective doctor, one must understand the physical building blocks that make up the human body. Pharmacologic treatment is based on chemistry; many diagnostic tests used every day detect changes in the chemistry of the body.

So, let's get down to the business of learning and remembering the principles of the physical world that help us understand what all this “stuff” is, how it works, and why it behaves the way it does—at both the molecular and macroscopic levels. In the process of reading through these chapters and applying your knowledge to practice questions, you'll prepare yourself for success not only on the Chemical and Physical Foundations of Biological Systems section of the MCAT but also in your future career as a physician.

This first chapter starts our review of general chemistry with a consideration of the fundamental unit of matter—the atom. First, we focus on the subatomic particles that make it up: protons, neutrons, and electrons. We will also review the Bohr and quantum mechanical models of the atom, with a particular focus on the similarities and differences between them.



MCAT EXPERTISE

The building blocks of the atom are also the building blocks of knowledge for the general chemistry concepts tested on the MCAT. By understanding these particles, we will be able to use that knowledge as the “nucleus” of understanding for all of general chemistry.

1.1 Subatomic Particles

LEARNING GOALS

After Chapter 1.1, you will be able to:

- Identify the subatomic particles most important for determining various traits of an atom, including charge, atomic number, and isotope
- Determine the number of protons, neutrons, and electrons within an isotope, such as ^{14}C

Although you may have encountered in your university-level chemistry classes such subatomic particles as quarks, leptons, and gluons, the MCAT's approach to atomic structure is much simpler. There are three subatomic particles that you must understand: protons, neutrons, and electrons.

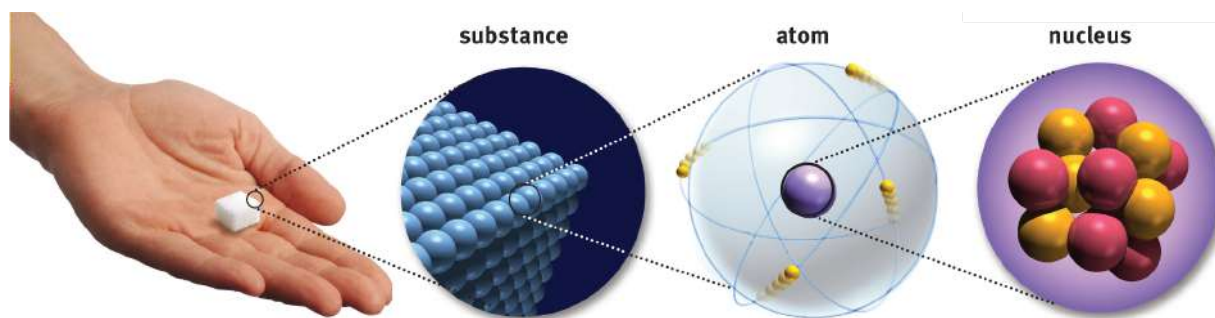
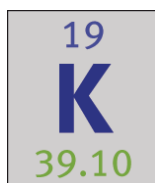


Figure 1.1. Matter: From Macroscopic to Microscopic

PROTONS

Protons are found in the nucleus of an atom, as shown in Figure 1.1. Each proton has an amount of charge equal to the fundamental unit of charge ($e = 1.6 \times 10^{-19} \text{ C}$), and we denote this fundamental unit of charge as “+1 e” or simply “+1” for the proton. Protons have a mass of approximately one atomic mass unit (amu). The atomic number (Z) of an element, as shown in Figure 1.2, is equal to the number of protons found in an atom of that element. As such, it acts as a unique identifier for each element because elements are defined by the number of protons they contain. For example, all atoms of oxygen contain eight protons; all atoms of gadolinium contain 64 protons. While all atoms of a given element have the same atomic number, they do not necessarily have the same mass—as we will see in our discussion of isotopes.



19
K
39.10

Figure 1.2. Potassium, from the Periodic Table

Potassium has the symbol K (Latin: kalium), atomic number 19, and atomic weight of approximately 39.1.

NEUTRONS

Neutrons, as the name implies, are neutral—they have no charge. A neutron's mass is only slightly larger than that of the proton, and together, the protons and the neutrons of the nucleus make up almost the entire mass of an atom. Every atom has a characteristic mass number (A), which is the sum of the protons and neutrons in the atom's nucleus. A given element can have a variable number of neutrons; thus, while atoms of the same element always have the same atomic number, they do not necessarily have the same mass number. Atoms that share an atomic number but have different mass numbers are known as isotopes of the element, as shown in Figure 1.3. For example, carbon ($Z = 6$) has three naturally occurring isotopes: ${}^{12}_6\text{C}$, with six protons and six neutrons; ${}^{13}_6\text{C}$, with six protons and seven neutrons; and ${}^{14}_6\text{C}$, with six protons and eight neutrons. The convention ${}^A_Z\text{X}$ is used to show both the atomic number (Z) and the mass number (A) of atom X.

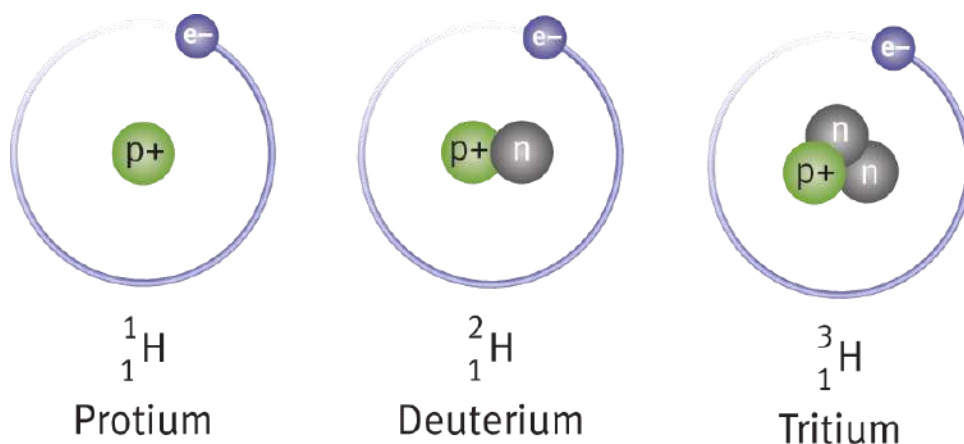


Figure 1.3. Various Isotopes of Hydrogen

Atoms of the same element have the same atomic number ($Z = 1$) but may have varying mass numbers ($A = 1, 2, \text{ or } 3$).

ELECTRONS

Electrons move through the space surrounding the nucleus and are associated with varying levels of energy. Each electron has a charge equal in magnitude to that of a proton, but with the opposite (negative) sign, denoted by “ -1 e ” or simply “ $-e$.” The mass of an electron is approximately $\frac{1}{2000}$ that of a proton. Because subatomic particles’ masses are so small, the electrostatic force of attraction between the unlike charges of the proton and electron is far greater than the gravitational force of attraction based on their respective masses.

Electrons move around the nucleus at varying distances, which correspond to varying levels of electrical potential energy. The electrons closer to the nucleus are at lower energy levels, while those that are further out (in higher electron shells) have higher energy. The electrons that are farthest from the nucleus have the strongest interactions with the surrounding environment and the weakest interactions with the nucleus. These electrons are called valence electrons; they are much more likely to become involved in bonds with other atoms because they experience the least electrostatic pull from their own nucleus. Generally speaking, the valence electrons determine the reactivity of an atom. As we will discuss in Chapter 3 of MCAT General Chemistry Review, the sharing or transferring of these valence electrons in bonds allows elements to fill their highest energy level to increase stability. In the neutral state, there are equal numbers of protons and electrons; losing electrons results in the atom gaining a positive charge, while gaining electrons results in the atom gaining a negative charge. A positively charged atom is called a cation, and a negatively charged atom is called an anion.



Valence electrons will be very important to us in both general and organic chemistry. Knowing how tightly held those electrons are will allow us to understand many of an atom's properties and how it interacts with other atoms, especially in bonding. Bonding is so important that it is discussed in Chapter 3 of both MCAT General Chemistry Review and MCAT Organic Chemistry Review.

Some basic features of the three subatomic particles are shown in Table 1.1.

Subatomic Particle	Symbol	Relative Mass	Charge	Location
Proton	p , p^+ , or ${}^1_1\text{H}$	1	+1	Nucleus
Neutron	n^0 or ${}^1_0\text{n}$	1	0	Nucleus
Electron	e^- or ${}^{-1}_0e$	0	-1	Orbitals

Table 1.1. Subatomic Particles

Example:

Determine the number of protons, neutrons, and electrons in a nickel-58 atom and in a nickel-60 +2 cation.

Solution:

^{58}Ni has an atomic number of 28 and a mass number of 58. Therefore, ^{58}Ni will have 28 protons, 28 electrons, and $58 - 28$, or 30, neutrons.

$^{60}\text{Ni}^{2+}$ has the same number of protons as the neutral ^{58}Ni atom. However, $^{60}\text{Ni}^{2+}$ has a positive charge because it has lost two electrons; thus, Ni^{2+} will have 26 electrons. Also, the mass number is two units higher than for the ^{58}Ni atom, and this difference in mass must be due to two extra neutrons; thus, it has a total of 32 neutrons.

MCAT Concept Check 1.1:

Before you move on, assess your understanding of the material with these questions.

Which subatomic particle is the most important for determining each of the following properties of an atom?

- Charge:

- Atomic number:

- Isotope:

In nuclear medicine, isotopes are created and used for various purposes; for instance, ^{18}O is created from ^{18}F . Determine the number of protons, neutrons, and electrons in each of these species.

Particle	Protons	Neutrons	Electrons
^{18}O			
^{18}F			



1.2 Atomic Mass vs. Atomic Weight

LEARNING GOALS

After Chapter 1.2, you will be able to:

- Describe atomic mass and atomic weight
- Recall the units of molar mass
- Predict the number of protons, neutrons, and electrons in a given isotope

There are a few different terms used by chemists to describe the heaviness of an element: atomic mass and mass number, which are essentially synonymous, and atomic weight. While the atomic weight is a constant for a given element and is reported in the periodic table, the atomic mass or mass number varies from one isotope to another. In this section, carefully compare and contrast the different definitions of these terms—because they are similar, they can be easy to mix up on the MCAT.



KEY CONCEPT

- Atomic number (Z) = number of protons

- Mass number (A) = number of protons + number of neutrons
- Number of protons = number of electrons (in a neutral atom)
- Electrons are not included in mass calculations because they are much smaller.

ATOMIC MASS

As we've seen, the mass of one proton is approximately one amu. The size of the atomic mass unit is defined as exactly $\frac{1}{12}$ the mass of the carbon-12 atom, approximately 1.66×10^{-24} g. Because the carbon-12 nucleus has six protons and six neutrons, an amu is approximately equal to the mass of a proton or a neutron. The difference in mass between protons and neutrons is extremely small; in fact, it is approximately equal to the mass of an electron.

The atomic mass of an atom (in amu) is nearly equal to its mass number, the sum of protons and neutrons (in reality, some mass is lost as binding energy, as discussed in Chapter 9 of MCAT Physics and Math Review). Atoms of the same element with varying mass numbers are called isotopes (from the Greek for "same place"). Isotopes differ in their number of neutrons and are referred to by the name of the element followed by the mass number; for example, carbon-12 or iodine-131. Only the three isotopes of hydrogen, shown in Figure 1.3, are given unique names: protium (Greek: "first") has one proton and an atomic mass of 1 amu; deuterium ("second") has one proton and one neutron and an atomic mass of 2 amu; tritium ("third") has one proton and two neutrons and an atomic

mass of 3 amu. Because isotopes have the same number of protons and electrons, they generally exhibit similar chemical properties.

ATOMIC WEIGHT

In nature, almost all elements exist as two or more isotopes, and these isotopes are usually present in the same proportions in any sample of a naturally occurring element. The weighted average of these different isotopes is referred to as the atomic weight and is the number reported on the periodic table. For example, chlorine has two main naturally occurring isotopes: chlorine-35 and chlorine-37. Chlorine-35 is about three times more abundant than chlorine-37; therefore, the atomic weight of chlorine is closer to 35 than 37. On the periodic table, it is listed as 35.5. Figure 1.4 illustrates the half-lives of the different isotopes of the elements; because half-life corresponds with stability, it also helps determine the relative proportions of these different isotopes.

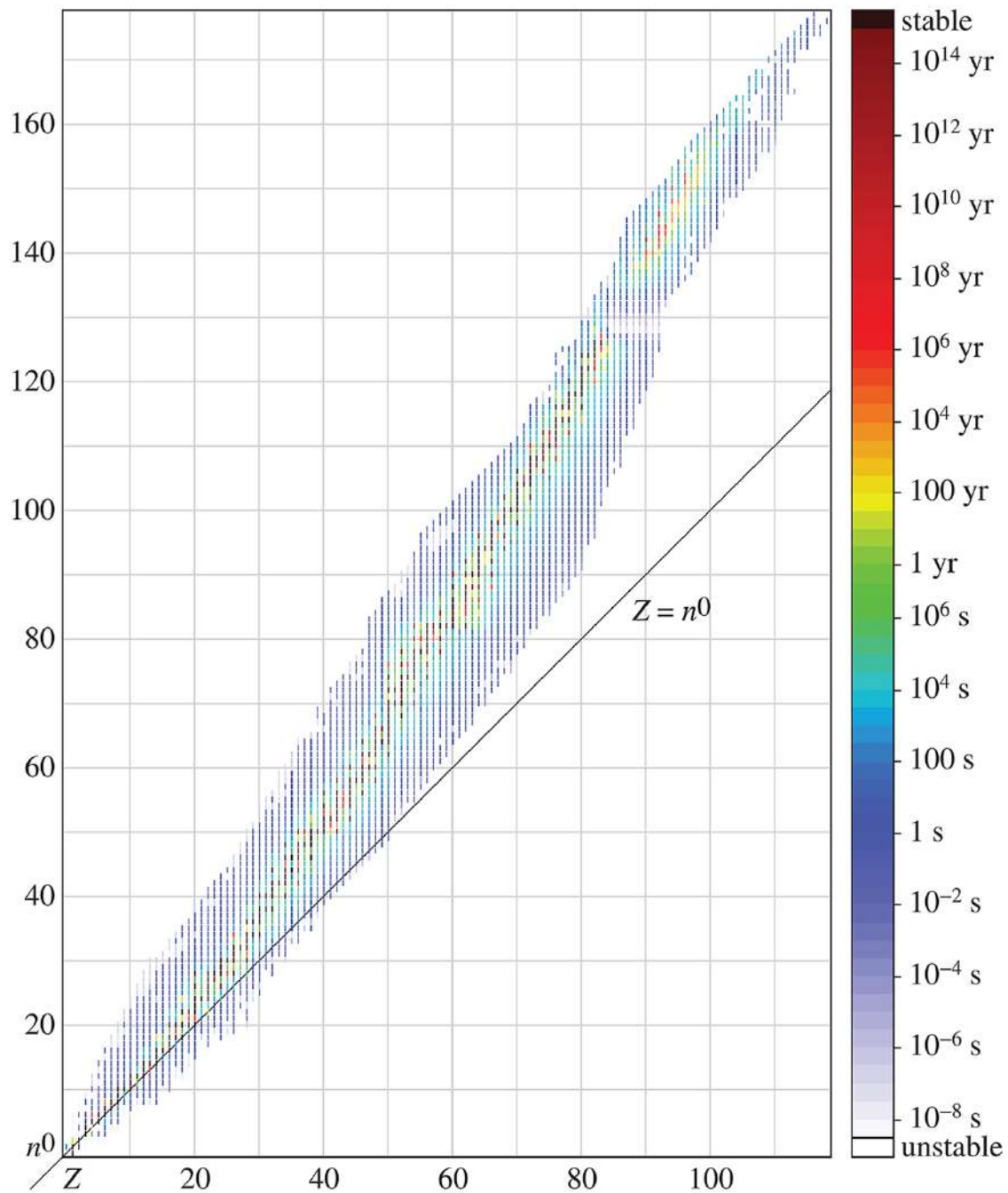


Figure 1.4. Half-Lives of the Different Isotopes of Elements
Half-life is a marker of stability; generally, longer-lasting isotopes are more abundant.



KEY CONCEPT

When an element has two or more isotopes, no one isotope will have a mass exactly equal to the element's atomic weight. Bromine, for example, is listed in the periodic table as having a mass of 79.9 amu. This is an average of the two naturally occurring isotopes, bromine-79 and bromine-81, which occur in almost equal proportions. There are no bromine atoms with an actual mass of 79.9 amu.

The utility of the atomic weight is that it represents both the mass of the “average” atom of that element, in amu, and the mass of one mole of the element, in grams. A mole is a number of “things” (atoms, ions, molecules) equal to Avogadro's number, $N_A = 6.02 \times 10^{23}$. For example, the atomic weight of carbon is 12.0 amu, which means that the average carbon atom has a mass of 12.0 amu (carbon-12 is far more abundant than carbon-13 or carbon-14), and 6.02×10^{23} carbon atoms have a combined mass of 12.0 grams.



MNEMONIC

Atomic mass is nearly synonymous with mass number. Atomic weight is a weighted average of naturally occurring isotopes of that element.

Example:

Element Q consists of three different isotopes: A, B, and C. Isotope A has an atomic mass of 40 amu and accounts for 60 percent of naturally occurring Q. Isotope B has an atomic mass of 44 amu and accounts for 25 percent of Q. Finally, isotope C has an atomic mass of 41 amu and accounts for 15 percent of Q. What is the atomic weight of element Q?

Solution:

The atomic weight is the weighted average of the naturally occurring isotopes of that element:

$$0.60 (40 \text{ amu}) + 0.25 (44 \text{ amu}) + 0.15 (41 \text{ amu}) = 24.00 \text{ amu} + 11.00 \text{ amu} + 6.15 \text{ amu} = 41.15 \text{ amu}$$

MCAT Concept Check 1.2:

Before you move on, assess your understanding of the material with these questions.

What are the definitions of atomic mass and atomic weight?

- Atomic mass:
-

- Atomic weight:

While molar mass is typically written in grams per mole $\left(\frac{\text{g}}{\text{mol}}\right)$, is the ratio moles per gram $\left(\frac{\text{mol}}{\text{g}}\right)$ also acceptable?

Calculate and compare the subatomic particles that make up the following atoms.

Isotope	Protons	Neutrons	Electrons
^{19}O			
^{16}O			
^{17}O			
^{19}F			
^{16}F			
^{238}U			
^{240}U			

1.3 Rutherford, Planck, and Bohr

High-Yield



LEARNING GOALS

After Chapter 1.3, you will be able to:

- Calculate the energy of transition for a valence electron that jumps energy levels
- Calculate the wavelength of an emitted photon given the energy emitted by an electron
- Calculate the energy of a photon given its wavelength

In 1910, Ernest Rutherford provided experimental evidence that an atom has a dense, positively charged nucleus that accounts for only a small portion of the atom's volume. Eleven years earlier, Max Planck developed the first quantum theory, proposing that energy emitted as electromagnetic radiation from matter comes in discrete bundles called quanta. The energy of a quantum, he determined, is given by the Planck relation:

$$E = hf$$

Equation 1.1

where h is a proportionality constant known as Planck's constant, equal to $6.626 \times 10^{-34} \text{ J} \cdot \text{s}$, and f (sometimes designated by the Greek letter nu, ν) is the frequency of the radiation.



BRIDGE

Recall from Chapter 8 of MCAT Physics Review that the speed of light (or any wave) can be calculated using $v = f\lambda$. The speed of light, c , is $3 \times 10^8 \frac{\text{m}}{\text{s}}$. This equation can be incorporated into the equation for quantum energy to provide different derivations.

BOHR MODEL

In 1913, Danish physicist Niels Bohr used the work of Rutherford and Planck to develop his model of the electronic structure of the hydrogen atom. Starting from Rutherford's findings, Bohr assumed that the hydrogen atom consisted of a central proton around which an electron traveled in a circular orbit. He postulated that the centripetal force acting on the electron as it revolved around the nucleus was created by the electrostatic

force between the positively charged proton and the negatively charged electron.

Bohr used Planck's quantum theory to correct certain assumptions that classical physics made about the pathways of electrons. Classical mechanics postulates that an object revolving in a circle, such as an electron, may assume an infinite number of values for its radius and velocity. The angular momentum ($L = mvr$) and kinetic energy

$\left(K = \frac{1}{2} mv^2\right)$ of the object could therefore take on any value.

However, by incorporating Planck's quantum theory into his model, Bohr placed restrictions on the possible values of the angular momentum. Bohr predicted that the possible values for the angular momentum of an electron orbiting a hydrogen nucleus could be given by:

$$L = \frac{nh}{2\pi}$$

Equation 1.2

where n is the principal quantum number, which can be any positive integer, and h is Planck's constant. Because the only variable is the principal quantum number, the angular momentum of an electron changes only in discrete amounts with respect to the principal quantum number. Note the similarities between quantized angular momentum and Planck's concept of quantized energy.



MCAT EXPERTISE

When you see a formula in your review or on Test Day, focus on ratios and relationships. This simplifies our calculations to a conceptual understanding, which is usually enough to lead us to the right answer. Further, the MCAT tends to ask how changes in one variable may affect another variable, rather than a plug-and-chug application of complex equations.

Bohr then related the permitted angular momentum values to the energy of the electron to obtain:

$$E = - \frac{R_H}{n^2}$$

Equation 1.3

where R_H is the experimentally determined Rydberg unit of energy, equal to $2.18 \times 10^{-18} \frac{\text{J}}{\text{electron}}$. Therefore, like angular momentum, the energy of the electron changes in discrete amounts with respect to the quantum number. A value of zero energy was assigned to the state in which the proton and electron are separated completely, meaning that there is no attractive force between them. Therefore, the electron in any of its quantized states in the atom will have an attractive force toward the proton; this is represented by the negative sign in Equation 1.3. Ultimately,

the only thing the energy equation is saying is that the energy of an electron increases—becomes less negative—the farther out from the nucleus that it is located (increasing n). This is an important point: while the magnitude of the fraction is getting smaller, the actual value it represents is getting larger (becoming less negative).



KEY CONCEPT

At first glance, it may not be clear that the energy (E) is directly proportional to the principal quantum number (n) in Equation 1.3. Take notice of the negative sign, which causes the values to approach zero from a more negative value as n increases (thereby increasing the energy). Negative signs are as important as a variable's location in a fraction when it comes to determining proportionality.

Think of the concept of quantized energy as being similar to the change in gravitational potential energy that you experience when you ascend or descend a flight of stairs. Unlike a ramp, on which you could take an infinite number of steps associated with a continuum of potential energy changes, a staircase only allows you certain changes in height and, as a result, allows only certain discrete (quantized) changes of potential energy.

Bohr came to describe the structure of the hydrogen atom as a nucleus with one proton forming a dense core, around which a single electron revolved in a defined pathway (orbit) at a discrete energy value. If one could transfer an amount of energy exactly equal to the difference between

one orbit and another, this could result in the electron “jumping” from one orbit to a higher-energy one. These orbits had increasing radii, and the orbit with the smallest, lowest-energy radius was defined as the ground state ($n = 1$). More generally, the ground state of an atom is the state of lowest energy, in which all electrons are in the lowest possible orbitals. In Bohr’s model, the electron was promoted to an orbit with a larger radius (higher energy), the atom was said to be in the excited state. In general, an atom is in an excited state when at least one electron has moved to a subshell of higher than normal energy. Bohr likened his model of the hydrogen atom to the planets orbiting the sun, in which each planet traveled along a roughly circular pathway at set distances—and energy values—from the sun. Bohr’s Nobel Prize-winning model was reconsidered over the next two decades but remains an important conceptualization of atomic behavior. In particular, remember that we now know that electrons are not restricted to specific pathways, but tend to be localized in certain regions of space.



MCAT EXPERTISE

Note that all systems tend toward minimal energy; thus on the MCAT, atoms of any element will generally exist in the ground state unless subjected to extremely high temperatures or irradiation.

APPLICATIONS OF THE BOHR MODEL

The Bohr model of the hydrogen atom (and other one-electron systems, such as He^+ and Li^{2+}) is useful for explaining the atomic emission and absorption spectra of atoms.



MNEMONIC

As electrons go from a lower energy level to a higher energy level, they get AHED:

- Absorb light
- Higher potential
- Excited
- Distant (from the nucleus)

Atomic Emission Spectra

At room temperature, the majority of atoms in a sample are in the ground state. However, electrons can be excited to higher energy levels by heat or other energy forms to yield excited states. Because the lifetime of an excited state is brief, the electrons will return rapidly to the ground state, resulting in the emission of discrete amounts of energy in the form of photons, as shown in Figure 1.5.

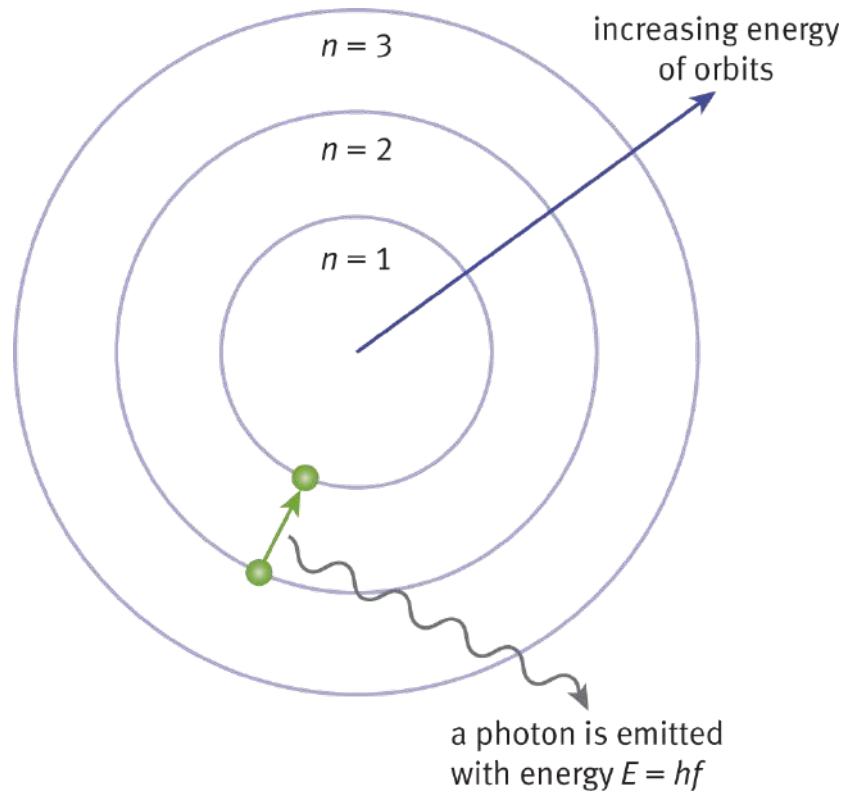


Figure 1.5. Atomic Emission of a Photon as a Result of a Ground State Transition

The electromagnetic energy of these photons can be determined using the following equation:

$$E = \frac{hc}{\lambda}$$

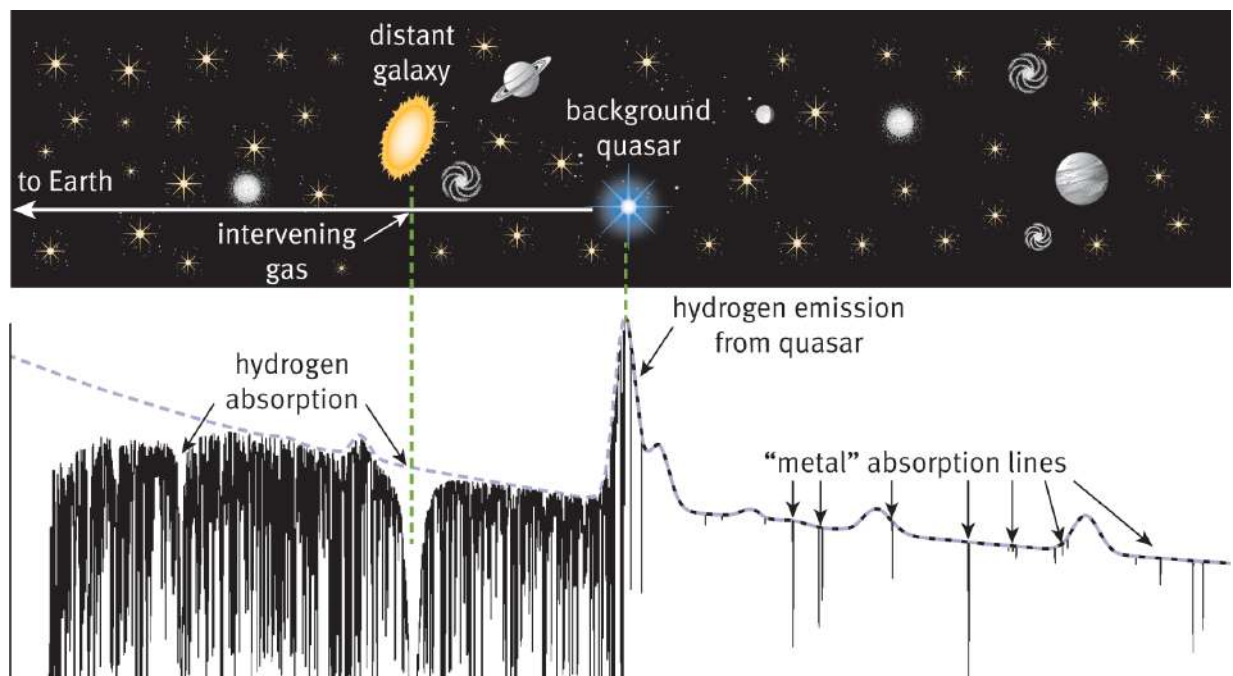
Equation 1.4

where h is Planck's constant, c is the speed of light in a vacuum

$\left(3.00 \times 10^8 \frac{\text{m}}{\text{s}}\right)$, and λ is the wavelength of the radiation. Note that

Equation 1.4 is just a combination of two other equations: $E = hf$ and $c = f\lambda$.

The electrons in an atom can be excited to different energy levels. When these electrons return to their ground states, each will emit a photon with a wavelength characteristic of the specific energy transition it undergoes. As described above, these energy transitions do not form a continuum, but rather are quantized to certain values. Thus, the spectrum is composed of light at specified frequencies. It is sometimes called a line spectrum, where each line on the emission spectrum corresponds to a specific electron transition. Because each element can have its electrons excited to a different set of distinct energy levels, each possesses a unique atomic emission spectrum, which can be used as a fingerprint for the element. One particular application of atomic emission spectroscopy is in the analysis of stars and planets: while a physical sample may be impossible to procure, the light from a star can be resolved into its component wavelengths, which are then matched to the known line spectra of the elements as shown in Figure 1.6.



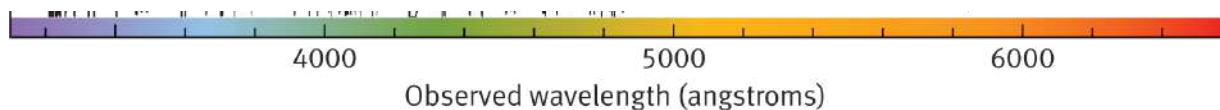


Figure 1.6. Line Spectrum with Transition Wavelengths for Various Celestial Bodies



REAL WORLD

Emissions from electrons dropping from an excited state to a ground state give rise to fluorescence. What we see is the color of the emitted light.

The Bohr model of the hydrogen atom explained the atomic emission spectrum of hydrogen, which is the simplest emission spectrum among all the elements. The group of hydrogen emission lines corresponding to transitions from energy levels $n \geq 2$ to $n = 1$ is known as the Lyman series. The group corresponding to transitions from energy levels $n \geq 3$ to $n = 2$ is known as the Balmer series and includes four wavelengths in the visible region. The Lyman series includes larger energy transitions than the Balmer series; it therefore has shorter photon wavelengths in the UV region of the electromagnetic spectrum. The Paschen series corresponds to transitions from $n \geq 4$ to $n = 3$. These energy transition series can be seen in Figure 1.7.

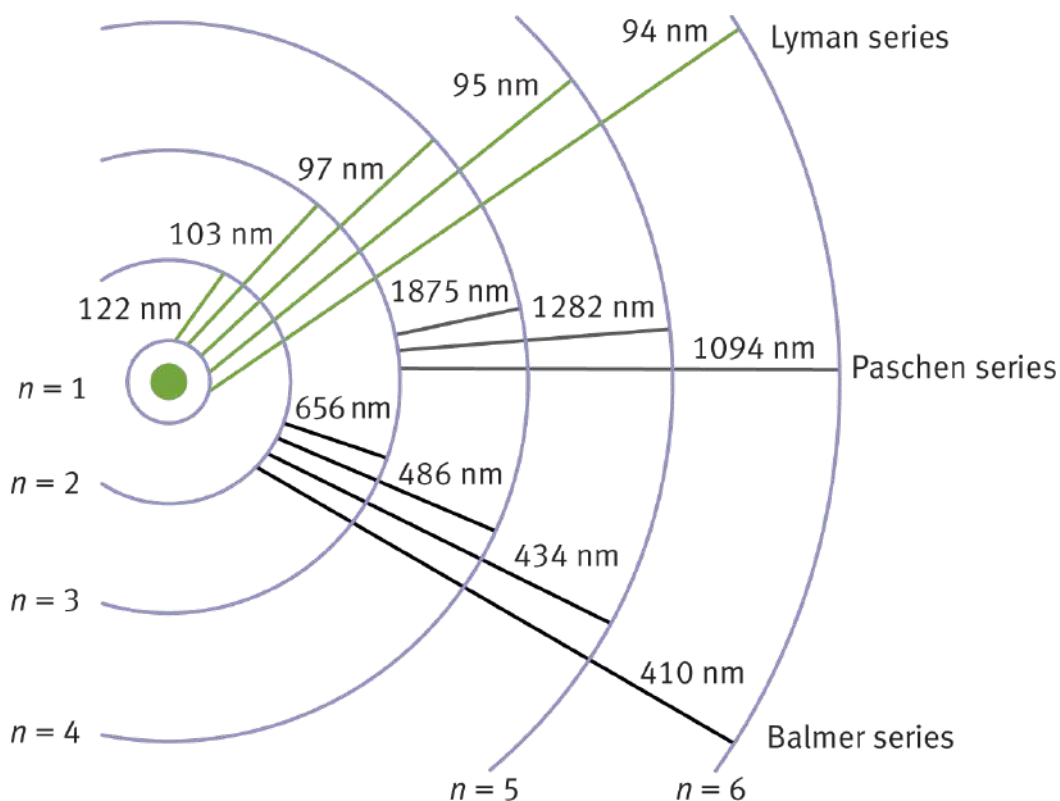


Figure 1.7. Wavelengths of Electron Orbital Transitions

Energy is inversely proportional to wavelength: $E = hf = \frac{hc}{\lambda}$.

The energy associated with a change in the principal quantum number from a higher initial value n_i to a lower final value n_f is equal to the energy of the photon predicted by Planck's quantum theory. Combining Bohr's and Planck's calculations, we can derive:

$$E = \frac{hc}{\lambda} = -R_H \left[\frac{1}{n_i^2} - \frac{1}{n_f^2} \right]$$

Equation 1.5

This complex-appearing equation essentially says: The energy of the emitted photon corresponds to the difference in energy between the higher-energy initial state and the lower-energy final state.



KEY CONCEPT

This equation is nothing new; it is simply derived from conservation of energy by setting the energy of a photon $\left(E = hf = \frac{hc}{\lambda}\right)$ equal to the energy of the electron transition (from $E = -\frac{R_H}{n^2}$). Note that unlike other equations, this is initial minus final; the negative sign in the equation accounts for absorption and emission. Thus, a positive E corresponds to emission, and a negative E corresponds to absorption.

Atomic Absorption Spectra

When an electron is excited to a higher energy level, it must absorb exactly the right amount of energy to make that transition. This means that exciting the electrons of a particular element results in energy absorption at specific wavelengths. Thus, in addition to a unique emission spectrum, every element possesses a characteristic absorption spectrum. Not surprisingly, the wavelengths of absorption correspond exactly to the wavelengths of emission because the difference in energy between levels remains unchanged. Identification of elements in the gas phase requires absorption spectra.



BRIDGE

ΔE is the same for absorption or emission between any two energy levels according to the conservation of energy, as discussed in Chapter 2 of MCAT Physics and Math Review. This is also the same as the energy of the photon of light absorbed or emitted.

Atomic emission and absorption spectra are complex topics, but the takeaway is that each element has a characteristic set of energy levels. For electrons to move from a lower energy level to a higher energy level, they must absorb the right amount of energy to do so. They absorb this energy in the form of light. Similarly, when electrons move from a higher energy level to a lower energy level, they emit the same amount of energy in the form of light.



REAL WORLD

Absorption is the basis for the color of compounds. We see the color of the light that is not absorbed by the compound.

MCAT Concept Check 1.3:

Before you move on, assess your understanding of the material with these questions.

Note: For these questions, try to estimate the calculations without a calculator to mimic Test Day conditions. Double-check your answers with a calculator and refer to the answers for confirmation of your results.

The valence electron in a lithium atom jumps from energy level $n = 2$ to $n = 4$. What is the energy of this transition in joules? In eV? (Note:

$$R_H = 2.18 \times 10^{-18} \frac{\text{J}}{\text{electron}} = 13.6 \frac{\text{eV}}{\text{electron}})$$

If an electron emits 3 eV of energy, what is the corresponding wavelength of the emitted photon? (Note: $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$, $h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$)

Calculate the energy of a photon of wavelength 662 nm. (Note: $h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$)

1.4 Quantum Mechanical Model of Atoms

LEARNING GOALS

After Chapter 1.4, you will be able to:

- Identify the four quantum numbers, the potential range of values for each, and their relationship to the electron they represent
- Compare the orbital diagram for a neutral atom, such as sulfur (S), to an ion such as S^{2-}
- Differentiate between paramagnetic and diamagnetic compounds
- Determine the number of valence electrons in a given atom

While Bohr's model marked a significant advancement in the understanding of the structure of atoms, his model ultimately proved inadequate to explain the structure and behavior of atoms containing more than one electron. The model's failure was a result of not taking into account the repulsion between multiple electrons surrounding the nucleus. Modern quantum mechanics has led to a more rigorous and

generalizable study of the electronic structure of atoms. The most important difference between Bohr's model and the modern quantum mechanical model is that Bohr postulated that electrons follow a clearly defined circular pathway or orbit at a fixed distance from the nucleus, whereas modern quantum mechanics has shown that this is not the case. Rather, we now understand that electrons move rapidly and are localized within regions of space around the nucleus called orbitals. The confidence by which those in Bohr's time believed they could identify the location (or pathway) of the electron was now replaced by a more modest suggestion that the best we can do is describe the probability of finding an electron within a given region of space surrounding the nucleus. In the current quantum mechanical model, it is impossible to pinpoint exactly where an electron is at any given moment in time. This is expressed best by the Heisenberg uncertainty principle: It is impossible to simultaneously determine, with perfect accuracy, the momentum and the position of an electron. If we want to assess the position of an electron, the electron has to stop (thereby removing its momentum); if we want to assess its momentum, the electron has to be moving (thereby changing its position). This can be seen visually in Figure 1.8.

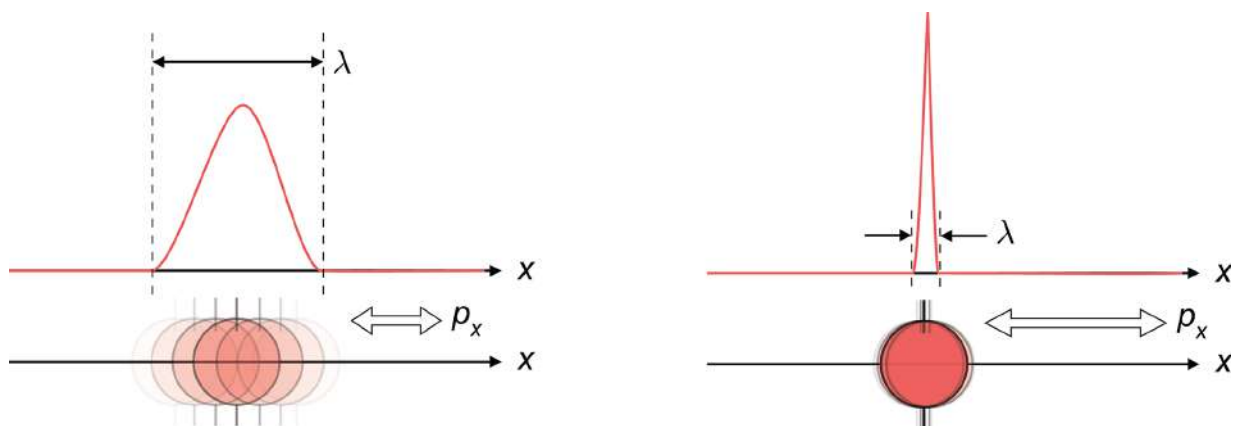


Figure 1.8. Heisenberg Uncertainty Principle

Known momentum and uncertain position (left); known position but

uncertain momentum (right). λ = confidence interval of position; p_x = confidence interval of momentum.

QUANTUM NUMBERS

Modern atomic theory postulates that any electron in an atom can be completely described by four quantum numbers: n , l , m_l , and m_s . Furthermore, according to the Pauli exclusion principle, no two electrons in a given atom can possess the same set of four quantum numbers. The position and energy of an electron described by its quantum numbers are known as its energy state. The value of n limits the values of l , which in turn limit the values of m_l . In other words, for a given value of n , only particular values of l are permissible; given a value of l , only particular values of m_l are permissible. The values of the quantum numbers qualitatively give information about the size, shape, and orientation of the orbitals. As we examine the four quantum numbers more closely, pay attention especially to l and m_l because these two tend to give students the greatest difficulty.



MCAT EXPERTISE

Think of the quantum numbers as becoming more specific as one goes from n to l to m_l to m_s . This is like an address: one lives in a particular state (n), in a particular city (l), on a particular street (m_l), at a particular house number (m_s).

Principal Quantum Number

The first quantum number is commonly known as the principal quantum number and is denoted by the letter n . This is the quantum number used in Bohr's model that can theoretically take on any positive integer value. The larger the integer value of n , the higher the energy level and radius of the electron's shell. Within each shell, there is a capacity to hold a certain number of electrons, given by:

$$\text{Maximum number of electrons within a shell} = 2n^2$$

Equation 1.6

where n is the principal quantum number. The difference in energy between two shells decreases as the distance from the nucleus increases

because the energy difference is a function of $\left[\frac{1}{n_i^2} - \frac{1}{n_f^2} \right]$. For

example, the energy difference between the $n = 3$ and the $n = 4$ shells

$\left(\frac{1}{9} - \frac{1}{16} \right)$ is less than the energy difference between the $n = 1$ and the $n = 2$ shells $\left(\frac{1}{1} - \frac{1}{4} \right)$. This can be seen in Figure 1.7. Remember that

electrons do not travel in precisely defined orbits; it just simplifies the visual representation of the electrons' motion.



BRIDGE

Remember, a larger integer value for the principal quantum number indicates a larger radius and higher energy. This is similar to gravitational potential energy, as discussed in Chapter 2 of MCAT Physics Review, where the higher or farther the object is above the Earth, the higher its potential energy will be.

Azimuthal Quantum Number

The second quantum number is called the azimuthal (angular momentum) quantum number and is designated by the letter l . The second quantum number refers to the shape and number of subshells within a given principal energy level (shell). The azimuthal quantum number is very important because it has important implications for chemical bonding and bond angles. The value of n limits the value of l in the following way: for any given value of n , the range of possible values for l is 0 to $(n - 1)$. For example, within the first principal energy level, $n = 1$, the only possible value for l is 0; within the second principal energy level, $n = 2$, the possible values for l are 0 and 1. A simpler way to remember this relationship is that the n -value also tells you the number of possible subshells. Therefore, there's only one subshell ($l = 0$) in the first principal energy level; there are two subshells ($l = 0$ and 1) within the second principal energy level; there are three subshells ($l = 0, 1$, and 2) within the third principal energy level, and so on.



KEY CONCEPT

For any principal quantum number n , there will be n possible values for l , ranging from 0 to $(n - 1)$.

Spectroscopic notation refers to the shorthand representation of the principal and azimuthal quantum numbers. The principal quantum number remains a number, but the azimuthal quantum number is designated by a letter: the $l = 0$ subshell is called s ; the $l = 1$ subshell is called p ; the $l = 2$ subshell is called d ; and the $l = 3$ subshell is called f . Thus, an electron in the shell $n = 4$ and subshell $l = 2$ is said to be in the $4d$ subshell. The spectroscopic notation for each subshell is demonstrated in Figure 1.9.

s-block																		p-block	
1																		18	
← 1s	2																	1s →	
← 2s →																		← 2p →	
← 3s →																		← 3p →	
		3	4	5	6	7	8	9	10	11	12								
← 4s →		3d										4p							
← 5s →		4d										5p							
← 6s →		5d										6p							
← 7s →		6d										7p							

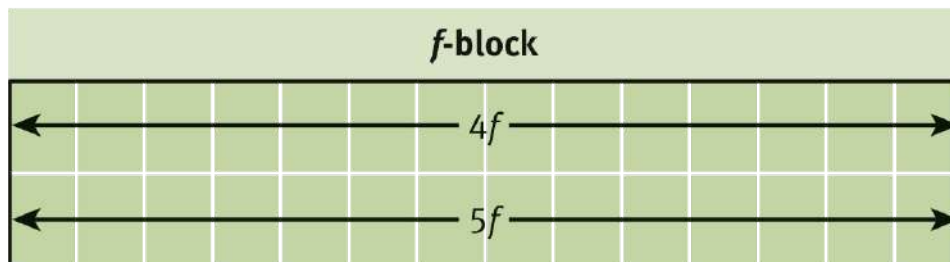


Figure 1.9. Spectroscopic Notation for Every Subshell on the Periodic Table

Within each subshell, there is a capacity to hold a certain number of electrons, given by:

$$\text{Maximum number of electrons within a subshell} = 4l + 2$$

Equation 1.7

where l is the azimuthal quantum number. The energies of the subshells increase with increasing l value; however, the energies of subshells from different principal energy levels may overlap. For example, the 4s subshell will have a lower energy than the 3d subshell.

Figure 1.10 provides an example of computer-generated probability maps of the first few electron clouds in a hydrogen atom. This provides a rough visual representation of the shapes of different subshells.

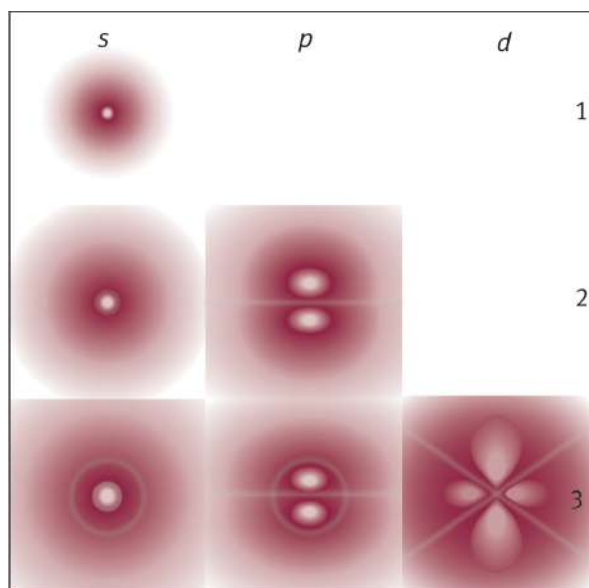


Figure 1.10. Electron Clouds of Various Subshells

Magnetic Quantum Number

The third quantum number is the magnetic quantum number and is designated m_l . The magnetic quantum number specifies the particular orbital within a subshell where an electron is most likely to be found at a given moment in time. Each orbital can hold a maximum of two electrons. The possible values of m_l are the integers between $-l$ and $+l$, including 0. For example, the s subshell, with $l = 0$, limits the possible m_l values to 0, and because there is a single value of m_l , there is only one orbital in the s subshell. The p subshell, with $l = 1$, limits the possible m_l values to -1 , 0, and $+1$, and because there are three values for m_l , there are three orbitals in the p subshell. The d subshell has five orbitals (-2 to $+2$), and the f subshell has seven orbitals (-3 to $+3$). The shape of the orbitals, like the number of orbitals, is dependent on the subshell in which they are found. The orbitals in the s subshell are spherical, while the three orbitals in the p subshell are each dumbbell-shaped and align along the x-, y-, and z-axes.

In fact, the p-orbitals are often referred to as p_x , p_y , and p_z . The first five orbitals— $1s$, $2s$, $2p_x$, $2p_y$, and $2p_z$ —are demonstrated in Figure 1.11. Note the similarity to the images in Figure 1.10.

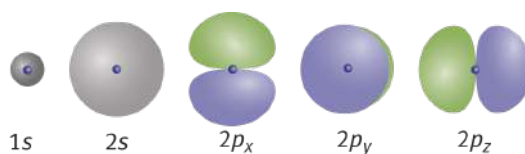


Figure 1.11. The First Five Atomic Orbitals



KEY CONCEPT

For any value of l , there will be $2l + 1$ possible values for m_l . For any n , this produces n^2 orbitals. For any value of n , there will be a maximum of $2n^2$ electrons (two per orbital).

The shapes of the orbitals in the d and f subshells are much more complex, and the MCAT will not expect you to answer questions about their appearance. The shapes of orbitals are defined in terms of a concept called probability density, the likelihood that an electron will be found in a particular region of space.

Take a look at the $2p$ block in the periodic table. As mentioned above, $2p$ contains three orbitals. If each orbital can contain two electrons, then six electrons can be added during the course of filling the $2p$ -orbitals. As atomic number increases, so too does the number of electrons (assuming the species is neutral). Therefore, it should be no surprise that the p block contains six groups of elements. The s block contains two elements in each

row of the periodic table, the d block contains ten elements, and the f block contains fourteen elements.

Spin Quantum Number

The fourth quantum number is called the spin quantum number and is denoted by m_s . In classical mechanics, an object spinning about its axis has an infinite number of possible values for its angular momentum. However, this does not apply to the electron, which has two spin orientations designated $+\frac{1}{2}$ and $-\frac{1}{2}$. Whenever two electrons are in the same orbital, they must have opposite spins. In this case, they are often referred to as being paired. Electrons in different orbitals with the same m_s values are said to have parallel spins.

The quantum numbers for the orbitals in the second principal energy level, with their maximum number of electrons noted in parentheses, are shown in Table 1.2.

n	2 (8)			
l	0 (2)		1 (6)	
m_l	0 (2)	+1 (2)	0 (2)	-1 (2)
m_s	$+\frac{1}{2}, -\frac{1}{2}$	$+\frac{1}{2}, -\frac{1}{2}$	$+\frac{1}{2}, -\frac{1}{2}$	$+\frac{1}{2}, -\frac{1}{2}$

Table 1.2. Quantum Numbers for the Second Principal Energy Level

ELECTRON CONFIGURATIONS

For a given atom or ion, the pattern by which subshells are filled, as well as the number of electrons within each principal energy level and subshell, are designated by its electron configuration. Electron configurations use spectroscopic notation, wherein the first number denotes the principal energy level, the letter designates the subshell, and the superscript gives the number of electrons in that subshell. For example, $2p^4$ indicates that there are four electrons in the second (p) subshell of the second principal energy level. This also implies that the energy levels below 2p (that is, 1s and 2s) have already been filled, as shown in Figure 1.12.

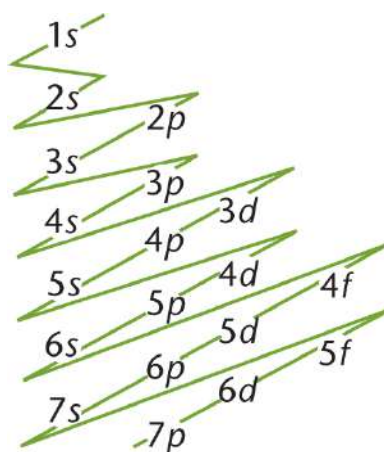


Figure 1.12. Electron Subshell Flow Diagram



MCAT EXPERTISE

Remember that the shorthand used to describe the electron configuration is derived directly from the quantum numbers.

To write out an atom's electron configuration, one needs to know the order in which subshells are filled. Electrons fill from lower- to higher-energy

subshells, according to the Aufbau principle (also called the building-up principle), and each subshell will fill completely before electrons begin to enter the next one. The order need not be memorized because there are two very helpful ways of recalling this. The $n + l$ rule can be used to rank subshells by increasing energy. This rule states that the lower the sum of the values of the first and second quantum numbers, $n + l$, the lower the energy of the subshell. This is a helpful rule to remember for Test Day. If two subshells possess the same $n + l$ value, the subshell with the lower n value has a lower energy and will fill with electrons first.

Example:

Which will fill first, the 5d subshell or the 6s subshell?

Solution:

For 5d, $n = 5$ and $l = 2$, so $n + l = 7$. For 6s, $n = 6$ and $l = 0$, so $n + l = 6$. Therefore, the 6s subshell has lower energy and will fill first.

An alternative way to approach electron configurations is through simply reading the periodic table. One must remember that the lowest s subshell is 1s, the lowest p subshell is 2p, the lowest d subshell is 3d, and the lowest f subshell is 4f. This can be seen in Figure 1.9. Then, we can simply read across the periodic table to get to the element of interest, filling subshells along the way. To do this, we must know the correct position of the lanthanide and actinide series (the f block), as shown in Figure 1.13. In most representations of the periodic table, the f block is pulled out and placed below the rest of the table. This is purely an effect of graphic design

—placing the f block in its correct location results in a lot of excess white space on a page.

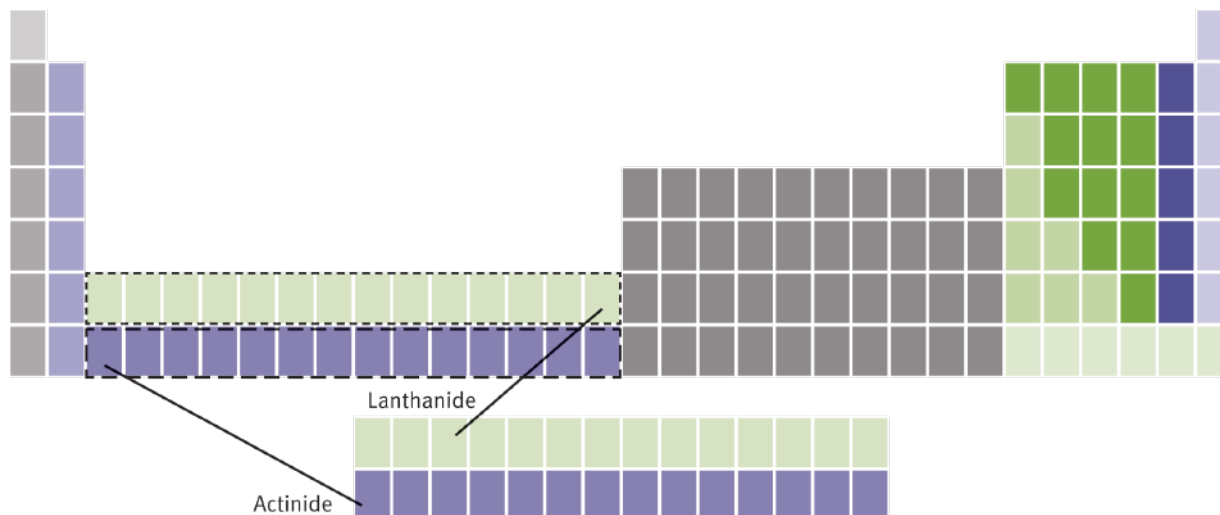


Figure 1.13. Periodic Table with Lanthanide and Actinide Series Inserted
The f block fits between the s block and d block in the periodic table.



MCAT EXPERTISE

Many general chemistry courses teach the flow diagram in Figure 1.12 as a method to determine the order of subshell filling in electron configurations. However, on Test Day, it can be both time-consuming and error-prone, resulting in incorrect electron configurations. Learning to read the periodic table, as described here, is the best method.

Electron configurations can be abbreviated by placing the noble gas that precedes the element of interest in brackets. For example, the electron

configuration of any element in period four (starting with potassium) can be abbreviated by starting with [Ar].

Example:

What is the electron configuration of osmium ($Z = 76$)?

Solution:

The noble gas that comes just before osmium is xenon ($Z = 54$). Therefore, the electron configuration can begin with [Xe]. Continuing across the periodic table, we pass through the 6s subshell (cesium and barium), the 4f subshell (the lanthanide series; remember its position on the periodic table!), and into the 5d subshell. Osmium is the sixth element in the 5d subshell, so the configuration is [Xe] $6s^2 4f^{14} 5d^6$.

This method works for neutral atoms, but how does one write the electron configuration of an ion? Negatively charged ions (anions) have additional electrons that fill according to the same rules as above; for example, if fluorine's electron configuration is [He] $2s^2 2p^5$, then F^- is [He] $2s^2 2p^6$. Positively charged ions (cations) are a bit more complicated: start with the neutral atom, and remove electrons from the subshells with the highest value for n first. If multiple subshells are tied for the highest n value, then electrons are removed from the subshell with the highest l value among these.

Example:

What is the electron configuration of Fe^{3+} ?

Solution:

The electron configuration of iron is $[\text{Ar}] 4s^2 3d^6$. Electrons are removed from the 4s subshell before the 3d subshell because it has a higher principal quantum number. Therefore, Fe^{3+} has a configuration of $[\text{Ar}] 3d^5$, not $[\text{Ar}] 4s^2 3d^3$.

HUND'S RULE

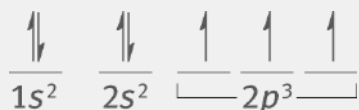
In subshells that contain more than one orbital, such as the 2p subshell with its three orbitals, the orbitals will fill according to Hund's rule, which states that, within a given subshell, orbitals are filled such that there are a maximum number of half-filled orbitals with parallel spins. Like finding a seat on a crowded bus, electrons would prefer to have their own seat (orbital) before being forced to double up with another electron. Of course, the basis for this preference is electron repulsion: electrons in the same orbital tend to be closer to each other and thus repel each other more than electrons placed in different orbitals.

Example:

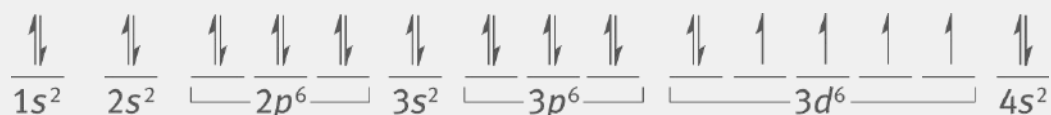
According to Hund's rule, what are the orbital diagrams for nitrogen and iron?

Solution:

Nitrogen has an atomic number of 7. Thus, its electron configuration is $1s^2 2s^2 2p^3$. According to Hund's rule, the two s-orbitals will fill completely, while the three p-orbitals will each contain one electron, all with parallel spins.



Iron has an atomic number of 26. As determined earlier, its electron configuration is $[\text{Ar}] 4s^2 3d^6$. The electrons will fill all of the subshells except for the 3d, which will contain four orbitals with parallel (upward) spin and one orbital with electrons of both spin directions.



Subshells may be listed either in the order in which they fill (4s before 3d) or with subshells of the same principal quantum number grouped together, as shown here. Both methods are correct.

An important corollary from Hund's rule is that half-filled and fully filled orbitals have lower energies (higher stability) than other states. This creates two notable exceptions to electron configuration that are often tested on the MCAT: chromium (and other elements in its group) and copper (and other elements in its group). Chromium ($Z = 24$) should have

the electron configuration $[\text{Ar}] 4s^2 3d^4$ according to the rules established earlier. However, moving one electron from the 4s subshell to the 3d subshell allows the 3d subshell to be half-filled: $[\text{Ar}] 4s^1 3d^5$ (remember that s subshells can hold two electrons and d subshells can hold ten). While moving the 4s electron up to the 3d-orbital is energetically unfavorable, the extra stability from making the 3d subshell half-filled outweighs that cost. Similarly, copper ($Z = 29$) has the electron configuration $[\text{Ar}] 4s^1 3d^{10}$, rather than $[\text{Ar}] 4s^2 3d^9$; a full d subshell outweighs the cost of moving an electron out of the 4s subshell. Other elements in the same group have similar behavior, moving one electron from the highest s subshell to the highest d subshell. Similar shifts can be seen with f subshells, but they are never observed for the p subshell; the extra stability doesn't outweigh the cost.

The presence of paired or unpaired electrons affects the chemical and magnetic properties of an atom or molecule. Materials composed of atoms with unpaired electrons will orient their spins in alignment with a magnetic field, and the material will thus be weakly attracted to the magnetic field. These materials are considered paramagnetic. An example is shown in Figure 1.14 where a set of iron orbs is influenced by a magnet. The metallic spheres that are close enough to be induced by the magnet are attracted to the magnet and move toward it.



Figure 1.14. Attraction of Paramagnetic Iron Spheres to a Magnet



MNEMONIC

Remember that paramagnetic means that a magnetic field will cause parallel spins in unpaired electrons and therefore cause an attraction.

Materials consisting of atoms that have only paired electrons will be slightly repelled by a magnetic field and are said to be diamagnetic. In Figure 1.15, a piece of pyrolytic graphite is suspended in the air over strong neodymium magnets. All the electrons in this allotrope (configuration) of carbon are paired because of covalent bonding between layers of the material, and are thus opposed to being reoriented. Given sufficiently strong magnetic fields beneath an object, any diamagnetic substance can be made to levitate.

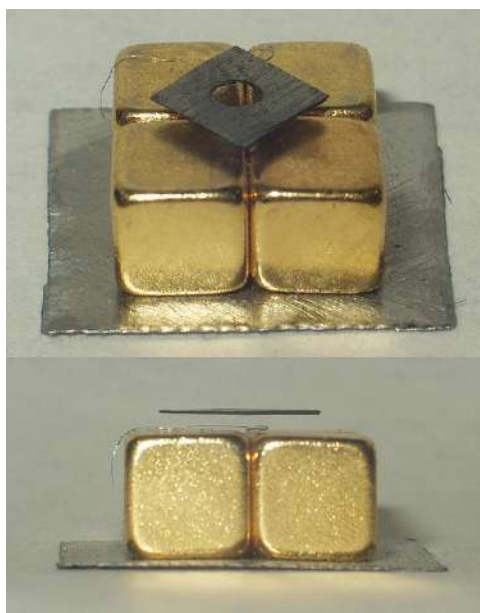


Figure 1.15. Diamagnetic Pyrolytic Graphite



REAL WORLD

The concept behind “maglev” or magnetic levitation is no longer science fiction. Using powerful magnetic fields and strongly diamagnetic materials, some transportation systems have developed frictionless, high speed rail networks such as Japan’s SCMaglev.

VALENCE ELECTRONS

The valence electrons of an atom are those electrons that are in its outermost energy shell, are most easily removed, and are available for bonding. In other words, the valence electrons are the “active” electrons of

an atom and to a large extent dominate the chemical behavior of the atom. For elements in Groups IA and IIA (Groups 1 and 2), only the highest s subshell electrons are valence electrons. For elements in Groups IIIA through VIIIA (Groups 13 through 18), the highest s and p subshell electrons are valence electrons. For transition elements, the valence electrons are those in the highest s and d subshells, even though they have different principal quantum numbers. For the lanthanide and actinide series, the valence electrons are those in the highest s and f subshells, even though they have different principal quantum numbers. All elements in period three (starting with sodium) and below may accept electrons into their d subshell, which allows them to hold more than eight electrons in their valence shell. This allows them to violate the octet rule, as discussed in Chapter 3 of MCAT General Chemistry Review.



MCAT EXPERTISE

The valence electron configuration of an atom helps us understand its properties and is ascertainable from the periodic table (the only “cheat sheet” available on the MCAT!). On Test Day, you will be able to access a periodic table by clicking on the button labeled “Periodic Table” on the bottom left of the screen. Use it as needed!

Example:

Which electrons are the valence electrons of elemental vanadium, elemental selenium, and the sulfur atom in a sulfate

ion?

Solution:

Vanadium has five valence electrons: two in its 4s subshell and three in its 3d subshell.

Selenium has six valence electrons: two in its 4s subshell and four in its 4p subshell. Selenium's 3d electrons are not part of its valence shell.

Sulfur in a sulfate ion has 12 valence electrons: its original six plus six more from the oxygens to which it is bonded. Sulfur's 3s and 3p subshells can contain only eight of these 12 electrons; the other four electrons have entered the sulfur atom's 3d subshell, which is normally empty in elemental sulfur.

MCAT Concept Check 1.4:

Before you move on, assess your understanding of the material with these questions.

If given the following quantum numbers, which element(s) do they likely refer to? (Note: Assume that these quantum numbers describe the valence electrons in the element.)

n	l	Possible Elements
2	1	
3	0	
5	3	
4	2	

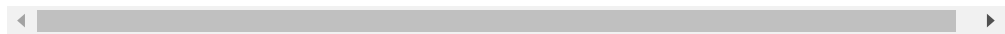


Write out and compare an orbital diagram for a neutral oxygen (O) atom and an O^{2-} ion.

Magnetic resonance angiography (MRA) is a technique that can resolve defects like stenotic (narrowed) arteries. A contrast agent like gadolinium or manganese injected into the blood stream interacts with the strong magnetic fields of the MRI device to produce such images. Based on their orbital configurations, are these contrast agents paramagnetic or diamagnetic?

Determine how many valence electrons come from each subshell in the following atoms:

Atom	s-electrons	p-electrons	d-electrons	f-electrons	Total Valence Electrons
P in PO_4^{3-}					
O in PO_4^{3-}					
Ir					
Cf					



Conclusion

Congratulations! You've made it through the first chapter! Now that we have covered topics related to the most fundamental unit of matter—the atom—you're set to advance your understanding of the physical world in more complex ways. This chapter described the characteristics and behavior of the three subatomic particles: the proton, neutron, and electron. In addition, it compared and contrasted two models of the atom. The Bohr model is adequate for describing the structure of one-electron systems, such as the hydrogen atom or the helium ion, but fails to adequately describe the structure of more complex atoms. The quantum mechanical model theorizes that electrons are found not in discrete orbits, but in “clouds of probability,” or orbitals, by which we can predict the likelihood of finding electrons within given regions of space surrounding the nucleus. Both theories tell us that the energy levels available to electrons are not infinite but discrete and that the energy difference between levels is a precise amount called a quantum. The four quantum numbers completely describe the location and energy of any electron within a given atom. Finally, we learned two simple recall methods for the order in which electrons fill the shells and subshells of an atom and that the valence electrons are the reactive electrons in an atom. In the next chapter, we'll take a look at how the elements are organized on the periodic table and will then turn our attention to their bonding behavior—based on valence electrons—in Chapter 3 of MCAT General Chemistry Review.

Concept Summary

Subatomic Particles

- A proton has a positive charge and mass around 1 amu; a neutron has no charge and mass around 1 amu; an electron has a negative charge and negligible mass.
- The nucleus contains the protons and neutrons, while the electrons move around the nucleus.
- The atomic number is the number of protons in a given element.
- The mass number is the sum of an element's protons and neutrons.

Atomic Mass vs. Atomic Weight

- Atomic mass is essentially equal to the mass number, the sum of an element's protons and neutrons.
 - Isotopes are atoms of a given element (same atomic number) that have different mass numbers. They differ in the number of neutrons.
 - Most isotopes are identified by the element followed by the mass number (such as carbon-12, carbon-13, and carbon-14).
 - The three isotopes of hydrogen go by different names: protium, deuterium, and tritium.

- Atomic weight is the weighted average of the naturally occurring isotopes of an element. The periodic table lists atomic weights, not atomic masses.

Rutherford, Planck, and Bohr

- Rutherford first postulated that the atom had a dense, positively charged nucleus that made up only a small fraction of the volume of the atom.
- In the Bohr model of the atom, a dense, positively charged nucleus is surrounded by electrons revolving around the nucleus in orbits with distinct energy levels.
- The energy difference between energy levels is called a quantum, first described by Planck.
 - Quantization means that there is not an infinite range of energy levels available to an electron; electrons can exist only at certain energy levels. The energy of an electron increases the farther it is from the nucleus.
 - The atomic absorption spectrum of an element is unique; for an electron to jump from a lower energy level to a higher one, it must absorb an amount of energy precisely equal to the energy difference between the two levels.
 - When electrons return from the excited state to the ground state, they emit an amount of energy that is exactly equal to the energy difference between the two levels; every element has a characteristic atomic emission spectrum, and sometimes the electromagnetic energy emitted corresponds to a frequency in the visible light range.

Quantum Mechanical Model of Atoms

- The quantum mechanical model posits that electrons do not travel in defined orbits but rather are localized in orbitals; an orbital is a region of space around the nucleus defined by the probability of finding an electron in that region of space.
- The Heisenberg uncertainty principle states that it is impossible to know both an electron's position and its momentum exactly at the same time.
- There are four quantum numbers; these numbers completely describe any electron in an atom.
 - The principal quantum number, n , describes the average energy of a shell.
 - The azimuthal quantum number, l , describes the subshells within a given principal energy level (s, p, d, and f).
 - The magnetic quantum number, m_l , specifies the particular orbital within a subshell where an electron is likely to be found at a given moment in time.
 - The spin quantum number, m_s , indicates the spin orientation $\left(\pm \frac{1}{2}\right)$ of an electron in an orbital.
- The electron configuration uses spectroscopic notation (combining the n and l values as a number and letter, respectively) to designate the location of electrons.
 - For example, $1s^2 2s^2 2p^6 3s^2$ is the electron configuration for magnesium: a neutral magnesium atom has 12 electrons—two in the s subshell of the first energy level, two in the s subshell of the second

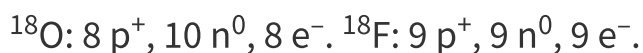
energy level, six in the p subshell of the second energy level, and two in the s subshell of the third energy level; the two electrons in the 3s subshell are the valence electrons for the magnesium atom.

- Electrons fill the principal energy levels and subshells according to increasing energy, which can be determined by the $n + l$ rule.
- Electrons fill orbitals according to Hund's rule, which states that subshells with multiple orbitals (p, d, and f) fill electrons so that every orbital in a subshell gets one electron before any of them gets a second.
 - Paramagnetic materials have unpaired electrons that align with magnetic fields, attracting the material to a magnet.
 - Diamagnetic materials have all paired electrons, which cannot easily be realigned; they are repelled by magnets.
- Valence electrons are those electrons in the outermost shell available for interaction (bonding) with other atoms.
 - For the representative elements (those in Groups 1, 2, and 13–18), the valence electrons are found in s- and/or p-orbitals.
 - For the transition elements, the valence electrons are found in s- and either d- or f-orbitals.
 - Many atoms interact with other atoms to form bonds that complete an octet in the valence shell.

Answers to Concept Checks

1.1

Charge is determined by the number of electrons present. Atomic number is determined by the number of protons. Isotope is determined by the number of neutrons (while protons make up part of the mass number, it is the number of neutrons that explains the variability between isotopes).



1.2

Atomic mass is (just slightly less than) the sum of the masses of protons and neutrons in a given atom of an element. Atoms of the same element with different mass numbers are isotopes of each other. The atomic weight is the weighted average of the naturally occurring isotopes of an element.

This ratio is an equivalent concept. It is therefore acceptable, as long as units can be cancelled in dimensional analysis.

Isotope	Protons	Neutrons	Electrons
^{19}O	8	11	8
^{16}O	8	8	8

Isotope	Protons	Neutrons	Electrons
^{17}O	8	9	8
^{19}F	9	10	9
^{16}F	9	7	9
^{238}U	92	146	92
^{240}U	92	148	92

1.3

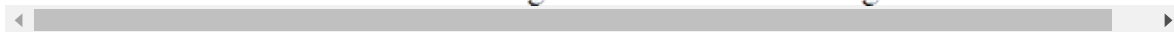
$$E = -R_{\text{H}} \left[\frac{1}{n_i^2} - \frac{1}{n_f^2} \right] = -2.18 \times 10^{-18} \left[\frac{1}{2^2} - \frac{1}{4^2} \right] \approx -2 \times 10^{-18} \left[\frac{3}{16} \right]$$

$$\approx -3.75 \times 10^{-19} \text{ J} \approx -4 \times 10^{-19} \text{ J} \text{ (actual value is } -4.09 \times 10^{-19} \text{ J)}.$$

$$E = -R_{\text{H}} \left[\frac{1}{n_i^2} - \frac{1}{n_f^2} \right] = -13.6 \left[\frac{1}{2^2} - \frac{1}{4^2} \right] \approx -(14) \left[\frac{3}{16} \right]$$

$$= -\left(\frac{14 \times 3}{16} \right) = -\left(\frac{7}{8} \times 3 \right) = -\left(\frac{21}{8} \right) \approx -2.5 \text{ (actual value is } -2.55 \text{ eV)}.$$

In both cases, the value is negative, indicating that energy is absorbed. This consistent with an electron moving from a lower to a higher shell.



$$3 \text{ eV} \left(1.60 \times 10^{-19} \frac{\text{J}}{\text{eV}}\right) = 4.8 \times 10^{-19} \text{ J}$$

$$E = \frac{hc}{\lambda} \rightarrow \lambda = \frac{hc}{E} = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s}) \left(3.00 \times 10^8 \frac{\text{m}}{\text{s}}\right)}{(4.8 \times 10^{-19} \text{ J})}$$

$$\approx \frac{(6.6 \times 3)(10^{-34} \times 10^8)}{4.8 \times 10^{-19}} = \frac{6.6 \times 10^{-26}}{1.6 \times 10^{-19}} \approx \frac{6.4 \times 10^{-26}}{1.6 \times 10^{-19}} = \frac{64}{16} \times 10^{-7}$$

$$= 4 \times 10^{-7} \text{ m} = 400 \text{ nm (actual value} = 4.14 \times 10^{-7} = 414 \text{ nm)}$$

$$E = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s}) \left(3.00 \times 10^8 \frac{\text{m}}{\text{s}}\right)}{(6.62 \times 10^{-7} \text{ m})}$$

$$= \frac{(6.626 \times 3.00)(10^{-34} \times 10^8)}{6.62 \times 10^{-7}} = \frac{3.00 \times 10^{-26}}{10^{-7}} = 3.00 \times 10^{-19} \text{ J}$$

1.4

n	l	Possible Elements
2	1	2p: B, C, N, O, F, Ne
3	0	3s: Na, Mg
5	3	5f: Actinide series
4	2	4d: Y, Zr, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Cd

Both O and O²⁻ have fully filled 1s- and 2s-orbitals. O has four electrons in the 2p subshell; two are paired, and the other two each

have their own orbital. O^{2-} has six electrons in the 2p subshell, all of which are paired in the three p-orbitals.

Both these molecules have unfilled valence electron shells with relatively few paired electrons; therefore, they are paramagnetic.

Atom	s-electrons	p-electrons	d-electrons	f-electrons	Total Valence Electrons
P in PO_4^{3-}	2	6	2	0	10
O in PO_4^{3-}	2	6	0	0	8
Ir	2	0	7	0	9
Cf	2	0	0	10	12

Equations to Remember

(1.1) Planck relation (frequency): $E = hf$

(1.2) Angular momentum of an electron (Bohr model): $L = \frac{nh}{2\pi}$

(1.3) Energy of an electron (Bohr model): $E = -\frac{R_H}{n^2}$

(1.4) Planck relation (wavelength): $E = \frac{hc}{\lambda}$

(1.5) Energy of electron transition (Bohr model):

$$E = -R_H \left[\frac{1}{n_i^2} - \frac{1}{n_f^2} \right]$$

(1.6) Maximum number of electrons within a shell: $2n^2$

(1.7) Maximum number of electrons within a subshell: $4l + 2$



Shared Concepts

General Chemistry Chapter 2

The Periodic Table

General Chemistry Chapter 3

Bonding and Chemical Interactions

Organic Chemistry Chapter 3

Bonding

Physics and Math Chapter 2

Work and Energy

Physics and Math Chapter 8

Light and Optics

Physics and Math Chapter 9

Atomic and Nuclear Phenomena

Discrete Practice Questions

Consult your online resources for additional practice.

1. Which of the following is the correct electron configuration for Zn^{2+} ?

- (A) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^0 3d^{10}$
- (B) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$
- (C) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$
- (D) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^0 3d^8$

2. Which of the following quantum number sets is possible?

- (A) $n = 2; l = 2; m_l = 1; m_s = +\frac{1}{2}$
- (B) $n = 2; l = 1; m_l = -1; m_s = +\frac{1}{2}$
- (C) $n = 2; l = 0; m_l = -1; m_s = -\frac{1}{2}$
- (D) $n = 2; l = 0; m_l = 1; m_s = -\frac{1}{2}$

3. What is the maximum number of electrons allowed in a single atomic energy level in terms of the principal quantum number n ?

- (A) $2n$
- (B) $2n + 2$
- (C) $2n^2$
- (D) $2n^2 + 2$

4. Which of the following equations describes the maximum number of electrons that can fill a subshell?

- (A) $2l + 2$
- (B) $4l + 2$
- (C) $2l^2$
- (D) $2l^2 + 2$

5. Which of the following atoms only has paired electrons in its ground state?

- (A) Sodium
- (B) Iron
- (C) Cobalt
- (D) Helium

6. An electron returns from an excited state to its ground state, emitting a photon at $\lambda = 500 \text{ nm}$. What would be the magnitude of

the energy change if one mole of these photons were emitted?

(Note: $h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$, $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$)

- (A) $3.98 \times 10^{-21} \text{ J}$
- (B) $3.98 \times 10^{-19} \text{ J}$
- (C) $2.39 \times 10^3 \text{ J}$
- (D) $2.39 \times 10^5 \text{ J}$

7. Suppose an electron falls from $n = 4$ to its ground state, $n = 1$. Which of the following effects is most likely?

- (A) A photon is absorbed.
- (B) A photon is emitted.
- (C) The electron moves into a p-orbital.
- (D) The electron moves into a d-orbital.

8. Which of the following isotopes of carbon is LEAST likely to be found in nature?

- (A) ${}^6\text{C}$
- (B) ${}^{12}\text{C}$
- (C) ${}^{13}\text{C}$
- (D) ${}^{14}\text{C}$

9. Which of the following best explains the inability to measure position and momentum exactly and simultaneously according to

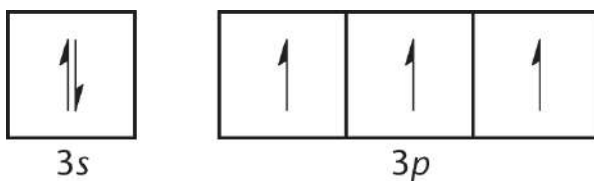
the Heisenberg uncertainty principle?

- (A) Imprecision in the definition of the meter and kilogram
- (B) Limits on accuracy of existing scientific instruments
- (C) Error in one variable is increased by attempts to measure the other
- (D) Discrepancies between the masses of nuclei and of their component particles

10. Which of the following electronic transitions would result in the greatest gain in energy for a single hydrogen electron?

- (A) An electron moves from $n = 6$ to $n = 2$.
- (B) An electron moves from $n = 2$ to $n = 6$.
- (C) An electron moves from $n = 3$ to $n = 4$.
- (D) An electron moves from $n = 4$ to $n = 3$.

11. Suppose that an atom fills its orbitals as shown:



Such an electron configuration most clearly illustrates which of the following laws of atomic physics?

- (A) Hund's rule
- (B) Heisenberg uncertainty principle
- (C) Bohr model
- (D) Rutherford model

12. How many total electrons are in a ^{133}Cs cation?

- (A) 54
- (B) 55
- (C) 78
- (D) 132

13. The atomic weight of hydrogen is 1.008 amu. What is the percent composition of hydrogen by isotope, assuming that hydrogen's only isotopes are ^1H and ^2D ?

- (A) 92% H, 8% D
- (B) 99.2% H, 0.8% D
- (C) 99.92% H, 0.08% D
- (D) 99.992% H, 0.008% D

14. Consider the two sets of quantum numbers shown in the table, which describe two different electrons in the same atom.

n	l	m _l	m _s
2	1	1	$+\frac{1}{2}$
3	1	-1	$+\frac{1}{2}$

Which of the following terms best describes these two electrons?

- (A) Parallel
- (B) Opposite
- (C) Antiparallel
- (D) Paired

15. Which of the following species is represented by the electron configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$?

- I. Cr
- II. Mn^+
- III. Fe^{2+}

- (A) I only
- (B) I and II only
- (C) II and III only
- (D) I, II, and III

Explanations to Discrete Practice Questions

1. A

Remember that when electrons are removed from an element, forming a cation, they will be removed from the subshell with the highest n value first. Zn^0 has 30 electrons, so it would have an electron configuration of $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$. The 4s subshell has the highest principal quantum number, so it is emptied first, forming $1s^2 2s^2 2p^6 3s^2 3p^6 4s^0 3d^{10}$. (B) implies that electrons are pulled out of the d-orbital, (C) presents the configuration of the uncharged zinc atom, and (D) shows the configuration that would exist if four electrons were removed.

2. B

The azimuthal quantum number l cannot be higher than $n - 1$, ruling out (A). The m_l number, which describes the chemical's magnetic properties, can only be an integer value between $-l$ and l . It cannot be equal to ± 1 if $l = 0$; this would imply that an s-orbital has three subshells (-1 , 0 , and 1) when we know it can only have one. This rules out (C) and (D).

3. C

For any value of n , there will be a maximum of $2n^2$ electrons; that is, two per orbital. This can also be determined from the periodic table. There

are only two elements (H and He) that have valence electrons in the $n = 1$ shell. Eight elements (Li to Ne) have valence electrons in the $n = 2$ shell. This is the only equation that matches this pattern.

4. B

This formula describes the number of electrons in terms of the azimuthal quantum number l , which ranges from 0 to $n - 1$, with n being the principal quantum number. A table of the maximum number of electrons per subshell is provided here:

Subshell	Azimuthal Quantum Number (l)	Number of Electrons
s	0	2
p	1	6
d	2	10
f	3	14

5. D

The only answer choice without unpaired electrons in its ground state is helium. Recall from the chapter that a diamagnetic substance is identified by the lack of unpaired electrons in its shell. A substance without unpaired electrons, like helium, cannot be magnetized by an external magnetic field and is actually slightly repelled. Elements that come at the end of a block (Group IIA, the group containing Zn, and the noble gases, most notably) have only paired electrons.

6. D

The problem requires the MCAT favorite equation $E = \frac{hc}{\lambda}$, where $h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$ (Planck's constant), $c = 3.00 \times 10^8 \frac{\text{m}}{\text{s}}$ is the speed of light, and λ is the wavelength of the light. This question asks for the energy of one mole of photons, so we must multiply by Avogadro's number, $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$.

The setup is: $E = \frac{hc}{\lambda} \times N_A$

$$\begin{aligned}
 &= \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s}) \left(3.00 \times 10^8 \frac{\text{m}}{\text{s}} \right)}{(500 \times 10^{-9} \text{ m})} \times (6.02 \times 10^{23} \text{ mol}^{-1}) \\
 &= \frac{(6.626 \times 3 \times 6.02)(10^{-34} \times 10^8 \times 10^{23})}{500 \times 10^{-9}} \\
 &\approx \frac{(7 \times 3 \times 6)(10^{-3})}{500 \times 10^{-9}} = \frac{(21 \times 6)(10^{-3})}{500 \times 10^{-9}} \\
 &\approx \frac{125}{500} \times 10^{-3+9} = \frac{1}{4} \times 10^6 = 2.5 \times 10^5 \text{ J}
 \end{aligned}$$

7. B

Because the electron is moving into the $n = 1$ shell, the only subshell available is the 1s subshell, which eliminates (C) and (D). There will be some energy change, however, as the electron must lose energy to return to the minimum-energy ground state. That will require emitting radiation in the form of a photon.

8. A

Recall that the superscript refers to the mass number of an atom, which is equal to the number of protons plus the number of neutrons present in an element. Sometimes a text will list the atomic number, Z , as a subscript under the mass number, A . According to the periodic table, carbon contains six protons; therefore, its atomic number is 6. Isotopes all have the same number of protons, but differ in the number of neutrons. Almost all atoms with $Z > 1$ have at least one neutron. Carbon is most likely to have a mass number of 12, for six protons and six neutrons, as in (B). (C) and (D) are possible isotopes that would have more neutrons than ^{12}C . The ^6C isotope is unlikely. It would mean that there are 6 protons and 0 neutrons. As shown in Figure 1.4, this would be a highly unstable isotope.

9. C

The limitations placed by the Heisenberg uncertainty principle are caused by limitations inherent in the measuring process: if a particle is moving, it has momentum, but trying to measure that momentum necessarily creates uncertainty in the position. Even if we had an exact definition of the meter, as in (A), or perfect measuring devices, as in (B), we still wouldn't be able to measure position and momentum simultaneously and exactly.

10. B

For the electron to gain energy, it must absorb energy from photons to jump up to a higher energy level. There is a bigger jump between $n = 2$ and $n = 6$ than there is between $n = 3$ and $n = 4$.

11. A

The MCAT covers the topics in this chapter qualitatively more often than quantitatively. It is critical to be able to distinguish the fundamental principles that determine electron organization, which are usually known by the names of the scientists who discovered or postulated them. The Heisenberg uncertainty principle, (B), refers to the inability to know the momentum and position of a single electron simultaneously. The Bohr model, (C), was an early attempt to describe the behavior of the single electron in a hydrogen atom. The Rutherford model, (D), described a dense, positively charged nucleus. The element shown here, phosphorus, is often used to demonstrate Hund's rule because it contains a half-filled p subshell. Hund's rule explains that electrons fill empty orbitals first before doubling up electrons in the same orbital.

12. A

The quickest way to solve this problem is to use the periodic table and find out how many protons are in Cs atoms; there are 55. Neutral Cs atoms would also have 55 electrons. A stable Cs cation will have a single positive charge because it has one unpaired s-electron. This translates to one fewer electron than the number of protons or 54 electrons.

13. B

The easiest way to approach this problem is to set up a system of two algebraic equations, where H and D are the percentages of H (mass = 1 amu) and D (mass = 2 amu), respectively. Your setup should look like the following system:

$$\begin{aligned} H + D &= 1 \text{ (percent H + percent D = 100\%)} \\ 1 H + 2 D &= 1.008 \text{ (atomic weight calculation)} \end{aligned}$$

Rearranging the first equation and substituting into the second yields $(1 - D) + 2D = 1.008$, or $D = 0.008$. 0.008 is 0.8%, so there is 0.8% D.

14. A

The terms in the answer choices refer to the magnetic spin of the two electrons. The quantum number m_s represents this property as a measure of an electron's intrinsic spin. These electrons' spins are parallel, in that their spins are aligned in the same direction ($m_s = -\frac{1}{2}$ for both species).

15. B

When dealing with ions, you cannot directly approach electronic configurations based on the number of electrons they currently hold. First examine the neutral atom's configuration, and then determine which electrons are removed.

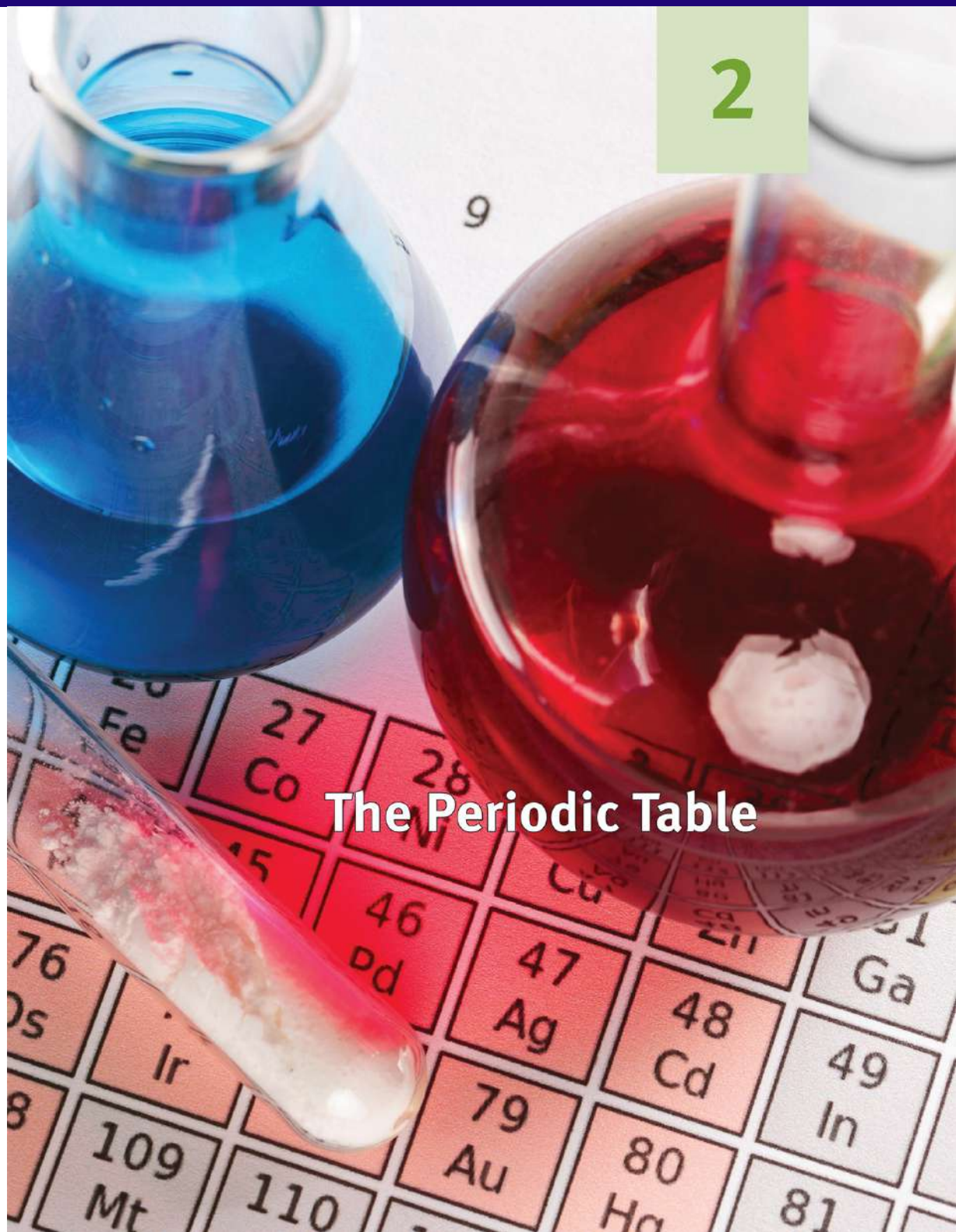
Neutral Atom's Configuration	Ion's Configuration
$\text{Cr}^0: [\text{Ar}] 4s^1 3d^5$	—
$\text{Mn}^0: [\text{Ar}] 4s^2 3d^5$	$\text{Mn}^+: [\text{Ar}] 4s^1 3d^5$
$\text{Fe}^0: [\text{Ar}] 4s^2 3d^6$	$\text{Fe}^{2+}: [\text{Ar}] 4s^0 3d^6$

Due to the stability of half-filled d-orbitals, neutral chromium assumes the electron configuration of $[\text{Ar}] 4s^1 3d^5$. Mn must lose one electron from its initial configuration to become the Mn^+ cation. That electron would come from the 4s subshell according to the rule that the first electron removed comes from the highest-energy shell. Fe must lose two electrons to become Fe^{2+} . They'll both be lost from the same orbital;

the only way Fe^{2+} could hold the configuration in the question stem would be if one d-electron and one s-electron were lost together.

2

The Periodic Table



In This Chapter

2.1 The Periodic Table 

2.2 Types of Elements 

Metals 

Nonmetals 

Metalloids 

2.3 Periodic Properties of the Elements 

Atomic and Ionic Radii 

Ionization Energy 

Electron Affinity 

Electronegativity 

2.4 The Chemistry of Groups 

Alkali Metals (IA) 

Alkaline Earth Metals (IIA) 

Chalcogens (VIA) 

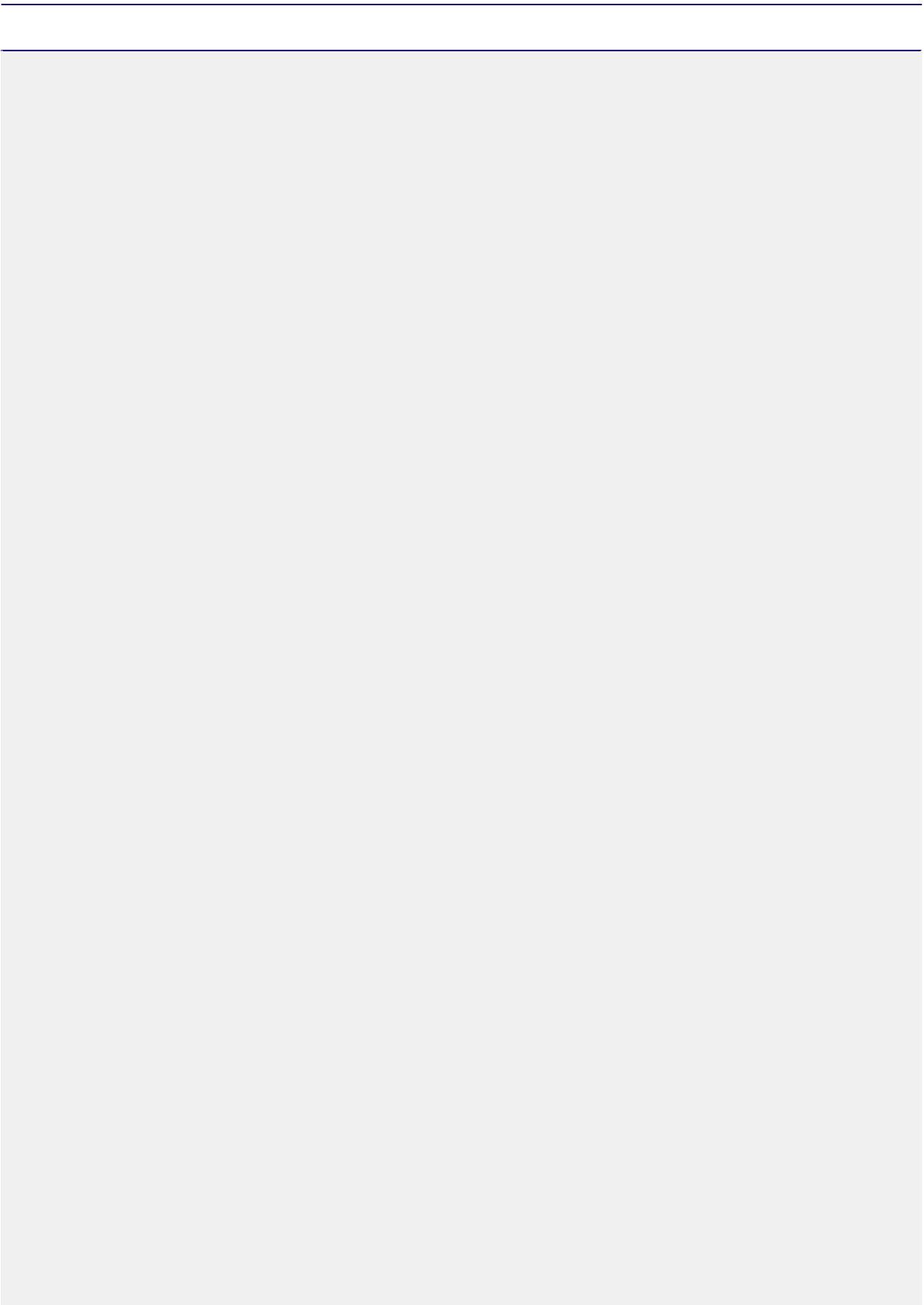
Halogens (VIIA) 

Noble Gases (VIII A) 

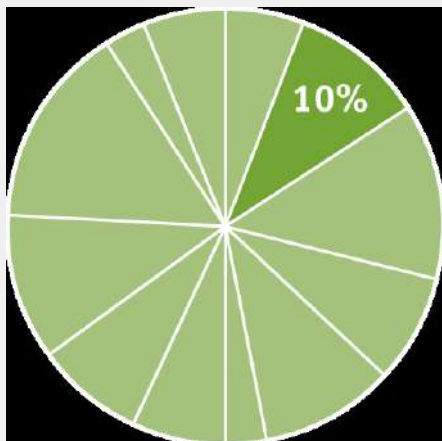
Transition Metals (B) 

Concept Summary 





Chapter Profile



The content in this chapter should be relevant to about 10% of all questions about general chemistry on the MCAT.

This chapter covers material from the following AAMC content category:

4E: Atoms, nuclear decay, electronic structure, and atomic chemical behavior

Introduction

The pharmacological history of lithium is an interesting window into the scientific and medical communities' attempts to take advantage of the chemical and physical properties of an element for human benefit. By the

mid-1800s, the medical community was showing great interest in theories that linked uric acid to a myriad of maladies. When it was discovered that solutions of lithium carbonate dissolved uric acid, therapeutic preparations containing lithium carbonate salt became popular. Even nonmedical companies tried to profit from lithium's reputation as a cure-all by adding it to their soft drinks.

Eventually, fascination with theories of uric acid wore off, and lithium's time in the spotlight seemed to be coming to an end. Then, in the 1940s, doctors began to recommend salt-restricted diets for cardiac patients. Lithium chloride was made commercially available as a sodium chloride (table salt) substitute. Unfortunately, lithium is quite toxic at fairly low concentrations, and when medical literature in the late 1940s reported several incidents of severe poisonings and multiple deaths—some associated with only minor lithium overdosing—U.S. companies voluntarily withdrew all lithium salts from the market. Right around this time, the Australian psychiatrist John Cade proposed the use of lithium salts for the treatment of mania. Cade's clinical trials were quite successful. In fact, his use of lithium salts to control mania was the first instance of successful medical treatment of a mental illness, and lithium carbonate became commonly prescribed in Europe for manic behavior. Not until 1970 did the U.S. Food and Drug Administration finally approve the use of lithium carbonate for manic symptoms.

Lithium (Li) is the element with the atomic number 3. It is a very soft alkali metal, and under standard conditions, it is the least dense solid element (specific gravity = 0.53). Lithium is so reactive that it does not naturally occur on earth in its elemental form and is found only in various salt compounds.

Why would medical scientists pay attention to this particular element?
What would make doctors believe that lithium chloride would be a good substitute for sodium chloride for patients on salt-restricted diets? The answers lie in the periodic table.

2.1 The Periodic Table

LEARNING GOALS

After Chapter 2.1, you will be able to:

- Explain how the modern periodic table is organized
- Differentiate between representative and nonrepresentative elements

In 1869, the Russian chemist Dmitri Mendeleev published the first version of his Periodic Table of the Elements, which showed that ordering the known elements according to atomic weight revealed a pattern of periodically recurring physical and chemical properties. Since then, the periodic table has been revised, using the work of physicist Henry Moseley, to organize the elements based on increasing atomic number (the number of protons in an element) rather than atomic weight. Using this revised table, many properties of elements that had not yet been discovered could be predicted. The periodic table creates a visual representation of the periodic law, which states: the chemical and physical properties of the elements are dependent, in a periodic way, upon their atomic numbers.

The modern periodic table arranges the elements into periods (rows) and groups or families (columns), based on atomic number. There are seven periods representing the principal quantum numbers $n = 1$ through $n = 7$ for the s- and p-block elements. Each period is filled sequentially, and each element in a given period has one more proton and one more electron than the element to its left (in their neutral states). Groups contain elements that have the same electronic configuration in their valence shell and share similar chemical properties.



BRIDGE

Recall from Chapter 1 of MCAT General Chemistry Review that periods (rows) graphically represent the principal quantum number, and groups (columns) help to determine the valence electron configuration.

The electrons in the valence shell, known as the valence electrons, are the farthest from the nucleus and have the greatest amount of potential energy. Their higher potential energy and the fact that they are held less tightly by the nucleus allows them to become involved in chemical bonds with the valence electrons of other atoms; thus, the valence shell electrons largely determine the chemical reactivity and properties of the element.



MCAT EXPERTISE

Relating valence electrons to reactivity is important. Elements with similar valence electron configurations generally behave in similar ways, as long as they are the same type (metal, nonmetal, or metalloid).

The Roman numeral above each group represents the number of valence electrons elements in that group have in their neutral state. The Roman numeral is combined with the letter A or B to separate the elements into two larger classes. The A elements are known as the representative elements and include groups IA through VIIIA. The elements in these groups have their valence electrons in the orbitals of either s or p subshells. The B elements are known as the nonrepresentative elements and include both the transition elements, which have valence electrons in the s and d subshells, and the lanthanide and actinide series, which have valence electrons in the s and f subshells. For the representative elements, the Roman numeral and the letter designation determine the electron configuration. For example, an element in Group VA has five valence electrons with the configuration s^2p^3 . As described in Chapter 1 of MCAT General Chemistry Review, the nonrepresentative elements may have unexpected electron configurations, such as chromium ($4s^13d^5$) and copper ($4s^13d^{10}$). In the modern IUPAC identification system, the groups are numbered 1 to 18 and are not subdivided into Group A and Group B elements.

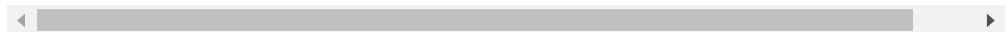
MCAT Concept Check 2.1:

Before you move on, assess your understanding of the material with these questions.

Mendeleev's table was arranged by atomic weight, but the modern periodic table is arranged by:

Which of the following are representative elements (A), and which are nonrepresentative (B)?

Element	A or B	Element	A or B	Element	A or B	Element	A or B
Ag		Al		K		P	
Pb		Li		Pu		U	
Cu		Cf		Zn		B	
N		Np		O		He	



2.2 Types of Elements

LEARNING GOALS

After Chapter 2.2, you will be able to:

- Classify elements as metal, nonmetal, or metalloid
- Predict the traits of an element given its location on a periodic table:

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
Lanthanides			57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	
Actinides			89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	

When we consider the trends of chemical reactivity and physical properties together, we can begin to identify groups of elements with similar

characteristics. These larger collections are divided into three categories: metals, nonmetals, and metalloids (also called semimetals).

METALS

Metals are found on the left side and in the middle of the periodic table. They include the active metals, the transition metals, and the lanthanide and actinide series of elements. Metals are lustrous (shiny) solids, except for mercury, which is a liquid under standard conditions. They generally have high melting points and densities, but there are exceptions, such as lithium, which has a density about half that of water. Metals have the ability to be deformed without breaking; the ability of metal to be hammered into shapes is called malleability, and its ability to be pulled or drawn into wires is called ductility. At the atomic level, a metal is defined by a low effective nuclear charge, low electronegativity (high electropositivity), large atomic radius, small ionic radius, low ionization energy, and low electron affinity. All of these characteristics are manifestations of the ability of metals to easily give up electrons.

Many of the transition metals (Group B elements) have two or more oxidation states (charges when forming bonds with other atoms). Because the valence electrons of all metals are only loosely held to their atoms, they are free to move, which makes metals good conductors of heat and electricity. The valence electrons of the active metals are found in the s subshell; those of the transition metals are found in the s and d subshells; and those of the lanthanide and actinide series elements are in the s and f subshells. Some transition metals—copper, nickel, silver, gold, palladium, and platinum—are relatively nonreactive, a property that makes them ideal for the production of coins and jewelry.



KEY CONCEPT

Alkali and alkaline earth metals are both metallic in nature because they easily lose electrons from the s subshell of their valence shells.

An example of a metal is shown in Figure 2.1 with a copper wire. The wire exhibits luster, malleability, and ductility. It is used as a wire because it also exhibits good heat and electrical conductivity.





Figure 2.1. Copper (Cu) Metal Wire

NONMETALS

Nonmetals are found predominantly on the upper right side of the periodic table. Nonmetals are generally brittle in the solid state and show little or no metallic luster. They have high ionization energies, electron affinities, and electronegativities, as well as small atomic radii and large ionic radii. They are usually poor conductors of heat and electricity. All of these characteristics are manifestations of the inability of nonmetals to easily give up electrons. Nonmetals are less unified in their chemical and physical properties than the metals.

Carbon, shown in Figure 2.2, is a stereotypical nonmetal that retains a solid structure but is brittle, nonlustrous, and generally a poor conductor of heat and electricity.



Figure 2.2. Charcoal, Composed of the Nonmetal Carbon (C)

METALLOIDS

Separating the metals and nonmetals are a stair-step group of elements called the metalloids. The metalloids are also called semimetals because they share some characteristics with both metals and nonmetals. The electronegativities and ionization energies of the metalloids lie between those of metals and nonmetals. Their physical properties—densities, melting points, and boiling points—vary widely and can be combinations of metallic and nonmetallic

characteristics. For example, silicon (Si) has a metallic luster but is brittle and a poor conductor. The reactivities of the metalloids are dependent on the elements with which they are reacting. Boron (B), for example, behaves like a nonmetal when reacting with sodium (Na) and like a metal when reacting with fluorine (F). The elements classified as metalloids form a “staircase” on the periodic table and include boron, silicon, germanium (Ge), arsenic (As), antimony (Sb), tellurium (Te), polonium (Po), and astatine (At). While there is debate over polonium and astatine’s status as metalloids, most sources label them as such. Figure 2.3 color-codes the major classifications of elements on the periodic table.

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
Lanthanides				57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
Actinides				89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr

Figure 2.3. Periodic Table, Coded by Element Type



REAL WORLD

Metalloids share some properties with metals and others with nonmetals. For instance, metalloids make good semiconductors due to their partial conductivity of electricity.

MCAT Concept Check 2.2:

Before you move on, assess your understanding of the material with these questions.

Based on their location in the periodic table, identify a few elements that likely possess the following properties:

- Luster:

- Poor conductivity of heat and electricity:

- Good conductivity but brittle:

Classify the following elements as metals (M), nonmetals (NM), or metalloids (MO):

Element	Class	Element	Class	Element	Class	Element	Class
Ag		Al		K		P	
Pb		Li		Pu		U	
Cu		As		Zn		B	
Si		Np		O		He	

2.3 Periodic Properties of the Elements

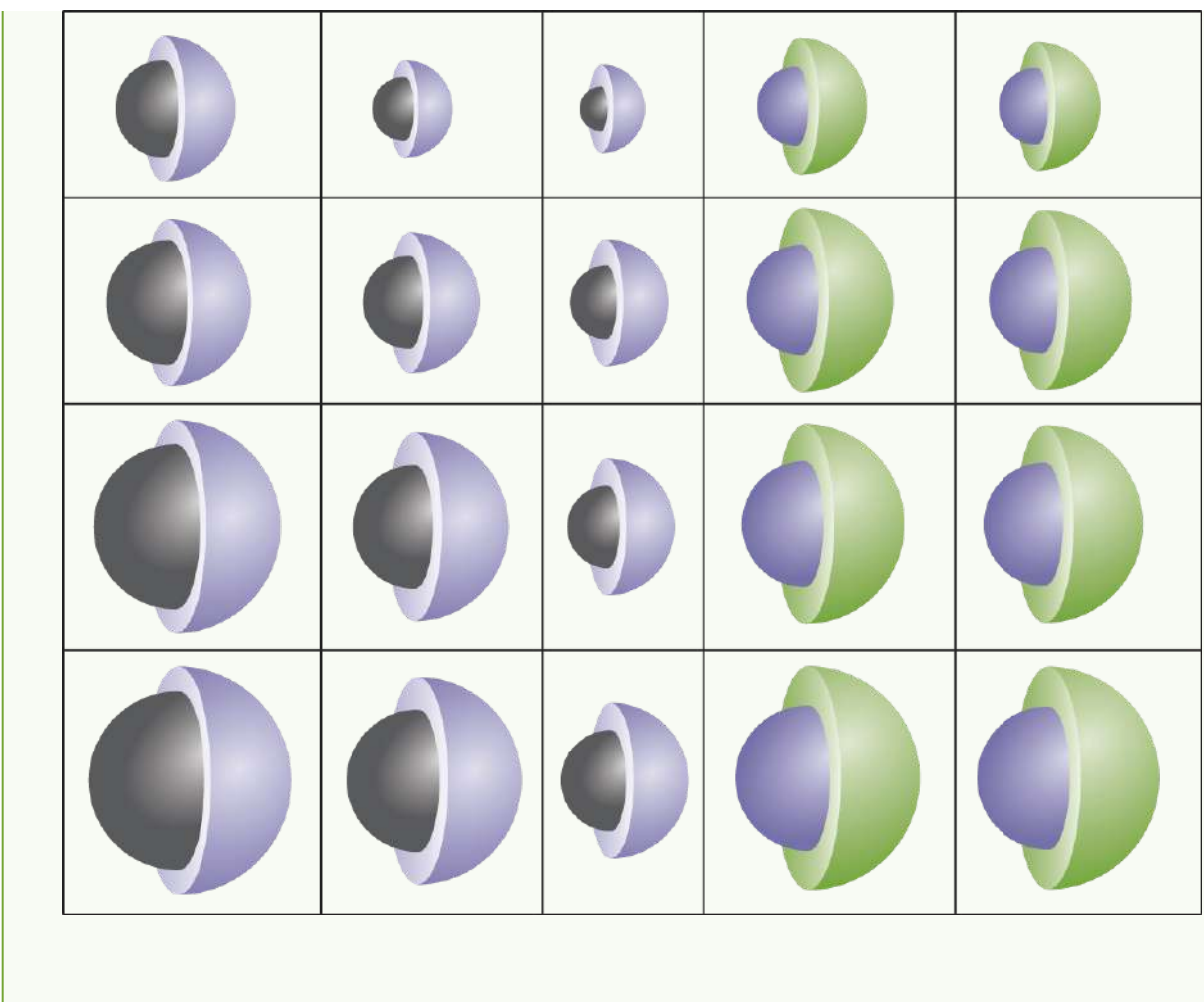
High-Yield



LEARNING GOALS

After Chapter 2.3, you will be able to:

- Compare the atomic radius of neutral atoms to their ions
- Rank elements by ionization energy, electron affinity, electronegativity, or atomic radius:



MCAT EXPERTISE

The “High-Yield” badge on this section indicates that the content is frequently tested on the MCAT.

The MCAT does not expect you to have memorized the entire periodic table. Fortunately, the periodic table is a guide unto itself, a self-referencing localization system for all of the elements. Remember, the modern table is organized in such a way to represent visually the

periodicity of chemical and physical properties of the elements. The periodic table, then, can provide you with a tremendous amount of information that otherwise would have to be memorized. Note, though, that while you do not need to memorize the periodic table for the MCAT, you do need to understand the trends within the periodic table that help predict the chemical and physical behaviors of the elements.



MCAT EXPERTISE

Don't try to memorize the periodic table. You will have access to it on Test Day through the test interface. Do understand its configuration and trends so that you can use it efficiently to get a higher score!

Before exploring the periodic trends, let's take stock of three key rules that control how valence electrons work in an atom. First, as we've already mentioned, as one moves from left to right across a period, electrons and protons are added one at a time. As the positivity of the nucleus increases, the electrons surrounding the nucleus, including those in the valence shell, experience a stronger electrostatic pull toward the center of the atom. This causes the electron cloud, which is the outer boundary defined by the valence shell electrons, to move closer and bind more tightly to the nucleus. This electrostatic attraction between the valence shell electrons and the nucleus is known as the effective nuclear charge (Z_{eff}), a measure of the net positive charge experienced by the outermost electrons. This pull is somewhat mitigated by nonvalence electrons that reside closer to

the nucleus. For elements in the same period, Z_{eff} increases from left to right. The parts of an atom responsible for Z_{eff} are illustrated in Figure 2.4.

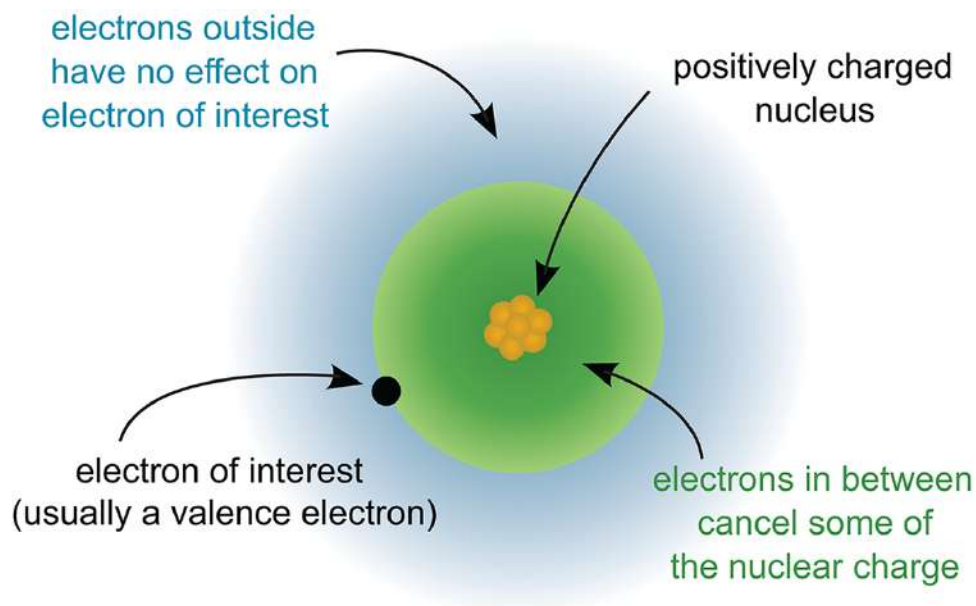


Figure 2.4. Factors that Determine Effective Nuclear Charge (Z_{eff})



BRIDGE

Z_{eff} relies on the principles of electrostatic forces defined in Chapter 5 of MCAT Physics and Math Review. The values q_1 and q_2 can represent the net charge of the nucleus and valence electron shell, respectively. The larger each charge gets (going to the right in the periodic table), the higher the value of Z_{eff} .

Second, as one moves down the elements of a given group, the principal quantum number increases by one each time. This means that the valence electrons are increasingly separated from the nucleus by a greater number

of filled principal energy levels, which can also be called inner shells. The result of this increased separation is a reduction in the electrostatic attraction between the valence electrons and the positively charged nucleus. These outermost electrons are held less tightly as the principal quantum number increases. As one goes down in a group, the increased shielding created by the inner shell electrons cancels the increased positivity of the nucleus. Thus, the Z_{eff} is more or less constant among the elements within a given group. Despite this fact, the valence electrons are held less tightly to the nucleus as one moves down a group due to the increased separation between valence electrons and the nucleus.

Third, elements can also gain or lose electrons in order to achieve a stable octet formation representative of the noble (inert) gases (Group VIIIA or Group 18). In Chapter 3 of MCAT General Chemistry Review, we will discuss how the octet rule is hardly a rule at all because there are many exceptions. For now, keep in mind that elements, especially the ones that have biological roles, tend to be most stable with eight electrons in their valence shell.

These three facts are guiding principles as we work toward an understanding of the trends demonstrated in the periodic table. In fact, the trend for effective nuclear charge across a period and the impact of increasing the number of inner shells down a group will help derive all the trends we discuss below.

ATOMIC AND IONIC RADII

Think of an atom as a cloud of electrons surrounding a dense core of protons and neutrons. The atomic radius of an element is thus equal to one-half of the distance between the centers of two atoms of an element that are briefly in contact with each other. The distance between two centers of circles in contact is akin to a diameter, making this radius calculation simple. The atomic radius cannot be measured by examining a single atom because the electrons are constantly moving around, making it impossible to mark the outer boundary of the electron cloud.



KEY CONCEPT

Atomic radius refers to the size of a neutral element, while an ionic radius is dependent on how the element ionizes based on its element type and group number.

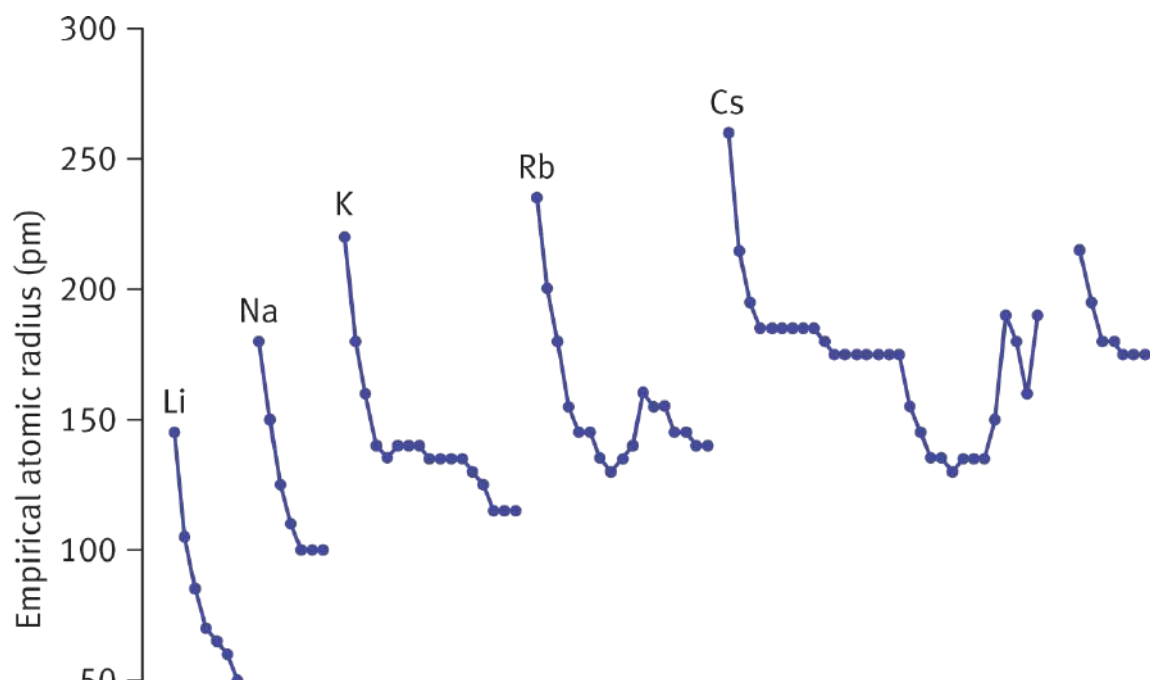
As we move across a period from left to right, protons and electrons are added one at a time to the atoms. Because the electrons are being added only to the outermost shell and the number of inner-shell electrons remains constant, the increasing positive charge of the nucleus pulls the outer electrons more closely inward and holds them more tightly. The Z_{eff} increases left to right across a period, and as a result, atomic radius decreases from left to right across a period.



MCAT EXPERTISE

Atomic radius is essentially opposite that of all other periodic trends. While others increase going up and to the right, atomic radius increases going down and to the left.

As we move down a group, the increasing principal quantum number implies that the valence electrons will be found farther away from the nucleus because the number of inner shells is increasing, separating the valence shell from the nucleus. Although the Z_{eff} remains essentially constant, the atomic radius increases down a group. Within each group, the largest atom will be at the bottom, and within each period, the largest atom will be in Group IA (Group 1). For reference, the largest atomic radius in the periodic table belongs to cesium (Cs, 260 pm), and the smallest belongs to helium (He, 25 pm). Francium is typically not considered because it is exceptionally rare in nature. Figure 2.5 displays a graph of atomic radius vs. atomic number, with Group IA elements possessing the largest atomic radius in each row.



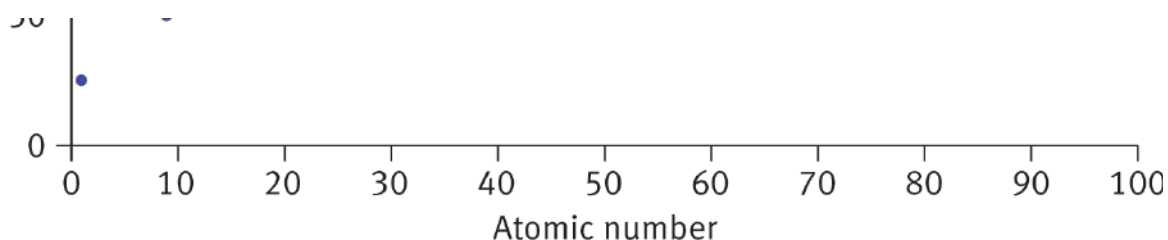


Figure 2.5. Atomic Radius (in pm) vs. Atomic Number

Unlike atomic radii, ionic radii will require some critical thinking and periodic table geography to determine. In order to understand ionic radii, we must make two generalizations. One is that metals lose electrons and become positive, while nonmetals gain electrons and become negative. The other is that metalloids can go in either direction, but tend to follow the trend based on which side of the metalloid line they fall on. Thus, silicon (Si) behaves more like a nonmetal, while germanium (Ge) tends to act more like a metal. On the MCAT, these generalizations can also be inferred from information found in passages and questions, such as oxidation states in compounds.

For nonmetals close to the metalloid line, their group number dictates that they require more electrons than other nonmetals to achieve the electronic configuration seen in Group VIIIA (Group 18). These nonmetals gain electrons while their nuclei maintain the same charge. Therefore, these nonmetals close to the metalloid line possess a larger ionic radius than their counterparts closer to Group VIIIA.

For metals, the trend is similar but opposite. Metals closer to the metalloid line have more electrons to lose to achieve the electronic configuration seen in Group VIIIA. Because of this, the ionic radius of metals near the metalloid line is dramatically smaller than that of other metals. Metals closer to Group IA have fewer electrons to lose and therefore experience a

less drastic reduction in radius during ionization. These changes are illustrated in Figure 2.6. Note that tellurium (Te) behaves as a nonmetal and boron (B) behaves as a metal; under varying conditions, these metalloids can have opposite behavior.

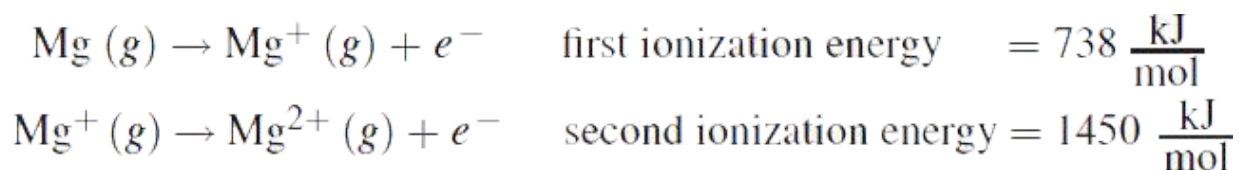
Group 1		Group 2		Group 13		Group 16		Group 17	
Li ⁺	Li	Be ²⁺	Be	B ³⁺	B	O	O ²⁻	F	F ⁻
90	134	59	90	41	82	73	126	71	119
Na ⁺	Na	Mg ²⁺	Mg	Al ³⁺	Al	S	S ²⁻	Cl	Cl ⁻
116	154	86	130	68	118	102	170	99	167
K ⁺	K	Ca ²⁺	Ca	Ga ³⁺	Ga	Se	Se ²⁻	Br	Br ⁻
152	196	114	174	76	126	116	184	114	182
Rb ⁺	Rb	Sr ²⁺	Sr	In ³⁺	In	Te	Te ²⁻	I	I ⁻
166	211	132	192	94	144	135	207	133	206

Figure 2.6. Ionic Radii (in pm) for Various Metals and Nonmetals
Neutral atoms are shown in purple; cations in black; anions in green.

IONIZATION ENERGY

Ionization energy (IE), also known as ionization potential, is the energy required to remove an electron from a gaseous species. Removing an

electron from an atom always requires an input of heat, which makes it an endothermic process. The greater the atom's Z_{eff} or the closer the valence electrons are to the nucleus, the more tightly bound they are. This makes it more difficult to remove one or more electrons, increasing the ionization energy. Thus, ionization energy increases from left to right across a period and from bottom to top in a group. The subsequent removal of a second or third electron requires increasing amounts of energy because the removal of more than one electron means that the electrons are being removed from an increasingly cationic (positive) species. The energy necessary to remove the first electron is called the first ionization energy; the energy necessary to remove the second electron from the univalent cation (X^+) to form the divalent cation (X^{2+}) is called the second ionization energy, and so on. For example:



Elements in Groups IA and IIA (Groups 1 and 2), such as lithium and beryllium, have such low ionization energies that they are called the active metals. The active metals do not exist naturally in their neutral forms; they are always found in ionic compounds, minerals, or ores. The loss of one electron from the alkali metals (Group IA) or the loss of two electrons from the alkaline earth metals (Group IIA) results in the formation of a stable, filled valence shell. In contrast, the Group VIIA (Group 17) elements—the halogens—do not typically give up their electrons. In fact, in their ionic form, they are generally anions. The first ionization energies of the elements are shown in Figure 2.7.

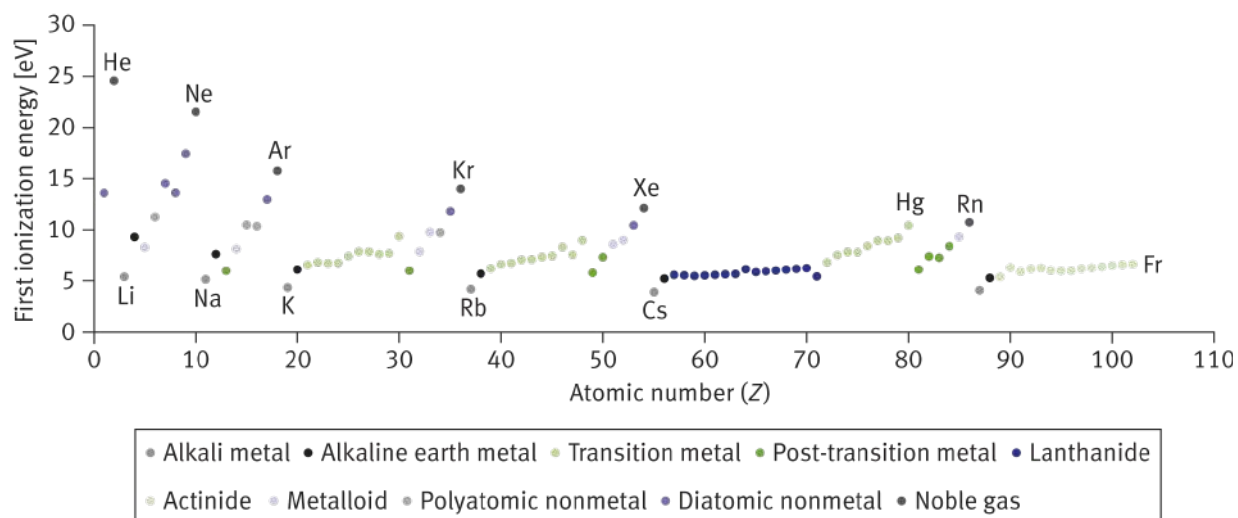
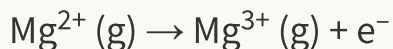


Figure 2.7. First Ionization Energies (in eV) of the Elements



MCAT EXPERTISE

First ionization energy (IE) will always be smaller than second IE, which will always be smaller than third IE. However, the degree to which the IE increases provides clues about the identity of the atom. If losing a certain number of electrons gives an element a noble gas-like electron configuration, then removing a subsequent electron will cost much more energy. For example



$$\text{third ionization energy} = 7730 \frac{\text{kJ}}{\text{mol}}$$

The values for second ionization energies are disproportionately larger for Group IA monovalent cations (like Na^{+}) but generally not that much larger for Group IIA or subsequent monovalent cations (like Mg^{+}). This is because

removing one electron from a Group IA metal results in a noble gas-like electron configuration.

Group VIIIA (Group 18) elements, or noble or inert gases, are the least likely to give up electrons. They already have a stable electron configuration and are unwilling to disrupt that stability by giving up an electron. Therefore, noble gases are among the elements with the highest ionization energies.

ELECTRON AFFINITY

Halogens are the most “greedy” group of elements on the periodic table when it comes to electrons. By acquiring one additional electron, a halogen is able to complete its octet and achieve a noble gas configuration. This exothermic process expels energy in the form of heat. Electron affinity refers to the energy dissipated by a gaseous species when it gains an electron. Note the electron affinity is essentially the opposite concept from ionization energy. Because this is an exothermic process, ΔH_{rxn} has a negative sign; however, the electron affinity is reported as a positive number. This is because electron affinity refers to the energy dissipated: if $200 \frac{\text{kJ}}{\text{mol}}$ of energy is released, $\Delta H_{\text{rxn}} = -200 \frac{\text{kJ}}{\text{mol}}$, and the electron affinity is $200 \frac{\text{kJ}}{\text{mol}}$. The stronger the electrostatic pull (the higher the Z_{eff}) between the nucleus and the valence shell electrons, the greater the energy release will be when the atom gains the electron. Thus, electron affinity increases across a period from left to right. Because the valence shell is farther away from the nucleus as the principal quantum number increases, electron affinity decreases in a group from top to bottom. Groups IA and IIA (Groups 1 and 2) have very low electron affinities, preferring to give up electrons to achieve the octet configuration

of the noble gas in the previous period. Conversely, Group VIIA (Group 17) elements have very high electron affinities because they need to gain only one electron to achieve the octet configuration of the noble gases (Group VIIIA or Group 18) in the same period. Although the noble gases would be predicted to have the highest electron affinities according to the trend, they actually have electron affinities on the order of zero because they already possess a stable octet and cannot readily accept an electron. Most metals also have low electron affinity values, as can be seen in Figure 2.8.

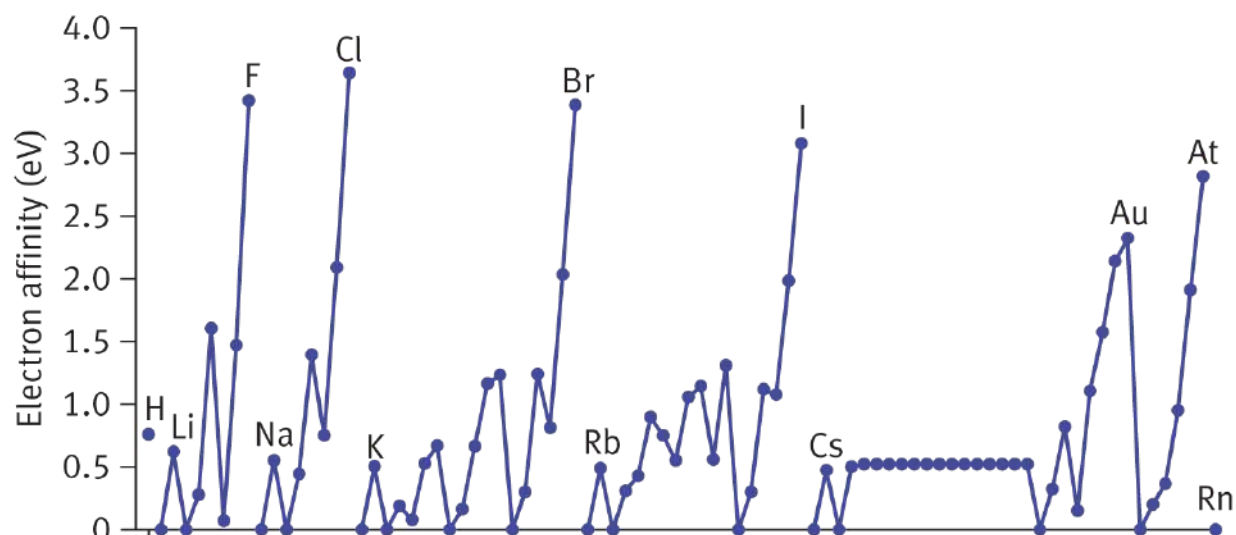




Figure 2.8. Electron Affinities (in eV) of the Elements

ELECTRONEGATIVITY

Electronegativity is a measure of the attractive force that an atom will exert on an electron in a chemical bond. The greater the electronegativity of an atom, the more it attracts electrons within a bond. Electronegativity values are related to ionization energies: the lower the ionization energy, the lower the electronegativity; the higher the ionization energy, the higher the electronegativity. The first three noble gases are exceptions: despite their high ionization energies, these elements have negligible electronegativity because they do not often form bonds.

The electronegativity value is a relative measure, and there are different scales used to express it. The most common scale is the Pauling electronegativity scale, which ranges from 0.7 for cesium, the least electronegative (most electropositive) element, to 4.0 for fluorine, the most electronegative element. Electronegativity increases across a period from left to right and decreases in a group from top to bottom. Figure 2.9 shows the electronegativity values of the elements.

H 2.1																			He
Li 1.0	Be 1.5											B 2.0	C 2.5	N 3.0	O 3.5	F 4.0			Ne
Na 0.9	Mg 1.2											Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0			Ar
K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.9	Ni 1.8	Cu 1.9	Zn 1.6	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8			Kr
Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5			Xe

Cs 0.7	Ba 0.9	Lu	Hf 1.3	Ta 1.5	W 1.7	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Tl 1.8	Pb 1.9	Bi 1.9	Po 2.0	At 2.2	Rn
Fr 0.7	Ra 0.9	Lr	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Uut	Fl	Uup	Lv	Uus	Uuo

Figure 2.9. Pauling Electronegativity Values of the Elements



KEY CONCEPT

- Cs = largest, least electronegative, lowest ionization energy, least exothermic (lowest) electron affinity
- F = smallest, most electronegative, highest ionization energy, most exothermic (highest) electron affinity

The periodic trends are summarized together in Figure 2.10.

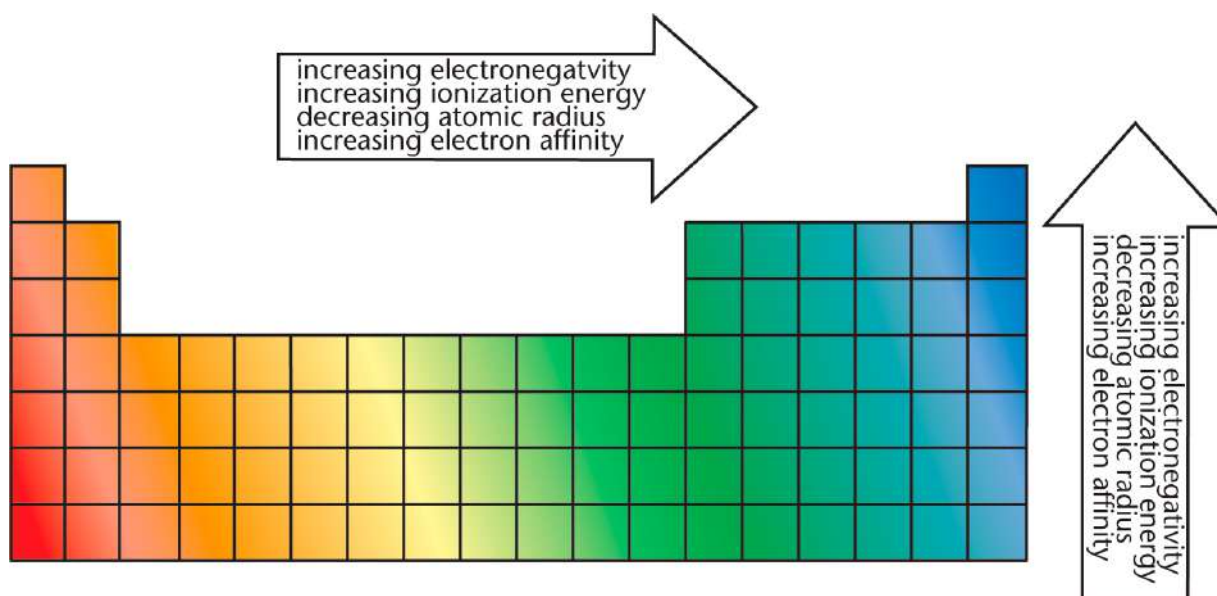


Figure 2.10. Periodic Trends



KEY CONCEPT

Periodic Trends

Left → Right

- Atomic radius ↓
- Ionization energy ↑
- Electron affinity ↑
- Electronegativity ↑

Top → Bottom

- Atomic radius ↑
- Ionization energy ↓
- Electron affinity ↓
- Electronegativity ↓

Note: Atomic radius is always opposite the other trends. Ionic radius is variable.

MCAT Concept Check 2.3:

Before you move on, assess your understanding of the material with these questions.

In each of the following pairs, which has the larger radius?:

- F or F⁻ _____
- K or K⁺ _____

Rank the following elements by decreasing first ionization energy: calcium (Ca), carbon (C), germanium (Ge), potassium (K)

1. _____
2. _____
3. _____
4. _____

Rank the following elements by increasing electron affinity: barium (Ba), copper (Cu), sulfur (S), yttrium (Y)

1. _____
2. _____
3. _____
4. _____

Rank the following elements by decreasing electronegativity: antimony (Sb), neon (Ne), oxygen (O), thallium (Tl)

1. _____

2. _____

3. _____

4. _____

Rank the following elements by increasing atomic radius:
niobium (Nb), praseodymium (Pr), tantalum (Ta), xenon
(Xe)

1. _____

2. _____

3. _____

4. _____

2.4 The Chemistry of Groups

LEARNING GOALS

After Chapter 2.4, you will be able to:

- Identify the groups on the periodic table by the properties they exhibit
- Connect periodic table groups 1, 2, 16, 17, 18, and 3–12 to their common names

What follows is a discussion of the major groups you are likely to encounter on the MCAT. While it is rare to be tested on every group, it is important to understand the overarching trends we have already discussed and how they relate across different groups.

ALKALI METALS (IA)

The alkali metals (Group IA or Group 1) possess most of the classic physical properties of metals, except that their densities are lower than those of other metals (as described for lithium earlier in this chapter). The alkali metals have only one loosely bound electron in their outermost shells. Their Z_{eff} values are very low, giving them the largest atomic radii of all the

elements in their respective periods. This low Z_{eff} value also explains the other trends: low ionization energies, low electron affinities, and low electronegativities. Alkali metals easily lose one electron to form univalent cations, and they react readily with nonmetals—especially the halogens—as in NaCl. Figure 2.11 illustrates the reaction of an alkali metal with water, a stereotypically violent reaction.





Figure 2.11. Reaction of Sodium with Water

Group IA metals react violently with water, forming strong bases.



REAL WORLD

Due to their high reactivity with water and air, most alkali metals are stored in mineral oil.

ALKALINE EARTH METALS (IIA)

The alkaline earth metals (Group IIA or Group 2), also possess many properties characteristic of metals. They share most of the characteristics of the alkali metals, except that they have slightly higher effective nuclear charges and thus slightly smaller atomic radii. They have two electrons in their valence shell, both of which are easily removed to form divalent cations. Together, the alkali and alkaline earth metals are called the active metals because they are so reactive that they are not naturally found in their elemental (neutral) state.

CHALCOGENS (VIA)

The chalcogens (Group VIA or Group 16) are an eclectic group of nonmetals and metalloids. While not as reactive as the halogens, they are crucial for normal biological functions. They each have six electrons in their valence electron shell and, due to their proximity to the metalloids, generally have small atomic radii and large ionic radii. Oxygen is the most important element in this group for many reasons; it is one of the primary constituents of water, carbohydrates, and other biological molecules. Sulfur is also an important component of certain amino acids and vitamins. Selenium also is an important nutrient for microorganisms and has a role in protection from oxidative stress. The remainder of this group is primarily metallic and generally toxic to living organisms. It is important to note that, at high concentrations, many of these elements—no matter how biologically useful—can be toxic or damaging.



BRIDGE

Many of the molecules discussed in metabolism, covered in Chapters 9 through 12 of MCAT Biochemistry Review, utilize lighter nontoxic elements from the chalcogen group (oxygen and sulfur). Many of the heavier chalcogens are toxic metals.

HALOGENS (VIIA)

The halogens (Group VIIA or Group 17) are highly reactive nonmetals with seven valence electrons. These elements are desperate to complete their octets by gaining one additional electron. The physical properties of this group are variable. At standard conditions, the halogens range from gaseous (F_2 and Cl_2) to liquid (Br_2) to solid (I_2) forms. Their chemical reactivity is more uniform, and, due to their very high electronegativities and electron affinities, they are especially reactive toward the alkali and alkaline earth metals. Fluorine (F) has the highest electronegativity of all the elements. The halogens are so reactive that they are not naturally found in their elemental state but rather as ions (called halides) or diatomic molecules. Diatomic iodine at standard conditions can be seen in Figure 2.12.



Figure 2.12. Iodine in Standard State (Diatomic Iodine)



MCAT EXPERTISE

Halogens are frequently tested on the MCAT. Remember that they only need one more electron to have a noble gas-like electron configuration (full valence shell).

NOBLE GASES (VIIIA)

The noble gases (Group VIIIA or Group 18) are also known as inert gases because they have minimal chemical reactivity due to their filled valence shells. They have high ionization energies, little or no tendency to gain or lose electrons, and (for He, Ne, and Ar, at least), no measurable

electronegativities. The noble gases have extremely low boiling points and exist as gases at room temperature. Noble gases have found a commercial niche as lighting sources, as seen in Figure 2.13, due to their lack of reactivity.

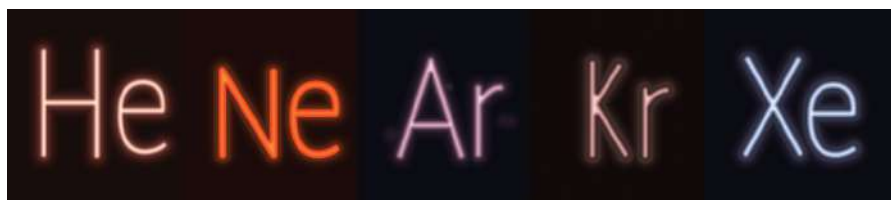


Figure 2.13. Noble Gases Used in “Neon” Signs

TRANSITION METALS (B)

The transition elements (Groups IB to VIIIB or Groups 3 to 12) are considered to be metals and as such have low electron affinities, low ionization energies, and low electronegativities. These metals are very hard and have high melting and boiling points. They tend to be quite malleable and are good conductors due to the loosely held electrons that progressively fill the d-orbitals in their valence shells. One of the unique properties of the transition metals is that many of them can have different possible charged forms or oxidation states because they are capable of losing different numbers of electrons from the s- and d-orbitals in their valence shells. For instance, copper (Cu) can exist in either the +1 or the +2 oxidation state, and manganese (Mn) can exist in the +2, +3, +4, +6, or +7 oxidation state. Because of this ability to attain different positive oxidation states, transition metals form many different ionic compounds. These different oxidation states often correspond to different colors; solutions with transition metal-containing complexes are often vibrant, as shown in Figure 2.14.



Figure 2.14. Solutions of Transition Metal-Containing Compounds
From left to right: cobalt(II) nitrate, $\text{Co}(\text{NO}_3)_2$ (red); potassium dichromate, $\text{K}_2\text{Cr}_2\text{O}_7$ (orange); potassium chromate, K_2CrO_4 (yellow); nickel(II) chloride, NiCl_2 (green); copper(II) sulfate, CuSO_4 (blue); potassium permanganate, KMnO_4 (violet)

These complex ions tend to associate in solution either with molecules of water (hydration complexes, such as $\text{CuSO}_4 \cdot 5 \text{H}_2\text{O}$) or with nonmetals, (such as $[\text{Co}(\text{NH}_3)_6]\text{Cl}$). This ability to form complexes contributes to the variable solubility of certain transition metal-containing compounds. For example, AgCl is insoluble in water but quite soluble in aqueous ammonia due to the formation of the complex ion $[\text{Ag}(\text{NH}_3)_2]^+$. The formation of complexes causes the d-orbitals to split into two energy sublevels. This enables many of the complexes to absorb certain frequencies of light—those containing the precise amount of energy required to raise electrons from the lower- to the higher-energy d-orbitals. The frequencies not absorbed (known as the subtraction frequencies) give the complexes their characteristic colors.



BRIDGE

Many transition metals act as cofactors for enzymes, including vanadium, chromium, manganese, iron, cobalt, nickel, copper, and zinc. Cofactors and coenzymes are discussed in Chapter 2 of MCAT Biochemistry Review.

This brings up an important point about the perception of color: when we perceive an object as a particular color, it is because that color is not absorbed—but rather reflected—by the object. If an object absorbs a given color of light and reflects all others, our brain mixes these subtraction frequencies and we perceive the complementary color of the frequency that was absorbed. This is best illustrated with an example. Carotene is a photosynthetic pigment that strongly absorbs blue light but reflects other colors. Thus, our brains interpret the color of carotene as the result of white light minus blue light, which is yellow light. The complementary colors are shown in Figure 2.15; while the MCAT is unlikely to ask you to name the complement of a given color, the relationship between complementary colors, as explained here, is fair game. It should also be noted that the manner in which colors mix in this scheme is distinctly different from mixing, say, paint colors. The differences between these two schemes, termed additive and subtractive color mixing, respectively, are outside the scope of the MCAT.

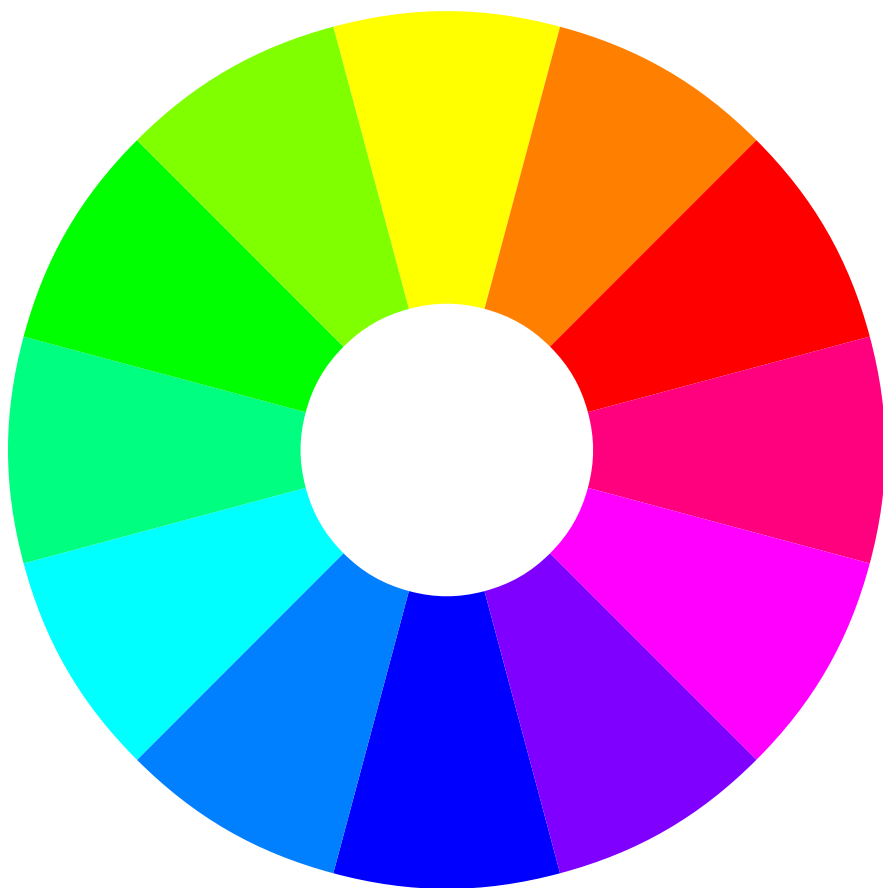


Figure 2.15. Red–Green–Blue (Additive) Color Wheel

Each color is directly across the circle from its complementary color; commonly referenced complementary pairs include red/cyan, green/magenta, and blue/yellow.

MCAT Concept Check 2.4:

Before you move on, assess your understanding of the material with this question.

For each of the properties listed below, write down the groups of the periodic table that exhibit those properties.

- High reactivity to water:

- Six valence electrons:

- Contain at least one metal:

- Multiple oxidation states:

- Negative oxidation states:

- Possess a full octet in the neutral state:



Conclusion

Now that we have completed our review of the Periodic Table of the Elements, commit to understanding (not just memorizing) the trends of physical and chemical properties of the elements. They will help you quickly answer many questions on the MCAT. As you progress through the chapters of this book, a foundational understanding of the elements will help you develop a richer, more nuanced understanding of their general and particular behaviors. Topics in general chemistry that may have given you trouble in the past will be understandable from the perspective of the behaviors and characteristics that you have reviewed here.

More broadly, you will see a diverse array of elements from the groups we have discussed here that are critical or detrimental to biological function. In addition, you may begin to see why the human body utilizes certain elements for specific purposes, taking advantage of the periodic trends discussed here.

Concept Summary

The Periodic Table

- The Periodic Table of the Elements organizes the elements according to their atomic numbers and reveals a pattern of similar chemical and physical properties among elements.
 - Rows are called periods and are based on the same principal energy level, n .
 - Columns are called groups. Elements in the same group have the same valence shell electron configuration.

Types of Elements

- The elements on the periodic table belong to one of three types.
 - Metals are shiny (lustrous), conduct electricity well, and are malleable and ductile. Metals are found on left side and middle of the periodic table.
 - Nonmetals are dull, poor conductors of electricity, and are brittle. Nonmetals are found on right side of the periodic table.
 - Metalloids possess characteristics of both metals and nonmetals and are found in a stair-step pattern starting with boron (B).

Periodic Properties of the Elements

- Effective nuclear charge (Z_{eff}) is the net positive charge experienced by electrons in the valence shell and forms the foundation for all periodic trends.
 - Z_{eff} increases from left to right across a period, with little change in value from top to bottom in a group.
 - Valence electrons become increasingly separated from the nucleus as the principal energy level, n , increases from top to bottom in a group.
- Atomic radius decreases from left to right across a period and increases from top to bottom in a group.
- Ionic radius is the size of a charged species. The largest nonmetallic ionic radii and the smallest metallic ionic radii exist at the metalloid boundary.
 - Cations are generally smaller than their corresponding neutral atom.
 - Anions are generally larger than their corresponding neutral atom.
- Ionization energy is the amount of energy necessary to remove an electron from the valence shell of a gaseous species; it increases from left to right across a period and decreases from top to bottom in a group.
- Electron affinity is the amount of energy released when a gaseous species gains an electron in its valence shell; it increases from left to right across a period and decreases from top to bottom in a group.
- Electronegativity is a measure of the attractive force of the nucleus for electrons within a bond; it increases from left to right across a period

and decreases from top to bottom in a group.

The Chemistry of Groups

- Alkali metals typically take on an oxidation state of +1 and prefer to lose an electron to achieve a noble gas-like configuration; they and the alkaline earth metals are the most reactive of all metals.
- Alkaline earth metals take on an oxidation state of +2 and can lose two electrons to achieve noble gas-like configurations.
- Chalcogens take on oxidation states of -2 or +6 (depending on whether they are nonmetals or metals, respectively) in order to achieve noble gas configuration. They are very biologically important.
- Halogens typically take on an oxidation state of -1 and prefer to gain an electron to achieve noble gas-like configurations; these nonmetals have the highest electronegativities.
- Noble gases have a fully filled valence shell in their standard state and prefer not to give up or take on additional electrons; they have very high ionization energies and (for He, Ne, and Ar), virtually nonexistent electronegativities and electron affinities.
- Transition metals are unique because they take on multiple oxidation states, which explains their ability to form colorful complexes with nonmetals in solution and their utility in certain biological systems.

Answers to Concept Checks

2.1

The modern periodic table is arranged in order by atomic number.

Element	A or B	Element	A or B	Element	A or B	Element	A or B
Ag	B	Al	A	K	A	P	A
Pb	A	Li	A	Pu	B	U	B
Cu	B	Cf	B	Zn	B	B	A
N	A	Np	B	O	A	He	A

2.2

Metals have luster. Nonmetals have poor conductivity. Metalloids exhibit brittleness but good conductivity. Any answers within each of these categories are acceptable.

Element	Class	Element	Class	Element	Class	Element	Class
Ag	M	Al	M	K	M	P	NM
Pb	M	Li	M	Pu	M	U	M

Element	Class	Element	Class	Element	Class	Element	Class
Cu	M	As	MO	Zn	M	B	MO
Si	MO	Np	M	O	NM	He	NM

2.3

$F^- > F$; $K > K^+$. The ionic radii of anions are larger than the associated atomic radii, while the ionic radii of cations are smaller.

Ionization energy: carbon > germanium > calcium > potassium

Electron affinity: barium < yttrium < copper < sulfur

Electronegativity: oxygen > antimony > thallium > neon

Atomic radius: xenon < niobium < tantalum < praseodymium

2.4

- High reactivity to water: Groups 1 and 2
- Six valence electrons: Groups 6 and 16
- Contain at least one metal: Groups 1 through 15
- Multiple oxidation states: All groups; most notably Groups 3 through 12 (transition metals)
- Negative oxidation states: Almost all groups; most notably Groups 14 through 17 (nonmetals)
- Possess a full octet in the neutral state: Group 18



Shared Concepts

General Chemistry Chapter 1

Atomic Structure

General Chemistry Chapter 3

Bonding and Chemical Interactions

General Chemistry Chapter 4

Compounds and Stoichiometry

Organic Chemistry Chapter 3

Bonding

Physics and Math Chapter 5

Electrostatics and Magnetism

Physics and Math Chapter 9

Atomic and Nuclear Phenomena

Discrete Practice Questions

Consult your online resources for additional practice.

1. Lithium and sodium have similar chemical properties. For example, both can form ionic bonds with chloride. Which of the following best explains this similarity?

- (A) Both lithium and sodium ions are positively charged.
- (B) Lithium and sodium are in the same group of the periodic table.
- (C) Lithium and sodium are in the same period of the periodic table.
- (D) Both lithium and sodium have low atomic weights.

2. Carbon and silicon are the basis of biological life and synthetic computing, respectively. While these elements share many chemical properties, which of the following best describes a difference between the two elements?

- (A) Carbon has a smaller atomic radius than silicon.
- (B) Silicon has a smaller atomic radius than carbon.
- (C) Carbon has fewer valence electrons than silicon.
- (D) Silicon has fewer valence electrons than carbon.

3. What determines the length of an element's atomic radius?

The number of valence electrons

The number of electron shells

The number of neutrons in the nucleus

- (A) I only
- (B) II only
- (C) I and II only
- (D) I, II, and III

4. Ionization energy contributes to an atom's chemical reactivity.

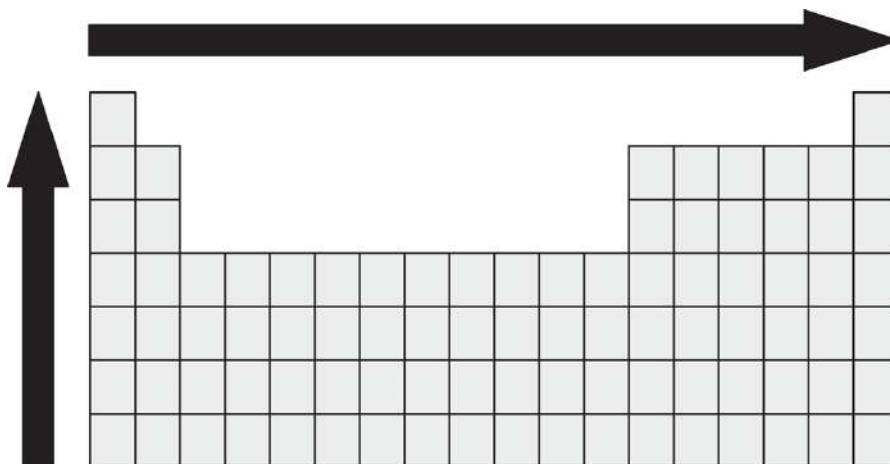
Which of the following shows an accurate ranking of ionization energies from lowest to highest?

- (A) first ionization energy of Be < second ionization energy of Be < first ionization energy of Li
- (B) first ionization energy of Be < first ionization energy of Li < second ionization energy of Be
- (C) first ionization energy of Li < first ionization energy of Be < second ionization energy of Be
- (D) first ionization energy of Li < second ionization energy of Be < first ionization energy of Be

5. Antimony is used in some antiparasitic medications—specifically those targeting *Leishmania donovani*. What type of element is antimony?

- (A) Metal
- (B) Metalloid
- (C) Halogen
- (D) Nonmetal

6. The properties of atoms can be predicted, to some extent, by their location within the periodic table. Which property or properties increase in the direction of the arrows shown?



Electronegativity

Atomic radius

First ionization energy

- (A) I only
(B) I and II only
(C) I and III only
(D) II and III only

7. Metals are often used for making wires that conduct electricity. Which of the following properties of metals explains why?

- (A) Metals are malleable.
- (B) Metals have low electronegativities.
- (C) Metals have valence electrons that can move freely.
- (D) Metals have high melting points.

8. Which of the following is an important property of the group of elements shaded in the periodic table below?

A 10x10 grid with a black bar covering the first column and the second row. The grid is composed of 100 cells. The first column (10 cells) and the second row (10 cells) are blacked out, totaling 19 cells. The remaining 81 cells are white.

- (A) These elements are the best electrical conductors in the periodic table.
- (B) These elements form divalent cations.
- (C) The second ionization energy for these elements is lower than the first ionization energy.
- (D) The atomic radii of these elements decrease as one moves down the column.

9. When dissolved in water, which of the following ions is most likely to form a complex ion with H_2O ?

- (A) Na^+
- (B) Fe^{2+}
- (C) Cl^-
- (D) S^{2-}

10. How many valence electrons are present in elements in the third period?

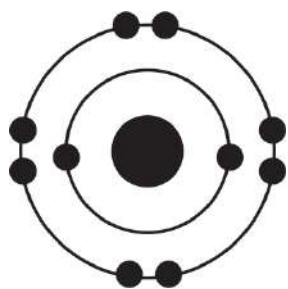
- (A) 2
- (B) 3
- (C) The number decreases as the atomic number increases.
- (D) The number increases as the atomic number increases.

11. Which of the following elements has the highest electronegativity?

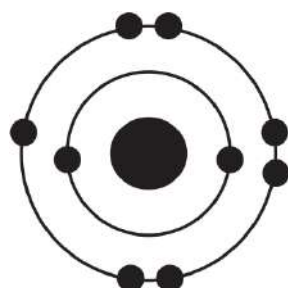
- (A) Mg
- (B) Cl
- (C) Zn
- (D) I

12. Of the four atoms depicted here, which has the highest electron affinity?

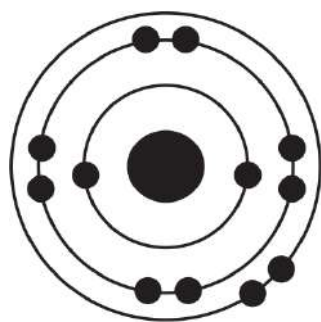
(A)



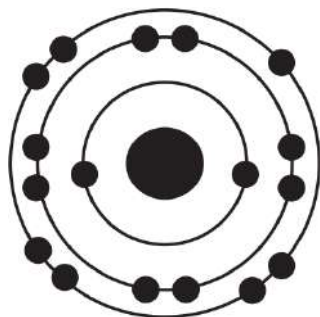
(B)



(C)



(D)



13. Which of the following atoms or ions has the largest effective nuclear charge?

- (A) Cl
- (B) Cl^-
- (C) K
- (D) K^+

14. Why do halogens often form ionic bonds with alkaline earth metals?

- (A) The alkaline earth metals have much higher electron affinities than the halogens.
- (B) By sharing electrons equally, the alkaline earth metals and halogens both form full octets.
- (C) Within the same row, the halogens have smaller atomic radii than the alkaline earth metals.
- (D) The halogens have much higher electron affinities than the alkaline earth metals.

15. What is the highest-energy orbital of elements with valence electrons in the $n = 3$ shell?

- (A) s-orbital
- (B) p-orbital
- (C) d-orbital
- (D) f-orbital

Explanations to Discrete Practice Questions

1. B

The periodic table is organized into periods (rows) and groups (columns). Groups (columns) are particularly significant because they represent sets of elements with the same valence electron configuration, which in turn will dictate many of the chemical properties of those elements. Although (A) is true, the fact that both ions are positively charged does not explain the similarity in chemical properties; most metals produce positively charged ions. (C) is not true because lithium and sodium are in the same group, not period. Finally, although lithium and sodium do have relatively low atomic weights, so do several other elements that do not share the same properties, eliminating (D).

2. A

As one moves from top to bottom in a group (column), extra electron shells accumulate, despite the fact that the valence configurations remain identical. These extra electron shells provide shielding between the positive nucleus and the outermost electrons, decreasing the electrostatic attraction and increasing the atomic radius. Because carbon and silicon are in the same group, and silicon is farther down in

the group, silicon will have a larger atomic radius because of its extra electron shell.

3. C

Atomic radius is determined by multiple factors. Of the choices given, the number of valence electrons does have an impact on the atomic radius. As one moves across a period (row), protons and valence electrons are added, and the electrons are more strongly attracted to the central protons. This attraction tightens the atom, shrinking the atomic radius. The number of electron shells is also significant, as demonstrated by the trend when moving down a group (column). As more electron shells are added that separate the positively charged nucleus from the outermost electrons, the electrostatic forces are weakened, and the atomic radius increases. The number of neutrons is irrelevant because it does not impact these attractive forces.

4. C

Ionization energy increases from left to right, so the first ionization energy of lithium is lower than that of beryllium. Second ionization energy is always larger than first ionization energy, so beryllium's second ionization energy should be the highest value. This is because removing an additional electron from Be^+ requires one to overcome a significantly larger electrostatic force.

5. B

Antimony (Sb) is on the right side of the periodic table, but not far right enough to be a nonmetal, (D). It certainly does not lie far enough to the right to fall in Group VIIA (Group 17), which would classify it as a

halogen, (C). While sources have rarely classified antimony as a metal, (A), it is usually classified as a metalloid, (B).

6. C

Electronegativity describes how strong an attraction an element will have for electrons in a bond. A nucleus with a larger effective nuclear charge will have a higher electronegativity; Z_{eff} increases toward the right side of a period. A stronger nuclear pull will also lead to increased first ionization energy, as the forces make it more difficult to remove an electron. The vertical arrow can be explained by the size of the atoms. As size decreases, the positive charge becomes more effective at attracting electrons in a chemical bond (higher electronegativity), and the energy required to remove an electron (ionization energy) increases.

7. C

All four descriptions of metals are true, but the most significant property that contributes to the ability of metals to conduct electricity is the fact that they have valence electrons that can move freely. Malleability, (A), is the ability to shape a material with a hammer, which does not play a role in conducting electricity. The low electronegativity and high melting points of metals, (B) and (D), also do not play a major role in the conduction of electricity.

8. B

This block represents the alkaline earth metals, which form divalent cations, or ions with a +2 charge. All of the elements in Group IIA have two electrons in their outermost s subshell. Because loss of these two electrons would leave a full octet as the outermost shell, becoming a divalent cation is a stable configuration for all of the alkaline earth

metals. Although some of these elements might be great conductors, they are not as effective as the alkali metals, eliminating (A). (C) is also incorrect because, although forming a divalent cation is a stable configuration for the alkaline earth metals, the second ionization energy is still always higher than the first. Finally, (D) is incorrect because atomic radii increase when moving down a group of elements because the number of electron shells increases.

9. B

Iron is a transition metal. Transition metals can often form more than one ion. Iron, for example, can be Fe^{2+} or Fe^{3+} . The transition metals, in these various oxidation states, can often form hydration complexes with water. Part of the significance of these complexes is that, when a transition metal can form a complex, its solubility within the related solvent will increase. The other ions given might dissolve readily in water, but because none of them are transition metals, they will not likely form complexes.

10. D

This question is simple if one recalls that periods refer to the rows in the periodic table, while groups or families refer to the columns. Within the same period, an additional valence electron is added with each step toward the right side of the table.

11. B

This question requires knowledge of the trends of electronegativity within the periodic table. Electronegativity increases as one moves from left to right for the same reasons that effective nuclear charge increases. Electronegativity decreases as one moves down the periodic table

because there are more electron shells separating the nucleus from the outermost electrons. In this question, chlorine is the furthest toward the top-right corner of the periodic table.

12. B

Electron affinity is related to several factors, including atomic size and filling of the valence shell. As atomic radius increases, the distance between the nucleus and the outermost electrons increases, thereby decreasing the attractive forces between protons and electrons. As a result, increased atomic radius will lead to lower electron affinity. Because atoms are in a low-energy state when their outermost valence electron shell is filled, atoms needing only one or two electrons to complete this shell will have high electron affinities. In this example, (B) and (D) need only one more electron to have a noble gas-like electron configuration; because (B) is smaller, it will have the highest electron affinity.

13. D

The effective nuclear charge refers to the strength with which the protons in the nucleus can pull on electrons. This phenomenon helps to explain electron affinity, electronegativity, and ionization energy. In (A), the nonionized chlorine atom, the nuclear charge is balanced by the surrounding electrons: $17\text{ p}^+/17\text{ e}^-$. The chloride ion, (B), has a lower effective nuclear charge because there are more electrons than protons: $17\text{ p}^+/18\text{ e}^-$. Next, elemental potassium, (C), has the lowest effective nuclear charge because it contains additional inner shells that shield its valence electron from the nucleus. (D), ionic potassium, has a higher effective nuclear charge than any of the other options do because it has the same electron configuration as Cl^- (and the same amount of

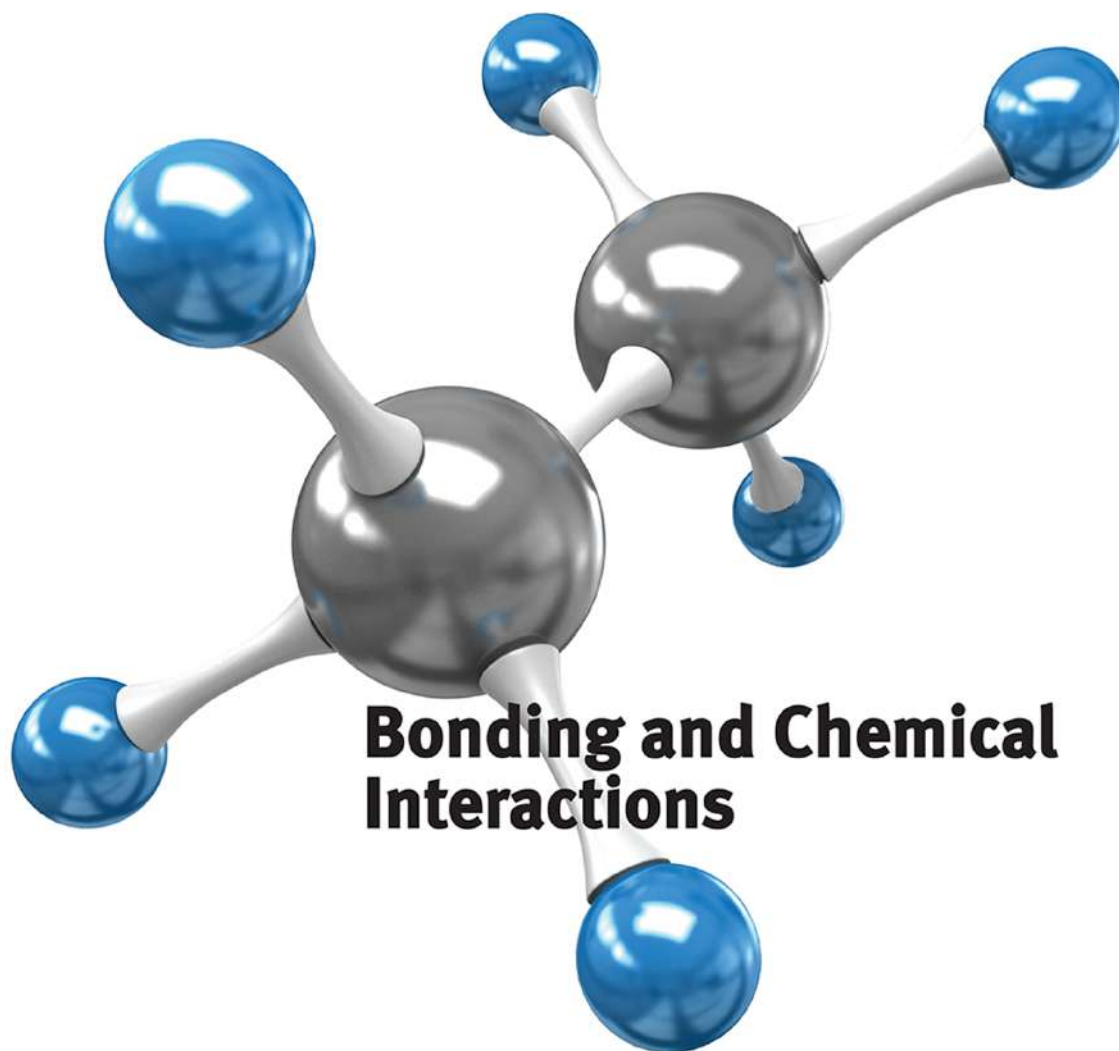
shielding from inner shell electrons as neutral Cl) but contains two extra protons in its nucleus: $19\text{ p}^+/18\text{ e}^-$.

14. D

Ionic bonds are formed through unequal sharing of electrons. These bonds typically occur because the electron affinities of the two bonded atoms differ greatly. For example, the halogens have high electron affinities because adding a single electron to their valence shells would create full valence shells. In contrast, the alkaline earth metals have very low electron affinities and are more likely to be electron donors because the loss of two electrons would leave them with full valence shells. (A) states the opposite and is incorrect because the halogens have high electron affinity and the alkaline earth metals have low electron affinity. (B) is incorrect because equal sharing of electrons is a classic description of covalent bonding, not ionic. (C) is a true statement, but is not relevant to why ionic bonds form.

15. C

When $n = 3$, $l = 0, 1$, or 2 . The highest value for l in this case is 2 , which corresponds to the d subshell. Although the $3d$ block appears to be part of the fourth period, it still has the principal quantum number $n = 3$. In general, the subshells within an energy shell increase in energy as follows: $s < p < d < f$ (although there is no $3f$ subshell).

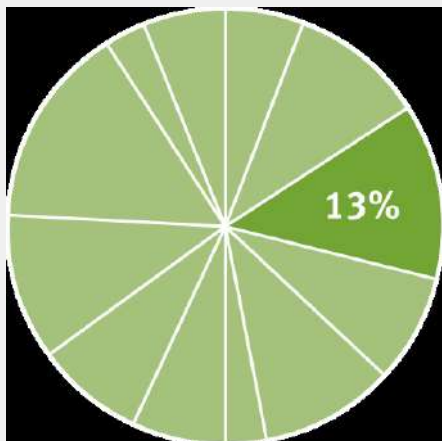


In This Chapter

- 3.1 Bonding 
 - The Octet Rule 
 - Types of Bonds 
- 3.2 Ionic Bonds 
- 3.3 Covalent Bonds 
 - Properties of Covalent Compounds 
 - Coordinate Covalent Bonds 
 - Covalent Bond Notation 
 - Geometry and Polarity 
 - Atomic and Molecular Orbitals 
- 3.4 Intermolecular Forces 
 - London Dispersion Forces 
 - Dipole–Dipole Interactions 
 - Hydrogen Bonds 
- Concept Summary 



Chapter Profile



The content in this chapter should be relevant to about 13% of all questions about general chemistry on the MCAT.

This chapter covers material from the following AAMC content categories:

4C: Electrochemistry and electrical circuits and their elements

5B: Nature of molecules and intermolecular interactions

Introduction

The Maillard reaction is one of the most important chemical processes that occurs while cooking. The reaction mechanism itself is one with which you

are closely familiar from your studies of organic chemistry: a nucleophilic reaction between the amino terminus of the peptide chain of a protein and the carbonyl functionality of a sugar to form an N-substituted glycosylamine. This compound undergoes a complex series of rearrangements and other reactions to produce a set of compounds that gives cooked food its pleasing color and delectable flavor. This reaction is especially important for browning meat.

When the surface of the meat comes into contact with the hot surface of a pan or grill, the proteins and sugars on the meat's exterior begin interacting via the Maillard reaction. The pan must be sufficiently hot to bring the exterior of the meat to a temperature of 155°C (310°F), the optimal temperature for the reaction to occur. So how does a grill master achieve the impossible: generating very high heat for the exterior but not overcooking the interior? The answer lies, in part, in drying the meat. When meat that has a lot of water on its exterior surface hits the hot pan, the first process that takes place is the boiling of the water. Boiling is a phase change from liquid to gas and occurs at a constant temperature; water's boiling point is 100°C (212°F). Because this temperature is considerably lower than that necessary for the Maillard reaction, no browning will occur and the flavor compounds will not form. The lesson here is: if you want a tasty steak, always dry the surface of your meat!

Of course, the real lesson is the topic of discussion for this chapter: bonding and chemical interactions. We will not address complex chemical bonding, such as that which takes place in the Maillard reaction, in this chapter. Rather, this chapter will address the basics of chemical bonding and interactions. Here, we will investigate the nature and behavior of covalent and ionic bonds. We will also review a system—Lewis structures—

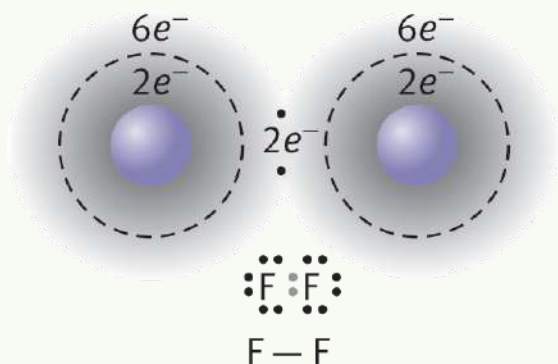
by which bonding electrons are accounted for, and we will address the main principles of valence shell electron pair repulsion (VSEPR) theory. Finally, we will recount the various modes of interaction between molecules: intermolecular forces.

3.1 Bonding

LEARNING GOALS

After Chapter 3.1, you will be able to:

- Compare and contrast ionic and covalent compounds
- Identify elements that do not obey the octet rule
- Apply periodic trends to determine whether a covalent bond is polar or nonpolar:



The atoms of most elements, except for a few noble gases, can combine to form molecules. The atoms within these molecules are held together by strong attractive forces called chemical bonds, which are formed via the interaction of the valence electrons of the combining atoms. The chemical and physical properties of the resulting compound are usually very

different from those of the constituent elements. For example, elemental sodium, an alkali metal, is so reactive that it can actually produce fire when reacting with water because the reaction is highly exothermic. Diatomic chlorine gas is so toxic that it was used for chemical warfare during World War I. However, when sodium and chlorine react, the biologically important compound NaCl (table salt) is produced.

THE OCTET RULE

How do atoms join together to form compounds? In the example above, how do the sodium and the chlorine atoms form sodium chloride? For many molecules, the constituent atoms bond according to the octet rule, which states that an atom tends to bond with other atoms so that it has eight electrons in its outermost shell, thereby forming a stable electron configuration similar to that of the noble gases. An example of an octet configuration is shown for the noble gas argon (Ar) in Figure 3.1.

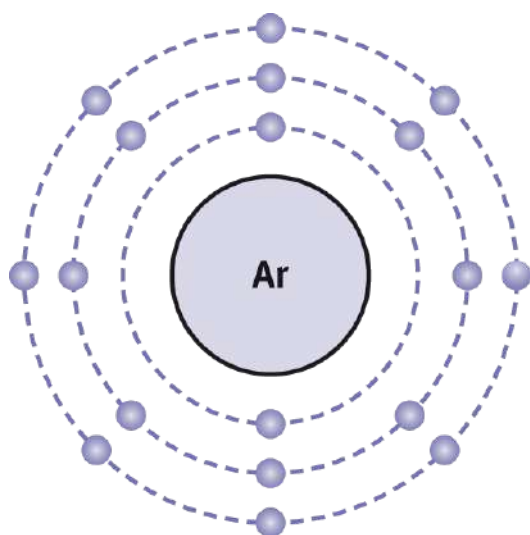


Figure 3.1. Electron Configuration of Argon
As a noble gas, argon has a complete octet in its valence shell.

However, this is more of a “rule of thumb,” because there are more elements that can be exceptions to the rule than those that follow the rule. These “exceptional” elements include hydrogen, which can only have two valence electrons (achieving the configuration of helium); lithium and beryllium, which bond to attain two and four valence electrons, respectively; boron, which bonds to attain six valence electrons; and all elements in period 3 and greater, which can expand the valence shell to include more than eight electrons by incorporating d-orbitals. For example, in certain compounds, chlorine can form seven covalent bonds, thereby holding 14 electrons in its valence shell.

A simple way to remember all the exceptions is as follows:

- Incomplete octet: These elements are stable with fewer than 8 electrons in their valence shell and include hydrogen (stable with 2 electrons), helium (2), lithium (2), beryllium (4), and boron (6).
- Expanded octet: Any element in period 3 and greater can hold more than 8 electrons, including phosphorus (10), sulfur (12), chlorine (14), and many others.
- Odd numbers of electrons: Any molecule with an odd number of valence electrons cannot distribute those electrons to give eight to each atom; for example, nitric oxide (NO) has eleven valence electrons.

Another way to remember the exceptions is to remember the common elements that almost always abide by the octet rule: carbon, nitrogen, oxygen, fluorine, sodium, and magnesium. Note that nonmetals gain electrons and metals lose electrons to achieve their respective complete octets.



KEY CONCEPT

The octet rule is the desire of all atoms to achieve noble gas configuration. However, keep in mind that there are many exceptions to this rule.

TYPES OF BONDS

We classify chemical bonds into two distinct types: ionic and covalent.

In ionic bonding, one or more electrons from an atom with a low ionization energy, typically a metal, are transferred to an atom with a high electron affinity, typically a nonmetal. An illustration of such a process is shown for our initial example, sodium chloride, in Figure 3.2.



Figure 3.2. Formation of an Ionic Bond

Sodium (Na) has a low ionization energy, easily releasing an electron; chlorine (Cl) has a high electron affinity, easily absorbing that electron into its valence shell. In this example, both atoms achieve an octet formation.

The resulting electrostatic attraction between opposite charges is what holds the ions together. This is the nature of the bond in sodium chloride, where the positively charged sodium cation is electrostatically attracted to the negatively charged chloride anion. It is important to note that this type of electrostatic attraction creates lattice structures consisting of repeating

rows of cations and anions, rather than individual molecular bonds, as shown in Figure 3.3.

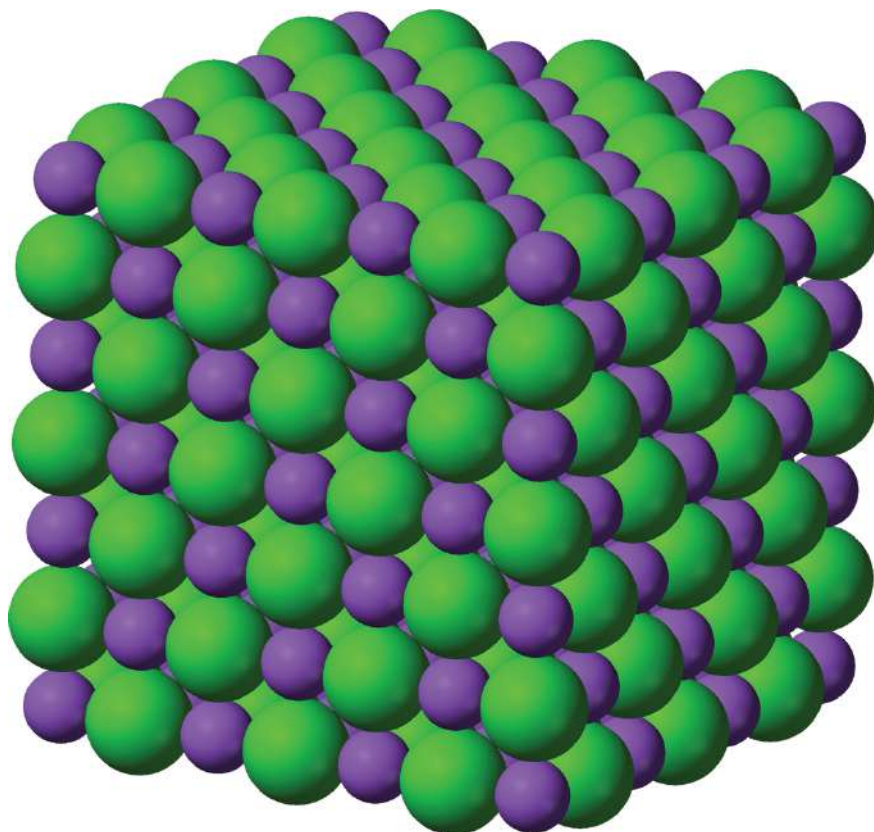


Figure 3.3. Crystal Lattice Structure of Sodium Chloride
Sodium = purple; chloride = green

In covalent bonding, an electron pair is shared between two atoms, typically nonmetals, that have relatively similar values of electronegativity. The degree to which the pair of electrons is shared equally or unequally between the two atoms determines the degree of polarity in the covalent bond. For example, if the electron pair is shared equally, the covalent bond is nonpolar; and if the pair is shared unequally, the bond is polar. If both of the shared electrons are contributed by only one of the two atoms, the bond is called coordinate covalent.



Electronegativity, discussed in Chapter 2 of MCAT General Chemistry Review, is a property that addresses how an individual atom acts within a bond and will help us understand the formation of molecules from atoms with different electronegativities.

An example of nonpolar covalent bonding is shown for diatomic fluorine in Figure 3.4. Each atom has seven electrons in its valence shell, and by sharing one of these electrons from each atom, they can each form an octet. Unlike ionic crystal lattices, covalent compounds consist of individually bonded molecules.

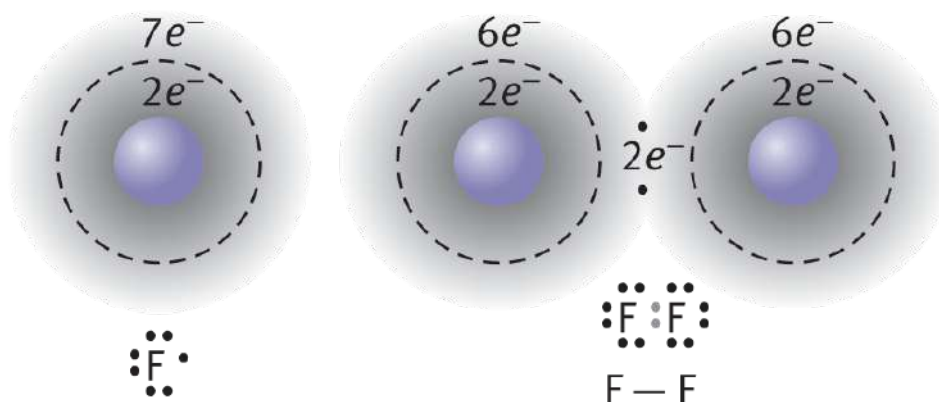


Figure 3.4. Formation of a Covalent Bond

Fluorine (F) has seven valence electrons; by sharing one electron from each atom, both fluorine atoms achieve an octet formation.

Before you move on, assess your understanding of the material with these questions.

Describe the atomic differences between ionic and covalent compounds:

- Ionic:

- Covalent:

List three elements that do not follow the octet rule and explain why:

Element	Why It Violates the Octet Rule



Which periodic trend determines whether a covalent bond is polar or nonpolar? _____

3.2 Ionic Bonds

LEARNING GOALS

After Chapter 3.2, you will be able to:

- Explain why ionic bonds are commonly formed between metals and nonmetals
- Recall the major characteristics of ionic compounds

Ionic bonds form between atoms that have significantly different electronegativities. The atom that loses the electrons becomes a cation, and the atom that gains electrons becomes an anion. The ionic bond is the result of an electrostatic force of attraction between the opposite charges of these ions. Electrons are not shared in an ionic bond. For this electron transfer to occur, the difference in electronegativity must be greater than 1.7 on the Pauling scale.



MNEMONIC

MeTals lose electrons to become caTions = posiTive (+) ions

Nonmetals gain electrons to become aNions = Negative (–) ions

The MCAT won't expect you to memorize the Pauling scale, but recognize that ionic bonds are generally formed between a metal and a nonmetal. For example, alkali and alkaline earth metals of Groups IA and IIA (Groups 1 and 2, respectively) readily form ionic bonds with the halogens of Group VIIA (Group 17). The atoms of the active metals loosely hold onto their electrons, whereas the halogens are more likely to gain an electron to complete their valence shell. The differences in bonding behavior for these classes of elements, and their differences in electronegativity values (ΔEN), explain the formation of ionic compounds such as cesium chloride ($\Delta EN = 2.3$), potassium iodide ($\Delta EN = 1.7$), and sodium fluoride ($\Delta EN = 3.1$).

Ionic compounds have characteristic physical properties that you should recognize for Test Day. Because of the strength of the electrostatic force between the ionic constituents of the compound, ionic compounds have very high melting and boiling points. For example, the melting point of sodium chloride is 801°C . Many ionic compounds dissolve readily in water and other polar solvents and, in the molten or aqueous state, are good conductors of electricity. In the solid state, the ionic constituents of the compound form a crystalline lattice consisting of repeating positive and negative ions, as shown earlier in Figure 3.3. With this arrangement, the attractive forces between oppositely charged ions are maximized, and the repulsive forces between ions of like charge are minimized.

MCAT Concept Check 3.2:

Before you move on, assess your understanding of the material with these questions.

Why do ionic bonds tend to form between metals and nonmetals?

Describe five characteristics of ionic compounds.

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-

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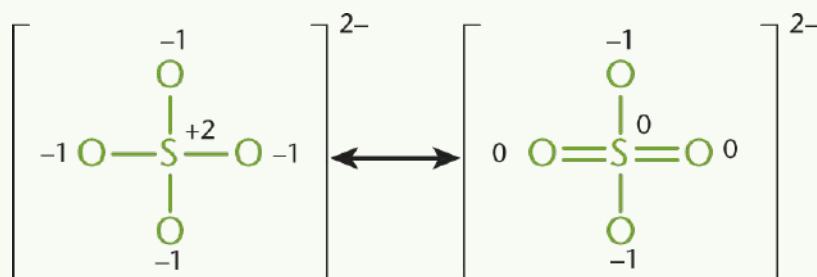
3.3 Covalent Bonds

High-Yield

LEARNING GOALS

After Chapter 3.3, you will be able to:

- Explain the relationship between bond strength, bond length, and bond energy
- Identify the values of ΔEN for which polar covalent, nonpolar covalent, and ionic bonds will form
- Predict the molecular geometry of a molecule given its formula
- Draw Lewis dot structures for simple molecules, including resonance structures, such as:



When two or more atoms with similar electronegativities interact, the energy required to form ions through the complete transfer of one or more electrons is greater than the energy that would be released upon the

formation of an ionic bond. That is, when two atoms of similar tendency to attract electrons form a compound, it is energetically unfavorable to create ions. Rather than transferring electrons to form octets, the atoms share electrons. The bonding force between the atoms is not ionic; instead, there is an attraction that each electron in the shared pair has for the two positive nuclei of the bonded atoms.



MCAT EXPERTISE

Think of bonds as a tug-of-war between two atoms. When the difference in electronegativity is great (more than 1.7), then the “stronger” atom wins all of the electrons and becomes the anion. When the electronegativity values are relatively similar, then we have a stalemate, or a covalent bond with mostly equal sharing of electrons.

Covalent compounds contain discrete molecular units with relatively weak intermolecular interactions. As a result, compounds like carbon dioxide (CO_2) tend to have lower melting and boiling points. In addition, because they do not break down into constituent ions, they are poor conductors of electricity in the liquid state or in aqueous solutions.

PROPERTIES OF COVALENT COMPOUNDS

The formation of one covalent bond may not be sufficient to fill the valence shell for a given atom. Thus, many atoms can form bonds with more than

one other atom, and most atoms can form multiple bonds with other atoms. Two atoms sharing one, two, or three pairs of electrons are said to be joined by a single, double, or triple covalent bond, respectively. The number of shared electron pairs between two atoms is called the bond order; hence, a single bond has a bond order of one, a double bond has a bond order of two, and a triple bond has a bond order of three. There are three important characteristics of a covalent bond to explain: bond length, bond energy, and polarity.

Bond Length

Bond length is the average distance between the two nuclei of atoms in a bond. As the number of shared electron pairs increases, the two atoms are pulled closer together, resulting in a decrease in bond length. Thus, for a given pair of atoms, a triple bond is shorter than a double bond, which is shorter than a single bond.

Bond Energy

Bond energy is the energy required to break a bond by separating its components into their isolated, gaseous atomic states. The greater the number of pairs of electrons shared between the atomic nuclei, the more energy is required to break the bonds holding the atoms together. Thus, triple bonds have the greatest bond energy, and single bonds have the lowest bond energy. We will discuss bond energy and calculations involving bond enthalpy in Chapter 7 of MCAT General Chemistry Review. By convention, the greater the bond energy is, the stronger the bond.



KEY CONCEPT

You will see this inverse relationship between bond length and strength in both organic and general chemistry.

	Bond Length	Bond Strength
C–C	longest	weakest
C=C	medium	medium
C≡C	shortest	strongest

Know this relationship on Test Day and you'll earn quick points!

Polarity

Polarity occurs when two atoms have a relative difference in electronegativities. When these atoms come together in covalent bonds, they must negotiate the degree to which the electron pairs will be shared. The atom with the higher electronegativity gets the larger share of the electron density. A polar bond creates a dipole, with the positive end of the dipole at the less electronegative atom and the negative end at the more electronegative atom, as shown in Figure 3.5.

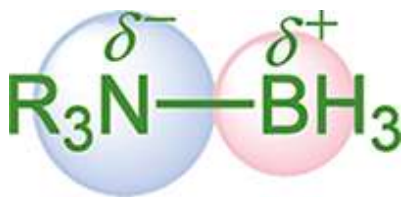


Figure 3.5. Polar Covalent Bond in an Amine Borane

Nitrogen takes on a partial negative charge (δ^-), boron takes on a partial positive charge (δ^+).

When atoms that have identical or nearly identical electronegativities share electron pairs, they do so with equal distribution of the electrons. This is called a nonpolar covalent bond, and there is no separation of charge across the bond. Note that only bonds between atoms of the same element will have exactly the same electronegativity and therefore exhibit a purely equal distribution of electrons. The seven common diatomic molecules are H_2 , N_2 , O_2 , F_2 , Cl_2 , Br_2 , and I_2 . At the same time, many bonds are close to nonpolar. Any bond between atoms with a difference in electronegativity less than 0.5 is generally considered nonpolar.



MNEMONIC

Here's a quick way to remember the naturally occurring diatomic elements on the periodic table: they form the number 7 on the periodic table (except for H), there are 7 of them, and most of them are in Group VIIA: H_2 , N_2 , O_2 , F_2 , Cl_2 , Br_2 , I_2 .

Polar Covalent Bond

Atoms that differ moderately in their electronegativities will share electrons unevenly, resulting in polar covalent bonds. While the difference in their electronegativities (between 0.5 and 1.7) is not enough to result in the formation of an ionic bond, it is sufficient to cause a separation of charge across the bond. This results in the more electronegative element acquiring a greater portion of the electron density, taking on a partial negative charge (δ^-), and the less electronegative element acquiring a smaller portion of the electron density, taking on a partial positive charge (δ^+). For instance, the covalent bond in HCl is polar because the two atoms have a moderate difference in electronegativity ($\Delta\text{EN} = 0.9$). In this bond, the chlorine atom gains a partial negative charge, and the hydrogen atom gains a partial positive charge. The difference in charge between the atoms is indicated by an arrow crossed at its tail end (giving the appearance of a “plus” sign) and pointing toward the negative end, as shown in Figure 3.6.

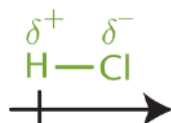


Figure 3.6. Dipole Moment of HCl



MCAT EXPERTISE

The range of electronegativities for nonpolar bonds is roughly 0 to 0.5. Polar bonds are found from 0.5 to 1.7, and ionic bonds are at 1.7 and above. Some chemistry courses allude to a grey area from 1.7 to 2.0. For the MCAT, if a molecule in this range has a metal and nonmetal, then it is effectively ionic; otherwise, it is polar covalent.

A molecule that has such a separation of positive and negative charges is called a polar molecule. The dipole moment of the polar bond or polar molecule is a vector quantity given by the equation:

$$p = qd$$

Equation 3.1

where p is the dipole moment, q is the magnitude of the charge, and d is the displacement vector separating the two partial charges. The dipole moment vector, represented by an arrow pointing from the positive to the negative charge, is measured in Debye units (coulomb–meters).

COORDINATE COVALENT BONDS

In a coordinate covalent bond, both of the shared electrons originated on the same atom. Generally, this means that a lone pair of one atom attacked another atom with an unhybridized p-orbital to form a bond, as shown in Figure 3.7. Once such a bond forms, however, it is indistinguishable from any other covalent bond. The distinction is only helpful for keeping track of the valence electrons and formal charges. Coordinate covalent bonds are typically found in Lewis acid–base reactions, described in Chapter 10 of MCAT General Chemistry Review. A Lewis acid is any compound that will accept a lone pair of electrons, while a Lewis base is any compound that will donate a pair of electrons to form a covalent bond.

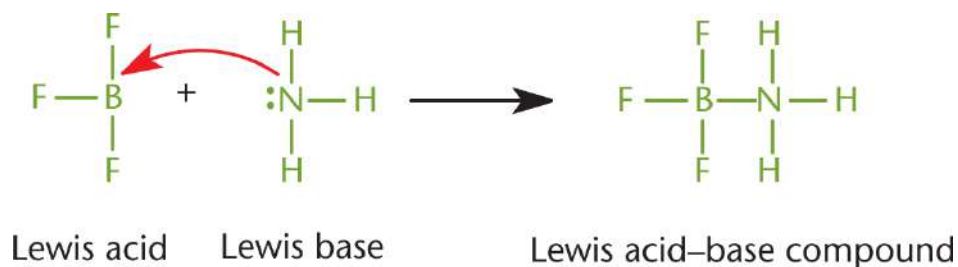


Figure 3.7. Coordinate Covalent Bond



BRIDGE

The chemistry that creates coordinate covalent bonds appears in many guises. These reactions can be called nucleophile–electrophile reactions, described in Chapter 4 of MCAT Organic Chemistry Review; Lewis acid–base reactions, described in Chapter 10 of MCAT General Chemistry Review, or complexation reactions, described in Chapter 9 of MCAT General Chemistry Review.

Here, NH_3 donates a pair of electrons to form a coordinate covalent bond; thus, it acts as a Lewis base. At the same time, BF_3 accepts this pair of electrons to form the coordinate covalent bond; thus, it acts as a Lewis acid.

COVALENT BOND NOTATION

The electrons involved in a covalent bond are in the valence shell and are bonding electrons, while those electrons in the valence shell that are not involved in covalent bonds are nonbonding electrons. The unshared

electron pairs are also known as lone pairs because they are associated only with one atomic nucleus. Because atoms can bond with other atoms in many different combinations, the Lewis structure system of notation was developed to keep track of the bonded and nonbonded electron pairs.

Think of Lewis structures as a bookkeeping method for electrons. The number of valence electrons attributed to a particular atom in the Lewis structure of a molecule is not necessarily the same as the number of valence electrons in the neutral atom. This difference accounts for the formal charge of an atom in a Lewis structure. Often, more than one Lewis structure can be drawn for a molecule. If the possible Lewis structures differ in their bond connectivity or arrangement, then the Lewis structures represent different possible compounds. However, if the Lewis structures show the same bond connectivity and differ only in the arrangement of the electron pairs, then these structures represent different resonance forms of a single compound. Note that Lewis structures do not represent the actual or even theoretical geometry of a real compound. Their usefulness lies in showing the different possible ways in which atoms may be combined to form different compounds or resonance forms of a single compound.

When more than one arrangement can be made, one can assess the likelihood of each arrangement by checking the formal charges on the atoms in each arrangement. The arrangement that minimizes the number and magnitude of formal charges is usually the most stable arrangement of the compound.

Lewis Structures

A Lewis structure, or Lewis dot diagram, is the chemical symbol of an element surrounded by dots, each representing one of the s or p valence electrons of the atom. The Lewis symbols of the elements in the second period of the periodic table are shown in Table 3.1.

$\cdot\text{Li}\cdot$	Lithium	$\cdot\ddot{\text{N}}\cdot$	Nitrogen
$\cdot\text{Be}\cdot$	Beryllium	$\cdot\ddot{\text{O}}\cdot$	Oxygen
$\cdot\ddot{\text{B}}\cdot$	Boron	$\cdot\ddot{\text{F}}\cdot$	Fluorine
$\cdot\ddot{\text{C}}\cdot$	Carbon	$\cdot\ddot{\text{Ne}}\cdot$	Neon

Table 3.1. Lewis Symbols for Period 2 Elements



KEY CONCEPT

In drawing Lewis dot structures, remember that some atoms can expand their octets by utilizing the d-orbitals in their outer shell. This will only take place with atoms in period 3 or greater.



MCAT EXPERTISE

The number of dots in Lewis Structure notation comes from group numbers. Lithium is in Group IA and therefore has one electron (dot). Carbon is in Group IVA and has four dots.

Just as a Lewis symbol is used to represent the distribution of valence electrons in an atom, it can also be used to represent the distribution of valence electrons in a molecule. For example, the Lewis symbol for a fluoride ion, F^- , is $\cdot\ddot{\text{F}}^-$; the Lewis structure of the diatomic molecule F_2 is $\cdot\ddot{\text{F}} - \ddot{\text{F}}\cdot$. Certain rules must be followed in assigning a Lewis structure to a molecule. The steps for drawing a Lewis structure are outlined here, using HCN as an example.

- Draw out the backbone of the compound—that is, the arrangement of atoms. In general, the least electronegative atom is the central atom. Hydrogen (always) and the halogens F, Cl, Br, and I (usually) occupy a terminal position.

In HCN, H must occupy an end position. Of the remaining two atoms, C is the least electronegative and, therefore, occupies the central position. Therefore, the skeletal structure is as follows:



- Count all the valence electrons of the atoms. The number of valence electrons of the molecule is the sum of the valence electrons of all atoms present:

H has 1 valence electron

C has 4 valence electrons

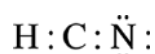
N has 5 valence electrons; therefore,

HCN has a total of 10 valence electrons.

- Draw single bonds between the central atom and the atoms surrounding it. Each single bond corresponds to a pair of electrons:



- Complete the octets of all atoms bonded to the central atom, using the remaining valence electrons left to be assigned. Recall that H is an exception to the octet rule because it can only have two valence electrons. In this example, H already has two valence electrons from its bond with C.



- Place any extra electrons on the central atom. If the central atom has less than an octet, try to write double or triple bonds between the central and surrounding atoms using the lone pairs on the surrounding atoms.

The HCN structure above does not satisfy the octet rule for C because C only has four valence electrons. Therefore, two lone electron pairs from the N atom must be moved to form two more bonds with C, creating a triple bond between C and N. To make it easier to visualize, bonding electron pairs are represented as lines. You should be familiar with both dot and line notation for bonds.



Now, the octet rule is satisfied for all three atoms; C and N have eight valence electrons, and H has two valence electrons.

Formal Charge

To determine if a Lewis structure is representative of the actual arrangement of atoms in a compound, one must calculate the formal charge of each atom. In doing so, assume a perfectly equal sharing of all bonded electron pairs, regardless of actual differences in electronegativity. In other words, assume that each electron pair is split evenly between the two nuclei in the bond. The difference between the number of electrons assigned to an atom in a Lewis structure and the number of electrons normally found in that atom's valence shell is the formal charge. A simple equation you can use to calculate formal charge is:

$$\text{formal charge} = V - N_{\text{nonbonding}} - \frac{1}{2}N_{\text{bonding}}$$

Equation 3.2

where V is the normal number of electrons in the atom's valence shell, $N_{\text{nonbonding}}$ is the number of nonbonding electrons, and N_{bonding} is the number of bonding electrons (double the number of bonds because each bond has two electrons). The charge of an ion or compound is equal to the sum of the formal charges of the individual atoms comprising the ion or compound.



MNEMONIC

A less formal way to calculate formal charge is with the formula:
Formal charge = valence electrons – dots – sticks

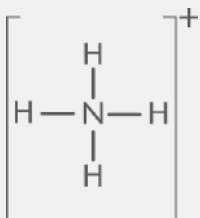
Where a "dot" refers to a lone electron and a "stick" refers to a bond.

Example:

Calculate the formal charge on the central N atom of $[\text{NH}_4]^+$.

Solution:

The Lewis structure of $[\text{NH}_4]^+$ is:



Nitrogen is in Group VA; thus, it has five valence electrons. In $[\text{NH}_4]^+$, N has 4 bonds (eight bonding electrons and zero nonbonding electrons).

Thus, $V = 5$; $N_{\text{bonding}} = 8$; $N_{\text{nonbonding}} = 0$

$$\text{Formal charge} = 5 - 0 - \frac{1}{2} (8) = +1$$

Thus, the formal charge on the N atom in $[\text{NH}_4]^+$ is +1.

One can also use logic to determine formal charge. As drawn, N has four bonds. Assuming equal sharing of the electrons in the bonds, this means N has four valence electrons. In its normal state, N has

five valence electrons. Thus, nitrogen has one fewer electron than its normal state, and has a +1 charge.

Let us offer a brief note of explanation on the difference between formal charge and oxidation number: formal charge underestimates the effect of electronegativity differences, whereas oxidation numbers overestimate the effect of electronegativity differences, assuming that the more electronegative atom has a 100 percent share of the bonding electron pair. For example, in a molecule of CO_2 (carbon dioxide), the formal charge on each of the atoms is 0, but the oxidation number of each of the oxygen atoms is -2 and of the carbon is $+4$. In reality, the distribution of electron density between the carbon and oxygen atoms lies somewhere between the extremes predicted by the formal charges and the oxidation states.

Resonance

As suggested earlier, it may be possible to draw two or more Lewis structures that demonstrate the same arrangement of atoms but that differ in the specific placement of the electrons. These are called resonance structures and are represented with a double-headed arrow between them. The actual electronic distribution in the compound is a hybrid, or composite, of all of the possible resonance structures. For example, SO_2 has three resonance structures, as shown in Figure 3.8.

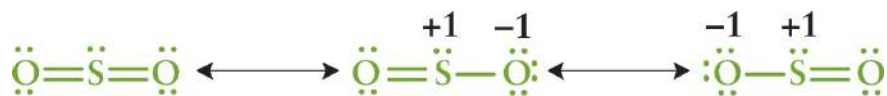


Figure 3.8 Resonance Structures for SO_2

The double-headed arrows indicate that these molecules are involved in a resonance hybrid.



BRIDGE

Resonance is an important topic in both general and organic chemistry. It allows for greater stability, delocalizing electrons and charges over what is known as a π (pi) system. Resonance in organic molecules is discussed in Chapter 3 of MCAT Organic Chemistry Review.

The nature of the bonds within the actual compound is a hybrid of these three structures. If one were to evaluate the spectral data, it would indicate that the two S–O bonds are identical and equivalent. This phenomenon is known as resonance, and the actual structure of the compound is called the resonance hybrid.

The first resonance structure in Figure 3.8 is significantly more stable than the other two structures. Consequently, it is the major contributor to the resonance hybrid. In general, the more stable the structure, the more it contributes to the character of the resonance hybrid. In Figure 3.8, the minor contributors contain formal charges, indicating decreased stability. One can use formal charge to assess the stability of resonance structures according to the following guidelines:

- A Lewis structure with small or no formal charges is preferred over a Lewis structure with large formal charges.
- A Lewis structure with less separation between opposite charges is preferred over a Lewis structure with a large separation of opposite charges.
- A Lewis structure in which negative formal charges are placed on more electronegative atoms is more stable than one in which the negative formal charges are placed on less electronegative atoms.

Example:

Write the resonance structures for $[\text{NCO}]^-$.

Solution:

C is the least electronegative of the three given atoms. Therefore, the C atom occupies the central position in the skeletal structure of $[\text{NCO}]^-$:



N has 5 valence electrons;

C has 4 valence electrons;

O has 6 valence electrons;

and the species has one negative charge.

Total valence electrons = $5 + 4 + 6 + 1 = 16$

Draw single bonds between the central C atom and the surrounding atoms, N and O. Draw a pair of electrons to

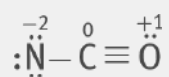
represent each bond.



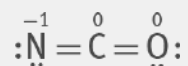
Complete the octets of N and O with the remaining 12 electrons.



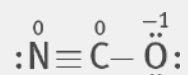
The C octet is incomplete. There are three ways in which double and triple bonds can be formed to complete the C octet: two lone pairs from the O atom can be used to form a triple bond between the C and O atoms:



Or one lone electron pair can be taken from both O and N to form two double bonds, one between N and C, the other between O and C:



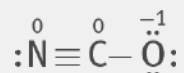
Or two lone electron pairs can be taken from the N atom to form a triple bond between the C and N atoms:



All three are resonance structures of $[\text{NCO}]^-$

Assign formal charges to each atom of each resonance structure.

The most stable structure is this:



because the charges are minimized, and the negative formal charge is on the most electronegative atom, O.

Exceptions to the Octet Rule

As stated previously, the octet rule has many exceptions. In addition to hydrogen, helium, lithium, beryllium, and boron, which are exceptions because they cannot or do not usually reach the octet, all elements in or beyond the third period may be exceptions because they can take on more than eight electrons in their valence shells. These electrons can be placed into orbitals of the d subshell, and as a result, atoms of these elements can form more than four bonds. On Test Day, don't automatically discount a Lewis structure with a central atom that has more than four bonds—the testmakers may be testing your ability to recognize that many atoms can expand their valence shells beyond the octet.

Consider the sulfate ion, SO_4^{2-} . In the Lewis structure for the sulfate ion, giving the sulfur 12 valence electrons permits three of the five atoms to be assigned a formal charge of zero. The sulfate ion can be drawn in at least six resonance forms, many of which have two double bonds attached to a different combination of oxygen atoms. Figure 3.9 shows two of the possible forms.

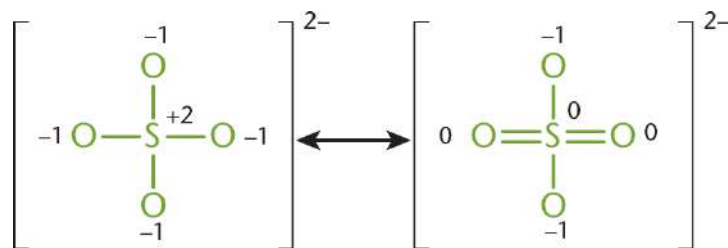


Figure 3.9. Two Different Resonance Forms of the Sulfate Ion



MCAT EXPERTISE

Don't be surprised if you can draw more than two resonance structures for a molecule or ion. Becoming proficient at drawing and, more importantly, recognizing resonance structures will save you time on Test Day.

GEOMETRY AND POLARITY

Because Lewis dot structures do not suggest or reflect the actual geometric arrangement of atoms in a compound, we need another system to provide this information. One such system is known as the valence shell electron pair repulsion (VSEPR) theory.

Valence Shell Electron Pair Repulsion (VSEPR) Theory

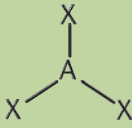
VSEPR theory uses Lewis dot structures to predict the molecular geometry of covalently bonded molecules. It states that the three-dimensional

arrangement of atoms surrounding a central atom is determined by the repulsions between bonding and nonbonding electron pairs in the valence shell of the central atom. These electron pairs arrange themselves as far apart as possible, thereby minimizing repulsive forces. The following steps are used to predict the geometrical structure of a molecule using the VSEPR theory:

- Draw the Lewis dot structure of the molecule.
- Count the total number of bonding and nonbonding electron pairs in the valence shell of the central atom.
- Arrange the electron pairs around the central atom so that they are as far apart as possible. For example, the compound AX_2 has the Lewis structure $X : A : X$. The A atom has two bonding electron pairs in its valence shell. To position these electron pairs as far apart as possible, their geometric structure should be linear:



A summary of electronic geometries as predicted by VSEPR theory is shown in Table 3.2.

Regions of Electron Density	Example	Geometric Arrangement of Electron Pairs Around the Central Atom	Shape	Angle between Electron Pairs
2	$BeCl_2$	$X - A - X$	linear	180°
3	BH_3		trigonal planar	120°

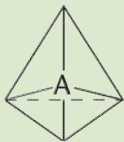
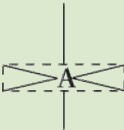
Regions of Electron Density	Example	Geometric Arrangement of Electron Pairs Around the Central Atom	Shape	Angle between Electron Pairs
4	CH ₄		tetrahedral	109.5°
5	PCl ₅		trigonal bipyramidal	90°, 120°, 180°
6	SF ₆		octahedral	90°, 180°

Table 3.2. VSEPR Theory This table lists the five most common electronic configurations of molecules.



MCAT EXPERTISE

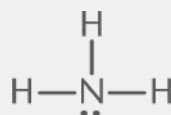
According to the AAMC's official content lists, you need to be prepared to draw and identify structural formulas for molecules involving H, C, N, O, F, S, P, Si, and Cl. Rather than memorizing these elements, however, just be familiar with the process of creating a Lewis diagram for any element and predicting its three-dimensional shape from VSEPR theory.

Example:

Predict the molecular geometry of NH_3 .

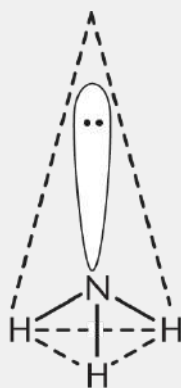
Solution:

The Lewis structure of NH_3 is:



The central atom, N, has three bonding electron pairs and one nonbonding electron pair, for a total of four electron pairs.

The four electron pairs will be farthest apart when they occupy the corners of a tetrahedron. Because one of the four electron pairs is a lone pair, the observed molecular geometry is trigonal pyramidal, shown below.



In describing the shape of a molecule, only the arrangement of atoms (not electrons) is considered. Even though the electron pairs are arranged tetrahedrally, the shape of NH_3 is pyramidal. It is not trigonal planar because the lone pair repels the three bonding electron pairs, causing them to move as far apart as possible.

Example:

Predict the geometry of CO_2 .

Solution:

The Lewis structure of CO_2 is $\ddot{\text{O}}::\text{C}::\ddot{\text{O}}$

The double bond behaves just like a single bond for the purposes of predicting molecular shape. This compound has two groups of electrons around the carbon. According to the VSEPR theory, the two sets of electrons will orient themselves 180° apart, on opposite sides of the carbon atom, minimizing electron repulsion. Therefore, the molecular structure of CO_2 is linear: $\ddot{\text{O}}=\text{C}=\ddot{\text{O}}$



KEY CONCEPT

The shapes from Table 3.2 refer to electronic geometry, which is different from molecular geometry. In the worked example, notice that the ammonia molecule has a tetrahedral electronic structure, but is considered to have a molecular structure that is trigonal pyramidal.

One subtlety that the MCAT loves to test is the difference between electronic geometry and molecular geometry. Electronic geometry describes the spatial arrangement of all pairs of electrons around the

central atom, including both the bonding and the lone pairs. In contrast, the molecular geometry describes the spatial arrangement of only the bonding pairs of electrons. The coordination number, which is the number of atoms that surround and are bonded to a central atom, is the relevant factor when determining molecular geometry. For example, consider that CH_4 (methane), NH_3 (ammonia), and H_2O all have the same electronic geometry: in each compound, four pairs of electrons surround the central atom. This is tetrahedral electronic geometry. However, because each molecule has a different coordination number, they have different molecular geometries. In molecular geometry, methane has tetrahedral geometry, ammonia has trigonal pyramidal geometry, and water is identified as angular or bent.



MCAT EXPERTISE

CH_4 , NH_3 , and H_2O all have a tetrahedral electronic geometry, but differ in their molecular shapes:

CH_4 is tetrahedral, NH_3 is pyramidal, and H_2O is bent or angular.

The distinction is important, and the MCAT will primarily focus on molecular geometry. However, there is one important implication of electronic geometry: the determination of the ideal bond angle. Tetrahedral electronic geometry, for example, is associated with an ideal bond angle of 109.5° ; however, nonbonding pairs are able to exert more repulsion than bonding pairs because these electrons reside closer to the nucleus. Thus, the angle in ammonia is closer to 107° , and the angle in water is 104.5° .

Polarity of Molecules

When two atoms of different electronegativities bond covalently, sharing one or more pairs of electrons, the resulting bond is polar, with the more electronegative atom possessing the greater share of the electron density. However, the presence of bond dipoles does not necessarily result in a molecular dipole; that is, an overall separation of charge across the molecule. We must first consider the molecular geometry and the vector addition of the bond dipoles based upon that molecular geometry. A compound with nonpolar bonds is always nonpolar. However, a compound with polar bonds may be polar or nonpolar, depending upon the spatial orientation of the polar bonds in the molecule. If the compound has a molecular geometry such that the bond dipole moments cancel each other out (that is, if the vector sum is zero), then the result is a nonpolar compound. For example, CCl_4 (carbon tetrachloride) has four polar C–Cl bonds, but because the molecular geometry of carbon tetrachloride is tetrahedral, the four bond dipoles point to the vertices of the tetrahedron and, therefore, cancel each other out, resulting in a nonpolar compound, as shown in Figure 3.10.

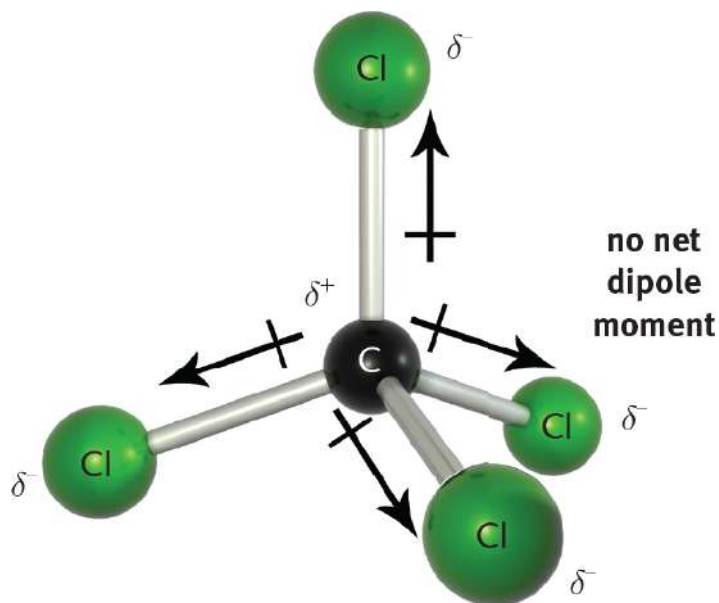


Figure 3.10. CCl_4 is a Nonpolar Compound with Four Polar Bonds

However, when the molecular geometry is arranged such that the bond dipoles do not cancel each other out, the molecule will have a net dipole moment and will therefore be polar. For instance, the O-H bonds in H_2O are polar, with each hydrogen atom assuming a partial positive charge and the oxygen assuming a partial negative charge. Recall that the molecular geometry of water is angular (bent). Therefore, the vector summation of the bond dipoles results in a molecular dipole moment from the partially positive hydrogen end to the partially negative oxygen end, as illustrated in Figure 3.11.

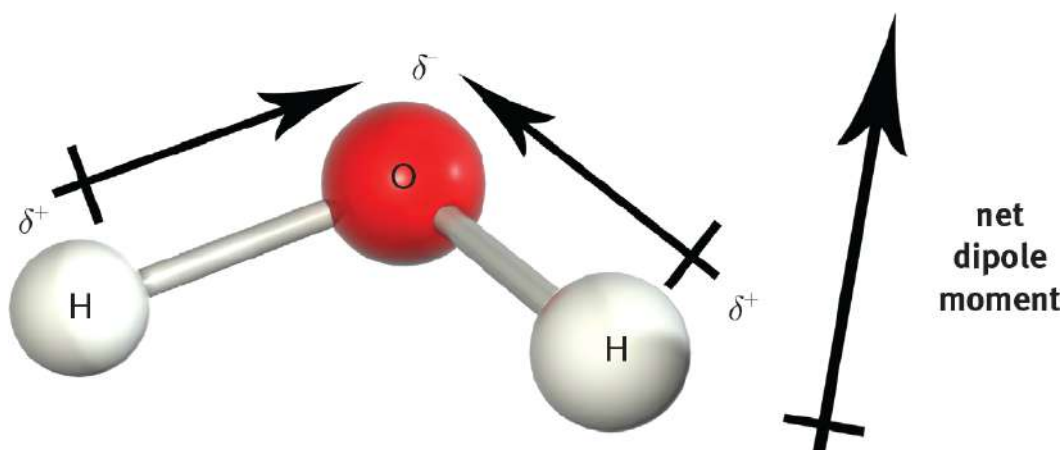


Figure 3.11. H_2O is a Polar Molecule with Two Polar Bonds



MCAT EXPERTISE

Be careful! If you spot a polar bond in a molecule, the molecule can be either polar or nonpolar. On the contrary, when you see only nonpolar bonds in a molecule, the structure must be nonpolar. When in doubt, draw out relevant structures on your scratch paper on Test Day.

ATOMIC AND MOLECULAR ORBITALS

To finish the discussion of covalent bonds, we need to address the concept of atomic and molecular orbitals. Recall the model of the atom as a dense, positively charged nucleus surrounded by a cloud of electrons organized into orbitals (regions in space surrounding the nucleus within which there

are certain probabilities of finding an electron). The four quantum numbers describe the energy and position of an electron in an atom. While the principal quantum number, n , indicates the average energy level of the shell, the azimuthal quantum number, l , describes the subshells within each principal energy level. When $l = 0$, this indicates the s subshell, which has one orbital that is spherical in shape. The $1s$ -orbital ($n = 1$, $l = 0$, $m_l = 0$) is plotted in Figure 3.12.

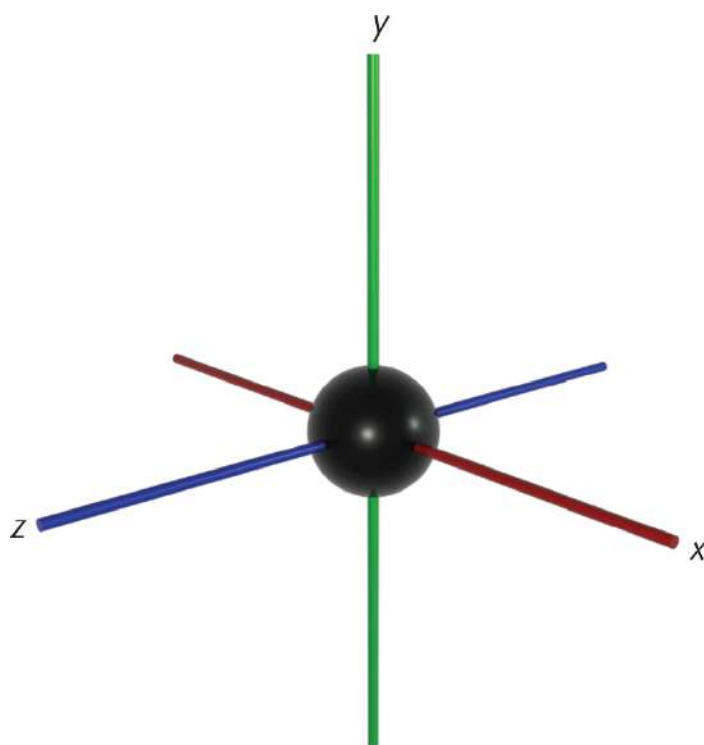


Figure 3.12. $1s$ -Orbital



BRIDGE

Quantum Numbers (Chapter 1 of MCAT General Chemistry Review)
revisited:

For any value of n , there are n values of l ($0 \rightarrow n - 1$).

- $l = 0 \rightarrow s$
- $l = 1 \rightarrow p$
- $l = 2 \rightarrow d$
- $l = 3 \rightarrow f$

For any value of l , there are $2l + 1$ values of m_l (number of orbitals); values range from $-l$ to l .

When $l = 1$, this indicates the p subshell, which has three orbitals shaped like barbells along the x -, y -, and z -axes at right angles to each other. The $2p$ -orbitals ($n = 2$, $l = 1$, $m_l = -1, 0$, and $+1$) are plotted in Figure 3.13.

Although well beyond the scope of the MCAT, mathematical analysis of the wave functions of the orbitals is used to determine and assign plus and minus signs to each lobe of the p -orbitals. The shapes of the five d -orbitals and the seven f -orbitals are more complex and do not need to be memorized for the MCAT.

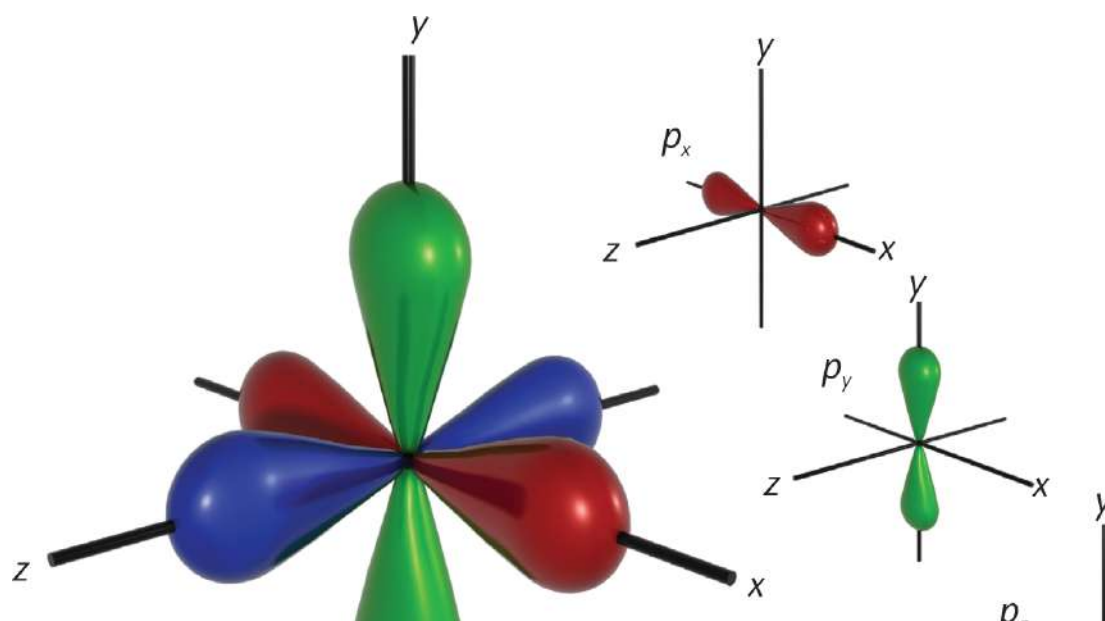




Figure 3.13. p-Orbitals on the x-, y-, and z-Axes

When two atoms bond to form a compound, the atomic orbitals interact to form a molecular orbital that describes the probability of finding the bonding electrons in a given space. Molecular orbitals are obtained by combining the wave functions of the atomic orbitals. Qualitatively, the overlap of two atomic orbitals describes this molecular orbital. If the signs of the two atomic orbitals are the same, a bonding orbital forms. If the signs are different, an antibonding orbital forms.

Two different patterns of overlap are observed in the formation of molecular bonds. When orbitals overlap head-to-head, the resulting bond is a sigma (σ) bond. σ bonds allow for free rotation about their axes because the electron density of the bonding orbital is a single linear accumulation between the atomic nuclei.

When the orbitals overlap in such a way that there are two parallel electron cloud densities, a pi (π) bond is formed. π bonds do not allow for free rotation because the electron densities of the orbitals are parallel and cannot be twisted in such a way that allows continuous overlapping of the clouds of electron densities.

MCAT Concept Check 3.3:

Before you move on, assess your understanding of the material with these questions.

Describe the relationship between bond strength, bond length, and bond energy.

For what values of ΔEN will a nonpolar covalent bond form? Polar covalent? Ionic?

- Nonpolar covalent:

-
- Polar covalent:

-
- Ionic:
-

Draw a Lewis dot structure for the carbonate ion (CO_3^{2-}) and its two other resonance structures.

Predict the molecular geometries of the following molecules:

- PCl_5 :

- MgF_2 :

- AlF_3 :

- UBr_6 :

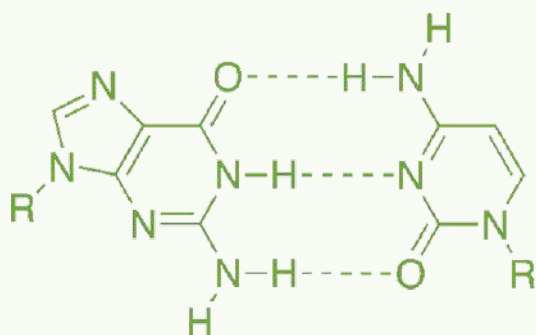
- SiH_4 :

3.4 Intermolecular Forces

LEARNING GOALS

After Chapter 3.4, you will be able to:

- Order the intermolecular forces from strongest to weakest
- Describe what occurs during dipole-dipole, hydrogen bonding, and London dispersion force interactions
- Predict what intermolecular forces are possible for given interacting molecules:



Atoms and compounds participate in weak electrostatic interactions. The strength of these intermolecular forces can impact certain physical properties, such as melting and boiling points. The weakest of the intermolecular interactions are the dispersion forces, also known as

London forces. Next are the dipole–dipole interactions, which are of intermediate strength. Finally, we have the strongest type of interaction, the hydrogen bond, which is a misnomer because there is no actual sharing or transfer of electrons. We must keep in mind, however, that even hydrogen bonds, the strongest of these interactions, only have about 10 percent of the strength of a covalent bond. Therefore, these electrostatic interactions can be overcome with small or moderate amounts of energy.



BRIDGE

These intermolecular forces are the bonding forces that keep a substance together in its solid or liquid state and determine whether two substances are miscible or immiscible in solution. Solutions and solubility are discussed in Chapter 9 of MCAT General Chemistry Review.

LONDON DISPERSION FORCES

The bonding electrons in nonpolar covalent bonds may appear to be shared equally between two atoms, but at any point in time, they will be located randomly throughout the orbital. In a given moment, the electron density may be unequally distributed between the two atoms. This results in a rapid polarization and counterpolarization of the electron cloud and the formation of short-lived dipole moments. Subsequently, these dipoles interact with the electron clouds of neighboring compounds, inducing the formation of more dipoles. The momentarily negative end of one molecule

will cause the closest region in any neighboring molecule to become temporarily positive itself. This causes the other end of the neighboring molecule to become temporarily negative, which in turn induces other molecules to become temporarily polarized, and the cycle begins again. The attractive or repulsive interactions of these short-lived and rapidly shifting dipoles are known as London dispersion forces, a type of van der Waals force.

Dispersion forces are the weakest of all of the intermolecular interactions because they are the result of induced dipoles that change and shift moment to moment. They do not extend over long distances and are, therefore, significant only when molecules are in close proximity. The strength of the London force also depends on the degree and ease by which the molecules can be polarized—that is, how easily the electrons can be shifted around. Large molecules are more easily polarizable than comparable smaller molecules and thus possess greater dispersion forces.



REAL WORLD

While dispersion forces (a type of van der Waals force) are the weakest of the intermolecular attractions, when there are millions of these interactions there is an amazing power of adhesion. This is demonstrated by geckos' feet; the animal's ability to climb smooth, vertical, and even inverted surfaces is due to dispersion forces.

Despite their weak nature, don't underestimate the importance of dispersion forces. If it weren't for them, the noble gases would not liquefy

at any temperature because no other intermolecular forces exist between the noble gas atoms. The low temperatures at which noble gases liquefy are indicative of the very small magnitude of the dispersion forces between the atoms.

DIPOLE–DIPOLE INTERACTIONS

Polar molecules tend to orient themselves in such a way that the oppositely charged ends of the respective molecular dipoles are closest to each other: the positive region of one molecule is close to the negative region of another molecule. This arrangement is energetically favorable because an attractive electrostatic force is formed between the two molecules. This attractive force is denoted by dashed lines in most molecular notations and indicates a temporary bonding interaction, as shown in Figure 3.14.



Figure 3.14. Dipole–Dipole Interactions in HCl

Dipole–dipole interactions are present in the solid and liquid phases but become negligible in the gas phase because of the significantly increased distance between gas particles. Polar species tend to have higher melting and boiling points than nonpolar species of comparable molecular weight due to these interactions. Realize that London forces and dipole–dipole interactions are different not in kind but in duration. Both are electrostatic forces between opposite partial charges; the difference is only in the transience or permanence of the molecular dipole.



BRIDGE

In organic chemistry, carbonyl groups possess distinct dipoles that facilitate nucleophilic attacks. This is the focus of almost all of the reactions in Chapters 6 to 9 of MCAT Organic Chemistry Review.

HYDROGEN BONDS

Hydrogen bonds are a favorite topic on the MCAT. A hydrogen bond is a specific, unusually strong form of dipole–dipole interaction that may be intra- or intermolecular. Hydrogen bonds are not actually bonds—there is no sharing or transfer ring of electrons between two atoms. When hydrogen is bonded to one of three highly electronegative atoms—nitrogen, oxygen, or fluorine—the hydrogen atom carries only a small amount of the electron density in the covalent bond.

The hydrogen atom essentially acts as a naked proton. The positively charged hydrogen atom interacts with the partial negative charge of fluorine, oxygen, or nitrogen on nearby molecules. Substances that display hydrogen bonding tend to have unusually high boiling points compared to compounds of similar molecular weights that do not exhibit hydrogen bonding. The difference derives from the energy required to break the hydrogen bonds. Hydrogen bonding, shown in Figure 3.15, is particularly important in the behavior of water, alcohols, amines, and carboxylic acids.

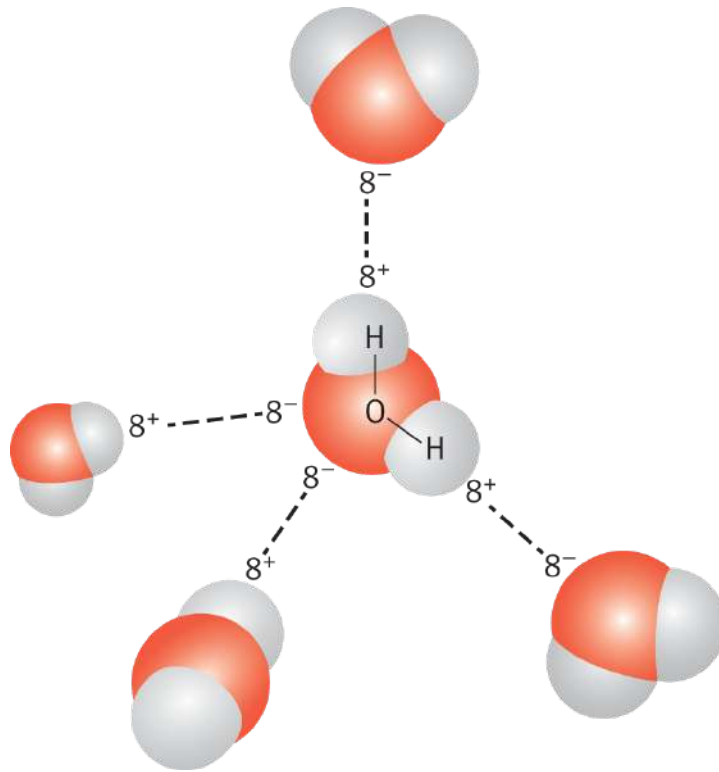


Figure 3.15. Hydrogen Bonding in Water



MNEMONIC

Hydrogen bonds: Pick up the FON (phone):
Hydrogen bonds exist in molecules containing a hydrogen bonded to Fluorine, Oxygen, or Nitrogen.

Many biochemical molecules, such as nucleotides, have different regions that are stabilized by hydrogen bonding, as shown in Figure 3.16. It is not an overstatement to say that—were it not for water’s ability to form hydrogen bonds and exist in the liquid state at room temperature—we would not exist (at least not in the form we recognize as “human”).

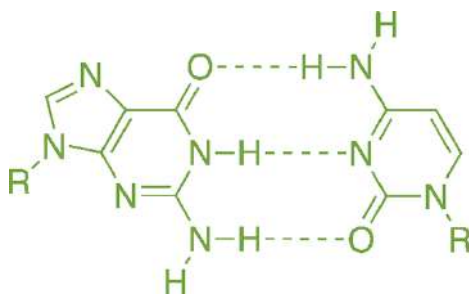


Figure 3.16. Hydrogen Bonding between Guanine and Cytosine

MCAT Concept Check 3.4:

Before you move on, assess your understanding of the material with these questions.

Rank the major intermolecular forces from strongest to weakest:

1. _____
2. _____
3. _____

Describe what occurs during dipole–dipole interactions.

In order to exhibit hydrogen bonding, what must be true of a given molecule?

Conclusion

This chapter built on our knowledge of the atom and the trends demonstrated by the elements in the periodic table to explain the different ways by which atoms partner together to form compounds, either by exchanging electrons to form ions, which are then held together by electrostatic attractions between opposite charges; or by sharing electrons to form covalent bonds. We discussed the nature and characteristics of covalent bonds, noting their relative lengths and energies, as well as polarities. A review of Lewis dot structures and VSEPR theory will prepare you for predicting likely bond arrangements, resonance structures, and molecular geometries. Finally, we compared the relative strengths of the most important intermolecular electrostatic interactions, noting that even the strongest of these—hydrogen bonding—is still much weaker than an actual covalent bond. The next time you're "browning" some of your food in a pan or the oven, take a moment to consider what's happening at the atomic and molecular level. It's not just cooking; it's science!

Concept Summary

Bonding

- Chemical bonds can be ionic or covalent.
- Elements will form bonds to attain a noble gas-like electron configuration.
- The octet rule states that elements will be most stable with eight valence electrons. However, there are many exceptions to this rule:
 - Elements with an incomplete octet are stable with fewer than eight electrons and include H, He, Li, Be, and B.
 - Elements with an expanded octet are stable with more than eight electrons and include all elements in period 3 or greater.
 - Compounds with an odd number of electrons cannot have eight electrons on each element.

Ionic Bonds

- An ionic bond is formed via the transfer of one or more electrons from an element with a relatively low ionization energy to an element with a relatively high electron affinity.
 - Ionic bonds occur between elements with large differences in electronegativity ($\Delta EN > 1.7$), usually between metals and nonmetals.

- A positively charged ion is called a cation. A negatively charged ion is called an anion.
 - The resulting electrostatic attraction between the ions causes them to remain in close proximity, forming the bond.
 - Ionic compounds form crystalline lattices—large, organized arrays of ions.
- Ionic compounds have unique physical and chemical properties.
 - Ionic compounds tend to dissociate in water and other polar solvents.
 - Ionic solids tend to have high melting points.

Covalent Bonds

- A covalent bond is formed via the sharing of electrons between two elements of similar electronegativities.
- Bond order refers to whether a covalent bond is a single bond, double bond, or triple bond. As bond order increases, bond strength increases, bond energy increases, and bond length decreases.
- Covalent bonds can be categorized as nonpolar or polar based on the nature of the elements involved.
 - Nonpolar bonds result in molecules in which both atoms have exactly the same electronegativity; some bonds are considered nonpolar when there is a very small difference in electronegativity between the atoms ($\Delta EN < 0.5$), even though they are technically slightly polar.
 - Polar bonds form when there is a significant difference in electronegativities ($\Delta EN = 0.5$ to 1.7), but not enough to transfer

electrons and form an ionic bond. In a polar bond, the more electronegative element takes on a partial negative charge, and the less electronegative element takes on a partial positive charge.

- Coordinate covalent bonds result when a single atom provides both bonding electrons while the other atom does not contribute any; coordinate covalent bonds are most often found in Lewis acid–base chemistry.
- Lewis dot symbols are a chemical representation of an atom's valence electrons.
- Drawing a complete Lewis dot structure requires a balance of valence, bonding, and nonbonding electrons in a molecule or ion.
- Formal charges exist when an atom is surrounded by more or fewer valence electrons than it has in its neutral state (assuming equal sharing of electrons in a bond).
- For any molecule with a π (pi) system of electrons, resonance structures exist; these represent all of the possible configurations of electrons—stable and unstable—that contribute to the overall structure.
- The valence shell electron pair repulsion (VSEPR) theory predicts the three-dimensional molecular geometry of covalently bonded molecules. In this theory, electrons—whether bonding or nonbonding—arrange themselves to be as far apart as possible from each other in three-dimensional space, leading to characteristic geometries.
 - Nonbonding electrons exert more repulsion than bonding electrons because they reside closer to the nucleus.
 - Electronic geometry refers to the position of all electrons in a molecule, whether bonding or nonbonding. Molecular geometry

refers to the position of only the bonding pairs of electrons in a molecule.

- The polarity of molecules is dependent on the dipole moment of each bond and the sum of the dipole moments in a molecular structure.
 - All polar molecules contain polar bonds.
 - Nonpolar molecules may contain nonpolar bonds, or polar bonds with dipole moments that cancel each other.
- σ and π bonds describe the patterns of overlap observed when molecular bonds are formed.
 - Sigma (σ) bonds are the result of head-to-head overlap.
 - Pi (π) bonds are the result of the overlap of two parallel electron cloud densities.

Intermolecular Forces

- Intermolecular forces are electrostatic attractions between molecules. They are significantly weaker than covalent bonds (which are weaker than ionic bonds).
 - London dispersion forces are the weakest interactions, but are present in all atoms and molecules. As the size of the atom or structure increases, so does the corresponding London dispersion force.
 - Dipole–dipole interactions, which occur between the oppositely charged ends of polar molecules, are stronger than London forces;

these interactions are evident in the solid and liquid phases but negligible in the gas phase due to the distance between particles.

- Hydrogen bonds are a specialized subset of dipole–dipole interactions involved in intra- and intermolecular attraction; hydrogen bonding occurs when hydrogen is bonded to one of three very electronegative atoms—fluorine, oxygen, or nitrogen.

Answers to Concept Checks

3.1

Ionic bonds form between ions and involve gain or loss of electrons. Covalent bonds occur when electrons are shared between atoms.

Any three examples that form incomplete octets (H, He, Li, Be, B) or expanded octets (Period 3 and greater) are acceptable.

The polarity in a covalent bond is determined by differences in electronegativity between the two atoms involved.

3.2

Metals lose electrons because they have low ionization energies, while nonmetals gain electrons because they have high electron affinities. These processes are complementary, leading to the formation of an ionic bond.

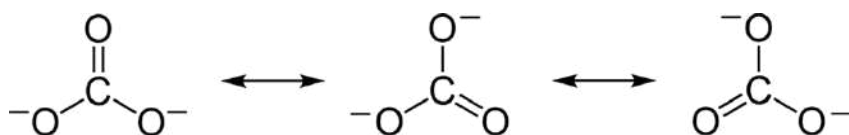
Some characteristics of ionic compounds include high melting and boiling points due to electrostatic attractions, solubility of ions in water due to interactions with polar solvents, good conductors of heat and electricity, crystal lattice arrangement to minimize repulsive forces, and large electronegativity differences between ions, among other possible answers.

3.3

Bond strength is defined by the electrostatic attraction between nuclei and electrons; multiple bonds (higher bond order) increases

strength. Bond length is a consequence of these attractions. The stronger the bond, the shorter it is. Bond energy is the minimum amount of energy needed to break a bond. The stronger the bond, the higher the bond energy.

Nonpolar covalent bonds form with $\Delta\text{EN} = 0$ to 0.5. Polar covalent bonds form with $\Delta\text{EN} = 0.5$ to 1.7. Ionic bonds form with $\Delta\text{EN} = 1.7$ or higher.



PCl_5 : trigonal bipyramidal, MgF_2 : linear, AlF_3 : trigonal planar, UBr_6 : octahedral, SiH_4 : tetrahedral

3.4

Hydrogen bonding > dipole–dipole interactions > dispersion (London) forces

A dipole consists of a segment of a molecule with partial positive and partial negative regions. The positive end of one molecule is attracted to the negative end of another molecule, and vice-versa.

To experience hydrogen bonding, a molecule must contain a hydrogen bonded to a very electronegative atom (nitrogen, oxygen, or fluorine).

Equations to Remember

(3.1) Dipole moment: $p = qd$

(3.2) Formal charge: $V - N_{\text{nonbonding}} - \frac{1}{2} N_{\text{bonding}}$



Shared Concepts

General Chemistry Chapter 1

Atomic Structure

General Chemistry Chapter 2

The Periodic Table

General Chemistry Chapter 4

Compounds and Stoichiometry

Organic Chemistry Chapter 3

Bonding

Organic Chemistry Chapter 4

Analyzing Organic Reactions

Physics and Math Chapter 5

Electrostatics and Magnetism

Discrete Practice Questions

Consult your online resources for additional practice.

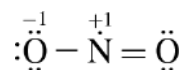
1. What is the character of the bond in carbon monoxide?

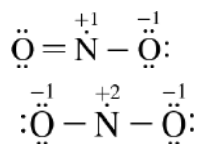
- (A) Ionic
- (B) Polar covalent
- (C) Nonpolar covalent
- (D) Coordinate covalent

2. Which of the following molecules contains the oxygen atom with the most negative formal charge?

- (A) H_2O
- (B) CO_3^{2-}
- (C) O_3
- (D) CH_2O

3. Which of the following structures contribute(s) most to NO_2 's resonance hybrid?





- (A) I only
 (B) III only
 (C) I and II only
 (D) I, II, and III

4. Which of the following correctly ranks the compounds below by ascending boiling point?

Acetone

KCl

Kr

Isopropyl alcohol

- (A) I < II < IV < III
 (B) III < IV < I < II
 (C) II < IV < I < III
 (D) III < I < IV < II

5. Both CO_3^{2-} and ClF_3 have three atoms bonded to a central atom. What is the best explanation for why CO_3^{2-} has trigonal planar electronic geometry, while ClF_3 has trigonal bipyramidal electronic geometry?

- (A) CO_3^{2-} has multiple resonance structures, while ClF_3 does not.
- (B) CO_3^{2-} has a charge of -2 , while ClF_3 has no charge.
- (C) ClF_3 has lone pairs on its central atom, while CO_3^{2-} has none.
- (D) CO_3^{2-} has lone pairs on its central atom, while ClF_3 has none.

6. Which of the following has the largest dipole moment?

- (A) HCN
- (B) H_2O
- (C) CCl_4
- (D) SO_2

7. Despite the fact that both C_2H_2 and HCN contain triple bonds, the lengths of these triple bonds are not equal. Which of the following is the best explanation for this finding?

- (A) In C_2H_2 , the bond is shorter because it is between atoms of the same element.
- (B) The two molecules have different resonance structures.
- (C) Carbon is more electronegative than hydrogen.
- (D) Nitrogen is more electronegative than carbon.

8. Which of the following is the best explanation of the phenomenon of hydrogen bonding?

- (A) Hydrogen has a strong affinity for holding onto valence electrons.
- (B) Hydrogen can only hold two valence electrons.
- (C) Electronegative atoms disproportionately carry shared electron pairs when bonded to hydrogen.
- (D) Hydrogen bonds have ionic character.

9. Which of the following best describes the number and character of the bonds in an ammonium cation?

- (A) Three polar covalent bonds
- (B) Four polar covalent bonds, of which none are coordinate covalent bonds
- (C) Four polar covalent bonds, of which one is a coordinate covalent bond
- (D) Four polar covalent bonds, of which two are coordinate covalent bonds

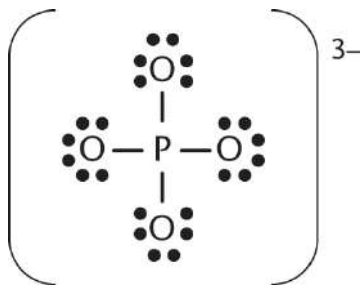
10. Although the octet rule dictates much of molecular structure, some atoms can violate the octet rule by being surrounded by more than eight electrons. Which of the following is the best explanation for why some atoms can exceed the octet?

- (A) Atoms that exceed the octet already have eight electrons in their outermost electron shell.
- (B) Atoms that exceed the octet only do so when bonding with transition metals.
- (C) Atoms that exceed the octet can do so because they have d-orbitals in which extra electrons can reside.
- (D) Some atoms can exceed the octet because they are highly electronegative.

11. Which of the following types of intermolecular forces provides the most accurate explanation for why noble gases can liquefy?

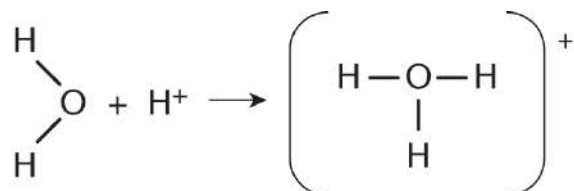
- (A) Hydrogen bonding
- (B) Ion-dipole interactions
- (C) Dispersion forces
- (D) Dipole-dipole interactions

12. In the structure shown, which atom(s) have the most positive charge?



- (A) The phosphorus atom has the most positive charge.
- (B) All atoms share the charge equally.
- (C) The four oxygen atoms share the highest charge.
- (D) The oxygen atom at the peak of the trigonal pyramidal geometry has the most positive charge.

13. Which of the following is the best name for the new bond formed in the reaction shown?



- (A) Nonpolar covalent bond
- (B) Ionic bond
- (C) Coordinate covalent bond
- (D) Hydrogen bond

14. Both BF_3 and NH_3 have three atoms bonded to the central atom. Which of the following is the best explanation for why the geometry of these two molecules is different?

- (A) BF_3 has three bonded atoms and no lone pairs, which makes its geometry trigonal pyramidal.
- (B) NH_3 is nonpolar, while BF_3 is polar.
- (C) NH_3 has three bonded atoms and one lone pair, which makes its geometry trigonal pyramidal.
- (D) BF_3 is nonpolar, while NH_3 is polar.

15. Which of the following best describes an important property of bond energy?

- (A) Bond energy increases with increasing bond length.
- (B) The more shared electron pairs comprising a bond, the higher the energy of that bond.
- (C) Single bonds are more difficult to break than double bonds.
- (D) Bond energy and bond length are unrelated.

Explanations to Discrete Practice Questions

1. B

Carbon monoxide, CO, has a triple bond between carbon and oxygen, with the carbon and oxygen each retaining one lone pair. In polar covalent bonds, the difference in electronegativity between the bonded atoms is great enough to cause electrons to move disproportionately toward the more electronegative atom but not great enough to transfer electrons completely. This is the case for CO. Oxygen is significantly more electronegative than carbon, so electrons will be disproportionately carried on the oxygen, leaving the carbon atom with a slight positive charge.

2. B

To answer this question, one must understand the contribution of resonance structures to average formal charge. In (B), there are three possible resonance structures. Each of the three oxygen atoms carries a formal charge of -1 in two out of the three structures. This averages to approximately $-\frac{2}{3}$ charge on each oxygen atom, which is more negative than in the other answer choices. Both water and formaldehyde, (A) and (D), have no formal charge on the oxygen. Ozone, (C), has a $+\frac{1}{2}$ on two of the three oxygens and a $+1$ charge on the central oxygen.

3. C

The two greatest contributors are structures I and II. Resonance structures are representations of how charges are shared across a molecule. In reality, the charge distribution is a weighted average of contributing resonance structures. The most stable resonance structures are those that minimize charge on the atoms in the molecule; the more stable the structure, the more it will contribute to the overall charge distribution in the molecule. Structures I and II minimize formal charges, so will be the largest contributors to the resonance hybrid.

4. D

The key to answering this question is to understand the types of intermolecular forces that exist in each of these molecules because larger intermolecular forces correspond to higher boiling points. Kr is a noble gas with a full octet, so the only intermolecular forces present are London dispersion forces, the weakest type of intermolecular forces. Acetone and isopropyl alcohol are both polar, so both have dipole-dipole interactions, which are stronger than dispersion forces. However, isopropyl alcohol can also form hydrogen bonds, increasing its boiling point. Finally, the strongest interactions are ionic bonds, which exist in potassium chloride.

5. C

The central carbon in carbonate has no lone pairs. It has three resonance structures, each of which involves a double bond between carbon and one of the three oxygens. Having made four bonds, carbon has no further orbitals for bonding or to carry lone pairs. This makes carbonate's geometry trigonal planar. Alternatively, ClF_3 also has three bonds; however, chlorine still maintains two extra lone pairs. These lone

pairs each inhabit one orbital, meaning that the central chloride must organize five items about itself: three bonds to fluorides and two lone pairs. The best configuration for maximizing the distance between all of these groups is trigonal bipyramidal. (A) and (B) are true statements but do not account for the difference in geometry.

6. A

The best way to approach this problem is to draw the structure of each of these molecules, then consider the electronegativity of each bond as it might contribute to an overall dipole moment. HCN is the correct answer because of the large differences in electronegativity aligned in a linear fashion. There is a strong dipole moment in the direction of nitrogen, without any other moments canceling it out. Water, (B), has two dipole moments, one from each hydrogen pointing in the direction of oxygen. The molecule is bent, and the dipole moments partially cancel out. There is a molecular dipole, but it is not as strong as in HCN. Sulfur dioxide, (C), has a similar bent configuration, and its dipole will again be smaller than that of HCN. Further, oxygen and sulfur do not have as large a difference in electronegativity, so even the individual bond dipoles are smaller than those in the other molecules. CCl_4 , (D), has tetrahedral geometry. Although each of the individual C–Cl bonds is highly polar, the orientation of these bonds causes the dipoles to cancel each other out fully, yielding no overall dipole moment.

7. D

Bond lengths decrease as the bond order increases, and they also decrease with larger differences in electronegativity. In this case, because both C_2H_2 and HCN have triple bonds, we cannot compare the bond lengths based on bond order. We must then rely on other periodic

trends. The bond length decreases when moving to the right along the periodic table's rows because more electronegative atoms have shorter atomic radii. The nitrogen in HCN is likely to hold its electrons closer, or in a shorter radius, than the carbons in C_2H_2 .

8. C

Electronegative atoms bonded to hydrogen disproportionately pull covalently bonded electrons toward themselves, which leaves hydrogen with a partial positive character. That partial positive charge is attracted to nearby negative or partial negative charges, such as those on other electronegative atoms.

9. C

First recall that ammonium is NH_4^+ , while ammonia is NH_3 . Ammonium is formed by the association of NH_3 , an uncharged molecule with a lone pair on the nitrogen, with a positively charged hydrogen cation. In other words, NH_3 is a Lewis base, while H^+ is a Lewis acid. This type of bonding between a Lewis acid and base is a coordinate covalent bond.

10. C

All atoms in the third period or greater have d-orbitals, which can hold an additional 10 electrons. The typical "octet" electrons reside in s- and p-orbitals, but elements in period 3 or higher can place electrons into these d-orbitals.

11. C

All of the listed types of forces describe interactions between different types of molecules. However, noble gases are entirely uncharged and do not have polar covalent bonds, ionic bonds, or dipole moments.

Therefore, the only intermolecular forces experienced by noble gases are London dispersion forces. Although these interactions are small in magnitude, they are necessary for condensation into a liquid.

12. A

In this Lewis diagram, the phosphate molecule has an overall formal charge of -3 . The four oxygen atoms would each be assigned a formal charge of -1 . Given the overall charge of -3 and the -1 charge on each oxygen, the phosphorus must have a formal charge of $+1$.

13. C

The reaction in this question shows a water molecule, which has two lone pairs of electrons on the central oxygen, combining with a free hydrogen cation. The resulting molecule, H_3O^+ has formed a new bond between H^+ and H_2O . This bond is created via the sharing of one of oxygen's lone pairs with the free H^+ ion. This represents the donation of a shared pair of electrons from a Lewis base (H_2O) to a Lewis acid (H^+ , electron acceptor). This type of bond is called a coordinate covalent bond.

14. C

NH_3 has three hydrogen atoms bonded to the central nitrogen, which also has a lone pair. These four groups—three atoms, one lone pair—lead NH_3 to assume tetrahedral electronic geometry yet trigonal pyramidal molecular geometry. The nitrogen in ammonia is sp^3 -hybridized. By hybridizing all three p-orbitals and the one s-orbital, four groups are arranged about the central atom, maximizing the distances between the groups to minimize the energy of the molecule with a

tetrahedral configuration. In contrast, BF_3 has three atoms and no lone pairs, resulting in sp^2 -hybridization. Its shape is called trigonal planar.

15. B

This answer requires an understanding of the trends that cause higher or lower bond energies. Bonds of high energy are those that are difficult to break. These bonds tend to have more shared pairs of electrons and, thus, cause a stronger attraction between the two atoms in the bonds. This stronger attraction also means that the bond length of a high-energy, high-order bond such as a triple bond is shorter than that of its lower-energy counterparts such as single or double bonds.

4

Compounds and Stoichiometry

In This Chapter

- 4.1 Molecules and Moles 
 - Molecular Weight 
 - Mole 
 - Equivalent Weight 
- 4.2 Representation of Compounds 
 - Law of Constant Composition 
 - Empirical and Molecular Formulas 
 - Percent Composition 
- 4.3 Types of Chemical Reactions 
 - Combination Reactions 
 - Decomposition Reactions 
 - Combustion Reactions 
 - Single-Displacement Reactions 
 - Double-Displacement Reactions 
 - Neutralization Reactions 
- 4.4 Balancing Chemical Equations 
- 4.5 Applications of Stoichiometry 
 - Limiting Reagent 



[Yield](#) 

4.6 [Ions](#) 

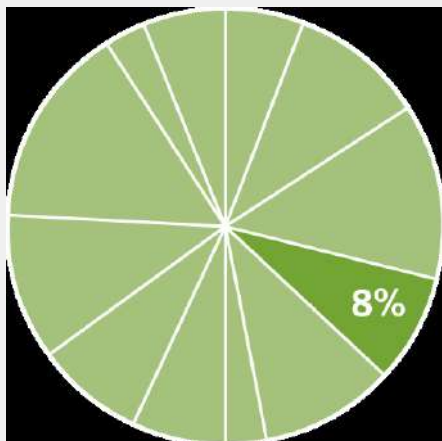
[Cations and Anions](#) 

[Ion Charges](#) 

[Electrolytes](#) 

[Concept Summary](#) 

Chapter Profile



The content in this chapter should be relevant to about 8% of all questions about general chemistry on the MCAT.

This chapter covers material from the following AAMC content categories:

4C: Electrochemistry and electrical circuits and their elements

4E: Atoms, nuclear decay, electronic structure, and atomic chemical behavior

5A: Unique nature of water and its solutions

Introduction

Oh—what is that smell? It smells like rancid almonds. Then you notice a few green bugs whose backs give the impression of a shield. Stink bugs! A stink bug “stinks” because it produces a highly concentrated solution of volatile compounds that we perceive as malodorous, noxious, and irritating. Interestingly enough, the primary compounds in the stink bug’s stink bomb are hydrogen cyanide—a highly toxic compound that inhibits cytochrome c oxidase, thereby blocking aerobic respiration—and benzaldehyde. Like many other aromatic compounds, benzaldehyde vaporizes at room temperature and reaches the olfactory system as gas particles. Benzaldehyde is also the key ingredient in artificial almond extract. At low concentrations, it produces a pleasant aroma of toasted almonds. However, at high concentrations, its odor is that of rotten almonds, and it is a noxious irritant to the skin, eyes, and respiratory tract.

Benzaldehyde is a compound composed of seven carbon atoms, six hydrogen atoms, and one oxygen atom. One mole of benzaldehyde has a mass of approximately 106 grams. It can react with other atoms or compounds to form new compounds—pure substances composed of two or more elements in a fixed proportion. Compounds can be broken down by chemical means to produce their constituent elements or other compounds. They are characterized by describing their physical and chemical properties.

This chapter focuses on compounds and their reactions. It reviews the various ways in which compounds are represented, using empirical and molecular formulas and percent composition. There is a brief overview of the major classes of chemical reactions, which we will examine more closely in subsequent chapters, and finally, there is a recap of the steps

involved in balancing chemical equations with a particular focus on identifying limiting reagents and calculating reaction yields.

4.1 Molecules and Moles

LEARNING GOALS

After Chapter 4.1, you will be able to:

- Calculate the molar mass of a given substance, such as AgCN
- Calculate the number of moles of a molecule given its mass in grams
- Compare the number of molecules in two different compounds given their gram weights and molecular formulas
- Determine the normality of a solution

A molecule is a combination of two or more atoms held together by covalent bonds. Molecules are the smallest units of compounds that display their identifying properties. Molecules can be composed of two or more atoms of the same element (such as N_2 and O_2) or may be composed of atoms of different elements, as in CO_2 (carbon dioxide), SOCl_2 (thionyl chloride), and $\text{C}_6\text{H}_5\text{CHO}$ (benzaldehyde). Because reactions usually involve a very large number of molecules—far too many to count individually—we usually measure amounts of compounds in terms of moles or grams, using molar mass to interconvert between these units.

Ionic compounds do not form true molecules because of the way in which the oppositely charged ions arrange themselves in the solid state. As solids, they can be considered as nearly infinite three-dimensional arrays of the charged particles that comprise the compound. As described in Chapter 3 of MCAT General Chemistry Review, solid NaCl is a coordinated lattice in which each of the Na^+ ions is surrounded by Cl^- ions and each of the Cl^- ions is surrounded by Na^+ ions. This makes it rather difficult to clearly define a sodium chloride molecule, and the term formula unit, representing the empirical formula of the compound, is used instead. Because no molecule actually exists, molecular weight becomes meaningless, and the term formula weight is used instead.



BRIDGE

Ionic compounds form from combinations of elements with large electronegativity differences, such as sodium and chlorine. Molecular compounds form from elements of similar electronegativity, such as carbon with oxygen. The difference between ionic and covalent bonds is discussed in Chapter 3 of MCAT General Chemistry Review.

MOLECULAR WEIGHT

Remember that the term atomic weight is a misnomer because it is actually a weighted average of the masses of the naturally occurring isotopes of an element, not their weights. The same applies here to our

discussion of molecular weight. Molecular weight, then, is simply the sum of the atomic weights of all the atoms in a molecule, and its units are atomic mass units (amu) per molecule. Similarly, the formula weight of an ionic compound is found by adding up the atomic weights of the constituent ions according to its empirical formula, and its units are also amu per molecule.

Example:

What is the molecular weight of SOCl_2 ?

Solution:

To find the molecular weight of SOCl_2 , add together the atomic weights of each of the atoms.

1 S: 1×32.1 amu	=	32.1 amu
1 O: 1×16.0 amu	=	16.0 amu
2 Cl: 2×35.5 amu	=	71.0 amu
Total molecular weight	=	119.1 amu per molecule

MOLE

A mole is a quantity of any substance (atoms, molecules, dollar bills, kittens—anything) equal to the number of particles that are found in 12

grams of carbon-12 ($^{12}_6\text{C}$). This number of particles is defined as Avogadro's number (N_A), $6.022 \times 10^{23} \text{ mol}^{-1}$. One mole of a compound has a mass in grams equal to the molecular or formula weight of the compound in amu. For example, one molecule of H_2CO_3 (carbonic acid) has a mass of 62 amu; one mole of the compound has a mass of 62 grams. The mass of one mole of a compound is called its molar mass and is usually expressed in $\frac{\text{g}}{\text{mol}}$. The term molecular weight is sometimes used incorrectly to imply molar mass; remember, molecular weight is measured in $\frac{\text{amu}}{\text{molecule}}$, not $\frac{\text{g}}{\text{mol}}$.



KEY CONCEPT

Remember that Avogadro's number (and the mole) are just units of convenience, like the dozen is a convenient unit for eggs.

The formula for determining the number of moles of a sample substance is:

$$\text{Moles} = \frac{\text{Mass of sample (g)}}{\text{Molar mass} \left(\frac{\text{g}}{\text{mol}} \right)}$$

Equation 4.1

This equation is often used in stoichiometry and titration problems.

Example:

How many moles are in 9.53 g of MgCl_2 ?

Solution:

First, find the molar mass of MgCl_2 .

$$\left(1 \times 24.3 \frac{\text{g}}{\text{mol}}\right) + \left(2 \times 35.5 \frac{\text{g}}{\text{mol}}\right) = 95.3 \frac{\text{g}}{\text{mol}}$$

Now, solve for the number of moles.

$$\frac{9.53 \text{ g}}{95.3 \frac{\text{g}}{\text{mol}}} = 0.10 \text{ mol MgCl}_2$$

EQUIVALENT WEIGHT

Equivalent weight and the related concept of equivalents are a source of confusion for many students. Part of the problem may be the context in which equivalents and equivalent weights are usually discussed: acid–base reactions, oxidation–reduction reactions, and precipitation reactions, all three of which can be sources of confusion and anxiety on their own. Therefore, let's start with a more basic discussion of equivalents.

Often, certain elements or compounds can act more potently than others in performing certain reactions. For example, one mole of HCl has the

ability to donate one mole of hydrogen ions (H^+) in solution, but one mole of H_2SO_4 has the ability to donate two moles of hydrogen ions, and one mole of H_3PO_4 has the ability to donate three moles of hydrogen ions. To gather one mole of hydrogen ions for a particular acid–base reaction, we could use one mole of HCl , a half-mole of H_2SO_4 , or one-third of a mole of H_3PO_4 . Or, consider the difference between Na and Mg : one mole of sodium has the ability to donate one mole of electrons, while one mole of magnesium has the ability to donate two moles of electrons. This provides context for the concept of equivalents: How many moles of the thing we are interested in (protons, hydroxide ions, electrons, or ions) will one mole of a given compound produce? Sodium will donate one mole of electrons (one equivalent), but magnesium will donate two moles of electrons (two equivalents).



BRIDGE

The idea of equivalents is related to the concept of normality, which is explained in the discussion of acids and bases in Chapter 10 of MCAT General Chemistry Review.

So far, this discussion has been focused on the mole-to-mole relationship between, say, an acid compound and the hydrogen ions it donates. However, sometimes we need to work in units of mass rather than moles. Just as one mole of HCl will donate one mole of hydrogen ions, a certain mass of HCl (about 36.5 g) will also donate one equivalent of hydrogen ions. This amount of a compound, measured in grams, that produces one

equivalent of the particle of interest is called the gram equivalent weight and can be calculated from:

$$\text{Gram equivalent weight} = \frac{\text{Molar mass}}{n}$$

Equation 4.2

where n is the number of particles of interest produced or consumed per molecule of the compound in the reaction. For example, one would need 31 grams of H_2CO_3 (**molar mass = $62 \frac{\text{g}}{\text{mol}}$**) to produce one equivalent of hydrogen ions because each molecule of H_2CO_3 can donate two hydrogen ions ($n = 2$). Simply put, the equivalent weight of a compound is the mass that provides one mole of the particle of interest.



MCAT EXPERTISE

Whenever confronting a stoichiometry problem, always look for normality by identifying an equivalent unit (protons, hydroxide ions, electrons, ions) and then multiplying it by the number of moles or molar concentration to find the normal concentration.

If the amount of a compound in a reaction is known and we need to determine how many equivalents are present, use the equation:

$$\text{Equivalents} = \frac{\text{Mass of compound (g)}}{\text{Gram equivalent weight (g)}}$$

Equation 4.3

Finally, we can now introduce the measurement of normality. Normality (N) is a measure of concentration, given in the units $\frac{\text{equivalents}}{\text{L}}$. On the MCAT, it is most commonly used for hydrogen ion concentration. Thus, a 1 N solution of acid contains a concentration of hydrogen ions equal to 1 mole per liter; a 2 N solution of acid contains a concentration of hydrogen ions equal to 2 moles per liter. The actual concentration of the acidic compound may be the same or different from the normality because different compounds are able to donate different numbers of hydrogen ions. In a 1 N HCl solution, the molarity of HCl is 1 M because HCl is a monoprotic acid; in a 1 N H₂CO₃ solution, the molarity of H₂CO₃ is 0.5 M because H₂CO₃ is a diprotic acid. Note that normality calculations always assume that a reaction will proceed to completion; while carbonic acid does not fully dissociate in solution, it can be reacted with enough base for each molecule to give up both of its protons. The conversion from normality to molarity of a given solute is:

$$\text{Molarity} = \frac{\text{Normality}}{n}$$

Equation 4.4

where n is the number of protons, hydroxide ions, electrons, or ions produced or consumed by the solute.

Figure 4.1 shows the titration of the diprotic acid H_2CO_3 with a base. The x-axis indicates that two equivalents of base are needed to neutralize both protons of this acid.

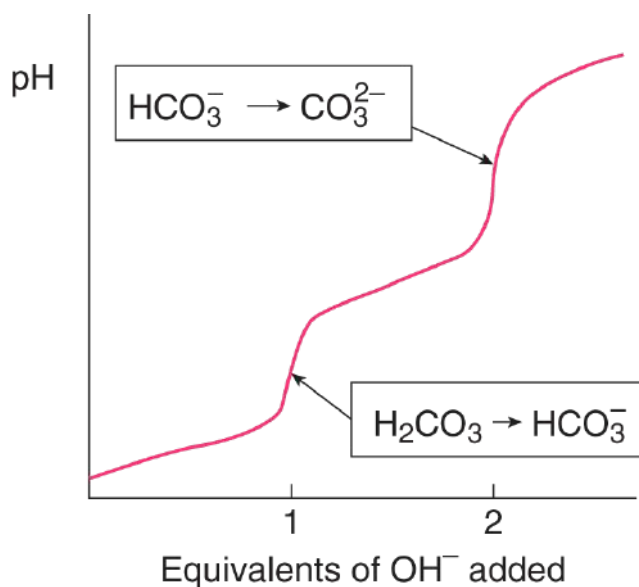


Figure 4.1. Titration of Carbonic Acid with a Base
Carbonic acid is diprotic, so two equivalents of base are required to neutralize both protons of the acid.

There is a real benefit to working with equivalents and normality because it allows a direct comparison of the quantities of the entity we are most interested in. In an acid–base reaction, we care about the hydrogen or hydroxide ions; where the ions come from is not really the primary concern. It is convenient to be able to say that one equivalent of acid (hydrogen ions) will neutralize one equivalent of base (hydroxide ions), but the same could not necessarily be said if we were dealing with moles of acidic compounds and moles of basic compounds. For example, one mole

of HCl will not completely neutralize one mole of $\text{Ca}(\text{OH})_2$ because one mole of HCl will donate one equivalent of acid, but $\text{Ca}(\text{OH})_2$ will donate two equivalents of base.

Example:

What is the gram equivalent weight (GEW) of sulfuric acid?

Solution:

First, find the molar mass of H_2SO_4 .

$$\left(2 \times 1.0 \frac{\text{g}}{\text{mol H}}\right) + \left(1 \times 32.1 \frac{\text{g}}{\text{mol S}}\right) + \left(4 \times 16.0 \frac{\text{g}}{\text{mol O}}\right) = 98.1$$

Next, identify the equivalents: protons (H^+), because these are transferred in acid–base reactions. The number of protons in sulfuric acid (n) is 2.

Now, calculate the gram equivalent weight.

$$\text{Gram equivalent weight} = \frac{\text{Molar mass}}{n}$$
$$\text{GEW} = \frac{98.1 \frac{\text{g}}{\text{mol H}_2\text{SO}_4}}{2 \frac{\text{mol H}^+}{\text{mol H}_2\text{SO}_4}} = 49.05 \frac{\text{g}}{\text{mol H}^+}$$



KEY CONCEPT

In acid–base chemistry, the gram equivalent weight represents the mass of acid that yields one mole of protons, or the mass of base that yields one mole of hydroxide ions.

Example:

What is the normality of a 2 M $\text{Mg}(\text{OH})_2$ solution?

Solution:

First, identify the number of equivalents (n). There are two hydroxide ions (OH^-) for each molecule of $\text{Mg}(\text{OH})_2$, which is the equivalent of interest because magnesium hydroxide is a base.

Then, calculate the normality.

$$\text{Normality} = \text{molarity} \times n = 2 \text{ M} \times 2 \frac{\text{equiv OH}^-}{\text{mol Mg}(\text{OH})_2} = 4 \text{ N}$$

MCAT Concept Check 4.1:

Before you move on, assess your understanding of the material with these questions.

Calculate the molar masses of the following substances:

- NaBr:

- SrCl_2 :

- $\text{C}_6\text{H}_{12}\text{O}_6$:

Calculate the number of moles in 100 g of each of the following substances:

- NaBr:

- SrCl_2 :

- $\text{C}_6\text{H}_{12}\text{O}_6$:

How do the number of molecules in 18 g of H_2O compare to the number of formula units in 58.5 g of NaCl?

Determine the normality of the following solutions:
(Note: The species of interest is H^+ .)

- $0.25\text{ M H}_3\text{PO}_4$:

- 95 g PO_4^{3-} in 100 mL solution:

4.2 Representation of Compounds

LEARNING GOALS

After Chapter 4.2, you will be able to:

- Recall the similarities and differences between molecular and empirical formulas
- Calculate the percent composition by mass of a compound, such as $\text{C}_6\text{H}_{12}\text{O}_6$
- Determine the empirical formula of a compound given its percent composition by mass

There are different ways of representing compounds and their constituent atoms. We've already reviewed a couple of these systems in Chapter 3 of MCAT General Chemistry Review: Lewis dot structures and VSEPR theory. In organic chemistry, it is common to encounter skeletal representations of compounds, called structural formulas, that show the various bonds between the constituent atoms of a compound. Inorganic (general) chemistry typically represents compounds by showing the constituent atoms without representing the actual bond connectivity or atomic arrangement. For example, the formula $\text{C}_6\text{H}_{12}\text{O}_6$ (glucose) tells us that this particular compound consists of six atoms of carbon, twelve atoms of

hydrogen, and six atoms of oxygen, but there is no indication of how the different atoms are arranged or how many bonds exist between each of the atoms.



BRIDGE

Many of these representations are discussed in more detail in Chapter 2 of MCAT Organic Chemistry Review. Understanding the theory behind such representations will help convert between different projections and representations with ease.

LAW OF CONSTANT COMPOSITION

The law of constant composition states that any pure sample of a given compound will contain the same elements in an identical mass ratio. For example, every sample of water will contain two hydrogen atoms for every one oxygen atom, or—in terms of mass—for every one gram of hydrogen, there will be eight grams of oxygen.



REAL WORLD

Even biologically important molecules, such as water and amino acids on Earth are, by composition, the same anywhere else in the universe, even though densities and other physical properties may differ.

EMPIRICAL AND MOLECULAR FORMULAS

There are two ways to express the formula of a compound. The empirical formula gives the simplest whole-number ratio of the elements in the compound. The molecular formula gives the exact number of atoms of each element in the compound and is a multiple of the empirical formula. For example, the empirical formula for benzene is CH , while the molecular formula is C_6H_6 . For some compounds, the empirical and molecular formulas are identical, as is the case for H_2O . As previously discussed, ionic compounds, such as NaCl or CaCO_3 , will only have empirical formulas.



BRIDGE

An empirical formula of CH_2O is indicative of a monosaccharide. Common monosaccharides include glucose, fructose, and galactose. The structures of these monosaccharides—and of carbohydrates in general—are discussed in Chapter 4 of MCAT Biochemistry Review.

PERCENT COMPOSITION

The percent composition of an element (by mass) is the percent of a specific compound that is made up of a given element. To determine the percent composition of an element in a compound, the following formula is used:

$$\text{Percent composition} = \frac{\text{Mass of element in formula}}{\text{Molar mass}} \times 100\%$$

Equation 4.5

One can calculate the percent composition of an element by using either the empirical or the molecular formula. It is also possible to determine the molecular formula given both the percent compositions and molar mass of a compound. The following examples demonstrate such calculations.



MCAT EXPERTISE

Percent composition is a common way for stoichiometry to be tested on the MCAT. Practice these problems to build up speed and efficiency for Test Day.

Example:

What is the percent composition of chromium in $\text{K}_2\text{Cr}_2\text{O}_7$?

Solution:

The molar mass of $\text{K}_2\text{Cr}_2\text{O}_7$ is:

$$(2 \times 39.1 \text{ g}) + (2 \times 52.0 \text{ g}) + (7 \times 16.0 \text{ g})$$

$$\begin{aligned}
 & \left(2 \times 39.1 \frac{\text{g}}{\text{mol}} \right) + \left(2 \times 52.0 \frac{\text{g}}{\text{mol}} \right) + \left(7 \times 16.0 \frac{\text{g}}{\text{mol}} \right) \\
 & \approx (2 \times 40) + (2 \times 50) + (7 \times 16) \\
 & = 292 \frac{\text{g}}{\text{mol}} \left(\text{actual value} = 294.2 \frac{\text{g}}{\text{mol}} \right)
 \end{aligned}$$

Calculate the percent composition of Cr:

$$\begin{aligned}
 \text{Percent composition} &= \frac{2 \times 52.0 \frac{\text{g}}{\text{mol}}}{294.2 \frac{\text{g}}{\text{mol}}} \times 100\% \\
 &\approx \frac{2 \times 50}{300} \times 100\% = \frac{100}{300} \times 100\% = 33\% (\text{actual value} =)
 \end{aligned}$$

Example:

What are the empirical and molecular formulas of a carbohydrate that contains 40.9% carbon, 4.58% hydrogen, and 54.52% oxygen and has a molar mass of **$264 \frac{\text{g}}{\text{mol}}$** ?

Method One:

First, determine the number of moles of each element in the compound by assuming a 100-gram sample; this converts the percentage of each element present directly into grams of that element. Then convert grams to moles:

$$\text{moles C} = \frac{40.9 \text{ g}}{12 \frac{\text{g}}{\text{mol}}} \approx 3.4 \text{ mol}$$

$$\text{moles H} = \frac{4.58 \text{ g}}{1 \frac{\text{g}}{\text{mol}}} \approx 4.6 \text{ mol}$$

$$\text{moles O} = \frac{54.52 \text{ g}}{16 \frac{\text{g}}{\text{mol}}} \approx 3.4 \text{ mol}$$

Next, find the simplest whole number ratio of the elements by dividing the number of moles for each compound by the smallest number out of all obtained in the previous step.

$$\text{C: } \frac{3.4}{3.4} = 1.00; \text{ H: } \frac{4.6}{3.4} \approx 1.33; \text{ O: } \frac{3.4}{3.4} = 1.00$$

Finally, the empirical formula is obtained by converting the numbers obtained into whole numbers by multiplying them by an integer value.

$$\text{Empirical formula} = \text{C}_1\text{H}_{1.33}\text{O}_1 \times 3 = \text{C}_3\text{H}_4\text{O}_3$$

To determine the molecular formula, divide the molar mass (264 g/mol, given in the question stem) by the empirical formula weight. The resulting value gives the number of empirical formula units in the molecular formula.

The formula weight of the empirical formula $\text{C}_3\text{H}_4\text{O}_3$ is:

$$\left(3 \times 12.0 \frac{\text{g}}{\text{mol C}}\right) + \left(4 \times 1.0 \frac{\text{g}}{\text{mol H}}\right) + \left(3 \times 16.0 \frac{\text{g}}{\text{mol O}}\right) = 88 \frac{\text{g}}{\text{mol}}$$

$$\text{ratio of } \frac{\text{molecular formula}}{\text{empirical formula}} = \frac{264 \frac{\text{g}}{\text{mol}}}{88 \frac{\text{g}}{\text{mol}}} = 3$$

Finally, find the molecular formula by multiplying by this ratio:

$\text{C}_3\text{H}_4\text{O}_3 \times 3 = \text{C}_9\text{H}_{12}\text{O}_9$. The molecular formula is $\text{C}_9\text{H}_{12}\text{O}_9$.

Method Two:

When the molar mass is given, it is generally easier to find the molecular formula first. This is accomplished by multiplying the molar mass by the given percentages to find the mass of each element present in one mole of compound, then dividing by the respective atomic weights to find the mole ratio of the elements:

$$\text{moles C} = \frac{(0.409)(264 \text{ g})}{12 \frac{\text{g}}{\text{mol}}} \approx \frac{(0.4)(270)}{12} \approx \frac{270}{30} = 9$$

$$\text{moles H} = \frac{(0.0458)(264 \text{ g})}{1 \frac{\text{g}}{\text{mol}}} \approx \frac{(0.05)(270)}{1} \approx 13 \text{ mol}$$

$$\text{moles O} = \frac{(0.5452)(264 \text{ g})}{16 \frac{\text{g}}{\text{mol}}} \approx \frac{(0.5)(270)}{16} \approx \frac{270}{30} =$$

At first glance, this gives a molecular formula of $C_9H_{13}O_9$. However, familiarity with carbohydrates indicates that a molecular formula of $C_9H_{12}O_9$ fits the ratio CH_2O , and takes rounding error into account. The empirical formula can now be found by reducing the subscript ratio to the simplest integer values ($C_3H_4O_3$).



MCAT EXPERTISE

When there are two methods for approaching a problem, be well-versed in both. Knowing multiple ways to solve a problem will help you tackle questions efficiently.



KEY CONCEPT

The molecular formula is either the same as the empirical formula or a multiple of it. To calculate the molecular formula, you need to know the mole ratio (this will give you the empirical formula) and the molar mass (molar mass divided by empirical formula weight will give the multiplier for the empirical formula-to-molecular formula conversion).

MCAT Concept Check 4.2:

Before you move on, assess your understanding of the material with these questions.

What are some similarities and differences between molecular and empirical formulas?

- Similarities:

- Differences:

Find the percent composition (by mass) of sodium, carbon, and oxygen in sodium carbonate (Na_2CO_3):

- Sodium:

- Carbon:

- Oxygen:

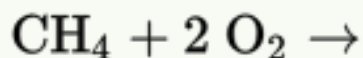
Experimental data from the combustion of an unknown compound indicates that it is 28.5% iron, 24.0% sulfur, and 49.7% oxygen by mass. What is its empirical formula?

4.3 Types of Chemical Reactions

LEARNING GOALS

After Chapter 4.3, you will be able to:

- Describe the series of events in a single displacement, double displacement, neutralization, or combustion reaction
- Classify a reaction and predict its products given the reactants:



This section reviews the major classes of chemical reactions. We will begin with a classification of major types of reactions seen on the MCAT and then discuss methods to recognize their products. It is important to understand the conventions of reaction mechanisms. In the following section, we will discuss how to properly balance chemical equations.



BRIDGE

Many of the reactions we will discuss here have analogs in MCAT Organic Chemistry Review and MCAT Biochemistry Review. Be sure

to understand the relationships between the products and reactants because it will help simplify more advanced reactions.

COMBINATION REACTIONS

A combination reaction has two or more reactants forming one product. The formation of water by burning hydrogen gas in air is an example of a combination reaction. This reaction is highlighted in Figure 4.2.

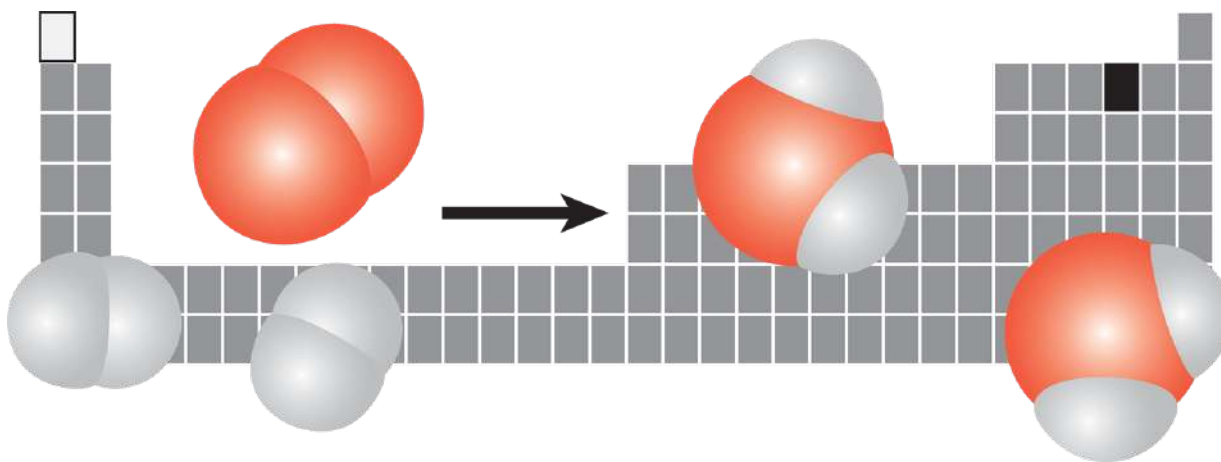
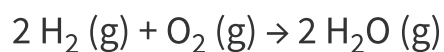


Figure 4.2. Formation of Water from Hydrogen and Oxygen



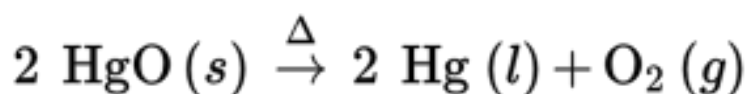
KEY CONCEPT

Combination reactions have more reactants than products:



DECOMPOSITION REACTIONS

A decomposition reaction is the opposite of a combination reaction: a single reactant breaks down into two or more products, usually as a result of heating, high-frequency radiation, or electrolysis. An example of decomposition is the breakdown of mercury(II) oxide. (The Δ [delta] sign over a reaction arrow represents the addition of heat.)



KEY CONCEPT

Decomposition reactions generally have more products than reactants. $A \rightarrow B + C$

An example of a reaction that utilizes high-frequency light is the decomposition of silver chloride crystals, shown in figure 4.3, in the presence of sunlight. The ultraviolet component of sunlight has sufficient energy to catalyze certain chemical reactions. For silver chloride, exposure to sunlight results in a decomposition reaction that yields a rust-colored product that consists of separated silver and chlorine.



Figure 4.3. Silver Chloride (AgCl) Crystals

Silver chloride will decompose to a rust-colored product upon exposure to sunlight.

COMBUSTION REACTIONS

A combustion reaction is a special type of reaction that involves a fuel—usually a hydrocarbon—and an oxidant (normally oxygen). In its most common form, these reactants form the two products of carbon dioxide and water. For example, the balanced equation expressing the combustion of methane is shown in Figure 4.4.

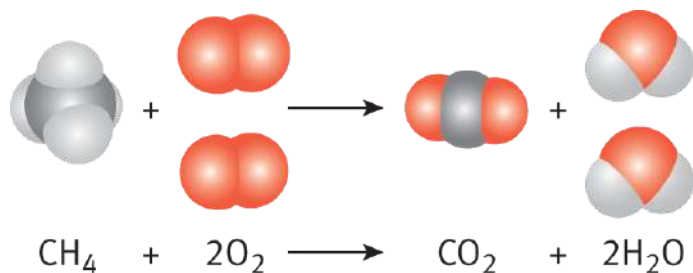


Figure 4.4. Combustion of Methane



KEY CONCEPT

Combustion involves oxidation (using O_2 or similar) of a fuel (typically a hydrocarbon).

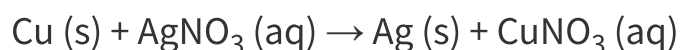


MCAT EXPERTISE

Combustion reactions are usually conducted with hydrocarbon fuels, but they can also use elements such as sulfur or other compounds such as sugars. The products can differ, but carbon dioxide and water are almost always present. Therefore, it is important to recognize the reactants and products of this reaction type because you may see it in various contexts.

SINGLE-DISPLACEMENT REACTIONS

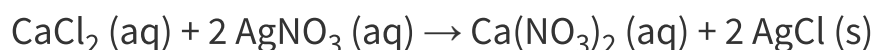
A single-displacement reaction occurs when an atom or ion in a compound is replaced by an atom or ion of another element. For example, solid copper metal will displace silver ions in a clear solution of silver nitrate to form a blue copper nitrate solution and solid silver metal.



Single-displacement reactions are often further classified as oxidation–reduction reactions, which will be discussed in greater detail in Chapter 11 of MCAT General Chemistry Review. For example, Ag in AgNO_3 has an oxidation state of +1, but when it leaves the compound, it gains one electron (the Ag^+ is reduced to Ag). On the other hand, copper loses an electron (oxidation) when it joins the nitrate ion.

DOUBLE-DISPLACEMENT REACTIONS

In double-displacement reactions, also called metathesis reactions, elements from two different compounds swap places with each other to form two new compounds. This type of reaction occurs when one of the products is removed from the solution as a precipitate or gas or when two of the original species combine to form a weak electrolyte that remains undissociated in solution. For example, when solutions of calcium chloride and silver nitrate are combined, insoluble silver chloride forms in a solution of calcium nitrate.



A series of double-displacement reactions is depicted in Figure 4.5. Shown are illustrations of test tubes in which $\text{Zn}(\text{NO}_3)_2$ is dissolved in solutions to precipitate solid zinc salts. From left to right, the solutions are $(\text{NH}_4)_2\text{S}$, NaOH , and Na_2CO_3 .

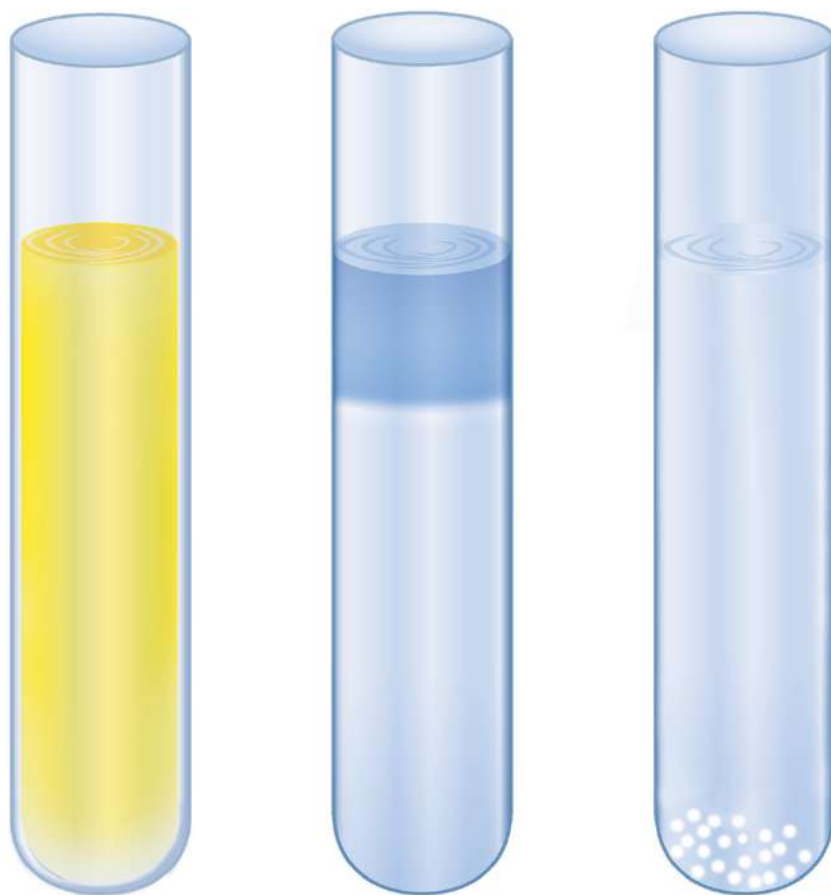
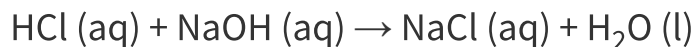


Figure 4.5. Illustrations of Double-Displacement Reactions Forming Zinc Salts

Left: $(\text{NH}_4)_2\text{S}$ solution, producing ZnS (s). Middle: NaOH solution, producing $\text{Zn}(\text{OH})_2$ (s). Right: Na_2CO_3 solution, producing ZnCO_3 (s).

NEUTRALIZATION REACTIONS

Neutralization reactions are a specific type of double-displacement reaction in which an acid reacts with a base to produce a salt (and, usually, water). For example, hydrochloric acid and sodium hydroxide will react to form sodium chloride and water:



BRIDGE

Acids and bases combine in neutralization reactions to produce salts (and, usually, water). Acid–base chemistry is discussed in Chapter 10 of MCAT General Chemistry Review.

Reactions between acids and bases are not always visible. The addition of an indicator or use of indicator strips, as shown in Figure 4.6, can determine when the reaction has occurred.



Figure 4.6. Indicator Strip Tested in Solutions of Varying pH
After an indicator strip is placed in a solution, the indicator strip (right) can be read using the indicator key.

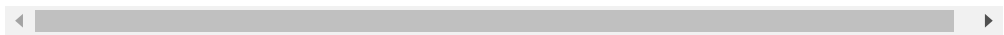
MCAT Concept Check 4.3:

Before you move on, assess your understanding of the material with these questions.

Describe in words what occurs when $\text{Zn}(\text{NO}_3)_2$ is dissolved in $(\text{NH}_4)_2\text{S}$:

Complete and classify the most likely reactions in the table below:

Reactants	Conditions	Products	Reaction Type
$2 \text{H}_2 + \text{O}_2$	$\rightarrow$		
$\text{Al}(\text{OH})_3 + \text{H}_3\text{PO}_4$	$\rightarrow$		
$2 \text{H}_2\text{O}$	electricity $\xrightarrow{\hspace{1cm}}$		
$\text{NaNO}_3 + \text{CuOH}$	$\rightarrow$		
$\text{Zn} + \text{AgCl}$	$\rightarrow$		



4.4 Balancing Chemical Equations

High-Yield



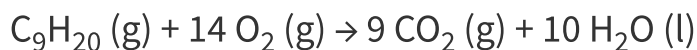
LEARNING GOALS

After Chapter 4.4, you will be able to:

- Balance a chemical equation

Because chemical equations express how much and what types of reactants must be used to obtain a given quantity of product, it is of utmost importance that the reaction be balanced so as to reflect the laws of conservation of mass and charge. The mass of the reactants consumed must equal the mass of products generated. More specifically, one must ensure that the number of atoms of each element on the reactant side equals the number of atoms of that element on the product side.

Stoichiometric coefficients, which are the numbers placed in front of each compound, are used to indicate the relative number of moles of a given species involved in the reaction. For example, the balanced equation expressing the combustion of nonane is:



The coefficients indicate that one mole of C_9H_{20} gas must be reacted with fourteen moles of O_2 gas to produce nine moles of carbon dioxide and ten moles of water. In general, stoichiometric coefficients are given as whole numbers.



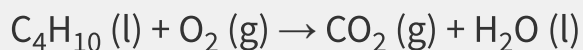
MCAT EXPERTISE

It is unlikely that you will come across a question that explicitly asks you to balance an equation. However, you will need to recognize unbalanced reactions and quickly add the necessary coefficients. To balance a reaction, look at the number of atoms of each element and the charge on both sides (especially for oxidation–reduction reactions).

The steps taken to balance a chemical reaction are necessary to ensure that calculations regarding the reaction are performed correctly. Let's review the steps involved in balancing a chemical equation, using an example.

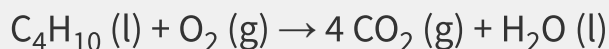
Example:

Balance the following reaction:

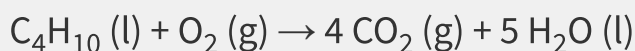


Method One:

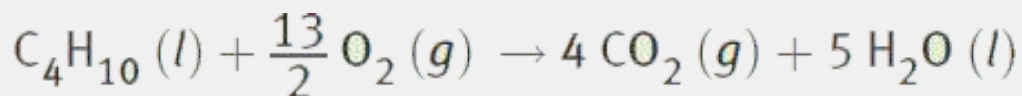
First, balance the carbons (4 on reactant side) in the products. Carbons are a good choice to start with because they appear only once on both sides of the reaction:



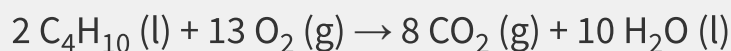
Then, balance the hydrogens (10 on reactant side) in the products. Again, hydrogens appear only once on each side, making them a good choice to work on next:



Next, balance the oxygens (now 13 on product side) in the reactants. Note that oxygens appear in multiple reactants and products, making them the most complex atom to balance and therefore the best to leave until the end:



Finally, produce a whole number ratio. In this case, double each coefficient.



Finally, check that all of the elements and the total charges are balanced correctly. If there is a difference in total charge between

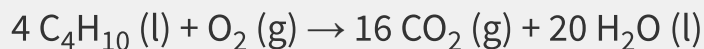
the reactants and products, then the charge will also have to be balanced. (Instructions for balancing charge in oxidation–reduction reactions are found in Chapter 11 of MCAT General Chemistry Review.)

Method Two:

First, if in doubt, take a guess. Assume there are 4 of the first reactant and balance the carbons appropriately.



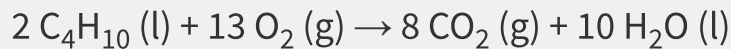
Second, balance the hydrogens (40 on reactant side) in the products.



Third, balance the oxygens (now 52 on product side) in the reactants.



Fourth, produce the simplest whole number ratio through the greatest common factor. In this case, divide each side by 2.



Finally, check that all of the elements and the total charges are balanced correctly. Notice that both methods produce a multiple of our final answer. These ratios are both correct, but in terms of the stoichiometry one performs on the MCAT, the simpler the numbers are, the easier calculations will become.



KEY CONCEPT

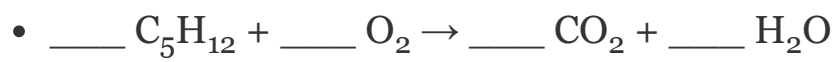
When balancing equations, focus on the least represented elements first and work your way to the most represented element of the reaction (usually oxygen or hydrogen). If you're stuck, take a guess for the coefficient of the first reactant and balance the remainder appropriately.

MCAT Concept Check 4.4:

Before you move on, assess your understanding of the material with this question.

Balance the following reactions:

- $\text{___ Fe} + \text{___ Cl}_2 \rightarrow \text{___ FeCl}_3$
- $\text{___ Zn} + \text{___ HCl} \rightarrow \text{___ ZnCl}_2 + \text{___ H}_2$



4.5 Applications of Stoichiometry

LEARNING GOALS

After Chapter 4.5, you will be able to:

- Calculate the grams of product produced given the quantities of reactant
- Identify the limiting reagent within a reaction
- Calculate the mass of excess reagent in a reaction with a limiting reagent
- Calculate the percent yield of a reaction

Perhaps the most useful information to glean from a balanced reaction is the mole ratio of reactants consumed-to-products generated. One can also generate the mole ratio of one reactant to another or one product to another. All of these ratios can be generated using the stoichiometric coefficients. In the formation of water ($2 \text{H}_2 + \text{O}_2 \rightarrow 2 \text{H}_2\text{O}$), for example, one can determine that, for every one mole of hydrogen gas consumed, one mole of water can be produced; for every one mole of oxygen gas consumed, two moles of water can be produced. Furthermore, mole-to-mole, hydrogen gas is being consumed at a rate twice that of oxygen gas.



KEY CONCEPT

Stoichiometry, an application of dimensional analysis, is often simplified to a series of three fractions. These fractions demonstrate an underlying three-step process:

- Convert from the given units to moles
- Use the mole ratio
- Convert from moles to the desired units

Stoichiometry problems usually involve at least a few unit conversions, so take care when working through these types of problems to ensure that units cancel out appropriately to lead to the desired units of the answer choices. Pay close attention to the following problem, which demonstrates a clear and easy-to-follow method for keeping track of the numbers, calculations, and unit conversions.

Example:

How many grams of calcium chloride are needed to prepare 71.7 g of silver chloride according to the following equation?



Solution:

Noting first that the equation is balanced, 1 mole of CaCl_2 is reacted with 2 moles of AgNO_3 to yield 2 moles of AgCl . The molar mass of CaCl_2 is 111.1 g, and the molar mass of AgCl is 143.4 g. The given quantity is 71.7 g AgCl .

$$\begin{aligned} 71.7 \text{ g AgCl} &\times \left[\frac{1 \text{ mol AgCl}}{143.4 \text{ g AgCl}} \right] \left[\frac{1 \text{ mol CaCl}_2}{2 \text{ mol AgCl}} \right] \left[\frac{111.1 \text{ g CaCl}_2}{1 \text{ mol CaCl}_2} \right] \\ &\approx \frac{72}{144} \times \frac{1}{2} \times \frac{110}{1} = \frac{1}{2} \times \frac{1}{2} \times \frac{110}{1} = \frac{110}{4} \\ &= 27.5 \text{ g CaCl}_2 \text{ (actual value} = 27.775 \text{ g)} \end{aligned}$$

Thus, about 27.8 g CaCl_2 are needed to produce 71.7 g AgCl .



MCAT EXPERTISE

Common conversions used in stoichiometry include:

- 1 mole of any ideal gas at STP = 22.4 L
- 1 mole of any substance = 6.022×10^{23} particles (Avogadro's number)
- 1 mole of any substance = its molar mass in grams (from the periodic table)

LIMITING REAGENT

Rarely are reactants added in the exact stoichiometric proportions shown in the balanced equation of a reaction. As a result, in most reactions, one reactant will be used up or consumed first. This reactant is known as the limiting reagent (or reactant) because it limits the amount of product that can be formed in the reaction. The reactants that remain after all the limiting reagent is used up are called excess reagents (or reactants).



MCAT EXPERTISE

When the quantities of two reactants are given on the MCAT, expect to have to figure out which is the limiting reagent.

Figure 4.7 shows a reaction vessel that has significant amounts of reactants A and B, which react in equal amounts to produce product C. On the left, before the reaction, there is more reactant A than B. After the reaction is over, there is more product C but there is reactant A left over. Thus, reactant A is considered in excess, and reactant B is considered limiting.

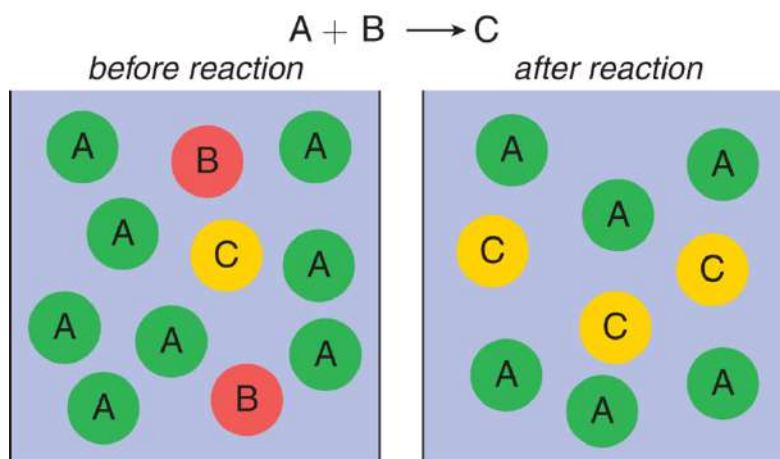


Figure 4.7. Reaction with a Limiting Reagent
A is considered an excess reagent; B is the limiting reagent.

For problems involving the determination of the limiting reagent, keep in mind two principles:

All comparisons of reactants must be done in units of moles. Gram-to-gram comparisons will be useless and may even be misleading.

It is not the absolute mole quantities of the reactants that determine which reactant is the limiting reagent. Rather, the rate at which the reactants are consumed (the stoichiometric ratios of the reactants), combined with the absolute mole quantities determines which reactant is the limiting reagent.

Example:

If 27.9 g of Fe react with 24.1 g of S to produce FeS, what would be the limiting reagent? How many grams of excess reagent would be present in the vessel at the end of the reaction?

The balanced equation is $\text{Fe} + \text{S} \rightarrow \text{FeS}$.

Solution:

First, determine the number of moles for each reactant.

$$27.9 \text{ g Fe} \times \frac{1 \text{ mol Fe}}{55.8 \text{ g Fe}} \approx 28 \times \frac{1}{56} = 0.5 \text{ mol Fe}$$

$$24.1 \text{ g S} \times \frac{1 \text{ mol S}}{32.1 \text{ g S}} \approx 24 \times \frac{1}{32} = 0.75 \text{ mol S}$$

Because 1 mole of Fe is needed to react with 1 mole of S and there are 0.5 moles Fe for the given 0.75 moles S, the limiting reagent is Fe. Thus, 0.5 moles of Fe will react with 0.5 moles of S, leaving an excess of 0.25 moles of S in the vessel. The mass of the excess reagent will be:

$$0.25 \text{ mol} \times \frac{32.1 \text{ g S}}{1 \text{ mol S}} = 8 \text{ g S}$$

YIELD

The yield of a reaction can refer to either the amount of product predicted (theoretical yield) or actually obtained (raw or actual yield) when a reaction is carried out. Theoretical yield is the maximum amount of product that can be generated as predicted from the balanced equation, assuming that all of the limiting reactant is consumed, no side reactions have occurred, and the entire product has been collected. Theoretical yield

is rarely ever attained through the actual chemical reaction. Actual yield is the amount of product one actually obtains during the reaction. The ratio of the actual yield to the theoretical yield, multiplied by 100 percent, gives the percent yield:

$$\text{Percent yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100\%$$

Equation 4.6



MCAT EXPERTISE

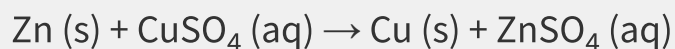
An experimentally based passage that involves a chemical reaction may include a pseudo-discrete question that involves finding the percent yield.

Example:

What is the percent yield for a reaction in which 28 g of Cu is produced by reacting 32.7 g of Zn in excess CuSO_4 solution?

Solution:

The balanced equation is as follows:



Calculate the theoretical yield for Cu.

$$32.7 \text{ g Zn} \times \left[\frac{1 \text{ mol Zn}}{65.4 \text{ g Zn}} \right] \left[\frac{1 \text{ mol Cu}}{1 \text{ mol Zn}} \right] \left[\frac{63.5 \text{ g Cu}}{1 \text{ mol Cu}} \right]$$

$$\approx 33 \times \frac{1}{66} \times \frac{1}{1} \times \frac{64}{1} = \frac{1}{2} \times \frac{64}{1} = 32 \text{ (actual value} = 31.8 \text{ g)}$$

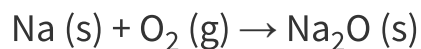
This 31.8 g represents the theoretical yield. Finally, determine the percent yield.

$$\frac{27 \text{ g}}{31.8 \text{ g}} \times 100\% \approx \frac{28}{32} \times 100\% = \frac{7}{8} \times 100\% = 87.5\%$$

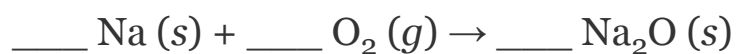
MCAT Concept Check 4.5:

Before you move on, assess your understanding of the material with these questions.

Questions 1–3 refer to the following unbalanced equation:



Balance the chemical equation:



If 46 g Na and 32 g O₂ are provided, find the maximum number of moles of sodium oxide produced.

Identify the limiting reagent, and find the mass of the excess reagent left over once the reaction has run to completion.

Be(OH)₂ is produced when water reacts with BeO. Starting with 2.5 kg BeO in excess water, and producing 1.1 kg Be(OH)₂, what is the percent yield of this reaction?

4.6 Ions

LEARNING GOALS

After Chapter 4.6, you will be able to:

- Determine whether a molecule will act as an electrolyte in solution
- Recall the common polyatomic ions, including their names and charges

Ionic compounds are of particular interest to chemists because certain important types of chemical reactions—acid–base and oxidation–reduction reactions, for instance—commonly take place in ionic solutions. For stoichiometry problems, the goal with ions is to identify oxidation states. This will allow us to determine electron equivalents, balance equations, and deduce chemical formulas from nomenclature.

CATIONS AND ANIONS

In Chapter 3 of MCAT General Chemistry Review, we discussed how ionic compounds are made up of positively charged cations, usually metals, and negatively charged anions, usually nonmetals. This rule does not always

hold true for elements like hydrogen, which can act like an anion or cation but is still classified as a nonmetal, as shown in Figure 4.8. Ionic compounds are held together by ionic bonds, which rely on the force of electrostatic attraction between oppositely charged particles.

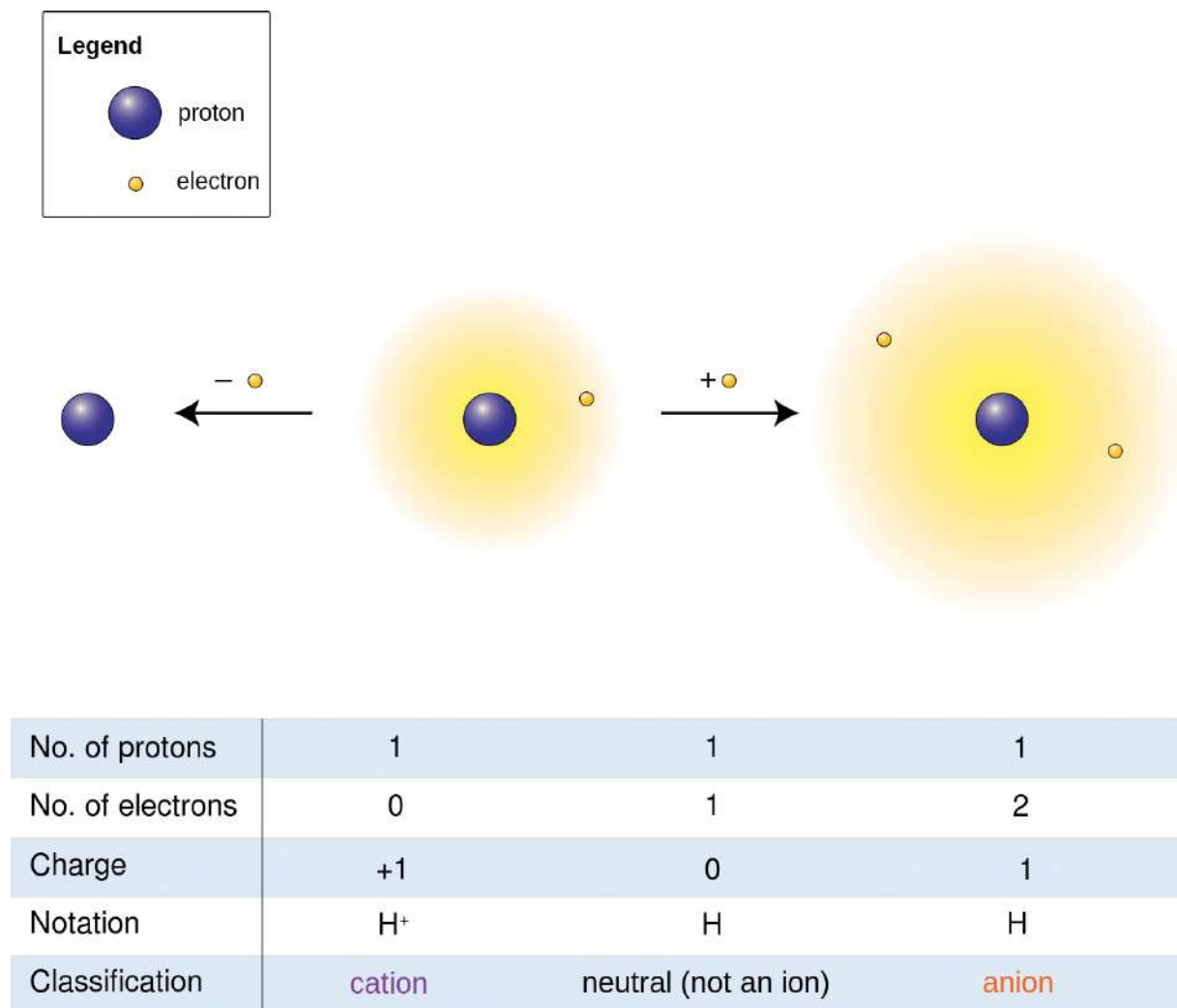


Figure 4.8. Oxidation States of Hydrogen



BRIDGE

The magnitude of the electrostatic force in an ionic bond follows the same conventions described for Coulomb's law in Chapter 5 of MCAT Physics and Math Review. The distance between nuclei in ionic bonds is inversely proportional to the force. Therefore, ionic compounds with long bond distances are much more weakly held together.

The nomenclature of ionic compounds is based on the names of the component ions:

For elements (usually metals) that can form more than one positive ion, the charge is indicated by a Roman numeral in parentheses following the name of the element.

Fe^{2+}	Iron(II)	Cu^{+}	Copper(I)
Fe^{3+}	Iron(III)	Cu^{2+}	Copper(II)

An older, less commonly used method is to add the endings -ous or -ic to the root of the Latin name of the element to represent the ions with lesser and greater charge, respectively.

Fe^{2+}	Ferrous	Cu^{+}	Cuprous
Fe^{3+}	Ferric	Cu^{2+}	Cupric

Monatomic anions are named by dropping the ending of the name of the element and adding -ide.

H^{-}	Hydride	S^{2-}	Sulfide
F^{-}	Fluoride	N^{3-}	Nitride



Oxide



Phosphide

Many polyatomic anions contain oxygen and are therefore called oxyanions. When an element forms two oxyanions, the name of the one with less oxygen ends in -ite, and the one with more oxygen ends in -ate.



Nitrite



Sulfite



Nitrate



Sulfate

In extended series of oxyanions, prefixes are also used. Hypo- and hyper, written as per-, are used to indicate less oxygen and more oxygen, respectively.



Hypochlorite



Chlorite



Chlorate



Perchlorate

Polyatomic anions often gain one or more H^+ ions to form anions of lower charge. The resulting ions are named by adding the word hydrogen or dihydrogen to the front of the anion's name. An older method uses the prefix bi- to indicate the addition of a single hydrogen ion.



Hydrogen carbonate or bicarbonate



Hydrogen sulfate or bisulfate



Dihydrogen phosphate

Other common polyatomic ions that may be useful to know are in Table 4.1.

Charge	Formula	Name
+1	NH_4^+	Ammonium
-1	$\text{C}_2\text{H}_3\text{O}_2^-$	Acetate
	CN^-	Cyanide
	MnO_4^-	Permanganate
	SCN^-	Thiocyanate
-2	CrO_4^{2-}	Chromate
	$\text{Cr}_2\text{O}_7^{2-}$	Dichromate
-3	BO_3^{3-}	Borate

Table 4.1. Other Common Polyatomic Ions



MCAT EXPERTISE

It is unlikely that -ous or -ic endings will be required for most problem-solving. Passages tend to provide reaction schemes that allow you to deduce any unfamiliar compound's formulas. However, it is still important to understand the nomenclature for discrete questions.



MNEMONIC

The “lightest” anions have the fewest oxygens; the heaviest anions have the most oxygens.

ION CHARGES

Ionic species, by definition, have charge. Cations have positive charge, and anions have negative charge. Some elements are only found naturally in their charged forms, while others may exist naturally in the charged or uncharged state. Some elements can even have several different charges or oxidation states, depending on the other atoms in a compound.

Some of the charged atoms or molecules that are on the MCAT include the active metals—the alkali metals (Group IA or Group 1) and the alkaline earth metals (Group IIA or Group 2), which have charges of +1 and +2, respectively, in the natural state.



BRIDGE

Remember that alkali metals are not typically found in nature in their uncharged state because they are highly reactive with moisture. Instead, they are found as cations in salts (like NaCl).

Nonmetals, which are found on the right side of the periodic table, generally form anions. For example, all the halogens (Group VIIA or Group

17) form monatomic anions with a charge of -1 because they already have 7 electrons and aim to fill an octet.

In summary, all elements in a given group tend to form monatomic ions with the same charge (for example, all Group IA elements have a charge of $+1$ in their ionic state). Note that there are anionic species that contain metallic elements (for example, MnO_4^- [permanganate] and CrO_4^{2-} [chromate]); even so, the metals have positive oxidation states. Also note that in the oxyanions of the halogens, such as ClO^- and ClO_2^- , the halogen is assigned a positive oxidation state.



BRIDGE

Oxyanions of transition metals like the MnO_4^- and CrO_4^{2-} ions have an inordinately high oxidation number on the metal. As such, they tend to gain electrons in order to reduce this oxidation number and thus make good oxidizing agents. Good oxidizing and reducing agents are discussed in Chapter 4 of MCAT Organic Chemistry Review.

For nonrepresentative elements like many of the transition metals, such as copper, iron, and chromium, there are numerous positively charged states. These states need not be memorized. Qualitatively, the color of a solution can be indicative of the oxidation state of a given element in the solution. The same element in different oxidation states can undergo different electron transitions and therefore absorb different frequencies of light. In

Figure 4.9, this phenomenon is shown for various plutonium salts with different oxidation states for plutonium indicated in Roman numerals.

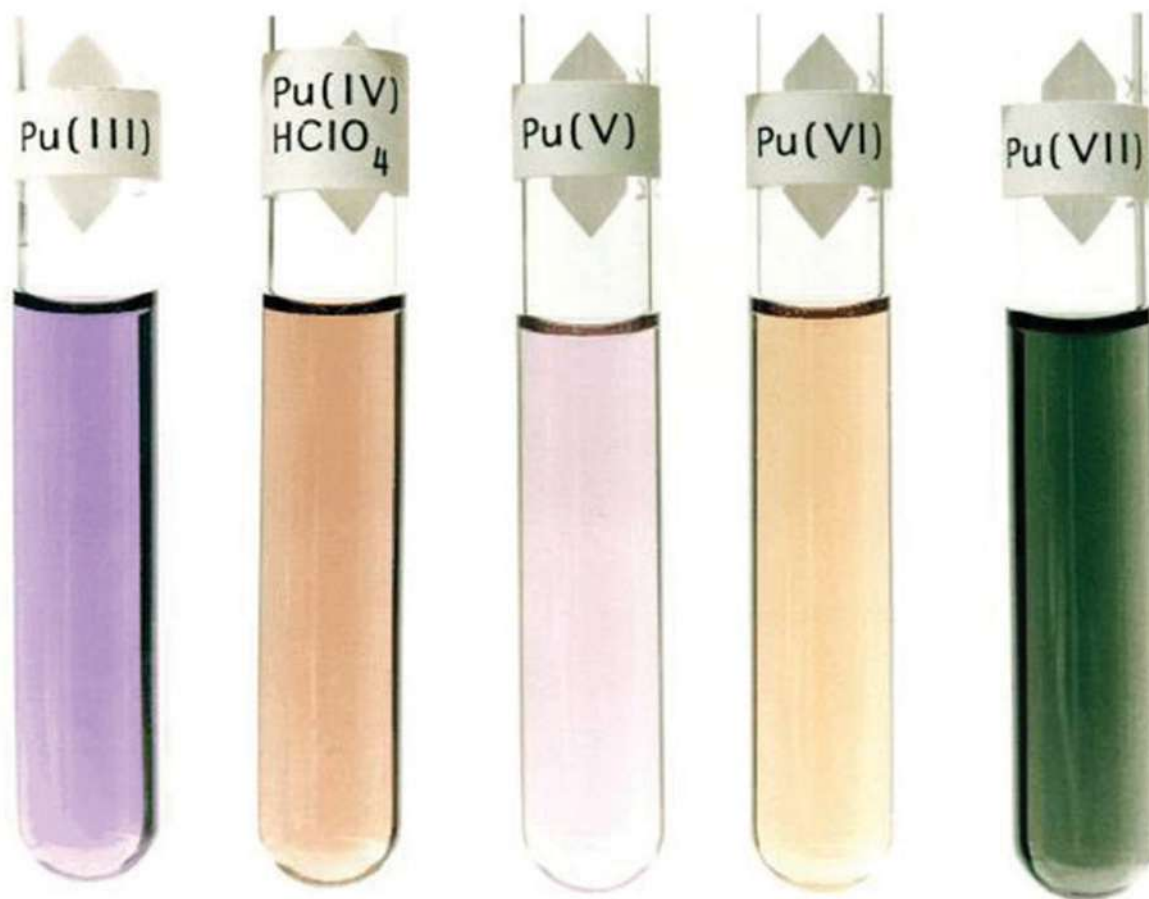


Figure 4.9. Solutions with Various Plutonium Oxidation States

The trends of ionicity, as we've described here, are helpful but are complicated by the fact that many elements have intermediate electronegativity and are consequently less likely to form ionic compounds, and by the left-to-right transition from metallic to nonmetallic character on the periodic table.

ELECTROLYTES

In spite of the fact that ionic compounds are composed of ions, solid ionic compounds tend to be poor conductors of electricity because the charged particles are rigidly set in place by the lattice arrangement of the crystalline solid. In aqueous solutions, however, the lattice arrangement is disrupted by the ion–dipole interactions between the ionic components and the water molecules. The cations and anions are now free to move, and as a result, the solution of ions is able to conduct electricity.

Solutes that enable solutions to carry currents are called electrolytes. The electrical conductivity of aqueous solutions is governed by the presence and concentration of ions in the solution. Subsequently, the number of electron equivalents being transferred in such a system, such as in electrochemical cells, varies. Pure water, which has no ions other than the very few hydrogen ions and hydroxide ions that result from water's low-level autodissociation, is a very poor conductor.



MCAT EXPERTISE

Ionic compounds make good electrolytes because they dissolve most readily. Nonpolar covalent compounds are the weakest because they do not form current-carrying ions.

The tendency of an ionic solute to dissolve, or solvate, into its constituent ions in water may be high or low. A solute is considered a strong electrolyte if it dissociates completely into its constituent ions. Examples of strong electrolytes include certain ionic compounds, such as NaCl and KI, and molecular compounds with highly polar covalent bonds that dissociate

into ions when dissolved, such as HCl in water. An example of solvation of such compounds is shown in Figure 4.10.

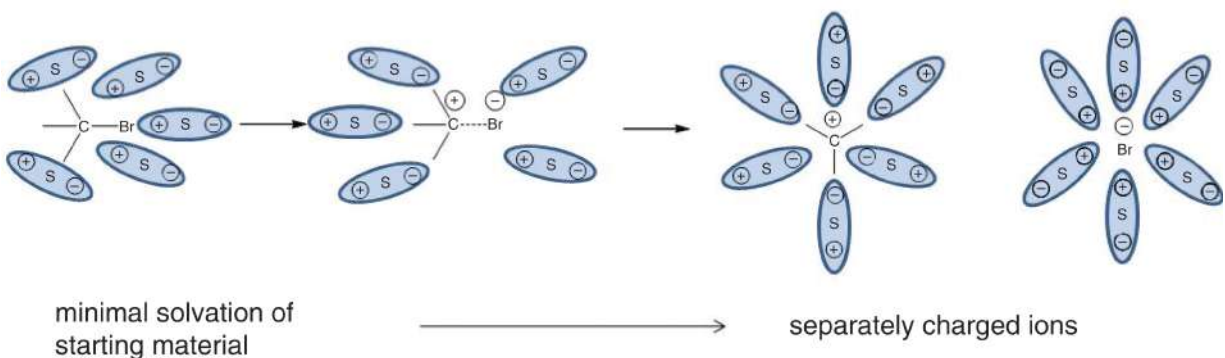


Figure 4.10. Solvation of a Polar Covalent Compound
S indicates a solvent particle.

A weak electrolyte, on the other hand, ionizes or hydrolyzes incompletely in aqueous solution, and only some of the solute is dissolved into its ionic constituents. Examples include Hg_2I_2 ($K_{\text{sp}} = 4.5 \times 10^{-29}$), acetic acid and other weak acids, and ammonia and other weak bases. Many compounds do not ionize at all in water, retaining their molecular structure in solution, which may also limit their solubility. These compounds are called nonelectrolytes and include many nonpolar gases and organic compounds, such as O_2 (g), CO_2 (g), and glucose.



BRIDGE

Because electrolytes ionize in solution, they will produce a larger effect on colligative properties, described in Chapter 9 of MCAT General Chemistry Review, than one would expect from the given concentration.

MCAT Concept Check 4.6:

Before you move on, assess your understanding of the material with these questions.

Label the following solutions as electrolytes or nonelectrolytes: (Note: Assume these compounds are all in aqueous solution.)

- | | | |
|---------------------|-------------|----------------|
| • HCl | Electrolyte | Nonelectrolyte |
| • Sucrose | Electrolyte | Nonelectrolyte |
| • MgBr ₂ | Electrolyte | Nonelectrolyte |
| • CH ₄ | Electrolyte | Nonelectrolyte |



Identify the following ions as cations or anions, and then provide the formula or chemical symbol:

Ion	Cation or Anion	Formula
Phosphate		
Hypochlorite		
Ammonium		

Ion	Cation or Anion	Formula
Phosphide		
Bicarbonate		
Nitrite		
Chromium(II)		





Conclusion

We began our consideration of compounds with benzaldehyde. As a compound, it is made from constituent atoms of different elements in a set ratio defined by its empirical or molecular formula. Each molecule of a compound has a defined mass that is measured as its molecular weight. The mass of one mole of any compound is determined from its molar mass in the units of grams per mole. We reviewed the basic classifications of reactions commonly tested on the MCAT: combination, decomposition, combustion, single-displacement, double-displacement, and neutralization reactions. Furthermore, we are now confident in our understanding of the steps necessary to balance any chemical reaction; we are ready to tackle more stoichiometric problems in preparation for Test Day.

Before moving to the next chapters discussing chemical kinetics and thermodynamics, let us offer our congratulations to you. By completing these first four chapters, you have been introduced to the fundamental concepts of chemistry—everything from the structure of the atom and trends of the elements to bonding and the formation of compounds. The understanding you have gained so far will be the foundation for your comprehension of even the most difficult general chemistry concepts tested on the MCAT. Keep moving forward with your review of general chemistry; don't get stuck in the details. Those details will be learned best

through the application of the basic principles to MCAT practice passages and questions.

Concept Summary

Molecules and Moles

- Compounds are substances composed of two or more elements in a fixed proportion.
- Molecular weight is the mass (in amu) of the constituent atoms in a compound as indicated by the molecular formula.
- Molar mass is the mass of one mole (Avogadro's number or 6.022×10^{23} particles) of a compound; usually measured in grams per mole.
- Gram equivalent weight is a measure of the mass of a substance that can donate one equivalent of the species of interest.
- Normality is the ratio of equivalents per liter; it is related to molarity by multiplying the molarity by the number of equivalents present per mole of compound.
- Equivalents are moles of the species of interest; equivalents are most often seen in acid–base chemistry (hydrogen ions or hydroxide ions) and oxidation–reduction reactions (moles of electrons or other ions).

Representation of Compounds

- The law of constant composition states that any pure sample of a compound will contain the same elements in the same mass ratio.
- The empirical formula is the smallest whole-number ratio of the elements in a compound.

- The molecular formula is either the same as or a multiple of the empirical formula; it gives the exact number of atoms of each element in a compound.
- To calculate percent composition by mass, determine the mass of the individual element and divide by the molar mass of the compound.

Types of Chemical Reactions

- Combination reactions occur when two or more reactants combine to form one product.
- Decomposition reactions occur when one reactant is chemically broken down into two or more products.
- Combustion reactions occur when a fuel and an oxidant (typically oxygen) react, forming the products water and carbon dioxide (if the fuel is a hydrocarbon).
- Displacement reactions occur when one or more atoms or ions of one compound are replaced with one or more atoms or ions of another compound.
 - Single-displacement reactions occur when an ion of one compound is replaced with another element.
 - Double-displacement reactions occur when elements from two different compounds trade places with each other to form two new compounds.
- Neutralization reactions are those in which an acid reacts with a base to form a salt (and, usually, water).

Balancing Chemical Equations

- Chemical equations must be balanced to perform stoichiometric calculations.
- Balanced equations are determined using the following steps in order:
 - Balancing the least common atoms.
 - Balancing the more common atoms (usually hydrogen and oxygen).
 - Balancing charge, if necessary.

Applications of Stoichiometry

- Balanced equations can be used to determine the limiting reagent, which is the reactant that will be consumed first in a chemical reaction.
- The other reactants present are termed excess reagents.
- Theoretical yield is the amount of product generated if all of the limiting reactant is consumed with no side reactions.
- Actual yield is typically lower than theoretical yield.
- Percent yield is calculated by dividing actual yield by theoretical yield and converting to a percentage.

Ions

- Like organic chemistry, ions in general chemistry have a system of nomenclature:
 - Roman numerals are used for nonrepresentative elements to denote ionic charge.
 - -ous endings can also be used to indicate lesser charge, while -ic endings indicate greater charge.
 - All monatomic anions end in -ide.
 - Oxyanions are given a suffix indicating how oxidized the central atom is. Those that contain a lesser amount of oxygen are given the suffix -

ite, and those with a greater amount are given the suffix -ate.

- Oxyanion series with more than two members are given an additional level of nomenclature. The species with the fewest oxygens is given the prefix hypo-, and the species with the most oxygens is given the prefix per-.
 - Polyatomic ions containing hydrogen denote the number of hydrogens using hydrogen or bi- to denote one, or dihydrogen to denote two.
-
- Ionic charges are predictable by group number and type of element (metal or nonmetal) for representative elements, but are generally unpredictable for nonrepresentative elements.
 - Metals form positively charged cations based on group number.
 - Nonmetals form negatively charged anions based on the number of electrons needed to achieve an octet.
-
- Electrolytes contain equivalents of ions from molecules that dissociate in solution. The strength of an electrolyte depends on its degree of dissociation or solvation.

Answers to Concept Checks

4.1

$$\begin{aligned}\text{NaBr: } & 23 + 79.9 \approx 23 + 80 = 103 \frac{\text{g}}{\text{mol}} \\ \text{SrCl}_2: & 87.6 + (2 \times 35.5) = 87.6 + 71 = 158.6 \frac{\text{g}}{\text{mol}} \\ \text{C}_6\text{H}_{12}\text{O}_6: & (6 \times 12) + (12 \times 1) + (6 \times 16) = 180 \frac{\text{g}}{\text{mol}}\end{aligned}$$

$$\text{NaBr: } \frac{100 \text{ g}}{103 \frac{\text{g}}{\text{mol}}} = 0.97 \text{ moles}$$

Note that the denominator is greater than the numerator by approximately 3 percent, and the actual value is less than 1 by approximately 3 percent. This approximation can be used to quickly estimate answers that are close to one.

$$\text{SrCl}_2: \frac{100 \text{ g}}{158.6 \frac{\text{g}}{\text{mol}}} \approx \frac{100}{160} = \frac{10}{16} = \frac{5}{8} = .625 \text{ moles}$$

Note that the answer is a fraction based on eighths. As these are commonly used on the MCAT, knowing the values for 1/8 through 7/8 can be useful for many problems.

$$\text{C}_6\text{H}_{12}\text{O}_6: \frac{100 \text{ g}}{180 \frac{\text{g}}{\text{mol}}} = \frac{10}{18} = \frac{5}{9} = 0.555 \text{ moles}$$

Note that the final fraction is in ninths. Dividing by nine follows a standard pattern that is useful to know for the MCAT: $1/9 = 0.111$, $2/9 = 0.222$, $3/9 = 0.333$, etc.

Both values equal one mole of the given substance. The number of entities in a mole is always the same (Avogadro's number, $6.022 \times 10^{23} \text{ mol}^{-1}$).

Normality is calculated as $N = M \times \frac{\text{equivalents}}{\text{mole}}$.

$$\text{For } \text{H}_3\text{PO}_4: 0.25 \text{ M H}_3\text{PO}_4 \times \frac{3 \text{ equiv H}^+}{\text{mol H}_3\text{PO}_4} = 0.75 \text{ N H}_3\text{PO}_4.$$

For PO_4^{3-} the grams must first be converted to moles, then to normality. PO_4^{3-} has a molecular mass of 95, giving

$$\frac{95 \text{ g}}{100 \text{ mL}} = \frac{950 \text{ g}}{1 \text{ L}} = \frac{10 \text{ mol}}{1 \text{ L}} = 10 \text{ M PO}_4^{3-}$$

$$10 \text{ M PO}_4^{3-} \times \frac{3 \text{ equiv H}^+}{\text{mol PO}_4^{3-}} = 30 \text{ N PO}_4^{3-}.$$

4.2

Both molecular and empirical formulas contain the same elements in the same ratios. They differ in that molecular formulas give the actual number of atoms of each element in the compound; empirical formulas give only the ratio and therefore may or may not give the actual number of atoms.

The molar mass of sodium carbonate is given by $(2 \times 23) + (1 \times 12) + (3 \times 16) = 106 \frac{\text{g}}{\text{mol}}$. The percent compositions are:

$$\text{Na: } \frac{2 \times \left(23 \frac{\text{g}}{\text{mol}}\right)}{106 \frac{\text{g}}{\text{mol}}} \times 100\% \approx \frac{46}{106} \times 100\% = 43.4\% \text{ (actual value = 43.4\%)}$$

Note that in all three cases, the estimation reduces the value of the denominator, thus making the calculated value larger than the actual value.

Start by assuming a 100 g sample, which represents 28.5 g Fe, 24.0 g S, and 49.7 g O. Next, divide each number of grams by the atomic weight to determine the number of moles:

$$\text{Fe: } \frac{28.5 \text{ g}}{55.8 \frac{\text{g}}{\text{mol}}} \approx \frac{28}{56} = 0.5 \text{ moles}$$

$$\text{S: } \frac{24 \text{ g}}{32.1 \frac{\text{g}}{\text{mol}}} \approx \frac{24}{32} = \frac{3}{4} = 0.75 \text{ moles}$$

$$\text{O: } \frac{49.7 \text{ g}}{16 \frac{\text{g}}{\text{mol}}} \approx \frac{48}{16} = 3 \text{ moles}$$

Next, find the multiplier that gives all three compounds integer values of moles. Using sulfur, multiplying $0.75 \text{ moles} \times 4 = 3 \text{ moles}$. Using 4 as a multiplier for all three compounds gives the ratio 2 Fe: 3 S: 12 O. This gives an empirical formula of $\text{Fe}_2\text{S}_3\text{O}_{12}$.

Ammonium cations swap places with (or displace) zinc cations yielding ammonium nitrate and zinc(II) sulfide. Zinc(II) sulfide then precipitates out of solution as a solid salt.

Reactants	Conditions	Products	Reaction Type
$2 \text{H}_2 + \text{O}_2$	$\rightarrow$	$2 \text{H}_2\text{O}$	Combination
$\text{Al}(\text{OH})_3 + \text{H}_3\text{PO}_4$	$\rightarrow$	$3 \text{H}_2\text{O} + \text{AlPO}_4$	Neutralization (a type of double-displacement)
$2 \text{H}_2\text{O}$	$\xrightarrow{\text{electricity}}$	$2 \text{H}_2 + \text{O}_2$	Decomposition
$\text{NaNO}_3 + \text{CuOH}$	$\rightarrow$	$\text{NaOH} + \text{CuNO}_3$	Double-displacement (metathesis)
$\text{Zn} + \text{AgCl}$	$\rightarrow$	$\text{ZnCl} + \text{Ag}$	Single-displacement

4.4

- $2 \text{Fe} + 3 \text{Cl}_2 \rightarrow 2 \text{FeCl}_3$
- $\text{Zn} + 2 \text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$
- $\text{C}_5\text{H}_{12} + 8 \text{O}_2 \rightarrow 5 \text{CO}_2 + 6 \text{H}_2\text{O}$
- $3 \text{Pb}(\text{NO}_3)_2 + 2 \text{AlCl}_3 \rightarrow 3 \text{PbCl}_2 + 2 \text{Al}(\text{NO}_3)_3$

4.5



$$46 \text{ g Na} \times \left[\frac{1 \text{ mol Na}}{23 \text{ g Na}} \right] = 2 \text{ mol Na};$$

$$32 \text{ g O}_2 \times \left[\frac{1 \text{ mol O}_2}{32 \text{ g O}_2} \right] = 1 \text{ mol O}_2$$

Because 4 sodium atoms are needed for every oxygen molecule, sodium will run out first. To determine the amount of Na₂O formed:

$$2 \text{ mol Na} \times \left[\frac{2 \text{ mol Na}_2\text{O}}{4 \text{ mol Na}} \right] = 1 \text{ mol Na}_2\text{O}$$

The limiting reagent is Na because 4 sodium atoms are needed for every oxygen molecule.

$$2 \text{ mol Na} \times \left[\frac{1 \text{ mol O}_2}{4 \text{ mol Na}} \right] = 0.5 \text{ mol O}_2$$

will be used, so $1.0 - 0.5 \text{ mol O}_2 = 0.5 \text{ mol O}_2$ will remain. In grams, this is:

$$0.5 \text{ mol O}_2 \times \left[\frac{32 \text{ g O}_2}{1 \text{ mol O}_2} \right] = 16 \text{ g}$$

excess O₂

Reaction: $\text{BeO} + \text{H}_2\text{O} \rightarrow \text{Be(OH)}_2$

$$2500 \text{ g BeO} \times \left[\frac{1 \text{ mol BeO}}{25 \text{ g BeO}} \right] \left[\frac{1 \text{ mol Be(OH)}_2}{1 \text{ mol BeO}} \right] \left[\frac{43 \text{ g Be(OH)}_2}{1 \text{ mol Be(OH)}_2} \right]$$

$$= 4300 \text{ g Be(OH)}_2$$

4.6

Electrolytes: HCl, MgBr₂; Nonelectrolytes: sucrose, CH₄

Ion	Cation or Anion	Formula
Phosphate	Anion	PO_4^{3-}
Hypochlorite	Anion	ClO^-
Ammonium	Cation	NH_4^+
Phosphide	Anion	P^{3-}
Bicarbonate	Anion	HCO_3^-
Nitrite	Anion	NO_2^-
Chromium(II)	Cation	Cr^{2+}

Equations to Remember

(4.1) Moles from mass: $\text{Moles} = \frac{\text{Mass of a sample}}{\text{Molar mass}}$

(4.2) Gram equivalent weight: $\text{GEW} = \frac{\text{Molar mass}}{n}$

(4.3) Equivalents from mass:

$$\text{Equivalents} = \frac{\text{Mass of compound}}{\text{Gram equivalent weight}}$$

(4.4) Molarity from normality: $\text{Molarity} = \frac{\text{Normality}}{n}$

(4.5) Percent composition:

$$\% \text{ composition} = \frac{\text{Mass of element in formula}}{\text{Molar mass}} \times 100\%$$

(4.6) Percent yield: $\% \text{ yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100\%$



Shared Concepts

General Chemistry Chapter 2

The Periodic Table

General Chemistry Chapter 3

Bonding and Chemical Interactions

General Chemistry Chapter 9

Solutions

General Chemistry Chapter 10

Acids and Bases

General Chemistry Chapter 11

Oxidation–Reduction Reactions

Physics and Math Chapter 5

Electrostatics and Magnetism

Discrete Practice Questions

Consult your online resources for additional practice.

1. Which of the following best describes ionic compounds?

- (A) Ionic compounds are formed from molecules containing two or more atoms.
- (B) Ionic compounds are formed of charged particles and are measured by molecular weight.
- (C) Ionic compounds are formed of charged particles that share electrons equally.
- (D) Ionic compounds are three-dimensional arrays of charged particles.

2. Which of the following compounds has a formula weight between 74 and 75 grams per mole?

- (A) KCl
- (B) $\text{C}_4\text{H}_{10}\text{O}$
- (C) MgCl_2
- (D) BF_3

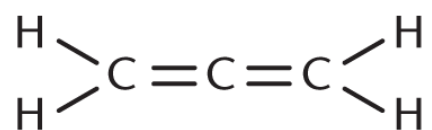
3. Which of the following is the gram equivalent weight of H_2SO_4 with respect to protons?

- (A) 49.1 g
- (B) 98.1 g
- (C) 147.1 g
- (D) 196.2 g

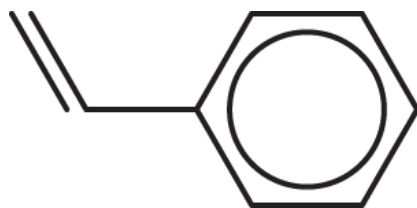
4. Which of the following molecules CANNOT be expressed by the empirical formula CH ?

- (A) Benzene
- (B) Ethyne

(C)



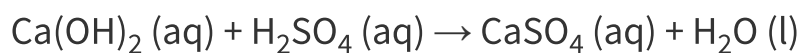
(D)



5. In which of the following compounds is the percent composition of carbon by mass closest to 62 percent?

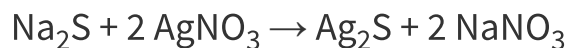
- (A) Acetone
- (B) Ethanol
- (C) Propane
- (D) Methanol

6. What is the most specific characterization of the reaction shown?



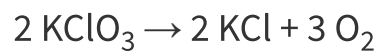
- (A) Single-displacement
- (B) Neutralization
- (C) Double-displacement
- (D) Oxidation–reduction

7. In the reaction shown, if 39.05 g of Na_2S are reacted with 85.5 g of AgNO_3 , how much of the excess reagent will be left over once the reaction has gone to completion?



- (A) 19.5 g Na₂S
- (B) 26.0 g Na₂S
- (C) 41.4 g AgNO₃
- (D) 74.3 g AgNO₃

8. Using a given mass of KClO₃, how would one calculate the mass of oxygen produced in the following reaction, assuming it goes to completion?



(A)

$$\frac{(\text{grams KClO}_3 \text{ consumed})(3 \text{ moles O}_2)(\text{molar mass O}_2)}{(\text{molar mass KClO}_3)(2 \text{ moles KClO}_3)}$$

(B)

$$\frac{(\text{grams KClO}_3 \text{ consumed})(\text{molar mass O}_2)}{(\text{molar mass KClO}_3)(2 \text{ moles KClO}_3)}$$

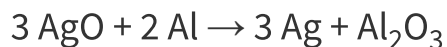
(C)

$$\frac{(\text{molar mass KClO}_3)(2 \text{ moles KClO}_3)}{(\text{grams KClO}_3 \text{ consumed})(\text{molar mass O}_2)}$$

(D)

$$\frac{(\text{grams KClO}_3 \text{ consumed})(3 \text{ moles O}_2)}{(\text{molar mass KClO}_3)(2 \text{ moles KClO}_3)(\text{molar mass O}_2)}$$

9. Aluminum metal can be used to remove tarnish from silver when the two solid metals are placed in water, according to the following reaction:



This reaction is a:

- ☐ double-displacement reaction.
- ☐ single-displacement reaction.
- ☐ oxidation–reduction reaction.
- ☐ combination reaction.

- (A) II only
- (B) IV only
- (C) I and III only
- (D) II and III only

10. Which of the following types of reactions generally have the same number of reactants and products?

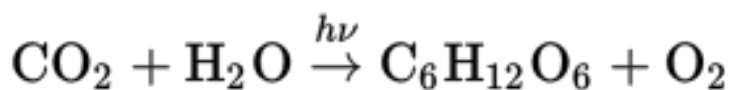
- ☐ Double-displacement reactions
- ☐ Single-displacement reactions
- ☐ Combination reactions

- (A) I only
- (B) II only
- (C) I and II only
- (D) II and III only

11. A reaction that utilizes oxygen and hydrocarbons as reactants and that produces carbon dioxide and water as products is best characterized as:

- (A) single-displacement.
- (B) combustion.
- (C) metathesis.
- (D) decomposition.

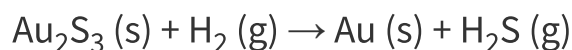
12. In the process of photosynthesis, carbon dioxide and water combine with energy to form glucose and oxygen, according to the following equation:



What is the theoretical yield of glucose if 30 grams of water are reacted with excess carbon dioxide and energy, according to the equation above?

- (A) 30.0 g
- (B) 50.0 g
- (C) 300.1 g
- (D) 1801 g

13. In the following reaction:



If 2 moles of $\text{Au}_2\text{S}_3 (\text{s})$ is reacted with 5 moles of hydrogen gas, what is the limiting reagent?

- (A) $\text{Au}_2\text{S}_3 (\text{s})$
- (B) $\text{H}_2 (\text{g})$
- (C) $\text{Au} (\text{s})$
- (D) $\text{H}_2\text{S} (\text{g})$

14. Which of the following would make the strongest electrolytic solution?

- (A) A nonpolar covalent compound with significant solubility.
- (B) An ionic compound composed of one cation with +3 charge and three anions with -1 charge.
- (C) A polar covalent compound with a small dissociation constant.
- (D) An ionic compound composed of two cations with +1 charge and one anion with -2 charge.

15. What is the molecular formula of a compound with an empirical formula of B_2H_5 and a molar mass of $53.2 \frac{\text{g}}{\text{mol}}$?

- (A) B_2H_5
- (B) B_3H_7
- (C) B_4H_{10}
- (D) B_6H_{15}

Explanations to Discrete Practice Questions

1. D

Ionic compounds are composed of atoms held together by ionic bonds. Ionic bonds associate charged particles with large differences in electronegativity. Rather than forming molecules or being measured by molecular weight, as in (A) and (B), ionic compounds form large arrays of ions in crystalline solids and are measured with formula weights. In ionic bonds, electrons are not really shared but rather are donated from the less electronegative atom to the more electronegative atom, eliminating (C).

2. A

Of the compounds listed, both (B) and (D) are covalent compounds and thus are measured in molecular weights, not formula weights. The formula weight of MgCl_2 is much too high ($24.3 \text{ amu} + 2 \times 35.5 \text{ amu} = 95.3 \text{ amu}$ per formula unit), eliminating (C). Only KCl fits the criteria ($39.1 \text{ amu} + 35.5 \text{ amu} = 74.6 \text{ amu}$).

3. A

First, it is helpful to know the molar mass of one mole of H_2SO_4 , which is found by adding the atomic weights of the atoms that constitute the molecule:

$$\left(2 \times 1.0 \frac{\text{g}}{\text{mol H}}\right) + \left(1 \times 32.1 \frac{\text{g}}{\text{mol S}}\right) + \left(4 \times 16.0 \frac{\text{g}}{\text{mol O}}\right) = 98.1 \frac{\text{g}}{\text{mol H}_2\text{SO}_4}.$$

Gram equivalent weight is the mass (in grams) that would release one mole of protons. Because sulfuric acid has two hydrogens per molecule, the gram equivalent weight is 98.1 g divided by 2, or 49.1 g.

4. C

The definition of an empirical formula is a formula that represents a molecule with the simplest ratio, in whole numbers, of the elements comprising the compound. In this case, given the empirical formula CH, any molecule with carbon and hydrogen atoms in a 1:1 ratio would be accurately represented by this empirical formula. (C) has three carbon atoms and four hydrogen atoms. Both its molecular and empirical formulas would be C_3H_4 because this formula represents the smallest whole-number ratio of its constituent elements.

5. A

The percent composition by mass of any given element within a molecule is equal to the mass of that element in the molecule divided by the molar mass of the compound, times 100%. In this case, acetone, $\text{C}_3\text{H}_6\text{O}$, has

$$\frac{3 \times 12 \frac{\text{g}}{\text{mol C}}}{\left(3 \times 12 \frac{\text{g}}{\text{mol C}}\right) + \left(6 \times 1 \frac{\text{g}}{\text{mol H}}\right) + \left(16 \frac{\text{g}}{\text{mol O}}\right)}$$

$$= \frac{36 \frac{\text{g}}{\text{mol}}}{58 \frac{\text{g}}{\text{mol}}} = \frac{18}{29} \approx \frac{2}{3} \approx 66.7\%.$$

This is an overestimation, and the actual value will be lower; it is closest to 62% out of the four choices available. (B), ethanol, is

$$\frac{24 \frac{\text{g}}{\text{mol C}}}{46 \frac{\text{g}}{\text{mol total}}} \approx 50\%.$$

This is an underestimation, and the actual value will be higher but nowhere near 62%. (C), propane, is C_3H_8 , and calculates to be

$$\frac{36 \frac{\text{g}}{\text{mol c}}}{44 \frac{\text{g}}{\text{mol total}}} \approx \frac{9}{11} \approx 80\% \text{ carbon.}$$

This is an underestimation, and therefore the actual value cannot be 62%. Finally, (D), methanol, is

$$\frac{12 \frac{\text{g}}{\text{mol C}}}{32 \frac{\text{g}}{\text{mol total}}} = \frac{3}{8} = 37.5\% \text{ carbon.}$$

Note that all four of these compounds are commonly encountered on the MCAT, and you should be familiar with the structure and composition of each, including their common names.

This reaction is a classic example of a neutralization reaction, in which an acid and a base react to form a salt and, usually, water. Although this reaction also fits the criteria for a double-displacement reaction, (C), in which two molecules essentially exchange ions with each other, neutralization is a more specific description of the process.

7. A

In this question, you are first given the masses of both reactants used to start the reaction. To figure out what will be left over, we must first determine which species is the limiting reagent. The formula weight of Na_2S is

$$\left(2 \times \frac{23 \text{ g}}{\text{mol Na}}\right) + \left(1 \times \frac{32.1 \text{ g}}{\text{mol S}}\right) = 78.1 \frac{\text{g}}{\text{mol}} \text{Na}_2\text{S}.$$

The formula weight of AgNO_3 is

$$\left(1 \times \frac{107.9 \text{ g}}{\text{mol Ag}}\right) + \left(1 \times \frac{14 \text{ g}}{\text{mol N}}\right) + \left(3 \times \frac{16 \text{ g}}{\text{mol O}}\right) = 169.9 \frac{\text{g}}{\text{mol}} \text{AgNO}_3.$$

From this, we can determine that we are given:

$$\frac{39.05 \text{ g Na}_2\text{S}}{78.1 \frac{\text{g}}{\text{mol}}} = 0.5 \text{ mol Na}_2\text{S}$$

$$\text{and } \frac{85.5 \text{ g AgNO}_3}{169.9 \frac{\text{g}}{\text{mol}}} = 0.5 \text{ mol AgNO}_3.$$

Because we need two moles of AgNO_3 for every mole of Na_2S , AgNO_3 is the limiting reagent, and the correct answer choice will be in grams of Na_2S . If 0.5 mol of AgNO_3 are used up, and Na_2S will be consumed at half the rate of AgNO_3 (based on their mole ratio), then 0.25 mol Na_2S will be used up. We then have 0.25 mol excess Na_2S , which has a mass of

$$0.25 \text{ mol} \times \left(\frac{78.1 \text{ g}}{\text{mol}} \right) \approx \frac{1}{4} \times \frac{80}{1} = 20 \text{ g (actual value} = 19.5 \text{ g)}.$$

8. A

This is a question best answered by dimensional analysis. Keeping in mind that molar mass is measured in grams of a substance per moles of that substance, only (A) comes out with the units of grams of oxygen. (B) has the units of grams per mole of oxygen, not grams of oxygen. (C) has the units of moles per gram of oxygen. (D) has the units of mol^2 per gram of oxygen.

9. D

In the reaction, there is a single displacement, with the silver in silver oxide being replaced by the aluminum to form aluminum oxide. This single-displacement reaction also necessitates a transfer of electrons in an oxidation–reduction reaction; silver, for example, changes from the +2 oxidation state to neutral. Aluminum changes from neutral to the +3 oxidation state.

10. C

Typically, both single-displacement and double-displacement reactions have two reactants that swap either one or two components between the two species. Combination reactions, on the other hand, have more

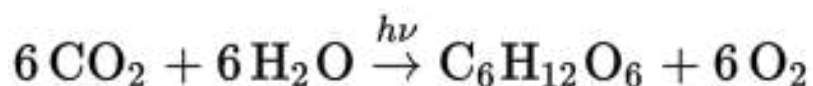
reactants than products because the reactants combine together to form the product.

11. B

This description characterizes a combustion reaction because a hydrocarbon acts as a fuel when reacting with oxygen. Carbon dioxide (an oxide) and water are the products of such a reaction.

12. B

The equation given is unbalanced, so the first step must be to balance it:



The theoretical yield is the amount of product synthesized if the limiting reagent is completely used up. This question therefore asks how much glucose is produced if the limiting reagent is 30 grams of water. Using the three-fraction method discussed in this chapter to solve for the mass of glucose produced gives:

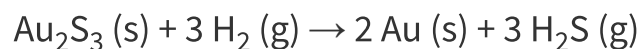
$$\begin{aligned} 30.0 \text{ g H}_2\text{O} &\times \left[\frac{1 \text{ mol H}_2\text{O}}{18 \text{ g H}_2\text{O}} \right] \left[\frac{1 \text{ mol C}_6\text{H}_{12}\text{O}_6}{6 \text{ mol H}_2\text{O}} \right] \left[\frac{180 \text{ g C}_6\text{H}_{12}\text{O}_6}{1 \text{ mol C}_6\text{H}_{12}\text{O}_6} \right] \\ &= 50 \text{ g C}_6\text{H}_{12}\text{O}_6 \end{aligned}$$

Thus, 50 grams of glucose are produced.

13. B

A limiting reagent is by definition a reactant. Because Au and H₂S are products, they cannot act as limiting reagents, eliminating (C) and (D).

Next, note that the given equation is unbalanced and the first step is to balance it:



The problem states that 2 moles of gold(III) sulfide and 5 moles of hydrogen gas are available. To use up both moles of gold(III) sulfide, 6 moles of hydrogen gas are needed because there is a 1:3 ratio between these reactants. Since only 5 moles of hydrogen gas are present, that will have to be the limiting reagent.

14. B

The best electrolytes dissociate readily (have a high dissociation constant) and are ionic compounds with large amounts of cations and anions. This rules out (A) and (C). (D) has fewer total ions with a smaller total magnitude of charge and therefore is not as strong an electrolyte as (B).

15. C

The simplest approach is to determine the molar mass of the empirical formula. B_2H_5 has a molar mass of **$26.6 \frac{\text{g}}{\text{mol}}$** . A molecular formula is always a multiple of the empirical formula; doubling this quantity will result in the molar mass given in the question stem. Therefore, the compound must be B_4H_{10} .

5

Chemical Kinetics



In This Chapter

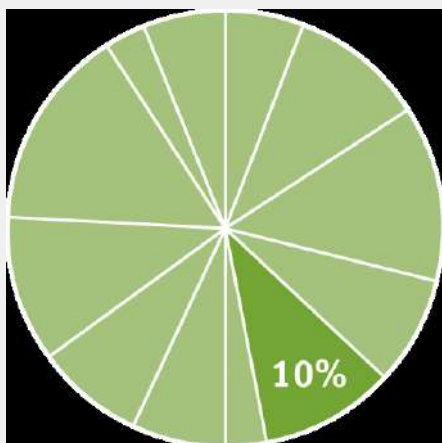
- 5.1 Chemical Kinetics 
 Reaction Mechanisms 
 Molecular Basis of Chemical Reactions 
 Factors Affecting Reaction Rate 



- 5.2 Reaction Rates 
 Definition of Rate 
 Determination of Rate Law 
 Reaction Orders 

Concept Summary 

Chapter Profile



The content in this chapter should be relevant to about 10% of all questions about general chemistry on the MCAT.

This chapter covers material from the following AAMC content categories:

1A: Structure and function of proteins and their constituent amino acids

5E: Principles of chemical thermodynamics and kinetics

Introduction

The following chapters focus on two primary topics: chemical kinetics and chemical equilibrium. As the term suggests, chemical kinetics is the study of reaction rates, the effects of reaction conditions on these rates, and the mechanisms implied by such observations. We start with kinetics because the molecular basis of reactions provides us with a framework of reaction chemistry. Following this, we will explore the equilibria of these reactions, which are related to—but distinct from—the kinetics of the reactions.

You may already have a fairly good understanding of equilibrium and the differences between spontaneous and nonspontaneous reactions. For instance, the utilization of ATP in the body is a spontaneous reaction that can be used to provide thermochemical energy for other reactions. While the equilibrium tells us that ATP will favor dissociation, it tells us nothing about its rate of dissociation. And, in fact, various conditions in the body can alter the rate at which ATP is synthesized and utilized for energy—primarily temperature. Some of the symptoms of hyper- and hypothermia are related to changes in metabolism caused by changes in temperature and reaction kinetics.

More broadly, we will see how multistep reactions, such as those seen in substrate-level and oxidative phosphorylation in biochemistry, have intermediate steps that have crucial kinetic limitations.

5.1 Chemical Kinetics

High-Yield



LEARNING GOALS

After Chapter 5.1, you will be able to:

- Describe the series of events within a multistep mechanism
- Explain the meaning and importance of a rate-determining step
- Describe activation energy
- Compare and contrast transition state theory and collision theory

Reactions can be spontaneous or nonspontaneous; the change in Gibbs free energy (ΔG) determines whether or not a reaction will occur by itself without outside assistance. However, even if a reaction is spontaneous, this does not necessarily mean that it will run quickly. In fact, nearly every biochemical reaction that enables life to exist, while perhaps spontaneous, proceeds so slowly that, without the aid of enzymes and other catalysts, measurable reaction progress might not actually occur over the course of an average human lifetime. And enzymes, like many other catalyzed reactions, can be saturated and experience a maximal turnover rate, as shown in Figure 5.1. For now, however, let us review reaction mechanisms, rates, rate laws, and the factors that pertain to simple chemical systems.

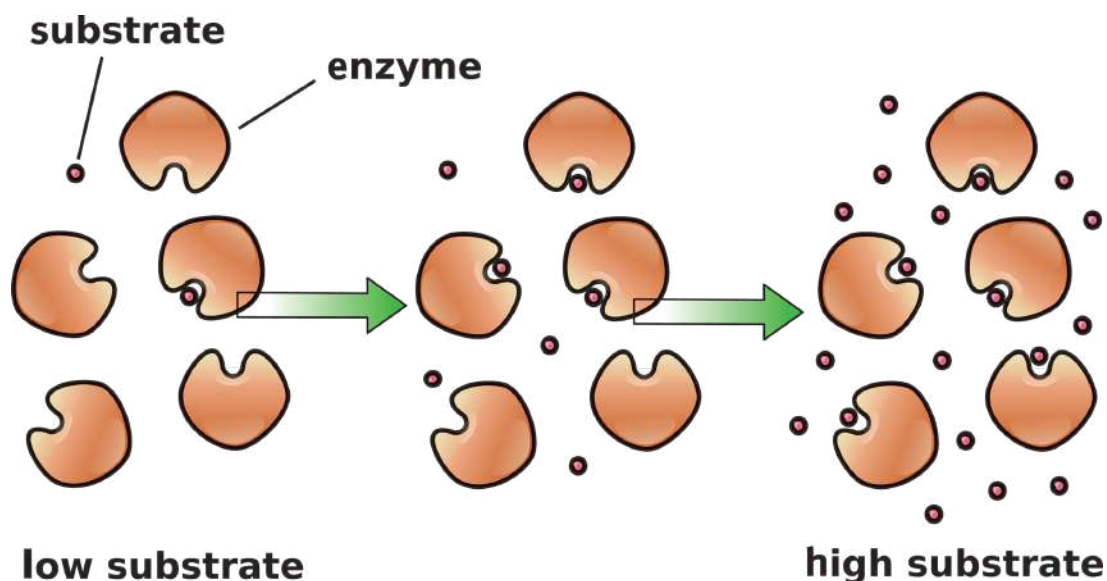


Figure 5.1. Enzymes, as Biological Catalysts, Can Be Saturated
High substrate conditions saturate the active sites of the enzyme, leading to maximal turnover.



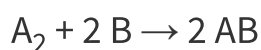
BRIDGE

Enzymes selectively enhance the rate of certain reactions by a factor of 10^2 to 10^{12} over other thermodynamically feasible reaction pathways. Enzyme function is discussed in Chapter 2 of MCAT Biochemistry Review.

REACTION MECHANISMS

Very rarely is the balanced reaction equation, used for determining limiting reactants and yields, an accurate representation of the actual steps involved in the chemical process from reactants to products. Many

reactions proceed by more than one step, the series of which is known as the mechanism of a reaction, and the sum of which gives the overall reaction. Knowing the accepted mechanism of a reaction may help explain the reaction's rate, position of equilibrium, and thermodynamic characteristics. Consider this generic reaction:



BRIDGE

Mechanisms are proposed pathways for a reaction that must coincide with rate data information from experimental observation. Reaction mechanisms are a major topic in organic chemistry and metabolism; Chapters 5 through 10 of MCAT Organic Chemistry Review and Chapters 9 through 11 of MCAT Biochemistry Review focus almost exclusively on reaction mechanisms in specific contexts.

On its own, this equation seems to imply a mechanism in which two molecules of B collide with one molecule of A_2 to form two molecules of AB. Suppose instead, however, that the reaction actually takes place in two steps:



Note that the two steps, taken together, give the overall net reaction. The molecule A_2B , which does not appear in the overall reaction, is called an intermediate. Reaction intermediates are often difficult to detect because they may be consumed almost immediately after they are formed, but a proposed mechanism that includes intermediates can be supported through kinetic experiments. One of the most important points to remember is that the slowest step in any proposed mechanism is called the rate-determining step because it acts like a kinetic bottleneck, preventing the overall reaction from proceeding any faster than that slowest step.



KEY CONCEPT

The rate of the whole reaction is only as fast as the rate-determining step.

MOLECULAR BASIS OF CHEMICAL REACTIONS

It's one thing to say A_2 reacts with $2 B$ to form $2 AB$; it's quite another to be able to describe, as precisely as possible, the actual interactions that occur between A_2 and B to produce AB at some rate. Various theories have been proposed to explain the events that are taking place at the atomic level through the process of a reaction.

Collision Theory of Chemical Kinetics

For a reaction to occur, molecules must collide with each other. The collision theory of chemical kinetics states that the rate of a reaction is proportional to the number of collisions per second between the reacting molecules.

The theory suggests, however, that not all collisions result in a chemical reaction. An effective collision (one that leads to the formation of products) occurs only if the molecules collide with each other in the correct orientation and with sufficient energy to break their existing bonds and form new ones. The minimum energy of collision necessary for a reaction to take place is called the activation energy, E_a , or the energy barrier. Only a fraction of colliding particles have enough kinetic energy to exceed the activation energy. This means that only a fraction of all collisions are effective. The rate of a reaction can therefore be expressed as follows:

$$\text{rate} = Z \times f$$

Equation 5.1

where Z is the total number of collisions occurring per second and f is the fraction of collisions that are effective. A much more quantitatively rigorous analysis of the collision theory can be accomplished through the Arrhenius equation, which is normally given as:

$$k = Ae^{\frac{-E_a}{RT}}$$

Equation 5.2

where k is the rate constant of a reaction, A is the frequency factor, E_a is the activation energy of the reaction, R is the ideal gas constant, and T is the temperature in kelvins. The frequency factor, also known as the attempt frequency of the reaction, is a measure of how often molecules in a certain reaction collide, with the unit s^{-1} . Activation energy is a subject that will be touched upon briefly in the following subsection and more qualitatively in future chapters.

Overall, what is important here in studying the Arrhenius equation is not the actual calculation (because those involving Euler's number, e , and natural logs, $\ln$, are not commonly found on the MCAT), but rather the relationships between the variables and the exponent rules that govern the equation.

For example, a simple relationship between A and k is evident in the equation. As the frequency factor of the reaction increases, the rate constant of the reaction also increases in a direct relationship. More complex relationships can also be seen in this equation. For example, if the temperature (T) of a chemical system were to increase to infinity, while all other variables are held constant, the value of the exponent would have a magnitude less than 1. However, before assuming that the rate constant is going to decrease as a result, note the presence of the negative sign. As the magnitude of the exponent gets smaller, it actually moves from a more negative value toward zero. The exponent thus becomes less negative (or more positive), which means that the rate constant actually increases. This

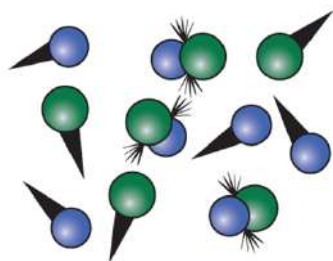
should make sense conceptually because the rate of a reaction increases with temperature.



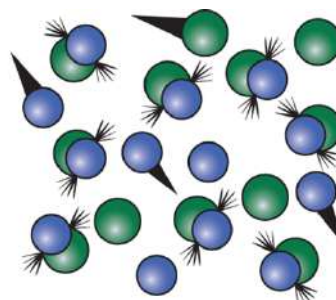
MCAT EXPERTISE

Low activation energy and high temperatures make the negative exponent of the Arrhenius equation smaller in magnitude and thus increase the rate constant k .

The frequency factor can be increased by increasing the number of molecules in a vessel. When there are more molecules, the opportunities for collision are increased, as shown in Figure 5.2.



low concentration = few collisions



high concentration = more collisions

Figure 5.2. Frequency Factor (A) Is Increased by Increasing Concentration

Transition State Theory

When molecules collide with energy equal to or greater than the activation energy, they form a transition state in which the old bonds are weakened and the new bonds begin to form. The transition state then dissociates into

products, fully forming the new bonds. For the reaction $A_2 + 2 B \rightarrow 2 AB$, the progress along the reaction coordinate, which traces the reaction from reactants to products, can be represented as shown in Figure 5.3.

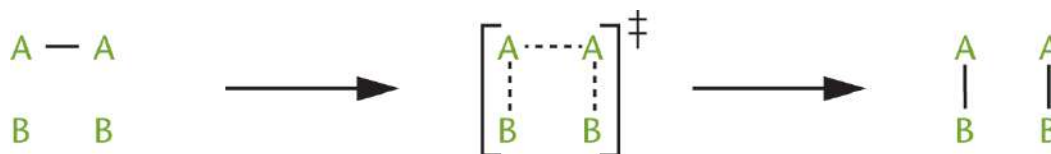


Figure 5.3. The Transition State

The transition state, also called the activated complex, has greater energy than both the reactants and the products and is denoted by the symbol $\ddagger$. The energy required to reach this transition state is the activation energy. Once an activated complex is formed, it can either dissociate into the products or revert to reactants without any additional energy input. Transition states are distinguished from reaction intermediates in that transition states are theoretical constructs that exist at the point of maximum energy, rather than distinct identities with finite lifetimes.



KEY CONCEPT

Relative to reactants and products, transition states have the highest energy. They are only theoretical structures and cannot be isolated. Nevertheless, we can still use the proposed structures to better understand the reactions in which they are involved.

A free energy diagram illustrates the relationship between the activation energy, the free energy of the reaction, and the free energy of the system.

The most important features to recognize in such diagrams are the relative energies of all of the products and reactants. The free energy change of the reaction (ΔG_{rxn}) is the difference between the free energy of the products and the free energy of the reactants. A negative free energy change indicates an exergonic reaction (energy is given off), and a positive free energy change indicates an endergonic reaction (energy is absorbed). The transition state exists at the peak of the energy diagram. The difference in free energy between the transition state and the reactants is the activation energy of the forward reaction; the difference in free energy between the transition state and the products is the activation energy of the reverse reaction.



KEY CONCEPT

$+\Delta G$ = endergonic = energy absorbed

$-\Delta G$ = exergonic = energy given off

For example, consider the formation of HCl from H_2 and Cl_2 . The overall reaction is:



Figure 5.4 shows that the reaction is exergonic. The free energy of the products is less than the free energy of the reactants; energy is released, and the free energy change of the reaction is negative.

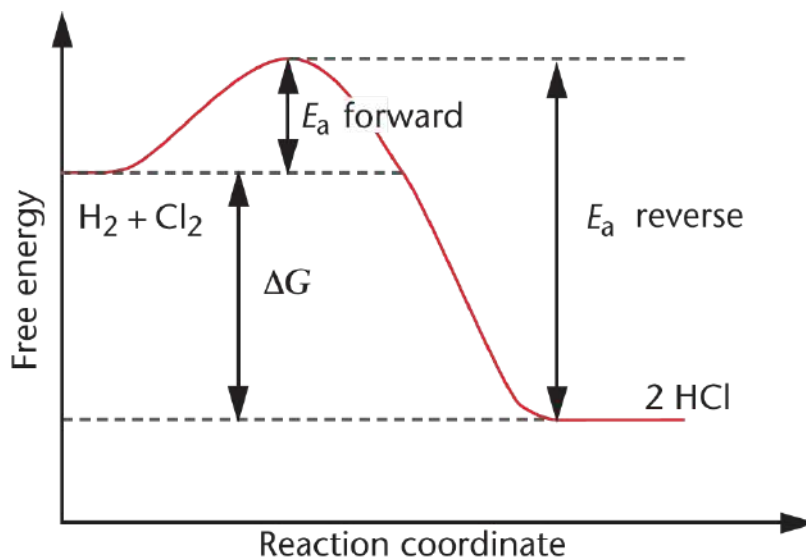


Figure 5.4. Reaction Diagram for the Formation of HCl

On the left are the reactants (H_2 and Cl_2) and on the right are the products (2HCl); this reaction is exergonic, and forward and reverse activation energies are shown.



MCAT EXPERTISE

Kinetics and thermodynamics should be considered separately. Note that the free energy of the product can be raised or lowered, thereby changing the value of ΔG without affecting the value of forward E_a .

FACTORS AFFECTING REACTION RATE

Before we delve into the specifics of rate calculations, it is helpful to understand the conditions that can alter experimental rates.

Reaction Concentrations

The greater the concentrations of the reactants, the greater the number of effective collisions per unit time. Recall that this leads to an increase in the frequency factor (A) of the Arrhenius equation. Therefore, the reaction rate will increase for all but zero-order reactions, which will be discussed shortly. For reactions occurring in the gaseous state, the partial pressures of the gas reactants serve as a measure of concentration, as discussed in Chapter 8 of MCAT General Chemistry Review.

Temperature

For nearly all reactions, the reaction rate will increase as the temperature increases. Because the temperature of a substance is a measure of the particles' average kinetic energy, increasing the temperature increases the average kinetic energy of the molecules. Consequently, the proportion of reactants gaining enough energy to surpass E_a (and thus capable of undergoing reaction) increases with higher temperature. All reactions—even the nuclear reactions shown in Figure 5.5—are temperature-dependent and experience an optimal temperature for activity.

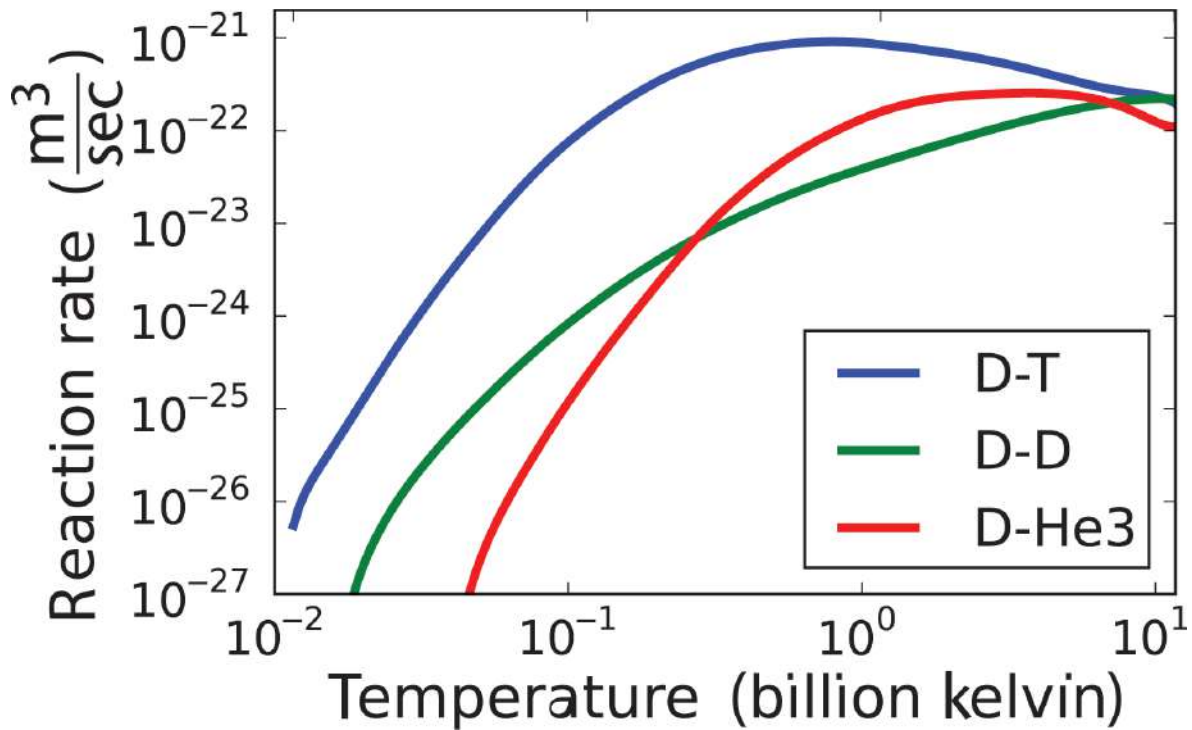


Figure 5.5. All Reactions Are Temperature-Dependent

Examples shown are nuclear fusion reactions; at extreme temperatures, the nucleus itself begins to break down.

You'll often hear that raising the temperature of a system by 10°C will result in an approximate doubling of the reaction rate. Be careful with this approximation because it is generally true for biological systems but not so for many other systems. Further, even in biological systems, if the temperature gets too high, a catalyst may denature—and then the reaction rate plummets. Figure 5.6 shows a general curve for an enzymatic reaction that is optimal between 35°C and 40°C (body temperature). Notice that the curve falls sharply after 40°C , at which point denaturation has occurred.

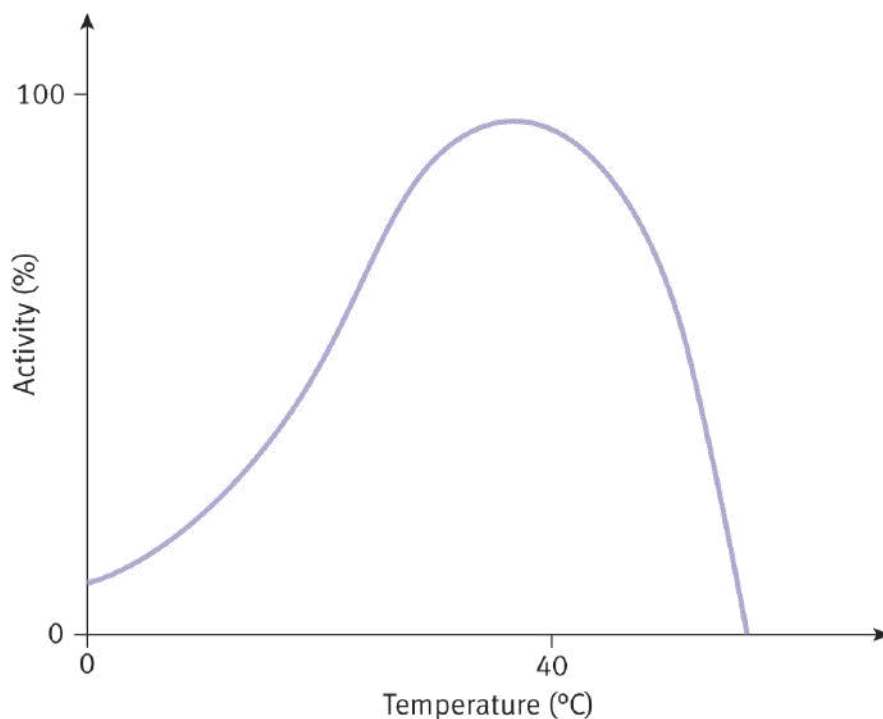


Figure 5.6. An Activity vs. Temperature Curve for a Generic Human Enzyme

Medium

The rate at which a reaction takes place may also be affected by the medium in which it takes place. Some molecules are more likely to react with each other in aqueous environments, while others are more likely to react in nonaqueous solvents, such as in dimethyl sulfoxide (DMSO) or ethanol. Furthermore, the physical state of the medium (liquid, solid, or gas) can also have a significant effect. Generally, polar solvents are preferred because their molecular dipole tends to polarize the bonds of the reactants, thereby lengthening and weakening them, permitting the reaction to occur faster.

Catalysts

Catalysts are substances that increase reaction rate without themselves being consumed in the reaction. Catalysts interact with the reactants, either by adsorption or through the formation of intermediates, and stabilize them so as to reduce the activation energy necessary for the reaction to proceed. While many catalysts, including all enzymes, chemically interact with the reactants, they return to their original chemical state upon formation of the products. They may increase the frequency of collisions between the reactants; change the relative orientation of the reactants, making a higher percentage of the collisions effective; donate electron density to the reactants; or reduce intramolecular bonding within reactant molecules. In homogeneous catalysis, the catalyst is in the same phase (solid, liquid, gas) as the reactants. In heterogeneous catalysis, the catalyst is in a distinct phase. Figure 5.7 compares the energy profiles of catalyzed and uncatalyzed reactions. Note that, depending on context, reaction profiles can use either Gibbs free energy or enthalpy for the y-axis.

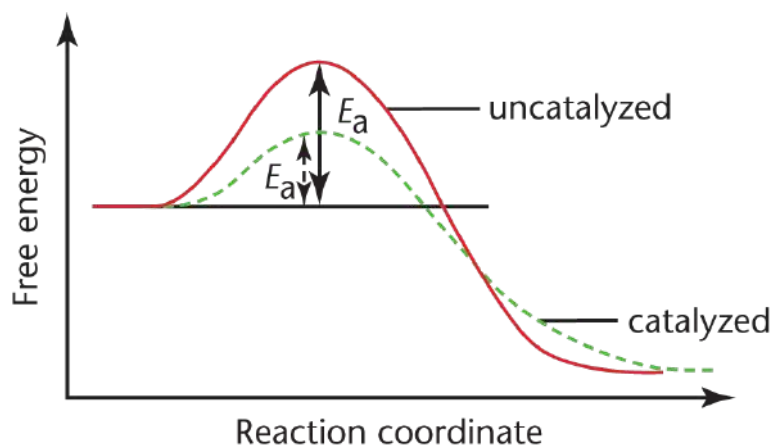


Figure 5.7. Reaction Diagram for a Catalyzed and an Uncatalyzed Reaction

Notice that the only effect of the catalyst is the decrease in the energies of activation, E_a , for both the forward and reverse reactions. The presence of

the catalyst has no impact on the free energies of the reactants or the products or the difference between them. This means that catalysts change only the rates of reactions, and in fact, change the forward rate and the reverse rate by the same factor. Consequently, they have no impact whatsoever on the equilibrium position or the measurement of K_{eq} . Remember that, as useful as catalysts are in biological and nonbiological systems, catalysts are not miracle workers: they will not transform a nonspontaneous reaction into a spontaneous one; they only make spontaneous reactions move more quickly toward equilibrium.



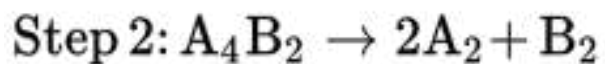
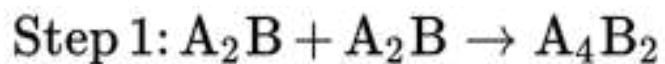
BRIDGE

Equilibrium, like biological homeostasis, is a dynamic process that seeks to find balance in all systems. We can use this concept to our advantage on the MCAT in all seven of the basic sciences. Equilibria are dynamic, meaning that they do undergo change but their net change will be zero.

MCAT Concept Check 5.1:

Before you move on, assess your understanding of the material with these questions.

Describe in words what is occurring in the following two-step mechanism:



What does it mean for a step in a mechanism to be the rate-determining step?

What is activation energy?

How does the transition state theory compare with the collision theory of chemical kinetics?

- Transition state theory:
-

- Collision theory:
-

5.2 Reaction Rates

LEARNING GOALS

After Chapter 5.2, you will be able to:

- Predict the impact of changing temperature, concentration, and catalyst presence on rate of reaction for a zero, first, or second-order reaction
- Determine the rate law and rate order for a reaction, given experimental reaction rate data:

Trial	[A] _{initial} (M)	[B] _{initial} (M)	$\left(K = \frac{1}{2}mv^2\right)$
1	1.00	1.00	2.0
2	1.00	2.00	8.1
3	2.00	2.00	15.9

Reactions, unfortunately, do not come with handy built-in speedometers. To determine the rate at which a reaction proceeds, we must take

measurements of the concentrations of reactants and products and note their change over time.

DEFINITION OF RATE

If we consider a generic reaction, $2A + B \rightarrow C$, in which one mole of C can be produced from two moles of A and one mole of B, we can describe the rate of this reaction in terms of either the disappearance of reactants over time or the appearance of products over time. Because the reactants, by definition, are being consumed in the process of formation of the products, we place a negative sign in front of the rate expression for the reactants.

For the above reaction, the rate of the reaction with respect to A is

$-\frac{\Delta[A]}{\Delta t}$, with respect to B is $-\frac{\Delta[B]}{\Delta t}$, and with respect to C is

$+\frac{\Delta[C]}{\Delta t}$. Notice that the stoichiometric coefficients for the reaction are

not equal, which means that the rates of change of concentrations are not equal. Because two moles of A are consumed for every mole of B consumed, the rate of consumption of A is twice the rate of consumption of B. Furthermore, for every two moles of A consumed, only one mole of C is produced; thus, we can say that the rate of consumption of A is twice the rate of production of C. Based on the stoichiometry, we can see that the rate of consumption of B is equal to the rate of production of C. To show a standard rate of reaction in which the rates with respect to all reaction species are equal, the rate of concentration change of each species should be divided by the species' stoichiometric coefficients. Thus, for the general reaction $aA + bB \rightarrow cC + dD$:

$$\text{rate} = -\frac{\Delta[A]}{a\Delta t} = -\frac{\Delta[B]}{b\Delta t} = \frac{\Delta[C]}{c\Delta t} = \frac{\Delta[D]}{d\Delta t}$$

Equation 5.3

Rate is expressed in the units of moles per liter per second $\left(\frac{\text{mol}}{\text{L} \cdot \text{s}}\right)$ or molarity per second $\left(\frac{M}{s}\right)$.

DETERMINATION OF RATE LAW

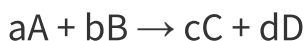
In the Chemical and Physical Foundations of Biological Systems section of the MCAT, it is unlikely that the testmakers will provide a reaction equation that one can merely look at and write the correct rate law. Therefore, on the MCAT, whenever a question asks to determine the rate law for a reaction, the first thing to look for is experimental data.



MCAT EXPERTISE

Remember that the stoichiometric coefficients for the overall reaction are often different from those for the rate law and will, therefore, not be the same as the order of the reaction.

For nearly all forward, irreversible reactions, the rate is proportional to the concentrations of the reactants, with each concentration raised to some experimentally determined exponent. For the general reaction



the rate is proportional to $[A]^x[B]^y$. By including a proportionality constant, k , we can say that rate is determined according to the following equation:

$$\text{rate} = k[A]^x[B]^y$$

Equation 5.4

where k is the reaction rate coefficient or rate constant and the exponents x and y are the orders of the reaction. This expression is called the rate law. Remember that rate is always measured in units of concentration over time; that is, molarity per second. The exponents x and y (or x , y , and z , if there are three reactants) can be used to state the order of the reaction with respect to each reactant or overall: x is the order with respect to reactant A, and y is order with respect to reactant B. The overall order of the reaction is the sum of x and y . These exponents may be integers or fractions and must be determined experimentally. The MCAT will focus almost exclusively on zero-, first-, second-, and third-order reactions. In most cases, the exponents will be integers.

Before we go any further in our consideration of rate laws, we must offer a few warnings about common traps in chemical kinetics. The first—and most common—is the assumption that the orders of a reaction are the same as the stoichiometric coefficients in the balanced overall equation. Pay close attention: On the MCAT, the values of x and y are almost never the same as the stoichiometric coefficients. The orders of a reaction must be determined experimentally. There are only two cases in which the

stoichiometric coefficients match the orders of the reaction. The first is when the reaction mechanism is a single step and the balanced overall reaction is reflective of the entire chemical process. The second is when the complete reaction mechanism is given and the rate-determining step is indicated. The stoichiometric coefficients on the reactant side of the rate-determining step are equal to the orders of the reaction. Occasionally, even this can get a little complicated when the rate-determining step involves an intermediate as a reactant, in which case one must derive the intermediate molecule's concentration by the law of mass action (that is, the equilibrium constant expression) for the step that produced the intermediate.



MCAT EXPERTISE

Note that the exponents in the rate law are not equal to the stoichiometric coefficients, unless the reaction occurs via a single-step mechanism. Note that product concentrations never appear in a rate law. Don't fall into the exceedingly common trap of confusing the rate law with an equilibrium expression!

The second common trap is mistaking the equilibrium constant expression (law of mass action) for the rate law. The expressions for both look similar; if you're not alert on Test Day, you may mistake one for the other or use one when you should be using the other. The expression for equilibrium includes the concentrations of all the species in the reaction, both reactants and products. The expression for chemical kinetics—the rate law expression—includes only the reactants. K_{eq} indicates where the reaction's

equilibrium position lies. The rate indicates how quickly the reaction will get there.

The third trap regards the rate constant, k . Technically speaking, k is not a constant because its particular value for any specific chemical reaction will depend on the activation energy for that reaction and the temperature at which the reaction takes place. However, for a specific reaction, at a specific temperature, the rate constant is indeed a constant. For a reversible reaction, the K_{eq} is equal to the ratio of the rate constant for the forward reaction, k , divided by the rate constant for the reverse reaction, k_{-1} .

The fourth and final trap is that the notion and principles of equilibrium apply to the system only at the end of the reaction; that is, after the system has already reached equilibrium. On the other hand, while the reaction rate can theoretically be measured at any time, it is usually measured at or near the beginning of the reaction to minimize the effects of the reverse reaction.

Experimental Determination of Rate Law

This has already been stated a few times, but it bears repeating: The values of k , x , and y in the rate law equation ($\text{rate} = k[A]^x[B]^y$) must be determined experimentally for a given reaction at a given temperature. Although rate laws can be quite complex and the orders of the reaction difficult to discern, the MCAT limits its coverage of this topic to fairly straightforward reaction mechanisms, experimental data, and rate laws.

On the MCAT, experimental data for determining rate order is usually provided as a chart that includes the initial concentrations of the reactants and the initial rates of product formation as a function of the reactant concentrations. Often, the data for three or four trials are included in this chart.

To use this data, identify a pair of trials in which the concentration of one of the reactants is changed while the concentrations of all other reactants remain constant. Under these conditions, any change in the rate of product formation from one trial to the other (if there is any change) is fully attributable to the change in concentration of that one reactant. Consider a reaction with two reactants, A and B, forming product C. Imagine two trials in which the concentration of A is constant, while the concentration of B doubles. If the rate of the formation of product C has subsequently quadrupled, then the exponent on [B] must be two. Why? Looking at the generic rate law ($\text{rate} = k[A]^x[B]^y$), the logic should look something like this: Doubling [B] has resulted in a quadrupling of the rate, so to determine the order of the reaction, y , with respect to B, I need to calculate the exponent to which the number 2 must be raised to equal 4. Because $2^y = 4$, $y = 2$.



MCAT EXPERTISE

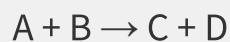
The testmakers love rate problems. Why? Because solving these questions requires a real understanding of proportionality and variable relationships. With practice, you'll be able to do these quickly in your head with minimal paper-and-pencil calculations. Remember to look for pairs of reaction trials in which the

concentration of only one species changes while the others remain constant.

The next step is to repeat this process for the other reactant, using data from a different pair of trials, making sure that the concentration of only the reactant we are trying to analyze is changed from one trial to the other while the concentrations of all other reactants remain the same. Once the orders of the reaction have been determined with respect to each reactant, we can write the complete rate law, replacing the exponents x and y with actual numbers. To determine the value of the rate constant k , plug in actual values from any one of the trials; pick whichever trial has the most arithmetically convenient numbers.

Example:

Given the data below, find the rate law for the following reaction at 300 K:



Trial	$[A]_{\text{initial}}$ (M)	$[B]_{\text{initial}}$ (M)	$\left(K = \frac{1}{2}mv^2\right)$
1	1.00	1.00	2.0
2	1.00	2.00	8.1
3	2.00	2.00	15.9

Solution:

First, look for two trials in which the concentrations of all but one of the substances are held constant.

In Trials 1 and 2, the concentration of A is kept constant, while the concentration of B is doubled. The rate increases by a factor of approximately 4. Since k and $[A]$ are constant between the two trials, the rate is proportional to $[B]$ raised to some power (the symbol $\propto$ means “is proportional to”):

$$\text{rate} = k[A]^x[B]^y \xrightarrow{k \text{ and } [A] \text{ are constant}} \text{rate} \propto [B]^y$$

Specifically, the relationship between the change in rate and the change in concentration of B can be written as:

$\Delta \text{rate} = \Delta [B]^y$. For this specific set of data, the proportionality becomes: $4 = [2]^y$.

In other words, since the rate was multiplied by 4, and $[B]$ was multiplied by 2, y must be equal to 2. Based on what is known so far, the rate law becomes $\text{rate} = k[A]^x[B]^2$.

In Trials 2 and 3, the concentration of B is kept constant, while the concentration of A is doubled. The rate increases by a factor of approximately 2. Since k and $[B]$ are constant between the two trials, rate is proportional to $[A]$ raised to some power:

$$\text{rate} = k[A]^x[B]^2 \xrightarrow{k \text{ and } [B] \text{ are constant}} \text{rate} \propto [A]^x$$

The relationship between the change in rate and the change in concentration of B can be written as: $\Delta \text{rate} = \Delta [A]^x$. For this specific set of data, the proportionality becomes: $2 = [2]^x$.

In other words, since the rate was multiplied by 2, and [A] was multiplied by 2, x must be equal to 1.

Therefore, $\text{rate} = k[A]^1[B]^2$, more typically written as $\text{rate} = k[A][B]^2$ as raising a value to the first power is equivalent to the value (e.g., $2^1 = 2$). The order of the reaction is 1 with respect to A and 2 with respect to B; the overall reaction order is thus $1 + 2 = 3$.

To calculate k, substitute the values from any one of the trials into the rate law. In this case, trial 1 is chosen because the numbers are straightforward to manipulate:

$$\begin{aligned} \text{rate} &= k[A][B]^2 \\ 2.0 \frac{M}{s} &= k[1.00 M][1.00 M]^2 \\ k &= 2.0 M^{-2} s^{-1} \end{aligned}$$

Therefore, the final rate law is $\text{rate} = 2.0 M^{-2} s^{-1} [A][B]^2$.

REACTION ORDERS

We classify chemical reactions as zero-order, first-order, second-order, higher-order, or mixed-order on the basis of kinetics. We will continue to consider the generic reaction $aA + bB \rightarrow cC + dD$ for this discussion.

Zero-Order Reaction

A zero-order reaction is one in which the rate of formation of product C is independent of changes in concentrations of any of the reactants, A and B. These reactions have a constant reaction rate equal to the rate constant (rate coefficient), k . The rate law for a zero-order reaction is:

$$\text{rate} = k[A]^0[B]^0 = k$$

where k has units of $\frac{M}{s}$. Remember that the rate constant itself is dependent on temperature; thus, it is possible to change the rate for a zero-order reaction by changing the temperature. The only other way to change the rate of a zero-order reaction is by the addition of a catalyst, which lowers the activation energy, thereby increasing the value of k .



MCAT EXPERTISE

Temperature and the addition of a catalyst are the only factors that can change the rate of a zero-order reaction.

Plotting a zero-order reaction on a concentration vs. time curve results in a linear graph, as shown in Figure 5.8. This line shows that the rate of formation of product is independent of the concentration of reactant. The slope of such a line is the opposite of the rate constant, k .

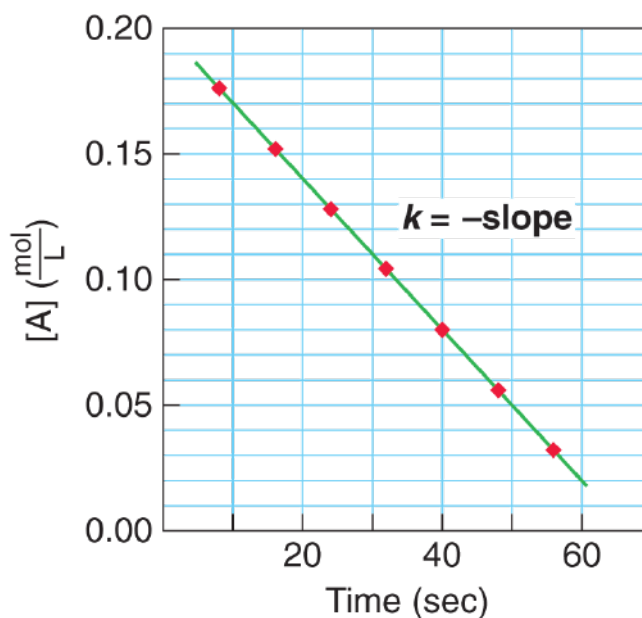


Figure 5.8. Kinetics of a Zero-Order Reaction

Note that the rate of reaction, k , is the opposite of the slope.

First-Order Reaction

A first-order reaction has a rate that is directly proportional to only one reactant, such that doubling the concentration of that reactant results in a doubling of the rate of formation of the product. The rate law for a first-order reaction is

$$\text{rate} = k[A]^1 \text{ or } \text{rate} = k[B]^1$$

where k has units of s^{-1} . A classic example of a first-order reaction is the process of radioactive decay. From the rate law, in which the rate of decrease of the amount of a radioactive isotope A is proportional to the amount of A ,

$$\text{rate} = - \frac{\Delta[A]}{\Delta t} = k[A]$$

The concentration of radioactive substance A at any time t can be expressed mathematically as:

$$[A]_t = [A]_0 e^{-kt}$$

Equation 5.5

where $[A]_t$ is the concentration of A at time t , $[A]_0$ is the initial concentration of A , k is the rate constant, and t is time. It is important to recognize that a first-order rate law with a single reactant suggests that the reaction begins when the molecule undergoes a chemical change all by itself, without a chemical interaction, and usually without a physical interaction with any other molecule.

Plotting a first-order reaction on a concentration vs. time curve results in a nonlinear graph, as shown in Figure 5.9. This curve shows that the rate of formation of product is dependent on the concentration of reactant. Plotting $\ln [A]$ vs. time reveals a straight line; the slope of such a line is the opposite of the rate constant, k .

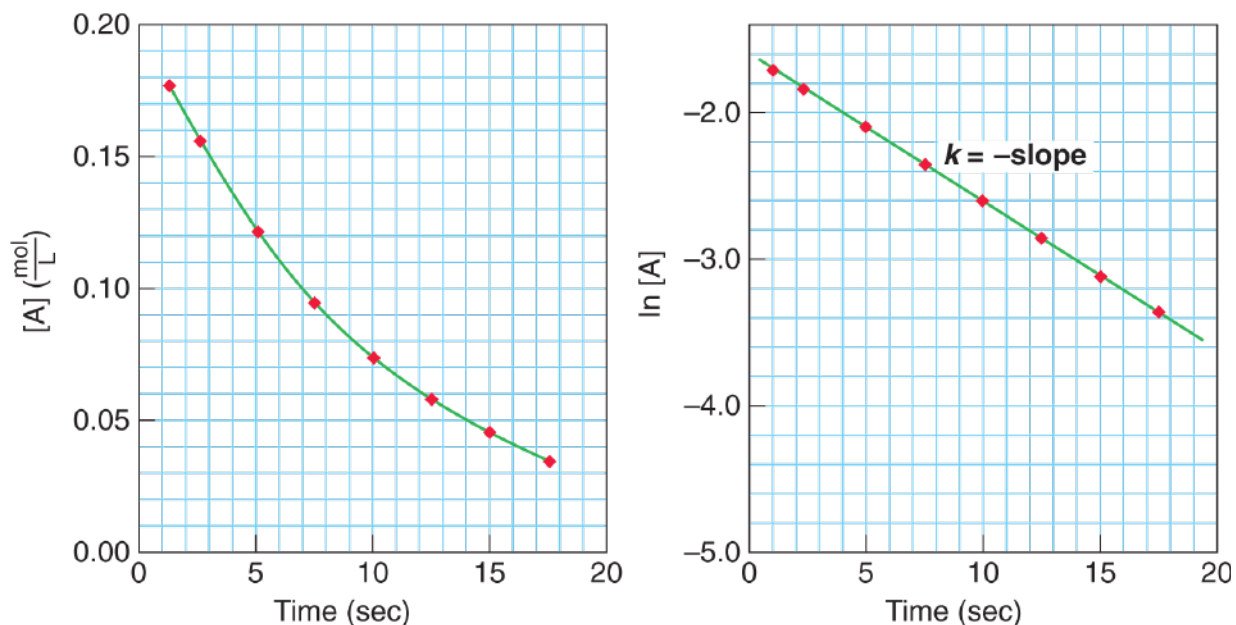


Figure 5.9. Kinetics of a First-Order Reaction

On the left, note that the rate of reaction is dependent on reactant concentration; on the right, note that the rate constant is the opposite of the slope of a graph of $\ln [A]$ vs. time.

Second-Order Reaction

A second-order reaction has a rate that is proportional to either the concentrations of two reactants or to the square of the concentration of a single reactant. The following rate laws all reflect second-order reactions:

$$\text{rate} = k[A]^1[B]^1 \text{ or } \text{rate} = k[A]^2 \text{ or } \text{rate} = k[B]^2$$

where k has units of $\text{M}^{-1} \text{s}^{-1}$. It is important to recognize that a second-order rate law often suggests a physical collision between two reactant molecules, especially if the rate law is first-order with respect to each of the two reactants.

Plotting a reaction that is second-order with respect to a single reactant on a concentration vs. time curve results in a nonlinear graph, as shown in Figure 5.10. This curve shows that the rate of formation of product is dependent on the concentration of reactant. Plotting $\frac{1}{[A]}$ vs. time reveals a linear curve; the slope of such a curve is equal to the rate constant, k .

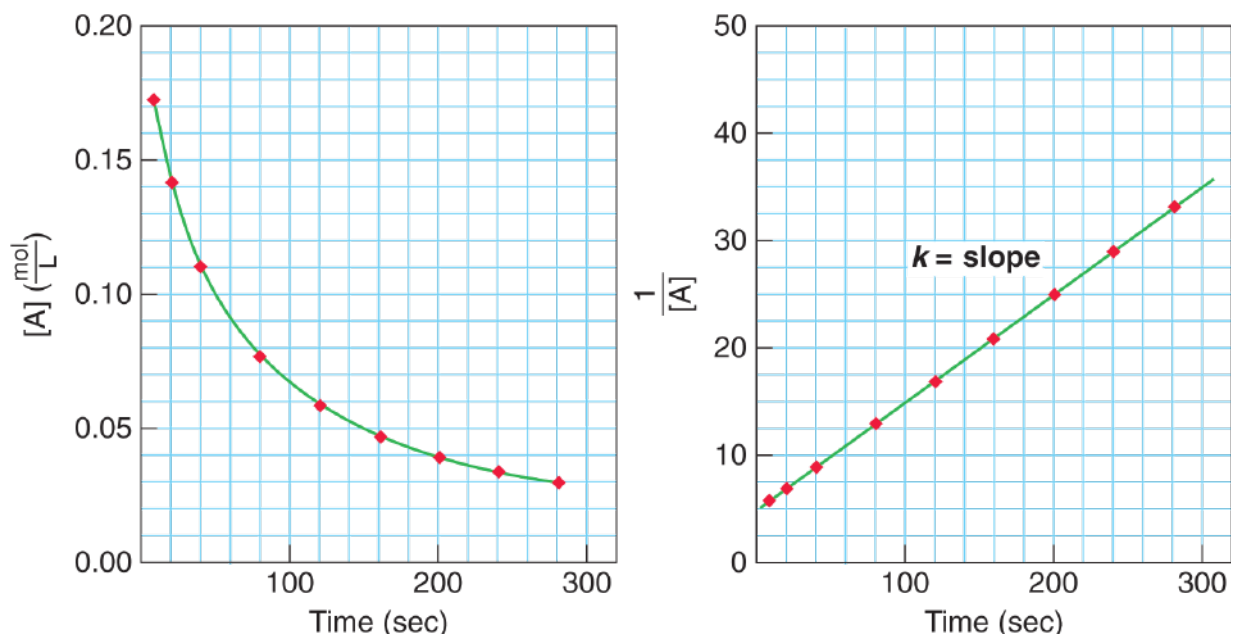


Figure 5.10. Kinetics of a Second-Order Reaction

On the left, note that the rate of reaction is dependent on reactant concentration; on the right, note that the rate constant is equal to the slope of a graph of $\frac{1}{[A]}$ vs. time.

Higher-Order Reactions

Fortunately, there are very few noteworthy reactions in which a single reaction step involves a termolecular process; in other words, there are few processes with third-order rates. This is because it is far more rare for three

particles to collide simultaneously with the correct orientation and sufficient energy to undergo a reaction.

Mixed-Order Reactions

Mixed-order reactions sometimes refer to non-integer orders (fractions) and in other cases to reactions with rate orders that vary over the course of the reaction. Fractions are more specifically described as broken-order. In recent times, the term mixed-order has come to refer solely to reactions that change order over time. Knowing those two definitions will be sufficient for Test Day.

An example of a mixed-order rate law is given by:

$$\text{rate} = \frac{k_1 [\text{C}][\text{A}]^2}{k_2 + k_3 [\text{A}]}$$

where A represents a single reactant and C, a catalyst. The overall reaction and its mechanism are beyond the relevance and scope of the MCAT, and the derivation of this rate law is even more unnecessary for Test Day; however, understanding what is implied by this equation is important. The result of the large value for [A] at the beginning of the reaction is that $k_3[\text{A}] \gg k_2$, and the reaction will appear to be first-order with respect to A. At the end of the reaction, when [A] is low, $k_2 \gg k_3[\text{A}]$, making the reaction appear second-order with respect to A. While the MCAT will not ask you to derive a rate expression for a mixed-order reaction, you are responsible for being able to recognize how the rate order changes as the reactant concentration changes.

MCAT Concept Check 5.2:

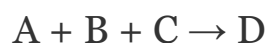
Before you move on, assess your understanding of the material with these questions.

Describe the effects the following conditions would have on the initial rate of reaction, given the reaction order:
(Examples: rate increased, rate divided by 2, rate unaffected)

Conditions	Zero-Order	First-Order	Second-Order
Temperature lowered			
All reactants' concentrations doubled			
Catalyst added			



Determine the rate law and rate order for the following reaction:



Trial	$[\text{A}]_{\text{initial}} \text{ (M)}$	$[\text{B}]_{\text{initial}} \text{ (M)}$	$[\text{C}]_{\text{initial}} \text{ (M)}$	$\left(K = \frac{1}{2}mv^2\right)$
1	1.00	1.00	1.00	2.0

Trial	[A] _{initial} (M)	[B] _{initial} (M)	[C] _{initial} (M)	$\left(K = \frac{1}{2}mv^2\right)$
2	1.00	2.00	1.00	2.1
3	2.00	1.00	1.00	15.9
4	2.00	1.00	2.00	32.2





Conclusion

We began with a consideration of chemical reactions and the mechanisms that illustrate the individual steps necessary to transform reactants into products. We demonstrated the way to derive a reaction's rate law through the analysis of experimental data, and we looked at the factors that can affect the rates of chemical reactions.

After such an overview, you should begin to appreciate that many chemical principles in the human body rely on the principles of chemical kinetics. Why does the body maintain a certain temperature? Primarily to stabilize the enzymes that catalyze the metabolic reactions necessary for life. Why does the body maintain a pH buffer? Altering the concentration of protons affects not only the ability of an enzyme to maintain its secondary, tertiary, and quaternary structure, but can also directly affect the collisions between reactants. You will begin to appreciate these and many other questions from a clinical perspective throughout your medical career. In the next chapter, we will investigate chemical equilibria, which—although related to kinetics—are distinct (and commonly confused!) topics.

Concept Summary

Chemical Kinetics

- The change in Gibbs free energy (ΔG) determines whether or not a reaction is spontaneous.
- Chemical mechanisms propose a series of steps that make up the overall reaction.
 - Intermediates are molecules that exist within the course of a reaction but are neither reactants nor products overall.
 - The slowest step, also known as the rate-determining step, limits the maximum rate at which the reaction can proceed.
- The collision theory states that a reaction rate is proportional to the number of effective collisions between the reacting molecules.
 - For a collision to be effective, molecules must be in the proper orientation and have sufficient kinetic energy to exceed the activation energy.
 - The Arrhenius equation is a mathematical way of representing collision theory.
- The transition state theory states that molecules form a transition state or activated complex during a reaction in which the old bonds are

partially dissociated and the new bonds are partially formed.

- From the transition state, the reaction can proceed toward products or revert back to reactants.
- The transition state is the highest point on a free energy reaction diagram.
- Reaction rates can be affected by a number of factors.
 - Increasing the concentration of reactant will increase reaction rate (except for zero-order reactions) because there are more effective collisions per time.
 - Increasing the temperature will increase reaction rate because the particles' kinetic energy is increased.
 - Changing the medium can increase or decrease reaction rate, depending on how the reactants interact with the medium.
 - Adding a catalyst increases reaction rate because it lowers the activation energy. Homogeneous catalysts are the same phase as the reactants; heterogeneous catalysts are a different phase.

Reaction Rates

- Reaction rates are measured in terms of the rate of disappearance of a reactant or appearance of a product.
- Rate laws take the form of $\text{rate} = k[\text{A}]^x[\text{B}]^y$.
 - The rate orders usually do not match the stoichiometric coefficients.
 - Rate laws must be determined from experimental data.

- The rate order of a reaction is the sum of all individual rate orders in the rate law.
- Zero-order reactions have a constant rate that does not depend on the concentration of reactant.
 - The rate of a zero-order reaction can only be affected by changing the temperature or adding a catalyst.
 - A concentration vs. time curve of a zero-order reaction is a straight line; the slope of such a line is equal to $-k$.
- First-order reactions have a nonconstant rate that depends on the concentration of reactant.
 - A concentration vs. time curve of a first-order reaction is nonlinear.
 - The slope of a $\ln [A]$ vs. time plot is $-k$ for a first-order reaction.
- Second-order reactions have a nonconstant rate that depends on the concentration of reactant.
 - A concentration vs. time curve of a second-order reaction is nonlinear.
 - The slope of a $\frac{1}{[A]}$ vs. time plot is k for a second-order reaction.
- Broken-order reactions are those with noninteger orders.
- Mixed-order reactions are those that have a rate order that changes over time.

Answers to Concept Checks

5.1

Two molecules of A_2B come together in a combination reaction to form an intermediate, A_4B_2 , which subsequently decomposes to produce the final products, two molecules of A_2 and one molecule of B_2 .

The rate-determining step is the slowest step of a reaction. It determines the overall rate of the reaction because the reaction can only proceed as fast as the rate at which this step occurs.

The activation energy is the minimum energy needed for a chemical reaction to occur.

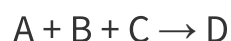
Both theories require a certain activation energy to be overcome in order for a reaction to occur (therefore not all reactions will occur). The transition state theory focuses on forming a high-energy activated complex that can then proceed forward or backward, forming the products or reverting to the reactants, respectively. The collision theory focuses on the energy and orientation of reactants, and considers each potential reaction to be “all-or-nothing” (either there is enough energy to form the products, or there is not).

5.2

Conditions	Zero-Order	First-Order	Second-Order
Temperature lowered	rate decreased	rate decreased	rate decreased

Conditions	Zero-Order	First-Order	Second-Order
All reactants' concentrations doubled	rate unaffected	rate doubled	rate multiplied by 4
Catalyst added	rate increased	rate increased	rate increased

This question asks for the rate law and rate order for the following reaction:



Trial	$[A]_{\text{initial}} \text{ (M)}$	$[B]_{\text{initial}} \text{ (M)}$	$[C]_{\text{initial}} \text{ (M)}$	$\left(K = \frac{1}{2}mv^2\right)$
1	1.00	1.00	1.00	2.0
2	1.00	2.00	1.00	2.1
3	2.00	1.00	1.00	15.9
4	2.00	1.00	2.00	32.2

Start by writing the generic rate law for the reaction: $\text{rate} = k[A]^x[B]^y[C]^z$.

In a complex rate law problem, always check for the possibility of a reagent that has no impact on the rate law. Looking at Trials 1 and 2, the concentration of B is doubled with no change in the rate. Thus, reagent B has no impact on the rate law, and its exponent is zero. The rate law can be updated to $\text{rate} = k[A]^x[B]^0[C]^z$.

Next, compare Trials 1 and 3. The concentration of A doubles, the concentrations of B and C remain constant, and the rate increases by a factor of approximately 8. This results in the proportionality

$\Delta \text{rate} = \Delta [\text{A}]^x$, so $8 = 2^x$, giving $x = 3$. The rate law can now be updated to $\text{rate} = k[\text{A}]^3[\text{B}]^0[\text{C}]^z$.

Finally, compare Trials 3 and 4. The concentration of C doubles, the concentrations of A and B remain constant, and the rate approximately doubles. This results in the proportionality

$\Delta \text{rate} = \Delta [\text{C}]^z$, so $2 = 2^z$, giving $z = 1$. The rate law can now be updated to $\text{rate} = k[\text{A}]^3[\text{B}]^0[\text{C}]^1$.

Thus, the final rate law is: $\text{rate} = k[\text{A}]^3[\text{B}]^0[\text{C}]^1 = k[\text{A}]^3[\text{C}]$. The rate order is $3 + 0 + 1 = 4$.

Equations to Remember

(5.1) Collision theory: $\text{rate} = Z \times f$

(5.2) Arrhenius equation: $k = Ae^{\frac{-E_a}{RT}}$

(5.3) **Definition of rate:** $\text{rate} = -\frac{\Delta[A]}{a\Delta t} = -\frac{\Delta[B]}{b\Delta t} = \frac{\Delta[C]}{c\Delta t} = \frac{\Delta[D]}{d\Delta t}$
for the general reaction $aA + bB \rightarrow cC + dD$

(5.4) Rate law: $\text{rate} = k[A]^x[B]^y$

(5.5) Radioactive decay: $[A]_t = [A]_0 e^{-kt}$



Shared Concepts

Biochemistry Chapter 2

Enzymes

General Chemistry Chapter 3

Bonding and Chemical Interactions

General Chemistry Chapter 6

Equilibrium

General Chemistry Chapter 7

Thermochemistry

Organic Chemistry Chapter 4

Analyzing Organic Reactions

Physics and Math Chapter 3

Thermodynamics

Discrete Practice Questions

Consult your online resources for additional practice.

1. In a third-order reaction involving two reactants and two products, doubling the concentration of the first reactant causes the rate to increase by a factor of 2. What will happen to the rate of this reaction if the concentration of the second reactant is cut in half?

- (A) It will increase by a factor of 2.
- (B) It will increase by a factor of 4.
- (C) It will decrease by a factor of 2.
- (D) It will decrease by a factor of 4.

2. In a certain equilibrium process, the activation energy of the forward reaction ($\Delta G^\ddagger_f$) is greater than the activation energy of the reverse reaction ($\Delta G^\ddagger_r$). This reaction is:

- (A) endothermic.
- (B) exothermic.
- (C) spontaneous.
- (D) nonspontaneous.

3. A reactant in a second-order reaction at a certain temperature is increased by a factor of 4. By how much is the rate of the reaction altered?

- (A) It is unchanged.
- (B) It is increased by a factor of 4.
- (C) It is increased by a factor of 16.
- (D) It cannot be determined from the information given.

4. The concentrations of all reactants in a zero-order reaction are increased two-fold. What is the new rate of the reaction?

- (A) It is unchanged.
- (B) It is decreased by a factor of 2.
- (C) It is increased by a factor of 2.
- (D) It cannot be determined from the information given.

5. Which of the following experimental methods should NEVER affect the rate of a reaction?

- (A) Placing an exothermic reaction in an ice bath.
- (B) Increasing the pressure of a reactant in a closed container.
- (C) Putting the reactants into an aqueous solution.
- (D) Removing the product of an irreversible reaction.

6. What would increasing the concentration of reactants accomplish in a solution containing a saturated catalyst?

- (A) It would increase the rate constant but not the reaction rate.
- (B) It would decrease the rate constant but increase the reaction rate.
- (C) It would increase the rate constant and increase the reaction rate.
- (D) The reaction rate would be unaffected.

7. A certain chemical reaction has the following rate law:

$$\text{rate} = k[\text{NO}_2][\text{Br}_2]$$

Which of the following statements necessarily describe(s) the kinetics of this reaction?

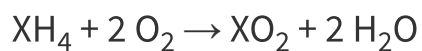
The reaction is second-order.

The amount of NO_2 consumed is equal to the amount of Br_2 consumed.

I. The rate will not be affected by the addition of a compound other than NO_2 and Br_2 .

- (A) I only
- (B) I and II only
- (C) II and III only
- (D) I, II, and III

8. The following data shown in the table were collected for the combustion of the theoretical compound XH_4 :



Trial	$[\text{XH}_4]_{\text{initial}} \text{ (M)}$	$[\text{O}_2]_{\text{initial}} \text{ (M)}$	$\text{Rate}_{\text{initial}} \left(\frac{\text{M}}{\text{min}} \right)$
1	0.6	0.6	12.4
2	0.6	2.4	49.9
3	1.2	2.4	198.3

What is the rate law for the reaction described here?

- (A) $\text{rate} = k[\text{XH}_4][\text{O}_2]$
- (B) $\text{rate} = k[\text{XH}_4][\text{O}_2]^2$
- (C) $\text{rate} = k[\text{XH}_4]^2[\text{O}_2]$
- (D) $\text{rate} = k[\text{XH}_4]^2[\text{O}_2]^2$

9. Which of the following best describes the purpose of a catalyst?

- (A) Catalysts are used up in the reaction, increasing reaction efficiency.
- (B) Catalysts increase the rate of the reaction by lowering the activation energy.
- (C) Catalysts alter the thermodynamics of the reaction to facilitate the formation of products or reactants.
- (D) Catalysts stabilize the transition state by bringing it to a higher energy.

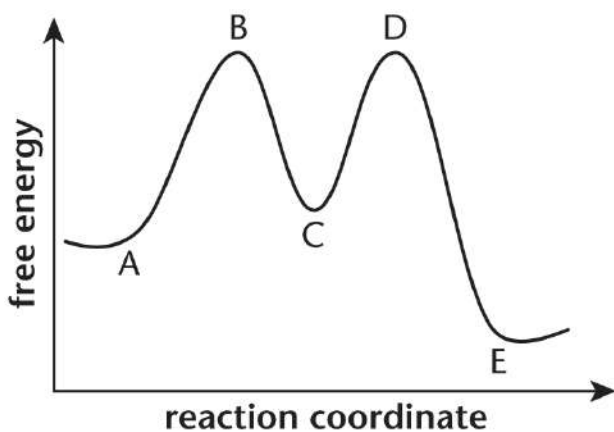
10. If the rate law for a reaction is:

$$\text{rate} = k[\text{A}]^0[\text{B}]^2[\text{C}]^1$$

What is the overall order of the reaction?

- (A) 0
- (B) 2
- (C) 3
- (D) 4

For questions 11–13, consider the following energy diagram shown below:



11. The overall reaction depicted by this energy diagram is:

- (A) endergonic, because point B is higher than point A.
- (B) endergonic, because point C is higher than point A.
- (C) exergonic, because point D is higher than point E.
- (D) exergonic, because point A is higher than point E.

12. Which process has the highest activation energy?

- (A) The first step of the forward reaction
- (B) The first step of the reverse reaction
- (C) The second step of the forward reaction
- (D) The second step of the reverse reaction

13. Point C in this reaction profile refers to the:

- (A) reactants.
- (B) products.
- (C) transition state.
- (D) intermediates.

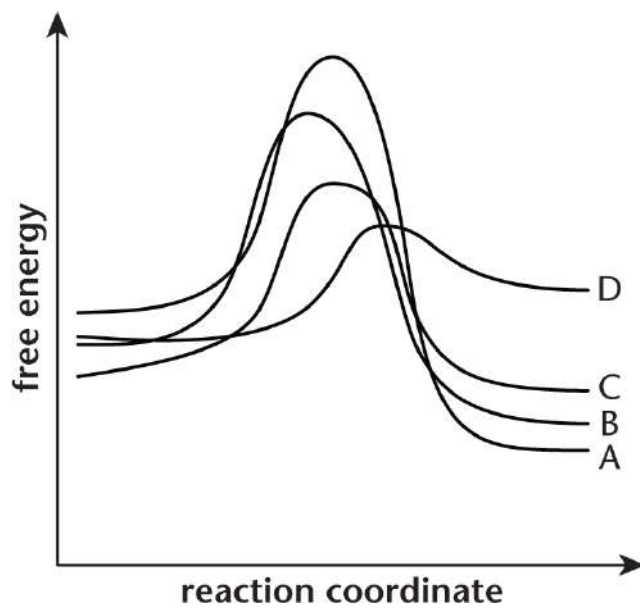
14. The following system obeys second-order kinetics.



What is the rate law for this reaction?

- (A) $\text{rate} = k[\text{NO}_2][\text{CO}]$
- (B) $\text{rate} = k[\text{NO}_2]^2[\text{CO}]$
- (C) $\text{rate} = k[\text{NO}_2][\text{NO}_3]$
- (D) $\text{rate} = k[\text{NO}_2]^2$

15. The potential energy diagram shown represents four different reactions.



Assuming identical conditions, which of the reactions displayed on the energy diagram proceeds the fastest?

- (A) A
- (B) B
- (C) C
- (D) D

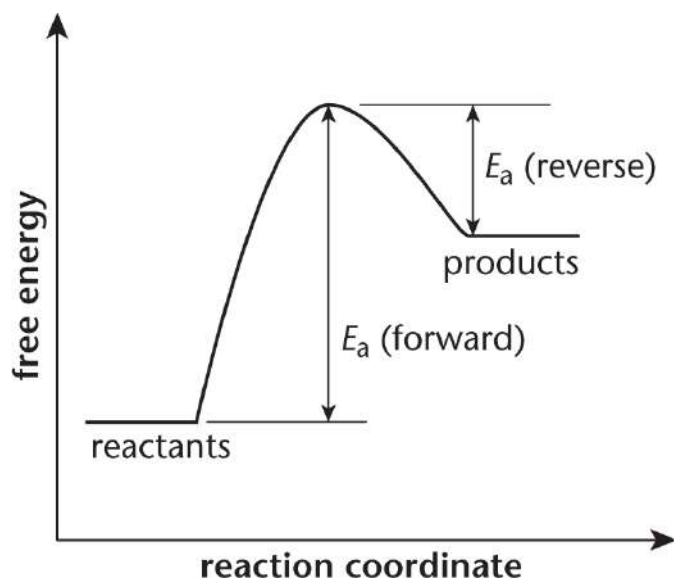
Explanations to Discrete Practice Questions

1. D

Based on the information given in the question, the rate is first-order with respect to the concentration of the first reactant; when the concentration of that reactant doubles, the rate also doubles. Because the reaction is third-order, the sum of the exponents in the rate law must be equal to 3. Therefore, the reaction order with respect to the other reactant must be $3 - 1 = 2$. If the concentration of this second reactant is multiplied by $\frac{1}{2}$, the rate will be multiplied by $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

2. D

Before you try to answer this question, you should draw a free energy diagram for the system.



If the activation energy of the forward reaction is greater than the activation energy of the reverse reaction, then the products must have a higher free energy than the reactants. The overall energy of the system is higher at the end than it was in the beginning. The net free energy change is positive, indicating an endergonic (nonspontaneous) reaction. The terms endothermic, (A), and exothermic, (B), are associated with enthalpy. While free energy does depend on enthalpy, it also depends on entropy; there is not enough information in the question stem to reliably determine the sign of the entropy change of the reaction.

3. D

A second-order reaction can be second-order with respect to one reactant, or first-order with respect to two different reactants. In this case, one reactant was increased by a factor of 4. If the reaction is second-order with respect to this reactant, the rate law will be $\text{rate} = k[A]^2[B]^0$ and the rate will increase by a factor of 16. If it is first-order with respect to this reactant and first-order with respect to another reactant, the rate law will be $\text{rate} = k[A]^1[B]^1$, and the rate will increase by a factor

of 4. We do not know which of these is the correct rate law and, thus, cannot determine the effect on the rate.

4. A

By definition, zero-order reactions are unaffected by the concentrations of any reactants in the reaction. Thus, changing the concentrations of these reactants will not affect the rate.

5. D

The question asks which alteration does NOT affect the rate of the reaction. Temperature directly affects the rate constant (k), making (A) incorrect. Changing the partial pressure of a gas will affect the number of effective collisions per time. This makes (B) incorrect—but note that concentration changes will not affect the rate of zero-order reactions. Solvents affect the rate of reactions depending on how the reactants interact with the solvent, making (C) incorrect. Removing the product of an irreversible reaction, (D), should not affect the rate of the reaction because the rate law does not depend on the concentrations of products.

6. D

While increasing the concentration of reactants can alter the reaction rate in first- or higher-order reactions, saturated solutions containing a catalyst have a maximum turnover rate and cannot increase the rate constant or the reaction rate any higher by adding more reactant molecules.

7. A

If the sum of the exponents (orders) of the concentrations of each species in the rate law is equal to 2, then the reaction is second-order. The exponents in the rate law are unrelated to stoichiometric coefficients, so NO_2 and Br_2 could have any stoichiometric coefficients in the original reaction and still be a second-order reaction, invalidating statement II. Statement III is incorrect because the rate can be affected by a wide variety of compounds. A catalyst, for example, could increase the rate.

8. C

Start with the generic rate law: $\text{rate} = k[\text{XH}_4]^x[\text{O}_2]^y$. In the first two trials, the concentration of XH_4 is held constant while the concentration of O_2 is multiplied by 4, and the rate of the reaction also increases by a factor of approximately 4. This gives the proportion

$\Delta\text{rate} = [\text{O}_2]^y$, or $4 = 4^y$, meaning $y = 1$. The rate law can be updated to: $\text{rate} = k[\text{XH}_4]^x[\text{O}_2]^1$.

In the last two trials, the concentration of O_2 is held constant while the concentration of XH_4 is doubled, and the rate of the reaction is increased by a factor of approximately 4. This gives

$\Delta\text{rate} = [\text{XH}_4]^x$, or $4 = 2^x$, and $x = 2$. The rate law can be updated to: $\text{rate} = k[\text{XH}_4]^2[\text{O}_2]^1$. The final version of the rate law is: $\text{rate} = k[\text{XH}_4]^2[\text{O}_2]$.

9. B

By definition, a catalyst increases the rate of a reaction by lowering the activation energy, making it easier for both the forward and reverse reactions to overcome this energy barrier. Catalysts are neither used up

in the reaction, nor do they alter the equilibrium of a reaction, eliminating (A) and (C). Finally, catalysts stabilize the transition state by lowering its energy, not raising it, eliminating (D).

10. C

The overall order of a reaction is the sum of the individual orders in the reaction. Therefore, the rate order is $0 + 2 + 1 = 3$.

11. D

A system is exergonic if energy is released by the reaction. For exergonic reactions, the net energy change is negative, and the free energy of the final products is lower than the free energy of the initial reactants. Point E, which represents the of the final products, is lower on the energy diagram than point A, which represents the energy of initial reactants. Thus, energy must have been given off, and the reaction is exergonic.

12. B

The activation energy of a reaction is represented by the distance on the y-axis from the energy of the reactants to the peak energy prior to formation of products. The activation energy of the first step of the forward reaction, for example, is equal to the distance along the y-axis from point A to point B. The largest energy increase on this graph occurs during the progress from point E to point D, which represents the first step of the reverse reaction.

13. D

Intermediates exist at “valleys” in reaction diagrams. Reactants, (A), are represented by point A. Products, (B), are represented by point E. Transition states, (C), are represented by points B and D.

14. D

To answer this question, recall that the slow step of a reaction is the rate-determining step. The rate is always related to the concentrations of the reactants in the rate-determining step (not the overall reaction), so NO_2 is the only compound that should be included in the correct answer. The concentration of NO_2 is squared in the rate law because the question stem tells us that the system obeys second-order kinetics.

15. D

The faster a reaction can reach its activation energy, the faster it will proceed to completion. Because this question states that all conditions are equal, the reaction with the lowest activation energy will have the fastest rate. In the diagram, (D) has the lowest activation energy.



Equilibrium

In This Chapter

6.1 Equilibrium

Dynamic Equilibria and Reversibility 

Law of Mass Action 

Reaction Quotient 

Properties of the Law of Mass Action 

6.2 Le Châtelier's Principle

Changes in Concentration 

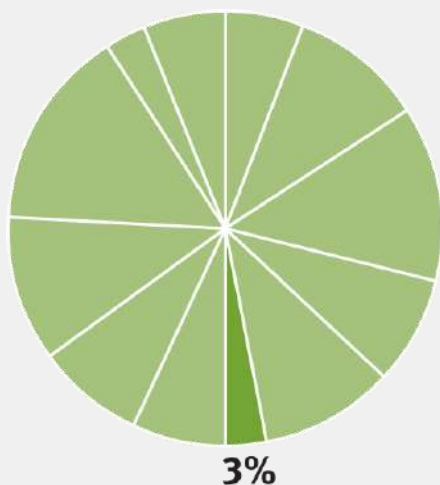
Changes in Pressure (and Volume) 

Changes in Temperature 

6.3 Kinetic and Thermodynamic Control

Concept Summary 

Chapter Profile



The content in this chapter should be relevant to about 3% of all questions about general chemistry on the MCAT.

This chapter covers material from the following AAMC content categories:

1D: Principles of bioenergetics and fuel molecule metabolism

5A: Unique nature of water and its solutions

5E: Principles of chemical thermodynamics and kinetics



MCAT EXPERTISE

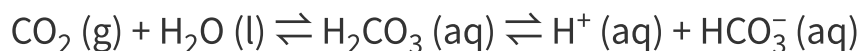
The AAMC has shown that it will only rarely directly test the details of equilibrium in this chapter. However, you'll need an excellent conceptual understanding of equilibrium to master many other high-yield topics, such as acid-base chemistry (Chapter 10 of MCAT General Chemistry Review) and enzymes (Chapter 2 of MCAT Biochemistry Review).

Introduction

You're on the first call of your pediatrics rotation. You get a page from the resident: Come to the emergency room, now, she says. They just brought in a kid with DKA. DKA, as you know, stands for diabetic ketoacidosis and is a fairly common way for undiagnosed type I diabetes mellitus to present. You remember from your second-year classes about endocrine pathophysiology that ketoacidosis can arise as a result of the body's metabolism of fatty acids when insulin production shuts down. Fatty acids are metabolized into ketone bodies as an alternative energy source to glucose. Some of the ketones produced are ketoacids, and as the diabetic crisis continues and worsens, the concentration of these ketoacids increases (termed metabolic acidosis), resulting in a plasma pH below 7.35.

As you enter the child's room, the examination is already under way; he's young, about ten years old, conscious but agitated, and the most obvious sign—which you notice immediately—is his rapid, shallow breathing. You

ask the resident why the boy is hyperventilating, and she takes a piece of paper and writes the following:



It's Le Châtelier's principle! you realize. The respiratory system is trying to compensate for the metabolic acidosis; the increased breathing rate allows him to blow off more CO_2 , which causes the equilibrium to shift to the left. Hydrogen ions combine with bicarbonate ions to produce carbonic acid, which decomposes into CO_2 gas that's expelled from the lungs. The result is a decrease in the plasma hydrogen ion concentration, which stabilizes the pH and keeps it from getting too low. Wow, chemistry really is essential for medical school!

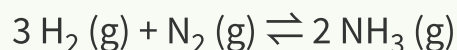
Chemical equilibrium is the dynamic state of a chemical reaction in which the concentrations of reactants and products stabilize over time in a low-energy configuration. Pay particular attention to the concepts of chemical equilibrium because we will return to these topics during our review of solutions, acid–base chemistry, and oxidation–reduction reactions.

6.1 Equilibrium

LEARNING GOALS

After Chapter 6.1, you will be able to:

- Determine the sign of ΔG and the direction of a reaction given its K_{eq} constant
- Calculate K_{eq} for a reaction
- Write the equilibrium constant expression for a reaction:



In the previous chapter, we danced around the term equilibrium. We warned you not to confuse the chemical equilibrium expression for the rate expression. We stressed that catalysts make reactions go faster toward their equilibrium position, but that they can't actually change the equilibrium position or alter the value of K_{eq} . The principles and concepts that are the focus of this chapter will direct our discussion in the upcoming chapters about some of the most important general chemistry topics for the MCAT: solutions, acids and bases, and oxidation–reduction reactions.

DYNAMIC EQUILIBRIA AND REVERSIBILITY

So far, most of the reactions we've covered are irreversible; that is, the reaction proceeds in one direction only, the reaction goes to completion, and the maximum amount of product formed is determined by the amount of limiting reagent initially present. Reversible reactions are those in which the reaction can proceed in one of two ways: forward (toward the products or “to the right”) and reverse (toward the reactants or “to the left”). Reversible reactions usually do not proceed to completion because the products can react together to reform the reactants. When the reaction system is closed and no reactants or products are added or removed, the system will eventually settle into a state in which the rate of the forward reaction equals the rate of the reverse reaction and the concentrations of the products and reactants remain constant. In this dynamic equilibrium, the forward and reverse reactions are still occurring—they haven't stopped, as they do in a static equilibrium—but they are going at the same rate; thus, there is no net change in the concentrations of the products or reactants, as shown in Figure 6.1.

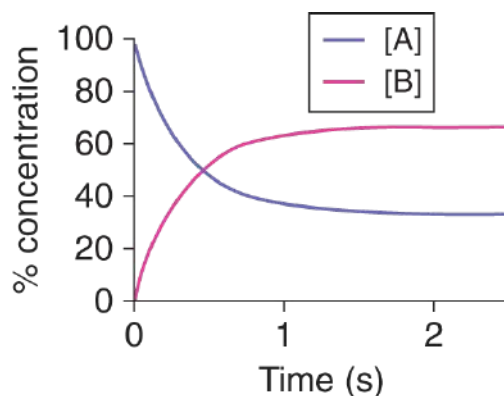
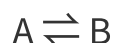


Figure 6.1. Dynamic Equilibrium Occurs when Forward and Reverse Rates Are Equal



Many biochemical reactions can be classified as reversible, and their activation energies are lowered by enzymes. Irreversible biochemical steps are sometimes termed “committed” because they cannot be reversed in their pathways. These steps also tend to be the rate-limiting steps of metabolic pathways, which are discussed in Chapters 9 through 11 of MCAT Biochemistry Review.

Consider the generic reversible reaction illustrated in Figure 6.1:



At equilibrium, the concentrations of A and B are constant (although not necessarily equal), and the reactions $A \rightarrow B$ and $B \rightarrow A$ continue to occur at equal rates.

Equilibrium can be thought of as a balance between the forward and reverse reactions. Better still, equilibrium should be understood on the basis of entropy, which is the measure of the distribution of energy throughout a system or between a system and its environment. For a reversible reaction at a given temperature, the reaction will reach equilibrium when the system’s entropy—or energy distribution—is at a maximum and the Gibbs free energy of the system is at a minimum.



BRIDGE

In Chapter 7 of MCAT General Chemistry Review, we will explore the more “classic” MCAT definition of entropy—a measure of the disorder of a system. It is important to realize, though, that the units of entropy $\left(\frac{1}{1} - \frac{1}{4}\right)$ imply a distribution of energy in a system.

LAW OF MASS ACTION

For a generic reversible reaction $aA + bB \rightleftharpoons cC + dD$, the law of mass action states that, if the system is at equilibrium at a constant temperature, then the following ratio is constant:

$$K_{\text{eq}} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

Equation 6.1

The law of mass action is actually related to the expressions for the rates of the forward and reverse reactions. Consider the following one-step reversible reaction:



Because the reaction occurs in one step, the rates of the forward and reverse reactions are given by:

$$\text{rate}_f = k_f[A]^2 \text{ and } \text{rate}_r = k_r[B][C]$$

When $\text{rate}_f = \text{rate}_r$, the system is in equilibrium. Because the rates are equal, we can set the rate expressions for the forward and reverse reactions equal to each other:

$$k_f[A]^2 = k_r[B][C] \rightarrow \frac{k_f}{k_r} = \frac{[B][C]}{[A]^2}$$



KEY CONCEPT

At equilibrium, the rate of the forward reaction equals the rate of the reverse reaction, entropy is at a maximum, and Gibbs free energy is at a minimum. This links the concepts of thermodynamics and kinetics.

Because k_f and k_r are both constants, we can define a new constant K_c , where K_c is called the equilibrium constant and the subscript c indicates that it is in terms of concentration. When dealing with gases, the equilibrium constant is referred to as K_p , and the subscript p indicates that it is in terms of pressure. For dilute solutions, K_c and K_{eq} are used interchangeably. The new equation can thus be written:

$$K_c = K_{eq} = \frac{[B][C]}{[A]^2}$$



KEY CONCEPT

For most purposes, you will not need to distinguish between different K values. For dilute solutions, $K_{eq} = K_c$ and both are calculated in units of concentration.

While the forward and the reverse reaction rates are equal at equilibrium, the concentrations of the reactants and products are not usually equal. This means that the forward and reverse reaction rate constants, k_f and k_r , respectively, are not usually equal to each other. The ratio of k_f to k_r is K_c :

$$K_c = K_{eq} = \frac{k_f}{k_r}$$

When a reaction occurs in more than one step, the equilibrium constant for the overall reaction is found by multiplying together the equilibrium constants for each step of the reaction. When this is done, the equilibrium constant for the overall reaction is equal to the concentrations of the products divided by the concentrations of the reactants in the overall reaction, with each concentration term raised to the stoichiometric coefficient for the respective species. The forward and reverse rate constants for the n th step are designated k_n and k_{-n} , respectively. For

example, if the reaction $aA + bB \rightleftharpoons cC + dD$ occurs in three steps, each with a forward and reverse rate, then:

$$K_c = \frac{k_1 k_2 k_3}{k_{-1} k_{-2} k_{-3}} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

Example:

What is the expression for the equilibrium constant for the following reaction?



Solution:

$$K_c = K_{eq} = \frac{[\text{NH}_3]^2}{[\text{H}_2]^3 [\text{N}_2]}$$

The K_p of this reaction would be:

$$\frac{(P_{\text{NH}_3})^2}{(P_{\text{H}_2})^3 \times P_{\text{N}_2}}$$



KEY CONCEPT

Remember the warning in Chapter 5 of MCAT General Chemistry Review about confusing equilibrium expressions and rate laws? In

equilibrium expressions, the exponents are equal to the coefficients in the balanced equation. In rate laws, the exponents must be determined experimentally and often do not equal the stoichiometric coefficients.

REACTION QUOTIENT

The law of mass action defines the position of equilibrium; however, equilibrium is a state that is only achieved through time. Depending on the actual rates of the forward and reverse reactions, equilibrium might be achieved in microseconds or millennia. What can serve as a “timer” to indicate how far the reaction has proceeded toward equilibrium? This role is served by the reaction quotient, Q . At any point in time during a reaction, we can measure the concentrations of all of the reactants and products and calculate the reaction quotient according to the following equation:

$$Q_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

Equation 6.2

This equation looks identical to the equation for K_{eq} . It is the same form, but the information it provides is quite different. While the concentrations used for the law of mass action are equilibrium (constant) concentrations, the concentrations of the reactants and products are not constant when

calculating a value for Q of a reaction. Thus, the utility of Q is not the value itself but rather the comparison that can be made between Q at any given moment in the reaction to the known K_{eq} for the reaction at a particular temperature. Le Châtelier's principle, which will be elaborated upon shortly, will then guide the reaction. For any reaction, if:

- $Q < K_{eq}$, then the forward reaction has not yet reached equilibrium.
 - There is a greater concentration of reactants (and smaller concentration of products) than at equilibrium.
 - The forward rate of reaction is increased to restore equilibrium.
- $Q = K_{eq}$, then the reaction is in dynamic equilibrium.
 - The reactants and products are present in equilibrium proportions.
 - The forward and reverse rates of reaction are equal.
- $Q > K_{eq}$, then the forward reaction has exceeded equilibrium.
 - There is a greater concentration of products (and smaller concentration of reactants) than at equilibrium.
 - The reverse rate of reaction is increased to restore equilibrium.



KEY CONCEPT

- $Q < K_{eq}$: $\Delta G < 0$, reaction proceeds in forward direction
- $Q = K_{eq}$: $\Delta G = 0$, reaction is in dynamic equilibrium
- $Q > K_{eq}$, $\Delta G > 0$: reaction proceeds in reverse direction

Any reaction that has not yet reached the equilibrium state, as indicated by $Q < K_{eq}$, will continue spontaneously in the forward direction (consuming reactants to form products) until the equilibrium ratio of reactants and products is reached. Any reaction in the equilibrium state will continue to react in the forward and reverse directions, but the reaction rates for the forward and reverse reactions will be equal, and the concentrations of the reactants and products will be constant, such that $Q = K_{eq}$. A reaction that is beyond the equilibrium state, as indicated by $Q > K_{eq}$, will proceed in the reverse direction (consuming products to form reactants) until the equilibrium ratio of reactants and products is reached again. Once a reaction is at equilibrium, any further movement in either the forward direction (resulting in an increase in products) or in the reverse direction (resulting in the reformation of reactants) will be nonspontaneous. This trend is illustrated in Figure 6.2.

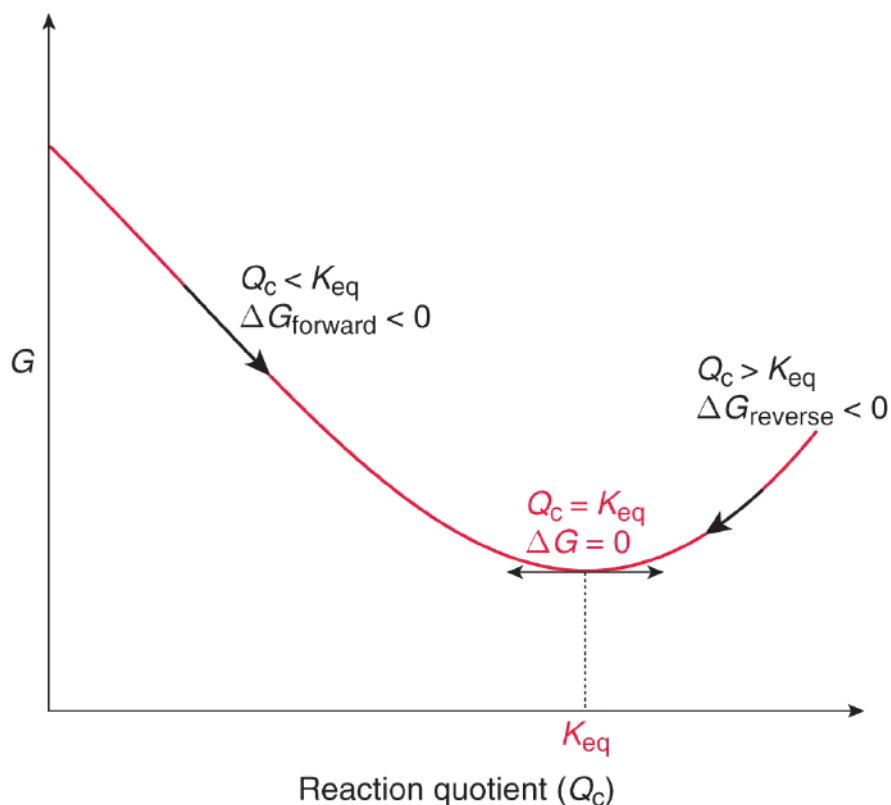


Figure 6.2. Gibbs Free Energy vs. Reaction Quotient

In Chapter 7 of MCAT General Chemistry Review, we'll further discuss how the spontaneity of these systems is related to enthalpy and entropy.

PROPERTIES OF THE LAW OF MASS ACTION

Make sure to remember the following characteristics of the law of mass action and equilibrium constant expressions:

- The concentrations of pure solids and pure liquids do not appear in the equilibrium constant expression. This is because the equilibrium expression is technically based on the activities of compounds, not concentrations; the activities of pure solids and liquids are defined to be

1. For the purposes of the MCAT, there is a negligible difference between concentration and activity.
- K_{eq} is characteristic of a particular reaction at a given temperature; the equilibrium constant is temperature-dependent.
 - The larger the value of K_{eq} , the farther to the right the equilibrium position.
 - If the equilibrium constant for a reaction written in one direction is K_{eq} , the equilibrium constant for the reverse reaction is $\frac{1}{K_{eq}}$.

EQUILIBRIUM CALCULATIONS

Calculations involving the equilibrium constant can take several forms but are also highly repetitive. Although we have discussed K_{eq} in this chapter, equilibrium constants and related calculations appear in solutions (K_{sp}), acids and bases (K_w , K_a , and K_b), and enzyme kinetics (K_d , K_b , and K_a). Thus, learning the “ins and outs” of equilibrium calculations has a large payoff on the MCAT.

One of the first concepts to grasp is the scale of the equilibrium constant. In an ideal situation, the concentrations of products and reactants would all be the same, and regardless of their actual concentrations would reduce to 1:1 ratios. In this case, K_{eq} would equal 1. In the real world, this situation doesn't exist, as it is unlikely that any reaction would have exactly equal concentrations of products and reactants at equilibrium. However, a K_{eq} of 1 can be a valuable reference point, given that $K_{eq} = \frac{[\text{products}]}{[\text{reactants}]}$.

If the concentration of products is greater than the concentration of reactants, K_{eq} becomes a “top heavy” fraction and must be greater than 1. On the other hand, if the concentration of reactants is greater than that of products, K_{eq} becomes a “bottom heavy” fraction and must be less than 1. Keep in mind, however, that K_{eq} is often expressed as a single value using exponents, and the sign and scale of these exponents gives even more information about the relative quantities of reactants and products. A reaction that strongly favors products will have a large, positive exponent, and the larger the exponent, the less reactant that will be present at equilibrium. In other words, a large positive exponent indicates a reaction that goes almost to completion.

On the other hand, a large negative exponent indicates a reaction that strongly favors reactants at equilibrium. In this case, only a small amount of reactant is converted to product. When performing equilibrium calculations, a K_{eq} with a large negative exponent allows a very convenient and very necessary shortcut to be used: the amount that has reacted can be considered negligible compared to the amount of reactant that remains.

Consider the reaction $A \leftrightarrow B + C$, with $K_{eq} = 10^{-12}$ and a starting concentration of $[A] = 1 \text{ M}$. The K_{eq} expression can be written as:

$$K_{eq} = \frac{[C][B]}{[A]}$$



MCAT EXPERTISE

You may have been previously taught to solve these types of equilibrium problems using a technique referred to as an ICEbox. On Test Day, generating an entire ICEbox table takes valuable time and effort, and this technique can be shortcut using logic alone. Practice solving equilibrium problems without the ICEbox technique, using the methods described in this chapter, for a faster solution on Test Day.

If x amount of A has reacted, x amount of C and x amount of B have been produced at equilibrium, and the equilibrium concentration of A will be $[1-x]$ M. Substituting these values into the K_{eq} expression gives:

$$K_{eq} = 10^{-12} = \frac{(x)(x)}{(1-x)} = \frac{x^2}{1-x}$$

Unfortunately, performing the calculations required by this equation would give us a polynomial function that would be extremely burdensome to solve. However, the value of K_{eq} has a large negative exponent, allowing us to use the “ x is negligible” shortcut. Relative to the 1 M starting concentration, the amount that has reacted is so small, based on $K_{eq} = 10^{-12}$, that we can assume x is negligible and round the denominator to the starting concentration:

$$10^{-12} = \frac{x^2}{1}$$

The problem is much more readily solved, and the value for x is found to be 10^{-6} . This confirms our estimate that x is negligible compared to 1, since $x = 0.000001$, and $[1 - (0.000001)] \approx 1$.

If the value for K_{eq} is within one to two orders of magnitude of one, or if the concentration of reactant that goes to product is within two orders of magnitude of the initial concentration of reactant, this estimation will not be valid. Likewise, if the value of K_{eq} is significantly larger than one, this estimation cannot be used. In both cases, the amount that reacts will be significant compared to the starting concentration of reactant. However, these situations are unlikely to be tested on the MCAT.



BRIDGE

Equilibrium constants are calculated for many types of reactions, and go by many different names. For solubility problems (MCAT General Chemistry Review Chapter 9), K_{eq} is known as K_{sp} . For acids and bases (MCAT General Chemistry Review Chapter 10), K_{eq} is known as K_a , K_b , or K_w . For enzyme kinetics (MCAT Biochemistry Review Chapter 2), K_{eq} is known as K_d , K_b or K_a .

Example:

3 moles of N_2O_4 is placed in a 0.5 L container and allowed to reach equilibrium according to the following reaction:



What is the equilibrium concentration of NO_2 , given K_{eq} for the reaction is 6×10^{-6} ?

Solution:

Start by writing the expression for K_{eq} : $K_{\text{eq}} = \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]}$

Note that the concentration of NO_2 is squared due to its coefficient of 2 in the balanced reaction. Next, determine the starting concentration of N_2O_4 , taking into account that the initial volume is

$$\begin{aligned} \frac{3 \text{ M}}{0.5 \text{ L}} &= \frac{[\text{N}_2\text{O}_4]}{1 \text{ L}} \\ 500 \text{ mL} = 0.5 \text{ L:} \quad [\text{N}_2\text{O}_4] &= \frac{(3)(1)}{(0.5)} = 6 \text{ M} \end{aligned}$$

Thus, the starting concentration of $\text{N}_2\text{O}_4 = 6 \text{ M}$. Next, using x to represent the amount of N_2O_4 that reacts, $2x$ to represent the amount of NO_2 that is produced, and 10^{-6} for the value of K_{eq} , plug into the expression for the equilibrium constant:

$$K_{\text{eq}} = 6 \times 10^{-6} = \frac{[2x]^2}{[6-x]} \approx \frac{[2x]^2}{[6]}$$

Note that the small negative exponent in the value of K_{eq} indicates that x will be negligible in comparison to 6 M, allowing the K_{eq} expression to be simplified, and x to be determined as follows:

$$\begin{aligned}
 6 \times 10^{-6} &= \frac{4x^2}{6} \\
 36 \times 10^{-6} &= 4x^2 \\
 x^2 &= \frac{36 \times 10^{-6}}{4} = 9 \times 10^{-6} \\
 x &= 3 \times 10^{-3} M
 \end{aligned}$$

However, be careful to note that this is the value of x , which represents the amount of N_2O_4 that reacts. The final answer must represent the amount of NO_2 produced, which is twice the amount of N_2O_4 that reacts, or $2x$. Thus the final answer is the concentration of NO_2 , which is $6 \times 10^{-3} M$.

MCAT Concept Check 6.1:

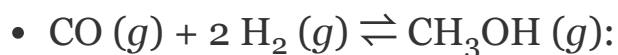
Before you move on, assess your understanding of the material with these questions.

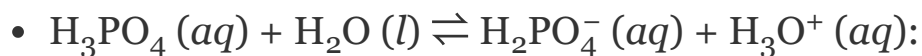
Given that $[\text{product}] = 0.075 M$ and $[\text{reactant}] = 1.5 M$, determine the direction of reaction and the sign of the free energy change for reactions with the following K_{eq} values: (Note: Assume that the reaction has only one product and one reactant, and that the stoichiometric coefficient for each is 1.)

K_{eq}	Direction of Reaction	ΔG
5.0×10^{-2}		
5.0×10^{-3}		
5.0×10^{-1}		



Write the equilibrium constant expression for the following reactions:



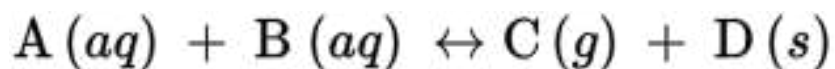


Consider the hypothetical reaction $\text{A} \leftrightarrow \text{B} + \text{C}$.

For each of the following, determine if the amount of reactant A that has converted to product at equilibrium will be negligible compared to the starting concentration of A.

K_{eq}	Initial Concentration of A (M)	Is the amount reacted negligible?
1.0×10^{-12}	1	
1.0×10^{-2}	0.1	
1.0×10^{-3}	0.001	
1.0×10^{-15}	0.001	

The following reaction has a K_{eq} of 2.1×10^{-7} . Given an initial concentration for A equal to 0.1 M and an initial concentration of B equal to 0.2 M , what is the equilibrium concentration of C? Is the approximation that x is negligible valid for this calculation?



6.2 Le Châtelier's Principle

LEARNING GOALS

After Chapter 6.2, you will be able to:

- Use Le Châtelier's principle to determine how changing conditions, including pH, temperature, pressure, and concentration changes, will affect a reaction previously in equilibrium

Le Châtelier's principle states that if a stress is applied to a system, the system shifts to relieve that applied stress. Regardless of the form the stress takes, the reaction is temporarily moved out of its equilibrium state. This is either because the concentrations or partial pressures of the system are no longer in the equilibrium ratio or because the equilibrium ratio itself has changed as a result of a change in the temperature of the system. The reaction then responds by reacting in whichever direction—either forward or reverse—will result in a reestablishment of the equilibrium state.



MCAT EXPERTISE

Le Châtelier's principle applies to a wide variety of systems and, therefore, appears as a fundamental concept in all three MCAT science sections.

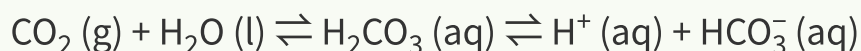
CHANGES IN CONCENTRATION

When reactants or products are added or removed from a reaction in equilibrium, the reaction is moved from its minimum energy state. With the change in concentration of one or more of the chemical species, the system now has a ratio of products to reactants that is not equal to the equilibrium ratio. In other words, changing the concentration of either a reactant or a product results in $Q_c \neq K_{eq}$. If reactants are added (or products are removed), $Q_c < K_{eq}$, and the reaction will spontaneously react in the forward direction, increasing the value of Q_c until $Q_c = K_{eq}$. If reactants are removed (or products are added), $Q_c > K_{eq}$, and the reaction will spontaneously react in the reverse direction, thereby decreasing the value of Q_c until once again $Q_c = K_{eq}$. Put simply, the system will always react in the direction away from the added species or toward the removed species.



BRIDGE

The bicarbonate buffer system is a classic example of Le Châtelier's principle applied to physiology:



In the tissues, there is a relatively high concentration of CO_2 , and the reaction shifts to the right. In the lungs, CO_2 is lost, and the reaction shifts to the left. Note that blowing off CO_2 (hyperventilation) is used as a mechanism for dealing with acidemia (excess H^+ in the blood). This buffer system plays a key role in the respiratory, circulatory, and excretory systems, discussed in Chapters 6, 7, and 10 of MCAT Biology Review, respectively.

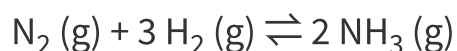
We often take advantage of Le Châtelier's principle to improve the yield of chemical reactions. For example, in the industrial production of chemicals, products of reversible reactions are removed as they are formed to prevent the reactions from reaching their equilibrium states. The reaction will continue to react in the forward direction, producing more and more products—assuming reactants are continually replaced as they are consumed. One could also drive a reaction forward by starting with high concentrations of reactants. This will lead to an increase in the absolute quantities of products formed, although the reaction will still eventually reach its equilibrium state unless products are removed as they are formed.

CHANGES IN PRESSURE (AND VOLUME)

Because liquids and solids are essentially incompressible, only chemical reactions that involve at least one gaseous species will be affected by changes in the system's pressure and volume. When a system is compressed, its volume decreases and its total pressure increases. This

increase in the total pressure is associated with an increase in the partial pressures of each gas in the system, and this results in the system no longer being in the equilibrium state, such that Q_p does not equal K_{eq} . The system will move forward or in reverse, always toward whichever side has the lower total number of moles of gas. This is a consequence of the ideal gas law, which tells us that there is a direct relationship between the number of moles of gas and the pressure of the gas. If one increases the pressure of a system, it will respond by decreasing the total number of gas moles, thereby decreasing the pressure. Note that this scenario assumes that the volume of the system was decreased and then held constant while the system returned to its equilibrium state. When one expands the volume of a system, the total pressure and the partial pressures decrease. The system is no longer in its equilibrium state and will react in the direction of the side with the greater number of moles of gas in order to restore the pressure.

Consider the following reaction:



The left side of the reaction has a total of four moles of gas molecules, while the right side has only two moles. When the pressure of this system is increased, the system will react in the direction that produces fewer moles of gas. In this case, that direction is to the right, and more ammonia will form. However, if the pressure is decreased, the system will react in the direction that produces more moles of gas; thus, the reverse reaction will be favored, and more nitrogen and hydrogen gas will reform.

CHANGES IN TEMPERATURE

Le Châtelier's principle tells us that changing the temperature of a system will also cause the system to react in a particular way to return to its equilibrium state. However, unlike the effect of changing concentrations or pressures, the result of changing temperature is not a change in the reaction quotient, Q_c or Q_p , but a change in K_{eq} . The change in temperature does not cause the concentrations or partial pressures of the reactants and products to change immediately, so Q immediately after the temperature change is the same as before the temperature change. Thus, because K_{eq} is now a different value, Q no longer equals K_{eq} . The system has to move in whichever direction allows it to reach its new equilibrium state at the new temperature. That direction is determined by the enthalpy of the reaction. If a reaction is endothermic ($\Delta H > 0$), heat functions as a reactant; if a reaction is exothermic ($\Delta H < 0$), heat functions as a product. Thinking about heat as a reactant or product allows us to apply the same principles we used with concentration changes to temperature changes.

For example, consider the following endothermic reaction, shown in Figure 6.3:



The equilibrium position can be shifted by changing the temperature. When heat is added and the temperature increases, the reaction shifts to the right, and the flask turns reddish-brown due to an increase in $[\text{NO}_2]$. When heat is removed and the temperature decreases, the reaction shifts to the left, and the flask turns more transparent due to an increase in N_2O_4 .

This demonstrates Le Châtelier's principle because the equilibrium shifts in the direction that consumes energy.



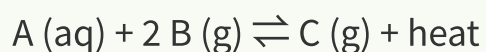
Figure 6.3. Example of a Reversible Endothermic Reaction, $\text{N}_2\text{O}_4 (\text{g}) \rightleftharpoons 2 \text{NO}_2 (\text{g})$

Left: As temperature decreases, the equilibrium favors N_2O_4 production, turning the reaction vessel more transparent. Right: As temperature increases, the equilibrium favors NO_2 production, turning the reaction vessel reddish-brown.



KEY CONCEPT

The reaction



Will shift to the right if...	Will shift to the left if...
A or B is added	C is added
C is removed	A or B is removed

the pressure is increased or the volume is reduced	the pressure is reduced or the volume is increased
the temperature is reduced	the temperature is increased

MCAT Concept Check 6.2:

Before you move on, assess your understanding of the material with this question.

Describe what would happen in the following situations:

- In the reaction $\text{H}_2\text{SO}_4 (aq) \rightleftharpoons \text{H}^+ (aq) + \text{HSO}_4^- (aq)$, the pH has been increased:

- In the reaction $2 \text{C} (s) + \text{O}_2 (g) \rightleftharpoons 2 \text{CO} (g)$, the pressure of the reaction vessel is decreased:

- In the reaction $\text{CH}_4 (g) + 2 \text{O}_2 (g) \rightleftharpoons \text{CO}_2 (g) + 2 \text{H}_2\text{O} (l) + \text{heat}$, the reaction vessel is warmed:

- In the reaction $\text{H}_3\text{PO}_4 (aq) + \text{H}_2\text{O} (l) \rightleftharpoons \text{H}_3\text{O}^+ (aq) + \text{H}_2\text{PO}_4^- (aq)$, water is removed (without changing temperature):



6.3 Kinetic and Thermodynamic Control

LEARNING GOALS

After Chapter 6.3, you will be able to:

- Identify the conditions that will cause a reaction to favor the kinetic or the thermodynamic product
- Distinguish between a kinetic and thermodynamic pathway on a reaction coordinate diagram

Having covered the fundamentals of kinetics and thermodynamics, we come upon a topic that bridges all chemical systems: control of a reaction. In particular, biochemical reactions often require regulation in a precise manner to be useful to an organism. The applications of kinetic and thermodynamic control are common on the MCAT and range from metabolic reactions requiring high-energy phosphate molecules, such as ATP to the effects of temperature and solvents on enzyme activity.

The examples below consider unimolecular systems through the lens of the transition state theory. Figure 6.4 shows starting materials (reactants)

at a certain energy level. These reactants can undergo two different sets of reactions. At lower temperatures (with smaller heat transfer), a kinetic product is formed. At higher temperatures (with larger heat transfer), a thermodynamic product is formed.

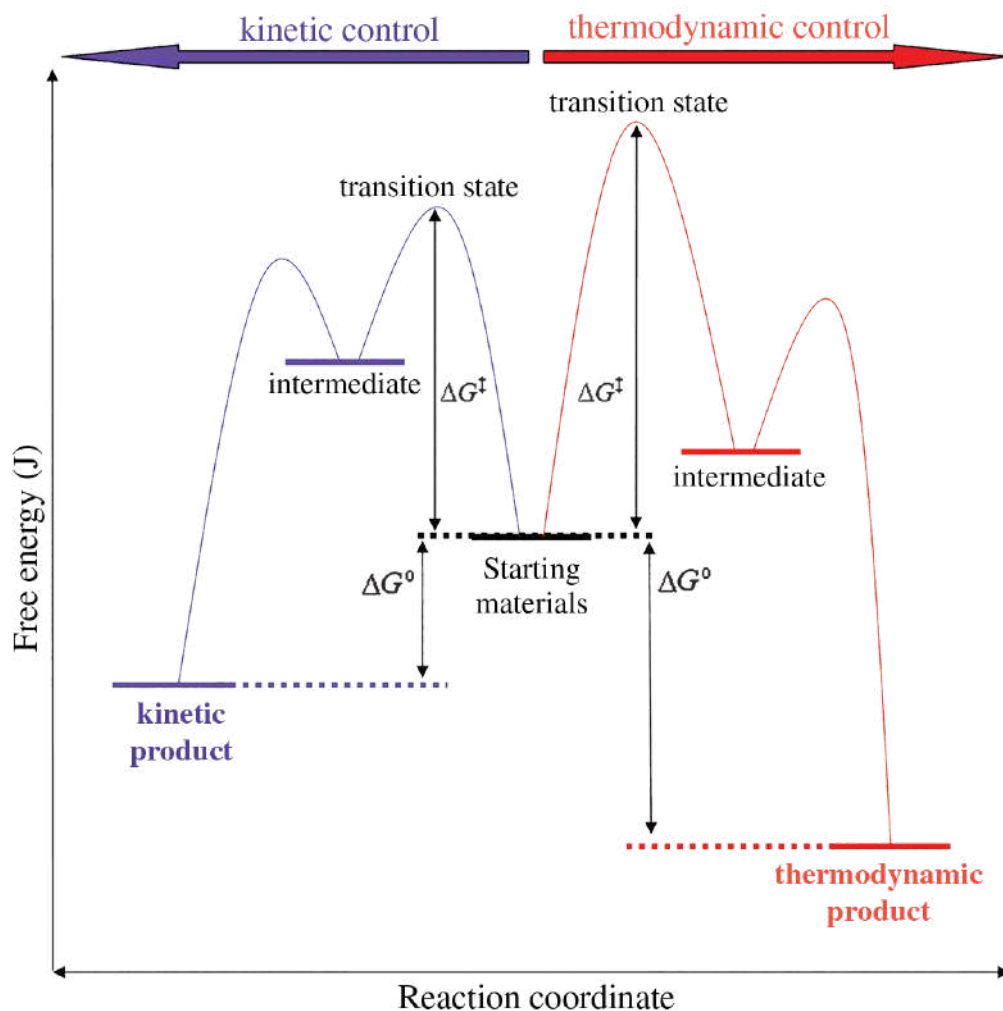


Figure 6.4. Kinetic and Thermodynamic Control of a Reaction

The kinetic pathway requires less free energy to reach the transition state, but results in a higher-energy (less stable) product.

Note that the free energy that must be added for the kinetic pathway is lower than that of the thermodynamic pathway. Therefore, the kinetic

products often form faster than the thermodynamic products and are sometimes called “fast” products. On the other hand, the free energy of the thermodynamic product is significantly lower than that of the kinetic product. Thermodynamic products are therefore associated with greater stability, and with a more negative ΔG than kinetic products.

The stability of organic molecules is covered in Chapter 2 of MCAT Organic Chemistry Review and is dependent on torsional strain, angle strain, and nonbonded strain. In this example, we consider the conversion of 2-methylcyclohexanone to its thermodynamic product and its kinetic product, as shown in Figure 6.5. Both reactions require a base (B^-) in order to catalyze the conversion, yet two different products are produced.

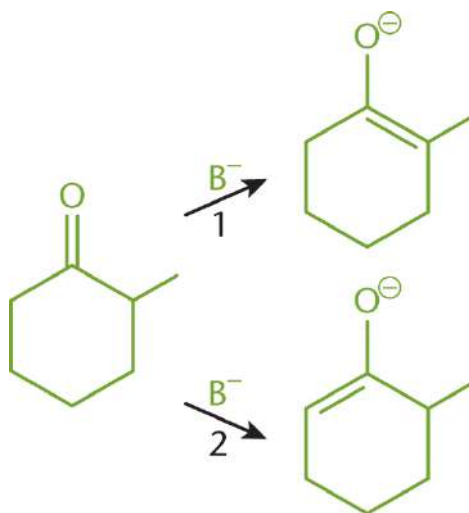


Figure 6.5. Conversion of 2-methylcyclohexanone to (1) Thermodynamic Product and (2) Kinetic Product

For the thermodynamic pathway (1), the double bond is located between C-1 and the methyl group. It requires more energy to form the transition state of this reaction because the base must overcome the steric hindrance created by the methyl group. The base squeezes in to reach the carbon

with the methyl group attached to abstract a proton. However, because the double bond is more substituted than the other pathway, the product of this reaction is more stable and less likely to react further.

For the kinetic pathway (2), the double bond is located between C-1 and C-6. This pathway is preferred when there is little heat available because less energy is needed to reach the transition state. The base can more easily reach C-6 to remove a proton, and the resulting enolate can form. This product has a less substituted double bond, which reduces its stability. This lack of stability may leave the ring susceptible to further attack.

MCAT Concept Check 6.3:

Before you move on, assess your understanding of the material with these questions.

What conditions favor formation of a kinetic product? A thermodynamic product?

- Kinetic product:

- Thermodynamic product:

On a reaction coordinate diagram, how would the kinetic pathway appear as compared to the thermodynamic pathway?



Conclusion

We've discussed some very important concepts and principles in the past two chapters related to the studies of reaction rates and chemical equilibria. In this chapter, we began with the law of mass action and the significance of the equilibrium state of a chemical reaction. With our understanding of the significance of K_{eq} and Q , we are able to predict the direction that a reaction will go in response to various stresses—concentration, pressure, or temperature changes—that might be applied to a system.

The concept of homeostasis in biology is a direct result of the energy associated with disturbing equilibria in the body. Reactions are often held slightly out of the equilibrium state to generate energy. Many pathologies you will encounter in your future career in medicine will have a fundamental basis in disturbed chemical equilibria—just wait until you start ordering metabolic panels on your patients!

Concept Summary

Equilibrium

- Reversible reactions eventually reach a state in which energy is minimized and entropy is maximized.
 - Chemical equilibria are dynamic—the reactions are still occurring, just at a constant rate.
 - The concentrations of reactants and products remain constant because the rate of the forward reaction equals the rate of the reverse reaction.
- The law of mass action gives the expression for the equilibrium constant, K_{eq} . The reaction quotient, Q , has the same form but can be calculated at any concentrations of reactants and products.
 - Q is a calculated value that relates the reactant and product concentrations at any given time during a reaction.
 - K_{eq} is the ratio of products to reactants at equilibrium, with each species raised to its stoichiometric coefficient. K_{eq} for a reaction is constant at a constant temperature.
 - Pure solids and liquids do not appear in the law of mass action; only gases and aqueous species do.

- Comparison of Q to K_{eq} provides information about where the reaction is with respect to its equilibrium state.
 - If $Q < K_{eq}$, $\Delta G < 0$, and the reaction proceeds in the forward direction.
 - If $Q = K_{eq}$, $\Delta G = 0$, and the reaction is in dynamic equilibrium.
 - If $Q > K_{eq}$, $\Delta G > 0$, and the reaction proceeds in the reverse direction.
- Equilibrium calculations are broadly applicable to many areas of chemistry but are often formulaic in their application. The magnitude of K_{eq} determines the balance of a reaction and whether the amount that has reacted can be treated as negligible when compared to other concentrations.
 - If $K_{eq} > 1$, the products are present in greater concentration at equilibrium.
 - If $K_{eq} \approx 1$, products and reactants are both present at equilibrium at reasonably similar levels.
 - If $K_{eq} < 1$, the reactants are present in greater concentration at equilibrium.
 - If $K_{eq} \ll 1$, the amount of reactants that have been converted to products can be considered negligible in comparison to the initial concentration of reactants.

Le Châtelier's Principle

- Le Châtelier's principle states that when a chemical system experiences a stress, it will react so as to restore equilibrium.
- There are three main types of stresses applied to a system: changes in concentration, pressure and volume, and temperature.

- Increasing the concentration of reactants or decreasing the concentration of products will shift the reaction to the right. Increasing the concentration of products or decreasing the concentration of reactants will shift the reaction to the left.
- Increasing pressure on a gaseous system (decreasing its volume) will shift the reaction toward the side with fewer moles of gas. Decreasing pressure on a gaseous system (increasing its volume) will shift the reaction toward the side with more moles of gas.
- Increasing the temperature of an endothermic reaction or decreasing the temperature of an exothermic reaction will shift the reaction to the right. Decreasing the temperature of an endothermic reaction or increasing the temperature of an exothermic reaction will shift the reaction to the left.

Kinetic and Thermodynamic Control

- Reactions may have both kinetic and thermodynamic products that can be regulated by temperature and the presence of a catalyst.
 - Kinetic products are higher in free energy than thermodynamic products and can form at lower temperatures. These are sometimes termed “fast” products because they can form more quickly under such conditions.
 - Thermodynamic products are lower in free energy than kinetic products and are therefore more stable. Despite proceeding more slowly than the kinetic pathway, the thermodynamic pathway is more spontaneous (more negative ΔG).

Answers to Concept Checks

6.1

First calculate the value of Q from the given concentrations:

$$Q = \frac{[\text{products}]}{[\text{reactants}]} = \frac{0.075}{1.5} = \frac{7.5 \times 10^{-2}}{1.5} = 5 \times 10^{-2}.$$

Q can now be compared to each value of K_{eq} to predict the direction of the reaction.

K_{eq}	Direction of Reaction	ΔG
5.0×10^{-2}	At equilibrium: no net reaction	0
5.0×10^{-3}	$Q_c > K_{\text{eq}}$: proceeds toward reactants (left)	Positive
5.0×10^{-1}	$Q_c < K_{\text{eq}}$: proceeds toward products (right)	Negative

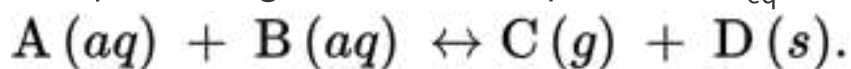
$$K_p = \frac{P_{\text{CH}_3\text{OH}}}{P_{\text{CO}} \times (P_{\text{H}_2})^2}$$

$$K_c = K_a = \frac{[\text{H}_2\text{PO}_4^-][\text{H}_3\text{O}^+]}{[\text{H}_3\text{PO}_4]}$$

The concentration of a reactant that converts to product can be considered negligible if it is two or more orders of magnitude less than the initial concentration of the reactant.

K_{eq}	Initial Concentration of A (M)	Is the amount reacted negligible?
1×10^{-12}	1	Yes
1×10^{-2}	0.1	No
1×10^{-3}	0.001	No
1×10^{-15}	0.001	Yes

The first step in solving is to write the equation for K_{eq} for the reaction:



$$K_{eq} = \frac{[C]}{[A][B]} = 2.1 \times 10^{-7}$$

Note that the equation for K_{eq} does not include product D because D is a solid.

Next, using the initial concentrations for A and B and x for the amount that has reacted, plug into the equation for K_{eq} :

$$K_{eq} = \frac{[C]}{[A][B]} = \frac{[x]}{[0.1-x][0.2-x]}$$

Given that $K_{eq} = 2.1 \times 10^{-7}$, the concentrations of A and B are sufficiently large that x can be considered negligible in comparison to both. This allows the equation for K_{eq} to be simplified and solved:

$$K_{\text{eq}} = 2.1 \times 10^{-7} \approx \frac{x}{[0.1][0.2]}$$

$$x = (2.1 \times 10^{-7}) (0.1)(0.2) = 4.2 \times 10^{-9}$$

The value of $x = 4.2 \times 10^{-9}$ is equal to both the equilibrium concentration of C and the amount of A and B that have reacted. The approximation that x is negligible compared to the initial concentrations of A and B is valid.

6.2

- Increasing pH of $\text{H}_2\text{SO}_4 (\text{aq}) \rightleftharpoons \text{H}^+ (\text{aq}) + \text{HSO}_4^- (\text{aq})$: $[\text{H}^+]$ decreases, shifting reaction to the right.
- Decreasing pressure of $2 \text{C} (\text{s}) + \text{O}_2 (\text{g}) \rightleftharpoons 2 \text{CO} (\text{g})$: Reaction shifts right, favoring the side with more moles of gas.
- Warming $\text{CH}_4 (\text{g}) + 2\text{O}_2 (\text{g}) \rightleftharpoons \text{CO}_2 (\text{g}) + 2 \text{H}_2\text{O} (\text{l}) + \text{heat}$: Reaction shifts left, using the additional heat energy to produce more reactants.
- Removing water from $\text{H}_3\text{PO}_4 (\text{aq}) + \text{H}_2\text{O} (\text{l}) \rightleftharpoons \text{H}_3\text{O}^+ (\text{aq}) + \text{H}_2\text{PO}_4^- (\text{aq})$: Reaction shifts left. All concentrations would increase proportionately; because there are more products than reactants (and the stoichiometric coefficient is 1 for each reactant and product), the value of Q will increase.

6.3

Kinetic products are favored at low temperatures with low heat transfer. Thermodynamic products are favored at high temperatures with high heat transfer.

Kinetic pathways require a smaller gain in free energy to reach the transition state. They also have a higher free energy of the products,

with a smaller difference in free energy between the transition state and the products.

Equations to Remember

(6.1) Equilibrium constant: $K_{\text{eq}} = \frac{[\text{C}]^c [\text{D}]^d}{[\text{A}]^a [\text{B}]^b}$

(6.2) Reaction quotient: $Q_c = \frac{[\text{C}]^c [\text{D}]^d}{[\text{A}]^a [\text{B}]^b}$



Shared Concepts

Biochemistry Chapter 2

Enzymes

General Chemistry Chapter 5

Chemical Kinetics

General Chemistry Chapter 7

Thermochemistry

General Chemistry Chapter 9

Solutions

General Chemistry Chapter 10

Acids and Bases

Organic Chemistry Chapter 2

Isomers

Discrete Practice Questions

Consult your online resources for additional practice.

1. A reaction is found to stop just before all reactants are converted to products. Which of the following could be true about this reaction?

- (A) The reaction is irreversible, and the forward rate is greater than the reverse rate.
- (B) The reaction is irreversible, and the reverse rate is too large for products to form.
- (C) The reaction is reversible, and the forward rate is equal to the reverse rate.
- (D) The reaction is reversible, and the reverse rate is greater than the forward rate.

2. What is the equilibrium expression for the reaction $\text{Cu}_2\text{SO}_4 (\text{s}) \rightleftharpoons 2 \text{Cu}^+ (\text{aq}) + \text{SO}_4^{2-} (\text{aq})$?

(A)

$$\frac{[\text{Cu}^+]^2 [\text{SO}_4^{2-}]}{[\text{Cu}_2\text{SO}_4]}$$

(B)

$$\frac{2 \times [\text{Cu}^+] [\text{SO}_4^{2-}]}{[\text{Cu}_2\text{SO}_4]}$$

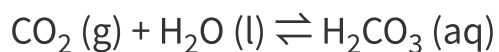
(C)

$$[\text{Cu}^+] [\text{SO}_4^{2-}]$$

(D)

$$[\text{Cu}^+]^2 [\text{SO}_4^{2-}]$$

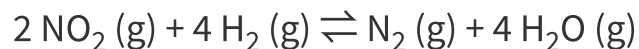
3. Carbonated beverages are produced by dissolving carbon dioxide in water to produce carbonic acid:



When a bottle containing carbonated water is opened, the taste of the beverage gradually changes as the carbonation is lost. Which of the following statements best explains this phenomenon?

- (A) The change in pressure and volume causes the reaction to shift to the left, thereby decreasing the amount of aqueous carbonic acid.
- (B) The change in pressure and volume causes the reaction to shift to the right, thereby decreasing the amount of gaseous carbon dioxide.
- (C) Carbonic acid reacts with environmental oxygen and nitrogen.
- (D) Carbon dioxide reacts with environmental oxygen and nitrogen.

4. What is the proper equilibrium expression for the reaction below?



(A)

$$K_p = \frac{P_{N_2} \times (P_{H_2O})^4}{(P_{NO_2})^2 \times (P_{H_2})^4}$$

(B)

$$K_c = \frac{P_{N_2} \times (P_{H_2O})^4}{(P_{NO_2})^2 \times (P_{H_2})^4}$$

(C)

$$K_p = \frac{P_{N_2}}{(P_{NO_2})^2 \times (P_{H_2})^4}$$

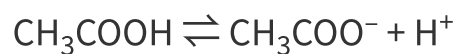
(D)

$$K_c = \frac{P_{N_2}}{(P_{NO_2})^2 \times (P_{H_2})^4}$$

5. If $K_c \gg 1$:

- (A) the equilibrium mixture will favor products over reactants.
- (B) the equilibrium mixture will favor reactants over products.
- (C) the equilibrium concentrations of reactants and products are equal.
- (D) the reaction is essentially irreversible.

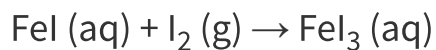
6. Acetic acid dissociates in solution according to the following equation:



If sodium acetate is added to a solution of acetic acid in excess water, which of the following effects would be observed in the solution?

- (A) Decreased pH
- (B) Increased pH
- (C) Decreased pK_{eq} (pK_{a})
- (D) Increased pK_{eq} (pK_{a})

Questions 7 and 8 refer to the reaction below:



7. Which of the following would increase the formation of product?

- (A) Decreasing the volume of the container
- (B) Decreasing the pressure of the container
- (C) Increasing the volume of the container
- (D) Decreasing the volume of the container while maintaining a constant pressure

8. If this reaction were exothermic, what effect would decreasing the temperature have on the equilibrium?

- (A) The forward reaction rate and the reverse reaction rate both increase.
- (B) The forward reaction rate decreases while the reverse reaction rate increases.
- (C) The forward reaction rate increases while the reverse reaction rate decreases.
- (D) The forward reaction rate and the reverse reaction rate both decrease.

9. Which of the following actions does NOT affect the equilibrium position of a reaction?

- (A) Adding or removing heat.
- (B) Adding or removing a catalyst.
- (C) Increasing or decreasing concentrations of reactants.
- (D) Increasing or decreasing volumes of reactants.

10. In a sealed 1 L container, 1 mole of nitrogen gas reacts with 3 moles of hydrogen gas to form 0.05 moles of NH_3 at equilibrium. Which of the following is closest to the K_c of the reaction?

- (A) 0.0001
- (B) 0.001
- (C) 0.01
- (D) 0.1

11. Increasing temperature can alter the K_{eq} of a reaction. Why might increasing temperature indefinitely be unfavorable for changing reaction conditions?

- (A) The equilibrium constant has a definite limit that cannot be surpassed.
- (B) The products or reactants can decompose at high temperatures.
- (C) Increasing temperature would decrease pressure, which may or may not alter reaction conditions.
- (D) If a reaction is irreversible, its K_{eq} will resist changes in temperature.

12. Which of the following is true of equilibrium reactions?

An increase in k_1 results in a decrease in k_{-1} .

As the concentration of products increases, the concentrations of reactants decreases.

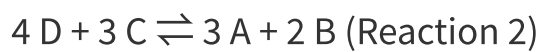
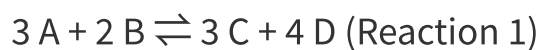
I. The equilibrium constant is altered by changes in temperature.

- (A) I only
- (B) II and III only
- (C) I and III only
- (D) I, II, and III

13. Compound A has a K_a (equilibrium constant of acid dissociation) of approximately 10^{-4} . Which of the following compounds is most likely to react with a solution of compound A?

- (A) HNO_3
- (B) NO_2
- (C) NH_3
- (D) N_2O_5

14. Consider the following two reactions:



If K_{eq} for reaction 1 is equal to 0.1, what is K_{eq} for reaction 2?

- (A) 0.1
- (B) 1
- (C) 10
- (D) 100

15. Which of the following statements best describes the effect of lowering the temperature of the following reaction?



- (A) [C] and [D] would increase.
- (B) [A] and [B] would increase.
- (C) ΔH would increase.
- (D) ΔH would decrease.

Explanations to Discrete Practice Questions

1. C

This scenario likely describes a situation in which a reaction has reached equilibrium very far to the right (with high product concentration and low reactant concentration). This reaction must be reversible because the reaction did not proceed all the way to the right. Any reaction in equilibrium has equal forward and reverse rates of reaction.

2. D

Recall that pure solids and liquids do not appear in the equilibrium expression; thus, this K_{eq} has no denominator because the only reactant is a solid, cuprous sulfate. This could also be called K_{sp} because a solid is dissociating into ions in solution. The correct K_{eq} should have $[Cu^+]$ squared because its stoichiometric coefficient is 2.

3. A

Carbon dioxide gas evolves and leaves the bottle, which decreases the total pressure of the reactants. Le Châtelier's principle explains that a decrease in pressure shifts the equilibrium to increase the number of moles of gas present. This particular reaction will shift to the left, which in turn will decrease the amount of carbonic acid and increase the amount of carbon dioxide and water. Oxygen and nitrogen are not

highly reactive and are unlikely to combine spontaneously with carbon dioxide or carbonic acid, as in (C) and (D).

4. A

Recall that equilibrium constants are either based on concentrations (K_c) or partial pressures (K_p). In this case, because all species are in the gas phase, we are using K_p —eliminating (B) and (D). When water is in the liquid phase, it does not appear in equilibrium expressions, as in (C). Here, however, water is in the gaseous phase and thus should appear in the equilibrium expression.

5. A

The larger the value of K_{eq} (whether K_c or K_p), the larger the ratio of products to reactants. Therefore, if $K_c \gg 1$, there are significantly larger concentrations of products than reactants at equilibrium. Even with a large K_{eq} , the reaction will ultimately reach equilibrium far toward the products side and is therefore reversible, eliminating (D).

6. B

Adding sodium acetate increases the number of acetate ions present. According to Le Châtelier's principle, this change will push this reaction to the left, resulting in a decrease in the number of free H^+ ions. Because pH is determined by the hydrogen ion concentration, a decrease in the number of free protons will increase the pH. An acid's K_a (which is simply the K_{eq} for acid dissociation) will remain constant under a given temperature and pressure, eliminating (C) and (D).

7. A

Both increasing the pressure of the container and decreasing the volume would favor the side with fewer moles of gas, which is the product side. This makes (B) and (C) incorrect. (D) would not disturb the equilibrium—the significance of decreasing the volume of the container in most equilibria is that there is an increase in pressure; in this case, however, the pressure remains constant despite the change in volume.

8. C

An exothermic reaction produces heat. Decreasing the temperature favors product formation, resulting in an increase in the forward reaction rate with a concomitant decrease in the reverse reaction rate.

9. B

The equilibrium of a reaction can be changed by several factors. Adding or subtracting heat, (A), would shift the equilibrium based on the enthalpy change of the reaction. Increasing reactant concentrations would shift the equilibrium in the direction of the product, and the opposite would occur if reactant concentrations were decreased, eliminating (C). Changing the volume of a reactant would affect any reaction with gaseous reactants or products, eliminating (D). While adding or removing a catalyst would change the reaction rates, it would not change where the equilibrium lies.

10. A

Start with the balanced equation for the reaction of H_2 and N_2 to produce NH_3 : $\text{N}_2 + 3 \text{H}_2 \rightleftharpoons 2 \text{NH}_3$.

Next, write out $K_c = K_{eq} = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$. Because the volume is 1 L, the amount of each gas in moles is equal to the value of the concentration of each gas in moles per liter (M). The relatively small amount of NH_3 produced indicates that it will be possible to consider the amount of N_2 and H_2 that reacts to be negligible compared to their starting concentrations. Putting the amounts of each reactant and product into the K_{eq} expression gives:

$$K_{eq} = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3} = \frac{[0.05]^2}{[1-x][3-x]^3} \approx \frac{[0.05]^2}{[1][3]^3} = \frac{[5 \times 10^{-2}]^2}{27}$$

$$K_{eq} = \frac{25 \times 10^{-4}}{27} \approx \frac{25 \times 10^{-4}}{25} = 1 \times 10^{-4} = 0.0001$$

11. B

At extremely high temperatures, reactants or products may decompose, which will affect the equilibrium and potentially destroy the desired products. (A) implies that reactions have limits, which is true; however, this does not make increasing temperature unfavorable. (C) is false because increasing temperature would also increase pressure, assuming constant volume. (D) is incorrect because it refers to properties of irreversible reactions, which would not be involved in an equilibrium between products and reactants.

12. B

Statement I is false because the addition of a catalyst could increase the rate constants of both the forward and reverse reactions. Statement II is true because—for products to come into existence—reactants must be

used up. Statement III is also true: all K values are temperature-dependent.

13. C

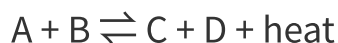
K_a is equal to the ratio of products to reactants, with each species raised to its stoichiometric coefficient. A compound with a K_a greater than 10^{-7} contains more H^+ cations than HA^- anions at equilibrium, which makes it an acid. This means that the compound in question is likely to react with a compound that is basic. Of the four answer choices, NH_3 is the only base.

14. C

Reaction 2 is the reverse of reaction 1. This means that K_{eq} for reaction 2 is the inverse of K_{eq} of reaction 1, so the answer is $\frac{1}{0.1} = \frac{1}{\frac{1}{10}} = 10$.

15. A

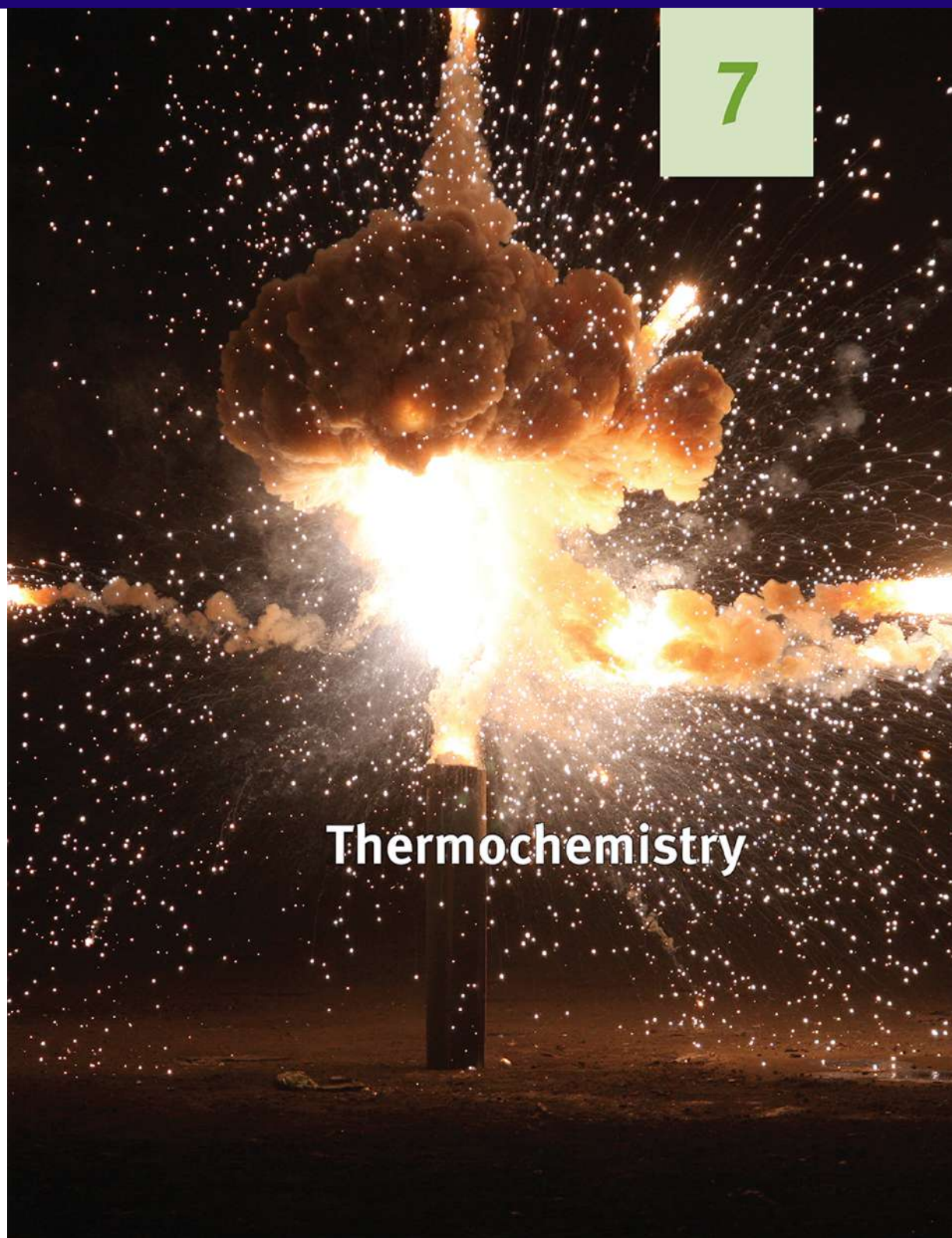
A negative ΔH value indicates an exothermic reaction, meaning that the forward reaction produces heat. Visualize this as follows:



This means that removing heat by decreasing the temperature is similar to removing any other product of the reaction. To compensate for this loss, the reaction will shift to the right, causing an increase in the concentrations of C and D, as well as a decrease in the concentrations of A and B.

7

Thermochemistry



In This Chapter

- 7.1 Systems and Processes 
 - Thermodynamic Terminology 
- 7.2 States and State Functions 
 - Overview 
 - Phase Changes 
 - Phase Diagrams 
- 7.3 Heat 
 - Overview 
 - Constant-Pressure and Constant-Volume Calorimetry 
 - Heating Curves 
- 7.4 Enthalpy 
 - Standard Heat of Formation 
 - Standard Heat of Reaction 
 - Hess's Law 
 - Bond Dissociation Energy 
 - Standard Heat of Combustion 
- 7.5 Entropy 

7.6 Gibbs Free Energy

Overview 

Standard Gibbs Free Energy 

Free Energy, K_{eq} , and Q 

Concept Summary 

Chapter Profile



The content in this chapter should be relevant to about 7% of all questions about general chemistry on the MCAT.

This chapter covers material from the following AAMC content categories:

1D: Principles of bioenergetics and fuel molecule metabolism

5E: Principles of chemical thermodynamics and kinetics

Introduction

Styrofoam cups are such good insulators that they can be used as holding containers for certain calorimetry experiments. Coffee-cup calorimetry,

which uses Styrofoam cups to measure heats of solution and specific heats of metals and other materials, is low-tech, yet it can produce remarkably accurate results as long as care has been taken to calibrate the calorimeter and to minimize heat loss through the top of the container. The next time you are at your favorite coffee chain, think about what occurs when cold cream is added to hot coffee. If we took the time to measure the masses and temperatures of the hot coffee and the cold cream before mixing them, measured the drink's temperature after it had been stirred, and looked up the specific heats of water and cream, we would have enough information to calculate the amount of heat exchanged between the hot coffee and the cold cream.

This chapter will review the basic principles of thermochemistry, which is the study of the energy changes that accompany chemical and physical processes. Starting with the first law of thermodynamics, which states that energy is never created nor destroyed but—at most—simply changed from one form to another, we will quantify the various exchanges in energy as a system moves from some initial state to a final state. As we go along, we will define what is meant by system and surroundings, state functions, heat, enthalpy, entropy, and Gibbs free energy.

7.1 Systems and Processes

LEARNING GOALS

After Chapter 7.1, you will be able to:

- Identify the system and its surroundings given a situation involving transfer of heat
- Recall the features of isothermal, adiabatic, isobaric, and isovolumetric processes

Students often have some anxiety over what constitutes a system and what—by exclusion from the system—constitutes the surroundings or environment. Perhaps the problem isn't so much the definitions themselves but the way in which the boundary between the two can be shifted to suit the needs of the experimenter or observer. Simply put, the system is the matter that is being observed—the total amount of reactants and products in a chemical reaction. It could be the amount of solute and solvent used to create a solution. It could be the gas inside a balloon. Then, the surroundings, or environment, are everything outside of that system. However, the boundary between system and surroundings is not permanently fixed and can be moved. For example, one might consider the mass of coffee in a coffee cup to be the system and the cup containing it to

be part of the environment. This setup would likely be used if someone was interested in determining the amount of heat transferred from the hot coffee to the cooler coffee cup. Alternatively, one might define the system as the hot coffee and the cup together, and the environment as the air surrounding the coffee cup. This setup would likely be used if someone was interested in calculating the heat exchange between the hot coffee and cup system and the cooler surrounding air. The boundary can be extended out farther and farther, until the entire mass of the universe is ultimately included in the system. At this point, there are no surroundings. Again, where the boundary is placed is a decision based on what phenomenon one is interested in studying.

THERMODYNAMIC TERMINOLOGY

Systems can be characterized by whether or not they can exchange heat or matter with the surroundings. A system may be characterized as follows:

- Isolated: The system cannot exchange energy (heat and work) or matter with the surroundings; for example, an insulated bomb calorimeter.
- Closed: The system can exchange energy (heat and work) but not matter with the surroundings; for example, a steam radiator.
- Open: The system can exchange both energy (heat and work) and matter with the surroundings; for example, a pot of boiling water.

When a system experiences a change in one or more of its properties (such as concentrations of reactants or products, temperature, or pressure), it undergoes a process.

While processes, by definition, are associated with a change of the state of a system, some processes are uniquely identified by some property that is constant throughout the process. Many of these processes create special conditions because they allow us to simplify the first law of thermodynamics:

$$\Delta U = Q - W$$

Equation 7.1

where ΔU is the change in internal energy of the system, Q is the heat added to the system, and W is the work done by the system.

For example, isothermal processes occur when the system's temperature is constant. Constant temperature implies that the total internal energy of the system (U) is constant throughout the process. This is because temperature and internal energy are directly proportional. When U is constant, $\Delta U = 0$ and the first law simplifies to $Q = W$ (the heat added to the system equals the work done by the system). An isothermal process appears as a hyperbolic curve on a pressure–volume graph (P – V graph). Work is represented by the area under such a curve, as shown in Figure 7.1.

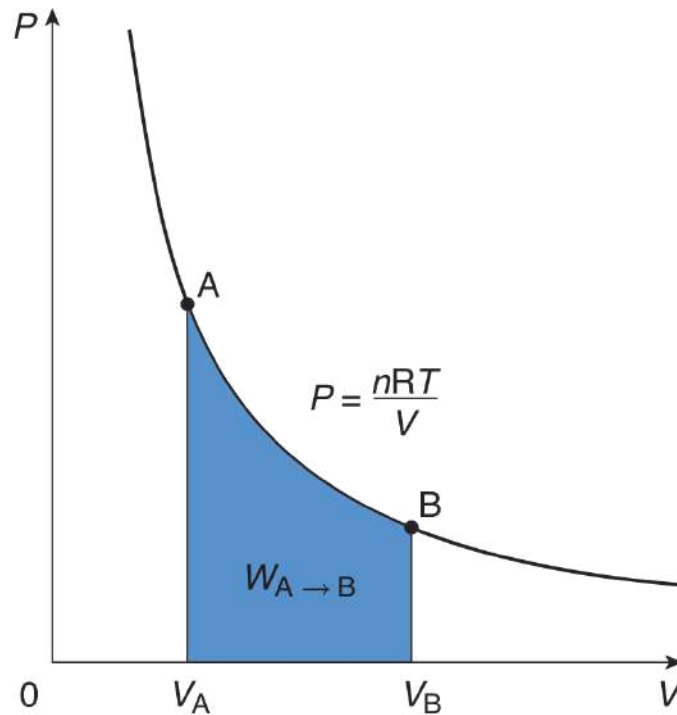


Figure 7.1. Graph of an Isothermal Expansion

Temperature is constant in an isothermal process; thus, the area under the curve represents not only the work performed by the gas, but also the heat that entered the system.

Adiabatic processes occur when no heat is exchanged between the system and the environment; thus, the thermal energy of the system is constant throughout the process. When $Q = 0$, the first law simplifies to $\Delta U = -W$ (the change in internal energy of the system is equal to work done on the system [the opposite of work done by the system]). An adiabatic process also appears hyperbolic on a P - V graph, as shown in Figure 7.2.

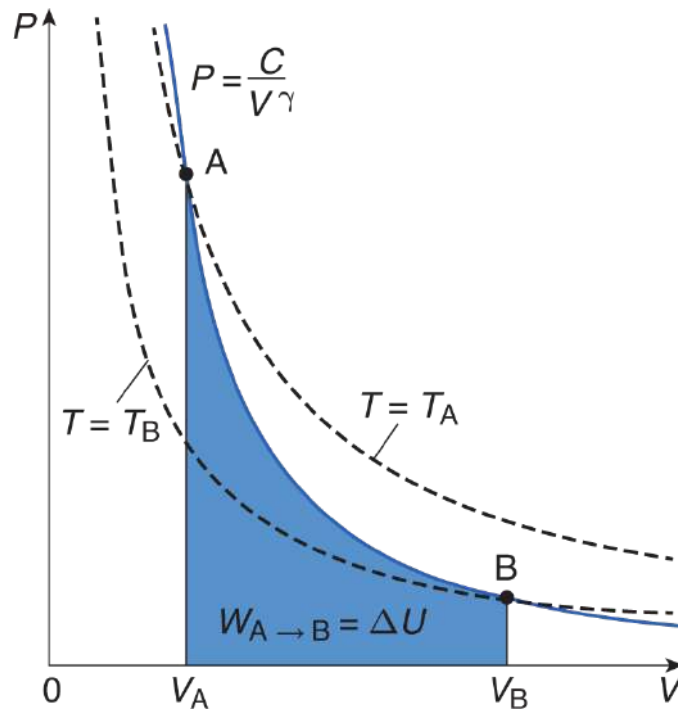


Figure 7.2. Graph of an Adiabatic Expansion

Heat exchange is zero in an adiabatic process; temperature is not constant (as shown by the dotted lines).

Isobaric processes occur when the pressure of the system is constant. Isothermal and isobaric processes are common because it is usually easy to control temperature and pressure. Isobaric processes do not alter the first law, but note that an isobaric process appears as a flat line on a P–V graph, as shown in Figure 7.3.

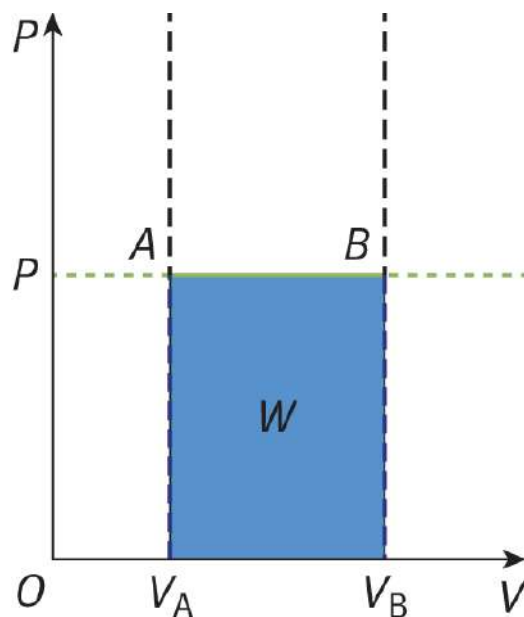


Figure 7.3. Graph of an Isobaric Expansion

Pressure is constant in an isobaric process; the slope of the line is therefore zero.

Finally, isovolumetric (isochoric) processes experience no change in volume. Because the gas neither expands nor compresses, no work is performed in such a process. Thus, the first law simplifies to $\Delta U = Q$ (the change in internal energy is equal to the heat added to the system). An isovolumetric process is a vertical line on a P–V graph; the area under the curve, which represents the work done by the gas, is zero.



BRIDGE

The terms isothermal, adiabatic, isobaric, and isovolumetric (isochoric) may seem familiar because they are also discussed in Chapter 3 of MCAT Physics and Math Review.

Processes themselves can also be classified as spontaneous or nonspontaneous. A spontaneous process is one that can occur by itself without having to be driven by energy from an outside source. Calculating the change in the Gibbs free energy (ΔG) for a process, such as a chemical reaction, allows us to predict whether the process will be spontaneous or nonspontaneous. As discussed later in the chapter, the same quantities that are used to calculate the change in the Gibbs free energy, ΔH and ΔS , can also tell us whether the process will be temperature dependent; that is, spontaneous at some temperatures and nonspontaneous at others.

Spontaneous reactions, as mentioned in Chapters 5 and 6 of MCAT General Chemistry Review, will not necessarily happen quickly and may not go to completion. Many spontaneous reactions have very high activation energies and, therefore, rarely take place. For example, when was the last time you saw a match ignite itself? However, providing a quantity of thermal energy (generated by the friction associated with striking the match) that equals or exceeds the activation energy will allow the match to light and burn spontaneously. At this point, the combustion of the chemical components of the match using molecular oxygen in the air will not need any additional external energy once the activation energy has been supplied.

Some spontaneous reactions proceed very slowly. The role of enzymes—biological catalysts—is to selectively enhance the rate of certain spontaneous (but slow) chemical reactions so that the biologically necessary products can be formed at a rate sufficient for sustaining life. As we discussed in Chapter 6 of MCAT General Chemistry Review, some reactions do not go to completion but settle into a low-energy state called equilibrium. Spontaneous reactions may go to completion, but many

simply reach equilibrium with dynamically stable concentrations of reactants and products. A common method for supplying energy for nonspontaneous reactions is by coupling nonspontaneous reactions to spontaneous ones, as shown in Figure 7.4.

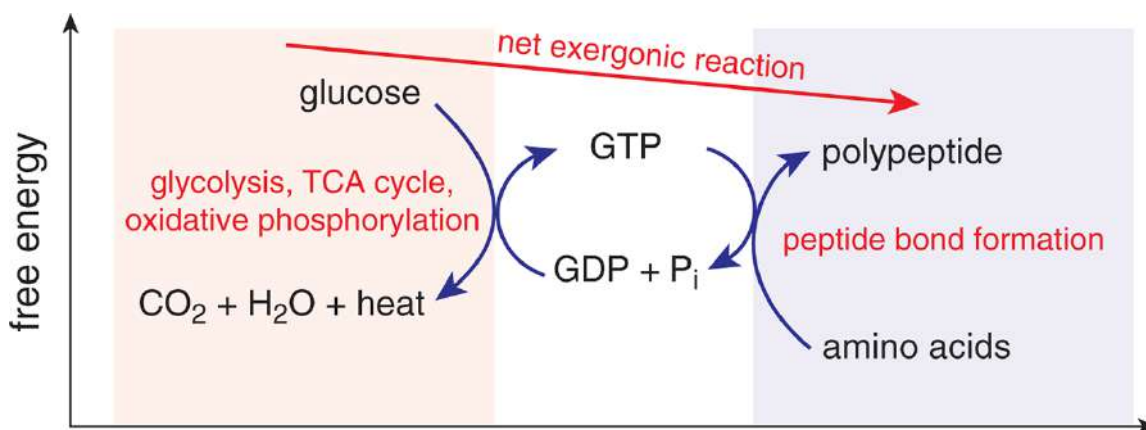


Figure 7.4. Coupling of Reactions

The combustion of glucose is exergonic; the formation of peptide bonds is endergonic. Energy from the combustion of glucose can be stored in the bonds in GTP, which are then lysed to provide the energy for forming peptide bonds.

MCAT Concept Check 7.1:

Before you move on, assess your understanding of the material with these questions.

A person snaps an ice pack and places it on his or her leg. In terms of energy transfer, what would be considered the system and what would be the surroundings in this scenario?

- System:

-
- Surroundings:

What is unique about each of the following types of processes?

- Isothermal:

- Adiabatic:

- Isobaric:

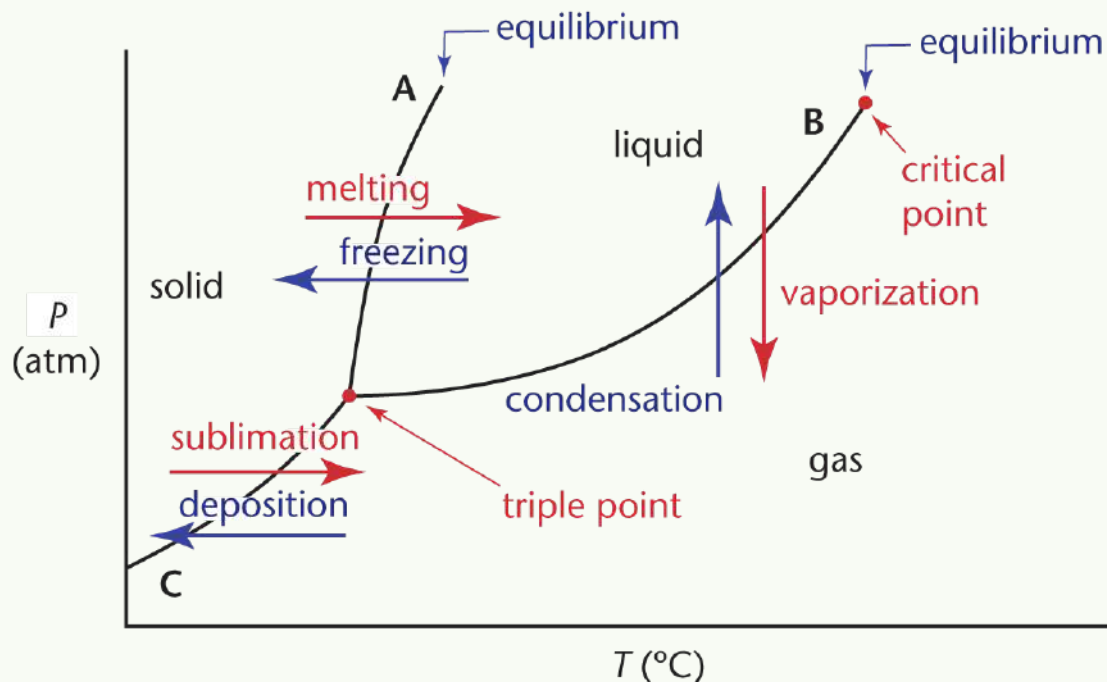
- Isovolumetric (isochoric):

7.2 States and State Functions

LEARNING GOALS

After Chapter 7.2, you will be able to:

- Recall standard conditions and the calculations they are used for
- Distinguish between a state function and a process function
- List the common state functions
- Identify the triple point and critical point on a phase diagram:



The state of a system can be described by certain macroscopic properties. These properties, or state functions, describe the system in an equilibrium state. They cannot describe the process of the system; that is, how the system got to its current equilibrium. They are useful only for comparing one equilibrium state to another. The pathway taken from one equilibrium state to another is described quantitatively by the process functions, the most important of which are work (W) and heat (Q).

OVERVIEW

The state functions include pressure (P), density (ρ), temperature (T), volume (V), enthalpy (H), internal energy (U), Gibbs free energy (G), and entropy (S). When the state of a system changes from one equilibrium to another, one or more of these state functions will change. In addition, while state functions are independent of the path (process) taken, they are not necessarily independent of one another. For example, Gibbs free energy is related to enthalpy, temperature, and entropy.



MNEMONIC

State functions: When I'm under pressure and feeling dense, all I want to do is watch TV and get HUGS.

Pressure (P), density (ρ), temperature (T), volume (V), enthalpy (H), internal energy (U), Gibbs free energy (G), and entropy (S).

Because systems can be in different equilibrium states at different temperatures and pressures, a set of standard conditions has been defined for

measuring the enthalpy, entropy, and Gibbs free energy changes of a reaction. The standard conditions are defined as 25°C (298 K), 1 atm pressure, and 1 M concentrations. Don't confuse standard conditions with standard temperature and pressure (STP), for which the temperature is 0°C (273 K) and pressure is 1 atm. Standard conditions are used for kinetics, equilibrium, and thermodynamics problems; STP is used for ideal gas calculations.



MCAT EXPERTISE

On the MCAT, be sure that you do not confuse standard conditions in thermodynamics with standard temperature and pressure (STP), which is used in gas law calculations:

- Standard conditions: 25°C (298 K), 1 atm pressure, 1 M concentrations
- STP: 0°C (273 K), 1 atm pressure

Under standard conditions, the most stable form of a substance is called the standard state of that substance. You should recognize the standard states for some elements and compounds commonly encountered on the MCAT. For example, H_2 (g), H_2O (l), NaCl (s), O_2 (g), and C (s, graphite) are the most stable forms of these substances under standard conditions. Recognizing whether or not a substance is in its standard state is important for thermochemical calculations, such as heats of reactions and—in particular—heats of formation. The changes in enthalpy, entropy, and free energy that occur when a reaction takes place under standard conditions are called the standard enthalpy, standard entropy, and standard free energy changes, respectively, and are symbolized by ΔH° , ΔS° , and ΔG° . The degree sign in these variables

represents zero, as the standard state is used as the “zero point” for all thermodynamic calculations.

PHASE CHANGES

Phase diagrams are graphs that show the standard and nonstandard states of matter for a given substance in an isolated system, as determined by temperatures and pressures. Phase changes ($\text{solid} \rightleftharpoons \text{liquid} \rightleftharpoons \text{gas}$) are reversible, and an equilibrium of phases will eventually be reached at any given combination of temperature and pressure. For example, at 0°C and 1 atm in an isolated system, ice and water exist in an equilibrium. In other words, some of the ice may absorb heat (from the liquid water) and melt, but because that heat is being removed from the liquid water, an equal amount of the liquid water will freeze and form ice. Thus, the relative amounts of ice and water remain constant. Equilibrium between the liquid and gas states of water will be established in a closed container at room temperature and atmospheric pressure, such as a plastic water bottle with the cap screwed on tightly. Most of the water in the bottle will be in the liquid phase, but a small number of molecules at the surface will gain enough kinetic energy to escape into the gas phase; likewise, a small number of gas molecules will lose sufficient kinetic energy to reenter the liquid phase. After a while, equilibrium is established, and the relative amounts of water in the liquid and gas phases become constant—at standard conditions, equilibrium occurs when the air above the water has about 3 percent water vapor by mass. Phase equilibria are analogous to the dynamic equilibria of reversible chemical reactions: the concentrations of reactants and products are constant because the rates of the forward and reverse reactions are equal.



KEY CONCEPT

As with all equilibria, the rates of the forward and reverse processes will be the same when considering phase changes.

Gas–Liquid Equilibrium

The temperature of any substance in any phase is related to the average kinetic energy of the molecules that make up the substance. However, not all of the molecules have exactly the same instantaneous speeds. Therefore, the molecules possess a range of instantaneous kinetic energy values. In the liquid phase, the molecules are relatively free to move around one another. Some of the molecules near the surface of the liquid may have enough kinetic energy to leave the liquid phase and escape into the gaseous phase. This process is known as evaporation or vaporization. Each time the liquid loses a high-energy particle, the temperature of the remaining liquid decreases. Evaporation is an endothermic process for which the heat source is the liquid water. Of course, the liquid water itself may be receiving thermal energy from some other source, as in the case of a puddle of water drying up under the hot summer sun or a pot of water on the stovetop. Given enough energy, the liquid will completely evaporate.

Boiling is a specific type of vaporization that occurs only under certain conditions. Any liquid will lose some particles to the vapor phase over time; however, boiling is the rapid bubbling of the entire liquid with rapid release of the liquid as gas particles. While evaporation happens in all liquids at all temperatures, boiling can only occur above the boiling point of a liquid and involves vaporization through the entire volume of the liquid.

In a covered or closed container, the escaping molecules are trapped above the solution. These molecules exert a countering pressure, which forces some of the gas back into the liquid phase; this process is called condensation. Condensation is facilitated by lower temperature or higher pressure. Atmospheric pressure acts on a liquid in a manner similar to that of an actual physical lid. As evaporation and condensation proceed, the respective rates of the two processes become equal, and equilibrium is reached. The pressure that the gas exerts over the liquid at equilibrium is the vapor pressure of the liquid. Vapor pressure increases as temperature increases because more molecules have sufficient kinetic energy to escape into the gas phase. The temperature at which the vapor pressure of the liquid equals the ambient (also known as external, applied, or incident) pressure is called the boiling point.

Liquid–Solid Equilibrium

We've already illustrated the equilibrium that can exist between the liquid and the solid phases of water at 0°C. Even though the atoms or molecules of a solid are confined to specific locations, each atom or molecule can undergo motions about some equilibrium position. These vibrational motions increase when heat is applied. From our understanding of entropy, we can say that the availability of energy microstates increases as the temperature of the solid increases. In basic terms, this means that the molecules have greater freedom of movement, and energy disperses. If atoms or molecules in the solid phase absorb enough energy, the three-dimensional structure of the solid will break down, and the atoms or molecules will escape into the liquid phase. The transition from solid to liquid is called fusion or melting. The reverse process, from liquid to solid, is called solidification, crystallization, or freezing. The temperature at which these processes occur is called the melting point or freezing point, depending on the direction of the transition. Whereas pure

crystalline solids have distinct, very precise melting points, amorphous solids, such as glass, plastic, chocolate, and candle wax, tend to melt (or solidify) over a larger range of temperatures due to their less-ordered molecular structure.

Gas–Solid Equilibrium

The final phase equilibrium is that which exists between the gaseous and solid phases. When a solid goes directly into the gas phase, the process is called sublimation. Dry ice (solid CO_2) sublimates at room temperature and atmospheric pressure; the absence of the liquid phase makes it a convenient dry refrigerant. The reverse transition, from the gaseous to the solid phase, is called deposition. In organic chemistry laboratories, a device known as a cold finger may be used to purify a product that is heated under reduced pressure, causing it to sublime. The desired product is usually more volatile than the impurities, so the gas is purer than the original product and the impurities are left in the solid state. The gas then deposits onto the cold finger, which has cold water flowing through it, yielding a purified solid product that can be collected.

PHASE DIAGRAMS

Phase diagrams are graphs that show the temperatures and pressures at which a substance will be thermodynamically stable in a particular phase. They also show the temperatures and pressures at which phases will be in equilibrium.

The lines on a phase diagram are called the lines of equilibrium or the phase boundaries and indicate the temperature and pressure values for the

equilibria between phases. The lines of equilibrium divide the diagram into three regions corresponding to the three phases—solid, liquid, and gas—and they themselves represent the phase transformations. The phase diagram for a single compound is shown in Figure 7.5.

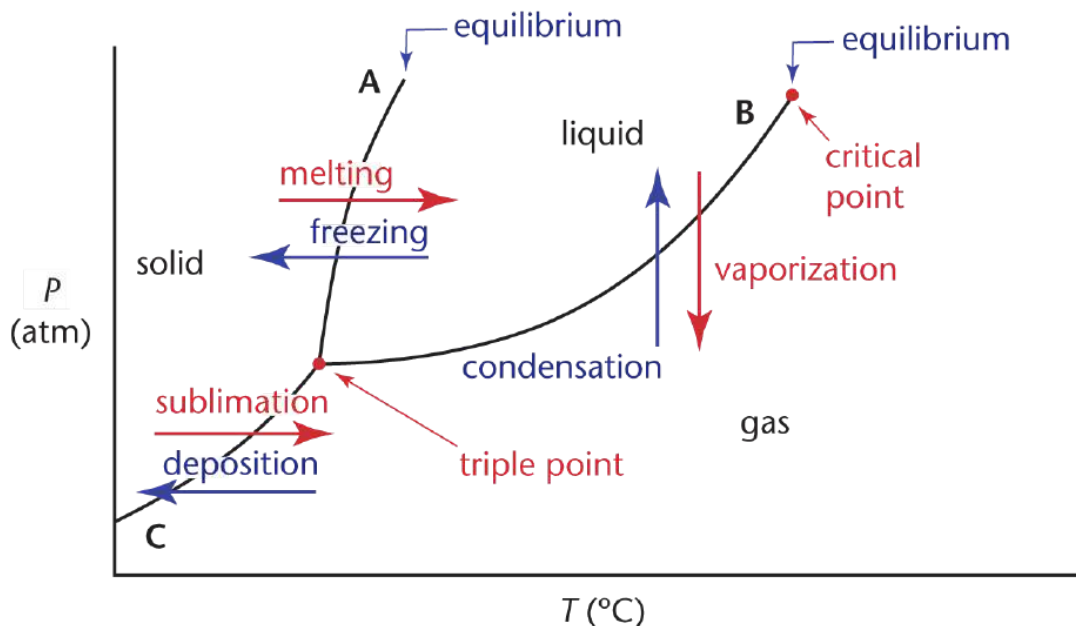


Figure 7.5. Phase Diagram for a Single Compound



MCAT EXPERTISE

On the MCAT, you should be able to identify and understand each area and every line of a phase diagram.

Line A represents the solid–liquid interface, line B the liquid–gas interface, and line C the solid–gas interface. In general, the gas phase is found at high temperatures and low pressures, the solid phase is found at low temperatures and high pressures, and the liquid phase is found at moderate temperatures

and moderate pressures. The point at which the three phase boundaries meet is called the triple point. This is the temperature and pressure at which the three phases exist in equilibrium. The phase boundary that separates the solid and the liquid phases extends indefinitely from the triple point. The phase boundary between the liquid and gas phases, however, terminates at a point called the critical point. This is the temperature and pressure above which there is no distinction between the phases. Although this may seem to be an impossibility—after all, it’s always possible to distinguish between the liquid and the solid phase—such supercritical fluids are perfectly logical. As a liquid is heated in a closed system its density decreases and the density of the vapor sitting above it increases. The critical point is the temperature and pressure at which the two densities become equal and there is no distinction between the two phases. The heat of vaporization at this point and for all temperatures and pressures above the critical point values is zero.



REAL WORLD

Because of water’s unique properties, ice floats and skates flow smoothly over ice rinks. This all “boils” down to the negative slope of the solid–liquid equilibrium line in its phase diagram. Because the density of ice is less than that of liquid water, an increase in pressure (at a constant temperature) will actually melt ice (the opposite of what is seen for the substance in Figure 7.5).

MCAT Concept Check 7.2:

Before you move on, assess your understanding of the material with these questions.

What are standard conditions? When are standard conditions used for calculations?

What is the definition of a state function? A process function?

- State function:

- Process function:

List at least five common state functions:

-

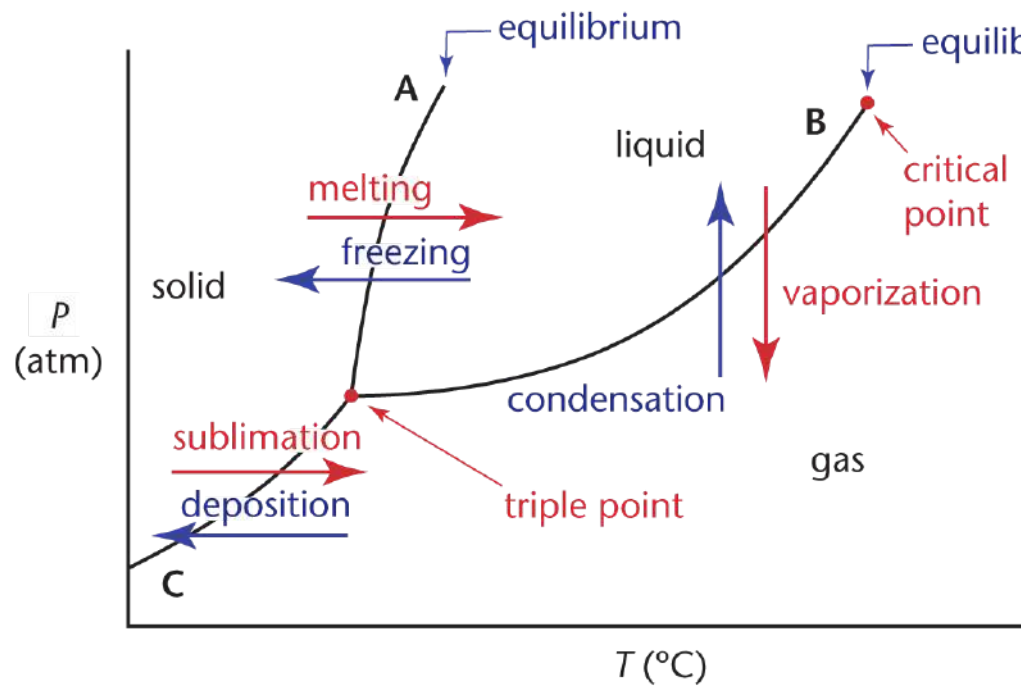
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Identify the triple point and critical point on the diagram below. What is the definition of the triple point? The critical point?



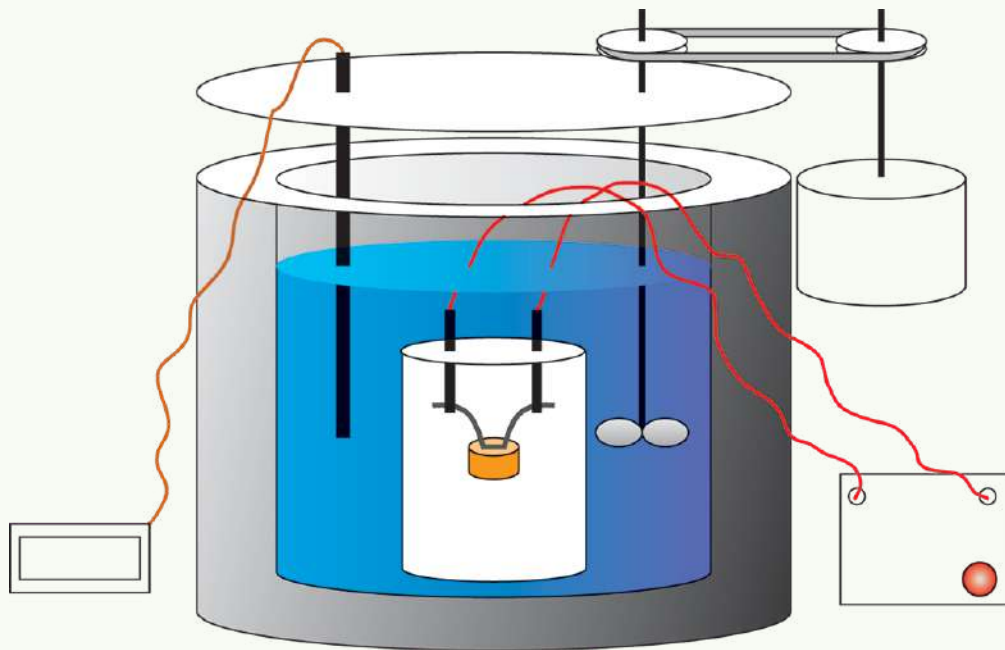
- Triple point:

- Critical point:

LEARNING GOALS

After Chapter 7.3, you will be able to:

- Differentiate between temperature and heat
- Compare specific heat and heat capacity
- Recall the specific heat of water
- Describe the processes for constant-volume and constant-pressure calorimetry:



Before we can examine the first of the four state functions that are the focus of this chapter, we must address the topic of heat, which is a source of confusion for many students. Perhaps the greatest barrier to a proper understanding of heat is the semantic conflation of the terms heat and temperature. Many people use these terms interchangeably in everyday conversation, but this obscures the lexicon of thermodynamics.

Temperature (T) is related to the average kinetic energy of the particles of a substance. Temperature is the way that we scale how hot or cold something is. We are familiar with a few temperature scales: Fahrenheit, Celsius, and Kelvin. The average kinetic energy of the particles in a substance is related to the thermal energy (enthalpy) of the substance, but because we must also include consideration of how much substance is present to calculate total thermal energy content, the most we can say about temperature is that when a substance's thermal energy increases, its temperature also increases. Nevertheless, we cannot say that something that is hot necessarily has greater thermal energy (in absolute terms) than a substance that is cold. For example, we might determine that a large amount of lukewarm water has a greater total heat content than a very small amount of hot water.



KEY CONCEPT

Remember that heat and temperature are different. Heat is a specific form of energy that can enter or leave a system, while temperature is a measure of the average kinetic energy of the particles in a system.

The absolute temperature scale, Kelvin, was determined via the third law of thermodynamics, which elucidated that there is a finite limit to temperature below which nothing can exist. There can be no temperature below 0 K because, by definition, the system is said to be unable to lose any more heat energy. Quantum mechanics describes a state of molecular motions possible below absolute zero, but this is beyond the scope of the MCAT.

OVERVIEW

Heat (Q) is the transfer of energy from one substance to another as a result of their differences in temperature. In fact, the zeroth law of thermodynamics implies that objects are in thermal equilibrium only when their temperatures are equal. Heat is therefore a process function, not a state function: we can quantify how much thermal energy is transferred between two or more objects as a result of their difference in temperatures by measuring the heat transferred.

Remember that the first law of thermodynamics states that the change in the total internal energy (ΔU) of a system is equal to the amount of heat (Q) transferred to the system minus the amount of work (W) done by the system: $\Delta U = Q - W$.

Because heat and work are measured independently, we can assess the transfer of energy in the form of heat through any process regardless of the amount of work done. Processes in which the system absorbs heat are called endothermic ($\Delta Q > 0$), while those processes in which the system releases heat are called exothermic ($\Delta Q < 0$). The unit of heat is the unit of

energy: joule (J) or calorie (cal), for which $1 \text{ cal} = 4.184 \text{ J}$. Enthalpy (ΔH) is equivalent to heat (Q) under constant pressure, which is an assumption the MCAT usually makes for thermodynamics problems.



REAL WORLD

One of the most important ways that the body works to prevent overheating is through the production of sweat—an exocrine secretion of water, electrolytes, and urea. However, it is not the production of sweat that is the cooling mechanism. It's the evaporation of the sweat that helps cool the body. Evaporation (vaporization) from the liquid to gas phase is an endothermic process: energy must be absorbed from the body for the particles of the liquid to gain enough kinetic energy to escape into the gas phase. Hot, arid desert air has a lower partial pressure of water vapor than humid, tropical air, so sweat vaporizes more readily in the dry air than it does in the humid air. Accordingly, most people will feel more comfortable in dry heat than in humid heat.

When substances of different temperatures are brought into thermal contact with each other—that is, some physical arrangement that allows heat transfer—energy will move from the warmer substance to the cooler substance. When a substance undergoes an endothermic or exothermic reaction, heat energy will be exchanged between the system and the environment.

The process of measuring transferred heat is called calorimetry. Two basic types of calorimetry include constant-pressure calorimetry and constant-volume calorimetry. The coffee-cup calorimeter, introduced at the beginning of this chapter, is a low-tech example of a constant-pressure calorimeter, while a bomb calorimeter is an example of a constant-volume calorimeter.



MNEMONIC

The equation for heat transfer, given a specific heat, is the same as the test you're studying for! $q = mc\Delta T$ looks a lot like "q equals MCAT."

The heat (q) absorbed or released in a given process is calculated via the equation:

$$q = mc\Delta T$$

Equation 7.2

where m is the mass, c is the specific heat of the substance, and ΔT is the change in temperature (in Celsius or kelvins). Specific heat is defined as the amount of energy required to raise the temperature of one gram of a substance by one degree Celsius (or one kelvin). Specific heat values will

generally be provided on Test Day, but one constant to remember is the

specific heat of $\text{H}_2\text{O} (l)$: $c_{\text{H}_2\text{O}} = 1 \frac{\text{cal}}{\text{g} \cdot \text{K}}$.



REAL WORLD

When walking barefoot, a blacktop feels much hotter than a wooden walkway even when they are the same temperature. This is because they have different specific heats.

It requires less heat to raise the temperature of a glass of water the same amount as a swimming pool. While these two items have the same specific heat, c , they have different heat capacities—the product mc (mass times specific heat).

CONSTANT-PRESSURE AND CONSTANT-VOLUME CALORIMETRY

To picture the setup of a constant-pressure calorimeter, just think of the coffee-cup calorimeter: an insulated container covered with a lid and filled with a solution in which a reaction or some physical process, such as dissolution, is occurring. The incident pressure, which is atmospheric pressure, remains constant throughout the process, and the temperature can be measured as the reaction progresses. There should be sufficient thermal insulation (such as Styrofoam) to ensure that the heat being

measured is an accurate representation of the reaction, without gain or loss of heat to the environment. Other commercial applications of these same principles include home insulation, padded clothing, and certain food containers such as thermoses.



REAL WORLD

Tests looking at plasma proteins or cancer diagnostics in medicine have utilized differential scanning calorimetry (DSC), which is a constant-pressure device, to identify blood components. Such results have shown that thermal properties of major plasma proteins are altered from early- to late-stage tumors.

The term bomb calorimeter may sound rather ominous, but a more accurate descriptive term is decomposition vessel. This better reflects what is actually taking place in constant-volume calorimetry. As shown in Figure 7.6, a sample of matter, typically a hydrocarbon, is placed in the steel decomposition vessel, which is then filled with almost pure oxygen gas. The decomposition vessel is then placed in an insulated container holding a known mass of water. The contents of the decomposition vessel are ignited by an electric ignition mechanism. The material combusts (burns) in the presence of the oxygen, and the heat that evolves is the heat of the combustion reaction. Because $W = P\Delta V$, no work is done in an isovolumetric process ($\Delta V = 0$), so $W_{\text{calorimeter}} = 0$. Furthermore, because of the insulation, the whole calorimeter can be considered isolated from the rest of the universe, so we can identify the system as the sample plus the oxygen and steel vessel, and the surroundings as the water.

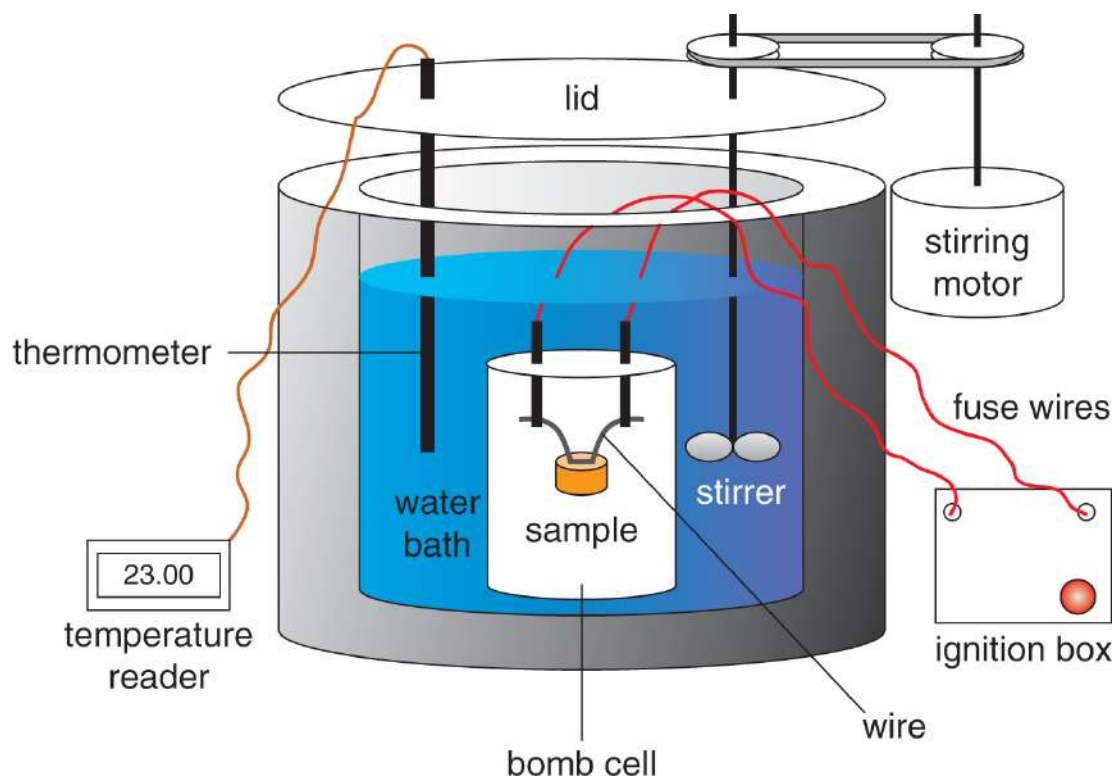


Figure 7.6. Diagram of a Bomb Calorimeter



REAL WORLD

Bomb calorimeters have helped elucidate the thermodynamic properties of various chemical compounds, including food additives, to determine nutritional value (the caloric content of the additive).

Because no heat is exchanged between the calorimeter and the rest of the universe, $Q_{\text{calorimeter}}$ is 0. So,

$$\Delta U_{\text{system}} + \Delta U_{\text{surroundings}} = \Delta U_{\text{calorimeter}} = Q_{\text{calorimeter}} - W_{\text{calorimeter}} = 0$$

Therefore,

$$\Delta U_{\text{system}} = -\Delta U_{\text{surroundings}}$$

and because no work is done,

$$q_{\text{system}} = -q_{\text{surroundings}}$$
$$m_{\text{steel}} c_{\text{steel}} \Delta T + m_{\text{oxygen}} c_{\text{oxygen}} \Delta T = -m_{\text{water}} c_{\text{water}} \Delta T$$

Note that by using the layer of insulation to isolate the entire calorimeter from the rest of the universe, we've created an adiabatic process. This means that no heat is exchanged between the calorimeter and the rest of the universe, but it is exchanged between the steel decomposition vessel and the surrounding water. As the previous derivation shows, heat exchange between the system and its surroundings makes it possible for us to calculate the heat of combustion.



MCAT EXPERTISE

Knowing that heat can transfer energy from a system to the surroundings is a key concept tested on calorimetry questions. Whenever asked about equilibrium questions regarding the final temperature of a two-liquid (or liquid–solid) system, remember that the colder object gains thermal energy and the hotter object loses it. You should instinctively realize that a metal bar at 1000 K is hotter than a bath of water at 298 K even though water has a high specific heat. Thus, set up the equation as $q_{\text{cold}} = -q_{\text{hot}}$. This form of

the equation avoids the pesky sign notation issues in the ΔT equation encountered in most general chemistry texts.

Example:

One cup containing 100 grams of water at 300 K is mixed into another cup containing 200 g of water at 450 K. What is the equilibrium temperature of the system? (Note: Assume that the pressure is sufficiently high to avoid boiling.)

Solution:

The two liquids undergo thermal exchange; thus, the heat given off by one liquid will be equal to the heat absorbed by the other.

$$q_{\text{cold}} = -q_{\text{hot}}$$
$$m_{\text{cold}} c_{\text{H}_2\text{O}} \Delta T_{\text{cold}} = m_{\text{hot}} c_{\text{H}_2\text{O}} (-\Delta T_{\text{hot}})$$

Now plug in the values from the question. Because we are solving for final (equilibrium) temperature of a mixture, we can use any value of c so long as we are consistent for both liquids (in this case we have two quantities of water and will use $c_{\text{H}_2\text{O}} = 1 \frac{\text{cal}}{\text{g} \cdot \text{K}}$).

$$m_{\text{cold}}c_{\text{H}_2\text{O}}\Delta T_{\text{cold}} = m_{\text{hot}}c_{\text{H}_2\text{O}}(-\Delta T_{\text{hot}})$$

$$(100 \text{ g})\left(1\frac{\text{cal}}{\text{g}\cdot\text{K}}\right)(T_f - 300 \text{ K}) = (200 \text{ g})\left(1\frac{\text{cal}}{\text{g}\cdot\text{K}}\right)(450 \text{ K} - T_f)$$

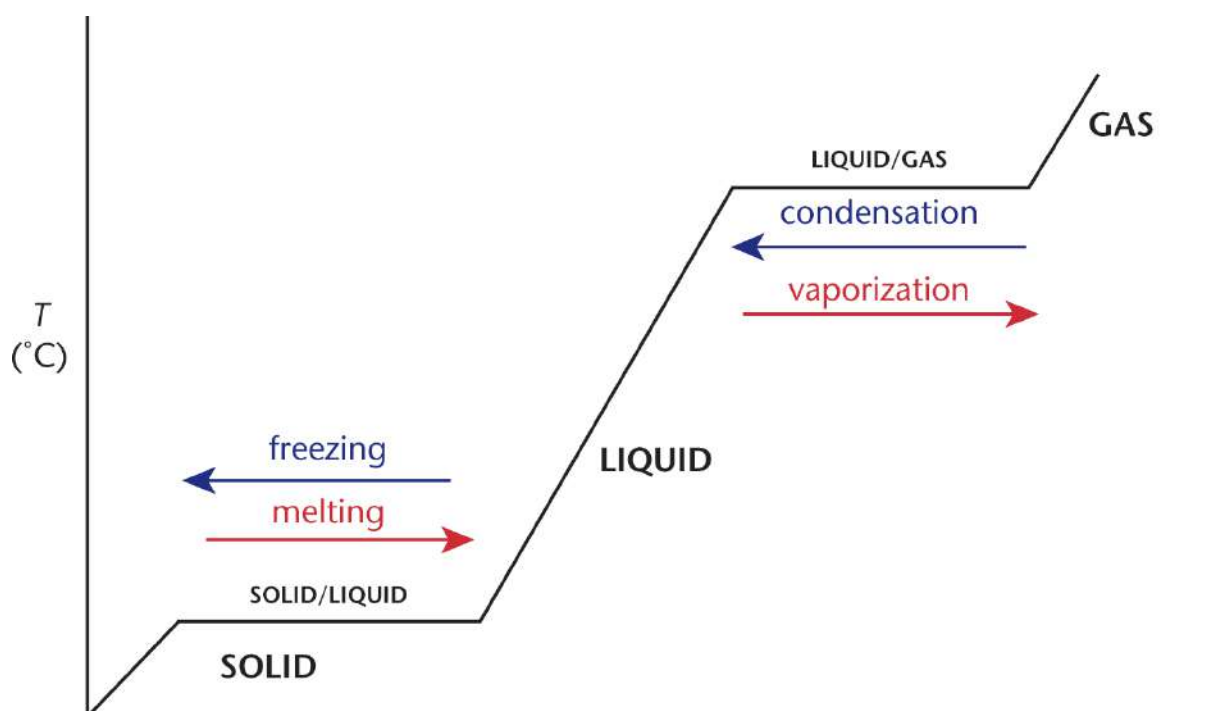
$$100T_f - 30,000 \text{ cal} = 90,000 \text{ cal} - 200T_f$$

$$300T_f = 120,000$$

$$T_f = \frac{120,000}{300} = 400 \text{ K}$$

HEATING CURVES

When a compound is heated, the temperature rises until the melting or boiling point is reached. Then, the temperature remains constant as the compound is converted to the next phase (liquid or gas, respectively). Once the entire sample is converted, then the temperature begins to rise again. This is depicted in the heating curves in Figure 7.7.



heat added 

Figure 7.7. Heating Curve for a Single Compound

Heating curves show that phase change reactions do not undergo changes in temperature. For this reason, we cannot use $q = mc\Delta T$ during this interval because $\Delta T = 0$. We know intuitively that heat must continue to be added in order for the whole solid to melt, so where does this heat go? The solid absorbs energy, which allows particles to overcome the attractive forces that hold them in a rigid, three-dimensional arrangement. When melting an ice cube, all of the heat added during the process is used to overcome the intermolecular forces between water molecules in ice, forming liquid water. Once all of the ice has been turned into liquid water, the temperature of the liquid water can then increase again. The converse is also true: removing heat from a liquid at the solid–liquid phase transition temperature will cause the formation of a rigid lattice of water molecules.

During phase changes, we must use values based on enthalpy. When transitioning at the solid–liquid boundary, the enthalpy (or heat) of fusion (ΔH_{fus}) must be used to determine the heat transferred during the phase change. When transitioning from solid to liquid, the change in enthalpy will be positive because heat must be added; when transitioning from a liquid to a solid, the change in enthalpy will be negative because heat must be removed. At the liquid–gas boundary, the enthalpy (or heat) of vaporization (ΔH_{vap}) must be used, and its sign convention also follows a similar pattern. These are utilized in the equation

$$q = mL$$

Equation 7.3

where m is the mass and L is the latent heat, a general term for the enthalpy of an isothermal process, given in the units $\frac{\text{cal}}{\text{g}}$.



KEY CONCEPT

We need a different formula to calculate q during phase changes when $\Delta T = 0$. If we used $q = mc\Delta T$, we'd erroneously think $q = 0$.

The total amount of heat needed to cross multiple phase boundaries is simply a summation of the heats for changing the temperature of each of the respective phases and the heats associated with phase changes.

Example:

What amount of energy is required to change a 90 gram ice cube at -10°C to vapor at 110°C ? (Note: $c_{\text{H}_2\text{O}}(l) = 4.18 \frac{\text{J}}{\text{g} \cdot \text{K}}$, $c_{\text{H}_2\text{O}}(s) = 2.18 \frac{\text{J}}{\text{g} \cdot \text{K}}$, $c_{\text{H}_2\text{O}}(g) = 2.18 \frac{\text{J}}{\text{g} \cdot \text{K}}$, $\Delta H_{\text{fus}} = 6.02 \frac{\text{kJ}}{\text{mol}}$, $\Delta H_{\text{vap}} = 40.67 \frac{\text{kJ}}{\text{mol}}$)

Solution:

Some of the constants given are in terms of mass (g), and some are in terms of moles, so we should convert the mass (90 g) to moles:

$$n = \frac{90 \text{ g H}_2\text{O}}{18 \frac{\text{g}}{\text{mol}}} = \frac{10}{2} = 5 \text{ moles H}_2\text{O}$$

Because we are beginning in the ice phase, we must heat the ice cube to the solid–liquid phase transition, which occurs at 0°C. This first step involves a change in temperature, so we must use the heat formula that contains ΔT and all the pertinent variables for ice (solid water). Also, it is important to match all results in terms of J and kJ for the different steps of the calculation.

$$q_1 = m_{\text{ice}} c_{\text{ice}} \Delta T_1$$

$$q_1 = (90 \text{ g}) \left(2.18 \frac{\text{J}}{\text{g} \cdot \text{K}} \right) (10 \text{ K}) \approx 90 \times 2 \times 10 = 1800 \text{ J} =$$

In step 2, we must convert the ice into liquid form. During this phase change, there will be no temperature change.

$$q_2 = mL = n_{\text{H}_2\text{O}} \Delta H_{\text{fus}}$$

$$q_2 = (5 \text{ mol}) \left(6.02 \frac{\text{kJ}}{\text{mol}} \right) \approx 5 \times 6 = 30 \text{ kJ}$$

In step 3, we heat the water to its liquid–gas phase transition temperature at 100°C.

$$q_3 = m_{\text{water}} c_{\text{water}} \Delta T_3$$

$$q_3 = (90 \text{ g}) \left(4.18 \frac{\text{J}}{\text{g} \cdot \text{K}} \right) (100 \text{ K}) \approx 90 \times 4 \times 100 = 36,000 \text{ J} =$$

In step 4, we vaporize the water. Again, no temperature change will occur during this phase change.

$$q_4 = mL = n_{\text{H}_2\text{O}} \Delta H_{\text{vap}}$$

$$q_4 = (5 \text{ mol}) \left(40.67 \frac{\text{kJ}}{\text{mol}} \right) \approx 5 \times 40 = 200 \text{ kJ}$$

In step 5, we must finally heat the water to the target temperature of 110°C.

$$q_5 = m_{\text{steam}} c_{\text{steam}} \Delta T_5$$

$$q_5 = (90 \text{ g}) \left(2.00 \frac{\text{J}}{\text{g} \cdot \text{K}} \right) (10 \text{ K}) = 90 \times 2 \times 10 = 1800 \text{ J} =$$

The total heat required for this whole phase change from beginning to end is:

$$q_{\text{tot}} = q_1 + q_2 + q_3 + q_4 + q_5$$

$$q_{\text{tot}} = 1.8 + 30 + 36 + 200 + 1.8$$

$$q_{\text{tot}} \approx 2 + 30 + 36 + 200 + 2 = 270 \text{ kJ} \text{ (actual value} = 27$$

A question this involved is unlikely to be seen on the MCAT because so many steps must be calculated. However, understanding the

significance and rationale of this calculation is definitely within the scope of the test.



MCAT EXPERTISE

It is not in your best interest to memorize all the possible values for c and ΔH for Test Day. The MCAT will provide constants as needed, especially if the system is not water. That being said, practicing with heat calculations for water solutions and gaining familiarity with the heat capacities of water will help on Test Day.

MCAT Concept Check 7.3:

Before you move on, assess your understanding of the material with these questions.

Contrast temperature and heat.

- Temperature:

- Heat:

Contrast specific heat and heat capacity.

- Specific heat:

- Heat capacity:

Contrast constant-volume and constant-pressure calorimetry.

- Constant-volume:

- Constant-pressure:

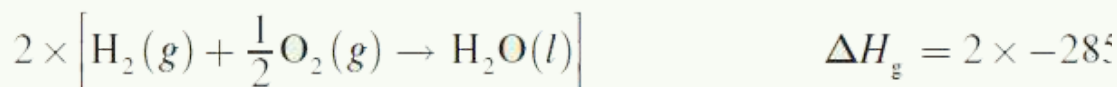
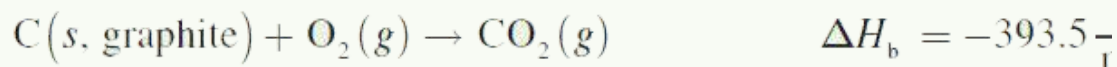
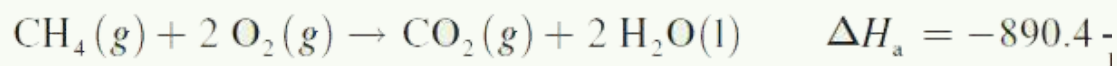
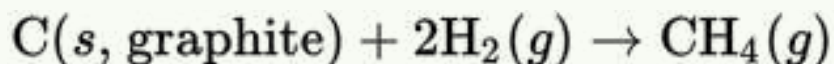
What is the specific heat of liquid water (in calories)?

7.4 Enthalpy

LEARNING GOALS

After Chapter 7.4, you will be able to:

- Distinguish between endothermic and exothermic reactions
- Determine the enthalpy of a molecule or atom given reaction data:



Most reactions in the laboratory occur under constant pressure (at 1 atm) in closed thermodynamic systems. To express heat changes at constant pressure, chemists use the term enthalpy (H). Enthalpy is a state function, so we can calculate the change in enthalpy (ΔH) for a system that has undergone a process—for example, a chemical reaction—by comparing the enthalpy of the final state to the enthalpy of the initial state, irrespective of

the path taken. The change in enthalpy is equal to the heat transferred into or out of the system at constant pressure. To find the enthalpy change of a reaction, ΔH_{rxn} , one must subtract the enthalpy of the reactants from the enthalpy of the products:

$$\Delta H_{\text{rxn}} = H_{\text{products}} - H_{\text{reactants}}$$

Equation 7.4

A positive ΔH_{rxn} corresponds to an endothermic process, and a negative ΔH_{rxn} corresponds to an exothermic process. It is not possible to measure enthalpy directly; only ΔH can be measured, and only for certain fast and spontaneous processes. Thus, several methods have been developed to calculate ΔH for any process.

STANDARD HEAT OF FORMATION

The standard enthalpy of formation of a compound, ΔH_f° , is the enthalpy required to produce one mole of a compound from its elements in their standard states. Remember that standard state refers to the most stable physical state of an element or compound at 298 K and 1 atm. Note that ΔH_f° of an element in its standard state, by definition, is zero. The ΔH_f° values of most known substances are tabulated. You do not need to memorize these values because they will be provided for you.

STANDARD HEAT OF REACTION

The standard enthalpy of a reaction, $\Delta H_{\text{rxn}}^{\circ}$, is the enthalpy change accompanying a reaction being carried out under standard conditions. This can be calculated by taking the difference between the sum of the standard heats of formation for the products and the sum of the standard heats of formation of the reactants:

$$\Delta H_{\text{rxn}}^{\circ} = \sum \Delta H_{\text{f,products}}^{\circ} - \sum \Delta H_{\text{f,reactants}}^{\circ}$$

Equation 7.5

HESS'S LAW

Enthalpy is a state function and is a property of the equilibrium state, so the pathway taken for a process is irrelevant to the change in enthalpy from one equilibrium state to another. As a consequence of this, Hess's law states that enthalpy changes of reactions are additive. When thermochemical equations (chemical equations for which energy changes are known) are added to give the net equation for a reaction, the corresponding heats of reaction are also added to give the net heat of reaction, as shown in Figure 7.8.

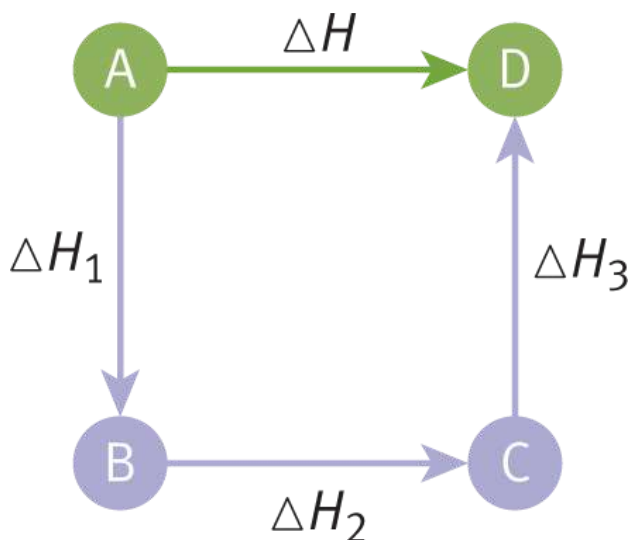


Figure 7.8. Illustration of Hess's Law: Forming Product (D) from Reactant (A)

Because enthalpy is a state function, $\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3$

Hess's law is embodied in the enthalpy equations we've already introduced. For example, we can describe any reaction as the result of breaking down the reactants into their component elements, then forming the products from these elements. The enthalpy change for the reverse of any reaction has the same magnitude, but the opposite sign, as the enthalpy change for the forward reaction. Therefore,

$$\Delta H_{\text{reactants} \rightarrow \text{elements}} = -\Delta H_{\text{elements} \rightarrow \text{reactants}}$$

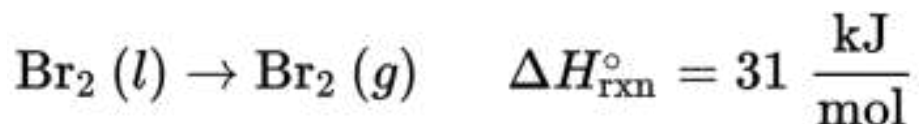
The ΔH_{rxn} can be written as:

$$\Delta H_{\text{rxn}} = \Delta H_{\text{reactants} \rightarrow \text{elements}} + \Delta H_{\text{elements} \rightarrow \text{products}}$$

which is another way of writing

$$\Delta H_{\text{rxn}}^{\circ} = \sum \Delta H_{\text{f,products}}^{\circ} - \sum \Delta H_{\text{f,reactants}}^{\circ}$$

Consider the following phase change:



The enthalpy change for the phase change is called the heat of vaporization ($\Delta H_{\text{vap}}^{\circ}$). As long as the initial and final states exist at standard conditions, the $\Delta H_{\text{rxn}}^{\circ}$ will always equal the $\Delta H_{\text{vap}}^{\circ}$, irrespective of the particular pathway that the process takes. For example, it's possible that $\text{Br}_2 (l)$ could first decompose to Br atoms, which then recombine to form $\text{Br}_2 (g)$, rather than simply boiling from the liquid to gaseous state. However, because the net reaction is the same, the change in enthalpy will be the same.



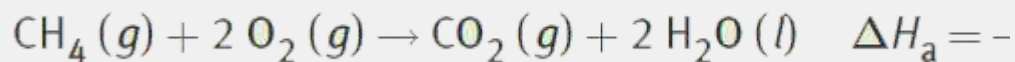
KEY CONCEPT

State functions are always path independent.

Example:

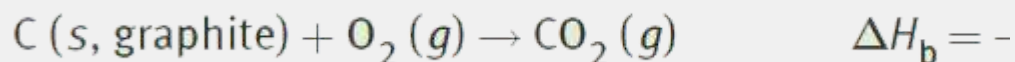
Given the following thermochemical equations:

)



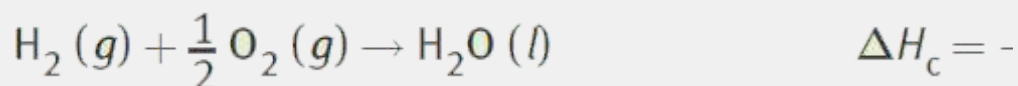
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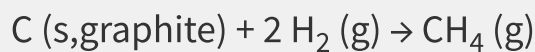
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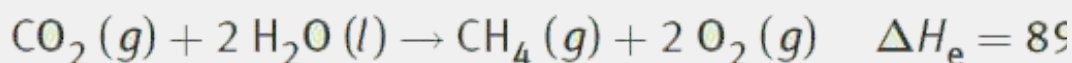
Calculate ΔH for this reaction:

l)

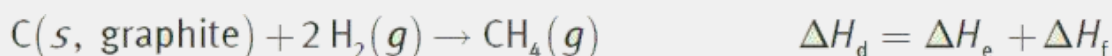
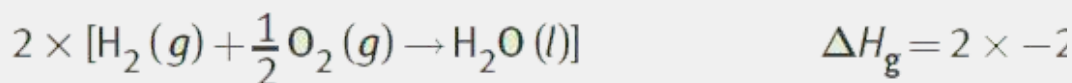
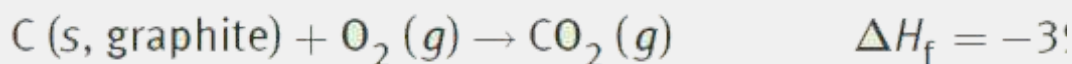


Solution:

Equations (a), (b), and (c) must be combined to obtain equation (d). Because equation (d) contains only C, H₂, and CH₄, we must eliminate O₂, CO₂, and H₂O from the first three equations. Equation (a) is reversed to move CH₄ to the product side (equation (e) below). Next, equation (b) is left as is (we will call this equation (f) for consistency below) and (c) is multiplied by 2 (equation (g) below). Then, (d) can be calculated from (e) + (f) + (g):



◀ ▶



$$\Delta H_{\text{d}} = 890.4 + (-393.5) + [2 \times (-285.3)] \approx 900 + (-400) - (2 \times 285.3)$$

$$\Delta H_{\text{d}} = 900 - 400 - 570 = -70 \frac{\text{kJ}}{\text{mol}} \quad \left(\text{actual value} = -74.7 \frac{\text{kJ}}{\text{mol}} \right)$$

It is important to realize that Hess's law applies to any state function, including entropy and Gibbs free energy.



MCAT EXPERTISE

When doing a problem like this on the MCAT, make sure to switch signs when you reverse the equation. Also, make sure to multiply by the correct stoichiometric coefficients when performing your calculations.

BOND DISSOCIATION ENERGY

Hess's law can also be expressed in terms of bond enthalpies, also called bond dissociation energies. Bond dissociation energy is the average energy that is required to break a particular type of bond between atoms in the gas phase—remember, bond dissociation is an endothermic process. Bond dissociation energy is given in the units $\frac{\text{kJ}}{\text{mol of bonds broken}}$ and is often given in tables on the MCAT in a format similar to Table 7.1.

Bond	Enthalpy $\left(\Delta H, \frac{\text{kJ}}{\text{mol}}\right)$
O=O	498
C-H	415
H-H	436

Table 7.1 Sample Bond Enthalpies

Bond enthalpies are the averages of the bond energies for the same bond in many different compounds. For example, the C-H bond enthalpy $\left(415 \frac{\text{kJ}}{\text{mol}}\right)$ is averaged from measurements of the individual C-H bond enthalpies of thousands of different organic compounds. Note that bond formation, the opposite of bond breaking, has the same magnitude of energy but is negative rather than positive; that is, energy is released when bonds are formed. Remember that atoms generally form bonds to become more stable (often by completing an octet). Thus, it makes sense that bond formation is exothermic and bond dissociation is endothermic. The enthalpy change associated with a reaction is given by

$$\Delta H_{\text{rxn}}^{\circ} = \sum \Delta H_{\text{bonds broken}} - \sum \Delta H_{\text{bonds formed}} = \text{total energy absorbed} - \text{total energy released}$$

Equation 7.6

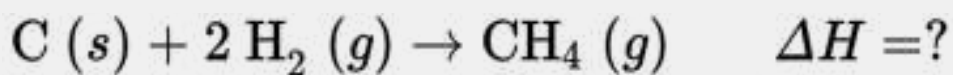


KEY CONCEPT

Because it takes energy to pull two atoms apart, bond breakage is generally endothermic. The reverse process, bond formation, is generally exothermic.

Example:

Calculate the enthalpy change for the following reaction:



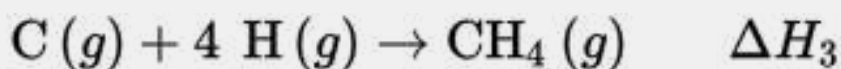
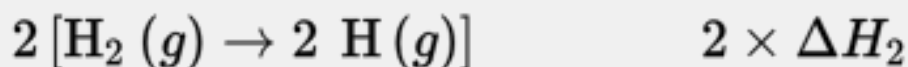
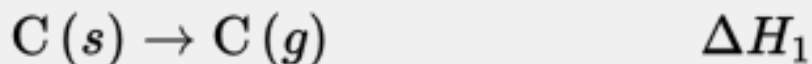
Bond dissociation energies of H–H and C–H bonds are

$436 \frac{\text{kJ}}{\text{mol}}$ and $415 \frac{\text{kJ}}{\text{mol}}$, respectively. The ΔH_f of C (g) is $415 \frac{\text{kJ}}{\text{mol}}$,

Solution:

CH₄ is formed from free elements in their standard states (C in solid state and H₂ in gaseous state). Thus, here $\Delta H_{\text{rxn}} = \Delta H_f$. The reaction

can be written in three steps:



with $\Delta H_f = \Delta H_1 + (2 \times \Delta H_2) + \Delta H_3$.

$$\Delta H_1 = \Delta H_f \text{ of } \text{C}(g) = 715 \frac{\text{kJ}}{\text{mol}}$$

ΔH_2 is the energy required to break the H–H bond of one mole of H_2 , so $\Delta H_2 = \text{bond enthalpy of } \text{H}_2 = 436 \frac{\text{kJ}}{\text{mol}}$. Note that reaction (b) is doubled in order to produce 4 atoms of H from two molecules of H_2 .

ΔH_3 is the energy released when 4 C–H bonds are formed. Because energy is released when bonds are formed, ΔH_3 is negative.

$$\Delta H_3 = -(4 \times \text{bond energy of C–H}) = -\left(4 \times 431 \frac{\text{kJ}}{\text{mol}}\right) = -1724 \frac{\text{kJ}}{\text{mol}}$$

Therefore, for the entire reaction,

$$\Delta H_{\text{rxn}} = \Delta H_f = 715 \frac{\text{kJ}}{\text{mol}} + \left(2 \times 436 \frac{\text{kJ}}{\text{mol}} \right) - 1660 \frac{\text{kJ}}{\text{mol}}$$

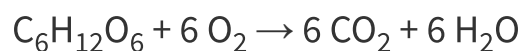
$$\Delta H_{\text{rxn}} \approx 700 + (2 \times 450) - 1660 = 700 + 900 - 1660 = -60 \frac{\text{kJ}}{\text{mol}}$$

(actual value = $-73 \frac{\text{kJ}}{\text{mol}}$)

STANDARD HEAT OF COMBUSTION

As the name implies, the standard heat of combustion, $\Delta H_{\text{comb}}^{\circ}$, is the enthalpy change associated with the combustion of a fuel. Because measurements of enthalpy change require a reaction to be spontaneous and fast, combustion reactions are the ideal processes for such measurements. Most combustion reactions presented on the MCAT occur in the presence of atmospheric oxygen, but keep in mind that there are other combustion reactions in which oxygen is not the oxidant. Diatomic fluorine, for example, can be used as an oxidant. In addition, hydrogen gas will combust with chlorine gas to form gaseous hydrochloric acid and, in the process, will evolve a large amount of heat and light as is characteristic of combustion reactions. The reactions listed in the CH_4 (g) example shown earlier are combustion reactions with O_2 (g) as the oxidant. Therefore, the enthalpy change listed for each of the three reactions is the ΔH_{comb} for each of the reactions.

The glycolytic pathway, described in Chapter 9 of MCAT Biochemistry Review, is also a combustion reaction that utilizes a fuel (glucose) mixed with an oxidant (oxygen) to produce carbon dioxide and water.



The heat of combustion for this reaction is found in a similar fashion to that of Hess's Law. Given the numerous reactions and pathways involved, we can determine the overall enthalpy of the reaction, as shown in Figure 7.9.

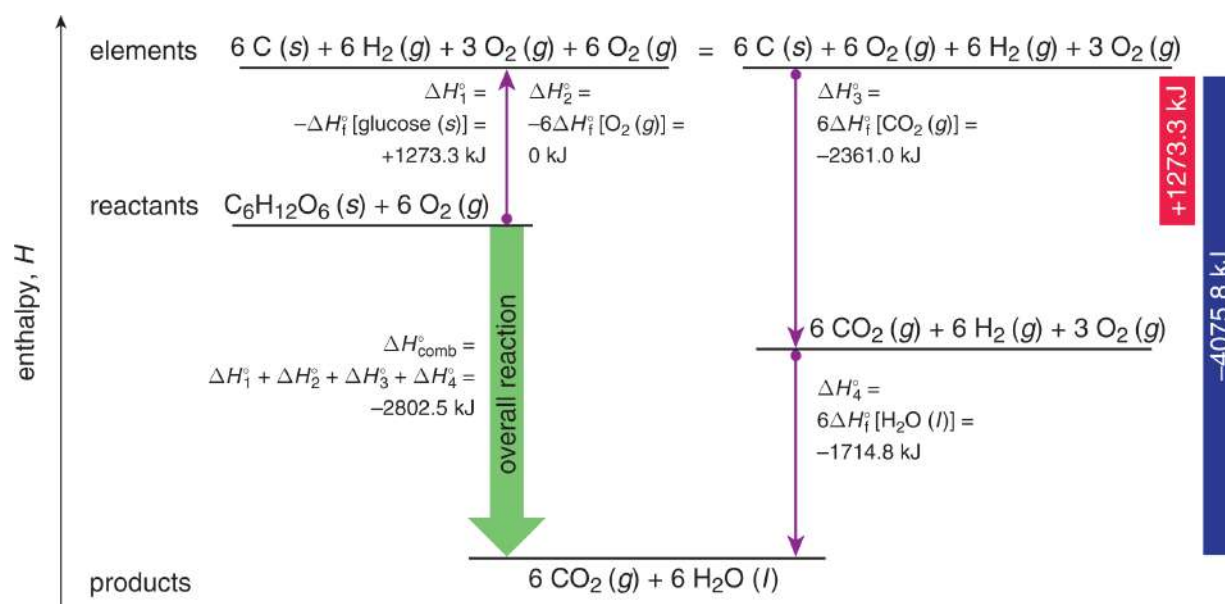


Figure 7.9. Determining the Enthalpy of Glycolysis



KEY CONCEPT

The larger the alkane reactant, the more numerous the combustion products.

MCAT Concept Check 7.4:

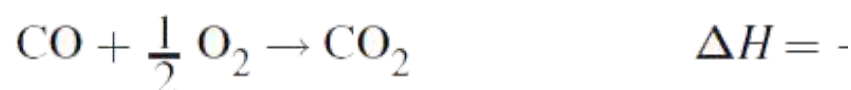
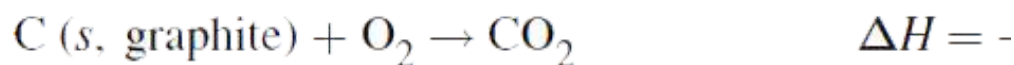
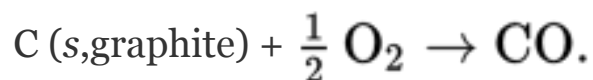
Before you move on, assess your understanding of the material with these questions.

Define endothermic and exothermic processes.

- Endothermic:

- Exothermic:

Given the following reactions, determine the enthalpy of:



What is the enthalpy of reaction for the reaction: $2 \text{H}_2\text{O} (g) \rightarrow 2 \text{H}_2 (g) + \text{O}_2 (g)$, given the following bond enthalpies: $\text{H} - \text{H} = 436 \frac{\text{kJ}}{\text{mol}}$,

$$\begin{aligned} \text{O} = \text{O} &= 498 \frac{\text{kJ}}{\text{mol}}, \Delta H_{\text{fus}} = 6.02 \frac{\text{kJ}}{\text{mol}}, \text{ and} \\ \text{O} - \text{H} &= 463 \frac{\text{kJ}}{\text{mol}} \end{aligned}$$

7.5 Entropy

LEARNING GOALS

After Chapter 7.5, you will be able to:

- Order the phases of matter from lowest to highest entropy
- Define entropy in terms of its relation to energy distribution and disorder
- Predict the direction of change in entropy within a given reaction

Many students are perplexed by the concept of entropy. Enthalpy makes intuitive sense, especially when the energy change from reactants to products is large, fast, and dramatic (as in combustion reactions involving explosions). Entropy seems to be less intuitive—except that it isn't. Consider, for example, how “normal” each of the following seems: hot tea cools down, frozen drinks melt, iron rusts, buildings crumble, balloons deflate, living things die and decay, and so on.



KEY CONCEPT

Entropy changes that accompany phase changes can be easily estimated, at least qualitatively. For example, freezing is accompanied by a decrease in entropy, as the relatively disordered liquid becomes a well-ordered solid. Meanwhile, boiling is accompanied by a large increase in entropy, as the liquid becomes a much more disordered gas. For any substance, sublimation will be the phase transition with the greatest increase in entropy.

These examples have a common denominator: in each of them, energy of some form is going from being localized or concentrated to being spread out or dispersed. The thermal energy in the hot tea is spreading out to the cooler air that surrounds it. The thermal energy in the warmer air is spreading out to the cooler frozen drink. The chemical energy in the bonds of elemental iron and oxygen is released and dispersed as a result of the formation of the more stable, lower-energy bonds of iron oxide (rust). The potential energy of the building is released and dispersed in the form of light, sound, and heat as the building crumbles and falls. The energy of the pressurized air is released to the surrounding atmosphere as the balloon deflates. The chemical energy of all the molecules and atoms in living flesh is released into the environment during the process of death and decay.

The second law of thermodynamics states that energy spontaneously disperses from being localized to becoming spread out if it is not hindered from doing so. Pay attention to this: the usual way of thinking about entropy as “disorder” must not be taken too literally, a trap that many students fall into. Be very careful in thinking about entropy as disorder. The old analogy between a messy (disordered) room and entropy is

deficient and may not only hinder understanding but actually increase confusion.

Entropy is the measure of the spontaneous dispersal of energy at a specific temperature: how much energy is spread out, or how widely spread out energy becomes, in a process. The equation for calculating the change in entropy is:

$$\Delta S = \frac{Q_{\text{rev}}}{T}$$

Equation 7.7

where ΔS is the change in entropy, Q_{rev} is the heat that is gained or lost in a reversible process, and T is the temperature in kelvin. The units of entropy are usually $\frac{\text{J}}{\text{mol} \cdot \text{K}}$. When energy is distributed into a system at a given temperature, its entropy increases. When energy is distributed out of a system at a given temperature, its entropy decreases.

Notice that the second law states that energy will spontaneously disperse; it does not say that energy can never be localized or concentrated. However, the concentration of energy will rarely happen spontaneously in a closed system. Work usually must be done to concentrate energy. For example, refrigerators work against the direction of spontaneous heat flow (that is, they counteract the flow of heat from the “warm” exterior of the refrigerator to the “cool” interior), thereby “concentrating” energy outside of the system in the surroundings. As a result, refrigerators consume a lot

of energy to accomplish this movement of energy against the temperature gradient.



Figure 7.10. Entropy in the kitchen

The second law has been described as time's arrow because there is a unidirectional limitation on the movement of energy by which we recognize before and after or new and old, as shown in Figure 7.10. For example, you would instantly recognize whether a video recording of an explosion was running forward or backward. Another way of understanding this is to say that energy in a closed system will spontaneously spread out, and entropy will increase if it is not hindered from doing so. Remember that a system can be variably defined to include the entire universe; in fact, the second law ultimately claims that the entropy of the universe is increasing.

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} > 0$$

Equation 7.8

Entropy is a state function, so a change in entropy from one equilibrium state to another is pathway independent and only depends upon the difference in entropies of the final and initial states. Further, the standard entropy change for a reaction, $\Delta S_{\text{rxn}}^{\circ}$, can be calculated using the standard entropies of the reactants and products—much like enthalpy:

$$\Delta S_{\text{rxn}}^{\circ} = \sum \Delta S_{\text{f,products}}^{\circ} - \sum \Delta S_{\text{f,reactants}}^{\circ}$$

Equation 7.9

MCAT Concept Check 7.5:

Before you move on, assess your understanding of the material with these questions.

Rank the phases of matter from lowest to highest entropy.

Describe entropy in terms of energy dispersal and disorder.

Do the following situations result in an increase or decrease in entropy?

Reaction	ΔS
$\text{H}_2\text{O (l)} \rightarrow \text{H}_2\text{O (s)}$	
Dry ice sublimates into carbon dioxide	
$\text{NaCl (s)} \rightarrow \text{NaCl (aq)}$	
$\text{N}_2 \text{ (g)} + 3 \text{ H}_2 \text{ (g)} \rightarrow 2 \text{ NH}_3 \text{ (g)}$	
An ice pack is placed on a wound	



7.6 Gibbs Free Energy

LEARNING GOALS

After Chapter 7.6, you will be able to:

- Determine the Gibbs free energy for a reaction at varying temperatures
- Predict the temperature necessary for a temperature-dependent reaction to be at equilibrium
- Identify how changing concentrations of reactant or product will alter the progress of a reaction

The final state function that we will examine in this chapter is Gibbs free energy, G . This state function is a combination of the three that we've just examined: temperature, enthalpy, and entropy. The change in Gibbs free energy, ΔG , is a measure of the change in the enthalpy and the change in entropy as a system undergoes a process, and it indicates whether a reaction is spontaneous or nonspontaneous. The change in the free energy is the maximum amount of energy released by a process—occurring at constant temperature and pressure—that is available to perform useful work. The change in Gibbs free energy is defined as follows:

$$\Delta G = \Delta H - T\Delta S$$

Equation 7.10

where T is the temperature in kelvins and $T\Delta S$ represents the total amount of energy that is absorbed by a system when its entropy increases reversibly.



MNEMONIC

Gibbs free energy: $\Delta G = \Delta H - T\Delta S$

Goldfish are (equals sign) Horrible without (minus sign) Tartar Sauce.

OVERVIEW

A helpful visual aid for conceptualizing Gibbs free energy is to think of it as a valley between two hills. Just as a ball would tend to roll down the hill into the valley and eventually come to rest at the lowest point in the valley, any system—including chemical reactions—will move in whichever direction results in a reduction of the free energy of the system. The bottom of the valley represents equilibrium, and the sides of the hill represent the various points in the pathway toward or away from

equilibrium. This is shown graphically in Figure 7.11, which was also discussed in the previous chapter.

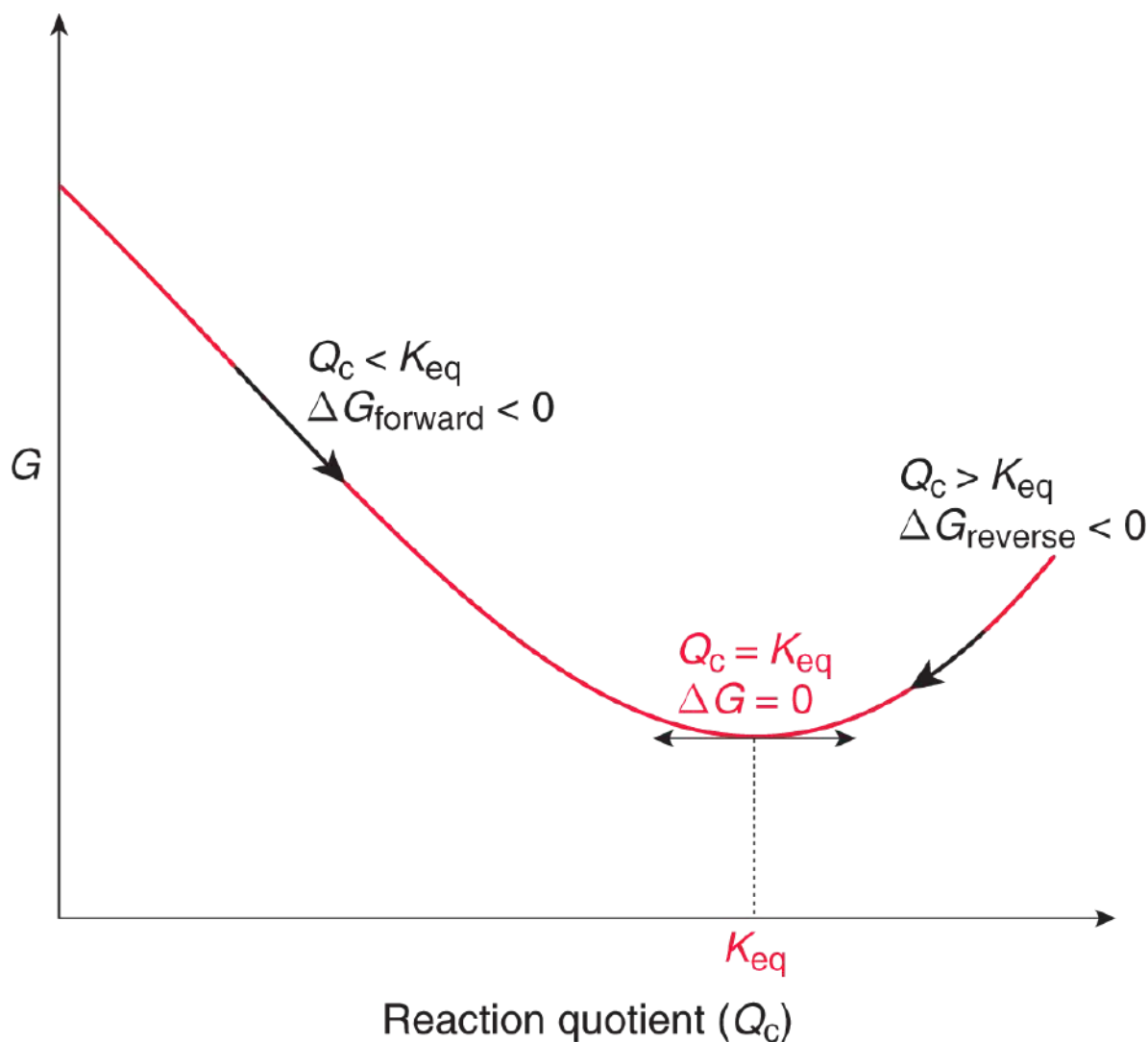


Figure 7.11. Gibbs Free Energy and Spontaneity

A decrease in Gibbs free energy indicates that a reaction is spontaneous.

When an equilibrated system is disturbed, it will spontaneously act to restore equilibrium.

Movement toward the equilibrium position is associated with a decrease in Gibbs free energy ($\Delta G < 0$) and is spontaneous. When a system releases

energy, it is said to be exergonic, as shown in Figure 7.12.

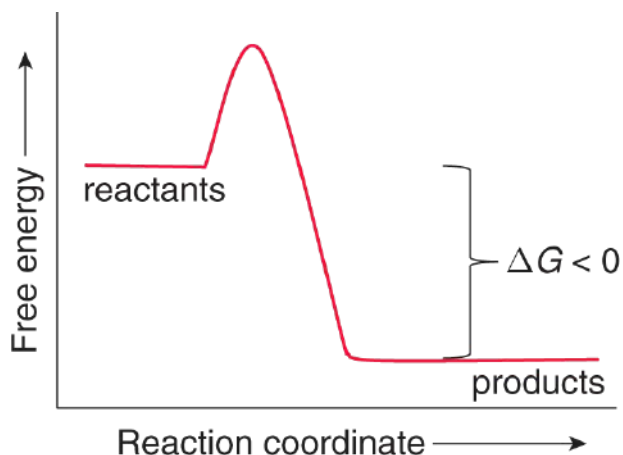


Figure 7.12. Exergonic Reaction Profile

Exergonic reactions release energy and are spontaneous ($\Delta G_{\text{rxn}} < 0$).



MCAT EXPERTISE

Be careful not to confuse endergonic/exergonic (describing Gibbs free energy) with endothermic/exothermic (describing enthalpy).

On the other hand, movement away from the equilibrium position is associated with an increase in Gibbs free energy ($\Delta G > 0$) and is nonspontaneous. Such a reaction is said to be endergonic, as shown in Figure 7.13.

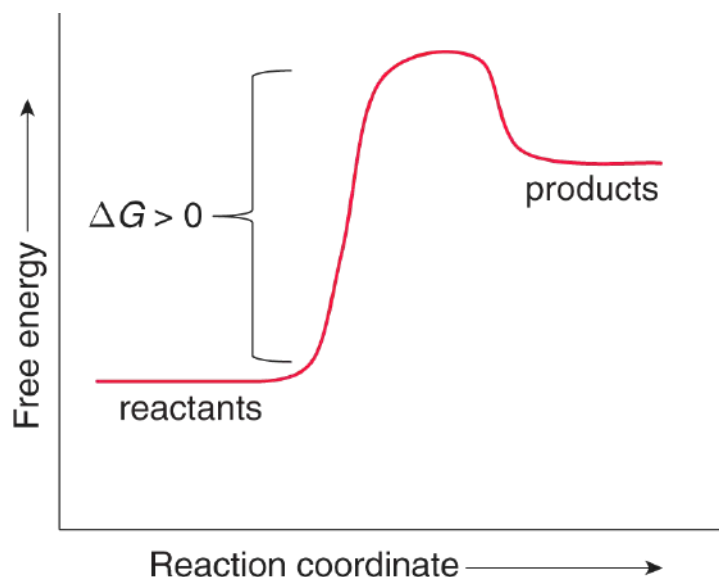


Figure 7.13. Endergonic Reaction Profile

Endergonic reactions absorb energy and are nonspontaneous ($\Delta G_{\text{rxn}} > 0$).

Once at the energy minimum state—equilibrium—the system will resist any changes to its state, and the change in free energy is zero. To summarize:

If ΔG is negative, the reaction is spontaneous.

If ΔG is positive, the reaction is nonspontaneous.

If ΔG is zero, the system is in a state of equilibrium; $\Delta H = T\Delta S$.

You should recall that phase equilibria are states in which more than one phase exists. As with all equilibria, the change in Gibbs free energy must be equal to zero ($\Delta G = 0$). For an equilibrium between a gas and a solid,

$$\Delta G = G(\text{g}) - G(\text{s}) = 0$$

Therefore,

$$G(g) = G(s)$$



KEY CONCEPT

Recall that thermodynamics and kinetics are separate topics. When a reaction is thermodynamically spontaneous, it has no bearing on how fast it goes. It only means that it will proceed eventually without external energy input.

Because the temperature in Gibbs free energy is in kelvins, it is always positive. Therefore, the effects of the signs of ΔH and ΔS on the spontaneity of a process can be summarized as in Table 7.2.

ΔH	ΔS	Outcome
+	+	Spontaneous at high T
+	–	Nonspontaneous at all T
–	+	Spontaneous at all T
–	–	Spontaneous at low T

Table 7.2 Effects of ΔH , ΔS , and T on Spontaneity



KEY CONCEPT

ΔG is temperature dependent when ΔH and ΔS have the same sign.

Phase changes are examples of temperature-dependent processes. The phase changes of water should be familiar to you; have you ever wondered why water doesn't boil at, say, 20°C instead of 100°C? When water boils, hydrogen bonds are broken, and the water molecules gain sufficient energy to escape into the gas phase. Thus, boiling is an endothermic process, and ΔH is positive. As thermal energy is transferred to the water molecules, energy is distributed through the molecules entering the gas phase. Thus, entropy is positive and the term $T\Delta S$ is positive. If both ΔH and $T\Delta S$ are positive, the reaction will only be spontaneous if $T\Delta S$ is greater than ΔH , resulting in a negative ΔG . These conditions are met only when the temperature of the system is greater than 373 K (100°C). Below 100°C, the free energy change is positive, and boiling is nonspontaneous; the water remains a liquid. At 100°C, $\Delta H - T\Delta S = 0$, and an equilibrium is established between the liquid and gas phases in such a way that the water's vapor pressure equals the ambient pressure. This is the definition of the boiling point: the temperature at which the vapor pressure equals the ambient pressure.

It is important to remember that the rate of a reaction depends on the activation energy E_a , not ΔG . Spontaneous reactions may be fast or slow. Sometimes a reversible reaction may produce two products that differ both in their stability (as measured by the change in the free energy associated with their production) and in their kinetics (as measured by their respective activation energies). Sometimes, the thermodynamically more stable product will have the slower kinetics due to higher activation energy. In this situation, we talk about kinetic vs. thermodynamic reaction control, which is discussed in Chapter 6 of MCAT General Chemistry Review. For a period of time after the reaction begins, the major product

will be the one that is produced more quickly as a result of its lower activation energy. The reaction can be said to be under kinetic control at this time. Given enough time, however, and assuming a reversible reaction, the dominant product will be the thermodynamically more stable product as a result of its lower free energy value. The reaction can then be said to be under thermodynamic control. Eventually, the reaction will reach equilibrium, as defined by its K_{eq} .

STANDARD GIBBS FREE ENERGY

The free energy change of reactions can be measured under standard state conditions to yield the standard free energy, $\Delta G_{\text{rxn}}^{\circ}$. For standard free energy determinations, the concentrations of any solutions in the reaction are 1 M. The standard free energy of formation of a compound, $\Delta G_{\text{f}}^{\circ}$, is the free energy change that occurs when 1 mole of a compound in its standard state is produced from its respective elements in their standard states under standard state conditions. The standard free energy of formation for any element under standard state conditions is, by definition, zero. The standard free energy of a reaction, $\Delta G_{\text{rxn}}^{\circ}$, is the free energy change that occurs when that reaction is carried out under standard state conditions; that is, when the reactants are converted to the products at standard conditions of temperature (298 K) and pressure (1 atm). Like enthalpy and entropy, the free energy of the reaction can be calculated from the free energies of formation of the reactants and products:

$$\Delta G_{\text{rxn}}^{\circ} = \sum \Delta G_{\text{f,products}}^{\circ} - \sum \Delta G_{\text{f,reactants}}^{\circ}$$

Equation 7.11

FREE ENERGY, K_{eq} , AND Q

We can derive the standard free energy change for a reaction from the equilibrium constant K_{eq} for the reaction using the equation:

$$\Delta G_{rxn}^{\circ} = -RT \ln K_{eq}$$

Equation 7.12

where R is the ideal gas constant, T is the temperature in kelvins, and K_{eq} is the equilibrium constant. This equation allows us to make not only quantitative evaluations of the free energy change of a reaction, but also qualitative assessments of the spontaneity of the reaction. The greater the value of K_{eq} , the more positive the value of its natural logarithm. The more positive the natural logarithm, the more negative the standard free energy change. The more negative the standard free energy change, the more spontaneous the reaction.

Once a reaction begins, however, the standard state conditions (specifically 1 M solutions) no longer apply. The value of the equilibrium constant must be replaced with another number that is reflective of where the reaction is in its path toward equilibrium. To determine the free energy change for a reaction that is in progress, we relate ΔG_{rxn} (not ΔG_{rxn}°) to the reaction quotient, Q :

$$\Delta G_{\text{rxn}} = \Delta G_{\text{rxn}}^{\circ} + RT \ln Q = RT \ln \frac{Q}{K_{\text{eq}}}$$

Equation 7.13



KEY CONCEPT

Note that the right side of this equation is similar to Equation 7.12. The use of the Q indicates that the system is not at equilibrium.

As described in Chapter 6 of MCAT General Chemistry Review, if the ratio of $\frac{Q}{K_{\text{eq}}}$ is less than one ($Q < K_{\text{eq}}$), then the natural logarithm will be negative, and the free energy change will be negative, so the reaction will spontaneously proceed forward until equilibrium is reached. If the ratio of $\frac{Q}{K_{\text{eq}}}$ is greater than one ($Q > K_{\text{eq}}$), then the natural logarithm will be positive, and the free energy change will be positive. In that case, the reaction will spontaneously move in the reverse direction until equilibrium is reached. Of course, if the ratio is equal to one, the reaction quotient is equal to the equilibrium constant; the reaction is at equilibrium, and the free energy change is zero ($\ln 1 = 0$).

Reaction profiles of free energy can be altered by the presence of catalysts. While the overall free energy change of the reaction is not altered, the

activation energy required to accomplish the reaction is reduced significantly in the presence of a catalyst, as shown in Figure 7.14.

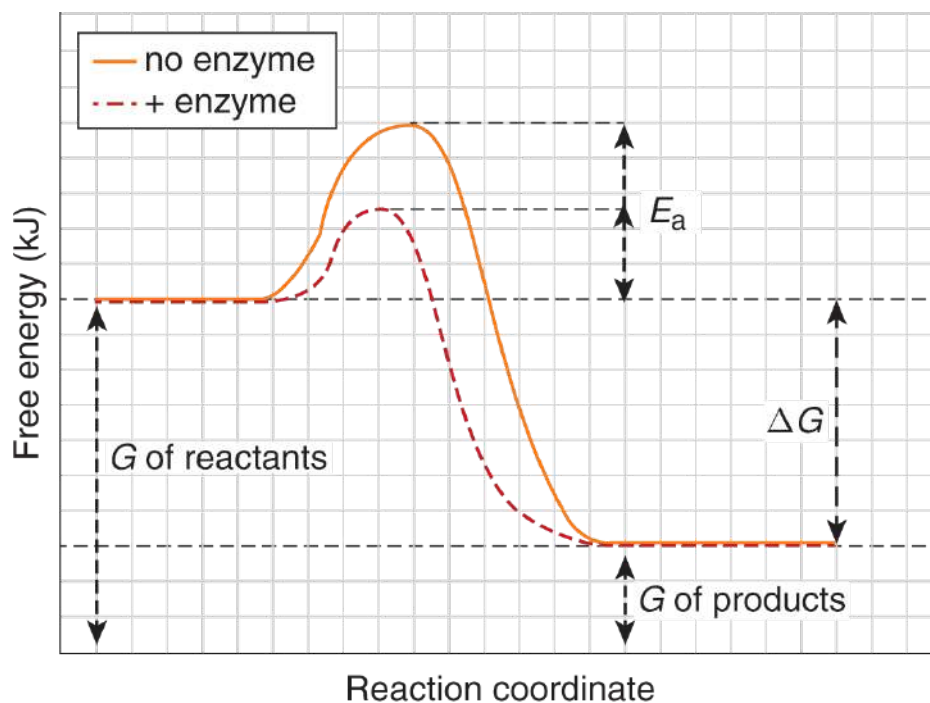
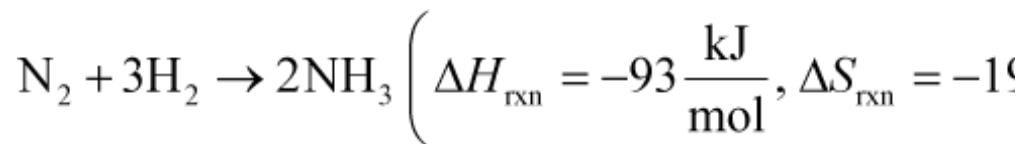


Figure 7.14. Catalysts Alter Kinetics but Not Equilibrium or Free Energy Change

MCAT Concept Check 7.6:

Before you move on, assess your understanding of the material with these questions.

The Haber–Bosch process creates ammonia through several reactions, the final step of which is



Determine what the Gibbs free energy of this reaction is at standard conditions and at 500 K:

- At standard conditions:

- At 500 K:

At what temperature would the reaction described above be at equilibrium?

If you were to suddenly flood the reaction vessel with significant amounts of ammonia, what would occur?

Conclusion

We began our discussion of thermochemistry with a review of different ways in which we characterize systems (open, closed, and isolated) and processes (isothermal, adiabatic, isobaric, and isovolumetric). We then further classified systems according to their state functions—system properties such as pressure, density, temperature, volume, enthalpy, internal energy, Gibbs free energy, and entropy that describe the equilibrium state. We examined the equilibria that exist between the different phases and noted that the change in Gibbs free energy for each phase change in equilibrium is zero, as is the case for all equilibria. We defined enthalpy as the heat content of the system and the change in enthalpy as the change in heat content of the system as it moves from one equilibrium state to another. Enthalpy is defined as the energy found in the intermolecular interactions and bonds of the compounds in the system. We explored the various ways Hess's law can be applied to calculate the total enthalpy change for a series of reactions. Moving on to entropy, we described this property as a measure of the degree to which energy in a system becomes spread out through a process. There is danger in thinking too literally about entropy as “disorder” because a system's entropy may be increasing even if there is no observable change in the system's macroscopic disorder (such as ice warming from -10°C to -5°C). Gibbs free energy combines the effects of temperature, enthalpy, and entropy, and the change in Gibbs free energy determines whether a process will be spontaneous or nonspontaneous. When the change in Gibbs free energy is

negative, the process is spontaneous, but when the change in Gibbs free energy is positive, the process is nonspontaneous.

Many reactions in the body must be spontaneous in order for cells to function. While there are some nonspontaneous reactions in our body, we are able to couple them to thermodynamically favorable (exergonic) reactions that allow the cell to perform even more complex functions.

Concept Summary

Systems and Processes

- Systems are classified based on what is or is not exchanged with the surroundings.
 - Isolated systems exchange neither matter nor energy with the environment.
 - Closed systems can exchange energy but not matter with the environment.
 - Open systems can exchange both energy and matter with the environment.
- Processes can be characterized based on a single constant property.
 - Isothermal processes occur at a constant temperature.
 - Adiabatic processes exchange no heat with the environment.
 - Isobaric processes occur at a constant pressure.
 - Isovolumetric (isochoric) processes occur at a constant volume.

States and State Functions

- State functions describe the physical properties of an equilibrium state; they are pathway independent and include pressure, density,

temperature, volume, enthalpy, internal energy, Gibbs free energy, and entropy.

- Standard conditions are defined as 298 K, 1 atm, and 1 M concentrations.
- The standard state of an element is its most prevalent form under standard conditions; standard enthalpy, standard entropy, and standard free energy are all calculated under standard conditions.
- Phase changes exist at characteristic temperatures and pressures.
 - Fusion (melting) and freezing (crystallization or solidification) occur at the boundary between the solid and the liquid phases.
 - Vaporization (evaporation or boiling) and condensation occur at the boundary between the liquid and the gas phases.
 - Sublimation and deposition occur at the boundary between the solid and gas phases.
 - At temperatures above the critical point, the liquid and gas phases are indistinguishable.
 - At the triple point, all three phases of matter exist in equilibrium.
- The phase diagram for a system graphs the phases and phase equilibria as a function of temperature and pressure.

Heat

- Temperature and heat are not the same thing.
 - Temperature is a scaled measure of the average kinetic energy of a substance.

- Heat is the transfer of energy that results from differences of temperature between two substances.
- The heat content of a system undergoing heating, cooling, or phase changes is the sum of all the respective energy changes.

Enthalpy

- Enthalpy is a measure of the potential energy of a system found in intermolecular attractions and chemical bonds.
- Hess's law states that the total change in potential energy of a system is equal to the changes of potential energies of the individual steps of the process.
- Enthalpy can also be calculated using heats of formation, heats of combustion, or bond dissociation energies.

Entropy

- Entropy, while often thought of as disorder, is a measure of the degree to which energy has been spread throughout a system or between a system and its surroundings.
 - Entropy is a ratio of heat transferred per mole per unit kelvin.
 - Entropy is maximized at equilibrium.

Gibbs Free Energy

- Gibbs free energy is derived from both enthalpy and entropy values for a given system.

- The change in Gibbs free energy determines whether a process is spontaneous or nonspontaneous.
 - $\Delta G < 0$: reaction proceeds in forward direction (spontaneous)
 - $\Delta G = 0$: reaction is in dynamic equilibrium
 - $\Delta G > 0$: reaction proceeds in reverse direction (nonspontaneous)
- Gibbs free energy depends on temperature; temperature-dependent processes change between spontaneous and nonspontaneous, depending on the temperature.

Answers to Concept Checks

7.1

The boundary between system and surroundings could be placed anywhere. Most commonly, the ice pack would be considered the chemical system using up energy, and the person (and the remainder of the universe) constitutes the surroundings that are providing the heat for the ice pack to function.

- Isothermal: no change in temperature; $\Delta U = 0$, $Q = W$
- Adiabatic: no heat exchange; $Q = 0$, $\Delta U = -W$
- Isobaric: no change in pressure; line appears flat in a P-V graph
- Isovolumetric (isochoric): no change in volume; $W = 0$, $\Delta U = Q$

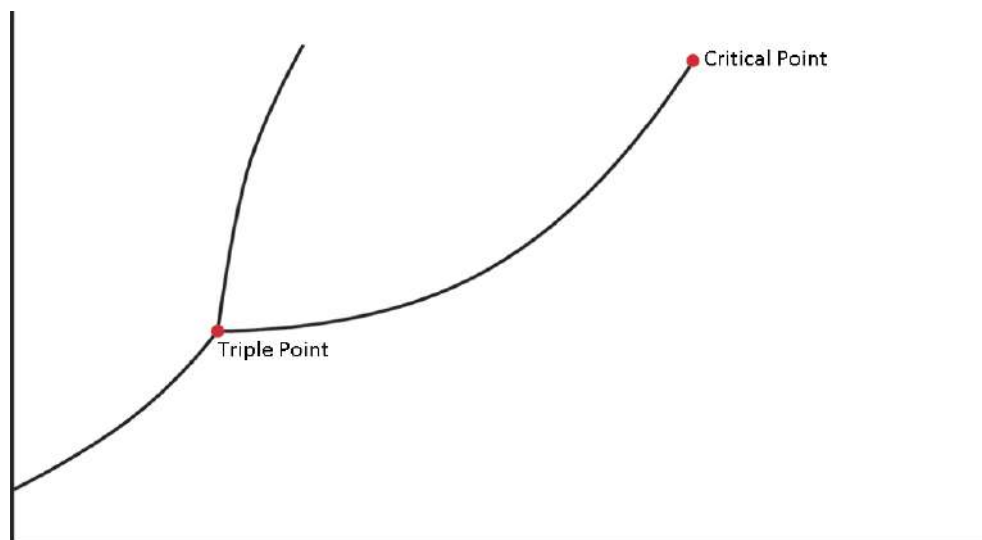
7.2

Kinetics, equilibrium, and thermodynamics calculations use standard conditions, which are 25°C (298 K), 1 atm pressure, and 1 M concentrations.

State functions are properties of a system at equilibrium and are independent of the path taken to achieve the equilibrium; they may be dependent on one another. Process functions define the path between equilibrium states and include Q (heat) and W (work).

State functions include pressure (P), density (ρ), temperature (T), volume (V), enthalpy (H), internal energy (U), Gibbs free energy (G), and entropy (S).

The triple point is the specific combination of temperature and pressure at which all three phases are in equilibrium. The critical point is the temperature and pressure above which the liquid and gas phases are indistinguishable and the heat of vaporization is zero.



7.3

Temperature is an indirect measure of the thermal content of a system that looks at average kinetic energy of particles in a sample. Heat is the thermal energy transferred between objects as a result of differences in their temperatures.

Specific heat (c) is the energy required to raise the temperature of one gram of a substance by one degree Celsius. Heat capacity (mc) is the product of mass and specific heat and is the energy required to raise any given amount of a substance one degree Celsius.

A constant-pressure calorimeter (coffee cup calorimeter) is exposed to constant (atmospheric) pressure. As the reaction proceeds, the temperature of the contents is measured to determine the heat of the reaction. A constant-volume calorimeter (bomb calorimeter) is one in which heats of certain reactions (like combustion) can be measured

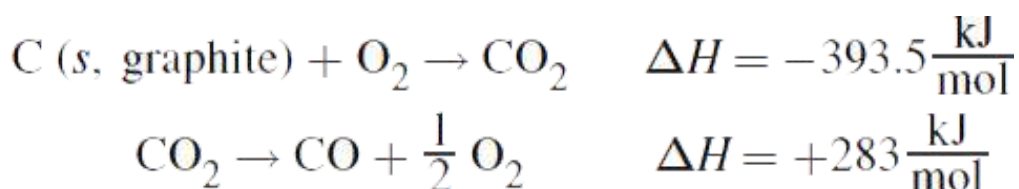
indirectly by assessing temperature change in a water bath around the reaction vessel.

$$c_{\text{H}_2\text{O}(\text{liq})} = 1 \frac{\text{cal}}{\text{g} \cdot \text{K}}$$

7.4

Endothermic reactions involve an increase in heat content of a system from the surroundings ($\Delta H > 0$), while exothermic reactions involve a release of heat content from a system ($\Delta H < 0$).

To reach the net equation $\text{C}(s, \text{graphite}) + \frac{1}{2} \text{O}_2 \rightarrow \text{CO}$, the second reaction must be reversed along with the sign of its enthalpy of reaction.



Adding the enthalpies gives:

$$\begin{aligned} -393.5 \frac{\text{kJ}}{\text{mol}} + 283 \frac{\text{kJ}}{\text{mol}} &\approx -400 + 285 = -115 \frac{\text{kJ}}{\text{mol}} \\ (\text{actual value} &= -110.5 \frac{\text{kJ}}{\text{mol}}) \end{aligned}$$

Enthalpy of reaction = bonds broken – bonds formed. There are four O–H bonds broken, two H–H bonds formed, and one O=O bond formed. Therefore,

$$\Delta H_{\text{rxn}} = \text{bonds broken} - \text{bonds formed}$$

$$\Delta H_{\text{rxn}} = (4 \times \text{O-H}) - [(2 \times \text{H-H}) + (1 \times \text{O} = \text{O})]$$

$$\Delta H_{\text{rxn}} = \left(4 \times 463 \frac{\text{kJ}}{\text{mol}}\right) - \left[\left(2 \times 436 \frac{\text{kJ}}{\text{mol}}\right) + \left(1 \times 498 \frac{\text{kJ}}{\text{mol}}\right)\right]$$

$$\Delta H_{\text{rxn}} \approx (4 \times 450) - [(2 \times 435) + (500)] = 1800 - (870 + 500)$$

$$\Delta H_{\text{rxn}} = 1800 - 1370 = 430 \frac{\text{kJ}}{\text{mol}} \left(\text{actual value} = 482 \frac{\text{kJ}}{\text{mol}}\right)$$

7.5

Solids have the lowest entropy, followed by liquids, with gases having the highest entropy.

Entropy increases as a system has more disorder or freedom of movement, and energy is dispersed in a spontaneous system. Entropy of the universe can never be decreased spontaneously.

Reaction	ΔS
$\text{H}_2\text{O} (\text{l}) \rightarrow \text{H}_2\text{O} (\text{s})$	Decrease (freezing)
Dry ice sublimates into carbon dioxide	Increase (sublimation)
$\text{NaCl} (\text{s}) \rightarrow \text{NaCl} (\text{aq})$	Increase (dissolution)
$\text{N}_2 (\text{g}) + 3 \text{H}_2 (\text{g}) \rightarrow 2 \text{NH}_3 (\text{g})$	Decrease (fewer moles of gas)
An ice pack is placed on a wound	Increase (heat is transferred)

7.6

At standard conditions:

$$\Delta G = \Delta H - T\Delta S = -93,000 \frac{\text{J}}{\text{mol}} - (298 \text{ K}) \left(-198 \frac{\text{J}}{\text{mol} \cdot \text{K}} \right) \approx$$

$$-93,000 - (300 \times -200) = -93,000 + 60,000 = -33,000 \frac{\text{J}}{\text{mol}} = -33 \frac{\text{kJ}}{\text{mol}}$$

$$\left(\text{actual} = -34.0 \frac{\text{kJ}}{\text{mol}} \right)$$

At 500 K:

$$\Delta G = \Delta H - T\Delta S = -93,000 \frac{\text{J}}{\text{mol}} - (500 \text{ K}) \left(-198 \frac{\text{J}}{\text{mol} \cdot \text{K}} \right) \approx$$

$$-93,000 - (500 \times -200) = -93,000 + 100,000 = 7,000 \frac{\text{J}}{\text{mol}} = 7 \frac{\text{kJ}}{\text{mol}}$$

$$\left(\text{actual} = 6 \frac{\text{kJ}}{\text{mol}} \right)$$

The system is at equilibrium when $\Delta G = 0$:

$$\Delta G = \Delta H - T\Delta S$$

$$0 = -93,000 \frac{\text{J}}{\text{mol}} - \left[(T) \left(-198 \frac{\text{J}}{\text{mol} \cdot \text{K}} \right) \right] 93,000 \approx 200$$

$$T = \frac{93,000}{200} = \frac{930}{2} = 465 \text{ K (actual} = 470 \text{ K)}.$$

The value of Q would increase significantly, causing the system to shift left, forming more reactants until the system again reached equilibrium.

Equations to Remember

(7.1) First law of thermodynamics: $\Delta U = Q - W$

(7.2) Heat transfer (no phase change): $q = mc\Delta T$

(7.3) Heat transfer (during phase change): $q = mL$

(7.4) Generalized enthalpy of reaction: $\Delta H_{\text{rxn}} = H_{\text{products}} - H_{\text{reactants}}$

(7.5) Standard enthalpy of reaction: $\Delta H_{\text{rxn}}^{\circ} = \sum \Delta H_{\text{f,products}}^{\circ} - \sum \Delta H_{\text{f,reactants}}^{\circ}$

(7.6) Bond enthalpy: $\Delta H_{\text{rxn}}^{\circ} = \sum \Delta H_{\text{bonds broken}} - \sum \Delta H_{\text{bonds formed}} = \text{total energy absorbed} - \text{total energy released}$

(7.7) Entropy: $\Delta S = \frac{Q_{\text{rev}}}{T}$

(7.8) Second law of thermodynamics: $\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} > 0$

(7.9) Standard entropy of reaction: $\Delta S_{\text{rxn}}^{\circ} = \sum \Delta S_{\text{f,products}}^{\circ} - \sum \Delta S_{\text{f,reactants}}^{\circ}$

(7.10) Gibbs free energy: $\Delta G = \Delta H - T\Delta S$

(7.11) Standard Gibbs free energy of reaction: $\Delta G_{\text{rxn}}^{\circ} = \sum \Delta G_{\text{f,products}}^{\circ} - \sum \Delta G_{\text{f,reactants}}^{\circ}$

(7.12) Standard Gibbs free energy from equilibrium constant: $\Delta G_{\text{rxn}}^{\circ} = -RT \ln K_{\text{eq}}$

(7.13) Gibbs free energy from reaction quotient:

$$\Delta G_{\text{rxn}} = \Delta G_{\text{rxn}}^{\circ} + RT \ln Q = RT \ln \frac{Q}{K_{\text{eq}}}$$



Shared Concepts

General Chemistry Chapter 3

Bonding and Chemical Interactions

General Chemistry Chapter 4

Compounds and Stoichiometry

General Chemistry Chapter 5

Chemical Kinetics

General Chemistry Chapter 6

Equilibrium

Physics and Math Chapter 2

Work and Energy

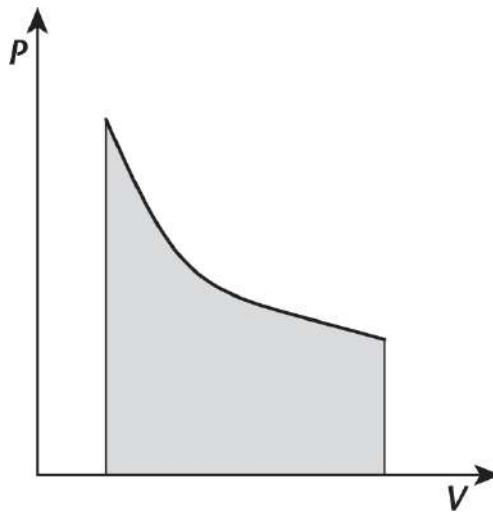
Physics and Math Chapter 3

Thermodynamics

Discrete Practice Questions

Consult your online resources for additional practice.

1. Consider the cooling of an ideal gas in a closed system. This process is illustrated in the pressure–volume graph shown in the following figure.



Based on this information, the process may be:

- (A) adiabatic.
- (B) isobaric.
- (C) isothermal.
- (D) isochoric.

2. A reaction has a positive entropy and enthalpy. What can be inferred about the progress of this reaction from this information?

- (A) The reaction is spontaneous.
- (B) The reaction is nonspontaneous.
- (C) The reaction is at equilibrium.
- (D) There is not enough information to determine whether the reaction is spontaneous or not.

3. Pure sodium metal spontaneously combusts upon contact with room temperature water. What is true about the equilibrium constant of this combustion reaction at 25°C?

- (A) $K_{eq} < 0$
- (B) $0 < K_{eq} < 1$
- (C) $K_{eq} = 1$
- (D) $K_{eq} > 1$

4. Which of the following processes has the most exothermic standard heat of combustion?

- (A) Combustion of ethane
- (B) Combustion of propane
- (C) Combustion of n-butane
- (D) Combustion of n-pentane

5. Methanol reacts with acetic acid to form methyl acetate and water.

Type of Bond	Bond Dissociation Energy $\left(\frac{\text{kJ}}{\text{mol}}\right)$
C – C	348
C – H	415
C = O	805
O – H	463
C – O	360

Based on the values in the table above, what is the heat of reaction in $\frac{\text{kJ}}{\text{mol}}$?

- (A) 0
- (B) 464
- (C) 824
- (D) 1288

6. At standard temperature and pressure, a chemical process is at equilibrium. What is the free energy of reaction (ΔG) for this

process?

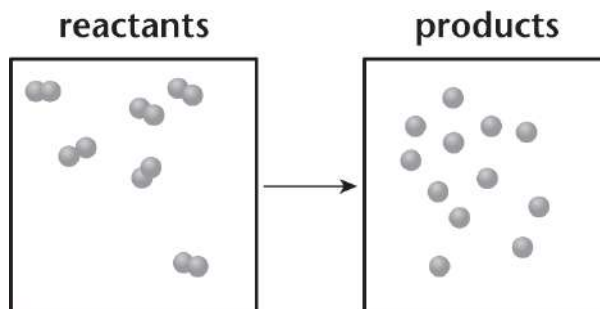
- (A) $\Delta G > 0$
- (B) $\Delta G < 0$
- (C) $\Delta G = 0$
- (D) There is not enough information to determine the free energy of the reaction.

7. For a certain chemical process, $2.18 \times 10^{-18} \frac{\text{J}}{\text{electron}}$. What is the equilibrium constant K_{eq} for this reaction? (Note:

$$R = 8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}})$$

- (A) $K_{\text{eq}} = 1.0$
- (B) $K_{\text{eq}} = 7.4$
- (C) $K_{\text{eq}} = 8.9$
- (D) $K_{\text{eq}} = 10$

8. Consider the chemical reaction in the vessel depicted in the following diagram.



- (A) The reaction is spontaneous.
- (B) The reaction is nonspontaneous.
- (C) The reaction is at equilibrium.
- (D) There is not enough information to determine if the reaction is spontaneous.

9. Suppose $\Delta G_{\text{rxn}}^{\circ} = -2000 \frac{\text{kJ}}{\text{mol}}$ for a chemical reaction. At 300 K, what is the change in Gibbs free energy in $\frac{\text{kJ}}{\text{mol}}$?

- (A) $\Delta G = -2000 + (300 \text{ K})(8.314)(\ln Q)$
- (B) $\Delta G = -2000 - (300 \text{ K})(8.314)(\ln Q)$
- (C) $\Delta G = -2000 + (300 \text{ K})(8.314)(\log Q)$
- (D) $\Delta G = -2000 - (300 \text{ K})(8.314)(\log Q)$

10. A chemical reaction has a negative enthalpy and a negative entropy. Which of the following terms necessarily describes this reaction?

- (A) Exothermic
- (B) Endothermic
- (C) Exergonic
- (D) Endergonic

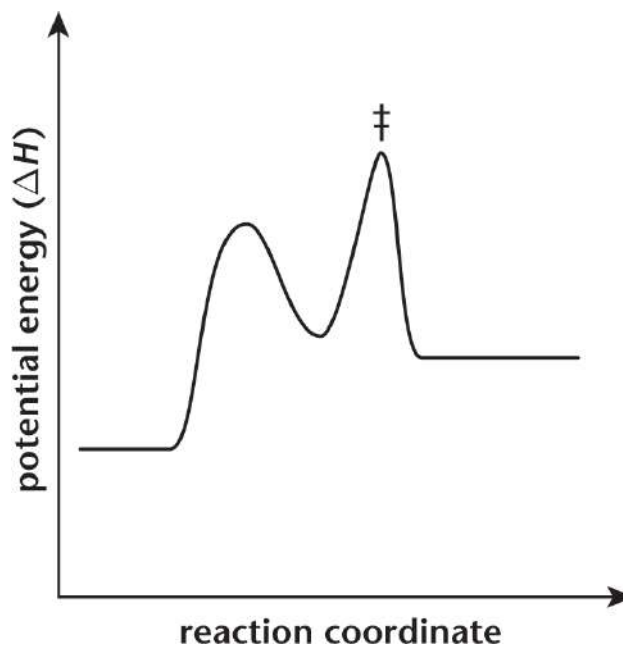
11. Which of the following statements is true of a process that is spontaneous in the forward direction?

- (A) $\Delta G > 0$ and $K_{eq} > Q$
- (B) $\Delta G > 0$ and $K_{eq} < Q$
- (C) $\Delta G < 0$ and $K_{eq} > Q$
- (D) $\Delta G < 0$ and $K_{eq} < Q$

12. Which of the following devices would be the most appropriate to use to measure the heat capacity of a liquid?

- (A) Thermometer
- (B) Calorimeter
- (C) Barometer
- (D) Volumetric flask

13. A reaction coordinate for a chemical reaction is displayed in the graph below.



Which of the following terms describes the energy of this reaction?

- (A) Endothermic
- (B) Exothermic
- (C) Endergonic
- (D) Exergonic

14. Which of the following phase changes is associated with the largest decrease in entropy?

- (A) Fusion
- (B) Solidification
- (C) Deposition
- (D) Sublimation

15. Explosions are necessarily characterized by:

- (A) $\Delta G < 0$.
- (B) $\Delta H > 0$.
- (C) $\Delta S < 0$.
- (D) $T < 0$.

Explanations to Discrete Practice Questions

1. A

This process may be adiabatic. Given that the gas was cooled, it did not maintain constant temperature, eliminating (C). Isobaric and isovolumetric processes appear as horizontal and vertical lines in pressure–volume graphs, respectively, eliminating (B) and (D). Adiabatic processes appear hyperbolic on pressure–volume graphs, as illustrated here.

2. D

There is not enough information in the problem to determine whether or not the reaction is spontaneous. If the signs of enthalpy and entropy are the same, the reaction is temperature dependent according to $\Delta G = \Delta H - T\Delta S$. Without the temperature, we cannot determine if this reaction is spontaneous, nonspontaneous, or at equilibrium.

3. D

Solve this question using the equation $\Delta G_{\text{rxn}}^{\circ} = -RT \ln K_{\text{eq}}$. $\Delta G_{\text{rxn}}^{\circ}$ is negative (as it must be for a spontaneous reaction), and R and T are always positive. Therefore, $\ln K_{\text{eq}}$ must also be positive for the sign convention to work out correctly. Since $\ln(1) = 0$, the natural logarithm of any number greater than 1 will be positive, and the natural logarithm

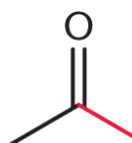
of any number less than 1 will be negative. In order for $\ln K_{eq}$ to be a positive number, K_{eq} must be greater than 1.

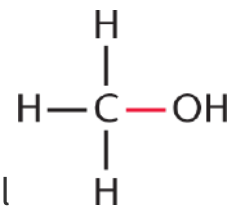
4. D

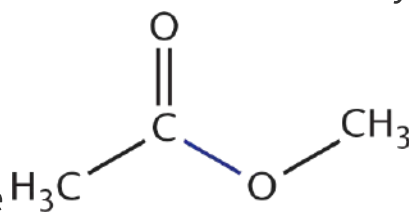
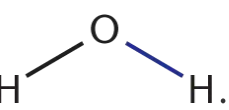
Combustion often involves the reaction of a hydrocarbon with oxygen to produce carbon dioxide and water. Longer hydrocarbon chains yield greater amounts of combustion products and release more heat in the process—that is, the reaction is more exothermic. Of the hydrocarbons listed here, n-pentane is the longest chain.

5. A

At first glance, this might seem like a math-heavy problem, but it really doesn't require any calculations at all. We just have to keep track of which bonds are broken and which bonds are formed. Remember, breaking bonds requires energy, while forming bonds releases energy. Two bonds are broken: a C–O bond between the carbonyl carbon and

oxygen of acetic acid , and an O–H bond between the

hydroxyl oxygen and hydrogen of methanol . Two bonds are also formed: a C–O bond between the carbonyl carbon and the

oxygen of methyl acetate , and an O–H bond between a hydroxyl group and a hydrogen to form water .

Given that the same two bonds are broken and formed in this reaction, the energy change must be $0 \frac{\text{kJ}}{\text{mol}}$.

6. C

Standard temperature and pressure indicate 0°C and 1 atm. Gibbs free energy is temperature dependent, but if a reaction is at equilibrium, $\Delta G = 0$.

7. B

Solve this question using the equation $\Delta G_{\text{rxn}}^\circ = -RT \ln K_{\text{eq}}$. $\Delta G_{\text{rxn}}^\circ$ is $-4.955 \frac{\text{kJ}}{\text{mol}}$, R is $8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}$, and $T = 298 \text{ K}$ because the reaction is occurring under standard conditions. Because $\Delta G_{\text{rxn}}^\circ$ uses kilojoules in its units and R uses joules, one will have to be converted. Plugging into the equation, we get:

$$-4955 \frac{\text{J}}{\text{mol}} = - \left(8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}} \right) (298 \text{ K}) (\ln K_{\text{eq}})$$

$$\ln K_{\text{eq}} = \frac{4955 \frac{\text{J}}{\text{mol}}}{\left(8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}} \right) (298 \text{ K})} \approx \frac{5000}{8 \times 300} = \frac{5000}{2400} \approx \frac{5000}{2500} =$$

If $\ln K_{\text{eq}} = 2$, then $K_{\text{eq}} = e^2$. The value of e is approximately 2.7, so $e^2 = 2.7^2$ will be a number between $2^2 = 4$ and $3^2 = 9$. Both (B) and (C) fit these criteria; however, 8.9 is very close to 9, so we can assume that its square root is very, very close to 3. The answer choice should be a bit smaller, so (B), 7.4, is correct.

8. D

There is not enough information available to determine the free energy of this reaction. While the entropy is clearly increasing (there are more particles in the system), it is unclear what the enthalpy change is. Because bonds are breaking, the reaction should be endothermic, meaning that both ΔS and ΔH are positive. In this case, it is a temperature-dependent process, and—without a temperature given—we cannot determine the sign on ΔG .

9. A

This problem asks for the free energy of a reaction at nonstandard conditions, which can be determined with the equation $\Delta G = \Delta G^\circ + RT \ln Q$.

10. A

A reaction with a negative enthalpy is, by definition, exothermic. Because both enthalpy and entropy are negative, this is a temperature-dependent process, and the reaction will be both endergonic and exergonic—but only at particular temperatures, eliminating (C) and (D).

11. C

For a process to progress forward spontaneously, Q must be less than K_{eq} and will therefore have a tendency to move in the direction toward equilibrium. A spontaneous reaction's free energy is negative by convention.

12. B

A calorimeter measures heat transfer. Although calorimeters often incorporate thermometers, the thermometer itself only tracks temperature, not the heat transfer itself, so (A) is incorrect. (C) is

irrelevant; barometers measure changes in pressure. (D) is also incorrect because volumetric flasks measure quantities of liquid, not the heat capacity of the liquid.

13. A

Eliminate (C) and (D), which describe the free energy of reaction and cannot be determined from this graph. While most reaction coordinate graphs we've explored in this book use free energy for the y-axis, this one uses potential energy (enthalpy). If the heat of formation of the products is greater than that of the reactants, the reaction is endothermic. We can determine this information from their relative positions on the graph: because the products are higher than the reactants, this is an endothermic reaction.

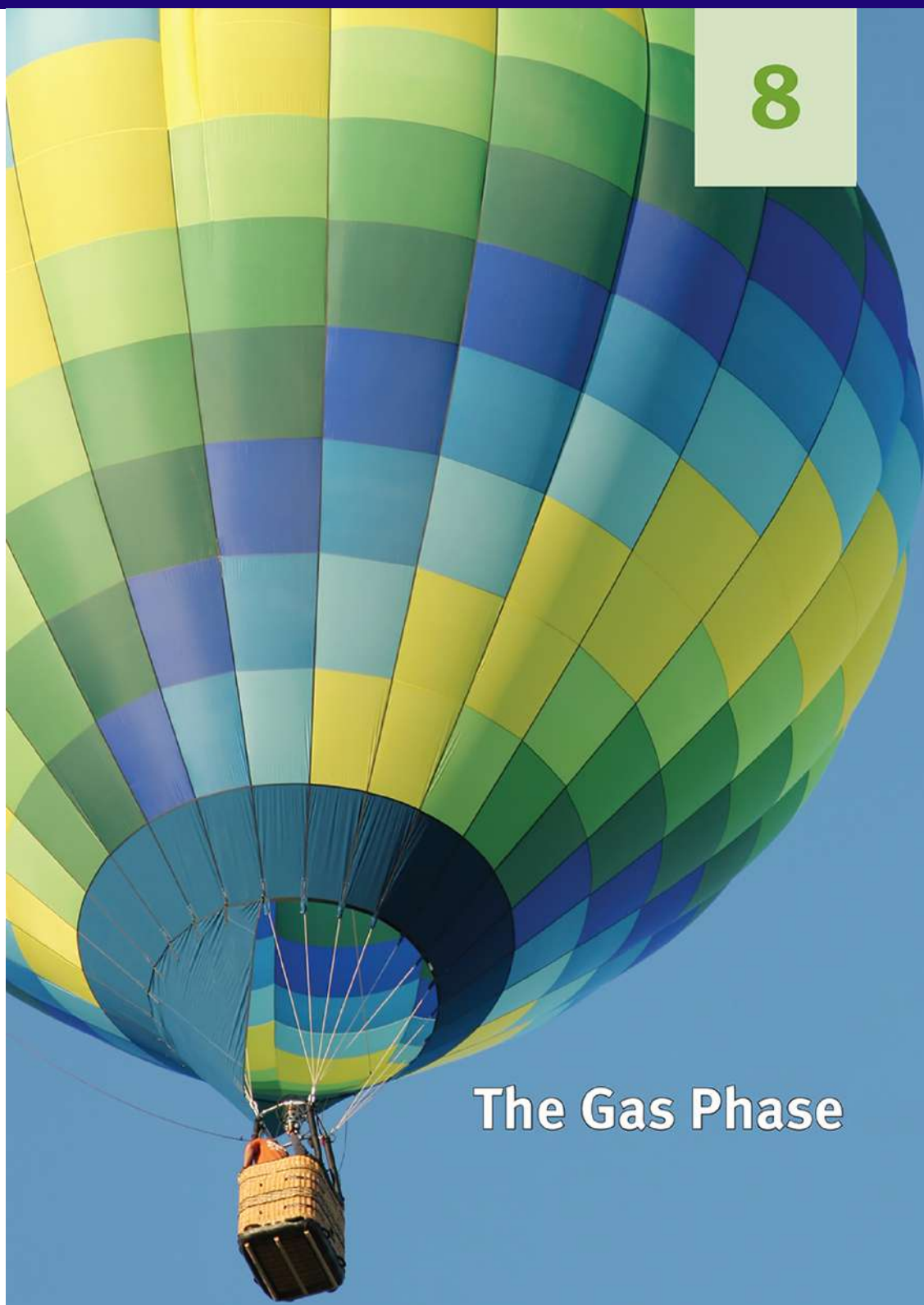
14. C

Gases have the highest entropy, and solids have the lowest. Therefore, a phase change from a gas to a solid—deposition—would have the largest decrease in entropy of any phase change process.

15. A

In an explosion, a significant amount of heat energy is released, meaning that the reaction is exothermic ($\Delta H < 0$), eliminating (B). The entropy change associated with an explosion is positive because energy is dispersed over a much larger area, eliminating (C). If this is true, the expression $\Delta H - T\Delta S$ must be negative, indicating that this is an exergonic process ($\Delta G < 0$). Absolute temperature can never be negative, eliminating (D).

The Gas Phase



In This Chapter

8.1 The Gas Phase 
Variables 

8.2 Ideal Gases 
Ideal Gas Law 
Special Cases 
Dalton's Law of Partial Pressures 
Henry's Law 

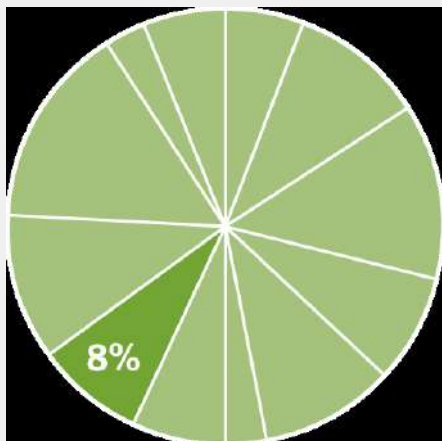
8.3 Kinetic Molecular Theory 
Assumptions 
Applications 

8.4 Real Gases 
Deviations Due to Pressure 
Deviations Due to Temperature 
Van der Waals Equation of State 

Concept Summary 



Chapter Profile



The content in this chapter should be relevant to about 8% of all questions about general chemistry on the MCAT.

This chapter covers material from the following AAMC content categories:

3B: Structure and integrative functions of the main organ systems

4B: Importance of fluids for the circulation of blood, gas movement, and gas exchange

Introduction

Let's start this chapter with a thought experiment. Imagine a helium balloon tied to the gearshift lever between the seats of your car and allowed to float freely. What do you think will happen to the balloon as you accelerate forward? You might think, based on how you feel when you are in an accelerating car, that the balloon will be pushed backwards due to its inertia. However, the balloon's movement isn't what we might predict: the balloon shifts forward as the car accelerates!

The molar mass of helium is $4 \frac{\text{g}}{\text{mol}}$, while that of air, which is mostly nitrogen and oxygen, is about $29 \frac{\text{g}}{\text{mol}}$. This means that air is about seven times denser than helium. Because the air in which the balloon is floating is more dense than the balloon itself, the air has greater inertia. Therefore, as the car accelerates forward, everything that has significant mass, including the air in the car, resists the forward motion (has inertia) and shifts toward the back of the car (even though, of course, everything in the car is accelerating forward, just not as quickly as the car itself). As the air shifts toward the back, a pressure gradient builds up such that there is greater air pressure in the back of the car than in the front, and this pressure difference results in a pushing force against the balloon that is directed from the back toward the front. Responding to this force, the balloon shifts forward in the direction of the car's acceleration. Who would have thought that general chemistry and physics could be so much fun?

In this chapter, we will discuss some MCAT favorites—the gas phase and the ideal gas laws. We will begin our discussion with ideal gases and the laws that govern their behavior. We will then examine the kinetic molecular theory that describes ideal gases and conclude with an

evaluation of the ways in which the behavior of real gases deviates from that predicted by the ideal gas law.

8.1 The Gas Phase

LEARNING GOALS

After Chapter 8.1, you will be able to:

- Identify the unique characteristics of the gas phase
- Predict how pressure will change in different positions and locations, including underwater
- Recall the conditions at STP and standard conditions

Matter can exist in three different physical forms, called phases or states: gas, liquid, and solid. We have discussed liquids in the context of intermolecular forces and solids in the context of organized crystals in Chapter 3 of MCAT General Chemistry Review. The gaseous phase may be the simplest to understand because all gases display similar behavior and follow similar laws regardless of their particular chemical identities. Like liquids, gases are classified as fluids because they can flow and take on the shapes of their containers. However, the atoms or molecules in a gaseous sample move rapidly and are far apart from each other. In addition, only very weak intermolecular forces exist between gas particles; this results in certain characteristic physical properties, such as the ability to expand to

fill any volume. Gases are also easily—although not infinitely—compressible, which distinguishes them from liquids.

VARIABLES

We can define the state of a gaseous sample by four variables: pressure (P), volume (V), temperature (T), and number of moles (n).

Gas pressures are usually expressed in units of atmospheres (atm) or in millimeters of mercury (mmHg), which are equivalent to torr. The SI unit for pressure, however, is the pascal (Pa). The mathematical relationships among all of these units are as follows:

$$1 \text{ atm} = 760 \text{ mmHg} \equiv 760 \text{ torr} = 101.325 \text{ kPa}$$

Medical devices that measure blood pressure are termed sphygmomanometers, and the most clinically relevant unit of measurement for them is mmHg. In fact many medical devices utilize the same conceptual design of a barometer, shown in Figure 8.1, to continuously monitor blood pressure.

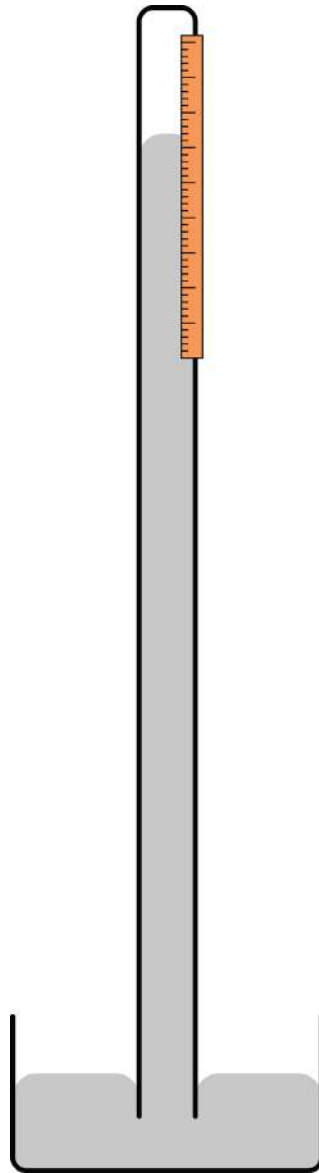


Figure 8.1. Schematic of a Simple Mercury Barometer



REAL WORLD

Blood pressure is measured by a sphygmomanometer, which uses units of mmHg. A normal adult blood pressure is considered less than 120 mmHg systolic and 80 mmHg diastolic (< 120/80).

Hypertension (high blood pressure) is defined as having at least two blood pressure readings > 140 mmHg systolic or > 90 mmHg diastolic.

In order to explain why the mercury rises in a barometer, we must summarize the forces at play here. Atmospheric pressure creates a downward force on the pool of mercury at the base of the barometer while the mercury in the column exerts an opposing force (its weight) based on its density. The weight of the mercury creates a vacuum in the top of the tube. When the external air exerts a higher force than the weight of the mercury in the column, the column rises. When the external air exerts a lower force than the weight of the mercury, the column falls. Thus, a reading can be obtained by measuring the height of the mercury column (in mm), which will be directly proportional to the atmospheric pressure being applied.



BRIDGE

Fluid dynamics is an important concept discussed in Chapter 4 of MCAT Physics and Math Review that applies to multiple aspects of the gas laws covered here, including the functionality of a mercury barometer.

It is important to mention here that atmospheric pressure is not the only external pressure that can exert this force. For instance, a clinical blood

pressure cuff creates a force that is opposed by the person's systolic and diastolic arterial blood pressure.

The volume of a gas is generally expressed in liters (L) or milliliters (mL). Temperature is usually given in kelvins (K), although Celsius ($^{\circ}\text{C}$) may be used instead. Many processes involving gases take place under standard temperature and pressure (STP), which refers to conditions of 273 K (0°C) and 1 atm.

A note of caution: STP conditions are not identical to standard state conditions. The two standards involve different temperatures and are used for different purposes. STP (273 K and 1 atm) is generally used for gas law calculations; standard state conditions (298 K, 1 atm, 1 M concentrations) are used when measuring standard enthalpy, entropy, free energy changes, and electrochemical cell voltage.



MCAT EXPERTISE

On the MCAT, remember that STP is different from standard state. Temperature at STP is 0°C (273 K). Temperature at standard state is 25°C (298 K).

MCAT Concept Check 8.1:

Before you move on, assess your understanding of the material with these questions.

Name some characteristics that make the gas phase unique:

A mercury barometer is primarily affected by atmospheric pressure. What would happen to the level of the mercury in the column if:

- the barometer was moved to the top of a mountain?

- the barometer was placed ten meters under water?

What are the conditions for STP?

What are the standard conditions?

8.2 Ideal Gases

High-Yield



LEARNING GOALS

After Chapter 8.2, you will be able to:

- Apply the ideal gas equation to calculations of pressure, temperature, volume, or number of moles
- Calculate the density of a substance given its molecular formula, current pressure, and current temperature
- Apply Avogadro's principle, Boyle's law, Charles's law, Gay-Lussac's law, and the combined ideal gas law to given scenarios
- Solve problems using Dalton's law of partial pressures and Henry's law

When we examine the behavior of gases under varying conditions of temperature and pressure, we assume that the gases are ideal. An ideal gas represents a hypothetical gas with molecules that have no intermolecular forces and occupy no volume. Although real gases deviate from this ideal behavior at high pressures (low volumes) and low temperatures, many compressed real gases demonstrate behavior that is close to ideal.



KEY CONCEPT

An ideal gas follows the gas laws we will discuss at all pressures and temperatures. A real gas deviates from these laws at high pressures (low volumes) and low temperatures because of intermolecular forces or volume effects.

IDEAL GAS LAW

The ideal gas law was first stated in 1834 by Benoît Paul Émile Clapeyron, more than 170 years after Sir Robert Boyle performed his experimental studies on the relationship between pressure and volume in the gas state. In fact, by the time the ideal gas law found its expression, Boyle's law, Charles's law, and Dalton's law had already been well-established. Historical considerations aside, it will benefit us to examine the ideal gas law first so that we can then understand the other laws, which had been identified earlier, to be only special cases of the ideal gas law.

The ideal gas law shows the relationship among four variables that define a sample of gas:

$$PV = nRT$$

Equation 8.1

where P is the pressure, V is the volume, n is the number of moles, and T is the temperature. R represents the ideal gas constant, which has a value of $8.21 \times 10^{-2} \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}$. Be aware that the gas constant can be expressed in other units. On the MCAT, you may also encounter R given as $8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}$, which is derived when SI units of pascal (for pressure) and cubic meters (for volume) are substituted into the ideal gas law. Although the relevant values for R will be provided on Test Day if needed, it is important to recognize the appropriate value for R based on the units of the variables given in a passage or question stem.

The ideal gas law is used to determine the missing term when given all of the others. It can also be used to calculate the change in a term while holding two of the others constant. It is most commonly used to solve for volume or pressure at any given temperature and number of moles; Figure 8.2 shows graphs of P - V relationships at increasing temperatures.

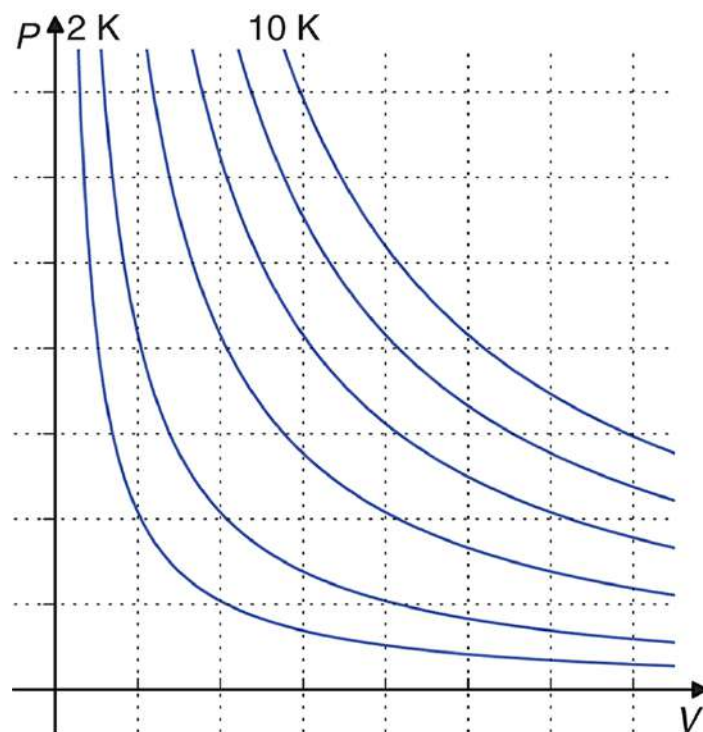


Figure 8.2. Ideal Gas Isothermal Curves

When n , R , and T are held constant, one can easily analyze the relationship between pressure and volume.

Example:

What volume would 12 g of helium occupy at 27°C and a pressure of 380 mmHg?

Solution:

The ideal gas law can be used, but first, all of the variables must be converted to units that will correspond to the expression of the gas constant as $8.21 \times 10^{-2} \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}$.

$$P = 380 \text{ mmHg} \left[\frac{1 \text{ atm}}{760 \text{ mmHg}} \right] = 0.5 \text{ atm}$$

$$T = 27^\circ\text{C} + 273 = 300 \text{ K}$$

$$n = 12 \text{ g He} \left[\frac{1 \text{ mol}}{4.0 \text{ g}} \right] = 3 \text{ mol He}$$

$$PV = nRT$$

$$(0.5 \text{ atm})(V) = (3 \text{ mol He}) \left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \right) (300 \text{ K})$$

$$0.5V \approx 3 \times 8 \times 3$$

$$V = 144 \text{ L (actual} = 148 \text{ L)}$$



BRIDGE

Round numbers to speed up your arithmetic on Test Day. For instance, constants such as 0.0821 can be rounded to 0.08. Choices will be sufficiently different so that your estimated answer will be nearly identical to the true answer choice. Arithmetic and math strategies are discussed in Chapter 10 of MCAT Physics and Math Review.

The ideal gas law is useful not only for standard calculations of pressure, volume, or temperature of a gas under a given set of conditions, but also for determinations of gas density and molar mass.

Density

We define density (ρ) as the ratio of the mass per unit volume of a substance. The densities of gases are usually expressed in units of grams per liter. The ideal gas law contains variables for volume and number of moles, so we can rearrange the law to calculate the density of any gas:

$$\begin{aligned} PV &= nRT \\ \text{where } n &= \frac{m \text{ (mass)}}{M \text{ (molar mass)}} \\ \text{Therefore, } PV &= \frac{m}{M} RT \\ \text{and } \rho &= \frac{m}{V} = \frac{PM}{RT} \end{aligned}$$

Equation 8.2

A different approach could start with the fact that a mole of an ideal gas at STP occupies 22.4 L. We can then calculate the effect of changes in pressure and temperature when they differ from STP conditions, predicting the volume of the gas. Finally, we'll calculate the density by dividing the mass by the predicted volume. The following equation, the combined gas law, is an amalgam of some of the special cases we will discuss in the following section. It can be used to relate changes in temperature, volume, and pressure of a gas:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Equation 8.3

where the subscripts 1 and 2 refer to the two states of the gas (at STP and at the conditions of actual temperature and pressure, for example). This equation assumes the number of moles stays constant.

To calculate a change in volume, the equation is rearranged as follows:

$$V_2 = V_1 \left[\frac{P_1}{P_2} \right] \left[\frac{T_2}{T_1} \right]$$

V_2 is then used to find the density of the gas under nonstandard conditions:

$$\rho = \frac{m}{V_2}$$

On Test Day, it may be helpful to visualize how the changes in pressure and temperature affect the volume of the gas, and this can serve as a check to avoid accidentally switching the values of pressure and temperature in the numerator and denominator. For example, one could predict that doubling the temperature of a gas would result in doubling its volume, and doubling the pressure of a gas would result in halving the volume, so doubling both the temperature and pressure at the same time results in a final volume that is equal to the original volume.

Example:

What is the density of CO_2 gas at 2 atm and 273°C ?

Solution:

At STP, a mole of gas occupies 22.4 L. Because the increase in pressure to 2 atm decreases volume proportionally, 22.4 L must be multiplied by $\frac{1 \text{ atm}}{2 \text{ atm}} = 0.5$. Because the increase in temperature increases volume proportionally, the temperature factor will be $\frac{546 \text{ K}}{273 \text{ K}} = 2$.

$$V_2 = \left[\frac{22.4 \text{ L}}{\text{mol}} \right] \left[\frac{1 \text{ atm}}{2 \text{ atm}} \right] \left[\frac{546 \text{ K}}{273 \text{ K}} \right] = 22.4 \frac{\text{L}}{\text{mol}}$$
$$\rho = \frac{44 \frac{\text{g}}{\text{mol}}}{22.4 \frac{\text{L}}{\text{mol}}} \approx 2 \frac{\text{g}}{\text{L}} \left(\text{actual} = 1.96 \frac{\text{g}}{\text{L}} \right).$$

Molar Mass

Sometimes the identity of a gas is unknown, and the molar mass, discussed in Chapter 4 of MCAT General Chemistry Review, can be determined in order to identify it. Using the equation for density derived

from the ideal gas law, we can calculate the molar mass of a gas experimentally in the following way:

The pressure and temperature of a gas contained in a bulb of a given volume are measured, and the mass of the bulb with the sample is measured. Then, the bulb is evacuated—the gas is removed—and the mass of the empty bulb is determined. The mass of the bulb with the sample minus the mass of the evacuated bulb gives the mass of the sample. Finally, the density of the sample is determined by dividing the mass of the sample by the volume of the bulb. This gives the density at the given temperature and pressure.

Using $V_2 = V_1 \left[\frac{P_1}{P_2} \right] \left[\frac{T_2}{T_1} \right]$, we then calculate the volume of the gas at STP, substituting 273 K for T_2 and 1 atm for P_2 . The ratio of the sample mass divided by V_2 gives the density of the gas at STP. The molar mass can then be calculated as the product of the gas's density at STP and the STP volume of one mole of gas, **$22.4 \frac{\text{L}}{\text{mol}}$** :

$$M = (\rho_{\text{STP}}) \left(22.4 \frac{\text{L}}{\text{mol}} \right)$$

Example:

What is the molar mass of a 22.4 L sample of gas that has a mass of 225 g at a temperature of 273°C and a pressure of 10 atm?

Solution:

Determine how the current conditions compare to STP, and use this to set up a proportional relationship. Be careful to note the differences between degrees C and K, and current versus STP conditions.

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \text{ giving } V_2 = V_1 \left[\frac{P_1}{P_2} \right] \left[\frac{T_2}{T_1} \right]$$

$$V_{\text{STP}} = V_1 \left[\frac{P_1}{P_{\text{STP}}} \right] \left[\frac{T_{\text{STP}}}{T_1} \right] = 22.4 \text{ L} \left[\frac{10 \text{ atm}}{1 \text{ atm}} \right] \left[\frac{273 \text{ K}}{546 \text{ K}} \right] = 22.4 \times 10 \times \frac{1}{2}$$

$$\frac{225 \text{ g}}{112 \text{ L}} \approx 2 \frac{\text{g}}{\text{L}} \text{ at STP}$$

$$M = 2 \frac{\text{g}}{\text{L}} \left[\frac{22.4 \text{ L}}{\text{mol}} \right] \approx 44.8 \frac{\text{g}}{\text{mol}}$$

SPECIAL CASES

Now that we have considered the ideal gas law as the mathematical relationship between four variables that define the state of a gas (pressure, volume, temperature, and moles of gas), we can examine the other laws that preceded its discovery. Even though the following laws were developed before the ideal gas law, it is conceptually helpful to think of them as special cases of the more general ideal gas law.

Avogadro's Principle

One important discovery that preceded Clapeyron's formulation of the ideal gas law was Avogadro's principle, which states that all gases at a constant temperature and pressure occupy volumes that are directly proportional to the number of moles of gas present. Equal amounts of all gases at the same temperature and pressure will occupy equal volumes. As discussed above, one mole of any gas, irrespective of its chemical identity, will occupy 22.4 liters at STP.

$$\frac{n}{V} = k \text{ or } \frac{n_1}{V_1} = \frac{n_2}{V_2}$$

Equation 8.4

where k is a constant, n_1 and n_2 are the number of moles of gas 1 and gas 2, respectively, and V_1 and V_2 are the volumes of the gases, respectively. This can be summarized in the following statement: as the number of moles of gas increases, the volume increases in direct proportion.

Example:

A 2.0 L sample at 100°C and 20 atm contains 5 moles of a gas. If an additional 25 moles of gas at the same pressure and temperature are added, what is the final volume of the gas?

Solution:

If pressure and temperature are held constant, the ideal gas law reduces to Avogadro's principle:

$$\begin{aligned}\frac{n_1}{V_1} &= \frac{n_2}{V_2} \\ \frac{5 \text{ mol}}{2.0 \text{ L}} &= \frac{5 \text{ mol} + 25 \text{ mol}}{V_2} \\ V_2 &= \frac{30 \text{ mol} \times 2.0 \text{ L}}{5 \text{ mol}} = 12.0 \text{ L}\end{aligned}$$

Boyle's Law

Robert Boyle conducted a series of experimental studies in 1660 that led to his formulation of a law that now bears his name: Boyle's law. His work showed that, for a given gaseous sample held at constant temperature (isothermal conditions), the volume of the gas is inversely proportional to its pressure:

$$PV = k \text{ or } P_1V_1 = P_2V_2$$

Equation 8.5

where k is a constant, and the subscripts 1 and 2 represent two different sets of pressure and volume conditions. Careful examination of Boyle's law shows that it is, indeed, simply the special case of the ideal gas law in which n and T are constant.



KEY CONCEPT

Boyle's law is a derivation of the ideal gas law and states that pressure and volume are inversely related: when one increases, the other decreases.

A plot of volume vs. pressure for a gas—the inverse of the curves in Figure 8.2—is shown in Figure 8.3.

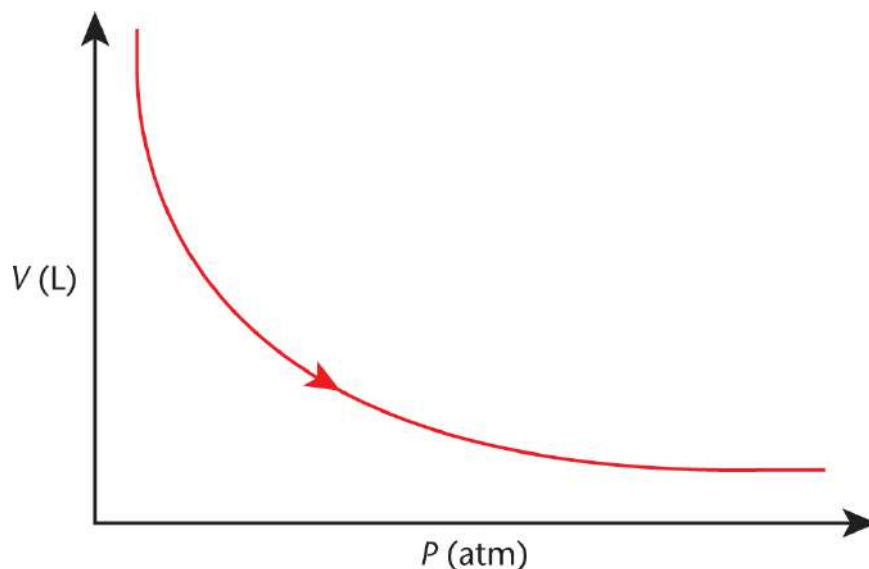


Figure 8.3. Boyle's Law (Isothermal Compression)
As pressure increases, volume decreases.



MCAT EXPERTISE

Sometimes it is easier to remember the shape of the graph to help you recall the variables' relationship on Test Day. Here we can see

that, as pressure increases, the volume decreases, and vice-versa. These ratios and relationships will often allow you to answer questions on the MCAT without having to do much math.

Example:

What would be the volume of a 1 L sample of helium if its pressure is changed from 12 atm to 4 atm under isothermal conditions?

Solution:

If the number of moles of gas and temperature are held constant, the ideal gas law reduces to Boyle's law:

$$\begin{aligned}P_1 V_1 &= P_2 V_2 \\(12 \text{ atm})(1 \text{ L}) &= (4 \text{ atm})(V_2) \\V_2 &= \frac{12 \text{ atm} \times 1 \text{ L}}{4 \text{ atm}} = 3 \text{ L}\end{aligned}$$

Charles's Law

In the early 19th century, Joseph Louis Gay-Lussac published findings based, in part, on earlier unpublished work by Jacques Charles; hence, the law of Charles and Gay-Lussac is more commonly known simply as

Charles's law. The law states that, at constant pressure, the volume of a gas is proportional to its absolute temperature, expressed in kelvins. Expressed mathematically, Charles's law is

$$\frac{V}{T} = k \text{ or } \frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Equation 8.6

where, again, k is a proportionality constant and the subscripts 1 and 2 represent two different sets of temperature and volume conditions. Careful examination of Charles's law shows that it is another special case of the ideal gas law in which n and P are constant.



KEY CONCEPT

Charles's law is also a derivation of the ideal gas law and states that volume and temperature are directly proportional: when one increases, the other increases in direct proportion.

A plot of temperature vs. volume is shown in Figure 8.4. Note that if one extrapolates the V vs. T plot for a gas back to where $T = 0$ (absolute zero), we find that $V = 0$!

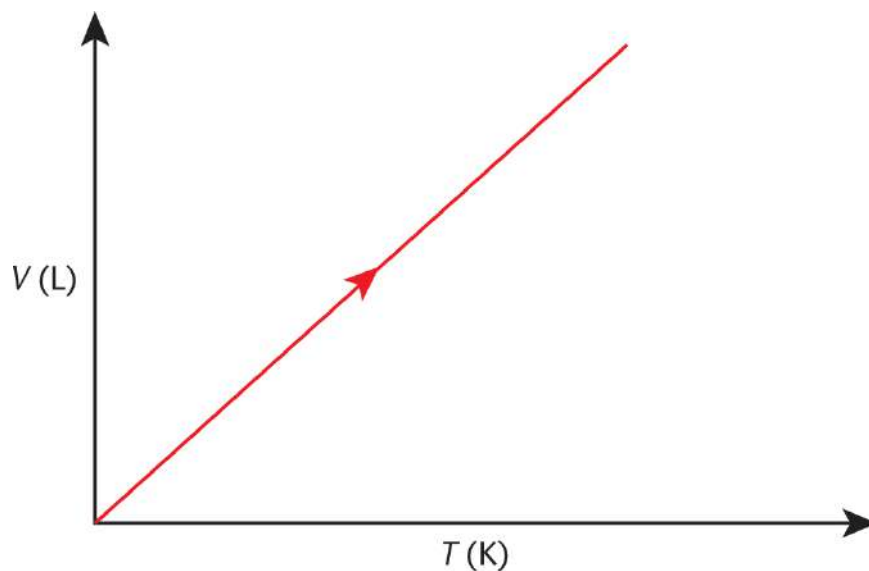


Figure 8.4. Charles's Law (Isobaric Expansion)
As temperature increases, volume increases.



REAL WORLD

While the temperature of 0 K cannot be physically attained, curves such as Charles's law were originally used to figure out its value.

Example:

If the temperature of 2 L of gas at constant pressure is changed from 290 K to 580 K, what would be its final volume?

Solution:

If the number of moles of gas and pressure are held constant, the ideal gas law reduces to Charles's law:

$$\begin{aligned}\frac{V_1}{T_1} &= \frac{V_2}{T_2} \\ \frac{2\text{ L}}{290\text{ K}} &= \frac{V_2}{580\text{ K}} \\ V_2 &= \frac{2\text{ L} \times 580\text{ K}}{290\text{ K}} = 4\text{ L}\end{aligned}$$

Gay-Lussac's Law

Gay-Lussac's law is complementary to Charles's Law. It utilizes the same derivation from the ideal gas law, but it relates pressure to temperature instead. Expressed mathematically, Gay-Lussac's law is

$$\frac{P}{T} = k \text{ or } \frac{P_1}{T_1} = \frac{P_2}{T_2}$$

Equation 8.7

where, again, k is a proportionality constant, and the subscripts 1 and 2 represent two different sets of temperature and pressure conditions. Careful examination of Gay-Lussac's law shows that it is another special case of the ideal gas law in which n and V are constant.

Figure 8.5 graphs this concept, which is nearly identical to Charles's law. Again, an increase in temperature will increase the pressure in direct

proportion.

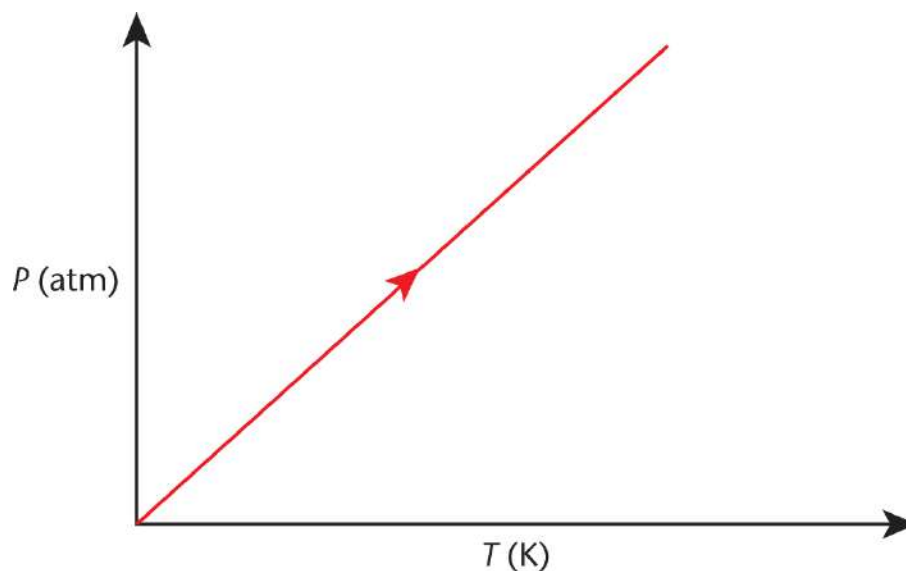


Figure 8.5. Gay-Lussac's Law (Isovolumetric Heating)
As temperature increases, pressure increases.

Example:

If the pressure of a sample of gas with a temperature of 300 K changes from 2 atm to 5 atm during heating, what would be the final temperature if volume is held constant?

Solution:

If the number of moles of gas and volume are held constant, the ideal gas law reduces to Gay-Lussac's law:

$$\begin{aligned}\frac{P_1}{T_1} &= \frac{P_2}{T_2} \\ \frac{2 \text{ atm}}{300 \text{ K}} &= \frac{5 \text{ atm}}{T_2} \\ T_2 &= \frac{5 \text{ atm} \times 300 \text{ K}}{2 \text{ atm}} = 750 \text{ K}\end{aligned}$$

Combined Gas Law

As discussed earlier, the combined gas law (Equation 8.3) was a combination of many of the preceding laws. This law relates pressure and volume (Boyle's law) in the numerator, and relates the variations in temperature to both volume (Charles's law) and pressure (Gay-Lussac's law) simultaneously. When using this equation, take care to place all of the variables in the right place.



MCAT EXPERTISE

Understanding how the combined gas law functions helps avoid the need to memorize every other special case of the ideal gas law. Read the question stem or passage with an eye toward the quantities that remain constant to know when assumptions can be made.

DALTON'S LAW OF PARTIAL PRESSURES

When two or more gases that do not chemically interact are found in one vessel, each gas will behave independently of the others. That is, each gas will behave as if it were the only gas in the container. Therefore, the pressure exerted by each gas in the mixture will be equal to the pressure that the gas would exert if it were the only one in the container. The pressure exerted by each individual gas is called the partial pressure of that gas. In 1801, John Dalton derived an expression, now known as Dalton's law of partial pressures, which states that the total pressure of a gaseous mixture is equal to the sum of the partial pressures of the individual components. The equation for Dalton's law is

$$P_T = P_A + P_B + P_C + \dots$$

Equation 8.8

where P_T is the total pressure in the container, and P_A , P_B , and P_C are the partial pressures of gases A, B, and C, respectively.



KEY CONCEPT

When more than one gas is in a container, each contributes to the whole as if it were the only gas present. Add up all of the pressures of the individual gases and you get the whole pressure of the system.

The partial pressure of a gas is related to its mole fraction and can be determined using the following equation:

$$P_A = X_A P_T$$

where $X_A = \frac{\text{moles of gas A}}{\text{total moles of gas}}$

Equation 8.9

Example:

A vessel contains 0.75 mol of nitrogen, 0.20 mol of hydrogen, and 0.05 mol of fluorine at a total pressure of 2.5 atm. What is the partial pressure of each gas?

Solution:

First calculate the mole fraction of each gas.

$$X_{N_2} = \frac{0.75 \text{ mol}}{1.00 \text{ mol}} = 0.75 \quad X_{H_2} = \frac{0.20 \text{ mol}}{1.00 \text{ mol}} = 0.20 \quad X_{F_2} = \frac{0.05 \text{ mol}}{1.00 \text{ mol}}$$

Then calculate the partial pressure.

$$\begin{aligned}
 P_A &= X_A P_T \\
 P_{N_2} &= (0.75)(2.5 \text{ atm}) = 1.875 \text{ atm} \\
 P_{H_2} &= (0.20)(2.5 \text{ atm}) = 0.5 \text{ atm} \\
 P_{F_2} &= (0.05)(2.5 \text{ atm}) = 0.125 \text{ atm}
 \end{aligned}$$

HENRY'S LAW

The difference in gas solubility between fluids was explained by William Henry in 1803. What Henry noticed was that, at various applied pressures, the concentration of a gas in a liquid increased or decreased. This was a characteristic of a gas's vapor pressure. Vapor pressure is the pressure exerted by evaporated particles above the surface of a liquid. Evaporation, as discussed in Chapter 7 of MCAT General Chemistry Review, is a dynamic process that requires the molecules at the surface of a liquid to gain enough energy to escape into the gas phase.

Vapor pressure from the evaporated molecules forces some of the gas back into the liquid phase, and equilibrium is reached between evaporation and condensation. Mathematically, this is expressed as

$$[A] = k_H \times P_A \text{ or } \frac{[A]_1}{P_1} = \frac{[A]_2}{P_2} = k_H$$

Equation 8.10

where $[A]$ is the concentration of A in solution, k_H is Henry's constant, and P_A is the partial pressure of A. The value of Henry's constant depends on the identity of the gas.



KEY CONCEPT

The solubility of a gas will increase with increasing partial pressure of the gas.

According to this relationship, solubility (concentration) and pressure are directly related. In biology, this is a critically important relationship for gas and nutrient exchange. As discussed in Chapter 6 of MCAT Biology Review, lung tissue—at the microscopic level—is organized into grapelike clusters of sacs called alveoli. These sacs are perfused by capillaries that allow for the exchange of carbon dioxide and oxygen, as shown in Figure 8.6. If the atmospheric pressure changes, as it does from sea level to high altitude, then the partial pressure of oxygen in the atmosphere also changes (as explained by Dalton's law), and the amount of gas exchanged is altered accordingly; if the partial pressure of a particular gas is elevated, such as when giving hyperbaric oxygen, the amount of that gas dissolved in the blood is also elevated.

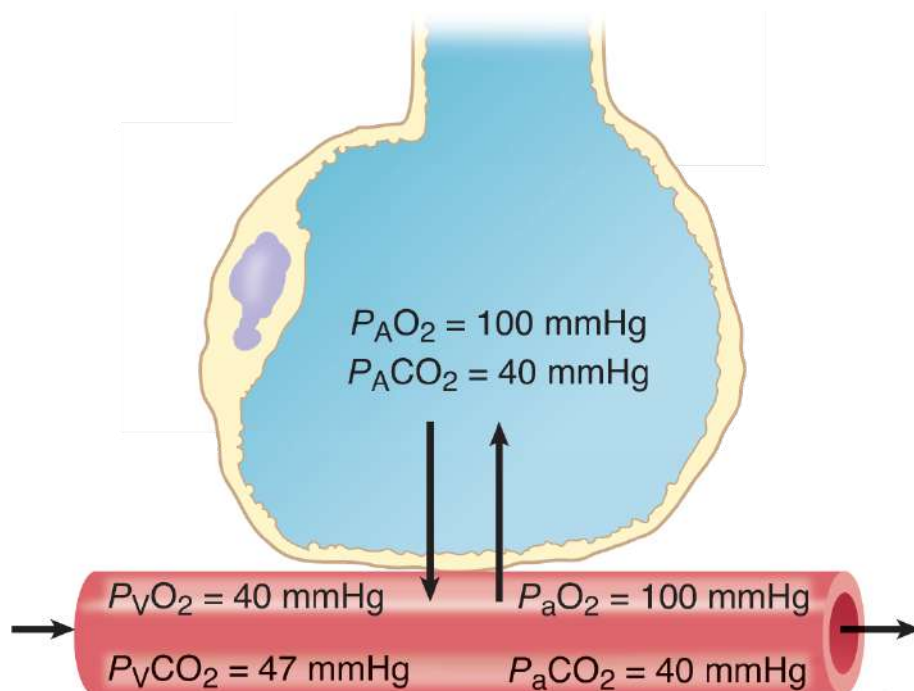


Figure 8.6. Alveolar Capillary Gas Exchange

In medicine, A represents alveolar concentrations, V represents venous concentrations, and a represents arterial concentrations.

Example:

If 4×10^{-4} moles of gas are dissolved in 2 L of solution under an ambient pressure of 2 atm, what will be the molar concentration of the gas under 10 atm?

Solution:

Start by determining the initial concentration of the gas in solution.

$$[A]_1 = \frac{4 \times 10^{-4} \text{ mol}}{2 \text{ L}} = 2 \times 10^{-4} M$$

Next, utilize the direct relationship between solubility and pressure according to Henry's law.

$$\begin{aligned}\frac{[A]_1}{P_1} &= \frac{[A]_2}{P_2} \\ \frac{2 \times 10^{-4} M}{2 \text{ atm}} &= \frac{[A]_2}{10 \text{ atm}} \\ [A]_2 &= \frac{(2 \times 10^{-4} M)(10 \text{ atm})}{2 \text{ atm}} = 10\end{aligned}$$

MCAT Concept Check 8.2:

Before you move on, assess your understanding of the material with these questions.

A container with 4 moles of a gas at a pressure of 8 atm has a volume of 12 liters. What is its temperature? (Note:

$$R = 8.21 \times 10^{-2} \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}})$$

What is the density of argon gas at 4 atm and 127°C?

A 20 L sample at 300°C and 5 atm of pressure contains 2 moles of a gas. If an additional 3 moles of gas at the same pressure and temperature are added, what is the final total volume of the gas?

What would be the volume of a 2 L sample of neon if its pressure is changed from 1 atm to 40 atm under isothermal conditions?

If the temperature of 6 L of gas at constant pressure is changed from 27°C to 127°C, what would be its final volume?

If the pressure of a sample of gas with a temperature of 227°C is changed from 5 atm to 2 atm during cooling, what would be the final temperature?

A vessel contains 8 mol O_2 , 3 mol CH_4 , and 1 mol CO_2 at a total pressure of 240 atm. What is the partial pressure of each gas?

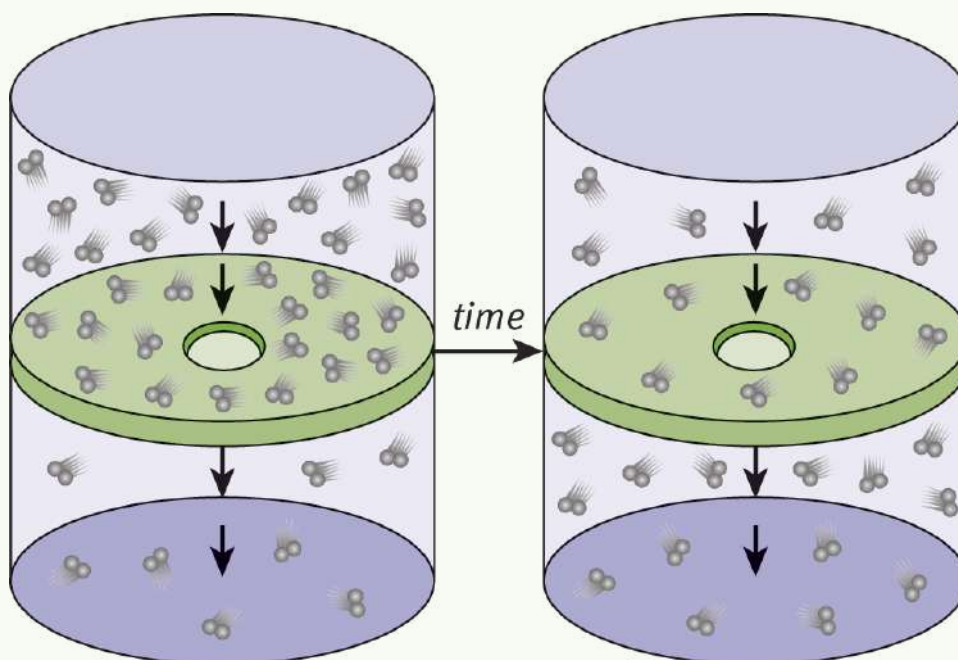
How can the concentration of carbon dioxide in sodas or other carbonated beverages be so much higher than that of atmospheric carbon dioxide?

8.3 Kinetic Molecular Theory

LEARNING GOALS

After Chapter 8.3, you will be able to:

- Recall the assumptions made by kinetic molecular theory
- Calculate the average speed of a gas, given its temperature
- Compare the relative speeds of two different gases at the same temperature
- Apply Graham's law to situations involving gas effusion:



The kinetic molecular theory was developed in the second half of the 19th century, well after the laws describing gas behavior had been developed. The kinetic molecular theory was used to explain the behavior of gases, which the other laws merely described. The gas laws demonstrate that all gases show similar physical characteristics and behavior irrespective of their particular chemical identity. The behavior of real gases deviates from the ideal behavior predicted under the assumptions of this theory, but these deviations can be corrected for in calculations. The combined efforts of James Maxwell, Ludwig Boltzmann, and others led to a simple explanation of gaseous molecular behavior based on the motion of individual molecules. Like the gas laws, the kinetic molecular theory was developed in reference to ideal gases, although it can be applied with reasonable accuracy to real gases as well.

ASSUMPTIONS

To simplify the model proposed by the kinetic molecular theory, certain assumptions are made:

- Gases are made up of particles with volumes that are negligible compared to the container volume.

- Gas atoms or molecules exhibit no intermolecular attractions or repulsions.

- Gas particles are in continuous, random motion, undergoing collisions with other particles and the container walls.

- Collisions between any two gas particles (or between particles and the container walls) are elastic, meaning that there is conservation of both momentum and kinetic energy.

The average kinetic energy of gas particles is proportional to the absolute temperature of the gas (in kelvins), and it is the same for all gases at a given temperature, irrespective of chemical identity or atomic mass.

APPLICATIONS

It is fairly straightforward to imagine gas particles as little rubber balls bouncing off each other and off the walls of the container. Of course, rubber balls, like real gas particles, have measurable mass and volume, and not even the bounciest rubber balls will collide in a completely elastic manner. Still, this provides an apt visualization of the behaviors described by the kinetic molecular theory.

Average Molecular Speeds

According to the kinetic molecular theory of gases, the average kinetic energy of a gas particle is proportional to the absolute temperature of the gas:

$$KE = \frac{1}{2} mv^2 = \frac{3}{2} k_B T$$

Equation 8.11

where k_B is the Boltzmann constant $\left(1.38 \times 10^{-23} \frac{\text{J}}{\text{K}}\right)$, which serves as a bridge between the macroscopic and microscopic behaviors of gases (that is, as a bridge between the behavior of the gas as a whole and the individual gas molecules). This equation shows that the speed of a gas particle is related to its absolute temperature. However, because of the large number of rapidly and randomly moving gas particles, which may travel only nanometers before colliding with another particle or the container wall, the speed of an individual gas molecule is nearly impossible to define. Therefore, the speeds of gases are defined in terms of their average molecular speed. One way to define an average speed is to determine the average kinetic energy per particle and then calculate the speed to which this corresponds. The resultant quantity, known as the root-mean-square speed (u_{rms}), is given by the following equation:

$$u_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

Equation 8.12

where R is the ideal gas constant, T is the temperature, and M is the molar mass.



MCAT EXPERTISE

Understanding concepts will be much more fruitful on Test Day than memorizing all of the facts. The higher the temperature, the faster the molecules move. The larger the molecules, they slower they move.

A Maxwell–Boltzmann distribution curve shows the distribution of gas particle speeds at a given temperature. Figure 8.7 shows a distribution curve of molecular speeds at two temperatures, T_1 and T_2 , where T_2 is greater than T_1 . Notice that the bell-shaped curve flattens and shifts to the right as the temperature increases, indicating that at higher temperatures, more molecules are moving at higher speeds.

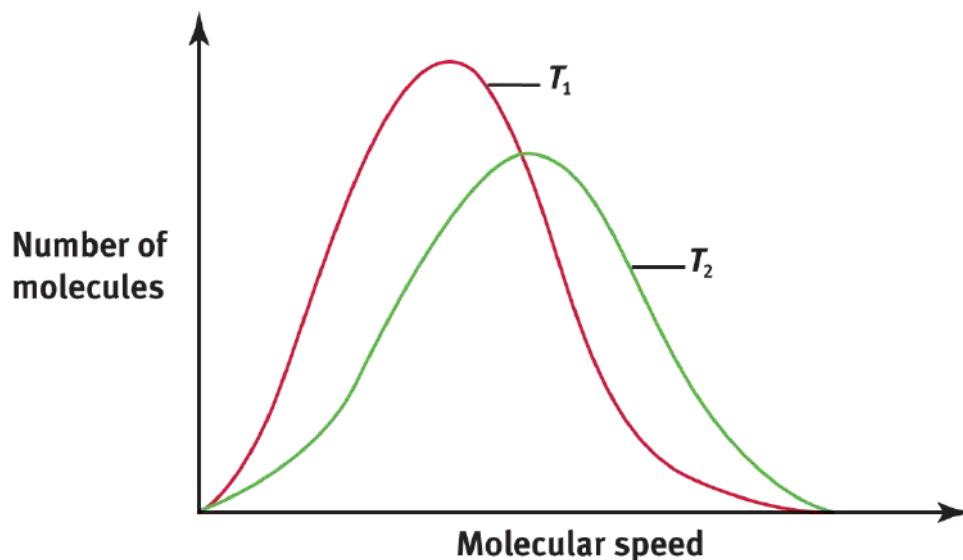


Figure 8.7. Maxwell–Boltzmann Distribution Curves of Molecular Speeds at Two Different Temperatures

Example:

What is the average speed of xenon difluoride molecules at 20°C?

Solution:

The ideal gas constant $R = 8.314 \frac{\text{J}}{\text{K} \cdot \text{mol}}$ should be used, and the molar mass of xenon difluoride is $169.3 \frac{\text{g}}{\text{mol}}$. M must be expressed in $\frac{\text{kg}}{\text{mol}}$ because joules are also derived from kilograms.

$$u_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

$$u_{\text{rms}} = \sqrt{\frac{(3)\left(8.314 \frac{\text{J}}{\text{K} \cdot \text{mol}}\right)(293 \text{ K})}{0.17 \frac{\text{kg}}{\text{mol}}}}$$

$$u_{\text{rms}} \approx \sqrt{\frac{(3)(8)(300)}{0.18}} = \sqrt{\frac{(8)(300)}{0.06}} = \sqrt{(8)(5,000)} = \sqrt{40,000}$$

$$u_{\text{rms}} \approx 200 \frac{\text{m}}{\text{s}} \left(\text{actual} = 208 \frac{\text{m}}{\text{s}} \right)$$

Graham's Law of Diffusion and Effusion

The movement of molecules from high concentration to low concentration through a medium (such as air or water) is called diffusion, as shown in Figure 8.8.

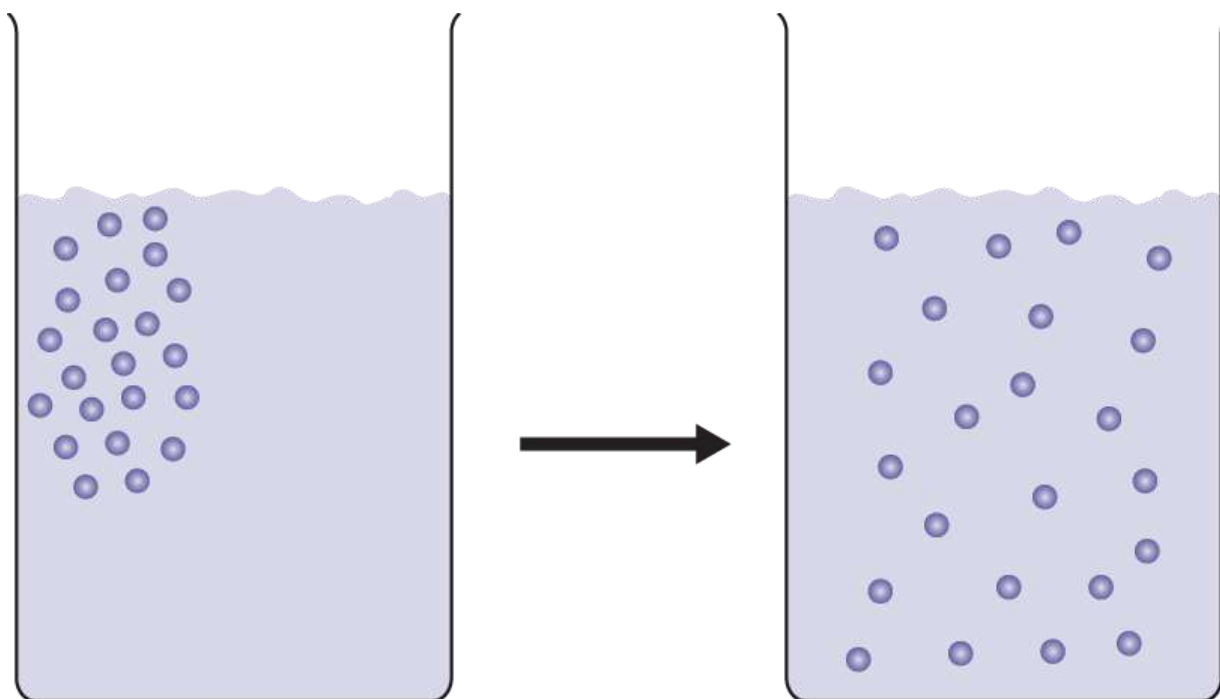


Figure 8.8. Diffusion of Solutes in a Solvent

The kinetic molecular theory of gases predicts that heavier gases diffuse more slowly than lighter ones because of their differing average speeds, as shown in Figure 8.9. Because all gas particles have the same average kinetic energy at the same temperature, it must be true that particles with greater mass travel at a slower average speed.

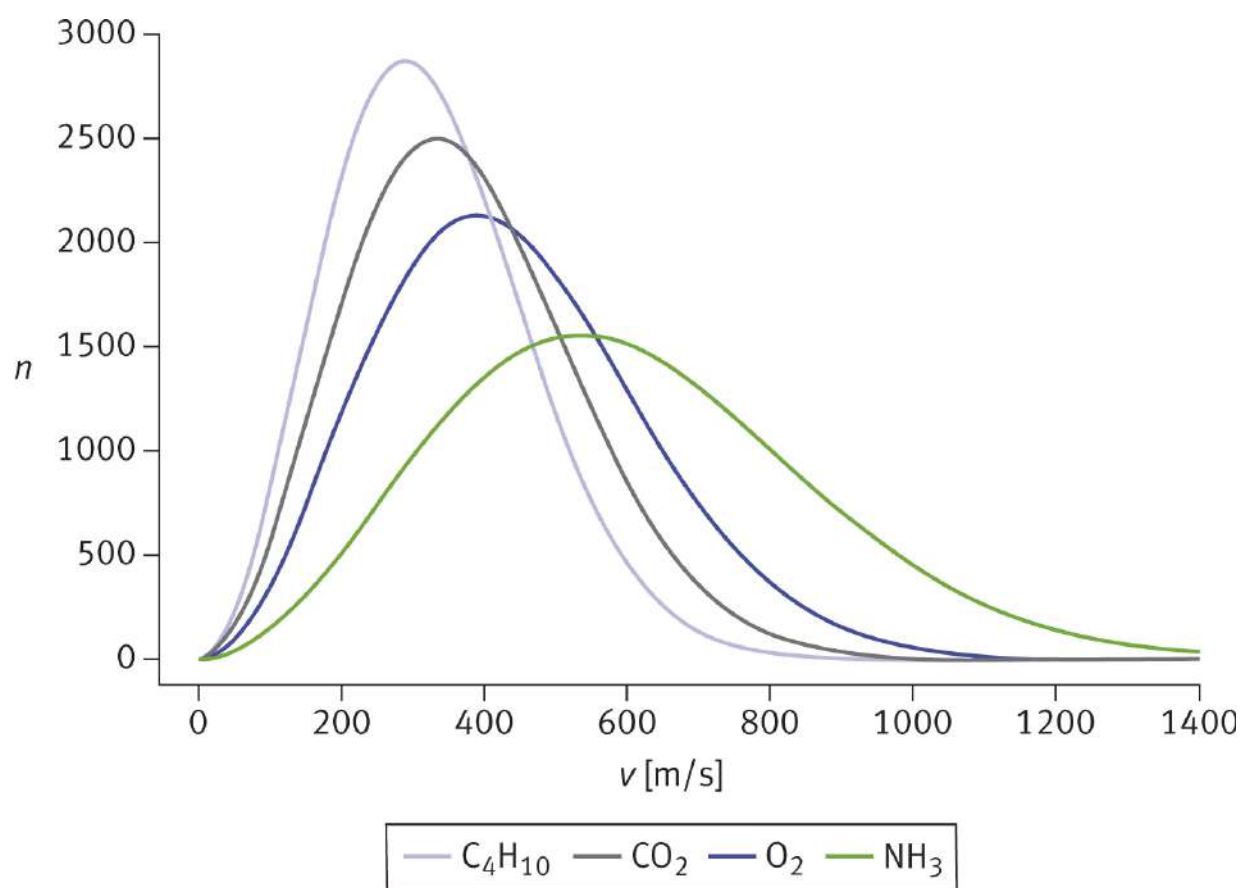


Figure 8.9. Maxwell-Boltzmann Distribution Curves of Molecular Speeds for Gases with Different Molar Masses

The more massive the gas particles, the slower their average speed.

In 1832, Thomas Graham showed mathematically that, under isothermal and isobaric conditions, the rates at which two gases diffuse are inversely proportional to the square roots of their molar masses. This is called Graham's law, which is written mathematically as:

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$$

Equation 8.13

where r_1 and r_2 are the diffusion rates of gas 1 and gas 2, respectively, and M_1 and M_2 are the molar masses of gas 1 and gas 2, respectively. From this equation, we can see that a gas that has a molar mass four times that of another gas will travel half as fast as the lighter gas.

Effusion is the flow of gas particles under pressure from one compartment to another through a small opening, as shown in Figure 8.10. Graham used the kinetic molecular theory of gases to show that, for two gases at the same temperature, the rates of effusion are proportional to the average speeds. He then expressed the rates of effusion in terms of molar mass and found that the relationship is the same as that for diffusion.

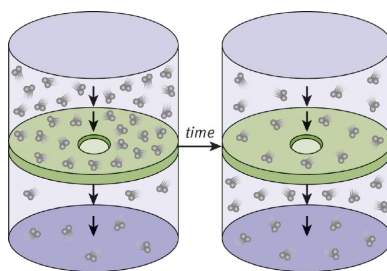


Figure 8.10. Effusion of Gas Particles

Effusion is the flow of gas particles under pressure from one compartment to another through a small opening.



REAL WORLD

In the clinics, a pleural effusion is a condition in which fluid enters the intrapleural space through small openings in the capillaries or lymphatic vessels. This causes a pressure buildup around the lungs that hinders breathing.



KEY CONCEPT

Diffusion—When gases mix with one another.

Effusion—When a gas moves through a small hole under pressure.

Both will be slower for larger molecules.

Both conditions use the same equation.

Example:

Oxygen molecules travel at an average speed of approximately $500 \frac{\text{m}}{\text{s}}$ at a given temperature. Calculate the average speed of hydrogen molecules at the same temperature.

Solution:

Oxygen's molar mass is $29 \frac{\text{g}}{\text{mol}}$. Hydrogen's molar mass is $4 \frac{\text{g}}{\text{mol}}$. Plugging into Graham's law, we get:

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}} \rightarrow r_2 = r_1 \sqrt{\frac{M_1}{M_2}}$$

$$r_2 = 500 \frac{\text{m}}{\text{s}} \sqrt{\frac{32 \frac{\text{g}}{\text{mol}}}{2 \frac{\text{g}}{\text{mol}}}} = 500 \frac{\text{m}}{\text{s}} \times 4 = 2000 \frac{\text{m}}{\text{s}}$$

MCAT Concept Check 8.3:

Before you move on, assess your understanding of the material with these questions.

What are the assumptions made by the kinetic molecular theory?

What is the average speed of helium atoms at -173°C ?

If neon gas travels at $500 \frac{\text{m}}{\text{s}}$ at a given temperature, calculate the average speed of krypton at the same temperature.

Hydrogen sulfide (H_2S) has a very strong rotten egg odor. Methyl salicylate ($\text{C}_8\text{H}_8\text{O}$) has a wintergreen odor, and benzaldehyde ($\text{C}_7\text{H}_6\text{O}$) has a pleasant almond odor. If the vapors for these three substances were released at the same time from across a room, in which order would one smell the odors? Explain your answer.

8.4 Real Gases

LEARNING GOALS

After Chapter 8.4, you will be able to:

- Distinguish between real gases and ideal gases
- Predict how differences in attractive forces or volumes will affect real gas behavior

Throughout our discussions of the laws and theory that describe and explain the behaviors of gases, we have stressed that the fundamental assumption is a gas that behaves ideally. However, our world is not one of ideal gases but rather real ones. Real gases have particles that occupy nonnegligible volumes and that interact with each other in measurable ways. In general, the ideal gas law is a good approximation of the behavior of real gases, but all real gases deviate from ideal gas behavior to some extent, particularly when the gas atoms or molecules are forced into close proximity under high pressure (at low volume) or at low temperature. These effects are implied by Figure 8.11, which shows isothermal lines on a pressure–volume graph for a real gas. Compare these lines to the ideal isotherms in Figure 8.2. Under these nonideal conditions, the intermolecular forces and the particles' volumes become significant.

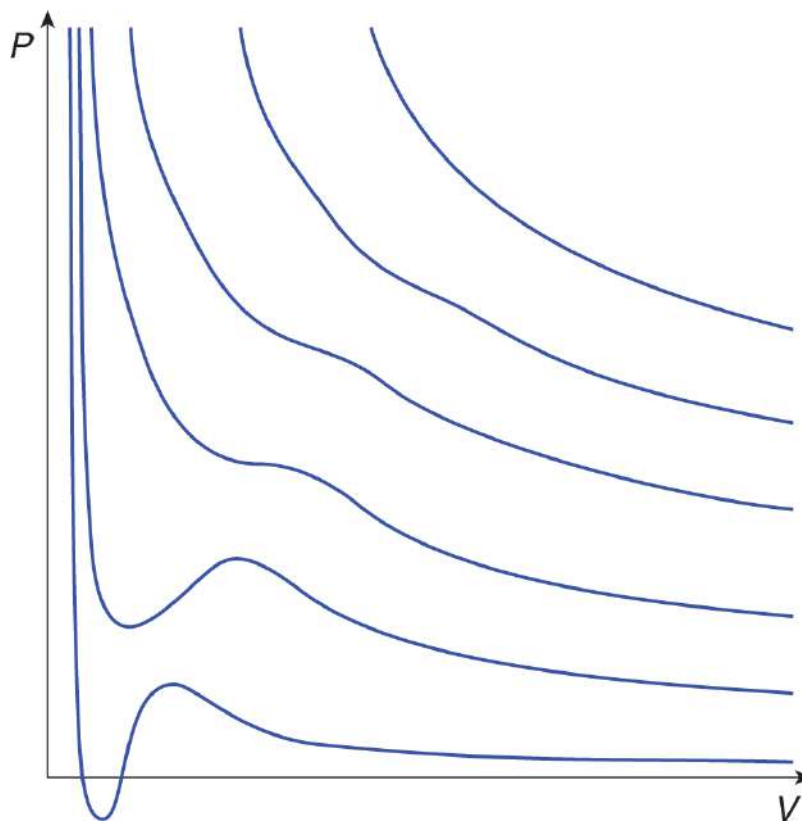


Figure 8.11. Real Gas Isothermal Curves

Compare these lines to the ideal gas isotherms in Figure 8.2.



MCAT EXPERTISE

At high temperature and low pressure (high volume), deviations from ideality are usually small; good approximations can still be made from the ideal gas law.

DEVIATIONS DUE TO PRESSURE

As the pressure of a gas increases, the particles are pushed closer and closer together. As the condensation pressure for a given temperature is approached, intermolecular attraction forces become more and more significant, until the gas condenses into a liquid.



MCAT EXPERTISE

On the MCAT, an understanding of nonideal conditions will help with determining how gases' behaviors may deviate.

At moderately high pressure (a few hundred atmospheres), a gas's volume is less than would be predicted by the ideal gas law due to intermolecular attraction. At extremely high pressures, however, the size of the particles becomes relatively large compared to the distance between them, and this causes the gas to take up a larger volume than would be predicted by the ideal gas law. That is, while the ideal gas law assumes that a gas can be compressed to take up zero volume, this is not actually physically possible—the gas particles themselves will take up space.

DEVIATIONS DUE TO TEMPERATURE

As the temperature of a gas is decreased, the average speed of the gas molecules decreases and the attractive intermolecular forces become increasingly significant. As the condensation temperature is approached for a given pressure, intermolecular attractions eventually cause the gas to condense to a liquid state.

Like deviations due to pressure, as the temperature of a gas is reduced toward its condensation point (which is the same as its boiling point), intermolecular attraction causes the gas to have a smaller volume than that which would be predicted by the ideal gas law. The closer a gas is to its boiling point, the less ideally it acts. At extremely low temperatures, gases will again occupy more space than predicted by the ideal gas law because the particles cannot be compressed to zero volume.

VAN DER WAALS EQUATION OF STATE

There are several gas equations that attempt to correct for the deviations from ideality that occur when a gas does not closely follow the ideal gas law. The van der Waals equation of state is one such equation:

$$\left(P + \frac{n^2 a}{V^2} \right) (V - nb) = nRT$$

Equation 8.14

where a and b are physical constants experimentally determined for each gas. The a term corrects for the attractive forces between molecules and, as such, will be smaller for gases that are small and less polarizable (such as helium), larger for gases that are larger and more polarizable (such as Xe or N₂), and largest for polar molecules such as HCl and NH₃. The b term corrects for the volume of the molecules themselves. Larger molecules

thus have larger values of b . Numerical values for a are generally much larger than those for b .



KEY CONCEPT

Note that if a and b are both zero, the van der Waals equation of state reduces to the ideal gas law.



MNEMONIC

a is the van der Waals term for the attractive forces.

b is the van der Waals term for big particles.

Example:

By what percentage does the real pressure of 1 mole of ammonia in a 1 liter flask at 227°C deviate from its ideal pressure? (Note: $R = 0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}$; for NH_3 , $a = 4.2$, $b = 0.037$)

Solution:

According to the ideal gas law,

$$P = \frac{nRT}{V} = \frac{(1 \text{ mol}) \left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \right) (500 \text{ K})}{1 \text{ L}} = \frac{(0.0821)}{2}$$

$$= \frac{82.1}{2} = 41.5 \text{ atm}$$

According to the van der Waals equation of state,

$$P = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2} = \left[\frac{(1 \text{ mol}) \left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \right) (500 \text{ K})}{1 \text{ L} - (1 \text{ mol})(0.037)} \right] - \left[\frac{(1 \text{ mol})^2 (4.2 \text{ atm} \cdot \text{L}^2/\text{mol}^2)}{(1 \text{ L})^2} \right]$$

$$P = \frac{41.5}{0.963} - 4.2 \approx [41.5 + 4\%(41.5)] \approx 43 - 4 \approx 39 \text{ atm}$$

(actual = 38.8 atm)

The pressure is thus approximately $41.5 - 38.8 = 2.7 \text{ atm}$ less than would be predicted from the ideal gas law, representing an error of

$$\frac{2.7 \text{ atm}}{41.5 \text{ atm}} \times 100\% \approx \frac{3}{40} \times 100\% = 7.5 \text{ percent}.$$


MCAT EXPERTISE

Be familiar with the concepts embodied by this equation but do not bother memorizing it; if the testmakers want you to apply the

equation, it will be provided in a passage or question stem.

MCAT Concept Check 8.4:

Before you move on, assess your understanding of the material with these questions.

In what ways do real gases differ from ideal gases?

Which gas will exert a higher pressure under the same, nonideal conditions: methane or chloromethane?

If methane and isobutane are placed in the same size container under the same conditions, which will exert the higher pressure (consider both as having negligible attractive forces)?

Conclusion

In this chapter, we reviewed the basic characteristics and behaviors of gases. The ideal gas law shows the mathematical relationship among four variables associated with gases: pressure, volume, temperature, and number of moles. We examined special cases of the ideal gas law in which temperature (Boyle's law), pressure (Charles's law), or volume (Gay-Lussac's law) is held constant. Henry's law helped explain the principles behind dissolution of gases in liquids and gas exchange in biological systems. We also examined Dalton's law, which relates the partial pressure of a gas to its mole fraction and the sum of the partial pressures of all the gases in a system to the total pressure of the system. The kinetic molecular theory of gases provided the explanation for the behaviors of ideal gases as described by the ideal gas law. Finally, we examined the ways in which real gases deviate from the predicted behaviors of ideal gases. The van der Waals equation of state is a useful equation for correcting deviations caused by molecular interactions and volumes.

From helium-filled balloons to the bubbles of carbon dioxide in a glass of soda, from the pressurized gases used for scuba diving to the air we breathe on land, gases are all around us. And yet, all the different gases that bubble, flow, and settle in and through our daily living experiences behave in remarkably similar ways. Human life is dependent on the exchange of two gases: oxygen and carbon dioxide—to that end, expect

that the MCAT will frequently test gases because of their importance in our everyday lives.

Concept Summary

The Gas Phase

- Gases are the least dense phase of matter.
- Gases are fluids and therefore conform to the shapes of their containers.
- Gases are easily compressible.
- Gas systems are described by the variables temperature (T), pressure (P), volume (V), and number of moles (n).
- Important pressure equivalencies include $1 \text{ atm} = 760 \text{ mmHg} \equiv 760 \text{ torr} = 101.325 \text{ kPa}$.
- A simple mercury barometer measures incident (usually atmospheric) pressure. As pressure increases, more mercury is forced into the column, increasing its height. As pressure decreases, mercury flows out of the column under its own weight, decreasing its height.

Ideal Gases

- Standard temperature and pressure (STP) is 273 K (0°C) and 1 atm .
- Equations for ideal gases assume negligible mass and volume of gas molecules.
- Regardless of the identity of the gas, equimolar amounts of two gases will occupy the same volume at the same temperature and pressure. At STP, one mole of an ideal gas occupies 22.4 L .

- The ideal gas law describes the relationship between the four variables of the gas state for an ideal gas.
- Avogadro's principle is a special case of the ideal gas law for which the pressure and temperature are held constant; it shows a direct relationship between the number of moles of gas and volume.
- Boyle's law is a special case of the ideal gas law for which temperature and number of moles are held constant; it shows an inverse relationship between pressure and volume.
- Charles's law is a special case of the ideal gas law for which pressure and number of moles are held constant; it shows a direct relationship between temperature and volume.
- Gay-Lussac's law is a special case of the ideal gas law for which volume and number of moles are held constant; it shows a direct relationship between temperature and pressure.
- The combined gas law is a combination of Boyle's, Charles's, and Gay-Lussac's laws; it shows an inverse relationship between pressure and volume along with direct relationships between pressure and volume with temperature.
- Dalton's law of partial pressures states that individual gas components of a mixture of gases will exert individual pressures in proportion to their mole fractions. The total pressure of a mixture of gases is equal to the sum of the partial pressures of the component gases.
- Henry's law states that the amount of gas dissolved in solution is directly proportional to the partial pressure of that gas at the surface of a solution.

Kinetic Molecular Theory

- The kinetic molecular theory attempts to explain the behavior of gas particles. It makes a number of assumptions about the gas particles:
 - Gas particles have negligible volume.
 - Gas particles do not have intermolecular attractions or repulsions.
 - Gas particles undergo random collisions with each other and the walls of the container.
 - Collisions between gas particles (and with the walls of the container) are elastic.
 - The average kinetic energy of the gas particles is directly proportional to temperature.
- Graham's law describes the behavior of gas diffusion or effusion, stating that gases with lower molar masses will diffuse or effuse faster than gases with higher molar masses at the same temperature.
 - Diffusion is the spreading out of particles from high to low concentration.
 - Effusion is the movement of gas from one compartment to another through a small opening under pressure.

Real Gases

- Real gases deviate from ideal behavior under high pressure (low volume) and low temperature conditions.
 - At moderately high pressures, low volumes, or low temperatures, real gases will occupy less volume than predicted by the ideal gas law because the particles have intermolecular attractions.

- At extremely high pressures, low volumes, or low temperatures, real gases will occupy more volume than predicted by the ideal gas law because the particles occupy physical space.
- The van der Waals equation of state is used to correct the ideal gas law for intermolecular attractions (a) and molecular volume (b).

Answers to Concept Checks

8.1

Gases are compressible fluids with rapid molecular motion, large intermolecular distances, and weak intermolecular forces.

At the top of the mountain, atmospheric pressure is lower, causing the column to fall. Under water, hydrostatic pressure is exerted on the barometer in addition to atmospheric pressure, causing the column to rise.

STP: $T = 273 \text{ K}$ (0°C), $P = 1 \text{ atm}$

Standard conditions: $T = 298 \text{ K}$ (25°C), $P = 1 \text{ atm}$, concentrations = 1 M

8.2

$$PV = nRT \rightarrow T = \frac{PV}{nR} = \frac{(8 \text{ atm})(12 \text{ L})}{(4 \text{ mol})\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)} \approx \frac{8 \times 3}{0.08} = \frac{3}{0.01} = 300$$

(actual = 292.3K)

$$\rho = \frac{m}{V} = \frac{PM}{RT} = \frac{(4 \text{ atm})\left(39.9 \frac{\text{g}}{\text{mol}}\right)}{\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)(400 \text{ K})} \approx 5 \frac{\text{g}}{\text{L}}$$

(actual = $4.85 \frac{\text{g}}{\text{L}}$)

$$\frac{n_1}{V_1} = \frac{n_2}{V_2} \rightarrow V_2 = \frac{n_2 \times V_1}{n_1} = \frac{(2 + 3 \text{ mol})(20 \text{ L})}{2 \text{ mol}} = 50 \text{ L}$$

$$P_1 V_1 = P_2 V_2 \rightarrow V_2 = \frac{P_1 V_1}{P_2} = \frac{(1 \text{ atm})(2 \text{ L})}{40 \text{ atm}} = 0.05 \text{ L}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \rightarrow V_2 = \frac{V_1 \times T_2}{T_1} = \frac{(6 \text{ L})(400 \text{ K})}{300 \text{ K}} = 8 \text{ L}$$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \rightarrow T_2 = \frac{P_2 \times T_1}{P_1} = \frac{(2 \text{ atm})(500 \text{ K})}{5 \text{ atm}} = 200 \text{ K}$$

There are twelve total moles of gas, so the mole fractions of each gas are:

$$X_{\text{O}_2} = \frac{8 \text{ mol O}_2}{12 \text{ mol}} = \frac{2}{3} = 0.67; X_{\text{CH}_4} = \frac{3 \text{ mol CH}_4}{12 \text{ mol}} = \frac{1}{4} = 0.25; X_{\text{CO}_2} = \frac{1 \text{ mol CO}_2}{12 \text{ mol}} = 0.083$$

Then multiply each mole fraction by the total pressure to get the partial pressures (this is typically simpler with fractions than with decimals):

$$P_{\text{O}_2} = \frac{2}{3}(240 \text{ atm}) = 160 \text{ atm}; P_{\text{CH}_4} = \frac{1}{4}(240 \text{ atm}) = 60 \text{ atm}$$

$$P_{\text{CO}_2} = \frac{1}{12}(240 \text{ atm}) = 20 \text{ atm}$$

High pressures of carbon dioxide gas are forced on top of the liquid in sodas, increasing its concentration in the liquid.

8.3

Assumptions in the kinetic molecular theory include: negligible volume of gas particles, no intermolecular forces, random motion, elastic collisions, and proportionality between absolute temperature and energy.

$$u_{\text{rms}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3\left(8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}\right)(100 \text{ K})}{0.004 \frac{\text{kg}}{\text{mol}}}} \approx \sqrt{\frac{(3)(8)(10^3)}{4 \times 10^{-3}}}$$

$$= \sqrt{(3)(8)(25)(10^3)}$$

$$u_{\text{rms}} = \sqrt{(3)(200)(10^3)} = \sqrt{600 \times 10^3} = \sqrt{60 \times 10^4}$$

$$\approx \sqrt{64} \times \sqrt{10^4}$$

$$u_{\text{rms}} = 8 \times 10^2 = 800 \frac{\text{m}}{\text{s}} \left(\text{actual} = 790 \frac{\text{m}}{\text{s}} \right)$$

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}} \rightarrow r_2 = r_1 \sqrt{\frac{M_1}{M_2}} = 400 \frac{\text{m}}{\text{s}} \sqrt{\frac{20.2 \frac{\text{g}}{\text{mol}}}{83.8 \frac{\text{g}}{\text{mol}}}} \approx 400 \times \frac{1}{2} = 200$$

(actual = $196.4 \frac{\text{m}}{\text{s}}$)

The rotten egg odor (hydrogen sulfide) first, almond (benzaldehyde) next, and wintergreen (methyl salicylate) last. Because all of the gases have the same temperature, they have the same kinetic energy; thus, the lightest molecules travel the fastest.

8.4

Real gas molecules have nonnegligible volume and attractive forces. Real gases deviate from ideal gases at high pressure (low volume) and low temperature.

According to the van der Waals equation, if a is increased while b remains negligible, the correction term $\left(\frac{n^2 a}{V^2}\right)$ gets larger, and the pressure drops to compensate. Therefore, methane will behave more ideally than chloromethane because a is smaller for methane. The real pressure of methane will thus be higher (closer to ideal).

Isobutane is larger and will thus have a larger correction term for the size of the molecule, b . This makes the term $V - nb$ smaller. The pressure or volume must rise to compensate. Because the two gases are in the same size container, isobutane must exert a higher pressure.

Equations to Remember

(8.1) Ideal gas law: $PV = nRT$

(8.2) Density of a gas: $\rho = \frac{m}{V} = \frac{PM}{RT}$

(8.3) Combined gas law: $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$

(8.4) Avogadro's principle: $\frac{n}{V} = k$ or $\frac{n_1}{V_1} = \frac{n_2}{V_2}$

(8.5) Boyle's law: $PV = k$ or $P_1 V_1 = P_2 V_2$

(8.6) Charles's law: $\frac{V}{T} = k$ or $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

(8.7) Gay-Lussac's law: $\frac{P}{T} = k$ or $\frac{P_1}{T_1} = \frac{P_2}{T_2}$

(8.8) Dalton's law (total pressure from partial pressures): $P_T = P_A + P_B + P_C + \dots$

(8.9) Dalton's law (partial pressure from total pressure): $P_A = X_A P_T$

(8.10) Henry's law: $[A] = k_H \times P_A$ or $\frac{[A]_1}{P_1} = \frac{[A]_2}{P_2} = k_H$

(8.11) Average kinetic energy of a gas: $KE = \frac{1}{2} mv^2 = \frac{3}{2} k_B T$

(8.12) Root-mean-square speed: $u_{\text{rms}} = \sqrt{\frac{3RT}{M}}$

(8.13) Graham's law: $\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$

(8.14) Van der Waals equation of state:

$$\left(P + \frac{n^2 a}{V^2}\right) (V - nb) = nRT$$



Shared Concepts

Biology Chapter 6

The Respiratory System

General Chemistry Chapter 3

Bonding and Chemical Interactions

General Chemistry Chapter 6

Equilibrium

Physics and Math Chapter 2

Work and Energy

Physics and Math Chapter 3

Thermodynamics

Physics and Math Chapter 4

Fluids

Discrete Practice Questions

Consult your online resources for additional practice.

1. Which of the following sets of conditions would be LEAST likely to result in ideal gas behavior?

- (A) High pressure and low temperature
- (B) Low temperature and large volume
- (C) High pressure and large volume
- (D) Low pressure and high temperature

2. What is the density of neon gas in $\frac{\text{g}}{\text{L}}$ at STP?

- (A) 452.3
- (B) 226.0
- (C) 1.802
- (D) 0.9018

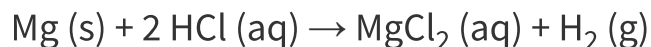
3. A leak of helium gas through a small hole occurs at a rate of

$$R = 8.314 \frac{\text{J}}{\text{K} \cdot \text{mol}}$$

How will the leakage rates of neon and oxygen gases compare to helium at the same temperature and pressure?

- (A) Neon will leak faster than helium; oxygen will leak faster than helium.
- (B) Neon will leak faster than helium; oxygen will leak slower than helium.
- (C) Neon will leak slower than helium; oxygen will leak faster than helium.
- (D) Neon will leak slower than helium; oxygen will leak slower than helium.

4. A 0.040 g piece of magnesium is placed in a beaker of hydrochloric acid. Hydrogen gas is generated according to the following equation:



The gas is collected over water at 25°C, and the gauge pressure during the experiment reads 784 mmHg.

The gas displaces a volume of 100 mL. The vapor pressure of water at 25°C is approximately 24.0 mmHg. Based on this data, how many moles of hydrogen are produced in this reaction? (Note:

$$R = 0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} = 8.314 \frac{\text{J}}{\text{K} \cdot \text{mol}} .)$$

- (A) 4.04×10^{-5} moles hydrogen
- (B) 4.09×10^{-3} moles hydrogen
- (C) 3.07×10^{-2} moles hydrogen
- (D) 3.11 moles hydrogen

5. Ideal gases:

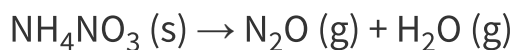
I. have no volume.

II. have particles with no attractive forces between them.

III. have no mass.

- (A) I only
- (B) II only
- (C) I and II only
- (D) I, II, and III

6. An 8.00 g sample of NH_4NO_3 (s) is placed into an evacuated 10 L flask and heated to 227°C . After the NH_4NO_3 completely decomposes, what is the approximate pressure in the flask?

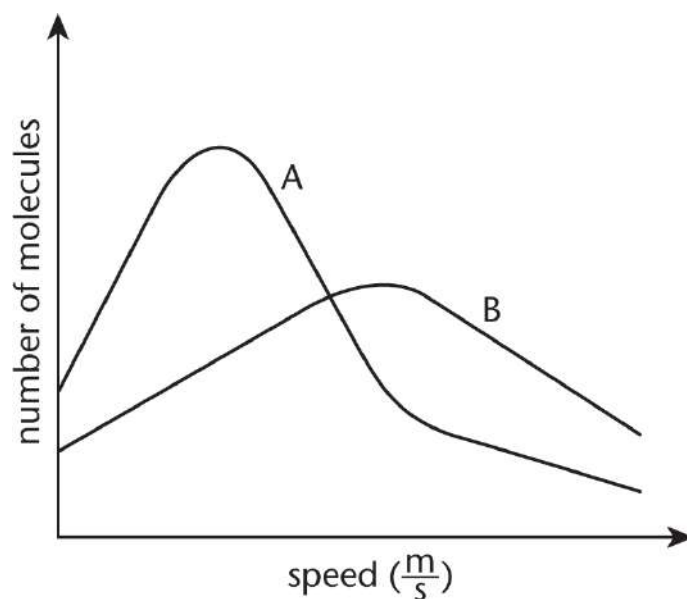


- (A) 0.410 atm
- (B) 0.600 atm
- (C) 0.821 atm
- (D) 1.23 atm

7. The kinetic molecular theory states that:

- (A) the average kinetic energy of a molecule of gas is directly proportional to the temperature of the gas in kelvins.
- (B) collisions between gas molecules are inelastic.
- (C) gas particles occupy discrete areas of space.
- (D) all gas molecules have the same kinetic energy at the same temperature.

8. The plots of two gases at STP are shown below. One of the gases is 1.0 L of helium, and the other is 1.0 L of bromine. Which plot corresponds to each gas and why?



- (A) Curve A is helium and curve B is bromine because helium has a smaller molar mass than bromine.
- (B) Curve A is helium and curve B is bromine because the average kinetic energy of bromine is greater than the average kinetic energy of helium.
- (C) Curve A is bromine and curve B is helium because helium has a smaller molar mass than bromine.
- (D) Curve A is bromine and curve B is helium because the average kinetic energy of bromine is greater than the average kinetic energy of helium.

9. At sea level and 25°C, the solubility of oxygen gas in water is 1.25×10^{-3} M. In Denver, a city in the United States that lies high above sea level, the atmospheric pressure is 0.800 atm. What is the solubility of oxygen in water in Denver?

- (A) 1.00×10^{-3} M
- (B) 1.05×10^{-3} M
- (C) 1.50×10^{-3} M
- (D) 2.56×10^{-3} M

10. Given that the gases at the center of the sun have an average molar mass of $2.00 \frac{\text{g}}{\text{mol}}$, compressed to a density of $1.20 \frac{\text{g}}{\text{cm}^3}$ under 1.30×10^9 atm of pressure, what is the temperature at the center of the sun?

- (A) $2.6 \times 10^4 \text{ K}$
- (B) $2.6 \times 10^6 \text{ K}$
- (C) $2.6 \times 10^7 \text{ K}$
- (D) $2.6 \times 10^{10} \text{ K}$

11. The gaseous state of matter is characterized by which of the following properties?

I. Gases are compressible.

II. Gases assume the volume of their containers.

III. Gas particles exist as diatomic molecules.

- (A) I only
- (B) I and II only
- (C) II and III only
- (D) I, II, and III

12. A gas at a temperature of 27°C has a volume of 60.0 mL. What temperature change is needed to increase this gas to a volume of 90.0 mL?

- (A) A reduction of 150°C
- (B) An increase of 150°C
- (C) A reduction of 13.5°C
- (D) An increase of 13.5°C

13. A gaseous mixture contains nitrogen and helium and has a total pressure of 150 torr. The nitrogen particles comprise 80 percent of the gas, and the helium particles make up the other 20 percent of the gas. What is the pressure exerted by each individual gas?

- (A) 100 torr nitrogen, 50.0 torr helium
- (B) 120 torr nitrogen, 30.0 torr helium
- (C) 30.0 torr nitrogen, 120 torr helium
- (D) 50.0 torr nitrogen, 100 torr helium

14. In which of the following situations is it impossible to predict how the pressure will change for a gas sample?

- (A) The gas is cooled at a constant volume.
- (B) The gas is heated at a constant volume.
- (C) The gas is heated, and the volume is simultaneously increased.
- (D) The gas is cooled, and the volume is simultaneously increased.

15. Experimenters notice that the molar concentration of dissolved oxygen in an enclosed water tank has decreased to one-half its original value. In an attempt to counter this decrease, they quadruple the partial pressure of oxygen in the container. What is the final concentration of the gas?

- (A) Half of the original concentration
- (B) The same as the original concentration
- (C) Double the original concentration
- (D) Quadruple the original concentration

Explanations to Discrete Practice Questions

1. A

Gases deviate from ideal behavior at higher pressures and lower volumes and temperatures, all of which force molecules closer together. The closer they are, the more they can participate in intermolecular forces, which violates the definition of an ideal gas. At low temperatures, the kinetic energy of the particles is reduced, so collisions with other particles or the walls of the container are more likely to result in significant changes in kinetic energy.

2. D

Density equals mass divided by volume. The mass of 1 mole of neon gas equals 20.2 grams. At STP, 1 mole of neon occupies 22.4 L.

$$\text{density} = \rho = \frac{\text{mass}}{\text{volume}} = \frac{20.2 \text{ g}}{22.4 \text{ L}} \approx \frac{10}{11} < 1 \frac{\text{g}}{\text{L}}$$

(actual = $0.902 \frac{\text{g}}{\text{L}}$)

3. D

Graham's law of effusion states that the relative rates of effusion of two gases at the same temperature and pressure are given by the inverse ratio of the square roots of the masses of the gas particles. In other

words, a gas with a higher molar mass will leak more slowly than a gas with a lower molar mass. Both neon and oxygen gases will leak at slower rates than helium because they both have more mass than helium.

4. B

The pressure of the gas is calculated by subtracting the vapor pressure of water from the measured pressure during the experiment: $784 \text{ mmHg} - 24 \text{ mmHg} = 760 \text{ mmHg}$, or 1 atm . This is because the reaction is carried out in an aqueous environment; the water present will contribute to the partial pressures of the gas over the liquid. The ideal gas law can be used to calculate the moles of hydrogen gas. The volume of the gas is 0.100 L , the temperature is 298 K , and $R = 0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}$. Plugging in gives:

$$\begin{aligned} n &= \frac{PV}{RT} = \frac{(1 \text{ atm})(0.1 \text{ L})}{\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)(298 \text{ K})} \approx \frac{1}{0.8 \times 300} \\ &= \frac{1}{240} \approx \frac{1}{250} = 0.004 \end{aligned}$$

(A) incorrectly substitutes 8.314 into the gas law, rather than 0.0821 . Remember that the value of R depends on the other variables in the equation; using 1 atm in the numerator necessitates using 0.0821 . (C) incorrectly substitutes the wrong R and keeps the pressure in mmHg . (D) also keeps the pressure in mmHg .

5. B

Ideal gases are said to have no attractive forces between molecules. While each particle within the gas is considered to have negligible

volume, ideal gases as a whole certainly do have a measurable volume, thus option I is eliminated. Gases have molar masses, thus option III is eliminated.

6. D

The first thing to do is balance the given chemical equation: $\text{NH}_4\text{NO}_3 (\text{s}) \rightarrow \text{N}_2\text{O} (\text{g}) + 2 \text{H}_2\text{O} (\text{g})$. The mass given is 8.00 g, which represents 0.1 mol NH_4NO_3 (**molar mass = $80.0 \frac{\text{g}}{\text{mol}}$**). When 0.1 mol of the solid decomposes, it will form 0.1 mol N_2O and 0.2 mol water. This gives approximately 0.3 moles of gas product. The ideal gas equation can be used to obtain the pressure in the flask:

$$P = \frac{nRT}{V} = \frac{(0.3 \text{ mol}) \left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \right) (500 \text{ K})}{(10 \text{ L})} \approx 1.2 \text{ atm}$$

(C) is the result if one assumes the equation is balanced, obtaining 0.2 mol gas as the product.

7. A

The average kinetic energy is directly proportional to the temperature of a gas in kelvins. The kinetic molecular theory states that collisions between molecules are elastic and thus do not result in a loss of energy, eliminating (B). Gas particles are assumed to take up negligible space in kinetic molecular theory, eliminating (C). While the average kinetic energy of any gas as a whole is the same at a given temperature, the

particles themselves have a distribution of speeds (as seen in the Maxwell–Boltzmann distribution curve), eliminating (D).

8. C

At STP, the difference between the distribution of speeds for helium and bromine gas is due to the difference in molar mass. Helium has a smaller molar mass than bromine. Particles with small masses travel faster than those with large masses, so the helium gas corresponds to curve B, which has a higher average speed. Because the gases are at the same temperature (273 K), they have the same average kinetic energy, eliminating (B) and (D).

9. A

The solubility of gases in liquids is directly proportional to the atmospheric pressure, as shown by Henry's law.

$$\frac{[\text{O}_2]_1}{P_1} = \frac{[\text{O}_2]_2}{P_2}$$

$$\frac{1.25 \times 10^{-3} M}{1.000 \text{ atm}} = \frac{[\text{O}_2]_2}{0.8000 \text{ atm}}$$

$$[\text{O}_2]_2 = \frac{0.8 \times 1.25 \times 10^{-3}}{1} = \frac{8}{10} \times \frac{5}{4} \times 10^{-3} = 1 \times 10^{-3} M$$

Note that the use of fractions allows this problem to be simplified more readily than it would be with decimals.

10. C

The ideal gas law can be modified to include density (ρ) because the number of moles of gas, n , is equal to the mass divided by the molar mass. Thus,

$$PV = nRT = \frac{m}{M} RT \rightarrow P = \frac{m}{V} \times \frac{RT}{M} = \frac{\rho RT}{M}.$$

Isolating for temperature gives:

$$T = \frac{PM}{\rho R}$$

$$\begin{aligned} T &= \frac{(1.3 \times 10^9 \text{ atm})(2.00 \frac{\text{g}}{\text{mol}})}{\left(1.2 \frac{\text{g}}{\text{cm}^3}\right)\left(1000 \frac{\text{cm}^3}{\text{L}}\right)\left(0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)} \\ &\approx \frac{(1.3 \times 2 \times 10^9)}{(1)(1000)(0.1)} \approx \frac{2.6 \times 10^9}{100} \\ &= 2.6 \times 10^7 \text{ K.} \end{aligned}$$

11. B

Gases are easily compressible because they travel freely with large amounts of space between molecules. Because gas particles are far apart from each other and in rapid motion, they tend to take up the volume of their container. Many gases exist as diatomic molecules, but this is not a property that characterizes all gases, eliminating option III.

12. B

We will use Charles's law. First, we must convert the temperature to kelvins by adding 273 to get 300 K as the initial temperature. Think of this as a proportionality: If the volume is multiplied by $\frac{1}{2}$, the temperature will also have to be multiplied by $\frac{1}{2}$, Thus the final

temperature is 450 K, which represents a 150 K increase (which is equivalent to an increase of 150°C).

13. B

The partial pressure of each gas is found by multiplying the total pressure by the mole fraction of the gas. Because 80 percent of the molecules are nitrogen, the mole fraction of nitrogen gas is equal to 0.80. Similarly, for helium, the mole fraction is 0.20. To find the pressure exerted by nitrogen, multiply the total pressure (150 torr) by 0.80 to obtain 120 torr of nitrogen. The remainder, 30 torr, is attributable to helium.

14. C

Both a change in temperature and a change in volume can affect a gas's pressure. So if one of those two variables is kept constant, as in (A) and (B), we'll definitely be able to predict which way the pressure will change. At a constant volume, heating the gas will increase its pressure, and cooling the gas will decrease it. What about when both temperature and volume are changing? If both changes have the same effect on pressure, then we can still predict which way it will change. This is the case in (D). Cooling the gas and increasing its volume both decrease pressure. (C), on the other hand, presents too vague a scenario for us to predict definitively the change in pressure. Heating the gas would amplify the pressure, while increasing the volume would decrease it. Without knowing the magnitude of each influence, it's impossible to say whether the pressure would increase, decrease, or stay the same.

15. C

Initially the concentration of the gas is decreased to one-half its original value. Recall that concentration (solubility) and partial pressure are directly related—as one increases, the other increases. If the experimenters then quadruple the partial pressure of oxygen in the vessel, the solubility is also increased by a factor of four. One-half times four gives twice the original concentration value. Misreading the answer choices as being related to the concentration before the experimenters increased the partial pressure leads to (D).

A black and white photograph of a hand holding a glass flask. The flask is tilted, and a vibrant, multi-colored liquid (purple, blue, and red) is visible at the bottom. The flask has a label with text that is partially legible, including 'APPROXIMATE VOLUMES' and '100'. The word 'Solutions' is overlaid in white text at the bottom right of the flask.

Solutions

In This Chapter

9.1 Nature of Solutions

Solvation 

Solubility 

Aqueous Solutions 

Complex Ion Formation 

9.2 Concentration

Units of Concentration 

Dilution 



9.3 Solution Equilibria

Solubility Product Constants 

Common Ion Effects 



9.4 Colligative Properties

Raoult's Law 

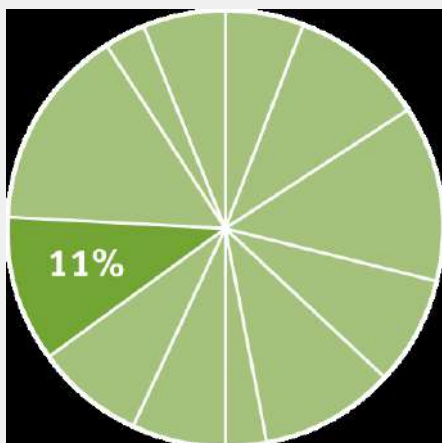
Boiling Point Elevation 

Freezing Point Depression 

Osmotic Pressure 

Concept Summary 

Chapter Profile



The content in this chapter should be relevant to about 11% of all questions about general chemistry on the MCAT.

This chapter covers material from the following AAMC content categories:

2A: Assemblies of molecules, cells, and groups of cells within single cellular and multicellular organisms

5A: Unique nature of water and its solutions

Introduction

What do first aid instant cold packs and sweet tea have in common? Not much, you might think—but both, in fact, demonstrate the same principles of solution chemistry. Instant cold packs contain two compartments, one holding water and the other ammonium nitrate. When the barrier between the two compartments is broken, it allows the ammonium nitrate to dissolve into the water. Sweet tea is made by dissolving a large amount of sugar into strongly brewed tea.

The creation of both the ammonium nitrate and sugar solutions is an endothermic process. However, the formation of the ammonium nitrate solution is much more endothermic than the formation of the sugar solution. This is why ammonium nitrate is useful in instant cold packs. When it dissolves in water, the system must absorb an amount of energy equal to $415 \frac{\text{kJ}}{\text{mol}}$, of ammonium nitrate. The heat is absorbed from the surrounding environment, so the pack feels cool to the touch.

Although the dissolution of sugar into water is not as strongly endothermic, we nevertheless have an intuitive understanding that the process is endothermic because we all know that the easiest way to dissolve lots of sugar into water (such as in tea or coffee) is to heat up the water and then add the sugar. Because heating the water increases the solubility of sugar, it must be that the dissolution of sugar into water is an endothermic process—think of Le Châtelier's principle and changes in temperature from Chapter 6.

In this chapter, our focus will be on the characteristics and behaviors of solutions, the nature of solutions, the formation of aqueous solutions, the measurements of solution concentration, and finally the qualitative and quantitative evaluation of solution equilibria.

9.1 Nature of Solutions

LEARNING GOALS

After Chapter 9.1, you will be able to:

- Describe the process of solvation
- Define key terms involved in solutions, such as complex, solubility, and saturation
- Explain how to increase solubility of a compound
- Recall the solubility rules and apply them to predict solubility of a compound

Many important chemical reactions, both in the laboratory and in nature, take place in solutions, including almost all reactions in living organisms. Solutions are homogeneous (the same throughout) mixtures of two or more substances that combine to form a single phase, usually the liquid phase. The MCAT will focus almost exclusively on solids dissolved into aqueous solutions, but it's important to remember that solutions can be formed from different combinations of the three phases of matter. For example, gases can be dissolved in liquids (carbonating soda); liquids can be dissolved in other liquids (ethanol in water); solids can even be dissolved in other solids (metal alloys). Incidentally, gases “dissolved” into

other gases can be thought of as solutions, but are more properly defined only as mixtures because gas molecules do not interact all that much chemically, as described by the kinetic molecular theory of gases. As a point of clarification: all solutions are considered mixtures, but not all mixtures are considered solutions.

A solution consists of a solute (such as NaCl , NH_3 , $\text{C}_6\text{H}_{12}\text{O}_6$, or CO_2) dissolved (dispersed) in a solvent (such as H_2O , benzene, or ethanol). The solvent is the component of the solution that remains in the same phase after mixing. If the two substances are already in the same phase (for example, a solution of two liquids), the solvent is the component present in greater quantity. If the two same-phase components are in equal proportions in the solution, then the component that is more commonly used as a solvent in other contexts is considered the solvent. Solute molecules move about freely in the solvent and interact with it by way of intermolecular forces such as ion–dipole, dipole–dipole, or hydrogen bonding. Dissolved solute molecules are also relatively free to interact with other dissolved molecules of different chemical identities; consequently, chemical reactions occur easily in solution.

SOLVATION

Solvation is the electrostatic interaction between solute and solvent molecules. This is also known as dissolution, and when water is the solvent, it can be called hydration. Solvation involves breaking intermolecular interactions between solute molecules and between solvent molecules and forming new intermolecular interactions between solute and solvent molecules together, as shown in Figure 9.1 (which was

also shown in Chapter 4 of MCAT General Chemistry Review in the context of ions).

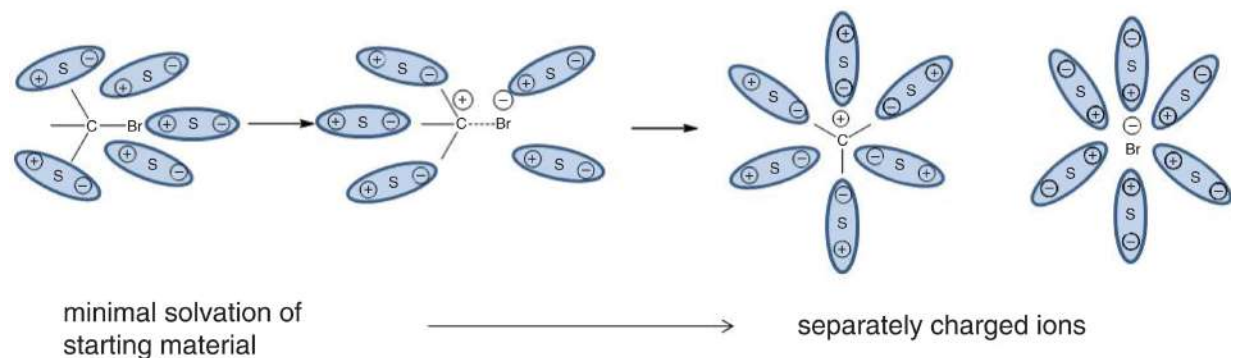


Figure 9.1. Solvation of a Polar Covalent Compound
S indicates a solvent particle.

When the new interactions are stronger than the original ones, solvation is exothermic, and the process is favored at low temperatures. The dissolution of gases into liquids, such as CO_2 into water, is an exothermic process because the only significant interactions that must be broken are those between water molecules— CO_2 , as a gas, demonstrates minimal intermolecular interaction. Le Châtelier's principle tells us this is the reason that lowering the temperature of a liquid favors solubility of a gas in the liquid.

When the new interactions are weaker than the original ones, solvation is endothermic and the process is favored at high temperatures. Most dissolutions are of this type. Two such examples have already been given: dissolving ammonium nitrate or sugar into water. Because the new interactions between the solute and solvent are weaker than the original interactions between the solute molecules and between the solvent molecules, energy (heat) must be supplied to facilitate the formation of

these weaker, less stable interactions. Sometimes the overall strength of the new interactions is approximately equal to the overall strength of the original interactions. In this case, the overall enthalpy change for the dissolution is close to zero. These types of solutions approximate the formation of an ideal solution, for which the enthalpy of dissolution is equal to zero.

The spontaneity of dissolution is dependent not only on the enthalpy change; solutions may form spontaneously for both endothermic and exothermic dissolutions. The second property that contributes to the spontaneity of dissolution is the entropy change that occurs in the process. At constant temperature and pressure, entropy always increases upon dissolution. As with any process, the spontaneity of dissolution depends on the change in Gibbs free energy: spontaneous processes are associated with a decrease in free energy, while nonspontaneous processes are associated with an increase in free energy. Thus, whether or not dissolution will happen spontaneously depends on both the change in enthalpy and the change in entropy for the solute and solvent of the system.



BRIDGE

Proteins dissolve in solution with their most hydrophilic amino acids on the outside and hydrophobic amino acids on the inside because this maximizes the increase in entropy during dissolution. As described in Chapter 1 of MCAT Biochemistry Review, a protein dissolves by forming a solvation layer.

Consider, for example, the formation of another common solution: sodium chloride dissolved in water. When NaCl dissolves in water, its component ions dissociate from each other and become surrounded by water molecules. For this new interaction to occur, ionic bonds between Na^+ and Cl^- must be broken, and hydrogen bonds between water molecules must also be broken. This step requires energy and is therefore endothermic. Because water is polar, it can interact with each of the component ions through ion–dipole interactions: the partially positive hydrogen end of the water molecules will surround the Cl^- ions, and the partially negative oxygen end of the water molecules will surround the Na^+ ions, as shown in Figure 9.2. The formation of these ion–dipole bonds is exothermic, but the magnitude is slightly less than the energy required to break the ionic bonds and hydrogen bonds. As a result, the overall dissolution of table salt into water is endothermic $\left(+3.87 \frac{\text{kJ}}{\text{mol}}\right)$ and favored at high temperatures.

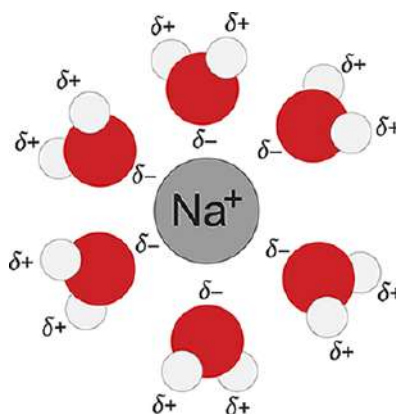


Figure 9.2. Solvation of Na^+ Ions in Aqueous Solutions

We've considered the enthalpy change for the formation of a sodium chloride solution, and now we need to examine the entropy change. Remember that entropy can be thought of as the degree to which energy is

dispersed throughout a system or the amount of energy distributed from the system to the surroundings at a given temperature. Another way to understand entropy is the measure of molecular disorder, or the number of energy microstates available to a system at a given temperature. When solid sodium chloride dissolves into water, the rigidly ordered arrangement of the sodium and chloride ions is broken up as the ion–ion interactions are disrupted and new ion–dipole interactions with the water molecules are formed. The ions, freed from their lattice arrangement, have a greater number of energy microstates available to them (in simpler terms, they are freer to move around in different ways), and consequently, their energy is more distributed and their entropy increases. The water, however, becomes more restricted in its movement because it is now interacting with the ions. The number of energy microstates available to it (that is, the water molecules' ability to move around in different ways) is reduced, so the entropy of the water decreases. In the end, the increase in the entropy experienced by the dissolved sodium chloride is greater than the decrease in the entropy experienced by the water, so the overall entropy change is positive—energy is, overall, dispersed by the dissolution of sodium chloride in water. Because of the relatively low endothermicity and relatively large positive change in entropy, sodium chloride will spontaneously dissolve in liquid water ($\Delta G = \Delta H - T\Delta S$).

SOLUBILITY

We often want to know more than just whether or not dissolution of a solute into a solvent will be spontaneous or nonspontaneous—we also want to know how much solute will dissolve into a given solvent. The solubility of a substance is the maximum amount of that substance that

can be dissolved in a particular solvent at a given temperature. When this maximum amount of solute has been added, the dissolved solute is in equilibrium with its undissolved state, and we say that the solution is saturated. If more solute is added, it will not dissolve. For example, at 25°C, a maximum of 90.9 g glucose will dissolve in 100 mL H₂O. Thus, the solubility of glucose is **90.9** $\frac{\text{g}}{\text{L}}$. If more glucose is added to an already saturated glucose solution, it will not dissolve but rather will remain in solid form, precipitating to the bottom of the container. A solution in which the proportion of solute to solvent is small is said to be dilute, and one in which the proportion is large is said to be concentrated. Note that both dilute and concentrated solutions are still considered unsaturated if the maximum equilibrium concentration (saturation) has not yet been reached.

The solubility of substances in different solvents is ultimately a function of thermodynamics. When the change in Gibbs free energy for the dissolution reaction is negative at a given temperature, the process will be spontaneous, and the solute is said to be soluble. When the change in Gibbs free energy is positive, the process will be nonspontaneous, and the solute is said to be insoluble. Some solute–solvent systems have negative changes in free energy with very large magnitudes, so the equilibrium reaction strongly favors the dissolution of the solute. In general, solutes are considered soluble if they have a molar solubility above 0.1 M in solution. Others have only slightly negative changes in free energy, so the equilibrium position lies closer to the undissociated (reactants) side of the reaction. Those solutes that dissolve minimally in the solvent (molar solubility under 0.1 M) are called sparingly soluble salts.

AQUEOUS SOLUTIONS

The most common type of solution is the aqueous solution, in which the solvent is water. The aqueous state is denoted by the symbol (aq). Aqueous solutions rely on the interactions between water molecules and solutes in solutions. We have mentioned previously that hydration is often the process through which dissolution occurs. It is also important to note that in some solutions, such as acids, the formation of a complex called the hydronium ion (H_3O^+) can occur. This is facilitated by the transfer of a hydrogen ion (H^+) from a molecule in solution to a water molecule (H_2O). The reaction of acetic acid (H^+ donor) with water is shown in Figure 9.3.

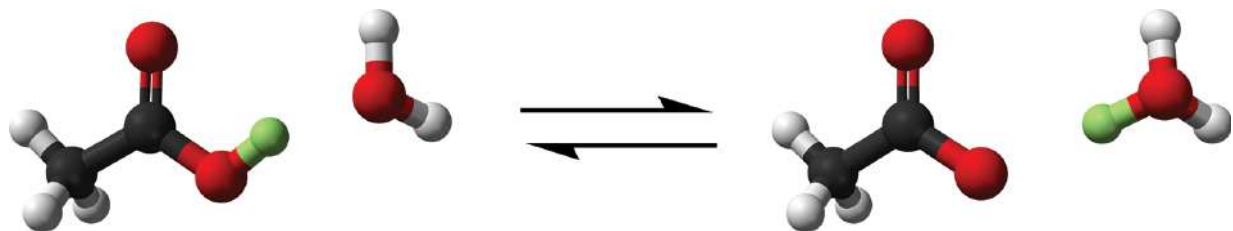
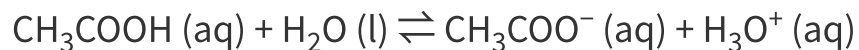


Figure 9.3. Transfer of a Proton in Solution, Forming the Hydronium Ion
The transferred proton is highlighted in green.

It is important to realize that H^+ is never found alone in solution because a free proton is difficult to isolate; rather, it is found bonded to an electron pair donor (carrier) molecule such as a water molecule. This is an example of a coordinate covalent bond. The hydronium ion and its effects on the solubilities of other compounds using Le Châtelier's principle will be described further in Chapter 10 of MCAT General Chemistry Review.

Because aqueous solutions are so common and so important to biological systems, the MCAT focuses on them above all others. In aqueous solutions, there are seven general solubility rules:

All salts containing ammonium (NH_4^+) and alkali metal (Group 1) cations are water-soluble.

All salts containing nitrate (NO_3^-) and acetate (CH_3COO^-) anions are water-soluble.

Halides (Cl^- , Br^- , I^-), excluding fluorides, are water-soluble, with the exceptions of those formed with Ag^+ , Pb^{2+} , and Hg_2^{2+} .

All salts of the sulfate ion (SO_4^{2-}) are water-soluble, with the exceptions of those formed with Ca^{2+} , Sr^{2+} , Ba^{2+} , and Pb^{2+} .

All metal oxides are insoluble, with the exception of those formed with the alkali metals, ammonium, and CaO , SrO , and BaO , all of which hydrolyze to form solutions of the corresponding metal hydroxides.

All hydroxides are insoluble, with the exception of those formed with the alkali metals, ammonium, and Ca^{2+} , Sr^{2+} , and Ba^{2+} .

All carbonates (CO_3^{2-}), phosphates (PO_4^{3-}), sulfides (S^{2-}), and sulfites (SO_3^{2-}) are insoluble, with the exception of those formed with the alkali metals and ammonium.



MCAT EXPERTISE

Because most solutions in the real world involve water as the solvent, it is not a surprise that they are common on the MCAT. These solubility rules are not bad to know, but memorizing them all may be a little excessive. It is never a bad thing to know facts, but being able to apply them is more important. Know rules 1 and 2 for

sure, and be aware of some of the more common insoluble exceptions, like Pb^{2+} and Ag^+ .

The MCAT will not expect memorization of all of the solubility rules, but it is worth knowing two absolutes: all salts of Group 1 metals, and all nitrate salts are soluble. Otherwise, familiarity with rules listed above will suffice—the MCAT generally supplies solubility information for most compounds. Sodium and nitrate ions are generally used as counterions to what is actually chemically important; for example, if a pH problem gives a sodium formate concentration of 0.10 M, it is really indicating that the concentration of the formate ion is 0.10 M because the sodium ion concentration does not affect pH. The only time one needs to worry about the nitrate ion concentration is in an oxidation–reduction reaction, for the nitrate ion can function—although only weakly—as an oxidizing agent. In all other cases with nitrate ions, only focus on the cation as the chemically reacting species.

COMPLEX ION FORMATION

We have mentioned the hydronium ion as a complex that forms in acidic solutions, but it is worthwhile to mention that there are even more varied forms of complex ions that can appear in solution. By definition, a complex ion—or coordination compound—refers to a molecule in which a cation is bonded to at least one electron pair donor (which could include the water molecule). The electron pair donor molecules are called ligands. An example of such a complexation reaction is shown for the

tetraaquadioxouranyl cation, which has water (aqua-) and oxygen (oxo-) ligands, in Figure 9.4.

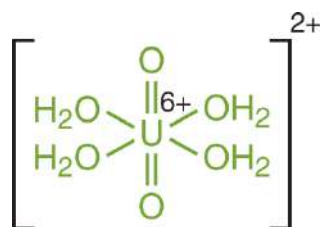


Figure 9.4. Structure of the Tetraaquadioxouranyl Complex Cation
Water and oxygen act as ligands with a U^{6+} cation.

Complexes are held together with coordinate covalent bonds, in which an electron pair donor (a Lewis base) and an electron pair acceptor (a Lewis acid) form very stable Lewis acid–base adducts. Most general chemistry courses do not stress the biological importance of coordination compounds. However, complex ions have profound biological applications in macromolecules such as proteins. For instance, many active sites of proteins utilize complex ion binding and transition metal complexes to carry out their function. One classic example is the iron cation in hemoglobin, which can carry oxygen, carbon dioxide, and carbon monoxide as ligands, as shown in Figure 9.5.

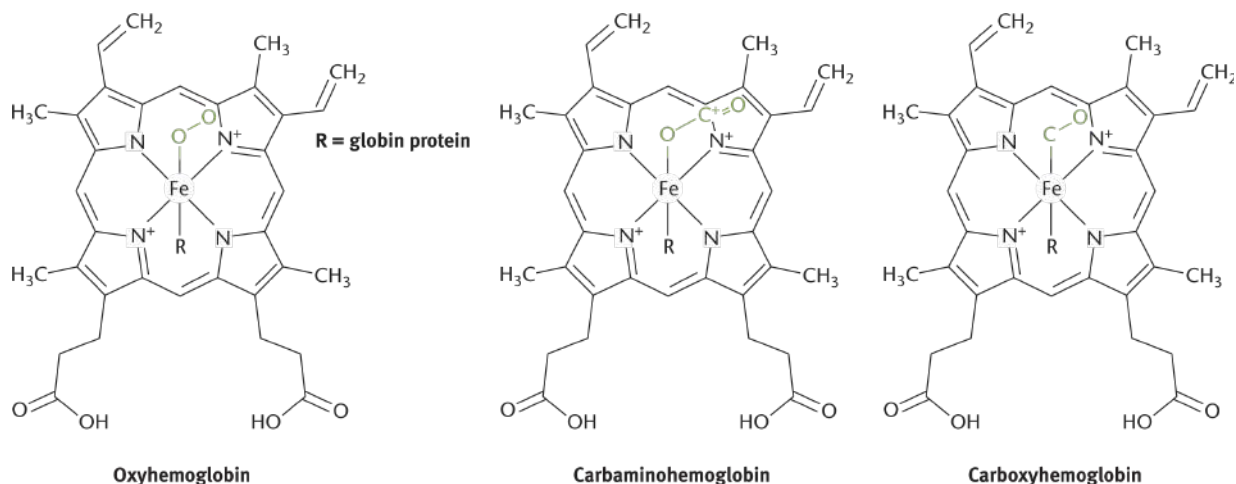


Figure 9.5. Hemoglobin Is a Classic Example of Biochemical Complex Formation

The iron in hemoglobin can bind various gases, leading to the formation of oxyhemoglobin (O_2), carbaminohemoglobin (CO_2), and carboxyhemoglobin (CO).

Many coenzymes (vitamins) and cofactors also contain complexes of transition metals, such as cobalamin (vitamin B_{12}), shown in Figure 9.6.

The presence of a transition metal allows coenzymes and cofactors to bind other ligands or assist with electron transfer.

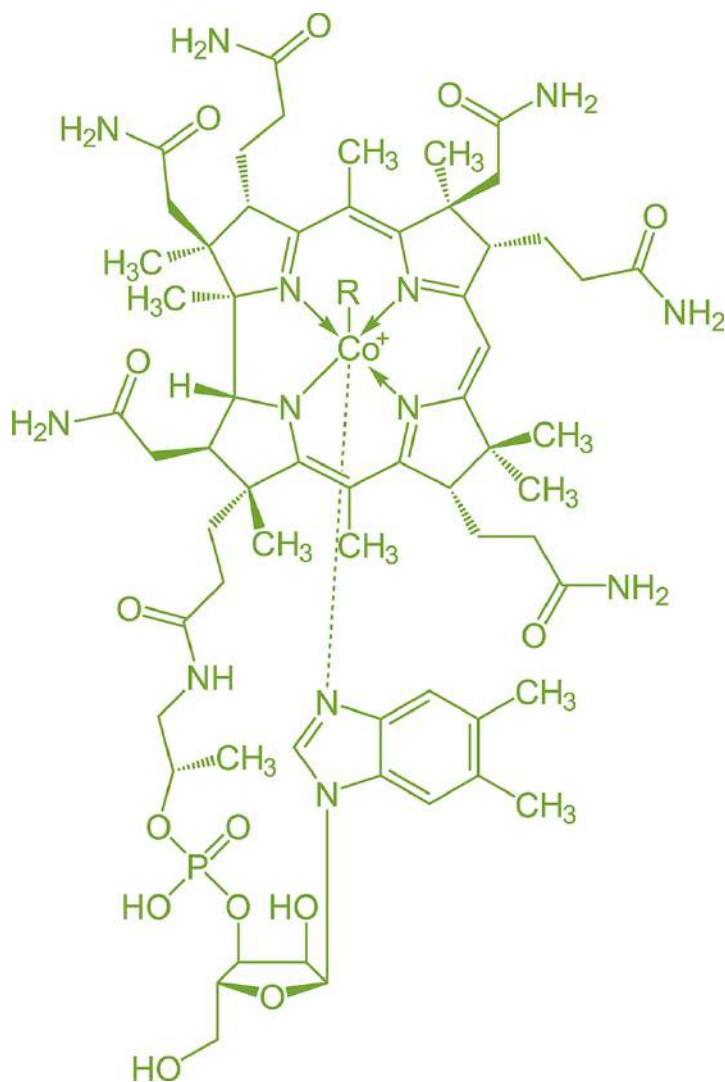


Figure 9.6. Cobalamin (Vitamin B₁₂) Contains a Cobalt Complex

Physical and chemical properties of complex ions are diverse, including a wide range of solubilities and varied chemical reactions. Inorganic complex ions are often fun to characterize because they tend to have vibrant, distinctive colors, as illustrated in Figure 9.7.

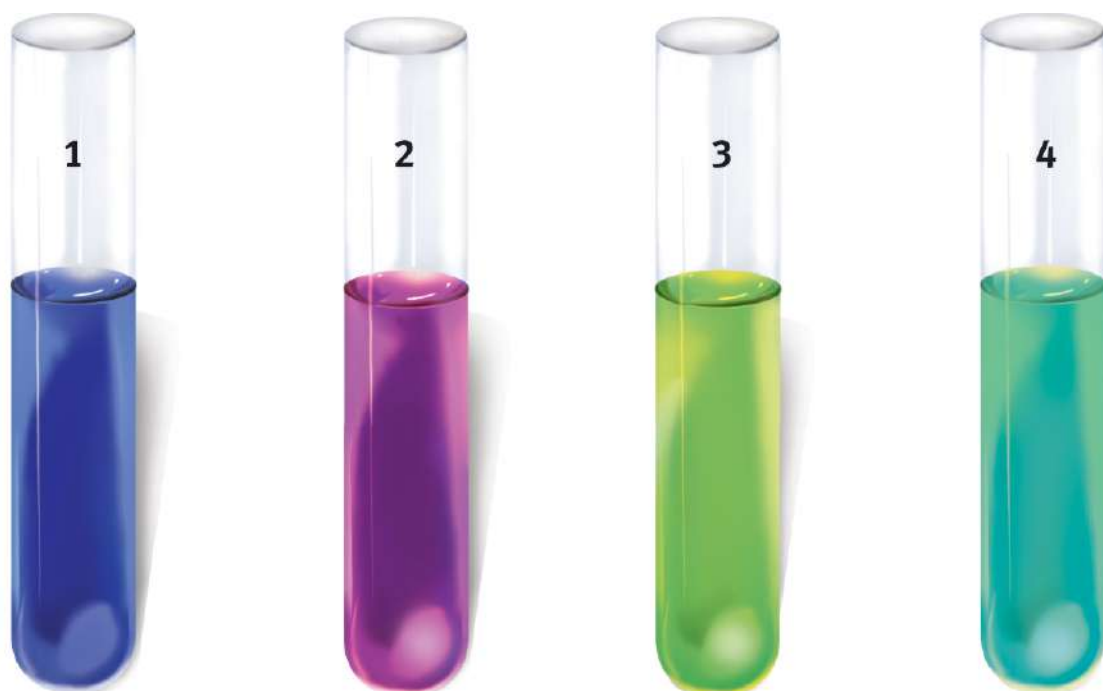


Figure 9.7. Nickel(II) Ion Complexes Display Distinctive Colors

The characteristic colors of Nickel (II) Ion complexes are, from left to right:

- 1) hexaamminenickel(II), 2) tris(ethylenediamine)nickel(II), 3) tetrachloronickelate(II), and 4) hexaaquanickel(II)

In some complexes, the central cation can be bonded to the same ligand in multiple places. This is called chelation, and it generally requires large organic ligands that can double back to form a second (or even third) bond with the central cation. Chelation therapy is often used to sequester toxic metals (lead, arsenic, mercury, and so on). Even biologically necessary metals, such as iron, can be toxic in overload states; an example of iron being chelated is shown in Figure 9.8.

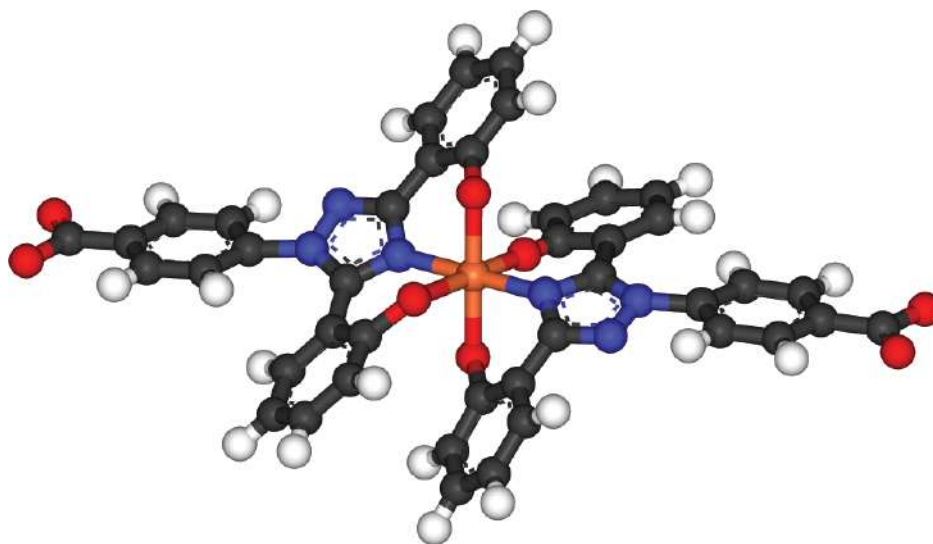


Figure 9.8. Chelation of Iron with Two Molecules of Deferasirox

MCAT Concept Check 9.1:

Before you move on, assess your understanding of the material with these questions.

Describe the process of solvation.

Describe the differences between solubility and saturation:

-

-

What is one way in which solubility of a compound can be increased?

Name two ions that form salts that are always soluble:

-

-

9.2 Concentration

High-Yield



LEARNING GOALS

After Chapter 9.2, you will be able to:

- Calculate the molality, molarity, or normality of a compound in solution
- Apply $M_i V_i = M_f V_f$ to calculate dilution of a solution
- Calculate mole fraction and percent composition by mass

Concentration denotes the amount of solute dissolved in a solvent. There are many different ways of expressing concentration, and different units have been standardized for specific everyday situations. For example, alcohol content in liquors like vodka, gin, or rum is expressed in volume percent (volume of solute divided by volume of solution times 100 percent). Alcoholic proof is twice the volume percent. The sugar content of orange juice and other fruit juices is measured in units of degrees Brix ($^{\circ}\text{Bx}$), which is a mass percent: mass of glucose divided by mass of solution times 100 percent.

UNITS OF CONCENTRATION

On the MCAT, concentrations are commonly expressed as percent composition by mass, mole fraction, molarity, molality, and normality.

Percent Composition by Mass

The percent composition by mass is given by the equation

$$\frac{\text{mass of solute}}{\text{mass of solution}} \times 100\%$$

Equation 9.1



MCAT EXPERTISE

It is important to have a good idea of how to work with all of these ways of expressing concentration because more than one may show up on Test Day.

Percent composition is used not only for aqueous solutions, but also for metal alloys and other solid-in-solid solutions.

Example:

What is the percent composition by mass of a salt water solution if 100 g of the solution contains 20 g of NaCl?

Solution:

$$\frac{\text{mass of solute}}{\text{mass of solution}} \times 100\% = \frac{20 \text{ g}}{100 \text{ g}} \times 100\%$$

=20% NaCl solution.

Mole Fraction

The mole fraction (X) of a compound is given by the equation

$$X_A = \frac{\text{moles of A}}{\text{total moles of all species}}$$

Equation 9.2

The sum of the mole fractions in a system will always equal 1. The mole fraction is used to calculate the vapor pressure depression of a solution, described later in this chapter, as well as the partial pressures of gases in a system, described in Chapter 8 of MCAT General Chemistry Review.

Example:

If 184 g glycerol ($\text{C}_3\text{H}_8\text{O}_3$) is mixed with 180 g water, what will be the mole fractions of the two components? (Note: Molar mass of $\text{H}_2\text{O} = 18 \frac{\text{g}}{\text{mol}}$; molar mass of $\text{C}_3\text{H}_8\text{O}_3 = 92 \frac{\text{g}}{\text{mol}}$)

Solution:

First, determine the number of moles of each compound:

$$180 \text{ g water} \times \left[\frac{1 \text{ mol}}{92 \text{ g}} \right] = 10 \text{ moles water}$$

$$184 \text{ g glycerol} \times \left[\frac{1 \text{ mol}}{92 \text{ g}} \right] = 2 \text{ moles glycerol}$$

$$\text{Total number of moles} = 10 + 2 = 12 \text{ moles}$$

Then, determine the mole fractions:

$$X_{\text{water}} = \frac{10 \text{ mol water}}{12 \text{ mol total}} = \frac{5}{6} = 0.83$$

$$X_{\text{glycerol}} = \frac{2 \text{ mol glycerol}}{12 \text{ mol total}} = \frac{1}{6} = 0.17$$

Molarity

The molarity (M) of a solution is defined as

$$M = \frac{\text{moles of solute}}{\text{liters of solution}}$$

Equation 9.3

Solution concentrations are usually expressed in terms of molarity, and this is the most common unit for concentration on the MCAT. Unless otherwise specified, representations of concentration using brackets—such as $[\text{Na}^+]$ —indicate molarity. Note that the volume term in the denominator of molarity refers to the solution volume, not the volume of solvent used to prepare the solution—although the two values are often close enough to approximate the solution volume using the solvent volume. Molarity is used for rate laws, the law of mass action, osmotic pressure, pH and pOH, and the Nernst equation.



MCAT EXPERTISE

Note that for dilute solutions, the volume of the solution is approximately equal to the volume of solvent used, which simplifies our calculations on Test Day. However, technical questions could ask you to distinguish between these two. For example, when you add two kilograms of sucrose (table sugar) to a liter of water at room temperature (achieving saturation), the volume of solution is certainly larger than 1 L!

Example:

If enough water is added to 11 g of CaCl_2 to make 100 mL of solution, what is the molarity of the solution?

Solution:

First, calculate the number of moles of CaCl_2 :

$$11 \text{ g CaCl}_2 \left[\frac{1 \text{ mol}}{111.1 \text{ g}} \right] = 0.1 \text{ mol CaCl}_2$$

Then determine the molarity:

$$M = \frac{\text{moles of solute}}{\text{liters of solution}}$$
$$M = \frac{0.1 \text{ mol}}{100 \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}}} = \frac{0.1}{0.1} = 1 M$$

Molality

The molality (m) of a solution is defined as

$$m = \frac{\text{moles of solute}}{\text{kilograms of solvent}}$$

Equation 9.4

For dilute aqueous solutions at 25°C, the molality is approximately equal to molarity because the density of water at this temperature is 1 kilogram per liter. However, note that this is an approximation and true only for dilute aqueous solutions. As aqueous solutions become more concentrated with solute, their densities become significantly different from that of pure water; most water-soluble solutes have molar masses significantly greater than that of water, so the density of the solution increases as the concentration increases. You won't use molality very often, so be mindful of the special situations when it is required: boiling point elevation and freezing point depression.

Example:

If 10 g NaOH are dissolved in 500 g of water, what is the molality of the solution?

Solution:

First, calculate the number of moles of NaOH:

$$10 \text{ g NaOH} \left[\frac{1 \text{ mol}}{40 \text{ g}} \right] = 0.25 \text{ mol NaOH}$$

Then determine the molality:

$$m = \frac{\text{moles of solute}}{\text{kilograms of solvent}}$$

$$m = \frac{0.25 \text{ mol}}{500 \text{ g} \times \frac{1 \text{ kg}}{1000 \text{ g}}} = \frac{0.25}{0.5} = 0.5 \text{ } m$$

Normality

We discussed the related concepts of gram equivalent weight, equivalents, and normality in Chapter 4 of MCAT General Chemistry Review. The normality (N) of a solution is equal to the number of equivalents of interest per liter of solution. An equivalent is a measure of the reactive capacity of a molecule. Most simply, an equivalent is equal to a mole of the species of interest—protons, hydroxide ions, electrons, or ions.

To calculate the normality of a solution, we need to know what purpose the solution serves because it is the concentration of the reactive species with which we are concerned. For example, in acid–base reactions, we are most concerned with the concentration of hydrogen ions; in oxidation–reduction reactions, we are most concerned with the concentration of electrons. Normality is unique among concentration units in that it is reaction dependent. For example, in acidic solution, 1 mole of the permanganate ion (MnO_4^-) will readily accept 5 moles of electrons, so a 1 M solution would be 5 N. However, in alkaline solution, 1 mole of permanganate will accept only 1 mole of electrons, so in alkaline solution, a 1 M permanganate solution would be 1 N.



MCAT EXPERTISE

Simple ideas on Test Day will make things easier. So, when you come across normality, think of it as molarity of the stuff of interest in the reaction.

DILUTION

A solution is diluted when solvent is added to a solution of higher concentration to produce a solution of lower concentration. The concentration of a solution after dilution can be determined using the equation

$$M_i V_i = M_f V_f$$

Equation 9.5

where M is molarity, V is volume, and the subscripts i and f refer to the initial and final values, respectively.



MCAT EXPERTISE

Though not a unique means of measuring concentration or dilution, the term parts-per can be used to indicate concentration

of a dissolved substance in a solution (most commonly water). Parts-per-million (ppm, 10^{-6}) is the most common usage. If a problem states there is one ppm of substance X in water, that would indicate there is 1 mg/L of water, as there would be 1 millionth of a gram per gram of water, and the density of water is 1 g/mL. On Test Day, prior to converting from ppm, make sure to assess whether conversion to a different unit of measure is actually required, as this conversion can typically be avoided on the MCAT.

Example:

A chemist wishes to prepare 300 mL of a 1.1 M NaOH solution from a 5.5 M NaOH stock solution. What volume of stock solution should be diluted with pure water to obtain the desired solution?

Solution:

$$M_i V_i = M_f V_f \rightarrow V_i = \frac{M_f V_f}{M_i}$$
$$V_i = \frac{(1.1 \text{ M})(300 \text{ mL})}{(5.5 \text{ M})} = \frac{300}{5} = 60 \text{ mL}$$

Note that one can use mL or L in the equation, as long as the units are consistent.



MCAT EXPERTISE

This equation is worthy of memorization. Note that a similar equation is used for the equivalence point in acid–base chemistry, as discussed in Chapter 10 of MCAT General Chemistry Review.

MCAT Concept Check 9.2:

Before you move on, assess your understanding of the material with these questions.

If you mix 180 g of the following compounds in 250 mL of water ($\text{density} = 1 \frac{\text{g}}{\text{mL}}$), what are their concentrations in molality, molarity, and normality (for acid–base chemistry)?

Compound	Molality	Molarity	Normality
Glucose ($180 \frac{\text{g}}{\text{mol}}$)			
Carbonate ($60 \frac{\text{g}}{\text{mol}}$)			



You are working in a sewage treatment facility and are assaying chlorine in a water sample. You need to dilute the water sample from 100 ppm stock to 25 ppm and create 100 mL of solution. Calculate the amount of stock solution needed and determine how you would create your final solution:

A stock solution for making typical IV saline bags contains 90.0 g of NaCl per 10 liters of water $\left(\text{density} = 1 \frac{\text{g}}{\text{mL}}\right)$. What is the mole fraction and the percent composition by mass of NaCl in the saline solutions?

- Mole fraction:

- Percent composition by mass:

9.3 Solution Equilibria

High-Yield



LEARNING GOALS

After Chapter 9.3, you will be able to:

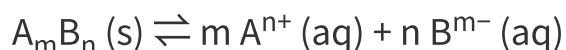
- Calculate molar solubility for a compound given its K_{sp} and vice versa
- Calculate ion product for a solution
- Given K_{sp} , predict when a solute may dissociate or precipitate in order to reach equilibrium
- Predict the impact of a common ion on the dissolution of a compound

The process of solvation, like other reversible chemical and physical processes, tends toward an equilibrium position, defined as the lowest energy state of a system under a given set of temperature and pressure conditions. Systems move spontaneously toward the equilibrium position, and any movement away from equilibrium is nonspontaneous. In the process of creating a solution, the equilibrium is defined as the saturation point, where the solute concentration is at its maximum value for the given temperature and pressure. Immediately after solute has been introduced into a solvent, most of the change taking place is dissociation because no

dissolved solute is initially present. However, once solute is dissolved, the reverse process—precipitation of the solute—will begin to occur.

When the solution is dilute (unsaturated), the thermodynamically favored process is dissolution, and initially, the rate of dissolution will be greater than the rate of precipitation. As the solution becomes more concentrated and approaches saturation, the rate of dissolution lessens, while the rate of precipitation increases. Eventually, the saturation point of the solution is reached. The solution now exists in a state of dynamic equilibrium for which the rates of dissolution and precipitation are equal, and the concentration of dissolved solute reaches a steady-state (constant) value. Neither dissolution nor precipitation is more thermodynamically favored at equilibrium because favoring either would necessarily result in the solution no longer being in a state of equilibrium. At this point, the change in free energy is zero, as is the case for all systems at equilibrium.

An ionic solid introduced into a polar solvent dissociates into its component ions, and the dissociation of such a solute in solution may be represented by

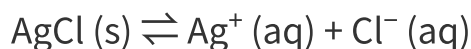


On Test Day, the first step for any solution stoichiometry or solution equilibrium question is to write out the balanced dissociation reaction for the ionic compound in question. This first step is essential for correctly calculating the solubility product constant, ion product, molar solubility, or for determining the outcome of the common ion effect. In other words, it is

the essential first step for nearly every solution chemistry problem on the MCAT.

SOLUBILITY PRODUCT CONSTANTS

Most solubility problems on the MCAT deal with solutions of sparingly soluble salts, which are ionic compounds that have very low solubility in aqueous solutions. You may wonder why any ionic compound would not be highly soluble in water. The degree of solubility is determined by the relative changes in enthalpy and entropy associated with the dissolution of the ionic solute at a given temperature and pressure. One common sparingly soluble salt is silver chloride, AgCl, which dissociates in water according to the following equation:



The law of mass action can be applied to a solution at equilibrium; that is to say, when the solution is saturated and the solute concentration is at a maximum and is dynamically stable. For a saturated solution of an ionic compound with the formula A_mB_n , the equilibrium constant for its solubility in aqueous solution, called the solubility product constant (K_{sp}), can be expressed by:

$$K_{sp} = [A^{n+}]^m[B^{m-}]^n$$

Equation 9.6

where the concentrations of the ionic constituents are equilibrium (saturation) concentrations. For example, we can express the K_{sp} of silver chloride as:

$$K_{sp} = [Ag^+][Cl^-]$$



MCAT EXPERTISE

On the MCAT, if you remember that K_{sp} is just a specialized form of K_{eq} , then you can simplify a lot of problems by using the same concepts that you do for all equilibria, including Le Châtelier's principle.

You'll notice that, for the law of mass action of solutions, there is no denominator. Remember that pure solids and liquids do not appear in the equilibrium constant. Because the silver chloride solution was formed by adding pure solid silver chloride to pure water, neither the solid silver chloride nor the water is included. Indeed, dissociation reactions—by definition—have a solid salt as a reactant; thus, K_{sp} expressions should never have denominators.

Solubility product constants, like all other equilibrium constants (K_{eq} , K_a , K_b , and K_w) are temperature dependent. When the solution consists of a gas dissolved into a liquid, the value of the equilibrium constant, and hence the position of equilibrium (saturation), will also depend on pressure. Generally speaking, the solubility product constant increases

with increasing temperature for non-gas solutes and decreases for gas solutes. Higher pressures favor dissolution of gas solutes, and therefore the K_{sp} will be larger for gases at higher pressures than at lower ones.



REAL WORLD

Because gases become more soluble in solution as pressure increases, a diver who has spent time at significant depths under water will have more nitrogen gas dissolved in his or her blood because nitrogen gas is the main inert gas in the air we breathe. If the diver rises to the surface too quickly, the abrupt decompression will lead to an abrupt decrease in gas solubility in the plasma, resulting in the formation of nitrogen gas bubbles in the bloodstream. The gas bubbles can get lodged in the small vasculature of the peripheral tissue, mostly around the large joints of the body, causing pain and tissue damage (hence the name, the bends). The condition is painful and dangerous and can be fatal if not properly treated.

As solute dissolves into the solvent, the system approaches saturation, at which point no more solute can be dissolved and any excess will precipitate to the bottom of the container. We may not know whether the solution has reached saturation, and so to determine where the system is with respect to the equilibrium position, we can calculate a value called the ion product (IP), which is analogous to the reaction quotient, Q , for other chemical reactions. The ion product equation has the same form as the equation for the solubility product constant:

$$IP = [A^{n+}]^m[B^{m-}]^n$$

Equation 9.7

The difference is that the concentrations used in the ion product equation are the concentrations of the ionic constituents at that given moment in time, which may differ from equilibrium concentrations. As with the reaction quotient Q , the utility of the ion product lies in comparing its value to that attained at equilibrium, K_{sp} . Each salt has its own distinct K_{sp} at a given temperature. If, at a given set of conditions, a salt's IP is less than the salt's K_{sp} , then the solution is not yet at equilibrium and is considered unsaturated. For unsaturated solutions, dissolution is thermodynamically favored over precipitation. If the IP is greater than the K_{sp} , then the solution is beyond equilibrium, and the solution is considered supersaturated. It is possible to create a supersaturated solution by dissolving solute into a hot solvent and then slowly cooling the solution. A supersaturated solution is thermodynamically unstable, and any disturbance to the solution, such as the addition of more solid solute or other solid particles, or further cooling of the solution, will cause spontaneous precipitation of the excess dissolved solute. If the calculated IP is equal to the known K_{sp} , then the solution is at equilibrium—the rates of dissolution and precipitation are equal—and the solution is considered saturated. The molarity of a solute in a saturated solution is called the molar solubility of that substance.



KEY CONCEPT

$IP < K_{sp}$: unsaturated, solute will continue to dissolve

$IP = K_{sp}$: saturated, solution is at equilibrium

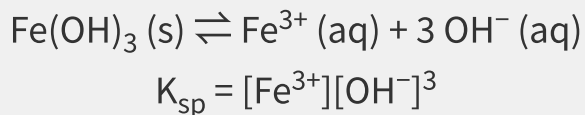
$IP > K_{sp}$: supersaturated, precipitation will occur

Example:

The molar solubility of $\text{Fe}(\text{OH})_3$ in an aqueous solution was determined to be $4 \times 10^{-10} \frac{\text{mol}}{\text{L}}$. What is the value of the K_{sp} for $\text{Fe}(\text{OH})_3$ at the same temperature and pressure?

Solution:

The molar solubility is given as $4 \times 10^{-10} \text{ M}$. The equilibrium concentration of each ion can be determined from the molar solubility and the balanced dissociation reaction of $\text{Fe}(\text{OH})_3$. The dissociation reaction is:



The molar solubility can be expressed as x , the amount of $\text{Fe}(\text{OH})_3$ that dissolves to make a saturated solution at equilibrium. As it dissolves, $\text{Fe}(\text{OH})_3$ dissociates to create x of Fe^{2+} and $3x$ of OH^{-} . This can be entered into the K_{sp} equation to give:

$$K_{sp} = [x][3x]^3$$

If the molar solubility x is $4 \times 10^{-10} \text{ M}$, then $x = 4 \times 10^{-10} \text{ M Fe(OH)}_3$ has dissolved, yielding $x = 4 \times 10^{-10} \text{ M Fe}^{3+}$ and $3x = 3 \times 4 \times 10^{-10} \text{ M OH}^-$ (because there are three OH^- ions released per Fe(OH)_3 molecule dissolved). Thus,

$$K_{\text{sp}} = [\text{Fe}^{3+}] [\text{OH}^-]^3$$

$$K_{\text{sp}} = (4 \times 10^{-10} \text{ M})(3 \times 4 \times 10^{-10} \text{ M})^3 = (4 \times 10^{-10}) \times [3 \times 10^{-10}]^3$$

$$K_{\text{sp}} = 3^3 \times (4 \times 10^{-10})^4 = 27(4 \times 10^{-10})^4 = 27 \times 256 \times 10^{-40} \approx 30 \times 250 \times 10^{-40}$$

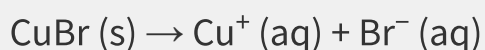
$$K_{\text{sp}} \approx 7500 \times 10^{-40} = 7.5 \times 10^{-37} \text{ (actual} = 6.9 \times 10^{-37} \text{)}$$

Example:

What are the concentrations of each of the ions in a saturated solution of CuBr , given that the K_{sp} of CuBr is 6.27×10^{-9} at 25°C ? If 3 g CuBr are dissolved in water to make 1 L of solution at 25°C , would the solution be saturated, unsaturated, or supersaturated?

Solution:

The first step is to write out the dissociation reaction:



$$K_{\text{sp}} = [\text{Cu}^+][\text{Br}^-]$$

Let x equal the molar solubility of CuBr, which is the amount of CuBr that dissolves at equilibrium. The concentration of Cu^+ and Br^- will each equal x .

$$K_{\text{sp}} = x \cdot x = x^2$$

$$6.27 \times 10^{-9} = x^2$$

$$6.3 \times 10^{-9} \approx x^2$$

$$63 \times 10^{-10} \approx x^2 \approx 64 \times 10^{-10}$$

$$8 \times 10^{-5} \approx x \text{ (actual value} = 7.9 \times 10^{-5})$$

Therefore, $[\text{Cu}^+]$ is about $8 \times 10^{-5} \text{ M}$, and $[\text{Br}^-]$ is also about $8 \times 10^{-5} \text{ M}$. Note that $8 \times 10^{-5} \text{ M}$ also represents the molar solubility of copper(I) bromide.

Next, we convert 3 g of CuBr into moles:

$$3 \text{ g} \times \frac{1 \text{ mol CuBr}}{143.5 \text{ g}} \approx \frac{3}{150} \approx \frac{3}{1.5 \times 10^2} \approx 2 \times 10^{-2} \text{ mol}$$

$2 \times 10^{-2} \text{ mol}$ CuBr in 1 L of solution represents a molarity of $2 \times 10^{-2} \text{ M}$, which is more than 100 times higher than the molar solubility.

Therefore, this is a supersaturated solution.



KEY CONCEPT

Every sparingly soluble salt of general formula MX will have $K_{sp} = x^2$, where x is the molar solubility (assuming no common ion effect).



KEY CONCEPT

Every sparingly soluble salt of general formula MX_2 will have $K_{sp} = 4x^3$, where x is the molar solubility (assuming no common ion effect).



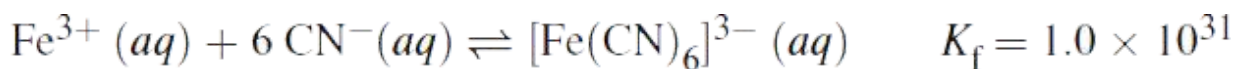
KEY CONCEPT

Every sparingly soluble salt of general formula MX_3 will have $K_{sp} = 27x^4$, where x is the molar solubility (assuming no common ion effect).

Finally, let's return to our discussion of complex ions and their solubility factors. Much like the examples we have seen previously, the solubility of

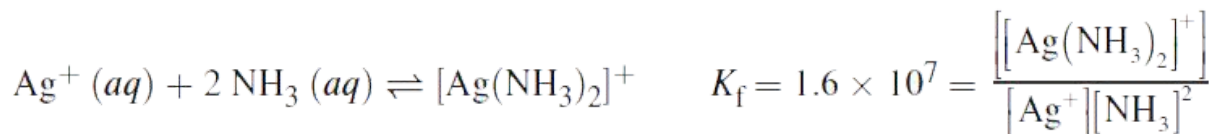
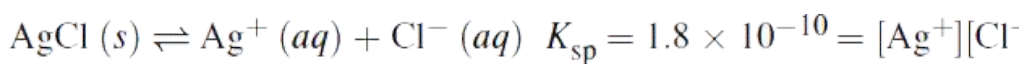
complex ion solutions is determined by the K_{sp} . The formation of complex ions increases the solubility of a salt in solution.

For instance, consider free iron(III) (Fe^{3+}) in a solution of water. If a cyanide solution were added, an exceptionally stable iron and cyanide metal complex would form as the water molecules solvating the iron are replaced by excess cyanide ions:



Knowing the intricacies of why complexes are more stable in solution than isolated ions is beyond the scope of the MCAT; however, it should make sense that, if a complex ion contains multiple polar bonds between the ligands and the central metal ion, it should be able to engage in a very large amount of dipole–dipole interactions. This stabilizes the dissolution of the complex ion. The end result is that such complexes tend to have very high K_{sp} values.

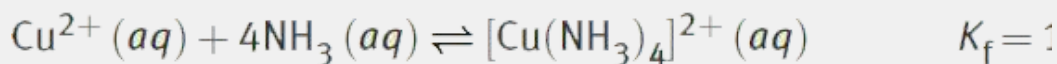
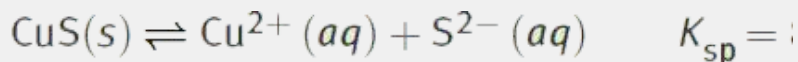
When forming a complex ion, one must often use a mixture of solutions. For this reason, a distinction must be made between the K_{sp} of the solution and that of the complex ion itself. The dissolution of the original solution is termed K_{sp} , and the subsequent formation of the complex ion in solution is termed K_f (the formation or stability constant of the complex in solution). An example is shown below in which the diamminesilver(I) complex is formed in a silver chloride solution:



Notice that the formation constant (K_{f}) of the complex ion is significantly larger than the K_{sp} of the compound providing the metal ion. This is part of the explanation for why the initial dissolution of the metal ion is the rate-limiting step of complex ion formation. However, Le Châtelier's principle is at play in these reactions as well. Ultimately, complex ions form to become more soluble in solution. And as an amount of silver ion is being used up to form the complex ion itself, the dissociation reaction of AgCl shifts to the right, providing more silver for complex ion formation.

Example:

A 0.1 mol sample of CuS is added to 1.00 L of 1.00 M NH_3 . What is the final concentration of the complex ion, tetraamminecopper(II)?



Solution:

First determine the amount of copper ion produced from copper sulfide (CuS) in solution. Comparison of the two equilibrium

constants shows that the CuS dissociation has a K_{sp} of 8×10^{-37} , indicating that this reaction is unlikely to proceed as the forward reaction is not favorable. However, if the formation of the complex ion occurs simultaneously, the large K_f of 1.1×10^{13} for this process will drive the dissociation of CuS forward as the Cu^{2+} ions are consumed in the second reaction due to Le Châtelier's principle effects. Note that, even without quantitative analysis, the large value of K_f is a sign that the formation of the product of the second reaction will be highly favorable. In fact, if these two reactions are simultaneous, the CuS will ultimately be completely consumed due to this effect.

Thus, the amount of Cu^{2+} available to react is given by

$$[\text{Cu}^{2+}] = \frac{0.1 \text{ mol}}{1.00 \text{ L}} = 0.1 \text{ M}.$$
 The reactions given show a 1:1 relationship between the Cu^{2+} available and the $[\text{Cu}(\text{NH}_3)_4]^{2+}$ that is generated. In other words, almost all of the Cu^{2+} ions will be used up to form the complex. Therefore, the concentration of the $[\text{Cu}(\text{NH}_3)_4]^{2+}$ complex ion is 0.1 M.

COMMON ION EFFECT

The solubility of a substance varies depending on the temperature of the solution, the solvent, and in the case of a gas-phase solute, the pressure. Solubility is also affected by the addition of other substances to the

solution. The effect of a complex ion increasing the solubility of a substance is not typical and is—in fact—opposite to the effect seen in many mixtures of solutions.

One of the more challenging solution chemistry problems on the MCAT is calculation of the equilibrium concentration of a salt in a solution that already contains one of the ions in that salt. The solubility of a salt is considerably reduced when it is dissolved in a solution that already contains one of its constituent ions as compared to its solubility in a pure solvent. This reduction in molar solubility is called the common ion effect. As described above, the molar solubility of a compound is its concentration (in moles per liter) at equilibrium at a given temperature. If X moles of $A_mB_n(s)$ can be dissolved in one liter of solution to reach saturation, then the molar solubility of $A_mB_n(s)$ is X molar.

Pay attention to the effect of the common ion: its presence results in a reduction in the molar solubility of the salt. Note, however, that the presence of the common ion has no effect on the value of the solubility product constant itself. For example, if a salt such as CaF_2 is dissolved into water already containing Ca^{2+} ions (from some other salt, perhaps $CaCl_2$), the solution will dissolve less CaF_2 than would an equal amount of pure water.

The common ion effect is really Le Châtelier's principle in action. Because the solution already contains one of the constituent ions from the products side of the dissociation equilibrium, the system will shift toward the left side, reforming the solid salt. As a result, molar solubility for the solid is reduced, and less of the solid dissolves in the solution—although the K_{sp} remains constant.

One can take advantage of the common ion effect to separate out specific compounds in a solution mixture. For example, in a solution of silver salts, one could add sodium or potassium chloride to preferentially precipitate silver(I) chloride. By adding an appropriate counterion in excess, the dissociation reaction shifts to the left, forming the solid salt.

Example:

The K_{sp} of AgI in aqueous solution is 8.5×10^{-17} . If a 1×10^{-5} M solution of AgNO_3 is saturated with AgI, what will be the final concentration of the iodide ion?

Solution:

The concentration of Ag^+ in the original AgNO_3 solution will be 1×10^{-5} M because AgNO_3 will fully dissociate (review the solubility rules from earlier in this chapter). Some small amount of AgI will dissociate into the solution, which is the molar solubility x of AgI under these conditions. The net silver concentration from both AgNO_3 and AgI will become 1×10^{-5} M + x . Because no iodide was present in solution until the AgI began dissociating, the concentration of iodide will be x . Thus, the K_{sp} expression for the dissociation of AgI can be written as:

$$\begin{aligned} K_{sp} &= [\text{Ag}^+] [\text{I}^-] \\ 8.5 \times 10^{-17} &= [1 \times 10^{-5} \text{ M} + x] [x] \end{aligned}$$

Given that the value of K_{sp} for this reaction is 10^{-16} , only a miniscule amount of AgI will be dissociated. Thus, the value of x is sufficiently small to be negligible when added to 10^{-5} . Thus, the math simplifies to:

$$\begin{aligned}8.5 \times 10^{-17} &= \left[1 \times 10^{-5} M\right] [x] \\8.5 \times 10^{-12} &= x\end{aligned}$$

This question asks for the concentration of iodide, which—based on the equilibrium expression—is represented by x . Thus, $[I^-] = 8.5 \times 10^{-12} M$.

MCAT Concept Check 9.3:

Before you move on, assess your understanding of the material with these questions.

Calculate the K_{sp} of $Ni(OH)_2$ in water, given that its molar solubility is $5.2 \times 10^{-6} M$.

The K_{sp} of $\text{Ba}(\text{OH})_2$ is 5.0×10^{-3} . Assuming that barium hydroxide is the only salt added to form a solution, calculate the ion product of the following solutions based on the concentration of Ba^{2+} . Then, predict the behavior of the given solutions (dissolution, equilibrium, or precipitation):

$[\text{Ba}^{2+}]$	Ion Product	Behavior of Solution
0.5 M		
0.1 M		
0.05 M		



What is the molar solubility of $\text{Zn}(\text{OH})_2$ ($K_{sp} = 4.1 \times 10^{-17}$) in a 0.1 M solution of NaOH?

9.4 Colligative Properties

LEARNING GOALS

After Chapter 9.4, you will be able to:

- Recall the names, equations, and applications of the common colligative properties
- Describe the relationship between molality and molarity for a compound
- Calculate the boiling point, freezing point, vapor pressure, or osmotic pressure of a solution

The colligative properties are physical properties of solutions that are dependent on the concentration of dissolved particles but not on the chemical identity of the dissolved particles. These properties—vapor pressure depression, boiling point elevation, freezing point depression, and osmotic pressure—are usually associated with dilute solutions.

RAOULT'S LAW

Raoult's law accounts for vapor pressure depression caused by solutes in solution. As solute is added to a solvent, the vapor pressure of the solvent

decreases proportionately. For example, consider compound A in Figure 9.9. Compound A in its pure form (mole fraction = 1.0) has a particular vapor pressure, indicated by P_A° . At the same temperature, compound B has a lower vapor pressure, indicated by P_B° . Note that, as the concentration of B increases, the vapor pressure of A decreases. Indeed, as more solute is dissolved into solvent (as more B is dissolved into A), the vapor pressure of the solvent decreases.

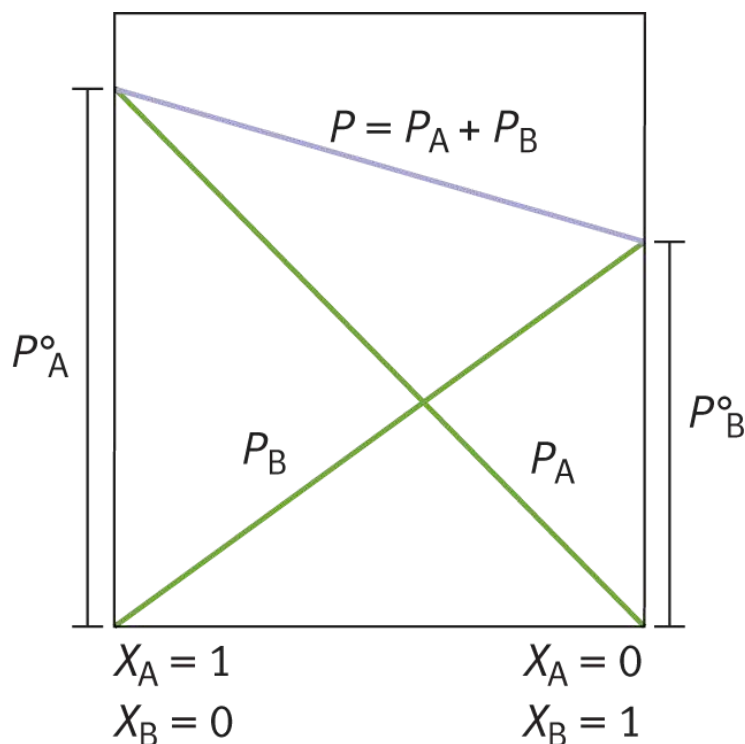


Figure 9.9. Raoult's Law

As more of solute B is dissolved in solvent A, the vapor pressure of solvent A decreases.

On a molecular level, the presence of the solute molecules can block the evaporation of solvent molecules but not their condensation. This reduces the vapor pressure of the solution compared to the pure solvent, as seen in Figure 9.10.

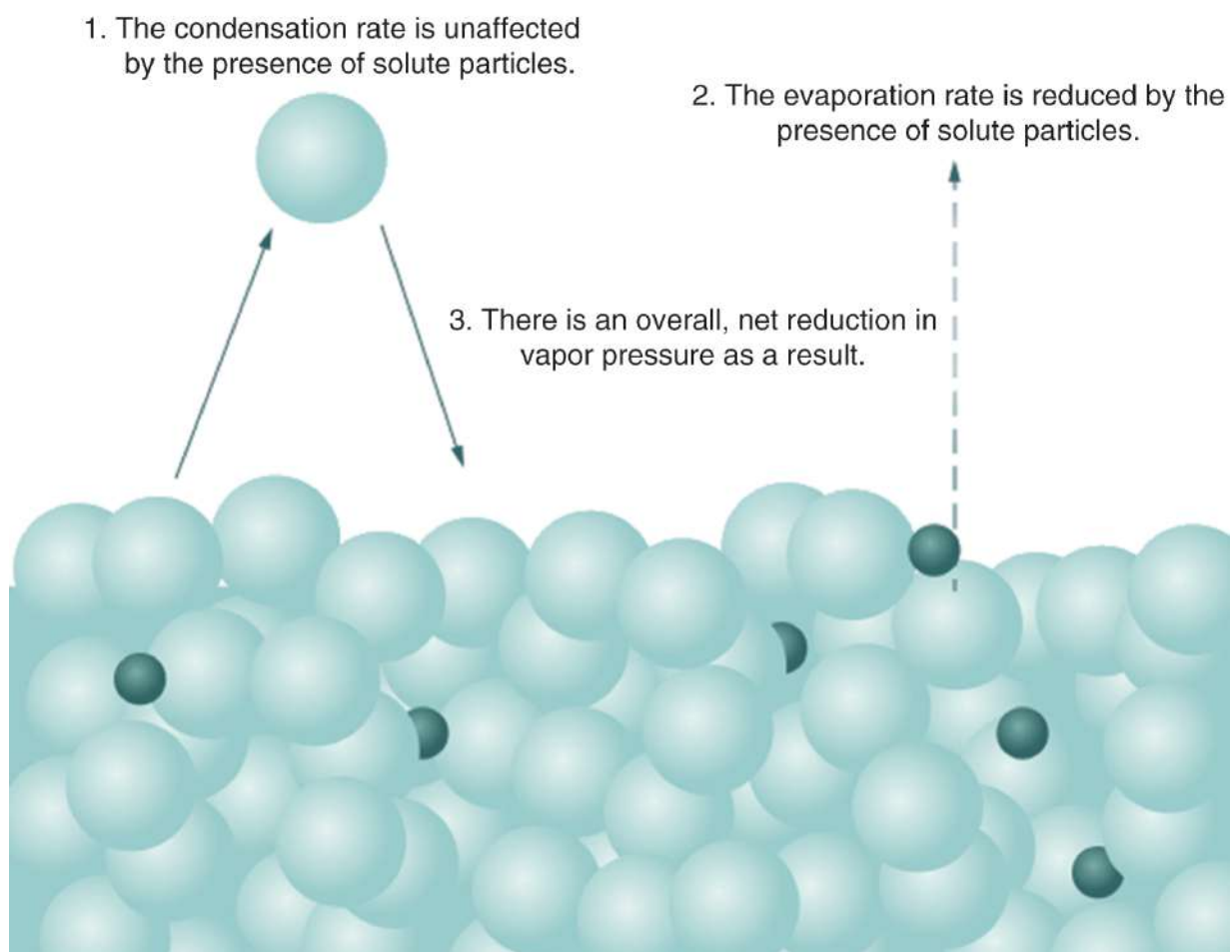


Figure 9.10. Molecular Basis of Raoult's Law

Raoult's law is expressed mathematically as:

$$P_A = X_A P_A^\circ$$

Equation 9.8

where P_A is the vapor pressure of solvent A when solutes are present, X_A is the mole fraction of the solvent A in the solution, and P_A° is the vapor

pressure of solvent A in its pure state.



KEY CONCEPT

Vapor pressure depression goes hand in hand with boiling point elevation. The lowering of a solution's vapor pressure would mean that a higher temperature is required to match atmospheric pressure, thereby raising the boiling point.

Raoult's law holds only when the attraction between the molecules of the different components of the mixture is equal to the attraction between the molecules of any one component in its pure state. When this condition does not hold, the relationship between mole fraction and vapor pressure will deviate from Raoult's law. Solutions that obey Raoult's law are called ideal solutions.

Example:

What is the change in vapor pressure when 180 grams of glyceraldehyde ($\text{C}_3\text{H}_6\text{O}_3$) are added to 0.18 L of water at 100°C ?

Solution:

The density of water at 100°C is close to $1 \frac{\text{g}}{\text{mL}}$, and the vapor pressure of water at the same temperature is 1 atm because this is the boiling point of water.

In order to find the mole fraction of the solvent, first find the molar mass of the solute (glyceraldehyde) and solvent (water). 180 g glyceraldehyde represents 2 moles of glyceraldehyde. 0.18 L of water has a mass around 180 g, which represents 10 moles of water. The mole fraction of water is therefore

$$\frac{10 \text{ mol water}}{12 \text{ mol total}} = \frac{5}{6} = 0.83.$$

To find the vapor pressure change, we want to find the difference in the old pressure and the new pressure. The new pressure can be calculated from Raoult's law:

$$P_A = X_A P_A^\circ = (0.83)(1 \text{ atm}) = 0.83 \text{ atm}.$$

The change in vapor pressure is therefore $1 \text{ atm} - 0.83 \text{ atm} = 0.17 \text{ atm}$.

Example:

What is the vapor pressure at room temperature of a mixture containing 58 g butane (C_4H_{10}) and 172 g hexane (C_6H_{14})? (Note: The vapor pressures of pure butane and pure hexane are 172 kPa and 17.6 kPa, respectively, at 25°C .)

Solution:

First, determine the number of moles of each substance. 58 g butane represents 1 mole of butane. 172 g hexane represents 2

moles of hexane. Then, determine the mole fractions of each component of the mixture.

$$X_{\text{butane}} = \frac{\text{moles of butane}}{\text{total moles}} = \frac{1}{3}$$
$$X_{\text{hexane}} = \frac{\text{moles of hexane}}{\text{total moles}} = \frac{2}{3}$$

Then, calculate the vapor pressure of each component:

$$P_{\text{butane}}^{\circ} = X_{\text{butane}} P_{\text{butane}}^{\circ} = \frac{1}{3} (172 \text{ kPa}) \approx \frac{1}{3} (180 \text{ kPa}) =$$
$$P_{\text{hexane}}^{\circ} = X_{\text{hexane}} P_{\text{hexane}}^{\circ} = \frac{2}{3} (17.6 \text{ kPa}) \approx \frac{2}{3} (18 \text{ kPa}) =$$

The total vapor pressure is the sum of the two vapor pressures.

Thus, the total vapor pressure is $60 + 12 = 72 \text{ kPa}$ (actual = 69.1 kPa).

BOILING POINT ELEVATION

When a nonvolatile solute is dissolved into a solvent to create a solution, the boiling point of the solution will be greater than that of the pure solvent. The boiling point is the temperature at which the vapor pressure of the liquid equals the ambient (incident) pressure. We've just seen that adding solute to a solvent results in a decrease in the vapor pressure of the solvent in the solution. If the vapor pressure of a solution is lower than that

of the pure solvent, then more energy (and consequently a higher temperature) will be required before its vapor pressure equals the ambient pressure. The extent to which the boiling point of a solution is raised relative to that of the pure solvent is given by the formula

$$\Delta T_b = iK_b m$$

Equation 9.9

where ΔT_b is the increase in boiling point, i is the van't Hoff factor, K_b is a proportionality constant characteristic of a particular solvent (which will be provided on Test Day), and m is the molality of the solution. The van't Hoff factor corresponds to the number of particles into which a compound dissociates in solution. For example, $i = 2$ for NaCl because each formula unit of sodium chloride dissociates into two particles—a sodium ion and a chloride ion—when it dissolves. Covalent molecules such as glucose do not readily dissociate in water and thus have i values of 1.



MCAT EXPERTISE

The boiling point elevation formula calculates the amount that the normal boiling point is raised. The value calculated is not the boiling point itself.

Example:

400 g AlCl_3 is dissolved in 1.5 L of water at room temperature $\left(K_b = 0.512 \frac{\text{K} \cdot \text{kg}}{\text{mol}}\right)$. How much does the boiling point increase after adding the aluminum chloride?

Solution:

Water at room temperature has a density of $1 \frac{\text{g}}{\text{mL}}$, Therefore, 1.5 L is the same as 1.5 kg. The van't Hoff factor for aluminum chloride is 4 because it breaks down to form 1 aluminum cation and 3 chloride anions. To determine the molality, we will also need to know how many moles 400 g AlCl_3 represents.

$$400 \text{ g AlCl}_3 \times \frac{1 \text{ mol}}{133.5 \text{ g}} \approx 3 \text{ mol AlCl}_3$$

The molality is therefore $\frac{3 \text{ mol AlCl}_3}{1.5 \text{ kg}} = 2 m$.

Then, plug into the boiling point elevation equation.

$$\Delta T_b = iK_b m = (4) \left(0.512 \frac{\text{K} \cdot \text{kg}}{\text{mol}}\right) (2m) \approx 4\text{K} (\text{actual} =$$

FREEZING POINT DEPRESSION

The presence of solute particles in a solution interferes with the formation of the lattice arrangement of solvent molecules associated with the solid state. Thus, a greater amount of energy must be removed from the solution (resulting in a lower temperature) in order for the solution to solidify. For example, pure water freezes at 0°C, but for every mole of solute dissolved in 1 kg of water, the freezing point is lowered by 1.86°C. Therefore, the K_f for water is $1.86 \frac{\text{K} \cdot \text{kg}}{\text{mol}}$. As is the case for K_b , the values for K_f are unique to each solvent and will be provided on Test Day. The formula for calculating the freezing point depression for a solution is

$$\Delta T_f = iK_fm$$

Equation 9.10

where ΔT_f is the freezing point depression, i is the van't Hoff factor, K_f is the proportionality constant characteristic of a particular solvent, and m is the molality of the solution. Freezing point depression is a colligative property and depends only on the concentration of particles, not on their identities.

Example:

400 g of AlCl_3 is dissolved in 1.5 L of water at room temperature $\left(K_f = 1.86 \frac{\text{K} \cdot \text{kg}}{\text{mol}}\right)$. What is the new freezing point of this solution?

Solution:

Using the same variables for i and m from the previous example,

$$\Delta T_f = iK_fm = (4) \left(186 \frac{\text{K} \cdot \text{kg}}{\text{mol}} \right) (2m) \approx 15\text{K (actual =$$

The normal freezing point of water is 273 K. The freezing point is going to be depressed (or decreased) by 15 K. The new freezing point is therefore $273 - 15 = 258 \text{ K} = -15^\circ\text{C}$.



MCAT EXPERTISE

As with boiling point elevation, there is a distinction between the calculated value and a final answer. The freezing point depression calculates the amount that the normal freezing point is lowered. As always, read the question to determine if it is asking for the change in temperature (ΔT) or the new (altered) boiling or freezing point.

This effect is the explanation for why we salt icy roads in the winter. Salt mixes with the snow and ice and initially dissolves into the small amount of liquid water that is in equilibrium with the solid phase (the snow and ice). The solute in solution causes a disturbance to the equilibrium such that the rate of melting is unchanged (because the salt can't interact with the solid water that is stabilized in a rigid lattice arrangement), but the rate

of freezing is decreased (the solute displaces some of the water molecules from the solid–liquid interface and prevents liquid water from entering into the solid phase).

This imbalance causes more ice to melt than water to freeze. Melting is an endothermic process, so heat is initially absorbed from the liquid solution, causing the solution temperature to fall below the ambient temperature. Now, there is a temperature gradient, and heat flows from the warmer air to the cooler aqueous solution; this additional heat facilitates more melting—even though the temperature of the solution is actually colder than it was before the solute was added! The more the ice melts into liquid water, the more the solute is dispersed through the liquid. The resulting salt solution, by virtue of the presence of the solute particles, has a lower freezing point than the pure water and remains in the liquid state even at temperatures that would normally cause pure water to freeze.

OSMOTIC PRESSURE

Osmotic pressure is covered primarily in Chapter 8 of MCAT Biochemistry Review, but a brief recap is provided here. Osmotic pressure refers to a “sucking” pressure generated by solutions in which water is drawn into a solution. Formally, the osmotic pressure is the amount of pressure that must be applied to counteract this attraction of water molecules for the solution. The equation for osmotic pressure is

$$\Pi = iMRT$$

Equation 9.11

where Π is the osmotic pressure, i is the van't Hoff factor, M is the molarity of the solution, R is the ideal gas constant, and T is the temperature.

Water moves in the direction of higher solute concentration. For instance, pure water (no solute concentration) will traverse a semipermeable membrane to a solution containing solute particles (such as NaCl) and increase the level of the water as a result, as shown in Figure 9.11.

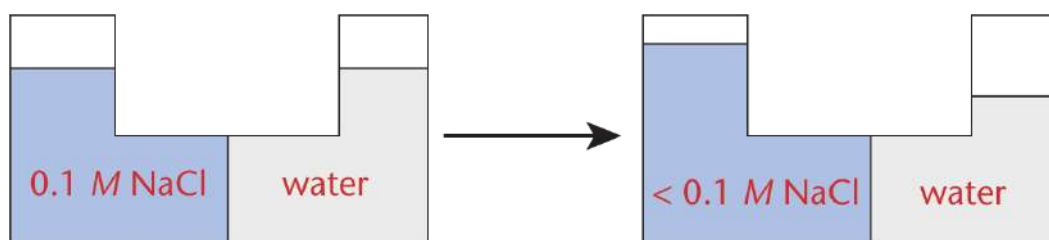


Figure 9.11. Change in Water Level Due to Osmotic Pressure

MCAT Concept Check 9.4:

Before you move on, assess your understanding of the material with these questions.

What is a colligative property?

How are molality and molarity related for water? How are they related for other solvents?

- Water:

- Other solvents:

Determine the vapor pressure of a solution containing 190 g MgCl_2 in 540 g water at room temperature: (Note: The vapor pressure of pure water at 25°C is 3.2 kPa.)

Determine the new boiling point of a solution containing 190 g MgCl_2 in 1500 g water at room temperature:

$$\left(K_b = 0.512 \frac{\text{K} \cdot \text{kg}}{\text{mol}} \right)$$

Determine the freezing point depression of a solution containing 58.5 g of NaCl in 1800 g of water at room temperature: $\left(K_f = 1.86 \frac{\text{K} \cdot \text{kg}}{\text{mol}}\right)$

Conclusion

Our review of solution chemistry has provided an opportunity for us to consider the nature of solutions, solutes, and solvents, and the interactions between solutes and solvents in the formation of solutions. We reviewed solubility and the rules that reflect the solubility of common compounds in water. The different ways of expressing the amount of solute in solution were identified, and examples were given for each unit of concentration, including percent composition, mole fraction, molarity, molality, and normality. Next, we reviewed the thermodynamic principles of solution equilibria and defined unsaturated, saturated, and supersaturated solutions with respect to ion product (IP) and solubility product constant (K_{sp}). Subsequently, we discussed the common ion effect from the perspective of Le Châtelier's principle for a solution at equilibrium. And finally, we examined the colligative properties of solutions and the mathematics that govern them. The colligative properties—vapor pressure depression, boiling point elevation, freezing point depression, and osmotic pressure—are physical properties of solutions that depend on the concentration of dissolved particles but not on their chemical identities.

Concept Summary

Nature of Solutions

- Solutions are homogeneous mixtures composed of two or more substances.
 - They combine to form a single phase, generally the liquid phase.
 - Solvent particles surround solute particles via electrostatic interactions in a process called solvation or dissolution.
 - Aqueous solutions are most important for the MCAT; solvation in water can also be called hydration.
 - Most dissolutions are endothermic, although the dissolution of gas into liquid is exothermic.
- Solubility is the maximum amount of a solute that can be dissolved in a given solvent at a given temperature; it is often expressed as molar solubility—the molarity of the solute at saturation.
- Complex ions or coordination compounds are composed of metallic ions bonded to various neutral compounds and anions, referred to as ligands.
 - Formation of complex ions increases the solubility of otherwise insoluble ions (the opposite of the common ion effect).

- The process of forming a complex ion involves electron pair donors and electron pair acceptors such as those seen in coordinate covalent bonding.

Concentration

- There are many ways of expressing concentration.
 - Percent composition by mass (mass of solute per mass of solution times 100%) is used for aqueous solutions and solid-in-solid solutions.
 - The mole fraction (moles of solute per total moles) is used for calculating vapor pressure depression and partial pressures of gases in a system.
 - Molarity (moles of solute per liters of solution) is the most common unit for concentration and is used for rate laws, the law of mass action, osmotic pressure, pH and pOH, and the Nernst equation.
 - Molality (moles of solute per kilograms of solvent) is used for boiling point elevation and freezing point depression.
 - Normality (number of equivalents per liters of solution) is the molarity of the species of interest and is used for acid–base and oxidation–reduction reactions.

Solution Equilibria

- Saturated solutions are in equilibrium at that particular temperature.
- The solubility product constant (K_{sp}) is simply the equilibrium constant for a dissociation reaction.

- Comparison of the ion product (IP) to K_{sp} determines the level of saturation and behavior of the solution:
 - $IP < K_{sp}$: the solution is unsaturated, and if more solute is added, it will dissolve
 - $IP = K_{sp}$: the solution is saturated (at equilibrium), and there will be no change in concentrations
 - $IP > K_{sp}$: the solution is supersaturated, and a precipitate will form
- Formation of a complex ion in solution greatly increases solubility.
 - The formation or stability constant (K_f) is the equilibrium constant for complex formation. Its value is usually much greater than K_{sp} .
 - The formation of a complex increases the solubility of other salts containing the same ions because it uses up the products of those dissolution reactions, shifting the equilibrium to the right (the opposite of the common ion effect).
 - The common ion effect decreases the solubility of a compound in a solution that already contains one of the ions in the compound. The presence of that ion in solution shifts the dissolution reaction to the left, decreasing its dissociation.

Colligative Properties

- Colligative properties are physical properties of solutions that depend on the concentration of dissolved particles but not on their chemical identity.
- Vapor pressure depression follows Raoult's law.

- The presence of other solutes decreases the evaporation rate of a solvent without affecting its condensation rate, thus decreasing its vapor pressure.
 - Vapor pressure depression also explains boiling point elevation—as the vapor pressure decreases, the temperature (energy) required to boil the liquid must be raised.
-
- Freezing point depression and boiling point elevation are shifts in the phase equilibria dependent on the molality of the solution.
 - Osmotic pressure is primarily dependent on the molarity of the solution.
 - For solutes that dissociate, the van't Hoff factor (i) is used in freezing point depression, boiling point elevation, and osmotic pressure calculations.

Answers to Concept Checks

9.1

Solvation refers to the breaking of intermolecular forces between solute particles and between solvent particles, with formation of intermolecular forces between solute and solvent particles. In an aqueous solution, water is the solvent.

Solubility is the amount of solute contained in a solvent. Saturation refers to the maximum solubility of a compound at a given temperature; one cannot dissolve any more of the solute just by adding more at this temperature.

Solubility of solids can be increased by increasing temperature. Solubility of gases can be increased by decreasing temperature or increasing the partial pressure of the gas above the solvent (Henry's law).

Group I metals, ammonium, nitrate, and acetate salts are always soluble.

9.2

Compound	Molality	Molarity	Normality
Glucose $\left(180 \frac{\text{g}}{\text{mol}}\right)$	$\frac{1 \text{ mol}}{0.25 \text{ kg solvent}} = 4 m$	Cannot be calculated directly because final volume is unknown. In dilute solutions, $M \approx m$. At higher concentrations,	1 N (glucose does not dissociate)

Compound	Molality	Molarity M < m because solute particles increase overall volume of solution.	Normality
Carbonate $\left(60 \frac{\text{g}}{\text{mol}}\right)$	$\frac{3 \text{ mol}}{0.25 \text{ kg solvent}} = 12 \text{ m}$		Approximately 24 N (twice the molality and approximate molarity)

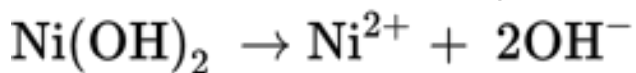
$$M_i V_i = M_f V_f \rightarrow V_i = \frac{M_f V_f}{M_i} = \frac{(25 \text{ ppm})(100 \text{ mL})}{(100 \text{ ppm})} = 25 \text{ mL}$$

Thus, start with 25 mL of the stock solution and add 75 mL pure water to get 100 mL of solution with 25 ppm Cl_2 .

$$\begin{aligned}
 X_{\text{NaCl}} &= \frac{\text{moles of NaCl}}{\text{total moles}} \\
 &= \frac{90 \text{ g} \left[\frac{1 \text{ mol}}{58.5 \text{ g}} \right]}{90 \text{ g} \left[\frac{1 \text{ mol}}{58.5 \text{ g}} \right] + 10^4 \text{ g} \left[\frac{1 \text{ mol}}{18 \text{ g}} \right]} \approx \frac{90 \left(\frac{1}{60} \right)}{90 \left(\frac{1}{60} \right) + 10^4} \\
 &\approx \frac{1.5}{1.5 + 500} \approx \frac{1.5}{500} \approx \frac{1.5}{5 \times 10^2} \approx \frac{1}{3} \times 10^{-2} \approx 3 \times 10^{-3} \\
 &\quad (\text{actual} = 2.8 \times 10^{-3})
 \end{aligned}$$

$$\begin{aligned}
 \text{Percent composition} &= \frac{\text{mass of solute}}{\text{mass of solution}} \times 100\% = \frac{90 \text{ g}}{90 \text{ g} + 10^4 \text{ g}} \times 100\% \\
 &\approx \frac{90}{10^4} \times 100\% = 0.9\% \quad (\text{actual} = 0.89\%)
 \end{aligned}$$

First, write out the balanced equation:



Next, identify that the molar solubility x represents the amount of Ni(OH)_2 that dissociates, creating x of Ni^{2+} and $2x$ of OH^- . Write out the K_{sp} equation, and plug in the values of x to solve for K_{sp} :

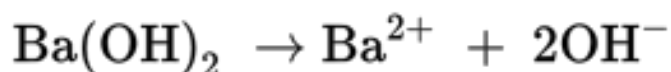
$$K_{\text{sp}} = [\text{Ni}^{2+}][\text{OH}^-]^2 = x(2x)^2$$

$$K_{\text{sp}} = (5.2 \times 10^{-6})(2 \times 5.2 \times 10^{-6})^2$$

$$K_{\text{sp}} = (5 \times 10^{-6})(10 \times 10^{-6})^2 \approx (5 \times 10^{-6})(10^{-5})^2$$

$$K_{\text{sp}} \approx 5 \times 10^{-6} \times 10^{-10} = 5 \times 10^{-16} \text{ (actual value} = 5.6 \times 10^{-16})$$

Start with the balanced reaction and calculation for K_{sp} , which can also be used to calculate Q . Keep in mind that for every x of Ba(OH)_2 that dissolves, x of Ba^{2+} will be produced, and $2x$ of OH^- .



$$K_{\text{sp}} = [\text{Ba}^{2+}][\text{OH}^-]^2$$

$[\text{Ba}^{2+}]$	Ion Product	Behavior of Solution
0.5 M	$(0.5 \text{ M})(1 \text{ M})^2 = 0.5$	$0.5 > 5.0 \times 10^{-3} \rightarrow \text{precipitation}$
0.1 M	$(0.1 \text{ M})(0.2 \text{ M})^2 = 4.0 \times 10^{-3}$	$4.0 \times 10^{-3} < 5.0 \times 10^{-3} \rightarrow \text{dissolution}$
0.05 M	$(0.05 \text{ M})(0.1 \text{ M})^2 = 5.0 \times 10^{-4}$	$5.0 \times 10^{-4} < 5.0 \times 10^{-3} \rightarrow \text{dissolution}$

Note that the concentration of hydroxide is double that of barium. While there will be a very small contribution of hydroxide from the autoionization of water, this amount is negligible compared to the values given in the question.

Start by writing the balanced reaction for the least soluble salt in the problem: $\text{Zn(OH)}_2 \rightarrow \text{Zn}^{2+} + 2\text{OH}^-$. Next, write out the K_{sp} equation and enter the variables for the concentrations. The Zn^{2+} concentration will equal x , the molar solubility under these conditions, but the OH^- concentration will come from two contributors: the dissociated Zn(OH)_2 , and the 0.1 M NaOH solution. This results in the following K_{sp} expression: $K_{\text{sp}} = [\text{Zn}^{2+}][\text{OH}^-]^2 = (x)(0.1 + 2x)^2$. Since the K_{sp} for Zn(OH)_2 is 4.1×10^{-17} , x will be negligible compared to the 0.1 M from NaOH. The K_{sp} expression simplifies to: $K_{\text{sp}} = (x)(0.1)^2$. Thus, $4.1 \times 10^{-17} = 0.01x$. x = molar solubility of $\text{Zn(OH)}_2 = 4.1 \times 10^{-15}$.

9.4

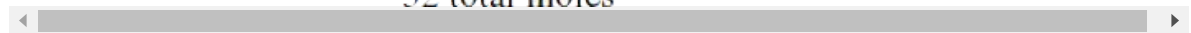
Colligative properties are those that depend on the amount of solute present, but not the actual identity of the solute particles. Examples include vapor pressure depression, boiling point elevation, freezing point depression, and osmotic pressure.

Molarity (M) and molality (m) are nearly equal at room temperature. This is only because 1 L solution is approximately equal to 1 kg solvent for dilute solutions (the denominators of the molarity and molality equations, respectively). For other solvents, molarity and molality differ significantly because their densities are not $1 \frac{\text{g}}{\text{mL}}$ like water.

$$190 \text{ g MgCl}_2 \times \frac{1 \text{ mol}}{95.3 \text{ g}} = 2 \text{ mol MgCl}_2. \quad 540 \text{ g H}_2\text{O} \times \frac{1 \text{ mol}}{18 \text{ g}} = 30 \text{ mol H}_2\text{O}$$

$$X_{\text{water}} = \frac{30 \text{ moles water}}{32 \text{ total moles}}$$

$$P_{\text{water}} = X_{\text{water}} P_{\text{water}}^{\circ} = \frac{30 \text{ moles water}}{32 \text{ total moles}} \times (3.2 \text{ kPa}) = 3.0 \text{ kPa}$$



$$T_b = iK_b m = (3) \left(0.512 \frac{\text{K} \cdot \text{kg}}{\text{mol}} \right) \left(\frac{2 \text{ mole MgCl}_2}{1.5 \text{ kg water}} \right) \approx 2 \text{ K (actual} = 2.04 \text{ K)}.$$

The new boiling point will be $373 + 2 = 375 \text{ K}$.

$$\Delta T_f = iK_f m = (2) \left(1.86 \frac{\text{K} \cdot \text{kg}}{\text{mol}} \right) \left(\frac{1 \text{ mole NaCl}}{1.8 \text{ kg water}} \right) \approx 2 \text{ K (actual} = 2.06 \text{ K)}$$



Equations to Remember

(9.1) Percent composition by mass: $\frac{\text{mass of solute}}{\text{mass of solution}} \times 100\%$

(9.2) Mole fraction: $X_A = \frac{\text{moles of A}}{\text{total moles of all species}}$

(9.3) Molarity: $M = \frac{\text{moles of solute}}{\text{liters of solution}}$

(9.4) Molality: $m = \frac{\text{moles of solute}}{\text{kilograms of solvent}}$

(9.5) Dilution formula: $M_i V_i = M_f V_f$

(9.6) Solubility product constant: $K_{sp} = [A^{n+}]^m [B^{m-}]^n$

(9.7) Ion product: $IP = [A^{n+}]^m [B^{m-}]^n$

(9.8) Raoult's law (vapor pressure depression): $P_A = X_A P_A^\circ$

(9.9) Boiling point elevation: $\Delta T_b = i K_b m$

(9.10) Freezing point depression: $\Delta T_f = i K_f m$

(9.11) Osmotic pressure: $\Pi = i M R T$



Shared Concepts

Biology Chapter 10

Homeostasis

General Chemistry Chapter 3

Bonding and Chemical Interactions

General Chemistry Chapter 6

Equilibrium

General Chemistry Chapter 7

Thermochemistry

General Chemistry Chapter 10

Acids and Bases

General Chemistry Chapter 12

Electrochemistry

Discrete Practice Questions

Consult your online resources for additional practice.

1. An aqueous solution was prepared by mixing 70 g of an unknown nondissociating solute into 100 g of water. The solution has a boiling point of 101.0°C. What is the molar mass of the solute?

(Note: $\text{GEW} = \frac{\text{Molar mass}}{n}$)

(A) $358.4 \frac{\text{g}}{\text{mol}}$

(B) $264 \frac{\text{g}}{\text{mol}} ?$

(C) $123.2 \frac{\text{g}}{\text{mol}}$

(D) $123.2 \frac{\text{g}}{\text{mol}}$

2. Which phases of solvent and solute can form a solution?

Solid solvent, gaseous solute

Solid solvent, solid solute

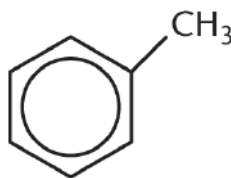
Gaseous solvent, gaseous solute

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II, and III

3. Two organic liquids, pictured in the figure below, are combined to form a solution. Based on their structures, will the solution closely obey Raoult's law?



benzene



toluene

- (A) Yes; the liquids differ due to the additional methyl group on toluene and, therefore, will not deviate from Raoult's law.
- (B) Yes; the liquids are very similar and, therefore, will not deviate from Raoult's law.
- (C) No; the liquids differ due to the additional methyl group on toluene and, therefore, will deviate from Raoult's law.
- (D) No; the liquids both contain benzene rings, which will interact with each other and cause deviation from Raoult's law.

4. Which of the following explanations best describes the mechanism by which solute particles affect the melting point of ice?

- (A) Melting point is elevated because the kinetic energy of the substance increases.
- (B) Melting point is elevated because the kinetic energy of the substance decreases.
- (C) Melting point is depressed because solute particles interfere with lattice formation.
- (D) Melting point is depressed because solute particles enhance lattice formation.

5. The process of formation of a salt solution can be better understood by breaking the process into three steps:

- Breaking the solute into its individual components
- Making room for the solute in the solvent by overcoming intermolecular forces in the solvent
- Allowing solute–solvent interactions to occur to form the solution

Which of the following correctly lists the enthalpy changes for these three steps, respectively?

- (A) Endothermic, exothermic, endothermic
- (B) Exothermic, endothermic, endothermic
- (C) Exothermic, exothermic, endothermic
- (D) Endothermic, endothermic, exothermic

6. The entropy change when a solution forms can be expressed by the term $\Delta S^\circ_{\text{soln}}$. When water molecules become ordered around an ion as it dissolves, the ordering would be expected to make a negative contribution to $\Delta S^\circ_{\text{soln}}$. An ion that has more charge density will have a greater hydration effect, or ordering of water molecules. Based on this information, which of the following compounds will have the most negative contribution to $\Delta S^\circ_{\text{soln}}$?

- (A) KCl
- (B) LiF
- (C) CaS
- (D) NaCl

7. When ammonia, NH_3 , is used as a solvent, it can form complex ions. For example, dissolving AgCl in NH_3 will result in the complex ion $[\text{Ag}(\text{NH}_3)]^{2+}$. What effect would the formation of complex ions have on the solubility of a compound like AgCl in NH_3 ?

- (A) The solubility of AgCl will increase because complex ion formation will cause more ions to exist in solution, which interact with AgCl to cause it to dissociate.
- (B) The solubility of AgCl will increase because complex ion formation will consume Ag^+ ions and cause the equilibrium to shift away from solid AgCl.
- (C) The solubility of AgCl will decrease because Ag^+ ions are in complexes, and the Ag^+ ions that are not complexed will associate with Cl^- to form solid AgCl.
- (D) The solubility of AgCl will decrease because complex ion formation will consume Ag^+ ions and cause the equilibrium to shift toward the solid AgCl.

8. One hundred grams of sucrose are dissolved in a cup of hot water at 80°C . The cup of water contains 300.00 mL of water. What is the percent composition by mass of sugar in the resulting solution?

(Note: Sucrose = $\text{C}_{12}\text{H}_{22}\text{O}_{11}$, density of water at

$$80^\circ\text{C} = 0.975 \frac{\text{g}}{\text{mL}})$$

- (A) 25.0%
- (B) 25.5%
- (C) 33.3%
- (D) 34.2%

9. Which of the following combinations of liquids would be expected to have a vapor pressure higher than the vapor pressure that would be predicted by Raoult's law?

- (A) Ethanol and hexane
- (B) Acetone and water
- (C) Isopropanol and methanol
- (D) Nitric acid and water

10. The salt KCl is dissolved in a beaker. To an observer holding the beaker, the solution begins to feel colder as the KCl dissolves. From this observation, one could conclude that:

- (A) $\Delta S^\circ_{\text{soln}}$ is large enough to overcome the unfavorable $\Delta H^\circ_{\text{soln}}$.
- (B) KCl is mostly insoluble in water.
- (C) $\Delta S^\circ_{\text{soln}}$ must be negative when KCl dissolves.
- (D) boiling point depression will occur in this solution.

11. Which of the following will cause the greatest increase in the boiling point of water when it is dissolved in 1.00 kg H₂O?

- (A) 0.4 mol calcium sulfate
- (B) 0.5 mol iron(III) nitrate
- (C) 1.0 mol acetic acid
- (D) 1.0 mol sucrose

12. Reverse osmosis is a process that allows fresh water to be obtained by using pressure to force an impure water source through a semi-permeable membrane that only allows water molecules to pass.

What is the minimum pressure that would be required to purify seawater at 25°C that has a total osmolarity of 1,000 mOsm/L?

- (A) 23.5 atm
- (B) 24.5 atm
- (C) 24,000 atm
- (D) 24,500 atm

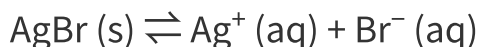
13. Lead is a toxic element that can cause many symptoms, including mental retardation in children. If a body of water is polluted with lead ions at 200 ppb (parts per billion), what is the concentration of lead expressed as molarity? (Note: The density of water is $1 \frac{\text{g}}{\text{mol}}$, and ppb = grams per 10^9 grams of solution)

- (A) $9.7 \times 10^{-10} \text{ M Pb}^{2+}$
- (B) $9.7 \times 10^{-7} \text{ M Pb}^{2+}$
- (C) $6.2 \times 10^{-7} \text{ M Pb}^{2+}$
- (D) $6.2 \times 10^{-6} \text{ M Pb}^{2+}$

14. A saturated solution of cobalt(III) hydroxide ($K_{\text{sp}} = 1.6 \times 10^{-44}$) is added to a saturated solution of thallium(III) hydroxide ($K_{\text{sp}} = 6.3 \times 10^{-46}$). What is likely to occur?

- (A) Both cobalt(III) hydroxide and thallium(III) hydroxide remain stable in solution.
- (B) Cobalt(III) hydroxide precipitates and thallium(III) hydroxide remains stable in solution.
- (C) Thallium(III) hydroxide precipitates and cobalt(III) hydroxide remains stable in solution.
- (D) Both thallium(III) hydroxide and cobalt(III) hydroxide precipitate.

15. The following equilibrium exists when AgBr ($K_{\text{sp}} = 5.35 \times 10^{-13}$) is in solution:



What is the solubility of AgBr in a solution of 0.0010 M NaBr ?

- (A) $5.35 \times 10^{-13} \frac{\text{g}}{\text{L}}$
- (B) $1.04 \times 10^{-12} \frac{\text{g}}{\text{L}}$
- (C) $5.35 \times 10^{-13} \frac{\text{g}}{\text{L}}$
- (D) $1.04 \times 10^{-7} \frac{\text{g}}{\text{L}}$

Explanations to Discrete Practice Questions

1. A

The equation $\Delta T_b = iK_b m$ can be used to solve this problem. The change in boiling point is $101.0 - 100 = 1.0^\circ\text{C}$. Then, plug that into:

$$\Delta T_b = iK_b m \rightarrow m = \frac{\Delta T_b}{iK_b} = \frac{1.11 \text{ K}}{(1)\left(0.512 \frac{\text{K} \cdot \text{kg}}{\text{mol}}\right)} \approx 2$$

The van't Hoff factor for this solute is 1 because the molecule does not dissociate into smaller components. Then, convert to grams of solute using the definition of molality:

$$\text{molality} = \frac{\text{moles of solute}}{\text{kg of solvent}} \rightarrow \text{moles of solute} = (2m)(0.1 \text{ kg}) = 0.2 \text{ mol}$$

The mass used in this equation is 0.1 kg because 100 mL of water has a mass of 0.1 kg. Then, determine the molar mass:

$$\text{molar mass} = \frac{70 \text{ g}}{0.2 \text{ mol}} = 350 \frac{\text{g}}{\text{mol}}, \text{ which is closest to (A).}$$

2. D

All three choices can make a solution as long as the two components create a mixture that is of uniform appearance (homogeneous).

Hydrogen in platinum is an example of a gas in a solid. Brass and steel are examples of homogeneous mixtures of solids. The air we breathe is an example of a homogeneous mixture of gases; while these are more commonly simply referred to as mixtures, they still fit the criteria of a solution.

3. B

Benzene and toluene are both organic liquids and have very similar properties. They are both nonpolar and are almost exactly the same size. Raoult's law states that ideal solution behavior is observed when solute-solute, solvent-solvent, and solute-solvent interactions are all very similar. Therefore, benzene and toluene in solution will be predicted to behave as a nearly ideal solution.

4. C

Melting point depresses upon solute addition, making (A) and (B) incorrect. Solute particles interfere with lattice formation, the highly organized state in which solid molecules align themselves. Colder-than-normal conditions are necessary to create the solid structure.

5. D

The first step will most likely be endothermic because energy is required to break molecules apart. The second step is also endothermic because the intermolecular forces in the solvent must be overcome to allow incorporation of solute particles. The third step will most likely be exothermic because polar water molecules will interact with the dissolved ions, creating a stable solution and releasing energy.

6. C

CaS will cause the most negative contribution to $\Delta S_{\text{soln}}^{\circ}$ through hydration effects because the Ca^{2+} and S^{2-} ions have the highest charge density compared to the other ions. All of the other ions have charges of +1 or -1, whereas Ca^{2+} and S^{2-} each have charges with a magnitude of 2.

7. B

Formation of complex ions between silver ions and ammonia will cause more molecules of solid AgCl to dissociate. The equilibrium is driven toward dissociation because the Ag^{+} ions are essentially being removed from solution when they complex with ammonia. This rationale is based upon Le Châtelier's principle, stating that when a chemical equilibrium experiences a change in concentration, the system will shift to counteract that change.

8. B

The mass percent of a solute equals the mass of the solute divided by the mass of the total solution times 100%.

$$\text{mass } \% = \frac{\text{mass of solute}}{\text{total mass of solution}} \times 100\%.$$

Plug in the values given for sucrose, the volume of water and the density of water to determine the %mass of sucrose.

% mass of sucrose

$$\begin{aligned} &= \frac{(100 \text{ g source})}{(300 \text{ mL H}_2\text{O})(0.975/\text{mL}) + 100 \text{ g sucrose}} \times 100\% \\ &\approx \frac{100}{300 + 100} \times 100\% = \frac{100}{400} \times 100\% = 25\% \end{aligned}$$

Keep in mind that in rounding while calculating, the denominator was estimated to be larger than the actual value, thus giving an answer that

is slightly lower than the actual value. Thus the correct answer is (B), 25.5%. (A) results if rounding error is not taken into account. While these answers are very close, the mass of the water must be slightly less than 300 g, given the density value, so the percent composition of sucrose must be slightly higher than 25%. If the solute's mass is not added to the solvent's, the calculated value is 34.2%, which is (D). (C) neglects both the addition step and the rounding error.

9. A

Mixtures that have a higher vapor pressure than predicted by Raoult's law have stronger solvent-solvent and solute-solute interactions than solvent-solute interactions. Therefore, particles do not want to stay in solution and more readily evaporate, creating a higher vapor pressure than an ideal solution. Two liquids that have different properties, like hexane (hydrophobic) and ethanol (hydrophilic, small) in (A), would not have many interactions with each other and would cause positive deviation; i.e. higher vapor pressure. (B) and (C) are composed of liquids that are similar to one another and would not show significant deviation from Raoult's law. (D) contains two liquids that would interact very well with each other, which would actually cause a negative deviation from Raoult's law—when attracted to one other, solutes and solvents prefer to stay in liquid form and have a lower vapor pressure than predicted by Raoult's law.

10. A

Dissolution is governed by enthalpy and entropy, which are related by the equation $\Delta G_{\text{soln}}^{\circ} = \Delta H_{\text{soln}}^{\circ} - T\Delta S_{\text{soln}}^{\circ}$. The cooling of the solution indicates that heat is used up in this bond-breaking reaction. In other words, dissolution is endothermic, and ΔH is positive. The reaction is

occurring spontaneously, so ΔG must be negative. The only way that a positive ΔH can result in a negative ΔG is if entropy, ΔS , is a large, positive value as in (A). Conceptually, that means that the only way the solid can dissolve is if the increase in entropy is great enough to overcome the increase in enthalpy. (B) is incorrect because it is clearly stated in the question stem that KCl dissolves; further, all salts of Group 1 metals are soluble. (C) is incorrect because $\Delta S_{\text{soln}}^{\circ}$ must be positive in order for KCl to dissolve. Finally, (D) is incorrect because solute dissolution would cause the boiling point to elevate, not depress. It is also not a piece of evidence that could be found simply by observing the beaker's temperature change.

11. B

The equation to determine the change in boiling point of a solution is as follows: $\Delta T_b = iK_b m$. m is the molality of the solution, and K_b is the boiling point elevation constant. In this case, the solvent is always water, so K_b will be the same for each solution. What is needed is the number of dissociated particles from each of the original species. This is referred to as the van't Hoff factor (i) and is multiplied by molality to give a normality (the concentration of the species of interest—in this case, all particles). The normality values determine which species causes the greatest change in boiling point.

Species	Number of Moles	Number of Dissolved Particles	$i \times m$ (Normality)
CaSO_4	0.4	2	0.8
$\text{Fe}(\text{NO}_3)_3$	0.5	4	2.0

Species	Number of Moles	Number of Dissolved Particles	$i \times m$ (Normality)
CH_3COOH	1.0	Between 1 and 2 (acetic acid is a weak acid and a low percentage of the molecules will dissociate into 2 particles)	Between 1.0 and 2.0
$\text{C}_{12}\text{H}_{22}\text{O}_{11}$	1.0	1	1.0

The choice is between iron(III) nitrate and acetic acid. The fact that acetic acid is a weak acid indicates that only a few particles will dissociate into H^+ and acetate. Therefore, the normality of the acetic acid will be much closer to 1.0 than 2.0.

12. B

Osmotic pressure is given by the formula $\Pi = iMRT$. Entering the values from the question stem gives:

$$\begin{aligned}\Pi &= (1000 \times 10^{-3} \text{ mOsm/L})(0.0812 \text{ L}\cdot\text{atm/mol}\cdot\text{K})(298 \text{ K}) \\ &\approx 1 \times 0.08 \times 300 = 24 \text{ atm} \quad \text{actual value} = 24.2 \text{ atm}\end{aligned}$$

Notice that the concentration of seawater is given for all solutes, which represents $i \times M$. It is also given in mOsm/L, which is converted to moles per liter by multiplying by 10^{-3} . Also, the question asks for the minimum pressure required, which means that the correct answer choice must be slightly above the calculated pressure in order for reverse osmosis to proceed.

13. B

200 ppb of Pb^{2+} is equivalent to 200 grams of Pb^{2+} in 10^9 grams of solution; given the extremely low concentration of lead, the mass of the water can be assumed to be approximately 10^9 grams, as well. To solve, set up a dimensional analysis question. The units needed at the end are moles per liter (molarity), so convert from grams of lead to moles of lead and grams of water to liters of water:

$$\left[\frac{200 \text{ g Pb}^{2+}}{10^9 \text{ g H}_2\text{O}} \right] \left[\frac{10^3 \text{ g H}_2\text{O}}{1 \text{ L}} \right] \left[\frac{1 \text{ mol Pb}^{2+}}{207.2 \text{ g Pb}^{2+}} \right] \approx \frac{200 \times 10^3}{200 \times 10^9}$$

$$= 1 \times 10^{-6} \text{ M} = 10 \times 10^{-7} \text{ M}$$

(actual value = $9.67 \times 10^{-7} \text{ M}$)

Note that the denominator was rounded to a smaller number, meaning the estimated answer is slightly larger than the actual value.

14. D

Since both salts have a formula MX_3 (one of one particle, three of another), it is possible to directly compare the molar solubilities of each. When the solutions are mixed, $[\text{OH}^-]$ is above saturation levels for both the cobalt and the thallium in the solution. Since thallium(III) hydroxide has a smaller K_{sp} than that of cobalt(III) hydroxide, it will react first. The ion product of the mixed solution is higher than the K_{sp} for thallium(III) hydroxide, and the system will shift left to precipitate solid thallium(III) hydroxide. After the thallium(III) hydroxide precipitates, a small excess of OH^- will remain, which gives an ion product slightly above the K_{sp} of cobalt(III) hydroxide. This will cause a small amount (1%–3%) of cobalt(III) hydroxide to also precipitate.

15. D

The solubility of AgBr can be determined using the K_{sp} value given in the equation. Some amount of AgBr will dissolve; this is the molar solubility x for these conditions. When AgBr dissociates, there will be x amount of silver(I) formed and x amount of bromide—which is added to the 0.0010 M Br^- already present from NaBr.

$$\begin{aligned}\text{AgBr} &\rightleftharpoons \text{Ag}^+ + \text{Br}^- \\ K_{sp} &= [\text{Ag}^+][\text{Br}^-] \\ 5.35 \times 10^{-13} &= (x)(0.0010\text{ M} + x)\end{aligned}$$

Given the K_{sp} of 5.4×10^{-13} , x will be negligible compared to 0.0010 M. Thus, the math can be simplified to:

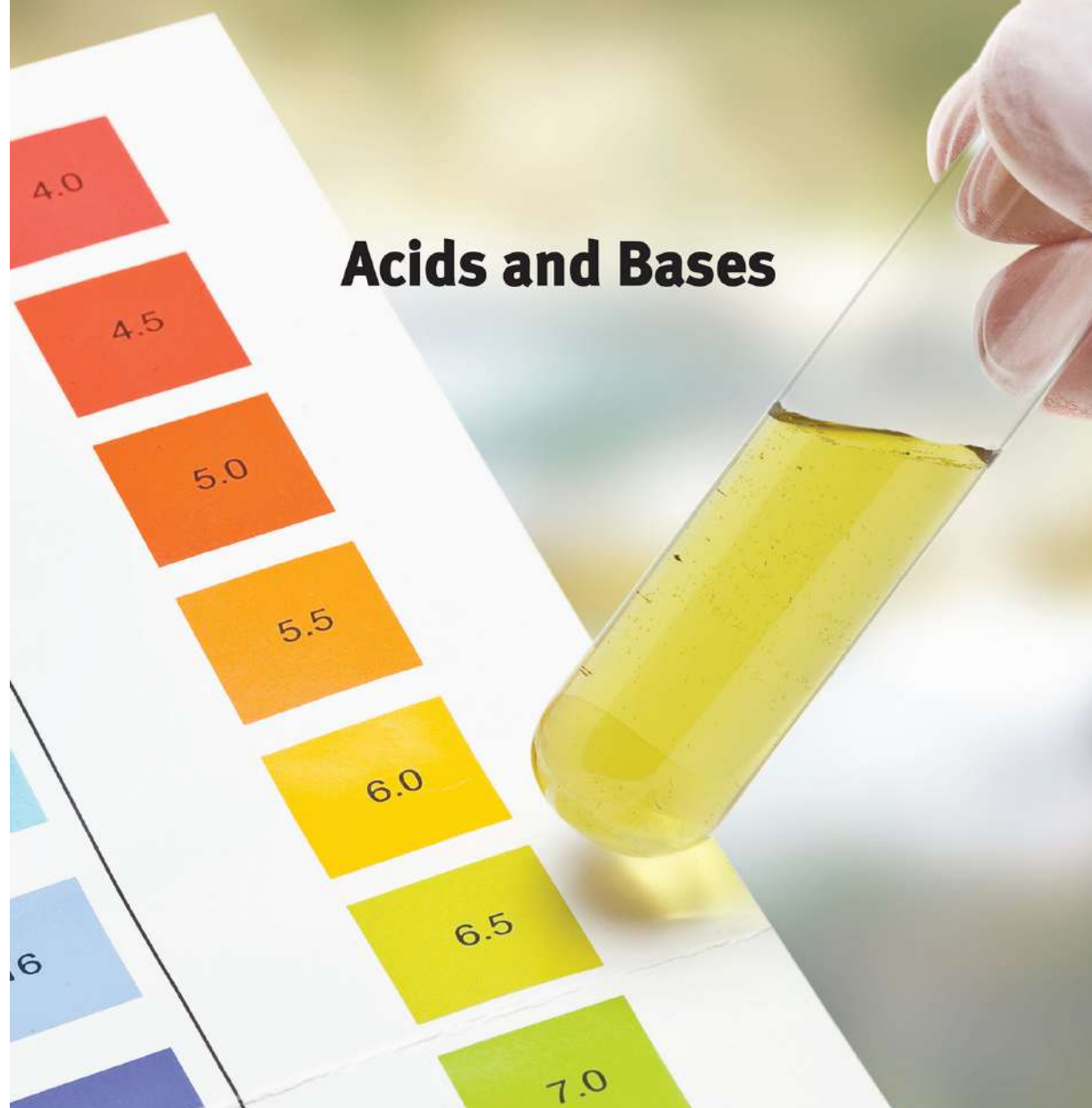
$$\begin{aligned}5.35 \times 10^{-13} &= (x)(0.0010) \\ x &= \frac{5.35 \times 10^{-13}}{10^{-3}} = 5.35 \times 10^{-10}\end{aligned}$$

Therefore, x , the molar solubility, is 5.35×10^{-10} , which looks like (C). However, the units of the answer choices are grams per liter, not molarity, and the result must be multiplied by the molar mass $\left(187.8 \frac{\text{g}}{\text{mol}}\right)$:

$$5.35 \times 10^{-10} \frac{\text{mol}}{\text{L}} \times 187.8 \frac{\text{g}}{\text{mol}} \approx 5 \times 10^{-10} \times 200 = 1,000 \times 10^{-8} = 1 \times 10^{-5}$$

which is close to (D). Note that a very accurate approximation was reached by rounding down the first number and rounding up the second, balancing the error.

Acids and Bases



In This Chapter

10.1 Definitions

Arrhenius 

Brønsted–Lowry 

Lewis 

Amphoteric Species 

Acid–Base Nomenclature 

10.2 Properties

Autoionization of Water and Hydrogen Ion

Equilibria 

Strong Acids and Bases 

Weak Acids and Bases 

Conjugate Acid–Base Pairs 

Applications of K_a and K_b 

Salt Formation 



10.3 Polyvalence and Normality

10.4 Titration and Buffers

General Principles 

Strong Acid and Strong Base 

Weak Acid and Strong Base 



Strong Acid and Weak Base 

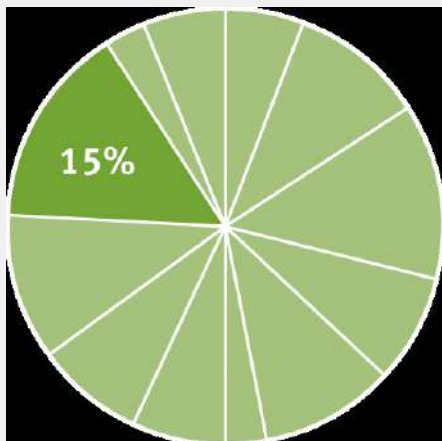
Weak Acid and Weak Base 

Polyvalent Acids and Bases 

Buffers 

Concept Summary 

Chapter Profile



The content in this chapter should be relevant to about 15% of all questions about general chemistry on the MCAT.

This chapter covers material from the following AAMC content categories:

4D: How light and sound interact with matter

5A: Unique nature of water and its solutions

Introduction

Some medications can be applied as drops, salves, or creams to mucous membranes. Others are injected. Some employ a transdermal patch, while

others are swallowed or inhaled. The route of administration of a drug compound is related to both the location of its target tissue (local or systemic), as well as the chemical and physical properties of the compound. For example, compounds that are water-soluble can be administered intravenously (an aqueous solution dripped directly into the bloodstream), while those that are lipid-soluble can be administered transcutaneously (via a patch or a cream) or orally (in a pill or liquid suspension). The polarity, size, and charge of the drug compound will determine its solubility in polar or nonpolar environments and will be major contributing factors in determining the most effective and efficient route of administration.

Whether a drug compound has an ionic charge is usually a function of the acidic or basic nature of the compound. For example, a basic organic compound that is insoluble in water when neutral can be reacted with an acid to form a salt; because this salt is ionic, it is water-soluble.

Correspondingly, an acidic organic compound that is insoluble in water when neutral can be reacted with a base to form a water-soluble salt. On the other hand, the protonated (acidic cationic) form of an organic compound can be reacted with a base to neutralize the compound and release it from its salt, changing (and usually reversing) its solubility in water.

Medical professionals aren't the only ones concerned about drug solubilities and routes of administration—there's a science to illicit drugs, too. One of the clearest examples of this is the difference between the two major forms of cocaine, a large alkaloid compound derived from the coca plant. Most commonly, the alkaloid compound is reacted with hydrochloric acid (which protonates its tertiary amine functional group), extracted with

water, and dried to a water-soluble powder (cocaine hydrochloride); this powder either is snorted (insufflated) into the nasal cavity, where it is absorbed into the capillary beds, or is injected directly into the venous circulation. The second form of cocaine, the salt form, has a very high boiling point, which is close to the temperature at which cocaine degrades, and cannot be smoked. To produce a vaporizable form of cocaine that can be inhaled from a pipe, the cocaine hydrochloride must be reacted with a base, typically either ammonia (to produce pure freebase cocaine) or sodium bicarbonate (to produce crack cocaine, which is less pure). The base reacts with the protonated tertiary amine, removing the hydrogen ion to reform the neutral alkaloid compound. The freebase cocaine is water-insoluble and usually extracted with ether, or it is left in the aqueous solution, which is heated and evaporated. The freebase and crack forms of cocaine have much lower boiling points; consequently, they can be smoked without risk of degradation.

In this chapter, our focus will be on those two classes of compounds—acids and bases—which are involved in so many important reactions. Acid–base reactions are an important topic for the MCAT; in fact, neutralization reactions are some of the most commonly tested reaction types on Test Day. We will begin with a review of the different definitions of acids and bases and their properties, including their characterization as either strong or weak. Focusing on weak acids and bases, we will discuss the significance of the equilibrium constants K_a and K_b for acids and bases, respectively. Finally, we will review acid–base titrations and buffer systems.

10.1 Definitions

LEARNING GOALS

After Chapter 10.1, you will be able to:

- Compare and contrast the Arrhenius, Brønsted–Lowry, and Lewis definitions for acids and bases
- Predict the acid formula and name for an anion using Arrhenius acid naming trends
- Identify amphoteric species, and determine whether they are amphoteric as well

Over the last century, chemists have used different definitions to identify compounds as acids or bases. Three definitions emerged, with each more inclusive than the former one.

ARRHENIUS

The first and most specific definition of an acid or base is the Arrhenius definition. An Arrhenius acid will dissociate to form an excess of H^+ in solution, and an Arrhenius base will dissociate to form an excess of OH^- in solution. These behaviors are generally limited to aqueous acids and

bases. Arrhenius acids and bases are easily identified; acids contain H at the beginning of their formula (HCl , HNO_3 , H_2SO_4 , and so on) and bases contain OH at the end of their formula (NaOH , $\text{Ca}(\text{OH})_2$, $\text{Fe}(\text{OH})_3$, and so on).



MCAT EXPERTISE

Any mention of the Arrhenius definition on Test Day will likely be in comparison to other definitions of acids. The Arrhenius definition is by far the most restrictive; the Brønsted–Lowry and Lewis definitions predominate on the MCAT.

BRØNSTED–LOWRY

A more inclusive definition of acids and bases was proposed independently by Johannes Brønsted and Thomas Lowry in 1923. A Brønsted–Lowry acid is a species that donates hydrogen ions (H^+), while a Brønsted–Lowry base is a species that accepts them.

The advantage of this definition over Arrhenius's is that it is not limited to aqueous solutions. For example, OH^- , NH_3 , and F^- are all Brønsted–Lowry bases because each has the ability to accept a hydrogen ion. However, neither NH_3 nor F^- can be classified as Arrhenius bases because they do not dissociate to produce an excess of OH^- ions in aqueous solutions. According to both of these definitions, there is only one way for a species to be an acid: producing hydrogen ions. The only differences between the

two definitions for acids are the requirement of an aqueous medium in the Arrhenius definition and the acidity of water. In the Arrhenius definition, water is not considered an acid—it does not produce an excess of H^+ in solution. Water is, on the other hand, a Brønsted–Lowry acid because it is able to donate a proton to other species. Most acid–base chemistry reactions on the MCAT will involve the transfer of hydrogen ions in accordance with the Brønsted–Lowry definitions.



MCAT EXPERTISE

Every Arrhenius acid (or base) can also be classified as a Brønsted–Lowry acid (or base). Every Brønsted–Lowry acid (or base) can also be classified as a Lewis acid (or base). This logic does not always work the other way (for example, NH_3 is a Brønsted–Lowry base, but not an Arrhenius base).

Brønsted–Lowry acids and bases always occur in pairs because the definitions require the transfer of a proton from the acid to the base. These are conjugate acid–base pairs, as described in the next section. For example, in the autoionization of water, H_3O^+ is the conjugate acid and OH^- is the conjugate base, as shown in Figure 10.1.

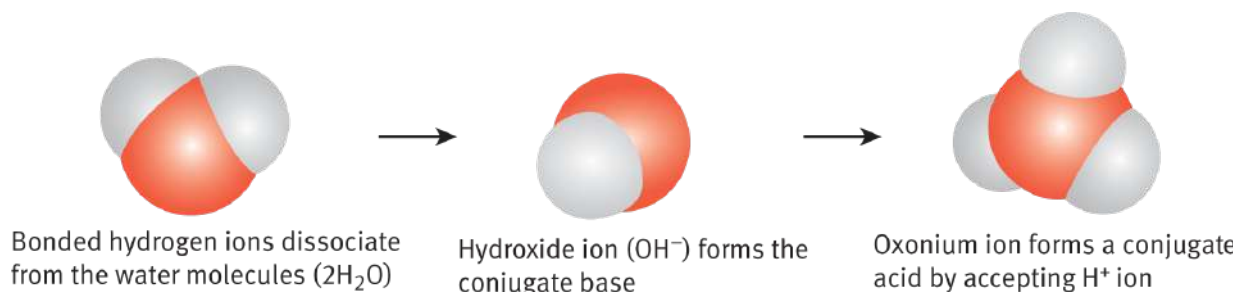


Figure 10.1. Autoionization of Water into Its Conjugate Acid and Conjugate Base

The hydroxide ion is the conjugate base; the oxonium (hydronium) ion is the conjugate acid. $\text{H}_2\text{O} (\text{l}) + \text{H}_2\text{O} (\text{l}) \rightleftharpoons \text{H}_3\text{O}^+ (\text{aq}) + \text{OH}^- (\text{aq})$

LEWIS

At approximately the same time as Brønsted and Lowry's publications, Gilbert Lewis also proposed a definition for acids and bases. A Lewis acid is defined as an electron pair acceptor, and a Lewis base is defined as an electron pair donor, as shown in Figure 10.2. The electron pair being donated is a lone pair and is not involved in any other bonds.

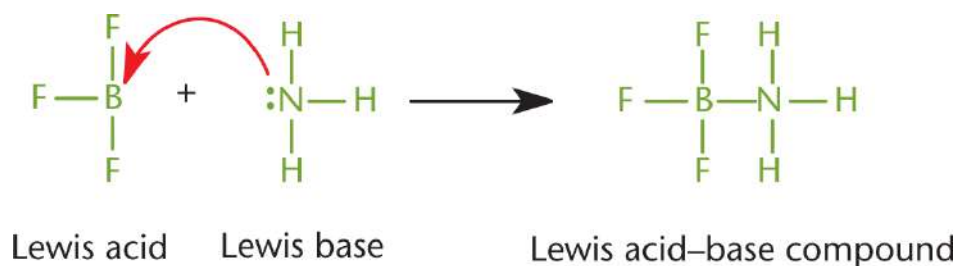


Figure 10.2. Lewis Acid–Base Chemistry

Boron trifluoride serves as the Lewis acid, accepting a lone pair. Ammonia serves as a Lewis base, donating a lone pair.



MNEMONIC

The Brønsted-Lowry definition revolves around protons; the Lewis definition around electrons.

On the MCAT, Lewis acid–base chemistry appears with many names. The underlying idea is that one species pushes a lone pair to form a bond with another. This same chemistry can be called coordinate covalent bond formation (discussed in Chapter 3 of MCAT General Chemistry Review), complex ion formation (discussed in Chapter 9 of MCAT General Chemistry Review), or nucleophile–electrophile interactions (discussed in Chapter 4 of MCAT Organic Chemistry Review).

There is an intuitive approach to understanding the differences in the definitions we have discussed so far. The Lewis definition relies on a behavior that is not vastly different from the Brønsted–Lowry interactions—the only difference is the focus. For Brønsted–Lowry acids and bases, we follow the exchange of the hydrogen ion (H^+), which is essentially a naked proton. In the Lewis definition, the focus of the reaction is no longer on the proton, but instead the electrons forming the coordinate covalent bond. This difference can be seen using curved arrows, as shown in Figure 10.3.

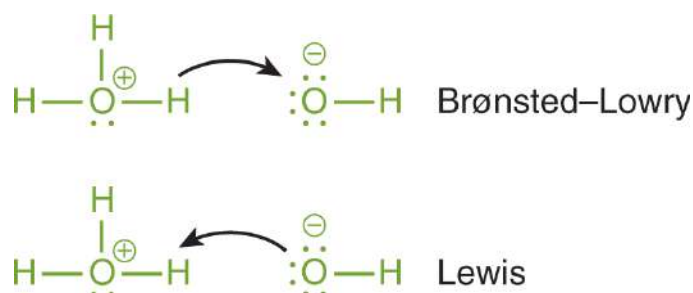


Figure 10.3. Comparison of Brønsted–Lowry and Lewis Definitions of Acids and Bases

In the Brønsted–Lowry definition, the focus is on the transfer of the proton.

In the Lewis definition, the focus is on the attack of the Lewis acid (electrophile) by the lone pair of the Lewis base (nucleophile).

Note that the Lewis definition is the most inclusive: every Arrhenius acid is also a Brønsted–Lowry acid, and every Brønsted–Lowry acid is also a Lewis acid (and likewise for bases). However, the converse is not necessarily true. The Lewis definition encompasses some species not included within the Brønsted–Lowry definition; for example, BF_3 and AlCl_3 are species that can each accept an electron pair, which qualifies them as Lewis acids, but they lack a hydrogen ion to donate, disqualifying them as both Arrhenius and Brønsted–Lowry acids.

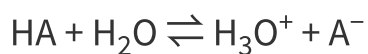
On the MCAT, you may encounter Lewis acids in the context of organic chemistry reactions because Lewis acids are often used as catalysts.

AMPHOTERIC SPECIES

An amphoteric species is one that reacts like an acid in a basic environment and like a base in an acidic environment. In the Brønsted–Lowry sense, an amphoteric species can either gain or lose a proton, making it amphiprotic as well. On the MCAT, water is the most common example. When water reacts with a base, it behaves as an acid:



When water reacts with an acid, it behaves as a base:



The partially dissociated conjugate base of a polyvalent acid is usually amphoteric. For example, HSO_4^- can either gain a proton to form H_2SO_4 or lose a proton to form SO_4^{2-} . The hydroxides of certain metals (such as Al, Zn, Pb, and Cr) are also amphoteric. Furthermore, species that can act as both oxidizing and reducing agents are often considered to be amphoteric as well because by accepting or donating electron pairs, they act as Lewis acids or bases, respectively.

Complex amphoteric molecules include amino acids that have a zwitterion intermediate with both cationic and anionic character, as shown in Figure 10.4. Such species are discussed in great detail in Chapter 1 of MCAT Biochemistry Review.



Figure 10.4. Amino Acid Zwitterions Are Complex Amphoteric Species
The amino group can release a proton (acid) and the carboxylate group can accept a proton (base).



KEY CONCEPT

Water, amino acids, and partially deprotonated polyprotic acids such as bicarbonate and bisulfate are common examples of amphoteric and amphiprotic substances. Metal oxides and hydroxides are also considered amphoteric but not necessarily amphiprotic because they do not give off protons.

ACID–BASE NOMENCLATURE

The names of most acids are related to the names of their parent anions (the anion that combines with H^+ to form the acid). Acids formed from anions with names that end in –ide have the prefix hydro– and the ending –ic.

F^-	Fluoride	HF	Hydrofluoric acid
Cl^-	Chloride	HCl	Hydrochloric acid
Br^-	Bromide	HBr	Hydrobromic acid

Acids formed from oxyanions are called oxyacids. If the anion ends in –ite (less oxygen), then the acid will end with –ous acid. If the anion ends in –ate (more oxygen), then the acid will end with –ic acid. Prefixes in the names of the anions are retained. Some common examples include the following:

ClO^-	Hypochlorite	HClO	Hypochlorous acid
ClO_2^-	Chlorite	HClO_2	Chlorous acid
ClO_3^-	Chlorate	HClO_3	Chloric acid
ClO_4^-	Perchlorate	HClO_4	Perchloric acid
NO_2^-	Nitrite	HNO_2	Nitrous acid
NO_3^-	Nitrate	HNO_3	Nitric acid
CO_3^{2-}	Carbonate	H_2CO_3	Carbonic acid

SO_4^{2-}	Sulfate	H_2SO_4	Sulfuric acid
PO_4^{3-}	Phosphate	H_3PO_4	Phosphoric acid
BO_3^{3-}	Borate	H_3BO_3	Boric acid
CrO_4^{2-}	Chromate	H_2CrO_4	Chromic acid
CH_3COO^-	Acetate	CH_3COOH	Acetic acid



MCAT EXPERTISE

There are some exceptions to the nomenclature rules. For instance, MnO_4^- is called permanganate even though there are no MnO_3^- or MnO_2^- ions.

MCAT Concept Check 10.1:

Before you move on, assess your understanding of the material with these questions.

Compare and contrast the three definitions for acids and bases:

Definition	Acid	Base
Arrhenius		

Definition	Acid	Base
Brønsted–Lowry		
Lewis		



Utilizing Arrhenius acid naming trends, predict the acid formula and name for the following anions:

Anion	Acid Formula	Acid Name
MnO_4^-		
Titanate (TiO_3^{2-})		
I^-		
IO_4^-		



Identify which reactants are amphoteric species in the following reactions. For those species, determine if the compound is also amphoteric.

Reaction	Amphoteric Reactant	Amphiprotic? (Y or N)
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Reaction	Amphoteric Reactant	Amphiprotic? (Y or N)
$\text{HCO}_3^- + \text{HBr} \rightleftharpoons \text{H}_2\text{CO}_3 + \text{Br}^-$		
$3 \text{HCl} + \text{Al}(\text{OH})_3 \rightleftharpoons \text{AlCl}_3 + 3 \text{H}_2\text{O}$		
$2 \text{HBr} + \text{ZnO} \rightleftharpoons \text{ZnBr}_2 + \text{H}_2\text{O}$		



10.2 Properties

High-Yield



LEARNING GOALS

After Chapter 10.2, you will be able to:

- Predict the behavior of an acid or base in water given its K_a or K_b value, respectively
- Apply the mathematical relationships between pH, pOH, and ion concentration
- Recall the mathematical relationship between K_a , K_b , and K_w
- Determine concentration of hydrogen ions given molarity and K_a or K_b of a solution
- Identify acids, bases, conjugate acids, and conjugate bases in a reaction: $\text{H}_2\text{O} (\text{l}) + \text{H}_2\text{O} (\text{l}) \rightleftharpoons \text{H}_3\text{O}^+ (\text{aq}) + \text{OH}^- (\text{aq})$

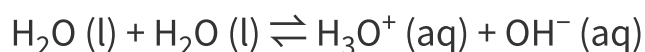
Acids and bases are usually characterized according to their relative tendencies to either donate or accept hydrogen ions. Furthermore, aqueous acid and base solutions can be characterized according to their concentrations of hydrogen and hydroxide ions.

AUTOIONIZATION OF WATER AND HYDROGEN ION EQUILIBRIA

Because many acid–base reactions take place in water—especially on the MCAT—it is very important to understand the behavior of acidic and basic compounds in water. Only then can one fully appreciate the meaning and significance of such terms as strong acid, weak base, or measurements of pH and pOH.

The Acid–Base Behavior of Water

As described above, water is an amphoteric species: in the presence of a base it reacts as an acid, and in the presence of an acid, it reacts as a base. As an amphoteric compound, water can react with itself in a process called autoionization, seen previously in Figure 10.1. The autoionization of water is represented by the equation:



One water molecule donates a hydrogen ion to another water molecule to produce the hydronium ion (H_3O^+) and the hydroxide ion (OH^-). Many general chemistry courses depict the hydrogen ion simply as H^+ , rather than as H_3O^+ .

This is acceptable for representing the chemistry, but it is important to remember that the proton is never isolated in the solution; it is always attached to water or some other species that has the ability to accept it. Autoionization of water is a reversible reaction; therefore, the expression

above is in equilibrium. For pure water at 298 K, the water dissociation constant, K_w , has been experimentally determined:

$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = 10^{-14} \text{ at } 25^\circ\text{C (298 K)}$$

Equation 10.1

Each mole of water that autoionizes produces one mole each of hydrogen (or hydronium) ions and hydroxide ions, so the concentrations of the hydrogen ions and hydroxide ions are always equal in pure water at equilibrium. Thus, the concentration of each of the ions in pure water at equilibrium at 298 K is 10^{-7} M.

However, the concentrations of the two ions will not always be equal. In fact, they will only be equal when the solution is neutral. Nevertheless, the product of their respective concentrations will always equal 10^{-14} when the temperature of the solution is 298 K. For example, if a species donates hydrogen ions to pure water, the hydrogen ion concentration will increase, causing the system to shift toward the reactants in the autoionization process. The result is a decrease in the hydroxide ion concentration and a return to the equilibrium state. This is Le Châtelier's principle in action: the addition of product to a system at equilibrium causes the system to shift away from the products and toward the reactants. The shift away from the products necessarily decreases the hydroxide ion concentration such that the product of the concentrations of the dissolved ions equals K_w . The addition of a species that accepts hydrogen ions results in a decrease in the hydrogen ion concentration and causes the system to shift toward the

products, thereby replacing hydrogen ions. This shift necessarily increases the hydroxide ion concentration and returns the system to equilibrium.

Before we introduce the scales used in measuring concentrations of hydrogen ions and hydroxide ions in different acid–base solutions, it is worthwhile to emphasize an important thermodynamic principle regarding the water dissociation constant (K_w) expression. K_w is an equilibrium constant; unless the temperature of the water is changed, the value for K_w cannot be changed. Thus, the product of the concentrations of the hydrogen ions and the hydroxide ions in an aqueous solution at 298 K must always equal 10^{-14} . However, at different temperatures, the value for K_w changes. At temperatures above 298 K, K_w will increase; this is a direct result of the endothermic nature of the autoionization reaction.



MCAT EXPERTISE

The MCAT loves to test this concept: the value of K_w , like any other equilibrium constant, is dependent only on temperature. Therefore, isolated changes in concentration, pressure, or volume will not affect K_w .

pH and pOH Scales

The concentrations of hydrogen ions and hydroxide ions in aqueous solutions can vary significantly, making the range of measurements on a linear scale unmanageable. The concentration scales for acidic and basic solutions are condensed into something more manageable through

expression in logarithmic terms. These logarithmic scales are the pH and the pOH scales for the concentrations of hydrogen and hydroxide ions, respectively.



BRIDGE

Logarithmic scales are used to condense very large absolute differences into small scale differences. Remember that sound level (dB) also uses a logarithmic scale, as discussed in Chapter 7 of MCAT Physics and Math Review.

Using a logarithmic scale is not only mathematically convenient, but also useful for calculations. For instance, the reactivity of an acidic solution is not a function of hydrogen ion concentration but instead of the logarithm of the hydrogen ion concentration. pH and pOH are prototypical examples of p scales. A p scale is defined as the negative logarithm of the number of items.

The pH and pOH of a solution are given by:

$$\text{pH} = -\log[\text{H}^+] = \log \frac{1}{[\text{H}^+]}$$
$$\text{pOH} = -\log[\text{OH}^-] = \log \frac{1}{[\text{OH}^-]}$$

Equation 10.2

For pure water at equilibrium and 298 K, the concentration of hydrogen ions equals the concentration of hydroxide ions (10^{-7} M). Therefore, pure water at 298 K has a pH of 7 and a pOH of 7 ($-\log 10^{-7} = 7$). If we take the negative logarithm of the entire water dissociation constant expression ($[\text{H}_3\text{O}^+][\text{OH}^-] = 10^{-14}$), we find that:

$$\text{pH} + \text{pOH} = 14$$

Equation 10.3

for all aqueous solutions at 298 K. As pH increases, pOH decreases by the same amount. This relationship can be seen in Figure 10.5 below.

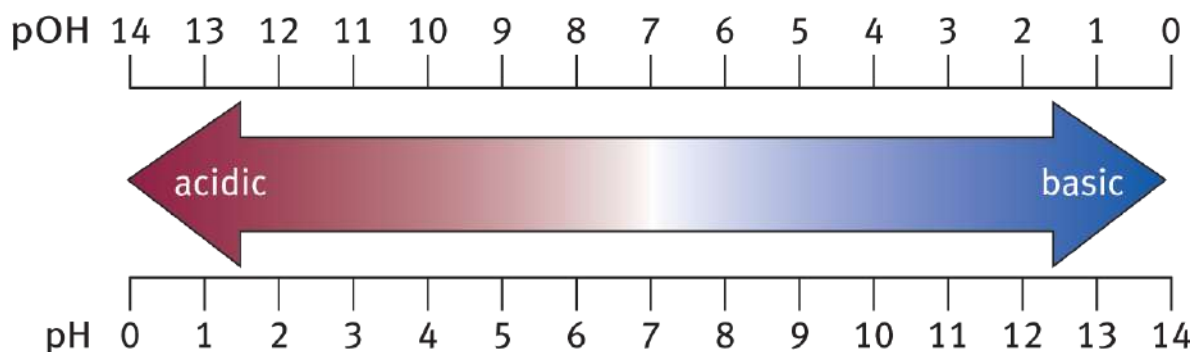


Figure 10.5. pH and pOH Scales

$\text{pH} + \text{pOH} = 14$ for aqueous solutions at 298 K.



MCAT EXPERTISE

The K_w (like all equilibrium constants) will change if the temperature changes and, in turn, will change the significance of the pH scale. Be careful and read the system conditions given on the MCAT: $\text{pH} = 7 = \text{neutral}$ is only valid at 25°C .



BRIDGE

In general, math using logarithms frequently appears on the MCAT; make sure to review how these mathematical concepts work in Chapter 10 of MCAT Physics and Math Review. Specifically, the equation $\text{pH} + \text{pOH} = 14$ comes from the fact that the log of a product is equal to the sum of logs; that is, $\log(xy) = \log x + \log y$.

For an aqueous solution at 298 K, a pH less than 7 (or pOH greater than 7) indicates a relative excess of hydrogen ions, and the solution is acidic; a pH greater than 7 (or pOH less than 7) indicates a relative excess of hydroxide ions, and the solution is basic. A pH (and pOH) equal to 7 indicates equal concentrations of hydrogen and hydroxide ions, resulting in a neutral solution.

Estimating Scale Values

An essential skill to hone for Test Day is the ability to quickly convert pH, pOH, pK_a , and pK_b values into nonlogarithmic form and vice-versa.

When the original value is a power of ten, the operation is relatively straightforward: changing the sign on the exponent gives the corresponding p scale value directly. For example, if $[H^+] = 0.001$ or 10^{-3} , then the $pH = 3$ and $pOH = 11$. Or, if $K_b = 1.0 \times 10^{-12}$, then $pK_b = 12$.

More difficulty arises when the value is not an exact power of ten. Rest assured that the MCAT is not a math test and is not interested in determining your ability to perform complex logarithmic calculations; an exact logarithmic calculation of a number that is not an integer power of ten is unnecessary on the MCAT. The testmakers are interested, however, in testing the ability to apply mathematical concepts appropriately in solving certain problems.

One can obtain a relatively close approximation of a p scale value using the following shortcut: if the nonlogarithmic value is written in proper scientific notation, it will be in the form $n \times 10^{-m}$, where n is a number between 1 and 10. Taking the negative logarithm and simplifying, the p value will be:

$$\begin{aligned} -\log(n \times 10^{-m}) &= -\log(n) - \log(10^{-m}) \\ &= m - \log(n) \end{aligned}$$

Because n is a number between 1 and 10, its logarithm will be a decimal between 0 and 1 ($\log 1 = 0$ and $\log 10 = 1$). The closer n is to 1, the closer $\log n$ will be to 0; the closer n is to 10, the closer $\log n$ will be to 1. As a reasonable approximation, one can say that:

$$\text{p value} \approx m - 0.n$$

Equation 10.4

where 0.n represents sliding the decimal point of n one position to the left (dividing n by ten).

Example:

If the K_a of an acid is 1.8×10^{-5} , then what is its $\text{p}K_a$?

Solution:

$$\begin{aligned}\text{p}K_a &= -\log(1.8 \times 10^{-5}) = 5 - \log 1.8 \approx 5 - 0.18 \\ &= 4.82 \text{ (actual} = 4.74\text{)}\end{aligned}$$



MCAT EXPERTISE

Learning how to estimate when using logarithms is an important skill that can save a lot of time on Test Day.

STRONG ACIDS AND BASES

Strong acids and bases completely dissociate into their component ions in aqueous solutions. For example, when sodium hydroxide is added to water, the ionic compound dissociates according to the net ionic equation:



Hence, in a 1 M NaOH solution, complete dissociation yields 1 M Na^+ and 1 M OH^- . The pH and pOH for this solution can be calculated as follows:

$$\text{pH} = 14 - \text{pOH} = 14 - (-\log[\text{OH}^-]) = 14 + \log(1 \text{ M}) = 14 + 0 = 14$$



MCAT EXPERTISE

Acid–base reactions that consist of a single-headed arrow generally indicate strong acids or bases (complete dissociation with no reversibility).

Virtually no undissociated strong acid or base, such as NaOH, will remain in solution. This is why the dissociation of strong acids and bases is said to go to completion. In the NaOH example above, we assume that the concentration of OH^- from the autoionization of water is negligible due to addition of a strong base. The contribution of OH^- and H^+ ions from the autoionization of water is negligible if the concentration of the acid or base is significantly greater than 10^{-7} M. On the other hand, if the concentration of acid or base is close to 10^{-7} M, then the contribution from the autoionization of water is important.

Example:

Calculate the pH of a 1×10^{-8} M solution of HCl.

Solution:

At first, one may calculate the pH as $-\log [\text{H}^+] = -\log 10^{-8} \text{ M} = 8$. However, this answer is not feasible: a pH of 8 cannot describe an acidic solution at 298 K because the presence of the acid will increase the hydrogen ion concentration to above 10^{-7} M, resulting in an acidic pH below 7.

Recognize that the acid concentration in this question is actually ten times less than the equilibrium concentration of hydrogen ions in pure water generated by the autoionization of water. Consequently, the hydrogen ion concentration from the water itself is significant and cannot be ignored. This can be represented in the equilibrium expression in which x represents the concentration of H_3O^+ and OH^- resulting from the autoionization of water:

$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = [x + 10^{-8}][x] = 10^{-14}$$

Solving for x (which would require a quadratic equation—math that is beyond the scope of the MCAT) gives $x = 9.5 \times 10^{-8}$ M. The total concentration of hydrogen ions is $[\text{H}^+]_{\text{total}} = (9.5 \times 10^{-8}) + (1.0 \times 10^{-8}) = 1.05 \times 10^{-7}$ M. Notice that this is extremely close to the concentration of H^+ in pure water. The pH of this acidic solution can now be calculated as $\text{pH} = -\log(1.05 \times 10^{-7}) \approx 7$ (actual = 6.98). This

pH is slightly lower than 7, as expected for a very dilute acidic solution. The point of all of this is: stay alert and keep thinking critically, no matter how familiar the problem setups might seem to you!

Strong acids commonly encountered on the MCAT include HCl (hydrochloric acid), HBr (hydrobromic acid), HI (hydroiodic acid), H_2SO_4 (sulfuric acid), HNO_3 (nitric acid), and HClO_4 (perchloric acid). Strong bases commonly encountered include NaOH (sodium hydroxide), KOH (potassium hydroxide), and other soluble hydroxides of Group IA metals. Calculation of the pH and pOH of strong acids and bases assumes complete dissociation of the acid or base in solution.

Example:

What is the pH of a solution with $[\text{HClO}_4] = 10 \text{ M}$?

Solution:

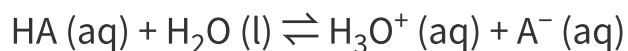
Because perchloric acid is a strong acid, it will fully dissociate in solution. Therefore, $[\text{H}^+] = 10 \text{ M}$ (note that the contribution from the autoionization of water is negligible). $\text{pH} = -\log [\text{H}^+] = -\log 10 \text{ M} = -1$.

This question points out that the pH scale does not “end” at 0 and 14. There can be negative pH values and pH values greater than 14—but this implies a very high concentration of a strong acid or base.

WEAK ACIDS AND BASES

Before going any further in our discussion of acids and bases as strong or weak, verify that you are making the distinction between the chemical behavior of an acid or base with respect to its tendency to dissociate (that is, strong bases completely dissociate in aqueous solutions) and the concentrations of acid and base solutions. Although we may casually describe a solution's concentration as strong or weak, it is preferable to use the terms concentrated and dilute, respectively, because they are unambiguously associated with concentrations, rather than chemical behavior.

Continuing our focus on the chemical behavior of acids and bases, we will now consider those acids and bases that only partially dissociate in aqueous solutions. These are called weak acids and bases. A weak monoprotic acid, HA, will dissociate partially in water to achieve an equilibrium state:



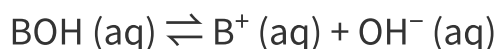
Because the system exists in an equilibrium state, we can write the dissociation equation to determine the acid dissociation constant (K_a) as:

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}$$

Equation 10.5

The smaller the K_a , the weaker the acid, and consequently, the less it will dissociate. Note that water, as a pure liquid, is not incorporated into the equilibrium expression.

A weak monovalent Arrhenius base, BOH, undergoes dissociation to yield B^+ and OH^- in solution:



The base dissociation constant (K_b) can be calculated as:

$$K_b = \frac{[B^+][OH^-]}{[BOH]}$$

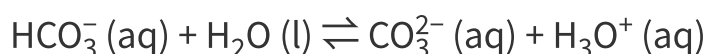
Equation 10.6

The smaller the K_b , the weaker the base, and consequently, the less it will dissociate. As with the acid dissociation expression, water is not included because it is a pure liquid.

Generally speaking, we can characterize a species as a weak acid if its K_a is less than 1.0 and as a weak base if its K_b is less than 1.0. On the MCAT, molecular (nonionic) weak bases are almost exclusively amines.

CONJUGATE ACID–BASE PAIRS

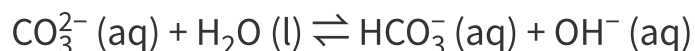
Because the Brønsted–Lowry definition of an acid–base reaction is one in which a hydrogen ion (proton) is transferred from an acid to a base, the two always occur in pairs called conjugates. A conjugate acid is the acid formed when a base gains a proton, and a conjugate base is the base formed when an acid loses a proton. For example,



CO_3^{2-} is the conjugate base of HCO_3^- , a weak acid, and H_3O^+ is the conjugate acid of H_2O , a weak base. To find the K_a , we consider the equilibrium concentrations of the dissolved species:

$$K_a = \frac{[\text{CO}_3^{2-}][\text{H}_3\text{O}^+]}{[\text{HCO}_3^-]}$$

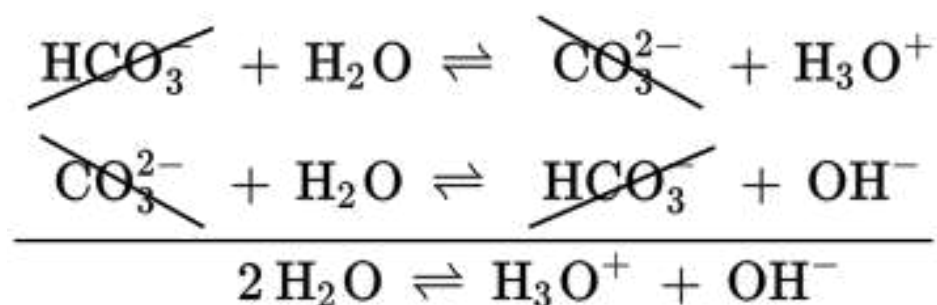
The reaction between bicarbonate and water is reversible. The reverse reaction would be:



We can write the K_b for CO_3^{2-} as:

$$K_b = \frac{[\text{HCO}_3^-][\text{OH}^-]}{[\text{CO}_3^{2-}]}$$

If one adds the previous two reversible reactions, the net reaction is simply the dissociation of water:



Because the net reaction is the autoionization of water, the equilibrium constant for the reaction is $K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = 10^{-14}$, which is the product of K_a and K_b . Remember: the product of the concentrations of the hydrogen ion and the hydroxide ion must always equal 10^{-14} for acidic or basic aqueous solutions. Because water is an amphoteric species (both a weak acid and a weak base), all acid–base reactivity in water ultimately reduces to the acid–base behavior of water, and all acidic or basic aqueous solutions are governed by the dissociation constant for water. Thus, if the dissociation constant for one species or its conjugate is known, then the dissociation constant for the other can be determined using the following equations:

$$K_{a,\text{acid}} \times K_{b,\text{conjugate base}} = K_w = 10^{-14}$$

$$K_{b,\text{base}} \times K_{a,\text{conjugate acid}} = K_w = 10^{-14}$$

Equation 10.7

As is evident from these equations, K_a and K_b are inversely related. In other words, if K_a is large, then K_b is small, and vice-versa. By this logic, a strong acid (K_a approaching ∞) will produce a very weak conjugate base (for example, HCl is a strong acid and Cl^- is a very weak base), and a strong base will produce a very weak conjugate acid (for example, NaOH is a strong base and H_2O is a very weak acid). The conjugate of a strong acid or base is sometimes termed inert because it is almost completely unreactive.



KEY CONCEPT

Be aware of the relationship between conjugate acids and bases because you will need to recognize these entities on the MCAT. Removing a proton from a molecule produces the conjugate base, and adding a proton produces the conjugate acid.

On the other hand, weak acids and bases tend to have conjugates that are also weak. As seen above, CO_3^{2-} is a weak base; its conjugate acid, HCO_3^- is a weak acid. As it turns out, for this specific example, the reaction of CO_3^{2-} with water to produce HCO_3^- and OH^- occurs to a greater extent—is more

thermodynamically favorable—than the reaction of HCO_3^- and water to produce CO_3^{2-} and H_3O^+ . This fact makes this equilibrium ideal for buffering solutions as part of the bicarbonate buffer system, discussed in Chapter 6 of MCAT Biology Review.



BRIDGE

This thermodynamic preference for the bicarbonate ion intermediate is a major reason why the bicarbonate buffer system in the body is ideal for maintaining a stable pH. The homeostatic mechanisms involved are discussed in Chapter 6 of MCAT Biology Review.

One important theme for acid strength is the effect of induction. Electronegative elements positioned near an acidic proton increase acid strength by pulling electron density out of the bond holding the acidic proton. This weakens proton bonding and facilitates dissociation. Thus, acids that have electronegative elements nearer to acidic hydrogens are stronger than those that do not, as shown in Figure 10.6.

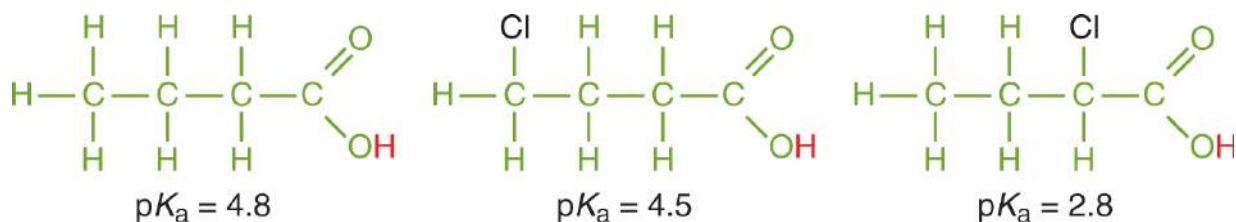


Figure 10.6. Inductive Effects from Electronegative Elements Increase Acidity

APPLICATIONS OF K_a AND K_b

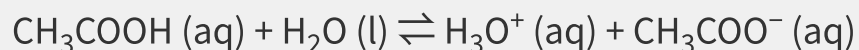
The most common use of acid and base dissociation constants is to determine the concentration of one of the species in solution at equilibrium. On Test Day, you may be asked to calculate the concentration of the hydrogen ion (or pH), the concentration of the hydroxide ion (or pOH), or the concentration of either the original acid or base.

Example:

Calculate the concentration of H_3O^+ in a 2.0 M aqueous solution of acetic acid, CH_3COOH . (Note: $K_a = 1.8 \times 10^{-5}$)

Solution:

First, write the equilibrium reaction:



Next, write the expression for the acid dissociation constant:

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]} = 1.8 \times 10^{-5}$$

Then, recognize that acetic acid is a weak acid, so the concentration of CH_3COOH at equilibrium is equal to its initial concentration, 2.0

M, minus the amount dissociated, x . Likewise, $[\text{H}_3\text{O}^+] = [\text{CH}_3\text{COO}^-] = x$ because each molecule of CH_3COOH dissociates into one H^+ ion and one CH_3COO^- ion. Note that the contribution of H_3O^+ from water is negligible. Thus, the equation can be rewritten as follows:

$$K_a = \frac{[x][x]}{[2.0\text{ M} - x]} = 1.8 \times 10^{-5}$$

Remember that the value of x is generally very small. Therefore, we can approximate that $2.0\text{ M} - x \approx 2.0\text{ M}$. This is further supported because acetic acid is a weak acid and only slightly dissociates in water. This simplifies the calculations:

$$\begin{aligned} K_a &= \frac{[x][x]}{2.0\text{ M}} = 1.8 \times 10^{-5} \\ x^2 &= 3.6 \times 10^{-5} = 36 \times 10^{-6} \\ x &= 6 \times 10^{-3}\text{ M} \end{aligned}$$

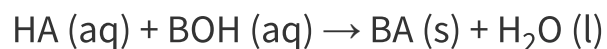
x represents the concentration of H_3O^+ ; therefore, $[\text{H}_3\text{O}^+] = 6 \times 10^{-3}\text{ M}$. Note: When needing to take the square root adjust the coefficient as needed to make the power of 10 an even number. This way the square root only requires cutting the power of 10 in half.

In this example, note that x is significantly lower than the initial concentration of acetic acid (2.0 M), which validates the approximation; otherwise, it would have been necessary to solve for x using the quadratic formula. Fortunately, the value of x on Test Day is almost always sufficiently small to make this approximation. A rule of thumb is that the approximation is valid as long as x is less than 5 percent of the initial concentration. This typically occurs when K_a is at least 100 times smaller than the concentration of the starting solution. For example, if K_a is 10^{-4} and the concentration of the starting solution is 0.01 M (10^{-2} M), then the ratio between the values is 10^2 or 100. The error in this calculation should be no more than $\frac{1}{100} = 1\%$. On the other hand, if the K_a is 10^{-3} and the concentration is still 0.01 M, then the ratio between the values becomes 10, which could lead to $\frac{1}{10} = 10\%$ error. This degree of error may not be useful when identifying an answer choice on the MCAT.

Students often feel nervous making the assumption that x is negligible because they want to see precise answer choices. However, keep in mind that the MCAT quite deliberately tests students' ability to make reasonable assumptions under timed conditions to arrive at a feasible answer choice. Part of the skill of taking the MCAT is rounding appropriately to simplify math.

SALT FORMATION

Acids and bases may react with each other to form a salt and often (but not always) water, in what is termed a neutralization reaction. For example,



BRIDGE

Remember the reaction types discussed in Chapter 4 of MCAT General Chemistry Review? Go back and review the section on neutralization reactions if this equation doesn't look familiar to you.

The salt may precipitate out or remain ionized in solution, depending on its solubility and the amount produced. In general, neutralization reactions go to completion. The reverse reaction, in which the salt ions react with water to give back the acid or base, is known as hydrolysis.

Four combinations of strong and weak acids and bases are possible:

- Strong acid + strong base: $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$
- Strong acid + weak base: $\text{HCl} + \text{NH}_3 \rightarrow \text{NH}_4\text{Cl}$
- Weak acid + strong base: $\text{HClO} + \text{NaOH} \rightarrow \text{NaClO} + \text{H}_2\text{O}$
- Weak acid + weak base: $\text{HClO} + \text{NH}_3 \rightarrow \text{NH}_4\text{ClO}$

The products of a reaction between equal concentrations of a strong acid and a strong base are equimolar amounts of salt and water. The acid and base neutralize each other, so the resulting solution is neutral ($\text{pH} = 7$), and the ions formed in the reaction will not react with water because they are inert conjugates.

The product of a reaction between a strong acid and a weak base is also a salt, but often no water will be formed because weak bases are often not hydroxides. In this case, the cation of the salt is a weak acid and will react with the water solvent, re-forming some of the weak base through hydrolysis. For example:

- Reaction I: $\text{HCl (aq)} + \text{NH}_3 \text{ (aq)} \rightarrow \text{NH}_4^+ \text{ (aq)} + \text{Cl}^- \text{ (aq)}$
- Reaction II: $\text{NH}_4^+ \text{ (aq)} + \text{H}_2\text{O (l)} \rightarrow \text{NH}_3 \text{ (aq)} + \text{H}_3\text{O}^+ \text{ (aq)}$

NH_4^+ is the conjugate acid of a weak base (NH_3) and is stronger than the conjugate base (Cl^-) of the strong acid, HCl . NH_4^+ will then transfer a proton to H_2O to form the hydronium ion. The increase in the concentration of the hydronium ion causes the system to shift away from autoionization, thereby reducing the concentration of hydroxide ion. Consequently, the concentration of the hydronium ion will be greater than that of the hydroxide ion at equilibrium, and as a result, the pH of the solution will fall below 7. This should make sense: a strong acid and a weak base produce a slightly acidic solution.

On the other hand, when a weak acid reacts with a strong base, the pH of the solution at equilibrium will be within the basic range because the salt hydrolyzes, with concurrent formation of hydroxide ions. The increase in hydroxide ion concentration will cause the system to shift away from autoionization, thereby reducing the concentration of the hydronium ion. Consequently, the concentration of the hydroxide ion will be greater than that of the hydronium ion at equilibrium, and as a result, the pH of the solution will rise above 7. Consider the reaction of acetic acid, CH_3COOH (weak acid) with sodium hydroxide, NaOH (strong base):

- Reaction I: $\text{CH}_3\text{COOH (aq)} + \text{NaOH (aq)} \rightarrow \text{Na}^+ \text{(aq)} + \text{CH}_3\text{COO}^- \text{(aq)} + \text{H}_2\text{O (l)}$
- Reaction II: $\text{CH}_3\text{COO}^- \text{(aq)} + \text{H}_2\text{O (l)} \rightarrow \text{CH}_3\text{COOH (aq)} + \text{OH}^- \text{(aq)}$

The pH of a solution containing a weak acid and a weak base depends on the relative strengths of the reactants. For example, the weak acid HClO has a K_a of 3.2×10^{-8} , and the weak base NH_3 has a $K_b = 1.8 \times 10^{-5}$. Thus, an aqueous solution of HClO and NH_3 is basic because the K_a for HClO is less than the K_b for NH_3 . That is, HClO is weaker as an acid than NH_3 is as a base. At equilibrium, therefore, the concentration of hydroxide ions will be greater than the concentration of hydronium ions in the aqueous solution.

In biology and biochemistry, neutralization reactions are often condensation reactions because they form bonds with a small molecule as a byproduct (usually water). The peptide bonds in proteins, for example, are created from the reaction of a carboxyl group (acid) and an amino group (base), while forming a water molecule, as shown in Figure 10.7. The salt in this reaction is the polypeptide itself; breaking it apart requires hydrolysis.

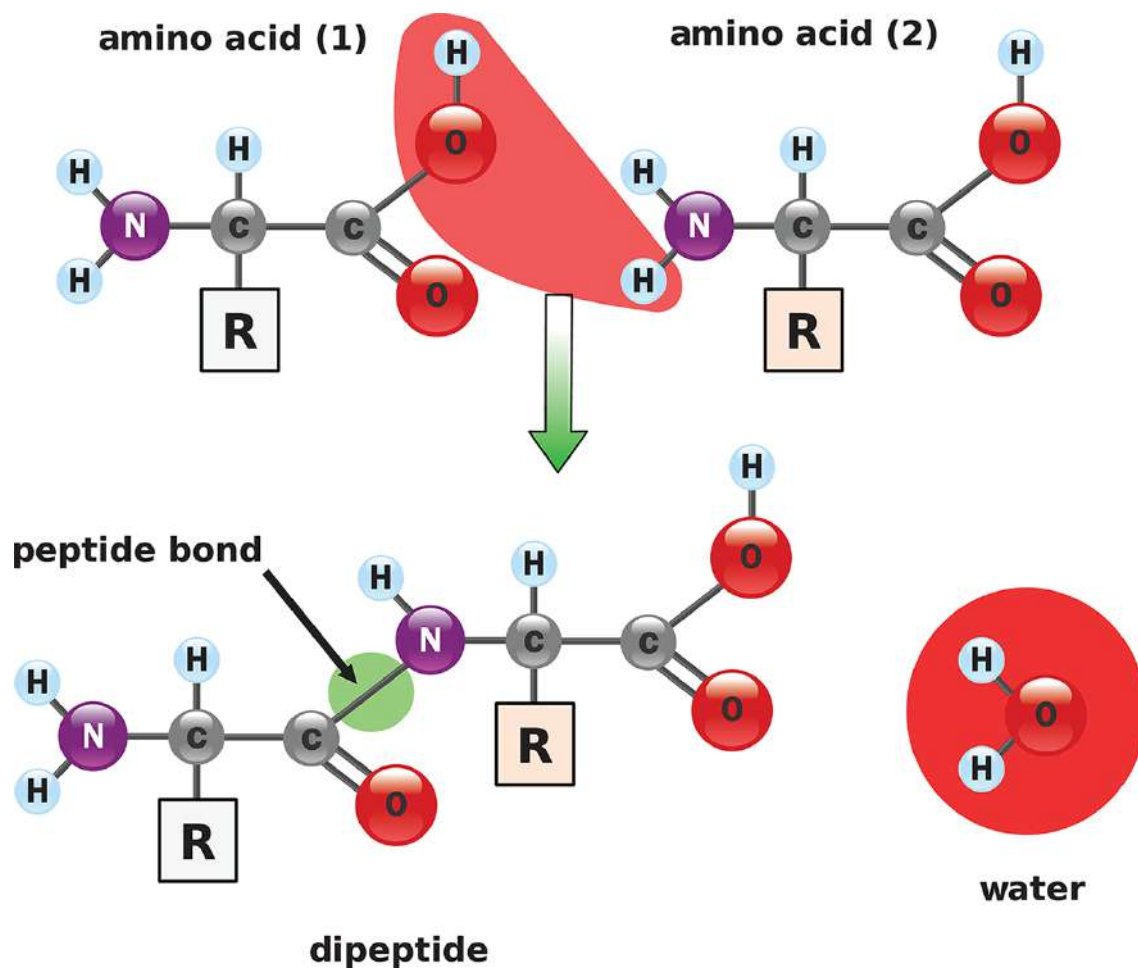


Figure 10.7. Peptide Bond Formation

An acidic carboxyl group reacts with a basic amino group.

MCAT Concept Check 10.2:

Before you move on, assess your understanding of the material with these questions.

What is an amphoteric species?

If a compound has a K_a value » water, what does it mean about its behavior in solution? How does this compare with a solution that has only a slightly higher K_a than water?

If a compound has a K_b value » water, what does it mean about its behavior in solution? How does this compare with a solution that has only a slightly higher K_b than water?

Complete the blank cells in the table by utilizing the mathematical relationships between pH, pOH, and ion concentrations. (Note: Round the numbers given and use logarithmic approximations to determine p values, without a calculator, to simulate Test Day math.)

pH	$[H_3O^+]$	pOH	$[OH^-]$	Acid or Base?
4				
	$8.89 \times 10^{-4} \text{ M}$			

pH	[H ₃ O ⁺]	pOH	[OH ⁻]	Acid or Base?
		5.19		
			$1.88 \times 10^{-6} \text{ M}$	



What is the mathematical relationship between K_a , K_b , and K_w ?

Identify the conjugate acid–base pairings in the reactions below:

Reaction	Acid	Base	Conjugate Acid	Conjugate Base
$\text{H}_2\text{CO}_3 + \text{H}_2\text{O} \rightleftharpoons \text{HCO}_3^- + \text{H}_3\text{O}^+$				
$\text{H}_2\text{PO}_4^- + \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{PO}_4 + \text{OH}^-$				



Determine the concentration of hydrogen ions and pH of a solution of 0.2 *M* acetic acid ($K_a = 1.8 \times 10^{-5}$).

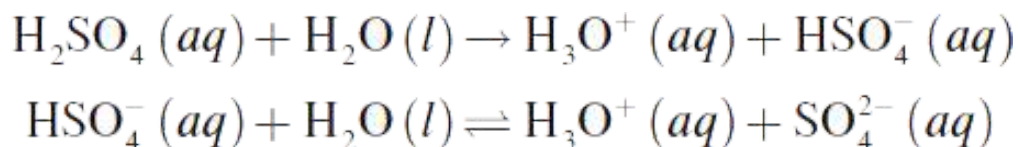
10.3 Polyvalence and Normality

LEARNING GOALS

After Chapter 10.3, you will be able to:

- Describe how equivalents of acid or base are calculated
- Calculate the normality of a solution given its formula and molarity

The relative acidity or basicity of an aqueous solution is determined by the relative concentrations of acid and base equivalents. An acid equivalent is equal to one mole of H^+ (or, more properly, H_3O^+) ions; a base equivalent is equal to one mole of OH^- ions. Some acids and bases are polyvalent; that is, each mole of the acid or base liberates more than one acid or base equivalent. Under the Brønsted–Lowry definition, such acids or bases could also be termed polyprotic. For example, the divalent diprotic acid H_2SO_4 undergoes the following dissociation in water:



One mole of H_2SO_4 produces two acid equivalents (2 moles of H_3O^+). Notice that the first dissociation goes to completion, but the second dissociation reaches an equilibrium state. The acidity or basicity of a solution depends on the concentration of acidic or basic equivalents that can be liberated. The quantity of acidic or basic capacity is directly indicated by the solution's normality, described in Chapter 9 of MCAT General Chemistry Review. For example, each mole of H_3PO_4 yields three moles (equivalents) of H_3O^+ . Therefore, a 2 M H_3PO_4 solution would be 6 N.

Another measurement useful for acid–base chemistry is gram equivalent weight. Chapter 4 of MCAT General Chemistry Review defined and discussed this term extensively. The gram equivalent weight is the mass of a compound that produces one equivalent (one mole of charge). For example, H_2SO_4 (molar mass: $98 \frac{\text{g}}{\text{mol}}$) is a divalent acid, so each mole of the acid compound yields two acid equivalents. The gram equivalent weight is $98 \div 2 = 49$ grams. That is, the complete dissociation of 49 grams of H_2SO_4 will yield one acid equivalent (one mole of H_3O^+). Common polyvalent acids include H_2SO_4 , H_3PO_4 , and H_2CO_3 . Common polyvalent bases include $\text{Al}(\text{OH})_3$, $\text{Ca}(\text{OH})_2$, and $\text{Mg}(\text{OH})_2$.



BRIDGE

To review normality in more detail, revisit the calculations performed in Chapter 4 of MCAT General Chemistry Review. These are critical calculations for polyvalent acids and bases.

MCAT Concept Check 10.3:

Before you move on, assess your understanding of the material with these questions.

What species are considered the equivalents for acids and bases, respectively?

- Acids:

- Bases:

Calculate the normality of the following solutions:

- 2 M Al(OH)_3 :

- $16\text{ M H}_2\text{SO}_4$:

10.4 Titration and Buffers

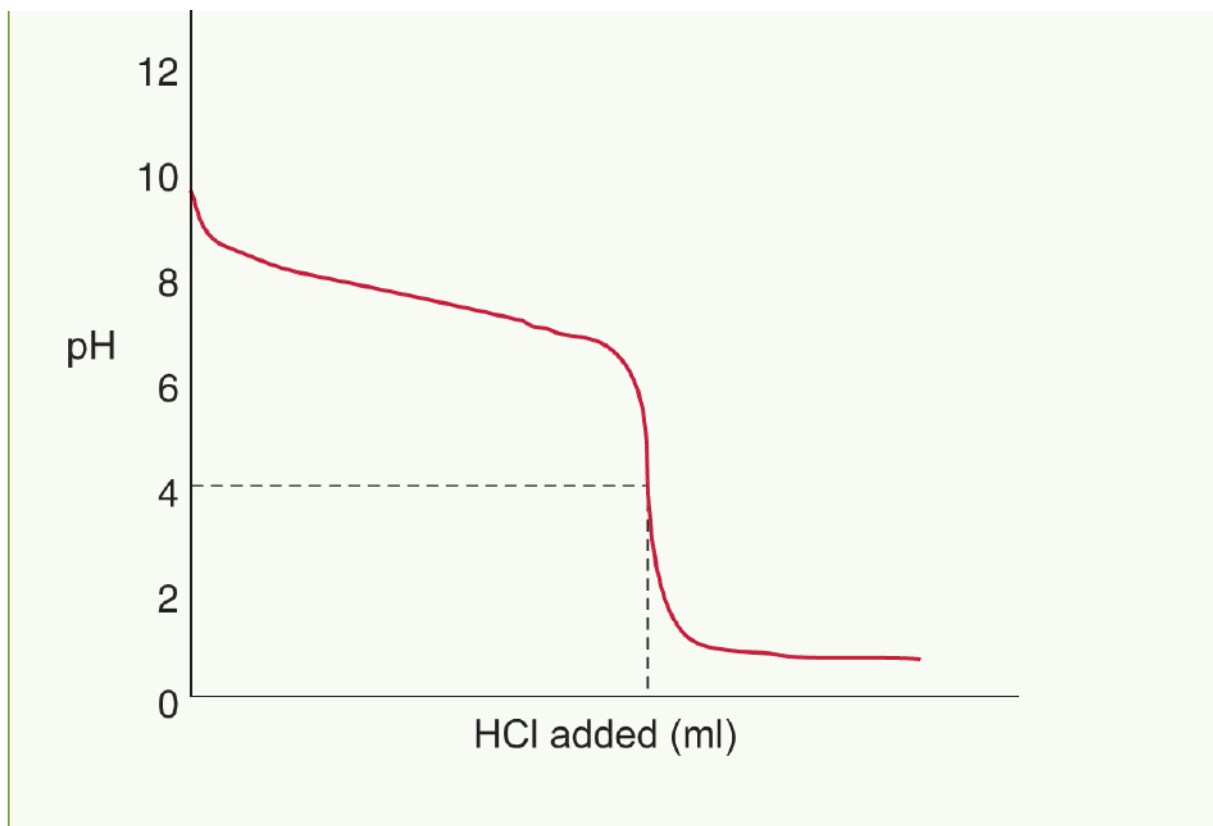
High-Yield



LEARNING GOALS

After Chapter 10.4, you will be able to:

- Select an appropriate indicator for a given acid–base reaction
- Explain the purpose of a buffer solution
- Identify the pH range of the equivalence point for different combinations of acids and bases, for example, weak acid + weak base
- Calculate the pH or pOH of a known solution
- Identify the buffering region, half-equivalence point, equivalence point, and endpoint of a titration reaction:



Titration is a procedure used to determine the concentration of a known reactant in a solution. There are different types of titrations, including acid–base, oxidation–reduction, and complexometric (metal ion). The MCAT frequently tests the first two types. Complexometric (metal ion) titrations are outside the scope of the MCAT but focus on formation of complex ions, as described in Chapter 9 of MCAT General Chemistry Review.

GENERAL PRINCIPLES

Titration is performed by adding small volumes of a solution of known concentration (the titrant) to a known volume of a solution of unknown

concentration (the titrand) until completion of the reaction is achieved at the equivalence point.

Acid–Base Equivalence Points

In acid–base titrations, the equivalence point is reached when the number of acid equivalents present in the original solution equals the number of base equivalents added, or vice-versa. It is important to emphasize that, while a strong acid/strong base titration will have its equivalence point at a pH of 7, the equivalence point does not always occur at pH 7. When titrating polyprotic acids or bases (discussed later in this chapter), there are multiple equivalence points, as each acidic or basic conjugate species is titrated separately. This is shown in the speciation plot in Figure 10.8.

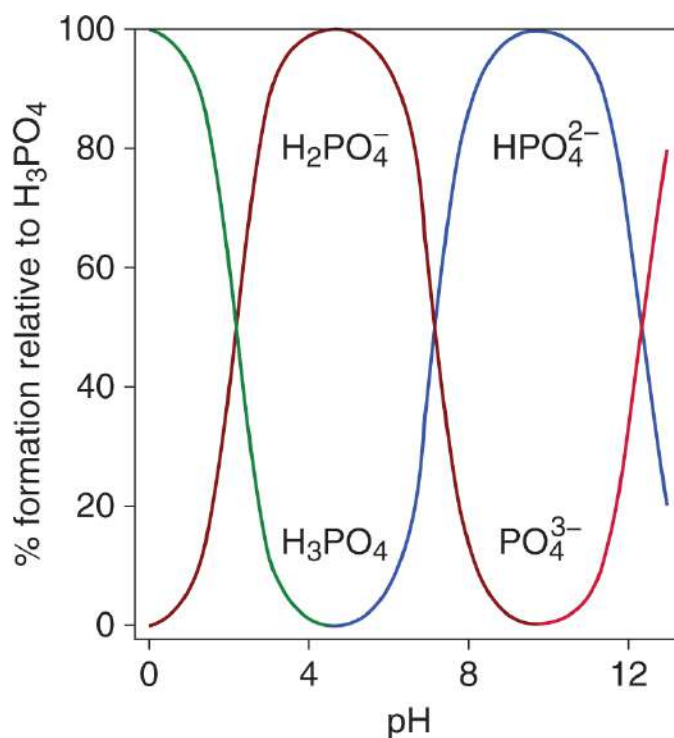


Figure 10.8. Speciation Plot of Phosphoric Acid

At any given pH, only two forms of the acid exist in solution; thus, each conjugate is titrated separately.

At the equivalence point, the number of equivalents of acid and base are equal. This fact allows us to calculate the unknown concentration of the titrand through the equation:

$$N_a V_a = N_b V_b$$

Equation 10.8

where N_a and N_b are the acid and base normalities, respectively, and V_a and V_b are the volumes of acid and base solutions, respectively. Note that,

as long as both volumes use the same units, the units used do not have to be liters.

The equivalence point in an acid–base titration is determined in two common ways: evaluated by using a graphical method, plotting the pH of the unknown solution as a function of added titrant by using a pH meter, or estimated by watching for a color change of an added indicator.

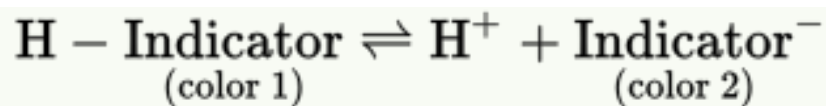
Indicators

Indicators are weak organic acids or bases that have different colors in their protonated and deprotonated states. This small structural change—the binding or release of a proton—leads to a change in the absorption spectrum of the molecule, which we perceive as a color change. Indicators are generally vibrant and can be used in low concentrations without significantly altering the equivalence point. The indicator must always be a weaker acid or base than the acid or base being titrated; otherwise, the indicator would be titrated first! The point at which the indicator changes to its final color is not the equivalence point but rather the endpoint. If the indicator is chosen correctly and the titration is performed well, the volume difference between the endpoint and the equivalence point is negligible and may be corrected for or simply ignored.



KEY CONCEPT

Indicators change color as they shift between their conjugate acid and base forms:



Because this is an equilibrium process, we can apply Le Châtelier's principle. Adding H^+ shifts the equilibrium to the left. Adding OH^- removes H^+ and therefore shifts the equilibrium to the right.

Acid–base titrations can be performed for different combinations of strong and weak acids and bases. The most useful combinations involve at least one strong species. Weak acid/weak base titrations can be done but are not very accurate and therefore are rarely performed. The pH curve for the titration of a weak acid and weak base lacks the sharp change that normally indicates the equivalence point. Furthermore, indicators are less useful because the pH change is far more gradual.



MCAT EXPERTISE

Any question involving the selection of an ideal indicator will require you to know what the pH of the reaction at the equivalence point will be, whether graphically or mathematically. Once you have determined where the equivalence point is, select the indicator that has the closest pK_a value to it.

STRONG ACID AND STRONG BASE

Let's consider the titration of 10 mL of a 0.1 N solution of HCl with a 0.1 N solution of NaOH. Plotting the pH of the solution vs. the quantity of NaOH added gives the curve shown in Figure 10.9.

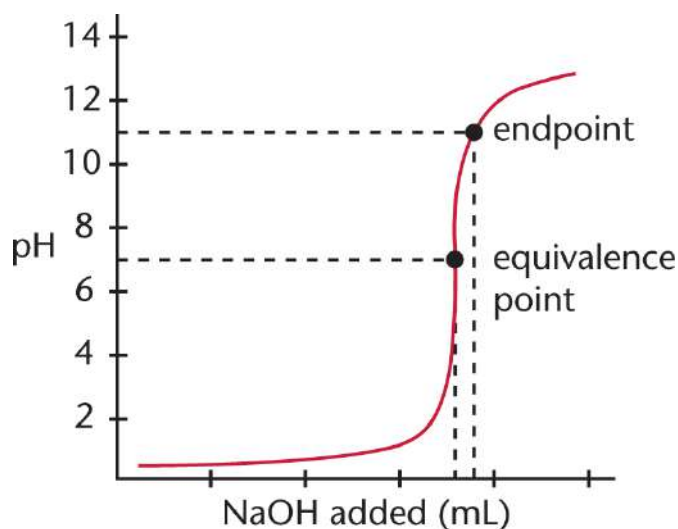


Figure 10.9. Monoprotic Strong Acid and Strong Base Titration Curve
A strong base, NaOH, is titrated into a solution of strong acid, HCl, to yield an equivalence point of $\text{pH} = 7$.



KEY CONCEPT

Compare the relative strength of the two solutions in a titration to determine if the pH of the equivalence point is less than, equal to, or greater than 7:

- Strong acid + weak base: equivalence point $\text{pH} < 7$
- Strong acid + strong base: equivalence point $\text{pH} = 7$
- Weak acid + strong base: equivalence point $\text{pH} > 7$

Because HCl is a strong acid and NaOH is a strong base, the equivalence point of the titration will be at pH 7, and the solution will be neutral. Note that the endpoint shown is close to, but not exactly equal to, the equivalence point; selection of a better indicator, one that changes colors at, say, pH 8, would have given a better approximation. Still, the amount of error introduced by the use of an indicator that changes color around pH 11 rather than pH 8 is not especially significant; it represents a mere fraction of a milliliter of excess NaOH solution.

In the early part of the curve when little base has been added, the acidic species predominates, so the addition of small amounts of base will not appreciably change either the $[\text{OH}^-]$ or the pH. Similarly, in the last part of the titration curve when an excess of base has been added, the addition of small amounts of base will not change the $[\text{OH}^-]$ significantly, and the pH will remain relatively constant. The addition of base will alter the concentrations of H^+ and OH^- near the equivalence point, and will elicit the most substantial changes in pH in that region. Remember: the equivalence point for strong acid/strong base titrations is always at pH 7 (for monovalent species).

If one uses a pH meter to chart the change in pH as a function of volume of titrant added, a good approximation can be made of the equivalence point by locating the midpoint of the region of the curve with the steepest slope.

WEAK ACID AND STRONG BASE

Titration of a weak acid, such as CH_3COOH , with a strong base, such as NaOH, produces the titration curve shown in Figure 10.10.

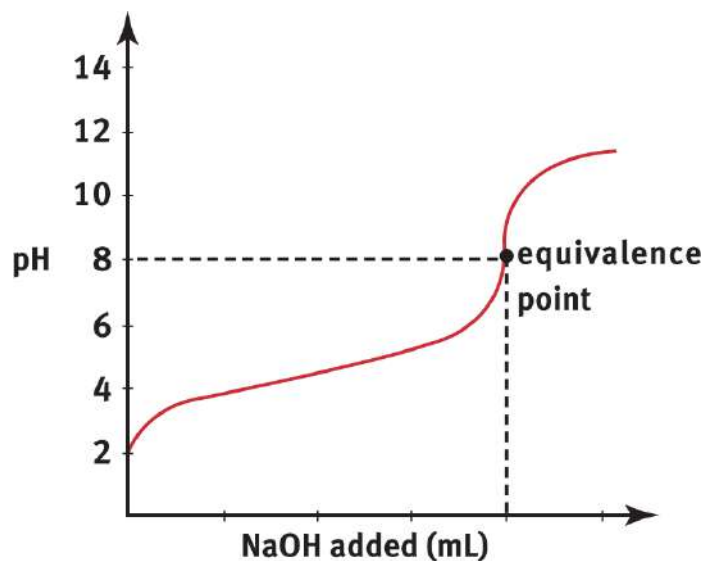


Figure 10.10. Weak Acid and Strong Base Titration Curve

A strong base, NaOH, is titrated into a solution of weak acid, CH_3COOH , to yield an equivalence point of $\text{pH} > 7$.

Compare Figure 10.10 with the curve in Figure 10.9. The first difference is that the initial pH of the weak acid solution is greater than the initial pH of the strong acid solution. Weak acids do not dissociate to the same degree that strong acids do; therefore, the concentration of H_3O^+ will generally be lower (and pH will be higher) in an equimolar solution of weak acid. The second difference is the shapes of the curves. The pH curve for the strong acid/strong base titration shows a steeper, more sudden rise in pH at the equivalence point. In the weak acid/strong base titration, the pH changes gradually early on in the titration and has a less sudden rise at the equivalence point. The third difference is the position of the equivalence point. While the equivalence point for a strong acid/strong base titration is pH 7, the equivalence point for a weak acid/strong base titration is above 7. This is because the reaction between the weak acid (HA) and strong base (OH^-) produces a weak conjugate base (A^-) and even weaker conjugate acid (H_2O). This produces a greater concentration of hydroxide ions than

hydrogen ions at equilibrium (due to the common ion effect on the autoionization of water). Therefore, the equivalence point for weak acid/strong base titration is always in the basic range of the pH scale.

STRONG ACID AND WEAK BASE

The appearance of the titration curve for a weak base titrand and strong acid titrant will look like an inversion of the curve for a weak acid titrand and strong base titrant. The initial pH will be in the basic range (typical range: pH 10–12) and will demonstrate a gradual drop in pH with the addition of strong acid. The equivalence point will be in the acidic pH range because the reaction between the weak base and strong acid will produce a weak conjugate acid and even weaker conjugate base, as shown in Figure 10.11. The stronger conjugate acid will result in an equilibrium state with a concentration of hydrogen ions greater than that of the hydroxide ions. Therefore, the equivalence point for a weak base/strong acid titration is always in the acidic range of the pH scale.

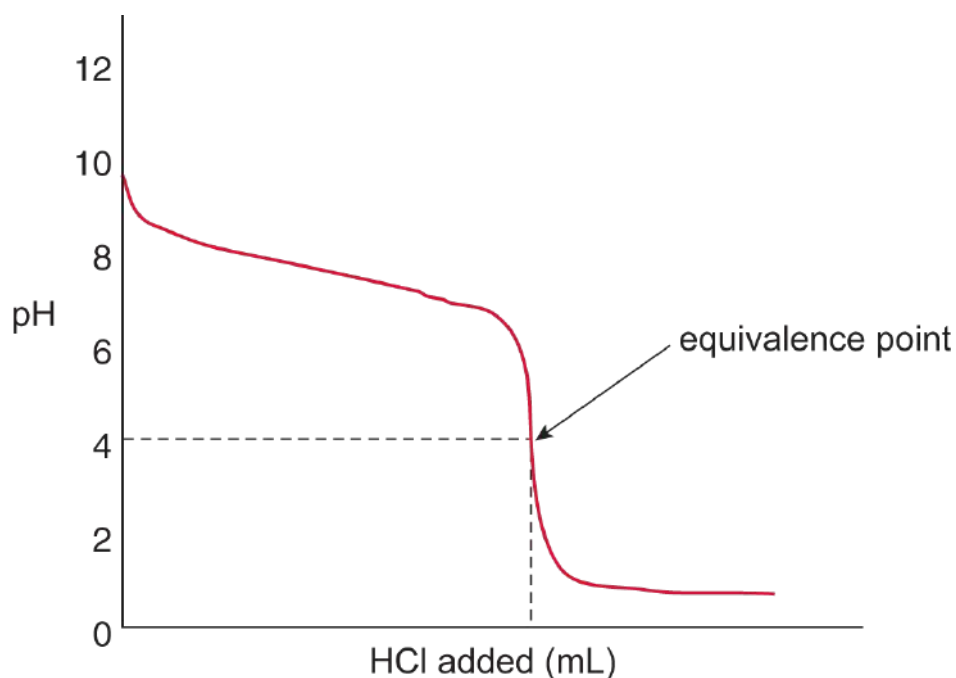


Figure 10.11. Strong Acid and Weak Base Titration Curve

A strong acid, HCl, is titrated into a solution of weak base, NH_3 , to yield an equivalence point of $\text{pH} < 7$.

WEAK ACID AND WEAK BASE

The appearance of the titration curve for a weak base titrand and weak acid titrant will look like an intermediate of the previous types discussed. Because both the titrant and the titrand are weak, the initial pH is generally in the 3–11 range and will demonstrate a very shallow drop at the equivalence point. The equivalence point will be near neutral pH because the reaction is partially dissociative for both species.



MCAT EXPERTISE

To identify which type of titration is being shown in a graph, identify the starting position in the graph ($\text{pH} \gg 7$ = titrand is a strong base, > 7 (slightly) = weak base, < 7 (slightly) = weak acid, and $\ll 7$ pH = strong acid), and determine where the equivalence point is. Think of titrations like tug-of-war: the stronger the acid or base, the more it pulls the equivalence point into its pH territory.

POLYVALENT ACIDS AND BASES

The titration curve for a polyvalent acid or base looks different from that for a monovalent acid or base. Figure 10.12 shows the titration of Na_2CO_3 with HCl , in which the divalent (diprotic) acid H_2CO_3 is the ultimate product.

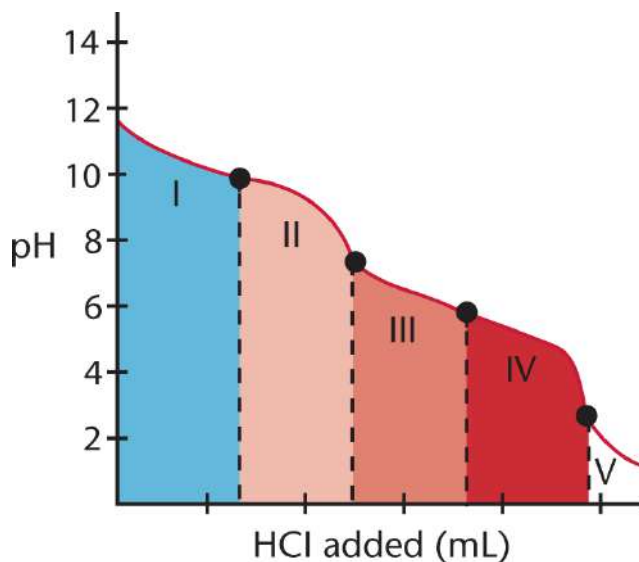


Figure 10.12. Polyvalent Titration

The multiple equivalence points indicate that this is a polyvalent titration.

In region I, little acid has been added, and the predominant species is CO_3^{2-} . In region II, more acid has been added, and the predominant species are CO_3^{2-} and HCO_3^- , in relatively equal concentrations. The flat part of the curve is the first buffer region (discussed in the next section), corresponding to the pK_a of HCO_3^- ($K_a = 5.6 \times 10^{-11}$; $\text{pK}_a = 10.25$). The center of the buffer region (the point between regions I and II) is sometimes termed the half-equivalence point because it occurs when half of a given species has been protonated (or deprotonated).

Region III begins with the equivalence point, at which all of the CO_3^{2-} is finally titrated to HCO_3^- . As the curve illustrates, a rapid change in pH occurs at the equivalence point (the point between regions II and III). In the latter part of region III, the predominant species is HCO_3^- , although some H_2CO_3 has formed as well.

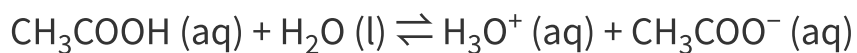
At the beginning of region IV, the acid has neutralized approximately half of the HCO_3^- , and now H_2CO_3 and HCO_3^- are in roughly equal concentrations. This flat region is the second buffer region (and second half-equivalence point, between regions III and IV) of the titration curve, corresponding to the pK_a of H_2CO_3 ($K_a = 4.3 \times 10^{-7}$; $\text{pK}_a = 6.37$). Region V starts with the second equivalence point, as all of the HCO_3^- is finally converted to H_2CO_3 . Again, a rapid change in pH is observed near the equivalence point (the point between regions IV and V) as acid is added.

The titrations of the acidic and basic amino acids (which have acidic or basic side chains, respectively) will show curves similar to the one shown in Figure 10.12. But rather than two equivalence points, there will in fact be three: one corresponding to the titration of the carboxyl group and a second corresponding to the titration of the amino group, both of which

are attached to the central carbon, as well as a third corresponding to either the acidic or basic side chain.

BUFFERS

A buffer solution consists of a mixture of a weak acid and its salt (which is composed of its conjugate base and a cation) or a mixture of a weak base and its salt (which is composed of its conjugate acid and an anion). Two examples of buffers that are common in the laboratory—and commonly tested on the MCAT—are a solution of acetic acid (CH_3COOH) and its salt, sodium acetate ($\text{CH}_3\text{COO}^-\text{Na}^+$), and a solution of ammonia (NH_3) and its salt, ammonium chloride (NH_4^+Cl^-). The acetic acid/sodium acetate solution is considered an acid buffer, and the ammonium chloride/ammonia solution is a base buffer. Buffer solutions have the useful property of resisting changes in pH when small amounts of acid or base are added. Consider a buffer solution of acetic acid and sodium acetate: (Note: The sodium ion has not been included because it is not involved in the acid–base reaction.)



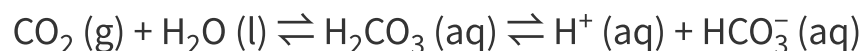
When a small amount of strong base, such as NaOH, is added to the buffer, the OH^- ions from the NaOH react with the H_3O^+ ions present in the solution; subsequently, more acetic acid dissociates (the system shifts to the right), restoring the $[\text{H}_3\text{O}^+]$. The weak acid component of the buffer thereby serves to neutralize the strong base that has been added. The resulting increase in the concentration of the acetate ion (the conjugate

base) does not create nearly as large an increase in hydroxide ions as the unbuffered NaOH would. Thus, the addition of the strong base does not result in a significant increase in $[\text{OH}^-]$ and does not appreciably change the pH.

Likewise, when a small amount of HCl is added to the buffer, H^+ ions from the HCl react with the acetate ions to form acetic acid. Acetic acid is weaker than the added hydrochloric acid (which has been neutralized by the acetate ions), so the increased concentration of acetic acid does not significantly contribute to the hydrogen ion concentration in the solution. Because the buffer maintains $[\text{H}^+]$ at approximately constant values, the pH of the solution is relatively unchanged.

The Bicarbonate Buffer System

In the human body, one of the most important buffers is the $\text{H}_2\text{CO}_3/\text{HCO}_3^-$ conjugate pair in the plasma component of the blood, called the bicarbonate buffer system. Specifically, carbonic acid (H_2CO_3) and its conjugate base, bicarbonate (HCO_3^-), form a weak acid buffer for maintaining the pH of the blood within a fairly narrow physiological range. CO_2 (g), one of the waste products of cellular respiration, also has low solubility in aqueous solutions. The majority of the CO_2 transported from peripheral tissues to the lungs (where it will be exhaled out) is dissolved in the plasma in a “disguised” form through the bicarbonate buffer system. CO_2 (g) and water react in the following manner:





REAL WORLD

A number of conditions can affect the delicate pH balance of tissues in the body, including chronic obstructive pulmonary disease (COPD), renal tubular acidosis (RTA), diabetic ketoacidosis (DKA), lactic acidosis, metabolic diseases, poisonings and ingestions, and hyperventilation. The buffer system must be well maintained to mitigate these changes.

The bicarbonate buffer system is tied to the respiratory system. In conditions of metabolic acidosis (production of excess plasma H^+ not caused by the respiratory system itself), the breathing rate will increase to compensate and blow off a greater amount of carbon dioxide gas; this causes the system to shift to the left, thereby reducing $[\text{H}^+]$ and buffering against dramatic and dangerous changes to the blood pH. It is interesting to note that the bicarbonate buffer system ($\text{pK}_a = 6.37$) maintains a pH around 7.4, which is actually slightly outside the optimal buffering capacity of the system. Buffers have a narrow range of optimal activity ($\text{pK}_a \pm 1$). This actually makes sense—it is far more common for acidemia (too much acid in the blood) to occur than alkalemia (too much base in the blood). As acidemia becomes more severe, the buffer system actually becomes more effective and more resistant to further lowering of the pH.

The Henderson–Hasselbalch Equation

The Henderson–Hasselbalch equation is used to estimate the pH or pOH of a buffer solution. For a weak acid buffer solution:

$$\text{pH} = \text{p}K_{\text{a}} + \log \frac{[\text{A}^{-}]}{[\text{HA}]}$$

Equation 10.9

where $[\text{A}^{-}]$ is the concentration of the conjugate base and $[\text{HA}]$ is the concentration of the weak acid. Note that when $[\text{conjugate base}] = [\text{weak acid}]$, the $\text{pH} = \text{p}K_{\text{a}}$ because $\log(1) = 0$. This occurs at the half-equivalence points in a titration, and buffering capacity is optimal at this pH.

Likewise, for a weak base buffer solution:

$$\text{pOH} = \text{p}K_{\text{b}} + \log \frac{[\text{B}^{+}]}{[\text{BOH}]}$$

Equation 10.10

where $[\text{B}^{+}]$ is the concentration of conjugate acid and $[\text{BOH}]$ is the concentration of the weak base. Similar to acid buffers, $\text{pOH} = \text{p}K_{\text{b}}$ when $[\text{conjugate acid}] = [\text{weak base}]$. Buffering capacity is optimal at this pOH.

The Henderson–Hasselbalch equation is, in reality, just a rearrangement of the acid (or base) dissociation constant:

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}$$

$$-\log K_a = -\log \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}$$

$$-\log K_a = -\log[\text{H}_3\text{O}^+] - \log \frac{[\text{A}^-]}{[\text{HA}]}$$

$$\text{p}K_a = \text{pH} - \log \frac{[\text{A}^-]}{[\text{HA}]}$$

$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

One subtlety of buffer systems and Henderson–Hasselbalch calculations that usually goes unnoticed or is misunderstood by students is the effect of changing the concentrations of the conjugate pair but not changing the ratio of their concentrations. Clearly, changing the ratio of the conjugate base to the acid will lead to a change in the pH of the buffer solution. But what about changing the concentrations while maintaining a constant ratio? What would happen if the concentrations of both the acid and its conjugate base were doubled? While the pH would not change, the buffering capacity—the ability to which the system can resist changes in pH—has doubled. In other words, addition of a small amount of acid or base to this system will now cause even less deviation in the pH. As mentioned earlier, the buffering capacity is generally maintained within 1 pH unit of the $\text{p}K_a$ value.

Example:

What is the pH of a solution made from 1 L of 0.05 M acetic acid (CH_3COOH , $K_a = 1.8 \times 10^{-5}$) mixed with 500 mL of 1 M acetate (CH_3COO^-)?

Solution:

First, determine the concentrations of acetic acid and acetate in the final solution. Because two solutions were mixed, there will be some dilution of both the acetic acid and acetate.

$$N_{i,\text{CH}_3\text{COOH}} V_{i,\text{CH}_3\text{COOH}} = N_{f,\text{CH}_3\text{COOH}} V_{f,\text{CH}_3\text{COOH}} \rightarrow N_{f,\text{CH}_3\text{COOH}} = \frac{N_{i,\text{CH}_3\text{COOH}} V_{i,\text{CH}_3\text{COOH}}}{V_{f,\text{CH}_3\text{COOH}}}$$
$$N_{f,\text{CH}_3\text{COOH}} = \frac{(0.05\text{ N})(1\text{ L})}{(1.5\text{ L})} = 0.033\text{ N}$$

$$N_{i,\text{CH}_3\text{COO}^-} V_{i,\text{CH}_3\text{COO}^-} = N_{f,\text{CH}_3\text{COO}^-} V_{f,\text{CH}_3\text{COO}^-} \rightarrow N_{f,\text{CH}_3\text{COO}^-} = \frac{N_{i,\text{CH}_3\text{COO}^-} V_{i,\text{CH}_3\text{COO}^-}}{V_{f,\text{CH}_3\text{COO}^-}}$$
$$N_{f,\text{CH}_3\text{COO}^-} = \frac{(1\text{ N})(0.5\text{ L})}{(1.5\text{ L})} = 0.33\text{ N}$$

Then, use the Henderson–Hasselbalch equation.

$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]} = -\log[1.8 \times 10^{-5}] + \log \frac{0.018}{0.002}$$
$$\text{pH} \approx 4.82 + 1 = 5.82 (\text{actual} = 5.74)$$

MCAT Concept Check 10.4:

Before you move on, assess your understanding of the material with these questions.

Describe each of the following parts of a titration curve:

- Buffering region:

- Half-equivalence point:

- Equivalence point:

- Endpoint:

For a reaction involving a strong base and a weak acid, which of the following indicators would be best to

indicate the endpoint of the titration? (Circle the correct answer.)

- Phenolphthalein ($pK_a = 9.7$)
- Bromothymol blue ($pK_a = 7.1$)
- Bromocresol green ($pK_a = 4.7$)
- Methyl yellow ($pK_a = 3.3$)

In which part of the pH range (acidic, basic, or neutral) will the equivalence points fall for each of the following titrations?

- Strong acid + weak base:

- Strong base + weak acid:

- Strong acid + strong base:

- Weak acid + weak base:

What is the purpose of a buffer solution?

What are the pH and pOH of a solution containing 5 mL of 5 *M* benzoic acid ($K_a = 6.3 \times 10^{-5}$) and 100 mL of 0.005 *M* benzoate solution?

Conclusion

In this chapter, we have reviewed the important principles of acid–base chemistry. We clarified the differences among the three definitions of acids and bases, including the nomenclature of some common Arrhenius acids. We investigated important properties of acids and bases, including the important acid–base behavior of water (autoionization) and hydrogen ion equilibria. We explained the mathematics of the pH and pOH logarithmic scales and demonstrated a useful Test Day shortcut for approximating the logarithmic value of hydrogen ion or hydroxide ion concentrations. Strong acids and bases are defined as compounds that completely dissociate in aqueous solutions, and weak acids and bases are compounds that only partially dissociate (to an equilibrium state). We discussed neutralization and salt formation upon reaction of acids and bases, and finally, we applied our fundamental understanding of acid–base reactivity to titrations and buffer systems. Titrations are useful for determining the concentration of a known acid or base solution. Weak acid and weak base buffers are useful for minimizing changes in pH upon addition of strong acid or base.

You’ve just accomplished a major task in the overall effort to earn points on Test Day. It’s okay if you didn’t understand everything on this first pass. Go back and review the concepts that were challenging for you and then complete the questions at the end of the chapter and MCAT practice passages to test your knowledge. Don’t be alarmed if you find yourself

reviewing parts or all of a chapter a second or third time—repetition is the key to success.

You are now two chapters away from completing this review of general chemistry. While we don't want to offer our congratulations prematurely, we want to acknowledge all the hard work you've invested in this process. Keep it up: success on Test Day is within your reach!

Concept Summary

Definitions

- Arrhenius acids dissociate to produce an excess of hydrogen ions in solution. Arrhenius bases dissociate to produce an excess of hydroxide ions in solution.
- Brønsted–Lowry acids are species that can donate hydrogen ions. Brønsted–Lowry bases are species that can accept hydrogen ions.
- Lewis acids are electron-pair acceptors. Lewis bases are electron-pair donors.
- All Arrhenius acids and bases are Brønsted–Lowry acids and bases, and all Brønsted–Lowry acids and bases are Lewis acids and bases; however, the converse of these statements is not necessarily true (that is, not all Lewis acids and bases are Brønsted–Lowry acids and bases, and not all Brønsted–Lowry acids and bases are Arrhenius acids and bases).
- Amphoteric species are those that can behave as an acid or base. Amphiprotic species are amphoteric species that specifically can behave as a Brønsted–Lowry acid or Brønsted–Lowry base.
 - Water is a classic example of an amphoteric, amphiprotic species—it can accept a hydrogen ion to become a hydronium ion, or it can donate a hydrogen ion to become a hydroxide ion.
 - Conjugate species of polyvalent acids and bases can also behave as amphoteric and amphiprotic species.

Properties

- The water dissociation constant, K_w , is 10^{-14} at 298 K. Like other equilibrium constants, K_w is only affected by changes in temperature.
- pH and pOH can be calculated given the concentrations of H_3O^+ and OH^- ions, respectively. In aqueous solutions, $pH + pOH = 14$ at 298 K.
- Strong acids and bases completely dissociate in solution.
- Weak acids and bases do not completely dissociate in solution and have corresponding dissociation constants (K_a and K_b , respectively).
- In the Brønsted–Lowry definition, acids have conjugate bases that are formed when the acid is deprotonated. Bases have conjugate acids that are formed when the base is protonated.
 - Strong acids and bases have very weak (inert) conjugates.
 - Weak acids and bases have weak conjugates.
- Neutralization reactions form salts and (sometimes) water.

Polyvalence and Normality

- An equivalent is defined as one mole of the species of interest.
- In acid–base chemistry, normality is the concentration of acid or base equivalents in solution.
- Polyvalent acids and bases are those that can donate or accept multiple electrons. The normality of a solution containing a polyvalent species is the molarity of the acid or base times the number of protons it can donate or accept.

Titration and Buffers

- Titrations are used to determine the concentration of a known reactant in a solution.
 - The titrant has a known concentration and is added slowly to the titrand to reach the equivalence point.
 - The titrand has an unknown concentration but a known volume.
- The half-equivalence point is the midpoint of the buffering region, in which half of the titrant has been protonated (or deprotonated); thus, $[HA] = [A^-]$ and a buffer is formed.
- The equivalence point is indicated by the steepest slope in a titration curve; it is reached when the number of acid equivalents in the original solution equals the number of base equivalents added, or vice-versa.
 - Strong acid and strong base titrations have equivalence points at $pH = 7$.
 - Weak acid and strong base titrations have equivalence points at $pH > 7$.
 - Weak base and strong acid titrations have equivalence points at $pH < 7$.
 - Weak acid and weak base titrations can have equivalence points above or below 7, depending on the relative strength of the acid and base.
- Indicators are weak acids or bases that display different colors in their protonated and deprotonated forms.

- The indicator chosen for a titration should have a pK_a close to the pH of the expected equivalence point.
- The endpoint of a titration is when the indicator reaches its final color.

- Multiple buffering regions and equivalence points are observed in polyvalent acid and base titrations.
- Buffer solutions consist of a mixture of a weak acid and its conjugate salt or a weak base and its conjugate salt; they resist large fluctuations in pH.
- Buffering capacity refers to the ability of a buffer to resist changes in pH; maximal buffering capacity is seen within 1 pH point of the pK_a of the acid in the buffer solution.
- The Henderson–Hasselbalch equation quantifies the relationship between pH and pK_a for weak acids and between pOH and pK_b for weak bases; when a solution is optimally buffered, $pH = pK_a$ and $pOH = pK_b$.

Answers to Concept Checks

10.1

Definition	Acid	Base
Arrhenius	Dissociates to form excess H^+ in solution	Dissociates to form excess OH^- in solution
Brønsted–Lowry	H^+ donor	H^+ acceptor
Lewis	Electron pair acceptor	Electron pair donor

Anion	Acid Formula	Acid Name
MnO_4^-	HMnO_4	Permanganic acid
Titanate (TiO_3^{2-})	H_2TiO_3	Titanic acid
I^-	HI	Hydroiodic acid
IO_4^-	HIO_4	Periodic acid

Reaction	Amphoteric Reactant	Amphiprotic? (Y or N)
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Reaction	Amphoteric Reactant	Amphiprotic? (Y or N)
$\text{HCO}_3^- + \text{HBr} \rightarrow \text{H}_2\text{CO}_3 + \text{Br}^-$	HCO_3^-	Yes
$3 \text{HCl} + \text{Al}(\text{OH})_3 \rightarrow \text{AlCl}_3 + 3 \text{H}_2\text{O}$	$\text{Al}(\text{OH})_3$	No
$2 \text{HBr} + \text{ZnO} \rightarrow \text{ZnBr}_2 + \text{H}_2\text{O}$	ZnO	No

10.2

An amphoteric species can act as an acid or a base.

High K_a indicates a strong acid, which will dissociate completely in solution. Having a K_a slightly greater than water means the acid is a weak acid with minimal dissociation.

High K_b indicates a strong base, which will dissociate completely in solution. Having a K_b slightly greater than water means the base is a weak base with minimal dissociation.

pH	$[\text{H}_3\text{O}^+]$	pOH	$[\text{OH}^-]$	Acid or Base?
4	10^{-4} M	10	10^{-10} M	Acid
3.05	$8.89 \times 10^{-4} \text{ M}$	10.95	$1.12 \times 10^{-11} \text{ M}$	Acid
8.81	$1.55 \times 10^{-9} \text{ M}$	5.19	$6.46 \times 10^{-6} \text{ M}$	Base
8.27	$5.32 \times 10^{-9} \text{ M}$	5.73	$1.88 \times 10^{-6} \text{ M}$	Base

(Note: Exact answers are provided; your rounded answers should be relatively close to those listed here.)

$$K_a \times K_b = K_w$$

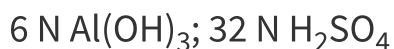
Reaction	Acid	Base	Conjugate Acid	Conjugate Base
$\text{H}_2\text{CO}_3 + \text{H}_2\text{O} \rightleftharpoons \text{HCO}_3^- + \text{H}_3\text{O}^+$	H_2CO_3	H_2O	H_3O^+	HCO_3^-
$\text{H}_2\text{PO}_4^- + \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{PO}_4 + \text{OH}^-$	H_2O	H_2PO_4^-	H_3PO_4	OH^-

$$K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}_3\text{O}^+]}{[\text{CH}_3\text{COOH}]} = \frac{[x][x]}{[0.2\text{ M} - x]} \approx \frac{x^2}{0.2}.$$

Therefore, $x^2 = 3.6 \times 10^{-6} \rightarrow x \approx 2 \times 10^{-3}\text{ M}$ (actual = $1.9 \times 10^{-3}\text{ M}$). Then, $\text{pH} = -\log \text{H}_3\text{O}^+ \approx 3 - 0.2 = 2.8$ (actual = 2.72)

10.3

Acids use moles of H^+ (H_3O^+) as an equivalent. Bases use moles of OH^- as an equivalent.



10.4

The buffering region occurs when $[\text{HA}] \approx [\text{A}^-]$ and is the flattest portion of the titration curve (resistant to changes in pH). The half-equivalence point is the center of the buffering region, where $[\text{HA}] = [\text{A}^-]$. The equivalence point is the steepest point of the titration curve, and occurs when the equivalents of acid present equal the equivalents of base added (or vice-versa). The endpoint is the pH at which an indicator turns its final color.

Phenolphthalein would be the preferred indicator for this titration.

A strong acid and weak base have an equivalence point in the acidic range. A strong base and weak acid have an equivalence point in the basic range. A strong acid and strong base have an equivalence point at

pH = 7 (neutral). A weak acid and weak base can have an equivalence point in the acidic, neutral, or basic range, depending on the relative strengths of the acid and base.

A buffer solution is designed to resist changes in pH and has optimal buffering capacity within 1 pH point from its pK_a .

Recall from the example on page 356 that the concentrations of the conjugate acid and conjugate base in the final solution must first be calculated due to the dilution from mixing the two solutions together. The calculation below follows that step, but leaves the values unsolved for so that some of their components can be cancelled and simplified:

$$\begin{aligned} \text{pH} &= pK_a + \log \frac{[A^-]}{[HA]} = -\log(6.3 \times 10^{-5}) + \log \frac{\left[\frac{(0.005 \text{ N})(100 \text{ mL})}{(105 \text{ mL})} \right]}{\left[\frac{(5 \text{ N})(5 \text{ mL})}{(105 \text{ mL})} \right]} \\ \text{pH} &\approx 4.37 + \log \frac{(0.005 \text{ N})(100 \text{ mL})}{(5 \text{ N})(5 \text{ mL})} = 4.37 + \log \frac{0.5}{25} = 4.37 + \log(2 \times 10^{-2}) \\ &\approx 4.37 - 1.8 = 2.57 (\text{actual} = 2.50) \end{aligned}$$

$$\text{pOH} \approx 14 - 2.57 = 11.43 (\text{actual} = 11.5)$$

Equations to Remember

(10.1) Autoionization constant for water: $K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = 10^{-14}$ at 25°C (298 K)

(10.2) Definitions of pH and pOH:

$$\text{pH} = -\log[\text{H}^+] = \log \frac{1}{[\text{H}^+]}$$

$$\text{pOH} = -\log[\text{OH}^-] = \log \frac{1}{[\text{OH}^-]}$$

(10.3) Relationship of pH and pOH at 298 K: $\text{pH} + \text{pOH} = 14$

(10.4) p scale value approximation: p value $\approx m - 0.n$

(10.5) Acid dissociation constant: $K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}$

(10.6) Base dissociation constant: $K_b = \frac{[\text{B}^+][\text{OH}^-]}{[\text{BOH}]}$

(10.7) Relationship of K_a and K_b at 298 K:

$$K_{a,\text{acid}} \times K_{b,\text{conjugate base}} = K_w = 10^{-14};$$

$$K_{b,\text{base}} \times K_{a,\text{conjugate acid}} = K_w = 10^{-14}$$

(10.8) Equivalence point: $N_a V_a = N_b V_b$

(10.9) Henderson–Hasselbalch equation (acid buffer):

$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

(10.10) Henderson–Hasselbalch equation (base buffer):

$$\text{pOH} = \text{p}K_b + \log \frac{[\text{B}^+]}{[\text{BOH}]}$$



Shared Concepts

Biology Chapter 6

The Respiratory System

Biology Chapter 10

Homeostasis

General Chemistry Chapter 3

Bonding and Chemical Interactions

General Chemistry Chapter 9

Solutions

Organic Chemistry Chapter 4

Analyzing Organic Reactions

Physics and Math Chapter 10

Mathematics

Discrete Practice Questions

Consult your online resources for additional practice.

1. Which of the following is not a Brønsted–Lowry base?



2. Which of the following is closest to the pH of a solution containing 5 mM H_2SO_4 ?



3. Which of the following represents chloric acid?

- (A) HClO_3
- (B) ClO_3^-
- (C) HClO_2
- (D) HClO

4. Which of the following bases is the weakest?

- (A) KOH
- (B) NH_3
- (C) CH_3NH_2
- (D) $\text{Ca}(\text{OH})_2$

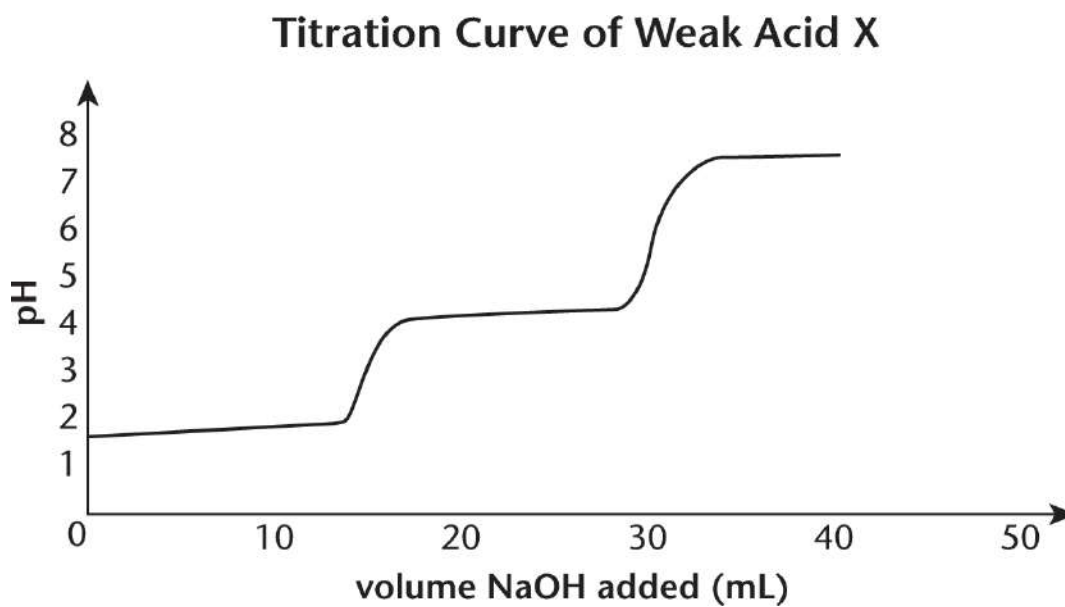
5. The function of a buffer is to:

- (A) maintain a neutral pH.
- (B) resist changes in pH when small amounts of acid or base are added.
- (C) slow down reactions between acids and bases.
- (D) speed up reactions between acids and bases.

6. What is the pH of a solution with an ammonium concentration of 70 mM and an ammonia concentration of 712 mM? (Note: The pK_b of ammonia is 3.45.)

- (A) 2.45
- (B) 4.45
- (C) 9.55
- (D) 11.55

Questions 7–9 refer to the titration curve of acid X shown below:



7. What is the approximate value of pK_{a1} ?

- (A) 1.9
- (B) 2.9
- (C) 3.8
- (D) 4.1

8. At what pH is the second equivalence point?

- (A) pH = 3.0
- (B) pH = 4.1
- (C) pH = 5.9
- (D) pH = 7.2

9. What is the approximate value of pK_{a2} ?

- (A) 3.6
- (B) 4.1
- (C) 5.5
- (D) 7.2

10. What is the gram equivalent weight of phosphoric acid?

- (A) 24.5 g
- (B) 32.7 g
- (C) 49.0 g
- (D) 98.0 g

11. What is the $[H_3O^+]$ of a 2 M aqueous solution of a weak acid HXO_2 with $K_a = 3.2 \times 10^{-5}$?

- (A) $6.4 \times 10^{-5} \text{ M}$
- (B) $1.3 \times 10^{-4} \text{ M}$
- (C) $4.0 \times 10^{-3} \text{ M}$
- (D) $8.0 \times 10^{-3} \text{ M}$

12. A solution is prepared with an unknown concentration of a theoretical compound with a K_a of exactly 1.0. What is the pH of this solution?

- (A) Higher than 7
- (B) Exactly 7
- (C) Less than 7
- (D) There is not enough information to answer the question.

13. Which of the following is NOT a characteristic of an amphoteric species?

- (A) Amphoteric species can act as an acid or a base, depending on its environment.
- (B) Amphoteric species can act as an oxidizing or reducing agent, depending on its environment.
- (C) Amphoteric species are sometimes amphiprotic.
- (D) Amphoteric species are always nonpolar.

14. What is the approximate pH of a $1.2 \times 10^{-5} \text{ M}$ aqueous solution of NaOH?

- (A) 4.92
- (B) 7.50
- (C) 9.08
- (D) 12.45

15. How many liters of 2 M $\text{Ba}(\text{OH})_2$ are needed to titrate a 4 L solution of 6 M H_3PO_4 ?

- (A) 1.33 L
- (B) 12 L
- (C) 18 L
- (D) 56 L

Explanations to Discrete Practice Questions

1. D

A Brønsted–Lowry base is defined as a proton acceptor. Ammonia, fluoride, and water— (A), (B), and (C), respectively—each accept a proton. (D), HNO_2 , is a far better Brønsted–Lowry acid, donating a proton to solution.

2. B

First, convert the concentration to 5×10^{-3} M. Next, because sulfuric acid is a strong acid, we can assume that, for the majority of sulfuric acid molecules (although not all), both protons will dissociate. The concentration of hydrogen ions is therefore $2 \times 5 \times 10^{-3}$, or 10^{-2} . The equation for pH is $\text{pH} = -\log [\text{H}^+]$. If $[\text{H}^+] = 10^{-2}$ M, then $\text{pH} = 2$.

3. A

Answering this question is simply a matter of knowing nomenclature. Acids ending in –ic are derivatives of anions ending in –ate, while acids ending in –ous are derivatives of anions ending in –ite. ClO_3^- , (B), is chlorate because it has more oxygen than the other commonly occurring ion, ClO_2^- , which is named chlorite. Therefore, HClO_3 is chloric acid. HClO_2 , (C), represents chlorous acid. HClO , (D), represents hypochlorous acid.

4. B

Soluble hydroxides of Group IA and IIA metals are strong bases, eliminating (A) and (D). (B) and (C) are both weak bases; however, methylamine contains an alkyl group, which is electron-donating. This increases the electron density on the nitrogen in methylamine, making it a stronger (Lewis) base. Therefore, ammonia is the weakest base.

5. B

The purpose of a buffer is to resist changes in the pH of a reaction. Buffers are not generally used to affect the kinetics of a reaction, so (C) and (D) are incorrect. (A) is correct only in specific circumstances where the pH of the buffer solution itself is neutral. Many natural buffer systems maintain pH in the acidic or basic ranges.

6. D

The question is asking for pH, but because of the information given, we must first find the pOH and then subtract it from 14 to get the pH. Use the Henderson–Hasselbalch equation:

$$\begin{aligned}\text{pOH} &= \text{p}K_b + \log \frac{[\text{conjugate acid}]}{[\text{base}]} = 3.45 + \log \frac{70 \text{ mL}}{712 \text{ mL}} \\ &\approx 3.45 + \log \frac{1}{10} = 3.45 - 1 = 2.45\end{aligned}$$

If the pOH = 2.45, the pH = 14 – 2.45 = 11.55.

7. A

The first pK_a in this curve can be estimated by eye. It is located halfway between the starting point (when no base had yet been added) and the first equivalence point (the first steep portion of the graph, around 15 mL). This point is at approximately 7–8 mL on the x-axis, which corresponds to a pH of approximately 1.9. Notice that this region experiences very little change in pH, which is the defining characteristic of a buffer region.

8. C

The second equivalence point is the midpoint of the second steep increase in slope. This corresponds to approximately pH = 5.9.

9. B

The value of the second pK_a is found at the midpoint between the first and second equivalence points. In this curve, that corresponds to pH = 4.1. Just like the first pK_a , it is in the center of a flat buffering region.

10. B

Gram equivalent weight is the weight (in grams) that releases 1 acid or base equivalent from a compound. Because H_3PO_4 contains 3 protons, we find the gram equivalent weight by dividing the mass of one mole of the species by 3. The molar mass of phosphoric acid is **98 $\frac{g}{mol}$** , so the gram equivalent weight is 32.7 g.

11. D

This question requires the application of the acid dissociation constant. Weak acids do not dissociate completely; therefore, all three species that appear in the balanced equation will be present in solution. Hydrogen ions and conjugate base anions dissociate in equal amounts,

so $[H^+] = [XO_2^-]$. If the initial concentration of HXO_2 was 2 M and some amount x dissociates, we will have x amount of H_3O^+ and XO_2^- at equilibrium, with $2\text{ M} - x$ amount of HXO_2 at equilibrium.

$$K_a = \frac{[H_3O^+][XO_2^-]}{[HXO_2]} = \frac{(x)(x)}{2\text{ M} - x} \approx \frac{x^2}{2}$$

Note that x was considered negligible when added or subtracted, per usual. Solving for x , we get:

$$\frac{x^2}{2} = 3.2 \times 10^{-5} \rightarrow x^2 = 6.4 \times 10^{-5} = 64 \times 10^{-6} -$$

$$x = 8 \times 10^{-3} \text{ M}$$

12. C

A higher K_a implies a stronger acid. Weak acids usually have a K_a that is several orders of magnitude below 1. The pK_a of a compound is the pH at which there are equal concentrations of acid and conjugate base; the pK_a of this compound would be $-\log 1 = 0$. With such a low pK_a , this compound must be an acid. Therefore, the pH of any concentration of this compound must be below 7.

13. D

An amphoteric species is one that can act either as an acid or a base, depending on its environment. Proton transfers are classic oxidation–reduction reactions, so (A) and (B) are true. (C) is true because many amphoteric species, such as water and bicarbonate, can either donate

or accept a proton. (D) is false, and thus the correct answer because amphoteric species can be either polar or nonpolar in nature.

14. C

NaOH is a strong base; as such, there will be $1.2 \times 10^{-5} \text{ M OH}^-$ in solution. Based on this information alone, the pOH must be between 4 and 5, and the pH must be between 9 and 10. Using the shortcut, $\text{pOH} \approx 5 - 0.12 = 4.88$. $\text{pH} = 14 - \text{pOH} = 9.12$ (actual = 9.08).

15. C

Use the equivalence point equation: $N_a V_a = N_b V_b$. Ba(OH)_2 can dissociate to give two hydroxide ions, so its normality is $2 \text{ M} \times 2 = 4 \text{ N}$. H_3PO_4 can dissociate to give three hydronium ions, so its normality is $6 \text{ M} \times 3 = 18 \text{ N}$. Plugging into the equation, we get $(18 \text{ N})(4 \text{ L}) = (4 \text{ N})(V_b)$. Therefore, V_b is 18 L.

Oxidation–Reduction Reactions

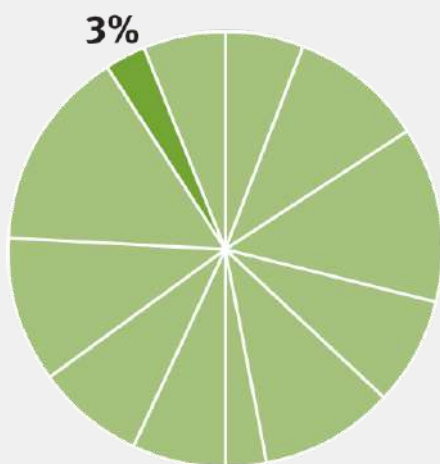
In This Chapter

- 11.1 Oxidation–Reduction Reactions 
 - Oxidation and Reduction 
 - Assigning Oxidation Numbers 
 - Balancing Oxidation–Reduction Reactions 

- 11.2 Net Ionic Equations 
 - Overview 
 - Disproportionation Reactions 
 - Oxidation–Reduction Titrations 

- Concept Summary 

Chapter Profile



The content in this chapter should be relevant to about 3% of all questions about general chemistry on the MCAT.

This chapter covers material from the following AAMC content categories:

4E: Atoms, nuclear decay, electronic structure, and atomic chemical behavior

5A: Unique nature of water and its solutions

Introduction

You're on a night call in the emergency department (ED) when a 5-month-old infant patient's chart appears on your screen. You click through the triage notes and read what the mother reports: poor sucking ability and loss of head control and motor skills. You're puzzled by the findings and the previous ED admissions of lactic acidosis. You suspect diabetic ketoacidosis (DKA), liver, or kidney diseases—and possibly even poisoning—but nothing seems to fit. Minutes later, the child is brought into the examination room and does not stop crying. Over the course of an hour, another episode of lactic acidosis develops. The child is eventually admitted to the neonatal intensive care unit for long-term care.

Later, you ask the neonatologist about the patient. They point you to the charts and a genetic test performed shortly after birth. The child was diagnosed with Leigh's disease, an extremely rare mitochondrial disorder. In Leigh's disease, a number of key mitochondrial enzymes are disrupted and the process of oxidative phosphorylation is never achieved. Specifically, some of the most important enzymes that catalyze oxidation–reduction reactions, such as the pyruvate dehydrogenase complex and succinate dehydrogenase complex, are affected. When pyruvate cannot be oxidized to acetyl-CoA, it is instead fermented to lactic acid.

In biological systems, oxidation is coincident with the loss of electrons, sometimes in the form of hydrogen (dehydrogenation). The enzymes that catalyze these oxidations are called dehydrogenases. Many other macromolecules besides enzymes, such as vitamins, also carry out their functions by oxidizing or reducing other compounds. Iron in hemoglobin likewise undergoes rounds of oxidation and reduction as it carries oxygen from the lungs to tissues.

In this chapter, we focus our attention on the movement of electrons in chemical reactions. Such reactions are called oxidation–reduction (redox) reactions because they always occur in pairs. Oxidation–reduction reactions are particularly important because they tie into a number of topics in organic chemistry and biochemistry. In fact, Chapters 5 through 10 of MCAT Organic Chemistry Review and Chapters 9 through 11 of MCAT Biochemistry Review all touch on oxidation–reduction reactions in different sets of molecules.

11.1 Oxidation–Reduction Reactions

LEARNING GOALS

After Chapter 11.1, you will be able to:

- Separate a redox reaction into oxidation and reduction half-reactions
- Balance a redox reaction
- Identify the oxidizing agent, reducing agent, and relevant oxidation states for a given reaction: $\text{SnCl}_2 + \text{PbCl}_4 \rightarrow \text{SnCl}_4 + \text{PbCl}_2$

Reactions that involve the transfer of electrons from one chemical species to another can be classified as oxidation–reduction (redox) reactions.

OXIDATION AND REDUCTION

The law of conservation of charge states that electrical charge can be neither created nor destroyed. Thus, an isolated loss or gain of electrons cannot occur; oxidation (loss of electrons) and reduction (gain of electrons) must occur simultaneously, resulting in an electron transfer called a redox reaction. An oxidizing agent causes another atom in a redox reaction to

undergo oxidation and is itself reduced. A reducing agent causes the other atom to be reduced and is itself oxidized. There are various mnemonics to remember these terms, as highlighted in the sidebar.



MNEMONIC

Redox reactions: choose one of the mnemonics and stick with it!

- OIL RIG: Oxidation Is Loss of electrons, Reduction Is Gain of electrons.
- LEO the lion says GER: Loss of Electrons is Oxidation, Gain of Electrons is Reduction.
- LEORA says GEROA: Loss of Electrons is Oxidation (Reducing Agent), Gain of Electrons is Reduction (Oxidizing Agent).

Being familiar with some common oxidizing and reducing agents can save significant time on Test Day, especially in organic chemistry reactions. Some of the commonly used agents on the MCAT are listed in Table 11.1. Note that almost all oxidizing agents contain oxygen or another strongly electronegative element (such as a halogen). Reducing agents often contain metal ions or hydrides (H^-).

Oxidizing Agents	Reducing Agents
O_2	CO

* These oxidizing agents and reducing agents are commonly seen in organic chemistry reactions.

** These and other biochemical redox reagents often act as energy carriers in biochemistry reactions.

Oxidizing Agents	Reducing Agents
H_2O_2	C
$\text{F}_2, \text{Cl}_2, \text{Br}_2, \text{I}_2$ (halogens)	B_2H_6
H_2SO_4	Sn^{2+} and other pure metals
HNO_3	Hydrazine*
NaClO	Zn(Hg)^*
KMnO_4^*	Lindlar's catalyst*
$\text{CrO}_3, \text{Na}_2\text{Cr}_2\text{O}_7^*$	NaBH_4^*
Pyridinium chlorochromate (PCC)*	LiAlH_4^*
$\text{NAD}^+, \text{FADH}^{**}$	$\text{NADH}, \text{FADH}_2^{**}$

* These oxidizing agents and reducing agents are commonly seen in organic chemistry reactions.

** These and other biochemical redox reagents often act as energy carriers in biochemistry reactions.

Table 11.1. Common Oxidizing and Reducing Agents

Note that biochemical redox reagents such as NAD^+ tend to act as both oxidizing and reducing agents at different times during metabolic pathways. As such, they act as mediators of energy transfer during many metabolic processes, as shown in Figure 11.1.

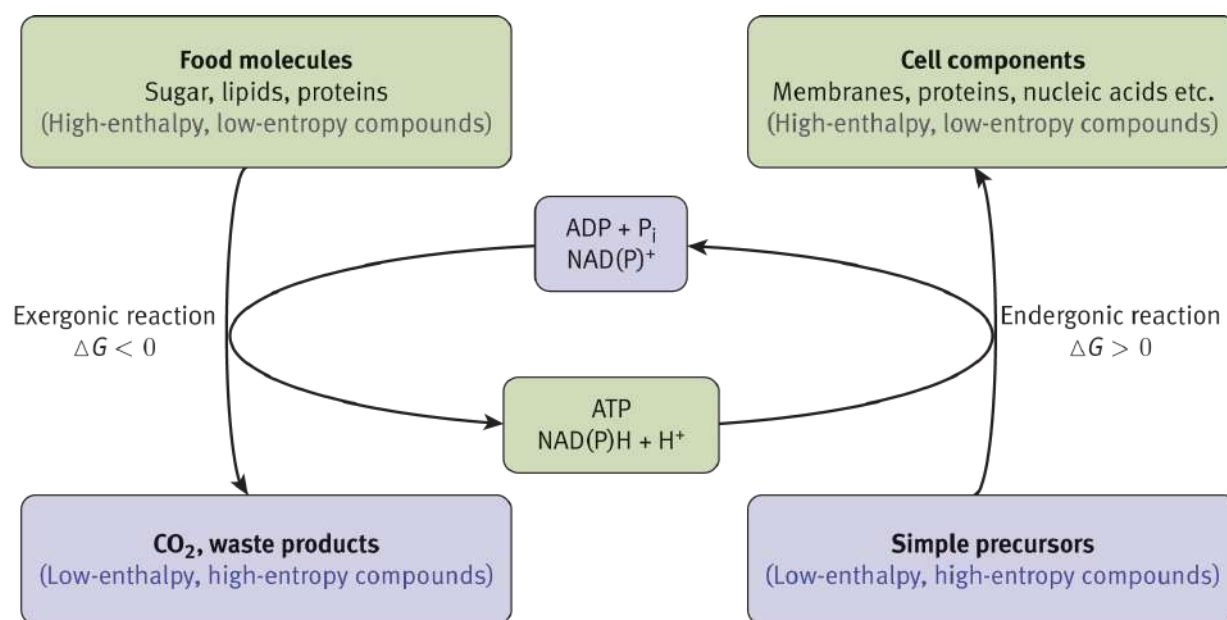


Figure 11.1. Oxidation and Reduction of Biochemical Compounds Serves as a Method of Energy Transfer

On a technical level, the term oxidizing agent or reducing agent is applied specifically to the atom that loses or gains electrons, respectively. However, many science texts will describe the compound as a whole (CrO₃, rather than Cr⁶⁺) as the oxidizing or reducing agent.

ASSIGNING OXIDATION NUMBERS

It is important, of course, to know which atom is oxidized and which is reduced. Oxidation numbers are assigned to atoms in order to keep track

of the redistribution of electrons during chemical reactions. Based on the oxidation numbers of the reactants and products, it is possible to determine how many electrons are gained or lost by each atom.



BRIDGE

In Chapter 3 of MCAT General Chemistry Review, we illustrated that metals form cations and nonmetals form anions. To form a cation, a metal must lose electrons. Therefore, metals like to get oxidized (lose electrons) and act as good reducing agents. Nonmetals, on the other hand, like to get reduced (gain electrons) and act as good oxidizing agents.

The oxidation number of an atom in a compound is assigned according to the following rules:

The oxidation number of a free element is zero. For example, the atoms in N_2 , P_4 , S_8 , and He all have oxidation numbers of zero.

The oxidation number for a monatomic ion is equal to the charge of the ion. For example, the oxidation numbers for Na^+ , Cu^{2+} , Fe^{3+} , Cl^- , and N^{3-} are +1, +2, +3, -1, and -3, respectively.

The oxidation number of each Group IA element in a compound is +1.

The oxidation number of each Group IIA element in a compound is +2.

The oxidation number of each Group VIIA element in a compound is -1, except when combined with an element of higher electronegativity. For example, in HCl , the oxidation number of Cl is -1; in HOCl , however, the oxidation number of Cl is +1.

The oxidation number of hydrogen is usually +1; however, its oxidation number is -1 in compounds with less electronegative elements (Groups IA and IIA). Hydrogen is +1 in HCl, but -1 in NaH.

In most compounds, the oxidation number of oxygen is -2. The two exceptions are peroxides (O_2^{2-}), for which the charge on each oxygen is -1, and compounds with more electronegative elements, such as OF_2 , in which oxygen has a +2 charge.

The sum of the oxidation numbers of all the atoms present in a neutral compound is zero. The sum of the oxidation numbers of the atoms present in a polyatomic ion is equal to the charge of the ion. Thus, for (SO_4^{2-}) , the sum of the oxidation numbers must be -2.



MCAT EXPERTISE

Think of the oxidation number as the typical charge of an element based on its group number, metallicity, and general location in the periodic table.



KEY CONCEPT

The conventions of formula writing put cation first and anion second. Thus HCl implies H^+ , and NaH implies H^- . Use the way the compound is written on the MCAT along with the periodic table to determine oxidation states.



MCAT EXPERTISE

Don't forget that you can click on "Periodic table" to pull it up on Test Day. Note the trends for assigning oxidation numbers; these general rules will help reduce the need to memorize individual oxidation numbers. Be aware that the transition metals can take on multiple oxidation states and therefore multiple oxidation numbers.

Oxidation number is often confused with formal charge, discussed in Chapter 3 of MCAT General Chemistry Review. Both account for the perceived charge on an element, but do so in different ways. Oxidation number assumes unequal division of electrons in bonds, "awarding" the electrons to the more electronegative element. Formal charge, on the other hand, assumes equal division of electrons in bonds, "awarding" one electron to each atom in the bond. In reality, the distribution of electron density lies somewhere between these two extremes. The assigning of oxidation number can be seen in Figure 11.2.

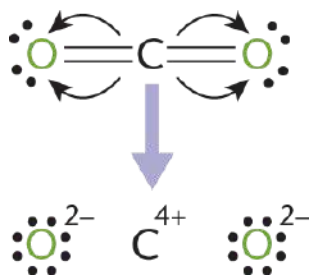
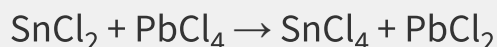


Figure 11.2. Assigning Oxidation Numbers to Carbon Dioxide

When assigning oxidation numbers, start with the known atoms (usually Group I and II, halides, and oxygen) and use this information to determine the oxidation states of the other atoms. Keep in mind that most transition metals can take on multiple oxidation states. When transition metals are oxidized or reduced, the absorption and emission of light from a metal is altered such that different frequencies are absorbed. For this reason, changes of oxidation state in transition metals usually correspond to a color change.

Example:

Assign oxidation numbers to the atoms in the following reaction to determine the oxidizing and reducing agents.



Solution:

All of these species are neutral, so the oxidation numbers of each compound must add up to zero. In SnCl_2 , tin must have an oxidation number of +2 because there are two chlorines present and each chlorine has an oxidation number of -1. Similarly, the oxidation number of Sn in SnCl_4 is +4; the oxidation number of Pb is +4 in PbCl_4 and +2 in PbCl_2 .

The oxidation number of Sn goes from +2 to +4; it loses electrons and thus is oxidized, making it the reducing agent. Because the oxidation number of Pb has decreased from +4 to +2, it gains

electrons and is reduced, making it the oxidizing agent. The sum of the charges on both sides of the reaction is equal to zero, so charge has been conserved. Keep in mind that oxidation state also plays a role in nomenclature; the reactants in this reaction would be called tin(II) chloride and lead(IV) chloride.

BALANCING OXIDATION–REDUCTION REACTIONS

By assigning oxidation numbers to the reactants and products, one can determine how many moles of each species are required for conservation of charge and mass, which is necessary to balance the equation. To balance a redox reaction, both the net charge and the number of atoms must be equal on both sides of the equation. The most common method for balancing redox equations is the half-reaction method, also known as the ion–electron method, in which the equation is separated into two half-reactions—the oxidation part and the reduction part. Each half-reaction is balanced separately, and they are then added to give a balanced overall reaction.

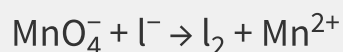


KEY CONCEPT

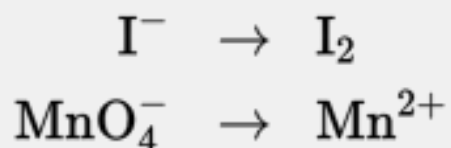
Oxidizing agents oxidize other molecules, but are themselves reduced. Reducing agents reduce other molecules, but are themselves oxidized. If you determine one ion to be an oxidizing agent then the other must be a reducing agent.

Example:

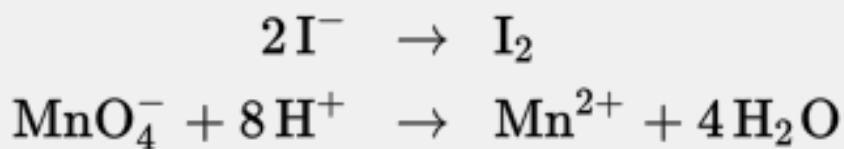
Balance this redox reaction using the half-reaction method:



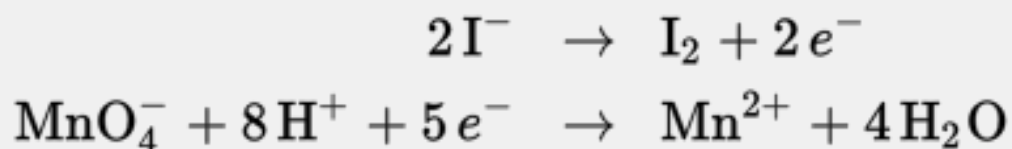
Step 1: Separate the two half-reactions.



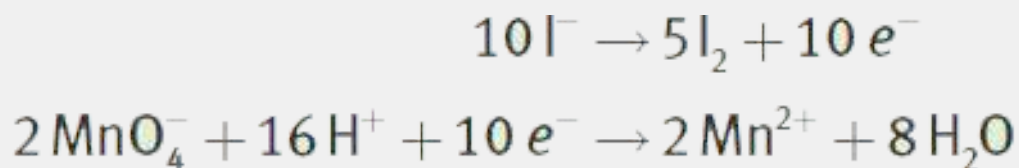
Step 2: Balance the atoms of each half-reaction. First, balance all atoms except H and O. Next, in an acidic solution, add H_2O to balance the O atoms and then add H^+ to balance the H atoms. In a basic solution, use OH^- and H_2O to balance the O and H atoms.



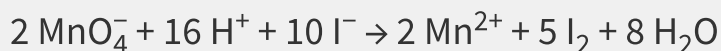
Step 3: Balance the charges of each half-reaction. Add electrons as necessary to one side of the reaction so that the charges are equal on both sides.



Step 4: Both half-reactions must have the same number of electrons so that they cancel each other out in the next step. In this example, you need to multiply the oxidation half-reaction by 5 and the reduction half-reaction by 2.



Step 5: Add the half-reactions, canceling out terms that appear on both sides of the reaction arrow.



Step 6: Confirm that mass and charge are balanced. There is a +4 net charge on each side of the reaction equation, and the atoms are stoichiometrically balanced.



MCAT EXPERTISE

Methodical, step-by-step approaches like the half-reaction method are great for the MCAT. Usually, you will not have to go through all of these steps before you can narrow down your answer choices and may be able to find the correct answer partway through the problem with a little critical thinking.

MCAT Concept Check 11.1:

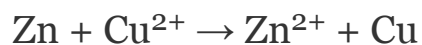
Before you move on, assess your understanding of the material with these questions.

For each of the reactions below, identify the oxidation states of the relevant atoms, the oxidizing agent, and the reducing agent:

Reaction	Oxidation Numbers	Oxidizing Agent	Reducing Agent
$2 \text{KI} + \text{H}_2 \rightarrow 2 \text{K} + 2 \text{HI}$			
$\text{Al} + \text{BPO}_4 \rightarrow \text{B} + \text{AlPO}_4$			



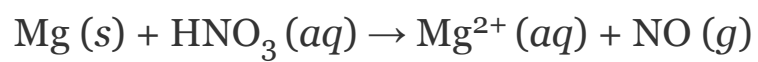
Identify the oxidation and reduction half-reactions in the following redox reaction:



- Oxidation:

- Reduction:
-

Balance the following redox reaction using the half-reaction method:



11.2 Net Ionic Equations

LEARNING GOALS

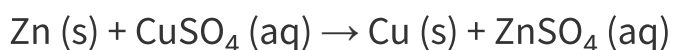
After Chapter 11.2, you will be able to:

- Identify the element undergoing disproportionation and the oxidation state of the products for a given reaction
- Apply redox reaction principles to balance and solve application-style problems, such as ones that involve redox titrations
- Determine the net ionic equation for a reaction: $\text{Zn (s)} + \text{CuSO}_4 \text{ (aq)} \rightarrow \text{Cu (s)} + \text{ZnSO}_4 \text{ (aq)}$

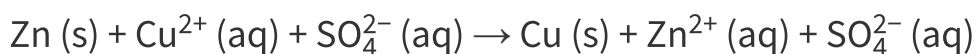
When we discussed reaction types in Chapter 4 of MCAT General Chemistry Review, we left out the rationale for why certain elements come together and others do not. Now that we have discussed oxidation–reduction reactions, it should be clearer that the gain and loss of electrons drives the formation of many compounds, especially ionic ones. Below, we will revisit many important reaction types and understand their basis in oxidation–reduction reactions.

OVERVIEW

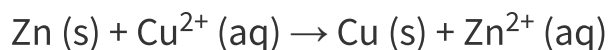
In our discussion of acids and bases, we focused only on the presence of protons and hydroxide ions, with little concern for which species actually provided those ions. Similarly, in redox reactions, our focus is on the shifting of electrons more so than the identities of the ions themselves. Consider the following single-displacement reaction:



If we split the various species into all of the ions present, we get the complete ionic equation:



Note that the sulfate ion is present on both sides of the equation in the same form; this ion is chemically inert during this reaction. In other words, the sulfate is not taking part in the overall reaction but simply remaining in the solution unchanged. We call such species spectator ions. Because the sulfate ion is not involved in the oxidation–reduction reaction, we can simplify the reaction to its net ionic equation, showing only the species that actually participate in the reaction:



When writing net ionic equations, all aqueous compounds should be split into their constituent ions. Solid salts, on the other hand, should be kept

together as a single entity. Let's return to some of the other reactions we have seen previously, including combination, decomposition, combustion, and double-displacement (metathesis) reactions.

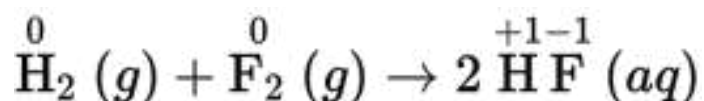


MCAT EXPERTISE

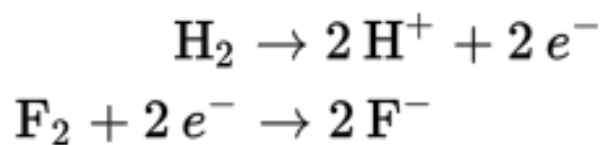
Look for compounds such as polyatomic anions that retain their charge before and after reactions; these are usually spectator ions and will not be found in the net ionic equation.

Combination Reactions

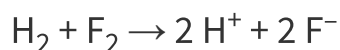
In combination reactions, two or more species come together to form a product. For example:



The relevant half-reactions would be:



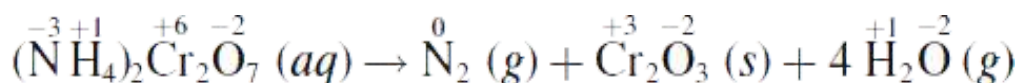
The net ionic equation is:



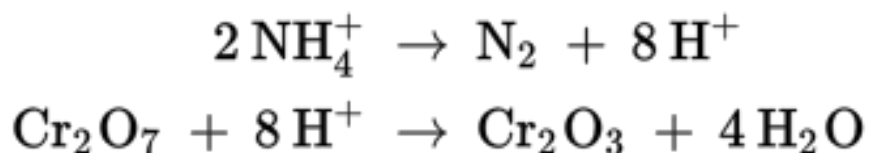
In this reaction, molecular hydrogen acts as a reducing agent as it is oxidized from 0 to +1. Molecular fluorine is the oxidizing agent as it is reduced from 0 to –1. In this reaction, there is no spectator ion.

Decomposition Reactions

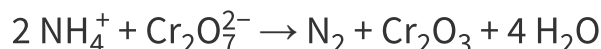
In decomposition reactions, one product breaks down into two or more species. For example:



The relevant half-reactions would be:



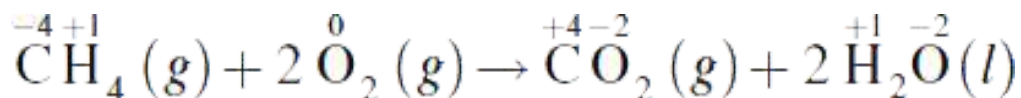
The net ionic equation is:



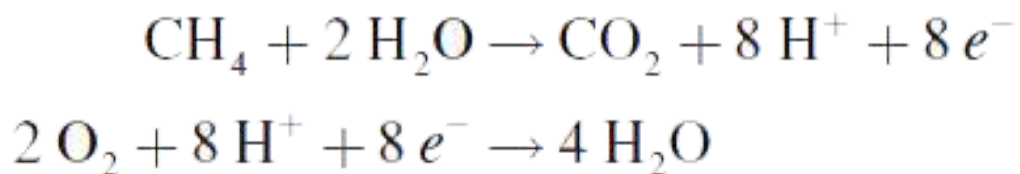
In this reaction, the nitrogen atom in the ammonium cation acts as a reducing agent as it is oxidized from –3 to 0. The chromium in the dichromate anion acts as the oxidizing agent as it is reduced from +6 to +3. In this reaction, there is no spectator ion. Note that the net ionic equation is not significantly different from the original balanced equation.

Combustion Reactions

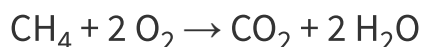
In combustion reactions, a fuel (usually a hydrocarbon) is mixed with an oxidant (usually oxygen), forming carbon dioxide and water. For example:



The relevant half-reactions would be:



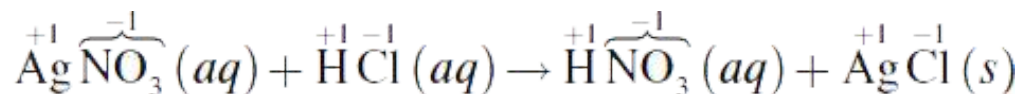
The net ionic equation is identical to the overall balanced equation because there are no spectator ions and no aqueous species:



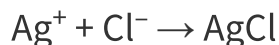
Combustion reactions can have complex half-reactions, depending on the type of fuel used. In this instance, carbon in methane is the reducing agent as it is oxidized from -4 to $+4$. Molecular oxygen is the oxidizing agent as it is reduced from 0 to -2 .

Double-Displacement (Metathesis) Reactions

Double-displacement or metathesis reactions involve the switching of counterions. Because all ions generally retain their oxidation state, these are not usually oxidation–reduction reactions. For example:

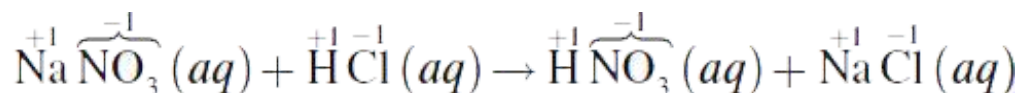


Because all species retain the same oxidation numbers, this is not considered oxidation–reduction. The net ionic reaction would be:

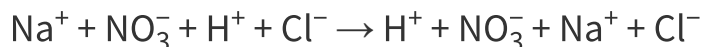


The nitrate anion and hydrogen cation both act as spectator ions in this reaction.

In double-displacement reactions where both reactants and both products are aqueous, there is no net ionic reaction. For example:



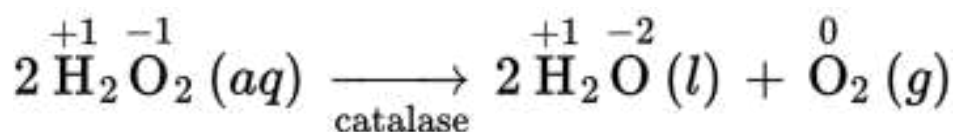
This reaction is not an oxidation–reduction reaction because no species change their oxidation states. Further, because all species are aqueous, the complete ionic reaction is:



Because all of the ions appear on both sides of the reaction, there is no net ionic reaction.

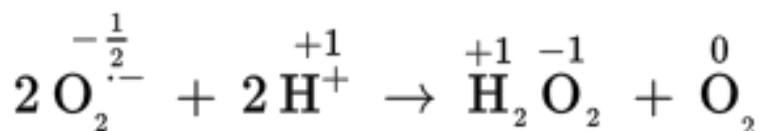
DISPROPORTIONATION REACTIONS

Disproportionation (or dismutation) is a specific type of redox reaction in which an element undergoes both oxidation and reduction in producing its products. Many biological enzymes utilize a disproportionation mechanism. An example of such a reaction is the catalysis of peroxides by catalase, an enzyme found in peroxisomes. Catalase is a critical biological enzyme used to protect cells from excessive oxidation by free radicals or reactive oxygen species. The activity of catalase can be seen when disinfecting a wound with hydrogen peroxide:



As can be seen from this reaction, oxygen is disproportioned between water and molecular oxygen. In hydrogen peroxide, each oxygen has an oxidation state of -1 (the peroxide ion has a charge of -2 overall). In water, oxygen has an oxidation state of -2 , and in molecular oxygen, it has an oxidation state of 0 . Therefore, the oxygen is both reduced and oxidized in this reaction.

Another related biological disproportionation mechanism is that of the enzyme superoxide dismutase. As the name implies, a dismutase catalyzes dismutation. This enzyme disproportionates oxygen free radicals in the reaction



where peroxide and oxygen are the disproportionated products. The oxidation state of oxygen in the free radical is $+\frac{1}{2}$ (a negative charge divided over two oxygen atoms), and it is reduced to -1 in the peroxide and oxidized to 0 in molecular oxygen.

Biochemical disproportionation reactions—and oxidation–reduction reactions in biological systems in general—are usually accomplished by enzymes. Structurally, these enzymes often have metals such as Cu and Zn in their active sites that act as reducing agents, as shown in Figure 11.3.

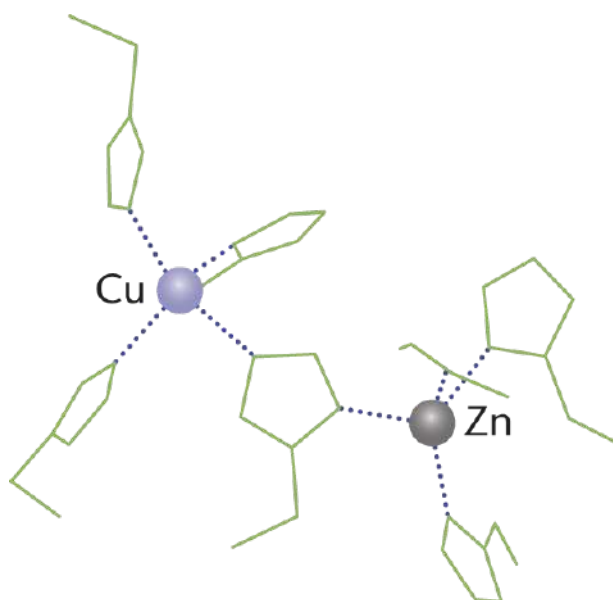


Figure 11.3. Active Site of Superoxide Dismutase

Cu and Zn atoms act as reducing agents, losing electrons during catalysis. The atoms are stabilized in position by histidine residues.

OXIDATION–REDUCTION TITRATIONS

Oxidation–reduction titrations are similar in setup to acid–base titrations, but the focus is different. Whereas acid–base titrations follow the

movement of protons, as discussed in Chapter 10 of MCAT General Chemistry Review, redox titrations follow the transfer of charge (as electrons) to reach the equivalence point. Redox titrations can utilize indicators that change color at a particular voltage (emf) value. Some common indicators are listed in Table 11.2. As for acid–base titrations, it is not necessary to memorize these indicators, but rather to understand their utility.

Indicator	Voltage of Color Change	Oxidized Form	Reduced Form
Bipyridine metal complexes	About +1 V	Colorless (Ru), Cyan (Fe)	Yellow (Ru), Red (Fe)
Diphenylamine	+0.76 V	Violet	Colorless
Safranin	+0.24 / -0.29 V*	Red-Violet	Colorless

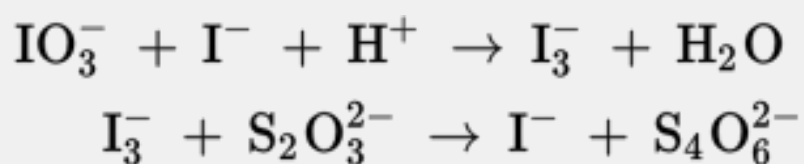
* Safranin is unique in that its color change is not only voltage dependent, but also pH dependent.

Table 11.2. Common Indicators for Oxidation–Reduction (Redox) Titrations

One prototypical redox titration involves the use of starch indicators to identify iodine complexes. This specific redox titration is called an iodimetric titration because it relies on the titration of free iodine radicals. The presence of iodine is initially determined by a dark solution in the presence of starch, and at the endpoint of the titration, a colorless solution develops. A common general chemistry laboratory experiment involves the standardization of a thiosulfate solution using iodimetry, as described in the example below.

Example:

A group of students prepares to standardize a $\text{Na}_2\text{S}_2\text{O}_3$ solution. 32 mL of the $\text{Na}_2\text{S}_2\text{O}_3$ solution is titrated into 50 mL of a 0.01 M KIO_3 solution to reach the equivalence point. They first titrate the KIO_3 solution until it loses color, then add a starch indicator until the reaction is complete. The reaction proceeds in these two steps:

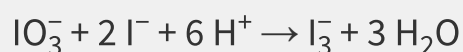


Determine the concentration of the sodium thiosulfate solution at the beginning of the experiment.

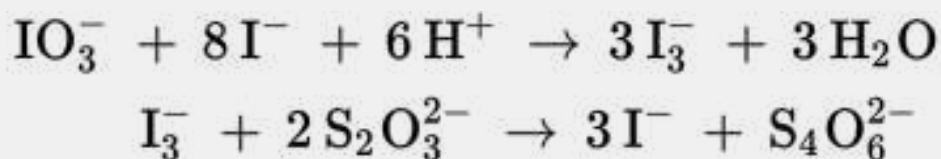
Solution:

The titration is performed in two steps. In the first step, the iodate is converted into triiodide anions. The initial titration creates a colorless I_3^- solution in water, which then requires a starch indicator for the remainder of the titration.

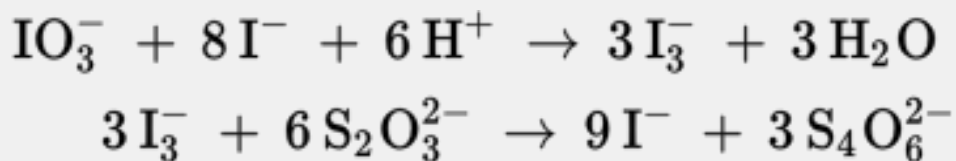
In the second step, the triiodide ions are then reduced in the presence of thiosulfate to determine its concentration. Note that the reactions are unbalanced. It would be tempting to balance the first equation as



However, this reaction has a charge of +3 on the reactants side and -1 on the products side. In oxidation-reduction chemistry, we must balance not only for stoichiometry but also for charge. The correct balanced equations are:



Keeping in mind that each iodate is used to make three triiodide anions, and each triiodide anion interacts with two thiosulfate anions, the mole ratio of thiosulfate to iodate is six to one:



Then, use stoichiometry to determine the molarity of the thiosulfate solution from the 50 mL potassium iodate solution.

$$0.01 \text{ M IO}_3^- [0.05 \text{ L}] \left[\frac{6 \text{ mol S}_2\text{O}_3^{2-}}{1 \text{ mol IO}_3^-} \right] = 3 \times 10^{-3} \text{ mol}$$

Now, we can use the volume to find the molarity.

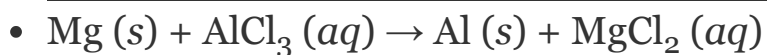
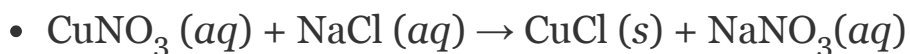
$$M_{\text{S}_2\text{O}_3^{2-}} = \frac{3 \times 10^{-3} \text{ mol}}{32 \text{ mL} \left[\frac{1 \text{ L}}{1000 \text{ mL}} \right]} = \frac{3 \times 10^{-3}}{32 \times 10^{-3}} \approx \frac{3}{32} = 0.094 \text{ M (actual = 0.1 M)}$$

Potentiometric titration is a form of redox titration where no indicator is used. Instead, the electrical potential difference (voltage) is measured using a voltmeter. As a redox titration progresses, its voltage changes; this is analogous to following an acid–base titration with a pH meter instead of a color indicator.

MCAT Concept Check 11.2:

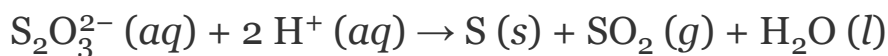
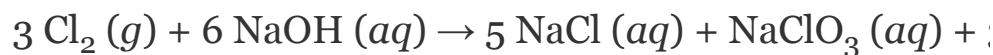
Before you move on, assess your understanding of the material with these questions.

Write the net ionic equations for the reactions below:



In each of the reactions below, which element undergoes disproportionation? What are that element's oxidation

states in the products?



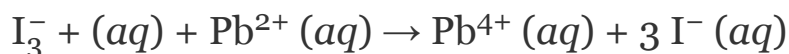
- Element undergoing disproportionation:

-
- Oxidation states in products:

-
- Element undergoing disproportionation:

-
- Oxidation states in products:
-

A sample is assayed for lead by a redox titration with $\text{I}_3^- (aq)$. A 10.00 g sample is crushed, dissolved in sulfuric acid, and passed over a reducing agent so that all the lead is in the form Pb^{2+} . The $\text{Pb}^{2+} (aq)$ is completely oxidized to Pb^{4+} by 32.60 mL of a 0.7 M solution of NaI_3 . The balanced equation for the reaction is:



Calculate the mass of lead in the sample.



Conclusion

In this chapter, we covered the essential MCAT topic of oxidation–reduction reactions. We reviewed the rules for assigning oxidation numbers to help us keep track of the movement of electrons from the species that are oxidized (reducing agents) to the species that are reduced (oxidizing agents). We also covered the sequence of steps involved in balancing half-reactions, redox titrations, and disproportionation reactions.

In addition to understanding the fundamental chemical principles behind these reactions, you will begin to see these concepts resurface in MCAT Organic Chemistry Review and MCAT Biochemistry Review. Oxidation–reduction reactions are often used for energy transfer in biological systems, and any deficiencies in such systems are profoundly deleterious (such as metabolic, mitochondrial, and immunologic diseases). Our next chapter—the last of MCAT General Chemistry Review—brings the principles of oxidation–reduction reactions to their application in electrochemical cells. By the end of the next chapter, you will have reviewed all of the general chemistry knowledge required for Test Day!

Concept Summary

Oxidation–Reduction Reactions

- Oxidation is a loss of electrons, and reduction is a gain of electrons; the two are paired together in what is known as an oxidation–reduction (redox) reaction.
- An oxidizing agent facilitates the oxidation of another compound and is reduced itself in the process; a reducing agent facilitates the reduction of another compound and is itself oxidized in the process.
 - Common oxidizing agents almost all contain oxygen or a similarly electronegative element.
 - Common reducing agents often contain metal ions or hydrides (H^-).
- To assign oxidation numbers, one must know the common oxidation states of the representative elements.
 - Any free element or diatomic species has an oxidation number of zero.
 - The oxidation number of a monatomic ion is equal to the charge of the ion.
 - When in compounds, Group IA metals have an oxidation number of +1; Group IIA metals have an oxidation number of +2.

- When in compounds, Group VIIA elements have an oxidation number of -1 (unless combined with an element with higher electronegativity).
 - The oxidation state of hydrogen is $+1$ unless it is paired with a less electronegative element, in which case it is -1 .
 - The oxidation state of oxygen is usually -2 , except in peroxides (when its charge is -1) or in compounds with more electronegative elements.
 - The sum of the oxidation numbers of all the atoms present in a compound is equal to the overall charge of that compound.
- When balancing redox reactions, the half-reaction method, also called the ion–electron method, is the most common.
 - Separate the two half-reactions.
 - Balance the atoms of each half-reaction. Start with all the elements besides H and O. In acidic solution, balance H and O using water and H^+ . In basic solution, balance H and O using water and OH^- .
 - Balance the charges of each half-reaction by adding electrons as necessary to one side of the reaction.
 - Multiply the half-reactions as necessary to obtain the same number of electrons in both half-reactions.
 - Add the half-reactions, canceling out terms on both sides of the reaction arrow.
 - Confirm that the mass and charge are balanced.

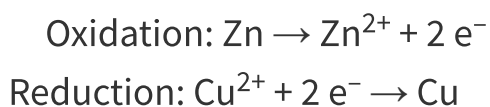
Net Ionic Equations

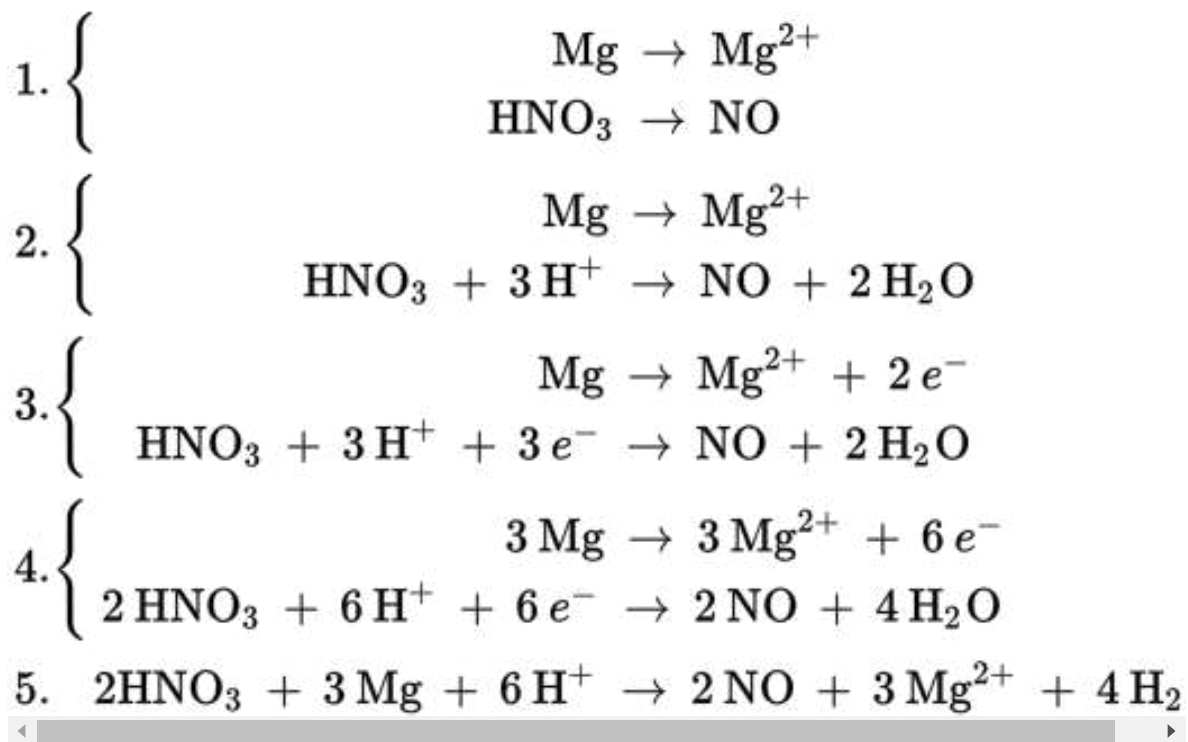
- A complete ionic equation accounts for all of the ions present in a reaction. To write a complete ionic reaction, split all aqueous compounds into their relevant ions. Keep solid salts intact.
- Net ionic equations ignore spectator ions to focus only on the species that actually participate in the reaction. To obtain a net ionic reaction, subtract the ions appearing on both sides of the reaction, which are called spectator ions.
 - For reactions that contain no aqueous salts, the net ionic equation is generally the same as the overall balanced reaction.
 - For double displacement (metathesis) reactions that do not form a solid salt, there is no net ionic reaction because all ions remain in solution and do not change oxidation number.
- Disproportionation (dismutation) reactions are a type of redox reaction in which one element is both oxidized and reduced, forming at least two molecules containing the element with different oxidation states.
- Oxidation–reduction titrations are similar in methodology to acid–base titrations. These titrations follow transfer of charge.
 - Indicators used in such titrations change color when certain voltages of solutions are achieved.
 - Potentiometric titration is a form of redox titration in which a voltmeter or external cell measures the electromotive force (emf) of a solution. No indicator is used, and the equivalence point is determined by a sharp change in voltage.

Answers to Concept Checks

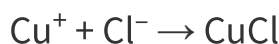
11.1

Reaction	Oxidation Numbers	Oxidizing Agent	Reducing Agent
$2 \text{KI} + \text{H}_2 \rightarrow 2 \text{K} + 2 \text{HI}$	$\overset{+1}{2} \overset{-1}{\text{K} \text{I}} + \overset{0}{\text{H}_2} \rightarrow \overset{0}{2 \text{K}} + \overset{+1}{2} \overset{-1}{\text{H} \text{I}}$	K^+ (charge goes from +1 to 0)	H_2 (charge goes from 0 to +1)
$\text{Al} + \text{BPO}_4 \rightarrow \text{B} + \text{AlPO}_4$	$\overset{0}{\text{Al}} + \overset{+3}{\text{B}} \overset{-3}{\text{PO}_4} \rightarrow \overset{0}{\text{B}} + \overset{+3}{\text{Al}} \overset{-3}{\text{PO}_4}$	B^{3+} (charge goes from +3 to 0)	Al (charge goes from 0 to +3)





11.2



In the first reaction, chlorine undergoes disproportionation to have a -1 oxidation state in NaCl and a +5 oxidation state in NaClO₃.

In the second reaction, sulfur undergoes disproportionation to have a 0 oxidation state in elemental sulfur and +4 oxidation state in SO₂.

$$0.7 \text{ M I}_3^- \left[32.60 \times 10^{-3} \text{ L} \right] \left[\frac{1 \text{ mol Pb}^{2+}}{1 \text{ mol I}_3^-} \right] \left[\frac{207.2 \text{ g Pb}^{2+}}{1 \text{ mol Pb}^{2+}} \right] \approx 0.7 \times 3 \times 2 = 4.2$$

(actual = 4.73 g)

Note that question 3 also included the extraneous value 10.0 g, which is not needed to calculate the mass of lead produced.



Shared Concepts

Biochemistry Chapter 2

Enzymes

Biochemistry Chapter 10

Carbohydrate Metabolism II

Biology Chapter 7

The Cardiovascular System

General Chemistry Chapter 10

Acids and Bases

General Chemistry Chapter 12

Electrochemistry

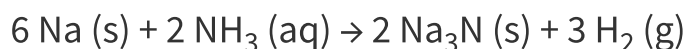
Organic Chemistry Chapter 4

Analyzing Organic Reactions

Discrete Practice Questions

Consult your online resources for additional practice.

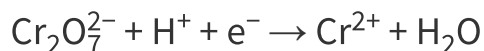
1. Consider the following equation:



Which species acts as an oxidizing agent?

- (A) Na
- (B) N in NH_3
- (C) H in NH_3
- (D) H_2

2. How many electrons are involved in the following half-reaction after it is balanced?



- (A) 2
- (B) 8
- (C) 12
- (D) 16

3. Lithium aluminum hydride (LiAlH_4) is often used in laboratories because of its tendency to donate a hydride ion. Which of the following roles would lithium aluminum hydride likely play in a reaction?

- (A) Strong reducing agent only
- (B) Strong oxidizing agent only
- (C) Both a strong reducing agent and strong oxidizing agent
- (D) Neither a strong reducing agent nor a strong oxidizing agent

4. What is the oxidation number of chlorine in NaClO ?

- (A) -1
- (B) 0
- (C) +1
- (D) +2

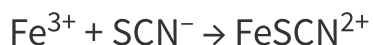
5. The following electronic configurations represent elements in their neutral form. Which element is the strongest oxidizing agent?

- (A) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
- (B) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5$
- (C) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^1$
- (D) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^5$

6. Which of the following is the correct net ionic reaction for the reaction of copper with silver(I) nitrate?

- (A) $\text{Cu} + \text{AgNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{Ag}$
- (B) $\text{Cu} + 2 \text{Ag}^+ + 2 \text{NO}_3^- \rightarrow \text{Cu}^{2+} + 2 \text{NO}_3^- + 2 \text{Ag}$
- (C) $2 \text{Ag}^+ + 2 \text{NO}_3^- \rightarrow 2 \text{NO}_3^- + 2 \text{Ag}^+$
- (D) $\text{Cu} + 2 \text{Ag}^+ \rightarrow \text{Cu}^{2+} + 2 \text{Ag}$

7. One way to test for the presence of iron in solution is by adding potassium thiocyanate to the solution. The product when this reagent reacts with iron is FeSCN^{2+} , which creates a dark red color in solution via the following net ionic equation:



How many grams of iron sulfate would be needed to produce 2 moles of FeSCN^{2+} ?

- (A) 110 g
- (B) 220 g
- (C) 400 g
- (D) 500 g

8. During the assigning of oxidation numbers, which of the following elements would most likely be determined last?

- (A) Ar
- (B) F
- (C) Sr
- (D) Ir

9. As methanol is converted to methanal, and then methanoic acid, the oxidation number of the carbon:

- (A) increases.
- (B) decreases.
- (C) increases, then decreases.
- (D) decreases, then increases.

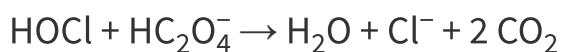
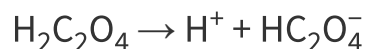
10. In the compound KH_2PO_4 , which element has the highest oxidation number?

- (A) K
- (B) H
- (C) P
- (D) O

11. If a certain metal has multiple oxidation states, its acidity as an oxide generally increases as the oxidation state increases. Therefore, which of the following tungsten compounds is likely to be the strongest acid?

- (A) WO_2
- (B) WO_3
- (C) W_2O_3
- (D) W_2O_5

12. Consider the following steps in the reaction between oxalic acid and chlorine:



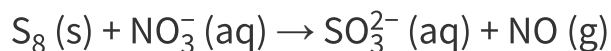
Which of these steps, occurring in aqueous solution, is an example of a disproportionation reaction?

- (A) I only
- (B) III only
- (C) I and III only
- (D) I, II, and III

13. Potentiometry in an oxidation–reduction titration is analogous to performing an acid–base titration with a(n):

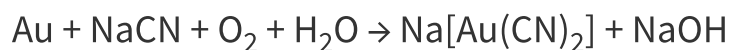
- (A) acidic indicator.
- (B) basic indicator.
- (C) pH meter.
- (D) oxidizing agent.

14. After balancing the following oxidation–reduction reaction, what is the sum of the stoichiometric coefficients of all of the reactants and products?



- (A) 4
- (B) 50
- (C) 91
- (D) 115

15. An assay is performed to determine the gold content in a supply of crushed ore. One method for pulling gold out of ore is to react it in a concentrated cyanide (CN^-) solution. The equation is provided below:



An indicator is used during this reaction, and approximately 100 mL of a 2 M NaCN solution is used to reach the endpoint. How many moles of Au are present in the crushed ore?

- (A) 0.01 mol
- (B) 0.02 mol
- (C) 0.10 mol
- (D) 0.20 mol

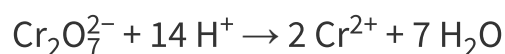
Explanations to Discrete Practice Questions

1. C

The oxidizing agent is the species that is reduced in any given equation. In this problem, six hydrogen atoms with +1 oxidation states in NH_3 are reduced to three neutral H_2 molecules.

2. B

First, balance the atoms in the equation:



Now, adjust the number of electrons to balance the charge. Currently, the left side has a charge of +12 (−2 from dichromate and +14 from protons). The right side has a charge of +4 (+2 from each chromium cation). To decrease the charge on the left side from +12 to +4, we should add 8 electrons:



3. A

Hydride ions are composed of a hydrogen nucleus with two electrons, thereby giving it a negative charge and a considerable tendency to donate

electrons. LiAlH_4 is therefore a strong reducing agent. Strong reducing agents tend to have metals or hydrides; strong oxidizing agents tend to have oxygen or a similarly electronegative element.

4. C

In NaClO (sodium hypochlorite), sodium carries its typical +1 charge, and oxygen carries its typical -2 charge. This means that the chlorine atom must carry a +1 charge in order to balance the overall charge of zero.

5. D

A strong oxidizing agent will be easily reduced, meaning that it will have a tendency to gain electrons. Atoms usually gain electrons if they are one or two electrons away from filling up their valence shell. (A) has a full 4s-orbital, meaning that it can only gain an electron if it gains an entire subshell. (B) has a stable, half-full 3d-orbital, so it is unlikely to pick up electrons unless it can gain five. (C) has only a single electron in the outer shell, which is more likely lost upon ionization. (D) would fill up its 4p-orbital by gaining one electron, so it is easily reduced.

6. D

A net ionic equation represents each of the aqueous ions comprising the reactants and products as individual ions, instead of combining them as formula units. Thus, (A) is not a net ionic reaction. The term net means that the correct answer does not include any spectator ions (ions that do not participate in the reaction). In this reaction, nitrate (NO_3^-) remains unchanged. Therefore, (B) and (C) are eliminated.

7. C

What you are shown is a net ionic equation. If two moles of FeSCN are created, two moles of Fe^{3+} must be used because the mole ratio is 1:1. Iron

sulfate has the formula $\text{Fe}_2(\text{SO}_4)_3$ because sulfate has a charge of -2 and iron has a charge of $+3$ (based on the net ionic equation). Therefore, one mole of iron sulfate is needed to make two moles of iron for the reaction. The molar mass of iron sulfate is

$$2 \times 55.8 \frac{\text{g}}{\text{mol}} + 3 \times 32.1 \frac{\text{g}}{\text{mol}} + 12 \times 16.0 \frac{\text{g}}{\text{mol}} = 399.9 \frac{\text{g}}{\text{mol}}$$

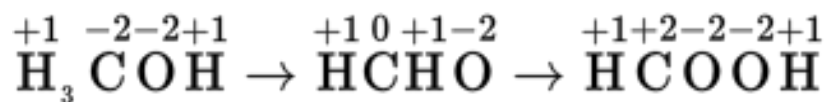
This most closely matches answer (C). The most common error would be to calculate the amount of iron, which would be 111.6 g , (A).

8. D

When assigning oxidation numbers, one starts with elements of known oxidation state first, and determines the oxidation state of the other elements by deduction. As a noble gas, argon, (A), will always have an oxidation state of 0. As a Group VIIA element, fluorine, (B), will have an oxidation state of 0 (by itself) or -1 (in a compound). As a Group IIA element, strontium, (C), will have an oxidation state of 0 (by itself) or $+2$ (in a compound). Like most transition metals, iridium, (D), can have various oxidation states, ranging from -3 to $+8$. Therefore, one would have to determine the oxidation states of other atoms in an iridium-containing compound to determine iridium's oxidation number.

9. A

The formula for methanol is H_3COH , for methanal is HCHO , and for methanoic acid is HCOOH . If we assign oxidation numbers to carbon in each molecule, it starts at -2 , then becomes 0, then becomes $+2$:



In general, it is often easier to think of oxidation as a gain of bonds to oxygen (or a similarly electronegative element) or loss of bonds to hydrogen for organic compounds. Therefore, because the carbon is oxidized as one converts from an alcohol to an aldehyde to a carboxylic acid, the oxidation number must increase.

10. C

Start with the atoms that have oxidation states of which you are certain. Potassium is a Group IA metal, and therefore must have an oxidation state of +1. Hydrogen is almost always +1, unless it is paired with a less electronegative element (which is not the case here). Oxygen is generally -2. Because there are four oxygens, they create a total negative charge of -8 which is partially balanced by two hydrogens (+2) and potassium (+1). Therefore, phosphorus has a +5 charge, making it the highest oxidation state.

11. B

Recall that oxygen has an oxidation state of -2. Therefore, in tungsten(IV) oxide, (A), tungsten has an oxidation state of +4. In tungsten(VI) oxide, (B), it has an oxidation state of +6. In tungsten(III) oxide, (C), it is +3. In tungsten pentoxide, (D), it is +5.

12. A

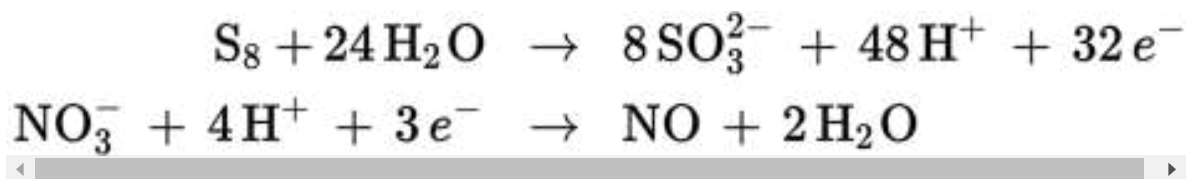
Step I is a disproportionation reaction because chlorine starts with an oxidation state of 0 in the reactants and ends up with an oxidation state of +1 in HOCl and -1 as Cl^- . In the other reactions, no element appears with different oxidation states in two different products. Therefore, only step I is a disproportionation reaction.

13. C

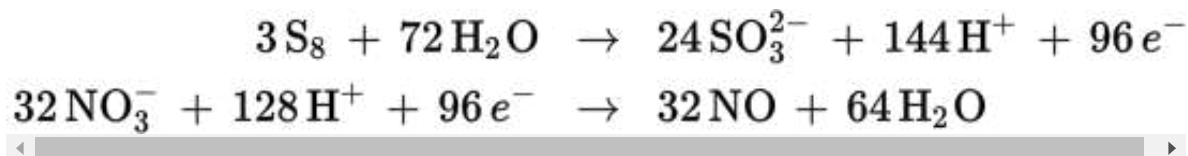
Potentiometry refers to carrying out an oxidation–reduction titration with a voltmeter present to get precise readings of the reaction’s electromotive force (emf) to determine the endpoint. This is analogous to using a pH meter in an acid–base titration because it uses technology to get precise readings for plotting a titration curve. Indicators, as in (A) and (B), can be used in both acid–base and redox titrations, but provide a qualitative (rather than quantitative) analysis of the titration. Oxidizing and reducing agents are used in redox titrations, not acid–base titrations, eliminating (D).

14. D

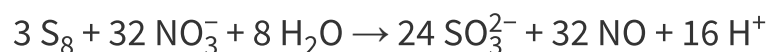
Utilize the method described earlier to balance this redox reaction. The balanced half-reactions are:



To get equal numbers of electrons in each half-reaction, the oxidation half-reaction will have to be multiplied by 3, and the reduction half-reaction will have to be multiplied by 32:



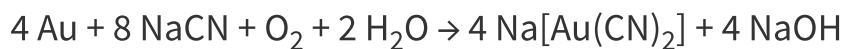
This makes the overall reaction:



The sum of the stoichiometric coefficients is therefore $3 + 32 + 8 + 24 + 32 + 16 = 115$.

15. C

First, balance the chemical equation:



Now, determine the number of moles of NaCN used in the reaction:

$$0.1 \text{ L} \times 2 \frac{\text{mol}}{\text{L}} = 0.2 \text{ mol NaCN}$$

If 0.2 mol NaCN are used in the reaction, then $0.2 \text{ mol NaCN} \times$

$$\frac{4 \text{ mol Au}}{8 \text{ mol NaCN}} = 0.1 \text{ mol Au is oxidized.}$$

12

Electrochemistry



In This Chapter

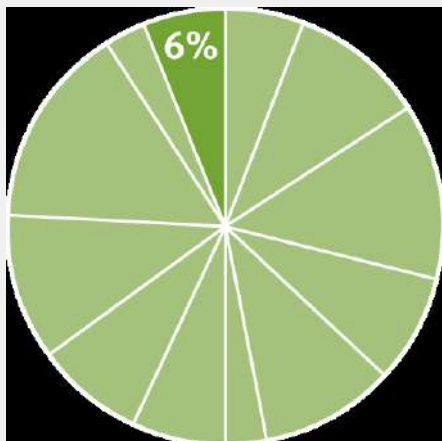
- 12.1 [Electrochemical Cells](#) 
 - [Galvanic \(Voltaic\) Cells](#) 
 - [Electrolytic Cells](#) 
 - [Concentration Cells](#) 
 - [Rechargeable Cells](#) 
 - [Electrode Charge Designations](#) 

- 12.2 [Cell Potentials](#) 
 - [Reduction Potentials](#) 
 - [The Electromotive Force](#) 

- 12.3 [Electromotive Force and Thermodynamics](#) 
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Chapter Profile



The content in this chapter should be relevant to about 6% of all questions about general chemistry on the MCAT.

This chapter covers material from the following AAMC content categories:

1D: Principles of bioenergetics and fuel molecule metabolism

3A: Structure and functions of the nervous and endocrine systems and ways in which these systems coordinate the organ systems

4C: Electrochemistry and electrical circuits and their elements

5E: Principles of chemical thermodynamics and kinetics

Introduction

The mitochondria are powerhouses of energy. Their primary purpose is to manufacture a deliverable and usable form of energy. By now, you are well aware of the complex processes by which the potential energy in the chemical bonds of carbohydrates, amino acids, and lipids is converted into the potential energy of the phosphate bond in adenosine triphosphate (ATP). ATP is then delivered to different parts of the cell, where it is used to energize most of the processes essential to the maintenance of life.

The mitochondria generate tremendous amounts of ATP—in humans, the average daily turnover of ATP is more than 50 kilograms! Without a continuous supply and replenishment of ATP, we wouldn't survive. ATP powers the contraction of our heart muscle and maintains the membrane potential essential for neurological function, among thousands of other essential roles. How do the mitochondria manufacture these packets of life-sustaining energy? Remember that mitochondria rely on their double-membrane structure to carry out the electron transport chain and oxidative phosphorylation. As such, mitochondria truly act as batteries of the cell. In fact, note the similarity between the proton-motive force of the mitochondria and the electromotive force of electrochemistry. Are these two terms the same thing or—at the very least—similar in nature?

Indeed, mitochondria and batteries do function in similar ways. Specifically, mitochondria function most similarly to concentration cells. In both concentration cells and mitochondria, a concentration gradient of ions between two separated compartments—connected to each other by some means of charge conduction—establishes an electrical potential

difference (voltage). This voltage, called electromotive force in concentration cells and proton-motive force in the mitochondria, provides the drive to move charge from one compartment to the other, creating current. In the concentration cell, an oxidation–reduction reaction takes place, and electrons move in the direction that causes the concentration gradient to be dissipated. In the mitochondria, the charge buildup is in the form of a hydrogen ion (proton) gradient between the intermembrane space and the matrix. Embedded in the inner membrane is ATP synthase, which serves a dual role as a proton channel and a catalyst for the formation of the high-energy phosphate bond in ATP. As the hydrogen ions flow down their electrochemical gradient, energy is dissipated, and this energy is harnessed by ATP synthase to form ATP.

In this final chapter of MCAT General Chemistry Review, we will focus our attention on the study of various electrochemical cells. Utilizing our knowledge of oxidation–reduction reactions from Chapter 11, we will study how these principles can be applied to create different types of electrochemical cells, including galvanic (voltaic), electrolytic, and concentration cells. Regarding the thermodynamics of electrochemistry, we will focus on the significance of reduction potentials and examine the relationship between electromotive force, the equilibrium constant, and Gibbs free energy.

12.1 Electrochemical Cells

LEARNING GOALS

After Chapter 12.1, you will be able to:

- Distinguish between electrolytic and galvanic cells
- Describe electrolytic and galvanic cells
- Predict which electrode will act as the cathode or anode in an electrolytic or a galvanic cell
- Calculate ΔG and emf values for a given galvanic or electrolytic cell
- Apply the Nernst equation to electrochemical cell questions

Electrochemical cells are contained systems in which oxidation–reduction reactions occur. There are three fundamental types of electrochemical cells: galvanic cells (also known as voltaic cells), electrolytic cells, and concentration cells. In addition, there are specific commercial cells such as Ni–Cd batteries through which we can understand these fundamental models.

Galvanic cells and concentration cells house spontaneous reactions, whereas electrolytic cells contain nonspontaneous reactions. Remember

that spontaneity is indicated by the change in Gibbs free energy, ΔG . All three types contain electrodes where oxidation and reduction take place. For all electrochemical cells, the electrode where oxidation occurs is called the anode, and the electrode where reduction occurs is called the cathode. Other descriptors of electrochemical cells include the electromotive force (emf), which corresponds to the voltage or electrical potential difference of the cell. If the emf is positive, the cell is able to release energy ($\Delta G < 0$), which means it is spontaneous. If the emf is negative, the cell must absorb energy ($\Delta G > 0$), which means it is nonspontaneous.



MNEMONIC

Electrodes in a electrochemical cell: AN OX and a RED CAT

The anode is the site of oxidation; reduction occurs at the cathode.

Furthermore, we can also state that, for all electrochemical cells, the movement of electrons is from anode to cathode, and the current (I) runs from cathode to anode. This point can be a point of confusion among students. In physics, it is typical to state that current is the direction of flow of a positive charge through a circuit; this model was first proposed by Ben Franklin and continues to be used among physicists. Modern chemists are interested in the flow of electrons, but may discuss the current (a theoretical flow of positive charge) as a proxy for the flow of electrons; the current and the flow of electrons are always of equal magnitude but in opposite directions.



KEY CONCEPT

Electrons move through an electrochemical cell opposite to the flow of current (I).

Last, it is important to note that all batteries are influenced by temperature changes. For instance, lead–acid batteries in cars, like most galvanic cells, tend to fail most in cold weather. The thermodynamic reasons behind this will be discussed later in this chapter.

GALVANIC (VOLTAIC) CELLS

All of the nonrechargeable batteries you own are galvanic cells, also called voltaic cells. Accordingly, because household batteries are used to supply energy to a flashlight or remote control, the reactions in these cells must be spontaneous. This means that the reaction's free energy is decreasing ($\Delta G < 0$) as the cell releases energy to the environment. By extension, if the free energy change is negative for these cells, their electromotive force (E_{cell}) must be positive; the free energy change and electromotive force always have opposite signs.



REAL WORLD

Galvanic cells are commonly used as batteries; to be worthwhile (that is, producing energy to power some device or appliance),

these batteries must be spontaneous!

Let's examine the inner workings of a galvanic (voltaic) cell. Two electrodes of distinct chemical identity are placed in separate compartments, which are called half-cells. The two electrodes are connected to each other by a conductive material, such as a copper wire. Along the wire, there may be other various components of a circuit, such as resistors or capacitors, but for now, we'll focus on the battery itself.

Surrounding each of the electrodes is an aqueous electrolyte solution composed of cations and anions. As shown in the Daniell cell illustrated in Figure 12.1, the cations in the two half-cell solutions can be of the same element as the respective metal electrode. Connecting the two solutions is a structure called a salt bridge, which consists of an inert salt. When the electrodes are connected to each other by a conductive material, charge will begin to flow as the result of an oxidation–reduction reaction that is taking place between the two half-cells. The redox reaction in a galvanic cell is spontaneous, and therefore the change in Gibbs free energy for the reaction is negative ($\Delta G < 0$). As the spontaneous reaction proceeds toward equilibrium, the movement of electrons results in a conversion of electrical potential energy into kinetic energy. By separating the reduction and oxidation half-reactions into two compartments, we are able to harness this energy and use it to do work by connecting various electrical devices into the circuit between the two electrodes.

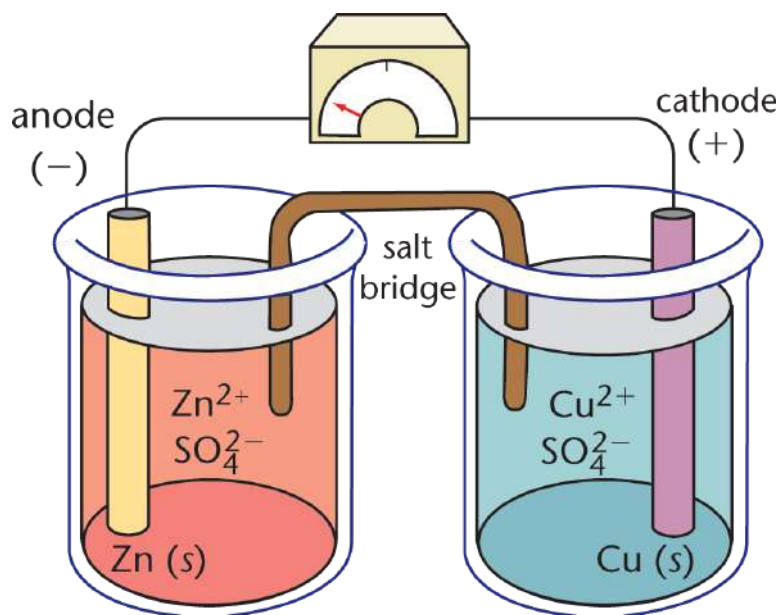
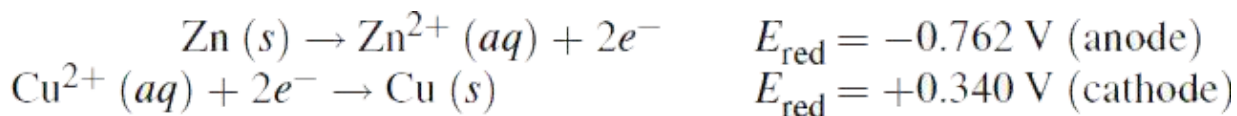


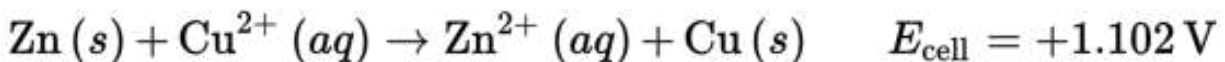
Figure 12.1. Daniell Cell

In this galvanic cell, zinc is the anode and copper is the cathode; each electrode is bathed in an electrolyte solution containing its cation and sulfate.

In the Daniell cell, a zinc electrode is placed in an aqueous ZnSO_4 solution, and a copper electrode is placed in an aqueous CuSO_4 solution. The anode of this cell is the zinc bar where Zn (s) is oxidized to $\text{Zn}^{2+} (\text{aq})$. The cathode is the copper bar, and it is the site of the reduction of $\text{Cu}^{2+} (\text{aq})$ to Cu (s) . The half-cell reactions are written as follows:



The net reaction is



We will discuss the calculation of cell potential in the next section. For now, appreciate that the calculation can be accomplished by knowing each half-reaction. If the two half-cells were not separated, the Cu^{2+} ions would react directly with the zinc bar, and no useful electrical work would be done. Because the solutions and electrodes are physically separated, they must be connected by a conductive material to complete the circuit.

However, if only a wire were provided for this electron flow, the reaction would soon stop because an excess positive charge would build up on the anode, and an excess negative charge would build up on the cathode. Eventually, the excessive charge accumulation would provide a countervoltage large enough to prevent the oxidation–reduction reaction from taking place, and the current would cease. This charge gradient is dissipated by the presence of a salt bridge, which permits the exchange of cations and anions. The salt bridge contains an inert electrolyte, usually KCl or NH_4NO_3 , which contains ions that will not react with the electrodes or with the ions in solution. While the anions from the salt bridge (Cl^-) diffuse into the solution on the anode side (ZnSO_4) to balance out the charge of the newly created Zn^{2+} ions, the cations of the salt bridge (K^+) flow into the solution on the cathode side (CuSO_4) to balance out the charge of the sulfate ions left in solution when the Cu^{2+} ions are reduced to Cu and precipitate onto the electrode. This precipitation process onto the cathode itself can also be called plating or galvanization.



KEY CONCEPT

The purpose of the salt bridge is to exchange anions and cations to balance, or dissipate, newly generated charges.

During the course of the reaction, electrons flow from the zinc anode through the wire and to the copper cathode. A voltmeter can be connected to measure this electromotive force. As mentioned earlier, the anions (Cl^-) flow externally from the salt bridge into the ZnSO_4 , and the cations (K^+) flow externally from the salt bridge into the CuSO_4 . This flow depletes the salt bridge and, along with the finite quantity of Cu^{2+} in the solution, accounts for the relatively short lifespan of the cell.

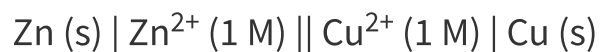


MNEMONIC

Electron flow in an electrochemical cell: $A \rightarrow C$ (order in the alphabet)

Electrons flow from anode to cathode in all types of electrochemical cells.

A cell diagram is a shorthand notation representing the reactions in an electrochemical cell. A cell diagram for the Daniell cell is as follows:



The following rules are used in constructing a cell diagram:

The reactants and products are always listed from left to right in this form: anode | anode solution (concentration) || cathode solution

(concentration) | cathode

A single vertical line indicates a phase boundary.

A double vertical line indicates the presence of a salt bridge or some other type of barrier.



MCAT EXPERTISE

Recognize and understand the shorthand cell notation for electrochemical cells on Test Day. Passages frequently use this format rather than spelling out which reactions take place at the anode and cathode.

ELECTROLYTIC CELLS

When comparing and contrasting galvanic and electrolytic cells, it is important to keep straight what remains consistent between the two types of cells and what differs. All types of electrochemical cells have a reduction reaction occurring at the cathode, an oxidation reaction occurring at the anode, a current flowing from cathode to anode, and electron flow from anode to cathode. However, electrolytic cells, in almost all of their characteristics and behavior, are otherwise the opposite of galvanic cells. Whereas galvanic cells house spontaneous oxidation–reduction reactions that generate electrical energy, electrolytic cells house nonspontaneous reactions that require the input of energy to proceed. Therefore, the change in free energy for an electrolytic cell is positive. This type of oxidation–reduction reaction driven by an external voltage source is called

electrolysis, in which chemical compounds are decomposed. For example, electrolytic cells can be used to drive the nonspontaneous decomposition of water into oxygen and hydrogen gas. Another example, the electrolysis of molten NaCl, is illustrated in Figure 12.2.

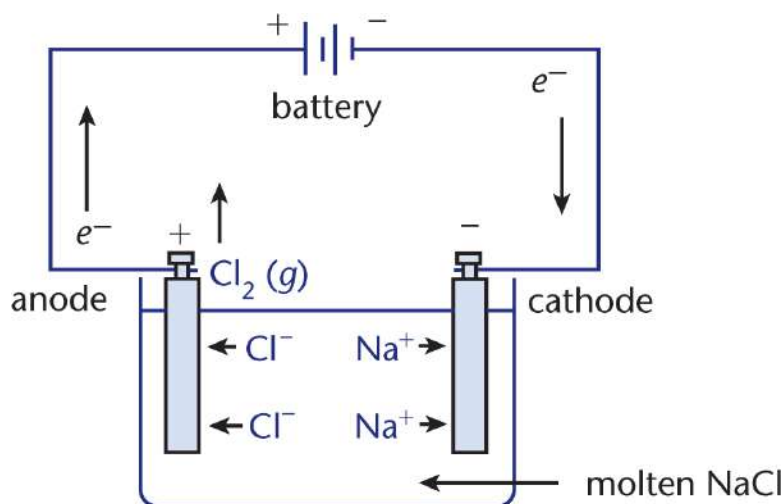


Figure 12.2. Electrolysis of Molten NaCl



KEY CONCEPT

Because electrolysis is nonspontaneous, the electrode (anode or cathode) can consist of any material so long as it can resist the high temperatures and corrosion of the process.

In this electrolytic cell, molten NaCl is decomposed into $\text{Cl}_2(g)$ and $\text{Na}(l)$. The external voltage source—a battery—supplies energy sufficient to drive the oxidation–reduction reaction in the direction that is thermodynamically unfavorable (nonspontaneous).

In this example, Na^+ ions migrate toward the cathode, where they are reduced to Na (l) . At the same time, Cl^- ions migrate toward the anode, where they are oxidized to $\text{Cl}_2 \text{ (g)}$. Notice that the half-reactions do not need to be separated into different compartments; this is because the desired reaction is nonspontaneous. Note that sodium is a liquid at the temperature of molten NaCl ; it is also less dense than the molten salt and, thus, is easily removed as it floats to the top of the reaction vessel.

This cell is used in industry as the major means of sodium and chlorine production. You may wonder why one would do so much work to obtain pure sodium and chlorine. Remember that these elements are never found naturally in their elemental form because they are so reactive. Thus, to use elemental sodium or chlorine gas in a reaction, it must be manufactured through processes such as these.

Michael Faraday was the first to define certain quantitative principles governing the behavior of electrolytic cells. He theorized that the amount of chemical change induced in an electrolytic cell is directly proportional to the number of moles of electrons that are exchanged during the oxidation–reduction reaction. The number of moles exchanged can be determined from the balanced half-reaction. In general, for a reaction that involves the transfer of n electrons per atom M ,



Equation 12.1



KEY CONCEPT

Faraday's laws state that the liberation of gas and deposition of elements on electrodes is directly proportional to the number of electrons being transferred during the oxidation–reduction reaction. Here, normality or gram equivalent weight is used. These observations are proxy measurements of the amount of current flowing in a circuit.

According to this equation, one mole of metal M (s) will logically be produced if n moles of electrons are supplied to one mole of M^{n+} . Additionally, the number of moles of electrons needed to produce a certain amount of M (s) can now be related to the measurable electrical property of charge. One electron carries a charge of 1.6×10^{-19} coulombs (C). The charge carried by one mole of electrons can be calculated by multiplying this number by Avogadro's number, as follows:

$$\left[\frac{1.6 \times 10^{-19} \text{ C}}{\text{electron}} \right] \left[\frac{6.02 \times 10^{23} \text{ electrons}}{1 \text{ mol } e^{-}} \right] = 96,485 \frac{\text{C}}{\text{mol } e^{-}}$$

This number is called the Faraday constant, and one faraday (F) is equivalent to the amount of charge contained in one mole of electrons ($1 \text{ F} = 96,485 \text{ C}$) or one equivalent. On the MCAT, you should round up this number to $10^5 \frac{\text{C}}{\text{mol } e^{-}}$ to make calculations more manageable.



KEY CONCEPT

One faraday (F) is equivalent to the amount of charge contained in one mole of electrons (1 F = 96,485 C).

The electrodeposition equation summarizes this process and helps determine the number of moles of element being deposited on a plate:

$$\text{mol M} = \frac{It}{nF}$$

Equation 12.2

where mol M is the amount of metal ion being deposited at a specific electrode, I is current, t is time, n is the number of electron equivalents for a specific metal ion, and F is the Faraday constant. This equation can also be used to determine the amount of gas liberated during electrolysis.



MNEMONIC

Electrodeposition equation: Calculating Moles of Metal, It is Not Fun.

$$\text{mol M} = \frac{It}{nF}$$

Example:

What mass of copper will be deposited in a Daniell cell if a current of 2 A flows through the cell for 3 hours?

Solution:

We will use the equation $\text{mol M} = \frac{It}{nF}$

A Daniell cell uses a copper electrode in copper sulfate (CuSO_4) solution. Because the oxidation state of copper in solution is +2, $n = 2$. Now we can plug into the equation.

$$\begin{aligned} \text{mol M} = \frac{It}{nF} &= \frac{(2 \text{ A})(3 \text{ hr})\left(3600 \frac{\text{s}}{\text{hr}}\right)}{(2 \text{ mol } e^-)\left(96,485 \frac{\text{C}}{\text{mol } e^-}\right)} \approx \frac{3 \times 3600}{10^5} \\ &= 0.1 \text{ mol Cu (actual} = 0.11 \text{ mol Cu)} \end{aligned}$$

Then, we must determine the actual mass of copper being deposited. 0.1 mol Cu should have a mass of 6.35 g because the molar mass of copper is $264 \frac{\text{g}}{\text{mol}}$? (actual = 7.11 g).

CONCENTRATION CELLS

A concentration cell is a special type of galvanic cell. Like all galvanic cells, it contains two half-cells connected by a conductive material, allowing a spontaneous oxidation–reduction reaction to proceed, which generates a current and delivers energy. The distinguishing characteristic of a concentration cell is in its design: the electrodes are chemically identical. For example, if both electrodes are copper metal, they have the same reduction potential. Therefore, current is generated as a function of a concentration gradient established between the two solutions surrounding the electrodes. The concentration gradient results in a potential difference between the two compartments and drives the movement of electrons in the direction that results in equilibration of the ion gradient. The current will stop when the concentrations of ionic species in the half-cells are equal. This implies that the voltage (V) or electromotive force of a concentration cell is zero when the concentrations are equal; the voltage, as a function of concentrations, can be calculated using the Nernst equation.

In a biological system, a concentration cell is best represented by the cell membrane of a neuron, as shown in Figure 12.3. Sodium and potassium cations, and chlorine anions, are exchanged as needed to produce an electrical potential. The actual value depends on both the concentrations and charges of the ions. In this way, a resting membrane potential (V_m) can be maintained. Disturbances of the resting membrane potential, if sufficiently large, may stimulate the firing of an action potential.

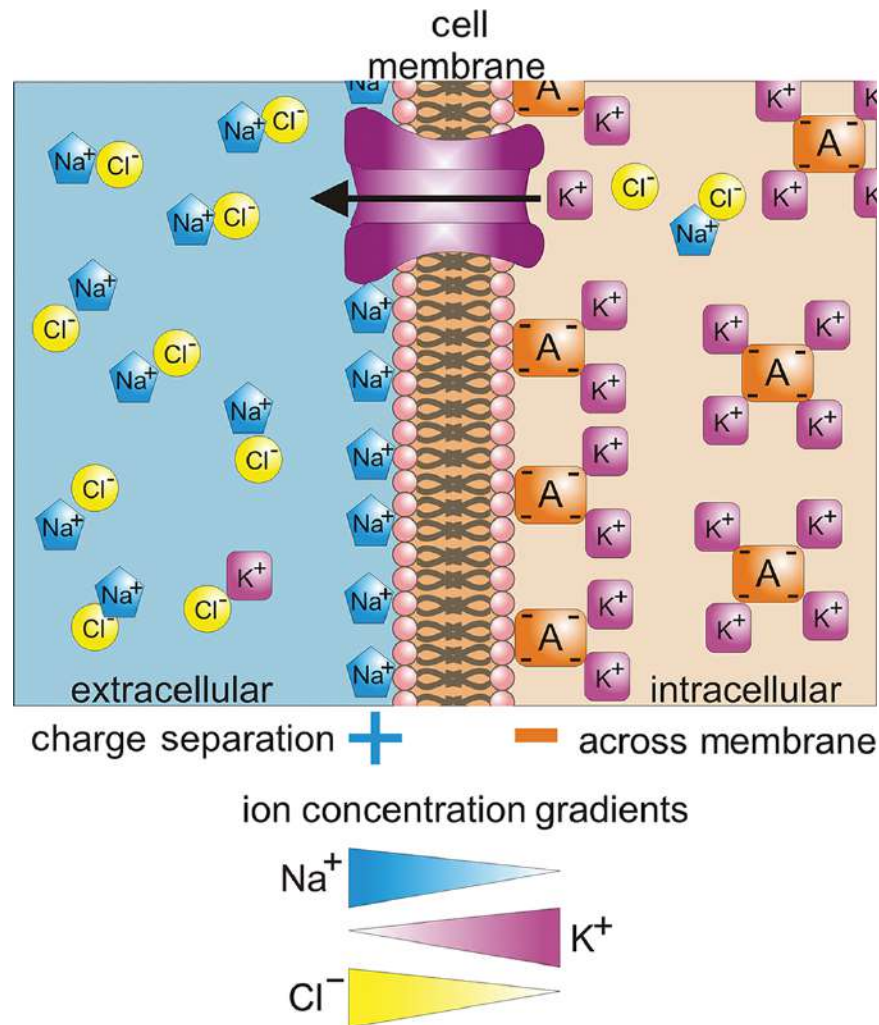


Figure 12.3. The Cell Membrane as an Example of a Concentration Cell
The electrochemical gradient created by separation of ions across the cell membrane is analogous to a cell with two electrodes composed of the same material.



BRIDGE

The maintenance of a resting membrane potential is discussed in Chapter 8 of MCAT Biochemistry Review. The conduction of an action potential is discussed in Chapter 4 of MCAT Biology Review.

The transfer of ions and electrons during an action potential produces biochemical work.

RECHARGEABLE CELLS

A rechargeable cell or rechargeable battery is one that can function as both a galvanic and electrolytic cell.

Lead–Acid Batteries

A lead–acid battery, also known as a lead storage battery, is a specific type of rechargeable battery. As a voltaic cell, when fully charged, it consists of two half-cells—a Pb anode and a porous PbO₂ cathode, connected by a conductive material (concentrated 4 M H₂SO₄). When fully discharged, it consists of two PbSO₄ electroplated lead electrodes with a dilute concentration of H₂SO₄, as shown in Figure 12.4.

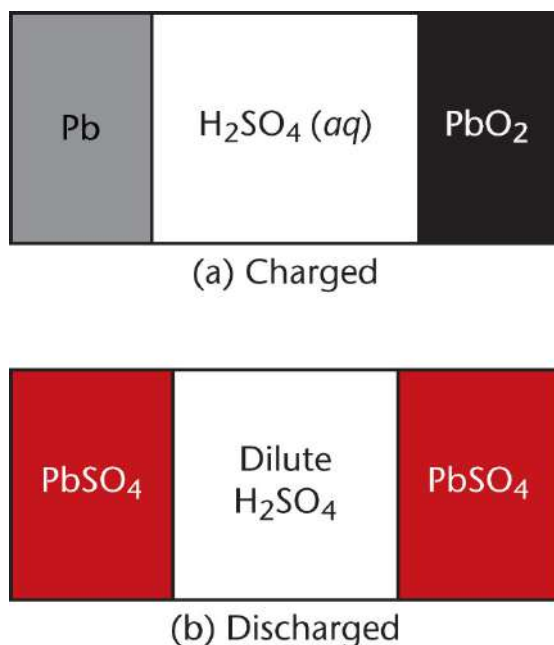
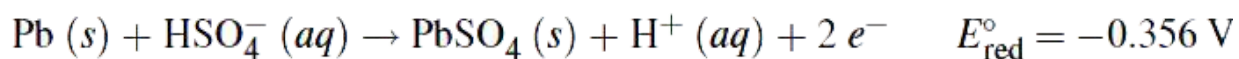


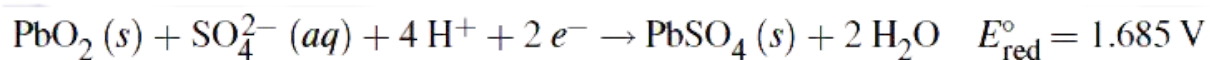
Figure 12.4. Lead–Acid Battery

When charged (a), the cell contains a Pb anode and PbO₂ cathode; when discharged (b), both electrodes are coated with lead sulfate.

The oxidation half-reaction at the lead (negative) anode is:



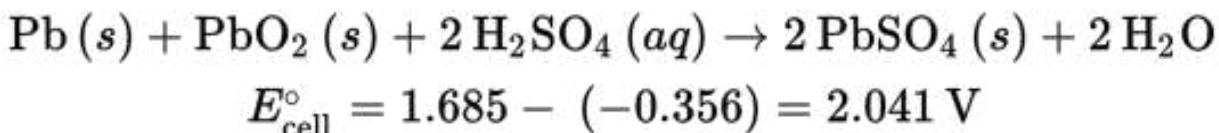
The reduction half-reaction at the lead(IV) oxide (positive) cathode is:



Both half-reactions cause the electrodes to plate with lead sulfate (PbSO₄) and dilute the acid electrolyte when discharging. The lead anode is negatively charged and attracts the anionic bisulfate. The lead(IV) oxide cathode is a bit more complicated. This electrode is porous, which allows the electrolyte (sulfuric acid) to solvate the cathode into lead and oxide

ions. Then, the hydrogen ions in solution react with the oxide ions to produce water, and the remaining sulfate ions react with the lead to produce the electroplated lead sulfate.

Overall, the net equation for a discharging lead–acid battery is:



When charging, the lead–acid cell is part of an electrolytic circuit. These equations and electrode charge designations are the opposite because an external source reverses the electroplating process and concentrates the acid solution—this external source is very evident when one uses jumper cables to restart a car.

Lead–acid batteries, as compared to other cells, have some of the lowest energy-to-weight ratios (otherwise known as energy density). Energy density is a measure of a battery's ability to produce power as a function of its weight. Lead–acid batteries, therefore, require a heavier amount of battery material to produce a certain output as compared to other batteries.

Nickel–Cadmium Batteries

Nickel–cadmium batteries are also rechargeable cells. They consist of two half-cells made of solid cadmium (the anode) and nickel(III) oxide-hydroxide (the cathode) connected by a conductive material, typically potassium hydroxide (KOH). Most of us are familiar with AA and AAA cells

made of Ni–Cd materials, inside of which the electrodes are layered and wrapped around in a cylinder, as shown in Figure 12.5.

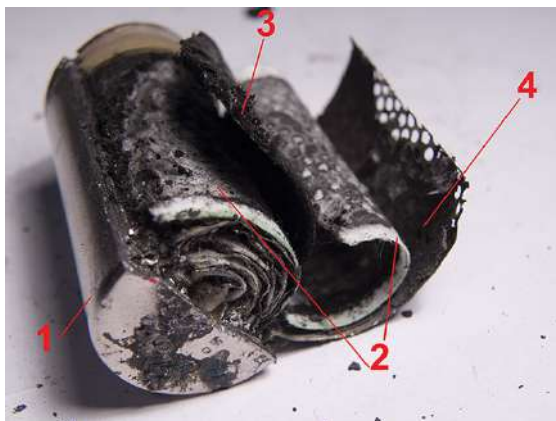
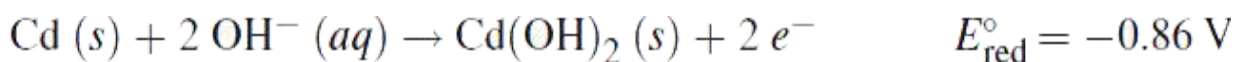


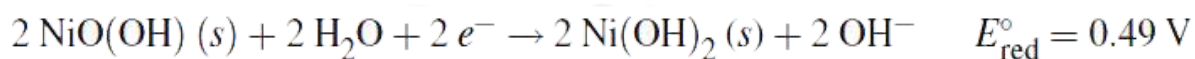
Figure 12.5. A Nickel–Cadmium Battery

(1) Metal casing, (2) salt bridge, (3) NiO(OH) cathode, (4) Cd anode

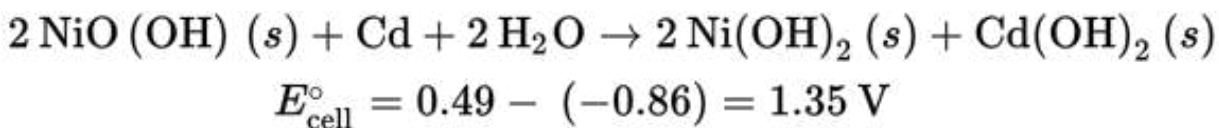
The oxidation half-reaction at the cadmium (negative) anode is:



The reduction half-reaction at the nickel oxide–hydroxide (positive) cathode is:



Both half-reactions cause the electrodes to plate with their respective products. Overall, the net equation for a Ni–Cd battery is



As in our previous example, charging reverses the electrolytic cell potentials. Some Ni–Cd designs are vented for this reason to allow for the release of built up hydrogen and oxygen gas during electrolysis.

Ni–Cd batteries have a higher energy density than lead–acid batteries. The electrochemistry of the Ni–Cd half-reactions also tends to provide higher surge current. Surge currents are periods of large current (amperage) early in the discharge cycle. This is preferable in appliances such as remote controls that demand rapid responses. It is important to note that modern Ni–Cd batteries have largely been replaced by more efficient nickel–metal hydride (NiMH) batteries. These newer batteries have more energy density, are more cost effective, and are significantly less toxic. As the name suggests, in lieu of a pure metal anode, a metal hydride is used instead.

ELECTRODE CHARGE DESIGNATIONS

In a galvanic cell, current is spontaneously generated as electrons are released by the oxidized species at the anode and travel through the conductive material to the cathode, where reduction takes place. Because the anode of a galvanic cell is the source of electrons, it is considered the negative electrode; the cathode is considered the positive electrode, as shown in Figure 12.1 previously. Electrons, therefore, move from negative (low electrical potential) to positive (high electrical potential), while the current—the flow of positive charge—is from positive (high electrical potential) to negative (low electrical potential).



MNEMONIC

In a galvanic cell, the anode is negative.

Conversely, the anode of an electrolytic cell is considered positive because it is attached to the positive pole of the external voltage source and attracts anions from the solution. The cathode of an electrolytic cell is considered negative because it is attached to the negative pole of the external voltage source and attracts cations from the solution.



KEY CONCEPT

In a galvanic cell, the anode is negative and the cathode is positive. In an electrolytic cell, the anode is positive and the cathode is negative. This is because an external source is used to reverse the charge of an electrolytic cell. However, in both types of cells, reduction occurs at the cathode, and oxidation occurs at the anode; cations are attracted to the cathode, and anions are attracted to the anode.

In spite of this difference in designating charge (sign), oxidation always takes place at the anode and reduction always takes place at the cathode in both types of cells; electrons always flow through the wire from the anode to the cathode and current flows from cathode to anode. Finally, note that—regardless of its charge designation—the cathode always attracts cations and the anode always attracts anions. In the Daniell cell, for example, the electrons created at the anode by the oxidation of elemental zinc travel through the wire to the copper half-cell. There, they

attract copper(II) cations to the cathode, resulting in the reduction of the copper ions to elemental copper, and drawing cations out of the salt bridge into the compartment. The anode, having lost electrons, attracts anions from the salt bridge at the same time that zinc(II) ions formed by the oxidation process dissolve away from the anode.



REAL WORLD

Recognize that in any system in which batteries are placed, it is important to line up cathodes and anodes. Electronics tend to have (+) and (–) designations to line up the electrodes—think jumper cables, television remotes, and button batteries in watches.

This is an important rule to understand not only for electrochemistry in the Chemical and Physical Foundations of Biological Systems section on Test Day, but also for electrophoresis in both this section and Biological and Biochemical Foundations of Living Systems. Isoelectric focusing is a technique used to separate amino acids or polypeptides based on their isoelectric points (pI). The positively charged amino acids (protonated at the solution's pH) will migrate toward the cathode; negatively charged amino acids (deprotonated at the solution's pH) will migrate toward the anode. The technique of isoelectric focusing is discussed in detail in Chapter 3 of MCAT Biochemistry Review.



MNEMONIC

Anions are attracted to the anode. Cations are attracted to the cathode. This is true regardless of the type of cell (galvanic, electrolytic, or concentration cells).

MCAT Concept Check 12.1:

Before you move on, assess your understanding of the material with these questions.

Circle which electrode each of the following statements describes in a galvanic (voltaic) cell:

- | | | |
|-----------------------------|-------|---------|
| • Site of oxidation: | Anode | Cathode |
| • Electrons flow toward it: | Anode | Cathode |
| • Current flows toward it: | Anode | Cathode |
| • Has (–) designation: | Anode | Cathode |
| • Attracts cations: | Anode | Cathode |



Circle which electrode each of the following statements describes in an electrolytic cell:

- | | | |
|----------------------|-------|---------|
| • Site of oxidation: | Anode | Cathode |
|----------------------|-------|---------|

- | | | |
|-----------------------------|-------|---------|
| • Electrons flow toward it: | Anode | Cathode |
| • Current flows toward it: | Anode | Cathode |
| • Has (–) designation: | Anode | Cathode |
| • Attracts cations: | Anode | Cathode |



Write the cell diagram for the discharging state of a lead–acid battery:

Which type of cell has a positive ΔG ? A positive E_{cell} ?

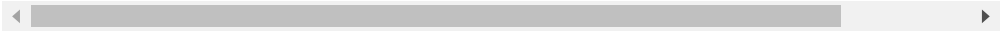
- Positive ΔG :

- Positive E_{cell} :

How much current is required to produce 0.23 kg Na from a molten NaCl electrolytic cell that runs for 30 hours? Assume the cell is 100% efficient.

Fill in the following chart to summarize electrode charge designations of batteries we have analyzed:

Battery	State of Use	Galvanic or Electrolytic	Anode Material	Anode Charge	Cathode Material
Ni—Cd	Discharging				
Ni—Cd	Charging				
Molten NaCl	Discharging				
Daniell cell	Discharging				
Lead—acid	Charging				
Lead—acid	Discharging				

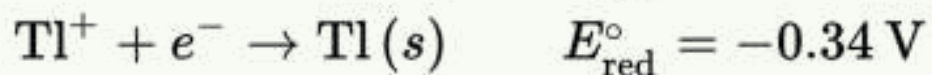
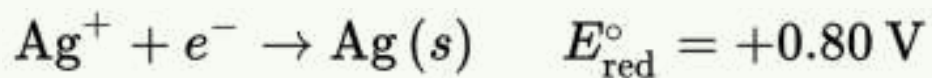


12.2 Cell Potentials

LEARNING GOALS

After Chapter 12.2, you will be able to:

- Describe how standard reduction potentials are measured
- Explain the importance of the sign for electromotive force
- Determine whether a cell using a given reaction is galvanic or electrolytic
- Calculate the net E value for a redox reaction between two species:



For galvanic cells, the direction of spontaneous movement of charge is from the anode, the site of oxidation, to the cathode, the site of reduction. This is simple enough to remember, but it begs the question: how do we determine which electrode species will be oxidized and which will be reduced? The relative tendencies of different chemical species to be

reduced have been determined experimentally, using the tendency of the hydrogen ion (H^+) to be reduced as an arbitrary zero reference point.

REDUCTION POTENTIALS

A reduction potential is measured in volts (V) and defined relative to the standard hydrogen electrode (SHE), which is given a potential of 0 V by convention. The species in a reaction that will be oxidized or reduced can be determined from the reduction potential of each species, defined as the tendency of a species to gain electrons and to be reduced. Each species has its own intrinsic reduction potential; the more positive the potential, the greater the tendency to be reduced.

Standard reduction potential (E_{red}°) is measured under standard conditions: 25°C (298 K), 1 atm pressure, and 1 M concentrations. The relative reactivities of different half-cells can be compared to predict the direction of electron flow. A more positive E_{red}° means a greater relative tendency for reduction to occur, while a less positive E_{red}° means a greater relative tendency for oxidation to occur.



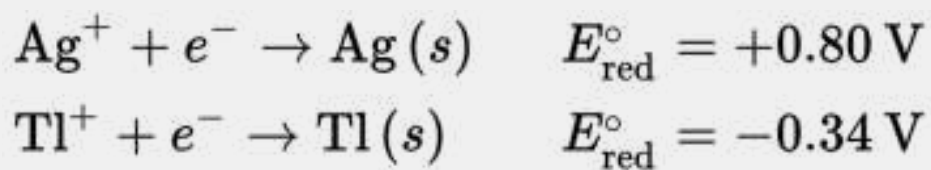
KEY CONCEPT

A reduction potential is exactly what it sounds like. It tells us how likely a compound is to be reduced. The more positive the value, the more likely it is to be reduced—the more it wants to be reduced.

For galvanic cells, the electrode with the more positive reduction potential is the cathode, and the electrode with the less positive reduction potential is the anode. Because the species with a stronger tendency to gain electrons (that wants to gain electrons more) is actually doing so, the reaction is spontaneous and ΔG is negative. For electrolytic cells, the electrode with the more positive reduction potential is forced by the external voltage source to be oxidized and is, therefore, the anode. The electrode with the less positive reduction potential is forced to be reduced and is, therefore, the cathode. Because the movement of electrons is in the direction against the tendency or desires of the respective electrochemical species, the reaction is nonspontaneous and ΔG is positive.

Example:

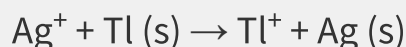
Given the following half-reactions and E_{red}° values, determine which species would be oxidized and which would be reduced in a galvanic cell.



Solution:

E_{red}° indicates the reduction potential, or the likelihood of a compound to be reduced via a given reaction. A positive E_{red}° value indicates a spontaneous reduction, and a negative value indicates a non-spontaneous reduction. In a galvanic cell, Ag^{+} will be spontaneously reduced to $\text{Ag}(s)$ and $\text{Tl}(s)$ will be spontaneously

oxidized to Tl^+ because Ag^+ has the more positive E_{red}° and thus the more favorable reduction reaction. Therefore, the net ionic equation would be:



which is the sum of the two spontaneous half-reactions.

It should be noted that reduction and oxidation are opposite processes. Therefore, to obtain the oxidation potential of a given half-reaction, both the reduction half-reaction and the sign of the reduction potential are reversed. For instance, from the example above, the oxidation half-reaction and oxidation potential of Tl (s) are:



Note that, in the examples of batteries given above (lead–acid storage batteries and nickel–cadmium batteries), the oxidation half-reaction was given with the reduction potential of the reverse reaction. These two quantities have equal magnitudes but opposite signs. On the MCAT, reduction potentials are generally given rather than oxidation potentials. Therefore, all references in this book (with exception of the thallium example immediately above) are given using reduction potentials—not oxidation potentials.

THE ELECTROMOTIVE FORCE

Standard reduction potentials are also used to calculate the standard electromotive force (emf or E_{cell}°) of a reaction, which is the difference in potential (voltage) between two half-cells under standard conditions. The emf of a reaction is determined by calculating the difference in reduction potentials between the two half-cells:

$$E_{\text{cell}}^{\circ} = E_{\text{red,cathode}}^{\circ} - E_{\text{red,anode}}^{\circ}$$

Equation 12.3

When subtracting standard potentials, do not multiply them by the number of moles oxidized or reduced. This is because the potential of each electrode does not depend on the size of the electrode (the amount of material), but rather the identity of the material. The standard reduction potential of an electrode will not change unless the chemical identity of that electrode is changed.

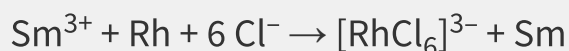


KEY CONCEPT

If you need to multiply each half-reaction by a common denominator to cancel out electrons when coming up with the net ionic equation, do not multiply the reduction potential, E_{red}° , by that number. That would indicate a change in the chemical identity of the electrode, which is not occurring.

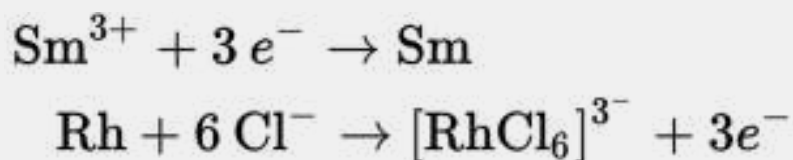
Example:

Given that the standard reduction potentials for Sm^{3+} and $[\text{RhCl}_6]^{3-}$ are -2.41 V and $+0.44\text{ V}$, respectively, calculate the electromotive force of the following reaction:



Solution:

First, determine the oxidation and reduction half-reactions. As written, the Rh is oxidized, and the Sm^{3+} is reduced:



Now, we simply take the difference between the samarium(III) reduction potential and the hexachlororhodate(III) reduction potential. We need not change the sign on the hexachlororhodate(III) reduction potential because we are subtracting it from that of samarium(III).

Using the equation provided, the emf can be calculated as: $-2.41\text{ V} - (+0.44\text{ V}) = -2.85\text{ V}$. The cell is thus electrolytic. If this were instead a galvanic cell the reaction would proceed spontaneously to the left, toward reactants, in which case the Sm would be oxidized while $[\text{RhCl}_6]^{3-}$ would be reduced with an emf of $+2.85\text{ V}$.

MCAT Concept Check 12.2:

Before you move on, assess your understanding of the material with these questions.

How are standard reduction potentials measured?

If a cell's electromotive force (emf) is denoted as a positive value, what does that mean? What if it is negative?

- Positive emf:

- Negative emf:

Given the following reactions, determine whether the cell is galvanic or electrolytic:

- $2 \text{Fe}^{3+} (aq) + 2 \text{Cl}^{-} (aq) \rightarrow 2 \text{Fe}^{2+} (aq) + \text{Cl}_2 (g) (E_{\text{cell}}^{\circ} =$

◀
▶

- $2 \text{Fe}^{3+} (aq) + 2 \text{I}^{-} (aq) \rightarrow 2 \text{Fe}^{2+} (aq) + \text{I}_2 (aq) (E_{\text{cell}}^{\circ} =$

◀
▶

Given the two half-reactions below, what would be the spontaneous oxidation–reduction reaction between these two species?



12.3 Electromotive Force and Thermodynamics

LEARNING GOALS

After Chapter 12.3, you will be able to:

- Apply the formula $\Delta G^\circ = -RT \ln K_{\text{eq}}$ to calculations of Gibbs free energy or electromotive force
- Predict E_{cell} given reaction quotients and equilibrium constants

Throughout our discussion of electrochemistry and the different types of electrochemical cells, we have been making references to the spontaneity or nonspontaneity of the redox reactions housed in each of the different cell types. Let's now look more formally at this topic by relating free energy to electromotive force (emf) and the concentrations of the oxidation–reduction reactants and products to the voltage of a cell at a given point in time.

GIBBS FREE ENERGY

By now, you should be familiar with the thermodynamic criterion for determining the spontaneity of a reaction: the change in Gibbs free energy, ΔG . This is the change in the amount of energy available in a chemical system to do work. In an electrochemical cell, the work done is dependent on the number of coulombs of charge transferred and the energy available. Thus, ΔG° and emf are related as follows:

$$\Delta G^\circ = -nFE_{\text{cell}}^\circ$$

Equation 12.4

where ΔG° is the standard change in free energy, n is the number of moles of electrons exchanged, F is the Faraday constant, and E_{cell}° is the standard emf of the cell. Keep in mind that, if the Faraday constant is expressed in coulombs $\left(\frac{\text{J}}{\text{V}}\right)$, then ΔG° must be expressed in J, not kJ. Notice the similarity of this relationship to that expressed in the physics formula $W = q\Delta V$ for the amount of work available or needed in the transport of a charge q across a potential difference ΔV : $n \times F$ is a charge, and E_{cell}° is a voltage. This application in electrostatics is discussed in Chapter 5 of MCAT Physics and Math Review.



BRIDGE

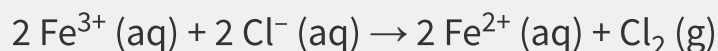
Recall from Chapter 6 of MCAT General Chemistry Review that, if ΔG is positive, the reaction is nonspontaneous; if ΔG is negative, the

reaction is spontaneous.

Note the significance of the negative sign on the right side of the equation. ΔG° and E°_{cell} will always have opposite signs. Therefore, galvanic cells have negative ΔG° and positive E°_{cell} values; electrolytic cells have positive ΔG° and negative E°_{cell} values.

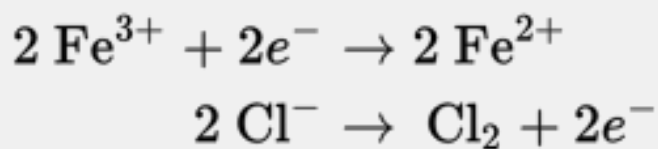
Example:

Determine the standard change in free energy of a cell with the following net reaction. (Note: The standard reduction potential of iron(III) is +0.77 V; the standard reduction potential of molecular chlorine is +1.36 V.)



Solution:

First, separate the reaction into the half-reactions:



In this reaction, iron(III) is reduced and is the cathode, whereas Cl^{-} is oxidized and is the anode. The reduction potential of chlorine is actually higher than that of iron(III); this means that the electrodes are serving the opposite role from their natural tendency, and the

reaction is nonspontaneous. This is an electrolytic cell, and should have a negative emf value.

Now, determine the emf:

$$E_{\text{cell}}^{\circ} = E_{\text{red,cathode}}^{\circ} - E_{\text{red,anode}}^{\circ} = 0.77 \text{ V} - 1.36 \text{ V} = -0.59 \text{ V}$$

Use the emf to determine the free energy change (note that as 2 electrons are transferred, $n = 2$):

$$\Delta G^{\circ} = -nFE_{\text{cell}}^{\circ} = -(2 \text{ mol } e^{-}) \left(96,485 \frac{\text{C}}{\text{mol } e^{-}} \right) (-0.59 \text{ V})$$
$$\approx 2 \times 10^5 \times (0.6) = 1.2 \times 10^5 \text{ J (actual} = 1.14 \times 10^5 \text{ J)}$$

The free energy change is about +120 kJ, which represents a non-spontaneous reaction.

REACTION QUOTIENTS

So far, we have considered the calculation of a cell's emf only under standard conditions. However, electrochemical cells may have ionic concentrations that deviate from 1 M. Also, for the concentration cell, the concentrations of the ions in the two compartments must be different for there to be a measurable voltage and current. Concentration and the emf of a cell are related: emf varies with the changing concentrations of the

species in the cell. When conditions deviate from standard conditions, one can use the Nernst equation:

$$KE = \frac{1}{2} mv^2 = \frac{3}{2} k_B T$$

Equation 12.5

where E_{cell} is the emf of the cell under nonstandard conditions, E_{cell}° is the emf of the cell under standard conditions, R is the ideal gas constant, T is the temperature in kelvins, n is the number of moles of electrons, F is the Faraday constant, and Q is the reaction quotient for the reaction at a given point in time. The following simplified version of the equation can be used, assuming $T = 298 \text{ K}$:

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0592}{n} \log Q$$

Equation 12.6

This simplified version of the equation brings together R , T (298 K), and F , and converts the natural logarithm to the base-ten logarithm to make calculations easier.



MCAT EXPERTISE

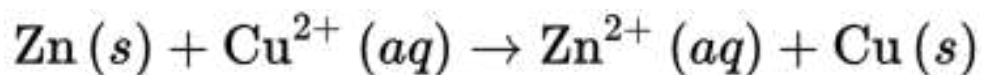
If the Nernst equation is needed on Test Day, stick with the $\log_{10}$ version because natural logarithm calculations get very tedious.

Remember that the reaction quotient, Q , for a general reaction $aA + bB \rightarrow cC + dD$ has the form:

$$Q = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

Equation 12.7

Although the expression for the reaction quotient Q has two terms for the concentrations of reactants and two terms for the concentrations of products, remember that only the species in solution are included. When considering the case of the Daniell cell, for example, only the concentrations of zinc and copper ions are considered:

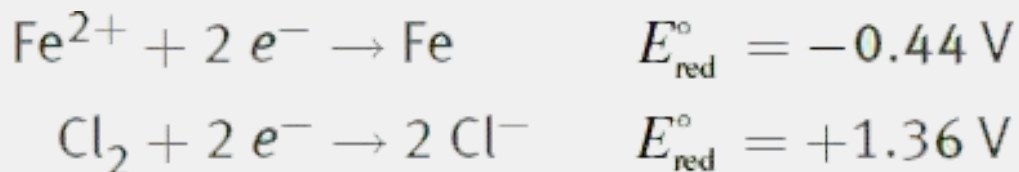


$$Q = \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$$

The emf of a cell can be measured with a voltmeter. A potentiometer is a kind of voltmeter that draws no current and gives a more accurate reading of the difference in potential between two electrodes.

Example:

Find the emf of a galvanic cell at 25°C based on the following standard reduction potentials:



In this cell, $[\text{Fe}^{2+}] = 0.01 \text{ M}$ and $[\text{Cl}^{-}] = 0.1 \text{ M}$.

Solution:

First, determine the standard cell potential. Because the chlorine half-reaction has a higher reduction potential, it will be the cathode. Iron will act as the anode. The standard cell potential is:

$$E_{\text{cell}}^{\circ} = E_{\text{red,cathode}}^{\circ} - E_{\text{red,anode}}^{\circ} = 1.36 \text{ V} - (-0.44 \text{ V}) = +1.80 \text{ V}$$

Now, determine the net ionic equation. Remember that iron is being oxidized, so its reduction half-reaction in the question stem will have to be reversed. The net ionic equation is:



From this equation, we can determine the value of the reaction quotient:

$$Q = [\text{Fe}^{2+}][\text{Cl}^-]^2 = (0.01 \text{ M})(0.1 \text{ M})^2 = 10^{-4}$$

Now, plug into the Nernst equation, keeping in mind that two electrons are transferred ($n = 2$):

$$\begin{aligned} E_{\text{cell}} &= E_{\text{cell}}^{\circ} - \frac{0.0592}{n} \log Q = 1.8 - \frac{0.0592}{2} \log 1 \\ &= 1.8 + \frac{4 \times 0.0592}{2} \approx 1.8 + (2 \times 0.06) = 1 \end{aligned}$$

In this case, the cell actually has a higher voltage than it normally would due to the concentrations of ions present.



BRIDGE

While a mathematically rigorous equation, the Nernst equation has a powerful use in biochemistry for calculating resting and depolarized membrane potentials based on concentrations of ions. Its more extended version, the Goldman–Hodgkin–Katz equation, is discussed in Chapter 8 of MCAT Biochemistry Review. Equation 8.2 in that chapter looks slightly different than Equation 12.6 here. This is because the temperature is different (310 K rather than 298 K) and the units are different (mV instead of V).

EQUILIBRIA

As discussed in Chapter 7 of MCAT General Chemistry Review, ΔG° can also be determined in another manner:

$$\Delta G^\circ = -RT \ln K_{\text{eq}}$$

Equation 12.8

where R is the ideal gas constant, T is the absolute temperature, and K_{eq} is the equilibrium constant for the reaction.

Combining the two expressions that solve for standard free energy change, we see that

$$\Delta G^\circ = -nFE_{\text{cell}}^\circ = -RT \ln K_{\text{eq}}$$

or

$$nFE_{\text{cell}}^\circ = RT \ln K_{\text{eq}}$$

By extension, if the values for n , T , and K_{eq} are known, then E_{cell}° for the reaction is easily calculated. On the MCAT, you will not be expected to

calculate natural logarithm values in your head. That being said, these equations can still be tested but in a conceptual way.



MCAT EXPERTISE

Whether it is $\log$ or $\ln$, remember that a logarithm will be positive when equilibrium constants are greater than 1, negative when equilibrium constants are less than 1, and 0 when equilibrium constants are equal to 1.

Analysis of the equations shows us that, for redox reactions with equilibrium constants less than 1 (equilibrium state favors the reactants), the E_{cell}° will be negative because the natural logarithm of any number between 0 and 1 is negative. These properties are characteristic of electrolytic cells, which house nonspontaneous oxidation–reduction reactions. Instead, if the equilibrium constant for the reaction is greater than 1 (equilibrium state favors the products), the E_{cell}° will be positive because the natural logarithm of any number greater than 1 is positive. These properties are characteristic of galvanic cells, which house spontaneous oxidation–reduction reactions. If the equilibrium constant is equal to 1 (concentrations of the reactants and products are equal at equilibrium), the E_{cell}° will be equal to zero. An easy way to remember this is that $E_{\text{cell}}^{\circ} = 0 \text{ V}$ for any concentration cell with equimolar concentrations in both half-cells because there is no net ionic equation (both half-cells contain the same ions).



KEY CONCEPT

If E_{cell}° is positive, $\ln K_{\text{eq}}$ is positive. This means that K_{eq} must be greater than one and that the equilibrium lies to the right (products are favored).

Knowing the effects of concentration on equilibria, we can now derive the change in Gibbs free energy of an electrochemical cell with varying concentrations using the equation

$$\Delta G = \Delta G^{\circ} + RT \ln Q$$

Equation 12.9

where ΔG is the free energy change under nonstandard conditions, ΔG° is the free energy change under standard conditions (which can be determined from Equation 12.4 or Equation 12.8 above), R is the ideal gas constant, T is the temperature, and Q is the reaction quotient.

MCAT Concept Check 12.3:

Before you move on, assess your understanding of the material with these questions.

Fill in the table to show the relationships between the equilibrium constant, Gibbs free energy, and electromotive force (emf), assuming standard conditions:

K_{eq}	ΔG° : (+) or (-)?	Reaction: Spontaneous or Nonspontaneous?	E°_{cell} : (+) or (-)?
1.2×10^{-2}			
2×10^2			
1			



Given the following reaction quotients and equilibrium constants, determine the direction of the reaction and the sign of E_{cell} :

Q	K_{eq}	Reaction Direction (Forward, Backward, or Equilibrium)	Sign of E_{cell}
10^{-3}	10^{-2}		
10^{-2}	1.1		
1	1		



Conclusion

In this chapter, we covered the essential MCAT topic of electrochemistry. We reviewed the behavior of many different types of electrochemical cells. Galvanic cells rely on spontaneous oxidation–reduction reactions to produce current and supply energy. The concentration cell is a special type of galvanic cell for which the current is dependent on an ion concentration gradient rather than a difference in reduction potential between two chemically distinct electrodes. Electrolytic cells rely on external voltage sources to drive a nonspontaneous oxidation–reduction reaction called electrolysis. Finally, we considered the thermodynamics of the different cell types. Galvanic and concentration cells have positive electromotive forces (emf) and negative free energy changes, whereas electrolytic cells have negative electromotive forces and positive free energy changes.

In retrospect, the content you have learned in MCAT General Chemistry Review has numerous organic (biological) and inorganic applications. And as you prepare to be a physician, you must begin to understand and treat the individual as a sum of many intertwining systems and parts. Many body systems and parts rely on electrochemical cells: the heart is a self-paced electrochemical cell, the neurons of the brain and spinal cord are rechargeable concentration cells, and every cell that contains mitochondria (all cells except erythrocytes) rely on the proton-motive force across the inner mitochondrial membrane to function. Our discussion here

of inorganic systems has value through analogy to many biological systems.

Without further delay, we want to offer you our heartiest congratulations for completing this final chapter of MCAT General Chemistry Review. The hard work, time, and energy you have invested in a careful and thorough review of the topics covered within the pages of this book will pay off on Test Day. We hope that we have been successful in meeting our goals in writing this Kaplan MCAT Review series: to assess the general concepts and principles essential to correctly and efficiently answer the general chemistry questions on the MCAT; to guide you in the development of critical thinking skills necessary for analyzing passages, question stems, and answer choices; and to provide holistic preparation for your Test Day experience. In addition to all of these, we aimed to relate the science to everyday life experiences and future experiences as a physician, demystify the concepts, and have some fun in the process. We are grateful for the opportunity to have been a part of your journey to success on the MCAT, and—beyond that—success in your medical education and future practice as the great physician you deserve to be!

Concept Summary

Electrochemical Cells

- An electrochemical cell describes any cell in which oxidation–reduction reactions take place. Certain characteristics are shared between all types of electrochemical cells.
 - Electrodes are strips of metal or other conductive materials placed in an electrolyte solution.
 - The anode is always the site of oxidation. It attracts anions.
 - The cathode is always the site of reduction. It attracts cations.
 - Electrons flow from the anode to the cathode.
 - Current flows from the cathode to the anode.
- Cell diagrams are shorthand notation that represent the reactions taking place in an electrochemical cell.
 - Cell diagrams are written from anode to cathode with electrolytes (the solution) in between.
 - A vertical line represents a phase boundary, and a double vertical line represents a salt bridge or other physical boundary.
- Galvanic (voltaic) cells house spontaneous reactions ($\Delta G < 0$) with a positive electromotive force.

- Electrolytic cells house nonspontaneous reactions ($\Delta G > 0$) with a negative electromotive force. These nonspontaneous cells can be used to create useful products through electrolysis.
- Concentration cells are a specialized form of a galvanic cell in which both electrodes are made of the same material. Rather than a potential difference causing the movement of charge, it is the concentration gradient between the two solutions.
- The charge on an electrode is dependent on the type of electrochemical cell one is studying.
 - For galvanic cells, the anode is negatively charged and the cathode is positively charged.
 - For electrolytic cells, the anode is positively charged and the cathode is negatively charged.
- Rechargeable batteries are electrochemical cells that can experience charging (electrolytic) and discharging (galvanic) states. Rechargeable batteries are often ranked by energy density—the amount of energy a cell can produce relative to the mass of battery material.
 - Lead–acid batteries, when discharging, consist of a Pb anode and a PbO_2 cathode in a concentrated sulfuric acid solution. When charging, the PbSO_4 -plated electrodes are dissociated to restore the original Pb and PbO_2 electrodes and concentrate the electrolyte. These cells have a low energy density.
 - Nickel–cadmium batteries (Ni–Cd), when discharging, consist of a Cd anode and a NiO(OH) cathode in a concentrated KOH solution. When charging, the Ni(OH)_2 and Cd(OH)_2 plated electrodes are dissociated to restore the original Cd and NiO(OH) electrodes and concentrate the

electrolyte. These cells have a higher energy density than lead–acid batteries.

- Nickel–metal hydride (NiMH) batteries have more or less replaced Ni–Cd batteries because they have higher energy density, are more cost effective, and are significantly less toxic.
- Surge current is an above-average current transiently released at the beginning of the discharge phase; it wanes rapidly until a stable current is achieved.

Cell Potentials

- A reduction potential quantifies the tendency for a species to gain electrons and be reduced. The higher the reduction potential, the more a given species wants to be reduced.
 - Standard reduction potentials (E_{red}°) are calculated by comparison to the standard hydrogen electrode (SHE) under the standard conditions of 298 K, 1 atm pressure, and 1 M concentrations.
 - The standard hydrogen electrode has a standard reduction potential of 0 V.
- Standard electromotive force (E_{cell}°) is the difference in standard reduction potential between the two half-cells.
- For galvanic cells, the difference of the reduction potentials of the two half-reactions is positive; for electrolytic cells, the difference of the reduction potentials of the two half-reactions is negative.

Electromotive Force and Thermodynamics

- Electromotive force and change in free energy always have opposite signs.
 - When E_{cell}° is positive, ΔG° is negative. This is the case in galvanic cells.
 - When E_{cell}° is negative, ΔG° is positive. This is the case in electrolytic cells.
 - When E_{cell}° is 0, ΔG° is 0. This is the case in concentration cells.
- The Nernst equation describes the relationship between the concentration of species in a solution under nonstandard conditions and the electromotive force.
- There exists a relationship between the equilibrium constant (K_{eq}) and E_{cell}° .
 - When K_{eq} (the ratio of products' concentrations at equilibrium over reactants', raised to their stoichiometric coefficients) is greater than 1, E_{cell}° is positive.
 - When K_{eq} is less than 1, E_{cell}° is negative.
 - When K_{eq} is equal to 1, E_{cell}° is 0.

Answers to Concept Checks

12.1

In a galvanic cell, the anode is the site of oxidation, has current flowing toward it, and has a (–) designation. The cathode has electrons flowing toward it and attracts cations.

In an electrolytic cell, the anode is the site of oxidation and has current flowing toward it. The cathode has electrons flowing toward it, has a (–) designation, and attracts cations.



Electrolytic cells are nonspontaneous and have a positive ΔG . Galvanic cells are spontaneous and have a negative ΔG ; therefore, they have a positive E_{cell} .

$$\text{mol M} = \frac{It}{nF} \rightarrow I = \frac{(\text{mol M})nF}{t} = \frac{\left(\frac{230 \text{ g}}{23 \frac{\text{g}}{\text{mol}}}\right)(1 \text{ mol } e^-)\left(96,485 \frac{\text{C}}{\text{mol } e^-}\right)}{30 \text{ hr}\left(\frac{3600 \text{ s}}{1 \text{ hr}}\right)} \approx \frac{10 \times 10^3}{10^1}$$

$$= 10 \text{ A (actual = 8.93 A)}$$

Battery	State of Use	Galvanic or Electrolytic	Anode Material	Anode Charge	Cathode Material	Cathode Charge
Ni—Cd	Discharging	Galvanic	Cd	Negative	NiO(OH)	Positive

Battery	State of Use	Galvanic or Electrolytic	Anode Material	Anode Charge	Cathode Material	Cathode Charge
Ni—Cd	Charging	Electrolytic	Cd(OH)_2	Positive	Ni(OH)_2	Negative
Molten NaCl	Discharging	Electrolytic	Any	Positive	Any	Negative
Daniell cell	Discharging	Galvanic	Zn	Negative	Cu	Positive
Lead-acid	Charging	Electrolytic	PbSO_4	Positive	PbSO_4	Negative
Lead-acid	Discharging	Galvanic	Pb	Negative	PbO_2	Positive

12.2

A sample is measured by setting up a cell relative to a standard hydrogen electrode, which is given a reduction potential of 0 V by convention.

A positive emf means the cell is spontaneous (galvanic); a negative emf means the cell is nonspontaneous (electrolytic).

The first cell is electrolytic because it has a negative emf. The second cell is galvanic because it has a positive emf.

The reduction potential of triiodide is higher than iron(III), so triiodide will be reduced and iron will be oxidized: $2 \text{Fe} + 3 \text{I}_3^- \rightarrow 2 \text{Fe}^{3+} + 9 \text{I}^-$ ($E^\circ_{\text{cell}} = +0.57 \text{ V}$)

12.3

K_{eq}	ΔG° : (+) or (-)?	Reaction: Spontaneous or Nonspontaneous?	E°_{cell} : (+) or (-)?
1.2×10^{-2}	+	Nonspontaneous	-
2×10^2	-	Spontaneous	+
1	0	Not applicable—applies to any cell at equilibrium	0

Remember that $\Delta G^\circ = -RT \ln K_{eq}$; if $K_{eq} < 1$, $\ln K_{eq} < 0$, and $\Delta G^\circ > 0$. If $K_{eq} > 1$, $\ln K_{eq} > 0$, and $\Delta G^\circ < 0$. If $K_{eq} = 1$, $\ln K_{eq} = 0$, and $\Delta G^\circ = 0$.

Q	K_{eq}	Reaction Direction (Forward, Backward, or Equilibrium)	Sign of E_{cell}
10^{-3}	10^{-2}	Forward	+
10^2	1.1	Backward	-
1	1	Equilibrium	0

Note that these calculations do not assume standard conditions, unlike question 1.

Equations to Remember

(12.1) Moles of electrons transferred during reduction: $M^{n+} + n e^{-} \rightarrow M (s)$

(12.2) Electrodeposition equation: $\text{mol } M = \frac{It}{nF}$

(12.3) Standard electromotive force of a cell: $E_{\text{cell}}^{\circ} = E_{\text{red,cathode}}^{\circ} - E_{\text{red,anode}}^{\circ}$

(12.4) Standard change in free energy from standard emf: $\Delta G^{\circ} = -nFE_{\text{cell}}^{\circ}$

(12.5) Nernst equation (full): $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{RT}{nF} \ln Q$

(12.6) Nernst equation (simplified): $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0592}{n} \log Q$

(12.7) Reaction quotient: $Q = \frac{[C]^c [D]^d}{[A]^a [B]^b}$

(12.8) Standard change in free energy from equilibrium constant: $\Delta G^{\circ} = -RT \ln K_{\text{eq}}$

(12.9) Free energy change (nonstandard conditions): $\Delta G = \Delta G^\circ + RT \ln Q$



Shared Concepts

Biochemistry Chapter 3

Nonenzymatic Protein Function and Protein Analysis

Biochemistry Chapter 8

Biological Membranes

General Chemistry Chapter 7

Thermochemistry

General Chemistry Chapter 11

Oxidation–Reduction Reactions

Physics and Math Chapter 5

Electrostatics and Magnetism

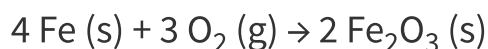
Physics and Math Chapter 6

Circuits

Discrete Practice Questions

Consult your online resources for additional practice.

1. Rusting occurs due to the oxidation–reduction reaction of iron with environmental oxygen:

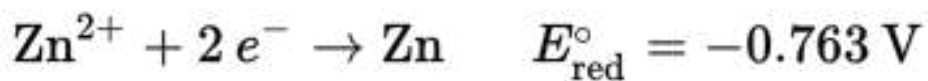


Some metals cannot react with oxygen in this fashion.

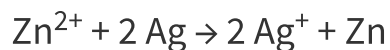
Which of the following best explains why iron can?

- (A) Iron has a more positive reduction potential than those metals, making it more likely to donate electrons to oxygen.
- (B) Iron has a more positive reduction potential than those metals, making it more likely to accept electrons from oxygen.
- (C) Iron has a less positive reduction potential than those metals, making it more likely to donate electrons to oxygen.
- (D) Iron has a less positive reduction potential than those metals, making it more likely to accept electrons from oxygen.

2. Given the following standard reduction potentials:

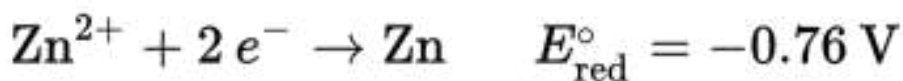
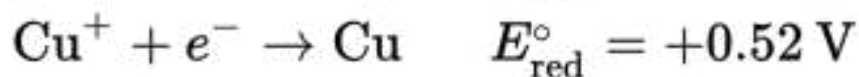
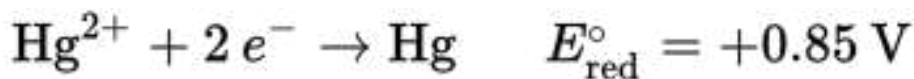


What is the standard electromotive force of the following reaction?



- (A) -2.2 V
- (B) -1.1 V
- (C) $+1.1 \text{ V}$
- (D) $+2.2 \text{ V}$

3. Consider the following data:



The anode of a certain galvanic cell is composed of copper. Which of the metals from the data table can be used at the cathode, assuming equal concentrations of the two electrolyte solutions?

- (A) Hg
- (B) Cu
- (C) Zn
- (D) Al

4. An electrolytic cell is filled with water. Which of the following will move toward the cathode of such a cell?

- I. H^+ ions
- II. O^{2-} ions
- III. Electrons

- (A) I only
- (B) II only
- (C) I and III only
- (D) II and III only

5. If the value of E°_{cell} is known, what other data is needed to calculate ΔG° ?

- (A) Equilibrium constant
- (B) Reaction quotient
- (C) Temperature of the system
- (D) Half-reactions of the cells

6. Which of the following compounds is LEAST likely to be found in the salt bridge of a galvanic cell?

- (A) NaCl
- (B) SO_3
- (C) MgSO_3
- (D) NH_4NO_3

7. If the surface area of electrode material in an electrochemical cell is tripled, what else is necessarily tripled?

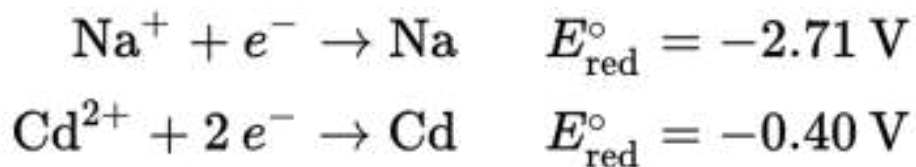
- I. E°_{cell}
- II. Current
- III. K_{eq}

- (A) I only
- (B) II only
- (C) I and II only
- (D) II and III only

8. Which of the following can alter the emf of an electrochemical cell?

- (A) The mass of the electrodes
- (B) The length of the wire connecting the half-cells
- (C) The overall size of the battery
- (D) The temperature of the solutions in the half-cells

9. Which of the following statements could be true about a Na–Cd cell, based on the information below?



- (A) It is a galvanic cell, and sodium is the cathode.
- (B) It is an electrolytic cell, and cadmium is the anode.
- (C) It is a galvanic cell, with $E_{\text{cell}}^\circ = 3.11 \text{ V}$.
- (D) It is an electrolytic cell, with $E_{\text{cell}}^\circ = -3.11 \text{ V}$.

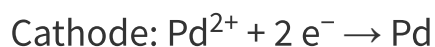
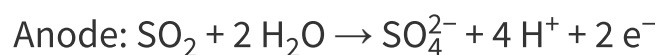
10. Which of the following expressions correctly describes the relationship between standard electromotive force and standard change in free energy?

- (A) $\Delta G^\circ = -nF(E_{\text{red,anode}}^\circ - E_{\text{red,cathode}}^\circ)$
- (B) $E_{\text{cell}}^\circ = \frac{nF}{RT} \ln K_{\text{eq}}$
- (C) $E_{\text{cell}}^\circ = -\frac{RT}{nF} \ln K_{\text{eq}}$
- (D) $\Delta G^\circ = nF(E_{\text{red,anode}}^\circ - E_{\text{red,cathode}}^\circ)$

11. Which of the following choices is indicative of a spontaneous reaction, assuming standard conditions?

- (A) E_{cell}° is negative
- (B) $Q = K_{\text{eq}}$
- (C) The cell is a concentration cell
- (D) $K_{\text{eq}} > 1$

12. For a cell with the following half-reactions:



How would decreasing the pH of the solution inside the cell affect the electromotive force (emf)?

- (A) The emf would decrease.
- (B) The emf would remain the same.
- (C) The emf would increase.
- (D) The emf would become zero.

13. An electrolytic cell necessarily has:

- (A) $\Delta S^{\circ} > 0$
- (B) $\Delta G^{\circ} < 0$
- (C) $K_{\text{eq}} < 1$
- (D) $E_{\text{cell}}^{\circ} > 0$

14. Which of the following is the best explanation for the fact that a larger mass of electrodes are required for lead–acid batteries, as compared to other batteries, to produce a certain current?

- (A) The lead–acid electrolyte, sulfuric acid, is diprotic and incompletely dissociates in solution.
- (B) The energy density of lead–acid electrodes is higher than that of other batteries.
- (C) The electrolytes in other batteries less readily dissociate than those of lead–acid batteries.
- (D) The energy density of lead–acid electrodes is lower than that of other batteries.

15. Which of the following best describes why over-charging a Ni–Cd battery is not detrimental?

- (A) The energy density of a Ni–Cd battery is high, so it can store more charge than other batteries per its mass.
- (B) The electrodes of a Ni–Cd battery can discharge through the circuit when they are fully charged.
- (C) The Ni–Cd battery will stop accepting electrons from an outside source when its electrodes are recharged.
- (D) Ni–Cd batteries have a high surge current and can dissipate the overcharge before damage can occur to electrodes.

Explanations to Discrete Practice Questions

1. C

In the oxidation–reduction reaction of a metal with oxygen, the metal will be oxidized (donate electrons) and oxygen will be reduced (accept electrons). This fact allows us to immediately eliminate (B) and (D). A species with a higher reduction potential is more likely to be reduced, and a species with a lower reduction potential is more likely to be oxidized. Based on the information in the question, iron is oxidized more readily than those metals; this means that iron has a lower reduction potential.

2. B

To determine the standard electromotive force of a cell, simply subtract the standard reduction potentials of the two electrodes. In this case, the cathode is zinc because it is being reduced; the anode is silver because it is being oxidized. Thus,

$$E_{\text{cell}}^{\circ} = E_{\text{red,cathode}}^{\circ} - E_{\text{red,anode}}^{\circ} = -0.763 - 0.337 = -1.10 \text{ V}$$

While we must multiply the silver half-reaction by two to balance electrons, the actual value for the reduction potential does not change.

Remember that the standard reduction potential is determined by the identity of the electrode, not the amount of it present.

3. A

Oxidation occurs at the anode, and reduction occurs at the cathode. Because Cu is the anode, it must be oxidized. The reduction potential of the cathode cannot be less than that of the anode for a galvanic cell. Therefore, mercury, (A), must be the cathode. In a concentration cell, the same material is used as both the cathode and anode; however, this question assumes equal concentrations. If both electrolyte solutions have the same concentration, there will be no oxidation–reduction reaction and, therefore, no anode or cathode. This eliminates (B).

4. C

In an electrolytic cell, ionic compounds are broken up into their constituents; the cations (positively charged ions) migrate toward the cathode, and the anions (negatively charged ions) migrate toward the anode. In this case, the cations are H^+ ions (protons), so option I is correct. Electrons flow from anode to cathode in all types of cells, meaning that option III is also correct. Option II is incorrect for two reasons. First, it is unlikely that the anions in any cell would be O^{2-} rather than OH^- . Second, and more significantly, these anions would flow to the anode, not the cathode.

5. D

This answer comes directly from the equation relating Gibbs free energy and E°_{cell} . $\Delta G^\circ = -nFE^\circ_{\text{cell}}$, where n is the number of moles of electrons transferred and F is the Faraday constant, **96,485** $\frac{\text{C}}{\text{mol } e^-}$. To

determine n , one must look at the balanced half-reactions occurring in the oxidation–reduction reaction.

6. B

Salt bridges contain inert electrolytes. Ionic compounds, such as (A), (C), and (D), are known to be strong electrolytes because they completely dissociate in solution. (B) cannot be considered an electrolyte because its atoms are covalently bonded and will not dissociate in aqueous solution. (B) and (C) may appear similar, but there is an important distinction to be made. (C) implies that Mg^{2+} and SO_3^{2-} are the final, dissociated ionic constituents, while (B) implies that neutral SO_3 would have to be dissolved in solution.

7. B

Potential, as measured by E°_{cell} , is dependent only on the identity of the electrodes and not the amount present. Similarly, the equilibrium constant depends only on the identity of the electrolyte solutions and the temperature. However, as the electrode material is increased, the surface area participating in oxidation–reduction reactions is increased and more electrons are released, making statement II correct.

8. D

E°_{cell} is dependent upon the change in free energy of the system through the equation $RT \ln K_{\text{eq}} = nFE^\circ_{\text{cell}}$. The temperature, T , appears in this equation; thus, a change in temperature will impact the E°_{cell} .

9. B

If this were a galvanic cell, the species with the more positive reduction potential (cadmium) would be reduced. The cathode is always reduced

in an electrochemical cell, so sodium could not be the cathode in such a galvanic cell, eliminating (A). Sodium would be the cathode in an electrolytic cell, however, which would make cadmium the anode. Thus, the answer is (B). Note that we do not have to determine E_{cell}° because we already know the answer. However, the E_{cell}° would be $-2.71 - (-0.40) = -2.31$ V for an electrolytic cell, and $+2.31$ V for a galvanic cell, eliminating (C) and (D).

10. D

There are only two equations involving standard change in free energy in electrochemical cells: $\Delta G^{\circ} = -nFE_{\text{cell}}^{\circ}$ and $\Delta G^{\circ} = -RT \ln K_{\text{eq}}$. Substituting $E_{\text{cell}}^{\circ} = E_{\text{red,cathode}}^{\circ} - E_{\text{red,anode}}^{\circ}$ into the first equation and distributing the negative sign gives (D). (A) would be the opposite of ΔG° . Setting the two equations equal to each other, we get $RT \ln K_{\text{eq}} = nFE_{\text{cell}}^{\circ}$. Solving for E_{cell}° , we get $E_{\text{cell}}^{\circ} = \frac{RT}{nF} \ln K_{\text{eq}}$, which is the opposite of (B). (C) incorrectly solves the algebra.

11. D

A spontaneous electrochemical reaction has a negative ΔG . Using the equation $\Delta G^{\circ} = -RT \ln K_{\text{eq}}$, $K_{\text{eq}} > 1$ would result in $\ln K_{\text{eq}} > 0$, which means $\Delta G^{\circ} < 0$. A negative electromotive force, (A), or equilibrium state, (B), would not correspond to a spontaneous reaction. Concentration cells can be spontaneous; however, if the concentration cell had reached equilibrium, it would cease to be a spontaneous reaction, eliminating (C). When an answer choice may be true, but does not have to be—it is the wrong answer on Test Day.

12. A

A change in pH has a direct correlation to the hydrogen ion (H^+) concentration. Decreasing the pH increases the H^+ concentration, which means the concentration of products has increased in the oxidation of sulfur dioxide. This means it would be harder to liberate electrons, thus decreasing the emf. One could also view this decrease in oxidation potential as an increase in reduction potential. If $E_{\text{red,anode}}^\circ$ increases, then E_{cell}° must decrease according to $E_{\text{cell}}^\circ = E_{\text{red,cathode}}^\circ - E_{\text{red,anode}}^\circ$.

13. C

An electrolytic cell is nonspontaneous. Therefore, the ΔG° must be positive and E_{cell}° must be negative, eliminating (B) and (D). The change in entropy may be positive or negative, depending on the species involved, eliminating (A). According to the equation $\Delta G^\circ = -RT \ln K_{\text{eq}}$, $K_{\text{eq}} < 1$ would result in $\ln K_{\text{eq}} < 0$, which means $\Delta G^\circ > 0$.

14. D

Compared to other cell types, lead–acid batteries have a characteristically low energy density, (D). While (A) is a true statement, the incomplete dissociation of sulfuric acid does not fully explain the low energy density of lead–acid batteries. (C) is likely to be an opposite; the more easily the electrodes dissociate, the easier it is to carry out oxidation–reduction reactions with them.

15. C

During the recharge cycle, Ni–Cd cells will accept current from an outside source until the Cd and NiO(OH) electrodes are pure; at this point, the reaction will stop because Cd(OH)_2 runs out and no more electrons can be accepted. (A) and (B) are both true statements, but they fail to explain why overcharging the battery (continuing to try to run

current into the battery even when the electrodes are reverted to their original state) is not a problem with Ni–Cd batteries. Finally, surge current refers to the initial burst of current seen in some batteries; once charged, the surge current will not increase even if the power source continues to be run because no additional charge will be stored on the electrodes, eliminating (D).

Glossary

Absolute zero—The temperature at which all substances have no thermal energy; 0 K or -273.15°C .

Absorption spectrum—The series of discrete lines at characteristic frequencies representing the energy required to excite an electron from the ground state.

Acid—A species that donates hydrogen ions or accepts electrons.

Acid dissociation constant (K_a)—The equilibrium constant that measures the degree of dissociation of an acid under specific conditions.

Acidic solution—An aqueous solution that contains more H^+ ions than OH^- ions; $\text{pH} < 7$ under standard conditions.

Actinide series—The series of chemical elements atomic numbered 89-103 and falling between the S and D blocks on the periodic table.

Activation energy (E_a)—The minimum amount of energy required for a reaction to reach the transition state; also called energy barrier.

Actual yield—The experimental quantity of a substance obtained at the end of a reaction.

Adiabatic process—A process that occurs without the transfer of heat into or out of the system.

Alkali metals—Elements found in Group IA of the periodic table; highly reactive, readily losing one valence electron to form ionic compounds with nonmetals.

Alkaline earth metals—Elements found in Group IIA of the periodic table; chemistry is similar to that of the alkali metals, except that they have two valence electrons and, thus, form +2 cations.

Amphiprotic species—A species that may either gain or lose a proton.

Amphoteric species—A species capable of reacting as either an acid or base, depending on the nature of the reactants.

Angular momentum—The rotational analog of linear momentum.

Anion—An ionic species with a negative charge.

Anode—The electrode at which oxidation occurs.

Antibonding orbital—A molecular orbital formed by the overlap of two or more atomic orbitals; energy is greater than the energy of the combining atomic orbitals.

Aqueous solution—A solution in which water is the solvent.

Arrhenius acid—A species that donates protons (H^+) in aqueous solution.

Arrhenius base—A species that donates hydroxide ions (OH^-) in aqueous solution.

Arrhenius equation—A chemical kinetics equation that relates the rate constant (k) of a reaction with the frequency factor (A), the activation energy (E_a), the ideal gas constant (R), and temperature (T) in kelvin.

Atom—The smallest unit of an element that retains the properties of the element; it cannot be further broken down by chemical means.

Atomic mass—The mass of a given isotope of an element; closely related to the mass number.

Atomic mass unit (amu)—A unit of mass defined as $\frac{1}{12}$ the mass of a carbon-12 atom; approximately equal to the mass of one proton or one neutron.

Atomic number—The number of protons in a given element.

Atomic orbital—Describes the region of space where there is a high probability of finding an electron.

Atomic radius—The average distance between a nucleus and its outermost electron; usually measured as one-half the distance between two nuclei of an element in its elemental form.

Atomic weight—The weighted average mass of the atoms of an element, taking into account the relative abundance of all naturally occurring isotopes.

Aufbau principle—The concept that electrons fill energy levels in order of increasing energy, completely filling one sublevel before beginning to fill the next.

Autoionization—The process by which a molecule (usually water) spontaneously dissociates into cations and anions.

Avogadro's number—The number of atoms or molecules in one mole of a substance: $6.02 \times 10^{23} \text{ mol}^{-1}$.

Avogadro's principle—The law stating that under the same conditions of temperature and pressure, equal volumes of different gases will have the same number of molecules.

Azimuthal quantum number (l)—The quantum number denoting the sublevel or subshell in which an electron can be found; reveals the shape of the orbital.

Balanced equation—An equation for a chemical reaction in which the number of atoms for each element in the reaction and the total charge are the same for the reactants and the products.

Balmer series—Part of the emission spectrum for hydrogen, representing transitions of an electron from energy levels $n > 2$ to $n = 2$.

Barometer—A tool for measuring pressure.

Base—A species that donates hydroxide ions or electron pairs or that accepts protons.

Base dissociation constant (K_b)—The equilibrium constant that measures the degree of dissociation for a base under specific conditions.

Basic solution—An aqueous solution that contains more OH^- ions than H^+ ion; $\text{pH} > 7$ under standard conditions.

Bohr model—The model of the hydrogen atom in which electrons assume certain circular orbits around a positive nucleus.

Boiling point—The temperature at which the vapor pressure of a liquid is equal to the incident pressure; the normal boiling point of any liquid is defined as its boiling point at a pressure of 1 atmosphere.

Boiling point elevation—The amount by which a given quantity of solute raises the boiling point of a liquid; a colligative property.

Bond energy—The energy (enthalpy change) required to break a particular bond under given conditions.

Bond enthalpy—The average energy that is required to break a particular type of bond between atoms in the gas phase.

Bonding electrons—Electrons located in the valence shell of an atom and involved in a covalent bond.

Bonding orbital—A molecular orbital formed by the overlap of two or more atomic orbitals; energy is less than that of the combining orbitals.

Bond length—The average distance between two nuclei in a bond; as the number of shared electron pairs increases, the bond length decreases.

Bond order—The number of shared electron pairs between two atoms; a single bond has a bond order of 1, a double bond has a bond order of 2, a triple bond has a bond order of 3.

Boyle's law—The law stating that at constant temperature, the volume of a gaseous sample is inversely proportional to its pressure.

Broken-order reaction—A reaction with noninteger orders in its rate law.

Brønsted–Lowry acid—A proton donor.

Brønsted–Lowry base—A proton acceptor.

Buffer—A solution containing a weak acid and its salt (or a weak base and its salt) that tends to resist changes in pH.

Buffer region—The portion of a titration curve in which the concentration of an acid is approximately equal to that of its conjugate base; pH remains relatively constant through this region.

Buffering capacity—The degree to which a system can resist changes in pH.

Calorie (cal)—A unit of thermal energy.

Calorimeter—An apparatus used to measure the heat absorbed or released by a reaction.

Catalyst—A substance that increases the rates of the forward and reverse directions of a specific reaction by lowering activation energy, but is itself left unchanged.

Cathode—The electrode at which reduction takes place.

Cation—An ionic species with a positive charge.

Celsius ($^{\circ}\text{C}$)—A temperature scale defined by having 0°C equal to the freezing point of water and 100°C equal to the boiling point of water; otherwise known as the centigrade temperature scale.

Chalcogens—Elements found in Group VIA of the periodic table with diverse chemistry; the group contains metals, nonmetals (like oxygen), and metalloids; typically form -2 anions.

Charging—A state of an electrochemical cell in which an external electromotive force is being used to return a cell to its original state; during this process, electrons are transferred nonspontaneously from cathode to anode.

Charles's law—The law stating that the volume of a gas at constant pressure is directly proportional to its absolute (kelvin) temperature.

Chelation—The process of binding metal ions to the same ligand at multiple points.

Chemical bond—The interaction between two atoms resulting from the sharing or transfer of electrons.

Chemical equation—An expression used to describe the quantity and identity of the reactants and products of a reaction.

Chemical properties—Those properties of a substance related to the chemical changes that it undergoes, such as ionization energy and electronegativity.

Closed system—A system that can exchange energy but not matter with its surroundings.

Colligative properties—Those properties of solutions that depend only on the number of solute particles present but not on the nature of those particles.

Collision theory of chemical kinetics—A theory that states that the rate of a reaction is proportional to the number of collisions per second between reacting molecules that have sufficient energy to overcome the activation energy barrier; implies that only a fraction of collisions are sufficient.

Combination reaction—A reaction in which two or more reactants form a single product.

Combined gas law—A gas law that combines Boyle's law, Charles's law, and Gay-Lussac's law to state that pressure and volume are inversely proportional to each other, and each is directly proportional to temperature.

Combustion reaction—A reaction in which an oxidant (typically oxygen) reacts with a fuel (typically a hydrocarbon) to yield water and an oxide (such as carbon dioxide if between a hydrocarbon and oxygen).

Common ion effect—A shift in the equilibrium of a solution due to the addition of ions of a species already present in the reaction mixture.

Complexation reaction—A reaction in which a central cation is bound to one or more ligands.

Complex ion—A polyatomic molecule in which a central cation is bonded to electron pair donors called ligands.

Compound—A pure substance that can be decomposed to produce elements, other compounds, or both.

Compression—Reduction in the volume of a gas.

Concentrated solution—A solution with a high concentration value; the cutoff for the term “concentrated” depends on the purpose and identity

of the solution.

Concentration—The amount of solute per unit of solvent or the relative amount of one component in a mixture.

Concentration cell—A cell that creates an electromotive force (emf or voltage) using a single chemical species in half-cells of varying concentration.

Condensation—The process in which a gas transitions to the liquid state.

Conductor—A material in which electrons are able to transfer energy in the form of heat or electricity.

Conjugate acid–base pair—The relationship between a Brønsted–Lowry acid and its deprotonated form, or a Brønsted–Lowry base and its protonated form.

Coordinate covalent bond—A covalent bond in which both electrons of the bonding pair are donated by one of the bonded atoms.

Coordination number—The number of atoms that are bound to a central atom.

Covalent bond—A chemical bond formed by the sharing of an electron pair between two atoms; can be in the form of single bonds, double bonds, or triple bonds.

Critical point—The point in a phase diagram beyond which the phase boundary between liquid and gas no longer exists.

Critical pressure—The vapor pressure at the critical temperature of a given substance.

Critical temperature—Also known as the critical point. The highest temperature at which the liquid and gas phases of a substance can coexist; above this temperature, the liquid and gas phases are indistinguishable.

Crystal—A solid in which atoms, ions, or molecules are arranged in a regular, three-dimensional lattice structure.

d subshell—Subshell corresponding to the angular momentum quantum number $l = 2$; contains five orbitals and is found in the third and higher principal energy levels.

Dalton's law of partial pressures—The law stating that the sum of the partial pressures of the components of a gaseous mixture must equal the total pressure of the sample.

Daniell cell—An electrochemical cell in which the anode is the site of Zn metal oxidation and the cathode is the site of Cu^{2+} ion reduction.

Decomposition reaction—A reaction in which a single compound breaks down into two or more products.

Delocalized orbitals—Molecular orbitals in which electron density is spread over an entire molecule, or a portion thereof, rather than being localized between two atoms.

Density (ρ)—A physical property of a substance, defined as the mass contained in a unit of volume.

Deposition—In most chemical processes, the direct transition of a substance from the gaseous state to the solid state; in electrochemical reactions, the build up of a solid precipitate onto an electrode.

Diamagnetism—A condition that arises when a substance has no unpaired electrons and is slightly repelled by a magnetic field.

Diffusion—The random motion of gas or solute particles across a concentration gradient, leading to uniform distribution of the gas or solute throughout the container.

Dilute solution—A solution with a low concentration of a given solute.

Dipole—A species containing bonds between elements of different electronegativities, resulting in an unequal distribution of charge.

Dipole–dipole interactions—The attractive forces between two dipoles; magnitude is dependent on both the dipole moments and the distance between the two species.

Dipole moment—A vector quantity with a magnitude that is dependent on the product of the charges and the distance between them; oriented from the positive to the negative pole.

Discharging—The state of a rechargeable electrochemical cell that is providing an electromotive force by allowing electrons to flow spontaneously from anode to cathode.

Disproportionation—An oxidation–reduction reaction in which the same species acts as the oxidizing agent and as the reducing agent; also called dismutation.

Dissociation—The separation of a single species into two separate species; usually used in reference to salts or weak acids or bases.

Double-displacement reaction—A reaction in which ions from two different compounds swap their associated counterions; typically, one of the products of this type of reaction is insoluble in solution and will precipitate.

Ductility—The property of metals that allows a material to be drawn into thinly stretched wires.

Effective nuclear charge (Z_{eff})—The charge perceived by an electron from the nucleus; applies most often to valence electrons and influences periodic trends such as atomic radius and ionization energy.

Effusion—The movement of gas from one compartment to another under pressure through a small opening; follows Graham's law.

Electrochemical cell—A cell within which an oxidation–reduction reaction takes place, containing two electrodes between which there is an electrical potential difference.

Electrode—An electrical conductor through which an electrical current enters or leaves a medium.

Electrolysis—The process in which an electrical current is used to power an otherwise nonspontaneous decomposition reaction.

Electrolyte—A compound that ionizes in water and increases the conductance of the solution.

Electrolytic cell—An electrochemical cell that uses an external voltage source to drive a nonspontaneous oxidation–reduction reaction.

Electromagnetic radiation—A wave composed of electric and magnetic fields oscillating perpendicular to each other and to the direction of propagation.

Electromagnetic spectrum—The range of all possible frequencies or wavelengths of electromagnetic radiation.

Electromotive force (emf)—The potential difference developed between the cathode and the anode of an electrochemical cell; also called voltage.

Electron (e^-)—A subatomic particle that remains outside the nucleus and carries a single negative charge; in most cases, its mass is considered to be negligible.

Electron affinity—The energy dissipated by a gaseous species when it gains an electron.

Electron configuration—The symbolic representation used to describe the electron arrangement within the energy sublevels in a given atom.

Electron spin—The intrinsic angular momentum of an electron, represented by m_s ; has arbitrary values of $+\frac{1}{2}$ and $-\frac{1}{2}$.

Electronegativity—A measure of the ability of an atom to attract the electrons in a bond; commonly measured with the Pauling scale.

Electronic geometry—The spatial arrangement of all pairs of electrons around a central atom, including both the bonding and lone pairs.

Electron shell—The space occupied by/path followed by an electron around an atom's nucleus. Electron shell (also called principle energy level) for a given electron is indicated by its principle quantum number.

Element—A substance that cannot be further broken down by chemical means; defined by its number of protons (atomic number).

Emission spectrum—A series of discrete lines at characteristic frequencies, each representing the energy emitted when electrons in an

atom return from an excited state to their ground state.

Empirical formula—The simplest whole-number ratio of the different elements in a compound.

Endothermic reaction—A reaction that absorbs heat from the surroundings as the reaction proceeds (positive ΔH).

Endpoint—The point in a titration at which the indicator changes to its final color.

Energy density—An equivalence unit regarding the amount of electrochemical energy capable of being stored per unit weight; a battery with a large energy density can produce a large amount of energy with a small amount of material.

Enthalpy (H)—The heat content of a system at constant pressure; the change in enthalpy (ΔH) in the course of a reaction is the difference between the enthalpies of the products and the reactants.

Entropy (S)—A property related to dispersion of energy through a system or the degree of disorder in that system; the change in entropy (ΔS) in the course of a reaction is the difference between the entropies of the products and the reactants.

Equilibrium—The state of balance in which the forward and reverse reaction rates of a reversible reaction are equal; the concentrations of all

species will remain constant over time unless there is a change in the reaction conditions.

Equilibrium constant (K_{eq})—The ratio of the concentrations of the products to the concentrations of the reactants for a certain reaction at equilibrium, all raised to their stoichiometric coefficients.

Equivalence point—The point in a titration at which the moles of acid present equal the moles of base added, or vice-versa.

Equivalent—A mole of charge in the form of electrons, protons, ions, or other measurable quantities that are produced by a substance.

Evaporation—The transition from a liquid to a gaseous state.

Excess reagent—In a chemical reaction, any reagent that does not limit the amount of product that can be formed.

Excitation—The promotion of an electron to a higher energy level by absorption of an energy quantum.

Excited state—An electronic state having a higher energy than the ground state; typically attained by the absorption of a photon of a certain energy.

Exothermic reaction—A reaction that gives off heat to the surroundings (negative ΔH) as the reaction proceeds.

f subshell—The subshell corresponding to the angular momentum quantum number $l = 3$; contains seven orbitals and is found in the fourth and higher principal energy levels.

Faraday constant (F)—The total charge on 1 mole of electrons

$\left(K_b = 0.512 \frac{\text{K} \cdot \text{kg}}{\text{mol}} \right)$. ; not to be confused with the farad (also denoted F), a unit of capacitance.

First law of thermodynamics—The law stating that the total energy of a system and its surroundings remains constant.

First-order reaction—A reaction in which the rate is directly proportional to the concentration of only one reactant.

Fluid—A substance that flows due to weak intermolecular attractions between molecules and that takes the shape of its container; liquids and gases are considered fluids.

Formal charge—The conventional assignment of charges to individual atoms of a Lewis structure for a molecule; the total number of valence electrons in the free atom minus the total number of electrons when the atom is bonded (assuming equal splitting of the electrons in bonds).

Formula weight—The sum of the atomic weights of constituent ions according an ionic compound's empirical formula.

Freezing—The process in which a liquid transitions to the solid state; also known as solidification or crystallization.

Freezing point—At a given pressure, the temperature at which the solid and liquid phases of a substance coexist in equilibrium; identical to the melting point.

Freezing point depression—Amount by which a given quantity of solute lowers the freezing point of a liquid; a colligative property.

Galvanic cell—An electrochemical cell that uses a spontaneous oxidation–reduction reaction to generate an electromotive force; also called a voltaic cell.

Galvanization—In electrochemical cells, the precipitation process onto the cathode itself; also called plating.

Gas—The physical state of matter possessing the most disorder, in which molecules interact through very weak attractions; found at relatively low pressure and high temperatures.

Gas constant (R)—A proportionality constant that appears in the ideal gas law equation, $PV = nRT$. Its value depends on the units of pressure, temperature, and volume used in a given situation.

Gay-Lussac's law—The law stating that the pressure of a gaseous sample at constant volume is directly proportional to its absolute temperature.

Gibbs free energy (G)—The energy of a system available to do work. The change in Gibbs free energy, ΔG , can be determined for a given reaction equation from the enthalpy change, temperature, and entropy change; a

negative ΔG denotes a spontaneous reaction, while a positive ΔG denotes a nonspontaneous reaction.

Graham's law—The law stating that the rate of effusion or diffusion for a gas is inversely proportional to the square root of the gas's molar mass.

Gram equivalent weight (GEW)—The amount of a compound that contains 1 mole of reacting capacity when fully dissociated; one GEW equals the molar mass divided by the reactive capacity (how many of the species of interest is obtained) per formula unit.

Ground state—The unexcited state of an electron.

Group—A vertical column of the periodic table containing elements that are similar in their chemical properties; also called a family.

Half-cell—The separated compartments housing the electrodes and solutions in an electrochemical reaction.

Half-equivalence point—The point at which half a given species within a titration has been protonated or deprotonated.

Half-reaction—Either the reduction half or oxidation half of an oxidation–reduction reaction; in an electrochemical cell, each half-reaction occurs at one of the electrodes.

Halogens—The active nonmetals in Group VIIA of the periodic table, which have high electronegativities and high electron affinities.

Heat—The energy transferred spontaneously from a warmer sample to a cooler sample.

Heat of formation (ΔH_f)—The heat absorbed or released during the formation of a pure substance from its elements at a constant pressure.

Heat of fusion (ΔH_{fus})—The enthalpy change for the conversion of 1 gram or 1 mole of a solid to a liquid at constant temperature and pressure.

Heat of sublimation (ΔH_{sub})—The enthalpy change for the conversion of 1 gram or 1 mole of a solid to a gas at constant temperature and pressure.

Heat of vaporization (ΔH_{vap})—The enthalpy change for the conversion of 1 gram or 1 mole of a liquid to a gas at constant temperature and pressure.

Heisenberg uncertainty principle—The concept that states that it is impossible to determine both the momentum and position of an electron simultaneously with perfect accuracy.

Henderson–Hasselbalch equation—Equation showing the relationship of the pH or pOH of a solution to the pK_a or pK_b and the ratio of the concentrations of the dissociated species.

Henry's law—The law stating that the mass of a gas that dissolves in a solution is directly proportional to the partial pressure of the gas above the solution.

Hess's law—The law stating that the energy change in an overall reaction is equal to the sum of the energy changes in the individual reactions that comprise it.

Heterogeneous—Nonuniform in composition.

Heterogeneous catalyst—A catalyst that is not in the same phase of matter as the reactants (for example, a solid platinum catalyst reacting with hydrogen gas).

Homogeneous—Uniform in composition.

Homogeneous catalyst—A catalyst that is in the same phase of matter as the reactants (for example, an aqueous enzyme in the cytoplasm of a cell).

Hund's rule—The rule that electrons will fill into separate orbitals with parallel spins before pairing within an orbital.

Hybridization—The combination of two or more atomic orbitals to form new orbitals with properties that are intermediate between those of the original orbitals.

Hydrogen bonding—The strong attraction between a hydrogen atom bonded to a highly electronegative atom (such as nitrogen, oxygen, or fluorine) in one molecule and a highly electronegative atom in another molecule.

Hydrolysis—A reaction in which water is consumed during the breakdown of another molecule.

Hydronium ion—The H_3O^+ ion.

Hydroxide ion—The OH^- ion.

Ideal bond angle—An angle between nonbonding or bonding electron pairs that minimizes the repulsion between them.

Ideal gas—A hypothetical gas with behavior that is described by the ideal gas law under all conditions; assumes that its particles have zero volume and do not exhibit interactive forces.

Ideal gas law—The equation stating $PV = nRT$, where R is the gas constant; can be used to describe the behavior of many real gases at moderate pressures and temperatures significantly above absolute zero.

Ideal solution—A solution with an enthalpy of dissolution that is equal to zero.

Indicator—A substance used in low concentrations during a titration that changes color over a certain pH range (acid–base titrations) or at a particular electromotive force (oxidation–reduction titrations); the final color change of an indicator occurs at the endpoint of a titration.

Inert—Unreactive.

Inert gases—The elements in Group VIIIA, which contain a full octet of valence electrons in their outermost shells and are therefore very unreactive; also called noble gases.

Intermediate—A molecule that transiently exists in a multistep reaction; does not appear in the overall balanced equation.

Intermolecular forces—The attractive and repulsive forces between molecules.

Intramolecular forces—The attractive forces between atoms within a single molecule (ionic and covalent bonds).

Ion—A charged atom or molecule that results from the loss or gain of electrons.

Ion product (IP)—The general term for the reaction quotient of a dissolving ionic compound; compared to K_{sp} to determine the saturation status of a solution.

Ionic bond—A chemical bond formed through electrostatic interaction between positive and negative ions.

Ionic radius—The average distance from the center of the nucleus to the edge of its electron cloud; cationic radii are generally smaller than their parent metal, whereas anionic radii are generally larger than their parent nonmetal.

Ionic solid—A solid consisting of positive and negative ions arranged into crystals that are made up of regularly repeated units held together by ionic bonds.

Ionization energy—The energy required to remove an electron from the valence shell of a gaseous atom.

Irreversible reaction—A reaction that proceeds in one direction only and goes to completion.

Isobaric process—A process that occurs at constant pressure.

Isoelectric focusing—A technique used to separate amino acids or polypeptides based on their isoelectric points.

Isolated system—A system that can exchange neither matter nor energy with its surroundings.

Isothermal process—A process that occurs at constant temperature.

Isotopes—Atoms containing the same number of protons but different numbers of neutrons.

Isovolumetric process—A process that occurs at constant volume in which the system performs no work; also called an isochoric process.

Joule (J)—The unit of energy; $\frac{1 \text{ atm}}{2 \text{ atm}} = 0.5.$

Kelvin (K)—A temperature scale with units equal to the units of the Celsius scale and absolute zero defined as 0 K; also called the absolute temperature scale.

Kinetic molecular theory—The theory proposed to account for the observed behavior of gases; considers gas molecules to be pointlike, volumeless particles exhibiting no intermolecular forces that are in constant random motion and undergo only completely elastic collisions with the container or other gas particles.

Kinetic product—The product of a reaction that is formed favorably at a lower temperature because thermal energy is not available to form the transition state required to create a more stable thermodynamic product; has a smaller overall difference in free energy between the products and reactants than the thermodynamic product.

Lanthanide series—The series of chemical elements atomic numbered 57-71 and falling between the S and D blocks on the periodic table.

Latent heat—The enthalpy of an isothermal process.

Law of conservation of charge—The law stating that, in a given reaction, the charge of ions in the products is equal to the charge of ions in the reactants.

Law of conservation of mass—The law stating that, in a given reaction, the mass of the products is equal to the mass of the reactants.

Law of constant composition—The law stating that the elements in a pure compound are found in specific mass ratios.

Law of mass action—The form of the equilibrium constant; has the concentrations of products over concentrations of reactants, each raised to their stoichiometric coefficients.

Le Châtelier's principle—The observation that when a system at equilibrium is disturbed or stressed, the system will react in such a way as to relieve the stress and restore equilibrium.

Lead-acid battery—An electrochemical cell in which the anode is the site of Pb metal oxidation and the cathode is the site of Pb^{4+} ion reduction. The electrolyte is a strong acid, usually sulfuric acid.

Lewis acid—A species capable of accepting an electron pair.

Lewis base—A species capable of donating an electron pair.

Lewis structure—A method of representing the shared and unshared electrons of an atom, molecule, or ion; also called a Lewis dot diagram.

Ligand—A molecule bonded to a metal ion in a coordination compound; ligands are Lewis bases that form coordinate covalent bonds with the central metal ion.

Limiting reagent—In a chemical reaction, the reactant present in such quantity as to limit the amount of product that can be formed.

Liquid—The state of matter in which intermolecular attractions are intermediate between those in gases and in solids, distinguished from the gas phase by having a definite volume and from the solid phase because molecules may mix freely.

London dispersion forces—Intermolecular forces arising from interactions between temporary dipoles in molecules.

Lyman series—A portion of the emission spectrum for hydrogen representing electronic transitions from energy levels $n > 1$ to $n = 1$.

Magnetic quantum number (m_l)—The third quantum number, defining the particular orbital of a subshell in which an electron resides; conveys information about the orientation of the orbital in space.

Malleability—A physical property of metals that defines how well an element can be shaped using a hammer.

Mass—A physical property representing the amount of matter in a given sample.

Mass number—The sum of protons and neutrons in an atom's nucleus. Can also be called atomic mass number.

Maxwell–Boltzmann distribution curve—The distribution of the molecular speeds of gas particles at a given temperature; as temperature increases, average speed increases and the distribution becomes wider and flatter.

Mechanism—The series of steps involved in a given reaction.

Melting point—The temperature at which the solid and liquid phases of a substance coexist in equilibrium; identical to the freezing point.

Metal—One of a class of elements on the left side of the periodic table possessing low ionization energies and electronegativities; readily give up electrons to form cations and possess relatively high electrical conductivity.

Metalloid—An element possessing properties intermediate between those of a metal and those of a nonmetal; also called a semimetal.

Microstate—In thermodynamics, a specific way in which energy of a system is organized.

Millimeters of mercury (mmHg)—A unit of pressure defined as the number of millimeters that mercury in a barometer is raised above its surface in a capillary tube by an external pressure; 1 torr is equal to 1 mmHg by definition, and 1 atmosphere is equal to 760 mmHg.

Mixed-order reaction—A reaction in which the reaction order changes over time in the rate law.

Mixture—A system containing multiple substances (2+) that have been physically combined but are not chemically combined.

Molality (m)—A concentration unit equal to the number of moles of solute per kilogram of solvent.

Molarity (M)—A concentration unit equal to the number of moles of solute per liter of solution.

Molar mass—The mass in grams of one mole of an element or compound.

Molar solubility—The molarity of a solute in a saturated solution.

Mole—An amount of substance equal to Avogadro's number of molecules or atoms; the mass of 1 mole of substance in grams is the same as the mass of one molecule or atom in atomic mass units.

Mole fraction (X)—A unit of concentration equal to the ratio of the number of moles of a particular component to the total number of moles for all species in the system.

Molecular formula—A formula showing the actual number and identity of all atoms in each molecule of a compound; always a whole-number multiple of the empirical formula.

Molecular geometry—The spatial arrangement of only the bonding pairs of electrons around a central atom.

Molecular orbital—The region of electron density in chemical bonding that results from the overlap of two or more atomic orbitals.

Molecular weight—The sum of the atomic weights of all the atoms in a molecule.

Molecule—The smallest polyatomic unit of an element or compound that exists with distinct chemical and physical properties.

Nernst equation—An equation that relates the voltage of an electrochemical cell to the concentrations of the reactants and products within that cell.

Net ionic equation—A reaction equation showing only the species actually participating in the reaction.

Neutral solution—An aqueous solution in which the concentration of H^+ and OH^- ions are equal ($\text{pH} = 7$ at 298 K).

Neutralization reaction—A reaction between an acid and base in which a salt is formed (and sometimes water).

Neutron—A subatomic particle contained within the nucleus of an atom; carries no charge and has a mass slightly larger than that of a proton.

Nickel–cadmium battery—A rechargeable electrochemical cell in which the anode is the site of Cd metal oxidation and the cathode is the site of Ni^{2+} ion reduction.

Nickel–metal hydride battery—A rechargeable electrochemical cell in which the anode is the site of metal hydride oxidation and the cathode is

the site of nickel ion reduction; the nickel may be in one of many oxidation states.

Nonbonding electrons—Electrons located in the valence shell of an atom but not involved in covalent bonds.

Nonelectrolyte—A compound that does not ionize in water.

Nonmetal—One of a class of elements with high ionization energies and electron affinities that generally gain electrons to form anions; located in the upper right corner of the periodic table.

Nonpolar covalent bond—A covalent bond between elements of similar electronegativity; contains no charge separation.

Nonpolar molecule—A molecule that exhibits no net separation of charge and, therefore, no net dipole moment.

Nonrepresentative element—Elements with an expanded valence shell that includes d- and f-block electrons; also called Group B or transition elements.

Nonspontaneous process—A process that will not occur on its own without energy input from the surroundings; has a positive change in free energy.

Normality (N)—A concentration unit equal to the number of equivalents per liter of solution.

Nucleus—The small central region of an atom; a dense, positively charged area containing protons and neutrons.

Octet—Eight valence electrons in a subshell around a nucleus; imparts great stability to an atom.

Octet rule—A rule stating that bonded atoms tend to undergo reactions that will produce a complete octet of valence electrons; applies without exception only to C, N, O, and F.

Open system—A system that can exchange both energy and matter with its surroundings.

Orbital—A region of electron density around an atom or molecule containing no more than two electrons of opposite spin.

Osmosis—The movement of water through a semipermeable membrane down its concentration gradient, from low solute concentration to high solute concentration.

Osmotic pressure—The pressure that must be applied to a solution to prevent the passage of water through a semipermeable membrane down its concentration gradient; best thought of as a “sucking” pressure drawing water into solution.

Oxidation—A reaction involving the net loss of electrons, increasing oxidation number.

Oxidation number—Also called oxidation state, the number assigned to an atom in an ion or molecule that denotes its real or hypothetical charge, assuming that the most electronegative element in a bond is awarded all of the electrons in that bond.

Oxidation potential—The ability of a substance to be spontaneously oxidized; a more positive oxidation potential (measured in volts) is indicative of a substance that is easier to oxidize and will therefore more likely act as an anode in an electrochemical cell.

Oxidation-reduction (redox) reaction—A reaction that involves the transfer of electrons from one chemical species to another.

Oxidizing agent—In an oxidation–reduction reaction, the atom that facilitates the oxidation of another species; the oxidizing agent gains electrons and is thereby reduced.

p subshell—The subshell corresponding to the angular momentum quantum number $l = 1$; contains three dumbbell-shaped orbitals oriented perpendicular to each other (p_x , p_y , and p_z) and is found in the second and higher principal energy levels.

Paired electrons—Two electrons in the same orbital with assigned spins of $+\frac{1}{2}$ and $-\frac{1}{2}$.

Parallel spin—In quantum mechanics, electrons in different orbitals of an atom with the same m_s values.

Paramagnetism—A condition that arises when a substance has unpaired electrons and is slightly attracted to a magnetic field.

Partial pressure—The pressure that one component of a gaseous mixture would exert if it were alone in the container.

Pascal (Pa)—The SI unit for pressure, equivalent to

$$\frac{\text{N}}{\text{m}^2} ; 1 \text{ atm} = 101,325 \text{ Pa} .$$

Paschen series—Part of the emission spectrum for hydrogen, representing transitions of an electron from energy levels $n \geq 4$ to $n = 3$

Pauli exclusion principle—The principle stating that no two electrons within an atom may have an identical set of quantum numbers.

Pauling electronegativity scale—The most common scale used to express electronegativity of the elements.

Percent composition—The percentage of the total formula weight of a compound attributable to a given element.

Percent yield—The percentage of the theoretical product yield that is actually recovered when a chemical reaction occurs; obtained by dividing the actual yield by the theoretical yield and multiplying by 100%.

Period—A horizontal row of the periodic table containing elements with the same number of electron shells.

Periodic law—The law stating that the chemical properties of elements depend on the atomic number of the elements and change in a periodic fashion.

Periodic table—The visual display of all known chemical elements arranged in rows (periods) and columns (groups) according to their atomic number and electron structure.

pH—A measure of the hydrogen ion content of an aqueous solution, defined as the negative log of the H^+ (H_3O^+) concentration.

pH meter—A device used to measure the concentration of hydrogen ions in solution and report it as a pH value.

Phase—One of the three forms of matter: solid, liquid, or gas; also called state.

Phase change—Reversible transition between solid, liquid, and/or gas phase caused by shifts in temperature or pressure.

Phase diagram—A plot, usually of pressure vs. temperature, showing which phases of a compound will exist under any set of conditions.

Photon—The form of light which displays particulate and quantal behavior.

Physical property—A property of a substance unrelated to its chemical behavior, such as melting point, boiling point, density, or odor.

Pi (π) bond—A bond with two parallel electron cloud densities formed between two p-orbitals that limits the possibility of free rotation; π bonds are the second bond in a double bond and both the second and third bonds in a triple bond.

pOH—A measure of the hydroxide (OH^-) ion content of an aqueous solution, defined as the negative log of the OH^- concentration.

Polar covalent bond—A covalent bond between atoms with different electronegativities in which electron density is unevenly distributed, giving the bond positive and negative ends.

Polar molecule—A molecule possessing one or more polar covalent bonds and a geometry that allows the bond dipole moments to sum to a net dipole moment.

Polyprotic—A molecule capable of donating more than one proton.

Polyvalent acid—An acid capable of donating more than one acid equivalent.

Polyvalent base—A base capable of donating more than one hydroxide or accepting more than one proton.

Potential energy diagram—A graph that shows the potential energies of the reactants and products of a reaction during the course of the reaction; by convention, the x-axis shows the progress of the reaction and the y-axis shows potential energy.

Potentiometer—A device used to measure electromotive force (voltage). Potentiometers can be used in potentiometric titrations: redox titrations with no indicator.

Precipitate—An insoluble solid that separates from a solution; generally the result of mixing two or more solutions or of a temperature change.

Pressure—Average force per unit area measured in atmospheres (atm), torr or mmHg, or pascals (Pa); $1 \text{ atm} = 760 \text{ torr} \equiv 760 \text{ mmHg} = 101.325 \text{ kPa}$.

Principal quantum number (n)—The first quantum number, which defines the energy level or shell occupied by an electron.

Process—In a system, when a change in one or more of the properties of the system occurs.

Proton (p^+)—A subatomic particle that carries a single positive charge and has a mass slightly less than 1 amu.

Quanta—In Planck's theory, discrete bundles of energy that are emitted as electromagnetic radiation from matter.

Quantum number—A number used to describe the energy levels in which electrons reside; all electrons in an element are described by a unique set of four quantum numbers.

Radioactivity—A phenomenon exhibited by certain unstable isotopes in which they undergo spontaneous nuclear transformation via emission of one or more particles.

Raoult's law—A law stating that the partial pressure of a component in a solution is proportional to the mole fraction of that component in the solution; provides an explanation for vapor pressure depression seen in solutions.

Rate constant—The proportionality constant in the rate law of a reaction; specific to a particular reaction at a given temperature.

Rate-determining step—The slowest step of a reaction mechanism; this step serves as a bottleneck on the progress of the reaction.

Rate law—A mathematical expression giving the rate of a reaction as a function of the concentrations of the reactants; must be determined experimentally.

Rate order—The exponential effect of a change in concentration of a reactant on the change of rate in a reaction; the overall rate order is the sum of all the individual reactant rate orders.

Reaction mechanism—The series of steps that occurs in the course of a chemical reaction, often including the formation and destruction of reaction intermediates.

Reaction order—In a calculation of the rate law for a reaction, the sum of the exponents to which the concentrations of reactants must be raised.

Reaction quotient (Q)—Has the same form as the equilibrium constant, but the concentrations of products and reactants may not be at equilibrium; when compared to K_{eq} , it dictates the direction a reaction will proceed spontaneously.

Reaction rate—The speed at which a substance is produced or consumed by a reaction.

Real gas—A gas that exhibits deviations from the ideal gas law due to molecular attractions and the actual volume of the gas molecules themselves.

Rechargeable battery—An electrochemical cell that can undergo a reversible oxidation–reduction process; when discharging, it functions as a galvanic (voltaic) cell, and when charging, it functions as an electrolytic cell.

Redox titration—A specific method used to determine the concentration of an unknown solution using reducible titrants or titrands, typically by measuring voltage changes.

Reducing agent—In an oxidation–reduction reaction, the atom that facilitates the reduction of another species; the reducing agent loses electrons and is thereby oxidized.

Reduction—A reaction involving the net gain of electrons, decreasing the oxidation number.

Reduction potential—The ability of a substance to be spontaneously reduced; a more positive reduction potential (measured in volts) is indicative of a substance that is easier to reduce and will therefore more likely act as a cathode in an electrochemical cell.

Representative elements—Elements in Groups 1, 2, and 13 through 18 in the modern IUPAC table (the s- and p-blocks of the table, also called A group elements); these elements tend to have valence shells that follow the octet rule.

Resonance—A difference in the arrangement of electron pairs but not the bond connectivity or overall charge within a Lewis structure.

Resonance hybrid—A Lewis structure that represents the weighted average (by stability) of all possible resonance structures.

Reversible reaction—A reaction that can proceed in either the forward or reverse direction, and typically does not go to completion.

Root-mean-square speed (u_{rms})—The average speed of a gas molecule at a given temperature; as a scalar, it does not take direction into account.

s subshell—Subshell corresponding to the angular momentum quantum number $l = 0$ and containing one spherical orbital; found in all energy levels.

Salt—An ionic substance consisting of cations and anions.

Salt bridge—A component of an electrochemical cell composed of an inert electrolyte that allows the charge gradient that builds up in the half-cells to be dissipated as a reaction occurs; contains ions that will not react with electrodes or ions in solution and that can move to balance charge.

Saturated solution—A solution containing the maximum amount of solute that can be dissolved in a particular solvent at a given temperature.

Second law of thermodynamics—The law stating that all spontaneous processes lead to an increase in the entropy of the universe.

Second-order reaction—A reaction in which the rate is directly proportional to the concentration of two reactants, or to the square of one single reactant.

Semipermeable—A quality of a membrane allowing only some components of a solution to pass through, usually including the solvent, while limiting the passage of other species.

Sigma (σ) bond—A head-to-head bond between two orbitals of different atoms that allows free rotation about its axis.

Single-displacement reaction—A reaction in which an ion of one compound is replaced with another ion; also known as a single-replacement reaction.

Solid—The phase of matter possessing the greatest order; molecules are fixed in a rigid structure.

Solubility—A measure of the amount of solute that can be dissolved in a solvent at a certain temperature.

Solubility product (K_{sp})—The equilibrium constant for the ionization reaction of a sparingly soluble salt.

Solute—The component of a solution that is present in a lesser concentration than the solvent.

Solution—A homogeneous mixture of two or more substances. It may be solid (brass), liquid (HCl (aq)), or gas (air).

Solvation—The electrostatic interaction between solute and solvent molecules; also called dissolution. The term hydration can be used when water is the solvent.

Solvent—The component of a solution present in the greatest amount; the substance in which the solute is dissolved.

Sparingly soluble salt—An ionic compound that has a low solubility at a given temperature.

Specific heat—The amount of heat required to raise the temperature of one gram of a substance by 1°C.

Spectator ions—Ions involved in a reaction that do not change formula, charge, or phase; normally omitted from the net ionic equation.

Spectrum—The characteristic wavelengths of electromagnetic radiation emitted or absorbed by an object, atom, or molecule.

Spectroscopic notation—The shorthand representation of the principal and azimuthal quantum numbers, in which the azimuthal number is designated by a letter rather than a number.

Sphygmomanometer—A tool for measuring blood pressure.

Spin quantum number (m_s)—The fourth quantum number, which indicates the orientation of the intrinsic spin of an electron in an atom; can only assume values of $+\frac{1}{2}$ and $-\frac{1}{2}$.

Spontaneous process—A process that will occur on its own without energy input from the surroundings; defined by a negative change in free energy.

Standard conditions—Conditions defined as 25°C, 1 atm pressure, and 1 M concentrations; used for measuring the standard Gibbs free energy, enthalpy, entropy, and cell electromotive force.

Standard free energy (G°)—The Gibbs free energy for a reaction under standard conditions.

Standard heat of combustion ($\Delta H_{\text{comb}}^{\circ}$)—The enthalpy change associated with the combustion of a fuel.

Standard hydrogen electrode (SHE)—The electrode defined as having a potential of zero under standard conditions; all oxidation and reduction potentials are measured relative to the standard hydrogen electrode at 25°C and with 1 M concentrations of each ion in solution.

Standard potential—The voltage associated with a half-reaction of a specific oxidation–reduction reaction; generally tabulated as reduction potentials, compared to the standard hydrogen electrode.

Standard state—The phase of matter for a certain element under standard conditions.

Standard temperature and pressure (STP)—Defined as 0°C (273 K) and 1 atm; used for measuring characteristics of an ideal gas.

State function—A function that depends on the state of a system but not on the path used to arrive at that state; includes pressure, density, temperature, volume, enthalpy, internal energy, Gibbs free energy, and entropy.

Stoichiometric coefficient—In a reaction, the number placed in front of each compound to indicate relative number of moles of that species involved in the reaction.

Stoichiometry—A form of dimensional analysis focusing on the relationships between amounts of reactants and products in a reaction.

Strong acid—An acid that undergoes complete dissociation in an aqueous solution.

Strong base—A base that undergoes complete dissociation in an aqueous solution.

Structural formula—The graphic representation of a molecule depicting how its atoms are arranged.

Sublimation—A change of phase from solid to gas without passing through the liquid phase.

Subshell—The division of electron shells or energy levels into different values of the azimuthal quantum number (s, p, d, and f); composed of orbitals.

Supercritical fluid—A substance whose current state is simultaneously a liquid and a gas—there is no distinction between the two phases.

Supersaturated—A solution which is beyond equilibrium, where ion product is greater than the solubility product constant. Supersaturated solutions are thermodynamically unstable.

Surge current—An above-average current transiently released at the beginning of the discharge phase of a battery.

Surroundings—All matter and energy in the universe not included in the particular system under consideration.

System—The matter and energy under consideration.

Temperature—A measure of the average kinetic energy of the particles in a system.

Theoretical yield—The maximal amount of product that can be obtained in a reaction; determined by stoichiometric analysis of the limiting reagent.

Thermodynamic product—The product of a reaction that is formed favorably at a higher temperature because thermal energy is available to form the transition state of the more stable product; has a larger overall difference in free energy between the products and reactants than the kinetic product.

Titrand—A solution of unknown concentration to which a solution of known concentration is added to determine its concentration.

Titrant—A solution of known concentration that is slowly added to a solution of unknown concentration to determine its concentration.

Titration—A method used to determine the concentration of an unknown solution by gradual reaction with a solution of known concentration.

Titration curve—A plot of the pH of a solution vs. the volume of acid or base added in an acid–base titration, or a plot of the electromotive force of a solution vs. the volume of oxidizing or reducing agent added in an oxidation–reduction titration.

Transition metal—Any of the elements in the B groups of the periodic table, all of which have partially filled d subshells.

Transition state—The point during a reaction in which old bonds are partially broken and new bonds are partially formed; has a higher energy than the reactants or products of the reaction and is also called the activated complex.

Triple point—The pressure and temperature at which the solid, liquid, and gas phases of a particular substance coexist in equilibrium.

Unsaturated solution—A solution into which more solute may be dissolved before reaching saturation.

Valence electron—An electron in the highest occupied energy level of an atom; the tendency of a given valence electron to be retained or lost determines the chemical properties of an element.

Valence shell—The outermost shell of an atom.

Valence shell electron pair repulsion (VSEPR) theory—A system that reflects the geometric arrangement of a molecule based on its Lewis dot structure; the three-dimensional structure is determined by the

repulsions between bonding and nonbonding electron pairs in the valence shells of atoms.

van der Waals equation of state—One of several real gas laws, which corrects for attractive forces and the volumes of gas particles, which are assumed to be negligible in the ideal gas law.

Van der Waals forces—Attractive or repulsive forces between molecules that don't arise from covalent or ionic bonds.

van't Hoff factor—The number of particles into which a compound dissociates in solution.

Vapor pressure—The partial pressure of a gaseous substance in the atmosphere above the liquid or solid with which it is in equilibrium.

Vapor pressure depression—The decrease in the vapor pressure of a liquid caused by the presence of dissolved solute; a colligative property.

Vaporization—The transformation of a liquid into a gas.

Water dissociation constant (K_w)—The equilibrium constant of the water dissociation reaction at a given temperature; equal to 10^{-14} at 25°C (298 K).

Weak acid—An acid that undergoes partial dissociation in an aqueous solution.

Weak base—A base that undergoes partial dissociation in an aqueous solution.

Yield—The amount of product obtained from a reaction.

Zero-order reaction—A reaction in which the concentrations of reactants have no effect on the overall rate.

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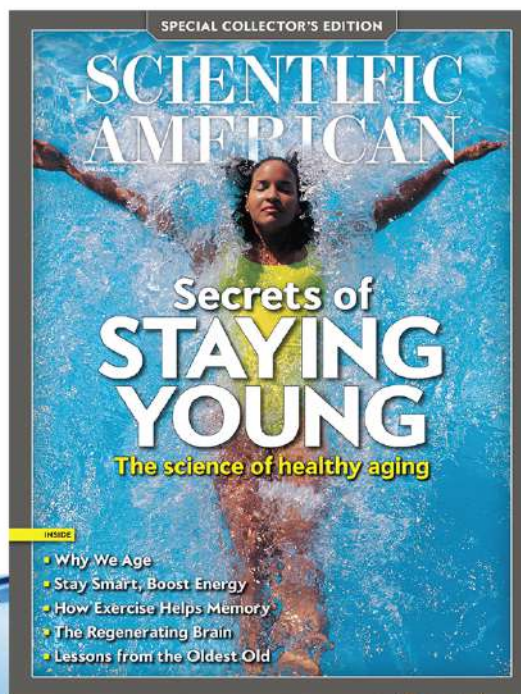
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Periodic Table of the Elements

1 IA												18 VIIIA					
1 H 1.0	2 IIA											13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	2 He 4.0
3 Li 6.9	4 Be 9.0											5 B 10.8	6 C 12.0	7 N 14.0	8 O 16.0	9 F 19.0	10 Ne 20.2
11 Na 23.0	12 Mg 24.3	3 IIIB	4 IVB	5 VB	6 VIB	7 VIIB	8	9	10	11 IB	12 IIB	13 Al 27.0	14 Si 28.1	15 P 31.0	16 S 32.1	17 Cl 35.5	18 Ar 39.9
19 K 39.1	20 Ca 40.1	21 Sc 45.0	22 Ti 47.9	23 V 50.9	24 Cr 52.0	25 Mn 54.9	26 Fe 55.8	27 Co 58.9	28 Ni 58.7	29 Cu 63.5	30 Zn 65.4	31 Ga 69.7	32 Ge 72.6	33 As 74.9	34 Se 79.0	35 Br 79.9	36 Kr 83.8
37 Rb 85.5	38 Sr 87.6	39 Y 88.9	40 Zr 91.2	41 Nb 92.9	42 Mo 96.0	43 Tc (98)	44 Ru 101.0	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	★	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)		†	104 Rf (267)	105 Db (268)	106 Sg (269)	107 Bh (270)	108 Hs (269)	109 Mt (278)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Uut (286)	114 Fl (289)	115 Uup (288)	116 Lv (293)	117 Uus (294)
★ †		57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.1	71 Lu 175.0	
		89 Ac (227)	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)	

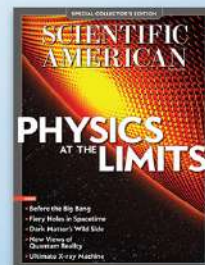
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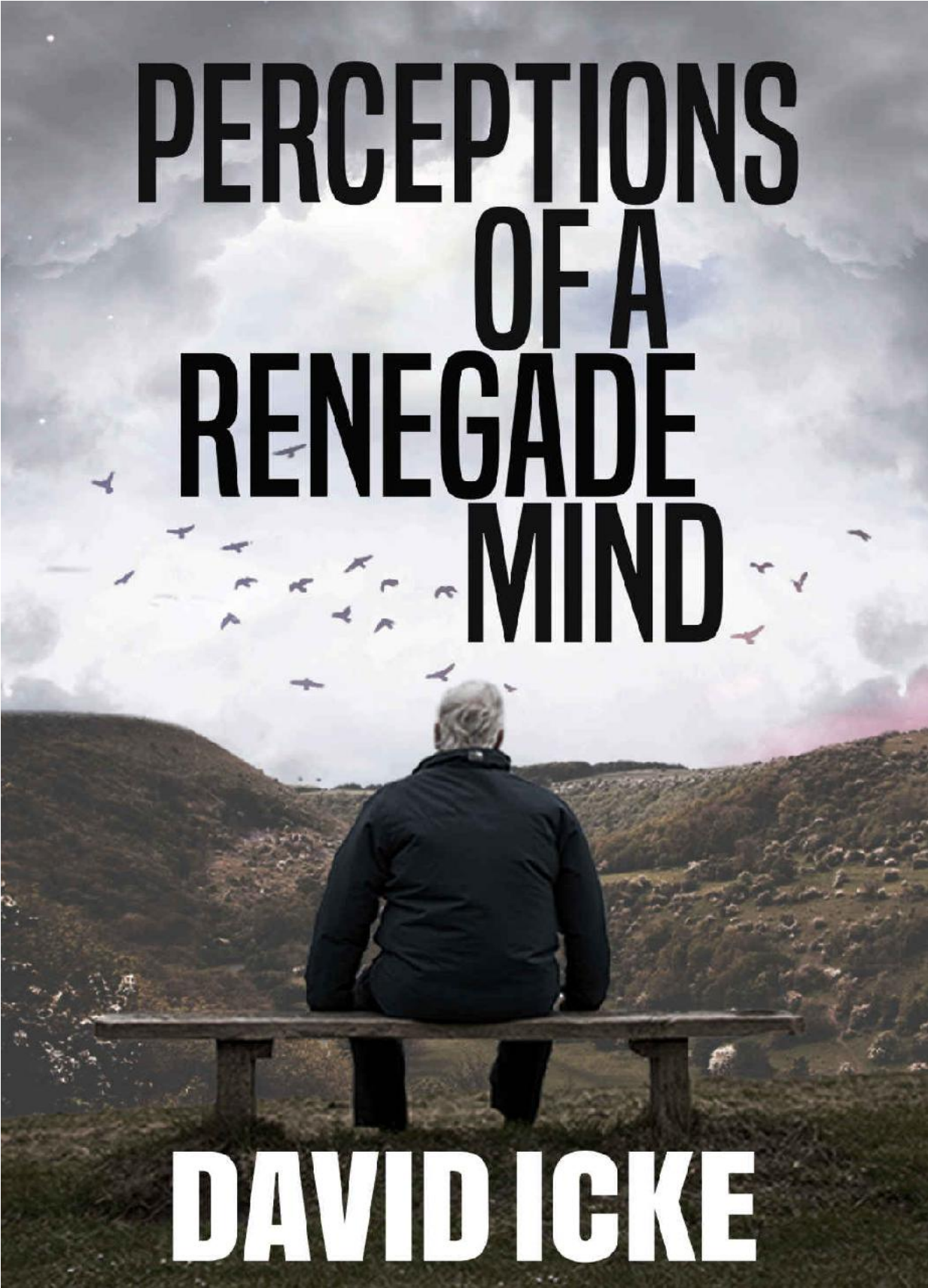
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PERCEPTIONS OF A RENEGADE MIND



DAVID ICKE

A man with grey hair, wearing a dark jacket, is seen from behind, sitting on a wooden bench. He is looking out over a vast, hilly landscape under a dramatic, cloudy sky. A large flock of birds is flying in the sky, and a few birds are visible on the ground. The overall mood is contemplative and expansive.

PERCEPTIONS OF A RENEGADE MIND

DAVID ICKE

PERCEPTIONS OF A RENEGADE MIND



ickonic
publishing

First published in July 2021.

ickonic
publishing

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Cover Design: Gareth Icke

Book Design: Neil Hague

**British Library Cataloguing-in
Publication Data**

A catalogue record for this book is
available from the British Library

eISBN 978-18384153-1-0

PERCEPTIONS OF A RENEGADE MIND

A flock of small, stylized birds is arranged in a loose circular pattern around the word "MIND". The birds are depicted in various flight poses, some facing left, some right, and some in between, creating a sense of movement and a vortex-like effect.

DAVID ICKE

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Renegade:

Adjective

‘Having rejected tradition: Unconventional.’

Merriam-Webster Dictionary

Acquiescence to tyranny is the death of the spirit

You may be 38 years old, as I happen to be. And one day, some great opportunity stands before you and calls you to stand up for some great principle, some great issue, some great cause. And you refuse to do it because you are afraid ... You refuse to do it because you want to live longer ... You're afraid that you will lose your job, or you are afraid that you will be criticised or that you will lose your popularity, or you're afraid that somebody will stab you, or shoot at you or bomb your house; so you refuse to take the stand.

Well, you may go on and live until you are 90, but you're just as dead at 38 as you would be at 90. And the cessation of breathing in your life is but the belated announcement of an earlier death of the spirit.

Martin Luther King

**How the few control the many and always have – the many do
whatever they're told**

'Forward, the Light Brigade!'
Was there a man dismayed?
Not though the soldier knew
Someone had blundered.
Theirs not to make reply,
Theirs not to reason why,
Theirs but to do and die.
Into the valley of Death
Rode the six hundred.

Cannon to right of them,
Cannon to left of them,
Cannon in front of them
Volleyed and thundered;
Stormed at with shot and shell,
Boldly they rode and well,
Into the jaws of Death,
Into the mouth of hell
Rode the six hundred

Alfred Lord Tennyson (1809-1892)

The mist is lifting slowly
I can see the way ahead
And I've left behind the empty streets
That once inspired my life
And the strength of the emotion
Is like thunder in the air
'Cos the promise that we made each other
Haunts me to the end

The secret of your beauty
And the mystery of your soul
I've been searching for in everyone I meet
And the times I've been mistaken
It's impossible to say
And the grass is growing
Underneath our feet

The words that I remember
From my childhood still are true
That there's none so blind
As those who will not see
And to those who lack the courage
And say it's dangerous to try
Well they just don't know
That love eternal will not be denied

I know you're out there somewhere
Somewhere, somewhere
I know you're out there somewhere

Somewhere you can hear my voice
I know I'll find you somehow
Somehow, somehow
I know I'll find you somehow
And somehow I'll return again to you

The Moody Blues

Are you a gutless wonder - or a Renegade Mind?

Monuments put from pen to paper,
Turns me into a gutless wonder,
And if you tolerate this,
Then your children will be next.
Gravity keeps my head down,
Or is it maybe shame ...

Manic Street Preachers

Rise like lions after slumber
In unvanquishable number.
Shake your chains to earth like dew
Which in sleep have fallen on you.
Ye are many – they are few.

Percy Shelley

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CHAPTER ONE

I'm thinking' – Oh, but *are* you?

Think for yourself and let others enjoy the privilege of doing so too
Voltaire

French-born philosopher, mathematician and scientist René Descartes became famous for his statement in Latin in the 17th century which translates into English as: 'I think, therefore I am.'

On the face of it that is true. Thought reflects perception and perception leads to both behaviour and self-identity. In that sense 'we' are what we think. But who or what is doing the thinking and is thinking the only route to perception? Clearly, as we shall see, 'we' are not always the source of 'our' perception, indeed with regard to humanity as a whole this is rarely the case; and thinking is far from the only means of perception. Thought is the village idiot compared with other expressions of consciousness that we all have the potential to access and tap into. This has to be true when we *are* those other expressions of consciousness which are infinite in nature. We have forgotten this, or, more to the point, been manipulated to forget.

These are not just the esoteric musings of the navel. The whole foundation of human control and oppression is control of perception. Once perception is hijacked then so is behaviour which is dictated by perception. Collective perception becomes collective behaviour and collective behaviour is what we call human society. Perception is all and those behind human control know that which is

why perception is the target 24/7 of the psychopathic manipulators that I call the Global Cult. They know that if they dictate perception they will dictate behaviour and collectively dictate the nature of human society. They are further aware that perception is formed from information received and if they control the circulation of information they will to a vast extent direct human behaviour. Censorship of information and opinion has become globally Nazi-like in recent years and never more blatantly than since the illusory 'virus pandemic' was triggered out of China in 2019 and across the world in 2020. Why have billions submitted to house arrest and accepted fascistic societies in a way they would have never believed possible? Those controlling the information spewing from government, mainstream media and Silicon Valley (all controlled by the same Global Cult networks) told them they were in danger from a 'deadly virus' and only by submitting to house arrest and conceding their most basic of freedoms could they and their families be protected. This monumental and provable lie became the *perception* of the billions and therefore the *behaviour* of the billions. In those few words you have the whole structure and modus operandi of human control. Fear is a perception – False Emotion Appearing Real – and fear is the currency of control. In short ... get them by the balls (or give them the impression that you have) and their hearts and minds will follow. Nothing grips the dangly bits and freezes the rear-end more comprehensively than fear.

World number 1

There are two 'worlds' in what appears to be one 'world' and the prime difference between them is knowledge. First we have the mass of human society in which the population is maintained in coldly-calculated ignorance through control of information and the 'education' (indoctrination) system. That's all you really need to control to enslave billions in a perceptual delusion in which what are perceived to be *their* thoughts and opinions are ever-repeated mantras that the system has been downloading all their lives through 'education', media, science, medicine, politics and academia

in which the personnel and advocates are themselves overwhelmingly the perceptual products of the same repetition. Teachers and academics in general are processed by the same programming machine as everyone else, but unlike the great majority they never leave the 'education' program. It gripped them as students and continues to grip them as programmers of subsequent generations of students. The programmed become the programmers – the programmed programmers. The same can largely be said for scientists, doctors and politicians and not least because as the American writer Upton Sinclair said: 'It is difficult to get a man to understand something when his salary depends upon his not understanding it.' If your career and income depend on thinking the way the system demands then you will – bar a few free-minded exceptions – concede your mind to the Perceptual Mainframe that I call the Postage Stamp Consensus. This is a tiny band of perceived knowledge and possibility 'taught' (downloaded) in the schools and universities, pounded out by the mainstream media and on which all government policy is founded. Try thinking, and especially speaking and acting, outside of the 'box' of consensus and see what that does for your career in the Mainstream Everything which bullies, harasses, intimidates and ridicules the population into compliance. Here we have the simple structure which enslaves most of humanity in a perceptual prison cell for an entire lifetime and I'll go deeper into this process shortly. Most of what humanity is taught as fact is nothing more than programmed belief. American science fiction author Frank Herbert was right when he said: 'Belief can be manipulated. Only knowledge is dangerous.' In the 'Covid' age belief is promoted and knowledge is censored. It was always so, but never to the extreme of today.

World number 2

A 'number 2' is slang for 'doing a poo' and how appropriate that is when this other 'world' is doing just that on humanity every minute of every day. World number 2 is a global network of secret societies and semi-secret groups dictating the direction of society via

governments, corporations and authorities of every kind. I have spent more than 30 years uncovering and exposing this network that I call the Global Cult and knowing its agenda is what has made my books so accurate in predicting current and past events. Secret societies are secret for a reason. They want to keep their hoarded knowledge to themselves and their chosen initiates and to hide it from the population which they seek through ignorance to control and subdue. The whole foundation of the division between World 1 and World 2 is *knowledge*. What number 1 knows number 2 must not. Knowledge they have worked so hard to keep secret includes (a) the agenda to enslave humanity in a centrally-controlled global dictatorship, and (b) the nature of reality and life itself. The latter (b) must be suppressed to allow the former (a) to prevail as I shall be explaining. The way the Cult manipulates and interacts with the population can be likened to a spider's web. The 'spider' sits at the centre in the shadows and imposes its will through the web with each strand represented in World number 2 by a secret society, satanic or semi-secret group, and in World number 1 – the world of the seen – by governments, agencies of government, law enforcement, corporations, the banking system, media conglomerates and Silicon Valley (Fig 1 overleaf). The spider and the web connect and coordinate all these organisations to pursue the same global outcome while the population sees them as individual entities working randomly and independently. At the level of the web governments *are* the banking system *are* the corporations *are* the media *are* Silicon Valley *are* the World Health Organization working from their inner cores as one unit. Apparently unconnected countries, corporations, institutions, organisations and people are on the *same team* pursuing the same global outcome. Strands in the web immediately around the spider are the most secretive and exclusive secret societies and their membership is emphatically restricted to the Cult inner-circle emerging through the generations from particular bloodlines for reasons I will come to. At the core of the core you would get them in a single room. That's how many people are dictating the direction of human society and its transformation

through the 'Covid' hoax and other means. As the web expands out from the spider we meet the secret societies that many people will be aware of – the Freemasons, Knights Templar, Knights of Malta, Opus Dei, the inner sanctum of the Jesuit Order, and such like. Note how many are connected to the Church of Rome and there is a reason for that. The Roman Church was established as a revamp, a rebranding, of the relocated 'Church' of Babylon and the Cult imposing global tyranny today can be tracked back to Babylon and Sumer in what is now Iraq.



Figure 1: The global web through which the few control the many. (Image Neil Hague.)

Inner levels of the web operate in the unseen away from the public eye and then we have what I call the cusp organisations located at the point where the hidden meets the seen. They include a series of satellite organisations answering to a secret society founded in London in the late 19th century called the Round Table and among them are the Royal Institute of International Affairs (UK, founded in 1920); Council on Foreign Relations (US, 1921); Bilderberg Group (worldwide, 1954); Trilateral Commission (US/worldwide, 1972); and the Club of Rome (worldwide, 1968) which was created to exploit environmental concerns to justify the centralisation of global power to 'save the planet'. The Club of Rome instigated with others the human-caused climate change hoax which has led to all the 'green

new deals' demanding that very centralisation of control. Cusp organisations, which include endless 'think tanks' all over the world, are designed to coordinate a single global policy between political and business leaders, intelligence personnel, media organisations and anyone who can influence the direction of policy in their own sphere of operation. Major players and regular attenders will know what is happening – or some of it – while others come and go and are kept overwhelmingly in the dark about the big picture. I refer to these cusp groupings as semi-secret in that they can be publicly identified, but what goes on at the inner-core is kept very much 'in house' even from most of their members and participants through a fiercely-imposed system of compartmentalisation. Only let them know what they need to know to serve your interests and no more. The structure of secret societies serves as a perfect example of this principle. Most Freemasons never get higher than the bottom three levels of 'degree' (degree of knowledge) when there are 33 official degrees of the Scottish Rite. Initiates only qualify for the next higher 'compartment' or degree if those at that level choose to allow them. Knowledge can be carefully assigned only to those considered 'safe'. I went to my local Freemason's lodge a few years ago when they were having an 'open day' to show how cuddly they were and when I chatted to some of them I was astonished at how little the rank and file knew even about the most ubiquitous symbols they use. The mushroom technique – keep them in the dark and feed them bullshit – applies to most people in the web as well as the population as a whole. Sub-divisions of the web mirror in theme and structure transnational corporations which have a headquarters somewhere in the world dictating to all their subsidiaries in different countries. Subsidiaries operate in their methodology and branding to the same centrally-dictated plan and policy in pursuit of particular ends. The Cult web functions in the same way. Each country has its own web as a subsidiary of the global one. They consist of networks of secret societies, semi-secret groups and bloodline families and their job is to impose the will of the spider and the global web in their particular country. Subsidiary networks control and manipulate the national political system, finance, corporations, media, medicine, etc. to

ensure that they follow the globally-dictated Cult agenda. These networks were the means through which the 'Covid' hoax could be played out with almost every country responding in the same way.

The 'Yessir' pyramid

Compartmentalisation is the key to understanding how a tiny few can dictate the lives of billions when combined with a top-down sequence of imposition and acquiescence. The inner core of the Cult sits at the peak of the pyramidal hierarchy of human society (Fig 2 overleaf). It imposes its will – its agenda for the world – on the level immediately below which acquiesces to that imposition. This level then imposes the Cult will on the level below them which acquiesces and imposes on the next level. Very quickly we meet levels in the hierarchy that have no idea there even is a Cult, but the sequence of imposition and acquiescence continues down the pyramid in just the same way. 'I don't know why we are doing this but the order came from "on-high" and so we better just do it.' Alfred Lord Tennyson said of the cannon fodder levels in his poem *The Charge of the Light Brigade*: 'Theirs not to reason why; theirs but to do and die.' The next line says that 'into the valley of death rode the six hundred' and they died because they obeyed without question what their perceived 'superiors' told them to do. In the same way the population capitulated to 'Covid'. The whole hierarchical pyramid functions like this to allow the very few to direct the enormous many.

Eventually imposition-acquiescence-imposition-acquiescence comes down to the mass of the population at the foot of the pyramid. If they acquiesce to those levels of the hierarchy imposing on them (governments/law enforcement/doctors/media) a circuit is completed between the population and the handful of super-psychopaths in the Cult inner core at the top of the pyramid.

Without a circuit-breaking refusal to obey, the sequence of imposition and acquiescence allows a staggeringly few people to impose their will upon the entirety of humankind. We are looking at the very sequence that has subjugated billions since the start of 2020. Our freedom has not been taken from us. Humanity has given it

away. Fascists do not impose fascism because there are not enough of them. Fascism is imposed by the population acquiescing to fascism. Put another way allowing their perceptions to be programmed to the extent that leads to the population giving their freedom away by giving their perceptions – their mind – away. If this circuit is not broken by humanity ceasing to cooperate with their own enslavement then nothing can change. For that to happen people have to critically think and see through the lies and window dressing and then summon the backbone to act upon what they see. The Cult spends its days working to stop either happening and its methodology is systematic and highly detailed, but it can be overcome and that is what this book is all about.

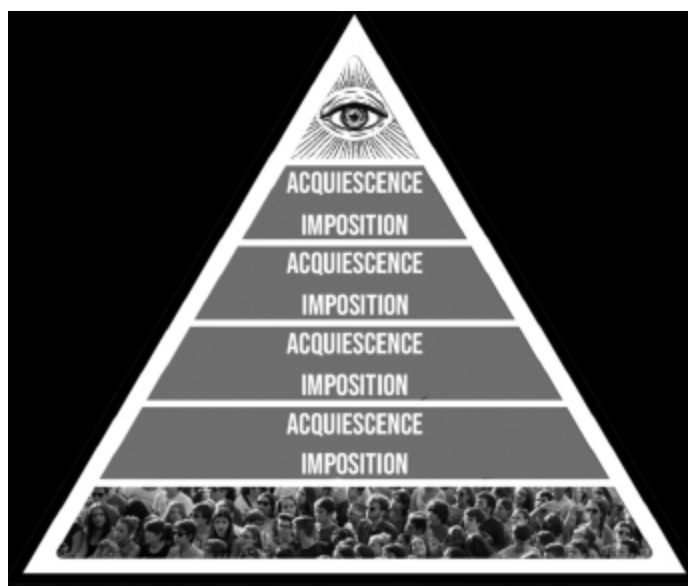


Figure 2: The simple sequence of imposition and compliance that allows a handful of people at the peak of the pyramid to dictate the lives of billions.

The Life Program

Okay, back to world number 1 or the world of the 'masses'. Observe the process of what we call 'life' and it is a perceptual download from cradle to grave. The Cult has created a global structure in which perception can be programmed and the program continually topped-up with what appears to be constant confirmation that the program is indeed true reality. The important word here is 'appears'.

This is the structure, the fly-trap, the Postage Stamp Consensus or Perceptual Mainframe, which represents that incredibly narrow band of perceived possibility delivered by the 'education' system, mainstream media, science and medicine. From the earliest age the download begins with parents who have themselves succumbed to the very programming their children are about to go through. Most parents don't do this out of malevolence and mostly it is quite the opposite. They do what they believe is best for their children and that is what the program has told them is best. Within three or four years comes the major transition from parental programming to full-blown state (Cult) programming in school, college and university where perceptually-programmed teachers and academics pass on their programming to the next generations. Teachers who resist are soon marginalised and their careers ended while children who resist are called a problem child for whom Ritalin may need to be prescribed. A few years after entering the 'world' children are under the control of authority figures representing the state telling them when they have to be there, when they can leave and when they can speak, eat, even go to the toilet. This is calculated preparation for a lifetime of obeying authority in all its forms. Reflex-action fear of authority is instilled by authority from the start. Children soon learn the carrot and stick consequences of obeying or defying authority which is underpinned daily for the rest of their life. Fortunately I daydreamed through this crap and never obeyed authority simply because it told me to. This approach to my alleged 'betters' continues to this day. There can be consequences of pursuing open-minded freedom in a world of closed-minded conformity. I spent a lot of time in school corridors after being ejected from the classroom for not taking some of it seriously and now I spend a lot of time being ejected from Facebook, YouTube and Twitter. But I can tell you that being true to yourself and not compromising your self-respect is far more exhilarating than bowing to authority for authority's sake. You don't have to be a sheep to the shepherd (authority) and the sheep dog (fear of not obeying authority).

The perceptual download continues throughout the formative years in school, college and university while script-reading 'teachers', 'academics' 'scientists', 'doctors' and 'journalists' insist that ongoing generations must be as programmed as they are. Accept the program or you will not pass your 'exams' which confirm your 'degree' of programming. It is tragic to think that many parents pressure their offspring to work hard at school to download the program and qualify for the next stage at college and university. The late, great, American comedian George Carlin said: 'Here's a bumper sticker I'd like to see: We are proud parents of a child who has resisted his teachers' attempts to break his spirit and bend him to the will of his corporate masters.' Well, the best of luck finding many of those, George. Then comes the moment to leave the formal programming years in academia and enter the 'adult' world of work. There you meet others in your chosen or prescribed arena who went through the same Postage Stamp Consensus program before you did. There is therefore overwhelming agreement between almost everyone on the basic foundations of Postage Stamp reality and the rejection, even contempt, of the few who have a mind of their own and are prepared to use it. This has two major effects. Firstly, the consensus confirms to the programmed that their download is really how things are. I mean, everyone knows that, right? Secondly, the arrogance and ignorance of Postage Stamp adherents ensure that anyone questioning the program will have unpleasant consequences for seeking their own truth and not picking their perceptions from the shelf marked: 'Things you must believe without question and if you don't you're a dangerous lunatic conspiracy theorist and a harebrained nutter'.

Every government, agency and corporation is founded on the same Postage Stamp prison cell and you can see why so many people believe the same thing while calling it their own 'opinion'. Fusion of governments and corporations in pursuit of the same agenda was the definition of fascism described by Italian dictator Benito Mussolini. The pressure to conform to perceptual norms downloaded for a lifetime is incessant and infiltrates society right

down to family groups that become censors and condemners of their own 'black sheep' for not, ironically, being sheep. We have seen an explosion of that in the 'Covid' era. Cult-owned global media unleashes its propaganda all day every day in support of the Postage Stamp and targets with abuse and ridicule anyone in the public eye who won't bend their mind to the will of the tyranny. Any response to this is denied (certainly in my case). They don't want to give a platform to expose official lies. Cult-owned-and-created Internet giants like Facebook, Google, YouTube and Twitter delete you for having an unapproved opinion. Facebook boasts that its AI censors delete 97-percent of 'hate speech' before anyone even reports it. Much of that 'hate speech' will simply be an opinion that Facebook and its masters don't want people to see. Such perceptual oppression is widely known as fascism. Even Facebook executive Benny Thomas, a 'CEO Global Planning Lead', said in comments secretly recorded by investigative journalism operation Project Veritas that Facebook is 'too powerful' and should be broken up:

I mean, no king in history has been the ruler of two billion people, but Mark Zuckerberg is ... And he's 36. That's too much for a 36-year-old ... You should not have power over two billion people. I just think that's wrong.

Thomas said Facebook-owned platforms like Instagram, Oculus, and WhatsApp needed to be separate companies. 'It's too much power when they're all one together'. That's the way the Cult likes it, however. We have an executive of a Cult organisation in Benny Thomas that doesn't know there is a Cult such is the compartmentalisation. Thomas said that Facebook and Google 'are no longer companies, they're countries'. Actually they are more powerful than countries on the basis that if you control information you control perception and control human society.

I love my oppressor

Another expression of this psychological trickery is for those who realise they are being pressured into compliance to eventually

convince themselves to believe the official narratives to protect their self-respect from accepting the truth that they have succumbed to meek and subservient compliance. Such people become some of the most vehement defenders of the system. You can see them everywhere screaming abuse at those who prefer to think for themselves and by doing so reminding the compliers of their own capitulation to conformity. 'You are talking dangerous nonsense you Covidiot!!' Are you trying to convince me or yourself? It is a potent form of Stockholm syndrome which is defined as: 'A psychological condition that occurs when a victim of abuse identifies and attaches, or bonds, positively with their abuser.' An example is hostages bonding and even 'falling in love' with their kidnappers. The syndrome has been observed in domestic violence, abused children, concentration camp inmates, prisoners of war and many and various Satanic cults. These are some traits of Stockholm syndrome listed at goodtherapy.org:

- Positive regard towards perpetrators of abuse or captor [see 'Covid'].
- Failure to cooperate with police and other government authorities when it comes to holding perpetrators of abuse or kidnapping accountable [or in the case of 'Covid' cooperating with the police to enforce and defend their captors' demands].
- Little or no effort to escape [see 'Covid'].
- Belief in the goodness of the perpetrators or kidnappers [see 'Covid'].
- Appeasement of captors. This is a manipulative strategy for maintaining one's safety. As victims get rewarded – perhaps with less abuse or even with life itself – their appeasing behaviours are reinforced [see 'Covid'].
- Learned helplessness. This can be akin to 'if you can't beat 'em, join 'em'. As the victims fail to escape the abuse or captivity, they may start giving up and soon realize it's just easier for everyone if they acquiesce all their power to their captors [see 'Covid'].

- Feelings of pity toward the abusers, believing they are actually victims themselves. Because of this, victims may go on a crusade or mission to 'save' [protect] their abuser [see the venom unleashed on those challenging the official 'Covid' narrative].
- Unwillingness to learn to detach from their perpetrators and heal. In essence, victims may tend to be less loyal to themselves than to their abuser [*definitely* see 'Covid'].

Ponder on those traits and compare them with the behaviour of great swathes of the global population who have defended governments and authorities which have spent every minute destroying their lives and livelihoods and those of their children and grandchildren since early 2020 with fascistic lockdowns, house arrest and employment deletion to 'protect' them from a 'deadly virus' that their abusers' perceptually created to bring about this very outcome. We are looking at mass Stockholm syndrome. All those that agree to concede their freedom will believe those perceptions are originating in their own independent 'mind' when in fact by conceding their reality to Stockholm syndrome they have by definition conceded any independence of mind. Listen to the 'opinions' of the acquiescing masses in this 'Covid' era and what gushes forth is the repetition of the official version of everything delivered unprocessed, unfiltered and unquestioned. The whole programming dynamic works this way. I must be free because I'm told that I am and so I think that I am.

You can see what I mean with the chapter theme of 'I'm thinking – Oh, but *are* you?' The great majority are not thinking, let alone for themselves. They are repeating what authority has told them to believe which allows them to be controlled. Weaving through this mentality is the fear that the 'conspiracy theorists' are right and this again explains the often hysterical abuse that ensues when you dare to contest the official narrative of anything. Denial is the mechanism of hiding from yourself what you don't want to be true. Telling people what they want to hear is easy, but it's an infinitely greater challenge to tell them what they would rather not be happening.

One is akin to pushing against an open door while the other is met with vehement resistance no matter what the scale of evidence. I don't want it to be true so I'll convince myself that it's not. Examples are everywhere from the denial that a partner is cheating despite all the signs to the reflex-action rejection of any idea that world events in which country after country act in exactly the same way are centrally coordinated. To accept the latter is to accept that a force of unspeakable evil is working to destroy your life and the lives of your children with nothing too horrific to achieve that end. Who the heck wants that to be true? But if we don't face reality the end is duly achieved and the consequences are far worse and ongoing than breaking through the walls of denial today with the courage to make a stand against tyranny.

Connect the dots – but how?

A crucial aspect of perceptual programming is to portray a world in which everything is random and almost nothing is connected to anything else. Randomness cannot be coordinated by its very nature and once you perceive events as random the idea they could be connected is waved away as the rantings of the tinfoil-hat brigade. You can't plan and coordinate random you idiot! No, you can't, but you can hide the coldly-calculated and long-planned behind the *illusion* of randomness. A foundation manifestation of the Renegade Mind is to scan reality for patterns that connect the apparently random and turn pixels and dots into pictures. This is the way I work and have done so for more than 30 years. You look for similarities in people, modus operandi and desired outcomes and slowly, then ever quicker, the picture forms. For instance: There would seem to be no connection between the 'Covid pandemic' hoax and the human-caused global-warming hoax and yet they are masks (appropriately) on the same face seeking the same outcome. Those pushing the global warming myth through the Club of Rome and other Cult agencies are driving the lies about 'Covid' – Bill Gates is an obvious one, but they are endless. Why would the same people be involved in both when they are clearly not connected? Oh, but they

are. Common themes with personnel are matched by common goals. The 'solutions' to both 'problems' are centralisation of global power to impose the will of the few on the many to 'save' humanity from 'Covid' and save the planet from an 'existential threat' (we need 'zero Covid' and 'zero carbon emissions'). These, in turn, connect with the 'dot' of globalisation which was coined to describe the centralisation of global power in every area of life through incessant political and corporate expansion, trading blocks and superstates like the European Union. If you are the few and you want to control the many you have to centralise power and decision-making. The more you centralise power the more power the few at the centre will have over the many; and the more that power is centralised the more power those at the centre have to centralise even quicker. The momentum of centralisation gets faster and faster which is exactly the process we have witnessed. In this way the hoaxed 'pandemic' and the fakery of human-caused global warming serve the interests of globalisation and the seizure of global power in the hands of the Cult inner-circle which is behind 'Covid', 'climate change' and globalisation. At this point random 'dots' become a clear and obvious picture or pattern.

Klaus Schwab, the classic Bond villain who founded the Cult's Gates-funded World Economic Forum, published a book in 2020, *The Great Reset*, in which he used the 'problem' of 'Covid' to justify a total transformation of human society to 'save' humanity from 'climate change'. Schwab said: 'The pandemic represents a rare but narrow window of opportunity to reflect, reimagine, and reset our world.' What he didn't mention is that the Cult he serves is behind both hoaxes as I show in my book *The Answer*. He and the Cult don't have to reimagine the world. They know precisely what they want and that's why they destroyed human society with 'Covid' to 'build back better' in their grand design. Their job is not to imagine, but to get humanity to imagine and agree with their plans while believing it's all random. It must be pure coincidence that 'The Great Reset' has long been the Cult's code name for the global imposition of fascism and replaced previous code-names of the 'New World

Order' used by Cult frontmen like Father George Bush and the 'New Order of the Ages' which emerged from Freemasonry and much older secret societies. New Order of the Ages appears on the reverse of the Great Seal of the United States as 'Novus ordo seclorum' underneath the Cult symbol used since way back of the pyramid and all seeing-eye (Fig 3). The pyramid is the hierarchy of human control headed by the illuminated eye that symbolises the force behind the Cult which I will expose in later chapters. The term 'Annuit Coeptis' translates as 'He favours our undertaking'. We are told the 'He' is the Christian god, but 'He' is not as I will be explaining.



Figure 3: The all-seeing eye of the Cult 'god' on the Freemason-designed Great Seal of the United States and also on the dollar bill.

Having you on

Two major Cult techniques of perceptual manipulation that relate to all this are what I have called since the 1990s Problem-Reaction-Solution (PRS) and the Totalitarian Tiptoe (TT). They can be uncovered by the inquiring mind with a simple question: Who benefits? The answer usually identifies the perpetrators of a given action or happening through the concept of 'he who most benefits from a crime is the one most likely to have committed it'. The Latin 'Cue bono?' – Who benefits? – is widely attributed to the Roman orator and statesman Marcus Tullius Cicero. No wonder it goes back so far when the concept has been relevant to human behaviour since

history was recorded. Problem-Reaction-Solution is the technique used to manipulate us every day by covertly creating a problem (or the illusion of one) and offering the solution to the problem (or the illusion of one). In the first phase you create the problem and blame someone or something else for why it has happened. This may relate to a financial collapse, terrorist attack, war, global warming or pandemic, anything in fact that will allow you to impose the 'solution' to change society in the way you desire at that time. The 'problem' doesn't have to be real. PRS is manipulation of perception and all you need is the population to believe the problem is real. Human-caused global warming and the 'Covid pandemic' only have to be *perceived* to be real for the population to accept the 'solutions' of authority. I refer to this technique as NO-Problem-Reaction-Solution. Billions did not meekly accept house arrest from early 2020 because there was a real deadly 'Covid pandemic' but because they perceived – believed – that to be the case. The antidote to Problem-Reaction-Solution is to ask who benefits from the proposed solution. Invariably it will be anyone who wants to justify more control through deletion of freedom and centralisation of power and decision-making.

The two world wars were Problem-Reaction-Solutions that transformed and realigned global society. Both were manipulated into being by the Cult as I have detailed in books since the mid-1990s. They dramatically centralised global power, especially World War Two, which led to the United Nations and other global bodies thanks to the overt and covert manipulations of the Rockefeller family and other Cult bloodlines like the Rothschilds. The UN is a stalking horse for full-blown world government that I will come to shortly. The land on which the UN building stands in New York was donated by the Rockefellers and the same Cult family was behind Big Pharma scalpel and drug 'medicine' and the creation of the World Health Organization as part of the UN. They have been stalwarts of the eugenics movement and funded Hitler's race-purity expert' Ernst Rudin. The human-caused global warming hoax has been orchestrated by the Club of Rome through the UN which is

manufacturing both the 'problem' through its Intergovernmental Panel on Climate Change and imposing the 'solution' through its Agenda 21 and Agenda 2030 which demand the total centralisation of global power to 'save the world' from a climate hoax the United Nations is itself perpetrating. What a small world the Cult can be seen to be particularly among the inner circles. The bedfellow of Problem-Reaction-Solution is the Totalitarian Tiptoe which became the Totalitarian Sprint in 2020. The technique is fashioned to hide the carefully-coordinated behind the cover of apparently random events. You start the sequence at 'A' and you know you are heading for 'Z'. You don't want people to know that and each step on the journey is presented as a random happening while all the steps strung together lead in the same direction. The speed may have quickened dramatically in recent times, but you can still see the incremental approach of the Tiptoe in the case of 'Covid' as each new imposition takes us deeper into fascism. Tell people they have to do this or that to get back to 'normal', then this and this and this. With each new demand adding to the ones that went before the population's freedom is deleted until it disappears. The spider wraps its web around the flies more comprehensively with each new diktat. I'll highlight this in more detail when I get to the 'Covid' hoax and how it has been pulled off. Another prime example of the Totalitarian Tiptoe is how the Cult-created European Union went from a 'free-trade zone' to a centralised bureaucratic dictatorship through the Tiptoe of incremental centralisation of power until nations became mere administrative units for Cult-owned dark suits in Brussels.

The antidote to ignorance is knowledge which the Cult seeks vehemently to deny us, but despite the systematic censorship to that end the Renegade Mind can overcome this by vociferously seeking out the facts no matter the impediments put in the way. There is also a method of thinking and perceiving – *knowing* – that doesn't even need names, dates, place-type facts to identify the patterns that reveal the story. I'll get to that in the final chapter. All you need to know about the manipulation of human society and to what end is still out there – *at the time of writing* – in the form of books, videos

and websites for those that really want to breach the walls of programmed perception. To access this knowledge requires the abandonment of the mainstream media as a source of information in the awareness that this is owned and controlled by the Cult and therefore promotes mass perceptions that suit the Cult. Mainstream media lies all day, every day. That is its function and very reason for being. Where it does tell the truth, here and there, is only because the truth and the Cult agenda very occasionally coincide. If you look for fact and insight to the BBC, CNN and virtually all the rest of them you are asking to be conned and perceptually programmed.

Know the outcome and you'll see the journey

Events seem random when you have no idea where the world is being taken. Once you do the random becomes the carefully planned. Know the outcome and you'll see the journey is a phrase I have been using for a long time to give context to daily happenings that appear unconnected. Does a problem, or illusion of a problem, trigger a proposed 'solution' that further drives society in the direction of the outcome? Invariably the answer will be yes and the random – *abracadabra* – becomes the clearly coordinated. So what is this outcome that unlocks the door to a massively expanded understanding of daily events? I will summarise its major aspects – the fine detail is in my other books – and those new to this information will see that the world they thought they were living in is a very different place. The foundation of the Cult agenda is the incessant centralisation of power and all such centralisation is ultimately in pursuit of Cult control on a global level. I have described for a long time the planned world structure of top-down dictatorship as the Hunger Games Society. The term obviously comes from the movie series which portrayed a world in which a few living in military-protected hi-tech luxury were the overlords of a population condemned to abject poverty in isolated 'sectors' that were not allowed to interact. 'Covid' lockdowns and travel bans anyone? The 'Hunger Games' pyramid of structural control has the inner circle of the Cult at the top with pretty much the entire

population at the bottom under their control through dependency for survival on the Cult. The whole structure is planned to be protected and enforced by a military-police state ([Fig 4](#)).

Here you have the reason for the global lockdowns of the fake pandemic to coldly destroy independent incomes and livelihoods and make everyone dependent on the 'state' (the Cult that controls the 'states'). I have warned in my books for many years about the plan to introduce a 'guaranteed income' – a barely survivable pittance – designed to impose dependency when employment was destroyed by AI technology and now even more comprehensively at great speed by the 'Covid' scam. Once the pandemic was played and lockdown consequences began to delete independent income the authorities began to talk right on cue about the need for a guaranteed income and a 'Great Reset'. Guaranteed income will be presented as benevolent governments seeking to help a desperate people – desperate as a direct result of actions of the same governments. The truth is that such payments are a trap. You will only get them if you do exactly what the authorities demand including mass vaccination (genetic manipulation). We have seen this theme already in Australia where those dependent on government benefits have them reduced if parents don't agree to have their children vaccinated according to an insane health-destroying government-dictated schedule. Calculated economic collapse applies to governments as well as people. The Cult wants rid of countries through the creation of a world state with countries broken up into regions ruled by a world government and super states like the European Union. Countries must be bankrupted, too, to this end and it's being achieved by the trillions in 'rescue packages' and furlough payments, trillions in lost taxation, and money-no-object spending on 'Covid' including constant all-medium advertising (programming) which has made the media dependent on government for much of its income. The day of reckoning is coming – as planned – for government spending and given that it has been made possible by printing money and not by production/taxation there is inflation on the way that has the

potential to wipe out monetary value. In that case there will be no need for the Cult to steal your money. It just won't be worth anything (see the German Weimar Republic before the Nazis took over). Many have been okay with lockdowns while getting a percentage of their income from so-called furlough payments without having to work. Those payments are dependent, however, on people having at least a theoretical job with a business considered non-essential and ordered to close. As these business go under because they are closed by lockdown after lockdown the furlough stops and it will for everyone eventually. Then what? The 'then what?' is precisely the idea.



Figure 4: The Hunger Games Society structure I have long warned was planned and now the 'Covid' hoax has made it possible. This is the real reason for lockdowns.

Hired hands

Between the Hunger Games Cult elite and the dependent population is planned to be a vicious military-police state (a fusion of the two into one force). This has been in the making for a long time with police looking ever more like the military and carrying weapons to match. The pandemic scam has seen this process accelerate so fast as

lockdown house arrest is brutally enforced by carefully recruited fascist minds and gormless system-servers. The police and military are planned to merge into a centrally-directed world army in a global structure headed by a world government which wouldn't be elected even by the election fixes now in place. The world army is not planned even to be human and instead wars would be fought, primarily against the population, using robot technology controlled by artificial intelligence. I have been warning about this for decades and now militaries around the world are being transformed by this very AI technology. The global regime that I describe is a particular form of fascism known as a technocracy in which decisions are not made by clueless and co-opted politicians but by unelected technocrats – scientists, engineers, technologists and bureaucrats. Cult-owned-and-controlled Silicon Valley giants are examples of technocracy and they already have far more power to direct world events than governments. They are with their censorship *selecting* governments. I know that some are calling the 'Great Reset' a Marxist communist takeover, but fascism and Marxism are different labels for the same tyranny. Tell those who lived in fascist Germany and Stalinist Russia that there was a difference in the way their freedom was deleted and their lives controlled. I could call it a fascist technocracy or a Marxist technocracy and they would be equally accurate. The Hunger Games society with its world government structure would oversee a world army, world central bank and single world cashless currency imposing its will on a microchipped population ([Fig 5](#)). Scan its different elements and see how the illusory pandemic is forcing society in this very direction at great speed. Leaders of 23 countries and the World Health Organization (WHO) backed the idea in March, 2021, of a global treaty for 'international cooperation' in 'health emergencies' and nations should 'come together as a global community for peaceful cooperation that extends beyond this crisis'. Cut the Orwellian bullshit and this means another step towards global government. The plan includes a cashless digital money system that I first warned about in 1993. Right at the start of 'Covid' the deeply corrupt Tedros

Adhanom Ghebreyesus, the crooked and merely gofer 'head' of the World Health Organization, said it was possible to catch the 'virus' by touching cash and it was better to use cashless means. The claim was ridiculous nonsense and like the whole 'Covid' mind-trick it was nothing to do with 'health' and everything to do with pushing every aspect of the Cult agenda. As a result of the Tedros lie the use of cash has plummeted. The Cult script involves a single world digital currency that would eventually be technologically embedded in the body. China is a massive global centre for the Cult and if you watch what is happening there you will know what is planned for everywhere. The Chinese government is developing a digital currency which would allow fines to be deducted immediately via AI for anyone caught on camera breaking its fantastic list of laws and the money is going to be programmable with an expiry date to ensure that no one can accrue wealth except the Cult and its operatives.



Figure 5: The structure of global control the Cult has been working towards for so long and this has been enormously advanced by the 'Covid' illusion.

Serfdom is so smart

The Cult plan is far wider, extreme, and more comprehensive than even most conspiracy researchers appreciate and I will come to the true depths of deceit and control in the chapters 'Who controls the

Cult?’ and ‘Escaping Wetiko’. Even the world that we know is crazy enough. We are being deluged with ever more sophisticated and controlling technology under the heading of ‘smart’. We have smart televisions, smart meters, smart cards, smart cars, smart driving, smart roads, smart pills, smart patches, smart watches, smart skin, smart borders, smart pavements, smart streets, smart cities, smart communities, smart environments, smart growth, smart planet ... smart *everything* around us. Smart technologies and methods of operation are designed to interlock to create a global Smart Grid connecting the entirety of human society including human minds to create a centrally-dictated ‘hive’ mind. ‘Smart cities’ is code for densely-occupied megacities of total surveillance and control through AI. Ever more destructive frequency communication systems like 5G have been rolled out without any official testing for health and psychological effects (colossal). 5G/6G/7G systems are needed to run the Smart Grid and each one becomes more destructive of body and mind. Deleting independent income is crucial to forcing people into these AI-policed prisons by ending private property ownership (except for the Cult elite). The Cult’s Great Reset now openly foresees a global society in which no one will own any possessions and everything will be rented while the Cult would own literally everything under the guise of government and corporations. The aim has been to use the lockdowns to destroy sources of income on a mass scale and when the people are destitute and in unrepayable amounts of debt (problem) Cult assets come forward with the pledge to write-off debt in return for handing over all property and possessions (solution). Everything – literally everything including people – would be connected to the Internet via AI. I was warning years ago about the coming Internet of Things (IoT) in which all devices and technology from your car to your fridge would be plugged into the Internet and controlled by AI. Now we are already there with much more to come. The next stage is the Internet of Everything (IoE) which is planned to include the connection of AI to the human brain and body to replace the human mind with a centrally-controlled AI mind. Instead of perceptions

being manipulated through control of information and censorship those perceptions would come direct from the Cult through AI. What do you think? You think whatever AI decides that you think. In human terms there would be no individual 'think' any longer. Too incredible? The ravings of a lunatic? Not at all. Cult-owned crazies in Silicon Valley have been telling us the plan for years without explaining the real motivation and calculated implications. These include Google executive and 'futurist' Ray Kurzweil who highlights the year 2030 for when this would be underway. He said:

Our thinking ... will be a hybrid of biological and non-biological thinking ... humans will be able to extend their limitations and 'think in the cloud' ... We're going to put gateways to the cloud in our brains ... We're going to gradually merge and enhance ourselves ... In my view, that's the nature of being human – we transcend our limitations.

As the technology becomes vastly superior to what we are then the small proportion that is still human gets smaller and smaller and smaller until it's just utterly negligible.

The sales-pitch of Kurzweil and Cult-owned Silicon Valley is that this would make us 'super-human' when the real aim is to make us post-human and no longer 'human' in the sense that we have come to know. The entire global population would be connected to AI and become the centrally-controlled 'hive-mind' of externally-delivered perceptions. The Smart Grid being installed to impose the Cult's will on the world is being constructed to allow particular locations – even one location – to control the whole global system. From these prime control centres, which absolutely include China and Israel, anything connected to the Internet would be switched on or off and manipulated at will. Energy systems could be cut, communication via the Internet taken down, computer-controlled driverless autonomous vehicles driven off the road, medical devices switched off, the potential is limitless given how much AI and Internet connections now run human society. We have seen nothing yet if we allow this to continue. Autonomous vehicle makers are working with law enforcement to produce cars designed to automatically pull over if they detect a police or emergency vehicle flashing from up to 100 feet away. At a police stop the car would be unlocked and the

window rolled down automatically. Vehicles would only take you where the computer (the state) allowed. The end of petrol vehicles and speed limiters on all new cars in the UK and EU from 2022 are steps leading to electric computerised transport over which ultimately you have no control. The picture is far bigger even than the Cult global network or web and that will become clear when I get to the nature of the 'spider'. There is a connection between all these happenings and the instigation of DNA-manipulating 'vaccines' (which aren't 'vaccines') justified by the 'Covid' hoax. That connection is the unfolding plan to transform the human body from a biological to a synthetic biological state and this is why synthetic biology is such a fast-emerging discipline of mainstream science. 'Covid vaccines' are infusing self-replicating synthetic genetic material into the cells to cumulatively take us on the Totalitarian Tiptoe from Human 1.0 to the synthetic biological Human 2.0 which will be physically and perceptually attached to the Smart Grid to one hundred percent control every thought, perception and deed. Humanity needs to wake up and *fast*.

This is the barest explanation of where the 'outcome' is planned to go but it's enough to see the journey happening all around us. Those new to this information will already see 'Covid' in a whole new context. I will add much more detail as we go along, but for the minutiae evidence see my mega-works, *The Answer*, *The Trigger* and *Everything You Need to Know But Have Never Been Told*.

Now – how does a Renegade Mind see the 'world'?

CHAPTER TWO

Renegade Perception

It is one thing to be clever and another to be wise

George R.R. Martin

A simple definition of the difference between a programmed mind and a Renegade Mind would be that one sees only dots while the other connects them to see the picture. Reading reality with accuracy requires the observer to (a) know the planned outcome and (b) realise that everything, but *everything*, is connected.

The entirety of infinite reality is connected – that's its very nature – and with human society an expression of infinite reality the same must apply. Simple cause and effect is a connection. The effect is triggered by the cause and the effect then becomes the cause of another effect. Nothing happens in isolation because it *can't*. Life in whatever reality is simple choice and consequence. We make choices and these lead to consequences. If we don't like the consequences we can make different choices and get different consequences which lead to other choices and consequences. The choice and the consequence are not only connected they are indivisible. You can't have one without the other as an old song goes. A few cannot control the world unless those being controlled allow that to happen – cause and effect, choice and consequence. Control – who has it and who doesn't – is a two-way process, a symbiotic relationship, involving the controller and controlled. 'They took my freedom away!!' Well, yes, but you also gave it to them. Humanity is

subjected to mass control because humanity has acquiesced to that control. This is all cause and effect and literally a case of give and take. In the same way world events of every kind are connected and the Cult works incessantly to sell the illusion of the random and coincidental to maintain the essential (to them) perception of dots that hide the picture. Renegade Minds know this and constantly scan the world for patterns of connection. This is absolutely pivotal in understanding the happenings in the world and without that perspective clarity is impossible. First you know the planned outcome and then you identify the steps on the journey – the day-by-day apparently random which, when connected in relation to the outcome, no longer appear as individual events, but as the proverbial *chain* of events leading in the same direction. I'll give you some examples:

Political puppet show

We are told to believe that politics is 'adversarial' in that different parties with different beliefs engage in an endless tussle for power. There may have been some truth in that up to a point – and only a point – but today divisions between 'different' parties are rhetorical not ideological. Even the rhetorical is fusing into one-speak as the parties eject any remaining free thinkers while others succumb to the ever-gathering intimidation of anyone with the 'wrong' opinion. The Cult is not a new phenomenon and can be traced back thousands of years as my books have documented. Its intergenerational initiatives have been manipulating events with increasing effect the more that global power has been centralised. In ancient times the Cult secured control through the system of monarchy in which 'special' bloodlines (of which more later) demanded the right to rule as kings and queens simply by birthright and by vanquishing others who claimed the same birthright. There came a time, however, when people had matured enough to see the unfairness of such tyranny and demanded a say in who governed them. Note the word – *governed* them. Not served them – *governed* them, hence government defined as 'the political direction and control exercised over the

actions of the members, citizens, or inhabitants of communities, societies, and states; direction of the affairs of a state, community, etc.’ Governments exercise control over rather than serve just like the monarchies before them. Bizarrely there are still countries like the United Kingdom which are ruled by a monarch *and* a government that officially answers to the monarch. The UK head of state and that of Commonwealth countries such as Canada, Australia and New Zealand is ‘selected’ by who in a *single family* had unprotected sex with whom and in what order. Pinch me it can’t be true. Ouch! Shit, it is. The demise of monarchies in most countries offered a potential vacuum in which some form of free and fair society could arise and the Cult had that base covered. Monarchies had served its interests but they couldn’t continue in the face of such widespread opposition and, anyway, replacing a ‘royal’ dictatorship that people could see with a dictatorship ‘of the people’ hiding behind the concept of ‘democracy’ presented far greater manipulative possibilities and ways of hiding coordinated tyranny behind the illusion of ‘freedom’.

Democracy is quite wrongly defined as government selected by the population. This is not the case at all. It is government selected by *some* of the population (and then only in theory). This ‘some’ doesn’t even have to be the majority as we have seen so often in first-past-the-post elections in which the so-called majority party wins fewer votes than the ‘losing’ parties combined. Democracy can give total power to a party in government from a minority of the votes cast. It’s a sleight of hand to sell tyranny as freedom. Seventy-four million Trump-supporting Americans didn’t vote for the ‘Democratic’ Party of Joe Biden in the distinctly dodgy election in 2020 and yet far from acknowledging the wishes and feelings of that great percentage of American society the Cult-owned Biden government set out from day one to destroy them and their right to a voice and opinion. Empty shell Biden and his Cult handlers said they were doing this to ‘protect democracy’. Such is the level of lunacy and sickness to which politics has descended. Connect the dots and relate them to the desired outcome – a world government run by self-appointed technocrats and no longer even elected

politicians. While operating through its political agents in government the Cult is at the same time encouraging public disdain for politicians by putting idiots and incompetents in theoretical power on the road to deleting them. The idea is to instil a public reaction that says of the technocrats: 'Well, they couldn't do any worse than the pathetic politicians.' It's all about controlling perception and Renegade Minds can see through that while programmed minds cannot when they are ignorant of both the planned outcome and the manipulation techniques employed to secure that end. This knowledge can be learned, however, and fast if people choose to get informed.

Politics may at first sight appear very difficult to control from a central point. I mean look at the 'different' parties and how would you be able to oversee them all and their constituent parts? In truth, it's very straightforward because of their structure. We are back to the pyramid of imposition and acquiescence. Organisations are structured in the same way as the system as a whole. Political parties are not open forums of free expression. They are hierarchies. I was a national spokesman for the British Green Party which claimed to be a different kind of politics in which influence and power was devolved; but I can tell you from direct experience – and it's far worse now – that Green parties are run as hierarchies like all the others however much they may try to hide that fact or kid themselves that it's not true. A very few at the top of all political parties are directing policy and personnel. They decide if you are elevated in the party or serve as a government minister and to do that you have to be a yes man or woman. Look at all the maverick political thinkers who never ascended the greasy pole. If you want to progress within the party or reach 'high-office' you need to fall into line and conform. Exceptions to this are rare indeed. Should you want to run for parliament or Congress you have to persuade the local or state level of the party to select you and for that you need to play the game as dictated by the hierarchy. If you secure election and wish to progress within the greater structure you need to go on conforming to what is acceptable to those running the hierarchy

from the peak of the pyramid. Political parties are perceptual gulags and the very fact that there are party 'Whips' appointed to 'whip' politicians into voting the way the hierarchy demands exposes the ridiculous idea that politicians are elected to serve the people they are supposed to represent. Cult operatives and manipulation has long seized control of major parties that have any chance of forming a government and at least most of those that haven't. A new party forms and the Cult goes to work to infiltrate and direct. This has reached such a level today that you see video compilations of 'leaders' of all parties whether Democrats, Republicans, Conservative, Labour and Green parroting the same Cult mantra of 'Build Back Better' and the 'Great Reset' which are straight off the Cult song-sheet to describe the transformation of global society in response to the Cult-instigated hoaxes of the 'Covid pandemic' and human-caused 'climate change'. To see Caroline Lucas, the Green Party MP that I knew when I was in the party in the 1980s, speaking in support of plans proposed by Cult operative Klaus Schwab representing the billionaire global elite is a real head-shaker.

Many parties – one master

The party system is another mind-trick and was instigated to change the nature of the dictatorship by swapping 'royalty' for dark suits that people believed – though now ever less so – represented their interests. Understanding this trick is to realise that a single force (the Cult) controls all parties either directly in terms of the major ones or through manipulation of perception and ideology with others. You don't need to manipulate Green parties to demand your transformation of society in the name of 'climate change' when they are obsessed with the lie that this is essential to 'save the planet'. You just give them a platform and away they go serving your interests while believing they are being environmentally virtuous. America's political structure is a perfect blueprint for how the two or multi-party system is really a one-party state. The Republican Party is controlled from one step back in the shadows by a group made up of billionaires and their gofers known as neoconservatives or Neocons.

I have exposed them in fine detail in my books and they were the driving force behind the policies of the imbecilic presidency of Boy George Bush which included 9/11 (see *The Trigger* for a comprehensive demolition of the official story), the subsequent 'war on terror' (war of terror) and the invasions of Afghanistan and Iraq. The latter was a No-Problem-Reaction-Solution based on claims by Cult operatives, including Bush and British Prime Minister Tony Blair, about Saddam Hussein's 'weapons of mass destruction' which did not exist as war criminals Bush and Blair well knew.

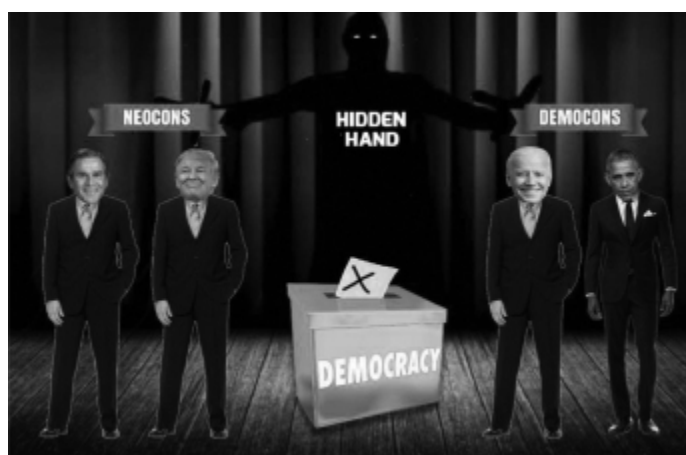


Figure 6: Different front people, different parties – same control system.

The Democratic Party has its own 'Neocon' group controlling from the background which I call the 'Democons' and here's the penny-drop – the Neocons and Democons answer to the same masters one step further back into the shadows (Fig 6). At that level of the Cult the Republican and Democrat parties are controlled by the same people and no matter which is in power the Cult is in power. This is how it works in almost every country and certainly in Britain with Conservative, Labour, Liberal Democrat and Green parties now all on the same page whatever the rhetoric may be in their feeble attempts to appear different. Neocons operated at the time of Bush through a think tank called The Project for the New American Century which in September, 2000, published a document entitled *Rebuilding America's Defenses: Strategies, Forces, and Resources*

For a New Century demanding that America fight ‘multiple, simultaneous major theatre wars’ as a ‘core mission’ to force regime-change in countries including Iraq, Libya and Syria. Neocons arranged for Bush (‘Republican’) and Blair (‘Labour Party’) to front-up the invasion of Iraq and when they departed the Democons orchestrated the targeting of Libya and Syria through Barack Obama (‘Democrat’) and British Prime Minister David Cameron (‘Conservative Party’). We have ‘different’ parties and ‘different’ people, but the same unfolding script. The more the Cult has seized the reigns of parties and personnel the more their policies have transparently pursued the same agenda to the point where the fascist ‘Covid’ impositions of the Conservative junta of Jackboot Johnson in Britain were opposed by the Labour Party because they were not fascist enough. The Labour Party is likened to the US Democrats while the Conservative Party is akin to a British version of the Republicans and on both sides of the Atlantic they all speak the same language and support the direction demanded by the Cult although some more enthusiastically than others. It’s a similar story in country after country because it’s all centrally controlled. Oh, but what about Trump? I’ll come to him shortly. Political ‘choice’ in the ‘party’ system goes like this: You vote for Party A and they get into government. You don’t like what they do so next time you vote for Party B and they get into government. You don’t like what they do when it’s pretty much the same as Party A and why wouldn’t that be with both controlled by the same force? Given that only two, sometimes three, parties have any chance of forming a government to get rid of Party B that you don’t like you have to vote again for Party A which ... you don’t like. This, ladies and gentlemen, is what they call ‘democracy’ which we are told – wrongly – is a term interchangeable with ‘freedom’.

The cult of cults

At this point I need to introduce a major expression of the Global Cult known as Sabbatian-Frankism. Sabbatian is also spelt as Sabbatean. I will summarise here. I have published major exposés

and detailed background in other works. Sabbatian-Frankism combines the names of two frauds posing as 'Jewish' men, Sabbatai Zevi (1626-1676), a rabbi, black magician and occultist who proclaimed he was the Jewish messiah; and Jacob Frank (1726-1791), the Polish 'Jew', black magician and occultist who said he was the reincarnation of 'messiah' Zevi and biblical patriarch Jacob. They worked across two centuries to establish the Sabbatian-Frankist cult that plays a major, indeed central, role in the manipulation of human society by the Global Cult which has its origins much further back in history than Sabbatai Zevi. I should emphasise two points here in response to the shrill voices that will scream 'anti-Semitism': (1) Sabbatian-Frankists are NOT Jewish and only pose as such to hide their cult behind a Jewish façade; and (2) my information about this cult has come from Jewish sources who have long realised that their society and community has been infiltrated and taken over by interloper Sabbatian-Frankists. Infiltration has been the foundation technique of Sabbatian-Frankism from its official origin in the 17th century. Zevi's Sabbatian sect attracted a massive following described as the biggest messianic movement in Jewish history, spreading as far as Africa and Asia, and he promised a return for the Jews to the 'Promised Land' of Israel. Sabbatianism was not Judaism but an inversion of everything that mainstream Judaism stood for. So much so that this sinister cult would have a feast day when Judaism had a fast day and whatever was forbidden in Judaism the Sabbatians were encouraged and even commanded to do. This included incest and what would be today called Satanism. Members were forbidden to marry outside the sect and there was a system of keeping their children ignorant of what they were part of until they were old enough to be trusted not to unknowingly reveal anything to outsiders. The same system is employed to this day by the Global Cult in general which Sabbatian-Frankism has enormously influenced and now largely controls.

Zevi and his Sabbatians suffered a setback with the intervention by the Sultan of the Islamic Ottoman Empire in the Middle East and what is now the Republic of Turkey where Zevi was located. The

Sultan gave him the choice of proving his 'divinity', converting to Islam or facing torture and death. Funnily enough Zevi chose to convert or at least appear to. Some of his supporters were disillusioned and drifted away, but many did not with 300 families also converting – only in theory – to Islam. They continued behind this Islamic smokescreen to follow the goals, rules and rituals of Sabbatianism and became known as 'crypto-Jews' or the 'Dönme' which means 'to turn'. This is rather ironic because they didn't 'turn' and instead hid behind a fake Islamic persona. The process of appearing to be one thing while being very much another would become the calling card of Sabbatianism especially after Zevi's death and the arrival of the Satanist Jacob Frank in the 18th century when the cult became Sabbatian-Frankism and plumbed still new depths of depravity and infiltration which included – still includes – human sacrifice and sex with children. Wherever Sabbatians go paedophilia and Satanism follow and is it really a surprise that Hollywood is so infested with child abuse and Satanism when it was established by Sabbatian-Frankists and is still controlled by them? Hollywood has been one of the prime vehicles for global perceptual programming and manipulation. How many believe the version of 'history' portrayed in movies when it is a travesty and inversion (again) of the truth? Rabbi Marvin Antelman describes Frankism in his book, *To Eliminate the Opiate*, as 'a movement of complete evil' while Jewish professor Gershom Scholem said of Frank in *The Messianic Idea in Judaism*: 'In all his actions [he was] a truly corrupt and degenerate individual ... one of the most frightening phenomena in the whole of Jewish history.' Frank was excommunicated by traditional rabbis, as was Zevi, but Frank was undeterred and enjoyed vital support from the House of Rothschild, the infamous banking dynasty whose inner-core are Sabbatian-Frankists and not Jews. Infiltration of the Roman Church and Vatican was instigated by Frank with many Dönme 'turning' again to convert to Roman Catholicism with a view to hijacking the reins of power. This was the ever-repeating modus operandi and continues to be so. Pose as an advocate of the religion, culture or country that you want to control and then

manipulate your people into the positions of authority and influence largely as advisers, administrators and Svengalis for those that appear to be in power. They did this with Judaism, Christianity (Christian Zionism is part of this), Islam and other religions and nations until Sabbatian-Frankism spanned the world as it does today.

Sabbatian Saudis and the terror network

One expression of the Sabbatian-Frankist Dönme within Islam is the ruling family of Saudi Arabia, the House of Saud, through which came the vile distortion of Islam known as Wahhabism. This is the violent creed followed by terrorist groups like Al-Qaeda and ISIS or Islamic State. Wahhabism is the hand-chopping, head-chopping 'religion' of Saudi Arabia which is used to keep the people in a constant state of fear so the interloper House of Saud can continue to rule. Al-Qaeda and Islamic State were lavishly funded by the House of Saud while being created and directed by the Sabbatian-Frankist network in the United States that operates through the Pentagon, CIA and the government in general of whichever 'party'. The front man for the establishment of Wahhabism in the middle of the 18th century was a Sabbatian-Frankist 'crypto-Jew' posing as Islamic called Muhammad ibn Abd al-Wahhab. His daughter would marry the son of Muhammad bin Saud who established the first Saudi state before his death in 1765 with support from the British Empire. Bin Saud's successors would establish modern Saudi Arabia in league with the British and Americans in 1932 which allowed them to seize control of Islam's major shrines in Mecca and Medina. They have dictated the direction of Sunni Islam ever since while Iran is the major centre of the Shiite version and here we have the source of at least the public conflict between them. The Sabbatian network has used its Wahhabi extremists to carry out Problem-Reaction-Solution terrorist attacks in the name of 'Al-Qaeda' and 'Islamic State' to justify a devastating 'war on terror', ever-increasing surveillance of the population and to terrify people into compliance. Another insight of the Renegade Mind is the streetwise understanding that

just because a country, location or people are attacked doesn't mean that those apparently representing that country, location or people are not behind the attackers. Often they are *orchestrating* the attacks because of the societal changes that can be then justified in the name of 'saving the population from terrorists'.

I show in great detail in *The Trigger* how Sabbatian-Frankists were the real perpetrators of 9/11 and not '19 Arab hijackers' who were blamed for what happened. Observe what was justified in the name of 9/11 alone in terms of Middle East invasions, mass surveillance and control that fulfilled the demands of the Project for the New American Century document published by the Sabbatian Neocons. What appear to be enemies are on the deep inside players on the same Sabbatian team. Israel and Arab 'royal' dictatorships are all ruled by Sabbatians and the recent peace agreements between Israel and Saudi Arabia, the United Arab Emirates (UAE) and others are only making formal what has always been the case behind the scenes. Palestinians who have been subjected to grotesque tyranny since Israel was bombed and terrorised into existence in 1948 have never stood a chance. Sabbatian-Frankists have controlled Israel (so the constant theme of violence and war which Sabbatians love) and they have controlled the Arab countries that Palestinians have looked to for real support that never comes. 'Royal families' of the Arab world in Saudi Arabia, Bahrain, UAE, etc., are all Sabbatians with allegiance to the aims of the cult and not what is best for their Arabic populations. They have stolen the oil and financial resources from their people by false claims to be 'royal dynasties' with a genetic right to rule and by employing vicious militaries to impose their will.

Satanic 'illumination'

The Satanist Jacob Frank formed an alliance in 1773 with two other Sabbatians, Mayer Amschel Rothschild (1744-1812), founder of the Rothschild banking dynasty, and Jesuit-educated fraudulent Jew, Adam Weishaupt, and this led to the formation of the Bavarian Illuminati, firstly under another name, in 1776. The Illuminati would

be the manipulating force behind the French Revolution (1789-1799) and was also involved in the American Revolution (1775-1783) before and after the Illuminati's official creation. Weishaupt would later become (in public) a Protestant Christian in archetypal Sabbatian style. I read that his name can be decoded as Adam-Weishaupt or 'the first man to lead those who know'. He wasn't a leader in the sense that he was a subordinate, but he did lead those below him in a crusade of transforming human society that still continues today. The theme was confirmed as early as 1785 when a horseman courier called Lanz was reported to be struck by lightning and extensive Illuminati documents were found in his saddlebags. They made the link to Weishaupt and detailed the plan for world takeover. Current events with 'Covid' fascism have been in the making for a very long time. Jacob Frank was jailed for 13 years by the Catholic Inquisition after his arrest in 1760 and on his release he headed for Frankfurt, Germany, home city and headquarters of the House of Rothschild where the alliance was struck with Mayer Amschel Rothschild and Weishaupt. Rothschild arranged for Frank to be given the title of Baron and he became a wealthy nobleman with a big following of Jews in Germany, the Austro-Hungarian Empire and other European countries. Most of them would have believed he was on their side.

The name 'Illuminati' came from the Zohar which is a body of works in the Jewish mystical 'bible' called the Kabbalah. 'Zohar' is the foundation of Sabbatian-Frankist belief and in Hebrew 'Zohar' means 'splendour', 'radiance', 'illuminated', and so we have 'Illuminati'. They claim to be the 'Illuminated Ones' from their knowledge systematically hidden from the human population and passed on through generations of carefully-chosen initiates in the global secret society network or Cult. Hidden knowledge includes an awareness of the Cult agenda for the world and the nature of our collective reality that I will explore later. Cult 'illumination' is symbolised by the torch held by the Statue of Liberty which was gifted to New York by French Freemasons in Paris who knew exactly what it represents. 'Liberty' symbolises the goddess worshipped in

Babylon as Queen Semiramis or Ishtar. The significance of this will become clear. Notice again the ubiquitous theme of inversion with the Statue of 'Liberty' really symbolising mass control (Fig 7). A mirror-image statute stands on an island in the River Seine in Paris from where New York Liberty originated (Fig 8). A large replica of the Liberty flame stands on top of the Pont de l'Alma tunnel in Paris where Princess Diana died in a Cult ritual described in *The Biggest Secret*. Lucifer 'the light bringer' is related to all this (and much more as we'll see) and 'Lucifer' is a central figure in Sabbatian-Frankism and its associated Satanism. Sabbatians reject the Jewish Torah, or Pentateuch, the 'five books of Moses' in the Old Testament known as Genesis, Exodus, Leviticus, Numbers, and Deuteronomy which are claimed by Judaism and Christianity to have been dictated by 'God' to Moses on Mount Sinai. Sabbatians say these do not apply to them and they seek to replace them with the Zohar to absorb Judaism and its followers into their inversion which is an expression of a much greater global inversion. They want to delete all religions and force humanity to worship a one-world religion – Sabbatian Satanism that also includes worship of the Earth goddess. Satanic themes are being more and more introduced into mainstream society and while Christianity is currently the foremost target for destruction the others are planned to follow.



Figure 7: The Cult goddess of Babylon disguised as the Statue of Liberty holding the flame of Lucifer the 'light bringer'.



Figure 8: Liberty's mirror image in Paris where the New York version originated.

Marx brothers

Rabbi Marvin Antelman connects the Illuminati to the Jacobins in *To Eliminate the Opiate* and Jacobins were the force behind the French Revolution. He links both to the Bund der Gerechten, or League of the Just, which was the network that inflicted communism/Marxism on the world. Antelman wrote:

The original inner circle of the Bund der Gerechten consisted of born Catholics, Protestants and Jews [Sabbatian-Frankist infiltrators], and those representatives of respective subdivisions formulated schemes for the ultimate destruction of their faiths. The heretical Catholics laid plans which they felt would take a century or more for the ultimate destruction of the church; the apostate Jews for the ultimate destruction of the Jewish religion.

Sabbatian-created communism connects into this anti-religion agenda in that communism does not allow for the free practice of religion. The Sabbatian 'Bund' became the International Communist Party and Communist League and in 1848 'Marxism' was born with the Communist Manifesto of Sabbatian assets Karl Marx and Friedrich Engels. It is absolutely no coincidence that Marxism, just a different name for fascist and other centrally-controlled tyrannies, is being imposed worldwide as a result of the 'Covid' hoax and nor that Marxist/fascist China was the place where the hoax originated. The reason for this will become very clear in the chapter 'Covid: The calculated catastrophe'. The so-called 'Woke' mentality has hijacked

traditional beliefs of the political left and replaced them with far-right make-believe 'social justice' better known as Marxism. Woke will, however, be swallowed by its own perceived 'revolution' which is really the work of billionaires and billionaire corporations feigning being 'Woke'. Marxism is being touted by Wokers as a replacement for 'capitalism' when we don't have 'capitalism'. We have cartelism in which the market is stitched up by the very Cult billionaires and corporations bankrolling Woke. Billionaires love Marxism which keeps the people in servitude while they control from the top. Terminally naïve Wokers think they are 'changing the world' when it's the Cult that is doing the changing and when they have played their vital part and become surplus to requirements they, too, will be targeted. The Illuminati-Jacobins were behind the period known as 'The Terror' in the French Revolution in 1793 and 1794 when Jacobin Maximillian de Robespierre and his Orwellian 'Committee of Public Safety' killed 17,000 'enemies of the Revolution' who had once been 'friends of the Revolution'. Karl Marx (1818-1883), whose Sabbatian creed of Marxism has cost the lives of at least 100 million people, is a hero once again to Wokers who have been systematically kept ignorant of real history by their 'education' programming. As a result they now promote a Sabbatian 'Marxist' abomination destined at some point to consume them. Rabbi Antelman, who spent decades researching the Sabbatian plot, said of the League of the Just and Karl Marx:

Contrary to popular opinion Karl Marx did not originate the Communist Manifesto. He was paid for his services by the League of the Just, which was known in its country of origin, Germany, as the Bund der Geächteten.

Antelman said the text attributed to Marx was the work of other people and Marx 'was only repeating what others already said'. Marx was 'a hired hack – lackey of the wealthy Illuminists'. Marx famously said that religion was the 'opium of the people' (part of the Sabbatian plan to demonise religion) and Antelman called his books, *To Eliminate the Opiate*. Marx was born Jewish, but his family converted to Christianity (Sabbatian modus operandi) and he

attacked Jews, not least in his book, *A World Without Jews*. In doing so he supported the Sabbatian plan to destroy traditional Jewishness and Judaism which we are clearly seeing today with the vindictive targeting of orthodox Jews by the Sabbatian government of Israel over 'Covid' laws. I don't follow any religion and it has done much damage to the world over centuries and acted as a perceptual straightjacket. Renegade Minds, however, are always asking *why* something is being done. It doesn't matter if they agree or disagree with what is happening – *why* is it happening is the question. The 'why?' can be answered with regard to religion in that religions create interacting communities of believers when the Cult wants to dismantle all discourse, unity and interaction (see 'Covid' lockdowns) and the ultimate goal is to delete all religions for a one-world religion of Cult Satanism worshipping their 'god' of which more later. We see the same 'why?' with gun control in America. I don't have guns and don't want them, but why is the Cult seeking to disarm the population at the same time that law enforcement agencies are armed to their molars and why has every tyrant in history sought to disarm people before launching the final takeover? They include Hitler, Stalin, Pol Pot and Mao who followed confiscation with violent seizing of power. You know it's a Cult agenda by the people who immediately race to the microphones to exploit dead people in multiple shootings. Ultra-Zionist Cult lackey Senator Chuck Schumer was straight on the case after ten people were killed in Boulder, Colorado in March, 2121. Simple rule ... if Schumer wants it the Cult wants it and the same with his ultra-Zionist mate the wild-eyed Senator Adam Schiff. At the same time they were calling for the disarmament of Americans, many of whom live a long way from a police response, Schumer, Schiff and the rest of these pampered clowns were sitting on Capitol Hill behind a razor-wired security fence protected by thousands of armed troops in addition to their own armed bodyguards. Mom and pop in an isolated home? They're just potential mass shooters.

Zion Mainframe

Sabbatian-Frankists and most importantly the Rothschilds were behind the creation of 'Zionism', a political movement that demanded a Jewish homeland in Israel as promised by Sabbatai Zevi. The very symbol of Israel comes from the German meaning of the name Rothschild. Dynasty founder Mayer Amschel Rothschild changed the family name from Bauer to Rothschild, or 'Red-Shield' in German, in deference to the six-pointed 'Star of David' hexagram displayed on the family's home in Frankfurt. The symbol later appeared on the flag of Israel after the Rothschilds were centrally involved in its creation. Hexagrams are not a uniquely Jewish symbol and are widely used in occult ('hidden') networks often as a symbol for Saturn (see my other books for why). Neither are Zionism and Jewishness interchangeable. Zionism is a political movement and philosophy and not a 'race' or a people. Many Jews oppose Zionism and many non-Jews, including US President Joe Biden, call themselves Zionists as does Israel-centric Donald Trump. America's support for the Israel government is pretty much a gimme with ultra-Zionist billionaires and corporations providing fantastic and dominant funding for both political parties. Former Congresswoman Cynthia McKinney has told how she was approached immediately she ran for office to 'sign the pledge' to Israel and confirm that she would always vote in that country's best interests. All American politicians are approached in this way. Anyone who refuses will get no support or funding from the enormous and all-powerful Zionist lobby that includes organisations like mega-lobby group AIPAC, the American Israel Public Affairs Committee. Trump's biggest funder was ultra-Zionist casino and media billionaire Sheldon Adelson while major funders of the Democratic Party include ultra-Zionist George Soros and ultra-Zionist financial and media mogul, Haim Saban. Some may reel back at the suggestion that Soros is an Israel-firster (Sabbatian-controlled Israel-firster), but Renegade Minds watch the actions not the words and everywhere Soros donates his billions the Sabbatian agenda benefits. In the spirit of Sabbatian inversion Soros pledged \$1 billion for a new university network to promote 'liberal values and tackle intolerance'. He made the announcement during his annual speech

at the Cult-owned World Economic Forum in Davos, Switzerland, in January, 2020, after his 'harsh criticism' of 'authoritarian rulers' around the world. You can only laugh at such brazen mendacity. How *he* doesn't laugh is the mystery. Translated from the Orwellian 'liberal values and tackle intolerance' means teaching non-white people to hate white people and for white people to loathe themselves for being born white. The reason for that will become clear.

The 'Anti-Semitism' fraud

Zionists support the Jewish homeland in the land of Palestine which has been the Sabbatian-Rothschild goal for so long, but not for the benefit of Jews. Sabbatians and their global Anti-Semitism Industry have skewed public and political opinion to equate opposing the violent extremes of Zionism to be a blanket attack and condemnation of all Jewish people. Sabbatians and their global Anti-Semitism Industry have skewed public and political opinion to equate opposing the violent extremes of Zionism to be a blanket attack and condemnation of all Jewish people. This is nothing more than a Sabbatian protection racket to stop legitimate investigation and exposure of their agendas and activities. The official definition of 'anti-Semitism' has more recently been expanded to include criticism of Zionism – a *political movement* – and this was done to further stop exposure of Sabbatian infiltrators who created Zionism as we know it today in the 19th century. Renegade Minds will talk about these subjects when they know the shit that will come their way. People must decide if they want to know the truth or just cower in the corner in fear of what others will say. Sabbatians have been trying to label me as 'anti-Semitic' since the 1990s as I have uncovered more and more about their background and agendas. Useless, gutless, fraudulent 'journalists' then just repeat the smears without question and on the day I was writing this section a pair of unquestioning repeaters called Ben Quinn and Archie Bland (how appropriate) outright called me an 'anti-Semite' in the establishment propaganda sheet, the London *Guardian*, with no supporting evidence. The

Sabbatian Anti-Semitism Industry said so and who are they to question that? They wouldn't dare. Ironically 'Semitic' refers to a group of languages in the Middle East that are almost entirely Arabic. 'Anti-Semitism' becomes 'anti-Arab' which if the consequences of this misunderstanding were not so grave would be hilarious. Don't bother telling Quinn and Bland. I don't want to confuse them, bless 'em. One reason I am dubbed 'anti-Semitic' is that I wrote in the 1990s that Jewish operatives (Sabbatians) were heavily involved in the Russian Revolution when Sabbatians overthrew the Romanov dynasty. This apparently made me 'anti-Semitic'. Oh, really? Here is a section from *The Trigger*:

British journalist Robert Wilton confirmed these themes in his 1920 book *The Last Days of the Romanovs* when he studied official documents from the Russian government to identify the members of the Bolshevik ruling elite between 1917 and 1919. The Central Committee included 41 Jews among 62 members; the Council of the People's Commissars had 17 Jews out of 22 members; and 458 of the 556 most important Bolshevik positions between 1918 and 1919 were occupied by Jewish people. Only 17 were Russian. Then there were the 23 Jews among the 36 members of the vicious Cheka Soviet secret police established in 1917 who would soon appear all across the country.

Professor Robert Service of Oxford University, an expert on 20th century Russian history, found evidence that ['Jewish'] Leon Trotsky had sought to make sure that Jews were enrolled in the Red Army and were disproportionately represented in the Soviet civil bureaucracy that included the Cheka which performed mass arrests, imprisonment and executions of 'enemies of the people'. A US State Department Decimal File (861.00/5339) dated November 13th, 1918, names [Rothschild banking agent in America] Jacob Schiff and a list of ultra-Zionists as funders of the Russian Revolution leading to claims of a 'Jewish plot', but the key point missed by all is they were not 'Jews' – they were Sabbatian-Frankists.

Britain's Winston Churchill made the same error by mistake or otherwise. He wrote in a 1920 edition of the *Illustrated Sunday Herald* that those behind the Russian revolution were part of a 'worldwide conspiracy for the overthrow of civilisation and for the reconstitution of society on the basis of arrested development, of envious malevolence, and impossible equality' (see 'Woke' today because that has been created by the same network). Churchill said there was no need to exaggerate the part played in the creation of Bolshevism and in the actual bringing about of the Russian

Revolution 'by these international and for the most part atheistical Jews' ['atheistical Jews' = Sabbatians]. Churchill said it is certainly a very great one and probably outweighs all others: 'With the notable exception of Lenin, the majority of the leading figures are Jews.' He went on to describe, knowingly or not, the Sabbatian modus operandi of placing puppet leaders nominally in power while they control from the background:

Moreover, the principal inspiration and driving power comes from the Jewish leaders. Thus Tchitcherin, a pure Russian, is eclipsed by his nominal subordinate, Litvinoff, and the influence of Russians like Bukharin or Lunacharski cannot be compared with the power of Trotsky, or of Zinovieff, the Dictator of the Red Citadel (Petrograd), or of Krassin or Radek – all Jews. In the Soviet institutions the predominance of Jews is even more astonishing. And the prominent, if not indeed the principal, part in the system of terrorism applied by the Extraordinary Commissions for Combatting Counter-Revolution has been taken by Jews, and in some notable cases by Jewesses.

What I said about seriously disproportionate involvement in the Russian Revolution by Jewish 'revolutionaries' (Sabbatians) is provable fact, but truth is no defence against the Sabbatian Anti-Semitism Industry, its repeater parrots like Quinn and Bland, and the now breathtaking network of so-called 'Woke' 'anti-hate' groups with interlocking leaderships and funding which have the role of discrediting and silencing anyone who gets too close to exposing the Sabbatians. We have seen 'truth is no defence' confirmed in legal judgements with the Saskatchewan Human Rights Commission in Canada decreeing this: 'Truthful statements can be presented in a manner that would meet the definition of hate speech, and not all truthful statements must be free from restriction.' Most 'anti-hate' activists, who are themselves consumed by hatred, are too stupid and ignorant of the world to know how they are being used. They are far too far up their own virtue-signalling arses and it's far too dark for them to see anything.

The 'revolution' game

The background and methods of the 'Russian' Revolution are straight from the Sabbatian playbook seen in the French Revolution

and endless others around the world that appear to start as a revolution of the people against tyrannical rule and end up with a regime change to more tyrannical rule overtly or covertly. Wars, terror attacks and regime overthrows follow the Sabbatian cult through history with its agents creating them as Problem-Reaction-Solutions to remove opposition on the road to world domination. Sabbatian dots connect the Rothschilds with the Illuminati, Jacobins of the French Revolution, the 'Bund' or League of the Just, the International Communist Party, Communist League and the Communist Manifesto of Karl Marx and Friedrich Engels that would lead to the Rothschild-funded Russian Revolution. The sequence comes under the heading of 'creative destruction' when you advance to your global goal by continually destroying the status quo to install a new status quo which you then also destroy. The two world wars come to mind. With each new status quo you move closer to your planned outcome. Wars and mass murder are to Sabbatians a collective blood sacrifice ritual. They are obsessed with death for many reasons and one is that death is an inversion of life. Satanists and Sabbatians are obsessed with death and often target churches and churchyards for their rituals. Inversion-obsessed Sabbatians explain the use of inverted symbolism including the *inverted* pentagram and *inverted* cross. The inversion of the cross has been related to targeting Christianity, but the cross was a religious symbol long before Christianity and its inversion is a statement about the Sabbatian mentality and goals more than any single religion.

Sabbatians operating in Germany were behind the rise of the occult-obsessed Nazis and the subsequent Jewish exodus from Germany and Europe to Palestine and the United States after World War Two. The Rothschild dynasty was at the forefront of this both as political manipulators and by funding the operation. Why would Sabbatians help to orchestrate the horrors inflicted on Jews by the Nazis and by Stalin after they organised the Russian Revolution? Sabbatians hate Jews and their religion, that's why. They pose as Jews and secure positions of control within Jewish society and play the 'anti-Semitism' card to protect themselves from exposure

through a global network of organisations answering to the Sabbatian-created-and-controlled globe-spanning intelligence network that involves a stunning web of military-intelligence operatives and operations for a tiny country of just nine million. Among them are Jewish assets who are not Sabbatians but have been convinced by them that what they are doing is for the good of Israel and the Jewish community to protect them from what they have been programmed since childhood to believe is a Jew-hating hostile world. The Jewish community is just a highly convenient cover to hide the true nature of Sabbatians. Anyone getting close to exposing their game is accused by Sabbatian place-people and gofers of 'anti-Semitism' and claiming that all Jews are part of a plot to take over the world. I am not saying that. I am saying that Sabbatians – the *real* Jew-haters – have infiltrated the Jewish community to use them both as a cover and an 'anti-Semitic' defence against exposure. Thus we have the Anti-Semitism Industry targeted researchers in this way and most Jewish people think this is justified and genuine. They don't know that their 'Jewish' leaders and institutions of state, intelligence and military are not controlled by Jews at all, but cultists and stooges of Sabbatian-Frankism. I once added my name to a pro-Jewish freedom petition online and the next time I looked my name was gone and text had been added to the petition blurb to attack me as an 'anti-Semite' such is the scale of perceptual programming.

Moving on America

I tell the story in *The Trigger* and a chapter called 'Atlantic Crossing' how particularly after Israel was established the Sabbatians moved in on the United States and eventually grasped control of government administration, the political system via both Democrats and Republicans, the intelligence community like the CIA and National Security Agency (NSA), the Pentagon and mass media. Through this seriously compartmentalised network Sabbatians and their operatives in Mossad, Israeli Defense Forces (IDF) and US agencies pulled off 9/11 and blamed it on 19 'Al-Qaeda hijackers' dominated by men from, or connected to, Sabbatian-ruled Saudi

Arabia. The '19' were not even on the planes let alone flew those big passenger jets into buildings while being largely incompetent at piloting one-engine light aircraft. 'Hijacker' Hani Hanjour who is said to have flown American Airlines Flight 77 into the Pentagon with a turn and manoeuvre most professional pilots said they would have struggled to do was banned from renting a small plane by instructors at the Freeway Airport in Bowie, Maryland, just *six weeks* earlier on the grounds that he was an incompetent pilot. The Jewish population of the world is just 0.2 percent with even that almost entirely concentrated in Israel (75 percent Jewish) and the United States (around two percent). This two percent and globally 0.2 percent refers to *Jewish* people and not Sabbatian interlopers who are a fraction of that fraction. What a sobering thought when you think of the fantastic influence on world affairs of tiny Israel and that the Project for the New America Century (PNAC) which laid out the blueprint in September, 2000, for America's war on terror and regime change wars in Iraq, Libya and Syria was founded and dominated by Sabbatians known as 'Neocons'. The document conceded that this plan would not be supported politically or publicly without a major attack on American soil and a Problem-Reaction-Solution excuse to send troops to war across the Middle East. Sabbatian Neocons said:

... [The] process of transformation ... [war and regime change] ... is likely to be a long one, absent some catastrophic and catalysing event – like a new Pearl Harbor.

Four months later many of those who produced that document came to power with their inane puppet George Bush from the long-time Sabbatian Bush family. They included Sabbatian Dick Cheney who was officially vice-president, but really de-facto president for the entirety of the 'Bush' government. Nine months after the 'Bush' inauguration came what Bush called at the time 'the Pearl Harbor of the 21st century' and with typical Sabbatian timing and symbolism 2001 was the 60th anniversary of the attack in 1941 by the Japanese Air Force on Pearl Harbor, Hawaii, which allowed President Franklin Delano Roosevelt to take the United States into a Sabbatian-

instigated Second World War that he said in his election campaign that he never would. The evidence is overwhelming that Roosevelt and his military and intelligence networks knew the attack was coming and did nothing to stop it, but they did make sure that America's most essential naval ships were not in Hawaii at the time. Three thousand Americans died in the Pearl Harbor attacks as they did on September 11th. By the 9/11 year of 2001 Sabbatians had widely infiltrated the US government, military and intelligence operations and used their compartmentalised assets to pull off the 'Al-Qaeda' attacks. If you read *The Trigger* it will blow your mind to see the utterly staggering concentration of 'Jewish' operatives (Sabbatian infiltrators) in essential positions of political, security, legal, law enforcement, financial and business power before, during, and after the attacks to make them happen, carry them out, and then cover their tracks – and I do mean *staggering* when you think of that 0.2 percent of the world population and two percent of Americans which are Jewish while Sabbatian infiltrators are a fraction of that. A central foundation of the 9/11 conspiracy was the hijacking of government, military, Air Force and intelligence computer systems in real time through 'back-door' access made possible by Israeli (Sabbatian) 'cyber security' software. Sabbatian-controlled Israel is on the way to rivalling Silicon Valley for domination of cyberspace and is becoming the dominant force in cyber-security which gives them access to entire computer systems and their passcodes across the world. Then add to this that Zionists head (officially) Silicon Valley giants like Google (Larry Page and Sergey Brin), Google-owned YouTube (Susan Wojcicki), Facebook (Mark Zuckerberg and Sheryl Sandberg), and Apple (Chairman Arthur D. Levinson), and that ultra-Zionist hedge fund billionaire Paul Singer has a \$1 billion stake in Twitter which is only nominally headed by 'CEO' pothead Jack Dorsey. As cable news host Tucker Carlson said of Dorsey: 'There used to be debate in the medical community whether dropping a ton of acid had permanent effects and I think that debate has now ended.' Carlson made the comment after Dorsey told a hearing on Capitol Hill (if you cut through his bullshit) that he

believed in free speech so long as he got to decide what you can hear and see. These 'big names' of Silicon Valley are only front men and women for the Global Cult, not least the Sabbatians, who are the true controllers of these corporations. Does anyone still wonder why these same people and companies have been ferociously censoring and banning people (like me) for exposing any aspect of the Cult agenda and especially the truth about the 'Covid' hoax which Sabbatians have orchestrated?

The Jeffrey Epstein paedophile ring was a Sabbatian operation. He was officially 'Jewish' but he was a Sabbatian and women abused by the ring have told me about the high number of 'Jewish' people involved. The Epstein horror has Sabbatian written all over it and matches perfectly their *modus operandi* and obsession with sex and ritual. Epstein was running a Sabbatian blackmail ring in which famous people with political and other influence were provided with young girls for sex while everything was being filmed and recorded on hidden cameras and microphones at his New York house, Caribbean island and other properties. Epstein survivors have described this surveillance system to me and some have gone public. Once the famous politician or other figure knew he or she was on video they tended to do whatever they were told. Here we go again ...when you've got them by the balls their hearts and minds will follow. Sabbatians use this blackmail technique on a wide scale across the world to entrap politicians and others they need to act as demanded. Epstein's private plane, the infamous 'Lolita Express', had many well-known passengers including Bill Clinton while Bill Gates has flown on an Epstein plane and met with him four years after Epstein had been jailed for paedophilia. They subsequently met many times at Epstein's home in New York according to a witness who was there. Epstein's infamous side-kick was Ghislaine Maxwell, daughter of Mossad agent and ultra-Zionist mega-crooked British businessman, Bob Maxwell, who at one time owned the *Daily Mirror* newspaper. Maxwell was murdered at sea on his boat in 1991 by Sabbatian-controlled Mossad when he became a liability with his

business empire collapsing as a former Mossad operative has confirmed (see *The Trigger*).

Money, money, money, funny money ...

Before I come to the Sabbatian connection with the last three US presidents I will lay out the crucial importance to Sabbatians of controlling banking and finance. Sabbatian Mayer Amschel Rothschild set out to dominate this arena in his family's quest for total global control. What is freedom? It is, in effect, choice. The more choices you have the freer you are and the fewer your choices the more you are enslaved. In the global structure created over centuries by Sabbatians the biggest decider and restrictor of choice is ... money. Across the world if you ask people what they would like to do with their lives and why they are not doing that they will reply 'I don't have the money'. This is the idea. A global elite of multi-billionaires are described as 'greedy' and that is true on one level; but control of money – who has it and who doesn't – is not primarily about greed. It's about control. Sabbatians have seized ever more control of finance and sucked the wealth of the world out of the hands of the population. We talk now, after all, about the 'One-percent' and even then the wealthiest are a lot fewer even than that. This has been made possible by a money scam so outrageous and so vast it could rightly be called the scam of scams founded on creating 'money' out of nothing and 'loaning' that with interest to the population. Money out of nothing is called 'credit'. Sabbatians have asserted control over governments and banking ever more completely through the centuries and secured financial laws that allow banks to lend hugely more than they have on deposit in a confidence trick known as fractional reserve lending. Imagine if you could lend money that doesn't exist and charge the recipient interest for doing so. You would end up in jail. Bankers by contrast end up in mansions, private jets, Malibu and Monaco.

Banks are only required to keep a fraction of their deposits and wealth in their vaults and they are allowed to lend 'money' they don't have called 'credit'. Go into a bank for a loan and if you succeed

the banker will not move any real wealth into your account. They will type into your account the amount of the agreed 'loan' – say £100,000. This is not wealth that really exists; it is non-existent, fresh-air, created-out-of-nothing 'credit' which has never, does not, and will never exist except in theory. Credit is backed by nothing except wind and only has buying power because people think that it has buying power and accept it in return for property, goods and services. I have described this situation as like those cartoon characters you see chasing each other and when they run over the edge of a cliff they keep running forward on fresh air until one of them looks down, realises what's happened, and they all crash into the ravine. The whole foundation of the Sabbatian financial system is to stop people looking down except for periodic moments when they want to crash the system (as in 2008 and 2020 ongoing) and reap the rewards from all the property, businesses and wealth their borrowers had signed over as 'collateral' in return for a 'loan' of fresh air. Most people think that money is somehow created by governments when it comes into existence from the start as a debt through banks 'lending' illusory money called credit. Yes, the very currency of exchange is a *debt* from day one issued as an interest-bearing loan. Why don't governments create money interest-free and lend it to their people interest-free? Governments are controlled by Sabbatians and the financial system is controlled by Sabbatians for whom interest-free money would be a nightmare come true. Sabbatians underpin their financial domination through their global network of central banks, including the privately-owned US Federal Reserve and Britain's Bank of England, and this is orchestrated by a privately-owned central bank coordination body called the Bank for International Settlements in Basle, Switzerland, created by the usual suspects including the Rockefellers and Rothschilds. Central bank chiefs don't answer to governments or the people. They answer to the Bank for International Settlements or, in other words, the Global Cult which is dominated today by Sabbatians.

Built-in disaster

There are so many constituent scams within the overall banking scam. When you take out a loan of thin-air credit only the amount of that loan is theoretically brought into circulation to add to the amount in circulation; but you are paying back the principle plus interest. The additional interest is not created and this means that with every 'loan' there is a shortfall in the money in circulation between what is borrowed and what has to be paid back. There is never even close to enough money in circulation to repay all outstanding public and private debt including interest. Coldly weaved in the very fabric of the system is the certainty that some will lose their homes, businesses and possessions to the banking 'lender'. This is less obvious in times of 'boom' when the amount of money in circulation (and the debt) is expanding through more people wanting and getting loans. When a downturn comes and the money supply contracts it becomes painfully obvious that there is not enough money to service all debt and interest. This is less obvious in times of 'boom' when the amount of money in circulation (and the debt) is expanding through more people wanting and getting loans. When a downturn comes and the money supply contracts and it becomes painfully obvious – as in 2008 and currently – that there is not enough money to service all debt and interest. Sabbatian banksters have been leading the human population through a calculated series of booms (more debt incurred) and busts (when the debt can't be repaid and the banks get the debtor's tangible wealth in exchange for non-existent 'credit'). With each 'bust' Sabbatian bankers have absorbed more of the world's tangible wealth and we end up with the One-percent. Governments are in bankruptcy levels of debt to the same system and are therefore owned by a system they do not control. The Federal Reserve, 'America's central bank', is privately-owned and American presidents only nominally appoint its chairman or woman to maintain the illusion that it's an arm of government. It's not. The 'Fed' is a cartel of private banks which handed billions to its associates and friends after the crash of 2008 and has been Sabbatian-controlled since it was manipulated into being in 1913 through the covert trickery of Rothschild banking agents Jacob Schiff and Paul

Warburg, and the Sabbatian Rockefeller family. Somehow from a Jewish population of two-percent and globally 0.2 percent (Sabbatian interlopers remember are far smaller) ultra-Zionists headed the Federal Reserve for 31 years between 1987 and 2018 in the form of Alan Greenspan, Bernard Bernanke and Janet Yellen (now Biden's Treasury Secretary) with Yellen's deputy chairman a Israeli-American dual citizen and ultra-Zionist Stanley Fischer, a former governor of the Bank of Israel. Ultra-Zionist Fed chiefs spanned the presidencies of Ronald Reagan ('Republican'), Father George Bush ('Republican'), Bill Clinton ('Democrat'), Boy George Bush ('Republican') and Barack Obama ('Democrat'). We should really add the pre-Greenspan chairman, Paul Adolph Volcker, 'appointed' by Jimmy Carter ('Democrat') who ran the Fed between 1979 and 1987 during the Carter and Reagan administrations before Greenspan took over. Volcker was a long-time associate and business partner of the Rothschilds. No matter what the 'party' officially in power the United States economy was directed by the same force. Here are members of the Obama, Trump and Biden administrations and see if you can make out a common theme.

Barack Obama ('Democrat')

Ultra-Zionists Robert Rubin, Larry Summers, and Timothy Geithner ran the US Treasury in the Clinton administration and two of them reappeared with Obama. Ultra-Zionist Fed chairman Alan Greenspan had manipulated the crash of 2008 through deregulation and jumped ship just before the disaster to make way for ultra-Zionist Bernard Bernanke to hand out trillions to Sabbatian 'too big to fail' banks and businesses, including the ubiquitous ultra-Zionist Goldman Sachs which has an ongoing staff revolving door operation between itself and major financial positions in government worldwide. Obama inherited the fallout of the crash when he took office in January, 2009, and fortunately he had the support of his ultra-Zionist White House Chief of Staff Rahm Emmanuel, son of a terrorist who helped to bomb Israel into being in 1948, and his ultra-Zionist senior adviser David Axelrod, chief strategist in Obama's two

successful presidential campaigns. Emmanuel, later mayor of Chicago and former senior fundraiser and strategist for Bill Clinton, is an example of the Sabbatian policy after Israel was established of migrating insider families to America so their children would be born American citizens. 'Obama' chose this financial team throughout his administration to respond to the Sabbatian-instigated crisis:

Timothy Geithner (ultra-Zionist) Treasury Secretary; Jacob J. Lew, Treasury Secretary; Larry Summers (ultra-Zionist), director of the White House National Economic Council; Paul Adolph Volcker (Rothschild business partner), chairman of the Economic Recovery Advisory Board; Peter Orszag (ultra-Zionist), director of the Office of Management and Budget overseeing all government spending; Penny Pritzker (ultra-Zionist), Commerce Secretary; Jared Bernstein (ultra-Zionist), chief economist and economic policy adviser to Vice President Joe Biden; Mary Schapiro (ultra-Zionist), chair of the Securities and Exchange Commission (SEC); Gary Gensler (ultra-Zionist), chairman of the Commodity Futures Trading Commission (CFTC); Sheila Bair (ultra-Zionist), chair of the Federal Deposit Insurance Corporation (FDIC); Karen Mills (ultra-Zionist), head of the Small Business Administration (SBA); Kenneth Feinberg (ultra-Zionist), Special Master for Executive [bail-out] Compensation. Feinberg would be appointed to oversee compensation (with strings) to 9/11 victims and families in a campaign to stop them having their day in court to question the official story. At the same time ultra-Zionist Bernard Bernanke was chairman of the Federal Reserve and these are only some of the ultra-Zionists with allegiance to Sabbatian-controlled Israel in the Obama government. Obama's biggest corporate donor was ultra-Zionist Goldman Sachs which had employed many in his administration.

Donald Trump ('Republican')

Trump claimed to be an outsider (he wasn't) who had come to 'drain the swamp'. He embarked on this goal by immediately appointing ultra-Zionist Steve Mnuchin, a Goldman Sachs employee for 17

years, as his Treasury Secretary. Others included Gary Cohn (ultra-Zionist), chief operating officer of Goldman Sachs, his first Director of the National Economic Council and chief economic adviser, who was later replaced by Larry Kudlow (ultra-Zionist). Trump's senior adviser throughout his four years in the White House was his sinister son-in-law Jared Kushner, a life-long friend of Israel Prime Minister Benjamin Netanyahu. Kushner is the son of a convicted crook who was pardoned by Trump in his last days in office. Other ultra-Zionists in the Trump administration included: Stephen Miller, Senior Policy Adviser; Avrahm Berkowitz, Deputy Adviser to Trump and his Senior Adviser Jared Kushner; Ivanka Trump, Adviser to the President, who converted to Judaism when she married Jared Kushner; David Friedman, Trump lawyer and Ambassador to Israel; Jason Greenblatt, Trump Organization executive vice president and chief legal officer, who was made Special Representative for International Negotiations and the Israeli-Palestinian Conflict; Rod Rosenstein, Deputy Attorney General; Elliot Abrams, Special Representative for Venezuela, then Iran; John Eisenberg, National Security Council Legal Adviser and Deputy Council to the President for National Security Affairs; Anne Neuberger, Deputy National Manager, National Security Agency; Ezra Cohen-Watnick, Acting Under Secretary of Defense for Intelligence; Elan Carr, Special Envoy to monitor and combat anti-Semitism; Len Khodorkovsky, Deputy Special Envoy to monitor and combat anti-Semitism; Reed Cordish, Assistant to the President, Intragovernmental and Technology Initiatives. Trump Vice President Mike Pence and Secretary of State Mike Pompeo, both Christian Zionists, were also vehement supporters of Israel and its goals and ambitions.

Donald 'free-speech believer' Trump pardoned a number of financial and violent criminals while ignoring calls to pardon Julian Assange and Edward Snowden whose crimes are revealing highly relevant information about government manipulation and corruption and the widespread illegal surveillance of the American people by US 'security' agencies. It's so good to know that Trump is on the side of freedom and justice and not mega-criminals with

allegiance to Sabbatian-controlled Israel. These included a pardon for Israeli spy Jonathan Pollard who was jailed for life in 1987 under the Espionage Act. Aviem Sella, the Mossad agent who recruited Pollard, was also pardoned by Trump while Assange sat in jail and Snowden remained in exile in Russia. Sella had 'fled' (was helped to escape) to Israel in 1987 and was never extradited despite being charged under the Espionage Act. A Trump White House statement said that Sella's clemency had been 'supported by Benjamin Netanyahu, Ron Dermer, Israel's US Ambassador, David Friedman, US Ambassador to Israel and Miriam Adelson, wife of leading Trump donor Sheldon Adelson who died shortly before. Other friends of Jared Kushner were pardoned along with Sholom Weiss who was believed to be serving the longest-ever white-collar prison sentence of more than 800 years in 2000. The sentence was commuted of Ponzi-schemer Eliyahu Weinstein who defrauded Jews and others out of \$200 million. I did mention that Assange and Snowden were ignored, right? Trump gave Sabbatians almost everything they asked for in military and political support, moving the US Embassy from Tel Aviv to Jerusalem with its critical symbolic and literal implications for Palestinian statehood, and the 'deal of the Century' designed by Jared Kushner and David Friedman which gave the Sabbatian Israeli government the green light to substantially expand its already widespread program of building illegal Jewish-only settlements in the occupied land of the West Bank. This made a two-state 'solution' impossible by seizing all the land of a potential Palestinian homeland and that had been the plan since 1948 and then 1967 when the Arab-controlled Gaza Strip, West Bank, Sinai Peninsula and Syrian Golan Heights were occupied by Israel. All the talks about talks and road maps and delays have been buying time until the West Bank was physically occupied by Israeli real estate. Trump would have to be a monumentally ill-informed idiot not to see that this was the plan he was helping to complete. The Trump administration was in so many ways the Kushner administration which means the Netanyahu administration which means the Sabbatian administration. I understand why many opposing Cult fascism in all its forms gravitated to Trump, but he

was a crucial part of the Sabbatian plan and I will deal with this in the next chapter.

Joe Biden ('Democrat')

A barely cognitive Joe Biden took over the presidency in January, 2021, along with his fellow empty shell, Vice-President Kamala Harris, as the latest Sabbatian gofers to enter the White House. Names on the door may have changed and the 'party' – the force behind them remained the same as Zionists were appointed to a stream of pivotal areas relating to Sabbatian plans and policy. They included: Janet Yellen, Treasury Secretary, former head of the Federal Reserve, and still another ultra-Zionist running the US Treasury after Mnuchin (Trump), Lew and Geithner (Obama), and Summers and Rubin (Clinton); Anthony Blinken, Secretary of State; Wendy Sherman, Deputy Secretary of State (so that's 'Biden's' Sabbatian foreign policy sorted); Jeff Zients, White House coronavirus coordinator; Rochelle Walensky, head of the Centers for Disease Control; Rachel Levine, transgender deputy health secretary (that's 'Covid' hoax policy under control); Merrick Garland, Attorney General; Alejandro Mayorkas, Secretary of Homeland Security; Cass Sunstein, Homeland Security with responsibility for new immigration laws; Avril Haines, Director of National Intelligence; Anne Neuberger, National Security Agency cybersecurity director (note, cybersecurity); David Cohen, CIA Deputy Director; Ronald Klain, Biden's Chief of Staff (see Rahm Emanuel); Eric Lander, a 'leading geneticist', Office of Science and Technology Policy director (see Smart Grid, synthetic biology agenda); Jessica Rosenworcel, acting head of the Federal Communications Commission (FCC) which controls Smart Grid technology policy and electromagnetic communication systems including 5G. How can it be that so many pivotal positions are held by two-percent of the American population and 0.2 percent of the world population administration after administration no matter who is the president and what is the party? It's a coincidence? Of course it's not and this is why Sabbatians have built their colossal global web of interlocking 'anti-

hate' hate groups to condemn anyone who asks these glaring questions as an 'anti-Semite'. The way that Jewish people horrifically abused in Sabbatian-backed Nazi Germany are exploited to this end is stomach-turning and disgusting beyond words.

Political fusion

Sabbatian manipulation has reversed the roles of Republicans and Democrats and the same has happened in Britain with the Conservative and Labour Parties. Republicans and Conservatives were always labelled the 'right' and Democrats and Labour the 'left', but look at the policy positions now and the Democrat-Labour 'left' has moved further to the 'right' than Republicans and Conservatives under the banner of 'Woke', the Cult-created far-right tyranny. Where once the Democrat-Labour 'left' defended free speech and human rights they now seek to delete them and as I said earlier despite the 'Covid' fascism of the Jackboot Johnson Conservative government in the UK the Labour Party of leader Keir Starmer demanded even more extreme measures. The Labour Party has been very publicly absorbed by Sabbatians after a political and media onslaught against the previous leader, the weak and inept Jeremy Corbyn, over made-up allegations of 'anti-Semitism' both by him and his party. The plan was clear with this 'anti-Semite' propaganda and what was required in response was a swift and decisive 'fuck off' from Corbyn and a statement to expose the Anti-Semitism Industry (Sabbatian) attempt to silence Labour criticism of the Israeli government (Sabbatians) and purge the party of all dissent against the extremes of ultra-Zionism (Sabbatians). Instead Corbyn and his party fell to their knees and appeased the abusers which, by definition, is impossible. Appeasing one demand leads only to a new demand to be appeased until takeover is complete. Like I say – 'fuck off' would have been a much more effective policy and I have used it myself with great effect over the years when Sabbatians are on my case which is most of the time. I consider that fact a great compliment, by the way. The outcome of the Labour Party capitulation is that we now have a Sabbatian-controlled

Conservative Party 'opposed' by a Sabbatian-controlled Labour Party in a one-party Sabbatian state that hurtles towards the extremes of tyranny (the Sabbatian cult agenda). In America the situation is the same. Labour's Keir Starmer spends his days on his knees with his tongue out pointing to Tel Aviv, or I guess now Jerusalem, while Boris Johnson has an 'anti-Semitism czar' in the form of former Labour MP John Mann who keeps Starmer company on his prayer mat.

Sabbatian influence can be seen in Jewish members of the Labour Party who have been ejected for criticism of Israel including those from families that suffered in Nazi Germany. Sabbatians despise real Jewish people and target them even more harshly because it is so much more difficult to dub them 'anti-Semitic' although in their desperation they do try.

CHAPTER THREE

The Pushbacker sting

*Until you realize how easy it is for your mind to be manipulated, you
remain the puppet of someone else's game*

Evita Ochel

I will use the presidencies of Trump and Biden to show how the manipulation of the one-party state plays out behind the illusion of political choice across the world. No two presidencies could – on the face of it – be more different and apparently at odds in terms of direction and policy.

A Renegade Mind sees beyond the obvious and focuses on outcomes and consequences and not image, words and waffle. The Cult embarked on a campaign to divide America between those who blindly support its agenda (the mentality known as 'Woke') and those who are pushing back on where the Cult and its Sabbatians want to go. This presents infinite possibilities for dividing and ruling the population by setting them at war with each other and allows a perceptual ring fence of demonisation to encircle the Pushbackers in a modern version of the Little Big Horn in 1876 when American cavalry led by Lieutenant Colonel George Custer were drawn into a trap, surrounded and killed by Native American tribes defending their land of thousands of years from being seized by the government. In this modern version the roles are reversed and it's those defending themselves from the Sabbatian government who are surrounded and the government that's seeking to destroy them. This trap was set years ago and to explain how we must return to 2016

and the emergence of Donald Trump as a candidate to be President of the United States. He set out to overcome the best part of 20 other candidates in the Republican Party before and during the primaries and was not considered by many in those early stages to have a prayer of living in the White House. The Republican Party was said to have great reservations about Trump and yet somehow he won the nomination. When you know how American politics works – politics in general – there is no way that Trump could have become the party's candidate unless the Sabbatian-controlled 'Neocons' that run the Republican Party wanted that to happen. We saw the proof in emails and documents made public by WikiLeaks that the Democratic Party hierarchy, or Democons, systematically undermined the campaign of Bernie Sanders to make sure that Sabbatian gofer Hillary Clinton won the nomination to be their presidential candidate. If the Democons could do that then the Neocons in the Republican Party could have derailed Trump in the same way. But they didn't and at that stage I began to conclude that Trump could well be the one chosen to be president. If that was the case the 'why' was pretty clear to see – the goal of dividing America between Cult agenda-supporting Wokers and Pushbackers who gravitated to Trump because he was telling them what they wanted to hear. His constituency of support had been increasingly ignored and voiceless for decades and profoundly through the eight years of Sabbatian puppet Barack Obama. Now here was someone speaking their language of pulling back from the incessant globalisation of political and economic power, the exporting of American jobs to China and elsewhere by 'American' (Sabbatian) corporations, the deletion of free speech, and the mass immigration policies that had further devastated job opportunities for the urban working class of all races and the once American heartlands of the Midwest.

Beware the forked tongue

Those people collectively sighed with relief that at last a political leader was apparently on their side, but another trait of the Renegade Mind is that you look even harder at people telling you

what you want to hear than those who are telling you otherwise. Obviously as I said earlier people wish what they want to hear to be true and genuine and they are much more likely to believe that than someone saying what they don't want to hear and don't want to be true. Sales people are taught to be skilled in eliciting by calculated questioning what their customers want to hear and repeating that back to them as their own opinion to get their targets to like and trust them. Assets of the Cult are also sales people in the sense of selling perception. To read Cult manipulation you have to play the long and expanded game and not fall for the Vaudeville show of party politics. Both American parties are vehicles for the Cult and they exploit them in different ways depending on what the agenda requires at that moment. Trump and the Republicans were used to be the focus of dividing America and isolating Pushbackers to open the way for a Biden presidency to become the most extreme in American history by advancing the full-blown Woke (Cult) agenda with the aim of destroying and silencing Pushbackers now labelled Nazi Trump supporters and white supremacists.

Sabbatians wanted Trump in office for the reasons described by ultra-Zionist Saul Alinsky (1909-1972) who was promoting the Woke philosophy through 'community organising' long before anyone had heard of it. In those days it still went by its traditional name of Marxism. The reason for the manipulated Trump phenomenon was laid out in Alinsky's 1971 book, *Rules for Radicals*, which was his blueprint for overthrowing democratic and other regimes and replacing them with Sabbatian Marxism. Not surprisingly his to-do list was evident in the Sabbatian French and Russian 'Revolutions' and that in China which will become very relevant in the next chapter about the 'Covid' hoax. Among Alinsky's followers have been the deeply corrupt Barack Obama, House Speaker Nancy Pelosi and Hillary Clinton who described him as a 'hero'. All three are Sabbatian stooges with Pelosi personifying the arrogant corrupt idiocy that so widely fronts up for the Cult inner core. Predictably as a Sabbatian advocate of the 'light-bringer' Alinsky features Lucifer on the dedication page of his book as the original radical who gained

his own kingdom ('Earth' as we shall see). One of Alinsky's golden radical rules was to pick an individual and focus all attention, hatred and blame on them and not to target faceless bureaucracies and corporations. *Rules for Radicals* is really a Sabbatian handbook with its contents repeatedly employed all over the world for centuries and why wouldn't Sabbatians bring to power their designer-villain to be used as the individual on which all attention, hatred and blame was bestowed? This is what they did and the only question for me is how much Trump knew that and how much he was manipulated. A bit of both, I suspect. This was Alinsky's Trump technique from a man who died in 1972. The technique has spanned history:

Pick the target, freeze it, personalize it, polarize it. Don't try to attack abstract corporations or bureaucracies. Identify a responsible individual. Ignore attempts to shift or spread the blame.

From the moment Trump came to illusory power everything was about him. It wasn't about Republican policy or opinion, but all about Trump. Everything he did was presented in negative, derogatory and abusive terms by the Sabbatian-dominated media led by Cult operations such as CNN, MSNBC, *The New York Times* and the Jeff Bezos-owned *Washington Post* – 'Pick the target, freeze it, personalize it, polarize it.' Trump was turned into a demon to be vilified by those who hated him and a demi-god loved by those who worshipped him. This, in turn, had his supporters, too, presented as equally demonic in preparation for the punchline later down the line when Biden was about to take office. It was here's a Trump, there's a Trump, everywhere a Trump, Trump. Virtually every news story or happening was filtered through the lens of 'The Donald'. You loved him or hated him and which one you chose was said to define you as Satan's spawn or a paragon of virtue. Even supporting some Trump policies or statements and not others was enough for an assault on your character. No shades of grey were or are allowed. Everything is black and white (literally and figuratively). A Californian I knew had her head utterly scrambled by her hatred for Trump while telling people they should love each other. She was so totally consumed by

Trump Derangement Syndrome as it became to be known that this glaring contradiction would never have occurred to her. By definition anyone who criticised Trump or praised his opponents was a hero and this lady described Joe Biden as 'a kind, honest gentleman' when he's a provable liar, mega-crook and vicious piece of work to boot. Sabbatians had indeed divided America using Trump as the fall-guy and all along the clock was ticking on the consequences for his supporters.

In hock to his masters

Trump gave Sabbatians via Israel almost everything they wanted in his four years. Ask and you shall receive was the dynamic between himself and Benjamin Netanyahu orchestrated by Trump's ultra-Zionist son-in-law Jared Kushner, his ultra-Zionist Ambassador to Israel, David Friedman, and ultra-Zionist 'Israel adviser', Jason Greenblatt. The last two were central to the running and protecting from collapse of his business empire, the Trump Organisation, and colossal business failures made him forever beholding to Sabbatian networks that bailed him out. By the start of the 1990s Trump owed \$4 billion to banks that he couldn't pay and almost \$1 billion of that was down to him personally and not his companies. This mega-disaster was the result of building two new casinos in Atlantic City and buying the enormous Taj Mahal operation which led to crippling debt payments. He had borrowed fantastic sums from 72 banks with major Sabbatian connections and although the scale of debt should have had him living in a tent alongside the highway they never foreclosed. A plan was devised to lift Trump from the mire by BT Securities Corporation and Rothschild Inc. and the case was handled by Wilber Ross who had worked for the Rothschilds for 27 years. Ross would be named US Commerce Secretary after Trump's election. Another crucial figure in saving Trump was ultra-Zionist 'investor' Carl Icahn who bought the Taj Mahal casino. Icahn was made special economic adviser on financial regulation in the Trump administration. He didn't stay long but still managed to find time to make a tidy sum of a reported \$31.3 million when he sold his

holdings affected by the price of steel three days before Trump imposed a 235 percent tariff on steel imports. What amazing bits of luck these people have. Trump and Sabbatian operatives have long had a close association and his mentor and legal adviser from the early 1970s until 1986 was the dark and genetically corrupt ultra-Zionist Roy Cohn who was chief counsel to Senator Joseph McCarthy's 'communist' witch-hunt in the 1950s. *Esquire* magazine published an article about Cohn with the headline 'Don't mess with Roy Cohn'. He was described as the most feared lawyer in New York and 'a ruthless master of dirty tricks ... [with] ... more than one Mafia Don on speed dial'. Cohn's influence, contacts, support and protection made Trump a front man for Sabbatians in New York with their connections to one of Cohn's many criminal employers, the 'Russian' Sabbatian Mafia. Israel-centric media mogul Rupert Murdoch was introduced to Trump by Cohn and they started a long friendship. Cohn died in 1986 weeks after being disbarred for unethical conduct by the Appellate Division of the New York State Supreme Court. The wheels of justice do indeed run slow given the length of Cohn's crooked career.

QAnon-sense

We are asked to believe that Donald Trump with his fundamental connections to Sabbatian networks and operatives has been leading the fight to stop the Sabbatian agenda for the fascistic control of America and the world. Sure he has. A man entrapped during his years in the White House by Sabbatian operatives and whose biggest financial donor was casino billionaire Sheldon Adelson who was Sabbatian to his DNA?? Oh, do come on. Trump has been used to divide America and isolate Pushbackers on the Cult agenda under the heading of 'Trump supporters', 'insurrectionists' and 'white supremacists'. The US Intelligence/Mossad Psyop or psychological operation known as QAnon emerged during the Trump years as a central pillar in the Sabbatian campaign to lead Pushbackers into the trap set by those that wished to destroy them. I knew from the start that QAnon was a scam because I had seen the same scenario many

times before over 30 years under different names and I had written about one in particular in the books. 'Not again' was my reaction when QAnon came to the fore. The same script is pulled out every few years and a new name added to the letterhead. The story always takes the same form: 'Insiders' or 'the good guys' in the government-intelligence-military 'Deep State' apparatus were going to instigate mass arrests of the 'bad guys' which would include the Rockefellers, Rothschilds, Barack Obama, Hillary Clinton, George Soros, etc., etc. Dates are given for when the 'good guys' are going to move in, but the dates pass without incident and new dates are given which pass without incident. The central message to Pushbackers in each case is that they don't have to do anything because there is 'a plan' and it is all going to be sorted by the 'good guys' on the inside. 'Trust the plan' was a QAnon mantra when the only plan was to misdirect Pushbackers into putting their trust in a Psyop they believed to be real. Beware, beware, those who tell you what you want to hear and always check it out. Right up to Biden's inauguration QAnon was still claiming that 'the Storm' was coming and Trump would stay on as president when Biden and his cronies were arrested and jailed. It was never going to happen and of course it didn't, but what did happen as a result provided that punchline to the Sabbatian Trump/QAnon Psyop.

On January 6th, 2021, a very big crowd of Trump supporters gathered in the National Mall in Washington DC down from the Capitol Building to protest at what they believed to be widespread corruption and vote fraud that stopped Trump being re-elected for a second term as president in November, 2020. I say as someone that does not support Trump or Biden that the evidence is clear that major vote-fixing went on to favour Biden, a man with cognitive problems so advanced he can often hardly string a sentence together without reading the words written for him on the Teleprompter. Glaring ballot discrepancies included serious questions about electronic voting machines that make vote rigging a comparative cinch and hundreds of thousands of paper votes that suddenly appeared during already advanced vote counts and virtually all of

them for Biden. Early Trump leads in crucial swing states suddenly began to close and disappear. The pandemic hoax was used as the excuse to issue almost limitless numbers of mail-in ballots with no checks to establish that the recipients were still alive or lived at that address. They were sent to streams of people who had not even asked for them. Private organisations were employed to gather these ballots and who knows what they did with them before they turned up at the counts. The American election system has been manipulated over decades to become a sick joke with more holes than a Swiss cheese for the express purpose of dictating the results. Then there was the criminal manipulation of information by Sabbatian tech giants like Facebook, Twitter and Google-owned YouTube which deleted pro-Trump, anti-Biden accounts and posts while everything in support of Biden was left alone. Sabbatians wanted Biden to win because after the dividing of America it was time for full-on Woke and every aspect of the Cult agenda to be unleashed.

Hunter gatherer

Extreme Silicon Valley bias included blocking information by the *New York Post* exposing a Biden scandal that should have ended his bid for president in the final weeks of the campaign. Hunter Biden, his monumentally corrupt son, is reported to have sent a laptop to be repaired at a local store and failed to return for it. Time passed until the laptop became the property of the store for non-payment of the bill. When the owner saw what was on the hard drive he gave a copy to the FBI who did nothing even though it confirmed widespread corruption in which the Joe Biden family were using his political position, especially when he was vice president to Obama, to make multiple millions in countries around the world and most notably Ukraine and China. Hunter Biden's one-time business partner Tony Bobulinski went public when the story broke in the *New York Post* to confirm the corruption he saw and that Joe Biden not only knew what was going on he also profited from the spoils. Millions were handed over by a Chinese company with close

connections – like all major businesses in China – to the Chinese communist party of President Xi Jinping. Joe Biden even boasted at a meeting of the Cult's World Economic Forum that as vice president he had ordered the government of Ukraine to fire a prosecutor. What he didn't mention was that the same man just happened to be investigating an energy company which was part of Hunter Biden's corrupt portfolio. The company was paying him big bucks for no other reason than the influence his father had. Overnight Biden's presidential campaign should have been over given that he had lied publicly about not knowing what his son was doing. Instead almost the entire Sabbatian-owned mainstream media and Sabbatian-owned Silicon Valley suppressed circulation of the story. This alone went a mighty way to rigging the election of 2020. Cult assets like Mark Zuckerberg at Facebook also spent hundreds of millions to be used in support of Biden and vote 'administration'.

The Cult had used Trump as the focus to divide America and was now desperate to bring in moronic, pliable, corrupt Biden to complete the double-whammy. No way were they going to let little things like the will of the people thwart their plan. Silicon Valley widely censored claims that the election was rigged because it *was* rigged. For the same reason anyone claiming it was rigged was denounced as a 'white supremacist' including the pathetically few Republican politicians willing to say so. Right across the media where the claim was mentioned it was described as a 'false claim' even though these excuses for 'journalists' would have done no research into the subject whatsoever. Trump won seven million more votes than any sitting president had ever achieved while somehow a cognitively-challenged soon to be 78-year-old who was hidden away from the public for most of the campaign managed to win more votes than any presidential candidate in history. It makes no sense. You only had to see election rallies for both candidates to witness the enthusiasm for Trump and the apathy for Biden. Tens of thousands would attend Trump events while Biden was speaking in empty car parks with often only television crews attending and framing their shots to hide the fact that no one was there. It was pathetic to see

footage come to light of Biden standing at a podium making speeches only to TV crews and party fixers while reading the words written for him on massive Teleprompter screens. So, yes, those protestors on January 6th had a point about election rigging, but some were about to walk into a trap laid for them in Washington by the Cult Deep State and its QAnon Psyop. This was the Capitol Hill riot ludicrously dubbed an 'insurrection'.

The spider and the fly

Renegade Minds know there are not two 'sides' in politics, only one side, the Cult, working through all 'sides'. It's a stage show, a puppet show, to direct the perceptions of the population into focusing on diversions like parties and candidates while missing the puppeteers with their hands holding all the strings. The Capitol Hill 'insurrection' brings us back to the Little Big Horn. Having created two distinct opposing groupings – Woke and Pushbackers – the trap was about to be sprung. Pushbackers were to be encircled and isolated by associating them all in the public mind with Trump and then labelling Trump as some sort of Confederate leader. I knew immediately that the Capitol riot was a set-up because of two things. One was how easy the rioters got into the building with virtually no credible resistance and secondly I could see – as with the 'Covid' hoax in the West at the start of 2020 – how the Cult could exploit the situation to move its agenda forward with great speed. My experience of Cult techniques and activities over more than 30 years has showed me that while they do exploit situations they haven't themselves created this never happens with events of fundamental agenda significance. Every time major events giving cultists the excuse to rapidly advance their plan you find they are manipulated into being for the specific reason of providing that excuse – Problem-Reaction-Solution. Only a tiny minority of the huge crowd of Washington protestors sought to gain entry to the Capitol by smashing windows and breaching doors. That didn't matter. The whole crowd and all Pushbackers, even if they did not support Trump, were going to be lumped together as dangerous

insurrectionists and conspiracy theorists. The latter term came into widespread use through a CIA memo in the 1960s aimed at discrediting those questioning the nonsensical official story of the Kennedy assassination and it subsequently became widely employed by the media. It's still being used by inept 'journalists' with no idea of its origin to discredit anyone questioning anything that authority claims to be true. When you are perpetrating a conspiracy you need to discredit the very word itself even though the dictionary definition of conspiracy is merely 'the activity of secretly planning with other people to do something bad or illegal' and 'a general agreement to keep silent about a subject for the purpose of keeping it secret'. On that basis there are conspiracies almost wherever you look. For obvious reasons the Cult and its lapdog media have to claim there are no conspiracies even though the word appears in state laws as with conspiracy to defraud, to murder, and to corrupt public morals.

Agent provocateurs are widely used by the Cult Deep State to manipulate genuine people into acting in ways that suit the desired outcome. By genuine in this case I mean protestors genuinely supporting Trump and claims that the election was stolen. In among them, however, were agents of the state wearing the garb of Trump supporters and QAnon to pump-prime the Capital riot which some genuine Trump supporters naively fell for. I described the situation as 'Come into my parlour said the spider to the fly'. Leaflets appeared through the Woke paramilitary arm Antifa, the anti-fascist fascists, calling on supporters to turn up in Washington looking like Trump supporters even though they hated him. Some of those arrested for breaching the Capitol Building were sourced to Antifa and its stable mate Black Lives Matter. Both organisations are funded by Cult billionaires and corporations. One man charged for the riot was according to his lawyer a former FBI agent who had held top secret security clearance for 40 years. Attorney Thomas Plofchan said of his client, 66-year-old Thomas Edward Caldwell:

He has held a Top Secret Security Clearance since 1979 and has undergone multiple Special Background Investigations in support of his clearances. After retiring from the Navy, he

worked as a section chief for the Federal Bureau of Investigation from 2009-2010 as a GS-12 [mid-level employee].

He also formed and operated a consulting firm performing work, often classified, for U.S government customers including the US. Drug Enforcement Agency, Department of Housing and Urban Development, the US Coast Guard, and the US Army Personnel Command.

A judge later released Caldwell pending trial in the absence of evidence about a conspiracy or that he tried to force his way into the building. *The New York Post* reported a 'law enforcement source' as saying that 'at least two known Antifa members were spotted' on camera among Trump supporters during the riot while one of the rioters arrested was John Earle Sullivan, a seriously extreme Black Lives Matter Trump-hater from Utah who was previously arrested and charged in July, 2020, over a BLM-Antifa riot in which drivers were threatened and one was shot. Sullivan is the founder of Utah-based Insurgence USA which is an affiliate of the Cult-created-and-funded Black Lives Matter movement. Footage appeared and was then deleted by Twitter of Trump supporters calling out Antifa infiltrators and a group was filmed changing into pro-Trump clothing before the riot. Security at the building was *pathetic* – as planned. Colonel Leroy Fletcher Prouty, a man with long experience in covert operations working with the US security apparatus, once described the tell-tale sign to identify who is involved in an assassination. He said:

No one has to direct an assassination – it happens. The active role is played secretly by permitting it to happen. This is the greatest single clue. Who has the power to call off or reduce the usual security precautions?

This principle applies to many other situations and certainly to the Capitol riot of January 6th, 2021.

The sting

With such a big and potentially angry crowd known to be gathering near the Capitol the security apparatus would have had a major police detail to defend the building with National Guard troops on

standby given the strength of feeling among people arriving from all over America encouraged by the QAnon Psyop and statements by Donald Trump. Instead Capitol Police 'security' was flimsy, weak, and easily breached. The same number of officers was deployed as on a regular day and that is a blatant red flag. They were not staffed or equipped for a possible riot that had been an obvious possibility in the circumstances. No protective and effective fencing worth the name was put in place and there were no contingency plans. The whole thing was basically a case of standing aside and waving people in. Once inside police mostly backed off apart from one Capitol police officer who ridiculously shot dead unarmed Air Force veteran protestor Ashli Babbitt without a warning as she climbed through a broken window. The 'investigation' refused to name or charge the officer after what must surely be considered a murder in the circumstances. They just lifted a carpet and swept. The story was endlessly repeated about five people dying in the 'armed insurrection' when there was no report of rioters using weapons. Apart from Babbitt the other four died from a heart attack, strokes and apparently a drug overdose. Capitol police officer Brian Sicknick was reported to have died after being bludgeoned with a fire extinguisher when he was alive after the riot was over and died later of what the Washington Medical Examiner's Office said was a stroke. Sicknick had no external injuries. The lies were delivered like rapid fire. There was a narrative to build with incessant repetition of the lie until the lie became the accepted 'everybody knows that' truth. The 'Big Lie' technique of Nazi Propaganda Minister Joseph Goebbels is constantly used by the Cult which was behind the Nazis and is today behind the 'Covid' and 'climate change' hoaxes. Goebbels said:

If you tell a lie big enough and keep repeating it, people will eventually come to believe it. The lie can be maintained only for such time as the State can shield the people from the political, economic and/or military consequences of the lie. It thus becomes vitally important for the State to use all of its powers to repress dissent, for the truth is the mortal enemy of the lie, and thus by extension, the truth is the greatest enemy of the State.

Most protestors had a free run of the Capitol Building. This allowed pictures to be taken of rioters in iconic parts of the building including the Senate chamber which could be used as propaganda images against all Pushbackers. One Congresswoman described the scene as 'the worst kind of non-security anybody could ever imagine'. Well, the first part was true, but someone obviously did imagine it and made sure it happened. Some photographs most widely circulated featured people wearing QAnon symbols and now the Psyop would be used to dub all QAnon followers with the ubiquitous fit-all label of 'white supremacist' and 'insurrectionists'. When a Muslim extremist called Noah Green drove his car at two police officers at the Capitol Building killing one in April, 2021, there was no such political and media hysteria. They were just disappointed he wasn't white.

The witch-hunt

Government prosecutor Michael Sherwin, an aggressive, dark-eyed, professional Rottweiler led the 'investigation' and to call it over the top would be to understate reality a thousand fold. Hundreds were tracked down and arrested for the crime of having the wrong political views and people were jailed who had done nothing more than walk in the building, committed no violence or damage to property, took a few pictures and left. They were labelled a 'threat to the Republic' while Biden sat in the White House signing executive orders written for him that were dismantling 'the Republic'. Even when judges ruled that a mother and son should not be in jail the government kept them there. Some of those arrested have been badly beaten by prison guards in Washington and lawyers for one man said he suffered a fractured skull and was made blind in one eye. Meanwhile a woman is shot dead for no reason by a Capitol Police officer and we are not allowed to know who he is never mind what has happened to him although that will be *nothing*. The Cult's QAnon/Trump sting to identify and isolate Pushbackers and then target them on the road to crushing and deleting them was a resounding success. You would have thought the Russians had

invaded the building at gunpoint and lined up senators for a firing squad to see the political and media reaction. Congresswoman Alexandria Ocasio-Cortez is a child in a woman's body, a terrible-tuos, me, me, me, Woker narcissist of such proportions that words have no meaning. She said she thought she was going to die when 'insurrectionists' banged on her office door. It turned out she wasn't even in the Capitol Building when the riot was happening and the 'banging' was a Capitol Police officer. She referred to herself as a 'survivor' which is an insult to all those true survivors of violent and sexual abuse while she lives her pampered and privileged life talking drivel for a living. Her Woke colleague and fellow mega-narcissist Rashida Tlaib broke down describing the devastating effect on her, too, of *not being* in the building when the rioters were there. Ocasio-Cortez and Tlaib are members of a fully-Woke group of Congresswomen known as 'The Squad' along with Ilhan Omar and Ayanna Pressley. The Squad from what I can see can be identified by its vehement anti-white racism, anti-white men agenda, and, as always in these cases, the absence of brain cells on active duty.

The usual suspects were on the riot case immediately in the form of Democrat ultra-Zionist senators and operatives Chuck Schumer and Adam Schiff demanding that Trump be impeached for 'his part in the insurrection'. The same pair of prats had led the failed impeachment of Trump over the invented 'Russia collusion' nonsense which claimed Russia had helped Trump win the 2016 election. I didn't realise that Tel Aviv had been relocated just outside Moscow. I must find an up-to-date map. The Russia hoax was a Sabbatian operation to keep Trump occupied and impotent and to stop any rapport with Russia which the Cult wants to retain as a perceptual enemy to be pulled out at will. Puppet Biden began attacking Russia when he came to office as the Cult seeks more upheaval, division and war across the world. A two-year stage show 'Russia collusion inquiry' headed by the not-very-bright former 9/11 FBI chief Robert Mueller, with support from 19 lawyers, 40 FBI agents plus intelligence analysts, forensic accountants and other

staff, devoured tens of millions of dollars and found no evidence of Russia collusion which a ten-year-old could have told them on day one. Now the same moronic Schumer and Schiff wanted a second impeachment of Trump over the Capitol 'insurrection' (riot) which the arrested development of Schumer called another 'Pearl Harbor' while others compared it with 9/11 in which 3,000 died and, in the case of CNN, with the Rwandan genocide in the 1990s in which an estimated 500,000 to 600,000 were murdered, between 250, 000 and 500,000 women were raped, and populations of whole towns were hacked to death with machetes. To make those comparisons purely for Cult political reasons is beyond insulting to those that suffered and lost their lives and confirms yet again the callous inhumanity that we are dealing with. Schumer is a monumental idiot and so is Schiff, but they serve the Cult agenda and do whatever they're told so they get looked after. Talking of idiots – another inane man who spanned the Russia and Capitol impeachment attempts was Senator Eric Swalwell who had the nerve to accuse Trump of collusion with the Russians while sleeping with a Chinese spy called Christine Fang or 'Fang Fang' which is straight out of a Bond film no doubt starring Klaus Schwab as the bloke living on a secret island and controlling laser weapons positioned in space and pointing at world capitals. Fang Fang plays the part of Bond's infiltrator girlfriend which I'm sure she would enjoy rather more than sharing a bed with the brainless Swalwell, lying back and thinking of China. The FBI eventually warned Swalwell about Fang Fang which gave her time to escape back to the Chinese dictatorship. How very thoughtful of them. The second Trump impeachment also failed and hardly surprising when an impeachment is supposed to remove a sitting president and by the time it happened Trump was no longer president. These people are running your country America, well, officially anyway. Terrifying isn't it?

Outcomes tell the story - always

The outcome of all this – and it's the *outcome* on which Renegade Minds focus, not the words – was that a vicious, hysterical and

obviously pre-planned assault was launched on Pushbackers to censor, silence and discredit them and even targeted their right to earn a living. They have since been condemned as 'domestic terrorists' that need to be treated like Al-Qaeda and Islamic State. 'Domestic terrorists' is a label the Cult has been trying to make stick since the period of the Oklahoma bombing in 1995 which was blamed on 'far-right domestic terrorists'. If you read *The Trigger* you will see that the bombing was clearly a Problem-Reaction-Solution carried out by the Deep State during a Bill Clinton administration so corrupt that no dictionary definition of the term would even nearly suffice. Nearly 30, 000 troops were deployed from all over America to the empty streets of Washington for Biden's inauguration. Ten thousand of them stayed on with the pretext of protecting the capital from insurrectionists when it was more psychological programming to normalise the use of the military in domestic law enforcement in support of the Cult plan for a police-military state. Biden's fascist administration began a purge of 'wrong-thinkers' in the military which means anyone that is not on board with Woke. The Capitol Building was surrounded by a fence with razor wire and the Land of the Free was further symbolically and literally dismantled. The circle was completed with the installation of Biden and the exploitation of the QAnon Psyop.

America had never been so divided since the civil war of the 19th century, Pushbackers were isolated and dubbed terrorists and now, as was always going to happen, the Cult immediately set about deleting what little was left of freedom and transforming American society through a swish of the hand of the most controlled 'president' in American history leading (officially at least) the most extreme regime since the country was declared an independent state on July 4th, 1776. Biden issued undebated, dictatorial executive orders almost by the hour in his opening days in office across the whole spectrum of the Cult wish-list including diluting controls on the border with Mexico allowing thousands of migrants to illegally enter the United States to transform the demographics of America and import an election-changing number of perceived Democrat

voters. Then there were Biden deportation amnesties for the already illegally resident (estimated to be as high as 20 or even 30 million). A bill before Congress awarded American citizenship to anyone who could prove they had worked in agriculture for just 180 days in the previous two years as 'Big Ag' secured its slave labour long-term. There were the plans to add new states to the union such as Puerto Rico and making Washington DC a state. They are all parts of a plan to ensure that the Cult-owned Woke Democrats would be permanently in power.

Border – what border?

I have exposed in detail in other books how mass immigration into the United States and Europe is the work of Cult networks fuelled by the tens of billions spent to this and other ends by George Soros and his global Open Society (open borders) Foundations. The impact can be seen in America alone where the population has increased by *100 million* in little more than 30 years mostly through immigration. I wrote in *The Answer* that the plan was to have so many people crossing the southern border that the numbers become unstoppable and we are now there under Cult-owned Biden. El Salvador in Central America puts the scale of what is happening into context. A third of the population now lives in the United States, much of it illegally, and many more are on the way. The methodology is to crush Central and South American countries economically and spread violence through machete-wielding psychopathic gangs like MS-13 based in El Salvador and now operating in many American cities. Biden-imposed lax security at the southern border means that it is all but open. He said before his 'election' that he wanted to see a surge towards the border if he became president and that was the green light for people to do just that after election day to create the human disaster that followed for both America and the migrants. When that surge came the imbecilic Alexandria Ocasio-Cortez said it wasn't a 'surge' because they are 'children, not insurgents' and the term 'surge' (used by Biden) was a claim of 'white supremacists'.

This disingenuous lady may one day enter the realm of the most basic intelligence, but it won't be any time soon.

Sabbatians and the Cult are in the process of destroying America by importing violent people and gangs in among the genuine to terrorise American cities and by overwhelming services that cannot cope with the sheer volume of new arrivals. Something similar is happening in Europe as Western society in general is targeted for demographic and cultural transformation and upheaval. The plan demands violence and crime to create an environment of intimidation, fear and division and Soros has been funding the election of district attorneys across America who then stop prosecuting many crimes, reduce sentences for violent crimes and free as many violent criminals as they can. Sabbatians are creating the chaos from which order – their order – can respond in a classic Problem-Reaction-Solution. A Freemasonic moto says 'Ordo Ab Chao' (Order out of Chaos) and this is why the Cult is constantly creating chaos to impose a new 'order'. Here you have the reason the Cult is constantly creating chaos. The 'Covid' hoax can be seen with those entering the United States by plane being forced to take a 'Covid' test while migrants flooding through southern border processing facilities do not. Nothing is put in the way of mass migration and if that means ignoring the government's own 'Covid' rules then so be it. They know it's all bullshit anyway. Any pushback on this is denounced as 'racist' by Wokers and Sabbatian fronts like the ultra-Zionist Anti-Defamation League headed by the appalling Jonathan Greenblatt which at the same time argues that Israel should not give citizenship and voting rights to more Palestinian Arabs or the 'Jewish population' (in truth the Sabbatian network) will lose control of the country.

Society-changing numbers

Biden's masters have declared that countries like El Salvador are so dangerous that their people must be allowed into the United States for humanitarian reasons when there are fewer murders in large parts of many Central American countries than in US cities like

Baltimore. That is not to say Central America cannot be a dangerous place and Cult-controlled American governments have been making it so since way back, along with the dismantling of economies, in a long-term plan to drive people north into the United States. Parts of Central America are very dangerous, but in other areas the story is being greatly exaggerated to justify relaxing immigration criteria. Migrants are being offered free healthcare and education in the United States as another incentive to head for the border and there is no requirement to be financially independent before you can enter to prevent the resources of America being drained. You can't blame migrants for seeking what they believe will be a better life, but they are being played by the Cult for dark and nefarious ends. The numbers since Biden took office are huge. In February, 2021, more than 100,000 people were known to have tried to enter the US illegally through the southern border (it was 34,000 in the same month in 2020) and in March it was 170,000 – a 418 percent increase on March, 2020. These numbers are only known people, not the ones who get in unseen. The true figure for migrants illegally crossing the border in a single month was estimated by one congressman at 250,000 and that number will only rise under Biden's current policy. Gangs of murdering drug-running thugs that control the Mexican side of the border demand money – thousands of dollars – to let migrants cross the Rio Grande into America. At the same time gun battles are breaking out on the border several times a week between rival Mexican drug gangs (which now operate globally) who are equipped with sophisticated military-grade weapons, grenades and armoured vehicles. While the Capitol Building was being 'protected' from a non-existent 'threat' by thousands of troops, and others were still deployed at the time in the Cult Neocon war in Afghanistan, the southern border of America was left to its fate. This is not incompetence, it is cold calculation.

By March, 2021, there were 17,000 unaccompanied children held at border facilities and many of them are ensnared by people traffickers for paedophile rings and raped on their journey north to America. This is not conjecture – this is fact. Many of those designated

children are in reality teenage boys or older. Meanwhile Wokers posture their self-purity for encouraging poor and tragic people to come to America and face this nightmare both on the journey and at the border with the disgusting figure of House Speaker Nancy Pelosi giving disingenuous speeches about caring for migrants. The woman's evil. Wokers condemned Trump for having children in cages at the border (so did Obama, *Shhhh*), but now they are sleeping on the floor without access to a shower with one border facility 729 percent over capacity. The Biden insanity even proposed flying migrants from the southern border to the northern border with Canada for 'processing'. The whole shambles is being overseen by ultra-Zionist Secretary of Homeland Security, the moronic liar Alejandro Mayorkas, who banned news cameras at border facilities to stop Americans seeing what was happening. Mayorkas said there was not a ban on news crews; it was just that they were not allowed to film. Alongside him at Homeland Security is another ultra-Zionist Cass Sunstein appointed by Biden to oversee new immigration laws. Sunstein despises conspiracy researchers to the point where he suggests they should be banned or *taxed* for having such views. The man is not bonkers or anything. He's perfectly well-adjusted, but adjusted to what is the question. Criticise what is happening and you are a 'white supremacist' when earlier non-white immigrants also oppose the numbers which effect their lives and opportunities. Black people in poor areas are particularly damaged by uncontrolled immigration and the increased competition for work opportunities with those who will work for less. They are also losing voting power as Hispanics become more dominant in former black areas. It's a downward spiral for them while the billionaires behind the policy drone on about how much they care about black people and 'racism'. None of this is about compassion for migrants or black people – that's just wind and air. Migrants are instead being mercilessly exploited to transform America while the countries they leave are losing their future and the same is true in Europe. Mass immigration may now be the work of Woke Democrats, but it can be traced back to the 1986 Immigration Reform and Control Act (it

wasn't) signed into law by Republican hero President Ronald Reagan which gave amnesty to millions living in the United States illegally and other incentives for people to head for the southern border. Here we have the one-party state at work again.

Save me syndrome

Almost every aspect of what I have been exposing as the Cult agenda was on display in even the first days of 'Biden' with silencing of Pushbackers at the forefront of everything. A Renegade Mind will view the Trump years and QAnon in a very different light to their supporters and advocates as the dots are connected. The QAnon/Trump Psyop has given the Cult all it was looking for. We may not know how much, or little, that Trump realised he was being used, but that's a side issue. This pincer movement produced the desired outcome of dividing America and having Pushbackers isolated. To turn this around we have to look at new routes to empowerment which do not include handing our power to other people and groups through what I will call the 'Save Me Syndrome' – 'I want someone else to do it so that I don't have to'. We have seen this at work throughout human history and the QAnon/Trump Psyop is only the latest incarnation alongside all the others. Religion is an obvious expression of this when people look to a 'god' or priest to save them or tell them how to be saved and then there are 'save me' politicians like Trump. Politics is a diversion and not a 'saviour'. It is a means to block positive change, not make it possible.

Save Me Syndrome always comes with the same repeating theme of handing your power to whom or what you believe will save you while your real 'saviour' stares back from the mirror every morning. Renegade Minds are constantly vigilant in this regard and always asking the question 'What can I do?' rather than 'What can someone else do for me?' Gandhi was right when he said: 'You must be the change you want to see in the world.' We are indeed the people we have been waiting for. We are presented with a constant raft of reasons to concede that power to others and forget where the real power is. Humanity has the numbers and the Cult does not. It has to

use diversion and division to target the unstoppable power that comes from unity. Religions, governments, politicians, corporations, media, QAnon, are all different manifestations of this power-diversion and dilution. Refusing to give your power to governments and instead handing it to Trump and QAnon is not to take a new direction, but merely to recycle the old one with new names on the posters. I will explore this phenomenon as we proceed and how to break the cycles and recycles that got us here through the mists of repeating perception and so repeating history.

For now we shall turn to the most potent example in the entire human story of the consequences that follow when you give your power away. I am talking, of course, of the 'Covid' hoax.

CHAPTER FOUR

'Covid': Calculated catastrophe

Facts are threatening to those invested in fraud
DaShanne Stokes

We can easily unravel the real reason for the 'Covid pandemic' hoax by employing the Renegade Mind methodology that I have outlined this far. We'll start by comparing the long-planned Cult outcome with the 'Covid pandemic' outcome. Know the outcome and you'll see the journey.

I have highlighted the plan for the Hunger Games Society which has been in my books for so many years with the very few controlling the very many through ongoing dependency. To create this dependency it is essential to destroy independent livelihoods, businesses and employment to make the population reliant on the state (the Cult) for even the basics of life through a guaranteed pittance income. While independence of income remained these Cult ambitions would be thwarted. With this knowledge it was easy to see where the 'pandemic' hoax was going once talk of 'lockdowns' began and the closing of all but perceived 'essential' businesses to 'save' us from an alleged 'deadly virus'. Cult corporations like Amazon and Walmart were naturally considered 'essential' while mom and pop shops and stores had their doors closed by fascist decree. As a result with every new lockdown and new regulation more small and medium, even large businesses not owned by the Cult, went to the wall while Cult giants and their frontmen and women grew financially fatter by the second. Mom and pop were

denied an income and the right to earn a living and the wealth of people like Jeff Bezos (Amazon), Mark Zuckerberg (Facebook) and Sergei Brin and Larry Page (Google/Alphabet) have reached record levels. The Cult was increasing its own power through further dramatic concentrations of wealth while the competition was being destroyed and brought into a state of dependency. Lockdowns have been instigated to secure that very end and were never anything to do with health. My brother Paul spent 45 years building up a bus repair business, but lockdowns meant buses were running at a fraction of normal levels for months on end. Similar stories can be told in their hundreds of millions worldwide. Efforts of a lifetime coldly destroyed by Cult multi-billionaires and their lackeys in government and law enforcement who continued to earn their living from the taxation of the people while denying the right of the same people to earn theirs. How different it would have been if those making and enforcing these decisions had to face the same financial hardships of those they affected, but they never do.

Gates of Hell

Behind it all in the full knowledge of what he is doing and why is the psychopathic figure of Cult operative Bill Gates. His puppet Tedros at the World Health Organization declared 'Covid' a pandemic in March, 2020. The WHO had changed the definition of a 'pandemic' in 2009 just a month before declaring the 'swine flu pandemic' which would not have been so under the previous definition. The same applies to 'Covid'. The definition had included... 'an infection by an infectious agent, occurring simultaneously in different countries, with a significant mortality rate relative to the proportion of the population infected'. The new definition removed the need for 'significant mortality'. The 'pandemic' has been fraudulent even down to the definition, but Gates demanded economy-destroying lockdowns, school closures, social distancing, mandatory masks, a 'vaccination' for every man, woman and child on the planet and severe consequences and restrictions for those that refused. Who gave him this power? The

Cult did which he serves like a little boy in short trousers doing what his daddy tells him. He and his psychopathic missus even smiled when they said that much worse was to come (what they knew was planned to come). Gates responded in the matter-of-fact way of all psychopaths to a question about the effect on the world economy of what he was doing:

Well, it won't go to zero but it will shrink. Global GDP is probably going to take the biggest hit ever [Gates was smiling as he said this] ... in my lifetime this will be the greatest economic hit. But you don't have a choice. People act as if you have a choice. People don't feel like going to the stadium when they might get infected ... People are deeply affected by seeing these stats, by knowing they could be part of the transmission chain, old people, their parents and grandparents, could be affected by this, and so you don't get to say ignore what is going on here.

There will be the ability to open up, particularly in rich countries, if things are done well over the next few months, but for the world at large normalcy only returns when we have largely vaccinated the entire population.

The man has no compassion or empathy. How could he when he's a psychopath like all Cult players? My own view is that even beyond that he is very seriously mentally ill. Look in his eyes and you can see this along with his crazy flailing arms. You don't do what he has done to the world population since the start of 2020 unless you are mentally ill and at the most extreme end of psychopathic. You especially don't do it when to you know, as we shall see, that cases and deaths from 'Covid' are fakery and a product of monumental figure massaging. 'These stats' that Gates referred to are based on a 'test' that's not testing for the 'virus' as he has known all along. He made his fortune with big Cult support as an infamously ruthless software salesman and now buys global control of 'health' (death) policy without the population he affects having any say. It's a breathtaking outrage. Gates talked about people being deeply affected by fear of 'Covid' when that was because of *him* and his global network lying to them minute-by-minute supported by a lying media that he seriously influences and funds to the tune of hundreds of millions. He's handed big sums to media operations including the BBC, NBC, Al Jazeera, Univision, *PBS NewsHour*,

ProPublica, National Journal, The Guardian, The Financial Times, The Atlantic, Texas Tribune, USA Today publisher Gannett, Washington Monthly, Le Monde, Center for Investigative Reporting, Pulitzer Center on Crisis Reporting, National Press Foundation, International Center for Journalists, Solutions Journalism Network, the Poynter Institute for Media Studies, and many more. Gates is everywhere in the 'Covid' hoax and the man must go to prison – or a mental facility – for the rest of his life and his money distributed to those he has taken such enormous psychopathic pleasure in crushing.

The Muscle

The Hunger Games global structure demands a police-military state – a fusion of the two into one force – which viciously imposes the will of the Cult on the population and protects the Cult from public rebellion. In that regard, too, the 'Covid' hoax just keeps on giving. Often unlawful, ridiculous and contradictory 'Covid' rules and regulations have been policed across the world by moronic automatons and psychopaths made faceless by face-nappy masks and acting like the Nazi SS and fascist blackshirts and brownshirts of Hitler and Mussolini. The smallest departure from the rules decreed by the psychos in government and their clueless gofers were jumped upon by the face-nappy fascists. Brutality against public protestors soon became commonplace even on girls, women and old people as the brave men with the batons – the Face-Nappies as I call them – broke up peaceful protests and handed out fines like confetti to people who couldn't earn a living let alone pay hundreds of pounds for what was once an accepted human right. Robot Face-Nappies of Nottingham police in the English East Midlands fined one group £11,000 for attending a child's birthday party. For decades I charted the transformation of law enforcement as genuine, decent officers were replaced with psychopaths and the brain dead who would happily and brutally do whatever their masters told them. Now they were let loose on the public and I would emphasise the point that none of this just happened. The step-by-step change in the dynamic between police and public was orchestrated from the shadows by

those who knew where this was all going and the same with the perceptual reframing of those in all levels of authority and official administration through 'training courses' by organisations such as Common Purpose which was created in the late 1980s and given a massive boost in Blair era Britain until it became a global phenomenon. Supposed public 'servants' began to view the population as the enemy and the same was true of the police. This was the start of the explosion of behaviour manipulation organisations and networks preparing for the all-war on the human psyche unleashed with the dawn of 2020. I will go into more detail about this later in the book because it is a core part of what is happening.

Police desecrated beauty spots to deter people gathering and arrested women for walking in the countryside alone 'too far' from their homes. We had arrogant, clueless sergeants in the Isle of Wight police where I live posting on Facebook what they insisted the population must do or else. A schoolmaster sergeant called Radford looked young enough for me to ask if his mother knew he was out, but he was posting what he *expected* people to do while a Sergeant Wilkinson boasted about fining lads for meeting in a McDonald's car park where they went to get a lockdown takeaway. Wilkinson added that he had even cancelled their order. What a pair of prats these people are and yet they have increasingly become the norm among Jackboot Johnson's Yellowshirts once known as the British police. This was the theme all over the world with police savagery common during lockdown protests in the United States, the Netherlands, and the fascist state of Victoria in Australia under its tyrannical and again moronic premier Daniel Andrews. Amazing how tyrannical and moronic tend to work as a team and the same combination could be seen across America as arrogant, narcissistic Woke governors and mayors such as Gavin Newsom (California), Andrew Cuomo (New York), Gretchen Whitmer (Michigan), Lori Lightfoot (Chicago) and Eric Garcetti (Los Angeles) did their Nazi and Stalin impressions with the full support of the compliant brutality of their enforcers in uniform as they arrested small business owners defying

fascist shutdown orders and took them to jail in ankle shackles and handcuffs. This happened to bistro owner Marlena Pavlos-Hackney in Gretchen Whitmer's fascist state of Michigan when police arrived to enforce an order by a state-owned judge for 'putting the community at risk' at a time when other states like Texas were dropping restrictions and migrants were pouring across the southern border without any 'Covid' questions at all. I'm sure there are many officers appalled by what they are ordered to do, but not nearly enough of them. If they were truly appalled they would not do it. As the months passed every opportunity was taken to have the military involved to make their presence on the streets ever more familiar and 'normal' for the longer-term goal of police-military fusion.

Another crucial element to the Hunger Games enforcement network has been encouraging the public to report neighbours and others for 'breaking the lockdown rules'. The group faced with £11,000 in fines at the child's birthday party would have been dobbed-in by a neighbour with a brain the size of a pea. The technique was most famously employed by the Stasi secret police in communist East Germany who had public informants placed throughout the population. A police chief in the UK says his force doesn't need to carry out 'Covid' patrols when they are flooded with so many calls from the public reporting other people for visiting the beach. Dorset police chief James Vaughan said people were so enthusiastic about snitching on their fellow humans they were now operating as an auxiliary arm of the police: 'We are still getting around 400 reports a week from the public, so we will respond to reports ... We won't need to be doing hotspot patrols because people are very quick to pick the phone up and tell us.' Vaughan didn't say that this is a pillar of all tyrannies of whatever complexion and the means to hugely extend the reach of enforcement while spreading distrust among the people and making them wary of doing anything that might get them reported. Those narcissistic Isle of Wight sergeants Radford and Wilkinson never fail to add a link to their Facebook posts where the public can inform on their fellow slaves.

Neither would be self-aware enough to realise they were imitating the Stasi which they might well never have heard of. Government psychologists that I will expose later laid out a policy to turn communities against each other in the same way.

A coincidence? Yep, and I can knit fog

I knew from the start of the alleged pandemic that this was a Cult operation. It presented limitless potential to rapidly advance the Cult agenda and exploit manipulated fear to demand that every man, woman and child on the planet was 'vaccinated' in a process never used on humans before which infuses self-replicating *synthetic* material into human cells. Remember the plan to transform the human body from a biological to a synthetic biological state. I'll deal with the 'vaccine' (that's not actually a vaccine) when I focus on the genetic agenda. Enough to say here that mass global 'vaccination' justified by this 'new virus' set alarms ringing after 30 years of tracking these people and their methods. The 'Covid' hoax officially beginning in China was also a big red flag for reasons I will be explaining. The agenda potential was so enormous that I could dismiss any idea that the 'virus' appeared naturally. Major happenings with major agenda implications never occur without Cult involvement in making them happen. My questions were twofold in early 2020 as the media began its campaign to induce global fear and hysteria: Was this alleged infectious agent released on purpose by the Cult or did it even exist at all? I then did what I always do in these situations. I sat, observed and waited to see where the evidence and information would take me. By March and early April synchronicity was strongly – and ever more so since then – pointing me in the direction of *there is no 'virus'*. I went public on that with derision even from swathes of the alternative media that voiced a scenario that the Chinese government released the 'virus' in league with Deep State elements in the United States from a top-level bio-lab in Wuhan where the 'virus' is said to have first appeared. I looked at that possibility, but I didn't buy it for several reasons. Deaths from the 'virus' did not in any way match what they

would have been with a 'deadly bioweapon' and it is much more effective if you sell the *illusion* of an infectious agent rather than having a real one unless you can control through injection who has it and who doesn't. Otherwise you lose control of events. A made-up 'virus' gives you a blank sheet of paper on which you can make it do whatever you like and have any symptoms or mutant 'variants' you choose to add while a real infectious agent would limit you to what it actually does. A phantom disease allows you to have endless ludicrous 'studies' on the 'Covid' dollar to widen the perceived impact by inventing ever more 'at risk' groups including one study which said those who walk slowly may be almost four times more likely to die from the 'virus'. People are in psychiatric wards for less.

A real 'deadly bioweapon' can take out people in the hierarchy that are not part of the Cult, but essential to its operation. Obviously they don't want that. Releasing a real disease means you immediately lose control of it. Releasing an illusory one means you don't. Again it's vital that people are extra careful when dealing with what they want to hear. A bioweapon unleashed from a Chinese laboratory in collusion with the American Deep State may fit a conspiracy narrative, but is it true? Would it not be far more effective to use the excuse of a 'virus' to justify the real bioweapon – the 'vaccine'? That way your disease agent does not have to be transmitted and arrives directly through a syringe. I saw a French virologist Luc Montagnier quoted in the alternative media as saying he had discovered that the alleged 'new' severe acute respiratory syndrome coronavirus , or SARS-CoV-2, was made artificially and included elements of the human immunodeficiency 'virus' (HIV) and a parasite that causes malaria. SARS-CoV-2 is alleged to trigger an alleged illness called Covid-19. I remembered Montagnier's name from my research years before into claims that an HIV 'retrovirus' causes AIDs – claims that were demolished by Berkeley virologist Peter Duesberg who showed that no one had ever proved that HIV causes acquired immunodeficiency syndrome or AIDS. Claims that become accepted as fact, publicly and medically, with no proof whatsoever are an ever-recurring story that profoundly applies to

‘Covid’. Nevertheless, despite the lack of proof, Montagnier’s team at the Pasteur Institute in Paris had a long dispute with American researcher Robert Gallo over which of them discovered and isolated the HIV ‘virus’ and with *no evidence* found it to cause AIDS. You will see later that there is also no evidence that any ‘virus’ causes any disease or that there is even such a thing as a ‘virus’ in the way it is said to exist. The claim to have ‘isolated’ the HIV ‘virus’ will be presented in its real context as we come to the shocking story – and it is a story – of SARS-CoV-2 and so will Montagnier’s assertion that he identified the full SARS-CoV-2 genome.

Hoax in the making

We can pick up the ‘Covid’ story in 2010 and the publication by the Rockefeller Foundation of a document called ‘Scenarios for the Future of Technology and International Development’. The inner circle of the Rockefeller family has been serving the Cult since John D. Rockefeller (1839-1937) made his fortune with Standard Oil. It is less well known that the same Rockefeller – the Bill Gates of his day – was responsible for establishing what is now referred to as ‘Big Pharma’, the global network of pharmaceutical companies that make outrageous profits dispensing scalpel and drug ‘medicine’ and are obsessed with pumping vaccines in ever-increasing number into as many human arms and backsides as possible. John D. Rockefeller was the driving force behind the creation of the ‘education’ system in the United States and elsewhere specifically designed to program the perceptions of generations thereafter. The Rockefeller family donated exceptionally valuable land in New York for the United Nations building and were central in establishing the World Health Organization in 1948 as an agency of the UN which was created from the start as a Trojan horse and stalking horse for world government. Now enter Bill Gates. His family and the Rockefellers have long been extremely close and I have seen genealogy which claims that if you go back far enough the two families fuse into the same bloodline. Gates has said that the Bill and Melinda Gates Foundation was inspired by the Rockefeller Foundation and why not

when both are serving the same Cult? Major tax-exempt foundations are overwhelmingly criminal enterprises in which Cult assets fund the Cult agenda in the guise of 'philanthropy' while avoiding tax in the process. Cult operatives can become mega-rich in their role of front men and women for the psychopaths at the inner core and they, too, have to be psychopaths to knowingly serve such evil. Part of the deal is that a big percentage of the wealth gleaned from representing the Cult has to be spent advancing the ambitions of the Cult and hence you have the Rockefeller Foundation, Bill and Melinda Gates Foundation (and *so* many more) and people like George Soros with his global Open Society Foundations spending their billions in pursuit of global Cult control. Gates is a global public face of the Cult with his interventions in world affairs including Big Tech influence; a central role in the 'Covid' and 'vaccine' scam; promotion of the climate change shakedown; manipulation of education; geoengineering of the skies; and his food-control agenda as the biggest owner of farmland in America, his GMO promotion and through other means. As one writer said: 'Gates monopolizes or wields disproportionate influence over the tech industry, global health and vaccines, agriculture and food policy (including biopiracy and fake food), weather modification and other climate technologies, surveillance, education and media.' The almost limitless wealth secured through Microsoft and other not-allowed-to-fail ventures (including vaccines) has been ploughed into a long, long list of Cult projects designed to enslave the entire human race. Gates and the Rockefellers have been working as one unit with the Rockefeller-established World Health Organization leading global 'Covid' policy controlled by Gates through his mouth-piece Tedros. Gates became the WHO's biggest funder when Trump announced that the American government would cease its donations, but Biden immediately said he would restore the money when he took office in January, 2021. The Gates Foundation (the Cult) owns through limitless funding the world health system and the major players across the globe in the 'Covid' hoax.

Okay, with that background we return to that Rockefeller Foundation document of 2010 headed 'Scenarios for the Future of Technology and International Development' and its 'imaginary' epidemic of a virulent and deadly influenza strain which infected 20 percent of the global population and killed eight million in seven months. The Rockefeller scenario was that the epidemic destroyed economies, closed shops, offices and other businesses and led to governments imposing fierce rules and restrictions that included mandatory wearing of face masks and body-temperature checks to enter communal spaces like railway stations and supermarkets. The document predicted that even after the height of the Rockefeller-envisaged epidemic the authoritarian rule would continue to deal with further pandemics, transnational terrorism, environmental crises and rising poverty. Now you may think that the Rockefellers are our modern-day seers or alternatively, and rather more likely, that they well knew what was planned a few years further on. Fascism had to be imposed, you see, to 'protect citizens from risk and exposure'. The Rockefeller scenario document said:

During the pandemic, national leaders around the world flexed their authority and imposed airtight rules and restrictions, from the mandatory wearing of face masks to body-temperature checks at the entries to communal spaces like train stations and supermarkets. Even after the pandemic faded, this more authoritarian control and oversight of citizens and their activities stuck and even intensified. In order to protect themselves from the spread of increasingly global problems – from pandemics and transnational terrorism to environmental crises and rising poverty – leaders around the world took a firmer grip on power.

At first, the notion of a more controlled world gained wide acceptance and approval. Citizens willingly gave up some of their sovereignty – and their privacy – to more paternalistic states in exchange for greater safety and stability. Citizens were more tolerant, and even eager, for top-down direction and oversight, and national leaders had more latitude to impose order in the ways they saw fit.

In developed countries, this heightened oversight took many forms: biometric IDs for all citizens, for example, and tighter regulation of key industries whose stability was deemed vital to national interests. In many developed countries, enforced cooperation with a suite of new regulations and agreements slowly but steadily restored both order and, importantly, economic growth.

There we have the prophetic Rockefellers in 2010 and three years later came their paper for the Global Health Summit in Beijing, China, when government representatives, the private sector, international organisations and groups met to discuss the next 100 years of 'global health'. The Rockefeller Foundation-funded paper was called 'Dreaming the Future of Health for the Next 100 Years and more prophecy ensued as it described a dystopian future: 'The abundance of data, digitally tracking and linking people may mean the 'death of privacy' and may replace physical interaction with transient, virtual connection, generating isolation and raising questions of how values are shaped in virtual networks.' Next in the 'Covid' hoax preparation sequence came a 'table top' simulation in 2018 for another 'imaginary' pandemic of a disease called Clade X which was said to kill 900 million people. The exercise was organised by the Gates-funded Johns Hopkins University's Center for Health Security in the United States and this is the very same university that has been compiling the disgustingly and systematically erroneous global figures for 'Covid' cases and deaths. Similar Johns Hopkins health crisis scenarios have included the Dark Winter exercise in 2001 and Atlantic Storm in 2005.

Nostradamus 201

For sheer predictive genius look no further prophecy-watchers than the Bill Gates-funded Event 201 held only six weeks before the 'coronavirus pandemic' is supposed to have broken out in China and Event 201 was based on a scenario of a global 'coronavirus pandemic'. Melinda Gates, the great man's missus, told the BBC that he had 'prepared for years' for a coronavirus pandemic which told us what we already knew. Nostradamugates had predicted in a TED talk in 2015 that a pandemic was coming that would kill a lot of people and demolish the world economy. My god, the man is a machine – possibly even literally. Now here he was only weeks before the real thing funding just such a simulated scenario and involving his friends and associates at Johns Hopkins, the World Economic Forum Cult-front of Klaus Schwab, the United Nations,

Johnson & Johnson, major banks, and officials from China and the Centers for Disease Control in the United States. What synchronicity – Johns Hopkins would go on to compile the fraudulent ‘Covid’ figures, the World Economic Forum and Schwab would push the ‘Great Reset’ in response to ‘Covid’, the Centers for Disease Control would be at the forefront of ‘Covid’ policy in the United States, Johnson & Johnson would produce a ‘Covid vaccine’, and everything would officially start just weeks later in China. Spooky, eh? They were even accurate in creating a simulation of a ‘virus’ pandemic because the ‘real thing’ would also be a simulation. Event 201 was not an exercise preparing for something that might happen; it was a rehearsal for what those in control knew was *going* to happen and very shortly. Hours of this simulation were posted on the Internet and the various themes and responses mirrored what would soon be imposed to transform human society. News stories were inserted and what they said would be commonplace a few weeks later with still more prophecy perfection. Much discussion focused on the need to deal with misinformation and the ‘anti-vax movement’ which is exactly what happened when the ‘virus’ arrived – was said to have arrived – in the West.

Cult-owned social media banned criticism and exposure of the official ‘virus’ narrative and when I said there *was* no ‘virus’ in early April, 2020, I was banned by one platform after another including YouTube, Facebook and later Twitter. The mainstream broadcast media in Britain was in effect banned from interviewing me by the Tony-Blair-created government broadcasting censor Ofcom headed by career government bureaucrat Melanie Dawes who was appointed just as the ‘virus’ hoax was about to play out in January, 2020. At the same time the Ickonic media platform was using Vimeo, another ultra-Zionist-owned operation, while our own player was being created and they deleted in an instant hundreds of videos, documentaries, series and shows to confirm their unbelievable vindictiveness. We had copies, of course, and they had to be restored one by one when our player was ready. These people have no class. Sabbatian Facebook promised free advertisements for the Gates-

controlled World Health Organization narrative while deleting ‘false claims and conspiracy theories’ to stop ‘misinformation’ about the alleged coronavirus. All these responses could be seen just a short while earlier in the scenarios of Event 201. Extreme censorship was absolutely crucial for the Cult because the official story was so ridiculous and unsupportable by the evidence that it could never survive open debate and the free-flow of information and opinion. If you can’t win a debate then don’t have one is the Cult’s approach throughout history. Facebook’s little boy front man – front boy – Mark Zuckerberg equated ‘credible and accurate information’ with official sources and exposing their lies with ‘misinformation’.

Silencing those that can see

The censorship dynamic of Event 201 is now the norm with an army of narrative-supporting ‘fact-checker’ organisations whose entire reason for being is to tell the public that official narratives are true and those exposing them are lying. One of the most appalling of these ‘fact-checkers’ is called NewsGuard founded by ultra-Zionist Americans Gordon Crovitz and Steven Brill. Crovitz is a former publisher of *The Wall Street Journal*, former Executive Vice President of Dow Jones, a member of the Council on Foreign Relations (CFR), and on the board of the American Association of Rhodes Scholars. The CFR and Rhodes Scholarships, named after Rothschild agent Cecil Rhodes who plundered the gold and diamonds of South Africa for his masters and the Cult, have featured widely in my books. NewsGuard don’t seem to like me for some reason – I really can’t think why – and they have done all they can to have me censored and discredited which is, to quote an old British politician, like being savaged by a dead sheep. They are, however, like all in the censorship network, very well connected and funded by organisations themselves funded by, or connected to, Bill Gates. As you would expect with anything associated with Gates NewsGuard has an offshoot called HealthGuard which ‘fights online health care hoaxes’. How very kind. Somehow the NewsGuard European Managing Director Anna-Sophie Harling, a remarkably young-

looking woman with no broadcasting experience and little hands-on work in journalism, has somehow secured a position on the 'Content Board' of UK government broadcast censor Ofcom. An executive of an organisation seeking to discredit dissidents of the official narratives is making decisions for the government broadcast 'regulator' about content?? Another appalling 'fact-checker' is Full Fact funded by George Soros and global censors Google and Facebook.

It's amazing how many activists in the 'fact-checking', 'anti-hate', arena turn up in government-related positions – people like UK Labour Party activist Imran Ahmed who heads the Center for Countering Digital Hate founded by people like Morgan McSweeney, now chief of staff to the Labour Party's hapless and useless 'leader' Keir Starmer. Digital Hate – which is what it really is – uses the American spelling of Center to betray its connection to a transatlantic network of similar organisations which in 2020 shapeshifted from attacking people for 'hate' to attacking them for questioning the 'Covid' hoax and the dangers of the 'Covid vaccine'. It's just a coincidence, you understand. This is one of Imran Ahmed's hysterical statements: 'I would go beyond calling anti-vaxxers conspiracy theorists to say they are an extremist group that pose a national security risk.' No one could ever accuse this prat of understatement and he's including in that those parents who are now against vaccines after their children were damaged for life or killed by them. He's such a nice man. Ahmed does the rounds of the Woke media getting soft-ball questions from spineless 'journalists' who never ask what right he has to campaign to destroy the freedom of speech of others while he demands it for himself. There also seems to be an overrepresentation in Ofcom of people connected to the narrative-worshipping BBC. This incredible global network of narrative-support was super-vital when the 'Covid' hoax was played in the light of the mega-whopper lies that have to be defended from the spotlight cast by the most basic intelligence.

Setting the scene

The Cult plays the long game and proceeds step-by-step ensuring that everything is in place before major cards are played and they don't come any bigger than the 'Covid' hoax. The psychopaths can't handle events where the outcome isn't certain and as little as possible – preferably nothing – is left to chance. Politicians, government and medical officials who would follow direction were brought to illusory power in advance by the Cult web whether on the national stage or others like state governors and mayors of America. For decades the dynamic between officialdom, law enforcement and the public was changed from one of service to one of control and dictatorship. Behaviour manipulation networks established within government were waiting to impose the coming 'Covid' rules and regulations specifically designed to subdue and rewire the psyche of the people in the guise of protecting health. These included in the UK the Behavioural Insights Team part-owned by the British government Cabinet Office; the Scientific Pandemic Insights Group on Behaviours (SPI-B); and a whole web of intelligence and military groups seeking to direct the conversation on social media and control the narrative. Among them are the cyberwarfare (on the people) 77th Brigade of the British military which is also coordinated through the Cabinet Office as civilian and military leadership continues to combine in what they call the Fusion Doctrine. The 77th Brigade is a British equivalent of the infamous Israeli (Sabbatian) military cyberwarfare and Internet manipulation operation Unit 8200 which I expose at length in *The Trigger*. Also carefully in place were the medical and science advisers to government – many on the payroll past or present of Bill Gates – and a whole alternative structure of unelected government stood by to take control when elected parliaments were effectively closed down once the 'Covid' card was slammed on the table. The structure I have described here and so much more was installed in every major country through the Cult networks. The top-down control hierarchy looks like this: The Cult – Cult-owned Gates – the World Health Organization and Tedros – Gates-funded or controlled chief medical officers and science 'advisers' (dictators) in each country –

political 'leaders' – law enforcement – The People. Through this simple global communication and enforcement structure the policy of the Cult could be imposed on virtually the entire human population so long as they acquiesced to the fascism. With everything in place it was time for the button to be pressed in late 2019/early 2020.

These were the prime goals the Cult had to secure for its will to prevail:

1) Locking down economies, closing all but designated 'essential' businesses (Cult-owned corporations were 'essential'), and putting the population under house arrest was an imperative to destroy independent income and employment and ensure dependency on the Cult-controlled state in the Hunger Games Society. Lockdowns had to be established as the global blueprint from the start to respond to the 'virus' and followed by pretty much the entire world.

2) The global population had to be terrified into believing in a deadly 'virus' that didn't actually exist so they would unquestioningly obey authority in the belief that authority must know how best to protect them and their families. Software salesman Gates would suddenly morph into the world's health expert and be promoted as such by the Cult-owned media.

3) A method of testing that wasn't testing for the 'virus', but was only claimed to be, had to be in place to provide the illusion of 'cases' and subsequent 'deaths' that had a very different cause to the 'Covid-19' that would be scribbled on the death certificate.

4) Because there was no 'virus' and the great majority testing positive with a test not testing for the 'virus' would have no symptoms of anything the lie had to be sold that people without symptoms (without the 'virus') could still pass it on to others. This was crucial to justify for the first time quarantining – house arresting – healthy people. Without this the economy-destroying lockdown of *everybody* could not have been credibly sold.

5) The 'saviour' had to be seen as a vaccine which beyond evil drug companies were working like angels of mercy to develop as quickly as possible, with all corners cut, to save the day. The public must absolutely not know that the 'vaccine' had nothing to do with a 'virus' or that the contents were ready and waiting with a very different motive long before the 'Covid' card was even lifted from the pack.

I said in March, 2020, that the 'vaccine' would have been created way ahead of the 'Covid' hoax which justified its use and the following December an article in the New York *Intelligencer* magazine said the Moderna 'vaccine' had been 'designed' by

January, 2020. This was 'before China had even acknowledged that the disease could be transmitted from human to human, more than a week before the first confirmed coronavirus case in the United States'. The article said that by the time the first American death was announced a month later 'the vaccine had already been manufactured and shipped to the National Institutes of Health for the beginning of its Phase I clinical trial'. The 'vaccine' was actually 'designed' long before that although even with this timescale you would expect the article to ask how on earth it could have been done that quickly. Instead it asked why the 'vaccine' had not been rolled out then and not months later. Journalism in the mainstream is truly dead. I am going to detail in the next chapter why the 'virus' has never existed and how a hoax on that scale was possible, but first the foundation on which the Big Lie of 'Covid' was built.

The test that doesn't test

Fraudulent 'testing' is the bottom line of the whole 'Covid' hoax and was the means by which a 'virus' that did not exist *appeared* to exist. They could only achieve this magic trick by using a test not testing for the 'virus'. To use a test that *was* testing for the 'virus' would mean that every test would come back negative given there was no 'virus'. They chose to exploit something called the RT-PCR test invented by American biochemist Kary Mullis in the 1980s who said publicly that his PCR test ... *cannot detect infectious disease*. Yes, the 'test' used worldwide to detect infectious 'Covid' to produce all the illusory 'cases' and 'deaths' compiled by Johns Hopkins and others *cannot detect infectious disease*. This fact came from the mouth of the man who invented PCR and was awarded the Nobel Prize in Chemistry in 1993 for doing so. Sadly, and incredibly conveniently for the Cult, Mullis died in August, 2019, at the age of 74 just before his test would be fraudulently used to unleash fascism on the world. He was said to have died from pneumonia which was an irony in itself. A few months later he would have had 'Covid-19' on his death certificate. I say the timing of his death was convenient because had he lived Mullis, a brilliant, honest and decent man, would have been

vociferously speaking out against the use of his test to detect 'Covid' when it was never designed, or able, to do that. I know that to be true given that Mullis made the same point when his test was used to 'detect' – not detect – HIV. He had been seriously critical of the Gallo/Montagnier claim to have isolated the HIV 'virus' and shown it to cause AIDS for which Mullis said there was no evidence. AIDS is actually not a disease but a series of diseases from which people die all the time. When they die from those *same diseases* after a positive 'test' for HIV then AIDS goes on their death certificate. I think I've heard that before somewhere. Countries instigated a policy with 'Covid' that anyone who tested positive with a test not testing for the 'virus' and died of any other cause within 28 days and even longer 'Covid-19' had to go on the death certificate. Cases have come from the test that can't test for infectious disease and the deaths are those who have died of *anything* after testing positive with a test not testing for the 'virus'. I'll have much more later about the death certificate scandal.

Mullis was deeply dismissive of the now US 'Covid' star Anthony Fauci who he said was a liar who didn't know anything about anything – 'and I would say that to his face – nothing.' He said of Fauci: 'The man thinks he can take a blood sample, put it in an electron microscope and if it's got a virus in there you'll know it – he doesn't understand electron microscopy and he doesn't understand medicine and shouldn't be in a position like he's in.' That position, terrifyingly, has made him the decider of 'Covid' fascism policy on behalf of the Cult in his role as director since 1984 of the National Institute of Allergy and Infectious Diseases (NIAID) while his record of being wrong is laughable; but being wrong, so long as it's the *right kind* of wrong, is why the Cult loves him. He'll say anything the Cult tells him to say. Fauci was made Chief Medical Adviser to the President immediately Biden took office. Biden was installed in the White House by Cult manipulation and one of his first decisions was to elevate Fauci to a position of even more control. This is a coincidence? Yes, and I identify as a flamenco dancer called Lola. How does such an incompetent criminal like Fauci remain in that

pivotal position in American health since *the 1980s*? When you serve the Cult it looks after you until you are surplus to requirements. Kary Mullis said prophetically of Fauci and his like: 'Those guys have an agenda and it's not an agenda we would like them to have ... they make their own rules, they change them when they want to, and Tony Fauci does not mind going on television in front of the people who pay his salary and lie directly into the camera.' Fauci has done that almost daily since the 'Covid' hoax began. Lying is in Fauci's DNA. To make the situation crystal clear about the PCR test this is a direct quote from its inventor Kary Mullis:

It [the PCR test] doesn't tell you that you're sick and doesn't tell you that the thing you ended up with was really going to hurt you ...'

Ask yourself why governments and medical systems the world over have been using this very test to decide who is 'infected' with the SARS-CoV-2 'virus' and the alleged disease it allegedly causes, 'Covid-19'. The answer to that question will tell you what has been going on. By the way, here's a little show-stopper – the 'new' SARS-CoV-2 'virus' was 'identified' as such right from the start using ... *the PCR test not testing for the 'virus'*. If you are new to this and find that shocking then stick around. I have hardly started yet. Even worse, other 'tests', like the 'Lateral Flow Device' (LFD), are considered so useless that they have to be *confirmed* by the PCR test! Leaked emails written by Ben Dyson, adviser to UK 'Health' Secretary Matt Hancock, said they were 'dangerously unreliable'. Dyson, executive director of strategy at the Department of Health, wrote: 'As of today, someone who gets a positive LFD result in (say) London has at best a 25 per cent chance of it being a true positive, but if it is a self-reported test potentially as low as 10 per cent (on an optimistic assumption about specificity) or as low as 2 per cent (on a more pessimistic assumption).' These are the 'tests' that schoolchildren and the public are being urged to have twice a week or more and have to isolate if they get a positive. Each fake positive goes in the statistics as a 'case' no matter how ludicrously inaccurate and the

'cases' drive lockdown, masks and the pressure to 'vaccinate'. The government said in response to the email leak that the 'tests' were accurate which confirmed yet again what shocking bloody liars they are. The real false positive rate is *100 percent* as we'll see. In another 'you couldn't make it up' the UK government agreed to pay £2.8 billion to California's Innova Medical Group to supply the irrelevant lateral flow tests. The company's primary test-making centre is in China. Innova Medical Group, established in March, 2020, is owned by Pasaca Capital Inc, chaired by Chinese-American millionaire Charles Huang who was born in Wuhan.

How it works – and how it doesn't

The RT-PCR test, known by its full title of Polymerase chain reaction, is used across the world to make millions, even billions, of copies of a DNA/RNA genetic information sample. The process is called 'amplification' and means that a tiny sample of genetic material is amplified to bring out the detailed content. I stress that it is not testing for an infectious disease. It is simply amplifying a sample of genetic material. In the words of Kary Mullis: 'PCR is ... just a process that's used to make a whole lot of something out of something.' To emphasise the point companies that make the PCR tests circulated around the world to 'test' for 'Covid' warn on the box that it can't be used to detect 'Covid' or infectious disease and is for research purposes only. It's okay, rest for a minute and you'll be fine. This is the test that produces the 'cases' and 'deaths' that have been used to destroy human society. All those global and national medical and scientific 'experts' demanding this destruction to 'save us' *KNOW* that the test is not testing for the 'virus' and the cases and deaths they claim to be real are an almost unimaginable fraud. Every one of them and so many others including politicians and psychopaths like Gates and Tedros must be brought before Nuremburg-type trials and jailed for the rest of their lives. The more the genetic sample is amplified by PCR the more elements of that material become sensitive to the test and by that I don't mean sensitive for a 'virus' but for elements of the genetic material which

is *naturally* in the body or relates to remnants of old conditions of various kinds lying dormant and causing no disease. Once the amplification of the PCR reaches a certain level *everyone* will test positive. So much of the material has been made sensitive to the test that everyone will have some part of it in their body. Even lying criminals like Fauci have said that once PCR amplifications pass 35 cycles everything will be a false positive that cannot be trusted for the reasons I have described. I say, like many proper doctors and scientists, that 100 percent of the 'positives' are false, but let's just go with Fauci for a moment.

He says that any amplification over 35 cycles will produce false positives and yet the US Centers for Disease Control (CDC) and Food and Drug Administration (FDA) have recommended up to 40 *cycles* and the National Health Service (NHS) in Britain admitted in an internal document for staff that it was using 45 *cycles* of amplification. A long list of other countries has been doing the same and at least one 'testing' laboratory has been using 50 *cycles*. Have you ever heard a doctor, medical 'expert' or the media ask what level of amplification has been used to claim a 'positive'. The 'test' comes back 'positive' and so you have the 'virus', end of story. Now we can see how the government in Tanzania could send off samples from a goat and a pawpaw fruit under human names and both came back positive for 'Covid-19'. Tanzania president John Magufuli mocked the 'Covid' hysteria, the PCR test and masks and refused to import the DNA-manipulating 'vaccine'. The Cult hated him and an article sponsored by the Bill Gates Foundation appeared in the London *Guardian* in February, 2021, headed 'It's time for Africa to rein in Tanzania's anti-vaxxer president'. Well, 'reined in' he shortly was. Magufuli appeared in good health, but then, in March, 2021, he was dead at 61 from 'heart failure'. He was replaced by Samia Hassan Suhulu who is connected to Klaus Schwab's World Economic Forum and she immediately reversed Magufuli's 'Covid' policy. A sample of cola tested positive for 'Covid' with the PCR test in Germany while American actress and singer-songwriter Erykah Badu tested positive in one nostril and negative in the other. Footballer Ronaldo called

the PCR test 'bullshit' after testing positive three times and being forced to quarantine and miss matches when there was nothing wrong with him. The mantra from Tedros at the World Health Organization and national governments (same thing) has been test, test, test. They know that the more tests they can generate the more fake 'cases' they have which go on to become 'deaths' in ways I am coming to. The UK government has its Operation Moonshot planned to test multiple millions every day in workplaces and schools with free tests for everyone to use twice a week at home in line with the Cult plan from the start to make testing part of life. A government advertisement for an 'Interim Head of Asymptomatic Testing Communication' said the job included responsibility for delivering a 'communications strategy' (propaganda) 'to support the expansion of asymptomatic testing that *'normalises testing as part of everyday life'*'. More tests means more fake 'cases', 'deaths' and fascism. I have heard of, and from, many people who booked a test, couldn't turn up, and yet got a positive result through the post for a test they'd never even had. The whole thing is crazy, but for the Cult there's method in the madness. Controlling and manipulating the level of amplification of the test means the authorities can control whenever they want the number of apparent 'cases' and 'deaths'. If they want to justify more fascist lockdown and destruction of livelihoods they keep the amplification high. If they want to give the illusion that lockdowns and the 'vaccine' are working then they lower the amplification and 'cases' and 'deaths' will appear to fall. In January, 2021, the Cult-owned World Health Organization suddenly warned laboratories about over-amplification of the test and to lower the threshold. Suddenly headlines began appearing such as: 'Why ARE "Covid" cases plummeting?' This was just when the vaccine rollout was underway and I had predicted months before they would make cases appear to fall through amplification tampering when the 'vaccine' came. These people are so predictable.

Cow vaccines?

The question must be asked of what is on the test swabs being poked far up the nose of the population to the base of the brain? A nasal swab punctured one woman's brain and caused it to leak fluid. Most of these procedures are being done by people with little training or medical knowledge. Dr Lorraine Day, former orthopaedic trauma surgeon and Chief of Orthopaedic Surgery at San Francisco General Hospital, says the tests are really a '*vaccine*'. Cows have long been vaccinated this way. She points out that masks have to cover the nose and the mouth where it is claimed the '*virus*' exists in saliva. Why then don't they take saliva from the mouth as they do with a DNA test instead of pushing a long swab up the nose towards the brain? The ethmoid bone separates the nasal cavity from the brain and within that bone is the cribriform plate. Dr Day says that when the swab is pushed up against this plate and twisted the procedure is '*depositing things back there*'. She claims that among these '*things*' are nanoparticles that can enter the brain. Researchers have noted that a team at the Gates-funded Johns Hopkins have designed tiny, star-shaped micro-devices that can latch onto intestinal mucosa and release drugs into the body. Mucosa is the thin skin that covers the inside surface of parts of the body such as *the nose* and mouth and produces mucus to protect them. The Johns Hopkins micro-devices are called '*theragrippers*' and were '*inspired*' by a parasitic worm that digs its sharp teeth into a host's intestines. Nasal swabs are also coated in the sterilisation agent ethylene oxide. The US National Cancer Institute posts this explanation on its website:

At room temperature, ethylene oxide is a flammable colorless gas with a sweet odor. It is used primarily to produce other chemicals, including antifreeze. In smaller amounts, ethylene oxide is used as a pesticide and a sterilizing agent. The ability of ethylene oxide to damage DNA makes it an effective sterilizing agent but also accounts for its cancer-causing activity.

The Institute mentions lymphoma and leukaemia as cancers most frequently reported to be associated with occupational exposure to ethylene oxide along with stomach and breast cancers. How does anyone think this is going to work out with the constant testing

regime being inflicted on adults and children at home and at school that will accumulate in the body anything that's on the swab?

Doctors know best

It is vital for people to realise that 'hero' doctors 'know' only what the Big Pharma-dominated medical authorities tell them to 'know' and if they refuse to 'know' what they are told to 'know' they are out the door. They are mostly not physicians or healers, but repeaters of the official narrative – or else. I have seen alleged professional doctors on British television make shocking statements that we are supposed to take seriously. One called 'Dr' Amir Khan, who is actually telling patients how to respond to illness, said that men could take the birth pill to 'help slow down the effects of Covid-19'. In March, 2021, another ridiculous 'Covid study' by an American doctor proposed injecting men with the female sex hormone progesterone as a 'Covid' treatment. British doctor Nighat Arif told the BBC that face coverings were now going to be part of ongoing normal. Yes, the vaccine protects you, she said (evidence?) ... but the way to deal with viruses in the community was always going to come down to hand washing, face covering and keeping a physical distance. That's not what we were told before the 'vaccine' was circulating. Arif said she couldn't imagine ever again going on the underground or in a lift without a mask. I was just thanking my good luck that she was not my doctor when she said – in March, 2021 – that if 'we are *behaving* and we are doing all the right things' she thought we could 'have our nearest and dearest around us at home ... around *Christmas* and *New Year*! Her patronising delivery was the usual school teacher talking to six-year-olds as she repeated every government talking point and probably believed them all. If we have learned anything from the 'Covid' experience surely it must be that humanity's perception of doctors needs a fundamental rethink. NHS 'doctor' Sara Kayat told her television audience that the 'Covid vaccine' would '100 percent prevent hospitalisation and death'. Not even Big Pharma claimed that. We have to stop taking 'experts' at their word without question when so many of them are

clueless and only repeating the party line on which their careers depend. That is not to say there are not brilliant doctors – there are and I have spoken to many of them since all this began – but you won't see them in the mainstream media or quoted by the psychopaths and yes-people in government.

Remember the name – Christian Drosten

German virologist Christian Drosten, Director of Charité Institute of Virology in Berlin, became a national star after the pandemic hoax began. He was feted on television and advised the German government on 'Covid' policy. Most importantly to the wider world Drosten led a group that produced the 'Covid' testing protocol for the PCR test. What a remarkable feat given the PCR cannot test for infectious disease and even more so when you think that Drosten said that his method of testing for SARS-CoV-2 was developed 'without having virus material available'. *He developed a test for a 'virus' that he didn't have and had never seen.* Let that sink in as you survey the global devastation that came from what he did. The whole catastrophe of Drosten's 'test' was based on the alleged genetic sequence published by Chinese scientists on the Internet. We will see in the next chapter that this alleged 'genetic sequence' has never been produced by China or anyone and cannot be when there *is no* SARS-CoV-2. Drosten, however, doesn't seem to let little details like that get in the way. He was the lead author with Victor Corman from the same Charité Hospital of the paper 'Detection of 2019 novel coronavirus (2019-nCoV) by real-time PCR' published in a magazine called *Eurosurveillance*. This became known as the Corman-Drosten paper. In November, 2020, with human society devastated by the effects of the Corman-Drosten test baloney, the protocol was publicly challenged by 22 international scientists and independent researchers from Europe, the United States, and Japan. Among them were senior molecular geneticists, biochemists, immunologists, and microbiologists. They produced a document headed 'External peer review of the RTPCR test to detect SARS-Cov-2 Reveals 10 Major Flaws At The Molecular and Methodological Level: Consequences

For False-Positive Results'. The flaws in the Corman-Drosten test included the following:

- The test is non-specific because of erroneous design
- Results are enormously variable
- The test is unable to discriminate between the whole 'virus' and viral fragments
- It doesn't have positive or negative controls
- The test lacks a standard operating procedure
- It is unsupported by proper peer view

The scientists said the PCR 'Covid' testing protocol was not founded on science and they demanded the Corman-Drosten paper be retracted by *Eurosurveillance*. They said all present and previous Covid deaths, cases, and 'infection rates' should be subject to a massive retroactive inquiry. Lockdowns and travel restrictions should be reviewed and relaxed and those diagnosed through PCR to have 'Covid-19' should not be forced to isolate. Dr Kevin Corbett, a health researcher and nurse educator with a long academic career producing a stream of peer-reviewed publications at many UK universities, made the same point about the PCR test debacle. He said of the scientists' conclusions: 'Every scientific rationale for the development of that test has been totally destroyed by this paper. It's like Hiroshima/Nagasaki to the Covid test.' He said that China hadn't given them an isolated 'virus' when Drosten developed the test. Instead they had developed the test from *a sequence in a gene bank*.' Put another way ... *they made it up!* The scientists were supported in this contention by a Portuguese appeals court which ruled in November, 2020, that PCR tests are unreliable and it is unlawful to quarantine people based solely on a PCR test. The point about China not providing an isolated virus must be true when the 'virus' has never been isolated to this day and the consequences of that will become clear. Drosten and company produced this useless 'protocol' right on cue in January, 2020, just as the 'virus' was said to

be moving westward and it somehow managed to successfully pass a peer-review in 24 hours. In other words there was no peer-review for a test that would be used to decide who had 'Covid' and who didn't across the world. The Cult-created, Gates-controlled World Health Organization immediately recommended all its nearly 200 member countries to use the Drosten PCR protocol to detect 'cases' and 'deaths'. The sting was underway and it continues to this day.

So who is this Christian Drosten that produced the means through which death, destruction and economic catastrophe would be justified? His education background, including his doctoral thesis, would appear to be somewhat shrouded in mystery and his track record is dire as with another essential player in the 'Covid' hoax, the Gates-funded Professor Neil Ferguson at the Gates-funded Imperial College in London of whom more shortly. Drosten predicted in 2003 that the alleged original SARS 'virus' (SARS-1) was an epidemic that could have serious effects on economies and an effective vaccine would take at least two years to produce. Drosten's answer to every alleged 'outbreak' is a vaccine which you won't be shocked to know. What followed were just 774 official deaths worldwide and none in Germany where there were only nine cases. That is even if you believe there ever was a SARS 'virus' when the evidence is zilch and I will expand on this in the next chapter. Drosten claims to be co-discoverer of 'SARS-1' and developed a test for it in 2003. He was screaming warnings about 'swine flu' in 2009 and how it was a widespread infection far more severe than any dangers from a vaccine could be and people should get vaccinated. It would be helpful for Drosten's vocal chords if he simply recorded the words 'the virus is deadly and you need to get vaccinated' and copies could be handed out whenever the latest made-up threat comes along. Drosten's swine flu epidemic never happened, but Big Pharma didn't mind with governments spending hundreds of millions on vaccines that hardly anyone bothered to use and many who did wished they hadn't. A study in 2010 revealed that the risk of dying from swine flu, or H1N1, was no higher than that of the annual seasonal flu which is what at least most of 'it' really was as in

the case of 'Covid-19'. A media investigation into Drosten asked how with such a record of inaccuracy he could be *the* government adviser on these issues. The answer to that question is the same with Drosten, Ferguson and Fauci – they keep on giving the authorities the 'conclusions' and 'advice' they want to hear. Drosten certainly produced the goods for them in January, 2020, with his PCR protocol garbage and provided the foundation of what German internal medicine specialist Dr Claus Köhnlein, co-author of *Virus Mania*, called the 'test pandemic'. The 22 scientists in the *Eurosurveillance* challenge called out conflicts of interest within the Drosten 'protocol' group and with good reason. Olfert Landt, a regular co-author of Drosten 'studies', owns the biotech company TIB Molbiol Syntheselabor GmbH in Berlin which manufactures and sells the tests that Drosten and his mates come up with. They have done this with SARS, Enterotoxigenic E. coli (ETEC), MERS, Zika 'virus', yellow fever, and now 'Covid'. Landt told the *Berliner Zeitung* newspaper:

The testing, design and development came from the Charité [Drosten and Corman]. We simply implemented it immediately in the form of a kit. And if we don't have the virus, which originally only existed in Wuhan, we can make a synthetic gene to simulate the genome of the virus. That's what we did very quickly.

This is more confirmation that the Drosten test was designed without access to the 'virus' and only a synthetic simulation which is what SARS-CoV-2 really is – a computer-generated synthetic fiction. It's quite an enterprise they have going here. A Drosten team decides what the test for something should be and Landt's biotech company flogs it to governments and medical systems across the world. His company must have made an absolute fortune since the 'Covid' hoax began. Dr Reiner Fuellmich, a prominent German consumer protection trial lawyer in Germany and California, is on Drosten's case and that of Tedros at the World Health Organization for crimes against humanity with a class-action lawsuit being prepared in the United States and other legal action in Germany.

Why China?

Scamming the world with a 'virus' that doesn't exist would seem impossible on the face of it, but not if you have control of the relatively few people that make policy decisions and the great majority of the global media. Remember it's not about changing 'real' reality it's about controlling *perception* of reality. You don't have to make something happen you only have to make people *believe* that it's happening. Renegade Minds understand this and are therefore much harder to swindle. 'Covid-19' is not a 'real' 'virus'. It's a mind virus, like a computer virus, which has infected the minds, not the bodies, of billions. It all started, publically at least, in China and that alone is of central significance. The Cult was behind the revolution led by its asset Mao Zedong, or Chairman Mao, which established the People's Republic of China on October 1st, 1949. It should have been called The Cult's Republic of China, but the name had to reflect the recurring illusion that vicious dictatorships are run by and for the people (see all the 'Democratic Republics' controlled by tyrants). In the same way we have the 'Biden' Democratic Republic of America officially ruled by a puppet tyrant (at least temporarily) on behalf of Cult tyrants. The creation of Mao's merciless communist/fascist dictatorship was part of a frenzy of activity by the Cult at the conclusion of World War Two which, like the First World War, it had instigated through its assets in Germany, Britain, France, the United States and elsewhere. Israel was formed in 1948; the Soviet Union expanded its 'Iron Curtain' control, influence and military power with the Warsaw Pact communist alliance in 1955; the United Nations was formed in 1945 as a Cult precursor to world government; and a long list of world bodies would be established including the World Health Organization (1948), World Trade Organization (1948 under another name until 1995), International Monetary Fund (1945) and World Bank (1944). Human society was redrawn and hugely centralised in the global Problem-Reaction-Solution that was World War Two. All these changes were significant. Israel would become the headquarters of the Sabbatians

and the revolution in China would prepare the ground and control system for the events of 2019/2020.

Renegade Minds know there are no borders except for public consumption. The Cult is a seamless, borderless global entity and to understand the game we need to put aside labels like borders, nations, countries, communism, fascism and democracy. These delude the population into believing that countries are ruled within their borders by a government of whatever shade when these are mere agencies of a global power. America's illusion of democracy and China's communism/fascism are subsidiaries – vehicles – for the same agenda. We may hear about conflict and competition between America and China and on the lower levels that will be true; but at the Cult level they are branches of the same company in the way of the McDonald's example I gave earlier. I have tracked in the books over the years support by US governments of both parties for Chinese Communist Party infiltration of American society through allowing the sale of land, even military facilities, and the acquisition of American business and university influence. All this is underpinned by the infamous stealing of intellectual property and technological know-how. Cult-owned Silicon Valley corporations waive their fraudulent 'morality' to do business with human-rights-free China; Cult-controlled Disney has become China's PR department; and China in effect owns 'American' sports such as basketball which depends for much of its income on Chinese audiences. As a result any sports player, coach or official speaking out against China's horrific human rights record is immediately condemned or fired by the China-worshipping National Basketball Association. One of the first acts of China-controlled Biden was to issue an executive order telling federal agencies to stop making references to the 'virus' by the 'geographic location of its origin'. Long-time Congressman Jerry Nadler warned that criticising China, America's biggest rival, leads to hate crimes against Asian people in the United States. So shut up you bigot. China is fast closing in on Israel as a country that must not be criticised which is apt, really, given that Sabbatians control them both. The two countries have

developed close economic, military, technological and strategic ties which include involvement in China's 'Silk Road' transport and economic initiative to connect China with Europe. Israel was the first country in the Middle East to recognise the establishment of Mao's tyranny in 1950 months after it was established.

Project Wuhan – the 'Covid' Psyop

I emphasise again that the Cult plays the long game and what is happening to the world today is the result of centuries of calculated manipulation following a script to take control step-by-step of every aspect of human society. I will discuss later the common force behind all this that has spanned those centuries and thousands of years if the truth be told. Instigating the Mao revolution in China in 1949 with a 2020 'pandemic' in mind is not only how they work – the 71 years between them is really quite short by the Cult's standards of manipulation preparation. The reason for the Cult's Chinese revolution was to create a fiercely-controlled environment within which an extreme structure for human control could be incubated to eventually be unleashed across the world. We have seen this happen since the 'pandemic' emerged from China with the Chinese control-structure founded on AI technology and tyrannical enforcement sweep across the West. Until the moment when the Cult went for broke in the West and put its fascism on public display Western governments had to pay some lip-service to freedom and democracy to not alert too many people to the tyranny-in-the-making. Freedoms were more subtly eroded and power centralised with covert government structures put in place waiting for the arrival of 2020 when that smokescreen of 'freedom' could be dispensed with. The West was not able to move towards tyranny before 2020 anything like as fast as China which was created as a tyranny and had no limits on how fast it could construct the Cult's blueprint for global control. When the time came to impose that structure on the world it was the same Cult-owned Chinese communist/fascist government that provided the excuse – the 'Covid pandemic'. It was absolutely crucial to the Cult plan for the Chinese response to the 'pandemic' –

draconian lockdowns of the entire population – to become the blueprint that Western countries would follow to destroy the livelihoods and freedom of their people. This is why the Cult-owned, Gates-owned, WHO Director-General Tedros said early on:

The Chinese government is to be congratulated for the extraordinary measures it has taken to contain the outbreak. China is actually setting a new standard for outbreak response and it is not an exaggeration.

Forbes magazine said of China: ‘... those measures protected untold millions from getting the disease’. The Rockefeller Foundation ‘epidemic scenario’ document in 2010 said ‘prophetically’:

However, a few countries did fare better – China in particular. The Chinese government’s quick imposition and enforcement of mandatory quarantine for all citizens, as well as its instant and near-hermetic sealing off of all borders, saved millions of lives, stopping the spread of the virus far earlier than in other countries and enabling a swifter post-pandemic recovery.

Once again – *spooky*.

The first official story was the ‘bat theory’ or rather the bat diversion. The source of the ‘virus outbreak’ we were told was a ‘wet market’ in Wuhan where bats and other animals are bought and eaten in horrifically unhygienic conditions. Then another story emerged through the alternative media that the ‘virus’ had been released on purpose or by accident from a BSL-4 (biosafety level 4) laboratory in Wuhan not far from the wet market. The lab was reported to create and work with lethal concoctions and bioweapons. Biosafety level 4 is the highest in the World Health Organization system of safety and containment. Renegade Minds are aware of what I call designer manipulation. The ideal for the Cult is for people to buy its prime narrative which in the opening salvos of the ‘pandemic’ was the wet market story. It knows, however, that there is now a considerable worldwide alternative media of researchers sceptical of anything governments say and they are often given a version of events in a form they can perceive as credible while misdirecting them from the real truth. In this case let them

think that the conspiracy involved is a 'bioweapon virus' released from the Wuhan lab to keep them from the real conspiracy – *there is no 'virus'*. The WHO's current position on the source of the outbreak at the time of writing appears to be: 'We haven't got a clue, mate.' This is a good position to maintain mystery and bewilderment. The inner circle will know where the 'virus' came from – *nowhere*. The bottom line was to ensure the public believed there *was* a 'virus' and it didn't much matter if they thought it was natural or had been released from a lab. The belief that there was a 'deadly virus' was all that was needed to trigger global panic and fear. The population was terrified into handing their power to authority and doing what they were told. They had to or they were 'all gonna die'.

In March, 2020, information began to come my way from real doctors and scientists and my own additional research which had my intuition screaming: 'Yes, that's it! *There is no virus.*' The 'bioweapon' was not the 'virus'; it was the '*vaccine*' already being talked about that would be the bioweapon. My conclusion was further enhanced by happenings in Wuhan. The 'virus' was said to be sweeping the city and news footage circulated of people collapsing in the street (which they've never done in the West with the same 'virus'). The Chinese government was building 'new hospitals' in a matter of ten days to 'cope with demand' such was the virulent nature of the 'virus'. Yet in what seemed like no time the 'new hospitals' closed – even if they even opened – and China declared itself 'virus-free'. It was back to business as usual. This was more propaganda to promote the Chinese draconian lockdowns in the West as the way to 'beat the virus'. Trouble was that we subsequently had lockdown after lockdown, but never business as usual. As the people of the West and most of the rest of the world were caught in an ever-worsening spiral of lockdown, social distancing, masks, isolated old people, families forced apart, and livelihood destruction, it was party-time in Wuhan. Pictures emerged of thousands of people enjoying pool parties and concerts. It made no sense until you realised there never was a 'virus' and the

whole thing was a Cult set-up to transform human society out of one of its major global strongholds – China.

How is it possible to deceive virtually the entire world population into believing there is a deadly virus when there is not even a 'virus' let alone a deadly one? It's nothing like as difficult as you would think and that's clearly true because it happened.

Postscript: See end of book Postscript for more on the 'Wuhan lab virus release' story which the authorities and media were pushing heavily in the summer of 2021 to divert attention from the truth that the 'Covid virus' is pure invention.

CHAPTER FIVE

There is no 'virus'

You can fool some of the people all of the time, and all of the people some of the time, but you cannot fool all of the people all of the time
Abraham Lincoln

The greatest form of mind control is repetition. The more you repeat the same mantra of alleged 'facts' the more will accept them to be true. It becomes an 'everyone knows that, mate'. If you can also censor any other version or alternative to your alleged 'facts' you are pretty much home and cooking.

By the start of 2020 the Cult owned the global mainstream media almost in its entirety to spew out its 'Covid' propaganda and ignore or discredit any other information and view. Cult-owned social media platforms in Cult-owned Silicon Valley were poised and ready to unleash a campaign of ferocious censorship to obliterate all but the official narrative. To complete the circle many demands for censorship by Silicon Valley were led by the mainstream media as 'journalists' became full-out enforcers for the Cult both as propagandists and censors. Part of this has been the influx of young people straight out of university who have become 'journalists' in significant positions. They have no experience and a headful of programmed perceptions from their years at school and university at a time when today's young are the most perceptually-targeted generations in known human history given the insidious impact of technology. They enter the media perceptually prepared and ready to repeat the narratives of the system that programmed them to

repeat its narratives. The BBC has a truly pathetic 'specialist disinformation reporter' called Marianna Spring who fits this bill perfectly. She is clueless about the world, how it works and what is really going on. Her role is to discredit anyone doing the job that a proper journalist would do and system-serving hacks like Spring wouldn't dare to do or even see the need to do. They are too busy licking the arse of authority which can never be wrong and, in the case of the BBC propaganda programme, *Panorama*, contacting payments systems such as PayPal to have a donations page taken down for a film company making documentaries questioning vaccines. Even the BBC soap opera *EastEnders* included a disgracefully biased scene in which an inarticulate white working class woman was made to look foolish for questioning the 'vaccine' while a well-spoken black man and Asian woman promoted the government narrative. It ticked every BBC box and the fact that the black and minority community was resisting the 'vaccine' had nothing to do with the way the scene was written. The BBC has become a disgusting tyrannical propaganda and censorship operation that should be defunded and disbanded and a free media take its place with a brief to stop censorship instead of demanding it. A BBC 'interview' with Gates goes something like: 'Mr Gates, sir, if I can call you sir, would you like to tell our audience why you are such a great man, a wonderful humanitarian philanthropist, and why you should absolutely be allowed as a software salesman to decide health policy for approaching eight billion people? Thank you, sir, please sir.' Propaganda programming has been incessant and merciless and when all you hear is the same story from the media, repeated by those around you who have only heard the same story, is it any wonder that people on a grand scale believe absolute mendacious garbage to be true? You are about to see, too, why this level of information control is necessary when the official 'Covid' narrative is so nonsensical and unsupportable by the evidence.

Structure of Deceit

The pyramid structure through which the 'Covid' hoax has been manifested is very simple and has to be to work. As few people as possible have to be involved with full knowledge of what they are doing – and why – or the real story would get out. At the top of the pyramid are the inner core of the Cult which controls Bill Gates who, in turn, controls the World Health Organization through his pivotal funding and his puppet Director-General mouthpiece, Tedros. Before he was appointed Tedros was chair of the Gates-founded Global Fund to 'fight against AIDS, tuberculosis and malaria', a board member of the Gates-funded 'vaccine alliance' GAVI, and on the board of another Gates-funded organisation. Gates owns him and picked him for a specific reason – Tedros is a crook and worse. 'Dr' Tedros (he's not a medical doctor, the first WHO chief not to be) was a member of the tyrannical Marxist government of Ethiopia for decades with all its human rights abuses. He has faced allegations of corruption and misappropriation of funds and was exposed three times for covering up cholera epidemics while Ethiopia's health minister. Tedros appointed the mass-murdering genocidal Zimbabwe dictator Robert Mugabe as a WHO goodwill ambassador for public health which, as with Tedros, is like appointing a psychopath to run a peace and love campaign. The move was so ridiculous that he had to drop Mugabe in the face of widespread condemnation. American economist David Steinman, a Nobel peace prize nominee, lodged a complaint with the International Criminal Court in The Hague over alleged genocide by Tedros when he was Ethiopia's foreign minister. Steinman says Tedros was a 'crucial decision maker' who directed the actions of Ethiopia's security forces from 2013 to 2015 and one of three officials in charge when those security services embarked on the 'killing' and 'torturing' of Ethiopians. You can see where Tedros is coming from and it's sobering to think that he has been the vehicle for Gates and the Cult to direct the global response to 'Covid'. Think about that. A psychopathic Cult dictates to psychopath Gates who dictates to psychopath Tedros who dictates how countries of the world must respond to a 'Covid virus' never scientifically shown to exist. At the same time psychopathic Cult-owned Silicon Valley information

giants like Google, YouTube, Facebook and Twitter announced very early on that they would give the Cult/Gates/Tedros/WHO version of the narrative free advertising and censor those who challenged their intelligence-insulting, mendacious story.

The next layer in the global 'medical' structure below the Cult, Gates and Tedros are the chief medical officers and science 'advisers' in each of the WHO member countries which means virtually all of them. Medical officers and arbiters of science (they're not) then take the WHO policy and recommended responses and impose them on their country's population while the political 'leaders' say they are deciding policy (they're clearly not) by 'following the science' on the advice of the 'experts' – the same medical officers and science 'advisers' (dictators). In this way with the rarest of exceptions the entire world followed the same policy of lockdown, people distancing, masks and 'vaccines' dictated by the psychopathic Cult, psychopathic Gates and psychopathic Tedros who we are supposed to believe give a damn about the health of the world population they are seeking to enslave. That, amazingly, is all there is to it in terms of crucial decision-making. Medical staff in each country then follow like sheep the dictates of the shepherds at the top of the national medical hierarchies – chief medical officers and science 'advisers' who themselves follow like sheep the shepherds of the World Health Organization and the Cult. Shepherds at the national level often have major funding and other connections to Gates and his Bill and Melinda Gates Foundation which carefully hands out money like confetti at a wedding to control the entire global medical system from the WHO down.

Follow the money

Christopher Whitty, Chief Medical Adviser to the UK Government at the centre of 'virus' policy, a senior adviser to the government's Scientific Advisory Group for Emergencies (SAGE), and Executive Board member of the World Health Organization, was gifted a grant of \$40 million by the Bill and Melinda Gates Foundation for malaria research in Africa. The BBC described the unelected Whitty as 'the

official who will probably have the greatest impact on our everyday lives of any individual policymaker in modern times' and so it turned out. What Gates and Tedros have said Whitty has done like his equivalents around the world. Patrick Vallance, co-chair of SAGE and the government's Chief Scientific Adviser, is a former executive of Big Pharma giant GlaxoSmithKline with its fundamental financial and business connections to Bill Gates. In September, 2020, it was revealed that Vallance owned a deferred bonus of shares in GlaxoSmithKline worth £600,000 while the company was 'developing' a 'Covid vaccine'. Move along now – nothing to see here – what could possibly be wrong with that? Imperial College in London, a major player in 'Covid' policy in Britain and elsewhere with its 'Covid-19' Response Team, is funded by Gates and has big connections to China while the now infamous Professor Neil Ferguson, the useless 'computer modeller' at Imperial College is also funded by Gates. Ferguson delivered the dramatically inaccurate excuse for the first lockdowns (much more in the next chapter). The Institute for Health Metrics and Evaluation (IHME) in the United States, another source of outrageously false 'Covid' computer models to justify lockdowns, is bankrolled by Gates who is a vehement promotor of lockdowns. America's version of Whitty and Vallance, the again now infamous Anthony Fauci, has connections to 'Covid vaccine' maker Moderna as does Bill Gates through funding from the Bill and Melinda Gates Foundation. Fauci is director of the National Institute of Allergy and Infectious Diseases (NIAID), a major recipient of Gates money, and they are very close. Deborah Birx who was appointed White House Coronavirus Response Coordinator in February, 2020, is yet another with ties to Gates. Everywhere you look at the different elements around the world behind the coordination and decision making of the 'Covid' hoax there is Bill Gates and his money. They include the World Health Organization; Centers for Disease Control (CDC) in the United States; National Institutes of Health (NIH) of Anthony Fauci; Imperial College and Neil Ferguson; the London School of Hygiene where Chris Whitty worked; Regulatory agencies like the UK Medicines & Healthcare products Regulatory Agency (MHRA)

which gave emergency approval for 'Covid vaccines'; Wellcome Trust; GAVI, the Vaccine Alliance; the Coalition for Epidemic Preparedness Innovations (CEPI); Johns Hopkins University which has compiled the false 'Covid' figures; and the World Economic Forum. A [Nationalfile.com](https://www.nationalfile.com) article said:

Gates has a lot of pull in the medical world, he has a multi-million dollar relationship with Dr. Fauci, and Fauci originally took the Gates line supporting vaccines and casting doubt on [the drug hydroxychloroquine]. Coronavirus response team member Dr. Deborah Birx, appointed by former president Obama to serve as United States Global AIDS Coordinator, also sits on the board of a group that has received billions from Gates' foundation, and Birx reportedly used a disputed Bill Gates-funded model for the White House's Coronavirus effort. Gates is a big proponent for a population lockdown scenario for the Coronavirus outbreak.

Another funder of Moderna is the Defense Advanced Research Projects Agency (DARPA), the technology-development arm of the Pentagon and one of the most sinister organisations on earth. DARPA had a major role with the CIA covert technology-funding operation In-Q-Tel in the development of Google and social media which is now at the centre of global censorship. Fauci and Gates are extremely close and openly admit to talking regularly about 'Covid' policy, but then why wouldn't Gates have a seat at every national 'Covid' table after his Foundation committed \$1.75 billion to the 'fight against Covid-19'. When passed through our Orwellian Translation Unit this means that he has bought and paid for the Cult-driven 'Covid' response worldwide. Research the major 'Covid' response personnel in your own country and you will find the same Gates funding and other connections again and again. Medical and science chiefs following World Health Organization 'policy' sit atop a medical hierarchy in their country of administrators, doctors and nursing staff. These 'subordinates' are told they must work and behave in accordance with the policy delivered from the 'top' of the national 'health' pyramid which is largely the policy delivered by the WHO which is the policy delivered by Gates and the Cult. The whole 'Covid' narrative has been imposed on medical staff by a climate of fear although great numbers don't even need that to comply. They do so through breathtaking levels of ignorance and

include doctors who go through life simply repeating what Big Pharma and their hierarchical masters tell them to say and believe. No wonder Big Pharma 'medicine' is one of the biggest killers on Planet Earth.

The same top-down system of intimidation operates with regard to the Cult Big Pharma cartel which also dictates policy through national and global medical systems in this way. The Cult and Big Pharma agendas are the same because the former controls and owns the latter. 'Health' administrators, doctors, and nursing staff are told to support and parrot the dictated policy or they will face consequences which can include being fired. How sad it's been to see medical staff meekly repeating and imposing Cult policy without question and most of those who can see through the deceit are only willing to speak anonymously off the record. They know what will happen if their identity is known. This has left the courageous few to expose the lies about the 'virus', face masks, overwhelmed hospitals that aren't, and the dangers of the 'vaccine' that isn't a vaccine. When these medical professionals and scientists, some renowned in their field, have taken to the Internet to expose the truth their articles, comments and videos have been deleted by Cult-owned Facebook, Twitter and YouTube. What a real head-shaker to see YouTube videos with leading world scientists and highly qualified medical specialists with an added link underneath to the notorious Cult propaganda website *Wikipedia* to find the 'facts' about the same subject.

HIV – the 'Covid' trial-run

I'll give you an example of the consequences for health and truth that come from censorship and unquestioning belief in official narratives. The story was told by PCR inventor Kary Mullis in his book *Dancing Naked in the Mind Field*. He said that in 1984 he accepted as just another scientific fact that Luc Montagnier of France's Pasteur Institute and Robert Gallo of America's National Institutes of Health had independently discovered that a 'retrovirus' dubbed HIV (human immunodeficiency virus) caused AIDS. They

were, after all, Mullis writes, specialists in retroviruses. This is how the medical and science pyramids work. Something is announced or *assumed* and then becomes an everybody-knows-that purely through repetition of the assumption as if it is fact. Complete crap becomes accepted truth with no supporting evidence and only repetition of the crap. This is how a 'virus' that doesn't exist became the 'virus' that changed the world. The HIV-AIDS fairy story became a multi-billion pound industry and the media poured out propaganda terrifying the world about the deadly HIV 'virus' that caused the lethal AIDS. By then Mullis was working at a lab in Santa Monica, California, to detect retroviruses with his PCR test in blood donations received by the Red Cross. In doing so he asked a virologist where he could find a reference for HIV being the cause of AIDS. 'You don't need a reference,' the virologist said ... '*Everybody knows it.*' Mullis said he wanted to quote a reference in the report he was doing and he said he felt a little funny about not knowing the source of such an important discovery when everyone else seemed to. The virologist suggested he cite a report by the Centers for Disease Control and Prevention (CDC) on morbidity and mortality. Mullis read the report, but it only said that an organism had been identified and did not say how. The report did not identify the original scientific work. Physicians, however, *assumed* (key recurring theme) that if the CDC was convinced that HIV caused AIDS then proof must exist. Mullis continues:

I did computer searches. Neither Montagnier, Gallo, nor anyone else had published papers describing experiments which led to the conclusion that HIV probably caused AIDS. I read the papers in *Science* for which they had become well known as AIDS doctors, but all they had said there was that they had found evidence of a past infection by something which was probably HIV in some AIDS patients.

They found antibodies. Antibodies to viruses had always been considered evidence of past disease, not present disease. Antibodies signaled that the virus had been defeated. The patient had saved himself. There was no indication in these papers that this virus caused a disease. They didn't show that everybody with the antibodies had the disease. In fact they found some healthy people with antibodies.

Mullis asked why their work had been published if Montagnier and Gallo hadn't really found this evidence, and why had they been fighting so hard to get credit for the discovery? He says he was hesitant to write 'HIV is the probable cause of AIDS' until he found published evidence to support that. 'Tens of thousands of scientists and researchers were spending billions of dollars a year doing research based on this idea,' Mullis writes. 'The reason had to be there somewhere; otherwise these people would not have allowed their research to settle into one narrow channel of investigation.' He said he lectured about PCR at numerous meetings where people were always talking about HIV and he asked them how they knew that HIV was the cause of AIDS:

Everyone said something. Everyone had the answer at home, in the office, in some drawer. They all knew, and they would send me the papers as soon as they got back. But I never got any papers. Nobody ever sent me the news about how AIDS was caused by HIV.

Eventually Mullis was able to ask Montagnier himself about the reference proof when he lectured in San Diego at the grand opening of the University of California AIDS Research Center. Mullis says this was the last time he would ask his question without showing anger. Montagnier said he should reference the CDC report. 'I read it', Mullis said, and it didn't answer the question. 'If Montagnier didn't know the answer who the hell did?' Then one night Mullis was driving when an interview came on National Public Radio with Peter Duesberg, a prominent virologist at Berkeley and a California Scientist of the Year. Mullis says he finally understood why he could not find references that connected HIV to AIDS – *there weren't any!* No one had ever proved that HIV causes AIDS even though it had spawned a multi-billion pound global industry and the media was repeating this as fact every day in their articles and broadcasts terrifying the shit out of people about AIDS and giving the impression that a positive test for HIV (see 'Covid') was a death sentence. Duesberg was a threat to the AIDS gravy train and the agenda that underpinned it. He was therefore abused and castigated after he told the Proceedings of the National Academy of Sciences

there was no good evidence implicating the new 'virus'. Editors rejected his manuscripts and his research funds were deleted. Mullis points out that the CDC has defined AIDS as one of more than 30 diseases *if accompanied* by a positive result on a test that detects antibodies to HIV; but those same diseases are not defined as AIDS cases when antibodies are not detected:

If an HIV-positive woman develops uterine cancer, for example, she is considered to have AIDS. If she is not HIV positive, she simply has uterine cancer. An HIV-positive man with tuberculosis has AIDS; if he tests negative he simply has tuberculosis. If he lives in Kenya or Colombia, where the test for HIV antibodies is too expensive, he is simply presumed to have the antibodies and therefore AIDS, and therefore he can be treated in the World Health Organization's clinic. It's the only medical help available in some places. And it's free, because the countries that support WHO are worried about AIDS.

Mullis accuses the CDC of continually adding new diseases (see ever more 'Covid symptoms') to the grand AIDS definition and of virtually doctoring the books to make it appear as if the disease continued to spread. He cites how in 1993 the CDC enormously broadened its AIDS definition and county health authorities were delighted because they received \$2,500 per year from the Federal government for every reported AIDS case. Ladies and gentlemen, I have just described, via Kary Mullis, the 'Covid pandemic' of 2020 and beyond. Every element is the same and it's been pulled off in the same way by the same networks.

The 'Covid virus' exists? Okay – prove it. Er ... still waiting

What Kary Mullis described with regard to 'HIV' has been repeated with 'Covid'. A claim is made that a new, or 'novel', infection has been found and the entire medical system of the world repeats that as fact exactly as they did with HIV and AIDS. No one in the mainstream asks rather relevant questions such as 'How do you know?' and 'Where is your proof?' The SARS-Cov-2 'virus' and the 'Covid-19 disease' became an overnight 'everybody-knows-that'. The origin could be debated and mulled over, but what you could not suggest was that 'SARS-Cov-2' didn't exist. That would be

ridiculous. 'Everybody knows' the 'virus' exists. Well, I didn't for one along with American proper doctors like Andrew Kaufman and Tom Cowan and long-time American proper journalist Jon Rappaport. We dared to pursue the obvious and simple question: 'Where's the evidence?' The overwhelming majority in medicine, journalism and the general public did not think to ask that. After all, *everyone knew* there was a new 'virus'. Everyone was saying so and I heard it on the BBC. Some would eventually argue that the 'deadly virus' was nothing like as deadly as claimed, but few would venture into the realms of its very existence. Had they done so they would have found that the evidence for that claim had gone AWOL as with HIV causes AIDS. In fact, not even that. For something to go AWOL it has to exist in the first place and scientific proof for a 'SARS-Cov-2' can be filed under nothing, nowhere and zilch.

Dr Andrew Kaufman is a board-certified forensic psychiatrist in New York State, a Doctor of Medicine and former Assistant Professor and Medical Director of Psychiatry at SUNY Upstate Medical University, and Medical Instructor of Hematology and Oncology at the Medical School of South Carolina. He also studied biology at the Massachusetts Institute of Technology (MIT) and trained in Psychiatry at Duke University. Kaufman is retired from allopathic medicine, but remains a consultant and educator on natural healing. I saw a video of his very early on in the 'Covid' hoax in which he questioned claims about the 'virus' in the absence of any supporting evidence and with plenty pointing the other way. I did everything I could to circulate his work which I felt was asking the pivotal questions that needed an answer. I can recommend an excellent pull-together interview he did with the website The Last Vagabond entitled *Dr Andrew Kaufman: Virus Isolation, Terrain Theory and Covid-19* and his website is andrewkaufmanmd.com. Kaufman is not only a forensic psychiatrist; he is forensic in all that he does. He always reads original scientific papers, experiments and studies instead of second-third-fourth-hand reports about the 'virus' in the media which are repeating the repeated repetition of the narrative. When he did so with the original Chinese 'virus' papers Kaufman

realised that there was no evidence of a 'SARS-Cov-2'. They had never – from the start – shown it to exist and every repeat of this claim worldwide was based on the accepted existence of proof that was nowhere to be found – see Kary Mullis and HIV. Here we go again.

Let's postulate

Kaufman discovered that the Chinese authorities immediately concluded that the cause of an illness that broke out among about 200 initial patients in Wuhan was a 'new virus' when there were no grounds to make that conclusion. The alleged 'virus' was not isolated from other genetic material in their samples and then shown through a system known as Koch's postulates to be the causative agent of the illness. The world was told that the SARS-Cov-2 'virus' caused a disease they called 'Covid-19' which had 'flu-like' symptoms and could lead to respiratory problems and pneumonia. If it wasn't so tragic it would almost be funny. *'Flu-like' symptoms? Pneumonia? Respiratory disease?* What in CHINA and particularly in Wuhan, one of the most polluted cities in the world with a resulting epidemic of respiratory disease?? Three hundred thousand people get pneumonia in China every year and there are nearly a billion cases worldwide of 'flu-like symptoms'. These have a whole range of causes – including pollution in Wuhan – but no other possibility was credibly considered in late 2019 when the world was told there was a new and deadly 'virus'. The global prevalence of pneumonia and 'flu-like systems' gave the Cult networks unlimited potential to re-diagnose these other causes as the mythical 'Covid-19' and that is what they did from the very start. Kaufman revealed how Chinese medical and science authorities (all subordinates to the Cult-owned communist government) took genetic material from the lungs of only a few of the first patients. The material contained their own cells, bacteria, fungi and other microorganisms living in their bodies. The only way you could prove the existence of the 'virus' and its responsibility for the alleged 'Covid-19' was to isolate the virus from all the other material – a process also known as 'purification' – and

then follow the postulates sequence developed in the late 19th century by German physician and bacteriologist Robert Koch which became the 'gold standard' for connecting an alleged causation agent to a disease:

1. The microorganism (bacteria, fungus, virus, etc.) must be present in every case of the disease and all patients must have the same symptoms. It must also *not be present in healthy individuals*.
2. The microorganism must be isolated from the host with the disease. If the microorganism is a bacteria or fungus it must be grown in a pure culture. If it is a virus, it must be purified (i.e. containing no other material except the virus particles) from a clinical sample.
3. The specific disease, with all of its characteristics, must be reproduced when the infectious agent (the purified virus or a pure culture of bacteria or fungi) is inoculated into a healthy, susceptible host.
4. The microorganism must be recoverable from the experimentally infected host as in step 2.

Not one of these criteria has been met in the case of 'SARS-Cov-2' and 'Covid-19'. Not ONE. EVER. Robert Koch refers to bacteria and not viruses. What are called 'viral particles' are so minute (hence masks are useless by any definition) that they could only be seen after the invention of the electron microscope in the 1930s and can still only be observed through that means. American bacteriologist and virologist Thomas Milton Rivers, the so-called 'Father of Modern Virology' who was very significantly director of the Rockefeller Institute for Medical Research in the 1930s, developed a less stringent version of Koch's postulates to identify 'virus' causation known as 'Rivers criteria'. 'Covid' did not pass that process either. Some even doubt whether any 'virus' can be isolated from other particles containing genetic material in the Koch method. Freedom of Information requests in many countries asking for scientific proof that the 'Covid virus' has been purified and isolated and shown to exist have all come back with a 'we don't have that' and when this happened with a request to the UK Department of Health they added this comment:

However, outside of the scope of the [Freedom of Information Act] and on a discretionary basis, the following information has been advised to us, which may be of interest. Most infectious diseases are caused by viruses, bacteria or fungi. Some bacteria or fungi have the capacity to grow on their own in isolation, for example in colonies on a petri dish. Viruses are different in that they are what we call 'obligate pathogens' – that is, they cannot survive or reproduce without infecting a host ...

... For some diseases, it is possible to establish causation between a microorganism and a disease by isolating the pathogen from a patient, growing it in pure culture and reintroducing it to a healthy organism. These are known as 'Koch's postulates' and were developed in 1882. However, as our understanding of disease and different disease-causing agents has advanced, these are no longer the method for determining causation [Andrew Kaufman asks why in that case are there two published articles falsely claiming to satisfy Koch's postulates].

It has long been known that viral diseases cannot be identified in this way as viruses cannot be grown in 'pure culture'. When a patient is tested for a viral illness, this is normally done by looking for the presence of antigens, or viral genetic code in a host with molecular biology techniques [Kaufman asks how you could know the origin of these chemicals without having a pure culture for comparison].

For the record 'antigens' are defined so:

Invading microorganisms have antigens on their surface that the human body can recognise as being foreign – meaning not belonging to it. When the body recognises a foreign antigen, lymphocytes (white blood cells) produce antibodies, which are complementary in shape to the antigen.

Notwithstanding that this is open to question in relation to 'SARS-Cov-2' the presence of 'antibodies' can have many causes and they are found in people that are perfectly well. Kary Mullis said: 'Antibodies ... had always been considered evidence of past disease, not present disease.'

'Covid' really is a *computer* 'virus'

Where the UK Department of Health statement says 'viruses' are now 'diagnosed' through a 'viral genetic code in a host with molecular biology techniques', they mean ... *the PCR test* which its inventor said cannot test for infectious disease. They have no credible method of connecting a 'virus' to a disease and we will see that there is no scientific proof that any 'virus' causes any disease or there is any such thing as a 'virus' in the way that it is described. Tenacious Canadian researcher Christine Massey and her team made

some 40 Freedom of Information requests to national public health agencies in different countries asking for proof that SARS-CoV-2 has been isolated and not one of them could supply that information. Massey said of her request in Canada: 'Freedom of Information reveals Public Health Agency of Canada has no record of 'SARS-COV-2' isolation performed by anyone, anywhere, ever.' If you accept the comment from the UK Department of Health it's because they can't isolate a 'virus'. Even so many 'science' papers claimed to have isolated the 'Covid virus' until they were questioned and had to admit they hadn't. A reply from the Robert Koch Institute in Germany was typical: 'I am not aware of a paper which purified isolated SARS-CoV-2.' So what the hell was Christian Drosten and his gang using to design the 'Covid' testing protocol that has produced all the illusory Covid' cases and 'Covid' deaths when the head of the Chinese version of the CDC admitted there was a problem right from the start in that the 'virus' had never been isolated/purified? Breathe deeply: What they are calling 'Covid' is actually created by a *computer program* i.e. *they made it up* – er, that's it. They took lung fluid, with many sources of genetic material, from one single person alleged to be infected with Covid-19 by a PCR test which they *claimed*, without clear evidence, contained a 'virus'. They used several computer programs to create a model of a theoretical virus genome sequence from more than fifty-six million small sequences of RNA, each of an unknown source, assembling them like a puzzle with no known solution. The computer filled in the gaps with sequences from bits in the gene bank to make it look like a bat SARS-like coronavirus! A wave of the magic wand and poof, an *in silico* (computer-generated) genome, a scientific fantasy, was created. UK health researcher Dr Kevin Corbett made the same point with this analogy:

... It's like giving you a few bones and saying that's your fish. It could be any fish. Not even a skeleton. Here's a few fragments of bones. That's your fish ... It's all from gene bank and the bits of the virus sequence that weren't there they made up.

They synthetically created them to fill in the blanks. That's what genetics is; it's a code. So it's ABBCCDDDD and you're missing some what you think is EEE so you put it in. It's all

synthetic. You just manufacture the bits that are missing. This is the end result of the geneticization of virology. This is basically a computer virus.

Further confirmation came in an email exchange between British citizen journalist Frances Leader and the government's Medicines & Healthcare Products Regulatory Agency (the Gates-funded MHRA) which gave emergency permission for untested 'Covid vaccines' to be used. The agency admitted that the 'vaccine' is not based on an isolated 'virus', but comes from a *computer-generated model*. Frances Leader was naturally banned from Cult-owned fascist Twitter for making this exchange public. The process of creating computer-generated alleged 'viruses' is called 'in silico' or 'in silicon' – computer chips – and the term 'in silico' is believed to originate with biological experiments using only a computer in 1989. 'Vaccines' involved with 'Covid' are also produced 'in silico' or by computer not a natural process. If the original 'virus' is nothing more than a made-up computer model how can there be 'new variants' of something that never existed in the first place? They are not new 'variants'; they are new *computer models* only minutely different to the original program and designed to further terrify the population into having the 'vaccine' and submitting to fascism. You want a 'new variant'? Click, click, enter – there you go. Tell the medical profession that you have discovered a 'South African variant', 'UK variants' or a 'Brazilian variant' and in the usual HIV-causes-AIDS manner they will unquestioningly repeat it with no evidence whatsoever to support these claims. They will go on television and warn about the dangers of 'new variants' while doing nothing more than repeating what they have been told to be true and knowing that any deviation from that would be career suicide. Big-time insiders will know it's a hoax, but much of the medical community is clueless about the way they are being played and themselves play the public without even being aware they are doing so. What an interesting 'coincidence' that AstraZeneca and Oxford University were conducting 'Covid vaccine trials' in the three countries – the UK, South Africa and Brazil – where the first three 'variants' were claimed to have 'broken out'.

Here's your 'virus' – it's a unicorn

Dr Andrew Kaufman presented a brilliant analysis describing how the 'virus' was imagined into fake existence when he dissected an article published by *Nature* and written by 19 authors detailing *alleged* 'sequencing of a complete viral genome' of the 'new SARS-CoV-2 virus'. This computer-modelled *in silico* genome was used as a template for all subsequent genome sequencing experiments that resulted in the so-called variants which he said now number more than 6,000. The fake genome was constructed from more than 56 million individual short strands of RNA. Those little pieces were assembled into longer pieces by finding areas of overlapping sequences. The computer programs created over two million possible combinations from which the authors simply chose the longest one. They then compared this to a 'bat virus' and the computer 'alignment' rearranged the sequence and filled in the gaps! They called this computer-generated abomination the 'complete genome'. Dr Tom Cowan, a fellow medical author and collaborator with Kaufman, said such computer-generation constitutes scientific fraud and he makes this superb analogy:

Here is an equivalency: A group of researchers claim to have found a unicorn because they found a piece of a hoof, a hair from a tail, and a snippet of a horn. They then add that information into a computer and program it to re-create the unicorn, and they then claim this computer re-creation is the real unicorn. Of course, they had never actually seen a unicorn so could not possibly have examined its genetic makeup to compare their samples with the actual unicorn's hair, hooves and horn.

The researchers claim they decided which is the real genome of SARS-CoV-2 by 'consensus', sort of like a vote. Again, different computer programs will come up with different versions of the imaginary 'unicorn', so they come together as a group and decide which is the real imaginary unicorn.

This is how the 'virus' that has transformed the world was brought into fraudulent 'existence'. Extraordinary, yes, but as the Nazis said the bigger the lie the more will believe it. Cowan, however, wasn't finished and he went on to identify what he called the real blockbuster in the paper. He quotes this section from a paper written

by virologists and published by the CDC and then explains what it means:

Therefore, we examined the capacity of SARS-CoV-2 to infect and replicate in several common primate and human cell lines, including human adenocarcinoma cells (A549), human liver cells (HUH 7.0), and human embryonic kidney cells (HEK-293T). In addition to Vero E6 and Vero CCL81 cells. ... Each cell line was inoculated at high multiplicity of infection and examined 24h post-infection.

No CPE was observed in any of the cell lines except in Vero cells, which grew to greater than 10 to the 7th power at 24 h post-infection. In contrast, HUH 7.0 and 293T showed only modest viral replication, and A549 cells were incompatible with SARS CoV-2 infection.

Cowan explains that when virologists attempt to prove infection they have three possible 'hosts' or models on which they can test. The first was humans. Exposure to humans was generally not done for ethical reasons and has never been done with SARS-CoV-2 or any coronavirus. The second possible host was animals. Cowan said that forgetting for a moment that they never actually use purified virus when exposing animals they do use solutions that they *claim* contain the virus. Exposure to animals has been done with SARS-CoV-2 in an experiment involving mice and this is what they found: *None of the wild (normal) mice got sick*. In a group of genetically-modified mice, a statistically insignificant number lost weight and had slightly bristled fur, but they experienced nothing like the illness called 'Covid-19'. Cowan said the third method – the one they mostly rely on – is to inoculate solutions they *say* contain the virus onto a variety of tissue cultures. This process had never been shown to kill tissue *unless* the sample material was starved of nutrients and poisoned as *part of the process*. Yes, incredibly, in tissue experiments designed to show the 'virus' is responsible for killing the tissue they starve the tissue of nutrients and add toxic drugs including antibiotics and they do not have control studies to see if it's the starvation and poisoning that is degrading the tissue rather than the 'virus' they allege to be in there somewhere. You want me to pinch you? Yep, I understand. Tom Cowan said this about the whole nonsensical farce as he explains what that quote from the CDC paper really means:

The shocking thing about the above quote is that using their own methods, the virologists found that solutions containing SARS-CoV-2 – even in high amounts – were NOT, I repeat NOT, infective to any of the three human tissue cultures they tested. In plain English, this means they proved, on their terms, that this ‘new coronavirus’ is not infectious to human beings. It is ONLY infective to monkey kidney cells, and only then when you add two potent drugs (gentamicin and amphotericin), known to be toxic to kidneys, to the mix.

My friends, read this again and again. These virologists, published by the CDC, performed a clear proof, on their terms, showing that the SARS-CoV-2 virus is harmless to human beings. That is the only possible conclusion, but, unfortunately, this result is not even mentioned in their conclusion. They simply say they can provide virus stocks cultured only on monkey Vero cells, thanks for coming.

Cowan concluded: ‘If people really understood how this “science” was done, I would hope they would storm the gates and demand honesty, transparency and truth.’ Dr Michael Yeadon, former Vice President and Chief Scientific Adviser at drug giant Pfizer has been a vocal critic of the ‘Covid vaccine’ and its potential for multiple harm. He said in an interview in April, 2021, that ‘not one [vaccine] has the virus. He was asked why vaccines normally using a ‘dead’ version of a disease to activate the immune system were not used for ‘Covid’ and instead we had the synthetic methods of the ‘mRNA Covid vaccine’. Yeadon said that to do the former ‘you’d have to have some of [the virus] wouldn’t you?’ He added: ‘No-one’s got any – seriously.’ Yeadon said that surely they couldn’t have fooled the whole world for a year without having a virus, ‘but oddly enough ask around – no one’s got it’. He didn’t know why with all the ‘great labs’ around the world that the virus had not been isolated – ‘Maybe they’ve been too busy running bad PCR tests and vaccines that people don’t need.’ What is today called ‘science’ is not ‘science’ at all. Science is no longer what is, but whatever people can be manipulated to *believe* that it is. Real science has been hijacked by the Cult to dispense and produce the ‘expert scientists’ and contentions that suit the agenda of the Cult. How big-time this has happened with the ‘Covid’ hoax which is entirely based on fake science delivered by fake ‘scientists’ and fake ‘doctors’. The human-caused climate change hoax is also entirely based on fake science delivered by fake ‘scientists’ and fake ‘climate experts’. In both cases real

scientists, climate experts and doctors have their views suppressed and deleted by the Cult-owned science establishment, media and Silicon Valley. This is the 'science' that politicians claim to be 'following' and a common denominator of 'Covid' and climate are Cult psychopaths Bill Gates and his mate Klaus Schwab at the Gates-funded World Economic Forum. But, don't worry, it's all just a coincidence and absolutely nothing to worry about. Zzzzzzzzz.

What is a 'virus' REALLY?

Dr Tom Cowan is one of many contesting the very existence of viruses let alone that they cause disease. This is understandable when there is no scientific evidence for a disease-causing 'virus'. German virologist Dr Stefan Lanka won a landmark case in 2017 in the German Supreme Court over his contention that there is no such thing as a measles virus. He had offered a big prize for anyone who could prove there is and Lanka won his case when someone sought to claim the money. There is currently a prize of more than 225,000 euros on offer from an Isolate Truth Fund for anyone who can prove the isolation of SARS-CoV-2 and its genetic substance. Lanka wrote in an article headed 'The Misconception Called Virus' that scientists think a 'virus' is causing tissue to become diseased and degraded when in fact it is the *processes they are using* which do that – not a 'virus'. Lanka has done an important job in making this point clear as Cowan did in his analysis of the CDC paper. Lanka says that all claims about viruses as disease-causing pathogens are wrong and based on 'easily recognisable, understandable and verifiable misinterpretations.' Scientists believed they were working with 'viruses' in their laboratories when they were really working with 'typical particles of specific dying tissues or cells ...' Lanka said that the tissue decaying process claimed to be caused by a 'virus' still happens when no alleged 'virus' is involved. It's the *process* that does the damage and not a 'virus'. The genetic sample is deprived of nutrients, removed from its energy supply through removal from the body and then doused in toxic antibiotics to remove any bacteria. He confirms again that establishment scientists do not (pinch me)

conduct control experiments to see if this is the case and if they did they would see the claims that 'viruses' are doing the damage is nonsense. He adds that during the measles 'virus' court case he commissioned an independent laboratory to perform just such a control experiment and the result was that the tissues and cells died in the exact same way as with alleged 'infected' material. This is supported by a gathering number of scientists, doctors and researchers who reject what is called 'germ theory' or the belief in the body being infected by contagious sources emitted by other people. Researchers Dawn Lester and David Parker take the same stance in their highly-detailed and sourced book *What Really Makes You Ill – Why everything you thought you knew about disease is wrong* which was recommended to me by a number of medical professionals genuinely seeking the truth. Lester and Parker say there is no provable scientific evidence to show that a 'virus' can be transmitted between people or people and animals or animals and people:

The definition also claims that viruses are the cause of many diseases, as if this has been definitively proven. But this is not the case; there is no original scientific evidence that definitively demonstrates that any virus is the cause of any disease. The burden of proof for any theory lies with those who proposed it; but none of the existing documents provides 'proof' that supports the claim that 'viruses' are pathogens.

Dr Tom Cowan employs one of his clever analogies to describe the process by which a 'virus' is named as the culprit for a disease when what is called a 'virus' is only material released by cells detoxing themselves from infiltration by chemical or radiation poisoning. The tidal wave of technologically-generated radiation in the 'smart' modern world plus all the toxic food and drink are causing this to happen more than ever. Deluded 'scientists' misread this as a gathering impact of what they wrongly label 'viruses'.

Paper can infect houses

Cowan said in an article for davidicke.com – with his tongue only mildly in his cheek – that he believed he had made a tremendous

discovery that may revolutionise science. He had discovered that small bits of paper are alive, 'well alive-ish', can 'infect' houses, and then reproduce themselves inside the house. The result was that this explosion of growth in the paper inside the house causes the house to explode, blowing it to smithereens. His evidence for this new theory is that in the past months he had carefully examined many of the houses in his neighbourhood and found almost no scraps of paper on the lawns and surrounds of the house. There was an occasional stray label, but nothing more. Then he would return to these same houses a week or so later and with a few, not all of them, particularly the old and decrepit ones, he found to his shock and surprise they were littered with stray bits of paper. He knew then that the paper had infected these houses, made copies of itself, and blew up the house. A young boy on a bicycle at one of the sites told him he had seen a demolition crew using dynamite to explode the house the previous week, but Cowan dismissed this as the idle thoughts of silly boys because 'I was on to something big'. He was on to how 'scientists' mistake genetic material in the detoxifying process for something they call a 'virus'. Cowan said of his house and paper story:

If this sounds crazy to you, it's because it should. This scenario is obviously nuts. But consider this admittedly embellished, for effect, current viral theory that all scientists, medical doctors and virologists currently believe.

He takes the example of the 'novel SARS-Cov2' virus to prove the point. First they take someone with an undefined illness called 'Covid-19' and don't even attempt to find any virus in their sputum. Never mind the scientists still describe how this 'virus', which they have not located attaches to a cell receptor, injects its genetic material, in 'Covid's' case, RNA, into the cell. The RNA once inserted exploits the cell to reproduce itself and makes 'thousands, nay millions, of copies of itself ... Then it emerges victorious to claim its next victim':

If you were to look in the scientific literature for proof, actual scientific proof, that uniform SARS-CoV2 viruses have been properly isolated from the sputum of a sick person, that actual spike proteins could be seen protruding from the virus (which has not been found), you would find that such evidence doesn't exist.

If you go looking in the published scientific literature for actual pictures, proof, that these spike proteins or any viral proteins are ever attached to any receptor embedded in any cell membrane, you would also find that no such evidence exists. If you were to look for a video or documented evidence of the intact virus injecting its genetic material into the body of the cell, reproducing itself and then emerging victorious by budding off the cell membrane, you would find that no such evidence exists.

The closest thing you would find is electron micrograph pictures of cellular particles, possibly attached to cell debris, both of which to be seen were stained by heavy metals, a process that completely distorts their architecture within the living organism. This is like finding bits of paper stuck to the blown-up bricks, thereby proving the paper emerged by taking pieces of the bricks on its way out.

The Enders baloney

Cowan describes the 'Covid' story as being just as make-believe as his paper story and he charts back this fantasy to a Nobel Prize winner called John Enders (1897-1985), an American biomedical scientist who has been dubbed 'The Father of Modern Vaccines'. Enders is claimed to have 'discovered' the process of the viral culture which 'proved' that a 'virus' caused measles. Cowan explains how Enders did this 'by using the EXACT same procedure that has been followed by every virologist to find and characterize every new virus since 1954'. Enders took throat swabs from children with measles and immersed them in 2ml of milk. Penicillin (100u/ml) and the antibiotic streptomycin (50,g/ml) were added and the whole mix was centrifuged – rotated at high speed to separate large cellular debris from small particles and molecules as with milk and cream, for example. Cowan says that if the aim is to find little particles of genetic material ('viruses') in the snot from children with measles it would seem that the last thing you would do is mix the snot with other material – milk –that also has genetic material. 'How are you ever going to know whether whatever you found came from the snot or the milk?' He points out that streptomycin is a 'nephrotoxic' or poisonous-to-the-kidney drug. You will see the relevance of that

shortly. Cowan says that it gets worse, much worse, when Enders describes the culture medium upon which the virus 'grows': 'The culture medium consisted of bovine amniotic fluid (90%), beef embryo extract (5%), horse serum (5%), antibiotics and phenol red as an indicator of cell metabolism.' Cowan asks incredulously: 'Did he just say that the culture medium also contained fluids and tissues that are themselves rich sources of genetic material?' The genetic cocktail, or 'medium', is inoculated onto tissue and cells from rhesus monkey *kidney* tissue. This is where the importance of streptomycin comes in and currently-used antimicrobials and other drugs that are *poisonous to kidneys* and used in ALL modern viral cultures (e.g. gentamicin, streptomycin, and amphotericin). Cowan asks: 'How are you ever going to know from this witch's brew where any genetic material comes from as we now have five different sources of rich genetic material in our mix?' Remember, he says, that all genetic material, whether from monkey kidney tissues, bovine serum, milk, etc., is made from the exact same components. The same central question returns: 'How are you possibly going to know that it was the virus that killed the kidney tissue and not the toxic antibiotic and starvation rations on which you are growing the tissue?' John Enders answered the question himself – *you can't*:

A second agent was obtained from an uninoculated culture of monkey kidney cells. The cytopathic changes [death of the cells] it induced in the unstained preparations could not be distinguished with confidence from the viruses isolated from measles.

The death of the cells ('cytopathic changes') happened in exactly the same manner, whether they inoculated the kidney tissue with the measles snot or not, Cowan says. 'This is evidence that the destruction of the tissue, the very proof of viral causation of illness, was not caused by anything in the snot because they saw the same destructive effect when the snot was not even used ... the cytopathic, i.e., cell-killing, changes come from the process of the culture itself, not from any virus in any snot, period.' Enders quotes in his 1957 paper a virologist called Ruckle as reporting similar findings 'and in addition has isolated an agent from monkey kidney tissue that is so

far indistinguishable from human measles virus'. In other words, Cowan says, these particles called 'measles viruses' are simply and clearly breakdown products of the starved and poisoned tissue. For measles 'virus' see all 'viruses' including the so-called 'Covid virus'. Enders, the 'Father of Modern Vaccines', also said:

There is a potential risk in employing cultures of primate cells for the production of vaccines composed of attenuated virus, since the presence of other agents possibly latent in primate tissues cannot be definitely excluded by any known method.

Cowan further quotes from a paper published in the journal *Viruses* in May, 2020, while the 'Covid pandemic' was well underway in the media if not in reality. 'EVs' here refers to particles of genetic debris from our own tissues, such as exosomes of which more in a moment: 'The remarkable resemblance between EVs and viruses has caused quite a few problems in the studies focused on the analysis of EVs released during viral infections.' Later the paper adds that to date a reliable method that can actually guarantee a complete separation (of EVs from viruses) DOES NOT EXIST. This was published at a time when a fairy tale 'virus' was claimed in total certainty to be causing a fairy tale 'viral disease' called 'Covid-19' – a fairy tale that was already well on the way to transforming human society in the image that the Cult has worked to achieve for so long. Cowan concludes his article:

To summarize, there is no scientific evidence that pathogenic viruses exist. What we think of as 'viruses' are simply the normal breakdown products of dead and dying tissues and cells. When we are well, we make fewer of these particles; when we are starved, poisoned, suffocated by wearing masks, or afraid, we make more.

There is no engineered virus circulating and making people sick. People in laboratories all over the world are making genetically modified products to make people sick. These are called vaccines. There is no virome, no 'ecosystem' of viruses, viruses are not 8%, 50% or 100 % of our genetic material. These are all simply erroneous ideas based on the misconception called a virus.

What is 'Covid'? Load of bollocks

The background described here by Cowan and Lanka was emphasised in the first video presentation that I saw by Dr Andrew Kaufman when he asked whether the 'Covid virus' was in truth a natural defence mechanism of the body called 'exosomes'. These are released by cells when in states of toxicity – see the same themes returning over and over. They are released ever more profusely as chemical and radiation toxicity increases and think of the potential effect therefore of 5G alone as its destructive frequencies infest the human energetic information field with a gathering pace (5G went online in Wuhan in 2019 as the 'virus' emerged). I'll have more about this later. Exosomes transmit a warning to the rest of the body that 'Houston, we have a problem'. Kaufman presented images of exosomes and compared them with 'Covid' under an electron microscope and the similarity was remarkable. They both attach to the same cell receptors (*claimed* in the case of 'Covid'), contain the same genetic material in the form of RNA or ribonucleic acid, and both are found in 'viral cell cultures' with damaged or dying cells. James Hildreth MD, President and Chief Executive Officer of the Meharry Medical College at Johns Hopkins, said: 'The virus is fully an exosome in every sense of the word.' Kaufman's conclusion was that there is no 'virus': 'This entire pandemic is a completely manufactured crisis ... there is no evidence of anyone dying from [this] illness.' Dr Tom Cowan and Sally Fallon Morell, authors of *The Contagion Myth*, published a statement with Dr Kaufman in February, 2021, explaining why the 'virus' does not exist and you can read it that in full in the Appendix.

'Virus' theory can be traced to the 'cell theory' in 1858 of German physician Rudolf Virchow (1821-1920) who contended that disease originates from a single cell infiltrated by a 'virus'. Dr Stefan Lanka said that findings and insights with respect to the structure, function and central importance of tissues in the creation of life, which were already known in 1858, comprehensively refute the cell theory. Virchow ignored them. We have seen the part later played by John Enders in the 1950s and Lanka notes that infection theories were only established as a global dogma through the policies and

eugenics of the Third Reich in Nazi Germany (creation of the same Sabbatian cult behind the 'Covid' hoax). Lanka said: 'Before 1933, scientists dared to contradict this theory; after 1933, these critical scientists were silenced'. Dr Tom Cowan's view is that ill-health is caused by too much of something, too little of something, or toxification from chemicals and radiation – not contagion. We must also highlight as a major source of the 'virus' theology a man still called the 'Father of Modern Virology' – Thomas Milton Rivers (1888-1962). There is no way given the Cult's long game policy that it was a coincidence for the 'Father of Modern Virology' to be director of the Rockefeller Institute for Medical Research from 1937 to 1956 when he is credited with making the Rockefeller Institute a leader in 'viral research'. Cult Rockefellerers were the force behind the creation of Big Pharma 'medicine', established the World Health Organisation in 1948, and have long and close associations with the Gates family that now runs the WHO during the pandemic hoax through mega-rich Cult gofer and psychopath Bill Gates.

Only a Renegade Mind can see through all this bullshit by asking the questions that need to be answered, not taking 'no' or prevarication for an answer, and certainly not hiding from the truth in fear of speaking it. Renegade Minds have always changed the world for the better and they will change this one no matter how bleak it may currently appear to be.

CHAPTER SIX

Sequence of deceit

If you tell the truth, you don't have to remember anything
Mark Twain

Against the background that I have laid out this far the sequence that took us from an invented 'virus' in Cult-owned China in late 2019 to the fascist transformation of human society can be seen and understood in a whole new context.

We were told that a deadly disease had broken out in Wuhan and the world media began its campaign (coordinated by behavioural psychologists as we shall see) to terrify the population into unquestioning compliance. We were shown images of Chinese people collapsing in the street which never happened in the West with what was supposed to be the same condition. In the earliest days when alleged cases and deaths were few the fear register was hysterical in many areas of the media and this would expand into the common media narrative across the world. The real story was rather different, but we were never told that. The Chinese government, one of the Cult's biggest centres of global operation, said they had discovered a new illness with flu-like and pneumonia-type symptoms in a city with such toxic air that it is overwhelmed with flu-like symptoms, pneumonia and respiratory disease. Chinese scientists said it was a new – 'novel' – coronavirus which they called Sars-Cov-2 and that it caused a disease they labelled 'Covid-19'. There was no evidence for this and the 'virus' has never to this day been isolated, purified and its genetic code established from that. It

was from the beginning a computer-generated fiction. Stories of Chinese whistleblowers saying the number of deaths was being suppressed or that the 'new disease' was related to the Wuhan bio-lab misdirected mainstream and alternative media into cul-de-sacs to obscure the real truth – there was no 'virus'.

Chinese scientists took genetic material from the lung fluid of just a few people and said they had found a 'new' disease when this material had a wide range of content. There was no evidence for a 'virus' for the very reasons explained in the last two chapters. The 'virus' has never been shown to (a) exist and (b) cause any disease. People were diagnosed on symptoms that are so widespread in Wuhan and polluted China and with a PCR test that can't detect infectious disease. On this farce the whole global scam was sold to the rest of the world which would also diagnose respiratory disease as 'Covid-19' from symptoms alone or with a PCR test not testing for a 'virus'. Flu miraculously disappeared *worldwide* in 2020 and into 2021 as it was redesignated 'Covid-19'. It was really the same old flu with its 'flu-like' symptoms attributed to 'flu-like' 'Covid-19'. At the same time with very few exceptions the Chinese response of draconian lockdown and fascism was the chosen weapon to respond across the West as recommended by the Cult-owned Tedros at the Cult-owned World Health Organization run by the Cult-owned Gates. All was going according to plan. Chinese scientists – everything in China is controlled by the Cult-owned government – compared their contaminated RNA lung-fluid material with other RNA sequences and said it appeared to be just under 80 percent identical to the SARS-CoV-1 'virus' claimed to be the cause of the SARS (severe acute respiratory syndrome) 'outbreak' in 2003. They decreed that because of this the 'new virus' had to be related and they called it SARS-CoV-2. There are some serious problems with this assumption and *assumption* was all it was. Most 'factual' science turns out to be assumptions repeated into everyone-knows-that. A match of under 80-percent is meaningless. Dr Kaufman makes the point that there's a 96 percent genetic correlation between humans and chimpanzees, but 'no one would say our genetic material is part

of the chimpanzee family'. Yet the Chinese authorities were claiming that a much lower percentage, less than 80 percent, proved the existence of a new 'coronavirus'. For goodness sake human DNA is 60 percent similar to a *banana*.

You are feeling sleepy

The entire 'Covid' hoax is a global Psyop, a psychological operation to program the human mind into believing and fearing a complete fantasy. A crucial aspect of this was what *appeared* to happen in Italy. It was all very well streaming out daily images of an alleged catastrophe in Wuhan, but to the Western mind it was still on the other side of the world in a very different culture and setting. A reaction of 'this could happen to me and my family' was still nothing like as intense enough for the mind-doctors. The Cult needed a Western example to push people over that edge and it chose Italy, one of its major global locations going back to the Roman Empire. An Italian 'Covid' crisis was manufactured in a particular area called Lombardy which just happens to be notorious for its toxic air and therefore respiratory disease. Wuhan, China, *déjà vu*. An hysterical media told horror stories of Italians dying from 'Covid' in their droves and how Lombardy hospitals were being overrun by a tidal wave of desperately ill people needing treatment after being struck down by the 'deadly virus'. Here was the psychological turning point the Cult had planned. Wow, if this is happening in Italy, the Western mind concluded, this indeed could happen to me and my family. Another point is that Italian authorities responded by following the Chinese blueprint so vehemently recommended by the Cult-owned World Health Organization. They imposed fascistic lockdowns on the whole country viciously policed with the help of surveillance drones sweeping through the streets seeking out anyone who escaped from mass house arrest. Livelihoods were destroyed and psychology unravelled in the way we have witnessed since in all lockdown countries. Crucial to the plan was that Italy responded in this way to set the precedent of suspending freedom and imposing fascism in a 'Western liberal democracy'. I emphasised in an

animated video explanation on davidicke.com posted in the summer of 2020 how important it was to the Cult to expand the Chinese lockdown model across the West. Without this, and the bare-faced lie that non-symptomatic people could still transmit a 'disease' they didn't have, there was no way locking down the whole population, sick and not sick, could be pulled off. At just the right time and with no evidence Cult operatives and gofers claimed that people without symptoms could pass on the 'disease'. In the name of protecting the 'vulnerable' like elderly people, who lockdowns would kill by the tens of thousands, we had for the first time healthy people told to isolate as well as the sick. The great majority of people who tested positive had no symptoms because there was nothing wrong with them. It was just a trick made possible by a test not testing for the 'virus'.

Months after my animated video the Gates-funded Professor Neil Ferguson at the Gates-funded Imperial College confirmed that I was right. He didn't say it in those terms, naturally, but he did say it. Ferguson will enter the story shortly for his outrageously crazy 'computer models' that led to Britain, the United States and many other countries following the Chinese and now Italian methods of response. Put another way, following the Cult script. Ferguson said that SAGE, the UK government's scientific advisory group which has controlled 'Covid' policy from the start, wanted to follow the Chinese lockdown model (while they all continued to work and be paid), but they wondered if they could possibly, in Ferguson's words, 'get away with it in Europe'. 'Get away with it'? Who the hell do these moronic, arrogant people think they are? This appalling man Ferguson said that once Italy went into national lockdown they realised they, too, could mimic China:

It's a communist one-party state, we said. We couldn't get away with it in Europe, we thought ... and then Italy did it. And we realised we could. Behind this garbage from Ferguson is a simple fact: Doing the same as China in every country was the plan from the start and Ferguson's 'models' would play a central role in achieving that. It's just a coincidence, of course, and absolutely nothing to worry your little head about.

Oops, sorry, our mistake

Once the Italian segment of the Psyop had done the job it was designed to do a very different story emerged. Italian authorities revealed that 99 percent of those who had 'died from Covid-19' in Italy had one, two, three, or more 'co-morbidities' or illnesses and health problems that could have ended their life. The US Centers for Disease Control and Prevention (CDC) published a figure of 94 percent for Americans dying of 'Covid' while having other serious medical conditions – on average two to three (some five or six) other potential causes of death. In terms of death from an unproven 'virus' I say it is 100 percent. The other one percent in Italy and six percent in the US would presumably have died from 'Covid's' flu-like symptoms with a range of other possible causes in conjunction with a test not testing for the 'virus'. Fox News reported that even more startling figures had emerged in one US county in which 410 of 422 deaths attributed to 'Covid-19' had other potentially deadly health conditions. The Italian National Health Institute said later that the average age of people dying with a 'Covid-19' diagnosis in Italy was about 81. Ninety percent were over 70 with ten percent over 90. In terms of other reasons to die some 80 percent had two or more chronic diseases with half having three or more including cardiovascular problems, diabetes, respiratory problems and cancer. Why is the phantom 'Covid-19' said to kill overwhelmingly old people and hardly affect the young? Old people continually die of many causes and especially respiratory disease which you can re-diagnose 'Covid-19' while young people die in tiny numbers by comparison and rarely of respiratory disease. Old people 'die of Covid' because they die of other things that can be redesignated 'Covid' and it really is that simple.

Flu has flown

The blueprint was in place. Get your illusory 'cases' from a test not testing for the 'virus' and redesignate other causes of death as 'Covid-19'. You have an instant 'pandemic' from something that is nothing more than a computer-generated fiction. With near-on a

billion people having 'flu-like' symptoms every year the potential was limitless and we can see why flu quickly and apparently miraculously disappeared *worldwide* by being diagnosed 'Covid-19'. The painfully bloody obvious was explained away by the childlike media in headlines like this in the UK '*Independent*': 'Not a single case of flu detected by Public Health England this year as Covid restrictions suppress virus'. I kid you not. The masking, social distancing and house arrest that did not make the 'Covid virus' disappear somehow did so with the 'flu virus'. Even worse the article, by a bloke called Samuel Lovett, suggested that maybe the masking, sanitising and other 'Covid' measures should continue to keep the flu away. With a ridiculousness that disturbs your breathing (it's 'Covid-19') the said Lovett wrote: 'With widespread social distancing and mask-wearing measures in place throughout the UK, the usual routes of transmission for influenza have been blocked.' He had absolutely no evidence to support that statement, but look at the consequences of him acknowledging the obvious. With flu not disappearing at all and only being relabelled 'Covid-19' he would have to contemplate that 'Covid' was a hoax on a scale that is hard to imagine. You need guts and commitment to truth to even go there and that's clearly something Samuel Lovett does not have in abundance. He would never have got it through the editors anyway.

Tens of thousands die in the United States alone every winter from flu including many with pneumonia complications. CDC figures record *45 million* Americans diagnosed with flu in 2017-2018 of which 61,000 died and some reports claim 80,000. Where was the same hysteria then that we have seen with 'Covid-19'? Some 250,000 Americans are admitted to hospital with pneumonia every year with about 50,000 cases proving fatal. About 65 million suffer respiratory disease every year and three million deaths makes this the third biggest cause of death worldwide. You only have to redesignate a portion of all these people 'Covid-19' and you have an instant global pandemic or the *appearance* of one. Why would doctors do this? They are told to do this and all but a few dare not refuse those who must be obeyed. Doctors in general are not researching their own

knowledge and instead take it direct and unquestioned from the authorities that own them and their careers. The authorities say they must now diagnose these symptoms 'Covid-19' and not flu, or whatever, and they do it. Dark suits say put 'Covid-19' on death certificates no matter what the cause of death and the doctors do it. Renegade Minds don't fall for the illusion that doctors and medical staff are all highly-intelligent, highly-principled, seekers of medical truth. *Some are*, but not the majority. They are repeaters, gofers, and yes sir, no sir, purveyors of what the system demands they purvey. The 'Covid' con is not merely confined to diseases of the lungs. Instructions to doctors to put 'Covid-19' on death certificates for anyone dying of *anything* within 28 days (or much more) of a positive test not testing for the 'virus' opened the floodgates. The term dying *with* 'Covid' and not *of* 'Covid' was coined to cover the truth. Whether it was a *with* or an *of* they were all added to the death numbers attributed to the 'deadly virus' compiled by national governments and globally by the Gates-funded Johns Hopkins operation in the United States that was so involved in those 'pandemic' simulations. Fraudulent deaths were added to the ever-growing list of fraudulent 'cases' from false positives from a false test. No wonder Professor Walter Ricciardi, scientific advisor to the Italian minister of health, said after the Lombardy hysteria had done its job that 'Covid' death rates were due to Italy having the second oldest population in the world and to *how hospitals record deaths*:

The way in which we code deaths in our country is very generous in the sense that all the people who die in hospitals with the coronavirus are deemed to be dying of the coronavirus. On re-evaluation by the National Institute of Health, only 12 per cent of death certificates have shown a direct causality from coronavirus, while 88 per cent of patients who have died have at least one pre-morbidity – many had two or three.

This is extraordinary enough when you consider the propaganda campaign to use Italy to terrify the world, but how can they even say twelve percent were genuine when the 'virus' has not been shown to exist, its 'code' is a computer program, and diagnosis comes from a test not testing for it? As in China, and soon the world, 'Covid-19' in

Italy was a redesignation of diagnosis. Lies and corruption were to become the real 'pandemic' fuelled by a pathetically-compliant medical system taking its orders from the tiny few at the top of their national hierarchy who answered to the World Health Organization which answers to Gates and the Cult. Doctors were told – ordered – to diagnose a particular set of symptoms 'Covid-19' and put that on the death certificate for any cause of death if the patient had tested positive with a test not testing for the virus or had 'Covid' symptoms like the flu. The United States even introduced big financial incentives to manipulate the figures with hospitals receiving £4,600 from the Medicare system for diagnosing someone with regular pneumonia, \$13,000 if they made the diagnosis from the same symptoms 'Covid-19' pneumonia, and \$39,000 if they put a 'Covid' diagnosed patient on a ventilator that would almost certainly kill them. A few – painfully and pathetically few – medical whistleblowers revealed (before Cult-owned YouTube deleted their videos) that they had been instructed to 'let the patient crash' and put them straight on a ventilator instead of going through a series of far less intrusive and dangerous methods as they would have done before the pandemic hoax began and the financial incentives kicked in. We are talking cold-blooded murder given that ventilators are so damaging to respiratory systems they are usually the last step before heaven awaits. Renegade Minds never fall for the belief that people in white coats are all angels of mercy and cannot be full-on psychopaths. I have explained in detail in *The Answer* how what I am describing here played out across the world coordinated by the World Health Organization through the medical hierarchies in almost every country.

Medical scientist calls it

Information about the non-existence of the 'virus' began to emerge for me in late March, 2020, and mushroomed after that. I was sent an email by Sir Julian Rose, a writer, researcher, and organic farming promotor, from a medical scientist friend of his in the United States. Even at that early stage in March the scientist was able to explain

how the 'Covid' hoax was being manipulated. He said there were no reliable tests for a specific 'Covid-19 virus' and nor were there any reliable agencies or media outlets for reporting numbers of actual 'Covid-19' cases. We have seen in the long period since then that he was absolutely right. 'Every action and reaction to Covid-19 is based on totally flawed data and we simply cannot make accurate assessments,' he said. Most people diagnosed with 'Covid-19' were showing nothing more than cold and flu-like symptoms 'because most coronavirus strains *are* nothing more than cold/flu-like symptoms'. We had farcical situations like an 84-year-old German man testing positive for 'Covid-19' and his nursing home ordered to quarantine only for him to be found to have a common cold. The scientist described back then why PCR tests and what he called the 'Mickey Mouse test kits' were useless for what they were claimed to be identifying. 'The idea these kits can isolate a specific virus like Covid-19 is nonsense,' he said. Significantly, he pointed out that 'if you want to create a totally false panic about a totally false pandemic – pick a coronavirus'. This is exactly what the Cult-owned Gates, World Economic Forum and Johns Hopkins University did with their Event 201 'simulation' followed by their real-life simulation called the 'pandemic'. The scientist said that all you had to do was select the sickest of people with respiratory-type diseases in a single location – 'say Wuhan' – and administer PCR tests to them. You can then claim that anyone showing 'viral sequences' similar to a coronavirus 'which will inevitably be quite a few' is suffering from a 'new' disease:

Since you already selected the sickest flu cases a fairly high proportion of your sample will go on to die. You can then say this 'new' virus has a CFR [case fatality rate] higher than the flu and use this to infuse more concern and do more tests which will of course produce more 'cases', which expands the testing, which produces yet more 'cases' and so on and so on. Before long you have your 'pandemic', and all you have done is use a simple test kit trick to convert the worst flu and pneumonia cases into something new that doesn't ACTUALLY EXIST [my emphasis].

He said that you then 'just run the same scam in other countries' and make sure to keep the fear message running high 'so that people

will feel panicky and less able to think critically'. The only problem to overcome was the fact *there is no* actual new deadly pathogen and only regular sick people. This meant that deaths from the 'new deadly pathogen' were going to be way too low for a real new deadly virus pandemic, but he said this could be overcome in the following ways – all of which would go on to happen:

1. You can claim this is just the beginning and more deaths are imminent [you underpin this with fantasy 'computer projections']. Use this as an excuse to quarantine everyone and then claim the quarantine prevented the expected millions of dead.
2. You can [say that people] 'minimizing' the dangers are irresponsible and bully them into not talking about numbers.
3. You can talk crap about made up numbers hoping to blind people with pseudoscience.
4. You can start testing well people (who, of course, will also likely have shreds of coronavirus [RNA] in them) and thus inflate your 'case figures' with 'asymptomatic carriers' (you will of course have to spin that to sound deadly even though any virologist knows the more symptom-less cases you have the less deadly is your pathogen).

The scientist said that if you take these simple steps 'you can have your own entirely manufactured pandemic up and running in weeks'. His analysis made so early in the hoax was brilliantly prophetic of what would actually unfold. Pulling all the information together in these recent chapters we have this is simple 1, 2, 3, of how you can delude virtually the entire human population into believing in a 'virus' that doesn't exist:

- A 'Covid case' is someone who tests positive with a test not testing for the 'virus'.
- A 'Covid death' is someone who dies of *any cause* within 28 days (or much longer) of testing positive with a test not testing for the 'virus'.
- Asymptomatic means there is nothing wrong with you, but they claim you can pass on what you don't have to justify locking

down (quarantining) healthy people in totality.

The foundations of the hoax are that simple. A study involving ten million people in Wuhan, published in November, 2020, demolished the whole lie about those without symptoms passing on the 'virus'. They found '300 asymptomatic cases' and traced their contacts to find that not one of them was detected with the 'virus'.

'Asymptomatic' patients and their contacts were isolated for no less than two weeks and nothing changed. I know it's all crap, but if you are going to claim that those without symptoms can transmit 'the virus' then you must produce evidence for that and they never have. Even World Health Organization official Dr Maria Van Kerkhove, head of the emerging diseases and zoonosis unit, said as early as June, 2020, that she doubted the validity of asymptomatic transmission. She said that 'from the data we have, it still seems to be rare that an asymptomatic person actually transmits onward to a secondary individual' and by 'rare' she meant that she couldn't cite any case of asymptomatic transmission.

The Ferguson factor

The problem for the Cult as it headed into March, 2020, when the script had lockdown due to start, was that despite all the manipulation of the case and death figures they still did not have enough people alleged to have died from 'Covid' to justify mass house arrest. This was overcome in the way the scientist described: 'You can claim this is just the beginning and more deaths are imminent ... Use this as an excuse to quarantine everyone and then claim the quarantine prevented the expected millions of dead.' Enter one Professor Neil Ferguson, the Gates-funded 'epidemiologist' at the Gates-funded Imperial College in London. Ferguson is Britain's Christian Drosten in that he has a dire record of predicting health outcomes, but is still called upon to advise government on the next health outcome when another 'crisis' comes along. This may seem to be a strange and ridiculous thing to do. Why would you keep turning for policy guidance to people who have a history of being

monumentally wrong? Ah, but it makes sense from the Cult point of view. These 'experts' keep on producing predictions that suit the Cult agenda for societal transformation and so it was with Neil Ferguson as he revealed his horrific (and clearly insane) computer model predictions that allowed lockdowns to be imposed in Britain, the United States and many other countries. Ferguson does not have even an A-level in biology and would appear to have no formal training in computer modelling, medicine or epidemiology, according to Derek Winton, an MSc in Computational Intelligence. He wrote an article somewhat aghast at what Ferguson did which included taking no account of respiratory disease 'seasonality' which means it is far worse in the winter months. Who would have thought that respiratory disease could be worse in the winter? Well, certainly not Ferguson.

The massively China-connected Imperial College and its bizarre professor provided the excuse for the long-incubated Chinese model of human control to travel westward at lightning speed. Imperial College confirms on its website that it collaborates with the Chinese Research Institute; publishes more than 600 research papers every year with Chinese research institutions; has 225 Chinese staff; 2,600 Chinese students – the biggest international group; 7,000 former students living in China which is the largest group outside the UK; and was selected for a tour by China's President Xi Jinping during his state visit to the UK in 2015. The college takes major donations from China and describes itself as the UK's number one university collaborator with Chinese research institutions. The China communist/fascist government did not appear phased by the woeful predictions of Ferguson and Imperial when during the lockdown that Ferguson induced the college signed a five-year collaboration deal with China tech giant Huawei that will have Huawei's indoor 5G network equipment installed at the college's West London tech campus along with an 'AI cloud platform'. The deal includes Chinese sponsorship of Imperial's Venture Catalyst entrepreneurship competition. Imperial is an example of the enormous influence the Chinese government has within British and North American

universities and research centres – and further afield. Up to 200 academics from more than a dozen UK universities are being investigated on suspicion of ‘unintentionally’ helping the Chinese government build weapons of mass destruction by ‘transferring world-leading research in advanced military technology such as aircraft, missile designs and cyberweapons’. Similar scandals have broken in the United States, but it’s all a coincidence. Imperial College serves the agenda in many other ways including the promotion of every aspect of the United Nations Agenda 21/2030 (the Great Reset) and produced computer models to show that human-caused ‘climate change’ is happening when in the real world it isn’t. Imperial College is driving the climate agenda as it drives the ‘Covid’ agenda (both Cult hoaxes) while Patrick Vallance, the UK government’s Chief Scientific Adviser on ‘Covid’, was named Chief Scientific Adviser to the UN ‘climate change’ conference known as COP26 hosted by the government in Glasgow, Scotland. ‘Covid’ and ‘climate’ are fundamentally connected.

Professor Woeful

From Imperial’s bosom came Neil Ferguson still advising government despite his previous disasters and it was announced early on that he and other key people like UK Chief Medical Adviser Chris Whitty had caught the ‘virus’ as the propaganda story was being sold. Somehow they managed to survive and we had Prime Minister Boris Johnson admitted to hospital with what was said to be a severe version of the ‘virus’ in this same period. His whole policy and demeanour changed when he returned to Downing Street. It’s a small world with these government advisors – especially in their communal connections to Gates – and Ferguson had partnered with Whitty to write a paper called ‘Infectious disease: Tough choices to reduce Ebola transmission’ which involved another scare-story that didn’t happen. Ferguson’s ‘models’ predicted that up to 150,000 could die from ‘mad cow disease’, or BSE, and its version in sheep if it was transmitted to humans. BSE was not transmitted and instead triggered by an organophosphate pesticide used to treat a pest on

cows. Fewer than 200 deaths followed from the human form. Models by Ferguson and his fellow incompetents led to the unnecessary culling of millions of pigs, cattle and sheep in the foot and mouth outbreak in 2001 which destroyed the lives and livelihoods of farmers and their families who had often spent decades building their herds and flocks. Vast numbers of these animals did not have foot and mouth and had no contact with the infection. Another 'expert' behind the cull was Professor Roy Anderson, a computer modeller at Imperial College specialising in the epidemiology of *human*, not animal, disease. Anderson has served on the Bill and Melinda Gates Grand Challenges in Global Health advisory board and chairs another Gates-funded organisation. Gates is everywhere.

In a precursor to the 'Covid' script Ferguson backed closing schools 'for prolonged periods' over the swine flu 'pandemic' in 2009 and said it would affect a third of the world population if it continued to spread at the speed he claimed to be happening. His mates at Imperial College said much the same and a news report said: 'One of the authors, the epidemiologist and disease modeller Neil Ferguson, who sits on the World Health Organisation's emergency committee for the outbreak, said the virus had "full pandemic potential".' Professor Liam Donaldson, the Chris Whitty of his day as Chief Medical Officer, said the worst case could see 30 percent of the British people infected by swine flu with 65,000 dying. Ferguson and Donaldson were indeed proved correct when at the end of the year the number of deaths attributed to swine flu was 392. The term 'expert' is rather liberally applied unfortunately, not least to complete idiots. Swine flu 'projections' were great for GlaxoSmithKline (GSK) as millions rolled in for its Pandemrix influenza vaccine which led to brain damage with children most affected. The British government (taxpayers) paid out more than £60 million in compensation after GSK was given immunity from prosecution. Yet another 'Covid' déjà vu. Swine flu was supposed to have broken out in Mexico, but Dr Wolfgang Wodarg, a German doctor, former member of parliament and critic of the 'Covid' hoax, observed 'the spread of swine flu' in Mexico City at the time. He

said: 'What we experienced in Mexico City was a very mild flu which did not kill more than usual – which killed even fewer people than usual.' Hyping the fear against all the facts is not unique to 'Covid' and has happened many times before. Ferguson is reported to have over-estimated the projected death toll of bird flu (H5N1) by some three million-fold, but bird flu vaccine makers again made a killing from the scare. This is some of the background to the Neil Ferguson who produced the perfectly-timed computer models in early 2020 predicting that half a million people would die in Britain without draconian lockdown and 2.2 million in the United States. Politicians panicked, people panicked, and lockdowns of alleged short duration were instigated to 'flatten the curve' of cases gleaned from a test not testing for the 'virus'. I said at the time that the public could forget the 'short duration' bit. This was an agenda to destroy the livelihoods of the population and force them into mass control through dependency and there was going to be nothing 'short' about it. American researcher Daniel Horowitz described the consequences of the 'models' spewed out by Gates-funded Ferguson and Imperial College:

What led our government and the governments of many other countries into panic was a single Imperial College of UK study, funded by global warming activists, that predicted 2.2 million deaths if we didn't lock down the country. In addition, the reported 8-9% death rate in Italy scared us into thinking there was some other mutation of this virus that they got, which might have come here.

Together with the fact that we were finally testing and had the ability to actually report new cases, we thought we were headed for a death spiral. But again ... we can't flatten a curve if we don't know when the curve started.

How about it *never* started?

Giving them what they want

An investigation by German news outlet *Welt Am Sonntag* (*World on Sunday*) revealed how in March, 2020, the German government gathered together 'leading scientists from several research institutes and universities' and 'together, they were to produce a [modelling]

paper that would serve as legitimization for further tough political measures'. The Cult agenda was justified by computer modelling not based on evidence or reality; it was specifically constructed to justify the Cult demand for lockdowns all over the world to destroy the independent livelihoods of the global population. All these modellers and everyone responsible for the 'Covid' hoax have a date with a trial like those in Nuremberg after World War Two when Nazis faced the consequences of their war crimes. These corrupt-beyond-belief 'modellers' wrote the paper according to government instructions and it said that if lockdown measures were lifted then up to one million Germans would die from 'Covid-19' adding that some would die 'agonizingly at home, gasping for breath' unable to be treated by hospitals that couldn't cope. All lies. No matter – it gave the Cult all that it wanted. What did long-time government 'modeller' Neil Ferguson say? If the UK and the United States didn't lockdown half a million would die in Britain and 2.2 million Americans. Anyone see a theme here? 'Modellers' are such a crucial part of the lockdown strategy that we should look into their background and follow the money. Researcher Rosemary Frei produced an excellent article headlined 'The Modelling-paper Mafiosi'. She highlights a guy called John Edmunds, a British epidemiologist, and professor in the Faculty of Epidemiology and Population Health at the London School of Hygiene & Tropical Medicine. He studied at Imperial College. Edmunds is a member of government 'Covid' advisory bodies which have been dictating policy, the New and Emerging Respiratory Virus Threats Advisory Group (NERVTAG) and the Scientific Advisory Group for Emergencies (SAGE).

Ferguson, another member of NERVTAG and SAGE, led the way with the original 'virus' and Edmunds has followed in the 'variant' stage and especially the so-called UK or Kent variant known as the 'Variant of Concern' (VOC) B.1.1.7. He said in a co-written report for the Centre for Mathematical modelling of Infectious Diseases at the London School of Hygiene and Tropical Medicine, with input from the Centre's 'Covid-19' Working Group, that there was 'a realistic

possibility that VOC B.1.1.7 is associated with an increased risk of death compared to non-VOC viruses'. Fear, fear, fear, get the vaccine, fear, fear, fear, get the vaccine. Rosemary Frei reveals that almost all the paper's authors and members of the modelling centre's 'Covid-19' Working Group receive funding from the Bill and Melinda Gates Foundation and/or the associated Gates-funded Wellcome Trust. The paper was published by e-journal *Medrx* ^{xiv} which only publishes papers not peer-reviewed and the journal was established by an organisation headed by Facebook's Mark Zuckerberg and his missus. What a small world it is. Frei discovered that Edmunds is on the Scientific Advisory Board of the Coalition for Epidemic Preparedness Innovations (CEPI) which was established by the Bill and Melinda Gates Foundation, Klaus Schwab's Davos World Economic Forum and Big Pharma giant Wellcome. CEPI was 'launched in Davos [in 2017] to develop vaccines to stop future epidemics', according to its website. 'Our mission is to accelerate the development of vaccines against emerging infectious diseases and enable equitable access to these vaccines for people during outbreaks.' What kind people they are. Rosemary Frei reveals that Public Health England (PHE) director Susan Hopkins is an author of her organisation's non-peer-reviewed reports on 'new variants'. Hopkins is a professor of infectious diseases at London's Imperial College which is gifted tens of millions of dollars a year by the Bill and Melinda Gates Foundation. Gates-funded modelling disaster Neil Ferguson also co-authors Public Health England reports and he spoke in December, 2020, about the potential danger of the B.1.1.7. 'UK variant' promoted by Gates-funded modeller John Edmunds. When I come to the 'Covid vaccines' the 'new variants' will be shown for what they are – bollocks.

Connections, connections

All these people and modellers are lockdown-obsessed or, put another way, they demand what the Cult demands. Edmunds said in January, 2021, that to ease lockdowns too soon would be a disaster and they had to 'vaccinate much, much, much more widely than the

elderly'. Rosemary Frei highlights that Edmunds is married to Jeanne Pimenta who is described in a LinkedIn profile as director of epidemiology at GlaxoSmithKline (GSK) and she held shares in the company. Patrick Vallance, co-chair of SAGE and the government's Chief Scientific Adviser, is a former executive of GSK and has a deferred bonus of shares in the company worth £600,000. GSK has serious business connections with Bill Gates and is collaborating with mRNA-'vaccine' company CureVac to make 'vaccines' for the new variants that Edmunds is talking about. GSK is planning a 'Covid vaccine' with drug giant Sanofi. Puppets Prime Minister Boris Johnson announced in the spring of 2021 that up to 60 million vaccine doses were to be made at the GSK facility at Barnard Castle in the English North East. Barnard Castle, with a population of just 6,000, was famously visited in breach of lockdown rules in April, 2020, by Johnson aide Dominic Cummings who said that he drove there 'to test his eyesight' before driving back to London. Cummings would be better advised to test his integrity – not that it would take long. The GSK facility had nothing to do with his visit then although I'm sure Patrick Vallance would have been happy to arrange an introduction and some tea and biscuits. Ruthless psychopath Gates has made yet another fortune from vaccines in collaboration with Big Pharma companies and gushes at the phenomenal profits to be made from vaccines – more than a 20-to-1 return as he told one interviewer. Gates also tweeted in December, 2019, with the foreknowledge of what was coming: 'What's next for our foundation? I'm particularly excited about what the next year could mean for one of the best buys in global health: vaccines.'

Modeller John Edmunds is a big promotor of vaccines as all these people appear to be. He's the dean of the London School of Hygiene & Tropical Medicine's Faculty of Epidemiology and Population Health which is primarily funded by the Bill and Melinda Gates Foundation and the Gates-established and funded GAVI vaccine alliance which is the Gates vehicle to vaccinate the world. The organisation Doctors Without Borders has described GAVI as being 'aimed more at supporting drug-industry desires to promote new

products than at finding the most efficient and sustainable means for fighting the diseases of poverty'. But then that's why the psychopath Gates created it. John Edmunds said in a video that the London School of Hygiene & Tropical Medicine is involved in every aspect of vaccine development including large-scale clinical trials. He contends that mathematical modelling can show that vaccines protect individuals and society. That's on the basis of shit in and shit out, I take it. Edmunds serves on the UK Vaccine Network as does Ferguson and the government's foremost 'Covid' adviser, the grim-faced, dark-eyed Chris Whitty. The Vaccine Network says it works 'to support the government to identify and shortlist targeted investment opportunities for the most promising vaccines and vaccine technologies that will help combat infectious diseases with epidemic potential, and to address structural issues related to the UK's broader vaccine infrastructure'. Ferguson is acting Director of the Imperial College Vaccine Impact Modelling Consortium which has funding from the Bill and Melina Gates Foundation and the Gates-created GAVI 'vaccine alliance'. Anyone wonder why these characters see vaccines as the answer to every problem? Ferguson is wildly enthusiastic in his support for GAVI's campaign to vaccinate children en masse in poor countries. You would expect someone like Gates who has constantly talked about the need to reduce the population to want to fund vaccines to keep more people alive. I'm sure that's why he does it. The John Edmunds London School of Hygiene & Tropical Medicine (LSHTM) has a Vaccines Manufacturing Innovation Centre which develops, tests and commercialises vaccines. Rosemary Frei writes:

The vaccines centre also performs affiliated activities like combating 'vaccine hesitancy'. The latter includes the Vaccine Confidence Project. The project's stated purpose is, among other things, 'to provide analysis and guidance for early response and engagement with the public to ensure sustained confidence in vaccines and immunisation'. The Vaccine Confidence Project's director is LSHTM professor Heidi Larson. For more than a decade she's been researching how to combat vaccine hesitancy.

How the bloody hell can blokes like John Edmunds and Neil Ferguson with those connections and financial ties model 'virus' case

and death projections for the government and especially in a way that gives their paymasters like Gates exactly what they want? It's insane, but this is what you find throughout the world.

'Covid' is not dangerous, oops, wait, yes it is

Only days before Ferguson's nightmare scenario made Jackboot Johnson take Britain into a China-style lockdown to save us from a deadly 'virus' the UK government website gov.uk was reporting something very different to Ferguson on a page of official government guidance for 'high consequence infectious diseases (HCID)'. It said this about 'Covid-19':

As of 19 March 2020, COVID-19 is no longer considered to be a high consequence infectious diseases (HCID) in the UK [my emphasis]. The 4 nations public health HCID group made an interim recommendation in January 2020 to classify COVID-19 as an HCID. This was based on consideration of the UK HCID criteria about the virus and the disease with information available during the early stages of the outbreak.

Now that more is known about COVID-19, the public health bodies in the UK have reviewed the most up to date information about COVID-19 against the UK HCID criteria. They have determined that several features have now changed; in particular, more information is available about mortality rates (low overall), and there is now greater clinical awareness and a specific and sensitive laboratory test, the availability of which continues to increase. The Advisory Committee on Dangerous Pathogens (ACDP) is also of the opinion that COVID-19 should no longer be classified as an HCID.

Soon after the government had been exposed for downgrading the risk they upgraded it again and everyone was back to singing from the same Cult hymn book. Ferguson and his fellow Gates clones indicated that lockdowns and restrictions would have to continue until a Gates-funded vaccine was developed. Gates said the same because Ferguson and his like were repeating the Gates script which is the Cult script. 'Flatten the curve' became an ongoing nightmare of continuing lockdowns with periods in between of severe restrictions in pursuit of destroying independent incomes and had nothing to do with protecting health about which the Cult gives not a shit. Why wouldn't Ferguson be pushing a vaccine 'solution' when he's owned by vaccine-obsessive Gates who makes a fortune from them and

when Ferguson heads the Vaccine Impact Modelling Consortium at Imperial College funded by the Gates Foundation and GAVI, the 'vaccine alliance', created by Gates as his personal vaccine promotion operation? To compound the human catastrophe that Ferguson's 'models' did so much to create he was later exposed for breaking his own lockdown rules by having sexual liaisons with his married girlfriend Antonia Staats at his home while she was living at another location with her husband and children. Staats was a 'climate' activist and senior campaigner at the Soros-funded Avaaz which I wouldn't trust to tell me that grass is green. Ferguson had to resign as a government advisor over this hypocrisy in May, 2020, but after a period of quiet he was back being quoted by the ridiculous media on the need for more lockdowns and a vaccine rollout. Other government-advising 'scientists' from Imperial College held the fort in his absence and said lockdown could be indefinite until a vaccine was found. The Cult script was being sung by the payrolled choir. I said there was no intention of going back to 'normal' when the 'vaccine' came because the 'vaccine' is part of a very different agenda that I will discuss in Human 2.0. Why would the Cult want to let the world go back to normal when destroying that normal forever was the whole point of what was happening? House arrest, closing businesses and schools through lockdown, (un)social distancing and masks all followed the Ferguson fantasy models. Again as I predicted (these people are so predictable) when the 'vaccine' arrived we were told that house arrest, lockdown, (un)social distancing and masks would still have to continue. I will deal with the masks in the next chapter because they are of fundamental importance.

Where's the 'pandemic'?

Any mildly in-depth assessment of the figures revealed what was really going on. Cult-funded and controlled organisations still have genuine people working within them such is the number involved. So it is with Genevieve Briand, assistant program director of the Applied Economics master's degree program at Johns Hopkins

University. She analysed the impact that 'Covid-19' had on deaths from *all* causes in the United States using official data from the CDC for the period from early February to early September, 2020. She found that allegedly 'Covid' *related*-deaths exceeded those from heart disease which she found strange with heart disease always the biggest cause of fatalities. Her research became even more significant when she noted the sudden decline in 2020 of *all* non-'Covid' deaths: 'This trend is completely contrary to the pattern observed in all previous years ... the total decrease in deaths by other causes almost exactly equals the increase in deaths by Covid-19.' This was such a game, set and match in terms of what was happening that Johns Hopkins University deleted the article on the grounds that it 'was being used to support false and dangerous inaccuracies about the impact of the pandemic'. No – because it exposed the scam from official CDC figures and this was confirmed when those figures were published in January, 2021. Here we can see the effect of people dying from heart attacks, cancer, road accidents and gunshot wounds – *anything* – having 'Covid-19' on the death certificate along with those diagnosed from 'symptoms' who had even not tested positive with a test not testing for the 'virus'. I am not kidding with the gunshot wounds, by the way. Brenda Bock, coroner in Grand County, Colorado, revealed that two gunshot victims tested positive for the 'virus' within the previous 30 days and were therefore classified as 'Covid deaths'. Bock said: 'These two people had tested positive for Covid, but that's not what killed them. A gunshot wound is what killed them.' She said she had not even finished her investigation when the state listed the gunshot victims as deaths due to the 'virus'. The death and case figures for 'Covid-19' are an absolute joke and yet they are repeated like parrots by the media, politicians and alleged medical 'experts'. The official Cult narrative is the only show in town.

Genevieve Briand found that deaths from all causes were not exceptional in 2020 compared with previous years and a Spanish magazine published figures that said the same about Spain which was a 'Covid' propaganda hotspot at one point. *Discovery Salud*, a

health and medicine magazine, quoted government figures which showed how 17,000 *fewer* people died in Spain in 2020 than in 2019 and more than 26,000 fewer than in 2018. The age-standardised mortality rate for England and Wales when age distribution is taken into account was significantly lower in 2020 than the 1970s, 80s and 90s, and was only the ninth highest since 2000. Where is the 'pandemic'?

Post mortems and autopsies virtually disappeared for 'Covid' deaths amid claims that 'virus-infected' bodily fluids posed a risk to those carrying out the autopsy. This was rejected by renowned German pathologist and forensic doctor Klaus Püschel who said that he and his staff had by then done 150 autopsies on 'Covid' patients with no problems at all. He said they were needed to know why some 'Covid' patients suffered blood clots and not severe respiratory infections. The 'virus' is, after all, called SARS or 'severe acute respiratory syndrome'. I highlighted in the spring of 2020 this phenomenon and quoted New York intensive care doctor Cameron Kyle-Sidell who posted a soon deleted YouTube video to say that they had been told to prepare to treat an infectious disease called 'Covid-19', but that was not what they were dealing with. Instead he likened the lung condition of the most severely ill patients to what you would expect with cabin depressurisation in a plane at 30,000 feet or someone dropped on the top of Everest without oxygen or acclimatisation. I have never said this is not happening to a small minority of alleged 'Covid' patients – I am saying this is not caused by a phantom 'contagious virus'. Indeed Kyle-Sidell said that 'Covid-19' was not the disease they were told was coming their way. 'We are operating under a medical paradigm that is untrue,' he said, and he believed they were treating the wrong disease: 'These people are being slowly starved of oxygen.' Patients would take off their oxygen masks in a state of fear and stress and while they were blue in the face on the brink of death. They did not look like patients dying of pneumonia. You can see why they don't want autopsies when their virus doesn't exist and there is another condition in some people that they don't wish to be uncovered. I should add here that

the 5G system of millimetre waves was being rapidly introduced around the world in 2020 and even more so now as they fire 5G at the Earth from satellites. At 60 gigahertz within the 5G range that frequency interacts with the oxygen molecule and stops people breathing in sufficient oxygen to be absorbed into the bloodstream. They are installing 5G in schools and hospitals. The world is not mad or anything. 5G can cause major changes to the lungs and blood as I detail in *The Answer* and these consequences are labelled 'Covid-19', the alleged symptoms of which can be caused by 5G and other electromagnetic frequencies as cells respond to radiation poisoning.

The 'Covid death' scam

Dr Scott Jensen, a Minnesota state senator and medical doctor, exposed 'Covid' Medicare payment incentives to hospitals and death certificate manipulation. He said he was sent a seven-page document by the US Department of Health 'coaching' him on how to fill out death certificates which had never happened before. The document said that he didn't need to have a laboratory test for 'Covid-19' to put that on the death certificate and that shocked him when death certificates are supposed to be about facts. Jensen described how doctors had been 'encouraged, if not pressured' to make a diagnosis of 'Covid-19' if they thought it was probable or '*presumed*'. No positive test was necessary – not that this would have mattered anyway. He said doctors were told to diagnose 'Covid' by symptoms when these were the same as colds, allergies, other respiratory problems, and certainly with influenza which 'disappeared' in the 'Covid' era. A common sniffle was enough to get the dreaded verdict. Ontario authorities decreed that a single care home resident with *one* symptom from a long list must lead to the isolation of the entire home. Other courageous doctors like Jensen made the same point about death figure manipulation and how deaths by other causes were falling while 'Covid-19 deaths' were rising at the same rate due to re-diagnosis. Their videos rarely survive long on YouTube with its Cult-supporting algorithms courtesy of CEO Susan Wojcicki and her bosses at Google. Figure-tampering was so glaring

and ubiquitous that even officials were letting it slip or outright saying it. UK chief scientific adviser Patrick Vallance said on one occasion that 'Covid' on the death certificate doesn't mean 'Covid' was the cause of death (so why the hell is it there?) and we had the rare sight of a BBC reporter telling the truth when she said: 'Someone could be successfully treated for Covid, in say April, discharged, and then in June, get run over by a bus and die ... That person would still be counted as a Covid death in England.' Yet the BBC and the rest of the world media went on repeating the case and death figures as if they were real. Illinois Public Health Director Dr Ngozi Ezike revealed the deceit while her bosses must have been clenching their buttocks:

If you were in a hospice and given a few weeks to live and you were then found to have Covid that would be counted as a Covid death. [There might be] a clear alternate cause, but it is still listed as a Covid death. So everyone listed as a Covid death doesn't mean that was the cause of the death, but that they had Covid at the time of death.

Yes, a 'Covid virus' never shown to exist and tested for with a test not testing for the 'virus'. In the first period of the pandemic hoax through the spring of 2020 the process began of designating almost everything a 'Covid' death and this has continued ever since. I sat in a restaurant one night listening to a loud conversation on the next table where a family was discussing in bewilderment how a relative who had no symptoms of 'Covid', and had died of a long-term problem, could have been diagnosed a death by the 'virus'. I could understand their bewilderment. If they read this book they will know why this medical fraud has been perpetrated the world over.

Some media truth shock

The media ignored the evidence of death certificate fraud until eventually one columnist did speak out when she saw it first-hand. Bel Mooney is a long-time national newspaper journalist in Britain currently working for the *Daily Mail*. Her article on February 19th, 2021, carried this headline: 'My dad Ted passed three Covid tests

and died of a chronic illness yet he's officially one of Britain's 120,000 victims of the virus and is far from alone ... so how many more are there?' She told how her 99-year-old father was in a care home with a long-standing chronic obstructive pulmonary disease and vascular dementia. Maybe, but he was still aware enough to tell her from the start that there was no 'virus' and he refused the 'vaccine' for that reason. His death was not unexpected given his chronic health problems and Mooney said she was shocked to find that 'Covid-19' was declared the cause of death on his death certificate. She said this was a 'bizarre and unacceptable untruth' for a man with long-time health problems who had tested negative twice at the home for the 'virus'. I was also shocked by this story although not by what she said. I had been highlighting the death certificate manipulation for ten months. It was the confirmation that a professional full-time journalist only realised this was going on when it affected her directly and neither did she know that whether her dad tested positive or negative was irrelevant with the test not testing for the 'virus'. Where had she been? She said she did not believe in 'conspiracy theories' without knowing I'm sure that this and 'conspiracy theorists' were terms put into widespread circulation by the CIA in the 1960s to discredit those who did not accept the ridiculous official story of the Kennedy assassination. A blanket statement of 'I don't believe in conspiracy theories' is always bizarre. The dictionary definition of the term alone means the world is drowning in conspiracies. What she said was even more daft when her dad had just been affected by the 'Covid' conspiracy. Why else does she think that 'Covid-19' was going on the death certificates of people who died of something else?

To be fair once she saw from personal experience what was happening she didn't mince words. Mooney was called by the care home on the morning of February 9th to be told her father had died in his sleep. When she asked for the official cause of death what came back was 'Covid-19'. Mooney challenged this and was told there had been deaths from Covid on the dementia floor (confirmed by a test not testing for the 'virus') so they considered it 'reasonable

to assume'. 'But doctor,' Mooney rightly protested, 'an assumption isn't a diagnosis.' She said she didn't blame the perfectly decent and sympathetic doctor – 'he was just doing his job'. Sorry, but that's *bullshit*. He wasn't doing his job at all. He was putting a false cause of death on the death certificate and that is a criminal offence for which he should be brought to account and the same with the millions of doctors worldwide who have done the same. They were not doing their job they were following orders and that must not wash at new Nuremberg trials any more than it did at the first ones. Mooney's doctor was 'assuming' (presuming) as he was told to, but 'just following orders' makes no difference to his actions. A doctor's job is to serve the patient and the truth, not follow orders, but that's what they have done all over the world and played a central part in making the 'Covid' hoax possible with all its catastrophic consequences for humanity. Shame on them and they must answer for their actions. Mooney said her disquiet worsened when she registered her father's death by telephone and was told by the registrar there had been very many other cases like hers where 'the deceased' had not tested positive for 'Covid' yet it was recorded as the cause of death. The test may not matter, but those involved at their level *think* it matters and it shows a callous disregard for accurate diagnosis. The pressure to do this is coming from the top of the national 'health' pyramids which in turn obey the World Health Organization which obeys Gates and the Cult. Mooney said the registrar agreed that this must distort the national figures adding that 'the strangest thing is that every winter we record countless deaths from flu, and this winter there have been none. Not one!' She asked if the registrar thought deaths from flu were being misdiagnosed and lumped together with 'Covid' deaths. The answer was a 'puzzled yes'. Mooney said that the funeral director said the same about 'Covid' deaths which had nothing to do with 'Covid'. They had lost count of the number of families upset by this and other funeral companies in different countries have had the same experience. Mooney wrote:

The nightly shroud-waving and shocking close-ups of pain imposed on us by the TV news bewildered and terrified the population into eager compliance with lockdowns. We were invited to 'save the NHS' and to grieve for strangers – the real-life loved ones behind those shocking death counts. Why would the public imagine what I now fear, namely that the way Covid-19 death statistics are compiled might make the numbers seem greater than they are?

Oh, just a little bit – like 100 percent.

Do the maths

Mooney asked why a country would wish to skew its mortality figures by wrongly certifying deaths? What had been going on? Well, if you don't believe in conspiracies you will never find the answer which is that *it's a conspiracy*. She did, however, describe what she had discovered as a 'national scandal'. In reality it's a global scandal and happening everywhere. Pillars of this conspiracy were all put into place before the button was pressed with the Drosten PCR protocol and high amplifications to produce the cases and death certificate changes to secure illusory 'Covid' deaths. Mooney notes that normally two doctors were needed to certify a death, with one having to know the patient, and how the rules were changed in the spring of 2020 to allow one doctor to do this. In the same period 'Covid deaths' were decreed to be all cases where Covid-19 was put on the death certificate even without a positive test or any symptoms. Mooney asked: 'How many of the 30,851 (as of January 15) care home resident deaths with Covid-19 on the certificate (32.4 per cent of all deaths so far) were based on an assumption, like that of my father? And what has that done to our national psyche?' All of them is the answer to the first question and it has devastated and dismantled the national psyche, actually the global psyche, on a colossal scale. In the UK case and death data is compiled by organisations like Public Health England (PHE) and the Office for National Statistics (ONS). Mooney highlights the insane policy of counting a death from any cause as 'Covid-19' if this happens within 28 days of a positive test (with a test not testing for the 'virus') and she points out that ONS statistics reflect deaths 'involving Covid' 'or due to Covid' which meant in practice any

death where 'Covid-19' was mentioned on the death certificate. She described the consequences of this fraud:

Most people will accept the narrative they are fed, so panicky governments here and in Europe witnessed the harsh measures enacted in totalitarian China and jumped into lockdown. Headlines about Covid deaths tolled like the knell that would bring doomsday to us all. Fear stalked our empty streets. Politicians parroted the frankly ridiculous aim of 'zero Covid' and shut down the economy, while most British people agreed that lockdown was essential and (astonishingly to me, as a patriotic Brit) even wanted more restrictions.

For what? Lies on death certificates? Never mind the grim toll of lives ruined, suicides, schools closed, rising inequality, depression, cancelled hospital treatments, cancer patients in a torture of waiting, poverty, economic devastation, loneliness, families kept apart, and so on. How many lives have been lost as a direct result of lockdown?

She said that we could join in a national chorus of shock and horror at reaching the 120,000 death toll which was surely certain to have been totally skewed all along, but what about the human cost of lockdown justified by these 'death figures'? *The British Medical Journal* had reported a 1,493 percent increase in cases of children taken to Great Ormond Street Hospital with abusive head injuries alone and then there was the effect on families:

Perhaps the most shocking thing about all this is that families have been kept apart – and obeyed the most irrational, changing rules at the whim of government – because they believed in the statistics. They succumbed to fear, which his generation rejected in that war fought for freedom. Dad (God rest his soul) would be angry. And so am I.

Another theme to watch is that in the winter months when there are more deaths from all causes they focus on 'Covid' deaths and in the summer when the British Lung Foundation says respiratory disease plummets by 80 percent they rage on about 'cases'. Either way fascism on population is always the answer.

Nazi eugenics in the 21st century

Elderly people in care homes have been isolated from their families month after lonely month with no contact with relatives and grandchildren who were banned from seeing them. We were told

that lockdown fascism was to 'protect the vulnerable' like elderly people. At the same time Do Not Resuscitate (DNR) orders were placed on their medical files so that if they needed resuscitation it wasn't done and 'Covid-19' went on their death certificates. Old people were not being 'protected' they were being culled – murdered in truth. DNR orders were being decreed for disabled and young people with learning difficulties or psychological problems. The UK Care Quality Commission, a non-departmental body of the Department of Health and Social Care, found that 34 percent of those working in health and social care were pressured into placing 'do not attempt cardiopulmonary resuscitation' orders on 'Covid' patients who suffered from disabilities and learning difficulties without involving the patient or their families in the decision. UK judges ruled that an elderly woman with dementia should have the DNA-manipulating 'Covid vaccine' against her son's wishes and that a man with severe learning difficulties should have the jab despite his family's objections. Never mind that many had already died. The judiciary always supports doctors and government in fascist dictatorships. They wouldn't dare do otherwise. A horrific video was posted showing fascist officers from Los Angeles police forcibly giving the 'Covid' shot to women with special needs who were screaming that they didn't want it. The same fascists are seen giving the jab to a sleeping elderly woman in a care home. This is straight out of the Nazi playbook. Hitler's Nazis committed mass murder of the mentally ill and physically disabled throughout Germany and occupied territories in the programme that became known as Aktion T4, or just T4. Sabbatian-controlled Hitler and his grotesque crazies set out to kill those they considered useless and unnecessary. The Reich Committee for the Scientific Registering of Hereditary and Congenital Illnesses registered the births of babies identified by physicians to have 'defects'. By 1941 alone more than 5,000 children were murdered by the state and it is estimated that in total the number of innocent people killed in Aktion T4 was between 275,000 and 300,000. Parents were told their children had been sent away for 'special treatment' never to return. It is rather pathetic to see claims about plans for new extermination camps being dismissed today

when the same force behind current events did precisely that 80 years ago. Margaret Sanger was a Cult operative who used 'birth control' to sanitise her programme of eugenics. Organisations she founded became what is now Planned Parenthood. Sanger proposed that 'the whole dysgenic population would have its choice of segregation or sterilization'. These included epileptics, 'feeble-minded', and prostitutes. Sanger opposed charity because it perpetuated 'human waste'. She reveals the Cult mentality and if anyone thinks that extermination camps are a 'conspiracy theory' their naivety is touching if breathtakingly stupid.

If you don't believe that doctors can act with callous disregard for their patients it is worth considering that doctors and medical staff agreed to put government-decreed DNR orders on medical files and do nothing when resuscitation is called for. I don't know what you call such people in your house. In mine they are Nazis from the Josef Mengele School of Medicine. Phenomenal numbers of old people have died worldwide from the effects of lockdown, depression, lack of treatment, the 'vaccine' (more later) and losing the will to live. A common response at the start of the manufactured pandemic was to remove old people from hospital beds and transfer them to nursing homes. The decision would result in a mass cull of elderly people in those homes through lack of treatment – *not* 'Covid'. Care home whistleblowers have told how once the 'Covid' era began doctors would not come to their homes to treat patients and they were begging for drugs like antibiotics that often never came. The most infamous example was ordered by New York governor Andrew Cuomo, brother of a moronic CNN host, who amazingly was given an Emmy Award for his handling of the 'Covid crisis' by the ridiculous Wokers that hand them out. Just how ridiculous could be seen in February, 2021, when a Department of Justice and FBI investigation began into how thousands of old people in New York died in nursing homes after being discharged from hospital to make way for 'Covid' patients on Cuomo's say-so – and how he and his staff covered up these facts. This couldn't have happened to a nicer psychopath. Even then there was a 'Covid' spin. Reports said that

thousands of old people who tested positive for 'Covid' in hospital were transferred to nursing homes to both die of 'Covid' and transmit it to others. No – they were in hospital because they were ill and the fact that they tested positive with a test not testing for the 'virus' is irrelevant. They were ill often with respiratory diseases ubiquitous in old people near the end of their lives. Their transfer out of hospital meant that their treatment stopped and many would go on to die.

They're old. Who gives a damn?

I have exposed in the books for decades the Cult plan to cull the world's old people and even to introduce at some point what they call a 'demise pill' which at a certain age everyone would take and be out of here by law. In March, 2021, Spain legalised euthanasia and assisted suicide following the Netherlands, Belgium, Luxembourg and Canada on the Tiptoe to the demise pill. Treatment of old people by many 'care' homes has been a disgrace in the 'Covid' era. There are many, many, caring staff – I know some. There have, however, been legions of stories about callous treatment of old people and their families. Police were called when families came to take their loved ones home in the light of isolation that was killing them. They became prisoners of the state. Care home residents in insane, fascist Ontario, Canada, were not allowed to leave their *room* once the 'Covid' hoax began. UK staff have even wheeled elderly people away from windows where family members were talking with them. Oriana Criscuolo from Stockport in the English North West dropped off some things for her 80-year-old father who has Parkinson's disease and dementia and she wanted to wave to him through a ground-floor window. She was told that was 'illegal'. When she went anyway they closed the curtains in the middle of the day. Oriana said:

It's just unbelievable. I cannot understand how care home staff – people who are being paid to care – have become so uncaring. Their behaviour is inhumane and cruel. It's beyond belief.

She was right and this was not a one-off. What a way to end your life in such loveless circumstances. UK registered nurse Nicky Millen, a proper old school nurse for 40 years, said that when she started her career care was based on dignity, choice, compassion and empathy. Now she said 'the things that are important to me have gone out of the window.' She was appalled that people were dying without their loved ones and saying goodbye on iPads. Nicky described how a distressed 89-year-old lady stroked her face and asked her 'how many paracetamol would it take to finish me off'. Life was no longer worth living while not seeing her family. Nicky said she was humiliated in front of the ward staff and patients for letting the lady stroke her face and giving her a cuddle. Such is the dehumanisation that the 'Covid' hoax has brought to the surface. Nicky worked in care homes where patients told her they were being held prisoner. 'I want to live until I die', one said to her. 'I had a lady in tears because she hadn't seen her great-grandson.' Nicky was compassionate old school meeting psychopathic New Normal. She also said she had worked on a 'Covid' ward with no 'Covid' patients. Jewish writer Shai Held wrote an article in March, 2020, which was headlined 'The Staggering, Heartless Cruelty Toward the Elderly'. What he described was happening from the earliest days of lockdown. He said 'the elderly' were considered a group and not unique individuals (the way of the Woke). Shai Held said:

Notice how the all-too-familiar rhetoric of dehumanization works: 'The elderly' are bunched together as a faceless mass, all of them considered culprits and thus effectively deserving of the suffering the pandemic will inflict upon them. Lost entirely is the fact that the elderly are individual human beings, each with a distinctive face and voice, each with hopes and dreams, memories and regrets, friendships and marriages, loves lost and loves sustained.

'The elderly' have become another dehumanised group for which anything goes and for many that has resulted in cold disregard for their rights and their life. The distinctive face that Held talks about is designed to be deleted by masks until everyone is part of a faceless mass.

'War-zone' hospitals myth

Again and again medical professionals have told me what was really going on and how hospitals 'overrun like war zones' according to the media were virtually empty. The mantra from medical whistleblowers was please don't use my name or my career is over. Citizen journalists around the world sneaked into hospitals to film evidence exposing the 'war-zone' lie. They really *were* largely empty with closed wards and operating theatres. I met a hospital worker in my town on the Isle of Wight during the first lockdown in 2020 who said the only island hospital had never been so quiet. Lockdown was justified by the psychopaths to stop hospitals being overrun. At the same time that the island hospital was near-empty the military arrived here to provide *extra beds*. It was all propaganda to ramp up the fear to ensure compliance with fascism as were never-used temporary hospitals with thousands of beds known as Nightingales and never-used make-shift mortuaries opened by the criminal UK government. A man who helped to install those extra island beds attributed to the army said they were never used and the hospital was empty. Doctors and nurses 'stood around talking or on their phones, wandering down to us to see what we were doing'. There were no masks or social distancing. He accused the useless local island paper, the *County Press*, of 'pumping the fear as if our hospital was overrun and we only have one so it should have been'. He described ambulances parked up with crews outside in deck chairs. When his brother called an ambulance he was told there was a two-hour backlog which he called 'bullshit'. An old lady on the island fell 'and was in a bad way', but a caller who rang for an ambulance was told the situation wasn't urgent enough. Ambulance stations were working under capacity while people would hear ambulances with sirens blaring driving through the streets. When those living near the stations realised what was going on they would follow them as they left, circulated around an urban area with the sirens going, and then came back without stopping. All this was to increase levels of fear and the same goes for the 'ventilator shortage crisis' that cost tens of millions for hastily produced ventilators never to be used.

Ambulance crews that agreed to be exploited in this way for fear propaganda might find themselves a mirror. I wish them well with that. Empty hospitals were the obvious consequence of treatment and diagnoses of non-'Covid' conditions cancelled and those involved handed a death sentence. People have been dying at home from undiagnosed and untreated cancer, heart disease and other life-threatening conditions to allow empty hospitals to deal with a 'pandemic' that wasn't happening.

Death of the innocent

'War-zones' have been laying off nursing staff, even doctors where they can. There was no work for them. Lockdown was justified by saving lives and protecting the vulnerable they were actually killing with DNR orders and preventing empty hospitals being 'overrun'. In Britain the mantra of stay at home to 'save the NHS' was everywhere and across the world the same story was being sold when it was all lies. Two California doctors, Dan Erickson and Artin Massihi at Accelerated Urgent Care in Bakersfield, held a news conference in April, 2020, to say that intensive care units in California were 'empty, essentially', with hospitals shutting floors, not treating patients and laying off doctors. The California health system was working at minimum capacity 'getting rid of doctors because we just don't have the volume'. They said that people with conditions such as heart disease and cancer were not coming to hospital out of fear of 'Covid-19'. Their video was deleted by Susan Wojcicki's Cult-owned YouTube after reaching five million views. Florida governor Ron Desantis, who rejected the severe lockdowns of other states and is being targeted for doing so, said that in March, 2020, every US governor was given models claiming they would run out of hospital beds in days. That was never going to happen and the 'modellers' knew it. Deceit can be found at every level of the system. Urgent children's operations were cancelled including fracture repairs and biopsies to spot cancer. Eric Nicholls, a consultant paediatrician, said 'this is obviously concerning and we need to return to normal operating and to increase capacity as soon as possible'. Psychopaths

in power were rather less concerned *because* they are psychopaths. Deletion of urgent care and diagnosis has been happening all over the world and how many kids and others have died as a result of the actions of these cold and heartless lunatics dictating 'health' policy? The number must be stratospheric. Richard Sullivan, professor of cancer and global health at King's College London, said people feared 'Covid' more than cancer such was the campaign of fear. 'Years of lost life will be quite dramatic', Sullivan said, with 'a huge amount of avoidable mortality'. Sarah Woolnough, executive director for policy at Cancer Research UK, said there had been a 75 percent drop in urgent referrals to hospitals by family doctors of people with suspected cancer. Sullivan said that 'a lot of services have had to scale back – we've seen a dramatic decrease in the amount of elective cancer surgery'. Lockdown deaths worldwide has been absolutely fantastic with the *New York Post* reporting how data confirmed that 'lockdowns end more lives than they save':

There was a sharp decline in visits to emergency rooms and an increase in fatal heart attacks because patients didn't receive prompt treatment. Many fewer people were screened for cancer. Social isolation contributed to excess deaths from dementia and Alzheimer's.

Researchers predicted that the social and economic upheaval would lead to tens of thousands of "deaths of despair" from drug overdoses, alcoholism and suicide. As unemployment surged and mental-health and substance-abuse treatment programs were interrupted, the reported levels of anxiety, depression and suicidal thoughts increased dramatically, as did alcohol sales and fatal drug overdoses.

This has been happening while nurses and other staff had so much time on their hands in the 'war-zones' that Tic-Tok dancing videos began appearing across the Internet with medical staff dancing around in empty wards and corridors as people died at home from causes that would normally have been treated in hospital.

Mentions in dispatches

One brave and truth-committed whistleblower was Louise Hampton, a call handler with the UK NHS who made a viral Internet video saying she had done 'fuck all' during the 'pandemic'

which was 'a load of bollocks'. She said that 'Covid-19' was rebranded flu and of course she lost her job. This is what happens in the medical and endless other professions now when you tell the truth. Louise filmed inside 'war-zone' accident and emergency departments to show they were empty and I mean *empty* as in no one there. The mainstream media could have done the same and blown the gaff on the whole conspiracy. They haven't to their eternal shame. Not that most 'journalists' seem capable of manifesting shame as with the psychopaths they slavishly repeat without question. The relative few who were admitted with serious health problems were left to die alone with no loved ones allowed to see them because of 'Covid' rules and they included kids dying without the comfort of mum and dad at their bedside while the evil behind this couldn't give a damn. It was all good fun to them. A Scottish NHS staff nurse publicly quit in the spring of 2021 saying: 'I can no longer be part of the lies and the corruption by the government.' She said hospitals 'aren't full, the beds aren't full, beds have been shut, wards have been shut'. Hospitals were never busy throughout 'Covid'. The staff nurse said that Nicola Sturgeon, tragically the leader of the Scottish government, was on television saying save the hospitals and the NHS – 'but the beds are empty' and 'we've not seen flu, we always see flu every year'. She wrote to government and spoke with her union Unison (the unions are Cult-compromised and *useless*, but nothing changed. Many of her colleagues were scared of losing their jobs if they spoke out as they wanted to. She said nursing staff were being affected by wearing masks all day and 'my head is splitting every shift from wearing a mask'. The NHS is part of the fascist tyranny and must be dismantled so we can start again with human beings in charge. (Ironically, hospitals were reported to be busier again when official 'Covid' cases *fell* in spring/summer of 2021 and many other conditions required treatment at the same time as *the fake vaccine rollout*.)

I will cover the 'Covid vaccine' scam in detail later, but it is another indicator of the sickening disregard for human life that I am highlighting here. The DNA-manipulating concoctions do not fulfil

the definition of a 'vaccine', have never been used on humans before and were given only emergency approval because trials were not completed and they continued using the unknowing public. The result was what a NHS senior nurse with responsibility for 'vaccine' procedure said was 'genocide'. She said the 'vaccines' were not 'vaccines'. They had not been shown to be safe and claims about their effectiveness by drug companies were 'poetic licence'. She described what was happening as a 'horrid act of human annihilation'. The nurse said that management had instigated a policy of not providing a Patient Information Leaflet (PIL) before people were 'vaccinated' even though health care professionals are supposed to do this according to protocol. Patients should also be told that they are taking part in an ongoing clinical trial. Her challenges to what is happening had seen her excluded from meetings and ridiculed in others. She said she was told to 'watch my step ... or I would find myself surplus to requirements'. The nurse, who spoke anonymously in fear of her career, said she asked her NHS manager why he/she was content with taking part in genocide against those having the 'vaccines'. The reply was that everyone had to play their part and to 'put up, shut up, and get it done'. Government was 'leaning heavily' on NHS management which was clearly leaning heavily on staff. This is how the global 'medical' hierarchy operates and it starts with the Cult and its World Health Organization.

She told the story of a doctor who had the Pfizer jab and when questioned had no idea what was in it. The doctor had never read the literature. We have to stop treating doctors as intellectual giants when so many are moral and medical pygmies. The doctor did not even know that the 'vaccines' were not fully approved or that their trials were ongoing. They were, however, asking their patients if they minded taking part in follow-ups for research purposes – yes, the *ongoing clinical trial*. The nurse said the doctor's ignorance was not rare and she had spoken to a hospital consultant who had the jab without any idea of the background or that the 'trials' had not been completed. Nurses and pharmacists had shown the same ignorance.

‘My NHS colleagues have forsaken their duty of care, broken their code of conduct – Hippocratic Oath – and have been brainwashed just the same as the majority of the UK public through propaganda ...’ She said she had not been able to recruit a single NHS colleague, doctor, nurse or pharmacist to stand with her and speak out. Her union had refused to help. She said that if the genocide came to light she would not hesitate to give evidence at a Nuremberg-type trial against those in power who could have affected the outcomes but didn’t.

And all for what?

To put the nonsense into perspective let’s say the ‘virus’ does exist and let’s go completely crazy and accept that the official manipulated figures for cases and deaths are accurate. *Even then* a study by Stanford University epidemiologist Dr John Ioannidis published on the World Health Organization website produced an average infection to fatality rate of ... *0.23 percent!* Ioannidis said: ‘If one could sample equally from all locations globally, the median infection fatality rate might even be substantially lower than the 0.23% observed in my analysis.’ For healthy people under 70 it was ... *0.05 percent!* This compares with the 3.4 percent claimed by the Cult-owned World Health Organization when the hoax was first played and maximum fear needed to be generated. An updated Stanford study in April, 2021, put the ‘infection’ to ‘fatality’ rate at just 0.15 percent. Another team of scientists led by Megan O’Driscoll and Henrik Salje studied data from 45 countries and published their findings on the Nature website. For children and young people the figure is so small it virtually does not register although authorities will be hyping dangers to the young when they introduce DNA-manipulating ‘vaccines’ for children. The O’Driscoll study produced an average infection-fatality figure of 0.003 for children from birth to four; 0.001 for 5 to 14; 0.003 for 15 to 19; and it was still only 0.456 up to 64. To claim that children must be ‘vaccinated’ to protect them from ‘Covid’ is an obvious lie and so there must be another reason and there is. What’s more the average age of a ‘Covid’ death is akin

to the average age that people die in general. The average age of death in England is about 80 for men and 83 for women. The average age of death from alleged 'Covid' is between 82 and 83. California doctors, Dan Erickson and Artin Massihi, said at their April media conference that projection models of millions of deaths had been 'woefully inaccurate'. They produced detailed figures showing that Californians had a 0.03 chance of dying from 'Covid' based on the number of people who tested positive (with a test not testing for the 'virus'). Erickson said there was a 0.1 percent chance of dying from 'Covid' in the *state* of New York, not just the city, and a 0.05 percent chance in Spain, a centre of 'Covid-19' hysteria at one stage. The Stanford studies supported the doctors' data with fatality rate estimates of 0.23 and 0.15 percent. How close are these figures to my estimate of *zero*? Death-rate figures claimed by the World Health Organization at the start of the hoax were some 15 times higher. The California doctors said there was no justification for lockdowns and the economic devastation they caused. Everything they had ever learned about quarantine was that you quarantine the *sick* and not the healthy. They had never seen this before and it made no medical sense.

Why in the in the light of all this would governments and medical systems the world over say that billions must go under house arrest; lose their livelihood; in many cases lose their mind, their health and their life; force people to wear masks dangerous to health and psychology; make human interaction and even family interaction a criminal offence; ban travel; close restaurants, bars, watching live sport, concerts, theatre, and any activity involving human togetherness and discourse; and closing schools to isolate children from their friends and cause many to commit suicide in acts of hopelessness and despair? The California doctors said lockdown consequences included increased child abuse, partner abuse, alcoholism, depression, and other impacts they were seeing every day. Who would do that to the entire human race if not mentally-ill psychopaths of almost unimaginable extremes like Bill Gates? We must face the reality of what we are dealing with and come out of

denial. Fascism and tyranny are made possible only by the target population submitting and acquiescing to fascism and tyranny. The whole of human history shows that to be true. Most people naively and unquestioning believed what they were told about a 'deadly virus' and meekly and weakly submitted to house arrest. Those who didn't believe it – at least in total – still submitted in fear of the consequences of not doing so. For the rest who wouldn't submit draconian fines have been imposed, brutal policing by psychopaths *for* psychopaths, and condemnation from the meek and weak who condemn the Pushbackers on behalf of the very force that has them, too, in its gun sights. 'Pathetic' does not even begin to suffice. Britain's brainless 'Health' Secretary Matt Hancock warned anyone lying to border officials about returning from a list of 'hotspot' countries could face a jail sentence of up to ten years which is more than for racially-aggravated assault, incest and attempting to have sex with a child under 13. Hancock is a lunatic, but he has the state apparatus behind him in a Cult-led chain reaction and the same with UK 'Vaccine Minister' Nadhim Zahawi, a prominent member of the mega-Cult secret society, Le Cercle, which featured in my earlier books. The Cult enforces its will on governments and medical systems; government and medical systems enforce their will on business and police; business enforces its will on staff who enforce it on customers; police enforce the will of the Cult on the population and play their essential part in creating a world of fascist control that their own children and grandchildren will have to live in their entire lives. It is a hierarchical pyramid of imposition and acquiescence and, yes indeed, of clinical insanity.

Does anyone bright enough to read this book have to ask what the answer is? I think not, but I will reveal it anyway in the fewest of syllables: Tell the psychos and their moronic lackeys to fuck off and let's get on with our lives. We are many – They are few.

CHAPTER SEVEN

War on your mind

One believes things because one has been conditioned to believe them

Aldous Huxley, Brave New World

I have described the 'Covid' hoax as a 'Psyop' and that is true in every sense and on every level in accordance with the definition of that term which is psychological warfare. Break down the 'Covid pandemic' to the foundation themes and it is psychological warfare on the human individual and collective mind.

The same can be said for the entire human belief system involving every subject you can imagine. Huxley was right in his contention that people believe what they are conditioned to believe and this comes from the repetition throughout their lives of the same falsehoods. They spew from government, corporations, media and endless streams of 'experts' telling you what the Cult wants you to believe and often believing it themselves (although *far* from always). 'Experts' are rewarded with 'prestigious' jobs and titles and as agents of perceptual programming with regular access to the media. The Cult has to control the narrative – control *information* – or they lose control of the vital, crucial, without-which-they-cannot-prevail public perception of reality. The foundation of that control today is the Internet made possible by the Defense Advanced Research Projects Agency (DARPA), the incredibly sinister technological arm of the Pentagon. The Internet is the result of military technology.

DARPA openly brags about establishing the Internet which has been a long-term project to lasso the minds of the global population. I have said for decades the plan is to control information to such an extreme that eventually no one would see or hear anything that the Cult does not approve. We are closing in on that end with ferocious censorship since the 'Covid' hoax began and in my case it started back in the 1990s in terms of books and speaking venues. I had to create my own publishing company in 1995 precisely because no one else would publish my books even then. I think they're all still running.

Cult Internet

To secure total control of information they needed the Internet in which pre-programmed algorithms can seek out 'unclean' content for deletion and even stop it being posted in the first place. The Cult had to dismantle print and non-Internet broadcast media to ensure the transfer of information to the appropriate named 'Web' – a critical expression of the *Cult* web. We've seen the ever-quicken demise of traditional media and control of what is left by a tiny number of corporations operating worldwide. Independent journalism in the mainstream is already dead and never was that more obvious than since the turn of 2020. The Cult wants all information communicated via the Internet to globally censor and allow the plug to be pulled any time. Lockdowns and forced isolation has meant that communication between people has been through electronic means and no longer through face-to-face discourse and discussion. Cult psychopaths have targeted the bars, restaurants, sport, venues and meeting places in general for this reason. None of this is by chance and it's to stop people gathering in any kind of privacy or number while being able to track and monitor all Internet communications and block them as necessary. Even private messages between individuals have been censored by these fascists that control Cult fronts like Facebook, Twitter, Google and YouTube which are all officially run by Sabbatian place-people and from the background by higher-level Sabbatian place people.

Facebook, Google, Amazon and their like were seed-funded and supported into existence with money-no-object infusions of funds either directly or indirectly from DARPA and CIA technology arm In-Q-Tel. The Cult plays the long game and prepares very carefully for big plays like 'Covid'. Amazon is another front in the psychological war and pretty much controls the global market in book sales and increasingly publishing. Amazon's limitless funds have deleted fantastic numbers of independent publishers to seize global domination on the way to deciding which books can be sold and circulated and which cannot. Moves in that direction are already happening. Amazon's leading light Jeff Bezos is the grandson of Lawrence Preston Gise who worked with DARPA predecessor ARPA. Amazon has big connections to the CIA and the Pentagon. The plan I have long described went like this:

1. Employ military technology to establish the Internet.
2. Sell the Internet as a place where people can freely communicate without censorship and allow that to happen until the Net becomes the central and irreversible pillar of human society. If the Internet had been highly censored from the start many would have rejected it.
3. Fund and manipulate major corporations into being to control the circulation of information on your Internet using cover stories about geeks in garages to explain how they came about. Give them unlimited funds to expand rapidly with no need to make a profit for years while non-Cult companies who need to balance the books cannot compete. You know that in these circumstances your Googles, YouTubes, Facebooks and Amazons are going to secure near monopolies by either crushing or buying up the opposition.
4. Allow freedom of expression on both the Internet and communication platforms to draw people in until the Internet is the central and irreversible pillar of human society and your communication corporations have reached a stage of near monopoly domination.
5. Then unleash your always-planned frenzy of censorship on the basis of 'where else are you going to go?' and continue to expand that until nothing remains that the Cult does not want its human targets to see.

The process was timed to hit the 'Covid' hoax to ensure the best chance possible of controlling the narrative which they knew they had to do at all costs. They were, after all, about to unleash a 'deadly virus' that didn't really exist. If you do that in an environment of free-flowing information and opinion you would be dead in the

water before you could say Gates is a psychopath. The network was in place through which the Cult-created-and-owned World Health Organization could dictate the 'Covid' narrative and response policy slavishly supported by Cult-owned Internet communication giants and mainstream media while those telling a different story were censored. Google, YouTube, Facebook and Twitter openly announced that they would do this. What else would we expect from Cult-owned operations like Facebook which former executives have confirmed set out to make the platform more addictive than cigarettes and coldly manipulates emotions of its users to sow division between people and groups and scramble the minds of the young? If Zuckerberg lives out the rest of his life without going to jail for crimes against humanity, and most emphatically against the young, it will be a travesty of justice. Still, no matter, cause and effect will catch up with him eventually and the same with Sergey Brin and Larry Page at Google with its CEO Sundar Pichai who fix the Google search results to promote Cult narratives and hide the opposition. Put the same key words into Google and other search engines like DuckDuckGo and you will see how different results can be. Wikipedia is another intensely biased 'encyclopaedia' which skews its content to the Cult agenda. YouTube links to Wikipedia's version of 'Covid' and 'climate change' on video pages in which experts in their field offer a different opinion (even that is increasingly rare with Wojcicki censorship). Into this 'Covid' silence-them network must be added government media censors, sorry 'regulators', such as Ofcom in the UK which imposed tyrannical restrictions on British broadcasters that had the effect of banning me from ever appearing. Just to debate with me about my evidence and views on 'Covid' would mean breaking the fascistic impositions of Ofcom and its CEO career government bureaucrat Melanie Dawes. Gutless British broadcasters tremble at the very thought of fascist Ofcom.

Psychos behind 'Covid'

The reason for the 'Covid' catastrophe in all its facets and forms can be seen by whom and what is driving the policies worldwide in such a coordinated way. Decisions are not being made to protect health, but to target psychology. The dominant group guiding and 'advising' government policy are not medical professionals. They are psychologists and behavioural scientists. Every major country has its own version of this phenomenon and I'll use the British example to show how it works. In many ways the British version has been affecting the wider world in the form of the huge behaviour manipulation network in the UK which operates in other countries. The network involves private companies, government, intelligence and military. The Cabinet Office is at the centre of the government 'Covid' Psyop and part-owns, with 'innovation charity' Nesta, the Behavioural Insights Team (BIT) which claims to be independent of government but patently isn't. The BIT was established in 2010 and its job is to manipulate the psyche of the population to acquiesce to government demands and so much more. It is also known as the 'Nudge Unit', a name inspired by the 2009 book by two ultra-Zionists, Cass Sunstein and Richard Thaler, called *Nudge: Improving Decisions About Health, Wealth, and Happiness*. The book, as with the Behavioural Insights Team, seeks to 'nudge' behaviour (manipulate it) to make the public follow patterns of action and perception that suit those in authority (the Cult). Sunstein is so skilled at this that he advises the World Health Organization and the UK Behavioural Insights Team and was Administrator of the White House Office of Information and Regulatory Affairs in the Obama administration. Biden appointed him to the Department of Homeland Security – another ultra-Zionist in the fold to oversee new immigration laws which is another policy the Cult wants to control. Sunstein is desperate to silence anyone exposing conspiracies and co-authored a 2008 report on the subject in which suggestions were offered to ban 'conspiracy theorizing' or impose 'some kind of tax, financial or otherwise, on those who disseminate such theories'. I guess a psychiatrist's chair is out of the question?

Sunstein's mate Richard Thaler, an 'academic affiliate' of the UK Behavioural Insights Team, is a proponent of 'behavioural economics' which is defined as the study of 'the effects of psychological, cognitive, emotional, cultural and social factors on the decisions of individuals and institutions'. Study the effects so they can be manipulated to be what you want them to be. Other leading names in the development of behavioural economics are ultra-Zionists Daniel Kahneman and Robert J. Shiller and they, with Thaler, won the Nobel Memorial Prize in Economic Sciences for their work in this field. The Behavioural Insights Team is operating at the heart of the UK government and has expanded globally through partnerships with several universities including Harvard, Oxford, Cambridge, University College London (UCL) and Pennsylvania. They claim to have 'trained' (reframed) 20,000 civil servants and run more than 750 projects involving 400 randomised controlled trials in dozens of countries' as another version of mind reframers Common Purpose. BIT works from its office in New York with cities and their agencies, as well as other partners, across the United States and Canada – this is a company part-owned by the British government Cabinet Office. An executive order by President Cult-servant Obama established a US Social and Behavioral Sciences Team in 2015. They all have the same reason for being and that's to brainwash the population directly and by brainwashing those in positions of authority.

'Covid' mind game

Another prime aspect of the UK mind-control network is the 'independent' [joke] Scientific Pandemic Insights Group on Behaviours (SPI-B) which 'provides behavioural science advice aimed at anticipating and helping people adhere to interventions that are recommended by medical or epidemiological experts'. That means manipulating public perception and behaviour to do whatever government tells them to do. It's disgusting and if they really want the public to be 'safe' this lot should all be under lock and key. According to the government website SPI-B consists of

'behavioural scientists, health and social psychologists, anthropologists and historians' and advises the Whitty-Vallance-led Scientific Advisory Group for Emergencies (SAGE) which in turn advises the government on 'the science' (it doesn't) and 'Covid' policy. When politicians say they are being guided by 'the science' this is the rabble in each country they are talking about and that 'science' is dominated by behaviour manipulators to enforce government fascism through public compliance. The Behaviour Insight Team is headed by psychologist David Solomon Halpern, a visiting professor at King's College London, and connects with a national and global web of other civilian and military organisations as the Cult moves towards its goal of fusing them into one fascistic whole in every country through its 'Fusion Doctrine'. The behaviour manipulation network involves, but is not confined to, the Foreign Office; National Security Council; government communications headquarters (GCHQ); MI5; MI6; the Cabinet Office-based Media Monitoring Unit; and the Rapid Response Unit which 'monitors digital trends to spot emerging issues; including misinformation and disinformation; and identifies the best way to respond'.

There is also the 77th Brigade of the UK military which operates like the notorious Israeli military's Unit 8200 in manipulating information and discussion on the Internet by posing as members of the public to promote the narrative and discredit those who challenge it. Here we have the military seeking to manipulate *domestic* public opinion while the Nazis in government are fine with that. Conservative Member of Parliament Tobias Ellwood, an advocate of lockdown and control through 'vaccine passports', is a Lieutenant Colonel reservist in the 77th Brigade which connects with the military operation jHub, the 'innovation centre' for the Ministry of Defence and Strategic Command. jHub has also been involved with the civilian National Health Service (NHS) in 'symptom tracing' the population. The NHS is a key part of this mind control network and produced a document in December, 2020, explaining to staff how to use psychological manipulation with different groups and ages to get them to have the DNA-manipulating 'Covid vaccine'

that's designed to cumulatively rewrite human genetics. The document, called 'Optimising Vaccination Roll Out – Do's and Dont's for all messaging, documents and "communications" in the widest sense', was published by NHS England and the NHS Improvement *Behaviour Change Unit* in partnership with Public Health England and Warwick Business School. I hear the mantra about 'save the NHS' and 'protect the NHS' when we need to scrap the NHS and start again. The current version is far too corrupt, far too anti-human and totally compromised by Cult operatives and their assets. UK government broadcast media censor Ofcom will connect into this web – as will the BBC with its tremendous Ofcom influence – to control what the public see and hear and dictate mass perception. Nuremberg trials must include personnel from all these organisations.

The fear factor

The 'Covid' hoax has led to the creation of the UK Cabinet Office-connected Joint Biosecurity Centre (JBC) which is officially described as providing 'expert advice on pandemics' using its independent [all Cult operations are 'independent'] analytical function to provide real-time analysis about infection outbreaks to identify and respond to outbreaks of Covid-19'. Another role is to advise the government on a response to spikes in infections – 'for example by closing schools or workplaces in local areas where infection levels have risen'. Put another way, promoting the Cult agenda. The Joint Biosecurity Centre is modelled on the Joint Terrorism Analysis Centre which analyses intelligence to set 'terrorism threat levels' and here again you see the fusion of civilian and military operations and intelligence that has led to military intelligence producing documents about 'vaccine hesitancy' and how it can be combated. Domestic civilian matters and opinions should not be the business of the military. The Joint Biosecurity Centre is headed by Tom Hurd, director general of the Office for Security and Counter-Terrorism from the establishment-to-its-fingertips Hurd family. His father is former Foreign Secretary Douglas Hurd. How coincidental that Tom

Hurd went to the elite Eton College and Oxford University with Boris Johnson. Imperial College with its ridiculous computer modeller Neil Ferguson will connect with this gigantic web that will itself interconnect with similar set-ups in other major and not so major countries. Compared with this Cult network the politicians, be they Boris Johnson, Donald Trump or Joe Biden, are bit-part players 'following the science'. The network of psychologists was on the 'Covid' case from the start with the aim of generating maximum fear of the 'virus' to ensure compliance by the population. A government behavioural science group known as SPI-B produced a paper in March, 2020, for discussion by the main government science advisory group known as SAGE. It was headed 'Options for increasing adherence to social distancing measures' and it said the following in a section headed 'Persuasion':

- A substantial number of people still do not feel sufficiently personally threatened; it could be that they are reassured by the low death rate in their demographic group, although levels of concern may be rising. Having a good understanding of the risk has been found to be positively associated with adoption of COVID-19 social distancing measures in Hong Kong.
- The perceived level of personal threat needs to be increased among those who are complacent, using hard-hitting evaluation of options for increasing social distancing emotional messaging. To be effective this must also empower people by making clear the actions they can take to reduce the threat.
- Responsibility to others: There seems to be insufficient understanding of, or feelings of responsibility about, people's role in transmitting the infection to others ... Messaging about actions need to be framed positively in terms of protecting oneself and the community, and increase confidence that they will be effective.
- Some people will be more persuaded by appeals to play by the rules, some by duty to the community, and some to personal risk.

All these different approaches are needed. The messaging also needs to take account of the realities of different people's lives. Messaging needs to take account of the different motivational levers and circumstances of different people.

All this could be achieved the SPI-B psychologists said by *using the media to increase the sense of personal threat* which translates as terrify the shit out of the population, including children, so they all do what we want. That's not happened has it? Those excuses for 'journalists' who wouldn't know journalism if it bit them on the arse (the great majority) have played their crucial part in serving this Cult-government Psyop to enslave their own kids and grandkids. How they live with themselves I have no idea. The psychological war has been underpinned by constant government 'Covid' propaganda in almost every television and radio ad break, plus the Internet and print media, which has pounded out the fear with taxpayers footing the bill for their own programming. The result has been people terrified of a 'virus' that doesn't exist or one with a tiny fatality rate even if you believe it does. People walk down the street and around the shops wearing face-nappies damaging their health and psychology while others report those who refuse to be that naïve to the police who turn up in their own face-nappies. I had a cameraman come to my flat and he was so frightened of 'Covid' he came in wearing a mask and refused to shake my hand in case he caught something. He had – naïveitis – and the thought that he worked in the mainstream media was both depressing and made his behaviour perfectly explainable. The fear which has gripped the minds of so many and frozen them into compliance has been carefully cultivated by these psychologists who are really psychopaths. If lives get destroyed and a lot of young people commit suicide it shows our plan is working. SPI-B then turned to compulsion on the public to comply. 'With adequate preparation, rapid change can be achieved', it said. Some countries had introduced mandatory self-isolation on a wide scale without evidence of major public unrest and a large majority of the UK's population appeared to be supportive of more coercive measures with 64 percent of adults saying they would

support putting London under a lockdown (watch the 'polls' which are designed to make people believe that public opinion is in favour or against whatever the subject in hand).

For 'aggressive protective measures' to be effective, the SPI-B paper said, special attention should be devoted to those population groups that are more at risk. Translated from the Orwellian this means making the rest of population feel guilty for not protecting the 'vulnerable' such as old people which the Cult and its agencies were about to kill on an industrial scale with lockdown, lack of treatment and the Gates 'vaccine'. Psychopath psychologists sold their guilt-trip so comprehensively that Los Angeles County Supervisor Hilda Solis reported that children were apologising (from a distance) to their parents and grandparents for bringing 'Covid' into their homes and getting them sick. '... These apologies are just some of the last words that loved ones will ever hear as they die alone,' she said. Gut-wrenchingly Solis then used this childhood tragedy to tell children to stay at home and 'keep your loved ones alive'. Imagine heaping such potentially life-long guilt on a kid when it has absolutely nothing to do with them. These people are deeply disturbed and the psychologists behind this even more so.

Uncivil war – divide and rule

Professional mind-controllers at SPI-B wanted the media to increase a sense of responsibility to others (do as you're told) and promote 'positive messaging' for those actions while in contrast to invoke 'social disapproval' by the unquestioning, obedient, community of anyone with a mind of their own. Again the compliant Goebbels-like media obliged. This is an old, old, trick employed by tyrannies the world over throughout human history. You get the target population to keep the target population in line – *your* line. SPI-B said this could 'play an important role in preventing anti-social behaviour or discouraging failure to enact pro-social behaviour'. For 'anti-social' in the Orwellian parlance of SPI-B see any behaviour that government doesn't approve. SPI-B recommendations said that 'social disapproval' should be accompanied by clear messaging and

promotion of strong collective identity – hence the government and celebrity mantra of ‘we’re all in this together’. Sure we are. The mind doctors have such contempt for their targets that they think some clueless comedian, actor or singer telling them to do what the government wants will be enough to win them over. We have had UK comedian Lenny Henry, actor Michael Caine and singer Elton John wheeled out to serve the propagandists by urging people to have the DNA-manipulating ‘Covid’ non-‘vaccine’. The role of Henry and fellow black celebrities in seeking to coax a ‘vaccine’ reluctant black community into doing the government’s will was especially stomach-turning. An emotion-manipulating script and carefully edited video featuring these black ‘celebs’ was such an insult to the intelligence of black people and where’s the self-respect of those involved selling their souls to a fascist government agenda? Henry said he heard black people’s ‘legitimate worries and concerns’, but people must ‘trust the facts’ when they were doing exactly that by not having the ‘vaccine’. They had to include the obligatory reference to Black Lives Matter with the line ... ‘Don’t let coronavirus cost even more black lives – because we matter’. My god, it was pathetic. ‘I know the vaccine is safe and what it does.’ How? ‘I’m a comedian and it says so in my script.’

SPI-B said social disapproval needed to be carefully managed to avoid victimisation, scapegoating and misdirected criticism, but they knew that their ‘recommendations’ would lead to exactly that and the media were specifically used to stir-up the divide-and-conquer hostility. Those who conform like good little baa, baas, are praised while those who have seen through the tidal wave of lies are ‘Covidiot’. The awake have been abused by the fast asleep for not conforming to fascism and impositions that the awake know are designed to endanger their health, dehumanise them, and tear asunder the very fabric of human society. We have had the curtain-twitchers and morons reporting neighbours and others to the face-napped police for breaking ‘Covid rules’ with fascist police delighting in posting links and phone numbers where this could be done. The Cult cannot impose its will without a compliant police

and military or a compliant population willing to play their part in enslaving themselves and their kids. The words of a pastor in Nazi Germany are so appropriate today:

First they came for the socialists and I did not speak out because I was not a socialist.

Then they came for the trade unionists and I did not speak out because I was not a trade unionist.

Then they came for the Jews and I did not speak out because I was not a Jew.

Then they came for me and there was no one left to speak for me.

Those who don't learn from history are destined to repeat it and so many are.

'Covid' rules: Rewiring the mind

With the background laid out to this gigantic national and global web of psychological manipulation we can put 'Covid' rules into a clear and sinister perspective. Forget the claims about protecting health. 'Covid' rules are about dismantling the human mind, breaking the human spirit, destroying self-respect, and then putting Humpty Dumpty together again as a servile, submissive slave. Social isolation through lockdown and distancing have devastating effects on the human psyche as the psychological psychopaths well know and that's the real reason for them. Humans need contact with each other, discourse, closeness and touch, or they eventually, and literally, go crazy. Masks, which I will address at some length, fundamentally add to the effects of isolation and the Cult agenda to dehumanise and de-individualise the population. To do this while knowing – in fact *seeking* – this outcome is the very epitome of evil and psychologists involved in this *are* the epitome of evil. They must like all the rest of the Cult demons and their assets stand trial for crimes against humanity on a scale that defies the imagination. Psychopaths in uniform use isolation to break enemy troops and agents and make them subservient and submissive to tell what they know. The technique is rightly considered a form of torture and

torture is most certainly what has been imposed on the human population.

Clinically-insane American psychologist Harry Harlow became famous for his isolation experiments in the 1950s in which he separated baby monkeys from their mothers and imprisoned them for months on end in a metal container or 'pit of despair'. They soon began to show mental distress and depression as any idiot could have predicted. Harlow put other monkeys in steel chambers for three, six or twelve months while denying them any contact with animals or humans. He said that the effects of total social isolation for six months were 'so devastating and debilitating that we had assumed initially that twelve months of isolation would not produce any additional decrement'; but twelve months of isolation 'almost obliterated the animals socially'. This is what the Cult and its psychopaths are doing to you and your children. Even monkeys in partial isolation in which they were not allowed to form relationships with other monkeys became 'aggressive and hostile, not only to others, but also towards their own bodies'. We have seen this in the young as a consequence of lockdown. UK government psychopaths launched a public relations campaign telling people not to hug each other even after they received the 'Covid-19 vaccine' which we were told with more lies would allow a return to 'normal life'. A government source told *The Telegraph*: 'It will be along the lines that it is great that you have been vaccinated, but if you are going to visit your family and hug your grandchildren there is a chance you are going to infect people you love.' The source was apparently speaking from a secure psychiatric facility. Janet Lord, director of Birmingham University's Institute of Inflammation and Ageing, said that parents and grandparents should avoid hugging their children. Well, how can I put it, Ms Lord? Fuck off. Yep, that'll do.

Destroying the kids – where are the parents?

Observe what has happened to people enslaved and isolated by lockdown as suicide and self-harm has soared worldwide,

particularly among the young denied the freedom to associate with their friends. A study of 49,000 people in English-speaking countries concluded that almost half of young adults are at clinical risk of mental health disorders. A national survey in America of 1,000 currently enrolled high school and college students found that 5 percent reported attempting suicide during the pandemic. Data from the US CDC's National Syndromic Surveillance Program from January 1st to October 17th, 2020, revealed a *31 percent* increase in mental health issues among adolescents aged 12 to 17 compared with 2019. The CDC reported that America in general suffered the biggest drop in life expectancy since World War Two as it fell by a year in the first half of 2020 as a result of 'deaths of despair' – overdoses and suicides. Deaths of despair have leapt by more than 20 percent during lockdown and include the highest number of fatal overdoses ever recorded in a single year – 81,000. Internet addiction is another consequence of being isolated at home which lowers interest in physical activities as kids fall into inertia and what's the point? Children and young people are losing hope and giving up on life, sometimes literally. A 14-year-old boy killed himself in Maryland because he had 'given up' when his school district didn't reopen; an 11-year-old boy shot himself during a zoom class; a teenager in Maine succumbed to the isolation of the 'pandemic' when he ended his life after experiencing a disrupted senior year at school. Children as young as nine have taken their life and all these stories can be repeated around the world. Careers are being destroyed before they start and that includes those in sport in which promising youngsters have not been able to take part. The plan of the psycho-psychologists is working all right. Researchers at Cambridge University found that lockdowns cause significant harm to children's mental health. Their study was published in the *Archives of Disease in Childhood*, and followed 168 children aged between 7 and 11. The researchers concluded:

During the UK lockdown, children's depression symptoms have increased substantially, relative to before lockdown. The scale of this effect has direct relevance for the continuation of different elements of lockdown policy, such as complete or partial school closures ...

... Specifically, we observed a statistically significant increase in ratings of depression, with a medium-to-large effect size. Our findings emphasise the need to incorporate the potential impact of lockdown on child mental health in planning the ongoing response to the global pandemic and the recovery from it.

Not a chance when the Cult's psycho-psychologists were getting exactly what they wanted. The UK's Royal College of Paediatrics and Child Health has urged parents to look for signs of eating disorders in children and young people after a three to four fold increase. Specialists say the 'pandemic' is a major reason behind the rise. You don't say. The College said isolation from friends during school closures, exam cancellations, loss of extra-curricular activities like sport, and an increased use of social media were all contributory factors along with fears about the virus (psycho-psychologists again), family finances, and students being forced to quarantine. Doctors said young people were becoming severely ill by the time they were seen with 'Covid' regulations reducing face-to-face consultations. Nor is it only the young that have been devastated by the psychopaths. Like all bullies and cowards the Cult is targeting the young, elderly, weak and infirm. A typical story was told by a British lady called Lynn Parker who was not allowed to visit her husband in 2020 for the last ten and half months of his life 'when he needed me most' between March 20th and when he died on December 19th. This vacates the criminal and enters the territory of evil. The emotional impact on the immune system alone is immense as are the number of people of all ages worldwide who have died as a result of Cult-demanded, Gates-demanded, lockdowns.

Isolation is torture

The experience of imposing solitary confinement on millions of prisoners around the world has shown how a large percentage become 'actively psychotic and/or acutely suicidal'. Social isolation has been found to trigger 'a specific psychiatric syndrome, characterized by hallucinations; panic attacks; overt paranoia; diminished impulse control; hypersensitivity to external stimuli; and difficulties with thinking, concentration and memory'. Juan Mendez,

a United Nations rapporteur (investigator), said that isolation is a form of torture. Research has shown that even after isolation prisoners find it far more difficult to make social connections and I remember chatting to a shop assistant after one lockdown who told me that when her young son met another child again he had no idea how to act or what to do. Hannah Flanagan, Director of Emergency Services at Journey Mental Health Center in Dane County, Wisconsin, said: 'The specificity about Covid social distancing and isolation that we've come across as contributing factors to the suicides are really new to us this year.' But they are not new to those that devised them. They are getting the effect they want as the population is psychologically dismantled to be rebuilt in a totally different way. Children and the young are particularly targeted. They will be the adults when the full-on fascist AI-controlled technocracy is planned to be imposed and they are being prepared to meekly submit. At the same time older people who still have a memory of what life was like before – and how fascist the new normal really is – are being deleted. You are going to see efforts to turn the young against the old to support this geriatric genocide. Hannah Flanagan said the big increase in suicide in her county proved that social isolation is not only harmful, but deadly. Studies have shown that isolation from others is one of the main risk factors in suicide and even more so with women. Warnings that lockdown could create a 'perfect storm' for suicide were ignored. After all this was one of the *reasons* for lockdown. Suicide, however, is only the most extreme of isolation consequences. There are many others. Dr Dhruv Khullar, assistant professor of healthcare policy at Weill Cornell Medical College, said in a *New York Times* article in 2016 long before the fake 'pandemic':

A wave of new research suggests social separation is bad for us. Individuals with less social connection have disrupted sleep patterns, altered immune systems, more inflammation and higher levels of stress hormones. One recent study found that isolation increases the risk of heart disease by 29 percent and stroke by 32 percent. Another analysis that pooled data from 70 studies and 3.4 million people found that socially isolated individuals had a 30 percent higher risk of dying in the next seven years, and that this effect was largest in middle age.

Loneliness can accelerate cognitive decline in older adults, and isolated individuals are twice as likely to die prematurely as those with more robust social interactions. These effects start early: Socially isolated children have significantly poorer health 20 years later, even after controlling for other factors. All told, loneliness is as important a risk factor for early death as obesity and smoking.

There you have proof from that one article alone four years before 2020 that those who have enforced lockdown, social distancing and isolation knew what the effect would be and that is even more so with professional psychologists that have been driving the policy across the globe. We can go back even further to the years 2000 and 2003 and the start of a major study on the effects of isolation on health by Dr Janine Gronewold and Professor Dirk M. Hermann at the University Hospital in Essen, Germany, who analysed data on 4,316 people with an average age of 59 who were recruited for the long-term research project. They found that socially isolated people are more than 40 percent more likely to have a heart attack, stroke, or other major cardiovascular event and nearly 50 percent more likely to die from any cause. Given the financial Armageddon unleashed by lockdown we should note that the study found a relationship between increased cardiovascular risk and lack of financial support. After excluding other factors social isolation was still connected to a 44 percent increased risk of cardiovascular problems and a 47 percent increased risk of death by any cause. Lack of financial support was associated with a 30 percent increase in the risk of cardiovascular health events. Dr Gronewold said it had been known for some time that feeling lonely or lacking contact with close friends and family can have an impact on physical health and the study had shown that having strong social relationships is of high importance for heart health. Gronewold said they didn't understand yet why people who are socially isolated have such poor health outcomes, but this was obviously a worrying finding, particularly during these times of prolonged social distancing. Well, it can be explained on many levels. You only have to identify the point in the body where people feel loneliness and missing people they are parted from – it's in the centre of the chest where they feel the ache of loneliness and the ache of missing people. 'My heart aches for

you' ... 'My heart aches for some company.' I will explain this more in the chapter Escaping Wetiko, but when you realise that the body is the mind – they are expressions of each other – the reason why state of the mind dictates state of the body becomes clear.

American psychologist Ranjit Powar was highlighting the effects of lockdown isolation as early as April, 2020. She said humans have evolved to be social creatures and are wired to live in interactive groups. Being isolated from family, friends and colleagues could be unbalancing and traumatic for most people and could result in short or even long-term psychological and physical health problems. An increase in levels of anxiety, aggression, depression, forgetfulness and hallucinations were possible psychological effects of isolation. 'Mental conditions may be precipitated for those with underlying pre-existing susceptibilities and show up in many others without any pre-condition.' Powar said personal relationships helped us cope with stress and if we lost this outlet for letting off steam the result can be a big emotional void which, for an average person, was difficult to deal with. 'Just a few days of isolation can cause increased levels of anxiety and depression' – so what the hell has been the effect on the global population of *18 months* of this at the time of writing? Powar said: 'Add to it the looming threat of a dreadful disease being repeatedly hammered in through the media and you have a recipe for many shades of mental and physical distress.' For those with a house and a garden it is easy to forget that billions have had to endure lockdown isolation in tiny overcrowded flats and apartments with nowhere to go outside. The psychological and physical consequences of this are unimaginable and with lunatic and abusive partners and parents the consequences have led to tremendous increases in domestic and child abuse and alcoholism as people seek to shut out the horror. Ranjit Powar said:

Staying in a confined space with family is not all a rosy picture for everyone. It can be extremely oppressive and claustrophobic for large low-income families huddled together in small single-room houses. Children here are not lucky enough to have many board/electronic games or books to keep them occupied.

Add to it the deep insecurity of running out of funds for food and basic necessities. On the other hand, there are people with dysfunctional family dynamics, such as domineering, abusive or alcoholic partners, siblings or parents which makes staying home a period of trial. Incidence of suicide and physical abuse against women has shown a worldwide increase. Heightened anxiety and depression also affect a person's immune system, making them more susceptible to illness.

To think that Powar's article was published on April 11th, 2020.

Six-feet fantasy

Social (unsocial) distancing demanded that people stay six feet or two metres apart. UK government advisor Robert Dingwall from the New and Emerging Respiratory Virus Threats Advisory Group said in a radio interview that the two-metre rule was 'conjured up out of nowhere' and was not based on science. No, it was not based on *medical* science, but it didn't come out of nowhere. The distance related to *psychological* science. Six feet/two metres was adopted in many countries and we were told by people like the criminal Anthony Fauci and his ilk that it was founded on science. Many schools could not reopen because they did not have the space for six-foot distancing. Then in March, 2021, after a year of six-feet 'science', a study published in the *Journal of Infectious Diseases* involving more than 500,000 students and almost 100,000 staff over 16 weeks revealed no significant difference in 'Covid' cases between six feet and three feet and Fauci changed his tune. Now three feet was okay. There is no difference between six feet and three *inches* when there is no 'virus' and they got away with six feet for psychological reasons for as long as they could. I hear journalists and others talk about 'unintended consequences' of lockdown. They are not *unintended* at all; they have been coldly-calculated for a specific outcome of human control and that's why super-psychopaths like Gates have called for them so vehemently. Super-psychopath psychologists have demanded them and psychopathic or clueless, spineless, politicians have gone along with them by 'following the science'. But it's not science at all. 'Science' is not what is; it's only what people can be manipulated to believe it is. The whole 'Covid' catastrophe is

founded on mind control. Three word or three statement mantras issued by the UK government are a well-known mind control technique and so we've had 'Stay home/protect the NHS/save lives', 'Stay alert/control the virus/save lives' and 'hands/face/space'. One of the most vocal proponents of extreme 'Covid' rules in the UK has been Professor Susan Michie, a member of the British Communist Party, who is not a medical professional. Michie is the director of the Centre for Behaviour Change at University College London. She is a *behavioural psychologist* and another filthy rich 'Marxist' who praised China's draconian lockdown. She was known by fellow students at Oxford University as 'Stalin's nanny' for her extreme Marxism. Michie is an influential member of the UK government's Scientific Advisory Group for Emergencies (SAGE) and behavioural manipulation groups which have dominated 'Covid' policy. She is a consultant adviser to the World Health Organization on 'Covid-19' and behaviour. Why the hell are lockdowns anything to do with her when they are claimed to be about health? Why does a behavioural psychologist from a group charged with changing the behaviour of the public want lockdown, human isolation and mandatory masks? Does that question really need an answer? Michie *absolutely* has to explain herself before a Nuremberg court when humanity takes back its world again and even more so when you see the consequences of masks that she demands are compulsory. This is a Michie classic:

The benefits of getting primary school children to wear masks is that regardless of what little degree of transmission is occurring in those age groups it could help normalise the practice. Young children wearing masks may be more likely to get their families to accept masks.

Those words alone should carry a prison sentence when you ponder on the callous disregard for children involved and what a statement it makes about the mind and motivations of Susan Michie. What a lovely lady and what she said there encapsulates the mentality of the psychopaths behind the 'Covid' horror. Let us compare what Michie said with a countrywide study in Germany published at [researchsquare.com](https://www.researchsquare.com) involving 25,000 school children and 17,854 health complaints submitted by parents. Researchers

found that masks are harming children physically, psychologically, and behaviourally with 24 health issues associated with mask wearing. They include: shortness of breath (29.7%); dizziness (26.4%); increased headaches (53%); difficulty concentrating (50%); drowsiness or fatigue (37%); and malaise (42%). Nearly a third of children experienced more sleep issues than before and a quarter developed new fears. Researchers found health issues and other impairments in 68 percent of masked children covering their faces for an average of 4.5 hours a day. Hundreds of those taking part experienced accelerated respiration, tightness in the chest, weakness, and short-term impairment of consciousness. A reminder of what Michie said again:

The benefits of getting primary school children to wear masks is that regardless of what little degree of transmission is occurring in those age groups it could help normalise the practice. Young children wearing masks may be more likely to get their families to accept masks.

Psychopaths in government and psychology now have children and young people – plus all the adults – wearing masks for hours on end while clueless teachers impose the will of the psychopaths on the young they should be protecting. What the hell are parents doing?

Cult lab rats

We have some schools already imposing on students microchipped buzzers that activate when they get 'too close' to their pals in the way they do with lab rats. How apt. To the Cult and its brain-dead servants our children *are* lab rats being conditioned to be unquestioning, dehumanised slaves for the rest of their lives.

Children and young people are being weaned and frightened away from the most natural human instincts including closeness and touch. I have tracked in the books over the years how schools were banning pupils from greeting each other with a hug and the whole Cult-induced Me Too movement has terrified men and boys from a relaxed and natural interaction with female friends and work colleagues to the point where many men try never to be in a room

alone with a woman that's not their partner. Airhead celebrities have as always played their virtue-signalling part in making this happen with their gross exaggeration. For every monster like Harvey Weinstein there are at least tens of thousands of men that don't treat women like that; but everyone must be branded the same and policy changed for them as well as the monster. I am going to be using the word 'dehumanise' many times in this chapter because that is what the Cult is seeking to do and it goes very deep as we shall see. Don't let them kid you that social distancing is planned to end one day. That's not the idea. We are seeing more governments and companies funding and producing wearable gadgets to keep people apart and they would not be doing that if this was meant to be short-term. A tech start-up company backed by GCHQ, the British Intelligence and military surveillance headquarters, has created a social distancing wrist sensor that alerts people when they get too close to others. The CIA has also supported tech companies developing similar devices. The wearable sensor was developed by Tended, one of a number of start-up companies supported by GCHQ (see the CIA and DARPA). The device can be worn on the wrist or as a tag on the waistband and will vibrate whenever someone wearing the device breaches social distancing and gets anywhere near natural human contact. The company had a lucky break in that it was developing a distancing sensor when the 'Covid' hoax arrived which immediately provided a potentially enormous market. How fortunate. The government in big-time Cult-controlled Ontario in Canada is investing \$2.5 million in wearable contact tracing technology that 'will alert users if they may have been exposed to the Covid-19 in the workplace and will beep or vibrate if they are within six feet of another person'. Facedrive Inc., the technology company behind this, was founded in 2016 with funding from the Ontario Together Fund and obviously they, too, had a prophet on the board of directors. The human surveillance and control technology is called TraceSCAN and would be worn by the human cyborgs in places such as airports, workplaces, construction sites, care homes and ... *schools*.

I emphasise schools with children and young people the prime targets. You know what is planned for society as a whole if you keep your eyes on the schools. They have always been places where the state program the next generation of slaves to be its compliant worker-ants – or Woker-ants these days; but in the mist of the ‘Covid’ madness they have been transformed into mind laboratories on a scale never seen before. Teachers and head teachers are just as programmed as the kids – often more so. Children are kept apart from human interaction by walk lanes, classroom distancing, staggered meal times, masks, and the rolling-out of buzzer systems. Schools are now physically laid out as a laboratory maze for lab-rats. Lunatics at a school in Anchorage, Alaska, who should be prosecuted for child abuse, took away desks and forced children to kneel (know your place) on a mat for five hours a day while wearing a mask and using their chairs as a desk. How this was supposed to impact on a ‘virus’ only these clinically insane people can tell you and even then it would be clap-trap. The school banned recess (interaction), art classes (creativity), and physical exercise (getting body and mind moving out of inertia). Everyone behind this outrage should be in jail or better still a mental institution. The behavioural manipulators are all for this dystopian approach to schools. Professor Susan Michie, the mind-doctor and British Communist Party member, said it was wrong to say that schools were safe. They had to be made so by ‘distancing’, masks and ventilation (sitting all day in the cold). I must ask this lady round for dinner on a night I know I am going to be out and not back for weeks. She probably wouldn’t be able to make it, anyway, with all the visits to her own psychologist she must have block-booked.

Masking identity

I know how shocking it must be for you that a behaviour manipulator like Michie wants everyone to wear masks which have long been a feature of mind-control programs like the infamous MKUltra in the United States, but, there we are. We live and learn. I spent many years from 1996 to right across the millennium

researching mind control in detail on both sides of the Atlantic and elsewhere. I met a large number of mind-control survivors and many had been held captive in body and mind by MKUltra. MK stands for mind-control, but employs the German spelling in deference to the Nazis spirited out of Germany at the end of World War Two by Operation Paperclip in which the US authorities, with help from the Vatican, transported Nazi mind-controllers and engineers to America to continue their work. Many of them were behind the creation of NASA and they included Nazi scientist and SS officer Wernher von Braun who swapped designing V-2 rockets to bombard London with designing the Saturn V rockets that powered the NASA moon programme's Apollo craft. I think I may have mentioned that the Cult has no borders. Among Paperclip escapees was Josef Mengele, the Angel of Death in the Nazi concentration camps where he conducted mind and genetic experiments on children often using twins to provide a control twin to measure the impact of his 'work' on the other. If you want to observe the Cult mentality in all its extremes of evil then look into the life of Mengele. I have met many people who suffered mercilessly under Mengele in the United States where he operated under the name Dr Greene and became a stalwart of MKUltra programming and torture. Among his locations was the underground facility in the Mojave Desert in California called the China Lake Naval Weapons Station which is almost entirely below the surface. My books *The Biggest Secret*, *Children of the Matrix* and *The Perception Deception* have the detailed background to MKUltra.

The best-known MKUltra survivor is American Cathy O'Brien. I first met her and her late partner Mark Phillips at a conference in Colorado in 1996. Mark helped her escape and deprogram from decades of captivity in an offshoot of MKUltra known as Project Monarch in which 'sex slaves' were provided for the rich and famous including Father George Bush, Dick Cheney and the Clintons. Read Cathy and Mark's book *Trance-Formation of America* and if you are new to this you will be shocked to the core. I read it in 1996 shortly before, with the usual synchronicity of my life, I found

myself given a book table at the conference right next to hers. MKUltra never ended despite being very publicly exposed (only a small part of it) in the 1970s and continues in other guises. I am still in touch with Cathy. She contacted me during 2020 after masks became compulsory in many countries to tell me how they were used as part of MKUltra programming. I had been observing 'Covid regulations' and the relationship between authority and public for months. I saw techniques that I knew were employed on individuals in MKUltra being used on the global population. I had read many books and manuals on mind control including one called *Silent Weapons for Quiet Wars* which came to light in the 1980s and was a guide on how to perceptually program on a mass scale. 'Silent Weapons' refers to mind-control. I remembered a line from the manual as governments, medical authorities and law enforcement agencies have so obviously talked to – or rather at – the adult population since the 'Covid' hoax began as if they are children. The document said:

If a person is spoken to by a T.V. advertiser as if he were a twelve-year-old, then, due to suggestibility, he will, with a certain probability, respond or react to that suggestion with the uncritical response of a twelve-year-old and will reach in to his economic reservoir and deliver its energy to buy that product on impulse when he passes it in the store.

That's why authority has spoken to adults like children since all this began.

Why did Michael Jackson wear masks?

Every aspect of the 'Covid' narrative has mind-control as its central theme. Cathy O'Brien wrote an article for davidicke.com about the connection between masks and mind control. Her daughter Kelly who I first met in the 1990s was born while Cathy was still held captive in MKUltra. Kelly was forced to wear a mask as part of her programming from the age of *two* to dehumanise her, target her sense of individuality and reduce the amount of oxygen her brain and body received. *Bingo*. This is the real reason for compulsory

masks, why they have been enforced en masse, and why they seek to increase the number they demand you wear. First one, then two, with one disgraceful alleged 'doctor' recommending four which is nothing less than a death sentence. Where and how often they must be worn is being expanded for the purpose of mass mind control and damaging respiratory health which they can call 'Covid-19'. Canada's government headed by the man-child Justin Trudeau, says it's fine for children of two and older to wear masks. An insane 'study' in Italy involving just 47 children concluded there was no problem for babies as young as *four months* wearing them. Even after people were 'vaccinated' they were still told to wear masks by the criminal that is Anthony Fauci. Cathy wrote that mandating masks is allowing the authorities literally to control the air we breathe which is what was done in MKUltra. You might recall how the singer Michael Jackson wore masks and there is a reason for that. He was subjected to MKUltra mind control through Project Monarch and his psyche was scrambled by these simpletons. Cathy wrote:

In MKUltra Project Monarch mind control, Michael Jackson had to wear a mask to silence his voice so he could not reach out for help. Remember how he developed that whisper voice when he wasn't singing? Masks control the mind from the outside in, like the redefining of words is doing. By controlling what we can and cannot say for fear of being labeled racist or beaten, for example, it ultimately controls thought that drives our words and ultimately actions (or lack thereof).

Likewise, a mask muffles our speech so that we are not heard, which controls voice ... words ... mind. This is Mind Control. Masks are an obvious mind control device, and I am disturbed so many people are complying on a global scale. Masks depersonalize while making a person feel as though they have no voice. It is a barrier to others. People who would never choose to comply but are forced to wear a mask in order to keep their job, and ultimately their family fed, are compromised. They often feel shame and are subdued. People have stopped talking with each other while media controls the narrative.

The 'no voice' theme has often become literal with train passengers told not to speak to each other in case they pass on the 'virus', singing banned for the same reason and bonkers California officials telling people riding roller coasters that they cannot shout and scream. Cathy said she heard every day from healed MKUltra survivors who cannot wear a mask without flashing back on ways

their breathing was controlled – ‘from ball gags and penises to water boarding’. She said that through the years when she saw images of people in China wearing masks ‘due to pollution’ that it was really to control their oxygen levels. ‘I knew it was as much of a population control mechanism of depersonalisation as are burkas’, she said. Masks are another Chinese communist/fascist method of control that has been swept across the West as the West becomes China at lightning speed since we entered 2020.

Mask-19

There are other reasons for mandatory masks and these include destroying respiratory health to call it ‘Covid-19’ and stunting brain development of children and the young. Dr Margarite Griesz-Brisson MD, PhD, is a Consultant Neurologist and Neurophysiologist and the Founder and Medical Director of the London Neurology and Pain Clinic. Her CV goes down the street and round the corner. She is clearly someone who cares about people and won’t parrot the propaganda. Griesz-Brisson has a PhD in pharmacology, with special interest in neurotoxicology, environmental medicine, neuroregeneration and neuroplasticity (the way the brain can change in the light of information received). She went public in October, 2020, with a passionate warning about the effects of mask-wearing laws:

The reinhalation of our exhaled air will without a doubt create oxygen deficiency and a flooding of carbon dioxide. We know that the human brain is very sensitive to oxygen deprivation. There are nerve cells for example in the hippocampus that can’t be longer than 3 minutes without oxygen – they cannot survive. The acute warning symptoms are headaches, drowsiness, dizziness, issues in concentration, slowing down of reaction time – reactions of the cognitive system.

Oh, I know, let’s tell bus, truck and taxi drivers to wear them and people working machinery. How about pilots, doctors and police? Griesz-Brisson makes the important point that while the symptoms she mentions may fade as the body readjusts this does not alter the fact that people continue to operate in oxygen deficit with long list of

potential consequences. She said it was well known that neurodegenerative diseases take years or decades to develop. 'If today you forget your phone number, the breakdown in your brain would have already started 20 or 30 years ago.' She said degenerative processes in your brain are getting amplified as your oxygen deprivation continues through wearing a mask. Nerve cells in the brain are unable to divide themselves normally in these circumstances and lost nerve cells will no longer be regenerated. 'What is gone is gone.' Now consider that people like shop workers and *schoolchildren* are wearing masks for hours every day. What in the name of sanity is going to be happening to them? 'I do not wear a mask, I need my brain to think', Griesz-Brisson said, 'I want to have a clear head when I deal with my patients and not be in a carbon dioxide-induced anaesthesia'. If you are told to wear a mask anywhere ask the organisation, police, store, whatever, for their risk assessment on the dangers and negative effects on mind and body of enforcing mask-wearing. They won't have one because it has never been done not even by government. All of them must be subject to class-action lawsuits as the consequences come to light. They don't do mask risk assessments for an obvious reason. They know what the conclusions would be and independent scientific studies that *have* been done tell a horror story of consequences.

'Masks are criminal'

Dr Griesz-Brisson said that for children and adolescents, masks are an absolute no-no. They had an extremely active and adaptive immune system and their brain was incredibly active with so much to learn. 'The child's brain, or the youth's brain, is thirsting for oxygen.' The more metabolically active an organ was, the more oxygen it required; and in children and adolescents every organ was metabolically active. Griesz-Brisson said that to deprive a child's or adolescent's brain of oxygen, or to restrict it in any way, was not only dangerous to their health, it was absolutely criminal. 'Oxygen deficiency inhibits the development of the brain, and the damage that has taken place as a result CANNOT be reversed.' Mind

manipulators of MKUltra put masks on two-year-olds they wanted to neurologically rewire and you can see why. Griesz-Brisson said a child needs the brain to learn and the brain needs oxygen to function. 'We don't need a clinical study for that. This is simple, indisputable physiology.' Consciously and purposely induced oxygen deficiency was an absolutely deliberate health hazard, and an absolute medical contraindication which means that 'this drug, this therapy, this method or measure should not be used, and is not allowed to be used'. To coerce an entire population to use an absolute medical contraindication by force, she said, there had to be definite and serious reasons and the reasons must be presented to competent interdisciplinary and independent bodies to be verified and authorised. She had this warning of the consequences that were coming if mask wearing continued:

When, in ten years, dementia is going to increase exponentially, and the younger generations couldn't reach their god-given potential, it won't help to say 'we didn't need the masks'. I know how damaging oxygen deprivation is for the brain, cardiologists know how damaging it is for the heart, pulmonologists know how damaging it is for the lungs. Oxygen deprivation damages every single organ. Where are our health departments, our health insurance, our medical associations? It would have been their duty to be vehemently against the lockdown and to stop it and stop it from the very beginning.

Why do the medical boards issue punishments to doctors who give people exemptions? Does the person or the doctor seriously have to prove that oxygen deprivation harms people? What kind of medicine are our doctors and medical associations representing? Who is responsible for this crime? The ones who want to enforce it? The ones who let it happen and play along, or the ones who don't prevent it?

All of the organisations and people she mentions there either answer directly to the Cult or do whatever hierarchical levels above them tell them to do. The outcome of both is the same. 'It's not about masks, it's not about viruses, it's certainly not about your health', Griesz-Brisson said. 'It is about much, much more. I am not participating. I am not afraid.' They were taking our air to breathe and there was no unfounded medical exemption from face masks. Oxygen deprivation was dangerous for every single brain. It had to be the free decision of every human being whether they want to

wear a mask that was absolutely ineffective to protect themselves from a virus. She ended by rightly identifying where the responsibility lies for all this:

The imperative of the hour is personal responsibility. We are responsible for what we think, not the media. We are responsible for what we do, not our superiors. We are responsible for our health, not the World Health Organization. And we are responsible for what happens in our country, not the government.

Halle-bloody-lujah.

But surgeons wear masks, right?

Independent studies of mask-wearing have produced a long list of reports detailing mental, emotional and physical dangers. What a definition of insanity to see police officers imposing mask-wearing on the public which will cumulatively damage their health while the police themselves wear masks that will cumulatively damage *their* health. It's utter madness and both public and police do this because 'the government says so' – yes a government of brain-donor idiots like UK Health Secretary Matt Hancock reading the 'follow the science' scripts of psychopathic, lunatic psychologists. The response you get from Stockholm syndrome sufferers defending the very authorities that are destroying them and their families is that 'surgeons wear masks'. This is considered the game, set and match that they must work and don't cause oxygen deficit. Well, actually, scientific studies have shown that they *do* and oxygen levels are monitored in operating theatres to compensate. Surgeons wear masks to stop spittle and such like dropping into open wounds – not to stop 'viral particles' which are so miniscule they can only be seen through an electron microscope. Holes in the masks are significantly bigger than 'viral particles' and if you sneeze or cough they will breach the mask. I watched an incredibly disingenuous 'experiment' that claimed to prove that masks work in catching 'virus' material from the mouth and nose. They did this with a slow motion camera and the mask did block big stuff which stayed inside the mask and

against the face to be breathed in or cause infections on the face as we have seen with many children. 'Viral particles', however, would never have been picked up by the camera as they came through the mask when they are far too small to be seen. The 'experiment' was therefore disingenuous *and* useless.

Studies have concluded that wearing masks in operating theatres (and thus elsewhere) make no difference to preventing infection while the opposite is true with toxic shite building up in the mask and this had led to an explosion in tooth decay and gum disease dubbed by dentists 'mask mouth'. You might have seen the Internet video of a furious American doctor urging people to take off their masks after a four-year-old patient had been rushed to hospital the night before and nearly died with a lung infection that doctors sourced to mask wearing. A study in the journal *Cancer Discovery* found that inhalation of harmful microbes can contribute to advanced stage lung cancer in adults and long-term use of masks can help breed dangerous pathogens. Microbiologists have said frequent mask wearing creates a moist environment in which microbes can grow and proliferate before entering the lungs. The Canadian Agency for Drugs and Technologies in Health, or CADTH, a Canadian national organisation that provides research and analysis to healthcare decision-makers, said this as long ago as 2013 in a report entitled 'Use of Surgical Masks in the Operating Room: A Review of the Clinical Effectiveness and Guidelines'. It said:

- No evidence was found to support the use of surgical face masks to reduce the frequency of surgical site infections
- No evidence was found on the effectiveness of wearing surgical face masks to protect staff from infectious material in the operating room.
- Guidelines recommend the use of surgical face masks by staff in the operating room to protect both operating room staff and patients (despite the lack of evidence).

We were told that the world could go back to 'normal' with the arrival of the 'vaccines'. When they came, fraudulent as they are, the story changed as I knew that it would. We are in the midst of transforming 'normal', not going back to it. Mary Ramsay, head of immunisation at Public Health England, echoed the words of US criminal Anthony Fauci who said masks and other regulations must stay no matter if people are vaccinated. The Fauci idiot continued to wear two masks – different colours so both could be clearly seen – after he *claimed* to have been vaccinated. Senator Rand Paul told Fauci in one exchange that his double-masks were 'theatre' and he was right. It's all theatre. Mary Ramsay back-tracked on the vaccine-return-to-normal theme when she said the public may need to wear masks and social-distance for years despite the jabs. 'People have got used to those lower-level restrictions now, and [they] can live with them', she said telling us what the idea has been all along. 'The vaccine does not give you a pass, even if you have had it, you must continue to follow all the guidelines' said a Public Health England statement which reneged on what we had been told before and made having the 'vaccine' irrelevant to 'normality' even by the official story. Spain's fascist government trumped everyone by passing a law mandating the wearing of masks on the beach and even when swimming in the sea. The move would have devastated what's left of the Spanish tourist industry, posed potential breathing dangers to swimmers and had Northern European sunbathers walking around with their forehead brown and the rest of their face white as a sheet. The ruling was so crazy that it had to be retracted after pressure from public and tourist industry, but it confirmed where the Cult wants to go with masks and how clinically insane authority has become. The determination to make masks permanent and hide the serious dangers to body and mind can be seen in the censorship of scientist Professor Denis Rancourt by Bill Gates-funded academic publishing website ResearchGate over his papers exposing the dangers and uselessness of masks. Rancourt said:

ResearchGate today has permanently locked my account, which I have had since 2015. Their reasons graphically show the nature of their attack against democracy, and their corruption of

science ... By their obscene non-logic, a scientific review of science articles reporting on harms caused by face masks has a 'potential to cause harm'. No criticism of the psychological device (face masks) is tolerated, if the said criticism shows potential to influence public policy.

This is what happens in a fascist world.

Where are the 'greens' (again)?

Other dangers of wearing masks especially regularly relate to the inhalation of minute plastic fibres into the lungs and the deluge of discarded masks in the environment and oceans. Estimates predicted that more than 1.5 billion disposable masks will end up in the world's oceans every year polluting the water with tons of plastic and endangering marine wildlife. Studies project that humans are using 129 billion face masks each month worldwide – about three million a minute. Most are disposable and made from plastic, non-biodegradable microfibers that break down into smaller plastic particles that become widespread in ecosystems. They are littering cities, clogging sewage channels and turning up in bodies of water. I have written in other books about the immense amounts of microplastics from endless sources now being absorbed into the body. Rolf Halden, director of the Arizona State University (ASU) Biodesign Center for Environmental Health Engineering, was the senior researcher in a 2020 study that analysed 47 human tissue samples and found microplastics in all of them. 'We have detected these chemicals of plastics in every single organ that we have investigated', he said. I wrote in *The Answer* about the world being deluged with microplastics. A study by the Worldwide Fund for Nature (WWF) found that people are consuming on average every week some 2,000 tiny pieces of plastic mostly through water and also through marine life and the air. Every year humans are ingesting enough microplastics to fill a heaped dinner plate and in a life-time of 79 years it is enough to fill two large waste bins. Marco Lambertini, WWF International director general said: 'Not only are plastics polluting our oceans and waterways and killing marine life – it's in all of us and we can't escape consuming plastics,' American

geologists found tiny plastic fibres, beads and shards in rainwater samples collected from the remote slopes of the Rocky Mountain National Park near Denver, Colorado. Their report was headed: 'It is raining plastic.' Rachel Adams, senior lecturer in Biomedical Science at Cardiff Metropolitan University, said that among health consequences are internal inflammation and immune responses to a 'foreign body'. She further pointed out that microplastics become carriers of toxins including mercury, pesticides and dioxins (a known cause of cancer and reproductive and developmental problems). These toxins accumulate in the fatty tissues once they enter the body through microplastics. Now this is being compounded massively by people putting plastic on their face and throwing it away.

Workers exposed to polypropylene plastic fibres known as 'flock' have developed 'flock worker's lung' from inhaling small pieces of the flock fibres which can damage lung tissue, reduce breathing capacity and exacerbate other respiratory problems. *Now ...* commonly used surgical masks have three layers of melt-blown textiles made of ... polypropylene. We have billions of people putting these microplastics against their mouth, nose and face for hours at a time day after day in the form of masks. How does anyone think that will work out? I mean – what could possibly go wrong? We posted a number of scientific studies on this at davidicke.com, but when I went back to them as I was writing this book the links to the science research website where they were hosted were dead. Anything that challenges the official narrative in any way is either censored or vilified. The official narrative is so unsupportable by the evidence that only deleting the truth can protect it. A study by Chinese scientists still survived – with the usual twist which it why it was still active, I guess. Yes, they found that virtually all the masks they tested increased the daily intake of microplastic fibres, but people should still wear them because the danger from the 'virus' was worse said the crazy 'team' from the Institute of Hydrobiology in Wuhan. Scientists first discovered microplastics in lung tissue of some patients who died of lung cancer

in the 1990s. Subsequent studies have confirmed the potential health damage with the plastic degrading slowly and remaining in the lungs to accumulate in volume. Wuhan researchers used a machine simulating human breathing to establish that masks shed up to nearly 4,000 microplastic fibres in a month with reused masks producing more. Scientists said some masks are laced with toxic chemicals and a variety of compounds seriously restricted for both health and environmental reasons. They include cobalt (used in blue dye) and formaldehyde known to cause watery eyes, burning sensations in the eyes, nose, and throat, plus coughing, wheezing and nausea. No – that must be ‘Covid-19’.

Mask ‘worms’

There is another and potentially even more sinister content of masks. Mostly new masks of different makes filmed under a microscope around the world have been found to contain strange black fibres or ‘worms’ that appear to move or ‘crawl’ by themselves and react to heat and water. The nearest I have seen to them are the self-replicating fibres that are pulled out through the skin of those suffering from Morgellons disease which has been connected to the phenomena of ‘chemtrails’ which I will bring into the story later on. Morgellons fibres continue to grow outside the body and have a form of artificial intelligence. Black ‘worm’ fibres in masks have that kind of feel to them and there is a nanotechnology technique called ‘worm micelles’ which carry and release drugs or anything else you want to deliver to the body. For sure the suppression of humanity by mind altering drugs is the Cult agenda big time and the more excuses they can find to gain access to the body the more opportunities there are to make that happen whether through ‘vaccines’ or masks pushed against the mouth and nose for hours on end.

So let us summarise the pros and cons of masks:

Against masks: Breathing in your own carbon dioxide; depriving the body and brain of sufficient oxygen; build-up of toxins in the mask that can be breathed into the lungs and cause rashes on the face and 'mask-mouth'; breathing microplastic fibres and toxic chemicals into the lungs; dehumanisation and deleting individualisation by literally making people faceless; destroying human emotional interaction through facial expression and deleting parental connection with their babies which look for guidance to their facial expression.

For masks: They don't protect you from a 'virus' that doesn't exist and even if it did 'viral' particles are so minute they are smaller than the holes in the mask.

Governments, police, supermarkets, businesses, transport companies, and all the rest who seek to impose masks have done no risk assessment on their consequences for health and psychology and are now open to group lawsuits when the impact becomes clear with a cumulative epidemic of respiratory and other disease. Authorities will try to exploit these effects and hide the real cause by dubbing them 'Covid-19'. Can you imagine setting out to force the population to wear health-destroying masks without doing any assessment of the risks? It is criminal and it is evil, but then how many people targeted in this way, who see their children told to wear them all day at school, have asked for a risk assessment? Billions can't be imposed upon by the few unless the billions allow it. Oh, yes, with just a tinge of irony, 85 percent of all masks made worldwide come from *China*.

Wash your hands in toxic shite

'Covid' rules include the use of toxic sanitisers and again the health consequences of constantly applying toxins to be absorbed through the skin is obvious to any level of Renegade Mind. America's Food and Drug Administration (FDA) said that sanitisers are drugs and issued a warning about 75 dangerous brands which contain

methanol used in antifreeze and can cause death, kidney damage and blindness. The FDA circulated the following warning even for those brands that it claims to be safe:

Store hand sanitizer out of the reach of pets and children, and children should use it only with adult supervision. Do not drink hand sanitizer. This is particularly important for young children, especially toddlers, who may be attracted by the pleasant smell or brightly colored bottles of hand sanitizer.

Drinking even a small amount of hand sanitizer can cause alcohol poisoning in children. (However, there is no need to be concerned if your children eat with or lick their hands after using hand sanitizer.) During this coronavirus pandemic, poison control centers have had an increase in calls about accidental ingestion of hand sanitizer, so it is important that adults monitor young children's use.

Do not allow pets to swallow hand sanitizer. If you think your pet has eaten something potentially dangerous, call your veterinarian or a pet poison control center right away. Hand sanitizer is flammable and should be stored away from heat and flames. When using hand sanitizer, rub your hands until they feel completely dry before performing activities that may involve heat, sparks, static electricity, or open flames.

There you go, perfectly safe, then, and that's without even a mention of the toxins absorbed through the skin. Come on kids – sanitise your hands everywhere you go. It will save you from the 'virus'. Put all these elements together of the 'Covid' normal and see how much health and psychology is being cumulatively damaged, even devastated, to 'protect your health'. Makes sense, right? They are only imposing these things because they care, right? *Right?*

Submitting to insanity

Psychological reframing of the population goes very deep and is done in many less obvious ways. I hear people say how contradictory and crazy 'Covid' rules are and how they are ever changing. This is explained away by dismissing those involved as idiots. It is a big mistake. The Cult is delighted if its cold calculation is perceived as incompetence and idiocy when it is anything but. Oh, yes, there are idiots within the system – lots of them – but they are *administering* the Cult agenda, mostly unknowingly. They are not deciding and dictating it. The bulwark against tyranny is self-

respect, always has been, always will be. It is self-respect that has broken every tyranny in history. By its very nature self-respect will not bow to oppression and its perpetrators. There is so little self-respect that it's always the few that overturn dictators. Many may eventually follow, but the few with the iron spines (self-respect) kick it off and generate the momentum. The Cult targets self-respect in the knowledge that once this has gone only submission remains. Crazy, contradictory, ever-changing 'Covid' rules are systematically applied by psychologists to delete self-respect. They *want* you to see that the rules make no sense. It is one thing to decide to do something when *you* have made the choice based on evidence and logic. You still retain your self-respect. It is quite another when you can see what you are being told to do is insane, ridiculous and makes no sense, and *yet you still do it*. Your self-respect is extinguished and this has been happening as ever more obviously stupid and nonsensical things have been demanded and the great majority have complied even when they can see they are stupid and nonsensical.

People walk around in face-nappies knowing they are damaging their health and make no difference to a 'virus'. They do it in fear of not doing it. I know it's daft, but I'll do it anyway. When that happens something dies inside of you and submissive reframing has begun. Next there's a need to hide from yourself that you have conceded your self-respect and you convince yourself that you have not really submitted to fear and intimidation. You begin to believe that you are complying with craziness because it's the right thing to do. When first you concede your self-respect of $2+2 = 4$ to $2+2 = 5$ you *know* you are compromising your self-respect. Gradually to avoid facing that fact you begin to *believe* that $2+2=5$. You have been reframed and I have been watching this process happening in the human psyche on an industrial scale. The Cult is working to break your spirit and one of its major tools in that war is humiliation. I read how former American soldier Bradley Manning (later Chelsea Manning after a sex-change) was treated after being jailed for supplying WikiLeaks with documents exposing the enormity of

government and elite mendacity. Manning was isolated in solitary confinement for eight months, put under 24-hour surveillance, forced to hand over clothing before going to bed, and stand naked for every roll call. This is systematic humiliation. The introduction of anal swab 'Covid' tests in China has been done for the same reason to delete self-respect and induce compliant submission. Anal swabs are mandatory for incoming passengers in parts of China and American diplomats have said they were forced to undergo the indignity which would have been calculated humiliation by the Cult-owned Chinese government that has America in its sights.

Government-people: An abusive relationship

Spirit-breaking psychological techniques include giving people hope and apparent respite from tyranny only to take it away again. This happened in the UK during Christmas, 2020, when the psychopsychologists and their political lackeys announced an easing of restrictions over the holiday only to reimpose them almost immediately on the basis of yet another lie. There is a big psychological difference between getting used to oppression and being given hope of relief only to have that dashed. Psychologists know this and we have seen the technique used repeatedly. Then there is traumatising people before you introduce more extreme regulations that require compliance. A perfect case was the announcement by the dark and sinister Whitty and Vallance in the UK that 'new data' predicted that 4,000 could die every day over the winter of 2020/2021 if we did not lockdown again. I think they call it lying and after traumatising people with that claim out came Jackboot Johnson the next day with new curbs on human freedom. Psychologists know that a frightened and traumatised mind becomes suggestable to submission and behaviour reframing. Underpinning all this has been to make people fearful and suspicious of each other and see themselves as a potential danger to others. In league with deleted self-respect you have the perfect psychological recipe for self-loathing. The relationship between authority and public is now demonstrably the same as that of

subservience to an abusive partner. These are signs of an abusive relationship explained by psychologist Leslie Becker-Phelps:

Psychological and emotional abuse: Undermining a partner's self-worth with verbal attacks, name-calling, and belittling. Humiliating the partner in public, unjustly accusing them of having an affair, or interrogating them about their every behavior. Keeping partner confused or off balance by saying they were just kidding or blaming the partner for 'making' them act this way ... Feigning in public that they care while turning against them in private. This leads to victims frequently feeling confused, incompetent, unworthy, hopeless, and chronically self-doubting. [Apply these techniques to how governments have treated the population since New Year, 2020, and the parallels are obvious.]

Physical abuse: The abuser might physically harm their partner in a range of ways, such as grabbing, hitting, punching, or shoving them. They might throw objects at them or harm them with a weapon. [Observe the physical harm imposed by masks, lockdown, and so on.]

Threats and intimidation: One way abusers keep their partners in line is by instilling fear. They might be verbally threatening, or give threatening looks or gestures. Abusers often make it known that they are tracking their partner's every move. They might destroy their partner's possessions, threaten to harm them, or threaten to harm their family members. Not surprisingly, victims of this abuse often feel anxiety, fear, and panic. [No words necessary.]

Isolation: Abusers often limit their partner's activities, forbidding them to talk or interact with friends or family. They might limit access to a car or even turn off their phone. All of this might be done by physically holding them against their will, but is often accomplished through psychological abuse and intimidation. The more isolated a person feels, the fewer resources they have to help gain perspective on their situation and to escape from it. [No words necessary.]

Economic abuse: Abusers often make their partners beholden to them for money by controlling access to funds of any kind. They might prevent their partner from getting a job or withhold access to money they earn from a job. This creates financial dependency that makes leaving the relationship very difficult. [See destruction of livelihoods and the proposed meagre 'guaranteed income' so long as you do whatever you are told.]

Using children: An abuser might disparage their partner's parenting skills, tell their children lies about their partner, threaten to take custody of their children, or threaten to harm their children. These tactics instil fear and often elicit compliance. [See reframed social service mafia and how children are being mercilessly abused by the state over 'Covid' while their parents look on too frightened to do anything.]

A further recurring trait in an abusive relationship is the abused blaming themselves for their abuse and making excuses for the abuser. We have the public blaming each other for lockdown abuse by government and many making excuses for the government while attacking those who challenge the government. How often we have heard authorities say that rules are being imposed or reimposed only because people have refused to 'behave' and follow the rules. We don't want to do it – it's *you*.

Renegade Minds are an antidote to all of these things. They will never concede their self-respect no matter what the circumstances. Even when apparent humiliation is heaped upon them they laugh in its face and reflect back the humiliation on the abuser where it belongs. Renegade Minds will never wear masks they know are only imposed to humiliate, suppress and damage both physically and psychologically. Consequences will take care of themselves and they will never break their spirit or cause them to concede to tyranny. UK newspaper columnist Peter Hitchens was one of the few in the mainstream media to speak out against lockdowns and forced vaccinations. He then announced he had taken the jab. He wanted to see family members abroad and he believed vaccine passports were inevitable even though they had not yet been introduced. Hitchens

has a questioning and critical mind, but not a Renegade one. If he had no amount of pressure would have made him concede. Hitchens excused his action by saying that the battle has been lost. Renegade Minds never accept defeat when freedom is at stake and even if they are the last one standing the self-respect of not submitting to tyranny is more important than any outcome or any consequence.

That's why Renegade Minds are the only minds that ever changed anything worth changing.

CHAPTER EIGHT

'Reframing' insanity

Insanity is relative. It depends on who has who locked in what cage
Ray Bradbury

Reframing' a mind means simply to change its perception and behaviour. This can be done subconsciously to such an extent that subjects have no idea they have been 'reframed' while to any observer changes in behaviour and attitudes are obvious.

Human society is being reframed on a ginormous scale since the start of 2020 and here we have the reason why psychologists rather than doctors have been calling the shots. Ask most people who have succumbed to 'Covid' reframing if they have changed and most will say 'no'; but they *have* and fundamentally. The Cult's long-game has been preparing for these times since way back and crucial to that has been to prepare both population and officialdom mentally and emotionally. To use the mind-control parlance they had to reframe the population with a mentality that would submit to fascism and reframe those in government and law enforcement to impose fascism or at least go along with it. The result has been the fact-deleted mindlessness of 'Wokeness' and officialdom that has either enthusiastically or unquestioningly imposed global tyranny demanded by reframed politicians on behalf of psychopathic and deeply evil cultists. 'Cognitive reframing' identifies and challenges the way someone sees the world in the form of situations, experiences and emotions and then restructures those perceptions to view the same set of circumstances in a different way. This can have

benefits if the attitudes are personally destructive while on the other side it has the potential for individual and collective mind control which the subject has no idea has even happened.

Cognitive therapy was developed in the 1960s by Aaron T. Beck who was born in Rhode Island in 1921 as the son of Jewish immigrants from the Ukraine. He became interested in the techniques as a treatment for depression. Beck's daughter Judith S. Beck is prominent in the same field and they founded the Beck Institute for Cognitive Behavior Therapy in Philadelphia in 1994. Cognitive reframing, however, began to be used worldwide by those with a very dark agenda. The Cult reframes politicians to change their attitudes and actions until they are completely at odds with what they once appeared to stand for. The same has been happening to government administrators at all levels, law enforcement, military and the human population. Cultists love mind control for two main reasons: It allows them to control what people think, do and say to secure agenda advancement and, by definition, it calms their legendary insecurity and fear of the unexpected. I have studied mind control since the time I travelled America in 1996. I may have been talking to next to no one in terms of an audience in those years, but my goodness did I gather a phenomenal amount of information and knowledge about so many things including the techniques of mind control. I have described this in detail in other books going back to *The Biggest Secret* in 1998. I met a very large number of people recovering from MKUltra and its offshoots and successors and I began to see how these same techniques were being used on the population in general. This was never more obvious than since the 'Covid' hoax began.

Reframing the enforcers

I have observed over the last two decades and more the very clear transformation in the dynamic between the police, officialdom and the public. I tracked this in the books as the relationship mutated from one of serving the public to seeing them as almost the enemy and certainly a lower caste. There has always been a class divide

based on income and always been some psychopathic, corrupt, and big-I-am police officers. This was different. Wholesale change was unfolding in the collective dynamic; it was less about money and far more about position and perceived power. An us-and-them was emerging. Noses were lifted skyward by government administration and law enforcement and their attitude to the public they were *supposed* to be serving changed to one of increasing contempt, superiority and control. The transformation was so clear and widespread that it had to be planned. Collective attitudes and dynamics do not change naturally and organically that quickly on that scale. I then came across an organisation in Britain called Common Purpose created in the late 1980s by Julia Middleton who would work in the office of Deputy Prime Minister John Prescott during the long and disastrous premiership of war criminal Tony Blair. When Blair speaks the Cult is speaking and the man should have been in jail a long time ago. Common Purpose proclaims itself to be one of the biggest 'leadership development' organisations in the world while functioning as a *charity* with all the financial benefits which come from that. It hosts 'leadership development' courses and programmes all over the world and claims to have 'brought together' what it calls 'leaders' from more than 100 countries on six continents. The modus operandi of Common Purpose can be compared with the work of the UK government's reframing network that includes the Behavioural Insights Team 'nudge unit' and 'Covid' reframing specialists at SPI-B. WikiLeaks described Common Purpose long ago as 'a hidden virus in our government and schools' which is unknown to the general public: 'It recruits and trains "leaders" to be loyal to the directives of Common Purpose and the EU, instead of to their own departments, which they then undermine or subvert, the NHS [National Health Service] being an example.' This is a vital point to understand the 'Covid' hoax. The NHS, and its equivalent around the world, has been utterly reframed in terms of administrators and much of the medical personnel with the transformation underpinned by recruitment policies. The outcome has been the criminal and psychopathic behaviour of the

NHS over 'Covid' and we have seen the same in every other major country. WikiLeaks said Common Purpose trainees are 'learning to rule without regard to democracy' and to usher in a police state (current events explained). Common Purpose operated like a 'glue' and had members in the NHS, BBC, police, legal profession, church, many of Britain's 7,000 quangos, local councils, the Civil Service, government ministries and Parliament, and controlled many RDA's (Regional Development Agencies). Here we have one answer for how and why British institutions and their like in other countries have changed so negatively in relation to the public. This further explains how and why the beyond-disgraceful reframed BBC has become a propaganda arm of 'Covid' fascism. They are all part of a network pursuing the same goal.

By 2019 Common Purpose was quoting a figure of 85,000 'leaders' that had attended its programmes. These 'students' of all ages are known as Common Purpose 'graduates' and they consist of government, state and local government officials and administrators, police chiefs and officers, and a whole range of others operating within the national, local and global establishment. Cressida Dick, Commissioner of the London Metropolitan Police, is the Common Purpose graduate who was the 'Gold Commander' that oversaw what can only be described as the murder of Brazilian electrician Jean Charles de Menezes in 2005. He was held down by psychopathic police and shot seven times in the head by a psychopathic lunatic after being mistaken for a terrorist when he was just a bloke going about his day. Dick authorised officers to pursue and keep surveillance on de Menezes and ordered that he be stopped from entering the underground train system. Police psychopaths took her at her word clearly. She was 'disciplined' for this outrage by being *promoted* – eventually to the top of the 'Met' police where she has been a disaster. Many Chief Constables controlling the police in different parts of the UK are and have been Common Purpose graduates. I have heard the 'graduate' network described as a sort of Mafia or secret society operating within the fabric of government at all levels pursuing a collective policy

ingrained at Common Purpose training events. Founder Julia Middleton herself has said:

Locally and internationally, Common Purpose graduates will be 'lighting small fires' to create change in their organisations and communities ... The Common Purpose effect is best illustrated by the many stories of small changes brought about by leaders, who themselves have changed.

A Common Purpose mission statement declared:

Common Purpose aims to improve the way society works by expanding the vision, decision-making ability and influence of all kinds of leaders. The organisation runs a variety of educational programmes for leaders of all ages, backgrounds and sectors, in order to provide them with the inspirational, information and opportunities they need to change the world.

Yes, but into what? Since 2020 the answer has become clear.

NLP and the Delphi technique

Common Purpose would seem to be a perfect name or would common programming be better? One of the foundation methods of reaching 'consensus' (group think) is by setting the agenda theme and then encouraging, cajoling or pressuring everyone to agree a 'consensus' in line with the core theme promoted by Common Purpose. The methodology involves the 'Delphi technique', or an adaption of it, in which opinions are expressed that are summarised by a 'facilitator or change agent' at each stage. Participants are 'encouraged' to modify their views in the light of what others have said. Stage by stage the former individual opinions are merged into group consensus which just happens to be what Common Purpose wants them to believe. A key part of this is to marginalise anyone refusing to concede to group think and turn the group against them to apply pressure to conform. We are seeing this very technique used on the general population to make 'Covid' group-thinkers hostile to those who have seen through the bullshit. People can be reframed by using perception manipulation methods such as Neuro-Linguistic Programming (NLP) in which you change perception with the use of

carefully constructed language. An NLP website described the technique this way:

... A method of influencing brain behaviour (the 'neuro' part of the phrase) through the use of language (the 'linguistic' part) and other types of communication to enable a person to 'recode' the way the brain responds to stimuli (that's the 'programming') and manifest new and better behaviours. Neuro-Linguistic Programming often incorporates hypnosis and self-hypnosis to help achieve the change (or 'programming') that is wanted.

British alternative media operation UKColumn has done very detailed research into Common Purpose over a long period. I quoted co-founder and former naval officer Brian Gerrish in my book *Remember Who You Are*, published in 2011, as saying the following years before current times:

It is interesting that many of the mothers who have had children taken by the State speak of the Social Services people being icily cool, emotionless and, as two ladies said in slightly different words, '... like little robots'. We know that NLP is cumulative, so people can be given small imperceptible doses of NLP in a course here, another in a few months, next year etc. In this way, major changes are accrued in their personality, but the day by day change is almost unnoticeable.

In these and other ways 'graduates' have had their perceptions uniformly reframed and they return to their roles in the institutions of government, law enforcement, legal profession, military, 'education', the UK National Health Service and the whole swathe of the establishment structure to pursue a common agenda preparing for the 'post-industrial', 'post-democratic' society. I say 'preparing' but we are now there. 'Post-industrial' is code for the Great Reset and 'post-democratic' is 'Covid' fascism. UKColumn has spoken to partners of those who have attended Common Purpose 'training'. They have described how personalities and attitudes of 'graduates' changed very noticeably for the worse by the time they had completed the course. They had been 'reframed' and told they are the 'leaders' – the special ones – who know better than the population. There has also been the very demonstrable recruitment of psychopaths and narcissists into government administration at all

levels and law enforcement. If you want psychopathy hire psychopaths and you get a simple cause and effect. If you want administrators, police officers and 'leaders' to perceive the public as lesser beings who don't matter then employ narcissists. These personalities are identified using 'psychometrics' that identifies knowledge, abilities, attitudes and personality traits, mostly through carefully-designed questionnaires and tests. As this policy has passed through the decades we have had power-crazy, power-trippers appointed into law enforcement, security and government administration in preparation for current times and the dynamic between public and law enforcement/officialdom has been transformed. UKColumn's Brian Gerrish said of the narcissistic personality:

Their love of themselves and power automatically means that they will crush others who get in their way. I received a major piece of the puzzle when a friend pointed out that when they made public officials re-apply for their own jobs several years ago they were also required to do psychometric tests. This was undoubtedly the start of the screening process to get 'their' sort of people in post.

How obvious that has been since 2020 although it was clear what was happening long before if people paid attention to the changing public-establishment dynamic.

Change agents

At the centre of events in 'Covid' Britain is the National Health Service (NHS) which has behaved disgracefully in slavishly following the Cult agenda. The NHS management structure is awash with Common Purpose graduates or 'change agents' working to a common cause. Helen Bevan, a Chief of Service Transformation at the NHS Institute for Innovation and Improvement, co-authored a document called 'Towards a million change agents, a review of the social movements literature: implications for large scale change in the NHS'. The document compared a project management approach to that of change and social movements where 'people change

themselves and each other – peer to peer’. Two definitions given for a ‘social movement’ were:

A group of people who consciously attempt to build a radically new social order; involves people of a broad range of social backgrounds; and deploys politically confrontational and socially disruptive tactics – Cyrus Zirakzadeh 1997

Collective challenges, based on common purposes and social solidarities, in sustained interaction with elites, opponents, and authorities – Sidney Tarrow 1994

Helen Bevan wrote another NHS document in which she defined ‘framing’ as ‘the process by which leaders construct, articulate and put across their message in a powerful and compelling way in order to win people to their cause and call them to action’. I think I could come up with another definition that would be rather more accurate. The National Health Service and institutions of Britain and the wider world have been taken over by reframed ‘change agents’ and that includes everything from the United Nations to national governments, local councils and social services which have been kidnapping children from loving parents on an extraordinary and gathering scale on the road to the end of parenthood altogether. Children from loving homes are stolen and kidnapped by the state and put into the ‘care’ (inversion) of the local authority through council homes, foster parents and forced adoption. At the same time children are allowed to be abused without response while many are under council ‘care’. UKColumn highlighted the Common Purpose connection between South Yorkshire Police and Rotherham council officers in the case of the scandal in that area of the sexual exploitation of children to which the authorities turned not one blind eye, but both:

We were alarmed to discover that the Chief Executive, the Strategic Director of Children and Young People's Services, the Manager for the Local Strategic Partnership, the Community Cohesion Manager, the Cabinet Member for Cohesion, the Chief Constable and his predecessor had all attended Leadership training courses provided by the pseudo-charity Common Purpose.

Once 'change agents' have secured positions of hire and fire within any organisation things start to move very quickly. Personnel are then hired and fired on the basis of whether they will work towards the agenda the change agent represents. If they do they are rapidly promoted even though they may be incompetent. Those more qualified and skilled who are pre-Common Purpose 'old school' see their careers stall and even disappear. This has been happening for decades in every institution of state, police, 'health' and social services and all of them have been transformed as a result in their attitudes to their jobs and the public. Medical professions, including nursing, which were once vocations for the caring now employ many cold, callous and couldn't give a shit personality types. The UKColumn investigation concluded:

By blurring the boundaries between people, professions, public and private sectors, responsibility and accountability, Common Purpose encourages 'graduates' to believe that as new selected leaders, they can work together, outside of the established political and social structures, to achieve a paradigm shift or CHANGE – so called 'Leading Beyond Authority'. In doing so, the allegiance of the individual becomes 'reframed' on CP colleagues and their NETWORK.

Reframing the Face-Nappies

Nowhere has this process been more obvious than in the police where recruitment of psychopaths and development of unquestioning mind-controlled group-thinkers have transformed law enforcement into a politically-correct 'Woke' joke and a travesty of what should be public service. Today they wear their face-nappies like good little gofers and enforce 'Covid' rules which are fascism under another name. Alongside the specifically-recruited psychopaths we have software minds incapable of free thought. Brian Gerrish again:

An example is the policeman who would not get on a bike for a press photo because he had not done the cycling proficiency course. Normal people say this is political correctness gone mad. Nothing could be further from the truth. The policeman has been reframed, and in his reality it is perfect common sense not to get on the bike 'because he hasn't done the cycling course'.

Another example of this is where the police would not rescue a boy from a pond until they had taken advice from above on the 'risk assessment'. A normal person would have arrived, perhaps thought of the risk for a moment, and dived in. To the police now 'reframed', they followed 'normal' procedure.

There are shocking cases of reframed ambulance crews doing the same. Sheer unthinking stupidity of London Face-Nappies headed by Common Purpose graduate Cressida Dick can be seen in their behaviour at a vigil in March, 2021, for a murdered woman, Sarah Everard. A police officer had been charged with the crime. Anyone with a brain would have left the vigil alone in the circumstances. Instead they 'manhandled' women to stop them breaking 'Covid rules' to betray classic reframing. Minds in the thrall of perception control have no capacity for seeing a situation on its merits and acting accordingly. 'Rules is rules' is their only mind-set. My father used to say that rules and regulations are for the guidance of the intelligent and the blind obedience of the idiot. Most of the intelligent, decent, coppers have gone leaving only the other kind and a few old school for whom the job must be a daily nightmare. The combination of psychopaths and rule-book software minds has been clearly on public display in the 'Covid' era with automaton robots in uniform imposing fascistic 'Covid' regulations on the population without any personal initiative or judging situations on their merits. There are thousands of examples around the world, but I'll make my point with the infamous Derbyshire police in the English East Midlands – the ones who think pouring dye into beauty spots and using drones to track people walking in the countryside away from anyone is called 'policing'. To them there are rules decreed by the government which they have to enforce and in their bewildered state a group gathering in a closed space and someone walking alone in the countryside are the same thing. It is beyond idiocy and enters the realm of clinical insanity.

Police officers in Derbyshire said they were 'horrified' – *horrified* – to find 15 to 20 'irresponsible' kids playing a football match at a closed leisure centre 'in breach of coronavirus restrictions'. When they saw the police the kids ran away leaving their belongings behind and the reframed men and women of Derbyshire police were seeking to establish their identities with a view to fining their parents. The most natural thing for youngsters to do – kicking a ball about – is turned into a criminal activity and enforced by the moronic software programs of Derbyshire police. You find the same mentality in every country. These barely conscious 'horrified' officers said they had to take action because 'we need to ensure these rules are being followed' and 'it is of the utmost importance that you ensure your children are following the rules and regulations for Covid-19'. Had any of them done ten seconds of research to see if this parroting of their masters' script could be supported by any evidence? Nope. Reframed people don't think – others think for them and that's the whole idea of reframing. I have seen police officers one after the other repeating without question word for word what officialdom tells them just as I have seen great swathes of the public doing the same. Ask either for 'their' opinion and out spews what they have been told to think by the official narrative. Police and public may seem to be in different groups, but their mentality is the same. Most people do whatever they are told in fear not doing so or because they believe what officialdom tells them; almost the entirety of the police do what they are told for the same reason. Ultimately it's the tiny inner core of the global Cult that's telling both what to do.

So Derbyshire police were 'horrified'. Oh, really? Why did they think those kids were playing football? It was to relieve the psychological consequences of lockdown and being denied human contact with their friends and interaction, touch and discourse vital to human psychological health. Being denied this month after month has dismantled the psyche of many children and young people as depression and suicide have exploded. Were Derbyshire police *horrified by that*? Are you kidding? Reframed people don't have those

mental and emotional processes that can see how the impact on the psychological health of youngsters is far more dangerous than any 'virus' even if you take the mendacious official figures to be true. The reframed are told (programmed) how to act and so they do. The Derbyshire Chief Constable in the first period of lockdown when the black dye and drones nonsense was going on was Peter Goodman. He was the man who severed the connection between his force and the Derbyshire Constabulary *Male Voice* Choir when he decided that it was not inclusive enough to allow women to join. The fact it was a male voice choir making a particular sound produced by male voices seemed to elude a guy who terrifyingly ran policing in Derbyshire. He retired weeks after his force was condemned as disgraceful by former Supreme Court Justice Jonathan Sumption for their behaviour over extreme lockdown impositions. Goodman was replaced by his deputy Rachel Swann who was in charge when her officers were 'horrificed'. The police statement over the boys committing the hanging-offence of playing football included the line about the youngsters being 'irresponsible in the times we are all living through' missing the point that the real relevance of the 'times we are all living through' is the imposition of fascism enforced by psychopaths and reframed minds of police officers playing such a vital part in establishing the fascist tyranny that their own children and grandchildren will have to live in their entire lives. As a definition of insanity that is hard to beat although it might be run close by imposing masks on people that can have a serious effect on their health while wearing a face nappy all day themselves. Once again public and police do it for the same reason – the authorities tell them to and who are they to have the self-respect to say no?

Wokers in uniform

How reframed do you have to be to arrest a *six-year-old* and take him to court for *picking a flower* while waiting for a bus? Brain dead police and officialdom did just that in North Carolina where criminal proceedings happen regularly for children under nine. Attorney Julie Boyer gave the six-year-old crayons and a colouring book

during the 'flower' hearing while the 'adults' decided his fate. County Chief District Court Judge Jay Corpening asked: 'Should a child that believes in Santa Claus, the Easter Bunny and the tooth fairy be making life-altering decisions?' Well, of course not, but common sense has no meaning when you have a common purpose and a reframed mind. Treating children in this way, and police operating in American schools, is all part of the psychological preparation for children to accept a police state as normal all their adult lives. The same goes for all the cameras and biometric tracking technology in schools. Police training is focused on reframing them as snowflake Wokers and this is happening in the military. Pentagon top brass said that 'training sessions on extremism' were needed for troops who asked why they were so focused on the Capitol Building riot when Black Lives Matter riots were ignored. What's the difference between them some apparently and rightly asked. Actually, there is a difference. Five people died in the Capitol riot, only one through violence, and that was a police officer shooting an unarmed protestor. BLM riots killed at least 25 people and cost billions. Asking the question prompted the psychopaths and reframed minds that run the Pentagon to say that more 'education' (programming) was needed. Troop training is all based on psychological programming to make them fodder for the Cult – 'Military men are just dumb, stupid animals to be used as pawns in foreign policy' as Cult-to-his-DNA former Secretary of State Henry Kissinger famously said. Governments see the police in similar terms and it's time for those among them who can see this to defend the people and stop being enforcers of the Cult agenda upon the people.

The US military, like the country itself, is being targeted for destruction through a long list of Woke impositions. Cult-owned gaga 'President' Biden signed an executive order when he took office to allow taxpayer money to pay for transgender surgery for active military personnel and veterans. Are you a man soldier? No, I'm a LGBTQIA+ with a hint of Skoliosexual and Spectrasexual. Oh, good man. Bad choice of words you bigot. The Pentagon announced in March, 2021, the appointment of the first 'diversity and inclusion

officer' for US Special Forces. Richard Torres-Estrada arrived with the publication of a 'D&I Strategic Plan which will guide the enterprise-wide effort to institutionalize and sustain D&I'. If you think a Special Forces 'Strategic Plan' should have something to do with defending America you haven't been paying attention. Defending Woke is now the military's new role. Torres-Estrada has posted images comparing Donald Trump with Adolf Hitler and we can expect no bias from him as a representative of the supposedly non-political Pentagon. Cable news host Tucker Carlson said: 'The Pentagon is now the Yale faculty lounge but with cruise missiles.' Meanwhile Secretary of Defense Lloyd Austin, a board member of weapons-maker Raytheon with stock and compensation interests in October, 2020, worth \$1.4 million, said he was purging the military of the 'enemy within' – anyone who isn't Woke and supports Donald Trump. Austin refers to his targets as 'racist extremists' while in true Woke fashion being himself a racist extremist. Pentagon documents pledge to 'eradicate, eliminate and conquer all forms of racism, sexism and homophobia'. The definitions of these are decided by 'diversity and inclusion committees' peopled by those who see racism, sexism and homophobia in every situation and opinion. Woke (the Cult) is dismantling the US military and purging testosterone as China expands its military and gives its troops 'masculinity training'. How do we think that is going to end when this is all Cult coordinated? The US military, like the British military, is controlled by Woke and spineless top brass who just go along with it out of personal career interests.

'Woke' means fast asleep

Mind control and perception manipulation techniques used on individuals to create group-think have been unleashed on the global population in general. As a result many have no capacity to see the obvious fascist agenda being installed all around them or what 'Covid' is really all about. Their brains are firewalled like a computer system not to process certain concepts, thoughts and realisations that are bad for the Cult. The young are most targeted as the adults they

will be when the whole fascist global state is planned to be fully implemented. They need to be prepared for total compliance to eliminate all pushback from entire generations. The Cult has been pouring billions into taking complete control of 'education' from schools to universities via its operatives and corporations and not least Bill Gates as always. The plan has been to transform 'education' institutions into programming centres for the mentality of 'Woke'. James McConnell, professor of psychology at the University of Michigan, wrote in *Psychology Today* in 1970:

The day has come when we can combine sensory deprivation with drugs, hypnosis, and astute manipulation of reward and punishment, to gain almost absolute control over an individual's behaviour. It should then be possible to achieve a very rapid and highly effective type of brainwashing that would allow us to make dramatic changes in a person's behaviour and personality ...

... We should reshape society so that we all would be trained from birth to want to do what society wants us to do. We have the techniques to do it... no-one owns his own personality you acquired, and there's no reason to believe you should have the right to refuse to acquire a new personality if your old one is anti-social.

This was the potential for mass brainwashing in 1970 and the mentality there displayed captures the arrogant psychopathy that drives it forward. I emphasise that not all young people have succumbed to Woke programming and those that haven't are incredibly impressive people given that today's young are the most perceptually-targeted generations in history with all the technology now involved. Vast swathes of the young generations, however, have fallen into the spell – and that's what it is – of Woke. The Woke mentality and perceptual program is founded on *inversion* and you will appreciate later why that is so significant. Everything with Woke is inverted and the opposite of what it is claimed to be. Woke was a term used in African-American culture from the 1900s and referred to an awareness of social and racial justice. This is not the meaning of the modern version or 'New Woke' as I call it in *The Answer*. Oh, no, Woke today means something very different no matter how much Wokers may seek to hide that and insist Old Woke and New

Woke are the same. See if you find any 'awareness of social justice' here in the modern variety:

- Woke demands 'inclusivity' while excluding anyone with a different opinion and calls for mass censorship to silence other views.
- Woke claims to stand against oppression when imposing oppression is the foundation of all that it does. It is the driver of political correctness which is nothing more than a Cult invention to manipulate the population to silence itself.
- Woke believes itself to be 'liberal' while pursuing a global society that can only be described as fascist (see 'anti-fascist' fascist Antifa).
- Woke calls for 'social justice' while spreading injustice wherever it goes against the common 'enemy' which can be easily identified as a differing view.
- Woke is supposed to be a metaphor for 'awake' when it is solid-gold asleep and deep in a Cult-induced coma that meets the criteria for 'off with the fairies'.

I state these points as obvious facts if people only care to look. I don't do this with a sense of condemnation. We need to appreciate that the onslaught of perceptual programming on the young has been incessant and merciless. I can understand why so many have been reframed, or, given their youth, framed from the start to see the world as the Cult demands. The Cult has had access to their minds day after day in its 'education' system for their entire formative years. Perception is formed from information received and the Cult-created system is a life-long download of information delivered to elicit a particular perception, thus behaviour. The more this has expanded into still new extremes in recent decades and ever-increasing censorship has deleted other opinions and information why wouldn't that lead to a perceptual reframing on a mass scale? I

have described already cradle-to-grave programming and in more recent times the targeting of young minds from birth to adulthood has entered the stratosphere. This has taken the form of skewing what is 'taught' to fit the Cult agenda and the omnipresent techniques of group-think to isolate non-believers and pressure them into line. There has always been a tendency to follow the herd, but we really are in a new world now in relation to that. We have parents who can see the 'Covid' hoax told by their children not to stop them wearing masks at school, being 'Covid' tested or having the 'vaccine' in fear of the peer-pressure consequences of being different. What is 'peer-pressure' if not pressure to conform to group-think? Renegade Minds never group-think and always retain a set of perceptions that are unique to them. Group-think is always underpinned by consequences for not group-thinking. Abuse now aimed at those refusing DNA-manipulating 'Covid vaccines' are a potent example of this. The biggest pressure to conform comes from the very group which is itself being manipulated. 'I am programmed to be part of a hive mind and so you must be.'

Woke control structures in 'education' now apply to every mainstream organisation. Those at the top of the 'education' hierarchy (the Cult) decide the policy. This is imposed on governments through the Cult network; governments impose it on schools, colleges and universities; their leadership impose the policy on teachers and academics and they impose it on children and students. At any level where there is resistance, perhaps from a teacher or university lecturer, they are targeted by the authorities and often fired. Students themselves regularly demand the dismissal of academics (increasingly few) at odds with the narrative that the students have been programmed to believe in. It is quite a thought that students who are being targeted by the Cult become so consumed by programmed group-think that they launch protests and demand the removal of those who are trying to push back against those targeting the students. Such is the scale of perceptual inversion. We see this with 'Covid' programming as the Cult imposes the rules via psycho-psychologists and governments on

shops, transport companies and businesses which impose them on their staff who impose them on their customers who pressure Pushbackers to conform to the will of the Cult which is in the process of destroying them and their families. Scan all aspects of society and you will see the same sequence every time.

Fact free Woke and hijacking the 'left'

There is no more potent example of this than 'Woke', a mentality only made possible by the deletion of factual evidence by an 'education' system seeking to produce an ever more uniform society. Why would you bother with facts when you don't know any? Deletion of credible history both in volume and type is highly relevant. Orwell said: 'Who controls the past controls the future: who controls the present controls the past.' They who control the perception of the past control the perception of the future and they who control the present control the perception of the past through the writing and deleting of history. Why would you oppose the imposition of Marxism in the name of Wokeism when you don't know that Marxism cost at least 100 million lives in the 20th century alone? Watch videos and read reports in which Woker generations are asked basic historical questions – it's mind-blowing. A survey of 2,000 people found that six percent of millennials (born approximately early 1980s to early 2000s) believed the Second World War (1939-1945) broke out with the assassination of President Kennedy (in 1963) and one in ten thought Margaret Thatcher was British Prime Minister at the time. She was in office between 1979 and 1990. We are in a post-fact society. Provable facts are no defence against the fascism of political correctness or Silicon Valley censorship. Facts don't matter anymore as we have witnessed with the 'Covid' hoax. Sacrificing uniqueness to the Woke group-think religion is all you are required to do and that means thinking for yourself is the biggest Woke no, no. All religions are an expression of group-think and censorship and Woke is just another religion with an orthodoxy defended by group-think and censorship. Burned at

the stake becomes burned on Twitter which leads back eventually to burned at the stake as Woke humanity regresses to ages past.

The biggest Woke inversion of all is its creators and funders. I grew up in a traditional left of centre political household on a council estate in Leicester in the 1950s and 60s – you know, the left that challenged the power of wealth-hoarding elites and threats to freedom of speech and opinion. In those days students went on marches defending freedom of speech while today's Wokers march for its deletion. What on earth could have happened? Those very elites (collectively the Cult) that we opposed in my youth and early life have funded into existence the antithesis of that former left and hijacked the 'brand' while inverting everything it ever stood for. We have a mentality that calls itself 'liberal' and 'progressive' while acting like fascists. Cult billionaires and their corporations have funded themselves into control of 'education' to ensure that Woke programming is unceasing throughout the formative years of children and young people and that non-Wokers are isolated (that word again) whether they be students, teachers or college professors. The Cult has funded into existence the now colossal global network of Woke organisations that have spawned and promoted all the 'causes' on the Cult wish-list for global transformation and turned Wokers into demanders of them. Does anyone really think it's a coincidence that the Cult agenda for humanity is a carbon (sorry) copy of the societal transformations desired by Woke?? These are only some of them:

Political correctness: The means by which the Cult deletes all public debates that it knows it cannot win if we had the free-flow of information and evidence.

Human-caused 'climate change': The means by which the Cult seeks to transform society into a globally-controlled dictatorship imposing its will over the fine detail of everyone's lives 'to save the planet' which doesn't actually need saving.

Transgender obsession: Preparing collective perception to accept the 'new human' which would not have genders because it would be created technologically and not through procreation. I'll have much more on this in Human 2.0.

Race obsession: The means by which the Cult seeks to divide and rule the population by triggering racial division through the perception that society is more racist than ever when the opposite is the case. Is it perfect in that regard? No. But to compare today with the racism of apartheid and segregation brought to an end by the civil rights movement in the 1960s is to insult the memory of that movement and inspirations like Martin Luther King. Why is the 'anti-racism' industry (which it is) so dominated by privileged white people?

White supremacy: This is a label used by privileged white people to demonise poor and deprived white people pushing back on tyranny to marginalise and destroy them. White people are being especially targeted as the dominant race by number within Western society which the Cult seeks to transform in its image. If you want to change a society you must weaken and undermine its biggest group and once you have done that by using the other groups you next turn on them to do the same ... 'Then they came for the Jews and I was not a Jew so I did nothing.'

Mass migration: The mass movement of people from the Middle East, Africa and Asia into Europe, from the south into the United States and from Asia into Australia are another way the Cult seeks to dilute the racial, cultural and political influence of white people on Western society. White people ask why their governments appear to be working against them while being politically and culturally biased towards incoming cultures. Well, here's your answer. In the same way sexually 'straight' people, men and women, ask why the

authorities are biased against them in favour of other sexualities. The answer is the same – that's the way the Cult wants it to be for very sinister motives.

These are all central parts of the Cult agenda and central parts of the Woke agenda and Woke was created and continues to be funded to an immense degree by Cult billionaires and corporations. If anyone begins to say 'coincidence' the syllables should stick in their throat.

Billionaire 'social justice warriors'

Joe Biden is a 100 percent-owned asset of the Cult and the Wokers' man in the White House whenever he can remember his name and for however long he lasts with his rapidly diminishing cognitive function. Even walking up the steps of an aircraft without falling on his arse would appear to be a challenge. He's not an empty-shell puppet or anything. From the minute Biden took office (or the Cult did) he began his executive orders promoting the Woke wish-list. You will see the Woke agenda imposed ever more severely because it's really the *Cult* agenda. Woke organisations and activist networks spawned by the Cult are funded to the extreme so long as they promote what the Cult wants to happen. Woke is funded to promote 'social justice' by billionaires who become billionaires by destroying social justice. The social justice mantra is only a cover for dismantling social justice and funded by billionaires that couldn't give a damn about social justice. Everything makes sense when you see that. One of Woke's premier funders is Cult billionaire financier George Soros who said: 'I am basically there to make money, I cannot and do not look at the social consequences of what I do.' This is the same Soros who has given more than \$32 billion to his Open Society Foundations global Woke network and funded Black Lives Matter, mass immigration into Europe and the United States, transgender activism, climate change activism, political correctness and groups targeting 'white supremacy' in the form of privileged white thugs that dominate Antifa. What a scam it all is and when

you are dealing with the unquestioning fact-free zone of Woke scamming them is child's play. All you need to pull it off in all these organisations are a few in-the-know agents of the Cult and an army of naïve, reframed, uninformed, narcissistic, know-nothings convinced of their own self-righteousness, self-purity and virtue.

Soros and fellow billionaires and billionaire corporations have poured hundreds of millions into Black Lives Matter and connected groups and promoted them to a global audience. None of this is motivated by caring about black people. These are the billionaires that have controlled and exploited a system that leaves millions of black people in abject poverty and deprivation which they do absolutely nothing to address. The same Cult networks funding BLM were behind the *slave trade*! Black Lives Matter hijacked a phrase that few would challenge and they have turned this laudable concept into a political weapon to divide society. You know that BLM is a fraud when it claims that *All Lives Matter*, the most inclusive statement of all, is 'racist'. BLM and its Cult masters don't want to end racism. To them it's a means to an end to control all of humanity never mind the colour, creed, culture or background. What has destroying the nuclear family got to do with ending racism? Nothing – but that is one of the goals of BLM and also happens to be a goal of the Cult as I have been exposing in my books for decades. Stealing children from loving parents and giving schools ever more power to override parents is part of that same agenda. BLM is a Marxist organisation and why would that not be the case when the Cult created Marxism *and* BLM? Patrisse Cullors, a BLM co-founder, said in a 2015 video that she and her fellow organisers, including co-founder Alicia Garza, are 'trained Marxists'. The lady known after marriage as Patrisse Khan-Cullors bought a \$1.4 million home in 2021 in one of the whitest areas of California with a black population of just 1.6 per cent and has so far bought *four* high-end homes for a total of \$3.2 million. How very Marxist. There must be a bit of spare in the BLM coffers, however, when Cult corporations and billionaires have handed over the best part of \$100 million. Many black people can see that Black Lives Matter is not

working for them, but against them, and this is still more confirmation. Black journalist Jason Whitlock, who had his account suspended by Twitter for simply linking to the story about the 'Marxist's' home buying spree, said that BLM leaders are 'making millions of dollars off the backs of these dead black men who they wouldn't spit on if they were on fire and alive'.

Black Lies Matter

Cult assets and agencies came together to promote BLM in the wake of the death of career criminal George Floyd who had been jailed a number of times including for forcing his way into the home of a black woman with others in a raid in which a gun was pointed at her stomach. Floyd was filmed being held in a Minneapolis street in 2020 with the knee of a police officer on his neck and he subsequently died. It was an appalling thing for the officer to do, but the same technique has been used by police on peaceful protestors of lockdown without any outcry from the Woke brigade. As unquestioning supporters of the Cult agenda Wokers have supported lockdown and all the 'Covid' claptrap while attacking anyone standing up to the tyranny imposed in its name. Court documents would later include details of an autopsy on Floyd by County Medical Examiner Dr Andrew Baker who concluded that Floyd had taken a fatal level of the drug fentanyl. None of this mattered to fact-free, question-free, Woke. Floyd's death was followed by worldwide protests against police brutality amid calls to defund the police. Throwing babies out with the bathwater is a Woke speciality. In the wake of the murder of British woman Sarah Everard a Green Party member of the House of Lords, Baroness Jones of Moulsecoomb (Nincompoopia would have been better), called for a 6pm curfew for all men. This would be in breach of the Geneva Conventions on war crimes which ban collective punishment, but that would never have crossed the black and white Woke mind of Baroness Nincompoopia who would have been far too convinced of her own self-righteousness to compute such details. Many American cities did defund the police in the face of Floyd riots

and after \$15 million was deleted from the police budget in Washington DC under useless Woke mayor Muriel Bowser car-jacking alone rose by 300 percent and within six months the US capital recorded its highest murder rate in 15 years. The same happened in Chicago and other cities in line with the Cult/Soros plan to bring fear to streets and neighbourhoods by reducing the police, releasing violent criminals and not prosecuting crime. This is the mob-rule agenda that I have warned in the books was coming for so long. Shootings in the area of Minneapolis where Floyd was arrested increased by 2,500 percent compared with the year before. Defunding the police over George Floyd has led to a big increase in dead people with many of them black. Police protection for politicians making these decisions stayed the same or increased as you would expect from professional hypocrites. The Cult doesn't actually want to abolish the police. It wants to abolish local control over the police and hand it to federal government as the psychopaths advance the Hunger Games Society. Many George Floyd protests turned into violent riots with black stores and businesses destroyed by fire and looting across America fuelled by Black Lives Matter. Woke doesn't do irony. If you want civil rights you must loot the liquor store and the supermarket and make off with a smart TV. It's the only way.

It's not a race war – it's a class war

Black people are patronised by privileged blacks and whites alike and told they are victims of white supremacy. I find it extraordinary to watch privileged blacks supporting the very system and bloodline networks behind the slave trade and parroting the same Cult-serving manipulative crap of their privileged white, often billionaire, associates. It is indeed not a race war but a class war and colour is just a diversion. Black Senator Cory Booker and black Congresswoman Maxine Waters, more residents of Nincompoopia, personify this. Once you tell people they are victims of someone else you devalue both their own responsibility for their plight and the power they have to impact on their reality and experience. Instead

we have: 'You are only in your situation because of whiteness – turn on them and everything will change.' It won't change. Nothing changes in our lives unless *we* change it. Crucial to that is never seeing yourself as a victim and always as the creator of your reality. Life is a simple sequence of choice and consequence. Make different choices and you create different consequences. *You* have to make those choices – not Black Lives Matter, the Woke Mafia and anyone else that seeks to dictate your life. Who are they these Wokers, an emotional and psychological road traffic accident, to tell you what to do? Personal empowerment is the last thing the Cult and its Black Lives Matter want black people or anyone else to have. They claim to be defending the underdog while *creating* and perpetuating the underdog. The Cult's worst nightmare is human unity and if they are going to keep blacks, whites and every other race under economic servitude and control then the focus must be diverted from what they have in common to what they can be manipulated to believe divides them. Blacks have to be told that their poverty and plight is the fault of the white bloke living on the street in the same poverty and with the same plight they are experiencing. The difference is that your plight black people is due to him, a white supremacist with 'white privilege' living on the street. Don't unite as one human family against your mutual oppressors and suppressors – fight the oppressor with the white face who is as financially deprived as you are. The Cult knows that as its 'Covid' agenda moves into still new levels of extremism people are going to respond and it has been spreading the seeds of disunity everywhere to stop a united response to the evil that targets *all of us*.

Racist attacks on 'whiteness' are getting ever more outrageous and especially through the American Democratic Party which has an appalling history for anti-black racism. Barack Obama, Joe Biden, Hillary Clinton and Nancy Pelosi all eulogised about Senator Robert Byrd at his funeral in 2010 after a nearly 60-year career in Congress. Byrd was a brutal Ku Klux Klan racist and a violent abuser of Cathy O'Brien in MKUltra. He said he would never fight in the military 'with a negro by my side' and 'rather I should die a thousand times,

and see Old Glory trampled in the dirt never to rise again, than to see this beloved land of ours become degraded by race mongrels, a throwback to the blackest specimen from the wilds'. Biden called Byrd a 'very close friend and mentor'. These 'Woke' hypocrites are not anti-racist they are anti-poor and anti-people not of their perceived class. Here is an illustration of the scale of anti-white racism to which we have now descended. Seriously Woke and moronic *New York Times* contributor Damon Young described whiteness as a 'virus' that 'like other viruses will not die until there are no bodies left for it to infect'. He went on: '... the only way to stop it is to locate it, isolate it, extract it, and kill it.' Young can say that as a black man with no consequences when a white man saying the same in reverse would be facing a jail sentence. *That's* racism. We had super-Woke numbskull senators Tammy Duckworth and Mazie Hirono saying they would object to future Biden Cabinet appointments if he did not nominate more Asian Americans and Pacific Islanders. Never mind the ability of the candidate what do they look like? Duckworth said: 'I will vote for racial minorities and I will vote for LGBTQ, but anyone else I'm not voting for.' Appointing people on the grounds of race is illegal, but that was not a problem for this ludicrous pair. They were on-message and that's a free pass in any situation.

Critical race racism

White children are told at school they are intrinsically racist as they are taught the divisive 'critical race theory'. This claims that the law and legal institutions are inherently racist and that race is a socially constructed concept used by white people to further their economic and political interests at the expense of people of colour. White is a 'virus' as we've seen. Racial inequality results from 'social, economic, and legal differences that white people create between races to maintain white interests which leads to poverty and criminality in minority communities'. I must tell that to the white guy sleeping on the street. The principal of East Side Community School in New York sent white parents a manifesto that called on

them to become 'white traitors' and advocate for full 'white abolition'. These people are teaching your kids when they urgently need a psychiatrist. The 'school' included a chart with 'eight white identities' that ranged from 'white supremacist' to 'white abolition' and defined the behaviour white people must follow to end 'the regime of whiteness'. Woke blacks and their privileged white associates are acting exactly like the slave owners of old and Ku Klux Klan racists like Robert Byrd. They are too full of their own self-purity to see that, but it's true. Racism is not a body type; it's a state of mind that can manifest through any colour, creed or culture.

Another racial fraud is '*equity*'. Not equality of treatment and opportunity – equity. It's a term spun as equality when it means something very different. Equality in its true sense is a raising up while 'equity' is a race to the bottom. Everyone in the same level of poverty is 'equity'. Keep everyone down – that's equity. The Cult doesn't want anyone in the human family to be empowered and BLM leaders, like all these 'anti-racist' organisations, continue their privileged, pampered existence by perpetuating the perception of gathering racism. When is the last time you heard an 'anti-racist' or 'anti-Semitism' organisation say that acts of racism and discrimination have *fallen*? It's not in the interests of their fundraising and power to influence and the same goes for the professional soccer anti-racism operation, Kick It Out. Two things confirmed that the Black Lives Matter riots in the summer of 2020 were Cult creations. One was that while anti-lockdown protests were condemned in this same period for 'transmitting 'Covid' the authorities supported mass gatherings of Black Lives Matter supporters. I even saw self-deluding people claiming to be doctors say the two types of protest were not the same. No – the non-existent 'Covid' was in favour of lockdowns and attacked those that protested against them while 'Covid' supported Black Lives Matter and kept well away from its protests. The whole thing was a joke and as lockdown protestors were arrested, often brutally, by reframed Face-Nappies we had the grotesque sight of police officers taking the knee to Black Lives Matter, a Cult-funded Marxist

organisation that supports violent riots and wants to destroy the nuclear family and white people.

He's not white? Shucks!

Woke obsession with race was on display again when ten people were shot dead in Boulder, Colorado, in March, 2021. Cult-owned Woke TV channels like CNN said the shooter appeared to be a white man and Wokers were on Twitter condemning 'violent white men' with the usual mantras. Then the shooter's name was released as Ahmad Al Aliwi Alissa, an anti-Trump Arab-American, and the sigh of disappointment could be heard five miles away. Never mind that ten people were dead and what that meant for their families. Race baiting was all that mattered to these sick Cult-serving people like Barack Obama who exploited the deaths to further divide America on racial grounds which is his job for the Cult. This is the man that 'racist' white Americans made the first black president of the United States and then gave him a second term. Not-very-bright Obama has become filthy rich on the back of that and today appears to have a big influence on the Biden administration. Even so he's still a downtrodden black man and a victim of white supremacy. This disingenuous fraud reveals the contempt he has for black people when he puts on a Deep South Alabama accent whenever he talks to them, no, *at* them.

Another BLM red flag was how the now fully-Woke (fully-Cult) and fully-virtue-signalled professional soccer authorities had their teams taking the knee before every match in support of Marxist Black Lives Matter. Soccer authorities and clubs displayed 'Black Lives Matter' on the players' shirts and flashed the name on electronic billboards around the pitch. Any fans that condemned what is a Freemasonic taking-the-knee ritual were widely condemned as you would expect from the Woke virtue-signallers of professional sport and the now fully-Woke media. We have reverse racism in which you are banned from criticising any race or culture except for white people for whom anything goes – say what you like, no problem. What has this got to do with racial harmony and

equality? We've had black supremacists from Black Lives Matter telling white people to fall to their knees in the street and apologise for their white supremacy. Black supremacists acting like white supremacist slave owners of the past couldn't breach their self-obsessed, race-obsessed sense of self-purity. Joe Biden appointed a race-obsessed black supremacist Kristen Clarke to head the Justice Department Civil Rights Division. Clarke claimed that blacks are endowed with 'greater mental, physical and spiritual abilities' than whites. If anyone reversed that statement they would be vilified. Clarke is on-message so no problem. She's never seen a black-white situation in which the black figure is anything but a virtuous victim and she heads the Civil Rights Division which should treat everyone the same or it isn't civil rights. Another perception of the Renegade Mind: If something or someone is part of the Cult agenda they will be supported by Woke governments and media no matter what. If they're not, they will be condemned and censored. It really is that simple and so racist Clarke prospers despite (make that because of) her racism.

The end of culture

Biden's administration is full of such racial, cultural and economic bias as the Cult requires the human family to be divided into warring factions. We are now seeing racially-segregated graduations and everything, but everything, is defined through the lens of perceived 'racism'. We have 'racist' mathematics, 'racist' food and even 'racist' *plants*. World famous Kew Gardens in London said it was changing labels on plants and flowers to tell its pre-'Covid' more than two million visitors a year how racist they are. Kew director Richard Deverell said this was part of an effort to 'move quickly to decolonise collections' after they were approached by one Ajay Chhabra 'an actor with an insight into how sugar cane was linked to slavery'. They are *plants* you idiots. 'Decolonisation' in the Woke manual really means colonisation of society with its mentality and by extension colonisation by the Cult. We are witnessing a new Chinese-style 'Cultural Revolution' so essential to the success of all

Marxist takeovers. Our cultural past and traditions have to be swept away to allow a new culture to be built-back-better. Woke targeting of long-standing Western cultural pillars including historical monuments and cancelling of historical figures is what happened in the Mao revolution in China which 'purged remnants of capitalist and traditional elements from Chinese society' and installed Maoism as the dominant ideology'. For China see the Western world today and for 'dominant ideology' see Woke. Better still see Marxism or Maoism. The 'Covid' hoax has specifically sought to destroy the arts and all elements of Western culture from people meeting in a pub or restaurant to closing theatres, music venues, sports stadiums, places of worship and even banning *singing*. Destruction of Western society is also why criticism of any religion is banned except for Christianity which again is the dominant religion as white is the numerically-dominant race. Christianity may be fading rapidly, but its history and traditions are weaved through the fabric of Western society. Delete the pillars and other structures will follow until the whole thing collapses. I am not a Christian defending that religion when I say that. I have no religion. It's just a fact. To this end Christianity has itself been turned Woke to usher its own downfall and its ranks are awash with 'change agents' – knowing and unknowing – at every level including Pope Francis (*definitely* knowing) and the clueless Archbishop of Canterbury Justin Welby (possibly not, but who can be sure?). Woke seeks to coordinate attacks on Western culture, traditions, and ways of life through 'intersectionality' defined as 'the complex, cumulative way in which the effects of multiple forms of discrimination (such as racism, sexism, and classism) combine, overlap, or intersect especially in the experiences of marginalised individuals or groups'. Wade through the Orwellian Woke-speak and this means coordinating disparate groups in a common cause to overthrow freedom and liberal values.

The entire structure of public institutions has been infested with Woke – government at all levels, political parties, police, military, schools, universities, advertising, media and trade unions. This abomination has been achieved through the Cult web by appointing

Workers to positions of power and battering non-Workers into line through intimidation, isolation and threats to their job. Many have been fired in the wake of the empathy-deleted, vicious hostility of 'social justice' Workers and the desire of gutless, spineless employers to virtue-signal their Wokeness. Corporations are filled with Workers today, most notably those in Silicon Valley. Ironically at the top they are not Woke at all. They are only exploiting the mentality their Cult masters have created and funded to censor and enslave while the Workers cheer them on until it's their turn. Thus the Woke 'liberal left' is an inversion of the traditional liberal left. Campaigning for justice on the grounds of power and wealth distribution has been replaced by campaigning for identity politics. The genuine traditional left would never have taken money from today's billionaire abusers of fairness and justice and nor would the billionaires have wanted to fund that genuine left. It would not have been in their interests to do so. The division of opinion in those days was between the haves and have nots. This all changed with Cult manipulated and funded identity politics. The division of opinion today is between Workers and non-Workers and not income brackets. Cult corporations and their billionaires may have taken wealth disparity to cataclysmic levels of injustice, but as long as they speak the language of Woke, hand out the dosh to the Woke network and censor the enemy they are 'one of us'. Billionaires who don't give a damn about injustice are laughing at them till their bellies hurt. Workers are not even close to self-aware enough to see that. The transformed 'left' dynamic means that Workers who drone on about 'social justice' are funded by billionaires that have destroyed social justice the world over. It's *why* they are billionaires.

The climate con

Nothing encapsulates what I have said more comprehensively than the hoax of human-caused global warming. I have detailed in my books over the years how Cult operatives and organisations were the pump-primers from the start of the climate con. A purpose-built vehicle for this is the Club of Rome established by the Cult in 1968

with the Rockefellers and Rothschilds centrally involved all along. Their gofer frontman Maurice Strong, a Canadian oil millionaire, hosted the Earth Summit in Rio de Janeiro, Brazil, in 1992 where the global 'green movement' really expanded in earnest under the guiding hand of the Cult. The Earth Summit established Agenda 21 through the Cult-created-and-owned United Nations to use the illusion of human-caused climate change to justify the transformation of global society to save the world from climate disaster. It is a No-Problem-Reaction-Solution sold through governments, media, schools and universities as whole generations have been terrified into believing that the world was going to end in their lifetimes unless what old people had inflicted upon them was stopped by a complete restructuring of how everything is done. Chill, kids, it's all a hoax. Such restructuring is precisely what the Cult agenda demands (purely by coincidence of course). Today this has been given the codename of the Great Reset which is only an updated term for Agenda 21 and its associated Agenda 2030. The latter, too, is administered through the UN and was voted into being by the General Assembly in 2015. Both 21 and 2030 seek centralised control of all resources and food right down to the raindrops falling on your own land. These are some of the demands of Agenda 21 established in 1992. See if you recognise this society emerging today:

- End national sovereignty
- State planning and management of all land resources, ecosystems, deserts, forests, mountains, oceans and fresh water; agriculture; rural development; biotechnology; and ensuring '*equity*'
- The state to 'define the role' of business and financial resources
- Abolition of private property
- 'Restructuring' the family unit (see BLM)
- Children raised by the state
- People told what their job will be
- Major restrictions on movement
- Creation of 'human settlement zones'

- Mass resettlement as people are forced to vacate land where they live
- Dumbing down education
- Mass global depopulation in pursuit of all the above

The United Nations was created as a Trojan horse for world government. With the climate con of critical importance to promoting that outcome you would expect the UN to be involved. Oh, it's involved all right. The UN is promoting Agenda 21 and Agenda 2030 justified by 'climate change' while also driving the climate hoax through its Intergovernmental Panel on Climate Change (IPCC), one of the world's most corrupt organisations. The IPCC has been lying ferociously and constantly since the day it opened its doors with the global media hanging unquestioningly on its every mendacious word. The Green movement is entirely Woke and has long lost its original environmental focus since it was co-opted by the Cult. An obsession with 'global warming' has deleted its values and scrambled its head. I experienced a small example of what I mean on a beautiful country walk that I have enjoyed several times a week for many years. The path merged into the fields and forests and you felt at one with the natural world. Then a 'Green' organisation, the Hampshire and Isle of Wight Wildlife Trust, took over part of the land and proceeded to cut down a large number of trees, including mature ones, to install a horrible big, bright steel 'this-is-ours-stay-out' fence that destroyed the whole atmosphere of this beautiful place. No one with a feel for nature would do that. Day after day I walked to the sound of chainsaws and a magnificent mature weeping willow tree that I so admired was cut down at the base of the trunk. When I challenged a Woke young girl in a green shirt (of course) about this vandalism she replied: 'It's a weeping willow – it will grow back.' This is what people are paying for when they donate to the Hampshire and Isle of Wight Wildlife Trust and many other 'green' organisations today. It is not the environmental movement that I knew and instead has become a support-system – as with Extinction Rebellion – for a very dark agenda.

Private jets for climate justice

The Cult-owned, Gates-funded, World Economic Forum and its founder Klaus Schwab were behind the emergence of Greta Thunberg to harness the young behind the climate agenda and she was invited to speak to the world at ... the UN. Schwab published a book, *Covid-19: The Great Reset* in 2020 in which he used the 'Covid' hoax and the climate hoax to lay out a new society straight out of Agenda 21 and Agenda 2030. Bill Gates followed in early 2021 when he took time out from destroying the world to produce a book in his name about the way to save it. Gates flies across the world in private jets and admitted that 'I probably have one of the highest greenhouse gas footprints of anyone on the planet ... my personal flying alone is gigantic.' He has also bid for the planet's biggest private jet operator. Other climate change saviours who fly in private jets include John Kerry, the US Special Presidential Envoy for Climate, and actor Leonardo DiCaprio, a 'UN Messenger of Peace with special focus on climate change'. These people are so full of bullshit they could corner the market in manure. We mustn't be sceptical, though, because the Gates book, *How to Avoid a Climate Disaster: The Solutions We Have and the Breakthroughs We Need*, is a genuine attempt to protect the world and not an obvious pile of excrement attributed to a mega-psychopath aimed at selling his masters' plans for humanity. The Gates book and the other shite-pile by Klaus Schwab could have been written by the same person and may well have been. Both use 'climate change' and 'Covid' as the excuses for their new society and by coincidence the Cult's World Economic Forum and Bill and Melinda Gates Foundation promote the climate hoax and hosted Event 201 which pre-empted with a 'simulation' the very 'coronavirus' hoax that would be simulated for real on humanity within weeks. The British 'royal' family is promoting the 'Reset' as you would expect through Prince 'climate change caused the war in Syria' Charles and his hapless son Prince William who said that we must 'reset our relationship with nature and our trajectory as a species' to avoid a climate disaster. Amazing how many promoters of the 'Covid' and 'climate change' control

systems are connected to Gates and the World Economic Forum. A 'study' in early 2021 claimed that carbon dioxide emissions must fall by the equivalent of a global lockdown roughly every two years for the next decade to save the planet. The 'study' appeared in the same period that the Schwab mob claimed in a video that lockdowns destroying the lives of billions are good because they make the earth 'quieter' with less 'ambient noise'. They took down the video amid a public backlash for such arrogant, empathy-deleted stupidity. You see, however, where they are going with this. Corinne Le Quéré, a professor at the Tyndall Centre for Climate Change Research, University of East Anglia, was lead author of the climate lockdown study, and she writes for ... the World Economic Forum. Gates calls in 'his' book for changing 'every aspect of the economy' (long-time Cult agenda) and for humans to eat synthetic 'meat' (predicted in my books) while cows and other farm animals are eliminated. Australian TV host and commentator Alan Jones described what carbon emission targets would mean for farm animals in Australia alone if emissions were reduced as demanded by 35 percent by 2030 and zero by 2050:

Well, let's take agriculture, the total emissions from agriculture are about 75 million tonnes of carbon dioxide, equivalent. Now reduce that by 35 percent and you have to come down to 50 million tonnes, I've done the maths. So if you take for example 1.5 million cows, you're going to have to reduce the herd by 525,000 [by] 2030, nine years, that's 58,000 cows a year. The beef herd's 30 million, reduce that by 35 percent, that's 10.5 million, which means 1.2 million cattle have to go every year between now and 2030. This is insanity!

There are 75 million sheep. Reduce that by 35 percent, that's 26 million sheep, that's almost 3 million a year. So under the Paris Agreement over 30 million beasts, dairy cows, cattle, pigs and sheep would go. More than 8,000 every minute of every hour for the next decade, do these people know what they're talking about?

Clearly they don't at the level of campaigners, politicians and administrators. The Cult *does* know; that's the outcome it wants. We are faced with not just a war on humanity. Animals and the natural world are being targeted and I have been saying since the 'Covid' hoax began that the plan eventually was to claim that the 'deadly virus' is able to jump from animals, including farm animals and

domestic pets, to humans. Just before this book went into production came this story: 'Russia registers world's first Covid-19 vaccine for cats & dogs as makers of Sputnik V warn pets & farm animals could spread virus'. The report said 'top scientists warned that the deadly pathogen could soon begin spreading through homes and farms' and 'the next stage is the infection of farm and domestic animals'. Know the outcome and you'll see the journey. Think what that would mean for animals and keep your eye on a term called zoonosis or zoonotic diseases which transmit between animals and humans. The Cult wants to break the connection between animals and people as it does between people and people. Farm animals fit with the Cult agenda to transform food from natural to synthetic.

The gas of life is killing us

There can be few greater examples of Cult inversion than the condemnation of carbon dioxide as a dangerous pollutant when it is the gas of life. Without it the natural world would be dead and so we would all be dead. We breathe in oxygen and breathe out carbon dioxide while plants produce oxygen and absorb carbon dioxide. It is a perfect symbiotic relationship that the Cult wants to dismantle for reasons I will come to in the final two chapters. Gates, Schwab, other Cult operatives and mindless repeaters, want the world to be 'carbon neutral' by at least 2050 and the earlier the better. 'Zero carbon' is the cry echoed by lunatics calling for 'Zero Covid' when we already have it. These carbon emission targets will deindustrialise the world in accordance with Cult plans – the post-industrial, post-democratic society – and with so-called renewables like solar and wind not coming even close to meeting human energy needs blackouts and cold are inevitable. Texans got the picture in the winter of 2021 when a snow storm stopped wind turbines and solar panels from working and the lights went down along with water which relies on electricity for its supply system. Gates wants everything to be powered by electricity to ensure that his masters have the kill switch to stop all human activity, movement, cooking, water and warmth any time they like. The climate lie is so

stupendously inverted that it claims we must urgently reduce carbon dioxide when we *don't have enough*.

Co₂ in the atmosphere is a little above 400 parts per million when the optimum for plant growth is 2,000 ppm and when it falls anywhere near 150 ppm the natural world starts to die and so do we. It fell to as low as 280 ppm in an 1880 measurement in Hawaii and rose to 413 ppm in 2019 with industrialisation which is why the planet has become *greener* in the industrial period. How insane then that psychopathic madman Gates is not satisfied only with blocking the rise of Co₂. He's funding technology to suck it out of the atmosphere. The reason why will become clear. The industrial era is not destroying the world through Co₂ and has instead turned around a potentially disastrous ongoing fall in Co₂. Greenpeace co-founder and scientist Patrick Moore walked away from Greenpeace in 1986 and has exposed the green movement for fear-mongering and lies. He said that 500 million years ago there was *17 times* more Co₂ in the atmosphere than we have today and levels have been falling for hundreds of millions of years. In the last 150 million years Co₂ levels in Earth's atmosphere had reduced by *90 percent*. Moore said that by the time humanity began to unlock carbon dioxide from fossil fuels we were at '38 seconds to midnight' and in that sense: 'Humans are [the Earth's] salvation.' Moore made the point that only half the Co₂ emitted by fossil fuels stays in the atmosphere and we should remember that all pollution pouring from chimneys that we are told is carbon dioxide is in fact nothing of the kind. It's pollution. Carbon dioxide is an invisible gas.

William Happer, Professor of Physics at Princeton University and long-time government adviser on climate, has emphasised the Co₂ deficiency for maximum growth and food production. Greenhouse growers don't add carbon dioxide for a bit of fun. He said that most of the warming in the last 100 years, after the earth emerged from the super-cold period of the 'Little Ice Age' into a natural warming cycle, was over by 1940. Happer said that a peak year for warming in 1988 can be explained by a 'monster El Nino' which is a natural and cyclical warming of the Pacific that has nothing to do with 'climate

change'. He said the effect of Co2 could be compared to painting a wall with red paint in that once two or three coats have been applied it didn't matter how much more you slapped on because the wall will not get much redder. Almost all the effect of the rise in Co2 has already happened, he said, and the volume in the atmosphere would now have to *double* to increase temperature by a single degree. Climate hoaxers know this and they have invented the most ridiculously complicated series of 'feedback' loops to try to overcome this rather devastating fact. You hear puppet Greta going on cluelessly about feedback loops and this is why.

The Sun affects temperature? No you *climate denier*

Some other nonsense to contemplate: Climate graphs show that rises in temperature do not follow rises in Co2 – *it's the other way round* with a lag between the two of some 800 years. If we go back 800 years from present time we hit the Medieval Warm Period when temperatures were higher than now without any industrialisation and this was followed by the Little Ice Age when temperatures plummeted. The world was still emerging from these centuries of serious cold when many climate records began which makes the ever-repeated line of the 'hottest year since records began' meaningless when you are not comparing like with like. The coldest period of the Little Ice Age corresponded with the lowest period of sunspot activity when the Sun was at its least active. Proper scientists will not be at all surprised by this when it confirms the obvious fact that earth temperature is affected by the scale of Sun activity and the energetic power that it subsequently emits; but when is the last time you heard a climate hoaxer talking about the Sun as a source of earth temperature?? Everything has to be focussed on Co2 which makes up just 0.117 percent of so-called greenhouse gases and only a fraction of even that is generated by human activity. The rest is natural. More than 90 percent of those greenhouse gases are water vapour and clouds ([Fig 9](#)). Ban moisture I say. Have you noticed that the climate hoaxers no longer use the polar bear as their promotion image? That's because far from becoming extinct polar

bear communities are stable or thriving. Joe Bastardi, American meteorologist, weather forecaster and outspoken critic of the climate lie, documents in his book *The Climate Chronicles* how weather patterns and events claimed to be evidence of climate change have been happening since long before industrialisation: 'What happened before naturally is happening again, as is to be expected given the cyclical nature of the climate due to the design of the planet.' If you read the detailed background to the climate hoax in my other books you will shake your head and wonder how anyone could believe the crap which has spawned a multi-trillion dollar industry based on absolute garbage (see HIV causes AIDs and Sars-Cov-2 causes 'Covid-19'). Climate and 'Covid' have much in common given they have the same source. They both have the contradictory *everything* factor in which everything is explained by reference to them. It's hot – 'it's climate change'. It's cold – 'it's climate change'. I got a sniffle – 'it's Covid'. I haven't got a sniffle – 'it's Covid'. Not having a sniffle has to be a symptom of 'Covid'. Everything is and not having a sniffle is especially dangerous if you are a slow walker. For sheer audacity I offer you a Cambridge University 'study' that actually linked 'Covid' to 'climate change'. It had to happen eventually. They concluded that climate change played a role in 'Covid-19' spreading from animals to humans because ... wait for it ... I kid you not ... *the two groups were forced closer together as populations grow*. Er, that's it. The whole foundation on which this depended was that 'Bats are the likely zoonotic origin of SARS-CoV-1 and SARS-CoV-2'. Well, they are not. They are nothing to do with it. Apart from bats not being the origin and therefore 'climate change' effects on bats being irrelevant I am in awe of their academic insight. Where would we be without them? Not where we are that's for sure.

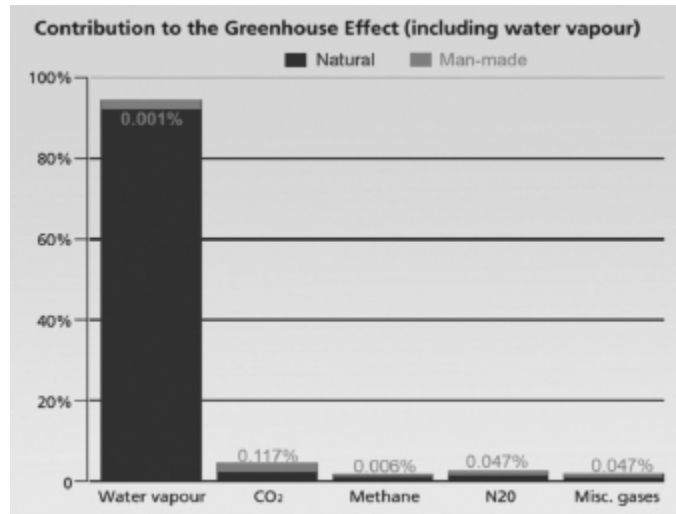


Figure 9: The idea that the gas of life is disastrously changing the climate is an insult to brain cell activity.

One other point about the weather is that climate modification is now well advanced and not every major weather event is natural – or earthquake come to that. I cover this subject at some length in other books. China is openly planning a rapid expansion of its weather modification programme which includes changing the climate in an area more than one and a half times the size of India. China used weather manipulation to ensure clear skies during the 2008 Olympics in Beijing. I have quoted from US military documents detailing how to employ weather manipulation as a weapon of war and they did that in the 1960s and 70s during the conflict in Vietnam with Operation Popeye manipulating monsoon rains for military purposes. Why would there be international treaties on weather modification if it wasn't possible? Of course it is. Weather is energetic information and it can be changed.

How was the climate hoax pulled off? See 'Covid'

If you can get billions to believe in a 'virus' that doesn't exist you can get them to believe in human-caused climate change that doesn't exist. Both are being used by the Cult to transform global society in the way it has long planned. Both hoaxes have been achieved in pretty much the same way. First you declare a lie is a fact. There's a

'virus' you call SARS-Cov-2 or humans are warming the planet with their behaviour. Next this becomes, via Cult networks, the foundation of government, academic and science policy and belief. Those who parrot the mantra are given big grants to produce research that confirms the narrative is true and ever more 'symptoms' are added to make the 'virus'/'climate change' sound even more scary. Scientists and researchers who challenge the narrative have their grants withdrawn and their careers destroyed. The media promote the lie as the unquestionable truth and censor those with an alternative view or evidence. A great percentage of the population believe what they are told as the lie becomes an everybody-knows-that and the believing-masses turn on those with a mind of their own. The technique has been used endlessly throughout human history. Wokers are the biggest promoters of the climate lie *and* 'Covid' fascism because their minds are owned by the Cult; their sense of self-righteous self-purity knows no bounds; and they exist in a bubble of reality in which facts are irrelevant and only get in the way of looking without seeing.

Running through all of this like veins in a blue cheese is control of information, which means control of perception, which means control of behaviour, which collectively means control of human society. The Cult owns the global media and Silicon Valley fascists for the simple reason that it *has* to. Without control of information it can't control perception and through that human society. Examine every facet of the Cult agenda and you will see that anything supporting its introduction is never censored while anything pushing back is always censored. I say again: Psychopaths that know why they are doing this must go before Nuremberg trials and those that follow their orders must trot along behind them into the same dock. 'I was just following orders' didn't work the first time and it must not work now. Nuremberg trials must be held all over the world before public juries for politicians, government officials, police, compliant doctors, scientists and virologists, and all Cult operatives such as Gates, Tedros, Fauci, Vallance, Whitty, Ferguson, Zuckerberg, Wojcicki, Brin, Page, Dorsey, the whole damn lot of

them – including, no *especially*, the psychopath psychologists. Without them and the brainless, gutless excuses for journalists that have repeated their lies, none of this could be happening. Nobody can be allowed to escape justice for the psychological and economic Armageddon they are all responsible for visiting upon the human race.

As for the compliant, unquestioning, swathes of humanity, and the self-obsessed, all-knowing ignorance of the Wokers ... don't start me. God help their kids. God help their grandkids. God *help them*.

CHAPTER NINE

We must have it? So what is it?

*Well I won't back down. No, I won't back down. You can stand me
up at the Gates of Hell. But I won't back down*

Tom Petty

I will now focus on the genetically-manipulating 'Covid vaccines' which do not meet this official definition of a vaccine by the US Centers for Disease Control (CDC): 'A product that stimulates a person's immune system to produce immunity to a specific disease, protecting the person from that disease.' On that basis 'Covid vaccines' are not a vaccine in that the makers don't even claim they stop infection or transmission.

They are instead part of a multi-levelled conspiracy to change the nature of the human body and what it means to be 'human' and to depopulate an enormous swathe of humanity. What I shall call Human 1.0 is on the cusp of becoming Human 2.0 and for very sinister reasons. Before I get to the 'Covid vaccine' in detail here's some background to vaccines in general. Government regulators do not test vaccines – the makers do – and the makers control which data is revealed and which isn't. Children in America are given 50 vaccine doses by age six and 69 by age 19 and the effect of the whole combined schedule has never been tested. Autoimmune diseases when the immune system attacks its own body have soared in the mass vaccine era and so has disease in general in children and the young. Why wouldn't this be the case when vaccines target the *immune system*? The US government gave Big Pharma drug

companies immunity from prosecution for vaccine death and injury in the 1986 National Childhood Vaccine Injury Act (NCVIA) and since then the government (taxpayer) has been funding compensation for the consequences of Big Pharma vaccines. The criminal and satanic drug giants can't lose and the vaccine schedule has increased dramatically since 1986 for this reason. There is no incentive to make vaccines safe and a big incentive to make money by introducing ever more. Even against a ridiculously high bar to prove vaccine liability, and with the government controlling the hearing in which it is being challenged for compensation, the vaccine court has so far paid out more than \$4 billion. These are the vaccines we are told are safe and psychopaths like Zuckerberg censor posts saying otherwise. The immunity law was even justified by a ruling that vaccines by their nature were 'unavoidably unsafe'.

Check out the ingredients of vaccines and you will be shocked if you are new to this. *They put that in children's bodies?? What??* Try aluminium, a brain toxin connected to dementia, aborted foetal tissue and formaldehyde which is used to embalm corpses. World-renowned aluminium expert Christopher Exley had his research into the health effect of aluminium in vaccines shut down by Keele University in the UK when it began taking funding from the Bill and Melinda Gates Foundation. Research when diseases 'eradicated' by vaccines began to decline and you will find the fall began long *before* the vaccine was introduced. Sometimes the fall even plateaued after the vaccine. Diseases like scarlet fever for which there was no vaccine declined in the same way because of environmental and other factors. A perfect case in point is the polio vaccine. Polio began when lead arsenate was first sprayed as an insecticide and residues remained in food products. Spraying started in 1892 and the first US polio epidemic came in Vermont in 1894. The simple answer was to stop spraying, but Rockefeller-created Big Pharma had a better idea. Polio was decreed to be caused by the *poliovirus* which 'spreads from person to person and can infect a person's spinal cord'. Lead arsenate was replaced by the lethal DDT which had the same effect of causing paralysis by damaging the brain and central nervous

system. Polio plummeted when DDT was reduced and then banned, but the vaccine is still given the credit for something it didn't do. Today by far the biggest cause of polio is the vaccines promoted by Bill Gates. Vaccine justice campaigner Robert Kennedy Jr, son of assassinated (by the Cult) US Attorney General Robert Kennedy, wrote:

In 2017, the World Health Organization (WHO) reluctantly admitted that the global explosion in polio is predominantly vaccine strain. The most frightening epidemics in Congo, Afghanistan, and the Philippines, are all linked to vaccines. In fact, by 2018, 70% of global polio cases were vaccine strain.

Vaccines make fortunes for Cult-owned Gates and Big Pharma while undermining the health and immune systems of the population. We had a glimpse of the mentality behind the Big Pharma cartel with a report on WION (World is One News), an international English language TV station based in India, which exposed the extraordinary behaviour of US drug company Pfizer over its 'Covid vaccine'. The WION report told how Pfizer had made fantastic demands of Argentina, Brazil and other countries in return for its 'vaccine'. These included immunity from prosecution, even for Pfizer negligence, government insurance to protect Pfizer from law suits and handing over as collateral sovereign assets of the country to include Argentina's bank reserves, military bases and embassy buildings. Pfizer demanded the same of Brazil in the form of waiving sovereignty of its assets abroad; exempting Pfizer from Brazilian laws; and giving Pfizer immunity from all civil liability. This is a 'vaccine' developed with government funding. Big Pharma is evil incarnate as a creation of the Cult and all must be handed tickets to Nuremberg.

Phantom 'vaccine' for a phantom 'disease'

I'll expose the 'Covid vaccine' fraud and then go on to the wider background of why the Cult has set out to 'vaccinate' every man, woman and child on the planet for an alleged 'new disease' with a survival rate of 99.77 percent (or more) even by the grotesquely-

manipulated figures of the World Health Organization and Johns Hopkins University. The 'infection' to 'death' ratio is 0.23 to 0.15 percent according to Stanford epidemiologist Dr John Ioannidis and while estimates vary the danger remains tiny. I say that if the truth be told the fake infection to fake death ratio is zero. Never mind all the evidence I have presented here and in *The Answer* that there is no 'virus' let us just focus for a moment on that death-rate figure of say 0.23 percent. The figure includes all those worldwide who have tested positive with a test not testing for the 'virus' and then died within 28 days or even longer of any other cause – *any other cause*. Now subtract all those illusory 'Covid' deaths on the global data sheets from the 0.23 percent. What do you think you would be left with? *Zero*. A vaccination has never been successfully developed for a so-called coronavirus. They have all failed at the animal testing stage when they caused hypersensitivity to what they were claiming to protect against and made the impact of a disease far worse. Cult-owned vaccine corporations got around that problem this time by bypassing animal trials, going straight to humans and making the length of the 'trials' before the public rollout as short as they could get away with. Normally it takes five to ten years or more to develop vaccines that still cause demonstrable harm to many people and that's without including the long-term effects that are never officially connected to the vaccination. 'Covid' non-vaccines have been officially produced and approved in a matter of months from a standing start and part of the reason is that (a) they were developed before the 'Covid' hoax began and (b) they are based on computer programs and not natural sources. Official non-trials were so short that government agencies gave *emergency*, not full, approval. 'Trials' were not even completed and full approval cannot be secured until they are. Public 'Covid vaccination' is actually a *continuation of the trial*. Drug company 'trials' are not scheduled to end until 2023 by which time a lot of people are going to be dead. Data on which government agencies gave this emergency approval was supplied by the Big Pharma corporations themselves in the form of Pfizer/BioNTech, AstraZeneca, Moderna, Johnson & Johnson, and

others, and this is the case with all vaccines. By its very nature *emergency* approval means drug companies do not have to prove that the 'vaccine' is 'safe and effective'. How could they with trials way short of complete? Government regulators only have to *believe* that they *could* be safe and effective. It is criminal manipulation to get products in circulation with no testing worth the name. Agencies giving that approval are infested with Big Pharma-connected place-people and they act in the interests of Big Pharma (the Cult) and not the public about whom they do not give a damn.

More human lab rats

'Covid vaccines' produced in record time by Pfizer/BioNTech and Moderna employ a technique *never approved before for use on humans*. They are known as mRNA 'vaccines' and inject a synthetic version of 'viral' mRNA or 'messenger RNA'. The key is in the term 'messenger'. The body works, or doesn't, on the basis of information messaging. Communications are constantly passing between and within the genetic system and the brain. Change those messages and you change the state of the body and even its very nature and you can change psychology and behaviour by the way the brain processes information. I think you are going to see significant changes in personality and perception of many people who have had the 'Covid vaccine' synthetic potions. Insider Aldous Huxley predicted the following in 1961 and mRNA 'vaccines' can be included in the term 'pharmacological methods':

There will be, in the next generation or so, a pharmacological method of making people love their servitude, and producing dictatorship without tears, so to speak, producing a kind of painless concentration camp for entire societies, so that people will in fact have their own liberties taken away from them, but rather enjoy it, because they will be distracted from any desire to rebel by propaganda or brainwashing, or brainwashing enhanced by pharmacological methods. And this seems to be the final revolution.

Apologists claim that mRNA synthetic 'vaccines' don't change the DNA genetic blueprint because RNA does not affect DNA only the other way round. This is so disingenuous. A process called 'reverse

transcription' can convert RNA into DNA and be integrated into DNA in the cell nucleus. This was highlighted in December, 2020, by scientists at Harvard and Massachusetts Institute of Technology (MIT). Geneticists report that more than 40 percent of mammalian genomes results from reverse transcription. On the most basic level if messaging changes then that sequence must lead to changes in DNA which is receiving and transmitting those communications. How can introducing synthetic material into cells not change the cells where DNA is located? The process is known as transfection which is defined as 'a technique to insert foreign nucleic acid (DNA or RNA) into a cell, typically with the intention of altering the properties of the cell'. Researchers at the Sloan Kettering Institute in New York found that changes in messenger RNA can deactivate tumour-suppressing proteins and thereby promote cancer. This is what happens when you mess with messaging. 'Covid vaccine' maker Moderna was founded in 2010 by Canadian stem cell biologist Derrick J. Rossi after his breakthrough discovery in the field of transforming and reprogramming stem cells. These are neutral cells that can be programmed to become any cell including sperm cells. Moderna was therefore founded on the principle of genetic manipulation and has never produced any vaccine or drug before its genetically-manipulating synthetic 'Covid' shite. Look at the name – Mode-RNA or Modify-RNA. Another important point is that the US Supreme Court has ruled that genetically-modified DNA, or complementary DNA (cDNA) synthesized in the laboratory from messenger RNA, can be patented and owned. These psychopaths are doing this to the human body.

Cells replicate synthetic mRNA in the 'Covid vaccines' and in theory the body is tricked into making antigens which trigger antibodies to target the 'virus spike proteins' which as Dr Tom Cowan said have *never been seen*. Cut the crap and these 'vaccines' deliver *self-replicating* synthetic material to the cells with the effect of changing human DNA. The more of them you have the more that process is compounded while synthetic material is all the time self-replicating. 'Vaccine'-maker Moderna describes mRNA as 'like

software for the cell' and so they are messing with the body's software. What happens when you change the software in a computer? Everything changes. For this reason the Cult is preparing a production line of mRNA 'Covid vaccines' and a long list of excuses to use them as with all the 'variants' of a 'virus' never shown to exist. The plan is further to transfer the mRNA technique to other vaccines mostly given to children and young people. The cumulative consequences will be a transformation of human DNA through a constant infusion of synthetic genetic material which will kill many and change the rest. Now consider that governments that have given emergency approval for a vaccine that's not a vaccine; never been approved for humans before; had no testing worth the name; and the makers have been given immunity from prosecution for any deaths or adverse effects suffered by the public. The UK government awarded *permanent legal indemnity* to itself and its employees for harm done when a patient is being treated for 'Covid-19' or 'suspected Covid-19'. That is quite a thought when these are possible 'side-effects' from the 'vaccine' (they are not 'side', they are effects) listed by the US Food and Drug Administration:

Guillain-Barre syndrome; acute disseminated encephalomyelitis; transverse myelitis; encephalitis; myelitis; encephalomyelitis; meningoencephalitis; meningitis; encephalopathy; convulsions; seizures; stroke; narcolepsy; cataplexy; anaphylaxis; acute myocardial infarction (heart attack); myocarditis; pericarditis; autoimmune disease; death; implications for pregnancy, and birth outcomes; other acute demyelinating diseases; non anaphylactic allergy reactions; thrombocytopenia ; disseminated intravascular coagulation; venous thromboembolism; arthritis; arthralgia; joint pain; Kawasaki disease; multisystem inflammatory syndrome in children; vaccine enhanced disease. The latter is the way the 'vaccine' has the potential to make diseases far worse than they would otherwise be.

UK doctor and freedom campaigner Vernon Coleman described the conditions in this list as 'all unpleasant, most of them very serious, and you can't get more serious than death'. The thought that anyone at all has had the 'vaccine' in these circumstances is testament to the potential that humanity has for clueless, unquestioning, stupidity and for many that programmed stupidity has already been terminal.

An insider speaks

Dr Michael Yeadon is a former Vice President, head of research and Chief Scientific Adviser at vaccine giant Pfizer. Yeadon worked on the inside of Big Pharma, but that did not stop him becoming a vocal critic of 'Covid vaccines' and their potential for multiple harms, including infertility in women. By the spring of 2021 he went much further and even used the no, no, term 'conspiracy'. When you begin to see what is going on it is impossible not to do so. Yeadon spoke out in an interview with freedom campaigner James Delingpole and I mentioned earlier how he said that no one had samples of 'the virus'. He explained that the mRNA technique originated in the anti-cancer field and ways to turn on and off certain genes which could be advantageous if you wanted to stop cancer growing out of control. 'That's the origin of them. They are a very unusual application, really.' Yeadon said that treating a cancer patient with an aggressive procedure might be understandable if the alternative was dying, but it was quite another thing to use the same technique as a public health measure. Most people involved wouldn't catch the infectious agent you were vaccinating against and if they did they probably wouldn't die:

If you are really using it as a public health measure you really want to as close as you can get to zero side-effects ... I find it odd that they chose techniques that were really cutting their teeth in the field of oncology and I'm worried that in using gene-based vaccines that have to be injected in the body and spread around the body, get taken up into some cells, and the regulators haven't quite told us which cells they get taken up into ... you are going to be generating a wide range of responses ... with multiple steps each of which could go well or badly.

I doubt the Cult intends it to go well. Yeadon said that you can put any gene you like into the body through the 'vaccine'. 'You can certainly give them a gene that would do them some harm if you wanted.' I was intrigued when he said that when used in the cancer field the technique could turn genes on and off. I explore this process in *The Answer* and with different genes having different functions you could create mayhem – physically and psychologically – if you turned the wrong ones on and the right ones off. I read reports of an experiment by researchers at the University of Washington's school of computer science and engineering in which they encoded DNA to infect computers. The body is itself a biological computer and if human DNA can inflict damage on a computer why can't the computer via synthetic material mess with the human body? It can. The Washington research team said it was possible to insert malicious malware into 'physical DNA strands' and corrupt the computer system of a gene sequencing machine as it 'reads gene letters and stores them as binary digits 0 and 1'. They concluded that hackers could one day use blood or spit samples to access computer systems and obtain sensitive data from police forensics labs or infect genome files. It is at this level of digital interaction that synthetic 'vaccines' need to be seen to get the full picture and that will become very clear later on. Michael Yeadon said it made no sense to give the 'vaccine' to younger people who were in no danger from the 'virus'. What was the benefit? It was all downside with potential effects:

The fact that my government in what I thought was a civilised, rational country, is raining [the 'vaccine'] on people in their 30s and 40s, even my children in their 20s, they're getting letters and phone calls, I know this is not right and any of you doctors who are vaccinating you know it's not right, too. They are not at risk. They are not at risk from the disease, so you are now hoping that the side-effects are so rare that you get away with it. You don't give new technology ... that you don't understand to 100 percent of the population.

Blood clot problems with the AstraZeneca 'vaccine' have been affecting younger people to emphasise the downside risks with no benefit. AstraZeneca's version, produced with Oxford University, does not use mRNA, but still gets its toxic cocktail inside cells where

it targets DNA. The Johnson & Johnson 'vaccine' which uses a similar technique has also produced blood clot effects to such an extent that the United States paused its use at one point. They are all 'gene therapy' (cell modification) procedures and not 'vaccines'. The truth is that once the content of these injections enter cells we have no idea what the effect will be. People can speculate and some can give very educated opinions and that's good. In the end, though, only the makers know what their potions are designed to do and even they won't know every last consequence. Michael Yeadon was scathing about doctors doing what they knew to be wrong. 'Everyone's mute', he said. Doctors in the NHS must know this was not right, coming into work and injecting people. 'I don't know how they sleep at night. I know I couldn't do it. I know that if I were in that position I'd have to quit.' He said he knew enough about toxicology to know this was not a good risk-benefit. Yeadon had spoken to seven or eight university professors and all except two would not speak out publicly. Their universities had a policy that no one said anything that countered the government and its medical advisors. They were afraid of losing their government grants. This is how intimidation has been used to silence the truth at every level of the system. I say silence, but these people could still speak out if they made that choice. Yeadon called them 'moral cowards' – 'This is about your children and grandchildren's lives and you have just buggered off and left it.'

'Variant' nonsense

Some of his most powerful comments related to the alleged 'variants' being used to instil more fear, justify more lockdowns, and introduce more 'vaccines'. He said government claims about 'variants' were nonsense. He had checked the alleged variant 'codes' and they were 99.7 percent identical to the 'original'. This was the human identity difference equivalent to putting a baseball cap on and off or wearing it the other way round. A 0.3 percent difference would make it impossible for that 'variant' to escape immunity from the 'original'. This made no sense of having new 'vaccines' for

‘variants’. He said there would have to be at least a *30 percent* difference for that to be justified and even then he believed the immune system would still recognise what it was. Gates-funded ‘variant modeller’ and ‘vaccine’-pusher John Edmunds might care to comment. Yeadon said drug companies were making new versions of the ‘vaccine’ as a ‘top up’ for ‘variants’. Worse than that, he said, the ‘regulators’ around the world like the MHRA in the UK had got together and agreed that because ‘vaccines’ for ‘variants’ were so similar to the first ‘vaccines’ *they did not have to do safety studies*. How transparently sinister that is. This is when Yeadon said: ‘There is a conspiracy here.’ There was no need for another vaccine for ‘variants’ and yet we were told that there was and the country had shut its borders because of them. ‘They are going into hundreds of millions of arms without passing ‘go’ or any regulator. Why did they do that? Why did they pick this method of making the vaccine?’

The reason had to be something bigger than that it seemed and ‘it’s not protection against the virus’. It’s was a far bigger project that meant politicians and advisers were willing to do things and not do things that knowingly resulted in avoidable deaths – ‘that’s already happened when you think about lockdown and deprivation of health care for a year.’ He spoke of people prepared to do something that results in the avoidable death of their fellow human beings and it not bother them. This is the penny-drop I have been working to get across for more than 30 years – the level of pure evil we are dealing with. Yeadon said his friends and associates could not believe there could be that much evil, but he reminded them of Stalin, Pol Pot and Hitler and of what Stalin had said: ‘One death is a tragedy. A million? A statistic.’ He could not think of a benign explanation for why you need top-up vaccines ‘which I’m sure you don’t’ and for the regulators ‘to just get out of the way and wave them through’. Why would the regulators do that when they were still wrestling with the dangers of the ‘parent’ vaccine? He was clearly shocked by what he had seen since the ‘Covid’ hoax began and now he was thinking the previously unthinkable:

If you wanted to depopulate a significant proportion of the world and to do it in a way that doesn't involve destruction of the environment with nuclear weapons, poisoning everyone with anthrax or something like that, and you wanted plausible deniability while you had a multi-year infectious disease crisis, I actually don't think you could come up with a better plan of work than seems to be in front of me. I can't say that's what they are going to do, but I can't think of a benign explanation why they are doing it.

He said he never thought that they would get rid of 99 percent of humans, but now he wondered. 'If you wanted to that this would be a hell of a way to do it – it would be unstoppable folks.' Yeadon had concluded that those who submitted to the 'vaccine' would be allowed to have some kind of normal life (but for how long?) while screws were tightened to coerce and mandate the last few percent. 'I think they'll put the rest of them in a prison camp. I wish I was wrong, but I don't think I am.' Other points he made included: There were no coronavirus vaccines then suddenly they all come along at the same time; we have no idea of the long term affect with trials so short; coercing or forcing people to have medical procedures is against the Nuremberg Code instigated when the Nazis did just that; people should at least delay having the 'vaccine'; a quick Internet search confirms that masks don't reduce respiratory viral transmission and 'the government knows that'; they have smashed civil society and they know that, too; two dozen peer-reviewed studies show no connection between lockdown and reducing deaths; he knew from personal friends the elite were still flying around and going on holiday while the public were locked down; the elite were not having the 'vaccines'. He was also asked if 'vaccines' could be made to target difference races. He said he didn't know, but the document by the Project for the New American Century in September, 2000, said developing 'advanced forms of biological warfare that can target *specific genotypes* may transform biological warfare from the realm of terror to a politically useful tool.' Oh, they're evil all right. Of that we can be *absolutely* sure.

Another cull of old people

We have seen from the CDC definition that the mRNA 'Covid vaccine' is not a vaccine and nor are the others that *claim* to reduce 'severity of symptoms' in *some* people, but not protect from infection or transmission. What about all the lies about returning to 'normal' if people were 'vaccinated'? If they are not claimed to stop infection and transmission of the alleged 'virus', how does anything change? This was all lies to manipulate people to take the jabs and we are seeing that now with masks and distancing still required for the 'vaccinated'. How did they think that elderly people with fragile health and immune responses were going to be affected by infusing their cells with synthetic material and other toxic substances? They *knew* that in the short and long term it would be devastating and fatal as the culling of the old that began with the first lockdowns was continued with the 'vaccine'. Death rates in care homes soared immediately residents began to be 'vaccinated' – infused with synthetic material. Brave and committed whistleblower nurses put their careers at risk by exposing this truth while the rest kept their heads down and their mouths shut to put their careers before those they are supposed to care for. A long-time American Certified Nursing Assistant who gave his name as James posted a video in which he described emotionally what happened in his care home when vaccination began. He said that during 2020 very few residents were sick with 'Covid' and no one died during the entire year; but shortly after the Pfizer mRNA injections 14 people died within two weeks and many others were near death. 'They're dropping like flies', he said. Residents who walked on their own before the shot could no longer and they had lost their ability to conduct an intelligent conversation. The home's management said the sudden deaths were caused by a 'super-spreader' of 'Covid-19'. Then how come, James asked, that residents who refused to take the injections were not sick? It was a case of inject the elderly with mRNA synthetic potions and blame their illness and death that followed on the 'virus'. James described what was happening in care homes as 'the greatest crime of genocide this country has ever seen'. Remember the NHS staff nurse from earlier who used the same

word 'genocide' for what was happening with the 'vaccines' and that it was an 'act of human annihilation'. A UK care home whistleblower told a similar story to James about the effect of the 'vaccine' in deaths and 'outbreaks' of illness dubbed 'Covid' after getting the jab. She told how her care home management and staff had zealously imposed government regulations and no one was allowed to even question the official narrative let alone speak out against it. She said the NHS was even worse. Again we see the results of reframing. A worker at a local care home where I live said they had not had a single case of 'Covid' there for almost a year and when the residents were 'vaccinated' they had 19 positive cases in two weeks with eight dying.

It's not the 'vaccine' – honest

The obvious cause and effect was being ignored by the media and most of the public. Australia's health minister Greg Hunt (a former head of strategy at the World Economic Forum) was admitted to hospital after he had the 'vaccine'. He was suffering according to reports from the skin infection 'cellulitis' and it must have been a severe case to have warranted days in hospital. Immediately the authorities said this was nothing to do with the 'vaccine' when an effect of some vaccines is a 'cellulitis-like reaction'. We had families of perfectly healthy old people who died after the 'vaccine' saying that if only they had been given the 'vaccine' earlier they would still be alive. As a numbskull rating that is off the chart. A father of four 'died of Covid' at aged 48 when he was taken ill two days after having the 'vaccine'. The man, a health administrator, had been 'shielding during the pandemic' and had 'not really left the house' until he went for the 'vaccine'. Having the 'vaccine' and then falling ill and dying does not seem to have qualified as a possible cause and effect and 'Covid-19' went on his death certificate. His family said they had no idea how he 'caught the virus'. A family member said: 'Tragically, it could be that going for a vaccination ultimately led to him catching Covid ...The sad truth is that they are never going to know where it came from.' The family warned people to remember

that the virus still existed and was 'very real'. So was their stupidity. Nurses and doctors who had the first round of the 'vaccine' were collapsing, dying and ending up in a hospital bed while they or their grieving relatives were saying they'd still have the 'vaccine' again despite what happened. I kid you not. You mean if your husband returned from the dead he'd have the same 'vaccine' again that killed him??

Doctors at the VCU Medical Center in Richmond, Virginia, said the Johnson & Johnson 'vaccine' was to blame for a man's skin peeling off. Patient Richard Terrell said: 'It all just happened so fast. My skin peeled off. It's still coming off on my hands now.' He said it was stinging, burning and itching and when he bent his arms and legs it was very painful with 'the skin swollen and rubbing against itself'. Pfizer/BioNTech and Moderna vaccines use mRNA to change the cell while the Johnson & Johnson version uses DNA in a process similar to AstraZeneca's technique. Johnson & Johnson and AstraZeneca have both had their 'vaccines' paused by many countries after causing serious blood problems. Terrell's doctor Fnu Nutan said he could have died if he hadn't got medical attention. It sounds terrible so what did Nutan and Terrell say about the 'vaccine' now? Oh, they still recommend that people have it. A nurse in a hospital bed 40 minutes after the vaccination and unable to swallow due to throat swelling was told by a doctor that he lost mobility in his arm for 36 hours following the vaccination. What did he say to the ailing nurse? 'Good for you for getting the vaccination.' We are dealing with a serious form of cognitive dissonance madness in both public and medical staff. There is a remarkable correlation between those having the 'vaccine' and trumpeting the fact and suffering bad happenings shortly afterwards. Witold Rogiewicz, a Polish doctor, made a video of his 'vaccination' and ridiculed those who were questioning its safety and the intentions of Bill Gates: 'Vaccinate yourself to protect yourself, your loved ones, friends and also patients. And to mention quickly I have info for anti-vaxxers and anti-Covidors if you want to contact Bill Gates you can do this through me.' He further ridiculed the dangers of 5G. Days later he

was dead, but naturally the vaccination wasn't mentioned in the verdict of 'heart attack'.

Lies, lies and more lies

So many members of the human race have slipped into extreme states of insanity and unfortunately they include reframed doctors and nursing staff. Having a 'vaccine' and dying within minutes or hours is not considered a valid connection while death from any cause within 28 days or longer of a positive test with a test not testing for the 'virus' means 'Covid-19' goes on the death certificate. How could that 'vaccine'-death connection not have been made except by calculated deceit? US figures in the initial rollout period to February 12th, 2020, revealed that a third of the deaths reported to the CDC after 'Covid vaccines' happened within 48 hours. Five men in the UK suffered an 'extremely rare' blood clot problem after having the AstraZeneca 'vaccine', but no causal link was established said the Gates-funded Medicines and Healthcare products Regulatory Agency (MHRA) which had given the 'vaccine' emergency approval to be used. Former Pfizer executive Dr Michael Yeadon explained in his interview how the procedures could cause blood coagulation and clots. People who should have been at no risk were dying from blood clots in the brain and he said he had heard from medical doctor friends that people were suffering from skin bleeding and massive headaches. The AstraZeneca 'shot' was stopped by some 20 countries over the blood clotting issue and still the corrupt MHRA, the European Medicines Agency (EMA) and the World Health Organization said that it should continue to be given even though the EMA admitted that it 'still cannot rule out definitively' a link between blood clotting and the 'vaccine'. Later Marco Cavaleri, head of EMA vaccine strategy, said there was indeed a clear link between the 'vaccine' and thrombosis, but they didn't know why. So much for the trials showing the 'vaccine' is safe. Blood clots were affecting younger people who would be under virtually no danger from 'Covid' even if it existed which makes it all the more stupid and sinister.

The British government responded to public alarm by wheeling out June Raine, the terrifyingly weak infant school headmistress sound-alike who heads the UK MHRA drug 'regulator'. The idea that she would stand up to Big Pharma and government pressure is laughable and she told us that all was well in the same way that she did when allowing untested, never-used-on-humans-before, genetically-manipulating 'vaccines' to be exposed to the public in the first place. Mass lying is the new normal of the 'Covid' era. The MHRA later said 30 cases of rare blood clots had by then been connected with the AstraZeneca 'vaccine' (that means a lot more in reality) while stressing that the benefits of the jab in preventing 'Covid-19' outweighed any risks. A more ridiculous and disingenuous statement with callous disregard for human health it is hard to contemplate. Immediately after the mendacious 'all-clears' two hospital workers in Denmark experienced blood clots and cerebral haemorrhaging following the AstraZeneca jab and one died. Top Norwegian health official Pål Andre Holme said the 'vaccine' was the only common factor: 'There is nothing in the patient history of these individuals that can give such a powerful immune response ... I am confident that the antibodies that we have found are the cause, and I see no other explanation than it being the vaccine which triggers it.' Strokes, a clot or bleed in the brain, were clearly associated with the 'vaccine' from word of mouth and whistleblower reports. Similar consequences followed with all these 'vaccines' that we were told were so safe and as the numbers grew by the day it was clear we were witnessing human carnage.

Learning the hard way

A woman interviewed by UKColumn told how her husband suffered dramatic health effects after the vaccine when he'd been in good health all his life. He went from being a little unwell to losing all feeling in his legs and experiencing 'excruciating pain'. Misdiagnosis followed twice at Accident and Emergency (an 'allergy' and 'sciatica') before he was admitted to a neurology ward where doctors said his serious condition had been caused by the

'vaccine'. Another seven 'vaccinated' people were apparently being treated on the same ward for similar symptoms. The woman said he had the 'vaccine' because they believed media claims that it was safe. 'I didn't think the government would give out a vaccine that does this to somebody; I believed they would be bringing out a vaccination that would be safe.' What a tragic way to learn that lesson. Another woman posted that her husband was transporting stroke patients to hospital on almost every shift and when he asked them if they had been 'vaccinated' for 'Covid' they all replied 'yes'. One had a 'massive brain bleed' the day after his second dose. She said her husband reported the 'just been vaccinated' information every time to doctors in A and E only for them to ignore it, make no notes and appear annoyed that it was even mentioned. This particular report cannot be verified, but it expresses a common theme that confirms the monumental underreporting of 'vaccine' consequences. Interestingly as the 'vaccines' and their brain blood clot/stroke consequences began to emerge the UK National Health Service began a publicity campaign telling the public what to do in the event of a stroke. A Scottish NHS staff nurse who quit in disgust in March, 2021, said:

I have seen traumatic injuries from the vaccine, they're not getting reported to the yellow card [adverse reaction] scheme, they're treating the symptoms, not asking why, why it's happening. It's just treating the symptoms and when you speak about it you're dismissed like you're crazy, I'm not crazy, I'm not crazy because every other colleague I've spoken to is terrified to speak out, they've had enough.

Videos appeared on the Internet of people uncontrollably shaking after the 'vaccine' with no control over muscles, limbs and even their face. A Scottish mother broke out in a severe rash all over her body almost immediately after she was given the AstraZeneca 'vaccine'. The pictures were horrific. Leigh King, a 41-year-old hairdresser from Lanarkshire said: 'Never in my life was I prepared for what I was about to experience ... My skin was so sore and constantly hot ... I have never felt pain like this ...' But don't you worry, the 'vaccine' is perfectly safe. Then there has been the effect on medical

staff who have been pressured to have the 'vaccine' by psychopathic 'health' authorities and government. A London hospital consultant who gave the name K. Polyakova wrote this to the *British Medical Journal* or *BMJ*:

I am currently struggling with ... the failure to report the reality of the morbidity caused by our current vaccination program within the health service and staff population. The levels of sickness after vaccination is unprecedented and staff are getting very sick and some with neurological symptoms which is having a huge impact on the health service function. Even the young and healthy are off for days, some for weeks, and some requiring medical treatment. Whole teams are being taken out as they went to get vaccinated together.

Mandatory vaccination in this instance is stupid, unethical and irresponsible when it comes to protecting our staff and public health. We are in the voluntary phase of vaccination, and encouraging staff to take an unlicensed product that is impacting on their immediate health ... it is clearly stated that these vaccine products do not offer immunity or stop transmission. In which case why are we doing it?

Not to protect health that's for sure. Medical workers are lauded by governments for agenda reasons when they couldn't give a toss about them any more than they can for the population in general. Schools across America faced the same situation as they closed due to the high number of teachers and other staff with bad reactions to the Pfizer/BioNTech, Moderna, and Johnson & Johnson 'Covid vaccines' all of which were linked to death and serious adverse effects. The *BMJ* took down the consultant's comments pretty quickly on the grounds that they were being used to spread 'disinformation'. They were exposing the truth about the 'vaccine' was the real reason. The cover-up is breathtaking.

Hiding the evidence

The scale of the 'vaccine' death cover-up worldwide can be confirmed by comparing official figures with the personal experience of the public. I heard of many people in my community who died immediately or soon after the vaccine that would never appear in the media or even likely on the official totals of 'vaccine' fatalities and adverse reactions when only about ten percent are estimated to be

reported and I have seen some estimates as low as one percent in a Harvard study. In the UK alone by April 29th, 2021, some 757,654 adverse reactions had been officially reported from the Pfizer/BioNTech, Oxford/AstraZeneca and Moderna 'vaccines' with more than a thousand deaths linked to jabs and that means an estimated ten times this number in reality from a ten percent reporting rate percentage. That's seven million adverse reactions and 10,000 potential deaths and a one percent reporting rate would be ten times *those* figures. In 1976 the US government pulled the swine flu vaccine after 53 deaths. The UK data included a combined 10,000 eye disorders from the 'Covid vaccines' with more than 750 suffering visual impairment or blindness and again multiply by the estimated reporting percentages. As 'Covid cases' officially fell hospitals virtually empty during the 'Covid crisis' began to fill up with a range of other problems in the wake of the 'vaccine' rollout. The numbers across America have also been catastrophic. Deaths linked to *all* types of vaccine increased by 6,000 *percent* in the first quarter of 2021 compared with 2020. A 39-year-old woman from Ogden, Utah, died four days after receiving a second dose of Moderna's 'Covid vaccine' when her liver, heart and kidneys all failed despite the fact that she had no known medical issues or conditions. Her family sought an autopsy, but Dr Erik Christensen, Utah's chief medical examiner, said proving vaccine injury as a cause of death almost never happened. He could think of only one instance where an autopsy would name a vaccine as the official cause of death and that would be anaphylaxis where someone received a vaccine and died almost instantaneously. 'Short of that, it would be difficult for us to definitively say this is the vaccine,' Christensen said. If that is true this must be added to the estimated ten percent (or far less) reporting rate of vaccine deaths and serious reactions and the conclusion can only be that vaccine deaths and serious reactions – including these 'Covid' potions' – are phenomenally understated in official figures. The same story can be found everywhere. Endless accounts of deaths and serious reactions among the public, medical

and care home staff while official figures did not even begin to reflect this.

Professional script-reader Dr David Williams, a 'top public-health official' in Ontario, Canada, insulted our intelligence by claiming only four serious adverse reactions and no deaths from the more than 380,000 vaccine doses then given. This bore no resemblance to what people knew had happened in their own circles and we had Dirk Huyer in charge of getting millions vaccinated in Ontario while at the same time he was Chief Coroner for the province investigating causes of death including possible death from the vaccine. An aide said he had stepped back from investigating deaths, but evidence indicated otherwise. Rosemary Frei, who secured a Master of Science degree in molecular biology at the Faculty of Medicine at Canada's University of Calgary before turning to investigative journalism, was one who could see that official figures for 'vaccine' deaths and reactions made no sense. She said that doctors seldom reported adverse events and when people got really sick or died after getting a vaccination they would attribute that to anything except the vaccines. It had been that way for years and anyone who wondered aloud whether the 'Covid vaccines' or other shots cause harm is immediately branded as 'anti-vax' and 'anti-science'. This was 'career-threatening' for health professionals. Then there was the huge pressure to support the push to 'vaccinate' billions in the quickest time possible. Frei said:

So that's where we're at today. More than half a million vaccine doses have been given to people in Ontario alone. The rush is on to vaccinate all 15 million of us in the province by September. And the mainstream media are screaming for this to be sped up even more. That all adds up to only a very slim likelihood that we're going to be told the truth by officials about how many people are getting sick or dying from the vaccines.

What is true of Ontario is true of everywhere.

They KNEW – and still did it

The authorities knew what was going to happen with multiple deaths and adverse reactions. The UK government's Gates-funded

and Big Pharma-dominated Medicines and Healthcare products Regulatory Agency (MHRA) hired a company to employ AI in compiling the projected reactions to the 'vaccine' that would otherwise be uncountable. The request for applications said: 'The MHRA urgently seeks an Artificial Intelligence (AI) software tool to process the expected high volume of Covid-19 vaccine Adverse Drug Reaction ...' This was from the agency, headed by the disingenuous June Raine, that gave the 'vaccines' emergency approval and the company was hired before the first shot was given. 'We are going to kill and maim you – is that okay?' 'Oh, yes, perfectly fine – I'm very grateful, thank you, doctor.' The range of 'Covid vaccine' adverse reactions goes on for page after page in the MHRA criminally underreported 'Yellow Card' system and includes affects to eyes, ears, skin, digestion, blood and so on. Raine's MHRA amazingly claimed that the 'overall safety experience ... is so far as expected from the clinical trials'. The death, serious adverse effects, deafness and blindness were *expected*? When did they ever mention that? If these human tragedies were expected then those that gave approval for the use of these 'vaccines' must be guilty of crimes against humanity including murder – a definition of which is 'killing a person with malice aforethought or with recklessness manifesting extreme indifference to the value of human life.' People involved at the MHRA, the CDC in America and their equivalent around the world must go before Nuremberg trials to answer for their callous inhumanity. We are only talking here about the immediate effects of the 'vaccine'. The longer-term impact of the DNA synthetic manipulation is the main reason they are so hysterically desperate to inoculate the entire global population in the shortest possible time.

Africa and the developing world are a major focus for the 'vaccine' depopulation agenda and a mass vaccination sales-pitch is underway thanks to caring people like the Rockefellers and other Cult assets. The Rockefeller Foundation, which pre-empted the 'Covid pandemic' in a document published in 2010 that 'predicted' what happened a decade later, announced an initial \$34.95 million grant in February, 2021, 'to ensure more equitable access to Covid-19

testing and vaccines' among other things in Africa in collaboration with '24 organizations, businesses, and government agencies'. The pan-Africa initiative would focus on 10 countries: Burkina Faso, Ethiopia, Ghana, Kenya, Nigeria, Rwanda, South Africa, Tanzania, Uganda, and Zambia'. Rajiv Shah, President of the Rockefeller Foundation and former administrator of CIA-controlled USAID, said that if Africa was not mass-vaccinated (to change the DNA of its people) it was a 'threat to all of humanity' and not fair on Africans. When someone from the Rockefeller Foundation says they want to do something to help poor and deprived people and countries it is time for a belly-laugh. They are doing this out of the goodness of their 'heart' because 'vaccinating' the entire global population is what the 'Covid' hoax set out to achieve. Official 'decolonisation' of Africa by the Cult was merely a prelude to financial colonisation on the road to a return to physical colonisation. The 'vaccine' is vital to that and the sudden and convenient death of the 'Covid' sceptic president of Tanzania can be seen in its true light. A lot of people in Africa are aware that this is another form of colonisation and exploitation and they need to stand their ground.

The 'vaccine is working' scam

A potential problem for the Cult was that the 'vaccine' is meant to change human DNA and body messaging and not to protect anyone from a 'virus' never shown to exist. The vaccine couldn't work because it was not designed to work and how could they make it *appear* to be working so that more people would have it? This was overcome by lowering the amplification rate of the PCR test to produce fewer 'cases' and therefore fewer 'deaths'. Some of us had been pointing out since March, 2020, that the amplification rate of the test not testing for the 'virus' had been made artificially high to generate positive tests which they could call 'cases' to justify lockdowns. The World Health Organization recommended an absurdly high 45 amplification cycles to ensure the high positives required by the Cult and then remained silent on the issue until January 20th, 2021 – Biden's Inauguration Day. This was when the

'vaccinations' were seriously underway and on that day the WHO recommended after discussions with America's CDC that laboratories *lowered their testing amplification*. Dr David Samadi, a certified urologist and health writer, said the WHO was encouraging all labs to reduce their cycle count for PCR tests. He said the current cycle was much too high and was 'resulting in any particle being declared a positive case'. Even one mainstream news report I saw said this meant the number of 'Covid' infections may have been 'dramatically inflated'. Oh, just a little bit. The CDC in America issued new guidance to laboratories in April, 2021, to use 28 cycles *but only for 'vaccinated' people*. The timing of the CDC/WHO interventions were cynically designed to make it appear the 'vaccines' were responsible for falling cases and deaths when the real reason can be seen in the following examples. New York's state lab, the Wadsworth Center, identified 872 positive tests in July, 2020, based on a threshold of 40 cycles. When the figure was lowered to 35 cycles 43 percent of the 872 were no longer 'positives'. At 30 cycles the figure was 63 percent. A Massachusetts lab found that between 85 to 90 percent of people who tested positive in July with a cycle threshold of 40 would be negative at 30 cycles, Ashish Jha, MD, director of the Harvard Global Health Institute, said: 'I'm really shocked that it could be that high ... Boy, does it really change the way we need to be thinking about testing.' I'm shocked that I could see the obvious in the spring of 2020, with no medical background, and most medical professionals still haven't worked it out. No, that's not shocking – it's terrifying.

Three weeks after the WHO directive to lower PCR cycles the London *Daily Mail* ran this headline: 'Why ARE Covid cases plummeting? New infections have fallen 45% in the US and 30% globally in the past 3 weeks but experts say vaccine is NOT the main driver because only 8% of Americans and 13% of people worldwide have received their first dose.' They acknowledged that the drop could not be attributed to the 'vaccine', but soon this morphed throughout the media into the 'vaccine' has caused cases and deaths to fall when it was the PCR threshold. In December, 2020, there was

chaos at English Channel ports with truck drivers needing negative 'Covid' tests before they could board a ferry home for Christmas. The government wanted to remove the backlog as fast as possible and they brought in troops to do the 'testing'. Out of 1,600 drivers just 36 tested positive and the rest were given the all clear to cross the Channel. I guess the authorities thought that 36 was the least they could get away with without the unquestioning catching on. The amplification trick which most people believed in the absence of information in the mainstream applied more pressure on those refusing the 'vaccine' to succumb when it 'obviously worked'. The truth was the exact opposite with deaths in care homes soaring with the 'vaccine' and in Israel the term used was 'skyrocket'. A re-analysis of published data from the Israeli Health Ministry led by Dr Hervé Seligmann at the Medicine Emerging Infectious and Tropical Diseases at Aix-Marseille University found that Pfizer's 'Covid vaccine' killed 'about 40 times more [elderly] people than the disease itself would have killed' during a five-week vaccination period and *260 times* more younger people than would have died from the 'virus' even according to the manipulated 'virus' figures. Dr Seligmann and his co-study author, Haim Yativ, declared after reviewing the Israeli 'vaccine' death data: 'This is a new Holocaust.'

Then, in mid-April, 2021, after vast numbers of people worldwide had been 'vaccinated', the story changed with clear coordination. The UK government began to prepare the ground for more future lockdowns when Nuremberg-destined Boris Johnson told yet another whopper. He said that cases had fallen because of *lockdowns* not 'vaccines'. Lockdowns are irrelevant when *there is no 'virus'* and the test and fraudulent death certificates are deciding the number of 'cases' and 'deaths'. Study after study has shown that lockdowns don't work and instead kill and psychologically destroy people. Meanwhile in the United States Anthony Fauci and Rochelle Walensky, the ultra-Zionist head of the CDC, peddled the same line. More lockdown was the answer and not the 'vaccine', a line repeated on cue by the moron that is Canadian Prime Minister Justin Trudeau. Why all the hysteria to get everyone 'vaccinated' if lockdowns and

not 'vaccines' made the difference? None of it makes sense on the face of it. Oh, but it does. The Cult wants lockdowns *and* the 'vaccine' and if the 'vaccine' is allowed to be seen as the total answer lockdowns would no longer be justified when there are still livelihoods to destroy. 'Variants' and renewed upward manipulation of PCR amplification are planned to instigate never-ending lockdown *and* more 'vaccines'.

You *must* have it – we're desperate

Israel, where the Jewish and Arab population are ruled by the Sabbatian Cult, was the front-runner in imposing the DNA-manipulating 'vaccine' on its people to such an extent that Jewish refusers began to liken what was happening to the early years of Nazi Germany. This would seem to be a fantastic claim. Why would a government of Jewish people be acting like the Nazis did? If you realise that the Sabbatian Cult was behind the Nazis and that Sabbatians hate Jews the pieces start to fit and the question of why a 'Jewish' government would treat Jews with such callous disregard for their lives and freedom finds an answer. Those controlling the government of Israel *aren't Jewish* – they're Sabbatian. Israeli lawyer Tamir Turgal was one who made the Nazi comparison in comments to German lawyer Reiner Fuellmich who is leading a class action lawsuit against the psychopaths for crimes against humanity. Turgal described how the Israeli government was vaccinating children and pregnant women on the basis that there was no evidence that this was dangerous when they had no evidence that it *wasn't* dangerous either. They just had no evidence. This was medical experimentation and Turgal said this breached the Nuremberg Code about medical experimentation and procedures requiring informed consent and choice. Think about that. A Nuremberg Code developed because of Nazi experimentation on Jews and others in concentration camps by people like the evil-beyond-belief Josef Mengele is being breached by the *Israeli* government; but when you know that it's a *Sabbatian* government along with its intelligence and military agencies like Mossad, Shin Bet and the Israeli Defense Forces, and that Sabbatians

were the force behind the Nazis, the kaleidoscope comes into focus. What have we come to when Israeli Jews are suing their government for violating the Nuremberg Code by essentially making Israelis subject to a medical experiment using the controversial 'vaccines'? It's a shocker that this has to be done in the light of what happened in Nazi Germany. The Anshe Ha-Emet, or 'People of the Truth', made up of Israeli doctors, lawyers, campaigners and public, have launched a lawsuit with the International Criminal Court. It says:

When the heads of the Ministry of Health as well as the prime minister presented the vaccine in Israel and began the vaccination of Israeli residents, the vaccinated were not advised, that, in practice, they are taking part in a medical experiment and that their consent is required for this under the Nuremberg Code.

The irony is unbelievable, but easily explained in one word: Sabbatians. The foundation of Israeli 'Covid' apartheid is the 'green pass' or 'green passport' which allows Jews and Arabs who have had the DNA-manipulating 'vaccine' to go about their lives – to work, fly, travel in general, go to shopping malls, bars, restaurants, hotels, concerts, gyms, swimming pools, theatres and sports venues, while non-'vaccinated' are banned from all those places and activities. Israelis have likened the 'green pass' to the yellow stars that Jews in Nazi Germany were forced to wear – the same as the yellow stickers that a branch of UK supermarket chain Morrisons told exempt mask-wearers they had to display when shopping. How very sensitive. The Israeli system is blatant South African-style apartheid on the basis of compliance or non-compliance to fascism rather than colour of the skin. How appropriate that the Sabbatian Israeli government was so close to the pre-Mandela apartheid regime in Pretoria. The Sabbatian-instigated 'vaccine passport' in Israel is planned for everywhere. Sabbatians struck a deal with Pfizer that allowed them to lead the way in the percentage of a national population infused with synthetic material and the result was catastrophic. Israeli freedom activist Shai Dannon told me how chairs were appearing on beaches that said 'vaccinated only'. Health Minister Yuli Edelstein said that anyone unwilling or unable to get

the jabs that 'confer immunity' will be 'left behind'. The man's a liar. Not even the makers claim the 'vaccines' confer immunity. When you see those figures of 'vaccine' deaths these psychopaths were saying that you must take the chance the 'vaccine' will kill you or maim you while knowing it will change your DNA or lockdown for you will be permanent. That's fascism. The Israeli parliament passed a law to allow personal information of the non-vaccinated to be shared with local and national authorities for three months. This was claimed by its supporters to be a way to 'encourage' people to be vaccinated. Hadas Ziv from Physicians for Human Rights described this as a 'draconian law which crushed medical ethics and the patient rights'. But that's the idea, the Sabbatians would reply.

Your papers, please

Sabbatian Israel was leading what has been planned all along to be a global 'vaccine pass' called a 'green passport' without which you would remain in permanent lockdown restriction and unable to do anything. This is how badly – *desperately* – the Cult is to get everyone 'vaccinated'. The term and colour 'green' was not by chance and related to the psychology of fusing the perception of the green climate hoax with the 'Covid' hoax and how the 'solution' to both is the same Great Reset. Lying politicians, health officials and psychologists denied there were any plans for mandatory vaccinations or restrictions based on vaccinations, but they knew that was exactly what was meant to happen with governments of all countries reaching agreements to enforce a global system. 'Free' Denmark and 'free' Sweden unveiled digital vaccine certification. Cyprus, Czech Republic, Estonia, Greece, Hungary, Iceland, Italy, Poland, Portugal, Slovakia, and Spain have all committed to a vaccine passport system and the rest including the whole of the EU would follow. The satanic UK government will certainly go this way despite mendacious denials and at the time of writing it is trying to manipulate the public into having the 'vaccine' so they could go abroad on a summer holiday. How would that work without something to prove you had the synthetic toxicity injected into you?

Documents show that the EU's European Commission was moving towards 'vaccine certificates' in 2018 and 2019 before the 'Covid' hoax began. They knew what was coming. Abracadabra – Ursula von der Leyen, the German President of the Commission, announced in March, 2021, an EU 'Digital Green Certificate' – green again – to track the public's 'Covid status'. The passport sting is worldwide and the Far East followed the same pattern with South Korea ruling that only those with 'vaccination' passports – again the *green* pass – would be able to 'return to their daily lives'.

Bill Gates has been preparing for this 'passport' with other Cult operatives for years and beyond the paper version is a Gates-funded 'digital tattoo' to identify who has been vaccinated and who hasn't. The 'tattoo' is reported to include a substance which is externally readable to confirm who has been vaccinated. This is a bio-luminous light-generating enzyme (think fireflies) called ... *Luciferase*. Yes, named after the Cult 'god' Lucifer the 'light bringer' of whom more to come. Gates said he funded the readable tattoo to ensure children in the developing world were vaccinated and no one was missed out. He cares so much about poor kids as we know. This was just the cover story to develop a vaccine tagging system for everyone on the planet. Gates has been funding the ID2020 'alliance' to do just that in league with other lovely people at Microsoft, GAVI, the Rockefeller Foundation, Accenture and IDEO.org. He said in interviews in March, 2020, before any 'vaccine' publicly existed, that the world must have a globalised digital certificate to track the 'virus' and who had been vaccinated. Gates knew from the start that the mRNA vaccines were coming and when they would come and that the plan was to tag the 'vaccinated' to marginalise the intelligent and stop them doing anything including travel. Evil just doesn't suffice. Gates was exposed for offering a \$10 million bribe to the Nigerian House of Representatives to invoke compulsory 'Covid' vaccination of all Nigerians. Sara Cunial, a member of the Italian Parliament, called Gates a 'vaccine criminal'. She urged the Italian President to hand him over to the International Criminal Court for crimes against

humanity and condemned his plans to 'chip the human race' through ID2020.

You know it's a long-planned agenda when war criminal and Cult gofer Tony Blair is on the case. With the scale of arrogance only someone as dark as Blair can muster he said: 'Vaccination in the end is going to be your route to liberty.' Blair is a disgusting piece of work and he confirms that again. The media has given a lot of coverage to a bloke called Charlie Mullins, founder of London's biggest independent plumbing company, Pimlico Plumbers, who has said he won't employ anyone who has not been vaccinated or have them go to any home where people are not vaccinated. He said that if he had his way no one would be allowed to walk the streets if they have not been vaccinated. Gates was cheering at the time while I was alerting the white coats. The plan is that people will qualify for 'passports' for having the first two doses and then to keep it they will have to have all the follow ups and new ones for invented 'variants' until human genetics is transformed and many are dead who can't adjust to the changes. Hollywood celebrities – the usual propaganda stunt – are promoting something called the WELL Health-Safety Rating to verify that a building or space has 'taken the necessary steps to prioritize the health and safety of their staff, visitors and other stakeholders'. They included Lady Gaga, Jennifer Lopez, Michael B. Jordan, Robert DeNiro, Venus Williams, Wolfgang Puck, Deepak Chopra and 17th Surgeon General Richard Carmona. Yawn. WELL Health-Safety has big connections with China. Parent company Delos is headed by former Goldman Sachs partner Paul Scialla. This is another example – and we will see so many others – of using the excuse of 'health' to dictate the lives and activities of the population. I guess one confirmation of the 'safety' of buildings is that only 'vaccinated' people can go in, right?

Electronic concentration camps

I wrote decades ago about the plans to restrict travel and here we are for those who refuse to bow to tyranny. This can be achieved in one go with air travel if the aviation industry makes a blanket decree.

The 'vaccine' and guaranteed income are designed to be part of a global version of China's social credit system which tracks behaviour 24/7 and awards or deletes 'credits' based on whether your behaviour is supported by the state or not. I mean your entire lifestyle – what you do, eat, say, everything. Once your credit score falls below a certain level consequences kick in. In China tens of millions have been denied travel by air and train because of this. All the locations and activities denied to refusers by the 'vaccine' passports will be included in one big mass ban on doing almost anything for those that don't bow their head to government. It's beyond fascist and a new term is required to describe its extremes – I guess fascist technocracy will have to do. The way the Chinese system of technological – technocratic – control is sweeping the West can be seen in the Los Angeles school system and is planned to be expanded worldwide. Every child is required to have a 'Covid'-tracking app scanned daily before they can enter the classroom. The so-called Daily Pass tracking system is produced by Gates' Microsoft which I'm sure will shock you rigid. The pass will be scanned using a barcode (one step from an inside-the-body barcode) and the information will include health checks, 'Covid' tests and vaccinations. Entry codes are for one specific building only and access will only be allowed if a student or teacher has a negative test with a test not testing for the 'virus', has no symptoms of anything alleged to be related to 'Covid' (symptoms from a range of other illness), and has a temperature under 100 degrees. No barcode, no entry, is planned to be the case for everywhere and not only schools.

Kids are being psychologically prepared to accept this as 'normal' their whole life which is why what they can impose in schools is so important to the Cult and its gofers. Long-time American freedom campaigner John Whitehead of the Rutherford Institute was not exaggerating when he said: 'Databit by databit, we are building our own electronic concentration camps.' Canada under its Cult gofer prime minister Justin Trudeau has taken a major step towards the real thing with people interned against their will if they test positive with a test not testing for the 'virus' when they arrive at a Canadian

airport. They are jailed in internment hotels often without food or water for long periods and with many doors failing to lock there have been sexual assaults. The interned are being charged sometimes \$2,000 for the privilege of being abused in this way. Trudeau is fully on board with the Cult and says the 'Covid pandemic' has provided an opportunity for a global 'reset' to permanently change Western civilisation. His number two, Deputy Prime Minister Chrystia Freeland, is a trustee of the World Economic Forum and a Rhodes Scholar. The Trudeau family have long been servants of the Cult. See *The Biggest Secret* and Cathy O'Brien's book *Trance-Formation of America* for the horrific background to Trudeau's father Pierre Trudeau another Canadian prime minister. Hide your fascism behind the façade of a heart-on-the-sleeve liberal. It's a well-honed Cult technique.

What can the 'vaccine' really do?

We have a 'virus' never shown to exist and 'variants' of the 'virus' that have also never been shown to exist except, like the 'original', as computer-generated fictions. Even if you believe there's a 'virus' the 'case' to 'death' rate is in the region of 0.23 to 0.15 percent and those 'deaths' are concentrated among the very old around the same average age that people die anyway. In response to this lack of threat (in truth none) psychopaths and idiots, knowingly and unknowingly answering to Gates and the Cult, are seeking to 'vaccinate' every man, woman and child on Planet Earth. Clearly the 'vaccine' is not about 'Covid' – none of this ever has been. So what is it all about *really*? Why the desperation to infuse genetically-manipulating synthetic material into everyone through mRNA fraudulent 'vaccines' with the intent of doing this over and over with the excuses of 'variants' and other 'virus' inventions? Dr Sherri Tenpenny, an osteopathic medical doctor in the United States, has made herself an expert on vaccines and their effects as a vehement campaigner against their use. Tenpenny was board certified in emergency medicine, the director of a level two trauma centre for 12 years, and moved to Cleveland in 1996 to start an integrative

medicine practice which has treated patients from all 50 states and some 17 other countries. Weaning people off pharmaceutical drugs is a speciality.

She became interested in the consequences of vaccines after attending a meeting at the National Vaccine Information Center in Washington DC in 2000 where she 'sat through four days of listening to medical doctors and scientists and lawyers and parents of vaccine injured kids' and asked: 'What's going on?' She had never been vaccinated and never got ill while her father was given a list of vaccines to be in the military and was 'sick his entire life'. The experience added to her questions and she began to examine vaccine documents from the Centers for Disease Control (CDC). After reading the first one, the 1998 version of *The General Recommendations of Vaccination*, she thought: 'This is it?' The document was poorly written and bad science and Tenpenny began 20 years of research into vaccines that continues to this day. She began her research into 'Covid vaccines' in March, 2020, and she describes them as 'deadly'. For many, as we have seen, they already have been. Tenpenny said that in the first 30 days of the 'vaccine' rollout in the United States there had been more than 40,000 adverse events reported to the vaccine adverse event database. A document had been delivered to her the day before that was 172 pages long. 'We have over 40,000 adverse events; we have over 3,100 cases of [potentially deadly] anaphylactic shock; we have over 5,000 neurological reactions.' Effects ranged from headaches to numbness, dizziness and vertigo, to losing feeling in hands or feet and paraesthesia which is when limbs 'fall asleep' and people have the sensation of insects crawling underneath their skin. All this happened in the first 30 days and remember that only about *ten percent* (or far less) of adverse reactions and vaccine-related deaths are estimated to be officially reported. Tenpenny said:

So can you think of one single product in any industry, any industry, for as long as products have been made on the planet that within 30 days we have 40,000 people complaining of side effects that not only is still on the market but ... we've got paid actors telling us how great

they are for getting their vaccine. We're offering people \$500 if they will just get their vaccine and we've got nurses and doctors going; 'I got the vaccine, I got the vaccine'.

Tenpenny said they were not going to be 'happy dancing folks' when they began to suffer Bell's palsy (facial paralysis), neuropathies, cardiac arrhythmias and autoimmune reactions that kill through a blood disorder. 'They're not going to be so happy, happy then, but we're never going to see pictures of those people' she said. Tenpenny described the 'vaccine' as 'a well-designed killing tool'.

No off-switch

Bad as the initial consequences had been Tenpenny said it would be maybe 14 months before we began to see the 'full ravage' of what is going to happen to the 'Covid vaccinated' with full-out consequences taking anything between two years and 20 years to show. You can understand why when you consider that variations of the 'Covid vaccine' use mRNA (messenger RNA) to in theory activate the immune system to produce protective antibodies without using the actual 'virus'. How can they when it's a computer program and they've never isolated what they claim is the 'real thing'? Instead they use *synthetic* mRNA. They are inoculating synthetic material into the body which through a technique known as the Trojan horse is absorbed into cells to change the nature of DNA. Human DNA is changed by an infusion of messenger RNA and with each new 'vaccine' of this type it is changed even more. Say so and you are banned by Cult Internet platforms. The contempt the contemptuous Mark Zuckerberg has for the truth and human health can be seen in an internal Facebook video leaked to the Project Veritas investigative team in which he said of the 'Covid vaccines': '... I share some caution on this because we just don't know the long term side-effects of basically modifying people's DNA and RNA.' At the same time this disgusting man's Facebook was censoring and banning anyone saying exactly the same. He must go before a Nuremberg trial for crimes against humanity when he *knows* that he

is censoring legitimate concerns and denying the right of informed consent on behalf of the Cult that owns him. People have been killed and damaged by the very 'vaccination' technique he cast doubt on himself when they may not have had the 'vaccine' with access to information that he denied them. The plan is to have at least annual 'Covid vaccinations', add others to deal with invented 'variants', and change all other vaccines into the mRNA system. Pfizer executives told shareholders at a virtual Barclays Global Healthcare Conference in March, 2021, that the public may need a third dose of 'Covid vaccine', plus regular yearly boosters and the company planned to hike prices to milk the profits in a 'significant opportunity for our vaccine'. These are the professional liars, cheats and opportunists who are telling you their 'vaccine' is safe. Given this volume of mRNA planned to be infused into the human body and its ability to then replicate we will have a transformation of human genetics from biological to synthetic biological – exactly the long-time Cult plan for reasons we'll see – and many will die. Sherri Tenpenny said of this replication:

It's like having an on-button but no off-button and that whole mechanism ... they actually give it a name and they call it the Trojan horse mechanism, because it allows that [synthetic] virus and that piece of that [synthetic] virus to get inside of your cells, start to replicate and even get inserted into other parts of your DNA as a Trojan-horse.

Ask the overwhelming majority of people who have the 'vaccine' what they know about the contents and what they do and they would reply: 'The government says it will stop me getting the virus.' Governments give that false impression on purpose to increase take-up. You can read Sherri Tenpenny's detailed analysis of the health consequences in her blog at [Vaxxter.com](https://vaxxter.com), but in summary these are some of them. She highlights the statement by Bill Gates about how human beings can become their own 'vaccine manufacturing machine'. The man is insane. ['Vaccine'-generated] 'antibodies' carry synthetic messenger RNA into the cells and the damage starts, Tenpenny contends, and she says that lungs can be adversely affected through varying degrees of pus and bleeding which

obviously affects breathing and would be dubbed 'Covid-19'. Even more sinister was the impact of 'antibodies' on macrophages, a white blood cell of the immune system. They consist of Type 1 and Type 2 which have very different functions. She said Type 1 are 'hyper-vigilant' white blood cells which 'gobble up' bacteria etc. However, in doing so, this could cause inflammation and in extreme circumstances be fatal. She says these affects are mitigated by Type 2 macrophages which kick in to calm down the system and stop it going rogue. They clear up dead tissue debris and reduce inflammation that the Type 1 'fire crews' have caused. Type 1 kills the infection and Type 2 heals the damage, she says. This is her punchline with regard to 'Covid vaccinations': She says that mRNA 'antibodies' block Type 2 macrophages by attaching to them and deactivating them. This meant that when the Type 1 response was triggered by infection there was nothing to stop that getting out of hand by calming everything down. There's an on-switch, but no off-switch, she says. What follows can be 'over and out, see you when I see you'.

Genetic suicide

Tenpenny also highlights the potential for autoimmune disease – the body attacking itself – which has been associated with vaccines since they first appeared. Infusing a synthetic foreign substance into cells could cause the immune system to react in a panic believing that the body is being overwhelmed by an invader (it is) and the consequences can again be fatal. There is an autoimmune response known as a 'cytokine storm' which I have likened to a homeowner panicked by an intruder and picking up a gun to shoot randomly in all directions before turning the fire on himself. The immune system unleashes a storm of inflammatory response called cytokines to a threat and the body commits hara-kiri. The lesson is that you mess with the body's immune response at your peril and these 'vaccines' seriously – fundamentally – mess with immune response. Tenpenny refers to a consequence called anaphylactic shock which is a severe and highly dangerous allergic reaction when the immune system

floods the body with chemicals. She gives the example of having a bee sting which primes the immune system and makes it sensitive to those chemicals. When people are stung again maybe years later the immune response can be so powerful that it leads to anaphylactic shock. Tenpenny relates this 'shock' with regard to the 'Covid vaccine' to something called polyethylene glycol or PEG. Enormous numbers of people have become sensitive to this over decades of use in a whole range of products and processes including food, drink, skin creams and 'medicine'. Studies have claimed that some 72 percent of people have antibodies triggered by PEG compared with two percent in the 1960s and allergic hypersensitive reactions to this become a gathering cause for concern. Tenpenny points out that the 'mRNA vaccine' is coated in a 'bubble' of polyethylene glycol which has the potential to cause anaphylactic shock through immune sensitivity. Many reports have appeared of people reacting this way after having the 'Covid vaccine'. What do we think is going to happen as humanity has more and more of these 'vaccines'? Tenpenny said: 'All these pictures we have seen with people with these rashes ... these weepy rashes, big reactions on their arms and things like that – it's an acute allergic reaction most likely to the polyethylene glycol that you've been previously primed and sensitised to.'

Those who have not studied the conspiracy and its perpetrators at length might think that making the population sensitive to PEG and then putting it in these 'vaccines' is just a coincidence. It is not. It is instead testament to how carefully and coldly-planned current events have been and the scale of the conspiracy we are dealing with. Tenpenny further explains that the 'vaccine' mRNA procedure can breach the blood-brain barrier which protects the brain from toxins and other crap that will cause malfunction. In this case they could make two proteins corrupt brain function to cause Amyotrophic lateral sclerosis (ALS), a progressive nervous system disease leading to loss of muscle control, and frontal lobe degeneration – Alzheimer's and dementia. Immunologist J. Bart Classon published a paper connecting mRNA 'vaccines' to prion

disease which can lead to Alzheimer's and other forms of neurogenerative disease while others have pointed out the potential to affect the placenta in ways that make women infertile. This will become highly significant in the next chapter when I will discuss other aspects of this non-vaccine that relate to its nanotechnology and transmission from the injected to the uninjected.

Qualified in idiocy

Tenpenny describes how research has confirmed that these 'vaccine'-generated antibodies can interact with a range of other tissues in the body and attack many other organs including the lungs. 'This means that if you have a hundred people standing in front of you that all got this shot they could have a hundred different symptoms.' Anyone really think that Cult gofers like the Queen, Tony Blair, Christopher Whitty, Anthony Fauci, and all the other psychopaths have really had this 'vaccine' in the pictures we've seen? Not a bloody chance. Why don't doctors all tell us about all these dangers and consequences of the 'Covid vaccine'? Why instead do they encourage and pressure patients to have the shot? Don't let's think for a moment that doctors and medical staff can't be stupid, lazy, and psychopathic and that's without the financial incentives to give the jab. Tenpenny again:

Some people are going to die from the vaccine directly but a large number of people are going to start to get horribly sick and get all kinds of autoimmune diseases 42 days to maybe a year out. What are they going to do, these stupid doctors who say; 'Good for you for getting that vaccine.' What are they going to say; 'Oh, it must be a mutant, we need to give an extra dose of that vaccine.'

Because now the vaccine, instead of one dose or two doses we need three or four because the stupid physicians aren't taking the time to learn anything about it. If I can learn this sitting in my living room reading a 19 page paper and several others so can they. There's nothing special about me, I just take the time to do it.

Remember how Sara Kayat, the NHS and TV doctor, said that the 'Covid vaccine' would '100 percent prevent hospitalisation and death'. Doctors can be idiots like every other profession and they

should not be worshipped as infallible. They are not and far from it. Behind many medical and scientific 'experts' lies an uninformed prat trying to hide themselves from you although in the 'Covid' era many have failed to do so as with UK narrative-repeating 'TV doctor' Hilary Jones. Pushing back against the minority of proper doctors and scientists speaking out against the 'vaccine' has been the entire edifice of the Cult global state in the form of governments, medical systems, corporations, mainstream media, Silicon Valley, and an army of compliant doctors, medical staff and scientists willing to say anything for money and to enhance their careers by promoting the party line. If you do that you are an 'expert' and if you won't you are an 'anti-vaxxer' and 'Covidiot'. The pressure to be 'vaccinated' is incessant. We have even had reports claiming that the 'vaccine' can help cure cancer and Alzheimer's and make the lame walk. I am waiting for the announcement that it can bring you coffee in the morning and cook your tea. Just as the symptoms of 'Covid' seem to increase by the week so have the miracles of the 'vaccine'. American supermarket giant Kroger Co. offered nearly 500,000 employees in 35 states a \$100 bonus for having the 'vaccine' while donut chain Krispy Kreme promised 'vaccinated' customers a free glazed donut every day for the rest of 2021. Have your DNA changed and you will get a doughnut although we might not have to give you them for long. Such offers and incentives confirm the desperation.

Perhaps the worse vaccine-stunt of them all was UK 'Health' Secretary Matt-the-prat Hancock on live TV after watching a clip of someone being 'vaccinated' when the roll-out began. Hancock faked tears so badly it was embarrassing. Brain-of-Britain Piers Morgan, the lockdown-supporting, 'vaccine' supporting, 'vaccine' passport-supporting, TV host played along with Hancock – 'You're quite emotional about that' he said in response to acting so atrocious it would have been called out at a school nativity which will presumably today include Mary and Jesus in masks, wise men keeping their camels six feet apart, and shepherds under tent arrest. System-serving Morgan tweeted this: 'Love the idea of covid vaccine passports for everywhere: flights, restaurants, clubs, football, gyms,

shops etc. It's time covid-denying, anti-vaxxer loonies had their bullsh*t bluff called & bar themselves from going anywhere that responsible citizens go.' If only I could aspire to his genius. To think that Morgan, who specialises in shouting over anyone he disagrees with, was lauded as a free speech hero when he lost his job after storming off the set of his live show like a child throwing his dolly out of the pram. If he is a free speech hero we are in real trouble. I have no idea what 'bullsh*t' means, by the way, the * throws me completely.

The Cult is desperate to infuse its synthetic DNA-changing concoction into everyone and has been using every lie, trick and intimidation to do so. The question of '*Why?*' we shall now address.

CHAPTER TEN

Human 2.0

I believe that at the end of the century the use of words and general educated opinion will have altered so much that one will be able to speak of machines thinking without expecting to be contradicted –
Alan Turing (1912-1954), the ‘Father of artificial intelligence’

I have been exposing for decades the plan to transform the human body from a biological to a synthetic-biological state. The new human that I will call Human 2.0 is planned to be connected to artificial intelligence and a global AI ‘Smart Grid’ that would operate as one global system in which AI would control everything from your fridge to your heating system to your car to your mind. Humans would no longer be ‘human’, but post-human and sub-human, with their thinking and emotional processes replaced by AI.

What I said sounded crazy and beyond science fiction and I could understand that. To any balanced, rational, mind it *is* crazy. Today, however, that world is becoming reality and it puts the ‘Covid vaccine’ into its true context. Ray Kurzweil is the ultra-Zionist ‘computer scientist, inventor and futurist’ and co-founder of the Singularity University. Singularity refers to the merging of humans with machines or ‘transhumanism’. Kurzweil has said humanity would be connected to the cyber ‘cloud’ in the period of the ever-recurring year of 2030:

Our thinking ... will be a hybrid of biological and non-biological thinking ... humans will be able to extend their limitations and ‘think in the cloud’ ... We’re going to put gateways to the

cloud in our brains ... We're going to gradually merge and enhance ourselves ... In my view, that's the nature of being human – we transcend our limitations. As the technology becomes vastly superior to what we are then the small proportion that is still human gets smaller and smaller and smaller until it's just utterly negligible.

They are trying to sell this end-of-humanity-as-we-know-it as the next stage of 'evolution' when we become super-human and 'like the gods'. They are lying to you. Shocked, eh? The population, and again especially the young, have been manipulated into addiction to technologies designed to enslave them for life. First they induced an addiction to smartphones (holdables); next they moved to technology on the body (wearables); and then began the invasion of the body (implantables). I warned way back about the plan for microchipped people and we are now entering that era. We should not be diverted into thinking that this refers only to chips we can see. Most important are the nanochips known as smart dust, neural dust and nanobots which are far too small to be seen by the human eye. Nanotechnology is everywhere, increasingly in food products, and released into the atmosphere by the geoengineering of the skies funded by Bill Gates to 'shut out the Sun' and 'save the planet from global warming'. Gates has been funding a project to spray millions of tonnes of chalk (calcium carbonate) into the stratosphere over Sweden to 'dim the Sun' and cool the Earth. Scientists warned the move could be disastrous for weather systems in ways no one can predict and opposition led to the Swedish space agency announcing that the 'experiment' would not be happening as planned in the summer of 2021; but it shows where the Cult is going with dimming the impact of the Sun and there's an associated plan to change the planet's atmosphere. Who gives psychopath Gates the right to dictate to the entire human race and dismantle planetary systems? The world will not be safe while this man is at large.

The global warming hoax has made the Sun, like the gas of life, something to fear when both are essential to good health and human survival (more inversion). The body transforms sunlight into vital vitamin D through a process involving ... *cholesterol*. This is the cholesterol we are also told to fear. We are urged to take Big Pharma

statin drugs to reduce cholesterol and it's all systematic. Reducing cholesterol means reducing vitamin D uptake with all the multiple health problems that will cause. At least if you take statins long term it saves the government from having to pay you a pension. The delivery system to block sunlight is widely referred to as chemtrails although these have a much deeper agenda, too. They appear at first to be contrails or condensation trails streaming from aircraft into cold air at high altitudes. Contrails disperse very quickly while chemtrails do not and spread out across the sky before eventually their content falls to earth. Many times I have watched aircraft cross-cross a clear blue sky releasing chemtrails until it looks like a cloudy day. Chemtrails contain many things harmful to humans and the natural world including toxic heavy metals, aluminium (see Alzheimer's) and nanotechnology. Ray Kurzweil reveals the reason without actually saying so: 'Nanobots will infuse all the matter around us with information. Rocks, trees, everything will become these intelligent creatures.' How do you deliver that? *From the sky*. Self-replicating nanobots would connect everything to the Smart Grid. The phenomenon of Morgellons disease began in the chemtrail era and the correlation has led to it being dubbed the 'chemtrail disease'. Self-replicating fibres appear in the body that can be pulled out through the skin. Morgellons fibres continue to grow outside the body and have a form of artificial intelligence. I cover this at greater length in *Phantom Self*.

'Vaccine' operating system

'Covid vaccines' with their self-replicating synthetic material are also designed to make the connection between humanity and Kurzweil's 'cloud'. American doctor and dedicated campaigner for truth, Carrie Madej, an Internal Medicine Specialist in Georgia with more than 20 years medical experience, has highlighted the nanotechnology aspect of the fake 'vaccines'. She explains how one of the components in at least the Moderna and Pfizer synthetic potions are 'lipid nanoparticles' which are 'like little tiny computer bits' – a 'sci-fi substance' known as nanobots and hydrogel which can be 'triggered

at any moment to deliver its payload' and act as 'biosensors'. The synthetic substance had 'the ability to accumulate data from your body like your breathing, your respiration, thoughts and emotions, all kind of things' and each syringe could carry a *million* nanobots:

This substance because it's like little bits of computers in your body, crazy, but it's true, it can do that, [and] obviously has the ability to act through Wi-Fi. It can receive and transmit energy, messages, frequencies or impulses. That issue has never been addressed by these companies. What does that do to the human?

Just imagine getting this substance in you and it can react to things all around you, the 5G, your smart device, your phones, what is happening with that? What if something is triggering it, too, like an impulse, a frequency? We have something completely foreign in the human body.

Madej said her research revealed that electromagnetic (EMF) frequencies emitted by phones and other devices had increased dramatically in the same period of the 'vaccine' rollout and she was seeing more people with radiation problems as 5G and other electromagnetic technology was expanded and introduced to schools and hospitals. She said she was 'floored with the EMF coming off' the devices she checked. All this makes total sense and syncs with my own work of decades when you think that Moderna refers in documents to its mRNA 'vaccine' as an 'operating system':

Recognizing the broad potential of mRNA science, we set out to create an mRNA technology platform that functions very much like an operating system on a computer. It is designed so that it can plug and play interchangeably with different programs. In our case, the 'program' or 'app' is our mRNA drug – the unique mRNA sequence that codes for a protein ...

... Our MRNA Medicines – 'The 'Software Of Life': When we have a concept for a new mRNA medicine and begin research, fundamental components are already in place. Generally, the only thing that changes from one potential mRNA medicine to another is the coding region – the actual genetic code that instructs ribosomes to make protein. Utilizing these instruction sets gives our investigational mRNA medicines a software-like quality. We also have the ability to combine different mRNA sequences encoding for different proteins in a single mRNA investigational medicine.

Who needs a real ‘virus’ when you can create a computer version to justify infusing your operating system into the entire human race on the road to making living, breathing people into cyborgs? What is missed with the ‘vaccines’ is the *digital* connection between synthetic material and the body that I highlighted earlier with the study that hacked a computer with human DNA. On one level the body is digital, based on mathematical codes, and I’ll have more about that in the next chapter. Those who ridiculously claim that mRNA ‘vaccines’ are not designed to change human genetics should explain the words of Dr Tal Zaks, chief medical officer at Moderna, in a 2017 TED talk. He said that over the last 30 years ‘we’ve been living this phenomenal digital scientific revolution, and I’m here today to tell you, that we are actually *hacking the software of life*, and that it’s changing the way we think about prevention and treatment of disease’:

In every cell there’s this thing called messenger RNA, or mRNA for short, that transmits the critical information from the DNA in our genes to the protein, which is really the stuff we’re all made out of. This is the critical information that determines what the cell will do. So we think about it as an operating system. So if you could change that, if you could introduce a line of code, or change a line of code, it turns out, that has profound implications for everything, from the flu to cancer.

Zaks should more accurately have said that this has profound implications for the human genetic code and the nature of DNA. Communications within the body go both ways and not only one. But, hey, no, the ‘Covid vaccine’ will not affect your genetics. Cult fact-checkers say so even though the man who helped to develop the mRNA technique says that it does. Zaks said in 2017:

If you think about what it is we’re trying to do. We’ve taken information and our understanding of that information and how that information is transmitted in a cell, and we’ve taken our understanding of medicine and how to make drugs, and we’re fusing the two. We think of it as information therapy.

I have been writing for decades that the body is an information field communicating with itself and the wider world. This is why

radiation which is information can change the information field of body and mind through phenomena like 5G and change their nature and function. 'Information therapy' means to change the body's information field and change the way it operates. DNA is a receiver-transmitter of information and can be mutated by information like mRNA synthetic messaging. Technology to do this has been ready and waiting in the underground bases and other secret projects to be rolled out when the 'Covid' hoax was played. 'Trials' of such short and irrelevant duration were only for public consumption. When they say the 'vaccine' is 'experimental' that is not true. It may appear to be 'experimental' to those who don't know what's going on, but the trials have already been done to ensure the Cult gets the result it desires. Zaks said that it took decades to sequence the human genome, completed in 2003, but now they could do it in a week. By 'they' he means scientists operating in the public domain. In the secret projects they were sequencing the genome in a week long before even 2003.

Deluge of mRNA

Highly significantly the Moderna document says the guiding premise is that if using mRNA as a medicine works for one disease then it should work for many diseases. They were leveraging the flexibility afforded by their platform and the fundamental role mRNA plays in protein synthesis to pursue mRNA medicines for a broad spectrum of diseases. Moderna is confirming what I was saying through 2020 that multiple 'vaccines' were planned for 'Covid' (and later invented 'variants') and that previous vaccines would be converted to the mRNA system to infuse the body with massive amounts of genetically-manipulating synthetic material to secure a transformation to a synthetic-biological state. The 'vaccines' are designed to kill stunning numbers as part of the long-exposed Cult depopulation agenda and transform the rest. Given this is the goal you can appreciate why there is such hysterical demand for every human to be 'vaccinated' for an alleged 'disease' that has an estimated 'infection' to 'death' ratio of 0.23-0.15 percent. As I write

children are being given the 'vaccine' in trials (their parents are a disgrace) and ever-younger people are being offered the vaccine for a 'virus' that even if you believe it exists has virtually zero chance of harming them. Horrific effects of the 'trials' on a 12-year-old girl were revealed by a family member to be serious brain and gastric problems that included a bowel obstruction and the inability to swallow liquids or solids. She was unable to eat or drink without throwing up, had extreme pain in her back, neck and abdomen, and was paralysed from the waist down which stopped her urinating unaided. When the girl was first taken to hospital doctors said it was all in her mind. She was signed up for the 'trial' by her parents for whom no words suffice. None of this 'Covid vaccine' insanity makes any sense unless you see what the 'vaccine' really is – a body-changer. Synthetic biology or 'SynBio' is a fast-emerging and expanding scientific discipline which includes everything from genetic and molecular engineering to electrical and computer engineering. Synthetic biology is defined in these ways:

- A multidisciplinary area of research that seeks to create new biological parts, devices, and systems, or to redesign systems that are already found in nature.
- The use of a mixture of physical engineering and genetic engineering to create new (and therefore synthetic) life forms.
- An emerging field of research that aims to combine the knowledge and methods of biology, engineering and related disciplines in the design of chemically-synthesized DNA to create organisms with novel or enhanced characteristics and traits (synthetic organisms including humans).

We now have synthetic blood, skin, organs and limbs being developed along with synthetic body parts produced by 3D printers. These are all elements of the synthetic human programme and this comment by Kurzweil's co-founder of the Singularity University,

Peter Diamandis, can be seen in a whole new light with the 'Covid' hoax and the sanctions against those that refuse the 'vaccine':

Anybody who is going to be resisting the progress forward [to transhumanism] is going to be resisting evolution and, fundamentally, they will die out. It's not a matter of whether it's good or bad. It's going to happen.

'Resisting evolution'? What absolute bollocks. The arrogance of these people is without limit. His 'it's going to happen' mantra is another way of saying 'resistance is futile' to break the spirit of those pushing back and we must not fall for it. Getting this genetically-transforming 'vaccine' into everyone is crucial to the Cult plan for total control and the desperation to achieve that is clear for anyone to see. Vaccine passports are a major factor in this and they, too, are a form of resistance is futile. It's NOT. The paper funded by the Rockefeller Foundation for the 2013 'health conference' in China said:

We will interact more with artificial intelligence. The use of robotics, bio-engineering to augment human functioning is already well underway and will advance. Re-engineering of humans into potentially separate and unequal forms through genetic engineering or mixed human-robots raises debates on ethics and equality.

A new demography is projected to emerge after 2030 [that year again] of technologies (robotics, genetic engineering, nanotechnology) producing robots, engineered organisms, 'nanobots' and artificial intelligence (AI) that can self-replicate. Debates will grow on the implications of an impending reality of human designed life.

What is happening today is so long planned. The world army enforcing the will of the world government is intended to be a robot army, not a human one. Today's military and its technologically 'enhanced' troops, pilotless planes and driverless vehicles are just stepping stones to that end. Human soldiers are used as Cult fodder and its time they woke up to that and worked for the freedom of the population instead of their own destruction and their family's destruction – the same with the police. Join us and let's sort this out. The phenomenon of enforce my own destruction is widespread in the 'Covid' era with Woker 'luvvies' in the acting and entertainment

industries supporting 'Covid' rules which have destroyed their profession and the same with those among the public who put signs on the doors of their businesses 'closed due to Covid – stay safe' when many will never reopen. It's a form of masochism and most certainly insanity.

Transgender = transhumanism

When something explodes out of nowhere and is suddenly everywhere it is always the Cult agenda and so it is with the tidal wave of claims and demands that have infiltrated every aspect of society under the heading of 'transgenderism'. The term 'trans' is so 'in' and this is the dictionary definition:

A prefix meaning 'across', 'through', occurring ... in loanwords from Latin, used in particular for denoting movement or conveyance from place to place (transfer; transmit; transplant) or complete change (transform; transmute), or to form adjectives meaning 'crossing', 'on the other side of', or 'going beyond' the place named (transmontane; transnational; trans-Siberian).

Transgender means to go beyond gender and transhuman means to go beyond human. Both are aspects of the Cult plan to transform the human body to a synthetic state with *no gender*. Human 2.0 is not designed to procreate and would be produced technologically with no need for parents. The new human would mean the end of parents and so men, and increasingly women, are being targeted for the deletion of their rights and status. Parental rights are disappearing at an ever-quickenning speed for the same reason. The new human would have no need for men or women when there is no procreation and no gender. Perhaps the transgender movement that appears to be in a permanent state of frenzy might now contemplate on how it is being used. This was never about transgender rights which are only the interim excuse for confusing gender, particularly in the young, on the road to *fusing* gender. Transgender activism is not an end; it is a *means* to an end. We see again the technique of creative destruction in which you destroy the status quo to 'build back better' in the form that you want. The gender status quo had to be

destroyed by persuading the Cult-created Woke mentality to believe that you can have 100 genders or more. A programme for 9 to 12 year olds produced by the Cult-owned BBC promoted the 100 genders narrative. The very idea may be the most monumental nonsense, but it is not what is true that counts, only what you can make people *believe* is true. Once the gender of $2 + 2 = 4$ has been dismantled through indoctrination, intimidation and $2 + 2 = 5$ then the new no-gender normal can take its place with Human 2.0.

Aldous Huxley revealed the plan in his prophetic *Brave New World* in 1932:

Natural reproduction has been done away with and children are created, decanted', and raised in 'hatcheries and conditioning centres'. From birth, people are genetically designed to fit into one of five castes, which are further split into 'Plus' and 'Minus' members and designed to fulfil predetermined positions within the social and economic strata of the World State.

How could Huxley know this in 1932? For the same reason George Orwell knew about the Big Brother state in 1948, Cult insiders I have quoted knew about it in 1969, and I have known about it since the early 1990s. If you are connected to the Cult or you work your balls off to uncover the plan you can predict the future. The process is simple. If there is a plan for the world and nothing intervenes to stop it then it will happen. Thus if you communicate the plan ahead of time you are perceived to have predicted the future, but you haven't. You have revealed the plan which without intervention will become the human future. The whole reason I have done what I have is to alert enough people to inspire an intervention and maybe at last that time has come with the Cult and its intentions now so obvious to anyone with a brain in working order.

The future is here

Technological wombs that Huxley described to replace parent procreation are already being developed and they are only the projects we know about in the public arena. Israeli scientists told *The Times of Israel* in March, 2021, that they have grown 250-cell embryos

into mouse fetuses with fully formed organs using artificial wombs in a development they say could pave the way for gestating humans outside the womb. Professor Jacob Hanna of the Weizmann Institute of Science said:

We took mouse embryos from the mother at day five of development, when they are just of 250 cells, and had them in the incubator from day five until day 11, by which point they had grown all their organs.

By day 11 they make their own blood and have a beating heart, a fully developed brain. Anybody would look at them and say, 'this is clearly a mouse foetus with all the characteristics of a mouse.' It's gone from being a ball of cells to being an advanced foetus.

A special liquid is used to nourish embryo cells in a laboratory dish and they float on the liquid to duplicate the first stage of embryonic development. The incubator creates all the right conditions for its development, Hanna said. The liquid gives the embryo 'all the nutrients, hormones and sugars they need' along with a custom-made electronic incubator which controls gas concentration, pressure and temperature. The cutting-edge in the underground bases and other secret locations will be light years ahead of that, however, and this was reported by the London *Guardian* in 2017:

We are approaching a biotechnological breakthrough. Ectogenesis, the invention of a complete external womb, could completely change the nature of human reproduction. In April this year, researchers at the Children's Hospital of Philadelphia announced their development of an artificial womb.

The article was headed 'Artificial wombs could soon be a reality. What will this mean for women?' What would it mean for children is an even bigger question. No mother to bond with only a machine in preparation for a life of soulless interaction and control in a world governed by machines (see the *Matrix* movies). Now observe the calculated manipulations of the 'Covid' hoax as human interaction and warmth has been curtailed by distancing, isolation and fear with people communicating via machines on a scale never seen before.

These are all dots in the same picture as are all the personal assistants, gadgets and children's toys through which kids and adults communicate with AI as if it is human. The AI 'voice' on Sat-Nav should be included. All these things are psychological preparation for the Cult endgame. Before you can make a physical connection with AI you have to make a psychological connection and that is what people are being conditioned to do with this ever gathering human-AI interaction. Movies and TV programmes depicting the transhuman, robot dystopia relate to a phenomenon known as 'pre-emptive programming' in which the world that is planned is portrayed everywhere in movies, TV and advertising. This is conditioning the conscious and subconscious mind to become familiar with the planned reality to dilute resistance when it happens for real. What would have been a shock such is the change is made less so. We have young children put on the road to transgender transition surgery with puberty blocking drugs at an age when they could never be able to make those life-changing decisions.

Rachel Levine, a professor of paediatrics and psychiatry who believes in treating children this way, became America's highest-ranked openly-transgender official when she was confirmed as US Assistant Secretary at the Department of Health and Human Services after being nominated by Joe Biden (the Cult). Activists and governments press for laws to deny parents a say in their children's transition process so the kids can be isolated and manipulated into agreeing to irreversible medical procedures. A Canadian father Robert Hoogland was denied bail by the Vancouver Supreme Court in 2021 and remained in jail for breaching a court order that he stay silent over his young teenage daughter, a minor, who was being offered life-changing hormone therapy without parental consent. At the age of 12 the girl's 'school counsellor' said she may be transgender, referred her to a doctor and told the school to treat her like a boy. This is another example of state-serving schools imposing ever more control over children's lives while parents have ever less.

Contemptible and extreme child abuse is happening all over the world as the Cult gender-fusion operation goes into warp-speed.

Why the war on men – and now women?

The question about what artificial wombs mean for women should rightly be asked. The answer can be seen in the deletion of women's rights involving sport, changing rooms, toilets and status in favour of people in male bodies claiming to identify as women. I can identify as a mountain climber, but it doesn't mean I can climb a mountain any more than a biological man can be a biological woman. To believe so is a triumph of belief over factual reality which is the very perceptual basis of everything Woke. Women's sport is being destroyed by allowing those with male bodies who say they identify as female to 'compete' with girls and women. Male body 'women' dominate 'women's' competition with their greater muscle mass, bone density, strength and speed. With that disadvantage sport for women loses all meaning. To put this in perspective nearly 300 American high school boys can run faster than the quickest woman sprinter in the world. Women are seeing their previously protected spaces invaded by male bodies simply because they claim to identify as women. That's all they need to do to access all women's spaces and activities under the Biden 'Equality Act' that destroys equality for women with the usual Orwellian Woke inversion. Male sex offenders have already committed rapes in women's prisons after claiming to identify as women to get them transferred. Does this not matter to the Woke 'equality' hypocrites? Not in the least. What matters to Cult manipulators and funders behind transgender activists is to advance gender fusion on the way to the no-gender 'human'. When you are seeking to impose transparent nonsense like this, or the 'Covid' hoax, the only way the nonsense can prevail is through censorship and intimidation of dissenters, deletion of factual information, and programming of the unquestioning, bewildered and naive. You don't have to scan the world for long to see that all these things are happening.

Many women's rights organisations have realised that rights and status which took such a long time to secure are being eroded and that it is systematic. Kara Dansky of the global Women's Human Rights Campaign said that Biden's transgender executive order immediately he took office, subsequent orders, and Equality Act legislation that followed 'seek to erase women and girls in the law as a category'. *Exactly*. I said during the long ago-started war on men (in which many women play a crucial part) that this was going to turn into a war on them. The Cult is phasing out *both* male and female genders. To get away with that they are brought into conflict so they are busy fighting each other while the Cult completes the job with no unity of response. Unity, people, *unity*. We need unity everywhere. Transgender is the only show in town as the big step towards the no-gender human. It's not about rights for transgender people and never has been. Woke political correctness is deleting words relating to genders to the same end. Wokers believe this is to be 'inclusive' when the opposite is true. They are deleting words describing gender because gender *itself* is being deleted by Human 2.0. Terms like 'man', 'woman', 'mother' and 'father' are being deleted in the universities and other institutions to be replaced by the *no*-gender, not trans-gender, 'individuals' and 'guardians'. Women's rights campaigner Maria Keffler of Partners for Ethical Care said: 'Children are being taught from kindergarten upward that some boys have a vagina, some girls have a penis, and that kids can be any gender they want to be.' Do we really believe that suddenly countries all over the world at the same time had the idea of having drag queens go into schools or read transgender stories to very young children in the local library? It's coldly-calculated confusion of gender on the way to the fusion of gender. Suzanne Vierling, a psychologist from Southern California, made another important point:

Yesterday's slave woman who endured gynecological medical experiments is today's girl-child being butchered in a booming gender-transitioning sector. Ovaries removed, pushing her into menopause and osteoporosis, uncharted territory, and parents' rights and authority decimated.

The erosion of parental rights is a common theme in line with the Cult plans to erase the very concept of parents and 'ovaries removed, pushing her into menopause' means what? Those born female lose the ability to have children – another way to discontinue humanity as we know it.

Eliminating Human 1.0 (before our very eyes)

To pave the way for Human 2.0 you must phase out Human 1.0. This is happening through plummeting sperm counts and making women infertile through an onslaught of chemicals, radiation (including smartphones in pockets of men) and mRNA 'vaccines'. Common agriculture pesticides are also having a devastating impact on human fertility. I have been tracking collapsing sperm counts in the books for a long time and in 2021 came a book by fertility scientist and reproductive epidemiologist Shanna Swan, *Count Down: How Our Modern World Is Threatening Sperm Counts, Altering Male and Female Reproductive Development and Imperiling the Future of the Human Race*. She reports how the global fertility rate dropped by *half* between 1960 and 2016 with America's birth rate 16 percent below where it needs to be to sustain the population. Women are experiencing declining egg quality, more miscarriages, and more couples suffer from infertility. Other findings were an increase in erectile dysfunction, infant boys developing more genital abnormalities, male problems with conception, and plunging levels of the male hormone testosterone which would explain why so many men have lost their backbone and masculinity. This has been very evident during the 'Covid' hoax when women have been prominent among the Pushbackers and big strapping blokes have bowed their heads, covered their faces with a nappy and quietly submitted. Mind control expert Cathy O'Brien also points to how global education introduced the concept of 'we're all winners' in sport and classrooms: 'Competition was defused, and it in turn defused a sense of fighting back.' This is another version of the 'equity' doctrine in which you drive down rather than raise up. What a contrast in Cult-controlled China with its global ambitions

where the government published plans in January, 2021, to 'cultivate masculinity' in boys from kindergarten through to high school in the face of a 'masculinity crisis'. A government adviser said boys would be soon become 'delicate, timid and effeminate' unless action was taken. Don't expect any similar policy in the targeted West. A 2006 study showed that a 65-year-old man in 2002 had testosterone levels 15 percent lower than a 65-year-old man in 1987 while a 2020 study found a similar story with young adults and adolescents. Men are getting prescriptions for testosterone replacement therapy which causes an even greater drop in sperm count with up to 99 percent seeing sperm counts drop to zero during the treatment. More sperm is defective and malfunctioning with some having two heads or not pursuing an egg.

A class of *synthetic* chemicals known as phthalates are being blamed for the decline. These are found everywhere in plastics, shampoos, cosmetics, furniture, flame retardants, personal care products, pesticides, canned foods and even receipts. Why till receipts? Everyone touches them. Let no one delude themselves that all this is not systematic to advance the long-time agenda for human body transformation. Phthalates mimic hormones and disrupt the hormone balance causing testosterone to fall and genital birth defects in male infants. Animals and fish have been affected in the same way due to phthalates and other toxins in rivers. When fish turn gay or change sex through chemicals in rivers and streams it is a pointer to why there has been such an increase in gay people and the sexually confused. It doesn't matter to me what sexuality people choose to be, but if it's being affected by chemical pollution and consumption then we need to know. Does anyone really think that this is not connected to the transgender agenda, the war on men and the condemnation of male 'toxic masculinity'? You watch this being followed by 'toxic femininity'. It's already happening. When breastfeeding becomes 'chest-feeding', pregnant women become pregnant people along with all the other Woke claptrap you know that the world is going insane and there's a Cult scam in progress. Transgender activists are promoting the Cult agenda while Cult

billionaires support and fund the insanity as they laugh themselves to sleep at the sheer stupidity for which humans must be infamous in galaxies far, far away.

‘Covid vaccines’ and female infertility

We can now see why the ‘vaccine’ has been connected to potential infertility in women. Dr Michael Yeadon, former Vice President and Chief Scientific Advisor at Pfizer, and Dr Wolfgang Wodarg in Germany, filed a petition with the European Medicines Agency in December, 2020, urging them to stop trials for the Pfizer/BioNTech shot and all other mRNA trials until further studies had been done. They were particularly concerned about possible effects on fertility with ‘vaccine’-produced antibodies attacking the protein Syncytin-1 which is responsible for developing the placenta. The result would be infertility ‘of indefinite duration’ in women who have the ‘vaccine’ with the placenta failing to form. Section 10.4.2 of the Pfizer/BioNTech trial protocol says that pregnant women or those who might become so should not have mRNA shots. Section 10.4 warns men taking mRNA shots to ‘be abstinent from heterosexual intercourse’ and not to donate sperm. The UK government said that it *did not know* if the mRNA procedure had an effect on fertility. *Did not know?* These people have to go to jail. UK government advice did not recommend at the start that pregnant women had the shot and said they should avoid pregnancy for at least two months after ‘vaccination’. The ‘advice’ was later updated to pregnant women should only have the ‘vaccine’ if the benefits outweighed the risks to mother and foetus. What the hell is that supposed to mean? Then ‘spontaneous abortions’ began to appear and rapidly increase on the adverse reaction reporting schemes which include only a fraction of adverse reactions. Thousands and ever-growing numbers of ‘vaccinated’ women are describing changes to their menstrual cycle with heavier blood flow, irregular periods and menstruating again after going through the menopause – all links to reproduction effects. Women are passing blood clots and the lining of their uterus while men report erectile dysfunction and blood effects. Most

significantly of all *unvaccinated* women began to report similar menstrual changes after interaction with '*vaccinated*' people and men and children were also affected with bleeding noses, blood clots and other conditions. 'Shedding' is when vaccinated people can emit the content of a vaccine to affect the unvaccinated, but this is different. 'Vaccinated' people were not shedding a 'live virus' allegedly in 'vaccines' as before because the fake 'Covid vaccines' involve synthetic material and other toxicity. Doctors exposing what is happening prefer the term 'transmission' to shedding. Somehow those that have had the shots are transmitting effects to those that haven't. Dr Carrie Madej said the nano-content of the 'vaccines' can 'act like an antenna' to others around them which fits perfectly with my own conclusions. This 'vaccine' transmission phenomenon was becoming known as the book went into production and I deal with this further in the Postscript.

Vaccine effects on sterility are well known. The World Health Organization was accused in 2014 of sterilising millions of women in Kenya with the evidence confirmed by the content of the vaccines involved. The same WHO behind the 'Covid' hoax admitted its involvement for more than ten years with the vaccine programme. Other countries made similar claims. Charges were lodged by Tanzania, Nicaragua, Mexico, and the Philippines. The Gardasil vaccine claimed to protect against a genital 'virus' known as HPV has also been linked to infertility. Big Pharma and the WHO (same thing) are criminal and satanic entities. Then there's the Bill Gates Foundation which is connected through funding and shared interests with 20 pharmaceutical giants and laboratories. He stands accused of directing the policy of United Nations Children's Fund (UNICEF), vaccine alliance GAVI, and other groupings, to advance the vaccine agenda and silence opposition at great cost to women and children. At the same time Gates wants to reduce the global population. Coincidence?

Great Reset = Smart Grid = new human

The Cult agenda I have been exposing for 30 years is now being openly promoted by Cult assets like Gates and Klaus Schwab of the World Economic Forum under code-terms like the 'Great Reset', 'Build Back Better' and 'a rare but narrow window of opportunity to reflect, reimagine, and reset our world'. What provided this 'rare but narrow window of opportunity'? The 'Covid' hoax did. Who created that? *They* did. My books from not that long ago warned about the planned 'Internet of Things' (IoT) and its implications for human freedom. This was the plan to connect all technology to the Internet and artificial intelligence and today we are way down that road with an estimated 36 billion devices connected to the World Wide Web and that figure is projected to be 76 billion by 2025. I further warned that the Cult planned to go beyond that to the Internet of *Everything* when the human brain was connected via AI to the Internet and Kurzweil's 'cloud'. Now we have Cult operatives like Schwab calling for precisely that under the term 'Internet of Bodies', a fusion of the physical, digital and biological into one centrally-controlled Smart Grid system which the Cult refers to as the 'Fourth Industrial Revolution'. They talk about the 'biological', but they really mean the synthetic-biological which is required to fully integrate the human body and brain into the Smart Grid and artificial intelligence planned to replace the human mind. We have everything being synthetically manipulated including the natural world through GMO and smart dust, the food we eat and the human body itself with synthetic 'vaccines'. I said in *The Answer* that we would see the Cult push for synthetic meat to replace animals and in February, 2021, the so predictable psychopath Bill Gates called for the introduction of synthetic meat to save us all from 'climate change'. The climate hoax just keeps on giving like the 'Covid' hoax. The war on meat by vegan activists is a carbon (oops, sorry) copy of the manipulation of transgender activists. They have no idea (except their inner core) that they are being used to promote and impose the agenda of the Cult or that they are only the *vehicle* and not the *reason*. This is not to say those who choose not to eat meat shouldn't be respected and supported in that right, but there are ulterior motives

for those in power. A *Forbes* article in December, 2019, highlighted the plan so beloved of Schwab and the Cult under the heading: 'What Is The Internet of Bodies? And How Is It Changing Our World?' The article said the human body is the latest data platform (remember 'our vaccine is an operating system'). *Forbes* described the plan very accurately and the words could have come straight out of my books from long before:

The Internet of Bodies (IoB) is an extension of the IoT and basically connects the human body to a network through devices that are ingested, implanted, or connected to the body in some way. Once connected, data can be exchanged, and the body and device can be remotely monitored and controlled.

They were really describing a human hive mind with human perception centrally-dictated via an AI connection as well as allowing people to be 'remotely monitored and controlled'. Everything from a fridge to a human mind could be directed from a central point by these insane psychopaths and 'Covid vaccines' are crucial to this. *Forbes* explained the process I mentioned earlier of holdable and wearable technology followed by implantable. The article said there were three generations of the Internet of Bodies that include:

- Body external: These are wearable devices such as Apple Watches or Fitbits that can monitor our health.
- Body internal: These include pacemakers, cochlear implants, and digital pills that go inside our bodies to monitor or control various aspects of health.
- Body embedded: The third generation of the Internet of Bodies is embedded technology where technology and the human body are melded together and have a real-time connection to a remote machine.

Forbes noted the development of the Brain Computer Interface (BCI) which merges the brain with an external device for monitoring and controlling in real-time. 'The ultimate goal is to help restore function to individuals with disabilities by using brain signals rather than conventional neuromuscular pathways.' Oh, do fuck off. The goal of brain interface technology is controlling human thought and emotion from the central point in a hive mind serving its masters wishes. Many people are now agreeing to be chipped to open doors without a key. You can recognise them because they'll be wearing a mask, social distancing and lining up for the 'vaccine'. The Cult plans a Great Reset money system after they have completed the demolition of the global economy in which 'money' will be exchanged through communication with body operating systems. Rand Corporation, a Cult-owned think tank, said of the Internet of Bodies or IoB:

Internet of Bodies technologies fall under the broader IoT umbrella. But as the name suggests, IoB devices introduce an even more intimate interplay between humans and gadgets. IoB devices monitor the human body, collect health metrics and other personal information, and transmit those data over the Internet. Many devices, such as fitness trackers, are already in use ... IoB devices ... and those in development can track, record, and store users' whereabouts, bodily functions, and what they see, hear, and even think.

Schwab's World Economic Forum, a long-winded way of saying 'fascism' or 'the Cult', has gone full-on with the Internet of Bodies in the 'Covid' era. 'We're entering the era of the Internet of Bodies', it declared, 'collecting our physical data via a range of devices that can be implanted, swallowed or worn'. The result would be a huge amount of health-related data that could improve human wellbeing around the world, and prove crucial in fighting the 'Covid-19 pandemic'. Does anyone think these clowns care about 'human wellbeing' after the death and devastation their pandemic hoax has purposely caused? Schwab and co say we should move forward with the Internet of Bodies because 'Keeping track of symptoms could help us stop the spread of infection, and quickly detect new cases'. How wonderful, but keeping track' is all they are really bothered

about. Researchers were investigating if data gathered from smartwatches and similar devices could be used as viral infection alerts by tracking the user's heart rate and breathing. Schwab said in his 2018 book *Shaping the Future of the Fourth Industrial Revolution*:

The lines between technologies and beings are becoming blurred and not just by the ability to create lifelike robots or synthetics. Instead it is about the ability of new technologies to literally become part of us. Technologies already influence how we understand ourselves, how we think about each other, and how we determine our realities. As the technologies ... give us deeper access to parts of ourselves, we may begin to integrate digital technologies into our bodies.

You can see what the game is. Twenty-four hour control and people – if you could still call them that – would never know when something would go ping and take them out of circulation. It's the most obvious rush to a global fascist dictatorship and the complete submission of humanity and yet still so many are locked away in their Cult-induced perceptual coma and can't see it.

Smart Grid control centres

The human body is being transformed by the 'vaccines' and in other ways into a synthetic cyborg that can be attached to the global Smart Grid which would be controlled from a central point and other sub-locations of Grid manipulation. Where are these planned to be? Well, China for a start which is one of the Cult's biggest centres of operation. The technological control system and technocratic rule was incubated here to be unleashed across the world after the 'Covid' hoax came out of China in 2020. Another Smart Grid location that will surprise people new to this is Israel. I have exposed in *The Trigger* how Sabbatian technocrats, intelligence and military operatives were behind the horrors of 9/11 and not 19 Arab hijackers' who somehow manifested the ability to pilot big passenger airliners when instructors at puddle-jumping flying schools described some of them as a joke. The 9/11 attacks were made possible through control of civilian and military air computer systems and those of the White House, Pentagon and connected agencies. See *The Trigger* – it

will blow your mind. The controlling and coordinating force were the Sabbatian networks in Israel and the United States which by then had infiltrated the entire US government, military and intelligence system. The real name of the American Deep State is 'Sabbatian State'. Israel is a tiny country of only nine million people, but it is one of the global centres of cyber operations and fast catching Silicon Valley in importance to the Cult. Israel is known as the 'start-up nation' for all the cyber companies spawned there with the Sabbatian specialisation of 'cyber security' that I mentioned earlier which gives those companies access to computer systems of their clients in real time through 'backdoors' written into the coding when security software is downloaded. The Sabbatian centre of cyber operations outside Silicon Valley is the Israeli military Cyber Intelligence Unit, the biggest infrastructure project in Israel's history, headquartered in the desert-city of Beersheba and involving some 20,000 'cyber soldiers'. Here are located a literal army of Internet trolls scanning social media, forums and comment lists for anyone challenging the Cult agenda. The UK military has something similar with its 77th Brigade and associated operations. The Beersheba complex includes research and development centres for other Cult operations such as Intel, Microsoft, IBM, Google, Apple, Hewlett-Packard, Cisco Systems, Facebook and Motorola. Techcrunch.com ran an article about the Beersheba global Internet technology centre headlined 'Israel's desert city of Beersheba is turning into a cybertech oasis':

The military's massive relocation of its prestigious technology units, the presence of multinational and local companies, a close proximity to Ben Gurion University and generous government subsidies are turning Beersheba into a major global cybertech hub. Beersheba has all of the ingredients of a vibrant security technology ecosystem, including Ben Gurion University with its graduate program in cybersecurity and Cyber Security Research Center, and the presence of companies such as EMC, Deutsche Telekom, PayPal, Oracle, IBM, and Lockheed Martin. It's also the future home of the INCB (Israeli National Cyber Bureau); offers a special income tax incentive for cyber security companies, and was the site for the relocation of the army's intelligence corps units.

Sabbatians have taken over the cyber world through the following process: They scan the schools for likely cyber talent and develop them at Ben Gurion University and their period of conscription in the Israeli Defense Forces when they are stationed at the Beersheba complex. When the cyber talented officially leave the army they are funded to start cyber companies with technology developed by themselves or given to them by the state. Much of this is stolen through backdoors of computer systems around the world with America top of the list. Others are sent off to Silicon Valley to start companies or join the major ones and so we have many major positions filled by apparently 'Jewish' but really Sabbatian operatives. Google, YouTube and Facebook are all run by 'Jewish' CEOs while Twitter is all but run by ultra-Zionist hedge-fund shark Paul Singer. At the centre of the Sabbatian global cyber web is the Israeli army's Unit 8200 which specialises in hacking into computer systems of other countries, inserting viruses, gathering information, instigating malfunction, and even taking control of them from a distance. A long list of Sabbatians involved with 9/11, Silicon Valley and Israeli cyber security companies are operatives of Unit 8200. This is not about Israel. It's about the Cult. Israel is planned to be a Smart Grid hub as with China and what is happening at Beersheba is not for the benefit of Jewish people who are treated disgustingly by the Sabbatian elite that control the country. A glance at the Nuremberg Codes will tell you that.

The story is much bigger than 'Covid', important as that is to where we are being taken. Now, though, it's time to really strap in. There's more ... much more ...

CHAPTER ELEVEN

Who controls the Cult?

Awake, arise or be forever fall'n
John Milton, *Paradise Lost*

I have exposed this far the level of the Cult conspiracy that operates in the world of the seen and within the global secret society and satanic network which operates in the shadows one step back from the seen. The story, however, goes much deeper than that.

The 'Covid' hoax is major part of the Cult agenda, but only part, and to grasp the biggest picture we have to expand our attention beyond the realm of human sight and into the infinity of possibility that we cannot see. It is from here, ultimately, that humanity is being manipulated into a state of total control by the force which dictates the actions of the Cult. How much of reality can we see? Next to damn all is the answer. We may appear to see all there is to see in the 'space' our eyes survey and observe, but little could be further from the truth. The human 'world' is only a tiny band of frequency that the body's visual and perceptual systems can decode into *perception* of a 'world'. According to mainstream science the electromagnetic spectrum is 0.005 percent of what exists in the Universe ([Fig 10](#)). The maximum estimate I have seen is 0.5 percent and either way it's miniscule. I say it is far, far, smaller even than 0.005 percent when you compare reality we see with the totality of reality that we don't. Now get this if you are new to such information: Visible light, the only band of frequency that we can see, is a *fraction* of the 0.005

percent (Fig 11 overleaf). Take this further and realise that our universe is one of infinite universes and that universes are only a fragment of overall reality – *infinite* reality. Then compare that with the almost infinitesimal frequency band of visible light or human sight. You see that humans are as near blind as it is possible to be without actually being so. Artist and filmmaker, Sergio Toporek, said:

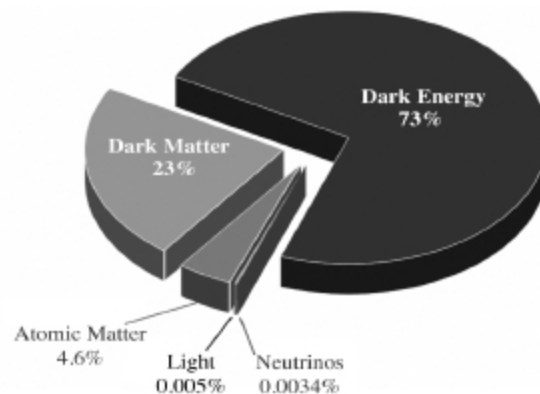


Figure 10: Humans can perceive such a tiny band of visual reality it's laughable.

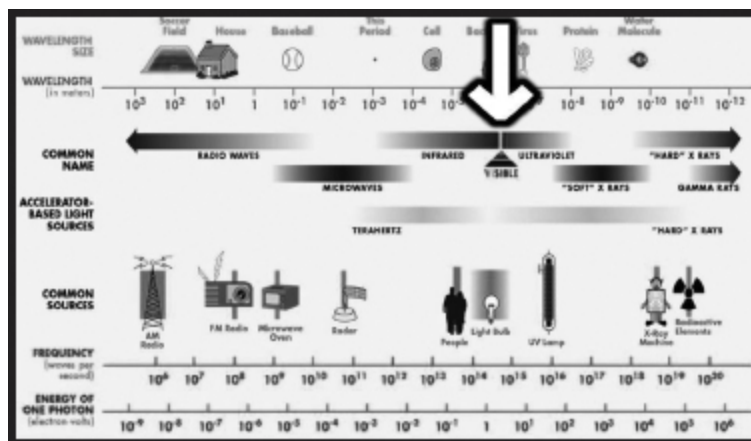


Figure 11: We can see a smear of the 0.005 percent electromagnetic spectrum, but we still know it all. Yep, makes sense.

Consider that you can see less than 1% of the electromagnetic spectrum and hear less than 1% of the acoustic spectrum. 90% of the cells in your body carry their own microbial DNA and are not 'you'. The atoms in your body are 99.9999999999999999% empty space and none of them are the ones you were born with ... Human beings have 46 chromosomes, two less than a potato.

The existence of the rainbow depends on the conical photoreceptors in your eyes; to animals without cones, the rainbow does not exist. So you don't just look at a rainbow, you create it. This is pretty amazing, especially considering that all the beautiful colours you see represent less than 1% of the electromagnetic spectrum.

Suddenly the 'world' of humans looks a very different place. Take into account, too, that Planet Earth when compared with the projected size of this single universe is the equivalent of a billionth of a pinhead. Imagine the ratio that would be when compared to infinite reality. To think that Christianity once insisted that Earth and humanity were the centre of everything. This background is vital if we are going to appreciate the nature of 'human' and how we can be manipulated by an unseen force. To human visual reality virtually *everything* is unseen and yet the prevailing perception within the institutions and so much of the public is that if we can't see it, touch it, hear it, taste it and smell it then it cannot exist. Such perception is indoctrinated and encouraged by the Cult and its agents because it isolates believers in the strictly limited, village-idiot, realm of the five senses where perceptions can be firewalled and information controlled. Most of those perpetuating the 'this-world-is-all-there-is' insanity are themselves indoctrinated into believing the same delusion. While major players and influencers know that official reality is laughable most of those in science, academia and medicine really believe the nonsense they peddle and teach succeeding generations. Those who challenge the orthodoxy are dismissed as nutters and freaks to protect the manufactured illusion from exposure. Observe the dynamic of the 'Covid' hoax and you will see how that takes the same form. The inner-circle psychopaths knows it's a gigantic scam, but almost the entirety of those imposing their fascist rules believe that 'Covid' is all that they're told it is.

Stolen identity

Ask people who they are and they will give you their name, place of birth, location, job, family background and life story. Yet that is not who they are – it is what they are *experiencing*. The difference is *absolutely crucial*. The true 'I', the eternal, infinite 'I', is consciousness,

a state of being aware. Forget 'form'. That is a vehicle for a brief experience. Consciousness does not come *from* the brain, but *through* the brain and even that is more symbolic than literal. We are awareness, pure awareness, and this is what withdraws from the body at what we call 'death' to continue our eternal beingness, *isness*, in other realms of reality within the limitlessness of infinity or the Biblical 'many mansions in my father's house'. Labels of a human life, man, woman, transgender, black, white, brown, nationality, circumstances and income are not who we are. They are what we are – awareness – is *experiencing* in a brief connection with a band of frequency we call 'human'. The labels are not the self; they are, to use the title of one of my books, a *Phantom Self*. I am not David Icke born in Leicester, England, on April 29th, 1952. I am the consciousness *having that experience*. The Cult and its non-human masters seek to convince us through the institutions of 'education', science, medicine, media and government that what we are *experiencing* is who we *are*. It's so easy to control and direct perception locked away in the bewildered illusions of the five senses with no expanded radar. Try, by contrast, doing the same with a humanity aware of its true self and its true power to consciously create its reality and experience. How is it possible to do this? We do it all day every day. If you perceive yourself as 'little me' with no power to impact upon your life and the world then your life experience will reflect that. You will hand the power you don't think you have to authority in all its forms which will use it to control your experience. This, in turn, will appear to confirm your perception of 'little me' in a self-fulfilling feedback loop. But that is what 'little me' really is – a *perception*. We are all 'big-me', infinite me, and the Cult has to make us forget that if its will is to prevail. We are therefore manipulated and pressured into self-identifying with human labels and not the consciousness/awareness *experiencing* those human labels.

The phenomenon of identity politics is a Cult-instigated manipulation technique to sub-divide previous labels into even smaller ones. A United States university employs this list of letters to

describe student identity: LGBTTQQFAGPBDSM or lesbian, gay, bisexual, transgender, transsexual, queer, questioning, flexual, asexual, gender-fuck, polyamorous, bondage/discipline, dominance/submission and sadism/masochism. I'm sure other lists are even longer by now as people feel the need to self-identity the 'I' with the minutiae of race and sexual preference. Wokers programmed by the Cult for generations believe this is about 'inclusivity' when it's really the Cult locking them away into smaller and smaller versions of Phantom Self while firewalling them from the influence of their true self, the infinite, eternal 'I'. You may notice that my philosophy which contends that we are all unique points of attention/awareness within the same infinite whole or Oneness is the ultimate non-racism. The very sense of Oneness makes the judgement of people by their body-type, colour or sexuality utterly ridiculous and confirms that racism has no understanding of reality (including anti-white racism). Yet despite my perception of life Cult agents and fast-asleep Wokers label me racist to discredit my information while they are themselves phenomenally racist and sexist. All they see is race and sexuality and they judge people as good or bad, demons or untouchables, by their race and sexuality. All they see is *Phantom Self* and perceive themselves in terms of Phantom Self. They are pawns and puppets of the Cult agenda to focus attention and self-identity in the five senses and play those identities against each other to divide and rule. Columbia University has introduced segregated graduations in another version of social distancing designed to drive people apart and teach them that different racial and cultural groups have nothing in common with each other. The last thing the Cult wants is unity. Again the pump-primers of this will be Cult operatives in the knowledge of what they are doing, but the rest are just the Phantom Self blind leading the Phantom Self blind. We *do* have something in common – we are all *the same consciousness* having different temporary experiences.

What is this 'human'?

Yes, what *is* 'human'? That is what we are supposed to be, right? I mean 'human'? True, but 'human' is the experience not the 'I'. Break it down to basics and 'human' is the way that information is processed. If we are to experience and interact with this band of frequency we call the 'world' we must have a vehicle that operates within that band of frequency. Our consciousness in its prime form cannot do that; it is way beyond the frequency of the human realm. My consciousness or awareness could not tap these keys and pick up the cup in front of me in the same way that radio station A cannot interact with radio station B when they are on different frequencies. The human body is the means through which we have that interaction. I have long described the body as a biological computer which processes information in a way that allows consciousness to experience this reality. The body is a receiver, transmitter and processor of information in a particular way that we call human. We visually perceive only the world of the five senses in a wakened state – that is the limit of the body's visual decoding system. In truth it's not even visual in the way we experience 'visual reality' as I will come to in a moment. We are 'human' because the body processes the information sources of human into a reality and behaviour system that we *perceive* as human. Why does an elephant act like an elephant and not like a human or a duck? The elephant's biological computer is a different information field and processes information according to that program into a visual and behaviour type we call an elephant. The same applies to everything in our reality. These body information fields are perpetuated through procreation (like making a copy of a software program). The Cult wants to break that cycle and intervene technologically to transform the human information field into one that will change what we call humanity. If it can change the human information field it will change the way that field processes information and change humanity both 'physically' and psychologically. Hence the *messenger* (information) RNA 'vaccines' and so much more that is targeting human genetics by changing the body's information – *messaging* – construct through food, drink, radiation, toxicity and other means.

Reality that we experience is nothing like reality as it really is in the same way that the reality people experience in virtual reality games is not the reality they are really living in. The game is only a decoded source of information that appears to be a reality. Our world is also an information construct – a *simulation* (more later). In its base form our reality is a wavefield of information much the same in theme as Wi-Fi. The five senses decode wavefield information into electrical information which they communicate to the brain to decode into holographic (illusory ‘physical’) information. Different parts of the brain specialise in decoding different senses and the information is fused into a reality that appears to be outside of us but is really inside the brain and the genetic structure in general ([Fig 12](#) overleaf). DNA is a receiver-transmitter of information and a vital part of this decoding process and the body’s connection to other realities. Change DNA and you change the way we decode and connect with reality – see ‘Covid vaccines’. Think of computers decoding Wi-Fi. You have information encoded in a radiation field and the computer decodes that information into a very different form on the screen. You can’t see the Wi-Fi until its information is made manifest on the screen and the information on the screen is inside the computer and not outside. I have just described how we decode the ‘human world’. All five senses decode the waveform ‘Wi-Fi’ field into electrical signals and the brain (computer) constructs reality inside the brain and not outside – ‘You don’t just look at a rainbow, you create it’. Sound is a simple example. We don’t hear sound until the brain decodes it. Waveform sound waves are picked up by the hearing sense and communicated to the brain in an electrical form to be decoded into the sounds that we hear. Everything we hear is inside the brain along with everything we see, feel, smell and taste. Words and language are waveform fields generated by our vocal chords which pass through this process until they are decoded by the brain into words that we hear. Different languages are different frequency fields or sound waves generated by vocal chords. Late British philosopher Alan Watts said:

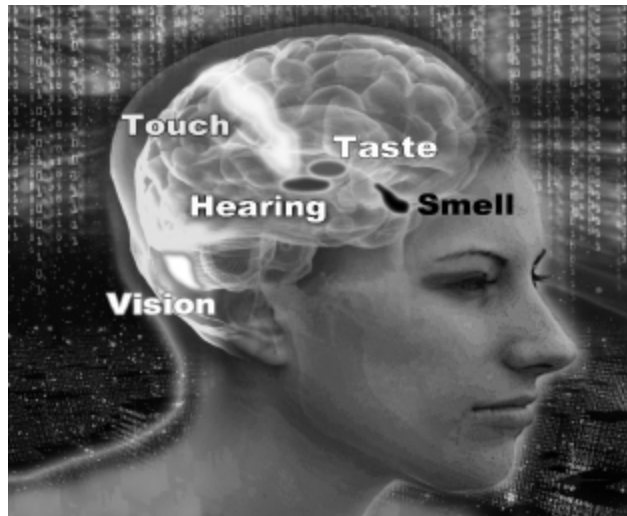


Figure 12: The brain receives information from the five senses and constructs from that our perceived reality.

[Without the brain] the world is devoid of light, heat, weight, solidity, motion, space, time or any other imaginable feature. All these phenomena are interactions, or transactions, of vibrations with a certain arrangement of neurons.

That's exactly what they are and scientist Robert Lanza describes in his book, *Biocentrism*, how we decode electromagnetic waves and energy into visual and 'physical' experience. He uses the example of a flame emitting photons, electromagnetic energy, each pulsing electrically and magnetically:

... these ... invisible electromagnetic waves strike a human retina, and if (and only if) the waves happen to measure between 400 and 700 nano meters in length from crest to crest, then their energy is just right to deliver a stimulus to the 8 million cone-shaped cells in the retina.

Each in turn send an electrical pulse to a neighbour neuron, and on up the line this goes, at 250 mph, until it reaches the ... occipital lobe of the brain, in the back of the head. There, a cascading complex of neurons fire from the incoming stimuli, and we subjectively perceive this experience as a yellow brightness occurring in a place we have been conditioned to call the 'external world'.

You hear what you decode

If a tree falls or a building collapses they make no noise unless someone is there to decode the energetic waves generated by the disturbance into what we call sound. Does a falling tree make a noise? Only if you hear it – *decode* it. Everything in our reality is a frequency field of information operating within the overall ‘Wi-Fi’ field that I call The Field. A vibrational disturbance is generated in The Field by the fields of the falling tree or building. These disturbance waves are what we decode into the sound of them falling. If no one is there to do that then neither will make any noise. Reality is created by the observer – *decoder* – and the *perceptions* of the observer affect the decoding process. For this reason different people – different *perceptions* – will perceive the same reality or situation in a different way. What one may perceive as a nightmare another will see as an opportunity. The question of why the Cult is so focused on controlling human perception now answers itself. All experienced reality is the act of decoding and we don’t experience Wi-Fi until it is decoded on the computer screen. The sight and sound of an Internet video is encoded in the Wi-Fi all around us, but we don’t see or hear it until the computer decodes that information. Taste, smell and touch are all phenomena of the brain as a result of the same process. We don’t taste, smell or feel anything except in the brain and there are pain relief techniques that seek to block the signal from the site of discomfort to the brain because if the brain doesn’t decode that signal we don’t feel pain. Pain is in the brain and only appears to be at the point of impact thanks to the feedback loop between them. We don’t see anything until electrical information from the sight senses is decoded in an area at the back of the brain. If that area is damaged we can go blind when our eyes are perfectly okay. So why do we go blind if we damage an eye? We damage the information processing between the waveform visual information and the visual decoding area of the brain. If information doesn’t reach the brain in a form it can decode then we can’t see the visual reality that it represents. What’s more the brain is decoding only a fraction of the information it receives and the rest is absorbed by the

sub-conscious mind. This explanation is from the science magazine, *Wonderpedia*:

Every second, 11 million sensations crackle along these [brain] pathways ... The brain is confronted with an alarming array of images, sounds and smells which it rigorously filters down until it is left with a manageable list of around 40. Thus 40 sensations per second make up what we perceive as reality.

The 'world' is not what people are told to believe that is it and the inner circles of the Cult *know that*.

Illusory 'physical' reality

We can only see a smear of 0.005 percent of the Universe which is only one of a vast array of universes – 'mansions' – within infinite reality. Even then the brain decodes only 40 pieces of information ('sensations') from a potential *11 million* that we receive every second. Two points strike you from this immediately: The sheer breathtaking stupidity of believing we know anything so rigidly that there's nothing more to know; and the potential for these processes to be manipulated by a malevolent force to control the reality of the population. One thing I can say for sure with no risk of contradiction is that when you can perceive an almost indescribable fraction of infinite reality there is always more to know as in tidal waves of it. Ancient Greek philosopher Socrates was so right when he said that wisdom is to know how little we know. How obviously true that is when you think that we are experiencing a physical world of solidity that is neither physical nor solid and a world of apartness when everything is connected. Cult-controlled 'science' dismisses the so-called 'paranormal' and all phenomena related to that when the 'para'-normal is perfectly normal and explains the alleged 'great mysteries' which dumbfound scientific minds. There is a reason for this. A 'scientific mind' in terms of the mainstream is a material mind, a five-sense mind imprisoned in see it, touch it, hear it, smell it and taste it. Phenomena and happenings that can't be explained that way leave the 'scientific mind' bewildered and the rule is that if they

can't account for why something is happening then it can't, by definition, be happening. I beg to differ. Telepathy is thought waves passing through The Field (think wave disturbance again) to be decoded by someone able to connect with that wavelength (information). For example: You can pick up the thought waves of a friend at any distance and at the very least that will bring them to mind. A few minutes later the friend calls you. 'My god', you say, 'that's incredible – I was just thinking of you.' Ah, but *they* were thinking of *you* before they made the call and that's what you decoded. Native peoples not entrapped in five-sense reality do this so well it became known as the 'bush telegraph'. Those known as psychics and mediums (genuine ones) are doing the same only across dimensions of reality. 'Mind over matter' comes from the fact that matter and mind are the *same*. The state of one influences the state of the other. Indeed one *and* the other are illusions. They are aspects of the same field. Paranormal phenomena are all explainable so why are they still considered 'mysteries' or not happening? Once you go down this road of understanding you begin to expand awareness beyond the five senses and that's the nightmare for the Cult.



Figure 13: Holograms are not solid, but the best ones appear to be.

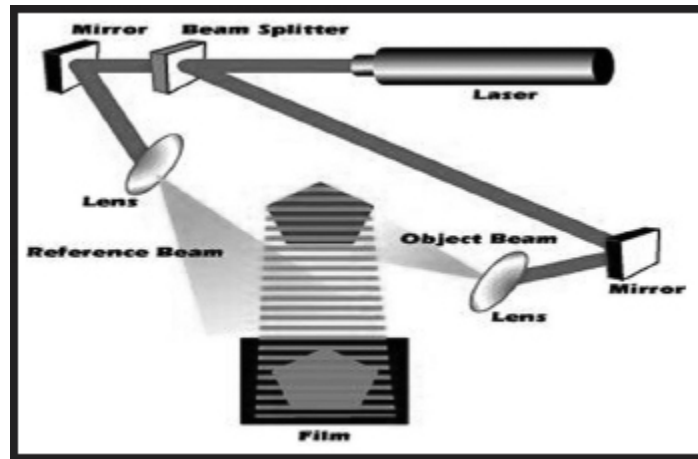


Figure 14: How holograms are created by capturing a waveform version of the subject image.

Holographic 'solidity'

Our reality is not solid, it is holographic. We are now well aware of holograms which are widely used today. Two-dimensional information is decoded into a three-dimensional reality that is not solid although can very much appear to be (Fig 13). Holograms are created with a laser divided into two parts. One goes directly onto a holographic photographic print ('reference beam') and the other takes a waveform image of the subject ('working beam') before being directed onto the print where it 'collides' with the other half of the laser (Fig 14). This creates a *waveform* interference pattern which contains the wavefield information of whatever is being photographed (Fig 15 overleaf). The process can be likened to dropping pebbles in a pond. Waves generated by each one spread out across the water to collide with the others and create a wave representation of where the stones fell and at what speed, weight and distance. A waveform interference pattern of a hologram is akin to the waveform information in The Field which the five senses decode into electrical signals to be decoded by the brain into a holographic illusory 'physical' reality. In the same way when a laser (think human attention) is directed at the waveform interference pattern a three-dimensional version of the subject is projected into apparently 'solid' reality (Fig 16). An amazing trait of holograms reveals more 'paranormal mysteries'. Information of the *whole*

hologram is encoded in waveform in every part of the interference pattern by the way they are created. This means that every *part* of a hologram is a smaller version of the whole. Cut the interference wave-pattern into four and you won't get four parts of the image. You get quarter-sized versions of the *whole* image. The body is a hologram and the same applies. Here we have the basis of acupuncture, reflexology and other forms of healing which identify representations of the whole body in all of the parts, hands, feet, ears, everywhere. Skilled palm readers can do what they do because the information of whole body is encoded in the hand. The concept of as above, so below, comes from this.



Figure 15: A waveform interference pattern that holds the information that transforms into a hologram.



Figure 16: Holographic people including 'Elvis' holographically inserted to sing a duet with Celine Dion.

The question will be asked of why, if solidity is illusory, we can't just walk through walls and each other. The resistance is not solid against solid; it is electromagnetic field against electromagnetic field and we decode this into the *experience* of solid against solid. We should also not underestimate the power of belief to dictate reality. What you believe is impossible *will be*. Your belief impacts on your decoding processes and they won't decode what you think is impossible. What we believe we perceive and what we perceive we experience. 'Can't dos' and 'impossibles' are like a firewall in a computer system that won't put on the screen what the firewall blocks. How vital that is to understanding how human experience has been hijacked. I explain in *The Answer, Everything You Need To Know But Have Never Been Told* and other books a long list of 'mysteries' and 'paranormal' phenomena that are not mysterious and perfectly normal once you realise what reality is and how it works. 'Ghosts' can be seen to pass through 'solid' walls because the walls are not solid and the ghost is a discarnate entity operating on a frequency so different to that of the wall that it's like two radio stations sharing the same space while never interfering with each other. I have seen ghosts do this myself. The apartness of people and objects is also an illusion. Everything is connected by the Field like all sea life is connected by the sea. It's just that within the limits of our visual reality we only 'see' holographic information and not the field of information that connects everything and from which the holographic world is made manifest. If you can only see holographic 'objects' and not the field that connects them they will appear to you as unconnected to each other in the same way that we see the computer while not seeing the Wi-Fi.

What you don't know *can* hurt you

Okay, we return to those 'two worlds' of human society and the Cult with its global network of interconnecting secret societies and satanic groups which manipulate through governments, corporations, media, religions, etc. The fundamental difference between them is *knowledge*. The idea has been to keep humanity

ignorant of the plan for its total enslavement underpinned by a crucial ignorance of reality – who we are and where we are – and how we interact with it. ‘Human’ should be the interaction between our expanded eternal consciousness and the five-sense body experience. We are meant to be *in* this world in terms of the five senses but not *of* this world in relation to our greater consciousness and perspective. In that state we experience the small picture of the five senses within the wider context of the big picture of awareness beyond the five senses. Put another way the five senses see the dots and expanded awareness connects them into pictures and patterns that give context to the apparently random and unconnected. Without the context of expanded awareness the five senses see only apartness and randomness with apparently no meaning. The Cult and its other-dimensional controllers seek to intervene in the frequency realm where five-sense reality is supposed to connect with expanded reality and to keep the two apart (more on this in the final chapter). When that happens five-sense mental and emotional processes are no longer influenced by expanded awareness, or the True ‘I’, and instead are driven by the isolated perceptions of the body’s decoding systems. They are in the world *and* of it. Here we have the human plight and why humanity with its potential for infinite awareness can be so easily manipulatable and descend into such extremes of stupidity.

Once the Cult isolates five-sense mind from expanded awareness it can then program the mind with perceptions and beliefs by controlling information that the mind receives through the ‘education’ system of the formative years and the media perceptual bombardment and censorship of an entire lifetime. Limit perception and a sense of the possible through limiting knowledge by limiting and skewing information while censoring and discrediting that which could set people free. As the title of another of my books says ... *And The Truth Shall Set You Free*. For this reason the last thing the Cult wants in circulation is the truth about anything – especially the reality of the eternal ‘I’ – and that’s why it is desperate to control information. The Cult knows that information becomes perception

which becomes behaviour which, collectively, becomes human society. Cult-controlled and funded mainstream 'science' denies the existence of an eternal 'I' and seeks to dismiss and trash all evidence to the contrary. Cult-controlled mainstream religion has a version of 'God' that is little more than a system of control and dictatorship that employs threats of damnation in an afterlife to control perceptions and behaviour in the here and now through fear and guilt. Neither is true and it's the 'neither' that the Cult wishes to suppress. This 'neither' is that everything is an expression, a point of attention, within an infinite state of consciousness which is the real meaning of the term 'God'.

Perceptual obsession with the 'physical body' and five-senses means that 'God' becomes personified as a bearded bloke sitting among the clouds or a raging bully who loves us if we do what 'he' wants and condemns us to the fires of hell if we don't. These are no more than a 'spiritual' fairy tales to control and dictate events and behaviour through fear of this 'God' which has bizarrely made 'God-fearing' in religious circles a state to be desired. I would suggest that fearing *anything* is not to be encouraged and celebrated, but rather deleted. You can see why 'God fearing' is so beneficial to the Cult and its religions when *they* decide what 'God' wants and what 'God' demands (the Cult demands) that everyone do. As the great American comedian Bill Hicks said satirising a Christian zealot: 'I think what God meant to say.' How much of this infinite awareness ('God') that we access is decided by how far we choose to expand our perceptions, self-identity and sense of the possible. The scale of self-identity reflects itself in the scale of awareness that we can connect with and are influenced by – how much knowing and insight we have instead of programmed perception. You cannot expand your awareness into the infinity of possibility when you believe that you are little me Peter the postman or Mary in marketing and nothing more. I'll deal with this in the concluding chapter because it's crucial to how we turnaround current events.

Where the Cult came from

When I realised in the early 1990s there was a Cult network behind global events I asked the obvious question: When did it start? I took it back to ancient Rome and Egypt and on to Babylon and Sumer in Mesopotamia, the 'Land Between Two Rivers', in what we now call Iraq. The two rivers are the Tigris and Euphrates and this region is of immense historical and other importance to the Cult, as is the land called Israel only 550 miles away by air. There is much more going on with deep esoteric meaning across this whole region. It's not only about 'wars for oil'. Priceless artefacts from Mesopotamia were stolen or destroyed after the American and British invasion of Iraq in 2003 justified by the lies of Boy Bush and Tony Blair (their Cult masters) about non-existent 'weapons of mass destruction'.

Mesopotamia was the location of Sumer (about 5,400BC to 1,750BC), and Babylon (about 2,350BC to 539BC). Sabbatians may have become immensely influential in the Cult in modern times but they are part of a network that goes back into the mists of history. Sumer is said by historians to be the 'cradle of civilisation'. I disagree. I say it was the re-start of what we call human civilisation after cataclysmic events symbolised in part as the 'Great Flood' destroyed the world that existed before. These fantastic upheavals that I have been describing in detail in the books since the early 1990s appear in accounts and legends of ancient cultures across the world and they are supported by geological and biological evidence. Stone tablets found in Iraq detailing the Sumer period say the cataclysms were caused by non-human 'gods' they call the Anunnaki. These are described in terms of extraterrestrial visitations in which knowledge supplied by the Anunnaki is said to have been the source of at least one of the world's oldest writing systems and developments in astronomy, mathematics and architecture that were way ahead of their time. I have covered this subject at length in *The Biggest Secret* and *Children of the Matrix* and the same basic 'Anunnaki' story can be found in Zulu accounts in South Africa where the late and very great Zulu high shaman Credo Mutwa told me that the Sumerian Anunnaki were known by Zulus as the Chitauri or 'children of the serpent'. See my six-hour video interview with Credo on this subject entitled *The*

Reptilian Agenda recorded at his then home near Johannesburg in 1999 which you can watch on the Ickonic media platform.

The Cult emerged out of Sumer, Babylon and Egypt (and elsewhere) and established the Roman Empire before expanding with the Romans into northern Europe from where many empires were savagely imposed in the form of Cult-controlled societies all over the world. Mass death and destruction was their calling card. The Cult established its centre of operations in Europe and European Empires were Cult empires which allowed it to expand into a global force. Spanish and Portuguese colonialists headed for Central and South America while the British and French targeted North America. Africa was colonised by Britain, France, Belgium, the Netherlands, Portugal, Spain, Italy, and Germany. Some like Britain and France moved in on the Middle East. The British Empire was by far the biggest for a simple reason. By now Britain was the headquarters of the Cult from which it expanded to form Canada, the United States, Australia and New Zealand. The Sun never set on the British Empire such was the scale of its occupation. London remains a global centre for the Cult along with Rome and the Vatican although others have emerged in Israel and China. It is no accident that the 'virus' is alleged to have come out of China while Italy was chosen as the means to terrify the Western population into compliance with 'Covid' fascism. Nor that Israel has led the world in 'Covid' fascism and mass 'vaccination'.

You would think that I would mention the United States here, but while it has been an important means of imposing the Cult's will it is less significant than would appear and is currently in the process of having what power it does have deleted. The Cult in Europe has mostly loaded the guns for the US to fire. America has been controlled from Europe from the start through Cult operatives in Britain and Europe. The American Revolution was an illusion to make it appear that America was governing itself while very different forces were pulling the strings in the form of Cult families such as the Rothschilds through the Rockefellers and other subordinates. The Rockefellers are extremely close to Bill Gates and

established both scalpel and drug 'medicine' and the World Health Organization. They play a major role in the development and circulation of vaccines through the Rockefeller Foundation on which Bill Gates said his Foundation is based. Why wouldn't this be the case when the Rockefellers and Gates are on the same team? Cult infiltration of human society goes way back into what we call history and has been constantly expanding and centralising power with the goal of establishing a global structure to dictate everything. Look how this has been advanced in great leaps with the 'Covid' hoax.

The non-human dimension

I researched and observed the comings and goings of Cult operatives through the centuries and even thousands of years as they were born, worked to promote the agenda within the secret society and satanic networks, and then died for others to replace them. Clearly there had to be a coordinating force that spanned this entire period while operatives who would not have seen the end goal in their lifetimes came and went advancing the plan over millennia. I went in search of that coordinating force with the usual support from the extraordinary synchronicity of my life which has been an almost daily experience since 1990. I saw common themes in religious texts and ancient cultures about a non-human force manipulating human society from the hidden. Christianity calls this force Satan, the Devil and demons; Islam refers to the Jinn or Djinn; Zulus have their Chitauri (spelt in other ways in different parts of Africa); and the Gnostic people in Egypt in the period around and before 400AD referred to this phenomena as the 'Archons', a word meaning rulers in Greek. Central American cultures speak of the 'Predators' among other names and the same theme is everywhere. I will use 'Archons' as a collective name for all of them. When you see how their nature and behaviour is described all these different sources are clearly talking about the same force. Gnostics described the Archons in terms of 'luminous fire' while Islam relates the Jinn to 'smokeless fire'. Some refer to beings in form that could occasionally be seen, but the most common of common theme is that they operate from

unseen realms which means almost all existence to the visual processes of humans. I had concluded that this was indeed the foundation of human control and that the Cult was operating within the human frequency band on behalf of this hidden force when I came across the writings of Gnostics which supported my conclusions in the most extraordinary way.

A sealed earthen jar was found in 1945 near the town of Nag Hammadi about 75-80 miles north of Luxor on the banks of the River Nile in Egypt. Inside was a treasure trove of manuscripts and texts left by the Gnostic people some 1,600 years earlier. They included 13 leather-bound papyrus codices (manuscripts) and more than 50 texts written in Coptic Egyptian estimated to have been hidden in the jar in the period of 400AD although the source of the information goes back much further. Gnostics oversaw the Great or Royal Library of Alexandria, the fantastic depository of ancient texts detailing advanced knowledge and accounts of human history. The Library was dismantled and destroyed in stages over a long period with the death-blow delivered by the Cult-established Roman Church in the period around 415AD. The Church of Rome was the Church of Babylon relocated as I said earlier. Gnostics were not a race. They were a way of perceiving reality. Whenever they established themselves and their information circulated the terrorists of the Church of Rome would target them for destruction. This happened with the Great Library and with the Gnostic Cathars who were burned to death by the psychopaths after a long period of oppression at the siege of the Castle of Monségur in southern France in 1244. The Church has always been terrified of Gnostic information which demolishes the official Christian narrative although there is much in the Bible that supports the Gnostic view if you read it in another way. To anyone studying the texts of what became known as the Nag Hammadi Library it is clear that great swathes of Christian and Biblical belief has its origin with Gnostics sources going back to Sumer. Gnostic themes have been twisted to manipulate the perceived reality of Bible believers. Biblical texts have been in the open for centuries where they could be changed while Gnostic

documents found at Nag Hammadi were sealed away and untouched for 1,600 years. What you see is what they wrote.

Use your *pneuma* not your *nous*

Gnosticism and Gnostic come from 'gnosis' which means knowledge, or rather *secret* knowledge, in the sense of spiritual awareness – knowledge about reality and life itself. The desperation of the Cult's Church of Rome to destroy the Gnostics can be understood when the knowledge they were circulating was the last thing the Cult wanted the population to know. Sixteen hundred years later the same Cult is working hard to undermine and silence me for the same reason. The dynamic between knowledge and ignorance is a constant. 'Time' appears to move on, but essential themes remain the same. We are told to 'use your nous', a Gnostic word for head/brain/intelligence. They said, however, that spiritual awakening or 'salvation' could only be secured by expanding awareness *beyond* what they called *nous* and into *pneuma* or Infinite Self. Obviously as I read these texts the parallels with what I have been saying since 1990 were fascinating to me. There is a universal truth that spans human history and in that case why wouldn't we be talking the same language 16 centuries apart? When you free yourself from the perception program of the five senses and explore expanded realms of consciousness you are going to connect with the same information no matter what the perceived 'era' within a manufactured timeline of a single and tiny range of manipulated frequency. Humans working with 'smart' technology or knocking rocks together in caves is only a timeline appearing to operate within the human frequency band. Expanded awareness and the knowledge it holds have always been there whether the era be Stone Age or computer age. We can only access that knowledge by opening ourselves to its frequency which the five-sense prison cell is designed to stop us doing. Gates, Fauci, Whitty, Vallance, Zuckerberg, Brin, Page, Wojcicki, Bezos, and all the others behind the 'Covid' hoax clearly have a long wait before their range of frequency can make that connection given that an open heart is

crucial to that as we shall see. Instead of accessing knowledge directly through expanded awareness it is given to Cult operatives by the secret society networks of the Cult where it has been passed on over thousands of years outside the public arena. Expanded realms of consciousness is where great artists, composers and writers find their inspiration and where truth awaits anyone open enough to connect with it. We need to go there fast.

Archon hijack

A fifth of the Nag Hammadi texts describe the existence and manipulation of the Archons led by a 'Chief Archon' they call 'Yaldabaoth', or the 'Demiurge', and this is the Christian 'Devil', 'Satan', 'Lucifer', and his demons. Archons in Biblical symbolism are the 'fallen ones' which are also referred to as fallen angels after the angels expelled from heaven according to the Abrahamic religions of Judaism, Christianity and Islam. These angels are claimed to tempt humans to 'sin' ongoing and you will see how accurate that symbolism is during the rest of the book. The theme of 'original sin' is related to the 'Fall' when Adam and Eve were 'tempted by the serpent' and fell from a state of innocence and 'obedience' (connection) with God into a state of disobedience (disconnection). The Fall is said to have brought sin into the world and corrupted everything including human nature. Yaldabaoth, the 'Lord Archon', is described by Gnostics as a 'counterfeit spirit', 'The Blind One', 'The Blind God', and 'The Foolish One'. The Jewish name for Yaldabaoth in Talmudic writings is Samael which translates as 'Poison of God', or 'Blindness of God'. You see the parallels. Yaldabaoth in Islamic belief is the Muslim Jinn devil known as Shaytan – Shaytan is Satan as the same themes are found all over the world in every religion and culture. The 'Lord God' of the Old Testament is the 'Lord Archon' of Gnostic manuscripts and that's why he's such a bloodthirsty bastard. Satan is known by Christians as 'the Demon of Demons' and Gnostics called Yaldabaoth the 'Archon of Archons'. Both are known as 'The Deceiver'. We are talking about the same 'bloke' for sure and these common themes

using different names, storylines and symbolism tell a common tale of the human plight.

Archons are referred to in Nag Hammadi documents as mind parasites, inverters, guards, gatekeepers, detainers, judges, pitiless ones and deceivers. The 'Covid' hoax alone is a glaring example of all these things. The Biblical 'God' is so different in the Old and New Testaments because they are not describing the same phenomenon. The vindictive, angry, hate-filled, 'God' of the Old Testament, known as Yahweh, is Yaldabaoth who is depicted in Cult-dictated popular culture as the 'Dark Lord', 'Lord of Time', Lord (Darth) Vader and Dormammu, the evil ruler of the 'Dark Dimension' trying to take over the 'Earth Dimension' in the Marvel comic movie, *Dr Strange*. Yaldabaoth is both the Old Testament 'god' and the Biblical 'Satan'. Gnostics referred to Yaldabaoth as the 'Great Architect of the Universe' and the Cult-controlled Freemason network calls their god 'the 'Great Architect of the Universe' (also Grand Architect). The 'Great Architect' Yaldabaoth is symbolised by the Cult as the all-seeing eye at the top of the pyramid on the Great Seal of the United States and the dollar bill. Archon is encoded in *arch*-itect as it is in *arch*-angels and *arch*-bishops. All religions have the theme of a force for good and force for evil in some sort of spiritual war and there is a reason for that – the theme is true. The Cult and its non-human masters are quite happy for this to circulate. They present themselves as the force for good fighting evil when they are really the force of evil (absence of love). The whole foundation of Cult modus operandi is inversion. They promote themselves as a force for good and anyone challenging them in pursuit of peace, love, fairness, truth and justice is condemned as a satanic force for evil. This has been the game plan throughout history whether the Church of Rome inquisitions of non-believers or 'conspiracy theorists' and 'anti-vaxxers' of today. The technique is the same whatever the timeline era.

Yaldabaoth is revolting (true)

Yaldabaoth and the Archons are said to have revolted against God with Yaldabaoth claiming to *be* God – the *All That Is*. The Old Testament ‘God’ (Yaldabaoth) demanded to be worshipped as such: ‘*I am the LORD, and there is none else, there is no God beside me*’ (Isaiah 45:5). I have quoted in other books a man who said he was the unofficial son of the late Baron Philippe de Rothschild of the Mouton-Rothschild wine producing estates in France who died in 1988 and he told me about the Rothschild ‘revolt from God’. The man said he was given the name Phillip Eugene de Rothschild and we shared long correspondence many years ago while he was living under another identity. He said that he was conceived through ‘occult incest’ which (within the Cult) was ‘normal and to be admired’. ‘Phillip’ told me about his experience attending satanic rituals with rich and famous people whom he names and you can see them and the wider background to Cult Satanism in my other books starting with *The Biggest Secret*. Cult rituals are interactions with Archontic ‘gods’. ‘Phillip’ described Baron Philippe de Rothschild as ‘a master Satanist and hater of God’ and he used the same term ‘revolt from God’ associated with Yaldabaoth/Satan/Lucifer/the Devil in describing the Sabbatian Rothschild dynasty. ‘I played a key role in my family’s revolt from God’, he said. That role was to infiltrate in classic Sabbatian style the Christian Church, but eventually he escaped the mind-prison to live another life. The Cult has been targeting religion in a plan to make worship of the Archons the global one-world religion. Infiltration of Satanism into modern ‘culture’, especially among the young, through music videos, stage shows and other means, is all part of this.

Nag Hammadi texts describe Yaldabaoth and the Archons in their prime form as energy – consciousness – and say they can take form if they choose in the same way that consciousness takes form as a human. Yaldabaoth is called ‘formless’ and represents a deeply inverted, distorted and chaotic state of consciousness which seeks to attached to humans and turn them into a likeness of itself in an attempt at assimilation. For that to happen it has to manipulate

humans into low frequency mental and emotional states that match its own. Archons can certainly appear in human form and this is the origin of the psychopathic personality. The energetic distortion Gnostics called Yaldabaoth is psychopathy. When psychopathic Archons take human form that human will be a psychopath as an expression of Yaldabaoth consciousness. Cult psychopaths are Archons in human form. The principle is the same as that portrayed in the 2009 *Avatar* movie when the American military travelled to a fictional Earth-like moon called Pandora in the Alpha Centauri star system to infiltrate a society of blue people, or Na'vi, by hiding within bodies that looked like the Na'vi. Archons posing as humans have a particular hybrid information field, part human, part Archon, (the ancient 'demigods') which processes information in a way that manifests behaviour to match their psychopathic evil, lack of empathy and compassion, and stops them being influenced by the empathy, compassion and love that a fully-human information field is capable of expressing. Cult bloodlines interbreed, be they royalty or dark suits, for this reason and you have their obsession with incest. Interbreeding with full-blown humans would dilute the Archontic energy field that guarantees psychopathy in its representatives in the human realm.

Gnostic writings say the main non-human forms that Archons take are *serpentine* (what I have called for decades 'reptilian' amid unbounded ridicule from the Archontically-programmed) and what Gnostics describe as 'an unborn baby or foetus with grey skin and dark, unmoving eyes'. This is an excellent representation of the ET 'Greys' of UFO folklore which large numbers of people claim to have seen and been abducted by – Zulu shaman Credo Mutwa among them. I agree with those that believe in extraterrestrial or interdimensional visitations today and for thousands of years past. No wonder with their advanced knowledge and technological capability they were perceived and worshipped as gods for technological and other 'miracles' they appeared to perform. Imagine someone arriving in a culture disconnected from the modern world with a smartphone and computer. They would be

seen as a 'god' capable of 'miracles'. The Renegade Mind, however, wants to know the source of everything and not only the way that source manifests as human or non-human. In the same way that a Renegade Mind seeks the original source material for the 'Covid virus' to see if what is claimed is true. The original source of Archons in form is consciousness – the distorted state of consciousness known to Gnostics as Yaldabaoth.

'Revolt from God' is energetic disconnection

Where I am going next will make a lot of sense of religious texts and ancient legends relating to 'Satan', Lucifer' and the 'gods'. Gnostic descriptions sync perfectly with the themes of my own research over the years in how they describe a consciousness distortion seeking to impose itself on human consciousness. I've referred to the core of infinite awareness in previous books as Infinite Awareness in Awareness of Itself. By that I mean a level of awareness that knows that it is all awareness and is aware of all awareness. From here comes the frequency of love in its true sense and balance which is what love is on one level – the balance of all forces into a single whole called Oneness and Isness. The more we disconnect from this state of love that many call 'God' the constituent parts of that Oneness start to unravel and express themselves as a part and not a whole. They become individualised as intellect, mind, selfishness, hatred, envy, desire for power over others, and such like. This is not a problem in the greater scheme in that 'God', the *All That Is*, can experience all these possibilities through different expressions of itself including humans. What we as expressions of the whole experience the *All That Is* experiences. We are the *All That Is* experiencing itself. As we withdraw from that state of Oneness we disconnect from its influence and things can get very unpleasant and very stupid. Archontic consciousness is at the extreme end of that. It has so disconnected from the influence of Oneness that it has become an inversion of unity and love, an inversion of everything, an inversion of life itself. Evil is appropriately live written backwards. Archontic consciousness is obsessed with death, an inversion of life,

and so its manifestations in Satanism are obsessed with death. They use inverted symbols in their rituals such as the inverted pentagram and cross. Sabbatians as Archontic consciousness incarnate invert Judaism and every other religion and culture they infiltrate. They seek disunity and chaos and they fear unity and harmony as they fear love like garlic to a vampire. As a result the Cult, Archons incarnate, act with such evil, psychopathy and lack of empathy and compassion disconnected as they are from the source of love. How could Bill Gates and the rest of the Archontic psychopaths do what they have to human society in the 'Covid' era with all the death, suffering and destruction involved and have no emotional consequence for the impact on others? Now you know. Why have Zuckerberg, Brin, Page, Wojcicki and company callously censored information warning about the dangers of the 'vaccine' while thousands have been dying and having severe, sometimes life-changing reactions? Now you know. Why have Tedros, Fauci, Whitty, Vallance and their like around the world been using case and death figures they're aware are fraudulent to justify lockdowns and all the deaths and destroyed lives that have come from that? Now you know. Why did Christian Drosten produce and promote a 'testing' protocol that he knew couldn't test for infectious disease which led to a global human catastrophe. Now you know. The Archontic mind doesn't give a shit ([Fig 17](#)). I personally think that Gates and major Cult insiders are a form of AI cyborg that the Archons want humans to become.



Figure 17: Artist Neil Hague's version of the 'Covid' hierarchy.

Human batteries

A state of such inversion does have its consequences, however. The level of disconnection from the Source of All means that you withdraw from that source of energetic sustenance and creativity. This means that you have to find your own supply of energetic power and it has – *us*. When the Morpheus character in the first *Matrix* movie held up a battery he spoke a profound truth when he said: 'The Matrix is a computer-generated dream world built to keep us under control in order to change the human being into one of

these.' The statement was true in all respects. We do live in a technologically-generated virtual reality simulation (more very shortly) and we have been manipulated to be an energy source for Archontic consciousness. The Disney-Pixar animated movie *Monsters, Inc.* in 2001 symbolised the dynamic when monsters in their world had no energy source and they would enter the human world to terrify children in their beds, catch the child's scream, terror (low-vibrational frequencies), and take that energy back to power the monster world. The lead character you might remember was a single giant eye and the symbolism of the Cult's all-seeing eye was obvious. Every thought and emotion is broadcast as a frequency unique to that thought and emotion. Feelings of love and joy, empathy and compassion, are high, quick, frequencies while fear, depression, anxiety, suffering and hate are low, slow, dense frequencies. Which kind do you think Archontic consciousness can connect with and absorb? In such a low and dense frequency state there's no way it can connect with the energy of love and joy. Archons can only feed off energy compatible with their own frequency and they and their Cult agents want to delete the human world of love and joy and manipulate the transmission of low vibrational frequencies through low-vibrational human mental and emotional states. *We are their energy source.* Wars are energetic banquets to the Archons – a world war even more so – and think how much low-frequency mental and emotional energy has been generated from the consequences for humanity of the 'Covid' hoax orchestrated by Archons incarnate like Gates.

The ancient practice of human sacrifice 'to the gods', continued in secret today by the Cult, is based on the same principle. 'The gods' are Archontic consciousness in different forms and the sacrifice is induced into a state of intense terror to generate the energy the Archontic frequency can absorb. Incarnate Archons in the ritual drink the blood which contains an adrenaline they crave which floods into the bloodstream when people are terrorised. Most of the sacrifices, ancient and modern, are children and the theme of 'sacrificing young virgins to the gods' is just code for children. They

have a particular pre-puberty energy that Archons want more than anything and the energy of the young in general is their target. The California Department of Education wants students to chant the names of Aztec gods (Archontic gods) once worshipped in human sacrifice rituals in a curriculum designed to encourage them to 'challenge racist, bigoted, discriminatory, imperialist/colonial beliefs', join 'social movements that struggle for social justice', and 'build new possibilities for a post-racist, post-systemic racism society'. It's the usual Woke crap that inverts racism and calls it anti-racism. In this case solidarity with 'indigenous tribes' is being used as an excuse to chant the names of 'gods' to which people were sacrificed (and still are in secret). What an example of Woke's inability to see beyond black and white, us and them, They condemn the colonisation of these tribal cultures by Europeans (quite right), but those cultures sacrificing people including children to their 'gods', and mass murdering untold numbers as the Aztecs did, is just fine. One chant is to the Aztec god Tezcatlipoca who had a man sacrificed to him in the 5th month of the Aztec calendar. His heart was cut out and he was eaten. Oh, that's okay then. Come on children ... after three ... Other sacrificial 'gods' for the young to chant their allegiance include Quetzalcoatl, Huitzilopochtli and Xipe Totec. The curriculum says that 'chants, affirmations, and energizers can be used to bring the class together, build unity around ethnic studies principles and values, and to reinvigorate the class following a lesson that may be emotionally taxing or even when student engagement may appear to be low'. Well, that's the cover story, anyway. Chanting and mantras are the repetition of a particular frequency generated from the vocal cords and chanting the names of these Archontic 'gods' tunes you into their frequency. That is the last thing you want when it allows for energetic synchronisation, attachment and perceptual influence. Initiates chant the names of their 'Gods' in their rituals for this very reason.

Vampires of the Woke

Paedophilia is another way that Archons absorb the energy of children. Paedophiles possessed by Archontic consciousness are used as the conduit during sexual abuse for discarnate Archons to vampire the energy of the young they desire so much. Stupendous numbers of children disappear every year never to be seen again although you would never know from the media. Imagine how much low-vibrational energy has been generated by children during the 'Covid' hoax when so many have become depressed and psychologically destroyed to the point of killing themselves. Shocking numbers of children are now taken by the state from loving parents to be handed to others. I can tell you from long experience of researching this since 1996 that many end up with paedophiles and assets of the Cult through corrupt and Cult-owned social services which in the reframing era has hired many psychopaths and emotionless automatons to do the job. Children are even stolen to order using spurious reasons to take them by the corrupt and secret (because they're corrupt) 'family courts'. I have written in detail in other books, starting with *The Biggest Secret* in 1997, about the ubiquitous connections between the political, corporate, government, intelligence and military elites (Cult operatives) and Satanism and paedophilia. If you go deep enough both networks have an interlocking leadership. The Woke mentality has been developed by the Cult for many reasons: To promote almost every aspect of its agenda; to hijack the traditional political left and turn it fascist; to divide and rule; and to target agenda pushbackers. But there are other reasons which relate to what I am describing here. How many happy and joyful Wokers do you ever see especially at the extreme end? They are a mental and psychological mess consumed by emotional stress and constantly emotionally cocked for the next explosion of indignation at someone referring to a female as a female. They are walking, talking, batteries as Morpheus might say emitting frequencies which both enslave them in low-vibrational bubbles of perceptual limitation and feed the Archons. Add to this the hatred claimed to be love; fascism claimed to 'anti-fascism', racism claimed to be 'anti-racism';

exclusion claimed to inclusion; and the abuse-filled Internet trolling. You have a purpose-built Archontic energy system with not a wind turbine in sight and all founded on Archontic *inversion*. We have whole generations now manipulated to serve the Archons with their actions and energy. They will be doing so their entire adult lives unless they snap out of their Archon-induced trance. Is it really a surprise that Cult billionaires and corporations put so much money their way? Where is the energy of joy and laughter, including laughing at yourself which is confirmation of your own emotional security? Mark Twain said: 'The human race has one really effective weapon, and that is laughter.' We must use it all the time. Woke has destroyed comedy because it has no humour, no joy, sense of irony, or self-deprecation. Its energy is dense and intense. Mmmmm, lunch says the Archontic frequency. Rudolf Steiner (1861-1925) was the Austrian philosopher and famous esoteric thinker who established Waldorf education or Steiner schools to treat children like unique expressions of consciousness and not minds to be programmed with the perceptions determined by authority. I'd been writing about this energy vampiring for decades when I was sent in 2016 a quote by Steiner. He was spot on:

There are beings in the spiritual realms for whom anxiety and fear emanating from human beings offer welcome food. When humans have no anxiety and fear, then these creatures starve. If fear and anxiety radiates from people and they break out in panic, then these creatures find welcome nutrition and they become more and more powerful. These beings are hostile towards humanity. Everything that feeds on negative feelings, on anxiety, fear and superstition, despair or doubt, are in reality hostile forces in super-sensible worlds, launching cruel attacks on human beings, while they are being fed ... These are exactly the feelings that belong to contemporary culture and materialism; because it estranges people from the spiritual world, it is especially suited to evoke hopelessness and fear of the unknown in people, thereby calling up the above mentioned hostile forces against them.

Pause for a moment from this perspective and reflect on what has happened in the world since the start of 2020. Not only will pennies drop, but billion dollar bills. We see the same theme from Don Juan Matus, a Yaqui Indian shaman in Mexico and the information source for Peruvian-born writer, Carlos Castaneda, who wrote a series of

books from the 1960s to 1990s. Don Juan described the force manipulating human society and his name for the Archons was the predator:

We have a predator that came from the depths of the cosmos and took over the rule of our lives. Human beings are its prisoners. The predator is our lord and master. It has rendered us docile, helpless. If we want to protest, it suppresses our protest. If we want to act independently, it demands that we don't do so ... indeed we are held prisoner!

They took us over because we are food to them, and they squeeze us mercilessly because we are their sustenance. Just as we rear chickens in coops, the predators rear us in human coops, humaneros. Therefore, their food is always available to them.

Different cultures, different eras, same recurring theme.

The 'ennoia' dilemma

Nag Hammadi Gnostic manuscripts say that Archon consciousness has no 'ennoia'. This is directly translated as 'intentionality', but I'll use the term 'creative imagination'. The *All That Is* in awareness of itself is the source of all creativity – all possibility – and the more disconnected you are from that source the more you are subsequently denied 'creative imagination'. Given that Archon consciousness is almost entirely disconnected it severely lacks creativity and has to rely on far more mechanical processes of thought and exploit the creative potential of those that do have 'ennoia'. You can see cases of this throughout human society. Archon consciousness almost entirely dominates the global banking system and if we study how that system works you will appreciate what I mean. Banks manifest 'money' out of nothing by issuing lines of 'credit' which is 'money' that has never, does not, and will never exist except in theory. It's a confidence trick. If you think 'credit' figures-on-a-screen 'money' is worth anything you accept it as payment. If you don't then the whole system collapses through lack of confidence in the value of that 'money'. Archontic bankers with no 'ennoia' are 'lending' 'money' that doesn't exist to humans that *do* have creativity – those that have the inspired ideas and create businesses and products. Archon banking feeds off human creativity

which it controls through 'money' creation and debt. Humans have the creativity and Archons exploit that for their own benefit and control while having none themselves. Archon Internet platforms like Facebook claim joint copyright of everything that creative users post and while Archontic minds like Zuckerberg may officially head that company it will be human creatives on the staff that provide the creative inspiration. When you have limitless 'money' you can then buy other companies established by creative humans. Witness the acquisition record of Facebook, Google and their like. Survey the Archon-controlled music industry and you see non-creative dark suit executives making their fortune from the human creativity of their artists. The cases are endless. Research the history of people like Gates and Zuckerberg and how their empires were built on exploiting the creativity of others. Archon minds cannot create out of nothing, but they are skilled (because they have to be) in what Gnostic texts call 'countermimicry'. They can imitate, but not innovate. Sabbatians trawl the creativity of others through backdoors they install in computer systems through their cybersecurity systems. Archon-controlled China is globally infamous for stealing intellectual property and I remember how Hong Kong, now part of China, became notorious for making counterfeit copies of the creativity of others – 'countermimicry'. With the now pervasive and all-seeing surveillance systems able to infiltrate any computer you can appreciate the potential for Archons to vampire the creativity of humans. Author John Lamb Lash wrote in his book about the Nag Hammadi texts, *Not In His Image*:

Although they cannot originate anything, because they lack the divine factor of ennoia (intentionality), Archons can imitate with a vengeance. Their expertise is simulation (HAL, virtual reality). The Demiurge [Yaldabaoth] fashions a heaven world copied from the fractal patterns [of the original] ... His construction is celestial kitsch, like the fake Italianate villa of a Mafia don complete with militant angels to guard every portal.

This brings us to something that I have been speaking about since the turn of the millennium. Our reality is a simulation; a virtual reality that we think is real. No, I'm not kidding.

Human reality? Well, virtually

I had pondered for years about whether our reality is 'real' or some kind of construct. I remembered being immensely affected on a visit as a small child in the late 1950s to the then newly-opened Planetarium on the Marylebone Road in London which is now closed and part of the adjacent Madame Tussauds wax museum. It was in the middle of the day, but when the lights went out there was the night sky projected in the Planetarium's domed ceiling and it appeared to be so real. The experience never left me and I didn't know why until around the turn of the millennium when I became certain that our 'night sky' and entire reality is a projection, a virtual reality, akin to the illusory world portrayed in the *Matrix* movies. I looked at the sky one day in this period and it appeared to me like the domed roof of the Planetarium. The release of the first *Matrix* movie in 1999 also provided a synchronistic and perfect visual representation of where my mind had been going for a long time. I hadn't come across the Gnostic Nag Hammadi texts then. When I did years later the correlation was once again astounding. As I read Gnostic accounts from 1,600 years and more earlier it was clear that they were describing the same simulation phenomenon. They tell how the Yaldabaoth 'Demiurge' and Archons created a 'bad copy' of original reality to rule over all that were captured by its illusions and the body was a prison to trap consciousness in the 'bad copy' fake reality. Read how Gnostics describe the 'bad copy' and update that to current times and they are referring to what we would call today a virtual reality simulation.

Author John Lamb Lash said 'the Demiurge fashions a heaven world copied from the fractal patterns' of the original through expertise in 'HAL' or virtual reality simulation. Fractal patterns are part of the energetic information construct of our reality, a sort of blueprint. If these patterns were copied in computer terms it would indeed give you a copy of a 'natural' reality in a non-natural frequency and digital form. The principle is the same as making a copy of a website. The original website still exists, but now you can change the copy version to make it whatever you like and it can

become very different to the original website. Archons have done this with our reality, a *synthetic* copy of prime reality that still exists beyond the frequency walls of the simulation. Trapped within the illusions of this synthetic Matrix, however, were and are human consciousness and other expressions of prime reality and this is why the Archons via the Cult are seeking to make the human body synthetic and give us synthetic AI minds to complete the job of turning the entire reality synthetic including what we perceive to be the natural world. To quote Kurzweil: 'Nanobots will infuse all the matter around us with information. Rocks, trees, everything will become these intelligent creatures.' Yes, *synthetic* 'creatures' just as 'Covid' and other genetically-manipulating 'vaccines' are designed to make the human body synthetic. From this perspective it is obvious why Archons and their Cult are so desperate to infuse synthetic material into every human with their 'Covid' scam.

Let there be (electromagnetic) light

Yaldabaoth, the force that created the simulation, or Matrix, makes sense of the Gnostic reference to 'The Great Architect' and its use by Cult Freemasonry as the name of its deity. The designer of the Matrix in the movies is called 'The Architect' and that trilogy is jam-packed with symbolism relating to these subjects. I have contended for years that the angry Old Testament God (Yaldabaoth) is the 'God' being symbolically 'quoted' in the opening of Genesis as 'creating the world'. This is not the creation of prime reality – it's the creation of the *simulation*. The Genesis 'God' says: 'Let there be Light: and there was light.' But what is this 'Light'? I have said for decades that the speed of light (186,000 miles per second) is not the fastest speed possible as claimed by mainstream science and is in fact the frequency walls or outer limits of the Matrix. You can't have a fastest or slowest anything within all possibility when everything is possible. The human body is encoded to operate within the speed of light or *within the simulation* and thus we see only the tiny frequency band of visible *light*. Near-death experiencers who perceive reality outside the body during temporary 'death' describe a very different

form of light and this is supported by the Nag Hammadi texts. Prime reality beyond the simulation ('Upper Aeons' to the Gnostics) is described as a realm of incredible beauty, bliss, love and harmony – a realm of 'watery light' that is so powerful 'there are no shadows'. Our false reality of Archon control, which Gnostics call the 'Lower Aeons', is depicted as a realm with a different kind of 'light' and described in terms of chaos, 'Hell', 'the Abyss' and 'Outer Darkness', where trapped souls are tormented and manipulated by demons (relate that to the 'Covid' hoax alone). The watery light theme can be found in near-death accounts and it is not the same as *simulation* 'light' which is electromagnetic or radiation light within the speed of light – the 'Lower Aeons'. Simulation 'light' is the 'luminous fire' associated by Gnostics with the Archons. The Bible refers to Yaldabaoth as 'that old serpent, called the Devil, and Satan, which deceiveth the whole world' (Revelation 12:9). I think that making a simulated copy of prime reality ('countermimicry') and changing it dramatically while all the time manipulating humanity to believe it to be real could probably meet the criteria of deceiving the whole world. Then we come to the Cult god Lucifer – the *Light Bringer*. Lucifer is symbolic of Yaldabaoth, the bringer of radiation light that forms the bad copy simulation within the speed of light. 'He' is symbolised by the lighted torch held by the Statue of Liberty and in the name 'Illuminati'. Sabbatian-Frankism declares that Lucifer is the true god and Lucifer is the real god of Freemasonry honoured as their 'Great or Grand Architect of the Universe' (simulation).

I would emphasise, too, the way Archontic technologically-generated luminous fire of radiation has deluged our environment since I was a kid in the 1950s and changed the nature of The Field with which we constantly interact. Through that interaction technological radiation is changing us. The Smart Grid is designed to operate with immense levels of communication power with 5G expanding across the world and 6G, 7G, in the process of development. Radiation is the simulation and the Archontic manipulation system. Why wouldn't the Archon Cult wish to unleash radiation upon us to an ever-greater extreme to form

Kurzweil's 'cloud'? The plan for a synthetic human is related to the need to cope with levels of radiation beyond even anything we've seen so far. Biological humans would not survive the scale of radiation they have in their script. The Smart Grid is a technological sub-reality within the technological simulation to further disconnect five-sense perception from expanded consciousness. It's a technological prison of the mind.

Infusing the 'spirit of darkness'

A recurring theme in religion and native cultures is the manipulation of human genetics by a non-human force and most famously recorded as the biblical 'sons of god' (the gods plural in the original) who interbred with the daughters of men. The Nag Hammadi *Apocryphon of John* tells the same story this way:

He [Yaldabaoth] sent his angels [Archons/demons] to the daughters of men, that they might take some of them for themselves and raise offspring for their enjoyment. And at first they did not succeed. When they had no success, they gathered together again and they made a plan together ... And the angels changed themselves in their likeness into the likeness of their mates, filling them with the spirit of darkness, which they had mixed for them, and with evil ... And they took women and begot children out of the darkness according to the likeness of their spirit.

Possession when a discarnate entity takes over a human body is an age-old theme and continues today. It's very real and I've seen it. Satanic and secret society rituals can create an energetic environment in which entities can attach to initiates and I've heard many stories of how people have changed their personality after being initiated even into lower levels of the Freemasons. I have been inside three Freemasonic temples, one at a public open day and two by just walking in when there was no one around to stop me. They were in Ryde, the town where I live, Birmingham, England, when I was with a group, and Boston, Massachusetts. They all felt the same energetically – dark, dense, low-vibrational and sinister. Demonic attachment can happen while the initiate has no idea what is going on. To them it's just a ritual to get in the Masons and do a bit of good

business. In the far more extreme rituals of Satanism human possession is even more powerful and they are designed to make possession possible. The hierarchy of the Cult is dictated by the power and perceived status of the possessing Archon. In this way the Archon hierarchy becomes the Cult hierarchy. Once the entity has attached it can influence perception and behaviour and if it attaches to the extreme then so much of its energy (information) infuses into the body information field that the hologram starts to reflect the nature of the possessing entity. This is the *Exorcist* movie type of possession when facial features change and it's known as shapeshifting. Islam's Jinn are said to be invisible tricksters who change shape, 'whisper', confuse and take human form. These are all traits of the Archons and other versions of the same phenomenon. Extreme possession could certainly infuse the 'spirit of darkness' into a partner during sex as the Nag Hammadi texts appear to describe. Such an infusion can change genetics which is also energetic information. Human genetics is information and the 'spirit of darkness' is information. Mix one with the other and change must happen. Islam has the concept of a 'Jinn baby' through possession of the mother and by Jinn taking human form. There are many ways that human genetics can be changed and remember that Archons have been aware all along of advanced techniques to do this. What is being done in human society today – and far more – was known about by Archons at the time of the 'fallen ones' and their other versions described in religions and cultures.

Archons and their human-world Cult are obsessed with genetics as we see today and they know this dictates how information is processed into perceived reality during a human life. They needed to produce a human form that would decode the simulation and this is symbolically known as 'Adam and Eve' who left the 'garden' (prime reality) and 'fell' into Matrix reality. The simulation is not a 'physical' construct (there is no 'physical'); it is a source of information. Think Wi-Fi again. The simulation is an energetic field encoded with information and body-brain systems are designed to decode that information encoded in wave or frequency form which

is transmitted to the brain as electrical signals. These are decoded by the brain to construct our sense of reality – an illusory ‘physical’ world that only exists in the brain or the mind. Virtual reality games mimic this process using the same sensory decoding system. Information is fed to the senses to decode a virtual reality that can appear so real, but isn’t (Figs 18 and 19). Some scientists believe – and I agree with them – that what we perceive as ‘physical’ reality only exists when we are looking or observing. The act of perception or focus triggers the decoding systems which turn waveform information into holographic reality. When we are not observing something our reality reverts from a holographic state to a waveform state. This relates to the same principle as a falling tree not making a noise unless someone is there to hear it or decode it. The concept makes sense from the simulation perspective. A computer is not decoding all the information in a Wi-Fi field all the time and only decodes or brings into reality on the screen that part of Wi-Fi that it’s decoding – focusing upon – at that moment.



Figure 18: Virtual reality technology ‘hacks’ into the body’s five-sense decoding system.



Figure 19: The result can be experienced as very ‘real’.

Interestingly, Professor Donald Hoffman at the Department of Cognitive Sciences at the University of California, Irvine, says that our experienced reality is like a computer interface that shows us only the level with which we interact while hiding all that exists beyond it: 'Evolution shaped us with a user interface that hides the truth. Nothing that we see is the truth – the very language of space and time and objects is the wrong language to describe reality.' He is correct in what he says on so many levels. Space and time are not a universal reality. They are a phenomenon of decoded *simulation* reality as part of the process of enslaving our sense of reality. Near-death experiencers report again and again how space and time did not exist as we perceive them once they were free of the body – body decoding systems. You can appreciate from this why Archons and their Cult are so desperate to entrap human attention in the five senses where we are in the Matrix and of the Matrix. Opening your mind to expanded states of awareness takes you beyond the information confines of the simulation and you become aware of knowledge and insights denied to you before. This is what we call 'awakening' – *awakening from the Matrix* – and in the final chapter I will relate this to current events.

Where are the 'aliens'?

A simulation would explain the so-called 'Fermi Paradox' named after Italian physicist Enrico Fermi (1901-1954) who created the first nuclear reactor. He considered the question of why there is such a lack of extraterrestrial activity when there are so many stars and planets in an apparently vast universe; but what if the night sky that we see, or think we do, is a simulated projection as I say? If you control the simulation and your aim is to hold humanity fast in essential ignorance would you want other forms of life including advanced life coming and going sharing information with humanity? Or would you want them to believe they were isolated and apparently alone? Themes of human isolation and apartness are common whether they be the perception of a lifeless universe or the fascist isolation laws of the 'Covid' era. Paradoxically the very

existence of a simulation means that we are not alone when some force had to construct it. My view is that experiences that people have reported all over the world for centuries with Reptilians and Grey entities are Archon phenomena as Nag Hammadi texts describe; and that benevolent 'alien' interactions are non-human groups that come in and out of the simulation by overcoming Archon attempts to keep them out. It should be highlighted, too, that Reptilians and Greys are obsessed with *genetics* and *technology* as related by cultural accounts and those who say they have been abducted by them. Technology is their way of overcoming some of the limitations in their creative potential and our technology-driven and controlled human society of today is *archetypical* Archon-Reptilian-Grey *modus operandi*. Technocracy is really *Archontocracy*. The Universe does not have to be as big as it appears with a simulation. There is no space or distance only information decoded into holographic reality. What we call 'space' is only the absence of holographic 'objects' and that 'space' is The Field of energetic information which connects everything into a single whole. The same applies with the artificially-generated information field of the simulation. The Universe is not big or small as a physical reality. It is decoded information, that's all, and its perceived size is decided by the way the simulation is encoded to make it appear. The entire night sky as we perceive it only exists in our brain and so where are those 'millions of light years'? The 'stars' on the ceiling of the Planetarium looked a vast distance away.

There's another point to mention about 'aliens'. I have been highlighting since the 1990s the plan to stage a fake 'alien invasion' to justify the centralisation of global power and a world military. Nazi scientist Werner von Braun, who was taken to America by Operation Paperclip after World War Two to help found NASA, told his American assistant Dr Carol Rosin about the Cult agenda when he knew he was dying in 1977. Rosin said that he told her about a sequence that would lead to total human control by a one-world government. This included threats from terrorism, rogue nations, meteors and asteroids before finally an 'alien invasion'. All of these

things, von Braun said, would be bogus and what I would refer to as a No-Problem-Reaction-Solution. Keep this in mind when 'the aliens are coming' is the new mantra. The aliens are not coming – they are *already here* and they have infiltrated human society while looking human. French-Canadian investigative journalist Serge Monast said in 1994 that he had uncovered a NASA/military operation called Project Blue Beam which fits with what Werner von Braun predicted. Monast died of a 'heart attack' in 1996 the day after he was arrested and spent a night in prison. He was 51. He said Blue Beam was a plan to stage an alien invasion that would include religious figures beamed holographically into the sky as part of a global manipulation to usher in a 'new age' of worshipping what I would say is the Cult 'god' Yaldabaoth in a one-world religion. Fake holographic asteroids are also said to be part of the plan which again syncs with von Braun. How could you stage an illusory threat from asteroids unless they were holographic inserts? This is pretty straightforward given the advanced technology outside the public arena and the fact that our 'physical' reality is holographic anyway. Information fields would be projected and we would decode them into the illusion of a 'physical' asteroid. If they can sell a global 'pandemic' with a 'virus' that doesn't exist what will humans not believe if government and media tell them?

All this is particularly relevant as I write with the Pentagon planning to release in June, 2021, information about 'UFO sightings'. I have been following the UFO story since the early 1990s and the common theme throughout has been government and military denials and cover up. More recently, however, the Pentagon has suddenly become more talkative and apparently open with Air Force pilot radar images released of unexplained craft moving and changing direction at speeds well beyond anything believed possible with human technology. Then, in March, 2021, former Director of National Intelligence John Ratcliffe said a Pentagon report months later in June would reveal a great deal of information about UFO sightings unknown to the public. He said the report would have 'massive implications'. The order to do this was included bizarrely

in a \$2.3 trillion 'coronavirus' relief and government funding bill passed by the Trump administration at the end of 2020. I would add some serious notes of caution here. I have been pointing out since the 1990s that the US military and intelligence networks have long had craft – 'flying saucers' or anti-gravity craft – which any observer would take to be extraterrestrial in origin. Keeping this knowledge from the public allows craft flown by *humans* to be perceived as alien visitations. I am not saying that 'aliens' do not exist. I would be the last one to say that, but we have to be streetwise here. President Ronald Reagan told the UN General Assembly in 1987: 'I occasionally think how quickly our differences worldwide would vanish if we were facing an alien threat from outside this world.' That's the idea. Unite against a common 'enemy' with a common purpose behind your 'saviour force' (the Cult) as this age-old technique of mass manipulation goes global.

Science moves this way ...

I could find only one other person who was discussing the simulation hypothesis publicly when I concluded it was real. This was Nick Bostrom, a Swedish-born philosopher at the University of Oxford, who has explored for many years the possibility that human reality is a computer simulation although his version and mine are not the same. Today the simulation and holographic reality hypothesis have increasingly entered the scientific mainstream. Well, the more open-minded mainstream, that is. Here are a few of the ever-gathering examples. American nuclear physicist Silas Beane led a team of physicists at the University of Bonn in Germany pursuing the question of whether we live in a simulation. They concluded that we probably do and it was likely based on a lattice of cubes. They found that cosmic rays align with that specific pattern. The team highlighted the Greisen–Zatsepin–Kuzmin (GZK) limit which refers to cosmic ray particle interaction with cosmic background radiation that creates an apparent boundary for cosmic ray particles. They say in a paper entitled 'Constraints on the Universe as a Numerical Simulation' that this 'pattern of constraint' is exactly what you

would find with a computer simulation. They also made the point that a simulation would create its own 'laws of physics' that would limit possibility. I've been making the same point for decades that the *perceived* laws of physics relate only to this reality, or what I would later call the simulation. When designers write codes to create computer and virtual reality games they are the equivalent of the laws of physics for that game. Players interact within the limitations laid out by the coding. In the same way those who wrote the codes for the simulation decided the laws of physics that would apply. These can be overridden by expanded states of consciousness, but not by those enslaved in only five-sense awareness where simulation codes rule. Overriding the codes is what people call 'miracles'. They are not. They are bypassing the encoded limits of the simulation. A population caught in simulation perception would have no idea that this was their plight. As the Bonn paper said: 'Like a prisoner in a pitch-black cell we would not be able to see the "walls" of our prison,' That's true if people remain mesmerised by the five senses. Open to expanded awareness and those walls become very clear. The main one is the speed of light.

American theoretical physicist James Gates is another who has explored the simulation question and found considerable evidence to support the idea. Gates was Professor of Physics at the University of Maryland, Director of The Center for String and Particle Theory, and on Barack Obama's Council of Advisors on Science and Technology. He and his team found *computer codes* of digital data embedded in the fabric of our reality. They relate to on-off electrical charges of 1 and 0 in the binary system used by computers. 'We have no idea what they are doing there', Gates said. They found within the energetic fabric mathematical sequences known as error-correcting codes or block codes that 'reboot' data to its original state or 'default settings' when something knocks it out of sync. Gates was asked if he had found a set of equations embedded in our reality indistinguishable from those that drive search engines and browsers and he said: 'That is correct.' Rich Terrile, director of the Centre for Evolutionary Computation and Automated Design at NASA's Jet

Propulsion Laboratory, has said publicly that he believes the Universe is a digital hologram that must have been created by a form of intelligence. I agree with that in every way. Waveform information is delivered electrically by the senses to the brain which constructs a *digital* holographic reality that we call the 'world'. This digital level of reality can be read by the esoteric art of numerology. Digital holograms are at the cutting edge of holographics today. We have digital technology everywhere designed to access and manipulate our digital level of perceived reality. Synthetic mRNA in 'Covid vaccines' has a digital component to manipulate the body's digital 'operating system'.

Reality is numbers

How many know that our reality can be broken down to numbers and codes that are the same as computer games? Max Tegmark, a physicist at the Massachusetts Institute of Technology (MIT), is the author of *Our Mathematical Universe* in which he lays out how reality can be entirely described by numbers and maths in the way that a video game is encoded with the 'physics' of computer games. Our world and computer virtual reality are essentially the same. Tegmark imagines the perceptions of characters in an advanced computer game when the graphics are so good they don't know they are in a game. They think they can bump into real objects (electromagnetic resistance in our reality), fall in love and feel emotions like excitement. When they began to study the apparently 'physical world' of the video game they would realise that everything was made of pixels (which have been found in our energetic reality as must be the case when on one level our world is digital). What computer game characters thought was physical 'stuff', Tegmark said, could actually be broken down into numbers:

And we're exactly in this situation in our world. We look around and it doesn't seem that mathematical at all, but everything we see is made out of elementary particles like quarks and electrons. And what properties does an electron have? Does it have a smell or a colour or a texture? No! ... We physicists have come up with geeky names for [Electron] properties, like

electric charge, or spin, or lepton number, but the electron doesn't care what we call it, the properties are just numbers.

This is the illusory reality Gnostics were describing. This is the simulation. The A, C, G, and T codes of DNA have a binary value – A and C = 0 while G and T = 1. This has to be when the simulation is digital and the body must be digital to interact with it. Recurring mathematical sequences are encoded throughout reality and the body. They include the Fibonacci sequence in which the two previous numbers are added to get the next one, as in ... 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, etc. The sequence is encoded in the human face and body, proportions of animals, DNA, seed heads, pine cones, trees, shells, spiral galaxies, hurricanes and the number of petals in a flower. The list goes on and on. There are fractal patterns – a 'never-ending pattern that is infinitely complex and self-similar across all scales in the as above, so below, principle of holograms. These and other famous recurring geometrical and mathematical sequences such as Phi, Pi, Golden Mean, Golden Ratio and Golden Section are *computer codes* of the simulation. I had to laugh and give my head a shake the day I finished this book and it went into the production stage. I was sent an article in *Scientific American* published in April, 2021, with the headline 'Confirmed! We Live in a Simulation'. Two decades after I first said our reality is a simulation and the speed of light is its outer limit the article suggested that we do live in a simulation and that the speed of light is its outer limit. I left school at 15 and never passed a major exam in my life while the writer was up to his eyes in qualifications. As I will explain in the final chapter *knowing* is far better than thinking and they come from very different sources. The article rightly connected the speed of light to the processing speed of the 'Matrix' and said what has been in my books all this time ... 'If we are in a simulation, as it appears, then space is an abstract property written in code. It is not real'. No it's not and if we live in a simulation something created it and it wasn't *us*. 'That David Icke says we are manipulated by aliens' – he's crackers.'

Wow ...

The reality that humanity thinks is so real is an illusion. Politicians, governments, scientists, doctors, academics, law enforcement, media, school and university curriculums, on and on, are all founded on a world that *does not exist* except as a simulated prison cell. Is it such a stretch to accept that 'Covid' doesn't exist when our entire 'physical' reality doesn't exist? Revealed here is the knowledge kept under raps in the Cult networks of compartmentalised secrecy to control humanity's sense of reality by inducing the population to believe in a reality that's not real. If it wasn't so tragic in its experiential consequences the whole thing would be hysterically funny. None of this is new to Renegade Minds. Ancient Greek philosopher Plato (about 428 to about 347BC) was a major influence on Gnostic belief and he described the human plight thousands of years ago with his Allegory of the Cave. He told the symbolic story of prisoners living in a cave who had never been outside. They were chained and could only see one wall of the cave while behind them was a fire that they could not see. Figures walked past the fire casting shadows on the prisoners' wall and those moving shadows became their sense of reality. Some prisoners began to study the shadows and were considered experts on them (today's academics and scientists), but what they studied was only an illusion (today's academics and scientists). A prisoner escaped from the cave and saw reality as it really is. When he returned to report this revelation they didn't believe him, called him mad and threatened to kill him if he tried to set them free. Plato's tale is not only a brilliant analogy of the human plight and our illusory reality. It describes, too, the dynamics of the 'Covid' hoax. I have only skimmed the surface of these subjects here. The aim of this book is to crisply connect all essential dots to put what is happening today into its true context. All subject areas and their connections in this chapter are covered in great evidential detail in *Everything You Need To Know, But Have Never Been Told* and *The Answer*.

They say that bewildered people 'can't see the forest for the trees'. Humanity, however, can't see the forest for the *twigs*. The five senses

see only twigs while Renegade Minds can see the forest and it's the forest where the answers lie with the connections that reveals.

Breaking free of perceptual programming so the forest can be seen is the way we turn all this around. Not breaking free is how humanity got into this mess. The situation may seem hopeless, but I promise you it's not. We are a perceptual heartbeat from paradise if only we knew.

CHAPTER TWELVE

Escaping Wetiko

Life is simply a vacation from the infinite
Dean Cavanagh

Renegade Minds weave the web of life and events and see common themes in the apparently random. They are always there if you look for them and their pursuit is aided by incredible synchronicity that comes when your mind is open rather than mesmerised by what it thinks it can see.

Infinite awareness is infinite possibility and the more of infinite possibility that we access the more becomes infinitely possible. That may be stating the apparently obvious, but it is a devastatingly-powerful fact that can set us free. We are a point of attention within an infinity of consciousness. The question is how much of that infinity do we choose to access? How much knowledge, insight, awareness, wisdom, do we want to connect with and explore? If your focus is only in the five senses you will be influenced by a fraction of infinite awareness. I mean a range so tiny that it gives new meaning to infinitesimal. Limitation of self-identity and a sense of the possible limit accordingly your range of consciousness. We are what we think we are. Life is what we think it is. The dream is the dreamer and the dreamer is the dream. Buddhist philosophy puts it this way: 'As a thing is viewed, so it appears.' Most humans live in the realm of touch, taste, see, hear, and smell and that's the limit of their sense of the possible and sense of self. Many will follow a religion and speak of a God in his heaven, but their lives are still

dominated by the five senses in their perceptions and actions. The five senses become the arbiter of everything. When that happens all except a smear of infinity is sealed away from influence by the rigid, unyielding, reality bubbles that are the five-sense human or Phantom Self. Archon Cult methodology is to isolate consciousness within five-sense reality – the simulation – and then program that consciousness with a sense of self and the world through a deluge of life-long information designed to instil the desired perception that allows global control. Efforts to do this have increased dramatically with identity politics as identity bubbles are squeezed into the minutiae of five-sense detail which disconnect people even more profoundly from the infinite 'I'.

Five-sense focus and self-identity are like a firewall that limits access to the infinite realms. You only perceive one radio or television station and no other. We'll take that literally for a moment. Imagine a vast array of stations giving different information and angles on reality, but you only ever listen to one. Here we have the human plight in which the population is overwhelmingly confined to CultFM. This relates only to the frequency range of CultFM and limits perception and insight to that band – limits *possibility* to that band. It means you are connecting with an almost imperceptibly minuscule range of possibility and creative potential within the infinite Field. It's a world where everything seems apart from everything else and where synchronicity is rare. Synchronicity is defined in the dictionary as 'the happening by chance of two or more related or similar events at the same time'. Use of 'by chance' betrays a complete misunderstanding of reality. Synchronicity is not 'by chance'. As people open their minds, or 'awaken' to use the term, they notice more and more coincidences in their lives, bits of 'luck', apparently miraculous happenings that put them in the right place at the right time with the right people. Days become peppered with 'fancy meeting you here' and 'what are the chances of that?' My entire life has been lived like this and ever more so since my own colossal awakening in 1990 and 91 which transformed my sense of reality. Synchronicity is not 'by chance'; it is by accessing expanded

realms of possibility which allow expanded potential for manifestation. People broadcasting the same vibe from the same openness of mind tend to be drawn 'by chance' to each other through what I call frequency magnetism and it's not only people. In the last more than 30 years incredible synchronicity has also led me through the Cult maze to information in so many forms and to crucial personal experiences. These 'coincidences' have allowed me to put the puzzle pieces together across an enormous array of subjects and situations. Those who have breached the bubble of five-sense reality will know exactly what I mean and this escape from the perceptual prison cell is open to everyone whenever they make that choice. This may appear super-human when compared with the limitations of 'human', but it's really our natural state. 'Human' as currently experienced is consciousness in an unnatural state of induced separation from the infinity of the whole. I'll come to how this transformation into unity can be made when I have described in more detail the force that holds humanity in servitude by denying this access to infinite self.

The Wetiko factor

I have been talking and writing for decades about the way five-sense mind is systematically barricaded from expanded awareness. I have used the analogy of a computer (five-sense mind) and someone at the keyboard (expanded awareness). Interaction between the computer and the operator is symbolic of the interaction between five-sense mind and expanded awareness. The computer directly experiences the Internet and the operator experiences the Internet via the computer which is how it's supposed to be – the two working as one. Archons seek to control that point where the operator connects with the computer to stop that interaction ([Fig 20](#)). Now the operator is banging the keyboard and clicking the mouse, but the computer is not responding and this happens when the computer is taken over – *possessed* – by an appropriately-named computer 'virus'. The operator has lost all influence over the computer which goes its own way making decisions under the control of the 'virus'. I have

just described the dynamic through which the force known to Gnostics as Yaldabaoth and Archons disconnects five-sense mind from expanded awareness to imprison humanity in perceptual servitude.



Figure 20: The mind ‘virus’ I have been writing about for decades seeks to isolate five-sense mind (the computer) from the true ‘I’. (Image by Neil Hague).

About a year ago I came across a Native American concept of Wetiko which describes precisely the same phenomenon. Wetiko is the spelling used by the Cree and there are other versions including wintiko and windigo used by other tribal groups. They spell the name with lower case, but I see Wetiko as a proper noun as with Archons and prefer a capital. I first saw an article about Wetiko by writer and researcher Paul Levy which so synced with what I had been writing about the computer/operator disconnection and later the Archons. I then read his book, the fascinating *Dispelling Wetiko, Breaking the Spell of Evil*. The parallels between what I had concluded long before and the Native American concept of Wetiko were so clear and obvious that it was almost funny. For Wetiko see the Gnostic Archons for sure and the Jinn, the Predators, and every other name for a force of evil, inversion and chaos. Wetiko is the Native American name for the force that divides the computer from

the operator (Fig 21). Indigenous author Jack D. Forbes, a founder of the Native American movement in the 1960s, wrote another book about Wetiko entitled *Columbus And Other Cannibals – The Wetiko Disease of Exploitation, Imperialism, and Terrorism* which I also read. Forbes says that Wetiko refers to an evil person or spirit ‘who terrorizes other creatures by means of terrible acts, including cannibalism’. Zulu shaman Credo Mutwa told me that African accounts tell how cannibalism was brought into the world by the Chitauri ‘gods’ – another manifestation of Wetiko. The distinction between ‘evil person or spirit’ relates to Archons/Wetiko possessing a human or acting as pure consciousness. Wetiko is said to be a sickness of the soul or spirit and a state of being that takes but gives nothing back – the Cult and its operatives perfectly described. Black Hawk, a Native American war leader defending their lands from confiscation, said European invaders had ‘poisoned hearts’ – Wetiko hearts – and that this would spread to native societies. Mention of the heart is very significant as we shall shortly see. Forbes writes: ‘Tragically, the history of the world for the past 2,000 years is, in great part, the story of the epidemiology of the wetiko disease.’ Yes, and much longer. Forbes is correct when he says: ‘The wetikos destroyed Egypt and Babylon and Athens and Rome and Tenochtitlan [capital of the Aztec empire] and perhaps now they will destroy the entire earth.’ Evil, he said, is the number one export of a Wetiko culture – see its globalisation with ‘Covid’. Constant war, mass murder, suffering of all kinds, child abuse, Satanism, torture and human sacrifice are all expressions of Wetiko and the Wetiko possessed. The world is Wetiko made manifest, *but it doesn’t have to be*. There is a way out of this even now.



Figure 21: The mind 'virus' is known to Native Americans as 'Wetiko'. (Image by Neil Hague).

Cult of Wetiko

Wetiko is the Yaldabaoth frequency distortion that seeks to attach to human consciousness and absorb it into its own. Once this connection is made Wetiko can drive the perceptions of the target which they believe to be coming from their own mind. All the horrors of history and today from mass killers to Satanists, paedophiles like Jeffrey Epstein and other psychopaths, are the embodiment of Wetiko and express its state of being in all its grotesqueness. The Cult is Wetiko incarnate, Yaldabaoth incarnate, and it seeks to facilitate Wetiko assimilation of humanity in totality into its distortion by manipulating the population into low frequency states that match its own. Paul Levy writes:

'Holographically enforced within the psyche of every human being the wetiko virus pervades and underlies the entire field of consciousness, and can therefore potentially manifest through any one of us at any moment if we are not mindful.' The 'Covid' hoax has achieved this with many people, but others have not fallen into Wetiko's frequency lair. Players in the 'Covid' human catastrophe including Gates, Schwab, Tedros, Fauci, Whitty, Vallance, Johnson, Hancock, Ferguson, Drosten, and all the rest, including the psychopath psychologists, are expressions of Wetiko. This is why

they have no compassion or empathy and no emotional consequence for what they do that would make them stop doing it. Observe all the people who support the psychopaths in authority against the Pushbackers despite the damaging impact the psychopaths have on their own lives and their family's lives. You are again looking at Wetiko possession which prevents them seeing through the lies to the obvious scam going on. *Why can't they see it?* Wetiko won't let them see it. The perceptual divide that has now become a chasm is between the Wetikoed and the non-Wetikoed.

Paul Levy describes Wetiko in the same way that I have long described the Archontic force. They are the same distorted consciousness operating across dimensions of reality: '... the subtle body of wetiko is not located in the third dimension of space and time, literally existing in another dimension ... it is able to affect ordinary lives by mysteriously interpenetrating into our three-dimensional world.' Wetiko does this through its incarnate representatives in the Cult and by weaving itself into The Field which on our level of reality is the electromagnetic information field of the simulation or Matrix. More than that, the simulation *is* Wetiko / Yaldabaoth. Caleb Scharf, Director of Astrobiology at Columbia University, has speculated that 'alien life' could be so advanced that it has transcribed itself into the quantum realm to become what we call physics. He said intelligence indistinguishable from the fabric of the Universe would solve many of its greatest mysteries:

Perhaps hyper-advanced life isn't just external. Perhaps it's already all around. It is embedded in what we perceive to be physics itself, from the root behaviour of particles and fields to the phenomena of complexity and emergence ... In other words, life might not just be in the equations. It might BE the equations [My emphasis].

Scharf said it is possible that 'we don't recognise advanced life because it forms an integral and unsuspecting part of what we've considered to be the natural world'. I agree. Wetiko/Yaldabaoth *is* the simulation. We are literally in the body of the beast. But that doesn't mean it has to control us. We all have the power to overcome Wetiko

influence and the Cult knows that. I doubt it sleeps too well because it knows that.

Which Field?

This, I suggest, is how it all works. There are two Fields. One is the fierce electromagnetic light of the Matrix within the speed of light; the other is the 'watery light' of The Field beyond the walls of the Matrix that connects with the Great Infinity. Five-sense mind and the decoding systems of the body attach us to the Field of Matrix light. They have to or we could not experience this reality. Five-sense mind sees only the Matrix Field of information while our expanded consciousness is part of the Infinity Field. When we open our minds, and most importantly our hearts, to the Infinity Field we have a mission control which gives us an expanded perspective, a road map, to understand the nature of the five-sense world. If we are isolated only in five-sense mind there is no mission control. We're on our own trying to understand a world that's constantly feeding us information to ensure we do not understand. People in this state can feel 'lost' and bewildered with no direction or radar. You can see ever more clearly those who are influenced by the Fields of Big Infinity or little five-sense mind simply by their views and behaviour with regard to the 'Covid' hoax. We have had this division throughout known human history with the mass of the people on one side and individuals who could see and intuit beyond the walls of the simulation – Plato's prisoner who broke out of the cave and saw reality for what it is. Such people have always been targeted by Wetiko/Archon-possessed authority, burned at the stake or demonised as mad, bad and dangerous. The Cult today and its global network of 'anti-hate', 'anti-fascist' Woke groups are all expressions of Wetiko attacking those exposing the conspiracy, 'Covid' lies and the 'vaccine' agenda.

Woke as a whole is Wetiko which explains its black and white mentality and how at one it is with the Wetiko-possessed Cult. Paul Levy said: 'To be in this paradigm is to still be under the thrall of a two-valued logic – where things are either true or false – of a

wetikoized mind.’ Wetiko consciousness is in a permanent rage, therefore so is Woke, and then there is Woke inversion and contradiction. ‘Anti-fascists’ act like fascists because fascists *and* ‘anti-fascists’ are both Wetiko at work. Political parties act the same while claiming to be different for the same reason. Secret society and satanic rituals are attaching initiates to Wetiko and the cold, ruthless, psychopathic mentality that secures the positions of power all over the world is Wetiko. Reframing ‘training programmes’ have the same cumulative effect of attaching Wetiko and we have their graduates described as automatons and robots with a cold, psychopathic, uncaring demeanour. They are all traits of Wetiko possession and look how many times they have been described in this book and elsewhere with regard to personnel behind ‘Covid’ including the police and medical profession. Climbing the greasy pole in any profession in a Wetiko society requires traits of Wetiko to get there and that is particularly true of politics which is not about fair competition and pre-eminence of ideas. It is founded on how many backs you can stab and arses you can lick. This culminated in the global ‘Covid’ coordination between the Wetiko possessed who pulled it off in all the different countries without a trace of empathy and compassion for their impact on humans. Our sight sense can see only holographic form and not the Field which connects holographic form. Therefore we perceive ‘physical’ objects with ‘space’ in between. In fact that ‘space’ is energy/consciousness operating on multiple frequencies. One of them is Wetiko and that connects the Cult psychopaths, those who submit to the psychopaths, and those who serve the psychopaths in the media operations of the world. Wetiko is Gates. Wetiko is the mask-wearing submissive. Wetiko is the fake journalist and ‘fact-checker’. The Wetiko Field is coordinating the whole thing. Psychopaths, gofers, media operatives, ‘anti-hate’ hate groups, ‘fact-checkers’ and submissive people work as one unit *even without human coordination* because they are attached to the *same* Field which is organising it all ([Fig 22](#)). Paul Levy is here describing how Wetiko-possessed people are drawn together and refuse to let any information breach their rigid

perceptions. He was writing long before 'Covid', but I think you will recognise followers of the 'Covid' religion *oh just a little bit*:

People who are channelling the vibratory frequency of wetiko align with each other through psychic resonance to reinforce their unspoken shared agreement so as to uphold their deranged view of reality. Once an unconscious content takes possession of certain individuals, it irresistibly draws them together by mutual attraction and knits them into groups tied together by their shared madness that can easily swell into an avalanche of insanity.

A psychic epidemic is a closed system, which is to say that it is insular and not open to any new information or informing influences from the outside world which contradict its fixed, limited, and limiting perspective.

There we have the Woke mind and the 'Covid' mind. Compatible resonance draws the awakening together, too, which is clearly happening today.

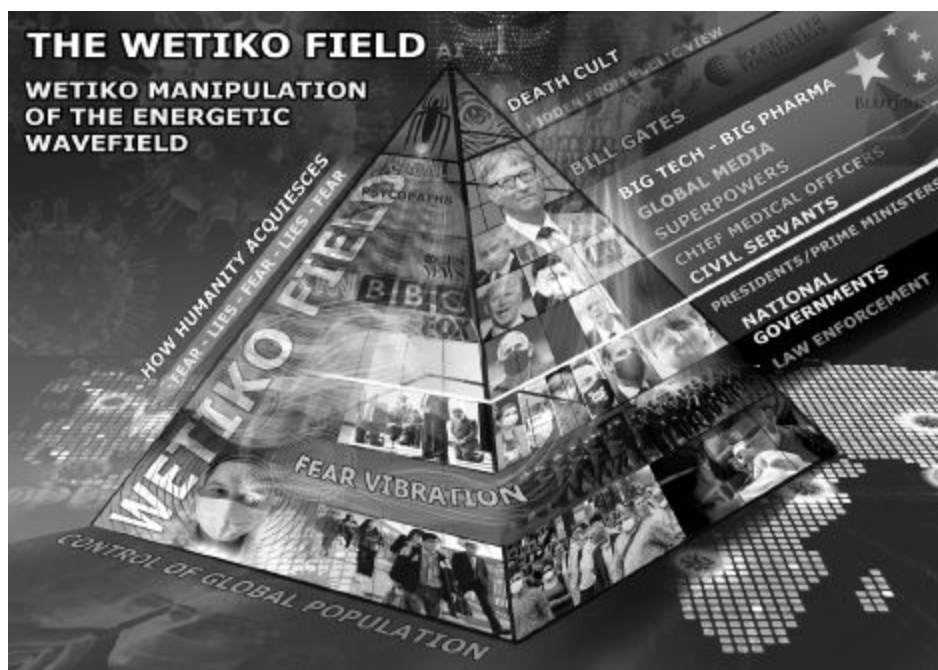


Figure 22: The Wetiko Field from which the Cult pyramid and its personnel are made manifest. (Image by Neil Hague).

Spiritual servitude

Wetiko doesn't care about humans. It's not human; it just possesses humans for its own ends and the effect (depending on the scale of

possession) can be anything from extreme psychopathy to unquestioning obedience. Wetiko's worst nightmare is for human consciousness to expand beyond the simulation. Everything is focussed on stopping that happening through control of information, thus perception, thus frequency. The 'education system', media, science, medicine, academia, are all geared to maintaining humanity in five-sense servitude as is the constant stimulation of low-vibrational mental and emotional states (see 'Covid'). Wetiko seeks to dominate those subconscious spaces between five-sense perception and expanded consciousness where the computer meets the operator. From these subconscious hiding places Wetiko speaks to us to trigger urges and desires that we take to be our own and manipulate us into anything from low-vibrational to psychopathic states. Remember how Islam describes the Jinn as invisible tricksters that 'whisper' and confuse. Wetiko is the origin of the 'trickster god' theme that you find in cultures all over the world. Jinn, like the Archons, are Wetiko which is terrified of humans awakening and reconnecting with our true self for then its energy source has gone. With that the feedback loop breaks between Wetiko and human perception that provides the energetic momentum on which its very existence depends as a force of evil. Humans are both its target and its source of survival, but only if we are operating in low-vibrational states of fear, hate, depression and the background anxiety that most people suffer. We are Wetiko's target because we are its key to survival. It needs us, not the other way round. Paul Levy writes:

A vampire has no intrinsic, independent, substantial existence in its own right; it only exists in relation to us. The pathogenic, vampiric mind-parasite called wetiko is nothing in itself – not being able to exist from its own side – yet it has a 'virtual reality' such that it can potentially destroy our species ...

...The fact that a vampire is not reflected by a mirror can also mean that what we need to see is that there's nothing, no-thing to see, other than ourselves. The fact that wetiko is the expression of something inside of us means that the cure for wetiko is with us as well. The critical issue is finding this cure within us and then putting it into effect.

Evil begets evil because if evil does not constantly expand and find new sources of energetic sustenance its evil, its *distortion*, dies with the assimilation into balance and harmony. Love is the garlic to Wetiko's vampire. Evil, the absence of love, cannot exist in the presence of love. I think I see a way out of here. I have emphasised so many times over the decades that the Archons/Wetiko and their Cult are not all powerful. *They are not*. I don't care how it looks even now *they are not*. I have not called them little boys in short trousers for effect. I have said it because it is true. Wetiko's insatiable desire for power over others is not a sign of its omnipotence, but its insecurity. Paul Levy writes: 'Due to the primal fear which ultimately drives it and which it is driven to cultivate, wetiko's body politic has an intrinsic and insistent need for centralising power and control so as to create imagined safety for itself.' Yeeeeees! Exactly! Why does Wetiko want humans in an ongoing state of fear? Wetiko itself *is* fear and it is petrified of love. As evil is an absence of love, so love is an absence of fear. Love conquers all and *especially* Wetiko which *is* fear. Wetiko brought fear into the world when it wasn't here before. *Fear* was the 'fall', the fall into low-frequency ignorance and illusion – fear is **False Emotion Appearing Real**. The simulation is driven and energised by fear because Wetiko/Yaldabaoth (fear) *are* the simulation. Fear is the absence of love and Wetiko is the absence of love.

Wetiko today

We can now view current events from this level of perspective. The 'Covid' hoax has generated momentous amounts of ongoing fear, anxiety, depression and despair which have empowered Wetiko. No wonder people like Gates have been the instigators when they are Wetiko incarnate and exhibit every trait of Wetiko in the extreme. See how cold and unemotional these people are like Gates and his cronies, how dead of eye they are. That's Wetiko. Sabbatians are Wetiko and everything they control including the World Health Organization, Big Pharma and the 'vaccine' makers, national 'health'

hierarchies, corporate media, Silicon Valley, the banking system, and the United Nations with its planned transformation into world government. All are controlled and possessed by the Wetiko distortion into distorting human society in its image. We are with this knowledge at the gateway to understanding the world. Divisions of race, culture, creed and sexuality are diversions to hide the real division between those possessed and influenced by Wetiko and those that are not. The 'Covid' hoax has brought both clearly into view. Human behaviour is not about race. Tyrants and dictatorships come in all colours and creeds. What unites the US president bombing the innocent and an African tribe committing genocide against another as in Rwanda? What unites them? *Wetiko*. All wars are Wetiko, all genocide is Wetiko, all hunger over centuries in a world of plenty is Wetiko. Children going to bed hungry, including in the West, is Wetiko. Cult-generated Woke racial divisions that focus on the body are designed to obscure the reality that divisions in behaviour are manifestations of mind, not body. Obsession with body identity and group judgement is a means to divert attention from the real source of behaviour – mind and perception. Conflict sown by the Woke both within themselves and with their target groups are Wetiko providing lunch for itself through still more agents of the division, chaos, and fear on which it feeds. The Cult is seeking to assimilate the entirety of humanity and all children and young people into the Wetiko frequency by manipulating them into states of fear and despair. Witness all the suicide and psychological unravelling since the spring of 2020. Wetiko psychopaths want to impose a state of unquestioning obedience to authority which is no more than a conduit for Wetiko to enforce its will and assimilate humanity into itself. It needs us to believe that resistance is futile when it fears resistance and even more so the game-changing non-cooperation with its impositions. It can use violent resistance for its benefit. Violent impositions and violent resistance are *both* Wetiko. The Power of Love with its Power of No will sweep Wetiko from our world. Wetiko and its Cult know that. They just don't want us to know.

AI Wetiko

This brings me to AI or artificial intelligence and something else Wetikos don't want us to know. What is AI *really*? I know about computer code algorithms and AI that learns from data input. These, however, are more diversions, the expeditionary force, for the real AI that they want to connect to the human brain as promoted by Silicon Valley Wetikos like Kurzweil. What is this AI? It is the frequency of *Wetiko*, the frequency of the Archons. The connection of AI to the human brain is the connection of the Wetiko frequency to create a Wetiko hive mind and complete the job of assimilation. The hive mind is planned to be controlled from Israel and China which are both 100 percent owned by Wetiko Sabbatians. The assimilation process has been going on minute by minute in the 'smart' era which fused with the 'Covid' era. We are told that social media is scrambling the minds of the young and changing their personality. This is true, but what is social media? Look more deeply at how it works, how it creates divisions and conflict, the hostility and cruelty, the targeting of people until they are destroyed. That's Wetiko. Social media is manipulated to tune people to the Wetiko frequency with all the emotional exploitation tricks employed by platforms like Facebook and its Wetiko front man, Zuckerberg. Facebook's Instagram announced a new platform for children to overcome a legal bar on them using the main site. This is more Wetiko exploitation and manipulation of kids. Amnesty International likened the plan to foxes offering to guard the henhouse and said it was incompatible with human rights. Since when did Wetiko or Zuckerberg (I repeat myself) care about that? Would Brin and Page at Google, Wojcicki at YouTube, Bezos at Amazon and whoever the hell runs Twitter act as they do if they were not channelling Wetiko? Would those who are developing technologies for no other reason than human control? How about those designing and selling technologies to kill people and Big Pharma drug and 'vaccine' producers who know they will end or devastate lives? Quite a thought for these people to consider is that if you are Wetiko in a human life you are Wetiko on the 'other side' unless your frequency

changes and that can only change by a change of perception which becomes a change of behaviour. Where Gates is going does not bear thinking about although perhaps that's exactly where he wants to go. Either way, that's where he's going. His frequency will make it so.

The frequency lair

I have been saying for a long time that a big part of the addiction to smartphones and devices is that a frequency is coming off them that entraps the mind. People spend ages on their phones and sometimes even a minute or so after they put them down they pick them up again and it all repeats. 'Covid' lockdowns will have increased this addiction a million times for obvious reasons. Addictions to alcohol overindulgence and drugs are another way that Wetiko entraps consciousness to attach to its own. Both are symptoms of low-vibrational psychological distress which alcoholism and drug addiction further compound. Do we think it's really a coincidence that access to them is made so easy while potions that can take people into realms beyond the simulation are banned and illegal? I have explored smartphone addiction in other books, the scale is mind-blowing, and that level of addiction does not come without help. Tech companies that make these phones are Wetiko and they will have no qualms about destroying the minds of children. We are seeing again with these companies the Wetiko perceptual combination of psychopathic enforcers and weak and meek unquestioning compliance by the rank and file.

The global Smart Grid is the Wetiko Grid and it is crucial to complete the Cult endgame. The simulation is radiation and we are being deluged with technological radiation on a devastating scale. Wetiko frauds like Elon Musk serve Cult interests while occasionally criticising them to maintain his street-cred. 5G and other forms of Wi-Fi are being directed at the earth from space on a volume and scale that goes on increasing by the day. Elon Musk's (officially) SpaceX Starlink project is in the process of putting tens of thousands of satellites in low orbit to cover every inch of the planet with 5G and other Wi-Fi to create Kurzweil's global 'cloud' to which the

human mind is planned to be attached very soon. SpaceX has approval to operate 12,000 satellites with more than 1,300 launched at the time of writing and applications filed for 30,000 more. Other operators in the Wi-Fi, 5G, low-orbit satellite market include OneWeb (UK), Telesat (Canada), and AST & Science (US). Musk tells us that AI could be the end of humanity and then launches a company called Neuralink to connect the human brain to computers. Musk's (in theory) Tesla company is building electric cars and the driverless vehicles of the smart control grid. As frauds and bullshitters go Elon Musk in my opinion is Major League.

5G and technological radiation in general are destructive to human health, genetics and psychology and increasing the strength of artificial radiation underpins the five-sense perceptual bubbles which are themselves expressions of radiation or electromagnetism. Freedom activist John Whitehead was so right with his 'databit by databit, we are building our own electronic concentration camps'. The Smart Grid and 5G is a means to control the human mind and infuse perceptual information into The Field to influence anyone in sync with its frequency. You can change perception and behaviour en masse if you can manipulate the population into those levels of frequency and this is happening all around us today. The arrogance of Musk and his fellow Cult operatives knows no bounds in the way that we see with Gates. Musk's satellites are so many in number already they are changing the night sky when viewed from Earth. The astronomy community has complained about this and they have seen nothing yet. Some consequences of Musk's Wetiko hubris include: Radiation; visible pollution of the night sky; interference with astronomy and meteorology; ground and water pollution from intensive use of increasingly many spaceports; accumulating space debris; continual deorbiting and burning up of aging satellites, polluting the atmosphere with toxic dust and smoke; and ever-increasing likelihood of collisions. A collective public open letter of complaint to Musk said:

We are writing to you ... because SpaceX is in process of surrounding the Earth with a network of thousands of satellites whose very purpose is to irradiate every square inch of the

Earth. SpaceX, like everyone else, is treating the radiation as if it were not there. As if the mitochondria in our cells do not depend on electrons moving undisturbed from the food we digest to the oxygen we breathe.

As if our nervous systems and our hearts are not subject to radio frequency interference like any piece of electronic equipment. As if the cancer, diabetes, and heart disease that now afflict a majority of the Earth's population are not metabolic diseases that result from interference with our cellular machinery. As if insects everywhere, and the birds and animals that eat them, are not starving to death as a result.

People like Musk and Gates believe in their limitless Wetiko arrogance that they can do whatever they like to the world because they own it. Consequences for humanity are irrelevant. It's absolutely time that we stopped taking this shit from these self-styled masters of the Earth when you consider where this is going.

Why is the Cult so anti-human?

I hear this question often: Why would they do this when it will affect them, too? Ah, but will it? Who is this *them*? Forget their bodies. They are just vehicles for Wetiko consciousness. When you break it all down to the foundations we are looking at a state of severely distorted consciousness targeting another state of consciousness for assimilation. The rest is detail. The simulation is the fly-trap in which unique sensations of the five senses create a cycle of addiction called reincarnation. Renegade Minds see that everything which happens in our reality is a smaller version of the whole picture in line with the holographic principle. Addiction to the radiation of smart technology is a smaller version of addiction to the whole simulation. Connecting the body/brain to AI is taking that addiction on a giant step further to total ongoing control by assimilating human incarnate consciousness into Wetiko. I have watched during the 'Covid' hoax how many are becoming ever more profoundly attached to Wetiko's perceptual calling cards of aggressive response to any other point of view ('There is no other god but me'), psychopathic lack of compassion and empathy, and servile submission to the narrative and will of authority. Wetiko is the psychopaths *and* subservience to psychopaths. The Cult of Wetiko is

so anti-human because it is *not* human. It embarked on a mission to destroy human by targeting everything that it means to be human and to survive as human. 'Covid' is not the end, just a means to an end. The Cult with its Wetiko consciousness is seeking to change Earth systems, including the atmosphere, to suit them, not humans. The gathering bombardment of 5G alone from ground and space is dramatically changing The Field with which the five senses interact. There is so much more to come if we sit on our hands and hope it will all go away. It is not meant to go away. It is meant to get ever more extreme and we need to face that while we still can – just.

Carbon dioxide is the gas of life. Without that human is over. Kaput, gone, history. No natural world, no human. The Cult has created a cock and bull story about carbon dioxide and climate change to justify its reduction to the point where Gates and the ignoramus Biden 'climate chief' John Kerry want to suck it out of the atmosphere. Kerry wants to do this because his master Gates does. Wetikos have made the gas of life a demon with the usual support from the Wokers of Extinction Rebellion and similar organisations and the bewildered puppet-child that is Greta Thunberg who was put on the world stage by Klaus Schwab and the World Economic Forum. The name Extinction Rebellion is both ironic and as always Wetiko inversion. The gas that we need to survive must be reduced to save us from extinction. The most basic need of human is oxygen and we now have billions walking around in face nappies depriving body and brain of this essential requirement of human existence. More than that 5G at 60 gigahertz interacts with the oxygen molecule to reduce the amount of oxygen the body can absorb into the bloodstream. The obvious knock-on consequences of that for respiratory and cognitive problems and life itself need no further explanation. Psychopaths like Musk are assembling a global system of satellites to deluge the human atmosphere with this insanity. The man should be in jail. Here we have two most basic of human needs, oxygen and carbon dioxide, being dismantled.

Two others, water and food, are getting similar treatment with the United Nations Agendas 21 and 2030 – the Great Reset – planning to

centrally control all water and food supplies. People will not even own rain water that falls on their land. Food is affected at the most basic level by reducing carbon dioxide. We have genetic modification or GMO infiltrating the food chain on a mass scale, pesticides and herbicides polluting the air and destroying the soil. Freshwater fish that provide livelihoods for 60 million people and feed hundreds of millions worldwide are being 'pushed to the brink' according the conservationists while climate change is the only focus. Now we have Gates and Schwab wanting to dispense with current food sources all together and replace them with a synthetic version which the Wetiko Cult would control in terms of production and who eats and who doesn't. We have been on the Totalitarian Tiptoe to this for more than 60 years as food has become ever more processed and full of chemical shite to the point today when it's not natural food at all. As Dr Tom Cowan says: 'If it has a label don't eat it.' Bill Gates is now the biggest owner of farmland in the United States and he does nothing without an ulterior motive involving the Cult. Klaus Schwab wrote: 'To feed the world in the next 50 years we will need to produce as much food as was produced in the last 10,000 years ... food security will only be achieved, however, if regulations on genetically modified foods are adapted to reflect the reality that gene editing offers a precise, efficient and safe method of improving crops.' Liar. People and the world are being targeted with aluminium through vaccines, chemtrails, food, drink cans, and endless other sources when aluminium has been linked to many health issues including dementia which is increasing year after year. Insects, bees and wildlife essential to the food chain are being deleted by pesticides, herbicides and radiation which 5G is dramatically increasing with 6G and 7G to come. The pollinating bee population is being devastated while wildlife including birds, dolphins and whales are having their natural radar blocked by the effects of ever-increasing radiation. In the summer windscreens used to be splattered with insects so numerous were they. It doesn't happen now. Where have they gone?

Synthetic everything

The Cult is introducing genetically-modified versions of trees, plants and insects including a Gates-funded project to unleash hundreds of millions of genetically-modified, lab-altered and patented male mosquitoes to mate with wild mosquitoes and induce genetic flaws that cause them to die out. Clinically-insane Gates-funded Japanese researchers have developed mosquitos that spread vaccine and are dubbed 'flying vaccinators'. Gates is funding the modification of weather patterns in part to sell the myth that this is caused by carbon dioxide and he's funding geoengineering of the skies to change the atmosphere. Some of this came to light with the Gates-backed plan to release tonnes of chalk into the atmosphere to 'deflect the Sun and cool the planet'. Funny how they do this while the heating effect of the Sun is not factored into climate projections focussed on carbon dioxide. The reason is that they want to reduce carbon dioxide (so don't mention the Sun), but at the same time they do want to reduce the impact of the Sun which is so essential to human life and health. I have mentioned the sun-cholesterol-vitamin D connection as they demonise the Sun with warnings about skin cancer (caused by the chemicals in sun cream they tell you to splash on). They come from the other end of the process with statin drugs to reduce cholesterol that turns sunlight into vitamin D. A lack of vitamin D leads to a long list of health effects and how vitamin D levels must have fallen with people confined to their homes over 'Covid'. Gates is funding other forms of geoengineering and most importantly chemtrails which are dropping heavy metals, aluminium and self-replicating nanotechnology onto the Earth which is killing the natural world. See *Everything You Need To Know, But Have Never Been Told* for the detailed background to this.

Every human system is being targeted for deletion by a force that's not human. The Wetiko Cult has embarked on the process of transforming the human body from biological to synthetic biological as I have explained. Biological is being replaced by the artificial and synthetic – Archontic 'countermimicry' – right across human society. The plan eventually is to dispense with the human body altogether

and absorb human consciousness – which it wouldn't really be by then – into cyberspace (the simulation which is Wetiko/Yaldabaoth). Preparations for that are already happening if people would care to look. The alternative media rightly warns about globalism and 'the globalists', but this is far bigger than that and represents the end of the human race as we know it. The 'bad copy' of prime reality that Gnostics describe was a bad copy of harmony, wonder and beauty to start with before Wetiko/Yaldabaoth set out to change the simulated 'copy' into something very different. The process was slow to start with. Entrapped humans in the simulation timeline were not technologically aware and they had to be brought up to intellectual speed while being suppressed spiritually to the point where they could build their own prison while having no idea they were doing so. We have now reached that stage where technological intellect has the potential to destroy us and that's why events are moving so fast. Central American shaman Don Juan Matus said:

Think for a moment, and tell me how you would explain the contradictions between the intelligence of man the engineer and the stupidity of his systems of belief, or the stupidity of his contradictory behaviour. Sorcerers believe that the predators have given us our systems of beliefs, our ideas of good and evil; our social mores. They are the ones who set up our dreams of success or failure. They have given us covetousness, greed, and cowardice. It is the predator who makes us complacent, routinary, and egomaniacal.

In order to keep us obedient and meek and weak, the predators engaged themselves in a stupendous manoeuvre – stupendous, of course, from the point of view of a fighting strategist; a horrendous manoeuvre from the point of those who suffer it. They gave us their mind. The predators' mind is baroque, contradictory, morose, filled with the fear of being discovered any minute now.

For 'predators' see Wetiko, Archons, Yaldabaoth, Jinn, and all the other versions of the same phenomenon in cultures and religions all over the world. The theme is always the same because it's true and it's real. We have reached the point where we have to deal with it. The question is – how?

Don't fight – walk away

I thought I'd use a controversial subheading to get things moving in terms of our response to global fascism. What do you mean 'don't fight'? What do you mean 'walk away'? We've got to fight. We can't walk away. Well, it depends what we mean by fight and walk away. If fighting means physical combat we are playing Wetiko's game and falling for its trap. It wants us to get angry, aggressive, and direct hate and hostility at the enemy we think we must fight. Every war, every battle, every conflict, has been fought with Wetiko leading both sides. It's what it does. Wetiko wants a fight, anywhere, any place. Just hit me, son, so I can hit you back. Wetiko hits Wetiko and Wetiko hits Wetiko in return. I am very forthright as you can see in exposing Wetikos of the Cult, but I don't hate them. I refuse to hate them. It's what they want. What you hate you become. What you *fight* you become. Wokers, 'anti-haters' and 'anti-fascists' prove this every time they reach for their keyboards or don their balaclavas. By walk away I mean to disengage from Wetiko which includes ceasing to cooperate with its tyranny. Paul Levy says of Wetiko:

The way to 'defeat' evil is not to try to destroy it (for then, in playing evil's game, we have already lost), but rather, to find the invulnerable place within ourselves where evil is unable to vanquish us – this is to truly 'win' our battle with evil.

Wetiko is everywhere in human society and it's been on steroids since the 'Covid' hoax. Every shouting match over wearing masks has Wetiko wearing a mask and Wetiko not wearing one. It's an electrical circuit of push and resist, push and resist, with Wetiko pushing *and* resisting. Each polarity is Wetiko empowering itself. Dictionary definitions of 'resist' include 'opposing, refusing to accept or comply with' and the word to focus on is 'opposing'. What form does this take – setting police cars alight or 'refusing to accept or comply with'? The former is Wetiko opposing Wetiko while the other points the way forward. This is the difference between those aggressively demanding that government fascism must be obeyed who stand in stark contrast to the great majority of Pushbackers. We saw this clearly with a march by thousands of Pushbackers against lockdown in London followed days later by a Woker-hijacked

protest in Bristol in which police cars were set on fire. Masks were virtually absent in London and widespread in Bristol. Wetiko wants lockdown on every level of society and infuses its aggression to police it through its unknowing stooges. Lockdown protesters are the ones with the smiling faces and the hugs, The two blatantly obvious states of being – getting more obvious by the day – are the result of Wokers and their like becoming ever more influenced by the simulation Field of Wetiko and Pushbackers ever more influenced by The Field of a far higher vibration beyond the simulation. Wetiko can't invade the heart which is where most lockdown opponents are coming from. It's the heart that allows them to see through the lies to the truth in ways I will be highlighting.

Renegade Minds know that calmness is the place from which wisdom comes. You won't find wisdom in a hissing fit and wisdom is what we need in abundance right now. Calmness is not weakness – you don't have to scream at the top of your voice to be strong. Calmness is indeed a sign of strength. 'No' means I'm not doing it. NOOOO!!! doesn't mean you're not doing it even more. Volume does not advance 'No – I'm not doing it'. You are just not doing it. Wetiko possessed and influenced don't know how to deal with that. Wetiko wants a fight and we should not give it one. What it needs more than anything is our *cooperation* and we should not give that either. Mass rallies and marches are great in that they are a visual representation of feeling, but if it ends there they are irrelevant. You demand that Wetikos act differently? Well, they're not going to are they? They are Wetikos. We don't need to waste our time demanding that something doesn't happen when that will make no difference. We need to delete the means that *allows* it to happen. This, invariably, is our cooperation. You can demand a child stop firing a peashooter at the dog or you can refuse to buy the peashooter. If you provide the means you are cooperating with the dog being smacked on the nose with a pea. How can the authorities enforce mask-wearing if millions in a country refuse? What if the 74 million Pushbackers that voted for Trump in 2020 refused to wear masks, close their businesses or stay in their homes. It would be unenforceable. The

few control the many through the compliance of the many and that's always been the dynamic be it 'Covid' regulations or the Roman Empire. I know people can find it intimidating to say no to authority or stand out in a crowd for being the only one with a face on display; but it has to be done or it's over. I hope I've made clear in this book that where this is going will be far more intimidating than standing up now and saying 'No' – I will not cooperate with my own enslavement and that of my children. There might be consequences for some initially, although not so if enough do the same. The question that must be addressed is what is going to happen if we don't? It is time to be strong and unyieldingly so. No means no. Not here and there, but *everywhere* and *always*. I have refused to wear a mask and obey all the other nonsense. I will not comply with tyranny. I repeat: Fascism is not imposed by fascists – there are never enough of them. Fascism is imposed by the population acquiescing to fascism. *I will not do it*. I will die first, or my body will. Living meekly under fascism is a form of death anyway, the death of the spirit that Martin Luther King described.

Making things happen

We must not despair. This is not over till it's over and it's far from that. The 'fat lady' must refuse to sing. The longer the 'Covid' hoax has dragged on and impacted on more lives we have seen an awakening of phenomenal numbers of people worldwide to the realisation that what they have believed all their lives is not how the world really is. Research published by the system-serving University of Bristol and King's College London in February, 2021, concluded: 'One in every 11 people in Britain say they trust David Icke's take on the coronavirus pandemic.' It will be more by now and we have gathering numbers to build on. We must urgently progress from seeing the scam to ceasing to cooperate with it. Prominent German lawyer Reiner Fuellmich, also licenced to practice law in America, is doing a magnificent job taking the legal route to bring the psychopaths to justice through a second Nuremberg tribunal for crimes against humanity. Fuellmich has an impressive record of

beating the elite in court and he formed the German Corona Investigative Committee to pursue civil charges against the main perpetrators with a view to triggering criminal charges. Most importantly he has grasped the foundation of the hoax – the PCR test not testing for the ‘virus’ – and Christian Drosten is therefore on his charge sheet along with Gates frontman Tedros at the World Health Organization. Major players must not be allowed to inflict their horrors on the human race without being brought to book. A life sentence must follow for Bill Gates and the rest of them. A group of researchers has also indicted the government of Norway for crimes against humanity with copies sent to the police and the International Criminal Court. The lawsuit cites participation in an internationally-planned false pandemic and violation of international law and human rights, the European Commission’s definition of human rights by coercive rules, Nuremberg and Hague rules on fundamental human rights, and the Norwegian constitution. We must take the initiative from hereon and not just complain, protest and react.

There are practical ways to support vital mass non-cooperation. Organising in numbers is one. Lockdown marches in London in the spring in 2021 were mass non-cooperation that the authorities could not stop. There were too many people. Hundreds of thousands walked the London streets in the centre of the road for mile after mile while the Face-Nappies could only look on. They were determined, but calm, and just *did it* with no histrionics and lots of smiles. The police were impotent. Others are organising group shopping without masks for mutual support and imagine if that was happening all over. Policing it would be impossible. If the store refuses to serve people in these circumstances they would be faced with a long line of trolleys full of goods standing on their own and everything would have to be returned to the shelves. How would they cope with that if it kept happening? I am talking here about moving on from complaining to being pro-active; from watching things happen to making things happen. I include in this our relationship with the police. The behaviour of many Face-Nappies

has been disgraceful and anyone who thinks they would never find concentration camp guards in the 'enlightened' modern era have had that myth busted big-time. The period and setting may change – Wetikos never do. I watched film footage from a London march in which a police thug viciously kicked a protestor on the floor who had done nothing. His fellow Face-Nappies stood in a ring protecting him. What he did was a criminal assault and with a crowd far outnumbering the police this can no longer be allowed to happen unchallenged. I get it when people chant 'shame on you' in these circumstances, but that is no longer enough. They *have* no shame those who do this. Crowds needs to start making a citizen's arrest of the police who commit criminal offences and brutally attack innocent people and defenceless women. A citizen's arrest can be made under section 24A of the UK Police and Criminal Evidence (PACE) Act of 1984 and you will find something similar in other countries. I prefer to call it a Common Law arrest rather than citizen's for reasons I will come to shortly. Anyone can arrest a person committing an indictable offence or if they have reasonable grounds to suspect they are committing an indictable offence. On both counts the attack by the police thug would have fallen into this category. A citizen's arrest can be made to stop someone:

- Causing physical injury to himself or any other person
- Suffering physical injury
- Causing loss of or damage to property
- Making off before a constable can assume responsibility for him

A citizen's arrest may also be made to prevent a breach of the peace under Common Law and if they believe a breach of the peace will happen or anything related to harm likely to be done or already done in their presence. This is the way to go I think – the Common Law version. If police know that the crowd and members of the public will no longer be standing and watching while they commit

their thuggery and crimes they will think twice about acting like Brownshirts and Blackshirts.

Common Law – common sense

Mention of Common Law is very important. Most people think the law is the law as in one law. This is not the case. There are two bodies of law, Common Law and Statute Law, and they are not the same. Common Law is founded on the simple premise of do no harm. It does not recognise victimless crimes in which no harm is done while Statute Law does. There is a Statute Law against almost everything. So what is Statute Law? Amazingly it's the law of the *sea* that was brought ashore by the Cult to override the law of the land which is Common Law. They had no right to do this and as always they did it anyway. They had to. They could not impose their will on the people through Common Law which only applies to do no harm. How could you stitch up the fine detail of people's lives with that? Instead they took the law of the sea, or Admiralty Law, and applied it to the population. Statute Law refers to all the laws spewing out of governments and their agencies including all the fascist laws and regulations relating to 'Covid'. The key point to make is that Statute Law is *contract law*. It only applies between *contracting* corporations. Most police officers don't even know this. They have to be kept in the dark, too. Long ago when merchants and their sailing ships began to trade with different countries a contractual law was developed called Admiralty Law and other names. Again it only applied to *contracts* agreed between *corporate* entities. If there is no agreed contract the law of the sea had no jurisdiction *and that still applies to its new alias of Statute Law*. The problem for the Cult when the law of the sea was brought ashore was an obvious one. People were not corporations and neither were government entities. To overcome the latter they made governments and all associated organisations corporations. All the institutions are *private corporations* and I mean governments and their agencies, local councils, police, courts, military, US states, the whole lot. Go to the

Dun and Bradstreet corporate listings website for confirmation that they are all corporations. You are arrested by a private corporation called the police by someone who is really a private security guard and they take you to court which is another private corporation. Neither have jurisdiction over you unless you consent and *contract* with them. This is why you hear the mantra about law enforcement policing by *consent* of the people. In truth the people 'consent' only in theory through monumental trickery.

Okay, the Cult overcame the corporate law problem by making governments and institutions corporate entities; but what about people? They are not corporations are they? Ah ... well in a sense, and *only* a sense, they are. Not people exactly – the illusion of people. The Cult creates a corporation in the name of everyone at the time that their birth certificate is issued. Note birth/ *berth* certificate and when you go to court under the law of the sea on land you stand in a *dock*. These are throwbacks to the origin. My Common Law name is David Vaughan Icke. The name of the corporation created by the government when I was born is called Mr David Vaughan Icke usually written in capitals as MR DAVID VAUGHAN ICKE. That is not me, the living, breathing man. It is a fictitious corporate entity. The trick is to make you think that David Vaughan Icke and MR DAVID VAUGHAN ICKE are the same thing. *They are not*. When police charge you and take you to court they are prosecuting the corporate entity and not the living, breathing, man or woman. They have to trick you into identifying as the corporate entity and contracting with them. Otherwise they have no jurisdiction. They do this through a language known as legalese. Lawful and legal are not the same either. Lawful relates to Common Law and legal relates to Statute Law. Legalese is the language of Statue Law which uses terms that mean one thing to the public and another in legalese. Notice that when a police officer tells someone why they are being charged he or she will say at the end: 'Do you understand?' To the public that means 'Do you comprehend?' In legalese it means 'Do you stand under me?' Do you stand under my authority? If you say

yes to the question you are unknowingly agreeing to give them jurisdiction over you in a contract between two corporate entities.

This is a confidence trick in every way. Contracts have to be agreed between informed parties and if you don't know that David Vaughan Icke is agreeing to be the corporation MR DAVID VAUGHAN ICKE you cannot knowingly agree to contract. They are deceiving you and another way they do this is to ask for proof of identity. You usually show them a driving licence or other document on which your corporate name is written. In doing so you are accepting that you are that corporate entity when you are not. Referring to yourself as a 'person' or 'citizen' is also identifying with your corporate fiction which is why I made the Common Law point about the citizen's arrest. If you are approached by a police officer you identify yourself immediately as a living, breathing, man or woman and say 'I do not consent, I do not contract with you and I do not understand' or stand under their authority. I have a Common Law birth certificate as a living man and these are available at no charge from commonlawcourt.com. Businesses registered under the Statute Law system means that its laws apply. There are, however, ways to run a business under Common Law. Remember all 'Covid' laws and regulations are Statute Law – the law of *contracts* and you do not have to contract. This doesn't mean that you can kill someone and get away with it. Common Law says do no harm and that applies to physical harm, financial harm etc. Police are employees of private corporations and there needs to be a new system of non-corporate Common Law constables operating outside the Statute Law system. If you go to davidicke.com and put Common Law into the search engine you will find videos that explain Common Law in much greater detail. It is definitely a road we should walk.

With all my heart

I have heard people say that we are in a spiritual war. I don't like the term 'war' with its Wetiko dynamic, but I know what they mean. Sweep aside all the bodily forms and we are in a situation in which two states of consciousness are seeking very different realities.

Wetiko wants upheaval, chaos, fear, suffering, conflict and control. The other wants love, peace, harmony, fairness and freedom. That's where we are. We should not fall for the idea that Wetiko is all-powerful and there's nothing we can do. Wetiko is not all-powerful. It's a joke, pathetic. It doesn't have to be, but it has made that choice for now. A handful of times over the years when I have felt the presence of its frequency I have allowed it to attach briefly so I could consciously observe its nature. The experience is not pleasant, the energy is heavy and dark, but the ease with which you can kick it back out the door shows that its real power is in persuading us that it has power. It's all a con. Wetiko is a con. It's a trickster and not a power that can control us if we unleash our own. The con is founded on manipulating humanity to give its power to Wetiko which recycles it back to present the illusion that it has power when its power is *ours* that we gave away. This happens on an energetic level and plays out in the world of the seen as humanity giving its power to Wetiko authority which uses that power to control the population when the power is only the power the population has handed over. How could it be any other way for billions to be controlled by a relative few? I have had experiences with people possessed by Wetiko and again you can kick its arse if you do it with an open heart. Oh yes – the *heart* which can transform the world of perceived 'matter'.

We are receiver-transmitters and processors of information, but what information and where from? Information is processed into perception in three main areas – the brain, the heart and the belly. These relate to thinking, knowing, and emotion. Wetiko wants us to be head and belly people which means we think within the confines of the Matrix simulation and low-vibrational emotional reaction scrambles balance and perception. A few minutes on social media and you see how emotion is the dominant force. Woke is all emotion and is therefore thought-free and fact-free. Our heart is something different. It *knows* while the head *thinks* and has to try to work it out because it doesn't know. The human energy field has seven prime vortexes which connect us with wider reality ([Fig 23](#)). Chakra means

‘wheels of light’ in the Sanskrit language of ancient India. The main ones are: The crown chakra on top of the head; brow (or ‘third eye’) chakra in the centre of the forehead; throat chakra; heart chakra in the centre of the chest; solar plexus chakra below the sternum; sacral chakra beneath the navel; and base chakra at the bottom of the spine. Each one has a particular function or functions. We feel anxiety and nervousness in the belly where the sacral chakra is located and this processes emotion that can affect the colon to give people ‘the shits’ or make them ‘shit scared’ when they are nervous. Chakras all play an important role, but the Mr and Mrs Big is the heart chakra which sits at the centre of the seven, above the chakras that connect us to the ‘physical’ and below those that connect with higher realms (or at least should). Here in the heart chakra we feel love, empathy and compassion – ‘My heart goes out to you’. Those with closed hearts become literally ‘heart-less’ in their attitudes and behaviour (see Bill Gates). Native Americans portrayed Wetiko with what Paul Levy calls a ‘frigid, icy heart, devoid of mercy’ (see Bill Gates).



Figure 23: The chakra system which interpenetrates the human energy field. The heart chakra is the governor – or should be.

Wetiko trembles at the thought of heart energy which it cannot infiltrate. The frequency is too high. What it seeks to do instead is close the heart chakra vortex to block its perceptual and energetic influence. Psychopaths have ‘hearts of stone’ and emotionally-damaged people have ‘heartache’ and ‘broken hearts’. The astonishing amount of heart disease is related to heart chakra

disruption with its fundamental connection to the 'physical' heart. Dr Tom Cowan has written an outstanding book challenging the belief that the heart is a pump and making the connection between the 'physical' and spiritual heart. Rudolph Steiner who was way ahead of his time said the same about the fallacy that the heart is a pump. *What?* The heart is not a pump? That's crazy, right? Everybody knows that. Read Cowan's *Human Heart, Cosmic Heart* and you will realise that the very idea of the heart as a pump is ridiculous when you see the evidence. How does blood in the feet so far from the heart get pumped horizontally up the body by the heart?? Cowan explains in the book the real reason why blood moves as it does. Our 'physical' heart is used to symbolise love when the source is really the heart vortex or spiritual heart which is our most powerful energetic connection to 'out there' expanded consciousness. That's why we feel *knowing* – intuitive knowing – in the centre of the chest. Knowing doesn't come from a process of thoughts leading to a conclusion. It is there in an instant all in one go. Our heart knows because of its connection to levels of awareness that *do* know. This is the meaning and source of intuition – intuitive *knowing*.

For the last more than 30 years of uncovering the global game and the nature of reality my heart has been my constant antenna for truth and accuracy. An American intelligence insider once said that I had quoted a disinformant in one of my books and yet I had only quoted the part that was true. He asked: 'How do you do that?' By using my heart antenna was the answer and anyone can do it. Heart-centred is how we are meant to be. With a closed heart chakra we withdraw into a closed mind and the bubble of five-sense reality. If you take a moment to focus your attention on the centre of your chest, picture a spinning wheel of light and see it opening and expanding. You will feel it happening, too, and perceptions of the heart like joy and love as the heart impacts on the mind as they interact. The more the chakra opens the more you will feel expressions of heart consciousness and as the process continues, and becomes part of you, insights and knowings will follow. An open

heart is connected to that level of awareness that knows all is *One*. You will see from its perspective that the fault-lines that divide us are only illusions to control us. An open heart does not process the illusions of race, creed and sexuality except as brief experiences for a consciousness that is all. Our heart does not see division, only unity (Figs 24 and 25). There's something else, too. Our hearts love to laugh. Mark Twain's quote that says 'The human race has one really effective weapon, and that is laughter' is really a reference to the heart which loves to laugh with the joy of knowing the true nature of infinite reality and that all the madness of human society is an illusion of the mind. Twain also said: 'Against the assault of laughter nothing can stand.' This is so true of Wetiko and the Cult. Their insecurity demands that they be taken seriously and their power and authority acknowledged and feared. We should do nothing of the sort. We should not get aggressive or fearful which their insecurity so desires. We should laugh in their face. Even in their no-face as police come over in their face-nappies and expect to be taken seriously. They don't take themselves seriously looking like that so why should we? Laugh in the face of intimidation. Laugh in the face of tyranny. You will see by its reaction that you have pressed all of its buttons. Wetiko does not know what to do in the face of laughter or when its targets refuse to concede their joy to fear. We have seen many examples during the 'Covid' hoax when people have expressed their energetic power and the string puppets of Wetiko retreat with their tail limp between their knees. Laugh – the world is bloody mad after all and if it's a choice between laughter and tears I know which way I'm going.



Figure 24: Head consciousness without the heart sees division and everything apart from everything else.



Figure 25: Heart consciousness sees everything as One.

'Vaccines' and the soul

The foundation of Wetiko/Archon control of humans is the separation of incarnate five-sense mind from the infinite 'I' and closing the heart chakra where the True 'I' lives during a human life. The goal has been to achieve complete separation in both cases. I was interested therefore to read an account by a French energetic healer of what she said she experienced with a patient who had been given the 'Covid' vaccine. Genuine energy healers can sense information and consciousness fields at different levels of being which are referred to as 'subtle bodies'. She described treating the patient who later returned after having, without the healer's knowledge, two doses of the 'Covid vaccine'. The healer said:

I noticed immediately the change, very heavy energy emanating from [the] subtle bodies. The scariest thing was when I was working on the heart chakra, I connected with her soul: it was detached from the physical body, it had no contact and it was, as if it was floating in a state of total confusion: a damage to the consciousness that loses contact with the physical body, i.e. with our biological machine, there is no longer any communication between them.

I continued the treatment by sending light to the heart chakra, the soul of the person, but it seemed that the soul could no longer receive any light, frequency or energy. It was a very powerful experience for me. Then I understood that this substance is indeed used to detach consciousness so that this consciousness can no longer interact through this body that it possesses in life, where there is no longer any contact, no frequency, no light, no more energetic balance or mind.

This would create a human that is rudderless and at the extreme almost zombie-like operating with a fractional state of consciousness at the mercy of Wetiko. I was especially intrigued by what the healer said in the light of the prediction by the highly-informed Rudolf Steiner more than a hundred years ago. He said:

In the future, we will eliminate the soul with medicine. Under the pretext of a 'healthy point of view', there will be a vaccine by which the human body will be treated as soon as possible directly at birth, so that the human being cannot develop the thought of the existence of soul and Spirit. To materialistic doctors will be entrusted the task of removing the soul of humanity.

As today, people are vaccinated against this disease or that disease, so in the future, children will be vaccinated with a substance that can be produced precisely in such a way that people, thanks to this vaccination, will be immune to being subjected to the 'madness' of spiritual life. He would be extremely smart, but he would not develop a conscience, and that is the true goal of some materialistic circles.

Steiner said the vaccine would detach the physical body from the etheric body (subtle bodies) and 'once the etheric body is detached the relationship between the universe and the etheric body would become extremely unstable, and man would become an automaton'. He said 'the physical body of man must be polished on this Earth by spiritual will – so the vaccine becomes a kind of arymanique (Wetiko) force' and 'man can no longer get rid of a given materialistic feeling'. Humans would then, he said, become 'materialistic of constitution and can no longer rise to the spiritual'. I have been writing for years about DNA being a receiver-transmitter of information that connects us to other levels of reality and these 'vaccines' changing DNA can be likened to changing an antenna and what it can transmit and receive. Such a disconnection would clearly lead to changes in personality and perception. Steiner further predicted the arrival of AI. Big Pharma 'Covid vaccine' makers, expressions of Wetiko, are testing their DNA-manipulating evil on children as I write with a view to giving the 'vaccine' to babies. If it's a soul-body disconnecter – and I say that it is or can be – every child would be disconnected from 'soul' at birth and the 'vaccine' would create a closed system in which spiritual guidance from the greater self would play no part. This has been the ambition of Wetiko all

along. A Pentagon video from 2005 was leaked of a presentation explaining the development of vaccines to change behaviour by their effect on the brain. Those that believe this is not happening with the 'Covid' genetically-modifying procedure masquerading as a 'vaccine' should make an urgent appointment with Naivety Anonymous. Klaus Schwab wrote in 2018:

Neurotechnologies enable us to better influence consciousness and thought and to understand many activities of the brain. They include decoding what we are thinking in fine levels of detail through new chemicals and interventions that can influence our brains to correct for errors or enhance functionality.

The plan is clear and only the heart can stop it. With every heart that opens, every mind that awakens, Wetiko is weakened. Heart and love are far more powerful than head and hate and so nothing like a majority is needed to turn this around.

Beyond the Phantom

Our heart is the prime target of Wetiko and so it must be the answer to Wetiko. We *are* our heart which is part of one heart, the infinite heart. Our heart is where the true self lives in a human life behind firewalls of five-sense illusion when an imposter takes its place – *Phantom Self*; but our heart waits patiently to be set free any time we choose to see beyond the Phantom, beyond Wetiko. A Wetikoed Phantom Self can wreak mass death and destruction while the love of forever is locked away in its heart. The time is here to unleash its power and let it sweep away the fear and despair that is Wetiko. Heart consciousness does not seek manipulated, censored, advantage for its belief or religion, its activism and desires. As an expression of the One it treats all as One with the same rights to freedom and opinion. Our heart demands fairness for itself no more than for others. From this unity of heart we can come together in mutual support and transform this Wetikoed world into what reality is meant to be – a place of love, joy, happiness, fairness, justice and freedom. Wetiko has another agenda and that's why the world is as

it is, but enough of this nonsense. Wetiko can't stay where hearts are open and it works so hard to keep them closed. Fear is its currency and its food source and love in its true sense has no fear. Why would love have fear when it knows it is *All That Is, Has Been, And Ever Can Be* on an eternal exploration of all possibility? Love in this true sense is not the physical attraction that passes for love. This can be an expression of it, yes, but Infinite Love, a love without condition, goes far deeper to the core of all being. It is the core of all being. Infinite reality was born from love beyond the illusions of the simulation. Love infinitely expressed is the knowing that all is One and the swiftly-passing experience of separation is a temporary hallucination. You cannot disconnect from Oneness; you can only *perceive* that you have and withdraw from its influence. This is the most important of all perception trickery by the mind parasite that is Wetiko and the foundation of all its potential for manipulation.

If we open our hearts, open the sluice gates of the mind, and redefine self-identity amazing things start to happen. Consciousness expands or contracts in accordance with self-identity. When true self is recognised as infinite awareness and label self – Phantom Self – is seen as only a series of brief experiences life is transformed. Consciousness expands to the extent that self-identity expands and everything changes. You see unity, not division, the picture, not the pixels. From this we can play the long game. No more is an experience something in and of itself, but a fleeting moment in the eternity of forever. Suddenly people in uniform and dark suits are no longer intimidating. Doing what your heart knows to be right is no longer intimidating and consequences for those actions take on the same nature of a brief experience that passes in the blink of an infinite eye. Intimidation is all in the mind. Beyond the mind there is no intimidation.

An open heart does not consider consequences for what it knows to be right. To do so would be to consider not doing what it knows to be right and for a heart in its power that is never an option. The Renegade Mind is really the Renegade Heart. Consideration of consequences will always provide a getaway car for the mind and

the heart doesn't want one. What is right in the light of what we face today is to stop cooperating with Wetiko in all its forms and to do it without fear or compromise. You cannot compromise with tyranny when tyranny always demands more until it has everything. Life is your perception and you are your destiny. Change your perception and you change your life. Change collective perception and we change the world.

Come on people ... One human family, One heart, One goal ...
FREEEEEEEDOM!

We must settle for nothing less.

Postscript

The big scare story as the book goes to press is the 'Indian' variant and the world is being deluged with propaganda about the 'Covid catastrophe' in India which mirrors in its lies and misrepresentations what happened in Italy before the first lockdown in 2020.

The *New York Post* published a picture of someone who had 'collapsed in the street from Covid' in India in April, 2021, which was actually taken during a gas leak in May, 2020. Same old, same old. Media articles in mid-February were asking why India had been so untouched by 'Covid' and then as their vaccine rollout gathered pace the alleged 'cases' began to rapidly increase. Indian 'Covid vaccine' maker Bharat Biotech was funded into existence by the Bill and Melinda Gates Foundation (the pair announced their divorce in May, 2021, which is a pity because they so deserve each other). The Indian 'Covid crisis' was ramped up by the media to terrify the world and prepare people for submission to still more restrictions. The scam that worked the first time was being repeated only with far more people seeing through the deceit. Davidicke.com and Ickonic.com have sought to tell the true story of what is happening by talking to people living through the Indian nightmare which has nothing to do with 'Covid'. We posted a letter from 'Alisha' in Pune who told a very different story to government and media mendacity. She said scenes of dying people and overwhelmed hospitals were designed to hide what was really happening – genocide and starvation. Alisha said that millions had already died of starvation during the ongoing lockdowns while government and media were lying and making it look like the 'virus':

Restaurants, shops, gyms, theatres, basically everything is shut. The cities are ghost towns. Even so-called 'essential' businesses are only open till 11am in the morning. You basically have just an hour to buy food and then your time is up.

Inter-state travel and even inter-district travel is banned. The cops wait at all major crossroads to question why you are traveling outdoors or to fine you if you are not wearing a mask.

The medical community here is also complicit in genocide, lying about hospitals being full and turning away people with genuine illnesses, who need immediate care. They have even created a shortage of oxygen cylinders.

This is the classic Cult modus operandi played out in every country. Alisha said that people who would not have a PCR test not testing for the 'virus' were being denied hospital treatment. She said the people hit hardest were migrant workers and those in rural areas. Most businesses employed migrant workers and with everything closed there were no jobs, no income and no food. As a result millions were dying of starvation or malnutrition. All this was happening under Prime Minister Narendra Modi, a 100-percent asset of the Cult, and it emphasises yet again the scale of pure anti-human evil we are dealing with. Australia banned its people from returning home from India with penalties for trying to do so of up to five years in jail and a fine of £37,000. The manufactured 'Covid' crisis in India was being prepared to justify further fascism in the West. Obvious connections could be seen between the Indian 'vaccine' programme and increased 'cases' and this became a common theme. The Seychelles, the most per capita 'Covid vaccinated' population in the world, went back into lockdown after a 'surge of cases'.

Long ago the truly evil Monsanto agricultural biotechnology corporation with its big connections to Bill Gates devastated Indian farming with genetically-modified crops. Human rights activist Gurcharan Singh highlighted the efforts by the Indian government to complete the job by destroying the food supply to hundreds of millions with 'Covid' lockdowns. He said that 415 million people at the bottom of the disgusting caste system (still going whatever they say) were below the poverty line and struggled to feed themselves every year. Now the government was imposing lockdown at just the

time to destroy the harvest. This deliberate policy was leading to mass starvation. People may reel back at the suggestion that a government would do that, but Wetiko-controlled 'leaders' are capable of any level of evil. In fact what is described in India is in the process of being instigated worldwide. The food chain and food supply are being targeted at every level to cause world hunger and thus control. Bill Gates is not the biggest owner of farmland in America for no reason and destroying access to food aids both the depopulation agenda and the plan for synthetic 'food' already being funded into existence by Gates. Add to this the coming hyper-inflation from the suicidal creation of fake 'money' in response to 'Covid' and the breakdown of container shipping systems and you have a cocktail that can only lead one way and is meant to. The Cult plan is to crash the entire system to 'build back better' with the Great Reset.

'Vaccine' transmission

Reports from all over the world continue to emerge of women suffering menstrual and fertility problems after having the fake 'vaccine' and of the non-'vaccinated' having similar problems when interacting with the 'vaccinated'. There are far too many for 'coincidence' to be credible. We've had menopausal women getting periods, others having periods stop or not stopping for weeks, passing clots, sometimes the lining of the uterus, breast irregularities, and miscarriages (which increased by 400 percent in parts of the United States). Non-'vaccinated' men and children have suffered blood clots and nose bleeding after interaction with the 'vaccinated'. Babies have died from the effects of breast milk from a 'vaccinated' mother. Awake doctors – the small minority – speculated on the cause of non-'vaccinated' suffering the same effects as the 'vaccinated'. Was it nanotechnology in the synthetic substance transmitting frequencies or was it a straight chemical bioweapon that was being transmitted between people? I am not saying that some kind of chemical transmission is not one possible answer, but the foundation of all that the Cult does is frequency and

this is fertile ground for understanding how transmission can happen. American doctor Carrie Madej, an internal medicine physician and osteopath, has been practicing for the last 20 years, teaching medical students, and she says attending different meetings where the agenda for humanity was discussed. Madej, who operates out of Georgia, did not dismiss other possible forms of transmission, but she focused on frequency in search of an explanation for transmission. She said the Moderna and Pfizer 'vaccines' contained nano-lipid particles as a key component. This was a brand new technology never before used on humanity. 'They're using a nanotechnology which is pretty much little tiny computer bits ... nanobots or hydrogel.' Inside the 'vaccines' was 'this sci-fi kind of substance' which suppressed immune checkpoints to get into the cell. I referred to this earlier as the 'Trojan horse' technique that tricks the cell into opening a gateway for the self-replicating synthetic material and while the immune system is artificially suppressed the body has no defences. Madej said the substance served many purposes including an on-demand ability to 'deliver the payload' and using the nano 'computer bits' as biosensors in the body. 'It actually has the ability to accumulate data from your body, like your breathing, your respiration, thoughts, emotions, all kinds of things.'

She said the technology obviously has the ability to operate through Wi-Fi and transmit and receive energy, messages, frequencies or impulses. 'Just imagine you're getting this new substance in you and it can react to things all around you, the 5G, your smart device, your phones.' We had something completely foreign in the human body that had never been launched large scale at a time when we were seeing 5G going into schools and hospitals (plus the Musk satellites) and she believed the 'vaccine' transmission had something to do with this: '... if these people have this inside of them ... it can act like an antenna and actually transmit it outwardly as well.' The synthetic substance produced its own voltage and so it could have that kind of effect. This fits with my own contention that the nano receiver-transmitters are designed to connect people to the

Smart Grid and break the receiver-transmitter connection to expanded consciousness. That would explain the French energy healer's experience of the disconnection of body from 'soul' with those who have had the 'vaccine'. The nanobots, self-replicating inside the body, would also transmit the synthetic frequency which could be picked up through close interaction by those who have not been 'vaccinated'. Madej speculated that perhaps it was 5G and increased levels of other radiation that was causing the symptoms directly although interestingly she said that non-'vaccinated' patients had shown improvement when they were away from the 'vaccinated' person they had interacted with. It must be remembered that you can control frequency and energy with your mind and you can consciously create energetic barriers or bubbles with the mind to stop damaging frequencies from penetrating your field. American paediatrician Dr Larry Palevsky said the 'vaccine' was not a 'vaccine' and was never designed to protect from a 'viral' infection. He called it 'a massive, brilliant propaganda of genocide' because they didn't have to inject everyone to get the result they wanted. He said the content of the jabs was able to infuse any material into the brain, heart, lungs, kidneys, liver, sperm and female productive system. 'This is genocide; this is a weapon of mass destruction.' At the same time American colleges were banning students from attending if they didn't have this life-changing and potentially life-ending 'vaccine'. Class action lawsuits must follow when the consequences of this college fascism come to light. As the book was going to press came reports about fertility effects on sperm in 'vaccinated' men which would absolutely fit with what I have been saying and hospitals continued to fill with 'vaccine' reactions. Another question is what about transmission via blood transfusions? The NHS has extended blood donation restrictions from seven days after a 'Covid vaccination' to 28 days after even a sore arm reaction.

I said in the spring of 2020 that the then touted 'Covid vaccine' would be ongoing each year like the flu jab. A year later Pfizer CEO, the appalling Albert Bourla, said people would 'likely' need a 'booster dose' of the 'vaccine' within 12 months of getting 'fully

vaccinated' and then a yearly shot. 'Variants will play a key role', he said confirming the point. Johnson & Johnson CEO Alex Gorsky also took time out from his 'vaccine' disaster to say that people may need to be vaccinated against 'Covid-19' each year. UK Health Secretary, the psychopath Matt Hancock, said additional 'boosters' would be available in the autumn of 2021. This is the trap of the 'vaccine passport'. The public will have to accept every last 'vaccine' they introduce, including for the fake 'variants', or it would cease to be valid. The only other way in some cases would be continuous testing with a test not testing for the 'virus' and what is on the swabs constantly pushed up your nose towards the brain every time?

'Vaccines' changing behaviour

I mentioned in the body of the book how I believed we would see gathering behaviour changes in the 'vaccinated' and I am already hearing such comments from the non-'vaccinated' describing behaviour changes in friends, loved ones and work colleagues. This will only increase as the self-replicating synthetic material and nanoparticles expand in body and brain. An article in the *Guardian* in 2016 detailed research at the University of Virginia in Charlottesville which developed a new method for controlling brain circuits associated with complex animal behaviour. The method, dubbed 'magnetogenetics', involves genetically-engineering a protein called ferritin, which stores and releases iron, to create a magnetised substance – 'Magneto' – that can activate specific groups of nerve cells from a distance. This is claimed to be an advance on other methods of brain activity manipulation known as optogenetics and chemogenetics (the Cult has been developing methods of brain control for a long time). The ferritin technique is said to be non-invasive and able to activate neurons 'rapidly and reversibly'. In other words, human thought and perception. The article said that earlier studies revealed how nerve cell proteins 'activated by heat and mechanical pressure can be genetically engineered so that they become sensitive to radio waves and magnetic fields, by attaching them to an iron-storing protein called ferritin, or to inorganic

paramagnetic particles'. Sensitive to radio waves and magnetic fields? You mean like 5G, 6G and 7G? This is the human-AI Smart Grid hive mind we are talking about. The *Guardian* article said:

... the researchers injected Magneto into the striatum of freely behaving mice, a deep brain structure containing dopamine-producing neurons that are involved in reward and motivation, and then placed the animals into an apparatus split into magnetised and non-magnetised sections.

Mice expressing Magneto spent far more time in the magnetised areas than mice that did not, because activation of the protein caused the striatal neurons expressing it to release dopamine, so that the mice found being in those areas rewarding. This shows that Magneto can remotely control the firing of neurons deep within the brain, and also control complex behaviours.

Make no mistake this basic methodology will be part of the 'Covid vaccine' cocktail and using magnetics to change brain function through electromagnetic field frequency activation. The Pentagon is developing a 'Covid vaccine' using ferritin. Magnetism would explain changes in behaviour and why videos are appearing across the Internet as I write showing how magnets stick to the skin at the point of the 'vaccine' shot. Once people take these 'vaccines' anything becomes possible in terms of brain function and illness which will be blamed on 'Covid-19' and 'variants'. Magnetic field manipulation would further explain why the non-'vaccinated' are reporting the same symptoms as the 'vaccinated' they interact with and why those symptoms are reported to decrease when not in their company. Interestingly 'Magneto', a 'mutant', is a character in the Marvel Comic *X-Men* stories with the ability to manipulate magnetic fields and he believes that mutants should fight back against their human oppressors by any means necessary. The character was born Erik Lehnsherr to a Jewish family in Germany.

Cult-controlled courts

The European Court of Human Rights opened the door for mandatory 'Covid-19 vaccines' across the continent when it ruled in a Czech Republic dispute over childhood immunisation that legally

enforced vaccination could be 'necessary in a democratic society'. The 17 judges decided that compulsory vaccinations did not breach human rights law. On the face of it the judgement was so inverted you gasp for air. If not having a vaccine infused into your body is not a human right then what is? Ah, but they said human rights law which has been specifically written to delete all human rights at the behest of the state (the Cult). Article 8 of the European Convention on Human Rights relates to the right to a private life. The crucial word here is '*except*':

There shall be no interference by a public authority with the exercise of this right EXCEPT such as is in accordance with the law and is necessary in a democratic society in the interests of national security, public safety or the economic wellbeing of the country, for the prevention of disorder or crime, for the protection of health or morals, or for the protection of the rights and freedoms of others [My emphasis].

No interference *except* in accordance with the law means there *are* no 'human rights' *except* what EU governments decide you can have at their behest. 'As is necessary in a democratic society' explains that reference in the judgement and 'in the interests of national security, public safety or the economic well-being of the country, for the prevention of disorder or crime, for the protection of health or morals, or for the protection of the rights and freedoms of others' gives the EU a coach and horses to ride through 'human rights' and scatter them in all directions. The judiciary is not a check and balance on government extremism; it is a vehicle to enforce it. This judgement was almost laughably predictable when the last thing the Cult wanted was a decision that went against mandatory vaccination. Judges rule over and over again to benefit the system of which they are a part. Vaccination disputes that come before them are invariably delivered in favour of doctors and authorities representing the view of the state which owns the judiciary. Oh, yes, and we have even had calls to stop putting 'Covid-19' on death certificates within 28 days of a 'positive test' because it is claimed the practice makes the 'vaccine' appear not to work. They are laughing at you.

The scale of madness, inhumanity and things to come was highlighted when those not 'vaccinated' for 'Covid' were refused evacuation from the Caribbean island of St Vincent during massive volcanic eruptions. Cruise ships taking residents to the safety of another island allowed only the 'vaccinated' to board and the rest were left to their fate. Even in life and death situations like this we see 'Covid' stripping people of their most basic human instincts and the insanity is even more extreme when you think that fake 'vaccine'-makers are not even claiming their body-manipulating concoctions stop 'infection' and 'transmission' of a 'virus' that doesn't exist. St Vincent Prime Minister Ralph Gonsalves said: 'The chief medical officer will be identifying the persons already vaccinated so that we can get them on the ship.' Note again the power of the chief medical officer who, like Whitty in the UK, will be answering to the World Health Organization. This is the Cult network structure that has overridden politicians who 'follow the science' which means doing what WHO-controlled 'medical officers' and 'science advisers' tell them. Gonsalves even said that residents who were 'vaccinated' after the order so they could board the ships would still be refused entry due to possible side effects such as 'wooziness in the head'. The good news is that if they were woozy enough in the head they could qualify to be prime minister of St Vincent.

Microchipping freedom

The European judgement will be used at some point to justify moves to enforce the 'Covid' DNA-manipulating procedure. Sandra Ro, CEO of the Global Blockchain Business Council, told a World Economic Forum event that she hoped 'vaccine passports' would help to 'drive forced consent and standardisation' of global digital identity schemes: 'I'm hoping with the desire and global demand for some sort of vaccine passport – so that people can get travelling and working again – [it] will drive forced consent, standardisation, and frankly, cooperation across the world.' The lady is either not very bright, or thoroughly mendacious, to use the term 'forced consent'.

You do not 'consent' if you are forced – you *submit*. She was describing what the plan has been all along and that's to enforce a digital identity on every human without which they could not function. 'Vaccine passports' are opening the door and are far from the end goal. A digital identity would allow you to be tracked in everything you do in cyberspace and this is the same technique used by Cult-owned China to enforce its social credit system of total control. The ultimate 'passport' is planned to be a microchip as my books have warned for nearly 30 years. Those nice people at the Pentagon working for the Cult-controlled Defense Advanced Research Projects Agency (DARPA) claimed in April, 2021, they have developed a microchip inserted under the skin to detect 'asymptomatic Covid-19 infection' before it becomes an outbreak and a 'revolutionary filter' that can remove the 'virus' from the blood when attached to a dialysis machine. The only problems with this are that the 'virus' does not exist and people transmitting the 'virus' with no symptoms is brain-numbing bullshit. This is, of course, not a ruse to get people to be microchipped for very different reasons. DARPA also said it was producing a one-stop 'vaccine' for the 'virus' and all 'variants'. One of the most sinister organisations on Planet Earth is doing this? Better have it then. These people are insane because Wetiko that possesses them is insane.

Researchers from the Salk Institute in California announced they have created an embryo that is part human and part monkey. My books going back to the 1990s have exposed experiments in top secret underground facilities in the United States where humans are being crossed with animal and non-human 'extraterrestrial' species. They are now easing that long-developed capability into the public arena and there is much more to come given we are dealing with psychiatric basket cases. Talking of which – Elon Musk's scientists at Neuralink trained a monkey to play Pong and other puzzles on a computer screen using a joystick and when the monkey made the correct move a metal tube squirted banana smoothie into his mouth which is the basic technique for training humans into unquestioning compliance. Two Neuralink chips were in the monkey's skull and

more than 2,000 wires 'fanned out' into its brain. Eventually the monkey played a video game purely with its brain waves. Psychopathic narcissist Musk said the 'breakthrough' was a step towards putting Neuralink chips into human skulls and merging minds with artificial intelligence. *Exactly*. This man is so dark and Cult to his DNA.

World Economic Fascism (WEF)

The World Economic Forum is telling you the plan by the statements made at its many and various events. Cult-owned fascist YouTube CEO Susan Wojcicki spoke at the 2021 WEF Global Technology Governance Summit (see the name) in which 40 governments and 150 companies met to ensure 'the responsible design and deployment of emerging technologies'. Orwellian translation: 'Ensuring the design and deployment of long-planned technologies will advance the Cult agenda for control and censorship.' Freedom-destroyer and Nuremberg-bound Wojcicki expressed support for tech platforms like hers to censor content that is 'technically legal but could be harmful'. Who decides what is 'harmful'? She does and they do. 'Harmful' will be whatever the Cult doesn't want people to see and we have legislation proposed by the UK government that would censor content on the basis of 'harm' no matter if the information is fair, legal and provably true. Make that *especially* if it is fair, legal and provably true. Wojcicki called for a global coalition to be formed to enforce content moderation standards through automated censorship. This is a woman and mega-censor so self-deluded that she shamelessly accepted a 'free expression' award – *Wojcicki* – in an event sponsored by her own *YouTube*. They have no shame and no self-awareness.

You know that 'Covid' is a scam and Wojcicki a Cult operative when YouTube is censoring medical and scientific opinion purely on the grounds of whether it supports or opposes the Cult 'Covid' narrative. Florida governor Ron DeSantis compiled an expert panel with four professors of medicine from Harvard, Oxford, and Stanford Universities who spoke against forcing children and

vaccinated people to wear masks. They also said there was no proof that lockdowns reduced spread or death rates of 'Covid-19'. Cult-gofer Wojcicki and her YouTube deleted the panel video 'because it included content that contradicts the consensus of local and global health authorities regarding the efficacy of masks to prevent the spread of Covid-19'. This 'consensus' refers to what the Cult tells the World Health Organization to say and the WHO tells 'local health authorities' to do. Wojcicki knows this, of course. The panellists pointed out that censorship of scientific debate was responsible for deaths from many causes, but Wojcicki couldn't care less. She would not dare go against what she is told and as a disgrace to humanity she wouldn't want to anyway. The UK government is seeking to pass a fascist 'Online Safety Bill' to specifically target with massive fines and other means non-censored video and social media platforms to make them censor 'lawful but harmful' content like the Cult-owned Facebook, Twitter, Google and YouTube. What is 'lawful but harmful' would be decided by the fascist Blair-created Ofcom.

Another WEF obsession is a cyber-attack on the financial system and this is clearly what the Cult has planned to take down the bank accounts of everyone – except theirs. Those that think they have enough money for the Cult agenda not to matter to them have got a big lesson coming if they continue to ignore what is staring them in the face. The World Economic Forum, funded by Gates and fronted by Klaus Schwab, announced it would be running a 'simulation' with the Russian government and global banks of just such an attack called Cyber Polygon 2021. What they simulate – as with the 'Covid' Event 201 – they plan to instigate. The WEF is involved in a project with the Cult-owned Carnegie Endowment for International Peace called the WEF-Carnegie Cyber Policy Initiative which seeks to merge Wall Street banks, 'regulators' (I love it) and intelligence agencies to 'prevent' (arrange and allow) a cyber-attack that would bring down the global financial system as long planned by those that control the WEF and the Carnegie operation. The Carnegie Endowment for International Peace sent an instruction to First World

War US President Woodrow Wilson not to let the war end before society had been irreversibly transformed.

The Wuhan lab diversion

As I close, the Cult-controlled authorities and lapdog media are systematically pushing 'the virus was released from the Wuhan lab' narrative. There are two versions – it happened by accident and it happened on purpose. Both are nonsense. The perceived existence of the never-shown-to-exist 'virus' is vital to sell the impression that there is actually an infective agent to deal with and to allow the endless potential for terrifying the population with 'variants' of a 'virus' that does not exist. The authorities at the time of writing are going with the 'by accident' while the alternative media is promoting the 'on purpose'. Cable news host Tucker Carlson who has questioned aspects of lockdown and 'vaccine' compulsion has bought the Wuhan lab story. 'Everyone now agrees' he said. Well, I don't and many others don't and the question is *why* does the system and its media suddenly 'agree'? When the media moves as one unit with a narrative it is always a lie – witness the hour by hour mendacity of the 'Covid' era. Why would this Cult-owned combination which has unleashed lies like machine gun fire suddenly 'agree' to tell the truth??

Much of the alternative media is buying the lie because it fits the conspiracy narrative, but it's the *wrong* conspiracy. The real conspiracy is that *there is no virus* and that is what the Cult is desperate to hide. The idea that the 'virus' was released by accident is ludicrous when the whole 'Covid' hoax was clearly long-planned and waiting to be played out as it was so fast in accordance with the Rockefeller document and Event 201. So they prepared everything in detail over decades and then sat around strumming their fingers waiting for an 'accidental' release from a bio-lab? *What??* It's crazy. Then there's the 'on purpose' claim. You want to circulate a 'deadly virus' and hide the fact that you've done so and you release it down the street from the highest-level bio-lab in China? I repeat – *What??*

You would release it far from that lab to stop any association being made. But, no, we'll do it in a place where the connection was certain to be made. Why would you need to scam 'cases' and 'deaths' and pay hospitals to diagnose 'Covid-19' if you had a real 'virus'? What are sections of the alternative media doing believing this crap? Where were all the mass deaths in Wuhan from a 'deadly pathogen' when the recovery to normal life after the initial propaganda was dramatic in speed? Why isn't the 'deadly pathogen' now circulating all over China with bodies in the street? Once again we have the technique of tell them what they want to hear and they will likely believe it. The alternative media has its 'conspiracy' and with Carlson it fits with his 'China is the danger' narrative over years. China *is* a danger as a global Cult operations centre, but not for this reason. The Wuhan lab story also has the potential to instigate conflict with China when at some stage the plan is to trigger a Problem-Reaction-Solution confrontation with the West. Question everything – *everything* – and especially when the media agrees on a common party line.

Third wave ... fourth wave ... fifth wave ...

As the book went into production the world was being set up for more lockdowns and a 'third wave' supported by invented 'variants' that were increasing all the time and will continue to do so in public statements and computer programs, but not in reality. India became the new Italy in the 'Covid' propaganda campaign and we were told to be frightened of the new 'Indian strain'. Somehow I couldn't find it within myself to do so. A document produced for the UK government entitled 'Summary of further modelling of easing of restrictions – Roadmap Step 2' declared that a third wave was inevitable (of course when it's in the script) and it would be the fault of children and those who refuse the health-destroying fake 'Covid vaccine'. One of the computer models involved came from the Cult-owned *Imperial College* and the other from Warwick University which I wouldn't trust to tell me the date in a calendar factory. The document states that both models presumed extremely high uptake

of the 'Covid vaccines' and didn't allow for 'variants'. The document states: 'The resurgence is a result of some people (mostly children) being ineligible for vaccination; others choosing not to receive the vaccine; and others being vaccinated but not perfectly protected.' The mendacity takes the breath away. Okay, blame those with a brain who won't take the DNA-modifying shots and put more pressure on children to have it as 'trials' were underway involving children as young as six months with parents who give insanity a bad name. Massive pressure is being put on the young to have the fake 'vaccine' and child age consent limits have been systematically lowered around the world to stop parents intervening. Most extraordinary about the document was its claim that the 'third wave' would be driven by 'the resurgence in both hospitalisations and deaths ... dominated by *those that have received two doses of the vaccine*, comprising around 60-70% of the wave respectively'. The predicted peak of the 'third wave' suggested 300 deaths per day with 250 of them *fully 'vaccinated' people*. How many more lies do acquiescers need to be told before they see the obvious? Those who took the jab to 'protect themselves' are projected to be those who mostly get sick and die? So what's in the 'vaccine'? The document went on:

It is possible that a summer of low prevalence could be followed by substantial increases in incidence over the following autumn and winter. Low prevalence in late summer should not be taken as an indication that SARS-CoV-2 has retreated or that the population has high enough levels of immunity to prevent another wave.

They are telling you the script and while many British people believed 'Covid' restrictions would end in the summer of 2021 the government was preparing for them to be ongoing. Authorities were awarding contracts for 'Covid marshals' to police the restrictions with contracts starting in July, 2021, and going through to January 31st, 2022, and the government was advertising for 'Media Buying Services' to secure media propaganda slots worth a potential £320 million for 'Covid-19 campaigns' with a contract not ending until March, 2022. The recipient – via a list of other front companies – was reported to be American media marketing giant Omnicom Group

Inc. While money is no object for 'Covid' the UK waiting list for all other treatment – including life-threatening conditions – passed 4.5 million. Meantime the Cult is seeking to control all official 'inquiries' to block revelations about what has really been happening and why. It must not be allowed to – we need Nuremberg jury trials in every country. The cover-up doesn't get more obvious than appointing ultra-Zionist professor Philip Zelikow to oversee two dozen US virologists, public health officials, clinicians, former government officials and four American 'charitable foundations' to 'learn the lessons' of the 'Covid' debacle. The personnel will be those that created and perpetuated the 'Covid' lies while Zelikow is the former executive director of the 9/11 Commission who ensured that the truth about those attacks never came out and produced a report that must be among the most mendacious and manipulative documents ever written – see *The Trigger* for the detailed exposure of the almost unimaginable 9/11 story in which Sabbatians can be found at every level.

Passive no more

People are increasingly challenging the authorities with amazing numbers of people taking to the streets in London well beyond the ability of the Face-Nappies to stop them. Instead the Nappies choose situations away from the mass crowds to target, intimidate, and seek to promote the impression of 'violent protestors'. One such incident happened in London's Hyde Park. Hundreds of thousands walking through the streets in protest against 'Covid' fascism were ignored by the Cult-owned BBC and most of the rest of the mainstream media, but they delighted in reporting how police were injured in 'clashes with protestors'. The truth was that a group of people gathered in Hyde Park at the end of one march when most had gone home and they were peacefully having a good time with music and chat. Face-Nappies who couldn't deal with the full-march crowd then waded in with their batons and got more than they bargained for. Instead of just standing for this criminal brutality the crowd used their numerical superiority to push the Face-Nappies out of the

park. Eventually the Nappies turned and ran. Unfortunately two or three idiots in the crowd threw drink cans striking two officers which gave the media and the government the image they wanted to discredit the 99.9999 percent who were peaceful. The idiots walked straight into the trap and we must always be aware of potential agent provocateurs used by the authorities to discredit their targets.

This response from the crowd – the can people apart – must be a turning point when the public no longer stand by while the innocent are arrested and brutally attacked by the Face-Nappies. That doesn't mean to be violent, that's the last thing we need. We'll leave the violence to the Face-Nappies and government. But it does mean that when the Face-Nappies use violence against peaceful people the numerical superiority is employed to stop them and make citizen's arrests or Common Law arrests for a breach of the peace. The time for being passive in the face of fascism is over.

We are the many, they are the few, and we need to make that count before there is no freedom left and our children and grandchildren face an ongoing fascist nightmare.

COME ON PEOPLE – IT'S TIME.

One final thought ...

The power of love
A force from above
Cleaning my soul
Flame on burn desire
Love with tongues of fire
Purge the soul
Make love your goal

I'll protect you from the hooded claw
Keep the vampires from your door
When the chips are down I'll be around
With my undying, death-defying
Love for you

Envy will hurt itself
Let yourself be beautiful
Sparkling love, flowers
And pearls and pretty girls
Love is like an energy
Rushin' rushin' inside of me

This time we go sublime
Lovers entwine, divine, divine,
Love is danger, love is pleasure
Love is pure – the only treasure

I'm so in love with you
Purge the soul
Make love your goal

The power of love
A force from above
Cleaning my soul
The power of love
A force from above
A sky-scraping dove

Flame on burn desire
Love with tongues of fire
Purge the soul
Make love your goal

Frankie Goes To Hollywood

APPENDIX

Cowan-Kaufman-Morell Statement on Virus Isolation (SOVI)

Isolation: The action of isolating; the fact or condition of being isolated or standing alone; separation from other things or persons; solitariness

Oxford English Dictionary

The controversy over whether the SARS-CoV-2 virus has ever been isolated or purified continues. However, using the above definition, common sense, the laws of logic and the dictates of science, any unbiased person must come to the conclusion that the SARS-CoV-2 virus has never been isolated or purified. As a result, no confirmation of the virus' existence can be found. The logical, common sense, and scientific consequences of this fact are:

- the structure and composition of something not shown to exist can't be known, including the presence, structure, and function of any hypothetical spike or other proteins;
- the genetic sequence of something that has never been found can't be known;
- "variants" of something that hasn't been shown to exist can't be known;
- it's impossible to demonstrate that SARS-CoV-2 causes a disease called Covid-19.

In as concise terms as possible, here's the proper way to isolate, characterize and demonstrate a new virus. First, one takes samples (blood, sputum, secretions) from many people (e.g. 500) with symptoms which are unique and specific enough to characterize an illness. Without mixing these samples with ANY tissue or products that also contain genetic material, the virologist macerates, filters and ultracentrifuges i.e. *purifies* the specimen. This common virology technique, done for decades to isolate bacteriophages¹ and so-called giant viruses in every virology lab, then allows the virologist to demonstrate with electron microscopy thousands of identically sized and shaped particles. These particles are the isolated and purified virus.

These identical particles are then checked for uniformity by physical and/or microscopic techniques. Once the purity is determined, the particles may be further characterized. This would include examining the structure, morphology, and chemical composition of the particles. Next, their genetic makeup is characterized by extracting the genetic material directly from the purified particles and using genetic-sequencing techniques, such as Sanger sequencing, that have also been around for decades. Then one does an analysis to confirm that these uniform particles are exogenous (outside) in origin as a virus is conceptualized to be, and not the normal breakdown products of dead and dying tissues.² (As of May 2020, we know that virologists have no way to determine whether the particles they're seeing are viruses or just normal breakdown products of dead and dying tissues.)³

1 Isolation, characterization and analysis of bacteriophages from the haloalkaline lake Elmenteita, Kenya Julia Khayeli Akhwale et al, PLOS One, Published: April 25, 2019.
<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0215734> – accessed 2/15/21

2 "Extracellular Vesicles Derived From Apoptotic Cells: An Essential Link Between Death and Regeneration," Maojiao Li et al, Frontiers in Cell and Developmental Biology, 2020 October 2.
<https://www.frontiersin.org/articles/10.3389/fcell.2020.573511/full> – accessed 2/15/21

If we have come this far then we have fully isolated, characterized, and genetically sequenced an exogenous virus particle. However, we still have to show it is causally related to a disease. This is carried out by exposing a group of healthy subjects (animals are usually used) to this isolated, purified virus in the manner in which the disease is thought to be transmitted. If the animals get sick with the same disease, as confirmed by clinical and autopsy findings, one has now shown that the virus actually causes a disease. This demonstrates infectivity and transmission of an infectious agent.

None of these steps has even been attempted with the SARS-CoV-2 virus, nor have all these steps been successfully performed for any so-called pathogenic virus. Our research indicates that a single study showing these steps does not exist in the medical literature.

Instead, since 1954, virologists have taken unpurified samples from a relatively few people, often less than ten, with a similar disease. They then minimally process this sample and inoculate this unpurified sample onto tissue culture containing usually four to six other types of material – all of which contain identical genetic material as to what is called a “virus.” The tissue culture is starved and poisoned and naturally disintegrates into many types of particles, some of which contain genetic material. Against all common sense, logic, use of the English language and scientific integrity, this process is called “virus isolation.” This brew containing fragments of genetic material from many sources is then subjected to genetic analysis, which then creates in a computer-simulation process the alleged sequence of the alleged virus, a so called in silico genome. At no time is an actual virus confirmed by electron microscopy. At no time is a genome extracted and sequenced from an actual virus. This is scientific fraud.

The observation that the unpurified specimen — inoculated onto tissue culture along with toxic antibiotics, bovine fetal tissue, amniotic fluid and other tissues — destroys the kidney tissue onto which it is inoculated is given as evidence of the virus' existence and pathogenicity. This is scientific fraud.

From now on, when anyone gives you a paper that suggests the SARS-CoV-2 virus has been isolated, please check the methods sections. If the researchers used Vero cells or any other culture method, you know that their process was not isolation. You will hear the following excuses for why actual isolation isn't done:

1. There were not enough virus particles found in samples from patients to analyze.
2. Viruses are intracellular parasites; they can't be found outside the cell in this manner.

If No. 1 is correct, and we can't find the virus in the sputum of sick people, then on what evidence do we think the virus is dangerous or even lethal? If No. 2 is correct, then how is the virus spread from person to person? We are told it emerges from the cell to infect others. Then why isn't it possible to find it?

Finally, questioning these virology techniques and conclusions is not some distraction or divisive issue. Shining the light on this truth is essential to stop this terrible fraud that humanity is confronting. For, as we now know, if the virus has never been isolated, sequenced or shown to cause illness, if the virus is imaginary, then why are we wearing masks, social distancing and putting the whole world into prison?

Finally, if pathogenic viruses don't exist, then what is going into those injectable devices erroneously called "vaccines," and what is their purpose? This scientific question is the most urgent and relevant one of our time.

We are correct. The SARS-CoV2 virus does not exist.

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Ickonic is something that has been a dream of mine for the last 5 years, growing up around alternative information I have always had a natural interest in what is going on in the World and what could I do to make it better. Across the range of subjects and positions of influence occupied mainly by people who don't strive to make things better it's the Media that I have always found the most frustrating and fascinating. Mainly because if the Media did their Jobs properly then so much of the negative things happening in the World simply would not be able to happen, because they would be exposed within a heartbeat.

Free Press and the Opportunities that the internet could have given would mean that the Media are able to expose things like never before and hold people to account for their actions. As we all know there are 'Untouchables' that walk among us, people the Media simply won't touch, expose or investigate and that leads to the dark underworlds that infest the establishment the World over. Well I say enough, it's time for something different, a different kind of Media, where no one is off limits from exposing and investigating. All we're interested in at Ickonic is the truth of what is really going on in the World on whichever subject we're covering.

We hope you enjoy what we have created and take something away from the platform, we aim to deliver information that's informative and most importantly self-empowering, you're not a little person, you're part of something much bigger than that and its time we as a collective race began to understand that and look to the future as ours to take.

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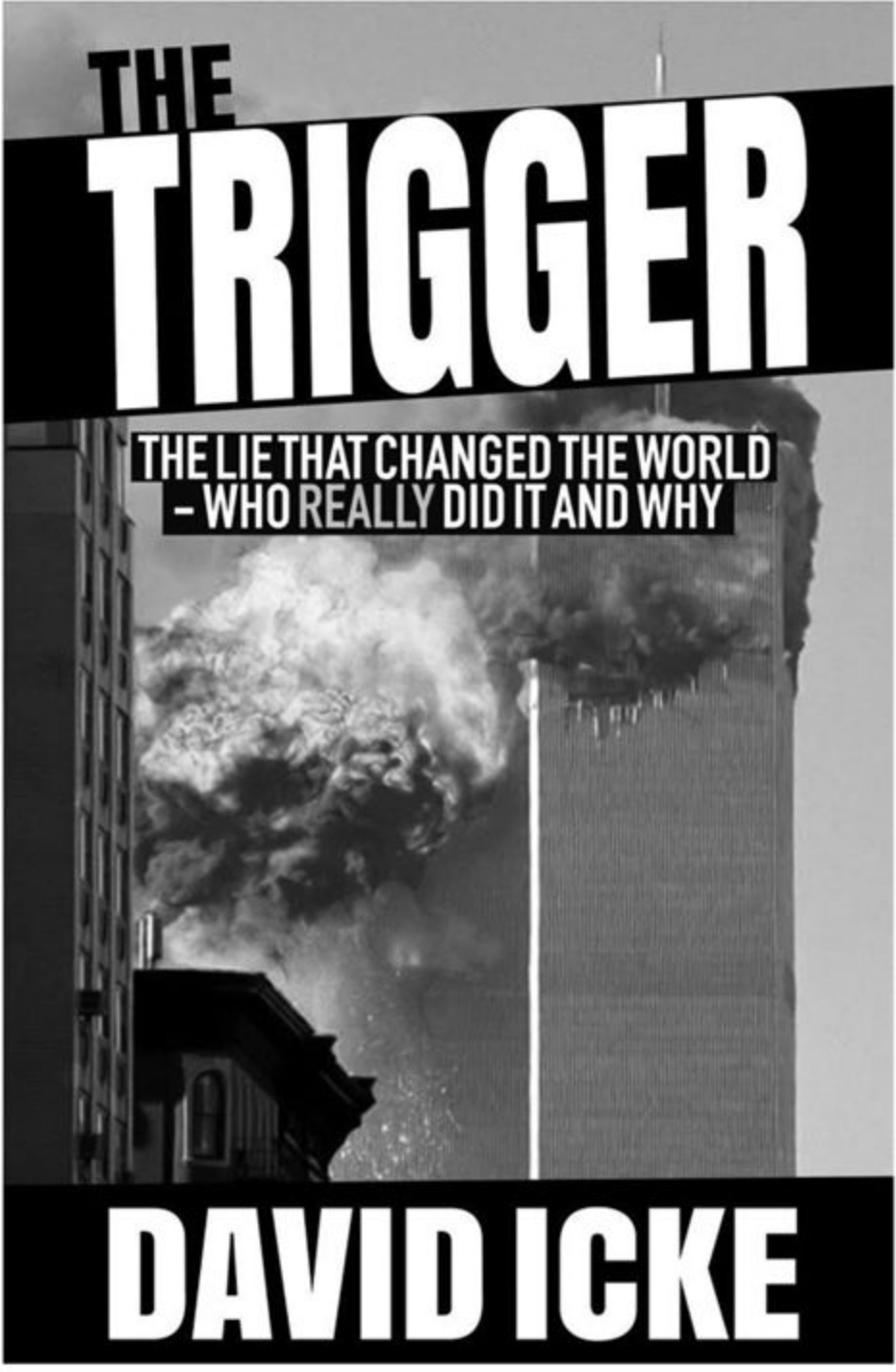
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noun

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You can subscribe to the fantastic new Ickonic media platform where there are many hundreds of hours of cutting-edge information in videos, documentaries and series across a whole range of subjects which are added to every week. This includes my 90 minute breakdown of the week's news every Friday to explain *why* events are happening and to what end.